

FIITJEE Computer Based All India Test Series for JEE-Main & JEE Advanced 2023
with Live Online Concept Strengthening Classroom Program (CSCP)
Schedule

S.No.	TEST / Live Online CSCP	Test Dates & Timings
1	PART TEST - I (Main)	November 12, 2022 (3:00 PM to 6:00 PM)
	Live Online CSCP by FIITJEE Delhi	November 15, 2022 (10:30 AM to 5:30 PM)
2	PART TEST - I (Advanced)	November 20, 2022 (Paper 1- 9:00 AM to 12:00 PM Paper 2- 2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	November 22, 2022 (10:30 AM to 5:30 PM)
3	PART TEST - II (Main)	December 3, 2022 (3:00 PM to 6:00 PM)
	Live Online CSCP by FIITJEE Delhi	December 6, 2022 (12:45 PM to 7:45 PM)
4	PART TEST - II (Advanced)	December 11, 2022 (Paper 1-9:00 AM to 12:00 PM Paper 2-2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	December 13, 2022 (12:45 PM to 7:45 PM)
5	PART TEST - III (Main)	December 17, 2022 (3:00 PM to 6:00 PM)
	Live Online CSCP by FIITJEE Delhi	December 20, 2022 (12:45 PM to 7:45 PM)
6	PART TEST - III (Advanced)	December 25, 2022 (Paper 1-9:00 AM to 12:00 PM Paper 2-2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	December 27, 2022 (12:45 PM to 7:45 PM)
7*, 8*	FULL TEST - I (Main) & Full Test - I (Advanced)	December 28, 2022 (Tests will be live for 7 days at fiitjee.glorifire.com)
9	FULL TEST - II (Main)	January 7, 2023 (3:00 PM to 6:00 PM)
	Live Online CSCP by FIITJEE Delhi	January 10, 2023 (12:45 PM to 7:45 PM)
10	FULL TEST - III (Main)	January 11, 2023 (3:00 PM to 6:00 PM)
	Live Online CSCP by FIITJEE Delhi	January 13, 2023 (12:45 PM to 7:45 PM)
11	OPEN TEST (Main)	January 15, 2023 (9:00 AM to 12:00 PM)
	Live Online CSCP by FIITJEE Delhi	January 17, 2023, 12:45 PM onwards
12*	FULL TEST V (Main)	January 18, 2023 (Tests will be live for 7 days at fiitjee.glorifire.com)
13	FULL TEST - VI (Main)	January 19, 2023 (9:00 AM to 12:00 PM)
14*, 15*	CONCEPT RECAPITULATION TEST – I, II (Main)	January 20, 2023 (Tests will be live for 7 days at fiitjee.glorifire.com)
16	OPEN TEST I (Advanced)	February 5, 2023 (Paper 1-9:00 AM to 12:00 PM Paper 2-2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	February 7, 2023 (12:45 PM to 7:45 PM)
17*, 18*, 19*	FULL TEST III, IV & V (Advanced)	February 8, 2023 (Tests will be live for 7 days at fiitjee.glorifire.com)
20	FULL TEST - VII (Main)	March 22, 2023 (9:00 AM to 12:00 PM)
	Live Online CSCP by FIITJEE Delhi	March 24, 2023 (12:45 PM to 7:45 PM)
21	FULL TEST - VIII (Main)	March 25, 2023 (3:00 PM to 6:00 PM)
	Live Online CSCP by FIITJEE Delhi	March 26, 2023 (12:45 PM to 7:45 PM)
22*, 23*	CONCEPT RECAPITULATION TEST PAPERS – SET III, IV (Main)	March 27, 2023 (Test will be live for 7 days at fiitjee.glorifire.com)
24	FULL TEST - IX (Main)	March 28, 2023 (9:00 AM to 12:00 PM)
	Live Online CSCP by FIITJEE Delhi	March 29, 2023 (12:45 PM to 7:45 PM)
25	FULL TEST - X (Main)	March 31, 2023 (3:00 PM to 6:00 PM)
	Live Online CSCP by FIITJEE Delhi	April 1, 2023 (12:45 PM to 7:45 PM)
26	FULL TEST - XI (Main)	April 2, 2023 (9:00 AM to 12:00 PM)
27	OPEN TEST II (Advanced)	April 16, 2023 (Paper 1-9:00 AM to 12:00 PM Paper 2-2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	April 18, 2023 (10:30 AM to 5:30 PM)
28	FULL TEST - VII (Advanced)	April 23, 2023 (Paper 1-9:00 AM to 12:00 PM Paper 2-2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	April 25, 2023 (10:30 AM to 5:30 PM)
29*, 30*, 31*, 32*	CONCEPT RECAPITULATION TEST PAPERS - SET I, II, III & IV (Advanced)	April 26, 2023 (Test will be live for 7 days at fiitjee.glorifire.com)
33	FULL TEST - VIII (Advanced)	April 30, 2023 (Paper 1-9:00 AM to 12:00 PM Paper 2-2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	May 2, 2023 (10:30 AM to 5:30 PM)
34	FULL TEST - IX (Advanced)	May 7, 2023 (Paper 1-9:00 AM to 12:00 PM Paper 2-2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	May 9, 2023 (10:30 AM to 5:30 PM)
35	FULL TEST - X (Advanced)	May 14, 2023 (Paper 1-9:00 AM to 12:00 PM Paper 2-2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	May 16, 2023 (10:30 AM to 5:30 PM)
36	FULL TEST - XI (Advanced)	May 21, 2023 (Paper 1-9:00 AM to 12:00 PM Paper 2-2:30 PM to 5:30 PM)
	Live Online CSCP by FIITJEE Delhi	May 23, 2023 (10:30 AM to 5:30 PM)

- Tests can be taken from CBT Centers located at cities namely Ahmedabad, Allahabad, Amritsar, Bangalore, Bhilai, Bhopal, Bhubaneswar, Bokaro, Chandigarh, Chennai, Cuttack, Delhi (Dwarka), Delhi (East), Delhi (North-West), Delhi (South), Delhi NCR (Faridabad), Delhi NCR (Ghaziabad), Delhi NCR (Gurugram), Delhi NCR (Noida), Dhanbad, Durgapur, Gorakhpur, Gwalior, Hyderabad, Indore, Jaipur, Jamshedpur, Jodhpur, Kanpur, Kharagpur, Kolkata, Kota, Lucknow, Meerut, Mumbai, Nagpur, Patna, Pune, Raipur, Ranchi, Vadodara, Varanasi, Vijayawada, Visakhapatnam.
- Tests can also be taken from home on one's own computer.
- * marked Tests can be taken only at <https://fiitjee.glorifire.com>. Tests will be live for 7 days from the given date.
- Link for Live Online CSCP will be sent by WhatsApp to your registered mobile number one/ two days before the session.

FIITJEE Computer Based All India Test Series for JEE Main, 2023

Syllabus

Part Test - I

CHEMISTRY	Atomic Structure, Chemical Kinetics, Chemical Equilibrium, Balancing of Redox Reaction, Ionic Equilibrium, Chemical Bonding, s-Block and Hydrogen, Boron and Carbon Family, Periodic Properties
MATHEMATICS	Sets, Relations and Functions, Limits, Continuity and Differentiability, Application of Derivatives, Indefinite Integration, Definite Integrals and their Properties, Area, Differential Equations, Volume & Surface
PHYSICS	Units and Measurement, Kinematics (Motion in One Dimension), Vectors, Kinematics (Motion in Two and Three Dimensions), Relative motion, Laws of Motion, Work, Energy and Power, Center of mass, Conservation of momentum Collisions, Circular motion, Rotational Motion

Part Test -II

CHEMISTRY	Organic Chemistry (Including GOC), Practical Organic Chemistry, Biomolecules, Polymers
MATHEMATICS	Trigonometric Ratios and Identities, Trigonometric Equations, Heights and distance, Solution of triangle, Straight Lines, Circles, Parabola, Ellipse, Hyperbola
PHYSICS	Kinetic theory of gases, Heat and Thermodynamics (second law+reversible and irreversible process, carnot engine+thermal expansion + calorimetry), Transfer of Heat + convection, Electrostatics, Current Electricity (color code of resistors), Thermal Effects of Current, Magnetic Effects of Current, Pure magnetism (current loop as magnetic dipole and its moment), bar magnet, magnetic field lines, earth magnetism, para-dia-ferro), Magnets, susceptibility and permeability, hysteresis, electromagnets and permanent magnets, Magnetostatics, Electromagnetics Induction and Alternating Currents (ac generator and transformer, wattless current, quality factor)

Part Test - III

CHEMISTRY	Group V to VIII, Transition Elements & Coordination Compounds, f-Block elements, Ores and Metallurgy, Liquid Solution, Surface Chemistry, Electrochemistry, Thermodynamics & Thermochemistry, Stoichiometry, Gaseous State, Qualitative Analysis, Solid State, Environmental Chemistry, Chemistry in Everyday Life
MATHEMATICS	Progression and Series, Quadratic Equations and Expressions, Complex Numbers, Binomial Theorem, Matrices, Determinants, Permutation and Combination, Statistics (Measures of Dispersion) and Probability, Probability Distribution, Binomial Distribution, Vector, Three Dimensional Geometry
PHYSICS	Oscillations (forced and damped oscillations), Wave & Sound, Ray Optics (microscope and astronomical telescope), Wave Optics (diffraction (single slit), resolving power of microscope and astronomical telescope, polarisation, brewsters law, uses of polaroids and polarised light), Electromagnetic Waves, Dual Nature of Matter & Radiation, Atoms, Molecules and Nuclei, Electronic Devices & Communication Systems, Experimental Skills in Physics, Gravitation, Properties of Solids and Fluids + streamline flow, turbulent flow, reynolds number, Geometrical Optics, Semi Conductor Devices

Note : Full Tests syllabi will be same as that of Part Test - I, II & III combined.

FIITJEE Computer Based All India Test Series for JEE Advanced, 2023

Syllabus

Part Test - I

CHEMISTRY	Atomic Structure, Chemical Kinetics, Chemical Equilibrium, Balancing of Redox Reaction, Ionic Equilibrium, Chemical Bonding, s-Block and Hydrogen, Boron and Carbon Family
MATHEMATICS	Functions, Limits, Continuity and Differentiability, Application of Derivatives, Indefinite Integration, Definite Integrals and their properties, Area, Differential Equations
PHYSICS	Kinematics, Laws of Motion, Work, Energy and Power, Conservation of Momentum, Rotation

Part Test - II

CHEMISTRY	Organic Chemistry (Including GOC), Practical Organic Chemistry, Biomolecules, Polymers
MATHEMATICS	Trigonometric ratios and Identities, Trigonometric equations, Solution of triangles, Straight Lines, Circles, Parabola, Ellipse, Hyperbola
PHYSICS	Heat and Thermodynamics, Electrostatics, Current Electricity, Magnetism, Electromagnetic Induction, A.C. Circuit

Part Test - III

CHEMISTRY	Group V to VIII, Transition Elements & Coordination Compounds, Ores and Metallurgy, Liquid Solution, Electrochemistry, Thermodynamics & Thermochemistry, Stoichiometry, Gaseous State, Qualitative Analysis, Solid State
MATHEMATICS	Progression and Series, Quadratic Equations and Expressions, Complex Numbers, Binomial Theorem, Matrices, Determinants, Permutation and Combination, Probability, Vector, Three Dimensional Geometry
PHYSICS	Gravitation, Fluids, Simple Harmonic Motion, Waves & Sound, Optics, Modern Physics, Error Analysis

Note : Full Tests syllabi will be same as that of Part Test - I, II & III combined.

FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Main)-2023

TEST DATE: 12-11-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

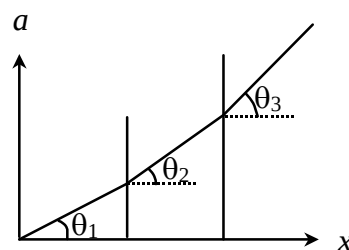
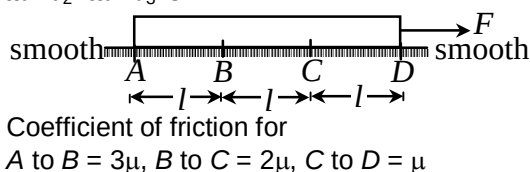
PART – A

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. In the diagram, the variation of coefficient of friction of the surface is shown. The external force F applied on board is equal to $2\mu Mg$. Now the board is gently pulled so it starts moving. The variation of acceleration of board with respect to displacement has been shown. The ratio $\tan \theta_1 : \tan \theta_2 : \tan \theta_3$ is



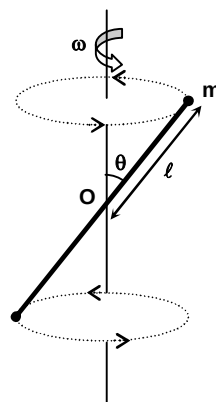
- (A) 3 : 2 : 1
 (B) 3 : 4 : 1
 (C) 3 : 4 : 5
 (D) 1 : 2 : 3

2. In terms of the basic units of mass M , length L , time T and charge Q , the dimensions of magnetic permeability of vacuum μ_0 are

- (A) $[MLQ^{-2}]$
 (B) $[ML^2T^{-1}Q^{-2}]$
 (C) $[LTQ^{-1}]$
 (D) $[LT^{-1}Q^{-1}]$

3. A thin massless rod of length 2ℓ has equal point masses m attached at its ends as shown in the figure. The rod is rotating about an axis passing through its centre and making angle θ with it. The magnitude of net torque about O is

- (A) $2m\ell^2\omega^2 \sin\theta \cos\theta$
 (B) $2m\ell^2\omega^2 \sin\theta$
 (C) $2m\ell^2\omega^2 \sin^2\theta$
 (D) $2m\ell^2\omega^2 \cos^2\theta$



4. A rain drop falling vertically under gravity gathers moisture from the atmosphere at a rate given by $\frac{dm}{dt} = kt^2$, where m is the instantaneous mass, t is time and k is a constant. The equation of

motion of the rain drop is $m \frac{dv}{dt} + v \frac{dm}{dt} = mg$

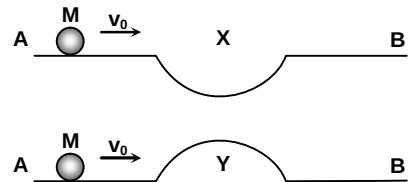
If the drop starts falling at $t = 0$, with zero initial velocity and initial mass m_0 ($m_0 = 2$ gm, $k = 12$ gm/s³ and $g = 1000$ cm/s²), the velocity v of the drop after one second is

- (A) 250 cm/s
 (B) 500 cm/s
 (C) 750 cm/s
 (D) 1000 cm/s

5. A particle is released at $x = 1$ in a force field $\vec{F}(x) = \left(\frac{2}{x^2} - \frac{x^2}{2} \right) \hat{x}$, $x \geq 0$. Which one of the following statement is FALSE?

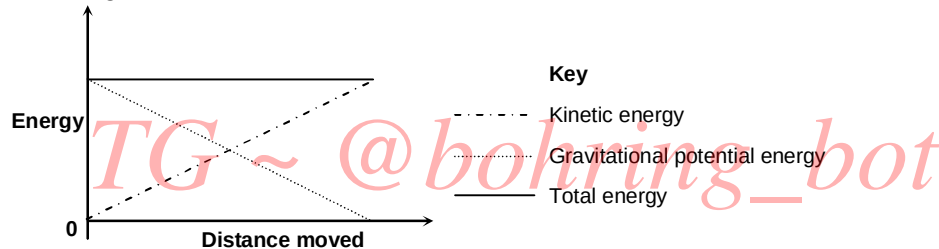
(A) $\vec{F}(x)$ is conservative
 (B) The angular momentum of the particle about the origin is constant
 (C) The particle moves towards $x = \sqrt{2}$
 (D) The particle moves towards the origin

6. A small mass M , is given an initial velocity v_0 , and it slides from A to B via two possible paths; either down the shallow dip X or over the hump Y, both of which are the same shape but inverted. Friction is to be ignored. Along which path does the mass take the shortest time to slide from A to B?



(A) Via X
 (B) Via Y
 (C) You cannot say
 (D) Same time taken

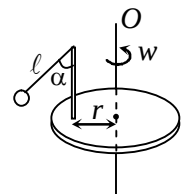
7. The graph below represents the kinetic energy, gravitational potential energy, and total energy of a moving block. Which best describes the motion of the block?



(A) Sliding down an incline with constant friction
 (B) Falling at a constant velocity
 (C) Acceleration on a flat horizontal surface
 (D) Falling freely under gravity

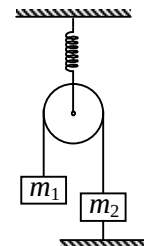
8. A ball of mass m is attached to the end of a thread fastened to the top of a vertical rod which is fitted to a horizontally revolving round table as shown. If the thread forms an angle α with the vertical, the angular velocity ω of table is

(A) $\sqrt{\frac{g}{\ell \cos \alpha}}$
 (B) $\sqrt{\frac{r}{g \tan \alpha}}$
 (C) $\sqrt{\frac{g \tan \alpha}{r + \ell \sin \alpha}}$
 (D) $\sqrt{\frac{g \tan \alpha}{r}}$

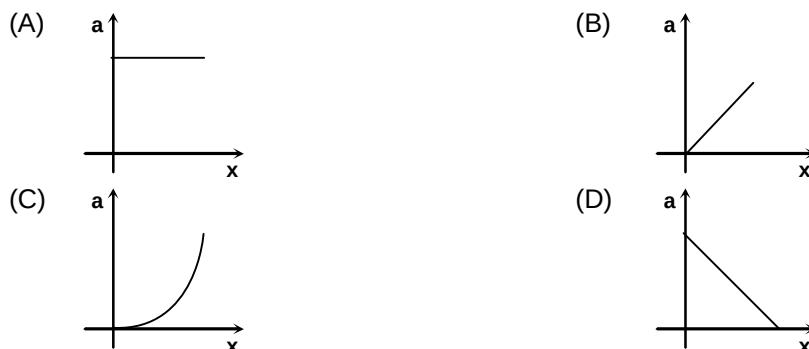
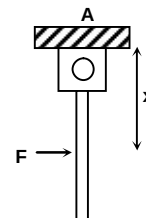


9. In the figure shown, pulley and spring are ideal. If k is spring constant of spring, the potential energy stored in it at equilibrium is ($m_1 > m_2$)

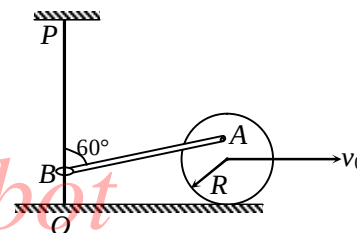
(A) $\frac{2m_1^2 g^2}{k}$
 (B) $\frac{2m_2^2 g^2}{k}$
 (C) $\frac{(m_1 + m_2)^2 g^2}{k}$
 (D) $\frac{1}{2} \frac{(m_1 - m_2)^2 g^2}{k}$



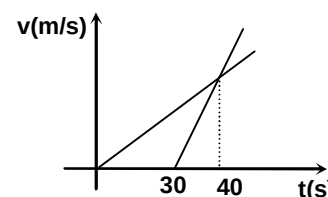
10. A rod of mass m and length ℓ is hinged at one of its end A as shown in the figure. A force F is applied at a distance ' x ' from A. The acceleration of centre of mass ' a ' varies with ' x ' as



11. A smooth rod OP is fixed vertically. A disc of mass m and radius R is rolling without slipping on a rough horizontal surface as shown. Velocity of centre of mass of disc is v_0 . A rod AB of length L is connected with disc at A, $R/2$ distance vertically up from centre of disc with pin joint (about which it can rotate freely), other end of the rod is attached to a small smooth ring B, which can move freely over rod OP . At an instant rod AB makes an angle 60° with the vertical. The velocity of ring on the vertical rod at this instant is

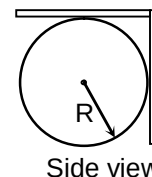


- (A) $\frac{3\sqrt{3}}{2}v_0$ (B) $\frac{3}{2}v_0$
(C) v_0 (D) $\frac{v_0}{2}$
12. The police is chasing a thief and both move on same straight road. After passing a particular point, their velocity v vary with time t as shown in the figure. When will the police catch the thief?
- (A) 90 sec (B) 80 sec
(C) 60 sec (D) 120 sec



13. Last mango in the mango tree is located at 30 m horizontal and 40 m vertical from a boy. The speed and angle with horizontal with which the boy should throw a stone so as to hit the last mango?
- (A) 50 m/s, $\tan^{-1}(2)$ (B) 30 m/s, $\tan^{-1}(3)$
(C) 30 m/s, $\tan^{-1}(2)$ (D) 20 m/s, $\tan^{-1}(3)$

14. A plate bent at right angles along its centre line, is placed onto a horizontal fixed cylinder of radius R as shown. The minimum co-efficient of friction between the cylinder and the plate so that plate does not slip off the cylinder will be (Assume mass of plate and cylinder to be same)



- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$
(C) $\sqrt{2} - 1$ (D) $1 - \frac{1}{\sqrt{2}}$

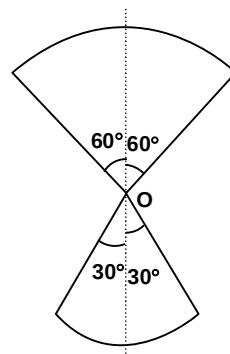
15. Two sectors made of same material and same radius R but angle 120° and 60° are arranged as shown with centre as origin. Find the location of centre of mass?

(A) $\frac{2R(\sqrt{3}-1)}{3}$

(B) $\frac{R(\sqrt{3}-1)}{3}$

(C) $\frac{3R(\sqrt{3}-1)}{3}$

(D) $\frac{4R(\sqrt{3}-1)}{3}$



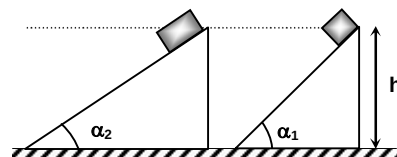
16. Two object of the same mass slide down on two different smooth fixed incline plane of same height but of different angles. When they reach the bottom then I_1 and I_2 are the impulse of the weight of first and second object. Choose the correct option.

(A) $\frac{I_1}{I_2} = \frac{\sin \alpha_1}{\sin \alpha_2}$

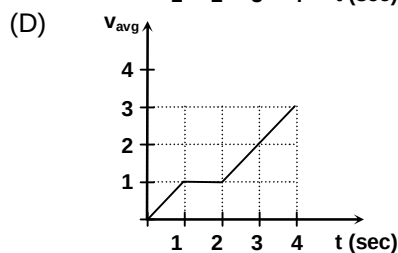
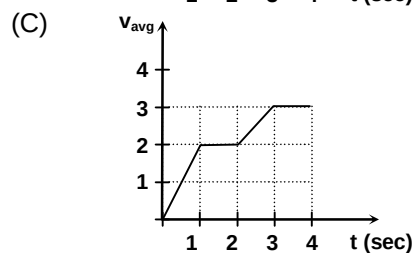
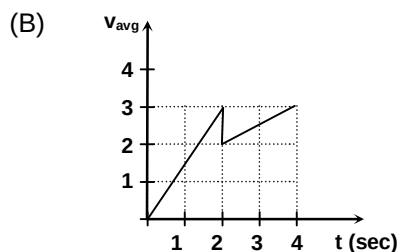
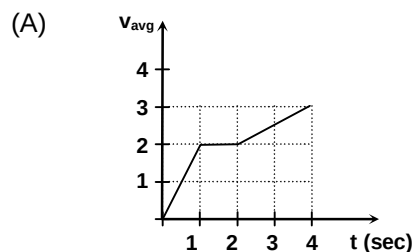
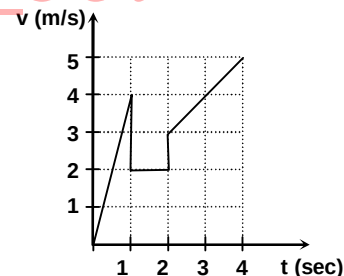
(B) $\frac{I_1}{I_2} = \frac{\sin \alpha_2}{\sin \alpha_1}$

(C) $\frac{I_1}{I_2} = \frac{\tan \alpha_1}{\tan \alpha_2}$

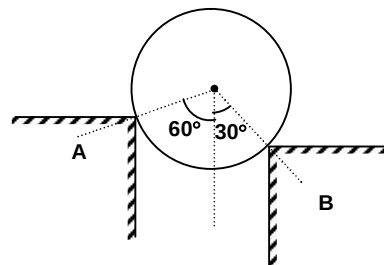
(D) $\frac{I_1}{I_2} = \frac{\tan \alpha_2}{\tan \alpha_1}$



17. The instantaneous velocity of a body moving in a straight line changes with time as shown in the figure. Choose the correct average velocity versus time depicting the same motion?

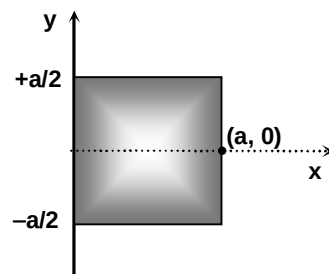


18. A cylinder of mass M and radius R is resting on two corner edges A and B as shown in the figure. The normal reaction at the edges A and B are: (Neglect friction)



- (A) $N_A = \sqrt{2}N_B$ (B) $N_B = \sqrt{2}N_A$
 (C) $N_A = \frac{Mg}{2}$ (D) $N_B = \frac{2\sqrt{3}Mg}{5}$

19. A square laminar sheet with side a and mass M , has mass per unit area given by $\sigma(x) = \sigma_0 \left[1 - \frac{x}{a} \right]$. Moment of inertia of the sheet about y axis is



- (A) $\frac{Ma^2}{2}$ (B) $\frac{Ma^2}{4}$
 (C) $\frac{Ma^2}{6}$ (D) $\frac{Ma^2}{12}$

20. A spring is compressed between two toy carts of masses m_1 and m_2 . When the toy carts are released, the spring exerts on each toy cart equal and opposite forces for the same time t . If the coefficients of friction μ between the ground and the toy carts are equal, then the displacements of the toy carts are in the ratio

- (A) $\frac{s_1}{s_2} = \frac{m_2}{m_1}$ (B) $\frac{s_1}{s_2} = \frac{m_1}{m_2}$
 (C) $\frac{s_1}{s_2} = \left(\frac{m_2}{m_1} \right)^2$ (D) $\frac{s_1}{s_2} = \left(\frac{m_1}{m_2} \right)^2$

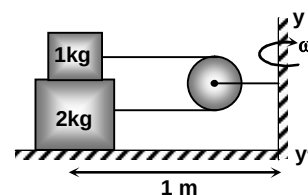
SECTION – B

(Numerical Answer Type)

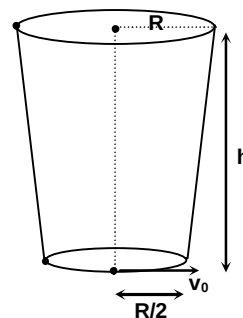
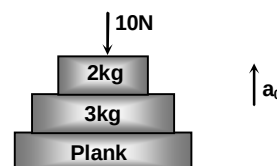
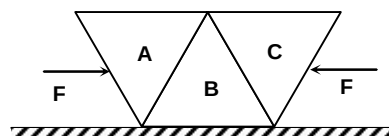
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A shell is fired at an angle θ with horizontal up a plane inclined at an angle $\frac{\pi}{6}$. The value of θ for maximum range is $\frac{\pi}{k}$. Find the value of k .

22. Two blocks A and B 1 kg and 2 kg respectively are kept on a turn table 1 m away from axis yy' as shown in the figure. Consider $\mu = 0.5$ between all contact surfaces and pulley to be massless and frictionless. The angular speed of the turn table for which the blocks just begin to slide is $\sqrt{5k}$. Find k . (Take $g = 10 \text{ m/s}^2$)



23. Two objects of masses m and $2m$ are moving at speeds of v and $v/2$, respectively. After undergoing a completely inelastic collision, they move together with a speed of $v/3$. The angle between the initial velocity of the two objects is K° . Find the value of k .
24. If there is a positive error of 50% in the measurement of velocity of a body, then the error in the measurement of kinetic energy is $k\%$. Find the value of k .
25. The motion of a point A on the rim of a wheel rolling uniformly along a horizontal path has the form
 $x = 2(5t - \sin 5t)$
 $y = 2(1 - \cos 5t)$
 The distance which it traverses between two successive contacts with the roadbed is k meter. Find the value of k .
26. Three identical equilateral prisms A, B and C are arranged as shown in the figure. Mass of each prism is $M = \sqrt{3}$ kg. Two horizontal forces of equal magnitude (F) are applied on A and C as shown in the figure. Find the minimum value of F in newton so that A and C loose contact with the floor. (Ignore friction and take $g = 10 \text{ m/s}^2$)
27. A modern machine gun fires 3600 bullets per minute with 600 km/hr. If each bullet weights 200 gm then the power of the gun is $\frac{500}{m}$ kilowatt. Find the value of m .
28. A solid cylinder rolls down an incline plane of angle θ . The least value of coefficient of friction so that the cylinder rolls without slipping is $\frac{\tan \theta}{k}$. Find the value of k .
29. Two blocks of masses 2 kg and 3 kg are placed on a plank. An external force of 10 N is acting on 2 kg block and the whole system including the plank is moving vertically up with a constant acceleration $a_0 = 2 \text{ m/s}^2$. The normal reaction force between the 3 kg block and Plank is α N. Find the value of $\frac{\alpha}{10}$. ($g = 10 \text{ m/s}^2$)
30. A smooth cylindrical glass lies on the ground. A small particle of mass m lies on the bottom surface touching circumference as shown in the figure. The minimum value of v_0 (along the circumference) such that particle just reaches at the top of glass is $\sqrt{\frac{Kgh}{3}}$. Find the value of K .



Chemistry

PART – B

SECTION – A

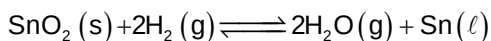
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

31. The CORRECT order of electron gain enthalpy of B, Be, N and C is (consider magnitude only)
 (A) $\text{Be} < \text{B} < \text{C} < \text{N}$
 (B) $\text{Be} < \text{N} < \text{B} < \text{C}$
 (C) $\text{N} < \text{Be} < \text{C} < \text{B}$
 (D) $\text{N} < \text{C} < \text{B} < \text{Be}$
32. Which of the following properties is INCORRECTLY matched?
 (A) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ (Oxidising character)
 (B) $\text{K} > \text{Mg} > \text{Al} > \text{B}$ (Metallic character)
 (C) $\text{C} > \text{Be} > \text{B} > \text{Li}$ (1^{st} Ionisation enthalpy)
 (D) $\text{Li} > \text{Na} > \text{K} > \text{Rb}$ (Chemical reactivity)
33. The minimum uncertainty in the velocity of a particle of mass $1.1 \times 10^{-28} \text{ kg}$ will be if its uncertainty in position is $3 \times 10^{-10} \text{ cm}$ ($h = 6.62 \times 10^{-34} \text{ Js}$).
 (A) $1.6 \times 10^6 \text{ ms}^{-1}$
 (B) $2.6 \times 10^5 \text{ ms}^{-1}$
 (C) $3.6 \times 10^5 \text{ ms}^{-1}$
 (D) $1.6 \times 10^5 \text{ ms}^{-1}$
34. The coefficients of BaCrO_4 , KI , CrCl_3 and KCl in the balanced equation of the following reaction are respectively
 $\text{BaCrO}_4 + \text{KI} + \text{HCl} \longrightarrow \text{BaCl}_2 + \text{I}_2 + \text{KCl} + \text{CrCl}_3 + \text{H}_2\text{O}$
- | | BaCrO_4 | KI | CrCl_3 | KCl |
|-----|------------------|-------------|-----------------|--------------|
| (A) | 2 | 6 | 2 | 6 |
| (B) | 6 | 2 | 6 | 6 |
| (C) | 2 | 2 | 6 | 6 |
| (D) | 6 | 6 | 2 | 2 |
35. At a given temperature, the rate constant for the decomposition of HI on a metal surface is 0.06 Ms^{-1}
 $2\text{HI}(\text{g}) \longrightarrow \text{H}_2(\text{g}) + \text{I}_2(\text{g})$
 The time taken for the concentration of HI to fall from 1.50 M to 0.30 M is
 (A) 10 s
 (B) 20 s
 (C) 15 s
 (D) 8 s

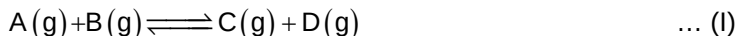
36. The increasing order of bond angles in OF_2 , OCl_2 , ClO_2 and H_2O is.....
- (A) $\text{OF}_2 < \text{H}_2\text{O} < \text{OCl}_2 < \text{ClO}_2$
 (B) $\text{H}_2\text{O} < \text{F}_2\text{O} < \text{Cl}_2\text{O} < \text{ClO}_2$
 (C) $\text{OF}_2 < \text{Cl}_2\text{O} < \text{H}_2\text{O} < \text{ClO}_2$
 (D) $\text{ClO}_2 < \text{Cl}_2\text{O} < \text{OF}_2 < \text{H}_2\text{O}$
37. Consider the following reactions in equilibrium
- $$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3 \quad (K_1)$$
- $$\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO} \quad (K_2)$$
- $$\text{H}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{H}_2\text{O} \quad (K_3)$$
- Then the equilibrium constant for the reaction
- $$2\text{NH}_3 + \frac{5}{2}\text{O}_2 \rightleftharpoons 2\text{NO} + 3\text{H}_2\text{O}$$
- (All the above reactions are in homogeneous equilibrium)
- (A) $\frac{K_2 K_3}{K_1}$
 (B) $\frac{K_2 K_1^3}{K_3}$
 (C) $\frac{K_2 K_3^3}{K_1}$
 (D) $\frac{K_2 K_1}{K_3}$
38. The pH of pure water at 25°C and 35°C are 7 and 6 respectively. The heat of formation of H^+ (aq) and OH^- (aq) ions in Kcal mol^{-1} is
- (A) -84.5
 (B) $+84.5$
 (C) -94.5
 (D) $+94.5$
39. A hypothetical small particle of mass 'm' moves in such a way that its potential energy $= -\frac{1}{2} mkr^4$. Where k is a constant and r is its distance from the nucleus (origin). Assuming Bohr's model r is directly proportional to
- (A) n^2
 (B) n
 (C) $\sqrt{n}$
 (D) $\sqrt[3]{n}$

40. Consider the reaction in equilibrium



at 300 K, the equilibrium mixture was found to contain 40% by volume of $\text{H}_2(\text{g})$. The K_p for the reaction is

- (A) 1.15
(B) 10.9
(C) 7.75
(D) 2.25
41. The three reactions are given below



Which of the following statement is INCORRECT?

- (A) Increase in pressure shifts the equilibrium (II)nd to the left
(B) Decrease in volume shifts the (III)rd equilibrium to right
(C) Addition of He at constant volume shifts the (II)nd equilibrium to left
(D) Addition of He at constant pressure shifts the (III)rd equilibrium to left
42. At high temperature and at a pressure of around 2 atm, nitrogen gas is bubbled over calcium carbide, the major product obtained is
(A) Calcium cyanide
(B) Calcium cyanamide
(C) Calcium nitride
(D) Calcium
43. Which of the following statement is NOT correct?
(A) Stability of peroxides and superoxides of alkali metals increases with increase in the size of metal ion.
(B) Sodium hydroxide does not absorb water from atmosphere.
(C) Lithium fluoride is less soluble in water due to its high lattice enthalpy.
(D) Caesium iodide is less soluble in water due to low hydration enthalpy.
44. In solid state, beryllium chloride has
(A) Three centre-four electron bonds.
(B) Three centre-three electron bonds.
(C) Two centre-three electron bonds.
(D) Two centre-two electron bonds.

45. Consider the following silicates

I. Pyro silicates

II. Cyclic silicates

III. Double chain silicates

IV. Single chain silicates

V. 3D-silicates

VI. Sheet silicates

The silicates in which **only two** corners **per tetrahedron** are shared

- (A) I, II, IV
(B) IV, VI
(C) I, VI
(D) II, IV

46. Select the CORRECT sequence of geometry of the given molecules/ions
 Borazon, Borazole, $B_3O_6^{3-}$, Al_2Cl_6
 (A) Non planar, non planar, non planar, planar
 (B) Planar, planar, non planar, non planar
 (C) Non planar, non planar, non planar, planar
 (D) Non planar, planar, planar, non planar
47. Consider the following species
 NO_2^+ , NO_2^- , NO_2 , NO_3^-
 Select the INCORRECT statement for the above molecule/ions?
 (A) NO_2 is paramagnetic
 (B) NO_2^+ is linear, NO_2^- is bent having bond angle more than 120°
 (C) NO_3^- has bond order 1.33
 (D) NO_2 has bond angle more than 120°
48. Consider the compounds LiH , NaH , Li_2CO_3 and Na_2CO_3 . Select the compound (s) which decomposes on heating around $400 - 500^\circ C$.
 (A) LiH , Li_2CO_3
 (B) NaH , Na_2CO_3
 (C) LiH , NaH
 (D) NaH , Li_2CO_3
49. Which of the following statement(s) is(are) NOT correct?
 I. Bond length order is $H_2^- = H_2^+ > H_2$.
 II. O_2^+ , NO , N_2^- have bond order 2.5.
 III. Bond order varies from 0 to 4.
 IV. In NO_3^- and BO_3^- the bond order is same between central atom and side atom.
 (A) I, II
 (B) II, III
 (C) I, IV
 (D) I, III
50. Select the INCORRECT statement regarding Diborane (B_2H_6)
 (A) Maximum 6 H-atoms lie in a plane
 (B) Maximum 6 atoms lie in a plane
 (C) There is hindered rotation between two boron atoms
 (D) Terminal H – B – H bond angle is greater than bridge H – B – H bond angle

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. A sample of solid $AgCl$ was treated with 5.00 mL of 1.5 M Na_2CO_3 solution to give Ag_2CO_3 . The remaining Cl^- ion concentration was found to be 0.0026 g L^{-1} . The solubility product of $AgCl$ is given by $x \times 10^{-y}$. Then the value of $(x + y)$ is.....(Rounded off to nearest integer).
 (K_{SP} of $Ag_2CO_3 = 8.2 \times 10^{-12}$)

52. A tiny particle moving with a velocity $\frac{1}{60}$ th times the velocity of light. If uncertainty in the position is $\frac{6.62}{3.14} \times 10^{-12}$ m, then the minimum uncertainty in kinetic energy is $x \times 10^{-y}$ J, then the value of xy is.....
53. Potassium permanganate reacts with sulfur dioxide in acidic medium to produce a product containing bisulphate ion(s) and same other product(s). The sum total of coefficients of potassium permanganate and the product containing bisulphate ions in the balanced equation is.....
54. For a reaction which following first order kinetics the half life is 693 s and the values of pre exponential factor and activation energy are 4×10^{13} Hz and 98.6 kJ mol^{-1} respectively, then the temperature in Kelvin when half life is 693 s, is..... ($\log 2 = 0.30$)
55. An equilibrium mixture at 300 K contains N_2O_4 and NO_2 at 0.28 and 1.1 atm respectively. If the volume of container get doubled, the equilibrium pressure of NO_2 (g) is $x \times 10^{-2}$ atm, then the value of x is.....
56. Total number of species having used all three P-orbitals for hybridisation of central atom and are non-polar also is.....
 XeO_2F_2 , SnCl_2 , IF_5 , I_3^+ , XeO_4 , $[\text{XeF}_7]^+$, SO_2
57. The number of correct order(s) in the following is.....
 1. $\text{CH}_4 < \text{CCl}_4 < \text{CF}_4$ (Electronegativity of carbon)
 2. $\text{Mg}^{++} < \text{K}^+ < \text{S}^{--} < \text{Se}^{--}$ (Ionic radius order)
 3. $\text{LiF} > \text{NaF} > \text{KF} > \text{RbF}$ (Lattice energy order)
 4. $\text{Li}^+ < \text{Mg}^{++} < \text{Al}^{3+}$ (Hydration energy order)
 5. $\text{Cl}_2 > \text{F}_2 > \text{Br}_2 > \text{I}_2$ (Bond energy order)
58.
$$\begin{array}{c} \text{Na}_2\text{O}_2 \xrightarrow[\text{(ice cold)}]{\text{H}_2\text{O}} \text{P} + \text{Q} \\ \downarrow \text{H}_2\text{O (Room temperature)} \\ \text{P} + \text{R} \uparrow \end{array}$$

 If molar mass of (Q) and (R) are M_1 and M_2 respectively then the value of $\left(\frac{M_1 + M_2}{2}\right)$ is.....
59. A 1.025 g sample containing a weak acid HA (MW = 82) is dissolved in 60 mL of water and titrated with 0.25 M NaOH. When half of the acid was neutralized, the pH was found to be 5.0 and at the equivalence point the pH is 9.0. The mass percent of weak acid HX in the sample is.....%.
60. The pH of the resultant solution of 25 mL of 0.1 M HCl, 50 mL of 0.2 M HNO_3 and 25 mL of 0.1 M NaOH is.....

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. Let f be a differentiable function satisfying $\int_0^x (x-p+1)f(p)dp = x^4 + x^2 \forall x \in [-2, \infty)$, then value of $\int_{-2}^2 (f'(x) + f(x))dx$ is
 (A) 0 (B) 36
 (C) 72 (D) 18
62. Let $P = \lim_{x \rightarrow 0} \frac{1 - (\cos 2x \cos 4x \cos 6x \cos 8x \cos 10x)^3}{(\tan^2 x)}$, then the value of $\left(\frac{P}{10}\right)$ is
 (A) 66 (B) 33
 (C) 22 (D) 11
63. If $A = \lim_{n \rightarrow \infty} \sum_{r=1}^n \sin\left(\frac{r}{n^2}\right)$ and $B = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sin\left(\frac{\pi r}{n}\right)$, then value of $2A + B$ is
 (A) $1 + \frac{2}{\pi}$ (B) $\frac{2}{\pi}$
 (C) $\frac{\pi-1}{2}$ (D) $\frac{\pi}{2} - 1$
64. Let $f(x) = \frac{1}{\sin^2 x - (x-a)^2}$, $x \in (0, 1)$, then which of the following is **INCORRECT** ?
 (A) $f(x)$ is always increasing in $(0, 1)$ (B) $f(x)$ is always decreasing in $(0, 1)$
 (C) $f(x)$ is increasing if $a \in (2, 3)$ (D) $f(x)$ is decreasing if $a \in (2, 3)$
65. If $A = \int_0^1 (1-x^{2023})^{\frac{1}{2022}} dx$ and $B = \int_0^1 (1-x^{2022})^{\frac{1}{2023}} dx$, then find the value of $\left(\frac{A^3 - B^3}{B^3}\right) + \left(\frac{A^4 + B^4}{B^4}\right)$ is
 (A) 0 (B) 1
 (C) 2 (D) none of these
66. The exhaustive set of domain of the function $f(x) = \operatorname{sgn}\left(\frac{x^2 + x + 4}{x^2 + x + 3}\right) + \tan\left(\left[\frac{x^2 + x + 3}{x^2 + x + 4}\right]\right)$, (where $[.]$ is G.I.F.) is
 (A) $(-\infty, 0)$ (B) $\left(\frac{1}{3}, 3\right)$
 (C) $\left[\frac{1}{3}, 3\right]$ (D) $(-\infty, \infty)$

67. If the area bounded between the curves $y = \tan x$, $y = \sin x$, $x = 0$ and $x = \frac{\pi}{4}$ is A, the A is equal to
- (A) $\frac{1}{2} \ln 2 + \frac{1}{\sqrt{2}}$ (B) $\frac{1}{2} \ln 2$
- (C) $\frac{1}{\sqrt{2}}$ (D) none of these
68. Let $f(x)$ be a function, such that $f(x) = x + \sin x$ and $I_1 = \int_0^{\pi} f(x) dx$, $I_2 = \int_0^{\pi} f^{-1}(x) dx$ and $I_3 = \int_0^1 e^{-x^2} dx$, then which of the following is true ?
- (A) $I_1 > I_2 > I_3$ (B) $I_3 > I_2 > I_1$
- (C) $I_1 > I_3 > I_2$ (D) $I_3 > I_1 > I_2$
69. If $f(x) = \int_0^x (\ln t)^2 dt$ and $g(x) = \int_0^x (\ln t) dt$, then area bounded between the curve $y = f'(x)$ and $y = g'(x)$ between $x = 1$ to $x = e$ is
- (A) $e - 3$ (B) 3
- (C) e (D) $3 - e$
70. Let $L_1 = \lim_{x \rightarrow \infty} \sec^{-1} x$, $L_2 = \lim_{x \rightarrow -\infty} \sec^{-1} x$, then the value of $\int_{L_1}^{L_2} \frac{x^3 + 1}{x^2 + 1} dx$
- (A) π (B) $\frac{\pi}{2}$
- (C) $\frac{\pi}{4}$ (D) 0
71. $\lim_{x \rightarrow 1} \left(\sum_{n=1}^{12} \frac{x}{n(n+1)x^2 + 3(2n+1)x + 9} \right)$ is equal to
- (A) $\frac{11}{60}$ (B) $\frac{3}{16}$
- (C) $\frac{13}{64}$ (D) $\frac{9}{44}$
72. Water is being filled at the rate of 100 litre/min in a conical vessel (vertex downwards) with axis vertical. Its semi vertical angle is $\tan^{-1}\left(\frac{5}{12}\right)$. The rate (in square meter per minute) at which the wet curved surface area of the tank is increasing, when the depth of water in tank is 3 meters is
- (A) $\frac{13}{75}$ (B) $\frac{13}{25}$
- (C) $\frac{1}{5}$ (D) $\frac{3}{11}$
73. The area of the region bounded by $A = \{(x, y) : x^2 - 4 \leq y \leq \min\{2x + 11, 9, 12 - 6x\}\}$ is
- (A) $\frac{511}{12}$ (B) $\frac{947}{12}$
- (C) $\frac{437}{8}$ (D) $\frac{240}{7}$

74. Let $f(t) = \int_0^t e^{x^2} \left(\frac{x^5}{(x^4 + 2x^2 + 2)^2} \right) dx$, then $f(1) + f'(1)$ is equal to
- (A) $\frac{3e}{10} - \frac{1}{4}$ (B) $\frac{7e}{50} - \frac{1}{4}$
 (C) $\frac{7e}{50} - \frac{1}{2}$ (D) $\frac{2e}{5} - \frac{1}{2}$
75. $\lim_{x \rightarrow \infty} \left(\sqrt{x^4 + kx^3 + 3x^2 + mx + 2} - \sqrt{x^4 + 2x^3 - nx^2 + 3x - p} \right) = 4$ then $k + n$ is equal to
- (A) 3 (B) 7
 (C) 5 (D) 0
76. If $I = \int_0^{\sqrt{2\pi}} \frac{x^3 \sin^{2022} x^2}{\cos^{2022} x^2 + \sin^{2022} x^2} dx$ is equal to
- (A) $\frac{\pi^2}{2}$ (B) $2\pi^2$
 (C) π^2 (D) 0
77. If the solution curve of the differential equation $\frac{dy}{dx} = \frac{x+y-4}{x-y}$ passes through the point (3, 2) and $(p+2, 3)$, $p > 0$, then
- (A) $2 \tan^{-1} \left(\frac{1}{p} \right) = \log_e (p^2 + 1)$ (B) $\tan^{-1} \left(\frac{1}{p} \right) = \log_e (p^2 + 1)$
 (C) $2 \tan^{-1} \left(\frac{1}{p+1} \right) = \log_e (p^2 + 2p + 2)$ (D) $2 \tan^{-1} \left(\frac{1}{p} \right) = \log_e \left(\frac{p^2 + 1}{p} \right)$
78. Let $f : [-4, 2] \rightarrow \mathbb{R}$ be given as $f(x) = \begin{cases} \min\{(x+6), x^2\} & -4 \leq x \leq 0 \\ \max\{\sqrt{x}, x^2\} & 0 \leq x \leq 2 \end{cases}$
- If the area bounded by $y = f(x)$ and x -axis in A then the value of $3A$ is equal to
- (A) 19 (B) 35
 (C) 37 (D) 32
79. If $f(t) = \int_1^t \frac{\log_{10} x}{1+x} dx$, $t > 0$, then $f(e^4) + f(e^{-4})$ is equal to
- (A) 8 (B) $\frac{16}{\log_e 10}$
 (C) $\frac{8}{\log_e 10}$ (D) $\frac{e^4 - 1}{2 \log_e 10}$
80. Let $f(x) = \begin{cases} x^3 - 2x^2 + 3x - 9 & x \leq 2 \\ -3x + \log_3(k^2 - 5) & x > 2 \end{cases}$, then the set of all values of k for which $f(x)$ has maximum value at $x = 2$ is
- (A) $[-4\sqrt{2}, 4\sqrt{2}]$ (B) $(-\sqrt{5}, 4\sqrt{2})$
 (C) $[-4\sqrt{2}, -\sqrt{5}) \cup (\sqrt{5}, 4\sqrt{2}]$ (D) $(\sqrt{5}, 2\sqrt{2}]$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. The number of solution of the equation $\sin^{-1}(x) = \left[\frac{x^2 + x + 2}{x^2 + x + 1} \right] = 0$, [.] is G.I.F.
82. The area bounded between the curves $y = \sin^{-1}(\sin x)$ and $y = \sqrt{\frac{\pi^2}{4} - \left(\pi - \frac{\pi}{2}\right)^2}$ is 'A', then the value of $\left[\frac{8A}{\pi^2} \right]$ (where [.] is G.I.F) is
83. Let $f(x)$ be a function defined as $f(x) = \begin{vmatrix} 0 & -x & -x^2 \\ x & 0 & x+1 \\ x^2 & -x-1 & 0 \end{vmatrix} + \left[\frac{x}{2} \right] \sin x + \sqrt{3(4-x^2)}$, then the number of integers in the range of $f(x)$ is (where [.] is G.I.F)
84. The number of points where the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x-2| \cos|x^2-3x| \sin|x-2| + (x-1)|x^3-8x^2+7x|$ is non differentiable is
85. Let $m = \begin{bmatrix} a & 0 & 1 \\ a & b & -b \\ 0 & 1 & c \end{bmatrix}$ and $\det(m) = 37$, if a, b, c are positive integers, then least value of $\text{trace}(m)$ is
86. If $y = y(x)$ is an implicit function of x such that $\log_e(x+y) = 6xy$, then $\frac{d^2y}{dx^2}$ at $x = 0$ is equal to _____
87. If the value of $\lim_{x \rightarrow 0} \left(3 - 2\cos x \sqrt{\cos 2x} \right)^{\left(\frac{x+3}{x^2} \right)}$ is equal to e^a , then a is equal to
88. If the tangent to the curve $y = x^3 - 2x^2 + 4x + 4$, at the point (a, b) (where $a > 0$) is also tangent to the curve $y = 3x^2 + 2x - 1$ at $(1, 4)$, then $a + b$ is equal to
89. If $f(x) = \int \frac{12x^{15} + 5x^{18}}{(2x^3 + 3 - 7x^8)^3} dx$ and $16f(1) - 1 = 0$ and $f(-1) = \frac{1}{k}$, then $|k| =$
90. Let $I_n = \int_1^e x^{23} (\log|x|)^n dx$ where $n \in \mathbb{N}$, if $24I_{15} = \alpha I_{14} + \beta I_{13}$, then $\alpha + \beta$ is equal to

ANSWERS, HINTS & SOLUTIONS

Physics *TG ~ @behring_bot* PART – A

SECTION – A

1. A

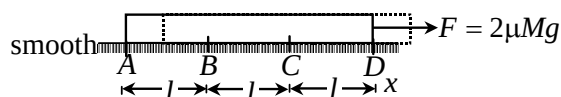
Sol. If the block has moved by distance x ,

$$Ma = 2\mu Mg - \left[\frac{\mu Mg}{3} + \frac{2\mu Mg}{3} + \frac{3\mu Mg}{3l}(l-x) \right]$$

$$a = 2\mu g - \left(\frac{\mu g}{3} + \frac{2\mu g}{3} + \frac{3\mu g}{3l}(l-x) \right)$$

$$\frac{da_1}{dx} = \frac{3\mu g}{3l} \quad [\text{for } 2^{\text{nd}} \text{ and } 3^{\text{rd}} \text{ cells similarly}]$$

$$\tan \theta_1 : \tan \theta_2 : \tan \theta_3 = 3 : 2 : 1$$



2. A

Sol. $\vec{F} = I \int d\vec{\ell} \times \vec{B}$, $B \approx \frac{\mu_0 I}{d} \Rightarrow F \propto I^2 \mu_0$

$$\Rightarrow [MLT^{-2}] = \left(\frac{Q}{T} \right)^2 \mu_0$$

$$\Rightarrow \mu_0 = [MLQ^{-2}]$$

3. A

Sol. $L = m[\vec{r} \times (\vec{\omega} \times \vec{r})] + m[\vec{r} \times (\vec{\omega} \times \vec{r})] = 2m[\vec{r} \times \ell \omega \sin \theta] = 2m\omega \ell^2 \sin \theta$

$$\text{Torque} = \frac{dL}{dt} = \vec{\omega} \times \vec{J} = 2m\omega \ell^2 \sin(90 - \theta) \sin \theta = 2m\omega \ell^2 \cos \theta \sin \theta$$

4. B

Sol. $\frac{dm}{dt} = kt^2 \Rightarrow m = \frac{kt^3}{3} + 2$

$$m \frac{dv}{dt} + v \frac{dm}{dt} = mg \Rightarrow m \frac{dv}{dt} + vkt^2 = mg$$

$$\frac{dv}{dt} + \frac{kt^2}{m}v = g \Rightarrow \frac{dv}{dt} + \frac{3kt^2}{kt^3 + 6}v = g \text{ which is linear differential equation}$$

$$v(kt^3 + 6) = \int g(kt^3 + 6)dt + c, t = 0, v = 0 \Rightarrow c = 0$$

$$v = \frac{g(kt^4 + 24t)}{4(kt^3 + 6)}$$

Put 2 gm , $k = 12 \text{ gm/s}^3$ and $g = 100 \text{ cm/s}^2$ and $t = 1$

$$v = \frac{1000(12 + 24)}{4(12 + 6)} = 500 \text{ cm/sec}$$

5. C

Sol. $\vec{F}(x) = \left(\frac{2}{x^2} - \frac{x^2}{2} \right) \hat{e}_x, x \geq 0$

Work done in a cycle is zero so conservative

$\vec{r} \times \vec{F} = 0$, so angular momentum is conserved

For equilibrium point $F = 0 \Rightarrow F = \left(\frac{2}{x^2} - \frac{x^2}{2} \right) = 0 \Rightarrow x = -\sqrt{2}, \sqrt{2}$ $x > 0$ so equilibrium point is $\sqrt{2}$

To check stable and unstable equilibrium point $\frac{dF}{dx} = 0 \Rightarrow \frac{-4}{x^3} - \frac{2x}{2} = 0$

$$\text{At } x = \sqrt{2}, \frac{dF}{dx} = 0 \Rightarrow \frac{-4}{x^3} - \frac{2x}{2} = -\frac{4}{2\sqrt{2}} - \sqrt{2} = -ve$$

So, it is unstable point so particle moves away from point $x = \sqrt{2}$

So, it is obvious that it will moves towards origin.

6. A

Sol. Speed is never less than v_0 in x whereas in y it is less than v_0 first and then increases $t v_0$ so, y take more time.

7. D

Sol. Friction will produce heat energy and so the total energy will not be PE + KE (or all of the PE will not turn into KE). At a constant velocity there is no gain of KE. On a horizontal surface there is no change of PE. Falling freely, the PE turns solely into KE.

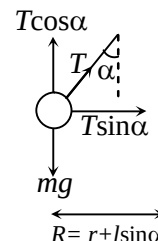
8. C

Sol. $T \sin \alpha = m\omega^2 R$

$$T \cos \alpha = mg$$

$$\tan \alpha = \frac{\omega^2 R}{g}$$

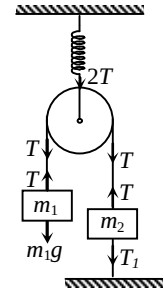
$$\omega = \sqrt{\frac{g \tan \alpha}{R}} = \sqrt{\frac{g \tan \alpha}{r + \ell \sin \alpha}}$$



9. A

Sol. $T = m_1 g$
 $kx = 2T = 2m_1 g$
 $x = \frac{2m_1 g}{k}$

$$\text{Energy stored} = \frac{1}{2} kx^2 = \frac{1}{2} k \frac{4m_1^2 g^2}{k^2} = \frac{2m_1^2 g^2}{k}$$



10. B

Sol. The rod will rotate about point A with angular acceleration:

$$\alpha = \frac{\tau}{I} = \frac{Fx}{\frac{m\ell^2}{3}} = \frac{3Fx}{m\ell^2}$$

$$\therefore a = \frac{\ell}{2} \alpha = \frac{3}{2} \frac{Fx}{m\ell} \text{ or } a \propto x$$

i.e., a-x graph is a straight line passing through origin.

11. A

Sol. $v \cos 60^\circ = \frac{3v_0}{2} \cos 30^\circ, v = \frac{3\sqrt{3}}{2} v_0$

12. C

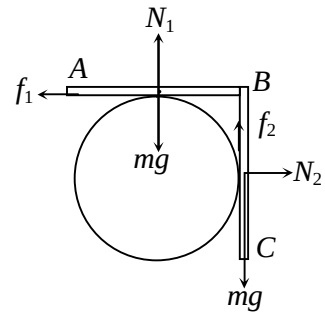
Sol. $v_1 = \frac{v_0 t}{40}, S_1 = \frac{v_0 t^2}{80} \Big|_{30}$
 $v_2 = \frac{v_0 t}{10} - 3v_0, S_2 = \frac{v_0 t^2}{20} - 3v_0 t \Big|_{30}$
 $S_1 = S_2$
 $\Rightarrow t = 60 \text{ sec}$

13. B

Sol. $v_{\min} = \sqrt{g(y + \sqrt{x^2 + y^2})} = 30 \text{ m/s}$
 $\tan \theta = \frac{v^2}{gx} = 3$

14. C

Sol. $N_1 + f_2 = 2mg \quad \dots(i)$
 $N_2 = f_1 \quad \dots(ii)$
 $f_1 = \mu N_1 \quad \dots(iii)$
 $f_2 = \mu N_2 \quad \dots(iv)$
 Taking moment about corner B of plate
 $(mg + N_2)R = N_1 R$
 $mg + N_2 = N_1 \quad \dots(v)$
 From (i), (ii) & (v),
 $f_1 + f_2 = mg \quad \dots(vi)$



From (iii), (iv) & (vi),

$$\mu^2 + 2\mu - 1 = 0 \text{ or}$$

$$\mu = \sqrt{2} - 1$$

15. A

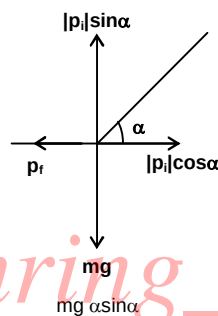
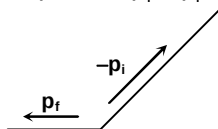
Sol. Centre of mass of arc is $\frac{2R \sin \alpha}{3\alpha}$

$$m_1 = 2m, m_2 = m$$

$$\text{centre of mass} = \frac{\frac{2m \left(2R \sin \left(\frac{\pi}{3} \right) \right)}{\left(3 \frac{\pi}{3} \right)} + m \left(\frac{-2R \sin \left(\frac{\pi}{6} \right)}{2 \left(\frac{\pi}{6} \right)} \right)}{3m} = \frac{2R(\sqrt{3} - 1)}{3}$$

16. A

Sol. impulse = $\vec{p}_f - \vec{p}_i$



TG ~ @bohring_bot

17. A

Sol. $t = 0$ to $t = 1$ sec

$$v = 4t \text{ m/s}$$

$$s = 2t^2 \text{ m}$$

$$s(t = 0) = 0 \text{ m}, s(t = 1) = 2 \text{ m}$$

$$v_{\text{average}} = \frac{\Delta s}{\Delta t} = 2 \text{ m/s}$$

$t = 1$ to $t = 2$ sec

$$v = 2 \text{ m/s}$$

$$s = 2t \text{ m}$$

$$s(t = 1) = 2 \text{ m}, s(t = 2) = 4 \text{ m}$$

$$v_{\text{average}} = 2 \text{ m/s}$$

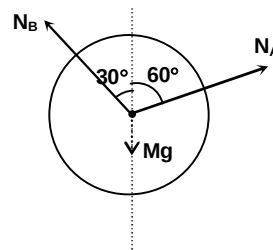
18. C

Sol. A cylinder of mass M

For equilibrium $N_A \cos 60^\circ + N_B \cos 30^\circ = Mg$

and $N_A \sin 60^\circ = N_B \sin 30^\circ$

$$\text{On solving } N_B = \sqrt{3}N_A; N_A = \frac{Mg}{2}$$



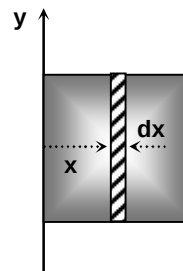
19. C

Sol. $dM = \sigma_0 \left(1 - \frac{x}{a}\right) dx$

$$M = \frac{\sigma_0 a}{2}$$

$$\Rightarrow d(MOI) = dm x^2$$

$$MOI = \int dM x^2 = \frac{Ma^2}{6}$$



20. C

Sol. Minimum stopping distance = s

Force of friction = μmg

Work done against the friction $W = \mu mgs$

Initial kinetic energy of the toy cart = $(p^2 / 2m)$

$$\therefore \mu mgs = (p^2 / 2m)$$

$$\frac{s_1}{s_2} = \left(\frac{m_2}{m_1}\right)^2$$

SECTION - B

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21. 3

Sol. Let the shell hit the plane at $p(x, y)$, the range being $AP = R$, as shown in the figure. The equation for the projectile's motion is

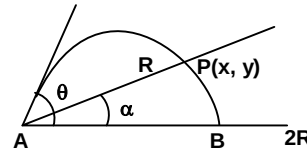
$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} \quad \dots(i)$$

$$\text{Now } y = R \sin \alpha \quad \dots(ii)$$

$$x = R \cos \alpha \quad \dots(iii)$$

Using equation (ii) and equation (iii) in equation (i) and simplifying

$$R = \frac{2u^2 \cos \theta \sin(\theta - \alpha)}{g \cos^2 \alpha}$$



The maximum range is obtained by setting $\frac{dR}{d\theta} = 0$, holding u , α and g constant. This gives

$$\cos(2\theta - \alpha) = 0 \text{ or } 2\theta - \alpha = \frac{\pi}{2}$$

22. 5

Sol. for block A

$$T = \mu m_A g + m_A \omega^2 R$$

For block B

$$T + \mu(m_A + m_B)g + \mu m_A g = m_B \omega^2 R$$

$$\omega = \sqrt{\frac{\mu g(3m_A + m_B)}{R(m_B - m_A)}}$$

23. 120

Sol. As $\Delta p_x = 0$, $\Delta p_y = 0$

For x direction,

$$\Rightarrow 2m \times \frac{v \cos \theta}{2} + mv \frac{\cos \theta}{2} = 3m \times \frac{v}{3}$$

$$\Rightarrow 2mv \frac{\cos \theta}{2} = mv \Rightarrow \frac{\cos \theta}{2} = \frac{1}{3} = \cos 60^\circ$$

$$\frac{\theta}{2} = 60^\circ \Rightarrow \theta = 120^\circ$$

24. 125

Sol. Kinetic energy $E = \frac{1}{2}mv^2$

$$\therefore \frac{\Delta E}{E} \times 100 = \frac{v'^2 - v^2}{v^2} \times 100 = [(1.5)^2 - 1] \times 100$$

$$\frac{\Delta E}{E} \times 100 = 125\%$$

25. 16

Sol. $v = \sqrt{v_x^2 + v_y^2}$

$$v_x = 10 - 10 \cos 5t, \quad v_y = +10 \sin 5t$$

$$v = 20 \sin\left(\frac{5t}{2}\right)$$

$$S = \int_0^t 20 \sin\left(\frac{5t}{2}\right) dt, \quad \omega t = 2\pi$$

$$\text{So, } S = 16 \text{ m}$$

26. 30

Sol. A begins to climb B if $F \cos 60^\circ \geq Mg \sin 60^\circ$

$$F \geq \sqrt{3} \text{ Mg}$$

$$F \geq 30 \text{ N}$$

27. 3

$$\begin{aligned} \text{Sol. Power} &= \frac{\text{K.E.}}{\text{time}} = \frac{n \frac{1}{2}mv^2}{t} = \frac{3600}{60} \left(\frac{1}{2}\right) (0.2) \left(\frac{500}{3}\right)^2 \\ &= \left(\frac{500}{3}\right) \times 10^3 \text{ watt} \end{aligned}$$

28. 3

Sol. $a = g(\sin \theta - \mu \cos \theta)$

$$\text{Also, } a = R\alpha, \quad \frac{Rz}{I} = \frac{R(\mu mg \cos \theta R)}{\frac{1}{2}mR^2} = 2\mu g \cos \theta$$

$$\therefore g(\sin \theta - \mu \cos \theta) = 2\mu g \cos \theta$$

$$\mu = \frac{\tan \theta}{3}$$

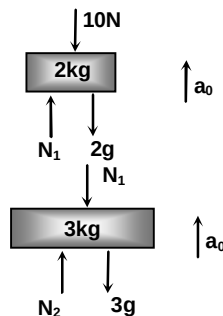
29. 7

Sol. $2a_0 = N_1 - 10 - 2g$

$$N_1 = 34 \text{ N}$$

$$3a_0 = N_2 - N_1 - 3g$$

$$N_2 = 70 \text{ N}$$



30. 8

Sol. By the conservation of energy

$$\frac{1}{2}mv_0^2 = mgh + \frac{1}{2}mv^2 \quad \dots(i)$$

By conservation of angular momentum

$$\frac{mv_0R}{2} = mvR$$

From equation (i) and equation (ii)

$$v_0 = \sqrt{\frac{8gh}{3}}$$

$$\Rightarrow k = 8$$

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Chemistry

PART – B

SECTION – A

31. B

 Sol. $\text{Be} \quad \text{B} \quad \text{C} \quad \text{N}$
 $2s^2 \quad 2s^2p^1 \quad 2s^2p^2 \quad 2s^2p^3$

Be and N have stable configurations, so do not accept electron easily, hence for these elements the process is Endothermic.

So, correct order is $\text{Be} < \text{N} < \text{B} < \text{C}$

32. D

Sol. On moving down the group the IE_1 of Gr(1) elements decreases continuously hence reactivity increases.

33. D

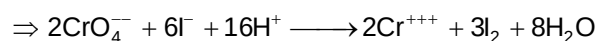
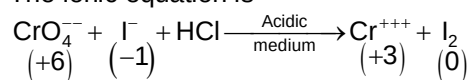
Sol. $\Delta v \times \Delta x = \frac{h}{4\pi m}$ (Using uncertainty principle)

$$\Rightarrow \Delta v = \frac{6.62 \times 10^{-34}}{4 \times 3.14 \times 1.1 \times 10^{-28} \times 3 \times 10^{-12}}$$

$$\Rightarrow \Delta v = 1.597 \times 10^5 \sim 1.6 \times 10^5 \text{ ms}^{-1}$$

34. A

Sol. The ionic equation is



or



35. A

Sol. $2\text{HI}(\text{g}) \longrightarrow \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ (follows zero order kinetics)

$$-\frac{1}{2} \frac{d[\text{R}]}{dt} = k[\text{R}]^0$$

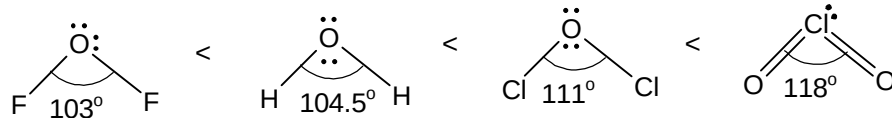
$$\Rightarrow t = \frac{[\text{R}_0] - [\text{R}]}{2k}$$

$$\Rightarrow t = \frac{1.50 - 0.30}{2 \times 0.06}$$

$$\Rightarrow t = \frac{1.20}{0.12} = 10 \text{ s}$$

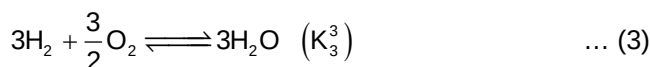
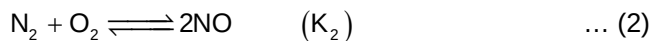
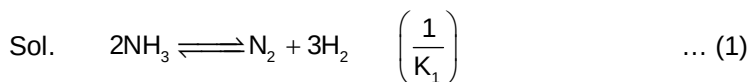
36. A

Sol.

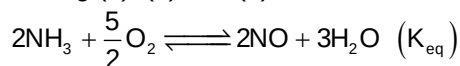


H_2O has larger bp – bp repulsion than OF_2 and OCl_2 has larger steric repulsion between two large Cl atoms while in ClO_2 the bond angle is 118° (close to 120°).

37. C

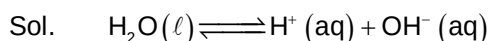


Adding (1), (2) and (3)



$$\Rightarrow K_{\text{eq}} = \frac{K_2 K_3^3}{K_1}$$

38. B



At 25°C , $[\text{H}^+] = [\text{OH}^-] = 10^{-7} \text{ M}$

$\Rightarrow K_{W_1} = 10^{-14}$

At 35°C , $[\text{H}^+] = [\text{OH}^-] = 10^{-6} \text{ M}$

$\Rightarrow K_{W_2} = 10^{-12}$

Now, $2.303 \log \frac{K_{W_2}}{K_{W_1}} = \frac{\Delta H}{R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$

$\Rightarrow 2.303 \log \frac{10^{-12}}{10^{-14}} = \frac{\Delta H}{2} \left(\frac{308 - 298}{308 \times 298} \right)$

$\Rightarrow 2.303 \times 2 = \frac{\Delta H}{2} \times \frac{10}{308 \times 298}$

$\Rightarrow \Delta H = 84.551 \text{ Kcal mol}^{-1}$

Since, $84.551 \text{ Kcal mol}^{-1}$ is the enthalpy change for the reaction

39. D

Sol. $\text{KE} = -\frac{1}{2}(\text{PE}) = -\frac{1}{2} \left(-\frac{1}{2} mkr^4 \right)$

$\text{KE} = \frac{1}{4} mkr^4 = \frac{1}{2} mv^2$

$\Rightarrow v^2 = \frac{mkr^4 \cdot 2}{4m} = \frac{1}{2} kr^4 \quad \dots (1)$

Since, $mvr = \frac{nh}{2\pi} \Rightarrow v^2 = \frac{n^2 h^2}{4\pi^2 m^2 r^2} \quad \dots (2)$

From (1) and (2)

$\frac{1}{2} kr^4 = \frac{n^2 h^2}{4\pi^2 m^2 r^2}$

$\Rightarrow r^6 \propto n^2 \Rightarrow r \propto \sqrt[3]{n}$

40. D

$$\text{Sol. } K_p = \frac{P_{\text{H}_2\text{O(g)}}^2}{P_{\text{H}_2\text{(g)}}^2} \dots (1)$$

$$P_{\text{H}_2} = \frac{40}{40 + 60} \times P_T = 0.4P_T$$

$$P_{\text{H}_2\text{O}} = \frac{60}{40 + 60} \times P_T = 0.6P_T$$

$$\Rightarrow \frac{P_{\text{H}_2\text{O}}}{P_{\text{H}_2}} = \frac{0.6P_T}{0.4P_T} = \frac{3}{2} = 1.5$$

$$\therefore \text{From Eq. (1)} K_p = (1.5)^2 = 2.25$$

41. C

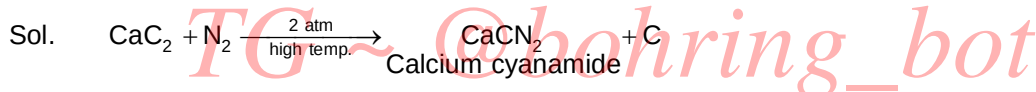
 Sol. (A) $\Rightarrow \Delta n_g > 0$, $\uparrow$ in P shifts the equilibrium $\leftarrow$

 (B) $\downarrow$ in volume in (III)rd equilibrium shifts the equilibrium $\rightarrow$ ($\Delta n_g < 0$)

(C) At constant volume no effect of addition of He

 (D) At constant pressure, addition of He shifts the equilibrium $\leftarrow$ ($\Delta n_g < 0$)

42. B



43. B

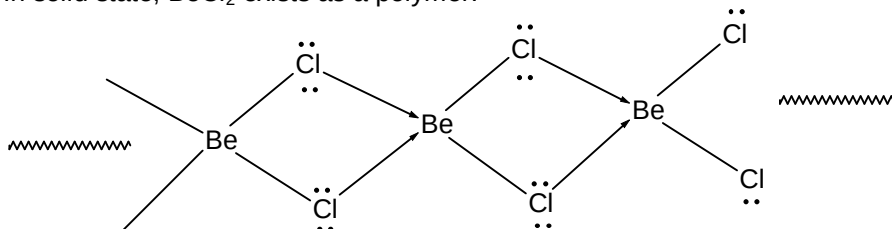
Sol. (A) On moving down the group, the size of metal ion increases, so stability of peroxides/superoxides increases.

i.e. Large cation – Large anion interactions increases.

(B) NaOH is hygroscopic in nature.

 (C), (D) Fact based on $\Delta_{\text{lattice}} \text{H}^\ominus$ and $\Delta_{\text{hyd}} \text{H}^\ominus$.

44. A

 Sol. In solid state, BeCl_2 exists as a polymer.


(four electrons are shared between three atoms, so 3c-4e type of bonds are present)

45. D

Sol. I. – 1 corner shared

II. – 2 corners shared

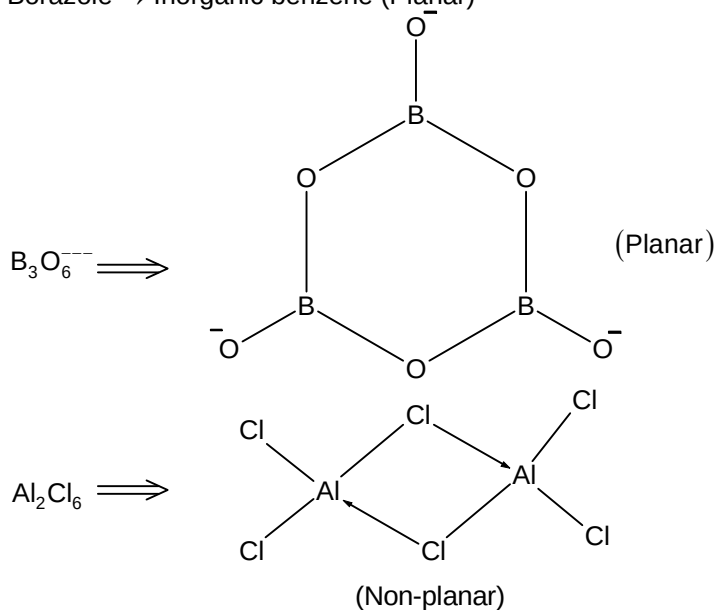
III. – One unit shares 2 corners while one unit shares 3-corners.

IV. – 2 corners shared

V. – 4 corners shared

VI – 3 corners shared

46. D
Sol. Borazon $\rightarrow$ Diamond like structure (Non-planar)
Borazole $\rightarrow$ Inorganic benzene (Planar)



47. B
Sol. $NO_2 \rightarrow$ odd electron species
 $\rightarrow$ Paramagnetic

NO_2 is linear while $\bar{NO}_2$ and NO_2^+ are bent.

$\bar{NO}_2 \rightarrow$ lp – bp repulsion

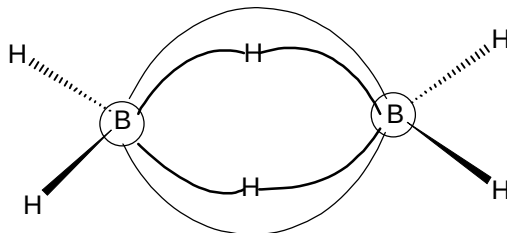
$NO_2 \rightarrow$ bond angle is around 134°

Bond order of $\bar{NO}_3$ is $\frac{4}{3} = 1.33$

Bond angles, $\bar{NO}_2 (115^\circ)$, $NO_2^+ (180^\circ)$, $NO_2 (134^\circ)$

48. D
Sol. $LiH \rightarrow$ Stable
 $NaH \rightarrow$ Unstable
 $Li_2CO_3 \rightarrow$ Unstable
 $Na_2CO_3 \rightarrow$ Stable
(Around $400 - 500^\circ C$)

49. C
Sol. I. $\rightarrow$ Bond length order is $H_2^- > H_2^+ > H_2$
II. $\rightarrow$ Isoelectronic species, so identical bond order.
III. $\rightarrow$ Fact.
IV $\rightarrow NO_3^-$ (Resonance), BO_3^- (no resonance).

50. A
Sol.


Maximum 4 – H atoms lie in a plane.

Maximum 6 – atoms (4H + 2B) lie in a plane.

Bridge (B – H) is stronger than terminal (B – H)

Terminal (H – B – H) bond angle > Bridge (H – B – H) bond angle.

SECTION – B

51. 12

 Sol. $\text{Ag}_2\text{CO}_3(\text{s}) \rightleftharpoons 2\text{Ag}^+(\text{aq}) + \text{CO}_3^{--}(\text{aq})$

$$K_{\text{SP}} = [\text{Ag}^+]^2 [\text{CO}_3^{--}]$$

$$\Rightarrow [\text{Ag}^+]^2 = \left[\frac{8.2 \times 10^{-12}}{7.5/5} \right]$$

$$\Rightarrow [\text{Ag}^+]^2 = \left[\frac{82}{15} \times 10^{-12} \right]$$

$$\Rightarrow [\text{Ag}^+] = 2.34 \times 10^{-6} \text{ M}$$

$$\text{Now } K_{\text{SP}}(\text{AgCl}) = [\text{Ag}^+][\text{Cl}^-]$$

$$K_{\text{SP}} = 2.34 \times 10^{-6} \times \left(\frac{0.0026}{35.5} \right)$$

$$K_{\text{SP}} = 1.71 \times 10^{-10}$$

$$\Rightarrow x = 1.71, \quad y = 10$$

$$\Rightarrow x + y = 1.71 + 10 = 11.71 \sim 12$$

52. 20

 Sol. $\text{KE} = \frac{1}{2}mv^2$

$$\Rightarrow d(\text{KE}) = d\left(\frac{1}{2}mv^2\right)$$

$$\Rightarrow d(\text{KE}) = mv(dv) \quad \dots (1)$$

$$\text{Since, } dv \times dx = \frac{h}{4\pi m}$$

$$\Rightarrow dv = \frac{h}{4\pi m(dx)} \quad \dots (2)$$

From (1) and (2)

$$d(\text{KE}) = mv \cdot \frac{h}{4\pi m(dx)}$$

$$\Rightarrow d(\text{KE}) = \frac{1}{60} \times \frac{3 \times 10^8 \times 6.62 \times 10^{-34}}{4 \times 3.14 \times \frac{6.62}{3.14} \times 10^{-12}}$$

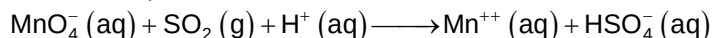
$$\Rightarrow d(\text{KE}) = \frac{1}{80} \times 10^{-14} = 1.25 \times 10^{-16} \text{ J}$$

$$\Rightarrow x = 1.25, y = 16$$

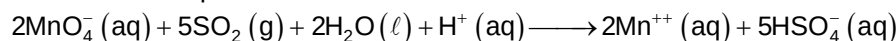
$$\Rightarrow xy = 1.25 \times 16 = 20$$

53. 7

Sol. The ionic equation involved is



The balanced equation is



$\therefore$ Sum total of coefficient(s) = 2 + 5 = 7.

54. 310

Sol. $k = Ae^{-E_a/RT}$

$$\log K = \log A - \frac{E_a}{2.303RT}$$

$$\Rightarrow \log \left(\frac{0.693}{693} \right) = \log (4 \times 10^{13}) - \frac{98.6 \times 1000}{2.303 \times 8.314 \times T}$$

or

$$\Rightarrow \log \frac{4 \times 10^{13}}{10^{-3}} = \frac{98.6 \times 1000}{2.303 \times 8.314 \times T}$$

$$\Rightarrow T = 310.1 \text{ K} \sim 310 \text{ K}$$

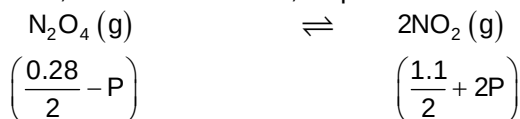
55. 64

Sol. $\text{N}_2\text{O}_4 (\text{g}) \rightleftharpoons 2\text{NO}_2 (\text{g})$

$$\begin{array}{cc} 0.28 & 1.1 \end{array}$$

$$K_p = \frac{(P_{\text{NO}_2})^2}{(P_{\text{N}_2\text{O}_4})} = \frac{(1.1)^2}{0.28} = 4.32$$

Now, volume is doubled, so pressure will become half and reaction proceeds in forward direction



$$\Rightarrow K_p = \frac{(0.55 + 2P)^2}{(0.14 - P)}$$

$$\Rightarrow 4.32 = \frac{(0.55)^2 + 2(0.55)(2P)}{0.14 - P}$$

$$= \frac{0.3025 + 2.20P}{0.14 - P}$$

(Ignoring P^2 terms)

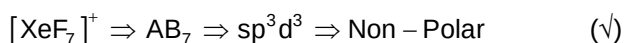
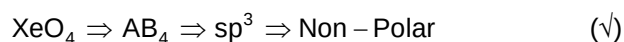
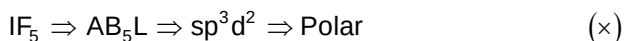
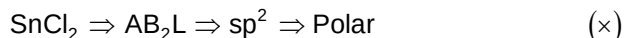
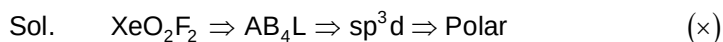
$$\Rightarrow P = 0.046 \text{ atm}$$

$$\therefore P_{\text{NO}_2} = 0.55 + 2 \times 0.046 \cong 0.642$$

$$= 64.2 \times 10^{-2} \text{ atm}$$

$$\Rightarrow x = 64.2 \sim 64$$

56. 2



57. 4

Sol. 1. Correct as EN of side atom increases, EN of carbon increases.

 2. Correct, $\text{Mg}^{++} \Rightarrow 3^{\text{rd}}$ period

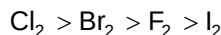
 $\text{K}^+ / \text{S}^{--} \Rightarrow 4^{\text{th}}$ period and isoelectronic

 $\text{Se}^{--} \Rightarrow 5^{\text{th}}$ period

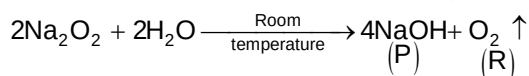
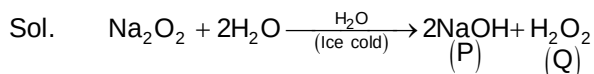
3. Correct as moving down the group cationic size increases so lattice energy decreases

 4. Correct as Hydration energy $\propto \frac{1}{\text{Size of cation}}$

5. Incorrect as the bond energy order among halogens is



58. 33



$$\text{Hence, } \frac{M_1 + M_2}{2} = \frac{34 + 32}{2} = 33$$

59. 80

 Sol. At half neutralization $|\text{S}| = |\text{A}|$

$$\text{pH} = \text{pK}_a + \log \frac{|\text{S}|}{|\text{A}|}$$

$$\Rightarrow \text{pH} = \text{pK}_a \Rightarrow \text{pK}_a = 5.0$$

Now, at equivalence point,

$$\text{PH} = \frac{1}{2}(\text{pK}_w + \text{pK}_a + \log C)$$

$$9 = \frac{1}{2}(14 + 5 + \log C)$$

$$\Rightarrow C = 0.1 \text{ M}$$

Let, V mL of NaOH is used

$$\text{Meq. (NaOH)} \Rightarrow \frac{0.25 \times V}{60 + V} = 0.1 = \text{Meq. (salt)}$$

$$\therefore \text{Volume (V)} = 40 \text{ mL}$$

$$\text{Meq. of acid} = \text{millimoles of acid} = 0.25 \times 40 = 10$$

$$\text{Mass of acid (HX)} = 10 \times 10^{-3} \times 82 = 0.829$$

$$\text{Mass\%} = \frac{0.82}{1.025} \times 100 = 80\%$$

60. 1

Sol. $[\text{H}^+]_{\text{total}} = 25 \times 0.1 + 50 \times 0.2 = 12.5 \text{ milimoles}$

$$[\text{OH}^-]_{\text{total}} = 25 \times 0.1 = 2.5 \text{ milimoles}$$

$$\therefore [\text{H}^+]_{\text{final}} = \frac{12.5 - 2.5}{25 + 50 + 25}$$

$$[\text{H}^+]_{\text{final}} = \frac{10}{100} \text{ M} = \frac{1}{10} \text{ M}$$

$$\therefore \text{pH} = -\log [\text{H}^+]_{\text{final}}$$

$$\text{pH} = -\log \left(\frac{1}{10} \right) = 1.00$$

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Mathematics

PART – C

SECTION – A

61. C

Sol. $f'(x) + f(x) = 12x^2 + 2$

62. B

Sol. Use expansion and collect the coefficient of x^2

$\Rightarrow P = 330$

63. A

Sol. For A $x - \frac{x^3}{6} < \sin x < x$ and for B use limit as a sum

$\Rightarrow A = \frac{1}{2}, B = \frac{2}{\pi}$

64. D

Sol. $f'(x) = \frac{-1}{(\sin^2 x - (x-a)^2)^2} (\sin 2x - 2(x-a)) < 0 \quad \forall x \in (0, 1) \text{ and } a \in (2, 3)$

65. C

Sol. $A - B = \int_0^1 f(x) dx + \int_{f(0)}^{f(1)} f^{-1}(x) dx = 0$, and also $A = B$.

66. D

Sol. Domain is $\mathbb{R}$.

67. D

Sol. $A = \int_0^{\pi/4} (\tan x - \sin x) dx = \left(\frac{1}{2} \ln 2 + \frac{1}{\sqrt{2}} - 1 \right)$

68. A

Sol. $I_1 = \frac{\pi^2}{2} + 2, I_2 = \frac{\pi^2}{2} - 2, I_3 < 1$

69. D

Sol. $f'(x) = (\ln x)^2, g'(x) = (\ln x)$, so area $= \int_1^e (\ln x - (\ln x)^2) dx = 3 - e$

70. D

Sol. $L_1 = L_2 = \frac{\pi}{2}$

71. B

Sol. $T_n = \frac{((n+1)x+3) - (nx+3)}{((n+1)x+3)(nx+3)} = \frac{1}{nx+3} - \frac{1}{(n+1)x+3}$

$S_{12} = V_1 - V_2 + V_2 - V_3 + \dots V_{12} - V_{13} = V_1 - V_{13}$

$$= \frac{1}{x+3} - \frac{1}{13x+3}$$

$$\lim_{x \rightarrow 14} \frac{1}{x+3} - \frac{1}{13x+3} = \frac{3}{16}$$

72. A

Sol. $\tan \theta = \frac{5}{12} = \frac{r}{h} \Rightarrow r = \frac{5h}{12}$

$$\frac{dv}{dt} = 0.1 \text{ m}^3/\text{min}$$

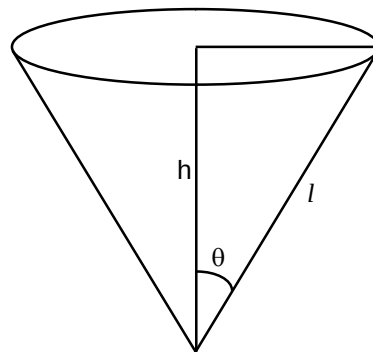
$$v = \frac{1}{3} \pi r^2 h = \frac{\pi 25 h^3}{432}$$

$$\Rightarrow \frac{dv}{dt} = \frac{25\pi h^2}{144} \frac{dh}{dt} \Rightarrow 0.1 = \frac{25}{144} \times \pi \times 3 \times 3 \times \frac{dh}{dt}$$

$$\Rightarrow \frac{16}{250} \pi = \frac{dh}{dt}$$

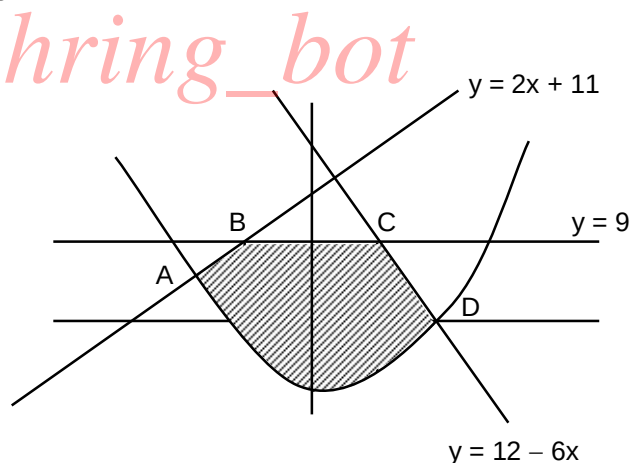
$$\text{Now } s = \pi r l = \pi r \sqrt{h^2 + r^2} = \frac{65\pi h^2}{144}$$

$$\frac{ds}{dt} = \frac{65\pi}{144} \times 2h \times \frac{dh}{dt} = \frac{65\pi}{144} \times 2 \times 3 \times \frac{16}{250\pi} = \frac{13}{75} \text{ m}^2/\text{min}$$



73. A

Sol. $x^2 - 4 = 2x + 11$
 $x = -3, 5 \therefore A(-3, 5)$
 $2x + 11 = 9$
 $x = 1 \therefore B(-1, 9)$
 $C\left(\frac{1}{2}, 9\right)$ and $D(2, 0)$



$$\text{Required area} = \int_{-3}^{-1} (2x + 11) - (x^2 - 4) dx + \int_{-1}^{1/2} 9 - (x^2 - 4) dx + \int_{1/2}^2 12 - 6x - (x^2 - 4) dx$$

$$= \int_{-3}^{-1} (2x - x^2 + 15) dx + \int_{-1}^{1/2} 13 - x^2 dx + \int_{1/2}^2 (16 - 6x - x^2) dx$$

$$= x^2 - \frac{x^3}{3} + 15x \Big|_{-3}^{-1} + 13x - \frac{x^3}{3} \Big|_{-1}^{1/2} + 16x - 3x^2 - \frac{x^3}{3} \Big|_{1/2}^2$$

$$= \frac{40}{3} + \frac{153}{8} + \frac{81}{8} = \frac{511}{12}$$

74. B

Sol. $f'(t) = \frac{e^{t^2} t^5}{(t^4 + 2t^2 + 2)^2} \Rightarrow f'(1) = \frac{e}{25}$

$$\begin{aligned} \text{Let } x^2 = v &\Rightarrow \frac{dv}{dx} = 2x \Rightarrow \frac{1}{2} \int \frac{e^v v^2}{(v^2 + 2v + 2)^2} dv \\ &\Rightarrow \frac{1}{2} \int e^v \left(\frac{1}{v^2 + 2v + 2} - \frac{2v + 2}{(v^2 + 2v + 2)^2} \right) = \frac{1}{2} \frac{e^v}{v^2 + 2v + 2} \\ &= \frac{1}{2} \frac{e^{x^2}}{x^4 + 2x^2 + 2} \\ \therefore f(t) &= \frac{1}{2} \frac{e^{x^2}}{x^4 + 2x^2 + 2} \Big|_0^t = \left(\frac{e^{t^2}}{t^4 + 2t^2 + 2} - \frac{1}{2} \right) \frac{1}{2} \\ \therefore f(1) &= \frac{e}{10} - \frac{1}{4} \\ \therefore f(1) + f'(1) &= \frac{e}{10} + \frac{e}{25} - \frac{1}{4} = \frac{7e}{50} - \frac{1}{4} \end{aligned}$$

75. B

Sol. $\lim_{x \rightarrow \infty} \frac{(k-2)x^3 + (3+n)x^2 + (m-3)x + (2-p)}{\sqrt{x^4 + kx^3 + 3x^2 + mx + 2} + \sqrt{x^4 + 2x^3 - nx^2 + 3x - p}}$

$k-2$ has to be 0 else limit wont be finite

$\Rightarrow k = 2$

Now dividing Num. & Den by x^2

$$\begin{aligned} &\lim_{x \rightarrow \infty} \frac{(3+n) + \frac{m-3}{x} + \frac{(2-p)}{x^2}}{\sqrt{1 + \frac{a}{x} + \frac{3}{x^2} + \frac{b}{x^3} + \frac{2}{x^4}} + \sqrt{1 + \frac{2}{x} - \frac{c}{x^2} + \frac{3}{x^3} - \frac{d}{x^4}}} \\ &= \frac{3+n}{2} \Rightarrow \frac{3+n}{2} = 4 \Rightarrow n = 5. \end{aligned}$$

76. A

Sol. Let $x^2 = t \Rightarrow \frac{dt}{dx} = 2x \Rightarrow I = \frac{1}{2} \int_0^{2\pi} \frac{t \sin^{2022} t \, dt}{\cos^{2022} t + \sin^{2022} t}$

$$\begin{aligned} \Rightarrow I &= \frac{1}{2} \int_0^{2\pi} \frac{(2\pi - t) \sin^{2022} t}{\cos^{2022} t + \sin^{2022} t} \Rightarrow 2I = \int_0^{2\pi} \frac{\pi \sin^{2022} t}{\cos^{2022} t + \sin^{2022} t} \\ &\Rightarrow 2I = 4 \int_0^{\frac{\pi}{2}} \frac{\pi \sin^{2022} t}{\cos^{2022} t + \sin^{2022} t} \\ &\Rightarrow 2I = 4 \int_0^{\frac{\pi}{2}} \frac{\pi \cos^{2022} t}{\cos^{2022} t + \sin^{2022} t} \\ &\Rightarrow 4I = 4 \int_0^{\frac{\pi}{2}} \pi \, dt \Rightarrow I = \frac{\pi^2}{2}. \end{aligned}$$

77. A

Sol. $\frac{dy}{dx} = \frac{(x-2)+(y-2)}{(x-2)-(y-2)}$

put $x-2 = h, y-2 = k$

$\frac{dk}{dh} = \frac{h+k}{h-k}$ put $k = vh$

$\Rightarrow v + h \frac{dv}{dh} = \frac{1+v}{1-v}$

$\Rightarrow \int \frac{1-v}{1+v^2} dv = \int \frac{dh}{h} \Rightarrow \tan^{-1} v = \frac{1}{2} \ln(1+v^2) + \ln h + c$

$\Rightarrow \tan^{-1} \left(\frac{y-2}{x-2} \right) = \frac{1}{2} \ln \left(1 + \frac{(y-2)^2}{(x-2)^2} \right) + \ln(x-2) + c$

$\therefore$ it passes through (3, 2)

$\Rightarrow \tan^{-1} 0 = \frac{1}{2} \ln 1 + \ln 1 + c \Rightarrow c = 0$

$\therefore$ it also passes through (p+2, 3)

$\tan^{-1} \left(\frac{1}{p} \right) = \frac{1}{2} \ln \left(1 + \frac{1}{p^2} \right) + \ln p$

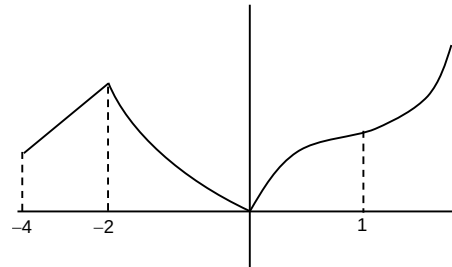
$\Rightarrow 2 \tan^{-1} \left(\frac{1}{p} \right) = \ln(1+p^2)$

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78. B

Sol. $A = \int_{-4}^{-2} (x+6) dx + \int_{-2}^0 x^2 dx + \int_0^1 \sqrt{x} dx + \int_1^2 x^2 dx$

$$= 6 + \frac{8}{3} + \frac{2}{3} + \frac{7}{3} = \frac{35}{3}$$



79. C

Sol. $f(e^4) = \int_1^{e^4} \frac{\log_{10} x}{1+x} dx \quad \dots(1)$

Let $x = \frac{1}{v} \Rightarrow \frac{dx}{dv} = \frac{-1}{v^2}$

$\Rightarrow f(t) = \int_1^{1/t} \frac{\log_{10}(1/v)}{1+\frac{1}{v}} - \frac{1}{v^2} dv = \int_1^{1/t} \frac{\log_{10} v}{v(v+1)} dv$

$\Rightarrow f(t) = \int_1^{1/t} \frac{\log_{10} v}{v(v+1)} dv \Rightarrow f(e^{-4}) = \int_1^{e^4} \frac{\log_{10} v}{v(v+1)} dv = \int_1^{e^4} \frac{\log_{10} x}{x(x+1)} dx \quad \dots(2)$

Add (1) and (2)

$f(e^{-4}) + f(e^4) = \frac{1}{\ln 10} \int_1^{e^4} \frac{\ln x}{(1+x)} \left(1 + \frac{1}{x} \right) dx$

$$= \frac{1}{\ln 10} \int_1^{e^4} \frac{\ln x}{x} = \frac{1}{\ln 10} \left. \frac{(\ln x)^2}{2} \right|_1^{e^4}$$

$$= \frac{1}{2 \ln 10} 16 = \frac{8}{\ln 10}$$

80. C

Sol. Let $g(x) = x^3 - 2x^2 + 3x - 9$
 $g'(x) = 3x^2 - 4x + 3 > 0 \quad \forall x \in \mathbb{R}$
 $\Rightarrow g(x)$ is always increasing
 $\therefore g(2) = 8 - 8 + 6 - 9 = -3$ is maximum value for $x \leq 2$
 If $-3x + \log^3(k^2 - 5) \leq -3$ for $x > 2$
 then we have maximum at $x = 2$
 $\Rightarrow -6 + \log^3(k^2 - 5) \leq -3 \Rightarrow \log^3(k^2 - 5) \leq 3$
 $\Rightarrow k^2 - 5 \leq 27 \Rightarrow (k - 4\sqrt{2})(k + 4\sqrt{2}) \leq 0$
 $k \in [-4\sqrt{2}, 4\sqrt{2}]$ and $k^2 - 5 > 0$
 $\Rightarrow k \in (-\infty, -\sqrt{5}) \cup (\sqrt{5}, \infty)$
 $\therefore k \in [-4\sqrt{2}, -\sqrt{5}) \cup (\sqrt{5}, 4\sqrt{2}]$

TG ~ @bohring_bot SECTION - B

81.

1
Sol. $\left[\frac{x^2 + x + 2}{x^2 + x + 1} \right] = 1$

82.

1
Sol. $A = \frac{\pi^3}{8} - \frac{\pi^2}{4}$

83.

4
Sol. $D_f = [-2, 2]$ as $f(x) = \sqrt{3}(\sqrt{4 - x^2})$
 $R_f = \in [0, 2\sqrt{3}]$

84.

2
Sol. $f(x) = \cos(x^2 - 3x) |x - 2| \sin|x - 2| + (x - 1) |x| |x - 1| |x - 7|$
 $\therefore f(x)$ non differentiable at $x = 0$ and 7 only.

85.

12
Sol. $a = 1, b = 6, c = 5$

86.

96
Sol. Put $x = 0$ we get $y = 1$
 $\ln(x + y) = 6xy$
 Differentiate w.r.t. x
 $\frac{1}{x + y} \left(1 + \frac{dy}{dx} \right) = 6 \left(\frac{xdy}{dx} + y \right)$

at $x = 0, y = 1 \Rightarrow \frac{dy}{dx} = 5$

$$\Rightarrow 1 + \frac{dy}{dx} = 6(x + y) \left(x \frac{dy}{dx} + y \right)$$

Differentiate w.r.t. x both side

$$\begin{aligned} \Rightarrow \frac{d^2y}{dx^2} &= 6 \left((x + y) \left(\frac{xd^2y}{dx^2} + \frac{dy}{dx} + \frac{dy}{dx} \right) + \left(\frac{xdy}{dx} + y \right) \left(1 + \frac{dy}{dx} \right) \right) \\ &= 6(1(0 + 5 + 5) + 1(6)) = 96 \end{aligned}$$

87. 9

Sol. $\therefore \lim_{x \rightarrow 0} \left(3 - 2\cos x \sqrt{\cos 2x} \right)^{\left(\frac{x+3}{x^2} \right)}$ is of the form 1^∞

$\therefore$ it is equal to $e^{\lim_{x \rightarrow 0} \left(\frac{2 - 2\cos x \sqrt{\cos 2x}}{x^2} \right) (x+3)}$

Now, $\lim_{x \rightarrow 0} \frac{2 - 2\cos x \sqrt{\cos 2x}}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{\frac{-2\cos x}{2\sqrt{\cos 2x}} \times -\sin 2x \times 2 - \sqrt{\cos 2x} (-\sin x)}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{2\cos x \sin 2x}{\sqrt{\cos 2x}} + 2\sin x \sqrt{\cos 2x}}{2x}$$

$$= 2 + 1 = 3$$

$\therefore e^{\lim_{x \rightarrow 0} \left(\frac{2 - 2\cos x \sqrt{\cos 2x}}{x^2} \right) (x+3)}$

$$= e^{3 \times 3} = e^9 \therefore a = 9$$

88. 14

Sol. $\frac{dy}{dx} = 3x^2 - 4x + 4 \Rightarrow \left(\frac{dy}{dx} \right)_{(a,b)} = 3a^2 - 4a + 4$

$$\frac{dy}{dx} = 6x + 2 \Rightarrow \left(\frac{dy}{dx} \right)_{(1,4)} = 8$$

$$\Rightarrow 3a^2 - 4a + 4 = 8 \Rightarrow a = 2, -\frac{2}{3}$$

$$\therefore b = 8 - 8 + 8 + 4 = 12$$

89. 144

Sol. $\int \frac{12x^{15} + 5x^{18}}{x^{24} (-7 + 3x^{-8} + 2x^{-5})^3} dx = \int \frac{12x^{-9} + 5x^{-6}}{(-7 + 3x^{-8} + 2x^{-5})^3}$

Let $-7 + 3x^{-8} + 2x^{-5} = t \Rightarrow \frac{dt}{dx} = -24x^{-9} - 10x^{-6}$

$$= -\frac{1}{2} \int \frac{1}{t^3} dt = -\frac{1}{2} \frac{t^{-2}}{-2} + c = \frac{1}{4(-7 + 3x^{-8} + 2x^{-5})^2} + c$$

$$\therefore f(1) = \frac{1}{16} \Rightarrow \frac{1}{16} = \frac{1}{4(-7 + 3 + 2)^2} + c \Rightarrow c = 0$$

$$\therefore f(x) = \frac{1}{4(-7 + 3x^{-8} + 2x^{-5})^2} \quad \therefore f(-1) = \frac{1}{4(-6)^2} = \frac{1}{144}$$

90. 23

Sol. $I_n = \int_1^e x^{23} \left(\log x \right)^n dx$

$$\Rightarrow I_n = (\ln x)^n \frac{x^{24}}{24} \Big|_1^e - \int_1^e n \frac{(\ln x)^{n-1}}{x} \frac{x^{24}}{24} dx$$

$$\Rightarrow I_n = \frac{e^{24}}{24} - \frac{n I_{n-1}}{24} \Rightarrow 24 I_n = e^{24} - n I_{n-1}$$

Put $n = 15$ $24 I_{15} = e^{24} - 15 I_{14}$... (1)
 Put $n = 14$ $24 I_{14} = e^{24} - 14 I_{13}$... (2)
 $24 I_{15} - 24 I_{14} = 14 I_{13} - 15 I_{14}$
 $24 I_{15} = 9 I_{14} + 14 I_{13}$

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FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – I
PAPER –1
TEST DATE: 20-11-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

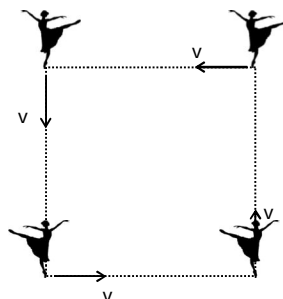
Physics

PART – I

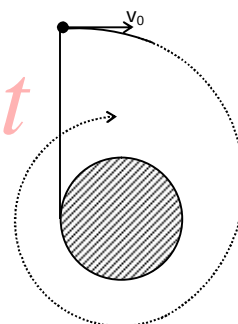
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

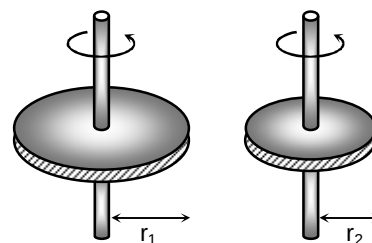
1. Four dancers start their dance by forming a square with side $\ell = 9\text{m}$. After that, they always move anticlockwise towards their neighbour with variable velocity $v = k\sqrt{x}$ where $k = 0.1 \text{ m}^{1/2}\text{sec}^{-1}$ and x is the distance of the closest neighbour in meter. Then choose the correct option(s).



- (A) All the dancers will meet with each other at $t = 60$ sec after they start their dance.
 (B) Angular velocity of the line joining the adjacent dancers increases with time.
 (C) All the dancers will meet with each other at $t = 63$ sec after they start their dance.
 (D) Magnitude of acceleration of each dancer is remains constant with time.
2. A point object of mass $m = 2\text{kg}$ is connected to a cylinder of radius $R = 1\text{m}$ via a massless inextensible rope. At time $t = 0$ the object is moving with an initial velocity $v_0 = 5 \text{ m/s}$ perpendicular to the rope as shown in the figure. The rope has initial length $L_0 = 10 \text{ m}$ and having non zero tension. Figure shows the top view and the motion of the object is taking place on the horizontal smooth surface and the cylinder is always fixed. The point object moves such that the rope wraps up around the cylinder. The rope will break when the tension in the rope exceeds 100 N . Then choose the correct option(s).

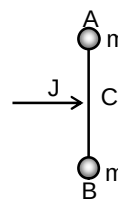


- (A) Kinetic energy of the object at the instant when the rope breaks is 25 J .
 (B) Angular momentum of the object about the axis of the cylinder at the instant when the rope breaks is less than $5 \text{ kg-m}^2/\text{sec}$.
 (C) Angular momentum of the object about the axis of the cylinder at the instant when the rope breaks is equal to $5 \text{ kg-m}^2/\text{sec}$.
 (D) Angular momentum of the object about the axis of the cylinder at the instant when the rope breaks is greater than to $5 \text{ kg-m}^2/\text{sec}$.
3. Two discs are mounted on thin, lightweight rods oriented through their centres and normal to the discs. These axes are constrained to be vertical at all times, and the discs can pivot frictionlessly on the rods. The discs have identical thickness and are made up of the same material, but having different radii r_1 and r_2 . The discs are given angular velocities ω_1 and ω_2 , respectively, and L_1 and L_2 represents angular momentum of the discs along their rotational axis respectively. Now they are brought into contact at the edges. After the discs interact via friction it is found that both discs come exactly to a halt. Which of the following(s) must hold? Ignore effects associated with the vertical rods.

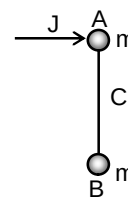


- (A) $\omega_1 r_1^4 = \omega_2 r_2^4$ (B) $\omega_1 r_1^3 = \omega_2 r_2^3$
 (C) $\frac{L_1}{L_2} = \frac{r_1}{r_2}$ (D) $\frac{L_1}{L_2} = \frac{\omega_1 r_1^2}{\omega_2 r_2^2}$

4. A dumbbell is formed by connecting two identical particles each of mass m with a massless rod. The system is placed on a horizontal smooth surface and impulse J is applied in two ways as shown in the figure as situation-(i) and situation-(ii). Then just after the application of impulse, choose the correct option(s).



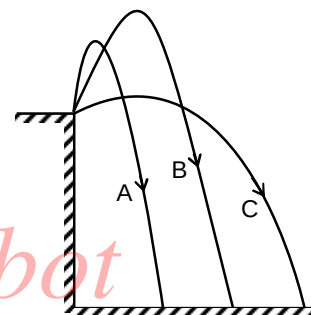
Situation-(i)



Situation-(ii)

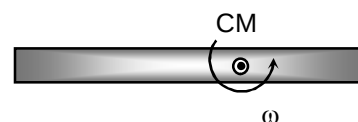
- (A) Velocity of centre of mass with respect to ground frame in situation-(i) and situation-(ii) are same.
 (B) Kinetic energy of the dumbbell with respect to centre of mass frame in situation-(i) and situation-(ii) are same.
 (C) Angular momentum of the dumbbell with respect to centre of mass frame in situation-(i) and situation-(ii) are different.
 (D) Velocity of particle A in situation-(ii) is double of velocity of particle A in situation-(i) just after the application of impulse.

5. Three particles A, B and C are simultaneously thrown from a cliff and their trajectories are shown in the given figure. Neglect air resistance and take gravity to be uniform during the motion. If u_A^y , u_B^y and u_C^y denotes the initial vertical components of velocity of particles A, B and C respectively and the times t_A , t_B and t_C , they take to hit the ground respectively. Then



- (A) $u_B^y > u_C^y > u_A^y$ (B) $u_B^y > u_A^y > u_C^y$
 (C) $t_A < t_C < t_B$ (D) $t_C < t_A < t_B$

6. A thin rod has an unknown non-uniform density. It is mounted on an axle passing perpendicular to it, through its centre of mass as shown and is then rotated about the axle. The axle divides the rod into two parts, one on each side of it. Which of the following must be true, no matter how the mass in the rod is distributed?

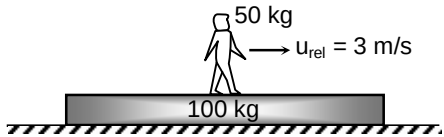
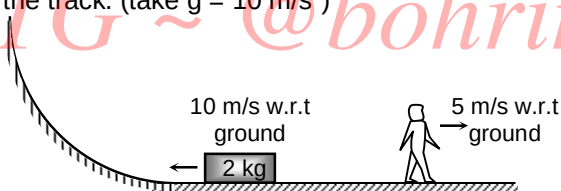
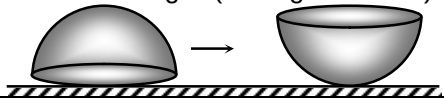


- (A) The two part must have the same mass.
 (B) The magnitudes of the linear momentum of the two parts must be equal.
 (C) The magnitudes of the angular momentum of the two parts about the centre of mass must be equal.
 (D) The kinetic energies of two parts must be equal.

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

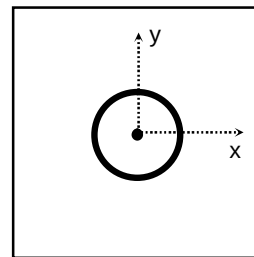
7. List –I describes for systems and List-II gives possible value of work done in Joules.

List –I		List -II	
(I)	A plank of mass 100 kg is placed on a smooth horizontal surface at rest along with a man. Now man starts moving on the plank with relative velocity of 3 m/s with respect to plank. The work done by man is 	(P)	+100J
(II)	A block of mass 2 kg is placed on a smooth track and imparted a velocity of 10 m/s to the left and observed by an observer moving with 5 m/s as shown in the figure. Work done by the normal force applied by the track on the block upto the instant when the block reaches the highest point on the track in the frame of observer. Assume that block never loose contact with the track. (take $g = 10 \text{ m/s}^2$) 	(Q)	–100J
(III)	A particle of mass 1 kg is projected from an unstable equilibrium point N towards a stable equilibrium point S with negligible velocity in a conservative force field having potential energy function $U = -\left(\frac{1}{100} \text{ J/m}^4\right)[x-10]^2[x-30]^2, \text{ where } x \text{ is in meters.}$ Then work done by conservative force during the displacement from N to S is	(R)	+150J
(IV)	A solid hemisphere of mass 4 kg and radius 10 m is slowly turned as shown in the figure. Then work done by external force in turning is (Take $g = 10 \text{ m/s}^2$) 	(S)	–150J
		(T)	Zero

Which one of the following options is correct ?

- (A) I $\rightarrow$ R, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ P
 (B) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ P
 (C) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ P
 (D) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ Q, IV $\rightarrow$ P

8. A disc placed on a frictionless horizontal plank is rotating with a constant angular velocity $\vec{\omega}$ about its vertical axis. Now the plank is made to move with a constant acceleration $\vec{a}$ on a straight path. Initially the centre of the disc is at the origin of xyz space which is fixed with the plank and xy plane is on the plank. List-I gives some possible $\vec{a}$, $\vec{\omega}$ and List-II gives trajectories of instantaneous centre of rotation of the disc.

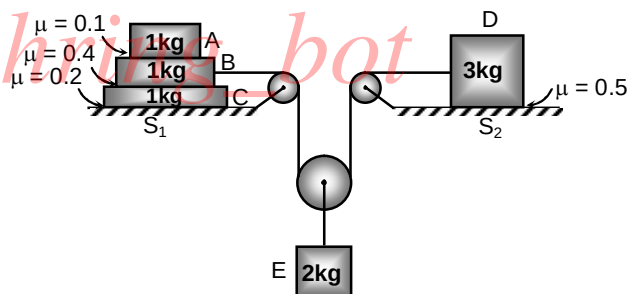


List -I		List -II	
(I)	$\vec{a} = -2\hat{j} \text{ m/s}^2, \vec{\omega} = 4\hat{k} \text{ rad/s}$	(P)	$y = 4$
(II)	$\vec{a} = -2\hat{j} \text{ m/s}^2, \vec{\omega} = -4\hat{k} \text{ rad/s}$	(Q)	$y = 4x^2 (x \leq 0)$
(III)	$\vec{a} = -2\hat{i} \text{ m/s}^2, \vec{\omega} = 4\hat{k} \text{ rad/s}$	(R)	$y = 4x^2 (x \geq 0)$
(IV)	$\vec{a} = -2\hat{i} \text{ m/s}^2, \vec{\omega} = -4\hat{k} \text{ rad/s}$	(S)	$x = 4y^2 (y \geq 0)$
		(T)	$x = 4y^2 (y \leq 0)$

Which one of the following options is correct ?

- (A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ S
 (B) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ T
 (C) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ T, IV $\rightarrow$ R
 (D) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ T

9. Figure shows a system of multiple blocks placed one on each other and connected through an inextensible string. All the pulleys are ideal and frictionless. List-I shows the surfaces between which the friction is asked and List-II gives the value of friction acting between the surfaces. (Take $g = 10 \text{ m/s}^2$)

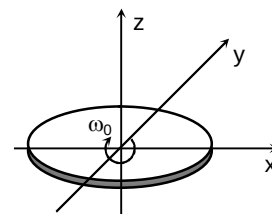


List -I		List -II	
(I)	Between A and B	(P)	15 N
(II)	Between B and C	(Q)	9.4 N
(III)	Between C and S_1	(R)	7.2 N
(IV)	Between D and S_2	(S)	6 N
		(T)	1 N

Which one of the following options is correct ?

- (A) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ Q
 (B) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ P
 (C) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ Q
 (D) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ P

10. A disc of mass m is placed over a horizontal smooth surface (xy plane) such that the centre of the disc is at the origin of the xyz space, as shown in the figure. Radius of the disc is $R = 1\text{m}$ and is rotating with an angular velocity $\vec{\omega}_0 = -6\hat{k}$ rad/s and having zero velocity of centre of mass. A particle of same mass m hits the disc with a speed $|\vec{v}| = 8\text{ m/s}$ in each case of list-I. List-I gives the trajectory and velocity direction of the particle before hitting the disc and the coefficient of restitutions (e) of the collision. List-II gives the angular velocity of the disc just after the collision (in rad/s)



List -I		List -II	
(I)	$x = 0, y = R$ and $\hat{v} = -\hat{k}, e = 0$	(P)	3
(II)	$y = \frac{x}{\sqrt{3}}, z = 0$ and $\hat{v} = \frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}, e = 1$	(Q)	5
(III)	$y = R, z = 0$ and $\hat{v} = \hat{i}, e = 0$	(R)	6
(IV)	$y = \frac{R}{2}, z = 0$ and $\hat{v} = \hat{i}, e = 0$	(S)	7
		(T)	8

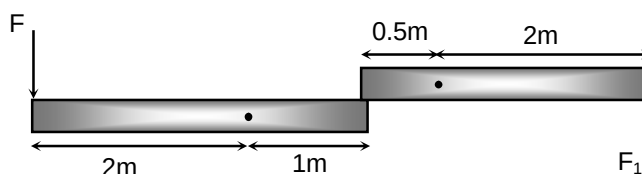
Which one of the following options is correct ?

- (A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ T
 (B) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ R
 (C) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ Q
 (D) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ T

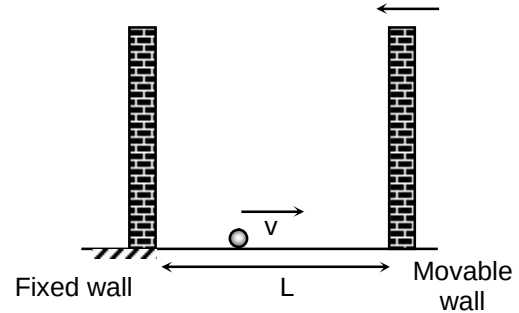
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

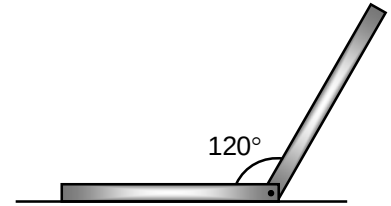
11. Two massless rods are attached to frictionless pivots, with their ends touching. The distances between the pivot points and the endpoints of the rods are shown in the figure. Neglect friction force between the rods, if a force $F = 9\text{N}$ is applied at the left end of the left rod, what force F_1 (in newton) must be applied at the right end of the right rod to keep the system in equilibrium.



12. A small sphere slides with a speed $v = 60 \text{ m/s}$ on a frictionless horizontal surface between two parallel massive walls. One of which is fixed and the other can be moved. Initial distance between walls is L . The sphere bounces between the walls perfectly elastically. Now the right wall is moved towards the left wall by a distance $0.01 L$ with speed $0.0001v$. Then what is the final speed of the sphere after the said movement of the wall in m/s .

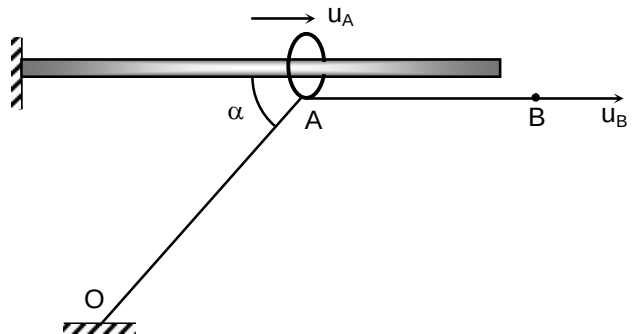


13. Figure shows a laptop made of two identical uniform plates, each of mass $m = 1 \text{ kg}$ connected by a hinge and placed on a rough horizontal surface as shown. The hinge is locked when the screen makes an angle of 120° from the keypad of the laptop. Assume that the laptop does not slide on the horizontal surface, then what is the minimum force (in newton) that can be applied on the top of the laptop, in the plane of the page to cause the bottom of the laptop to lift off the ground? (Take $g = 10 \text{ m/s}^2$)

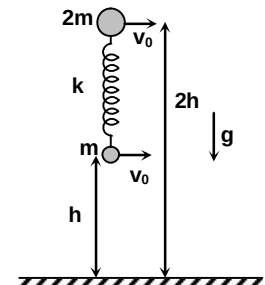


14. A block of mass $m = 1 \text{ kg}$ is dropped vertically onto a fixed ramp from a height of $h = 2.5 \text{ m}$ with angle $\theta = 37^\circ$ with horizontal and having coefficient of static friction and kinetic friction $\mu = 0.5$. The block is dropped in such a way that it does not rotate after colliding with the ramp. Through this problem assume that the time of the collision is negligible. The block slides down the ramp immediately after impact. What is the speed (in m/s) of the block right after the collision? (Take $g = 10 \text{ m/s}^2$)

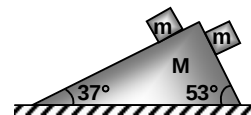
15. One end of the non stretchable rope is tied to a fixed knob O , while other end B of the rope is moving horizontally with a constant velocity $u_B = 1 \text{ m/s}$. That rope goes through a small ring A , which can move freely along a horizontal straight rod as shown in the figure. Determine velocity u_A of the ring (in m/s), when inclined section of the rope makes an angle $\alpha = 30^\circ$ with the rod.



16. Two small bodies with masses m and $2m$ are attached to a spring of negligible mass. Initially the compressed spring forms a straight vertical line, when both objects are launched with same velocities $v_0 = 1 \text{ m/s}$ in the horizontal direction as shown in the figure. Initial height of the lower body is $h = 1 \text{ m}$ from the ground, while distance of heavier body is $2h = 2 \text{ m}$. After $t = 0.1 \text{ sec}$ from the launch, lower body hits the ground. Determine height H (in m) of the upper body from the ground at that moment of time. (assume no air drag and $g = 10 \text{ m/s}^2$)



17. A right triangular wooden block of mass $M = 4 \text{ kg}$ is at rest on a smooth table as shown in the figure. Two smaller wooden cubes both of mass $m = 2 \text{ kg}$, initially at rest on the two sides of the larger block start sliding down. What is the normal force (in newton) applied by the system on the table? (take $g = 9.8 \text{ m/s}^2$)



18. The 'wall of death' ride consists of a drum with diameter $d = 10 \text{ m}$ which rotates at an angular velocity ω as shown in the figure-(a). Passengers stand against the inside vertical rough wall and as the ride speeds up to angular velocity ω , they are held against the wall and can not slide down due to friction with the wall. The rotating drum is now tilted so that its axis of rotation is $\theta = 37^\circ$ above the horizontal as shown in the figure-(b). If the coefficient of friction between the wall and the passenger is $\mu = 0.5$, then find the minimum angular velocity ω_0 (in rad/s) of the drum so that passengers are held against the wall at the highest point. (Take $g = 10 \text{ m/s}^2$)

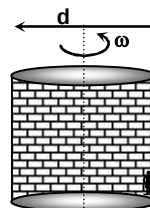


Figure-(a)

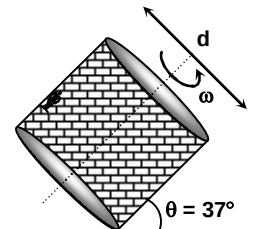


Figure-(b)

TG ~ @bohring_bot

Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

19. Which of the following species are polar as well as non-planar?
 (A) XeF_4
 (B) H_2O_2
 (C) $\text{XeF}_5^\ominus$
 (D) $\text{XeF}_5^\oplus$
20. All bonds are equivalent in
 (A) $\text{Cr}_2\text{O}_7^{2-}$
 (B) HCO_3^-
 (C) XeO_6^{4-}
 (D) SF_6
21. The radial part $R(r) = \frac{1}{9\sqrt{\sigma a_0^{3/2}}} \left(\frac{6r}{a_0} - \frac{r^2}{a_0^2} \right) e^{-r/3a_0}$ represent for an orbital. (a_0 is Bohr's radius of H-atom)
 Choose the correct statement about the orbital
 (A) this orbital may have one nodal plane in yz plane
 (B) it is 3p orbital
 (C) one node will be found at the $r = 6a_0$ distance from nucleus
 (D) this orbital can accommodate maximum 6 electrons
22. Which of the following geometry is most likely form from sp^3d – hybridisation ?
 (A) Linear
 (B) Tetrahedral
 (C) Bent T-shaped
 (D) See-Saw
23. Find the incorrect statement
 (A) Iodine exists as I_8^{2-} ion in Cs_2I_8
 (B) Oxidation state of Cr in CrO_5 and CrO_8^{3-} is +6
 (C) FeCl_2 is more ionic than FeCl_3
 (D) LiO_2 and NaO_2 are yellow coloured compound but KO_2 is colourless
24. Consider the following options and choose the correct one
 (A) In the given equilibrium $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$
 If O_2 is added and the volume of the reaction vessel is reduced the equilibrium is shift in product side.
 (B) If we add CrO_4^{2-} ion in saturated solution of Ag_2CrO_4 it will decreases $[\text{Ag}^+]$.

- (C) Three sparingly soluble salt A_2B , AB and AB_3 are given. If all the three having same value of solubility product (K_{SP}). In the saturated solution the correct order of their solubilities is $AB > A_2B > AB_3$.
- (D) If in option (A) inert gas is added at constant pressure and temperature. Partial pressure of SO_3 will be remain same.

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. If t_x represents time required for a fraction x of a reactant to be reacted during a reaction involving only one reactant. Then match List – I with List – II.

List – I		List – II	
(I)	Zero order reaction	(P)	$\frac{t_{3/4}}{t_{1/2}} = 3$
(II)	First order reaction	(Q)	$\frac{t_{1/4}}{t_{1/8}} = 2$
(III)	Second order reaction	(R)	$\frac{t_{7/8}}{t_{3/4}} = \frac{3}{2}$
(IV)	Order of reaction ≥ 1	(S)	$\alpha = (1 - e^{-kt})$
		(T)	$\frac{t_{100\%}}{t_{50\%}} = 2$

($\alpha \rightarrow$ degree of dissociation, $k \rightarrow$ rate constant)

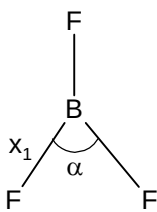
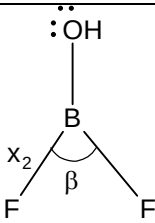
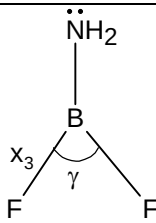
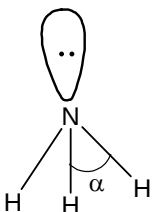
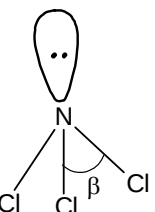
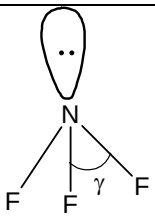
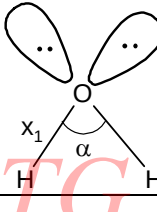
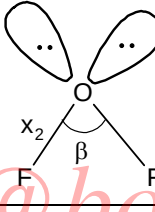
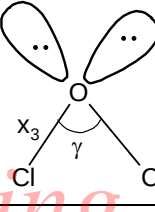
- (A) I $\rightarrow$ Q, T; II $\rightarrow$ R, S; III $\rightarrow$ P; IV $\rightarrow$ P, R, S
 (B) I $\rightarrow$ Q; II $\rightarrow$ R, T; III $\rightarrow$ P; IV $\rightarrow$ P, R
 (C) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ R, S; IV $\rightarrow$ Q, T
 (D) I $\rightarrow$ R, T; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ P, Q

26. List – I contains properties and List – II contains moles or ions

List – I		List – II	
(I)	2 lone pair on central atom	(P)	XeF_5^-
(II)	Zero dipole moment	(Q)	NF_3
(III)	Planar	(R)	ICl_3
(IV)	All adjacent bond angle are equal	(S)	XeF_4
		(T)	PCl_3F_2

- (A) I $\rightarrow$ P, R, S; II $\rightarrow$ S, T; III $\rightarrow$ P, Q, R, S; IV $\rightarrow$ P, Q, R, S, T
 (B) I $\rightarrow$ P, R, S; II $\rightarrow$ P, S, T; III $\rightarrow$ P, R, S; IV $\rightarrow$ P, Q, R, S
 (C) I $\rightarrow$ P, R, T; II $\rightarrow$ R, S, T; III $\rightarrow$ R, S; IV $\rightarrow$ S, T
 (D) I $\rightarrow$ P, S, T; II $\rightarrow$ P, R, S; III $\rightarrow$ P, Q, R, S, T; IV $\rightarrow$ P, R, S

27. List – I contains compounds with bond parameter and List – II contains bond parameter decreasing order (x_1, x_2, x_3 are bond length while α, β, γ are bond angle)

List – I			List – II	
(I)	  	(P)	$x_1 > x_3 > x_2$	
(II)	  	(Q)	$\gamma > \alpha > \beta$	
(III)	  	(R)	$\alpha > \beta > \gamma$	
(IV)	(S – S bond length) S_8 S_2F_2 S_2Cl_2 x_1 x_2 x_3	(S)	$x_3 > x_2 > x_1$	
		(T)	$\gamma > \beta > \alpha$	

- (A) I $\rightarrow$ S, T; II $\rightarrow$ R; III $\rightarrow$ Q, S; IV $\rightarrow$ S
 (B) I $\rightarrow$ R, S; II $\rightarrow$ Q; III $\rightarrow$ S, T; IV $\rightarrow$ P
 (C) I $\rightarrow$ R, S; II $\rightarrow$ R; III $\rightarrow$ Q, S; IV $\rightarrow$ P
 (D) I $\rightarrow$ R, S; II $\rightarrow$ T; III $\rightarrow$ S, T; IV $\rightarrow$ S

28. List – I contains elements and compounds in respective properties and List – II contains decreasing order of properties.

List – I					List – II	
(I)	B	Al	Ga	In	(P)	$4 > 3 > 2 > 1$
	1	2	3	4		(Atomic size)
(II)	NaOH	KOH	RbOH	CsOH	(Q)	$4 > 2 > 3 > 1$
	1	2	3	4		(Thermal stability)
(III)	SiO ₂	CO ₂	CaO	Cs ₂ O	(R)	$1 > 2 > 3 > 4$
	1	2	3	4		(Basic strength)
(IV)	BeF ₂	BaF ₂	CaF ₂	MgF ₂	(S)	$4 > 3 > 1 > 2$
	1	2	3	4		(Water solubility)
					(T)	$2 > 3 > 4 > 1$

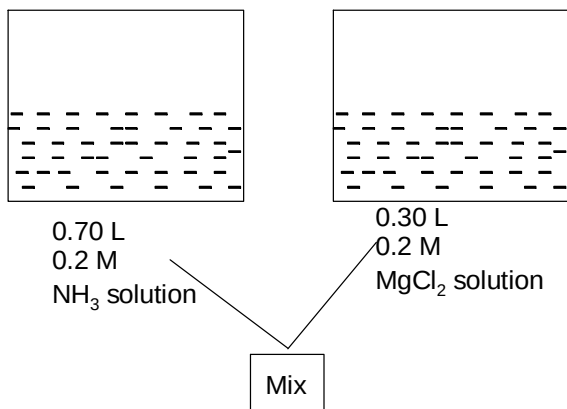
- (A) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ T
 (B) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ T
 (C) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ R
 (D) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ R

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

29. Fauzosite, a zeolite has formula $\text{Na}_x \left[(\text{AlO}_2)_y (\text{SiO}_2)_{136} \right] \cdot 250\text{H}_2\text{O}$
The value of $\frac{x}{y}$ is....
30. Calculate the pH at equilibrium point when a solution of 0.6 M CH_3COOH is titrated with a solution of 0.20 M NaOH (K_a of CH_3COOH 10^{-5} at 25°C ($\log 2 = 0.30$, $\log 3 = 0.48$))
31. In cyanuric triazide $[\text{C}_3\text{N}_3(\text{N}_3)_3]$. The number of sp^2 hybridised centre atoms is/are 'x' and number of π bond that can lie in same plane is/are 'y'. Calculate the value of $\frac{x}{y}$?
32. For reaction $\text{I}^- + \text{ClO}_3^- + \text{H}_2\text{SO}_4 \longrightarrow \text{Cl}^- + \text{HSO}_4^- + \text{I}_2$
If stoichiometric coefficient of HSO_4^- is 'x' and total number of electron gain in the balance reduction half reaction is 'y'. Calculate the value of $\frac{x+z}{y}$?
(z – Stoichiometric coefficient of I_2).
33. 2 mole of $\text{NH}_4\text{HS}(\text{s})$ is taken in close container of volume 2 litre. If 1 mole $\text{NH}_4\text{HS}(\text{s})$ taken part in reaction to attain the equilibrium state and at equilibrium moles of $\text{H}_2(\text{g})$ is 0.75.
 $\text{NH}_4\text{HS}(\text{s}) \rightleftharpoons \text{NH}_3(\text{g}) + \text{H}_2\text{S}(\text{g}) \quad K_{C_1} = x$
 $\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g}) \quad K_{C_2} = y$
At equilibrium $[\text{NH}_3] = a \text{ M}$
 $[\text{N}_2] = b \text{ M}$ calculate the value of $\frac{y_1}{x_1(a+b)}$ is...?
34. The mechanism of the reaction $\text{X} + 2\text{Y} \xrightarrow{\text{K}, \text{E}_a, \text{A}} \text{P}$ is
Step 1. $\text{X} + \text{Y} \xrightleftharpoons[\text{K}_2, \text{E}_{a_2}, \text{A}_2]{\text{K}_1, \text{E}_{a_1}, \text{A}_1} \text{I} \quad (\text{Fast})$
Step 2. $\text{I} + \text{Y} \xrightarrow{\text{K}_3, \text{E}_{a_3}, \text{A}_3} \text{P} \quad (\text{Slow})$
If the value of $\text{K}_1, \text{K}_2, \text{K}_3, \text{E}_{a_1}, \text{E}_{a_2}, \text{E}_{a_3}, \text{A}_1, \text{A}_2$ and A_3 are given below
- | K_1 | K_2 | K_3 | A_1 | A_2 | A_3 | E_{a_1} | E_{a_2} | E_{a_3} |
|--------------|--------------|--------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|
| 10^{-3} | 10^{-4} | 10^{-5} | 2×10^8 | 2×10^9 | 2×10^6 | 20 | 40 | 35 |
- Rate constant unit and pre-exponential factors for the reaction is in sec^{-1} and activation energy is in kJ. Calculate the value of $\frac{\text{K} \times \text{A}}{\text{E}_a}$ is....?

35.



What mass (in g) of $(\text{NH}_4)_2\text{SO}_4$ should be added to the above mixture cause the $\text{Mg}(\text{OH})_2$ to re-dissolve? (K_{SP} for $\text{Mg}(\text{OH})_2 = 6.0 \times 10^{-12}$ and K_b for $\text{NH}_3 = 2 \times 10^{-5}$)

36.

In a elementary chemical reaction $\text{A}(\text{aq}) \xrightarrow{t_{1/2}=10 \text{ min}} 2\text{B}(\text{aq}) + \text{C}(\text{aq})$

Reaction mixture react with another reactant T and n-factor of A, B and C react with T in the ratio 2 : 1 : 2 respectively.

If volume of 'T' used at 10 min is 40 ml. Calculated volume of T used at $t = 0$ is x ml, at $t = 20$ min, is y ml at $t = \infty$ is z ml.

Find out the value of $\frac{y+z}{x}$ is.....?

TG ~ @bohring_bot

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. If maximum value of $f(x) = \frac{3\sqrt{2x^2 + x}}{2x^2 + x + 2\sin^2 \theta}$ (where $x > 0$) is $\frac{3}{\sqrt{2}}$, then θ may lie in
- (A) $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$ (B) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$
 (C) $\left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$ (D) $(-\pi, \pi)$
38. If $\int_0^x f(t) dt = e^x - ae^{2x} \int_0^1 f(t) e^{-t} dt$, then
- (A) $a = \frac{1}{3-2e}$ (B) $f(x) = e^x - 2e^{2x}$
 (C) $a = \frac{1}{e}$ (D) $f(x) = e^x - e^{-x}$
39. If a function satisfies $(x-y)f(x+y) - (x+y)f(x-y) = 2(x^2y - y^3) \forall x, y \in \mathbb{R}$ and $f(1) = 2$, then
- (A) $|f(x)|$ is non-differentiable at two points (B) $|f(x)|$ is continuous everywhere
 (C) $f(-4) = 12$ (D) $f(4) = 18$
40. The value(s) 'a' for which the area of the triangle included between the axes and any tangent to the curve $x^a y = \lambda^a$ is constant is/are
- (A) $-\frac{1}{2}$ (B) -1
 (C) $\frac{1}{2}$ (D) 1
41. Let $f(x)$ be non-positive continuous function and $F(x) = \int_0^x f(t) dt \forall x \geq 0$ and $f(x) \geq c F(x)$ where $c > 0$ and let $g : [0, \infty) \rightarrow \mathbb{R}$ be a function such that $\frac{d(g(x))}{dx} < g(x) \forall x > 0$ and $g(0) = 0$, then
- (A) the total number of roots of the equation $f(x) = g(x)$ is 1
 (B) the total number of roots of the equation $f(x) = g(x)$ is 0
 (C) the number of solution of equation $|x^2 + x - 6| = f(x) + g(x)$ is 1
 (D) the number of solution of equation $|x^2 + x - 6| = f(x) + g(x)$ is 0
42. If $y(\sin 2x + y \cos x) + \sin x \frac{dy}{dx} (\sin x + 2y) = \sin x (1 + 2 \cos x)$ and $\left(\frac{\pi}{2}, 0\right)$ is a point on the curve, then which of the following is a point on the curve also
- (A) $\left(\frac{3\pi}{2}, -1\right)$ (B) $\left(\frac{3\pi}{2}, 0\right)$
 (C) $\left(\frac{\pi}{2}, -1\right)$ (D) none of these

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Match the following List-I with List-II

List-I		List-II	
(I)	If function f satisfies $\int_0^1 (f^4(x) - 8x^2 f^2(x)) dx + \frac{16}{5} = 0$, then the value of $f^2(1)$ is	(P)	0
(II)	If the point of inflection for the curve $y = \int_0^x \sqrt{1-t^4} dt$ is (α, β) , then $\alpha + \beta$ is equal to	(Q)	1
(III)	Let f be continuous and function g is defined as $g(x) = \int_0^x \left(t^2 \cdot \int_1^x f(u) du \right) dt$ also $f(1) = 3$, and $g'(1) + g''(1)$ is equal to $n_1 + \frac{f'(1)}{n_2}$ (where $n_1, n_2 \in \mathbb{N}$), then $n_1 - n_2$ is	(R)	2
(IV)	Let $I_1 = \int_0^1 \frac{e^x}{1+x} dx$ and $I_2 = \int_0^1 \frac{x^2}{e^{x^3}(2-x^3)} dx$, then $\frac{I_1}{e I_2}$ is	(S)	3
		(T)	4

The correct option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (B) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (P); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)

44. Match the following List-I with List-II

List-I		List-II	
(I)	Number of points of discontinuity of $f(x) = \left[\frac{2x}{\pi} \right] + \{\sin x\}$ in $x \in [0, \pi]$ is/are (where $[.]$ and $\{.\}$ represents greatest integer function and fractional part function)	(P)	0
(II)	If the area included between the curve $y^2 = x^3$, tangent to it at point (1, 1) and x -axis is A , then $[A + 1]$ (where $[.]$ represents greatest integer function)	(Q)	1
(III)	If the area bounded by $f : [1, \infty) \rightarrow [1, \infty)$ given by $f(x) = x^2 - 2x + 2$ and its inverse is A , then $3A$ is equal to	(R)	2
(IV)	Let $I_n = \int_0^1 x^n \sqrt{1-x^2} dx$, then $\lim_{n \rightarrow \infty} \frac{I_n}{I_{n-2}}$ is equal to	(S)	3
		(T)	4

The correct option is:

- (A) (I) $\rightarrow$ (R); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)
 (B) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (R); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (Q)
 (D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)

45. Match the following List-I with List-II

List-I		List-II	
(I)	Let $f(x) = x^3 - \frac{3}{2}x^2 + x + \frac{1}{4}$, then the value of $\int_{1/4}^{3/4} f(f(x))dx$ is	(P)	5
(II)	Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x - x^2$ and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^n$, $n \in \mathbb{N}$. If area between $f(x)$ and $g(x)$ in first quadrant is $\frac{1}{2}$, then n is	(Q)	$\frac{11}{2}$
(III)	If $f(x) = \frac{1}{1+e^x}$ and $S = f(5) + f(4) + f(3) + \dots + f(-3) + f(-4) + f(-5)$, then the value of S is	(R)	$\frac{13}{2}$
(IV)	If $f(x) = \left(4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7\right)^{\frac{1}{7}}$, then $f(f(4))$	(S)	$\frac{1}{4}$
		(T)	4

The correct option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (B) (I) $\rightarrow$ (S); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (T)
 (C) (I) $\rightarrow$ (S); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (Q)
 (D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (S)

46. Match the following List-I with List-II

List-I		List-II	
(I)	If $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable function such that $f(x + 2y) = f(x) + f(2y) + 4xy$ $\forall x, y \in \mathbb{R}$, then $f'(1) - f'(0)$ is equal to	(P)	1
(II)	If $\lim_{x \rightarrow \infty} f(x)$ exist and is finite and non-zero and $\lim_{x \rightarrow \infty} \left(f(x) + \frac{3f(x)-1}{f^2(x)} \right) = 3$, then value of $\lim_{x \rightarrow \infty} f(x)$ is	(Q)	2
(III)	Let $f(x) = 1 + 2\cos x + 3\sin x$. If real number a, b, c are such that $af(x) + bf(x+c) = 1$ holds for any $x \in \mathbb{R}$, then $\left \frac{bc \cos c}{a} \right $ is	(R)	3
(IV)	Define $f : \mathbb{R} \rightarrow \mathbb{R}$, such that $f(x+y) = 3^y f(x) + 2^x f(y) \forall x, y \in \mathbb{R}$ and $f(1) = 1$, then $f(2)$ is	(S)	4
		(T)	5

The correct option is:

- (A) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (P)
 (B) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (P); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. Function $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is differentiable and satisfies the condition $f(x) - f(y) \geq \ln\left(\frac{x}{y}\right) + x - y$, for all $x, y \in \mathbb{R}^+$. If g denote the derivative of f , then the value of the $g\left(\frac{1}{2022}\right)$
48. Consider a function $f(x) = \prod_{r=1}^{\infty} \left(\frac{1 + 2\cos\left(\frac{2x}{3^r}\right)}{3} \right)$. Then two number of points where $|x f(x)| + ||x - 2| - 1|$ is non differentiable on interval $(0, 3\pi)$
49. If the value of definite integral $I = \int_0^2 (3x^2 - 3x + 1)\cos(x^3 - 3x^2 + 4x - 2)dx$ can be expressed in the form of $\alpha(\sin \beta)$, where $\alpha, \beta \in \mathbb{N}$, then the value of $\frac{\alpha^2 + \beta^2 + 1}{2}$
50. If $S = \sum_{r=1}^{502} \left[\frac{305r}{503} \right]$, then $\frac{S}{10,000}$ is (where $[.]$ represents greatest integer function)
51. The value of $I = \frac{2^{2022} \int_0^1 x^{1010} (1-x)^{1010} dx}{\int_0^1 x^{1010} (1-x^{2022})^{1010} dx}$ is
52. Let f be a differentiable function satisfying the functional equation $f(xy) = f(x) + f(y) + \frac{x+y-1}{xy}$
 $\forall x, y > 0$ and $f'(1) = 2$, then the value of $[f(e^{200})]$ is equal to (where $[.]$ represents greatest integer function)
53. Consider the following area:
 A_1 = Area of region common to $y \leq f(x)$ and $y \geq g(x)$
 A_2 = Area of region common to $y \geq f(x)$ and $y \leq g(x)$
 A = Area of region bounded by $y = f(x)$, $y = g(x)$ and $x = \pm\pi$
 where $f(x) = |\cos |x||$, $g(x) = |\sin |x||$ for $x \in [-\pi, \pi]$. Then the value of $(\sqrt{2} + 1)(A_1 + A_2 + A_3)$ is
54. If $I = \int_0^{\pi/2} \sin x \log_e (\sin x) dx$ and A is area of the region bounded by the set $\left\{ (x, y) : 0 \leq y \leq \min\left\{ 3 - x, x + \frac{1}{x} \right\} \right\}$, then the value of $A - I$ is

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – I
PAPER –2
TEST DATE: 20-11-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

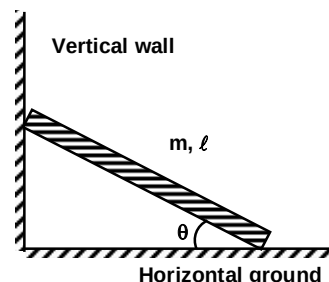
1. A uniform rod of mass m and length ℓ is released from the position shown in the diagram. If all the surfaces are smooth then find the Normal reaction applied by ground on the rod just after the release. $\left(\sin \theta = \frac{3}{5}\right)$.

(A) $\frac{9mg}{25}$

(B) $\frac{12mg}{25}$

(C) $\frac{16mg}{25}$

(D) $\frac{13mg}{25}$



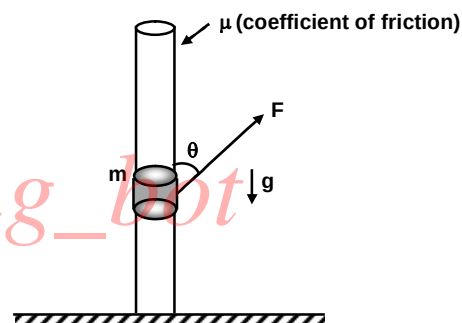
2. A collar of mass m is to be moved along a long fixed rough straight vertical rod by applying a force F which is constant in magnitude and whose direction is continuously changing such that θ is increased from 0° to 90° linearly with time. Initially the collar is at rest at ground and again comes to rest when the force becomes horizontal then

(A) $F = \frac{mg\pi}{2(1-\mu)}$

(B) $F = \frac{mg\pi}{(1-\mu)}$

(C) $F = \frac{3mg\pi}{2(1-\mu)}$

(D) $F = \frac{2mg\pi}{(1-\mu)}$



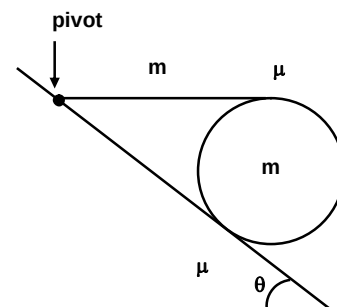
3. A horizontal stick of mass m has its left end attached to a pivot on the inclined plane while its right end rests on the top of a cylinder of same mass m as shown. The coefficient of friction between the cylinder and both the stick and plane is μ . Then the smallest value of μ for which the system doesn't slip anywhere is

(A) $\frac{\sin \theta}{2 + \cos \theta}$

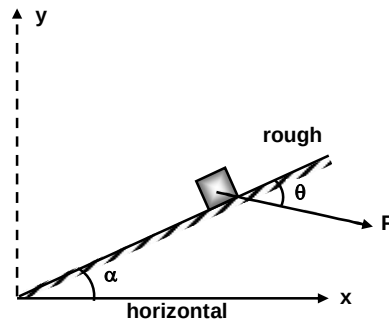
(B) $\frac{3 \sin \theta}{1 + \cos \theta}$

(C) $\frac{\sin \theta}{1 + \cos \theta}$

(D) $\frac{3 \sin \theta}{2 + \cos \theta}$



4. A block is placed on a fixed rough inclined plane ($\mu = \frac{1}{\sqrt{3}}$) as shown. A force F is applied on the block in the direction shown. Find the range of θ so that any value of the force F is not able to slide the block along the inclined plane. (Assuming that $\vec{F}$ always lies in x-y plane). (Also Given $\alpha < 30^\circ$).

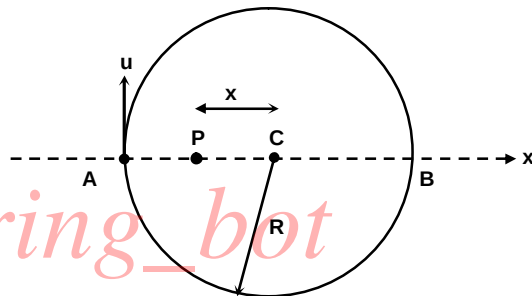


- (A) $[60^\circ, 120^\circ]$ (B) $[30^\circ, 120^\circ]$
 (C) $[30^\circ, 90^\circ]$ (D) $[60^\circ, 90^\circ]$

Section – A (Maximum Marks: 24)

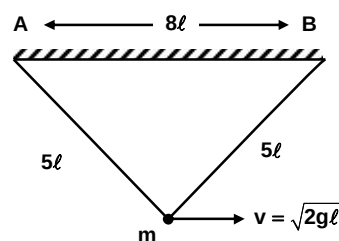
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A bead of mass m is given a velocity u from point A along a fixed smooth circular wire of radius R which passes inside the bead as shown in gravity free space. A force F (constant in magnitude) always acting on bead towards a fixed point P. If u_0 is the minimum value of u so that bead completes the entire circle and if $\frac{u}{u_0} = \alpha$ and $\frac{x}{R} = \beta$ and $\vec{N}$ is the normal reaction experienced by bead at point B. Then choose the correct option(s).



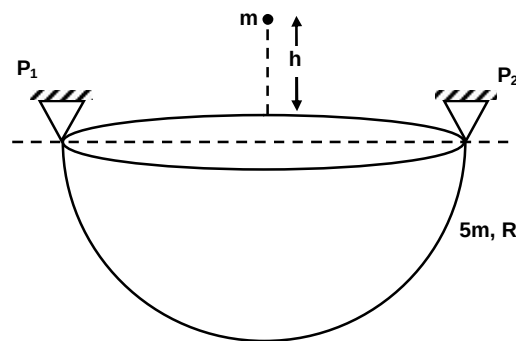
- (A) If $\alpha = \sqrt{2}, \beta = \frac{1}{8}$ then $\vec{N} = \frac{F}{2} \hat{i}$ (B) If $\alpha = \sqrt{2}, \beta = \frac{1}{3}$ then $\vec{N} = -\frac{F}{3} \hat{i}$
 (C) If $\alpha = \sqrt{3}, \beta = \frac{1}{8}$ then $\vec{N} = \frac{5F}{2} \hat{i}$ (D) If $\alpha = \sqrt{3}, \beta = \frac{1}{3}$ then $\vec{N} = -\frac{5F}{3} \hat{i}$

6. An ideal smooth massless inextensible string passes through the bead as shown whose ends are fixed at A and B. Initially the system is in equilibrium, now bead is given a velocity $\sqrt{2gl}$ in horizontal direction and in the plane of the string as shown. Let T_0 be the tension in the string and r_0 be the radius of curvature of particle just after the velocity is given. Then choose the correct option(s).



- (A) $T_0 = \frac{31mg}{30}$ (B) $r_0 = \frac{35l}{3}$
 (C) $T_0 = \frac{41mg}{40}$ (D) $r_0 = \frac{25l}{3}$

7. A small particle of mass m is dropped from height h from the flat surface of a uniform solid hemisphere of mass $5m$ and radius R which is held at rest with the help of two frictionless pivots on diametrically opposite points and the hemisphere is free to rotate about the axis passing through the pivots. The particle strikes on a point on circumference which is equidistant from the pivots and sticks to it. After the collision, the flat surface of hemisphere just able to reach in vertical plane. Then choose the correct option(s)



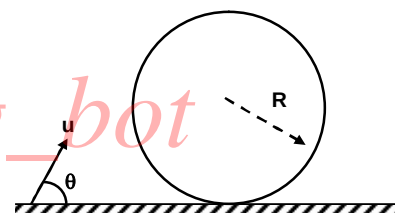
(A) $h = \frac{21R}{8}$

(B) $h = 3R$

(C) Angular velocity of the system just after collision is $\sqrt{\frac{7g}{12R}}$

(D) Angular velocity of the system just after collision is $\sqrt{\frac{8g}{12R}}$

8. A particle is thrown from ground with speed u at an angle θ with horizontal as shown. If u_0 is the minimum value of u so that the particle will cross the horizontal cylinder just grazing at two points on it and θ_0 is the corresponding angle of projection. Then choose the correct option(s).



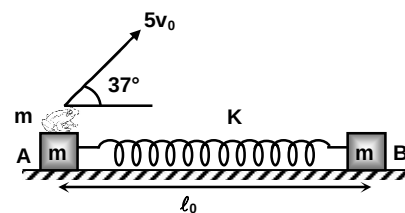
(A) $u_0^2 = Rg[2 + 2\sqrt{2}]$

(B) $\tan^2 \theta_0 = [3 + 2\sqrt{3}]$

(C) $u_0^2 = Rg[2 + 2\sqrt{3}]$

(D) $\tan^2 \theta_0 = [3 + 2\sqrt{2}]$

9. The system of two blocks and a frog (each having mass m) is placed on a smooth horizontal surface as shown. Initially the spring is in natural length (ℓ_0) and all are at rest. Now the frog jumps with velocity $5v_0$ at an angle of 37° with horizontal w.r.t. the block A and lands directly on block B. If R is the horizontal Range of frog in ground frame during the jump. If stiffness of the spring is



$K = \frac{\pi^2 mg^2}{288v_0^2}$, then choose the correct option(s).

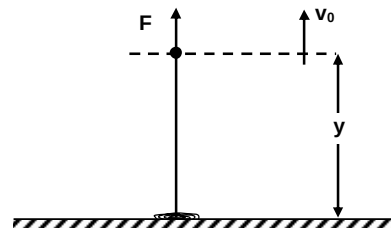
(A) $R = \frac{12v_0^2}{g}$

(B) $R = \frac{24v_0^2}{g}$

(C) $\ell_0 = \frac{12v_0^2}{g} \left[1 + \frac{2}{\pi} \right]$

(D) $\ell_0 = \frac{6v_0^2}{g} \left[3 - \frac{2}{\pi} \right]$

10. A uniform rope of linear mass density λ is coiled on a smooth horizontal table as shown. One end is pulled up with constant speed v_0 . Then choose the correct option(s)

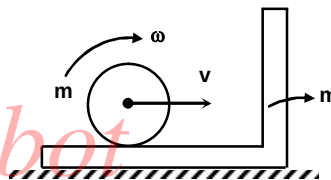


- (A) Force exerted (F) on the end of the rope is $\lambda y g + \frac{\lambda v_0^2}{2}$
 (B) Force exerted (F) on the end of the rope is $\lambda y g + \lambda v_0^2$
 (C) Heat loss per unit time is $\frac{\lambda v_0^3}{2}$
 (D) Heat loss per unit time is λv_0^3

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

11. An L shaped wedge of mass m having long horizontal base is placed on smooth horizontal surface as shown. A thin Ring of mass m and Radius R is set in motion on the rough horizontal part of the wedge with a linear velocity v and angular velocity ω such that $v = \omega R$. Ring collides with the smooth vertical part of wedge such that collision is perfectly elastic. After a sufficiently

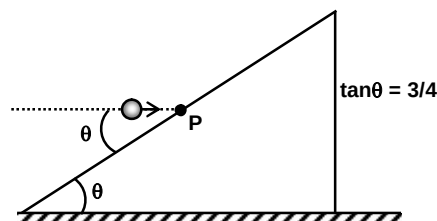


long time, the final velocity of wedge is $\frac{v}{K}$. Find the value of K .

(Assume wedge does not topple due to collision)

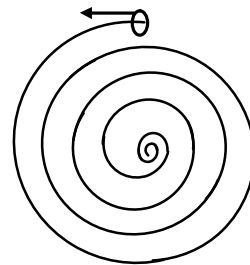
12. A projectile is thrown from ground with velocity 8 m/s. When it is moving with 6 m/s, it is at maximum separation from the point of projection corresponding to the speed 6 m/s. Then this maximum separation corresponding to speed 6 m/s is k meter. Find the value of k . (Take $g = 10 \text{ ms}^{-2}$)

13. A small elastic ball moving horizontally with 40 m/s collides with the smooth stationary wedge ($e = 1$) at point P which is placed on smooth horizontal surface as shown. After that it again strikes on the wedge at same point P after t_0 seconds. Find the value of t_0 . (Take $g = 10 \text{ ms}^{-2}$)

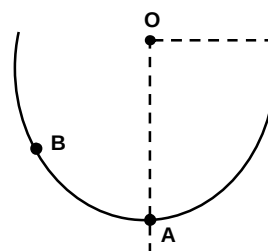


14. Bob of mass m of simple pendulum is given a velocity from the lower most position in horizontal direction in uniform vertical gravitational field (g) such that $\frac{T_{\max}}{T_{\min}} = 2$ during the subsequent motion. Then the tension in the string when it becomes horizontal is $K mg$. Find the value of K . (T represents tension in the string)

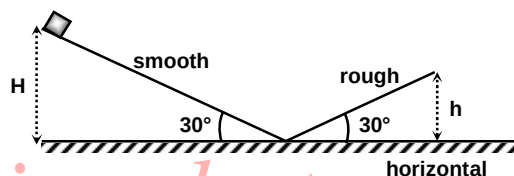
15. A bead of mass m is given a velocity along a smooth rigid spiral wire (mass $M = 2m$ and outer radius R) such that bead slides along the wire. When bead will reach the center of spiral and sticks to it, at that moment velocity of bead is v and angular velocity of spiral is ω . (Assume that the center of mass of loop lies at its centre and given that its moment of Inertia about an axis passing through centre and perpendicular to its plane is $I = \frac{MR^2}{2}$). Find $\frac{\omega R}{v}$ (Neglect gravity)



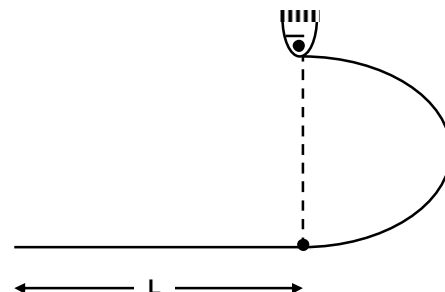
16. In a fixed hemispherical smooth bowl of Radius R , a particle is dropped at point B somewhere else other than point A. Collision with the bowl is perfectly elastic. For second collision to take place at point A, the maximum value of $\angle BOA$ is $\frac{\pi}{K}$ radian. Find the value of K .



17. A particle is released from the top of a fixed smooth inclined surface as shown from height H from the ground. The particle will reach upto height h from the ground on a rough fixed inclined plane which is directly connected to the smooth inclined plane. If coefficient of friction for rough inclined plane is $\frac{1}{2\sqrt{3}}$ the find $\frac{H}{4h}$. (Assume that particle does not jump.)



18. A uniform rod of length L is rigidly attached with a uniform semi-circular ring of same density as of rod to form a rigid body in a plane. The system is hanged in vertical plane in equilibrium with the help of hinge as shown such that rod is in horizontal position. Then the distance of centre of mass of the system from hinge is $\frac{L}{2} \left[\frac{\pi + \alpha}{\pi + \beta} \right]$. Find the value of $\alpha\beta$.



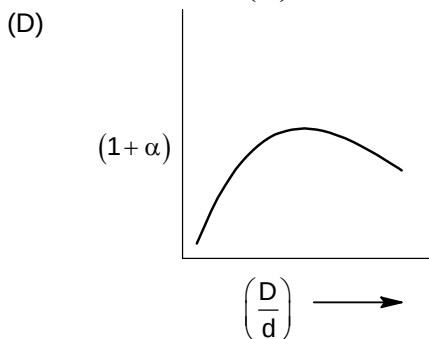
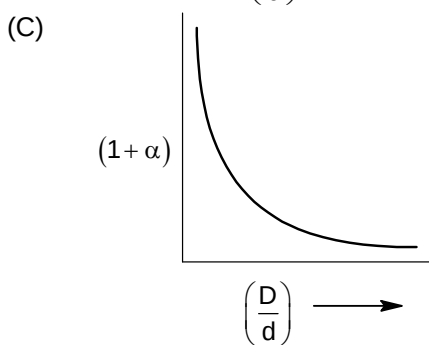
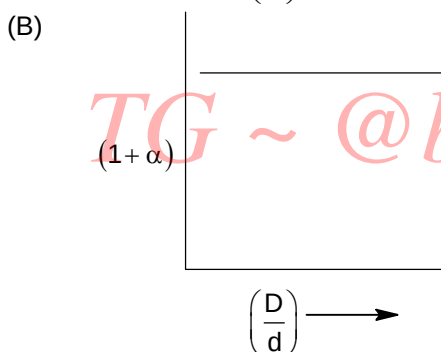
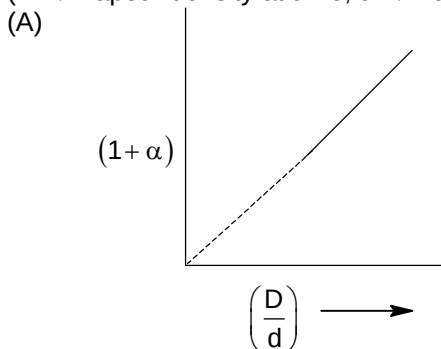
Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

19. For the dissociation of PCl_5 (g) the variation of $(1 + \alpha)$ against (D/d) is represented as
 (D $\rightarrow$ Vapour density at $t = 0$, $d \rightarrow$ Vapour density at equilibrium)



20. For an reaction $2 - \text{pentyne} \rightleftharpoons 1 - \text{pentyne}$. The temperature dependence of equilibrium constant is given by $\ln K = 8.0 - \frac{8000}{T}$. The value of ΔS° at 300 K. is....
- (A) $4R$
 (B) $8R$
 (C) $8000R$
 (D) $\frac{R}{8}$
21. An electron in an atom jumps in such a way that its kinetic energy changes from x to $\frac{x}{8}$. The change in its potential energy will be?
- (A) $+\frac{7}{8}x$
 (B) $+\frac{7}{4}x$
 (C) $-\frac{7}{4}x$
 (D) $\left(-\frac{9x}{4}\right)$
22. The time required for 50% completion of a first order reaction at 298 K is equal to that required for 75% completion of a reaction at 308 K. If the pre-exponential factor for the reaction is $3.5 \times 10^6 \text{ sec}^{-1}$ in the given temperature range. Find the ratio of K_{298} to K_{308} .
- (A) 0.5
 (B) 2
 (C) $\frac{3}{4}$
 (D) $\frac{3}{2}$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

23. In the emission spectrum of H-atom from energy level 'n' to ground state in one or more step and no line belongs to the Breckett series is observed. The wave length of lines belonging to Balmer series may be (R = Rydberg constant)
- (A) $\frac{36}{5R}$
 (B) $\frac{16}{3R}$
 (C) $\frac{R}{3R}$
 (D) $\frac{35}{3R}$

24. For the reaction started with 1 mol $A_2B_4(g)$ is
- $$A_2B_4(g) \longrightarrow A_2(g) + 2B_2(g)$$
- The vapour densities are measured at various time of dissociation and observed value are
- | | |
|----------------------------|---------|
| (i) at time $t = 0$ min | VD = 50 |
| (ii) at time $t = 10$ min | VD = 25 |
| (iii) at time $t = 20$ min | VD = 20 |
- If $\frac{V.D.(A_2B_4)}{V.D.(A_2)} = 5$ and $t_{100\%}$ is 40 min.
- Choose the correct option(s)
- (A) The ratio of atomic mass of 'A' and atomic mass of 'B' is 1 : 2
- (B) Mole% of B_2 at time $t = 10$ min is 50%
- (C) Assuming uniform dissociation the ratio of rate of disappearance of A_2B_4 and rate of appearance of $A_2(g)$ between $t = 10$ min to $t = 20$ min is 1 : 1
- (D) Mole % of A_2 at $t = 10$ min is 50%
25. A closed rigid container of volume 1 litre, initially contains 4 mole of HI (g) and 4.25 moles of $N_2(g)$ at $27^\circ C$. When the mixture is heated to $127^\circ C$ two reversible reactions takes place till equilibrium is established.
- $$2HI(g) \rightleftharpoons H_2(g) + I_2(g) \quad K_{C_1}$$
- $$2N_2(g) + 6H_2(g) \rightleftharpoons 4NH_3(g) \quad K_{C_2}$$
- At equilibrium the concentration of I_2 and NH_3 are 1.0 M and 0.5 M respectively. Choose the correct option
- (A) $K_{C_1} = \frac{5}{32}$
- (B) $K_{C_2} = \frac{16}{125}$
- (C) $K_{C_1} \times \sqrt{K_{C_2}} = \frac{1}{25}$
- (D) Conc. of N_2 at equilibrium = 4.0 mol/litre
26. How many of the following compounds has more than one $p\pi - d\pi$ bond
- (A) SO_2
- (B) SO_3
- (C) XeO_4
- (D) ClO_3^-
27. Which of the following statement(s) is/are true for $CH_2 = CH_2$ and $CH_2 = SF_4$ (Methyl sulphur tetrafluoride)?
- (A) In ethylene nodal plane lies in molecular plane
- (B) In methyl sulphur tetrafluoride nodal plane lies in axial plane
- (C) In methyl sulphur tetrafluoride nodal plane lies in equatorial plane
- (D) If the nodal plane lies in equatorial plane its stability will be less than if its lies in axial plane
28. How many of the following compounds has more than two $(3C - 2e)$ bond.
- (A) $(AlCl_3)_n$
- (B) B_2H_6
- (C) $Al(BH_4)_3$
- (D) $Be(BH_4)_2$

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

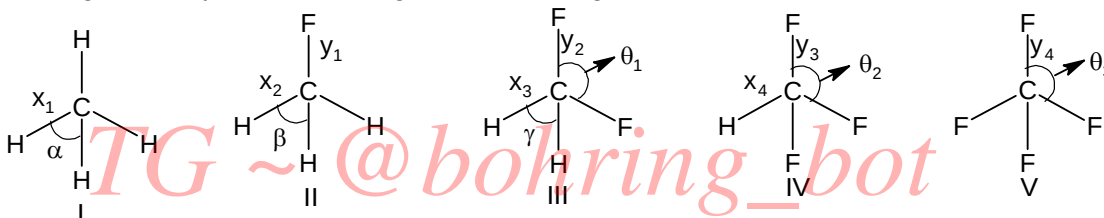
29. Not considering the electronic spin the degeneracy of the second excited state of H-atom is 'x' degeneracy of the second excited state of H^- is 'y'. And number of angular node in $d_{x^2-y^2}$ orbital is 'z'. Calculate the value of $\frac{x}{y} + z$?
30. The given reaction how many of them can give sp^2 hybridised boron centre. In case of the product(s)
- Borax $\xrightarrow{\Delta}$
 - Borax + $NH_4Cl \xrightarrow{\Delta}$
 - Borax $\xrightarrow{HCl}$
 - Borax $\xrightarrow{H_2O}$
 - $B_2H_6 + NH_3 \xrightarrow{\Delta}$
 - $B_3N_3H_3Cl_3 + LiBH_4 \xrightarrow{THF}$
 - $LiH + B_2H_6 \longrightarrow$
 - $K_3[AlF_6] + BF_3 \longrightarrow$
31. (i) If the sum of the number of lone pairs of electrons on each central atom in the given species is/are 'x'
- $[TeBr_6]^{2-}, [IF_2]^+, SNF_3$ and $[XeF_3]^-$
- (ii) and among the following species the number of diamagnetic spin is/are 'y'
- $B_2, Li_2, C_2, C_2^-, O_2^{2-}, O_2^+, He_2^+, Be_2, KO_2, K_2O, N_2O_4, B_3N_3H_6$
- (Atomic number of Te = 52, Br = 35, Xe = 54, K = 39, I = 127).
- Calculate the value of $\frac{x+y}{2}$?
32. Equilibrium constant of SO_3 (g) decomposition and formations was calculated by following two methods
- Method (1)
- $$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$$
- I
- at $t = 0$, 10 moles of SO_3 is added 5(l) in container. If at equilibrium mixture is titrated with 0.2 M acidic $KMnO_4$ its volume 4 litre. K_C for the reaction is 'x' (mol/litre)
- The value of $\frac{y}{x}$ is?
- Method (2)
- $$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$
- II
- Initially 2 : 1 molar ratio SO_2 and O_2 is added respectively. At the equilibrium total pressure of mixture 20 atm. Mole % of O_2 at equilibrium is 25%. K_P for the reaction is 'y' atm^{-1} .

33. Consider the following combination of atomic orbitals

Sr. No.	Combining orbitals	Internuclear axis
i.	$s + p_x$	(x)
ii.	$d_{xy} + d_{xy}$	(x)
iii.	$d_{yz} + p_z$	(z)
iv.	$p_y + p_y$	(y)
v.	$d_{yz} + d_{yz}$	(x)
vi.	$s + s$	(z)
vii.	$d_{z^2} + d_{z^2}$	(z)
viii.	$d_{xy} + d_{xy}$	(z)
ix.	$p_x + p_x$	(y)
x.	$s + p_z$	(x)

Calculate the value of $\frac{a^2 + b^2 + 2d}{4}$ (Where $a = \sigma$, M.O. $b = \pi$, M.O. $d = \text{non-bonding M.O.}$)

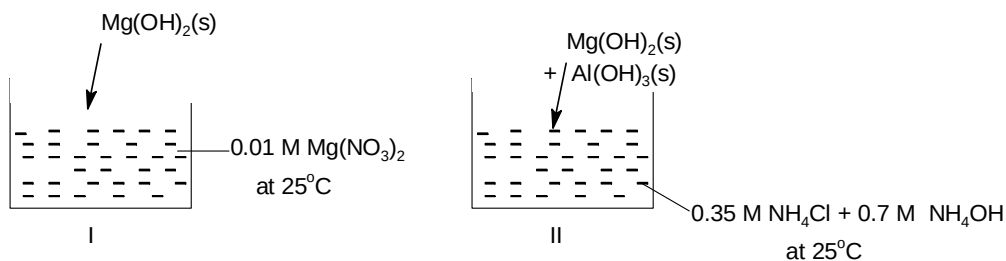
34. In the given compounds bond angle and bond length are defined



Order of bond parameter is/are given

- (I) $\gamma > \beta > \alpha$
 (II) $y_1 > y_2 > y_3 > y_4$
 (III) $\theta_3 > \theta_2 > \theta_1$
 (IV) $x_1 > x_2 > x_3 > x_4$
 (V) Dipole moment $\text{IV} > \text{II} > \text{III} > \text{I} = \text{V}$
 If the number of correct order(s) is 'x' and incorrect order (s) is 'y' about molecules I, II, III, IV and V. Calculate the value of $(2x + y)$ Is?

35. The solubility of Mg(OH)_2 in pure water is 5.8×10^{-4} g/litre at 25°C .



$$K_b \text{ for } \text{NH}_4\text{OH} = 1.0 \times 10^{-5}$$

$$K_{\text{SP}} \text{ Al(OH)}_3 = 1.6 \times 10^{-20}$$

If the $[\text{Mg}^{+2}]$ in container (I) is 'x M' and concentration of $[\text{Mg}^{+2}]$ and $[\text{Al}^{3+}]$. In container II is 'y M' and 'z M' respectively. Calculate the value of $\frac{yz}{25x} = \dots$? ($\log 2 = 0.30$)

36. Choose the correct and incorrect statement in the following given statements
1. Hydrogen peroxide. In the pure state is non-planar and almost colourless.
 2. $S_8(s) + aOH^-(aq) \longrightarrow bS^{2-}(aq) + cS_2O_3^{2-}(aq) + dH_2O(l)$
The value of $a + b$ is 16.
 3. Hydrogen peroxide. In its reaction with KIO_4 and NH_2OH respectively is acting reducing agent and oxidizing agent.
 4. $MgCl_2 \cdot 6H_2O$ on heating give anhydrous $MgCl_2$.
 5. H_3BO_3 is mono basic weak Lewis acid.
 6. According to MOT B_2 and C_2 both do not have σ – bond.
 7. 'Be' cannot be qualitatively confirmed by the flame test.
 8. Correct order of density of alkali metals is $Li < K < Na < Rb < Cs$.
 9. The silicates in which three oxygen atoms of $[SiO_4]^{4-}$ are shared known as sheet silicate.
 10. Pb^{+4} compounds are stronger oxidizing agent than Sn^{+4} compounds.
 11. Graphite is better lubricant on the earth than on moon.
 12. The hydrolysis of trialkylchlorosilane R_3SiCl yields trialkylchlorosilanol.
- If the correct number of statement(s) is/are x and incorrect statement(s) is/are y . Value of $\frac{3x + y}{6}$ is....

TG ~ @bohring_bot

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. If $f(x)$ is a function such that $f(x+y) + \frac{f(x) \cdot f(y)}{1+y^2} + \frac{\tan^{-1}x \cdot \tan^{-1}y}{1+x^2} = 0 \forall x, y \in \mathbb{R}$ and $f'(0) = 1$, then the function $f(x)$ is
 (A) $1 - \tan^{-1}x - 2e^{-\tan^{-1}x}$ (B) $-(1 + \tan^{-1}x)$
 (C) $e^{1-\tan^{-1}x}$ (D) $1 - \tan^{-1}x - e^{-\tan^{-1}x}$
38. A function $f(x)$ is such that the area enclosed by $|y| = f(x)$ between $x = 1$ and $x = a$ ($a > 1$) is given by $g(a) = a^4 + 6a^2 - 16a + 9$, then the area formed by the curve $k(x) = \frac{f(x)}{g(x)+5}$ between the lines $x = 2$ and $x = 3$ with the x -axis is
 (A) $\frac{1}{2} \ln\left(\frac{101}{22}\right)$ (B) $\frac{1}{2} \ln\left(\frac{131}{22}\right)$
 (C) $\frac{1}{2} \ln\left(\frac{141}{12}\right)$ (D) $\frac{1}{2} \ln\left(\frac{101}{12}\right)$
39. The value of $\int e^{x^2} (2x - \cot x)(x^2 - \ln(\sin x))(\operatorname{cosec} x) dx$ is equal to
 (A) $e^{x^2} (2x - \ln(\sin x) + 1)(\operatorname{cosec} x) + c$ (B) $e^{x^2} (2x - \ln(\sin x) - 1)(\operatorname{cosec} x) + c$
 (C) $e^{x^2} (x^2 - \ln(\sin x) + 1)(\operatorname{cosec} x) + c$ (D) $e^{x^2} (x^2 - \ln(\sin x) - 1)(\operatorname{cosec} x) + c$
40. Let C be a curve $f(x, y) = 0$ passing through $(1, 1)$ such that it is orthogonal to the family of circles $x^2 + y^2 - ax = 0$, where 'a' being parameter, then the shortest distance between the curve C and the parabola $x^2 = 2y + 1$ is
 (A) $\sqrt{2} + 1$ (B) $\sqrt{2} - 1$
 (C) $\sqrt{3} + 1$ (D) $\sqrt{3} - 1$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. Let $f(x) = (\cos^2 x) \cdot (e^{\tan x})$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then which of the following options is/are true?
 (A) $f'''(x) = 0$ has atleast one root in $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 (B) $y = f'(x)$ has a point of local maxima in $x \in \left(-\frac{\pi}{4}, 0\right)$
 (C) $f''(x) = 0$ has exactly one root in $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 (D) $\lim_{x \rightarrow \alpha} \frac{x^3 - 6x^2 + 11x - 6}{x^2 - 3x + 2} = -1$
 where ' α ' is the total number of points of maxima and minima of $y = f'(x)$

42. Let $f(x)$ be a function satisfying $f'(x) > 0$, $f''(x) > 0 \forall x \in (0, 1)$ and $f(0) = 0$, $f(1) = 1$, then which of the following options is/are true?
- (A) $f\left(\frac{1}{2}\right) \cdot f\left(\frac{1}{3}\right) < 1$ (B) $f^{-1}\left(\frac{1}{2}\right) \cdot f^{-1}\left(\frac{1}{3}\right) > 1$
- (C) $f\left(\frac{1}{3}\right) \cdot f^{-1}\left(\frac{1}{3}\right) < \frac{1}{3}$ (D) $f\left(\frac{1}{2}\right) \cdot f^{-1}\left(\frac{1}{2}\right) > \frac{1}{2}$
43. In a question, a student was given to find the derivative of the product of two functions 'f' and 'g'. The student by mistake thought $(fg)' = f' \cdot g'$ for his question $f(x) = \sqrt{\frac{x-1}{x-3}}$ and he got the correct answer. Given that $g(1) = e$. Which of the following options is/are correct?
- (A) $g(x)$ is discontinuous at one point
- (B) $\lim_{x \rightarrow \infty} g(x) = 1$
- (C) $\lim_{x \rightarrow \alpha} \frac{x^3 - 8}{x - 2} = 12$ where α is the number of points of discontinuity of $g(x)$
- (D) $g(x)$ is discontinuous at two points
44. Let $f(x) = \cos^{-1}\left\{\frac{1-x^2}{1+x^2}\right\}$ and $g(x) = \{f(x)\} \forall x \in \mathbb{R}$ (where $\{x\}$ denotes fractional part function of x), then which of the following options is/are true?
- (A) Total number of points where $f(x)$ is not continuous is 2
- (B) Total number of points where $g(x)$ is not continuous is 7
- (C) $f'(1) = f'(-1) = 1$ and $g(x)$ is not differentiable at $x = 0$
- (D) Total number of points where $f(x)$ is not continuous is 3
45. Let $f(x)$ be a function defined by $f(x) = \int_0^{3\pi/4} |x \cos t - 3 \sin t + \sqrt{2}x| \left(\frac{1 + \sqrt{2} \cos t}{(\cos t + \sqrt{2})^3} \right) dt$ where x is any real number, then which of the following options is/are true?
- (A) $f(x)$ is continuous and differentiable everywhere
- (B) $f(x)$ is continuous for all x but non-differentiable at two values of x
- (C) Minimum value of $f(x)$ is $\frac{3}{2}$ for $x \in \mathbb{R}$
- (D) Minimum value of $f(x)$ is $\frac{3}{4}$ for $x \in \mathbb{R}$
46. Let the function $f: (-\infty, \infty) \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ be given by $f(x) = \sin^{-1}\left(\log_3\left(\frac{x^2 - x + 1}{x^2 + x + 1}\right)\right)$, then which of the following options is/are true for $f(x)$?
- (A) $f(x)$ is injective function
- (B) $f(x)$ is surjective function
- (C) $f\left(\frac{1}{x}\right) = -f(-x)$
- (D) The equation $f(x) = x - 1$ has two distinct roots

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

47. If $f(x)$ is a function satisfying $3f^2(x) - 4xf(x) + 2 = 0$ and the value of integral $\int \frac{(6x-2)f(x) - 4x^2}{(3f(x) - 2x)(x^2 - f(x))^2} dx$ is equal to $\frac{ax + bf(x)}{3x^2f(x) - 4x^3 + 2} + c$ (where a, b, c are constants), then the value of $(a + b)$ is
48. If the area bounded between the curves $|x| + y = 4$ and $\sqrt{|x|} + \sqrt{y} = 2$ is $\frac{16\lambda}{3}$ sq. units, then the value of λ is
49. The value of $\lim_{n \rightarrow \infty} \left(\lim_{m \rightarrow \infty} \left(\sum_{r=1}^n \left(\sum_{k=1}^{mr} \frac{mn^2}{(m^2n^2 + k^2)(n^2 + r^2)} \right) \right) \right)$ is $\frac{\lambda\pi^2}{160}$, then the value of λ is
50. Consider $f(x) = \frac{(x^2 - 1)(x^2 - 9)}{g(x)}$, $x \in \mathbb{R} - \{a, b\}$, where $g(x)$ is a polynomial of degree ≤ 4 . $f(x)$ has a removable type of discontinuity at $x = a$ and $x = b$. The following three informations are also known
 (i) $\lim_{x \rightarrow \pm\infty} f(x) = \infty$ (ii) $\lim_{x \rightarrow 1} f(x) = -4$
 (iii) $\lim_{x \rightarrow -3} f(x) = a$ non-zero finite number
 Let L and M denotes the number of points of discontinuity and the number of points of non-differentiability of $y = |f(|x|)|$ respectively in $\mathbb{R}$ (the set of all real numbers), then the value of $(L + M)$ is
51. If $\int_0^{4\pi} \frac{dx}{2 + \sin 4x} = \lambda \int_0^{3\pi/8} \frac{dx}{7 + \cos\left(\frac{8x}{3}\right)}$, then the value of $\frac{3\lambda}{16}$ is
52. Let $f(x)$ be a polynomial of degree 6 such that $f(2) = 85$ and attains its local minimum value 4 at points $x = -1$, $x = 1$ and $x = 5$. The local maximum value of $f(x)$ lies at $x = \alpha$ and $x = \beta$.
 If $\lim_{x \rightarrow (\alpha+\beta)} \left(\frac{x^2 - 9}{x^2 - 5x + 6} \right) = \lambda$, then the value of $[\lambda]$ is (where $[\cdot]$ represents greatest integer function)
53. If the value of $\lim_{x \rightarrow 0} \left(\frac{1}{x^{2022}} \int_{x^{2023}}^{x^{2022}} \frac{(1+t)^{\frac{1}{2021}} - 1}{t} dt \right) = \frac{1}{\lambda}$, then the number of positive divisors of λ is
54. If the area bounded by the curves $x^2 = y$, $x^2 = -y$ and $y^2 = 4x - 3$ is $\frac{m}{n}$, where m and n are relatively prime positive integer, then $|m - n|$ is

FIITJEE
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JEE (Advanced)-2023
PART TEST – I
PAPER –1
TEST DATE: 20-11-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A, B, D

Sol.

$$-\frac{dx}{dt} = 0.1\sqrt{x}$$

$$\int_9^0 \frac{dx}{\sqrt{x}} = -0.1 \int_0^{\tau} dt \Rightarrow \tau = 60 \text{ sec}$$

$$\omega = \frac{0.1\sqrt{x}}{x} = \frac{0.1}{\sqrt{x}} \text{ rad/s}$$

$$a = v\omega = 0.1\sqrt{x} \frac{0.1}{\sqrt{x}} = 0.01 \text{ m/s}^2$$

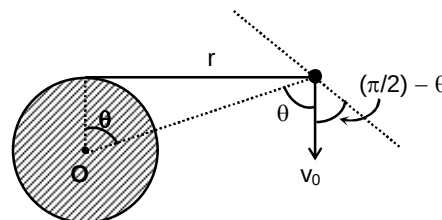
2. A, C

Sol.

Speed of the object will remain constant because tension is always perpendicular to the velocity.

$$\text{So angular momentum about O} = mv_0 \sin\theta \frac{r}{\sin\theta} = mv_0 r$$

$$\frac{mv_0 mv_0^2}{T_{\max}} = 5 \text{ kg-m}^2/\text{sec}$$



3. B, C

Sol.

$$\text{For the bigger disc } C\omega_1 r_1^4 - \int f dr_1 = 0$$

$$\text{For the smaller disc } C\omega_2 r_2^4 - \int f dr_2 = 0$$

$$\text{So, } \omega_1 r_1^3 = \omega_2 r_2^3 \text{ and } \frac{L_1}{L_2} = \frac{r_1}{r_2}$$

4. A, C, D

Sol. In both situations $v_C = \frac{J}{m}$

In situation-(ii), $\omega = \frac{J}{m\ell}$

5. B, D

Sol. $H_B^{\max} > H_A^{\max} > H_C^{\max}$ (as shown clearly)

So, $u_B^y > u_A^y > u_C^y$

Vertical component of initial velocity of each particle will decide the time taken.

So, $t_C < t_A < t_B$

6. B

Sol. For COM position, $m_1x_1 = m_2x_2$

Where m_1 and m_2 are masses of left and right part of the rod and x_1, x_2 are their respective position of centre of mass.

Simultaneously, $m_1x_1\omega = m_2x_2\omega$

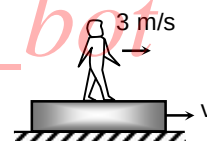
and $I_1\omega \neq I_2\omega$

and $\frac{1}{2}I_1\omega^2 \neq \frac{1}{2}I_2\omega^2$

7. C

Sol. (I) $0 = 100v + 50(v + 3)$ (COLM)
 $v = -1 \text{ m/s}$

So, work done by man = $\frac{1}{2}(100)(1)^2 + \frac{1}{2}(50)(2)^2 = 150 \text{ J}$



(II) $W_N + W_{mg} = \Delta \text{ K.E. (WET)}$

$$W_N - \frac{1}{2}(2)(10)^2 = \frac{1}{2}(2)[(5)^2 - (15)^2]$$

$$W_N = -100 \text{ J}$$

(III) Unstable equilibrium points are $x = 10 \text{ m}, 30 \text{ m}$

Stable equilibrium point is $x = 20 \text{ m}$

$$\text{So, } U_i = 0, \text{ and } U_f = -\frac{1}{100}(10)^4 = -100 \text{ J}$$

$$W_{\text{conservative force}} = U_i - U_f = 100 \text{ J}$$

$$(IV) W_{\text{ext}} = U_f - U_i = mg\left(\frac{5R}{8} - \frac{3R}{8}\right) = \frac{mgR}{4} = 100 \text{ J}$$

8. B

Sol. In case-(I)

$$at = \omega x$$

$$y = \frac{1}{2}at^2$$

$$y = \frac{1}{2}a \frac{\omega^2 x^2}{a^2}$$

$$y = \frac{\omega^2 x^2}{2a}$$

9. A
 Sol. 3 kg block will be at rest. (so friction between S_2 and D = tension in the string connect to block D)
 Considering A + B + C in a system
 $T - 6 = 3a$, $20 - 2T = 2a_1$ and $a = 2a_1$
 So, $a = \frac{8}{7} \text{ m/s}^2$
 So, there will be slipping between blocks A and B.
 Now consider (B + C) in a system
 $T' - 6 - 1 = 2a'$
 $20 - 2T' = 2a_2$ and $a' = 2a_2$
 So, $T' - 7 = 4a_2$
 $20 - 2T' = 2a_2$
 So, $a' = 1.2 \text{ m/s}^2$

10. C

Sol. **Case -I :** $\frac{mR^2}{2} \omega_0 = \left(\frac{mR^2}{2} + \frac{mR^2}{4} + \frac{mR^2}{4} \right) \omega$

$$\omega = \frac{\omega_0}{2} = 3 \text{ rad/s}$$

Case -II: $\omega = \omega_0 = 6 \text{ rad/s}$

Case -III: $\frac{mR^2}{2} \omega_0 + mv \frac{R}{2} = (mR^2) \omega$

So, $3 + \frac{8}{2} = \omega = 7 \text{ rad/s}$

Case -IV: $\frac{mR^2}{2} \omega_0 + mv \frac{R}{2} \left(\frac{1}{2} \right) = mR^2 \omega$

$3 + \frac{8}{4} = \omega = 5 \text{ rad/s}$

Section - B

11. 4.50 (range 4.40 to 4.60)
 Sol. (F) (2) = (N) (1) $\Rightarrow N = 2F$ (N is normal between rods)

and $(N) \left(\frac{1}{2} \right) = (F_1)(2)$

So, $F_1 = \frac{F}{2}$

12. 60.60 (range 60.00 to 61.00)

Sol. Total time taken = $\frac{0.01L}{0.0001v} = \frac{100L}{v}$

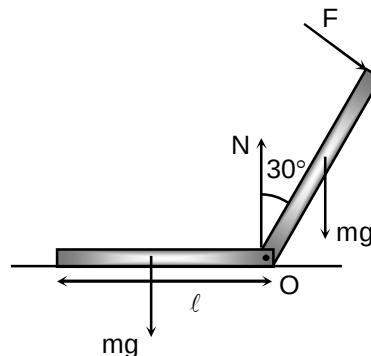
Total number of collision with moving wall in the said movement = $\frac{100L}{v \left(\frac{2L}{v} \right)} = 50$

In each collision with the movable wall speed of the sphere increases by $0.0002 v$.
 So, final speed of sphere = $v + 50 \times 0.0002v = 1.01 v = 60.60 \text{ m/s}$

13. 2.50 (range 2.40 to 2.60)
 Sol. from torque equilibrium about O

$$F\ell + mg \frac{\ell}{2} = mg \frac{\ell}{2}$$

$$F = \frac{mg}{4}$$



14. 1.41 (range 1.35 to 1.45)
 Sol. $\int N dt = mv \cos \theta$ ($v = \sqrt{2gh}$)

$$\text{So, } v_f = \frac{mv \sin \theta - \mu mv \cos \theta}{m} = \sqrt{2gh} \left(\frac{3}{5} - \frac{1}{2} \frac{4}{5} \right) = \frac{\sqrt{50}}{5} = \sqrt{2} \text{ m/s}$$

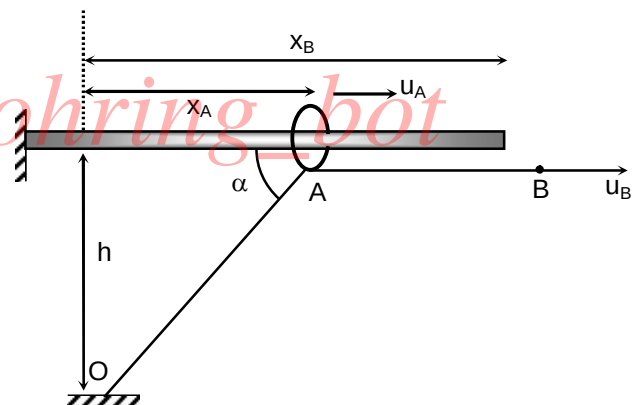
15. 7.46 (range 7.40 to 7.50)

Sol. $x_B - x_A + \sqrt{x_A^2 + h^2} = L$

$$\frac{dx_B}{dt} - \frac{dx_A}{dt} + \frac{x_A}{\sqrt{x_A^2 + h^2}} \frac{dx_A}{dt} = 0$$

$$\text{So, } u_A = \frac{u_B}{1 - \cos \alpha} = \frac{1}{1 - \frac{\sqrt{3}}{2}} = \frac{2}{2 - \sqrt{3}}$$

$$\Rightarrow u_A = 7.46 \text{ m/s}$$



16. 2.43 (range 2.35 to 2.55)

Sol. $\Delta y_{CM} = \frac{1}{2}(10) \left(\frac{1}{10} \right)^2 = 0.05 \text{ m}$

$$\text{So, } h_{CM} = 1 + \frac{2}{3} - 0.05 = 1.62 \text{ m}$$

$$\text{So, } y_{2m} = \frac{1.62 \times 3}{2} = 2.43 \text{ m}$$

17. 58.80 (range 57.00 to 59.00)

Sol. Assume the block M at rest, we see that both smaller cubes are applying net zero resultant horizontal force on the larger block.

$$\text{So, } N = Mg + mg (\cos^2 37^\circ + \cos^2 53^\circ) = Mg + mg$$

18. 2.00 (range 1.95 to 2.05)

Sol. $mg \cos 53^\circ = 0.5 N$

$$mg \sin 53^\circ + N = m\omega^2 r$$

$$\text{So, } mg(0.8) + \frac{mg(0.6)}{0.5} = m\omega^2 r$$

$$\text{So, } g\left(\frac{8}{10} + \frac{6}{5}\right) = \omega^2 r$$

$$\text{So, } 10 \times 2 = \omega^2 r$$

$$\text{So, } \omega = \sqrt{\frac{20}{5}} = 2.00 \text{ rad/s}$$

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Chemistry

PART – II

Section – A

19. B, D

Sol. $\text{XeF}_4 \rightarrow \text{sp}^3\text{d}^2 \rightarrow$ Square planar and nonplanar

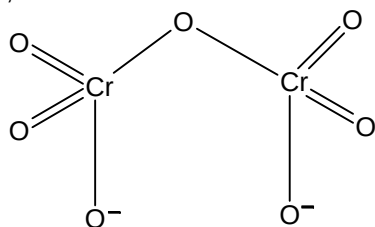
$\text{H}_2\text{O}_2 \rightarrow \text{sp}^3 \rightarrow$ open book structure. Nonplanar and polar

$\text{XeF}_5^- \rightarrow$ Pentagonal planar and non – polar

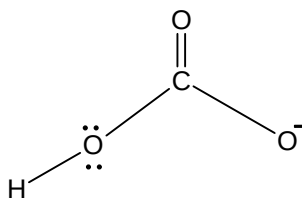
$\text{XeF}_5^+ \rightarrow$ Square pyramidal non – planar polar

20. C, D

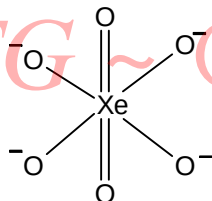
Sol.



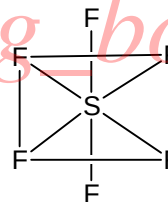
All Cr - O bonds are not equivalent



All C - O bonds are not equivalent



All Xe - O bonds are equivalent due to resonance



sp^3d^2 hybridisation

21. A, B, C

Sol. $r = 0$, $R(r) = 0$ so its true for p-orbital, Azimuthal quantum number is 1.

$$R(r) = 0, \frac{6r}{a_0} - \frac{r^2}{a_0^2} = 0$$

One radial node is observed at $r = 6a_0$ distance.

So, its true for 3p-orbital.

In P_x orbital nodal plane is in yz plane and p-orbital accommodate only 2-electron.

22. A, C, D

Sol. Tetrahedral will be formed by sp^3 -hybridisation. Other three can be formed by distorted sp^3d hybridisation. i.e.

XeF_2 and I_3^- are sp^3d but linear shape.

BrF_3 is sp^3d but bent T-shaped.

SF_4 is sp^3d but see-saw shaped.

23. A, C

Sol. In CrO_5^{3-} Cr has +5 oxidation state.

KO_2 is also colorful.

24. A, B, C

Sol. In option (D) if we add inert gas at constant pressure and temperature. Equilibrium will get disturbed and move in backward direction.

25. A

Sol. For reaction $A \longrightarrow \text{Product}$

Using the given formula we can solve all the options

For zero order reaction $t = \frac{[A]_0 - [A]_t}{k}$

For first order reaction $t = \frac{1}{k} \ln \frac{[A]_0}{[A]_t}$

For second order reaction $t = \frac{1}{k} \left[\frac{1}{[A]_t} - \frac{1}{[A]_0} \right]$

26. B

Sol. Correct answer is $I \rightarrow P, R, S$; $II \rightarrow P, S, T$; $III \rightarrow P, R, S$; $IV \rightarrow P, Q, R, S$

27. C

Sol. Correct order is $I \rightarrow R, S$; $II \rightarrow R$; $III \rightarrow Q, S$; $IV \rightarrow P$

28. D

Sol. Correct answer is $I \rightarrow Q$; $II \rightarrow P$; $III \rightarrow S$; $IV \rightarrow R$

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Section – B

29. 1.00

Sol. SiO_2 is neutral and Na has unit +ve charge AlO_2 should have unit – ve charge to balance the charge of molecule.

$$x(1) + y(-1) + 136(0) + 250(0) = 0$$

$$x = y$$

$$\frac{x}{y} = 1.00$$

30. 9.09

Sol. $\text{CH}_3\text{COOH} + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

$$0.60 \text{ V} \quad 0.20 \times 3 \text{ V} \quad 0.60 \text{ V}$$

$$[\text{CH}_3\text{COONa}] = \frac{0.60 \text{ V}}{4 \text{ V}} = \frac{3}{20} \text{ M}$$

pH at 25°C for salt of WA + SB is

$$\text{pH} = 7 + \frac{1}{2} (\text{p}K_a + \log(\text{salt}))$$

$$\text{pH} = 7 + \frac{1}{2} \left(5 + \log \frac{3}{20} \right)$$

$$= 7 + \frac{1}{2} (5 - 0.82)$$

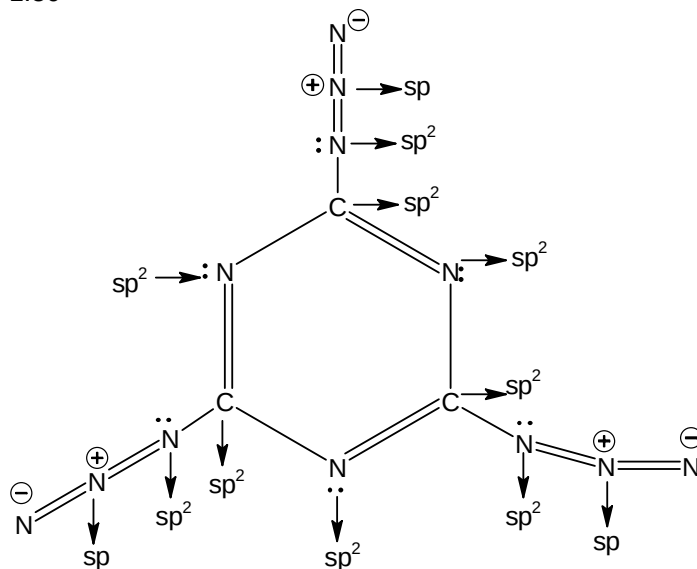
$$= 7 + \frac{1}{2} (4.18)$$

$$= 7 + 2.09$$

$$= 9.09$$

31. 1.50

Sol.



Maximum 6π bond can lie in same plane and sp^2 centre are 9.

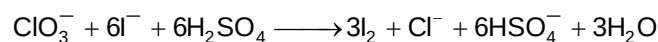
$$x = 9, y = 6$$

$$\Rightarrow \frac{x}{y} = \frac{9}{6} = 1.50$$

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32. 1.50

Sol. Balance reaction is

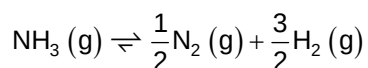


$$x = 6, y = 6, z = 3$$

33. 6.93

 Sol. $\text{NH}_4\text{HS}(s) \rightleftharpoons \text{NH}_3(g) + \text{H}_2\text{S}(g)$

$$2 - x \qquad x - y \qquad x$$



$$x - y \qquad \frac{y}{2} \qquad \frac{3y}{2}$$

Given,

$$2 - x = 1$$

$$x = 1$$

$$\frac{3y}{2} = 0.75$$

$$y = \left(\frac{1}{2}\right)$$

$$x_1 = K_{c1} \left(\frac{1}{2}\right) \cdot \left(\frac{1}{4}\right) = \left(\frac{1}{8}\right)$$

$$y_1 = K_{c2} = \frac{\left(\frac{3}{8}\right)^{\frac{3}{2}} \left(\frac{1}{8}\right)^{\frac{1}{2}}}{\left(\frac{1}{4}\right)} = \left(\frac{3\sqrt{3}}{16}\right)$$

$$a = [\text{NH}_3] = \frac{1}{4} \text{ M}$$

$$[\text{H}_2\text{S}] = \frac{1}{2} \text{ M}$$

$$b = [\text{N}_2] = \frac{1}{8} \text{ M}$$

$$[\text{H}_2] = \frac{3}{8} \text{ M}$$

$$\begin{aligned} \frac{y_1}{x_1(a+b)} &= \frac{3\sqrt{3}}{16} \times \frac{8}{1} \times \frac{8}{3} = 4\sqrt{3} \\ &= 4 \times 1.732 \\ &= 6.928 \\ &= 6.93 \end{aligned}$$

34. 1.33

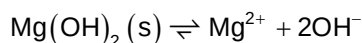
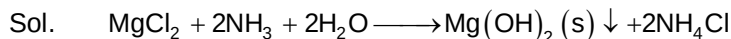
Sol. $K(\text{overall}) = \frac{K_1 K_3}{K_2} = 10^{-4} \text{ sec}^{-1}$

$$A(\text{overall}) = \frac{A_1 A_3}{A_2} = 2 \times 10^5 \text{ sec}^{-1}$$

$$E_a(\text{overall}) = E_{a_1} + E_{a_3} - E_{a_2} = 15 \text{ kJ}$$

$$\frac{K.A}{E_a} = \frac{2 \times 10^{-4} \times 10^5}{15} = 1.33$$

35. 18.48



$(\text{NH}_4)_2\text{SO}_4$ is acidic so it will dissolve $\text{Mg}(\text{OH})_2$ solid which is formed in mix. When all $\text{Mg}(\text{OH})_2$ will re-dissolve the conc. of Mg^{2+} ions in the final solution will become equal to its initial conc.

$$[\text{NH}_3] \text{ in mix initially} = \frac{0.70 \times 0.2}{1.0} = 0.14 \text{ M}$$

$$[\text{Mg}^{2+}] \text{ in mix initially} = \frac{0.30 \times 0.2}{1.0} = 0.06 \text{ M}$$

$$\text{When } [\text{Mg}^{2+}] = 0.06$$

$$[\text{OH}^-] = \sqrt{\frac{K_{\text{SP}}}{[\text{Mg}^{2+}]}} = \sqrt{\frac{6 \times 10^{-12}}{0.06}} = 10^{-5} \text{ M}$$

$$\text{For } \text{NH}_3 \quad K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} \Rightarrow [\text{NH}_4^+] = \frac{K_b [\text{NH}_3]}{[\text{OH}^-]}$$

$$[\text{NH}_4^+] = \frac{2 \times 10^{-5} \times 0.14}{10^{-5}} = 0.28 \text{ M}$$

$$\text{Mole of } [\text{NH}_4^+] = 0.28$$

$$\text{Mole of } [\text{NH}_4]_2 \text{SO}_4 \text{ needed} = \frac{0.28}{2} = 0.14$$

$$\begin{aligned} \text{Mass of } [\text{NH}_4]_2 \text{SO}_4 &= 132 \times 0.14 \\ &= 18.48 \text{ g.} \end{aligned}$$

36. 37.50

Sol.

	$\text{A (aq)} \longrightarrow 2\text{B (aq)} + \text{C (aq)}$		
$t = 0$	a	0	0
$t = 10 \text{ min}$	$a - \frac{a}{2}$	a	$\frac{a}{2}$
$t = 20 \text{ min}$	$a - \frac{3a}{4}$	$\frac{3a}{2}$	$\frac{3a}{4}$
$t = \infty$	0	$2a$	a

Let the normality of 'T' is N and initial volume of 'T' is used x ml

At $t = 0$ Meq. of A = Meq. of T

$$a \times 2n = N \times x \quad \dots (1)$$

At $t = 10 \text{ min}$ Meq. of (A + B + C) = Meq. of T

$$\frac{a}{2} \times 2n + a \times n + \frac{a}{2} \times 2n = 40 \times N \quad \dots (2)$$

At $t = 20 \text{ min}$ Meq. of (A + B + C) = Meq. of T

$$\frac{a}{4} \times 2n + \frac{3a}{2} \times n + \frac{3a}{4} \times 2n = N \times y \quad \dots (3)$$

At $t = \infty \text{ min}$ Meq. of (B + C) = Meq. of T

$$2a \times n + a \times 2n = N \times z \quad \dots (4)$$

$$\text{Solving these equilibrium } x = \frac{80}{3} \text{ ml, } y = \frac{140}{3} \text{ ml, } z = \frac{160}{3} \text{ ml}$$

$$\text{Value of } \frac{y+z}{x} = 37.5 .$$

Mathematics

PART – III

Section – A

37. B, C, D

Sol. $f(x) = \frac{3}{\sqrt{2x^2 + x} + \left(\frac{2\sin^2 \theta}{\sqrt{2x^2 + x}}\right)}$

Using AM $\geq$ GM

$$\Rightarrow \frac{\sqrt{2x^2 + x} + \frac{2\sin^2 \theta}{\sqrt{2x^2 + x}}}{2} \geq (2\sin^2 \theta)^{\frac{1}{2}} \Rightarrow \frac{3}{2f(x)} \geq 2^{\frac{1}{2}} |\sin \theta| \Rightarrow f(x) \leq \frac{3}{2^{\frac{1}{2}}} |\csc \theta|$$

$$\therefore f(x) \leq \frac{3}{\sqrt{2}} \Rightarrow |\csc \theta| = 2 \Rightarrow |\sin \theta| = \frac{1}{2}$$

38. A, B

Sol. Put $x = 0 \Rightarrow \int_0^1 f(t) e^{-t} dt = \frac{1}{a}$

$$\Rightarrow \int_0^x f(t) dt = e^x - ae^{2x} \times \frac{1}{a} = e^x - 2e^{2x}, \text{ differentiating we get, } f(x) = e^x - 2e^{2x}$$

39. A, B, C

Sol. $(x - y) f(x + y) - (x + y) f(x - y) = 2y(x - y)(x + y)$

$$\text{Let } x - y = u, x + y = v \Rightarrow uf(v) - v(f(u)) = (v - u)uv \Rightarrow \frac{f(v)}{v} - v = \frac{f(u)}{u} - u = k$$

$$\Rightarrow f(x) = kx + x^2 \therefore f(1) = 2 \Rightarrow k = 1 \Rightarrow f(x) = x^2 + x$$

40. D

Sol. $x^a y = \lambda^a$, on differentiating $ax^{a-1}y + x^a \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{ay}{x}$

$$\text{Tangent at point } (x_1, y_1) \text{ is } y - y_1 = -\frac{ay_1}{x_1}(x - x_1)$$

$$\text{Area} = \frac{1}{2} \lambda^a \cdot \frac{(a+1)^2}{a} (x_1)^{1-a} \Rightarrow \text{area is constant if } a = 1$$

41. A, D

Sol. $f(x) \leq 0, F'(x) = f(x)$ as $f(x) \geq c F(x) \Rightarrow F'(x) - c F(x) \geq 0$

$$\Rightarrow e^{-cx} \cdot F'(x) - c \cdot e^{-cx} F(x) \geq 0 \Rightarrow \frac{d}{dx}(e^{-cx} \cdot F(x)) \geq 0$$

$$\Rightarrow e^{-cx} \cdot F(x) \text{ is an increasing function}$$

$$\Rightarrow e^{-cx} \cdot F(x) \geq e^{-c(0)} \cdot f(0) \Rightarrow F(x) \geq 0 \Rightarrow f(x) \geq 0 \Rightarrow f(x) = 0$$

$$\text{Also, } \frac{d}{dx}(e^{-x}g(x)) < 0 \Rightarrow e^x g(x) \text{ is a decreasing function}$$

$$\Rightarrow e^{-x}g(x) < e^{-(0)}g(0) \Rightarrow g(x) < 0 \text{ (as } g(0) = 0)$$

$$\text{Thus, } f(x) = g(x) \text{ has one solution } x = 0 \text{ also, } |x^2 + x - 6| = f(x) + g(x) = g(x) \Rightarrow \text{no solution}$$

42. B, C

Sol. Solution of differential equation is $(\sin^2 x)y + (\sin x)y^2 + (\cos x + \cos^2 x) = 0$

43. D

Sol. (I) $\int_0^1 (f^4(x) - 8x^2 \cdot f^2(x) + 16x^4) dx \Rightarrow \int_0^1 (f^2(x) - 4x^2)^2 = 0 \Rightarrow f^2(x) = 4x^2$

(II) $f''(x) = -\frac{4x^3}{2\sqrt{1-x^4}} \Rightarrow f''(x) = 0$ at $x = 0$ and $f''(x)$ changes sign around it

(III) $g'(x) = x^2 \left(\int_1^x f(u) du \right) + f(x) \cdot \left(\int_0^x t^2 dt \right)$ and $g'(1) = 1 \Rightarrow g''(1) = 6 + \frac{f'(1)}{3}$

(IV) $I_2 = -\frac{1}{3} \int_0^{-1} \frac{dt}{e^{-t}(2+t)} = -\frac{1}{3} \int_1^0 \frac{e^{t-1}}{1+t} dt = \frac{1}{3e} I_1$ (Let $-x^3 = t$)

44. C

Sol. (I) $f(x)$ is discontinuous at $x = \frac{\pi}{2}, \pi$

(II) $A = \int_0^1 \left(x^{3/2} - \left(\frac{3x-1}{2} \right) \right) dx = \frac{3}{20}$

(III) Required area $= 2 \left| \int_1^2 x dx - \int_1^2 (x^2 - 2x + 2) dx \right| = \frac{1}{3}$

(IV) $I_n = \int_0^1 x^n \sqrt{1-x^2} dx = \int_0^1 x^{n-1} \cdot x \sqrt{1-x^2} dx = 0 + \frac{n-1}{3} \int_0^1 x^{n-2} (1-x^2) \sqrt{1-x^2} dx$
 $= \frac{n-1}{3} I_{n-2} - \frac{n-1}{3} I_n \Rightarrow \frac{I_n}{I_{n-2}} = \frac{n-1}{n+2}$

45. B

Sol. (I) For $f(x) = x^3 - \frac{3}{2}x^2 + x + \frac{1}{4} \Rightarrow f(x) + f(1-x) = 1 \Rightarrow f(f(x)) + f(1-f(x)) = 1$

$\Rightarrow I = \int_{1/4}^{3/4} f(f(x)) dx \dots (1)$

$I = \int_{1/4}^{3/4} f(f(1-x)) dx \dots (2)$

Adding equation (1) and (2), we get $2I = \int_{1/4}^{3/4} f(f(x)) + f(1-f(x)) = \frac{1}{2} \therefore I = \frac{1}{4}$

(II) Area (A) $= \int_0^1 (2x - x^2 - x^n) dx = \frac{1}{2} \Rightarrow n = 5$

(III) $f(x) + f(-x) = 1 \Rightarrow S = \frac{11}{2}$

(IV) $f(x) = \left(\frac{3}{2} - x^7 \right)^{1/7}$ and $f(f(x)) = x$

46. D

Sol. (I) Put $2y = y \Rightarrow f(x+y) = f(x) + f(y) + 2xy$ partially differentiate with respect to x
 $f'(x+y) = f'(x) + 2y$ put $x = 1, y = -1 \Rightarrow f'(1) - f'(0) = 2$

(II) Let $\lim_{x \rightarrow \infty} f(x) = \alpha$, then $\alpha + \frac{3\alpha-1}{\alpha^2} = 3 \Rightarrow \alpha = 1$

(III) $f(x) + f(x+\pi) = 2 \therefore \frac{1}{2}f(x) + \frac{1}{2}f(x+\pi) = 1$ also $af(x) + bf(x+c) = 1$

On comparing $a = b = \frac{1}{2}, c = \pi$

(IV) $f(x+y) = 3^y f(x) + 2^x f(y)$ (1)

Put $x = 1$ in equation (1) $f(1+y) = 3^y f(1) + 2f(y)$

$y \rightarrow x \Rightarrow f(1+x) = 3^x f(1) + 2f(x)$ (2)

Now, put $y = 1$ in equation (1)

$f(1+x) = 3f(x) + 2^x f(1)$ (3)

From equation (2) and (3), we get $f(x) = 3^x - 2^x$

Section - B

47. 2023.00

Sol. $f(x) - f(y) \geq \ln x - \ln y + x - y$ (1)

$x \rightarrow y$ and $y \rightarrow x$

$f(y) - f(x) \geq \ln y - \ln x + y - x$

$\Rightarrow f(x) - f(y) \leq \ln x - \ln y + x - y$ (2)

From equation (1) and (2), $f(x) - f(y) = (\ln x + x) - (\ln y + y)$

$\Rightarrow f(x) = \ln x + x \Rightarrow g(x) = 1 + \frac{1}{x} \Rightarrow g\left(\frac{1}{2022}\right) = 2023$

48. 5.00

Sol. $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta = \sin \theta (1 + 2 \cos 2\theta) \Rightarrow \sin x = \sin \frac{x}{3} \left(1 + 2 \cos \frac{2x}{3}\right)$

$\sin \frac{x}{3} = \sin \left(\frac{x}{3^2}\right) \left(1 + 2 \cos \frac{2x}{3^2}\right)$ and so on multiplying all

$$\sin x = \sin \left(\frac{x}{3^n}\right) \prod_{r=1}^n \left(1 + 2 \cos \frac{2x}{3^r}\right) \Rightarrow \frac{\sin x}{x} = \lim_{n \rightarrow \infty} \frac{\sin \left(\frac{x}{3^n}\right)}{\left(\frac{x}{3^n}\right)} \cdot \prod_{r=1}^n \frac{\left(1 + 2 \cos \frac{2x}{3^r}\right)}{3} \Rightarrow f(x) = \frac{\sin x}{x}$$

Thus $|x f(x)| + ||x - 2| - 1| = |\sin x| + ||x - 2| - 1|$ is not differentiable at 5 points in $(0, 3\pi)$

49. 4.50

Sol. $I = \int_0^2 (3x^2 - 6x + 4) \cos(x^3 - 3x^2 + 4x - 2) dx + \int_0^2 (3x - 3) \cos(x^3 - 3x^2 + 4x - 2) dx$

Let $I = I_1 + I_2$

$I_1 = \int_{-2}^2 (\cos t) dt = 2 \sin 2$ (where $t = x^3 - 3x^2 + 4x - 2$)

$I_2 = 3 \int_{-1}^1 u \cos(u^3 + u) du = 0$ (where $u = x - 1$)

50. 7.63

Sol. If each of two positive real number x, y is not integer but $x + y$ is an integer, then $[x] + [y] = x + y - 1$

$$(\because \{x\} + \{y\} = 1)$$

$$\text{Since, } \frac{305k}{503} + \frac{305(503-k)}{503} = 305 \text{ for } 1 \leq k \leq 502$$

$$\Rightarrow S = \left[\frac{305}{503} \right] + \left[\frac{305 \cdot 2}{503} \right] + \dots + \left[\frac{305 \cdot 502}{503} \right] = 304.251 = 76304$$

51. 4044.00

$$\text{Sol. } I_2 = \int_0^1 x^{1010} (1-x^{2022})^{1010} dx \text{ let } x^{1011} = t$$

$$\Rightarrow I_2 = \frac{1}{1011} \int_0^1 (1-t^2)^{1010} dt = \frac{1}{1011} \int_0^1 (1-(1-t^2))^{1010} dt = \frac{1}{1011} \int_0^1 t^{1010} (2-t)^{1010} dt$$

Put $t = 2y$

$$I_2 = \frac{2^{2021}}{1011} \int_0^{1/2} y^{1010} (1-y)^{1010} dy \text{ and } I_1 = \int_0^1 x^{1010} (1-x)^{1010} dx = 2 \int_0^{1/2} x^{1010} (1-x)^{1010} dx$$

$$\Rightarrow I = 2^{2022} \frac{I_1}{I_2} = 4044$$

52. 199.00

$$\text{Sol. } f(x) = \ln x - \frac{1}{x} \Rightarrow f(e^{200}) = 200 - \frac{1}{e^{200}}$$

53. 16.00

$$\text{Sol. } A_1 = 4 \int_0^{\pi/4} (\cos x - \sin x) dx = 4(\sqrt{2} - 1) \text{ also } A_1 = A_2 = \frac{A}{2}$$

54. 4.75

$$\text{Sol. } I = \frac{1}{2} \int_0^{\pi/2} \sin x (\ln(\sin^2 x)) dx, \text{ put } \cos x = t$$

$$I = \frac{1}{2} \int_0^1 \ln(1-t^2) dt = \frac{1}{2} \int_0^1 \left(-t^2 - \frac{(-t^2)^2}{2} + \frac{(-t^2)^3}{3} + \dots \right) dt = -\frac{1}{2} \left[\frac{1}{3} + \frac{1}{10} + \frac{1}{21} + \dots \right]$$

$$= -\left[\frac{1}{6} + \frac{1}{20} + \frac{1}{42} + \dots \right] = -\left[\left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \dots \right] = \log_e 2 - 1$$

$$\text{and } A = \frac{15}{4} + \ln 2$$

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – I
PAPER –2
TEST DATE: 20-11-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1.

Sol.

D
 $\tau = I\alpha$ (about ICOR)
 $mg\left(\frac{\ell}{5}\right)\frac{1}{2} = \left(\frac{m\ell^2}{12} + m\left(\frac{\ell}{2}\right)^2\right)\alpha$

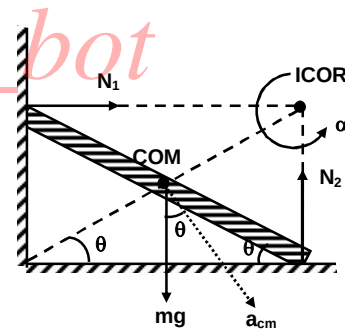
$$\alpha = \frac{6g}{5\ell}$$

$$\therefore a_{cm} = \frac{\ell}{2}\alpha = \frac{3g}{5}$$

$$mg - N_2 = (ma_{cm})\cos\theta$$

$$mg - N_2 = \frac{3mg}{5}\left(\frac{4}{5}\right)$$

$$\Rightarrow N_2 = \frac{13mg}{25}$$



2.

Sol.

A
 $\theta = \omega t$ ($\omega = \text{constant}$)

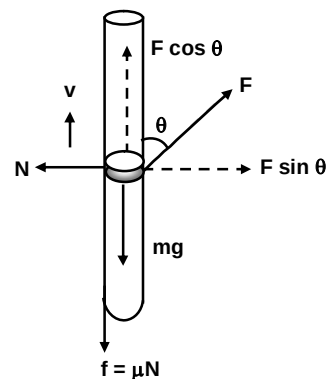
$$N = F \sin \omega t$$

$$\frac{mdv}{dt} = (F \cos \omega t - \mu N - mg)$$

$$\int_0^T mdv = \int_0^T (F(\cos \omega t - \mu \sin \omega t) - mg) dt$$

$$\frac{F}{\omega} [\sin \omega t + \mu \cos \omega t]_0^T = mgT$$

$$\therefore \text{Also } \omega T = \frac{\pi}{2}$$



$$\therefore F \left[\sin \frac{\pi}{2} + \mu \cos \frac{\pi}{2} - \mu \right] = mg \omega T$$

$$F[1 - \mu] = mg \frac{\pi}{2}$$

$$\therefore F = \frac{mg\pi}{2(1-\mu)}$$

3. B

Sol. F.B.D. of rod

$$\bar{\tau}_0 = 0 \quad (\text{about pivot})$$

$$\Rightarrow N_1 = \frac{mg}{2} \quad \dots(i)$$

F.B.D. of cylinder

$$\bar{\tau}_C = 0$$

$$\Rightarrow f_1 R = f_2 R$$

$$f_1 = f_2$$

(say f)

$$\text{Also } \bar{\tau}_0 = 0$$

$$(N_1 + mg)\ell = N_2 \ell \Rightarrow N_2 = N_1 + mg$$

$$N_2 = \frac{3mg}{2} \quad \dots(ii)$$

$$\sum F_x = 0 \quad (\text{on the cylinder})$$

$$f(1 + \cos \theta) = N_2 \sin \theta$$

$$\therefore f = \frac{3mg \sin \theta}{2(1 + \cos \theta)}$$

$$\text{Also } f_{1\max} < f_{2\max} \quad (\because N_1 < N_2)$$

$$\therefore f \leq \mu N_1 \Rightarrow \frac{3mg \sin \theta}{2(1 + \cos \theta)} \leq \mu \frac{mg}{2}$$

$$\mu \geq \frac{3 \sin \theta}{1 + \cos \theta}$$

4. A

Sol. F.B.D. of block is as shown ($\theta \leq 90^\circ$)

$$N = F \sin \theta + mg \cos \alpha$$

$$f = F \cos \theta - mg \sin \alpha$$

for No sliding $f \leq \mu N$

$$F \cos \theta - mg \sin \alpha \leq \frac{1}{\sqrt{3}} (F \sin \theta + mg \cos \alpha)$$

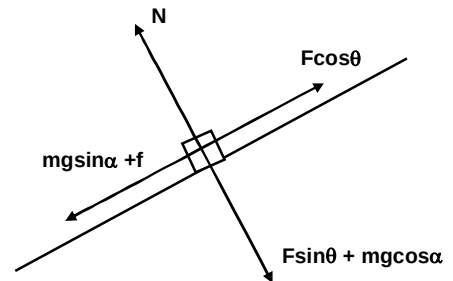
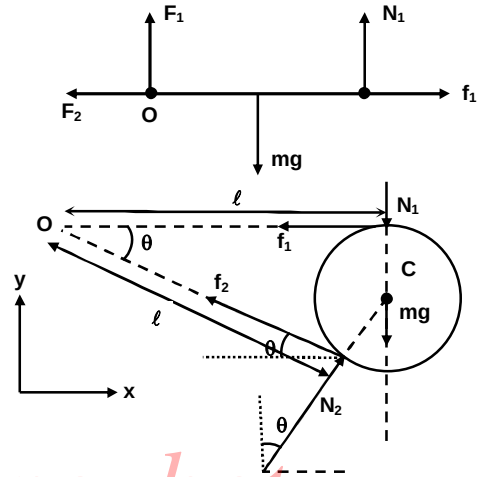
$$\cos \theta - \frac{\sin \theta}{\sqrt{3}} \leq \frac{mg}{F} \left[\sin \alpha + \frac{\cos \alpha}{\sqrt{3}} \right]$$

$$\cos \theta - \frac{\sin \theta}{\sqrt{3}} \leq 0 \quad (\because \text{no sliding even when } F \rightarrow \infty)$$

$$\tan \theta \geq \sqrt{3} \Rightarrow \theta \geq 60^\circ$$

For $\theta > 90^\circ$

$$N = F \sin \theta + mg \cos \alpha$$



$$f = mg \sin \alpha - F \cos \theta$$

$$f \leq \mu N \Rightarrow \theta \leq 120^\circ$$

$$\therefore \text{Range} \Rightarrow \theta \in [60^\circ, 120^\circ]$$

5. A, B, D

Sol. Applying WET on bead from A to B $W_N + W_F = \Delta K$

$$O + F[(R-x) - (R+x)] = \frac{m}{2}[v^2 - u^2]$$

conservative

$$v^2 = u^2 - \frac{4Fx}{m} \Rightarrow u_0 = 2\sqrt{\frac{Fx}{m}}$$

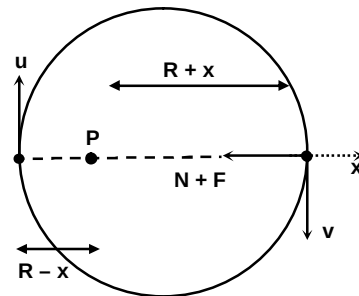
$$\text{Also } N + F = \frac{mv^2}{R}$$

$$N = \frac{m}{R}\left[u^2 - \frac{4Fx}{m}\right] - F = \frac{m}{R}[u^2 - u_0^2] - F$$

$$N = \frac{m}{R}[\alpha^2 - 1]u_0^2 - F = \frac{m}{R}[\alpha^2 - 1]\frac{4Fx}{m} - F$$

$$N = F[4\beta(\alpha^2 - 1) - 1]$$

$$\vec{N} = F[1 + 4\beta(1 - \alpha^2)]\hat{i}$$



6. A, D

Sol. As $PA + PB = 10\ell = \text{Constant}$

The bead will move along the elliptical path in vertical plane.

Such that $2a = 10\ell$ and $2ae = 8\ell \Rightarrow a = 5\ell, b = 3\ell$

$\therefore$ Equation of ellipse is

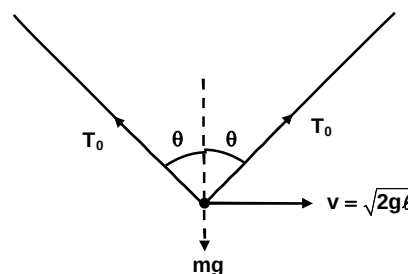
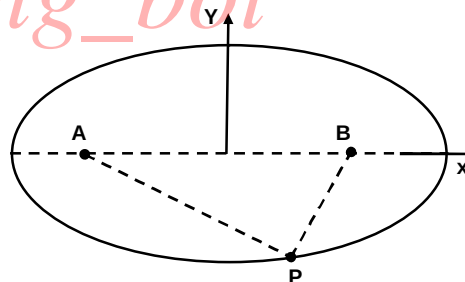
$$\frac{x^2}{(5\ell)^2} + \frac{y^2}{(3\ell)^2} = 1$$

$$r_0 = \frac{[1 + (y')^2]^{3/2}}{|y''|} \text{ at } (0, -3\ell) \quad (\because y' = 0, y'' = \frac{3}{25\ell})$$

$$\therefore r_0 = \frac{25\ell}{3}$$

$$2T_0 \cos \theta - mg = \frac{m}{r_0}(\sqrt{2g\ell})^2 \quad (\because \cos \theta = \frac{3}{5})$$

$$T_0 = \frac{31mg}{30}$$



7. A, C

Sol. Angular momentum of system is conserved about the axis passing through the pivots

$$\vec{L}_i \text{ (Just before collision)} = \vec{L}_f \text{ (Just after collision)}$$

$$mv_0 R = \left(mR^2 + \frac{2}{5}(5m)R^2\right)\omega$$

$$\Rightarrow \omega = \frac{v_0}{3R}$$

Now, applying conservation of mechanical energy after the collision

$$\frac{1}{2} \left[mR^2 + \frac{2}{5}(5m)R^2 \right] \omega^2 + mgR = 5mg \frac{3R}{8}$$

$$\frac{mR^2}{2} (3) \frac{v_0^2}{9R^2} = \frac{7mgR}{8}$$

$$\text{Also } v_0 = \sqrt{2gh} \Rightarrow h = \frac{21R}{8} \Rightarrow \omega = \sqrt{\frac{7g}{12R}}$$

8. A, D

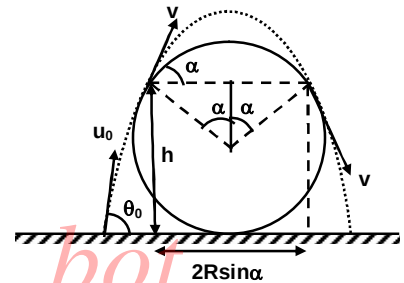
Sol. For u to be minimum, it just grazes the cylinder at two points as shown.

From COME

$$\frac{mu^2}{2} = \frac{mv^2}{2} + mgh$$

$$u^2 = v^2 + 2gh$$

$$u^2 = v^2 + 2gR(1 + \cos \alpha)$$



$$\text{Also } 2R \sin \alpha = v^2 \frac{\sin 2\alpha}{g} \Rightarrow v^2 = \frac{Rg}{\cos \alpha} \quad \dots(2)$$

From (1) and (2)

$$u^2 = Rg \left[2 + 2\cos \alpha + \frac{1}{\cos \alpha} \right]$$

$$\text{For } u \rightarrow \min. \quad \frac{d}{d\alpha}(u^2) = 0 \Rightarrow \cos \alpha = \frac{1}{\sqrt{2}} \Rightarrow \alpha = 45^\circ$$

$$u_0^2 = Rg [2 + 2\sqrt{2}]$$

$$\text{Also } u \cos \theta = v \cos \alpha$$

$$\tan^2 \theta_0 = [3 + 2\sqrt{2}]$$

9. A, D

Sol. Just after the jump by the frog.

According to COLM of (A + frog)

$$(4v_0 - v)m - mv = 0 \quad v = 2v_0$$

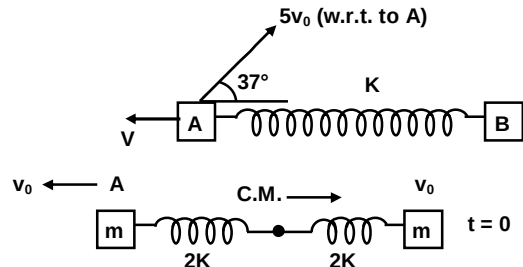
Now consider the motion of blocks in their COM frame.

At any time t

$$v_{BC} = v_B - v_{CM} = v_0 \cos(\omega t), \quad \omega = \sqrt{\frac{2k}{m}}$$

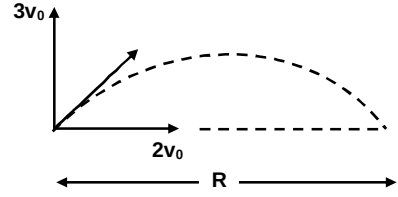
$$v_B = v_{CM} + v_0 \cos \omega t$$

$$v_B = -v_0 + v_0 \cos \omega t$$



Also the frog is in projectile motion from ground frame

$$R = 2 \left(\frac{3v_0}{g} \right) \frac{2v_0}{g} = \frac{12v_0^2}{g} \quad \left(T = \frac{6v_0}{g} \right)$$



Also frog land on B

Then $\ell_0 + \int_0^T v_B dt = R$

$$\ell_0 = R - \int_0^T v_0 (\cos \omega t - 1) dt$$

$$\ell_0 = R + v_0 T - \frac{v_0}{\omega} \sin \omega T$$

$$\therefore \ell_0 = \frac{12v_0^2}{g} + \frac{6v_0^2}{g} - v_0 \sqrt{\frac{m}{2K}} \sin \left(\sqrt{\frac{2K}{m}} \frac{6v_0}{g} \right)$$

$$\therefore \ell_0 = \frac{6v_0^2}{g} \left[3 - \frac{2}{\pi} \right].$$

10. B, C

Sol. $F = mg + v_0 \frac{dm}{dt}$ also $m = \lambda y$ TG ~ @bohring_bot

$$F = \lambda y g + v_0 \lambda \frac{dy}{dt}$$

$$= \lambda y g + \lambda v_0^2$$

$$\text{Heat loss per unit time} = F v_0 - \frac{dE}{dt}$$

$$= (\lambda y g + \lambda v_0^2) v_0 - \frac{d}{dt} \left[\frac{1}{2} (\lambda y) v_0^2 + \lambda y g \frac{y}{2} \right]$$

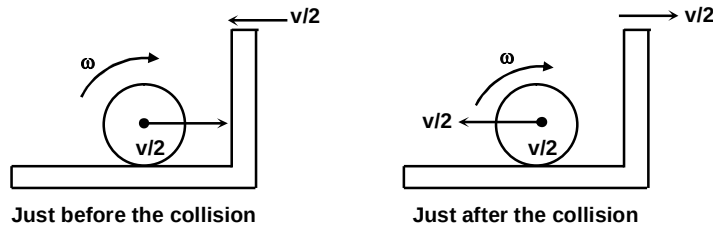
$$\lambda y g v_0 + \lambda v_0^3 - \frac{\lambda v_0^3}{2} - \lambda y g v_0$$

$$= \frac{\lambda v_0^3}{2}.$$

Section – B

11. 2

Sol. Observing the motion in COM frame. Then system will look like



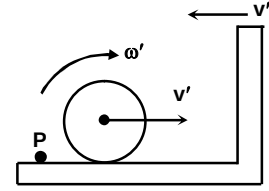
After the collision, ring starts sliding and then starts pure rolling on wedge as shown.

$$\omega' R = 2v' \text{ (pure rolling)}$$

Applying COAM of ring in centre of mass frame (after the collision) about point P on wedge

$$mR^2\omega + \left(-mR\frac{v}{2}\right) = mRv' + mR^2\omega'$$

$$mR\frac{v}{2} = 3mRv' \Rightarrow v' = \frac{v}{6}$$



Therefore the velocity of wedge is $\frac{v}{6}$ (in centre of mass frame) when pure rolling starts after the first collision. Then they again collide and ring again starts pure rolling on wedge and velocity of wedge is $\frac{v}{18}$ (in centre of mass frame) after 2nd collision.

The process goes on and there will be infinite collisions and finally wedge comes at rest in centre of mass frame.

$$\therefore \text{Velocity of wedge} = \frac{v}{2} \text{ (in ground frame)}$$

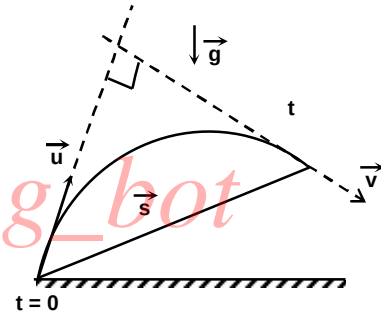
12. 5

Sol. $\vec{v} = \vec{u} + \vec{g}t$

$$\text{also } \vec{s} = (\vec{v}_{\text{ave}})t = \left(\frac{\vec{v} + \vec{u}}{2}\right)t$$

$$\therefore \vec{v} - \vec{u} = \vec{g}t \quad \dots (i)$$

$$\vec{v} + \vec{u} = \frac{2\vec{s}}{t} \quad \dots (ii)$$



Squaring and multiplying

$$(\vec{v} \cdot \vec{v} + \vec{u} \cdot \vec{u} + 2\vec{u} \cdot \vec{v})(\vec{v} \cdot \vec{v} + \vec{u} \cdot \vec{u} - 2\vec{u} \cdot \vec{v}) = 4g^2s^2$$

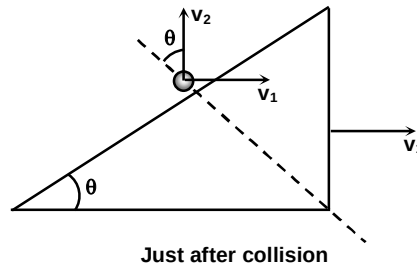
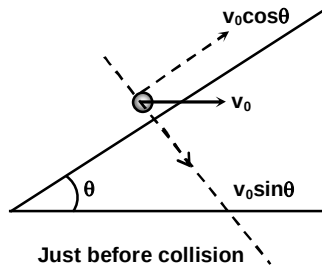
$$s^2 = \frac{(100 + 2\vec{u} \cdot \vec{v})(100 - 2\vec{u} \cdot \vec{v})}{4g^2}$$

s is max. when $\vec{u} \cdot \vec{v} = 0 \Rightarrow \vec{u} \perp \vec{v}$

$$\therefore s_{\text{max}} = \frac{100}{2(10)} = 5\text{m}$$

13. 6

Sol. If it strike again at same point, then the horizontal component of their velocities are same after the first collision.



As $e = 1$

$$v_{\text{sep}} = v_{\text{app}}$$

$$v_2 \cos \theta = v_0 \sin \theta$$

$$\therefore v_2 = v_0 \tan \theta = (40) \frac{3}{4} = 30 \text{ m/s}$$

$$t_0 = \frac{2v_2}{g} = 6 \text{ sec.}$$

14. 9

Sol.

$$T_{\max} = T_1$$

$$T_{\min} = T_2$$

$$T_1 - mg = \frac{mu^2}{l} \quad \dots(1)$$

$$T_2 + mg = \frac{mv^2}{l} \quad \dots(2)$$

From COME (of pendulum + earth system)

$$\frac{1}{2} mu^2 = \frac{1}{2} mv^2 + mg(2l)$$

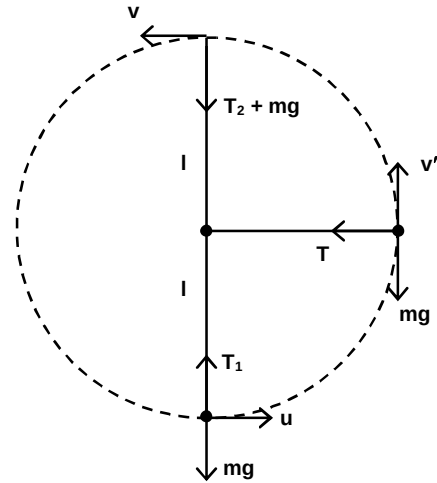
$$\therefore u^2 = v^2 + 4gl \quad \dots(3)$$

$$\text{also } \frac{T_1}{T_2} = 2 \quad \dots(4)$$

from (1), (2) (3) and (4)

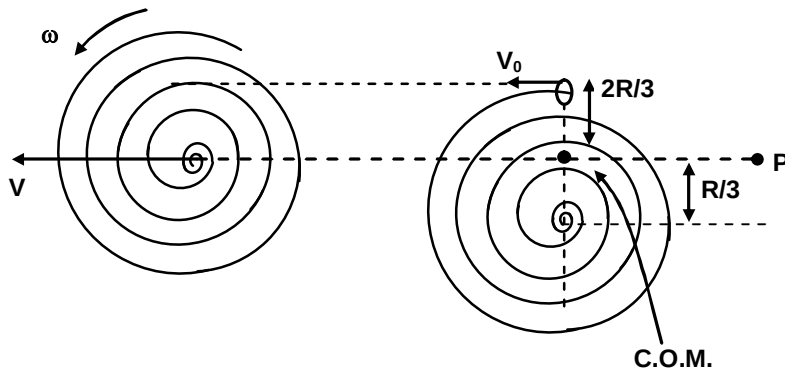
$$u = \sqrt{11gl}, \quad v = \sqrt{7gl}$$

$$\therefore v' = \sqrt{9gl}, \quad T = \frac{mv'^2}{l} = 9mg \quad K = 9$$



15. 2

Sol.



Applying COLM

$$mv_0 = (m + 2m)v \quad v = \frac{v_0}{3}$$

Applying COAM about point P

$$mv_0 \frac{2R}{3} = \left(\frac{MR^2}{2} \right) \omega \Rightarrow 2m \frac{v_0 R}{3} = mR^2 \omega$$

$$\therefore \omega R = \frac{2v_0}{3}$$

$$\therefore \frac{\omega R}{v} = 2$$

16. 3

Sol.
$$T = \frac{2v_0 \cos\left(\frac{3\theta}{2}\right)}{g \cos\left(\frac{\theta}{2}\right)}$$

$$= \frac{2v_0}{g} \left[4 \cos^2\left(\frac{\theta}{2}\right) - 3 \right]$$

Also $R \sin \theta = V_0 \sin 2\theta T$

$$R \sin \theta = \frac{2v_0^2}{g} (2 \sin \theta \cos \theta) [2(1 + \cos \theta) - 3]$$

As $v_0^2 = 2gh$

$$\therefore h = \frac{R}{8[\cos \theta (2 \cos \theta - 1)]}$$

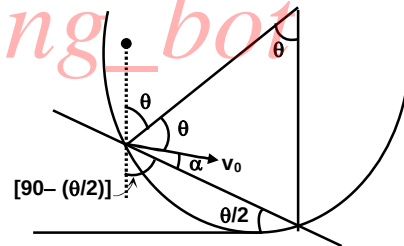
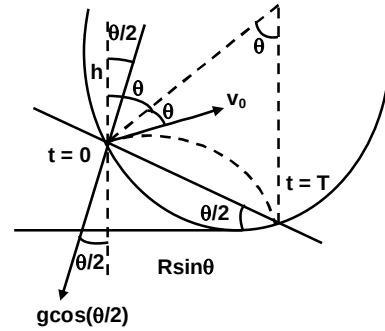
$$h \geq 0 \quad 2 \cos^2 \theta - \cos \theta \geq 0$$

$$\cos \theta (2 \cos \theta - 1) \geq 0$$

$$\cos \theta \geq \frac{1}{2}$$

$$\theta \leq \frac{\pi}{3} \quad K = 3.$$

2nd Method:



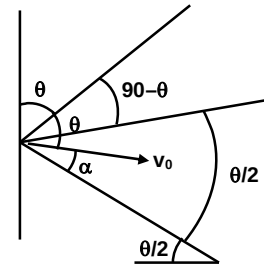
$$2\theta + \alpha + \left(90 - \frac{\theta}{2}\right) = 180^\circ$$

$$\alpha = 90 - \frac{3\theta}{2}$$

Now $\alpha > 0$

$$\theta < 60^\circ$$

$$\theta_{\max} = \frac{\pi}{3} \text{ radian}$$



17. 6

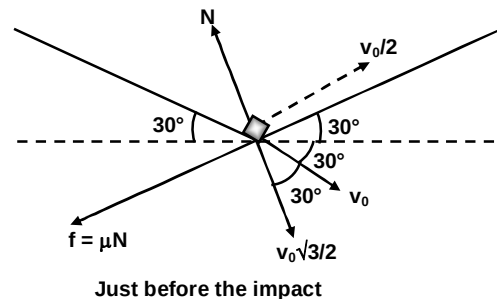
Sol. Applying impulse momentum theorem.

$$\int N dt = mv_0 \frac{\sqrt{3}}{2}$$

$$\int \mu N dt = m \left(\frac{v_0}{2} - v \right)$$

$$\therefore v = \frac{v_0}{2} (1 - \sqrt{3}\mu)$$

Also $v_0^2 = 2gH$ and

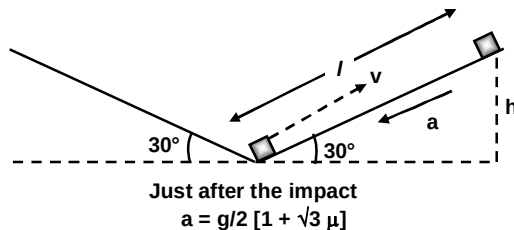


$$\therefore 2gh(1 + \sqrt{3}\mu) = \frac{2gH}{4}(1 - \sqrt{3}\mu)^2$$

Also $\mu = \frac{1}{2\sqrt{3}}$

$$h\left(1 + \frac{1}{2}\right) = \frac{H}{4}\left(1 - \frac{1}{2}\right)^2$$

$$\Rightarrow \frac{H}{4h} = 6$$



18.

8

Sol.

In equilibrium, the centre of mass of the system must lie on the vertical line passing through hinge.

$$\therefore X_{CM} = 0$$

$$-m_1 \frac{L}{2} + m_2 \left(\frac{2R}{\pi} \right) = 0$$

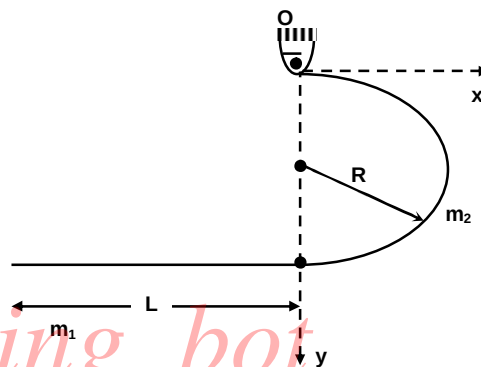
$$\frac{\lambda L^2}{2} = \lambda (\pi R) \frac{2R}{\pi}$$

$$\therefore L = 2R$$

$$Y_{CM} = \frac{m_1(2R) + m_2 R}{m_1 + m_2}$$

$$= \frac{\lambda L^2 + \lambda \pi R^2}{\lambda L + \lambda \pi R} = \left(\frac{\pi + 4}{\pi + 2} \right) \frac{L}{2} \Rightarrow \alpha = 4, \beta = 2$$

$$\therefore \alpha\beta = 8.$$

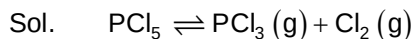


Chemistry

PART – II

Section – A

19. A



$$\alpha = \frac{D-d}{(n-1)d}, \quad n=2$$

$$(1+\alpha) = \left(\frac{D}{d}\right)$$

Thus, $(1+\alpha)$ will linearly increases with increase of $\left(\frac{D}{d}\right)$.

20. A

Sol. $\ell n K = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{R}$

$$\Delta S^\circ = 8R$$

21. B

Sol. Change in potential energy $= -\frac{2x}{8} - (-2x) = -\frac{2x}{8} + 2x = \frac{-2x + 16x}{8}$
 $= +\frac{14x}{8} = +\frac{7x}{4}$

22. A

Sol. $K_{298} = \frac{2.303}{t_{1/2}} \log \frac{100}{50}$

$$K_{308} = \frac{2.303}{t_{1/2}} \log \frac{100}{25}$$

$$\frac{K_{298}}{K_{308}} = \frac{\log 2}{\log 4} = \frac{1}{2} = 0.5$$

23. A, B

Sol. Possible transition $3 \rightarrow 2$ and $4 \rightarrow 2$

$$\lambda_{3 \rightarrow 2} \Rightarrow \frac{36}{5R}$$

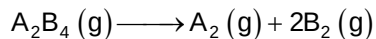
$$\lambda_{4 \rightarrow 2} \Rightarrow \frac{16}{3R}$$

24. A, B, C

Sol. Molecular mass of $\text{A}_2\text{B}_4 = 100$

$$2A + 4B = 100, \quad \frac{2A + 4B}{2A} = 5 \text{ (given)}$$

$$A = 10, B = 20, \quad \frac{A}{B} = \frac{10}{20} = 1:2$$



$$1 - \alpha \qquad \qquad \alpha \qquad \qquad 2\alpha$$

At 10 min. V.D. = 25

Molecular mass = 50

$$50 = \frac{100}{1 + 2\alpha}$$

$$\alpha = \frac{1}{2} \quad \%B_2 \text{ (by mole)} = \frac{1}{2} \times 100 = 50\%$$

$$\%A_2 \text{ (by mole)} = \frac{1}{4} \times 100 = 25\%$$

At t = 20 min.

$$40 = \frac{100}{1 + 2\alpha}$$

$$\alpha = 0.75$$

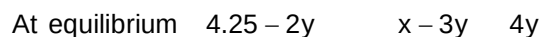
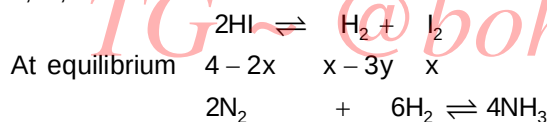
$$-\frac{d(A_2B_4)}{dt} = \frac{0.5 - 0.25}{10} = \frac{0.25}{10}$$

$$+\frac{d(A_2)}{dt} = \frac{0.75 - 0.50}{10} = \frac{0.25}{10}$$

Ratio of rate = 1 : 1

25.
Sol.

A, C, D



Given $x = 1.0 \text{ M}$

$$4y = 0.5 \text{ M}$$

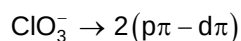
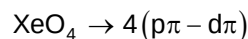
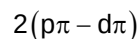
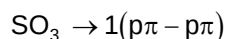
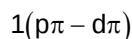
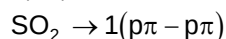
Using these value $K_{C_1} = \frac{5}{32}$

$$K_{C_2} = \frac{4^5}{5^6}$$

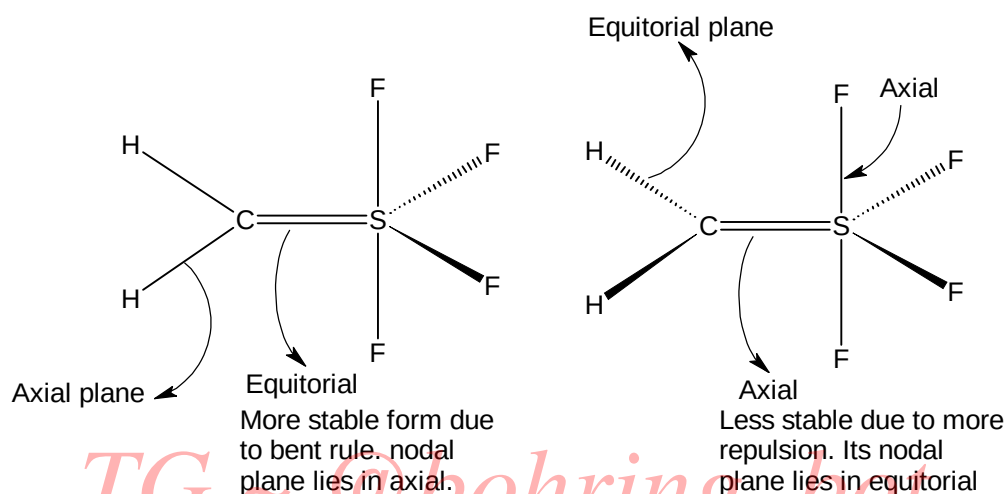
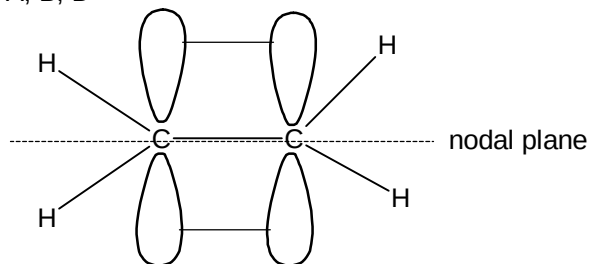
$$\sqrt{K_{C_2}} = \left(\frac{32}{125} \right)$$

26.
Sol.

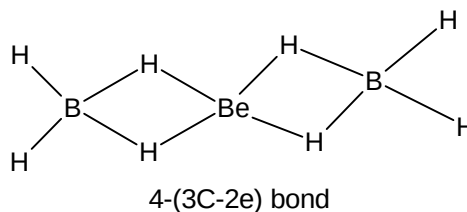
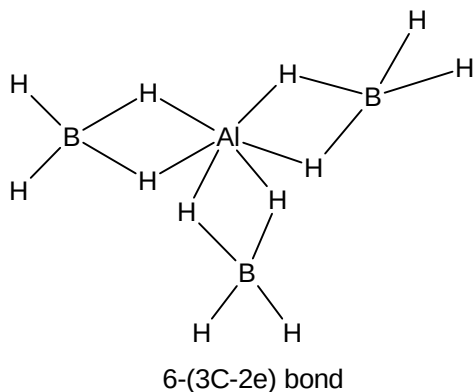
B, C, D



27. A, B, D
Sol.



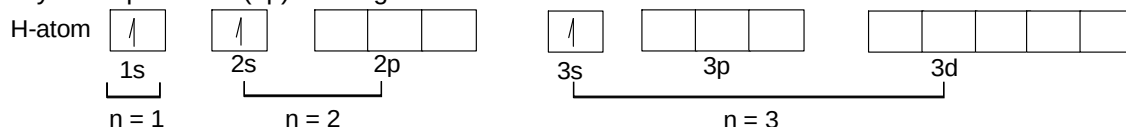
28. C, D
Sol. $(\text{AlCl}_3)_n \rightarrow$ has $(3C - 4e)$ bond not $(3C - 2e)$
 $(\text{B}_2\text{H}_6) \rightarrow$ Two $(3C - 2e)$ bond

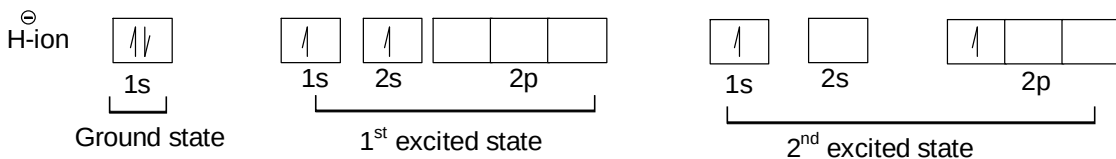


Section - B

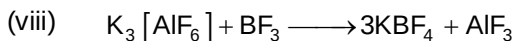
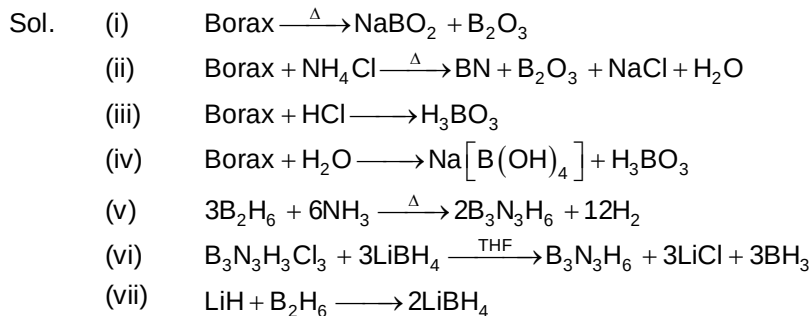
29. 5
Sol. $x = 9, y = 3, z = 2$

In one electron system all orbitals of a shell are degenerate (same energy level). In case of many electron system different orbitals of a shell are non-degenerate. Hence, in second excited state only three p-orbitals (2p) are degenerate.





30. 6



Except (vii) and (viii) all options can give sp^2 hybridised Boron centre as one of the product.

31. 6

Sol. Lone pair in $[\text{TeBr}_6]^{2-}$, $[\text{IF}_2]^+$, SNF_3 and $[\text{XeF}]_3^-$ are 1, 2, 0 and 3 respectively. Hence, $x = 6$.

Diamagnetic species are Li_2 , C_2 , O_2^{2-} , K_2O , N_2O_4 , $\text{B}_3\text{N}_3\text{H}_6$.

Note: Be_2 does not exist so we can't say it is paramagnetic or diamagnetic species.

Hence, $y = 6$.

$$\frac{x+y}{2} = 6.$$

32. 4

Sol.

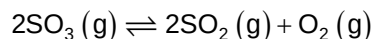
In container (I)

In container (II)

KMnO_4 will oxidized only SO_2
 Moles of O.A. = Moles of R.A.
 $\frac{2}{2} = \frac{5}{5}$

$$\frac{0.2 \times 4}{2} = \frac{\text{Moles of } \text{SO}_2}{5}$$

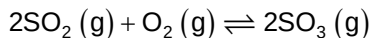
$$\text{Mole of } [\text{SO}_2]_{\text{equilibrium}} = 2$$



$$10 - 2x_1 \quad 2x_1 \quad x_1$$

$$2x_1 = 2$$

$$x_1 = 1.0$$



$$2x_1 \quad x_1 \quad 0$$

$$2x_1 - 2a \quad x_1 - a \quad 2a$$

$$\text{Total moles } n_T = (3x_1 - a)$$

$$\text{Given } \frac{x_1 - a}{3x_1 - a} = \frac{1}{4} \quad a = \frac{x_1}{3}$$

$$y = K_p = \frac{\left(\frac{2a}{n_T} \times 20\right)^2}{\left(\frac{2x_1 - 2a}{n_T} \times 20\right)^2 \left(\frac{x_1 - a}{n_T} \times 20\right)} = \left(\frac{1}{20}\right) \text{ atm}^{-1}$$

$$x = K_c \frac{\left(\frac{2}{5}\right)^2 \left(\frac{1}{5}\right)}{\left(\frac{8}{5}\right)^2} = \frac{1}{80} (\text{mol} / \ell)$$

$$\text{Value of } \frac{y}{x} = \frac{80}{20} = 4$$

33. 6

Sol. $a = 4 \rightarrow \text{i, iv, vi, vii}$
 $b = 2 \rightarrow \text{ii, ix}$
 $v \text{ and viii forms } \delta \text{ M.O.}$
 $d = 2 \rightarrow \text{iii, x}$
 $\frac{a^2 + b^2 + 2d}{4} = \frac{24}{4} = 6$

34. 9

Sol. I, II, III and IV are correct due to bent rule and V is incorrect. Correct order in case of V is
 $\text{II} > \text{III} > \text{IV} > \text{I} = \text{V}.$
 $x = 4$
 $y = 1$
 $2x + y = 9$

35. 8

Sol. K_{sp} of $\text{Mg}(\text{OH})_2$ is $= \left(\frac{5.8}{58} \times 10^{-3}\right)^3 \cdot 4 = 4 \times 10^{-12}$



$$(x + 0.01)(4x^2) = 4 \times 10^{-12}$$

$$4x^2 \times 0.01 = 4 \times 10^{-12}$$

$$x = 10^{-5} \text{ mole} / \ell$$

$$[\text{Mg}^{+2}] \text{ in container II} = 1.0 \times 10^{-5}$$

$$\text{For basic buffer } \text{pOH} = \text{pK}_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_4\text{OH}]}$$

$$\text{pOH} = 5 + \log 5$$

$$= 5.70$$

$$\text{OH}^- = 2 \times 10^{-6}$$

$$[\text{Mg}^{+2}] = \frac{4 \times 10^{-12}}{4 \times 10^{-12}} = 1.0 \text{ M}$$

$$[\text{Al}^{3+}] = \frac{1.6 \times 10^{-20}}{8 \times 10^{-18}} = 2 \times 10^{-3} \text{ M}$$

$$x = 10^{-5}$$

$$y = 1.0$$

$$z = 2 \times 10^{-3}$$

$$\text{Value of } \frac{yz}{25x} = \frac{200}{25} = 8.$$

36. 5

Sol. Statement 4, 11 and 12 are incorrect and other statements are correct.
 $x = 9, y = 3.$

Mathematics

PART – III

Section – A

37. A

Sol. By differentiating, $f'(x) + \frac{1}{1+x^2}f(x) + \frac{\tan^{-1}x}{1+x^2} = 0$

$$f(x) = 1 - \tan^{-1}x - 2e^{-\tan^{-1}x}$$

38. A

Sol. $\int_1^a (f(x) + |f(x)|) dx = g(a)$; $f(x) = 2x^3 + 6x - 8$

$$\int_2^3 \left| \frac{f(x)}{g(x)+5} \right| dx = \frac{1}{2} \ln \left(\frac{101}{22} \right)$$

39. D

Sol. $\int e^{x^2} \left(\frac{2x - \cot x}{\sin x} \right) \cdot \ln \left(\frac{e^{x^2}}{\sin x} \right) dx$

$$\text{Let } \frac{e^{x^2}}{\sin x} = t \Rightarrow \int \ln(t) dt = t(\ln t - 1) + c$$

40. B

Sol. $x^2 + y^2 - ax = 0 \Rightarrow x^2 + 2xy \cdot y' - y^2 = 0$

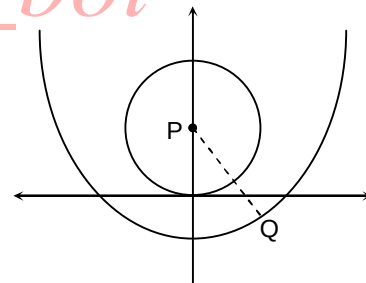
$$\text{Orthogonal curve : } x^2 - \frac{2xy}{y'} - y^2 = 0 ; y' = \frac{2xy}{x^2 - y^2}$$

$$\text{By solving, } x^2 + y^2 = \frac{y}{c}$$

$$\text{Passing through } (1, 1) \text{ so } x^2 + y^2 = 2y$$

$$\text{For shortest distance, } Q = (1, 0)$$

$$D = PQ - r = \sqrt{2} - 1$$



41. A, B, D

Sol. $f''(x) = f(x) \cdot (\tan x - 1)(\tan^3 x - \tan^2 x + 3 \tan x + 1)$

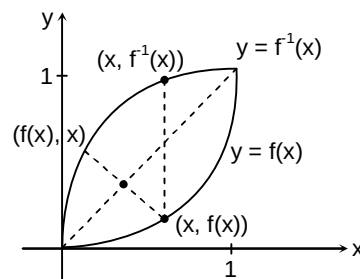
$g(x) = \tan^3 x - \tan^2 x + 3 \tan x + 1 = 0$ has exactly one root in $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ which lies in

$$x \in \left(-\frac{\pi}{4}, 0\right)$$

$$\text{So, } f''(x) = 0 \text{ at } x = \frac{\pi}{4} \text{ and } k \text{ where } k \in \left(-\frac{\pi}{4}, 0\right)$$

42. A, C

Sol. $f(x) \cdot f^{-1}(x) < x$

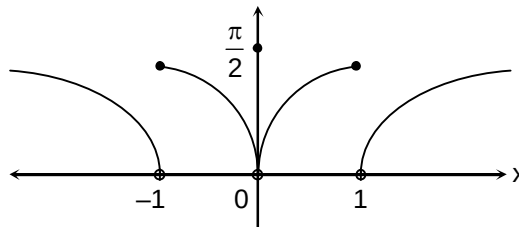


43. A, B

Sol. $f'g' = f'g + fg'$ by solving $g(x) = e^{\frac{1}{x-2}}$
 $g(x)$ is discontinuous at one point at $x = 2$

44. B, D

Sol. Graph of $y = f(x)$



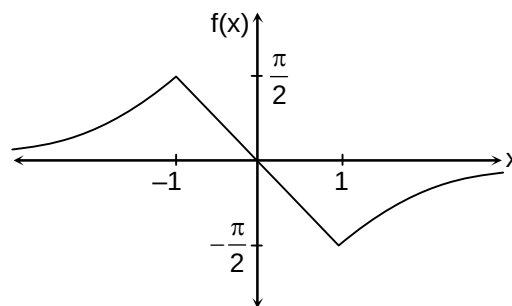
45. A, D

Sol. $f(x) = \int_0^{3\pi/4} \left| x - \frac{3 \sin t}{\cos t + \sqrt{2}} \right| \left(\frac{1 + \sqrt{2} \cos t}{(\cos t + \sqrt{2})^2} \right) dt$
 $f(x) = \int_0^1 |x - 3u| du \Rightarrow f(x) = \begin{cases} \frac{3}{2} - x & ; x \leq 0 \\ \frac{x^2}{3} - x + \frac{3}{2} & ; 0 \leq x \leq 3 \\ x - \frac{3}{2} & ; x \geq 3 \end{cases}$

TG ~ @bohring_bot

46. B, C

Sol. $\frac{x^2 - x + 1}{x^2 + x + 1} \in \left[\frac{1}{3}, 3 \right] \forall x \in \mathbb{R}$
 Maxima at $x = -1$ and minima at $x = 1$



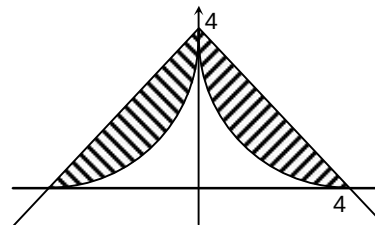
Section - B

47. 1

Sol. $\int \frac{(6xf(x) - 4x^2) - 2f(x)}{(3f(x) - 2x)(x^2 - f(x))^2} dx = \int \frac{2x}{(x^2 - f(x))^2} dx - \int \frac{f'(x)}{(x^2 - f(x))^2} dx = -\frac{1}{x^2 - f(x)} + c$
 Substitute $f(x) = \frac{2}{4x - 3f(x)}$

48. 2

Sol. $\frac{16\lambda}{3} = 2 \left[8 - \int_0^4 (x - 4\sqrt{x} + 4) dx \right]$



49. 5

Sol.
$$\lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \left(\frac{n^2}{n^2 + r^2} \left(\lim_{m \rightarrow \infty} \sum_{k=1}^m \left(\frac{1}{n^2 + \frac{k^2}{m^2}} \right) \frac{1}{m} \right) \right) \right) = \lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \frac{n^2}{n^2 + r^2} \int_0^1 \frac{1}{n^2 + x^2} dx \right)$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \frac{n}{n^2 + r^2} \tan^{-1} \left(\frac{r}{n} \right) \right) = \int_0^1 \frac{\tan^{-1} x}{1 + x^2} dx = \frac{\pi^2}{32}$$

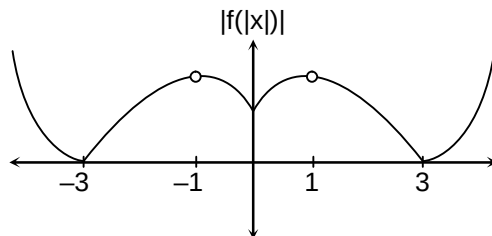
50. 7

Sol. $f(x) = \frac{(x^2 - 1)(x^2 - 9)}{(x - 1)(x + 3)} \quad x \in \mathbb{R} - \{-3, 1\}$

Graph of $y = |f(x)|$

Discontinuity at $x = \pm 1$

Non-differentiable at $x = 0, \pm 1, \pm 3$



51. 8

Sol. $I = \int_0^{4\pi} \frac{dx}{2 + \sin 4x} = \int_0^{4\pi} \frac{dx}{2 - \sin 4x} \Rightarrow I = 2 \int_0^{4\pi} \frac{1}{4 - \sin^2 4x} dx = 64 \int_0^{\pi/8} \frac{1}{4 - \sin^2 4x} dx$

$I = 64 \int_0^{\pi/8} \frac{2}{7 + \cos 8x} dx ; 3x = t \Rightarrow I = \frac{128}{3} \int_0^{3\pi/8} \frac{1}{7 + \cos \left(\frac{8t}{3} \right)} dt$

IG ~ @bohring_bot

52. 4

Sol. $f(x) = k(x + 1)^2(x - 1)^2(x - 5)^2 + 4 \Rightarrow f(2) = 81k + 4 = 85 \Rightarrow k = 1$

$f(x) = (x + 1)^2(x - 1)^2(x - 5)^2 + 4 \Rightarrow f'(x) = 2(x + 1)(x - 1)(x - 5)(3x^2 - 10x - 1)$

$\alpha + \beta = \frac{10}{3} \Rightarrow \lim_{x \rightarrow \frac{10}{3}} \left(\frac{x^2 - 9}{x^2 - 5x + 6} \right) = \frac{19}{4}$

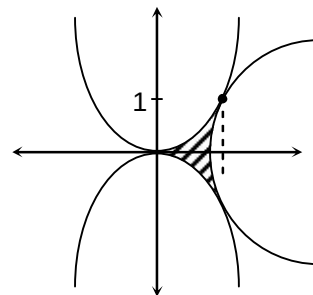
53. 4

Sol.
$$\lim_{x \rightarrow 0} \frac{\frac{2022((1 + x^{2022})^{\frac{1}{2021}} - 1)}{x} - \frac{2023((1 + x^{2023})^{\frac{1}{2021}} - 1)}{x}}{2022x^{2021}} = \frac{1}{2021}$$

$\lambda = 2021 = 43 \times 47$ number of divisors $= 2 \times 2 = 4$

54. 2

Sol. Area $= 2 \int \left(\frac{y^2 + 3}{4} - \sqrt{y} \right) dy = \frac{1}{3}$



FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Main)-2023

TEST DATE: 03-12-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

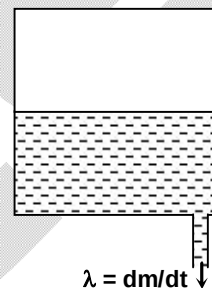
SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. A metal vessel of negligible heat capacity contains 0.5 kg of water (of specific heat capacity $4200 \text{ J kg}^{-1} \text{ K}^{-1}$). It is heated from 15°C to 85°C by 750 W immersion heater in 4 minutes. The average loss of heat in watts to the surroundings from the vessel during this time is nearly:
 (A) 70 (B) 123
 (C) 138 (D) 212

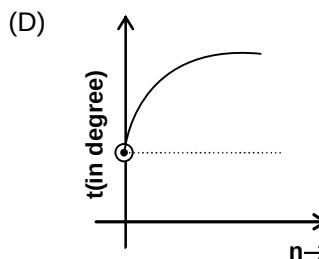
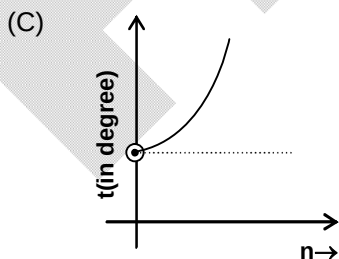
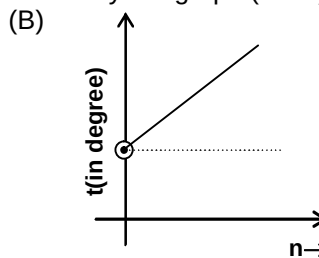
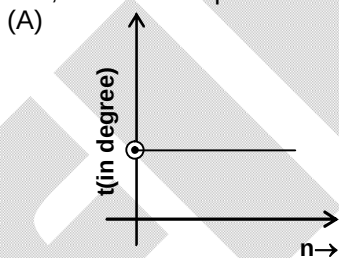
2. A well insulated box contains water (specific heat = C and very large conductivity) of mass m_0 and temperature T_0 at time $t = 0$. If heat is being added to it uniformly at a constant rate $\frac{dQ}{dt} = R$ and water is leaking from it at a constant rate $\frac{dm}{dt} = \lambda$, then the temperature of the water at time t



$\left(t < \frac{m_0}{\lambda}\right)$ will be:

- (A) $T = \frac{R}{2\lambda C} \ln\left(\frac{m_0}{m_0 - \lambda t}\right) + T_0$ (B) $T = \frac{R}{\lambda C} \ln\left(\frac{m_0}{m_0 - \lambda t}\right) + T_0$
 (C) $T = \frac{R}{\lambda C} e^{(m_0 - \lambda t)}$ (D) $T = \frac{R}{2\lambda C} e^{(m_0 - \lambda t)}$

3. A slab X of thickness 't' thermal conductivity 'k' and area of cross-section 'A' is placed in thermal contact with another slab Y which is $2n^2$ times thicker, $4n$ times conductive and having n times larger cross-section area. If the outside face of X is maintained at 100°C , the outside face of Y at 0°C , then the temperature of the junction is represented by the graph ($n > 0$):



4. Two identical metallic sheets each behaves like a black body are kept parallel to each other with small separation between them in vacuum. Thermal energy is generated at a constant rate P in one of the sheets by passing electricity through it. In steady state, temperature of the other sheet is found to be constant at 300K . Find the value of P in kilo-watt. (given area of plates =

$$A = \frac{10^3}{17 \times 27} \text{m}^2 \text{ and } \sigma = \frac{17}{3} \times 10^{-8} \text{ W/m}^2\text{K}^4$$

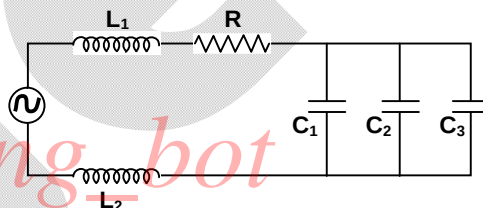
- (A) 3 (B) 4
(C) 5 (D) 6

5. A copper sphere is suspended in an evacuated chamber maintained at 300K . The sphere is maintained at a constant temperature of 500K by heating it electrically. A total of 300W of electric power is needed to do it. When half of the surface of the copper sphere is completely blackened 600W is needed to maintain the same temperature of the sphere. Calculate the emissivity of copper.

- (A) $e = \frac{1}{3}$ (B) $e = \frac{2}{3}$
(C) $e = \frac{1}{2}$ (D) $e = \frac{1}{6}$

6. A generator with an adjustable frequency of oscillation is connected to resistance, $R = 100 \Omega$, inductances, $L_1 = 1.7 \text{ mH}$ and $L_2 = 2.3 \text{ mH}$ and capacitances, $C_1 = 4 \mu\text{F}$, $C_2 = 2.5 \mu\text{F}$ and $C_3 = 3.5 \mu\text{F}$. The resonant angular frequency of the circuit is

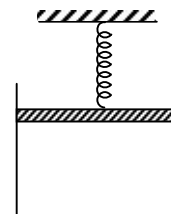
- (A) 0.5 rad/s
(B) $0.5 \times 10^4 \text{ rad/s}$
(C) 2 rad/s
(D) $2 \times 10^{-4} \text{ rad/s}$



7. Power factor of an L-R series circuit is 0.6 and that of a C-R series circuit is 0.5 . If the elements (L, C and R) of the two circuit are joined in series the power factor of this circuit is found to be 1 . The ratio of the resistance in L-R circuit to the resistance in the C-R circuit is

- (A) $\frac{6}{5}$ (B) $\frac{5}{6}$
(C) $\frac{4}{3\sqrt{3}}$ (D) $\frac{3\sqrt{3}}{4}$

8. One mole of an ideal gas is kept enclosed under a light and frictionless piston (area = 10^{-4} m^2) connected by a compressed spring (spring constant 100 N/m). The volume of gas is 0.83 m^3 and its temperature is 100 K . The gas is heated so that it compresses the spring further by 0.1 m . The work done by the gas in the process is X joules. Find the value of X . (Take $R = 8.3 \text{ J-mol}^{-1}\text{K}^{-1}$ and there is no atmosphere)



- (A) $\frac{1}{2}$ (B) $\frac{3}{2}$
(C) $\frac{5}{2}$ (D) $\frac{3}{4}$

9. A charged particle of specific charge α is released from origin at $t = 0$ with velocity $\vec{v} = v_0 \hat{i} + v_0 \hat{j}$ in magnetic field $\vec{B}_0 = B_0 \hat{i}$. The coordinate of the particle at time $t = \frac{\pi}{B_0 \alpha}$ are (specific charge $\alpha = q/m$)

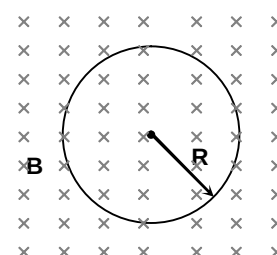
(A) $\left(\frac{v_0}{2B_0\alpha}, \frac{\sqrt{2}v_0}{\alpha B_0}, \frac{-v_0}{B_0\alpha} \right)$ (B) $\left(\frac{-v_0}{2B_0\alpha}, 0, 0 \right)$
 (C) $\left(0, \frac{2v_0}{\alpha B_0}, \frac{v_0\pi}{2B_0\alpha} \right)$ (D) $\left(\frac{v_0\pi}{B_0\alpha}, 0, \frac{-2v_0}{B_0\alpha} \right)$

10. An electron is released from the origin at a place where a uniform electric field $\vec{E}$ and uniform magnetic field $\vec{B}$ exists along the negative y-axis and negative z-axis respectively. Then which of the following are INCORRECT.

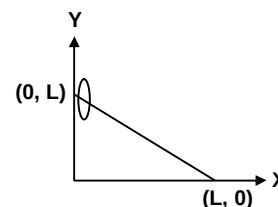
- (A) At time t the y-component of velocity of the electron becomes $v_y = \frac{E}{B} \sin \omega t$, where $\omega = \frac{eB}{m}$
 (B) At $t = \frac{\pi m}{eB}$ the electron will have only x-component of velocity.
 (C) At $t = \frac{2\pi m}{3eB}$, the y-component of velocity becomes zero.
 (D) The displacement along y-axis is $\frac{2Em}{eB^2}$, when the velocity of electron becomes perpendicular to the y-axis.

11. A conducting loop of radius R is present in a uniform magnetic field B perpendicular to the plane of the ring. If the radius R varies as a function of time ' t ', as $R = R_0 + t$, the emf induced in the loop is

- (A) $2\pi(R_0 + t)B$ clockwise
 (B) $\pi(R_0 + t)B$ clockwise
 (C) $2\pi(R_0 + t)B$ anticlockwise
 (D) zero



12. Electric field given by the vector $\vec{E} = x\hat{i} + y\hat{j}$ present in the XY plane. A small ring carrying charge $+Q$, which can freely slide on smooth non-conducting rod, is projected along the rod from the point $(0, L)$ such that it can reach the other end of the rod. What minimum velocity should be given to the ring? (Assume z-axis is the vertical axis)



(A) $\left(\frac{QL^2}{m} \right)^{1/2}$ (B) $2 \left(\frac{QL^2}{m} \right)^{1/2}$
 (C) $4 \left(\frac{QL^2}{m} \right)^{1/2}$ (D) $\left(\frac{QL^2}{2m} \right)^{1/2}$

13. Current density in a cylindrical wire of radius R is given as

$$J = \begin{cases} J_0 \left(\frac{x}{R} - 1 \right) & \text{for } 0 \leq x < \frac{R}{2} \\ J_0 \left(\frac{x}{R} \right) & \text{for } \frac{R}{2} \leq x \leq R \end{cases}$$

The current flowing in the wire is

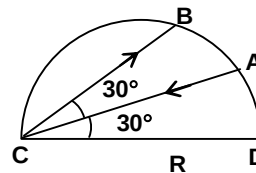
- (A) $\frac{7}{24} \pi J_0 R^2$ (B) $\frac{1}{6} \pi J_0 R^2$
 (C) $\frac{7}{12} \pi J_0 R^2$ (D) $\frac{5}{12} \pi J_0 R^2$
14. A container encloses 32 g of O_2 at 127°C and of pressure P_0 . A small hole is made in the container and the gas leaks out. At a certain state, pressure of gas becomes $\frac{P_0}{2}$ and its temperature 27°C . Mass of the gas that has leaked out till this stage is nearly:
 (A) 6.3g (B) 7.4g
 (C) 16.3g (D) 10.7g

15. A long straight wire along the z -axis carries I in the negative z -direction. The magnetic vector field B at a point having coordinates (x, y) in the $z = 0$ plane is

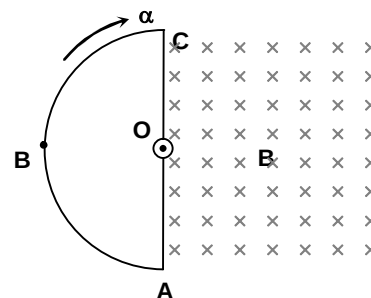
(A) $\frac{\mu_0 I (y\hat{i} - x\hat{j})}{2\pi(x^2 + y^2)}$ (B) $\frac{\mu_0 I (y\hat{i} + x\hat{j})}{2\pi(x^2 + y^2)}$
 (C) $\frac{\mu_0 I (x\hat{i} - y\hat{j})}{2\pi(x^2 + y^2)}$ (D) $\frac{\mu_0 I (x\hat{i} + y\hat{j})}{2\pi(x^2 + y^2)}$

16. A current carrying wire is placed in the grooves of an insulating semi-circular disc of radius R , as shown. The current enters at point A and leaves from point B. The magnetic field at point D, has magnitude:

(A) $\frac{\mu_0 I}{8\pi R\sqrt{3}}$ (B) $\frac{\mu_0 I}{4\pi R\sqrt{3}}$
 (C) $\frac{\sqrt{3}\mu_0 I}{4\pi R}$ (D) $\frac{\sqrt{3}\mu_0 I}{2\pi R}$

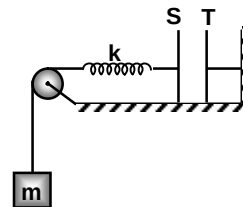


17. What is the emf induced in the loop OABCO (radius = a) at time t placed in uniform magnetic field B as shown when its angular acceleration is β ? n = number of half revolutions of the loop in a given time t .



(A) $\frac{Ba^2\beta t}{2}$ (B) $\frac{(-1)^n Ba^2\beta t}{2}$
 (C) $\frac{(-1)^n Ba^2\beta t}{4}$ (D) $(-1)^n Ba\beta t$

18. The plate S and T of an uncharged parallel plate capacitor are connected across a battery. The battery is then disconnected and the charged plates are now connected in a system as shown in the figure. The system shown is in equilibrium. All the strings are massless. The magnitude of charge on one of the capacitor plate is: (consider the area of plates to be A)



- (A) $\sqrt{2mgA\epsilon_0}$ (B) $\sqrt{\frac{4mgA\epsilon_0}{k}}$
 (C) $\sqrt{mgA\epsilon_0}$ (D) $\sqrt{\frac{2mgA\epsilon_0}{k}}$
19. Power radiated by a black body is P_0 and the wavelength corresponding to maximum energy is around λ_0 . On changing the temperature of the black body, it was observed that the power radiated becomes $\frac{256P_0}{81}$. The shift in wavelength corresponding to the maximum energy will be
- (A) $\frac{\lambda_0}{4}$ (B) $\frac{\lambda_0}{2}$
 (C) $-\frac{\lambda_0}{4}$ (D) $-\frac{\lambda_0}{2}$
20. One mole of an ideal gas contained in a cylinder undergoes a thermodynamic process during which pressure relates to volume as $P = \frac{A}{1 + \left(\frac{B}{V}\right)^2}$, where A and B are constants. As the volume of the gas is changed from $V = B$ to $V = 2B$, its change of temperature can be expressed as:
- (A) $\frac{6AB}{5R}$ (B) $\frac{AB}{2R}$
 (C) $-\frac{3AB}{5R}$ (D) $\frac{11AB}{10R}$

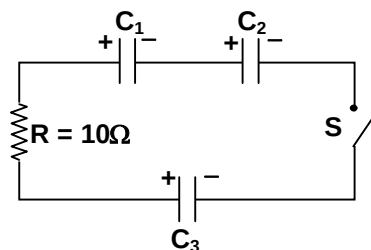
SECTION – B

(Numerical Answer Type)

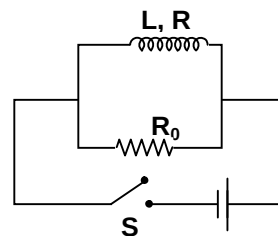
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A moving coil galvanometer 'A' has 100 turns and resistance 50Ω . Another moving coil galvanometer 'B' has number of turns 500 and resistance 200Ω . The other quantities are same in both cases. They are connected to a battery through a resistance of 50Ω . The ratio of deflection of B to A is
22. Three concentric thin spherical shells of radii R, 2R and 3R are given 2Q, Q and $-Q$ respectively. The innermost and the outermost shells are earthed. The final charge on the inner most sphere is $\left(-\frac{Q}{n}\right)$. Find the value of n.
23. A galvanometer having a resistance of 50Ω , gives a full scale deflection for a current of 0.05 A. Find the length (in meter) of a resistance wire of area of cross-section $2.97 \times 10^{-2} \text{ cm}^2$ that can be used to convert the galvanometer into an ammeter which can read a maximum of 5A current. (specific resistance of the wire = $5 \times 10^{-7} \Omega\text{-m}$)

24. Three charged capacitors C_1 , C_2 and C_3 of capacitances $3 \mu\text{F}$, $6 \mu\text{F}$ and $6 \mu\text{F}$ respectively and having charge $30 \mu\text{C}$ each are connected with a resistor of resistance $R = 10\Omega$ through a switch as shown in the figure. The switch S is closed at $t = 0$. Find the total heat (in μJ) dissipated through the resistor.



25. Two identical containers A and B with frictionless pistons contain the same ideal gas at the same temperature and the same volume V . The mass of the gas in A is m_A and that in B is m_B . The gas in each cylinder is now allowed to expand isothermally to the same final volume $2V$. The change in pressure in A and B are found to be ΔP and $1.5\Delta P$ respectively. If $\frac{m_A}{m_B} = \frac{2}{\lambda}$ then find the value of λ .
26. An ideal gas has a molar heat capacity $C = \frac{9R}{2}$ in the process $PT = \text{constant}$. Find the degree of freedom of the gas molecule. (R = ideal gas constant)
27. When an ideal diatomic gas is heated at constant pressure, during a process. If during the given process, the ratio $\frac{\Delta Q}{\Delta W} = \frac{n}{2}$. Find the value of n . (where ΔQ = Heat supplied to the gas during the process, ΔW = work done by the gas during the process)
28. If P - V diagram of a diatomic gas during a process is plotted, it is a straight line (when extended it passes through the origin). If molar heat capacity of the gas in the process is nR . Find the value of n . (where R = ideal gas constant)
29. In a region of space, the magnetic field is changing at a constant rate $\frac{dB}{dt}$. A given mass ' m ' of copper of resistivity ρ and density d drawn into a wire of radius r and formed into a circular loop of radius R_1 which is placed perpendicular to the given magnetic field region. If the current induced in the loop in the terms of given quantities is $\frac{nm}{4\pi\rho d} \frac{dB}{dt}$. Find n .
30. A coil of inductance $L = 2\mu\text{H}$ and $R = 1\Omega$ is connected to a source of constant emf, $\varepsilon = 3\text{V}$. A resistance $R_0 = 2\Omega$ is connected in parallel with the coil. The heat generated in the coil after the switch S is disconnected after a long time is $P \times 10^{-6} \text{ J}$. The internal resistance of the source is negligible. Find the value of P .



Chemistry

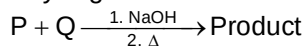
PART – B

SECTION – A

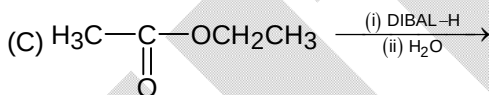
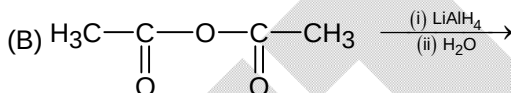
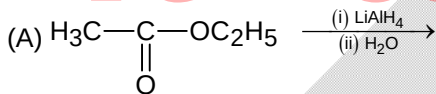
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

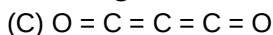
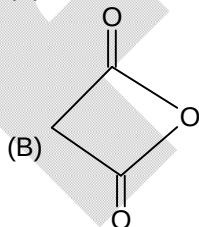
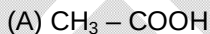
31. Molecular formula of P and Q is C_2H_4O and C_3H_6O . If P and Q both give (+ve) $NaHSO_3$ test and only P gives haloform test. Choose the INCORRECT option:



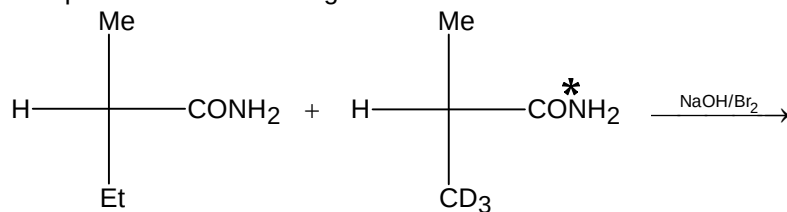
- (A) Total 4 products will be obtained excluding stereoisomers.
 (B) Total 4 products will be obtained including stereoisomers.
 (C) Reaction is known as cross-Aldol condensation.
 (D) All products obtained will give (+ve) Tollen's test as well as Fehling test.
32. If Q is benzal chloride, then choose the INCORRECT statement about this commercial method
- $$P \xrightarrow[h\nu]{Cl_2} Q \xrightarrow[373\text{ K}]{H_2O} R$$
- (A) R will give (–ve) Fehling test.
 (B) Total 10 benzenoid structural isomers for Q are possible.
 (C) (P) can have molecular formula C_7H_8 .
 (D) Free radical addition involves in conversion of P to Q.
33. The reaction in which two moles of (same or different) alcohol is not obtained as a product is



34. The product obtained on reaction of Malonic acid with P_2O_5 / Δ is

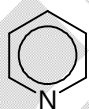


35. The product of the following reaction is:



- (A) $\begin{array}{c} \text{Et} \\ | \\ \text{H} - \text{C} - \text{NH}_2 \\ | \\ \text{Me} \end{array}$ and $\begin{array}{c} \text{CD}_3 \\ | \\ \text{H} - \text{C} - \text{NH}_2^* \\ | \\ \text{CH}_3 \end{array}$
- (B) $\begin{array}{c} \text{Me} \\ | \\ \text{H}_2\text{N} - \text{C} - \text{Et} \\ | \\ \text{H} \end{array}$ and $\begin{array}{c} \text{Me} \\ | \\ \text{H}_2\text{N} - \text{C} - \text{H} \\ | \\ \text{CD}_3 \end{array}$
- (C) $\begin{array}{c} \text{Me} \\ | \\ \text{H} - \text{C} - \text{NH}_2 \\ | \\ \text{Et} \end{array}$, $\begin{array}{c} \text{Me} \\ | \\ \text{H} - \text{C} - \text{NH}_2^* \\ | \\ \text{CD}_3 \end{array}$, $\begin{array}{c} \text{Me} \\ | \\ \text{H} - \text{C} - \text{NH}_2^* \\ | \\ \text{Et} \end{array}$ and $\begin{array}{c} \text{Me} \\ | \\ \text{H} - \text{C} - \text{NH}_2 \\ | \\ \text{CD}_3 \end{array}$
- (D) $\begin{array}{c} \text{Me} \\ | \\ \text{H} - \text{C} - \text{NH}_2 \\ | \\ \text{Et} \end{array}$ and $\begin{array}{c} \text{Me} \\ | \\ \text{H} - \text{C} - \text{NH}_2^* \\ | \\ \text{CD}_3 \end{array}$

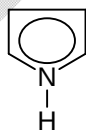
36. The correct order of rate of electrophilic aromatic substitution is



(I)



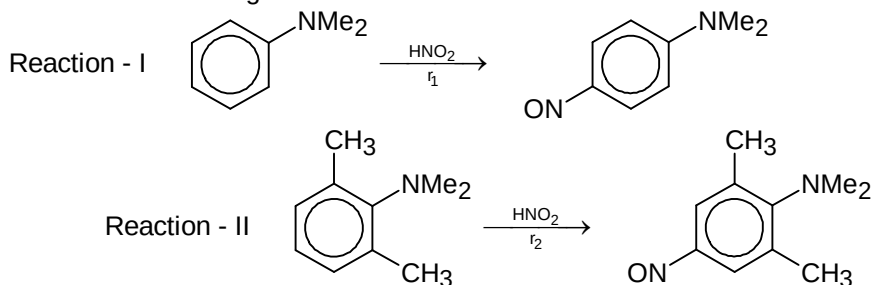
(II)



(III)

- (A) I > II > III
 (B) I < II < III
 (C) II > I > III
 (D) III > I > II

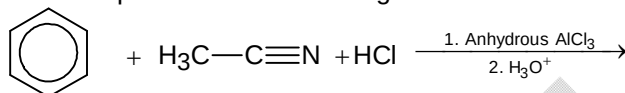
37. Consider the following reaction



Which of the following is a CORRECT comparison of rate of reaction?

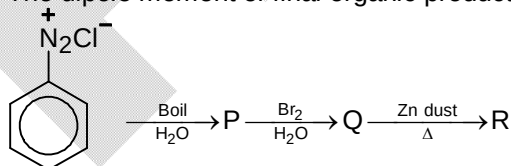
- (A) $r_1 > r_2$
 (B) $r_2 > r_1$
 (C) $r_1 = r_2$
 (D) Reactions are not possible

38. The final product of the following reaction is



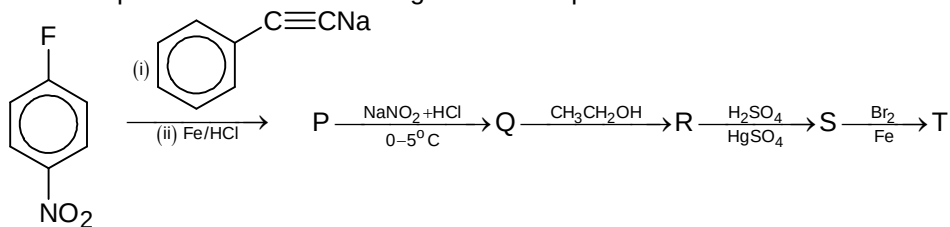
- (A)
- (B)
- (C)
- (D)

39. The dipole moment of final organic product obtained in the following reaction is:



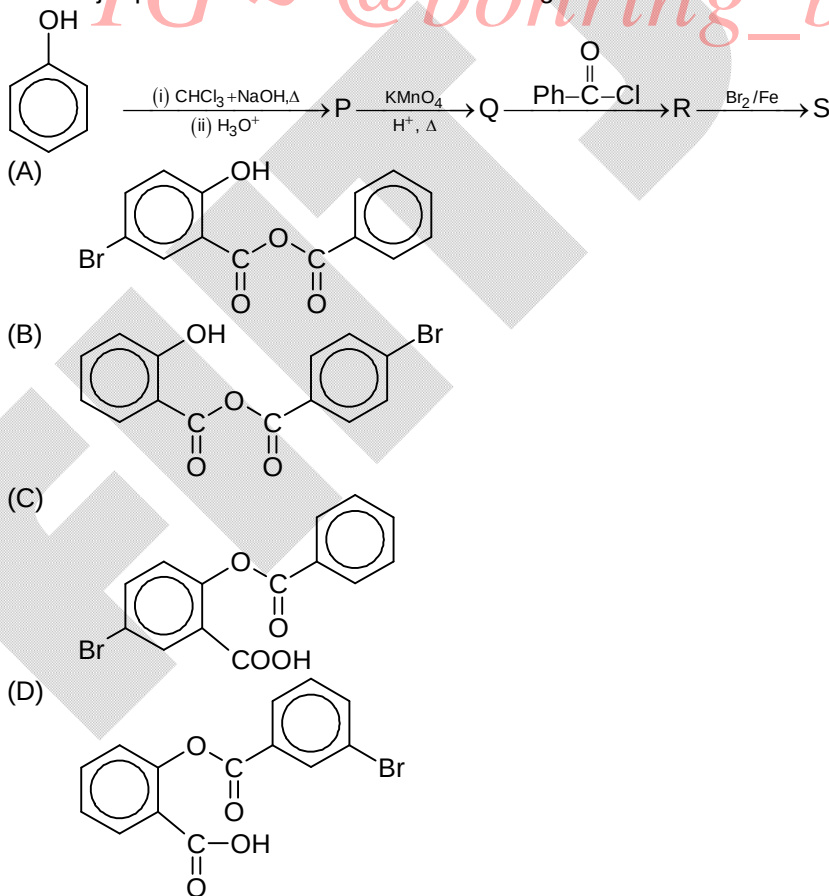
- (A) $\mu_R \neq 0$
 (B) $\mu_Q = 0$
 (C) $\mu_R < \mu_Q$
 (D) $\mu_R > \mu_Q$

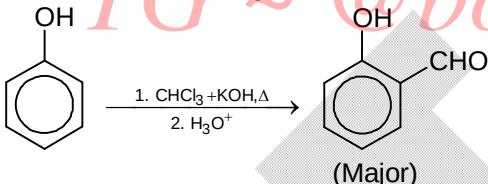
40. The final product 'T' in the following reaction sequence is



- (A)
- (B)
- (C)
- (D)

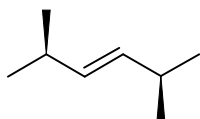
41. The major product 'S' obtained in the following reaction is



42. Compare van't Hoff factor for compounds $\text{CH}_3\text{CH}_2\text{OH}$ (A) and Ph_3COH (B) in cold conc. H_2SO_4 determined from measurement of freezing point.
 (A) $i_A > i_B$
 (B) $i_B > i_A$
 (C) $i_A = i_B$
 (D) $i_A < i_B < 1$
43. 896 mL vapour of a hydrocarbon 'P' having carbon 87.80% and hydrogen 12.19% weighs 3.28 g at STP. Hydrogenation of 'P' gives 2-methyl pentane. Also 'P' on hydration in the presence of $\text{H}_2\text{SO}_4 / \text{HgSO}_4$ gives a ketone 'Q' having molecular formula $\text{C}_6\text{H}_{12}\text{O}$. The ketone 'Q' gives a positive iodoform test. The INCORRECT statement is:
 (A) P gives white ppt. with Ammonical AgNO_3 .
 (B) Q gives orange colour with 2,4-DNP.
 (C) Q gives crystalline compound with NaHSO_3 .
 (D) P gives red colour of Cu_2O with Ammonical Cu_2Cl_2 .
44. Phenol and formaldehyde undergo condensation to give a polymer (P) which on heating with formaldehyde gives a thermosetting polymer (Q). The polymer (P) is:
 (A) Novolac
 (B) Bakelite
 (C) Glyptal
 (D) Polyster
45. Consider the following reaction
- 

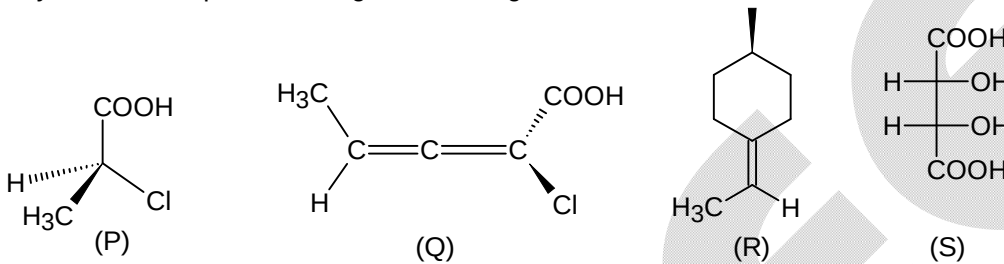
(Major)
- The intermediate formed in above reaction is
 (A) Singlet dichloro carbene
 (B) CCl_3^-
 (C) Chloronium ion
 (D) CCl_2^+
46. The basic strength order of the following in H_2O is
 MeNH_2 , Me_2NH , Me_3N
 (I) (II) (III)
 (A) $\text{I} > \text{II} > \text{III}$
 (B) $\text{II} > \text{I} > \text{III}$
 (C) $\text{III} > \text{II} > \text{I}$
 (D) $\text{I} > \text{III} > \text{II}$
47. Most acidic compound among the following is
 (A) Picric acid
 (B) Squaric acid
 (C) Carbonic acid
 (D) Carbolic acid

48. The total number of stereoisomers of following compound 'X' is



- (A) 2
(B) 6
(C) 4
(D) 8

49. Asymmetric compound among the following is



- (A) P, Q, R, S
(B) P, Q, S
(C) P, Q, R
(D) Q, S, R

50. A compound $\begin{matrix} & G \\ & | \\ G & - C - & G \\ & | \\ & G \end{matrix}$ contain $\frac{\sigma}{\pi}$ ratio 1 : 1, if G is

- (A) CHO
(B) CN
(C) COOH
(D) $C \equiv CH$

SECTION - B

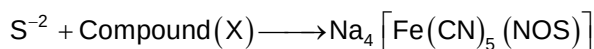
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51.
How many sp^2 hybridised carbons are present in [R]?

52. In soda lime decarboxylation, soda lime is made of x mole of NaOH and y moles of CaO, then $\frac{x}{y}$ is....
53. 0.5 g of an organic compound having molecular formula $C_{12}H_{24}N_x$ was analysed for nitrogen using Duma's method. The nitrogen evolved was collected over water and found to be 110 mL at 300 K and 802 mm Hg pressure. If aqueous tension of H_2O at 300 K is 42 mm Hg, the value of x is.....

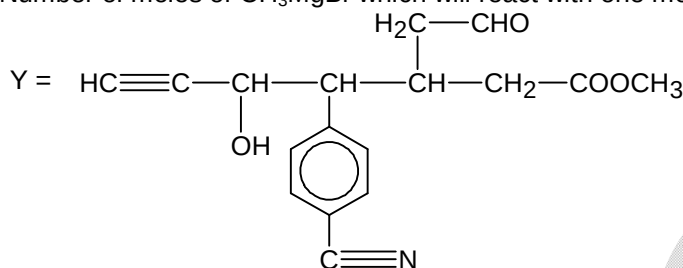
54. In the qualitative analysis of sulphide ions (S^{2-}), the following reaction is used



What is magnetic moment (in B. M.) for compound 'X'?

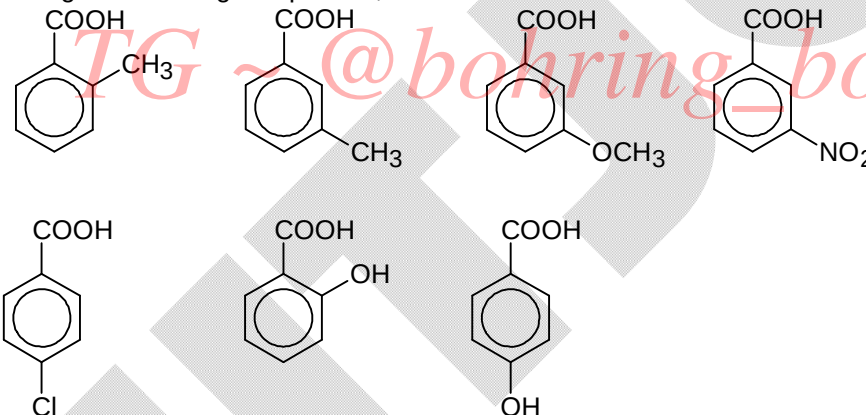
55. How many alcoholic structural isomers of molecular formula $C_6H_{14}O$ are possible which on oxidation with PCC give aldehyde product?

56. Number of moles of CH_3MgBr which will react with one mole of compound 'Y':



57. An organic compound 'A' having molecular formula $C_5H_{10}O_6$ on acetylation forms a compound having molecular weight 334. Number of hydroxyl groups ($-OH$) in the compound 'A' is:

58. Among the following compounds, which are more acidic than benzoic acid



59. Total number of alkenes of molecular formula C_6H_{12} which will show geometrical isomerism

60. Total number of meso structures in $CH_3(CHOH)_5CH_3$ is

Mathematics

PART – C

SECTION – A

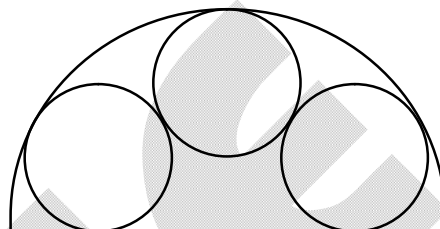
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. Three identical circles, each of radius a , are drawn as shown in figure and tangent to a semi-circle of radius r , express a in terms of r ?

(A) $\frac{r}{4}$
(C) $\frac{r}{3}$

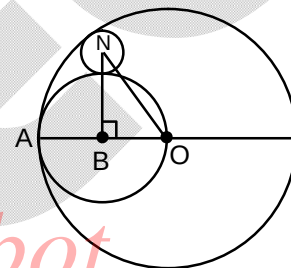
(B) $\frac{r}{2}$
(D) $\frac{2r}{3}$



62. In the figure below is a circle with centre O , and another circle with centre at the midpoint B of the radius OA and passing through O . A small circle is drawn as shown, tangent to both the larger circles, such that its centre N is directly above B (that is, $BN \perp AB$). If the perimeter of the triangle NBA is 8 cm, find the value (in cm) of the radius of the small circle

(A) $\frac{2}{3}$
(C) 1

(B) 2
(D) $\sqrt{2}$



63. In a $\triangle ABC$
 $3\sin A + 4\cos B = 6$
 and $4\sin B + 3\cos A = 1$
 then angle C is equal to

(A) $\frac{\pi}{6}$
(C) $\frac{\pi}{4}$

(B) $\frac{\pi}{3}$
(D) $\frac{2\pi}{3}$

64. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$, then $f_4(x) - f_6(x) = \frac{\lambda}{24}$, then λ is

(A) 1
(C) -2

(B) 2
(D) -1

65. In a triangle ABC , $\frac{a}{b} = 2 + \sqrt{3}$ and $\angle C = 60^\circ$, then

(A) $\angle A = 15^\circ$
(C) $\angle B = 105^\circ$

(B) $\angle A = 105^\circ$
(D) $\angle B = 30^\circ$

66. Let $A(\alpha, 2)$, $B(\alpha, -2)$ and $C(\beta, 2)$ be the vertices of a triangle ABC . If the circumcentre is origin and orthocentre is $(3, \gamma)$, then which of the following is **NOT CORRECT** about $\triangle ABC$

(A) area is 12

(B) perimeter is $10 + 2\sqrt{13}$

(C) inradius is $\frac{5 - \sqrt{13}}{2}$

(D) circum radius is $2\sqrt{13}$

67. Let PQ be a chord of length 8 of the circle $4x^2 + 4y^2 + 24x - 32y - 189 = 0$. If tangents drawn to the circle at point P and Q intersect at the point R, then perpendicular distance of R from chord PQ is
- (A) $\frac{15}{2}$ (B) $\frac{32}{15}$
(C) $\frac{68}{15}$ (D) $\frac{36}{17}$
68. Let P_1 be a parabola with vertex (1, 8) and focus (3, -1) and P_2 is the mirror image of P_1 with respect to the line $3x - y + 10 = 0$. If the focus of P_2 is (α, β) and directrix of P_2 is $7x + 6y = \gamma$, then $\alpha + \beta + \gamma$ is equal to
- (A) 125 (B) 137
(C) 52 (D) 119
69. If two tangents can be drawn to the different branches of hyperbola $x^2 - 9y^2 = 9$ from the point (α, β) then (α, β) can be
- (A) $\left(2, \frac{4}{9}\right)$ (B) $\left(-1, \frac{1}{4}\right)$
(C) $\left(1, \frac{1}{2}\right)$ (D) (5, 1)
70. Let the line $y = mx$ and the ellipse $4x^2 + 9y^2 = 1$ intersect at point P in the first quadrant. If the normal to the ellipse at P meets the coordinate axes at $A(\alpha, 0)$ and $B\left(0, \frac{5}{8\sqrt{3}}\right)$, then area of triangle OAB is equal to (where O is origin)
- (A) $\frac{1}{16\sqrt{3}}$ (B) $\frac{25}{576\sqrt{3}}$
(C) $\frac{9}{144\sqrt{3}}$ (D) $\frac{25}{288\sqrt{3}}$
71. Consider the circle $x^2 + y^2 = 25$ and $y^2 = 24x$. They intersect at P and Q in the first and fourth quadrant. Tangents at P and Q to the circle intersect the x-axis at R and tangent at P and Q to the parabola intersect x-axis at S, then radius of circumcircle of ΔPRS is
- (A) $20\sqrt{7}$ (B) $5\sqrt{7}$
(C) $2\sqrt{7}$ (D) $7\sqrt{3}$
72. Perpendicular tangents are drawn from the point P to an ellipse (whose axes are parallel to coordinate axes). The locus of point P is $x^2 + y^2 - 4x - 6y - 5 = 0$ and if the product of length of perpendicular from foci upon any one of these tangents is 6, then eccentricity is
- (A) $\frac{1}{\sqrt{6}}$ (B) $\frac{1}{\sqrt{3}}$
(C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$
73. If the directrix of hyperbola (having positive x intercept) $xy = 8 + x$ is the tangent to the hyperbola $x^2 - ky^2 = 30$, then k is equal to
- (A) $\sqrt{5}$ (B) 6
(C) $\sqrt{6}$ (D) $\sqrt{15}$

74. Given points P, Q on an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b > 0$) satisfying $OP \perp OQ$, the minimum of $|OP| \times |OQ|$ is
- (A) $\frac{2a^2b^2}{a^2 + b^2}$ (B) $\frac{2a^2b^2}{a^2 - b^2}$
 (C) $\frac{4a^2b^2}{a^2 - b^2}$ (D) $\frac{4a^2b^2}{a^2 + b^2}$
75. The straight line $\frac{x}{4} + \frac{y}{3} = 1$ intersect the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at two points A and B, there is a point P on the ellipse such that the area of $\triangle PAB$ is equal to 3. Then the number of possible point P is/are
- (A) 1 (B) 2
 (C) 4 (D) 3
76. Suppose points F_1, F_2 are the foci of ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, P is a point on the ellipse such that $\frac{|PF_1|}{|PF_2|} = \frac{2}{1}$, then the area of $\triangle PF_1F_2$ is equal to
- (A) 16 (B) 20
 (C) 4 (D) 32
77. The curve represented by the equation $\frac{x^2}{\sin\sqrt{2} - \sin\sqrt{3}} + \frac{y^2}{\cos\sqrt{2} - \cos\sqrt{3}} = 1$ is
- (A) an ellipse with the foci on x-axis (B) an ellipse with the foci on y-axis
 (C) an hyperbola with foci on x-axis (D) an hyperbola with foci on y-axis
78. If one side of square ABCD is on the line $y = 2x - 17$ and the other two vertices lies on the parabola $y = x^2$, then the minimum area of the square is
- (A) 160 (B) 1200
 (C) 80 (D) 60
79. Let a line with the inclination of 60° from positive x-axis be drawn through the focus F of the parabola $y^2 = 8(x + 2)$. If the two intersection points of the line and the parabola are A and B and the perpendicular bisector of the chord AB intersect the x-axis at the point P, then the length of segment PF is
- (A) $\frac{16}{3}$ (B) $\frac{8}{3}$
 (C) $\frac{16\sqrt{3}}{3}$ (D) $8\sqrt{3}$
80. $\triangle ABC$ is inscribed in a unit circle. The three bisectors of the angles A, B, C are extended to intersect the circle at A_1, B_1, C_1 respectively, then the value of $\frac{AA_1 \cos \frac{A}{2} + BB_1 \cos \frac{B}{2} + CC_1 \cos \frac{C}{2}}{\sin A + \sin B + \sin C}$ is
- (A) 2 (B) 4
 (C) 6 (D) 8

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $x_0 = 2022$ and let $x_{n+1} = \frac{1+x_n}{1-x_n}$ for $n \geq 1$, find x_{2024}
82. Let $x + y + z = 1$, $x, y, z \in \mathbb{R}^+$ then determine the least value of $\frac{1}{x} + \frac{4}{y} + \frac{9}{z}$.
83. The value of $(1 - \cot 23^\circ)(1 - \cot 22^\circ)$ is equal to
84. The number of values of x in the interval $\left(0, \frac{\pi}{2}\right)$ such that $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ is
85. If $\alpha = \frac{\pi}{2023}$, then value of $\cos \alpha \cdot \cos 2\alpha \cdot \cos 3\alpha \cdot \cos 4\alpha \dots \cos(1011\alpha) = 2^{-\lambda}$, then λ is
86. A horizontal park is in the shape of a rectangle ABCD with AB = 30m. A vertical lamp post BP is created at the point B such that $\angle PCB = 30^\circ$ and $\angle PDB = \tan^{-1}\left(\frac{8}{17\sqrt{3}}\right)$. If height of the lamp post is h, then [h] is equal to (where [.] denotes G.I.F.)
87. If the asymptotes of the hyperbola $9x^2 - 4y^2 - 18x - 24y - 63 = 0$ are tangents to the circle $13x^2 + 13y^2 - 26x + k = 0$ at point $A(\alpha_1, \beta_1)$ and $B(\alpha_2, \beta_2)$ then $|\alpha_1 + \alpha_2 + k|$ is equal to
88. Let α and β be real numbers in the interval $\left[0, \frac{\pi}{2}\right]$, then of $\sin^6 \alpha + 3\sin^2 \alpha \cos^2 \beta + \cos^6 \beta = 1$, then $\frac{\alpha}{\beta}$ is equal to
89. The points (0, 0), (a, 11) and (b, 37) are the vertices of an equilateral triangle, then ab is equal to
90. Let P is a point, lies on circle $(x + 5)^2 + (y - 12)^2 = 196$, then the minimum value of sum of squares of abscissa and ordinate of point P is

ALL INDIA TEST SERIES

PART TEST – II

JEE (Main)-2023

TEST DATE: 03-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics *TG ~ @bohring_bot*

PART – A

SECTION – A

1.

C

Sol.

$$Pt = Q_{\text{system}} + Q_{\text{lost}}$$
$$750 \times 4 \times 60 = 0.5 \times 4200 \times 70 + Q_{\text{lost}}$$

$$\therefore P_{\text{lost}} = \frac{Q_{\text{lost}}}{\text{time}}$$

2.

B

Sol.

$$(m_0 - \lambda t) CdT = Rdt$$

3.

A

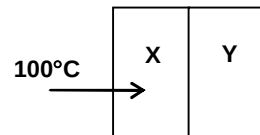
Sol.

At steady state

Let the temperature at the junction be T

$$\therefore \frac{kA}{\ell} (T - 100) = \frac{4nknA}{2n^2\ell} (0 - T)$$

$\therefore T$ is independent of n.



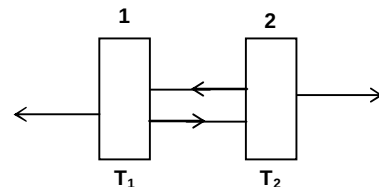
4.

A

Sol.

$$P_1 = 2\sigma AT_1^4 - \sigma AT_2^4$$

$$\sigma AT_1^4 = 2\sigma AT_2^4$$



5. A

Sol.
$$\frac{300}{600} = \frac{2(2\pi R^2)\sigma e(500^4 - 300^4)}{2\pi R^2\sigma e[500^4 - 300^4] + 2\pi R^2\sigma[500^4 - 300^4]}$$

6. B

Sol.
$$\omega^2 = \frac{1}{LC}$$

7. D

Sol.
$$\begin{aligned} \cos\phi_1 &= 0.6 & \cos\phi_2 &= 0.5 & \cos\phi_3 &= 1 \\ \tan\phi_1 &= \frac{X_L}{R_1} & \tan\phi_2 &= \frac{X_C}{R_2} & \therefore X_L &= X_C \\ R_1 \tan\phi_1 &= R_2 \tan\phi_2 \end{aligned}$$

8. B

Sol. The force exerted by gas on the piston when spring has compression x is
 $F = P_0 A + kx$

$$W = \int_0^{\ell} F dx$$

9. D

Sol. Pitch (along x-axis) = $V_0 t$
 Magnetic force is towards negative z-axis
 $\therefore \theta = \omega t = \pi$
 $\therefore$ at this instant $y = 0$ and $z = -2\text{radius}$

10. C

Sol.
$$\begin{aligned} \vec{u} &= u_x \hat{i} + u_y \hat{j} \\ \vec{E} &= -E \hat{j} \\ \vec{B} &= -B \hat{k} \\ \vec{F} &= -e[\vec{E} + \vec{u} \times \vec{B}] = -e[-E \hat{j} + (u_x \hat{i} + u_y \hat{j}) \times (-B \hat{k})] \\ F_x &= eu_y B, F_y = e(E - u_x B) \\ \therefore a_x &= \frac{eB}{m} u_y \\ a_y &= \frac{e}{m} [E - u_x B] \\ \therefore u_y &= A \sin(\omega t + \delta), a_y = A \omega \cos(\omega t + \delta) \\ \text{At } t = 0, u_y &= 0 \therefore \delta = 0 \\ a_y &= \frac{F_y}{m} = \frac{eE}{m} \\ \therefore A &= \frac{eE}{m\omega} = \frac{E}{B} \\ \therefore u_y &= \frac{E}{B} \sin \omega t \end{aligned}$$

The velocity is along x-axis implies final velocity along y-axis is zero.

$$u_y = \frac{dy}{dt} \Rightarrow \int dy = \int_0^{\pi/\omega} u_y dt.$$

11. C

Sol. $\varepsilon = -\frac{d\phi}{dt}$

12. D

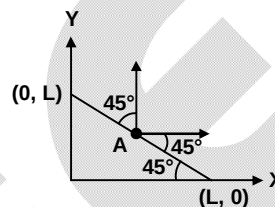
Sol. At point A, E along the rod = 0

$$x \cos 45^\circ = y \sin 45^\circ$$

$$\Rightarrow x = y$$

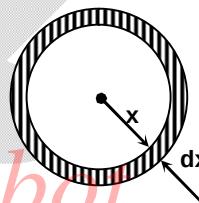
If the particle just reaches point A with velocity tending toward zero, then it will be able to reach the other end.

Apply work energy theorem



13. D

Sol. $\int_0^i di = \int_0^{R/2} J_0 \left(\frac{x}{R} - 1 \right) 2\pi x dx + \int_{R/2}^R J_0 \frac{x}{R} (2\pi x) dx$



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14. D

Sol. Apply $PV = nRT$

$$\frac{P}{nT} = \frac{R}{V}$$

15. A

Sol. Magnetic field is along the tangent

16. B

Sol. $\vec{B} = \frac{\mu_0 I [\sin 60^\circ] (+\hat{k})}{4\pi(2R \sin 30^\circ)} + \frac{\mu_0 I [\sin 30^\circ] (-\hat{k})}{4\pi(2R \sin 60^\circ)}$

17. B

Sol. $\theta = \frac{1}{2} \beta t^2$

$$\varepsilon = -\frac{d(B\theta a^2 / 2)}{dt}$$

18. A

Sol. $mg = kx$

$$\text{and } kx = \frac{q^2}{2\varepsilon_0 A}$$

19. C

Sol. $T_0 \propto \frac{1}{\lambda_0}$

$$P_0 \propto T_0^4$$

$$\therefore P_0 \propto \frac{1}{\lambda_0^4} \text{ and } \frac{256P_0}{81} \propto \frac{1}{\lambda^4}$$

$$\therefore \frac{P_0}{\frac{256P_0}{81}} = \frac{\lambda_0^4}{\lambda^4}$$

20. D

Sol. $PV = RT$

$$\text{and } P = \frac{A}{1 + \frac{B^2}{V^2}};$$

$$\frac{RT}{V} = \frac{A}{1 + \frac{B^2}{V^2}}$$

$$T = \frac{VA}{R \left(1 + \frac{B^2}{V^2} \right)}$$

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SECTION - B

21. 2

Sol. $V = I_1 R_1$ and $V = I_2 R_2$

$$NIAB = k\theta$$

$$I_1 = \frac{k\theta_1}{N_1 AB}$$

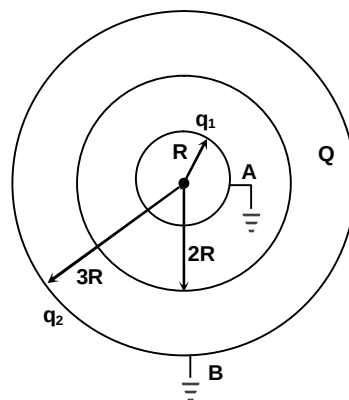
$$\therefore I_2 = \frac{k\theta_2}{N_2 AB}$$

$$V = \frac{k\theta_1 R_1}{N_1 AB} = \frac{k\theta_2 R_2}{N_2 AB}$$

22. 4

Sol. $V_A = 0 = \frac{kq_1}{R} + \frac{kQ}{2R} + \frac{k(q_2)}{3R}$

$$V_B = 0 = \frac{k(q_1 + Q + q_2)}{3R}$$



23. 3

Sol. $S = \frac{I_g G}{I - I_g} = \frac{\rho \ell}{A}$

24. 75

Sol. $\Delta H = \frac{(30)^2}{2} \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{6} \right) - \left[\frac{(15)^2}{2} \left(\frac{1}{3} + \frac{1}{6} \right) + \frac{(45)^2}{2 \times 6} \right] = 75 \mu\text{J}$

25. 3

Sol. $\Delta P = \frac{n_A RT}{2V} - \frac{n_A RT}{V} \quad \dots(i)$

$1.5\Delta P = \frac{n_B RT}{2V} - \frac{n_B RT}{V} \quad \dots(ii)$

Solve equation (i) and (ii)

26. 5

Sol. $PT = \text{constant}$

$PV^{1/2} = \text{constant}$

$x = \frac{1}{2}$

$C = C_v + \frac{R}{1-x}$

$\frac{9R}{2} = \frac{fR}{2} + 2R$

$\therefore f = 5$

27. 7

Sol. $\Delta Q = nC_p \Delta T$

$\Delta Q = \frac{7}{2} nR \Delta T$

$\Delta W = P \Delta V = nR \Delta T$

$\frac{\Delta Q}{\Delta W} = \frac{7}{2}$

28. 3

Sol. $P \propto V$

$PV^{-1} = \text{constant}$

$\Rightarrow x = -1$

$C = C_v + \frac{R}{1-x}$

$= \frac{5R}{2} + \frac{R}{2} = \frac{6R}{2} = 3R$

29. 1

Sol. Resistance, $R = \frac{\rho 2\pi R_1}{\pi r^2}$

$m = d(\pi r^2) 2\pi R_1$

$R = \rho (2\pi R_1)^2 \frac{d}{m}$

$$\text{Induced current in the loop, } I = \frac{dB}{dt} \frac{\pi R_1^2}{R}$$

30. 3

Sol. $I = \frac{\varepsilon}{R} = 3 \text{ A}$

$$H = \frac{1}{2} LI^2 = \frac{1}{2} \times 2 \times 10^{-6} \times (3)^2 = 9 \mu\text{J}$$

$$H_1 = \frac{H}{3} = \frac{9}{3} = 3 \mu\text{J}$$

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Chemistry

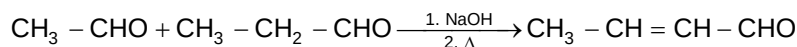
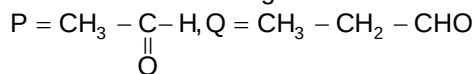
PART – B

SECTION – A

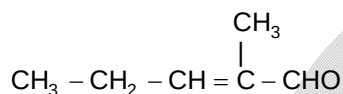
31.

B

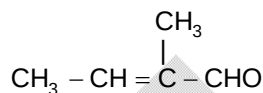
Sol. $P(C_2H_4O)$ and $Q(C_3H_6O)$ must be carbonyl compounds as they both show +ve $NaHSO_3$ test. P must be ethanal as it gives haloform test.



+



+



+

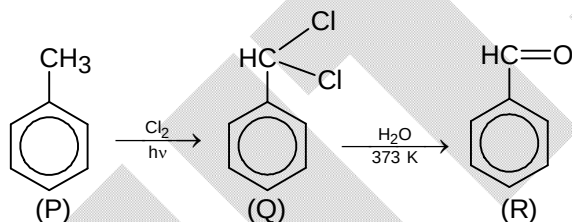


Total 4 products obtained if we exclude stereoisomerism.

32.

D

Sol.

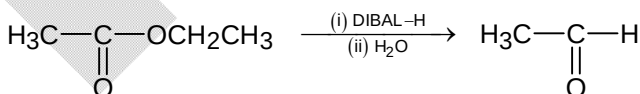
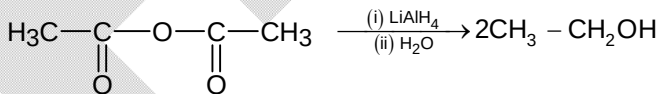
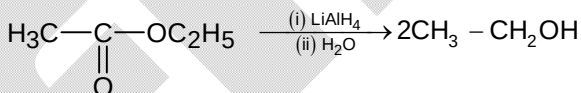


Free-radical substitution is involved in conversion of P to Q.

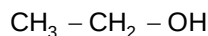
33.

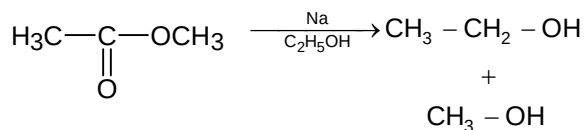
C

Sol.

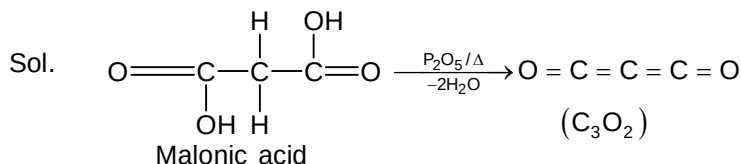


+



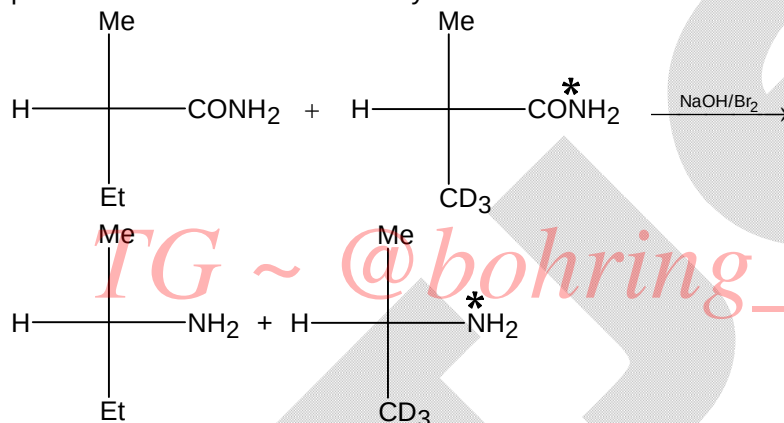


34. C



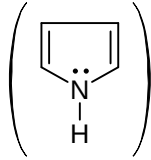
35. D

Sol. This is Hoffmann bromamide reaction which involved intramolecular migration. No cross-over product is formed. Stereochemistry is retained.



36. B

Sol. The lone pair on N-atom in pyridine  is localised but -I effect of N-atom deactivates

benzene ring. On the other hand, the lone pair on N-atom in pyrrole  is de-localized and hence activates the ring towards EAS.

∴ Order of reactivity is : III > II > I.

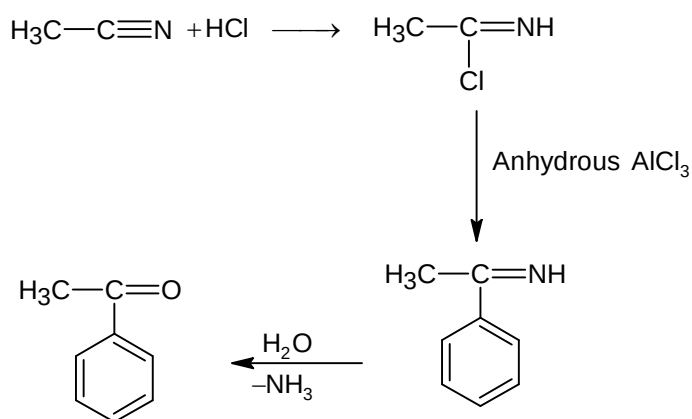
37. A

Sol. In the first reaction, the lone pair on nitrogen atom takes part in resonance with benzene ring and hence activates benzene ring towards EAS. But in second reaction, the lone pair on nitrogen atom does not take part in resonance with benzene ring due to SIR-effect and hence does not activate benzene ring.

∴ $r_1 > r_2$.

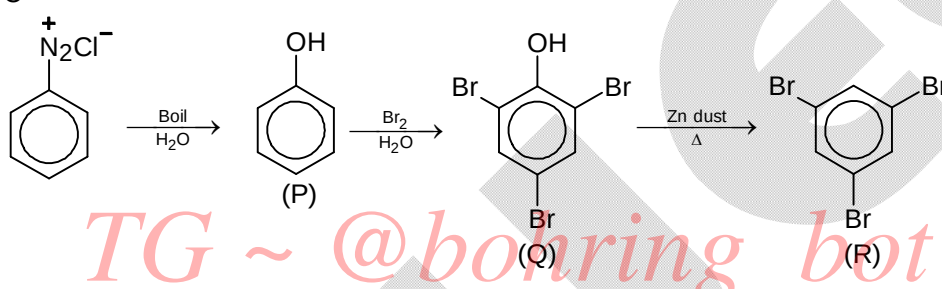
38. D

Sol.



39. C

Sol.

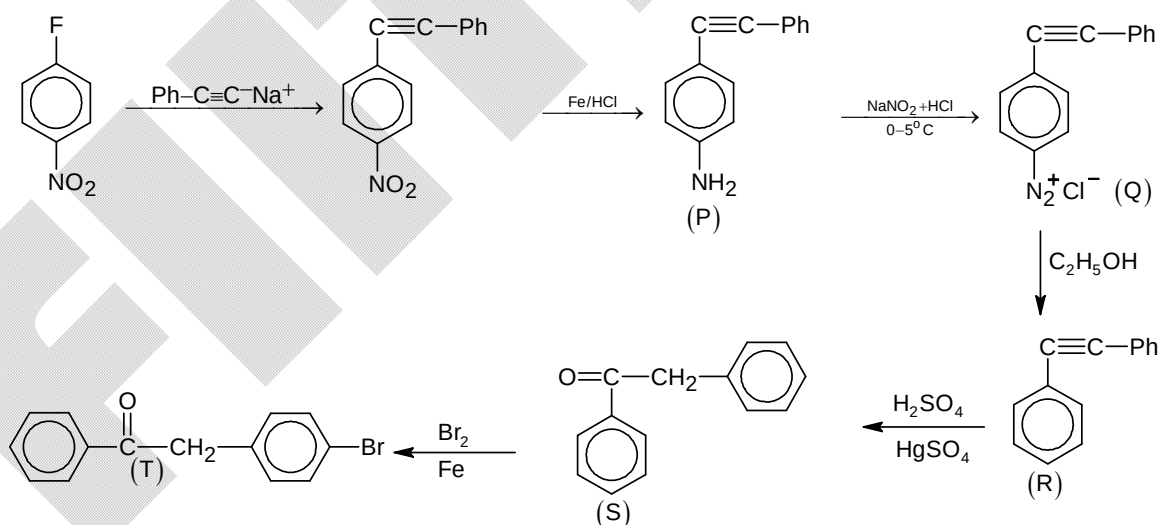


Q is polar while R is non-polar.

$\therefore \mu_Q > \mu_R$

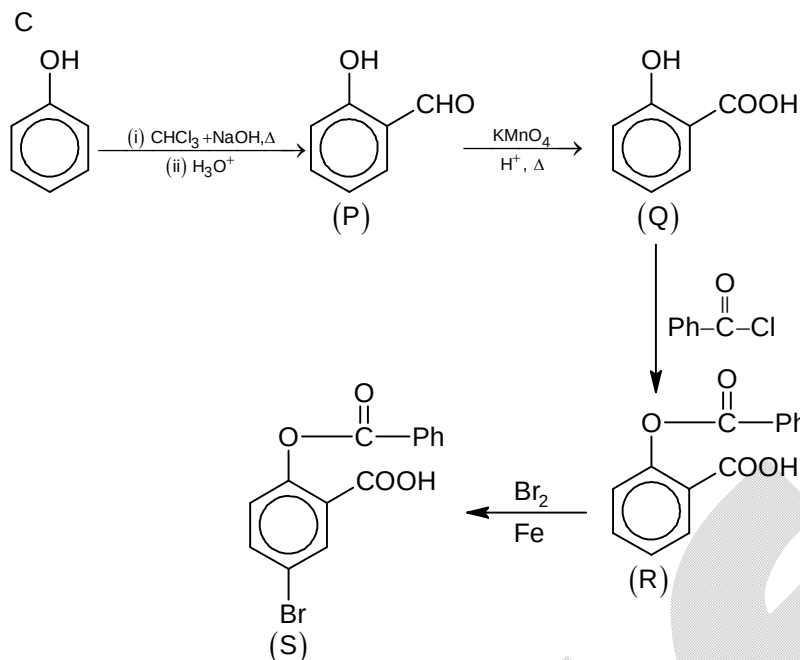
40. B

Sol.



41.

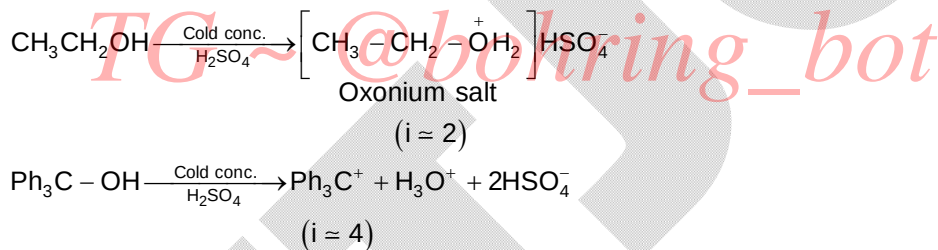
Sol.



42.

B

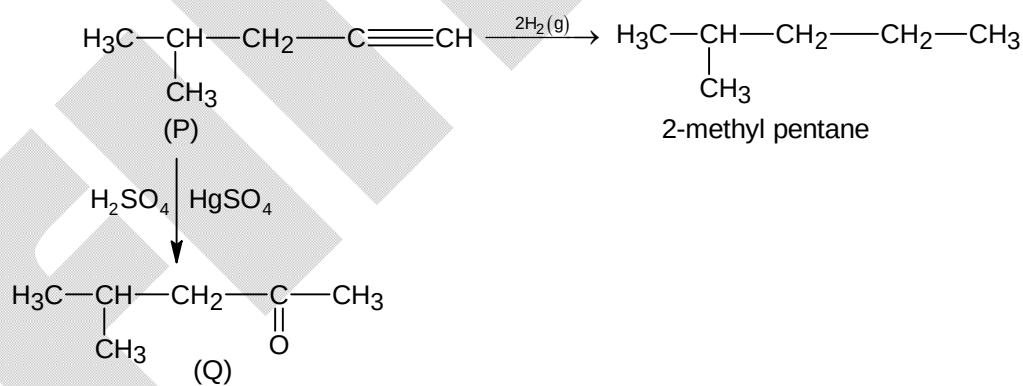
Sol.



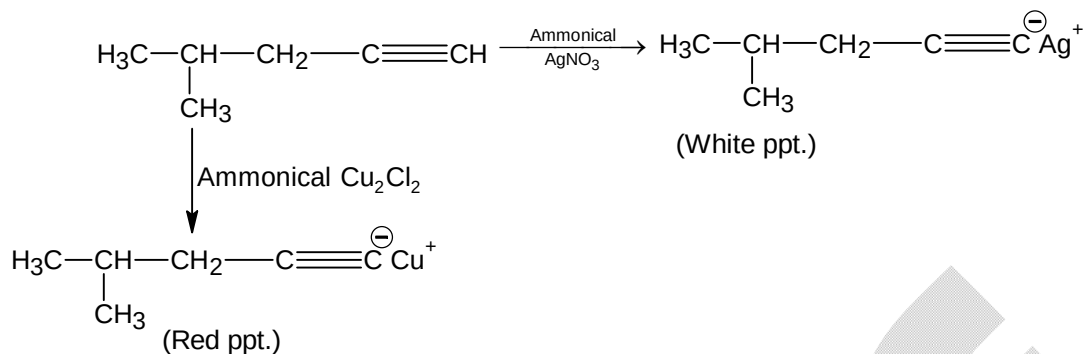
43.

D

Sol.



The ketone (Q) gives positive Iodoform test and shows 2,4-DNP test. It also forms white crystalline product with NaHSO_3 .



44. A

Sol. Phenol + H-CHO $\xrightarrow{\Delta}$ Novolac (P) $\xrightarrow[\text{H-CHO}]{\Delta}$ Bakelite (Q)

45. A

Sol. It's a Reimer-Tiemann reaction. Dichloro carbene (Singlet) is formed as an intermediate in this reaction.

46. B

Sol. Order of basic strength in aqueous medium is



(II) (I) (III)

47. A

Sol.

Picric acid	$\text{pK}_a = 0.4$
Squaric acid	$\text{pK}_{a_1} = 1.5$
Phenol (Carbolic acid)	$\text{pK}_a = 9.98$
Carbonic acid	$\text{pK}_{a_1} = 6.37$

48. A

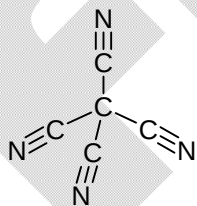
Sol. There is only geometrical isomerism in the given molecule around C = C. There are two geometrical isomers possible for the given molecule.

49. C

Sol. Assymmetric compounds are P, Q and R as these do not contain POS (Plane of Symmetry).

50. B

Sol.

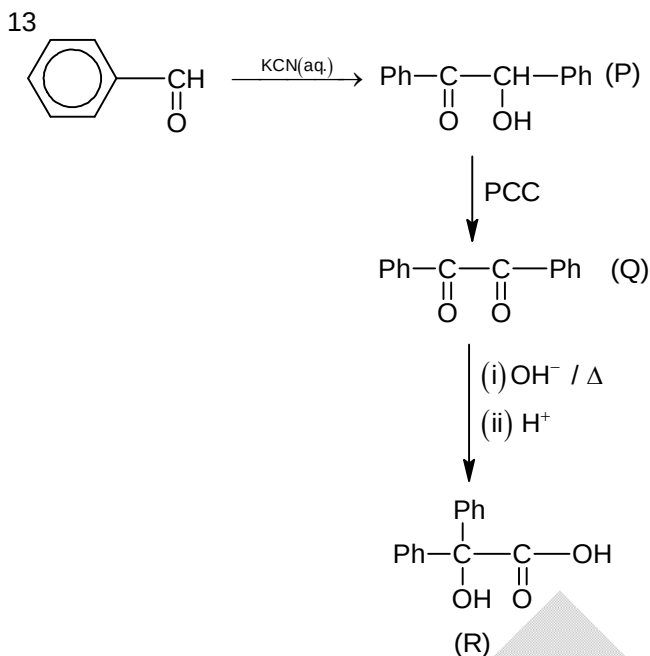


No. of σ bonds = 8

No. of π bonds = 8

$$\therefore \frac{\sigma}{\pi} = 1:1.$$

SECTION – B

 51.
Sol.

 Number of sp^2 hybridised carbons in R = 13.

52.

Sol. Soda lime is a mixture of 3 moles of NaOH and 1 mole of CaO.

53.

 Sol. Pressure = $802 - 42 = 760 \text{ mm Hg}$
 $= 1 \text{ atmosphere}$

 Volume = $110 \text{ mL} = 0.11 \text{ L}$

$$n = \frac{PV}{RT} = \frac{1 \times 0.11}{0.0821 \times 300} = 4.47 \times 10^{-3} \text{ moles}$$

 Weight of $\text{N}_2 = n \times 28$

$$= 4.47 \times 10^{-3} \times 28 = 0.125 \text{ g}$$

$$\% \text{ of nitrogen} = \frac{\text{Wt. of nitrogen}}{\text{Wt. of organic compound}} \times 100$$

$$= \frac{0.125}{0.5} \times 100 = 25\%$$

 Molecular formula = $\text{C}_{12}\text{H}_{24}\text{N}_x$

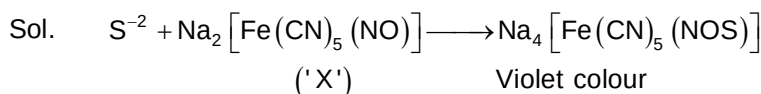
 Molecular weight = $168 + 14x$

$$\% \text{ of nitrogen} = \frac{14x}{168 + 14x} \times 100 = 25$$

$$x = 4$$

$$\therefore \text{Molecular formula} = \text{C}_{12}\text{H}_{24}\text{N}_4$$

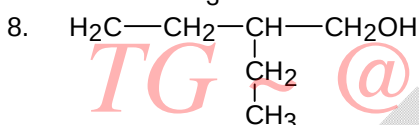
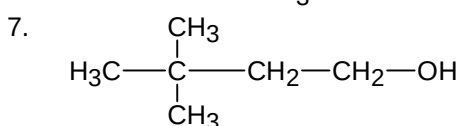
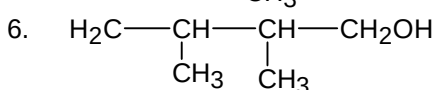
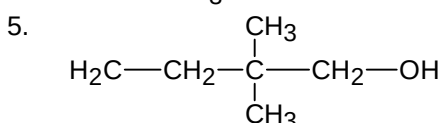
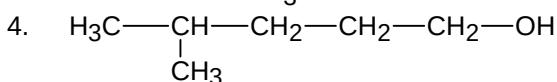
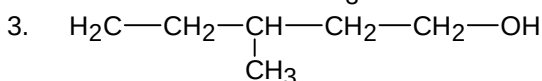
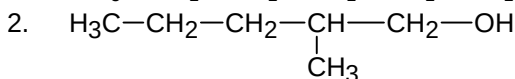
54.



Compound 'X' is sodium nitroprusside and its magnetic moment is (0.0 B.M.)

55. 8

Sol. Only primary alcohols give aldehydes on oxidation with PCC. 8 structural isomers are possible for such alcohol.



56. 6

Sol. With $\text{C} \equiv \text{CH} \rightarrow$ 1 mole G.R. reacts
 With $-\text{OH} \rightarrow$ 1 mole G.R. reacts
 With $\text{CN} \rightarrow$ 1 mole G.R. reacts
 With $\text{CHO} \rightarrow$ 1 mole G.R. reacts
 With $\text{COOCH}_3 \rightarrow$ 2 mole of G.R. reacts
 Thus, total 6 moles of G.R. are consumed

57. 4

Sol. On acetylation, $-\text{OH}$ group gets converted into $-\text{OCOCH}_3$ per $-\text{OH}$ group. It means that molecular weight increases by 42.

Molecular weight of 'A' = 166

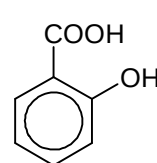
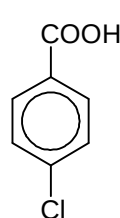
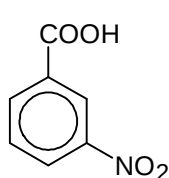
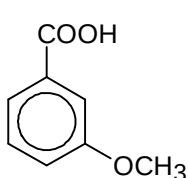
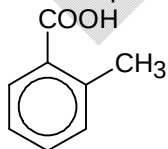
Molecular weight after acetylation = 334

Increase in molecular weight = 168

$$\therefore \text{Number of OH group} = \frac{168}{42} = 4$$

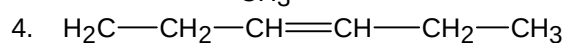
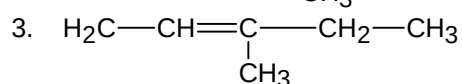
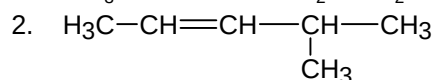
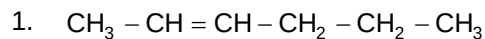
58. 5

Sol. The compounds which are more acidic than benzoic acid are



59. 4

Sol.



60. 4

Sol. Number of meso-isomers for a symmetrical molecule having odd number of chiral carbon

$$= 2^{\left(\frac{n-1}{2}\right)}, \text{ where } n = \text{number of chiral carbon}$$

$$\therefore \text{Number of meso isomers} = 2^{\left(\frac{5-1}{2}\right)} = 2^2 = 4.$$

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Mathematics

PART – C

SECTION – A

61. C

Sol. From the diagram, $\angle AOB = \angle BOC = \angle COD = 60^\circ$

$$\Rightarrow \angle YOD = \frac{\pi}{6}$$

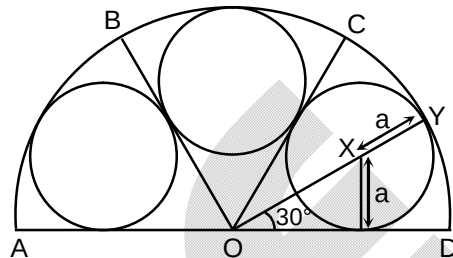
$\Rightarrow$ Let X be the centre of right hand circle,
 $OX \sin 30^\circ = a$

$$\Rightarrow OX = 2a$$

Now

$$r = OY = 2a + a$$

$$a = \frac{r}{3}$$



62. A

Sol. Let $AB = BO = R$ and radius of smallest circle is r , $OA = 2R$, $ON = 2R - r$ and $BN = R + r$

In $\triangle OBN$

$$BN^2 + OB^2 = ON^2$$

$$(R + r)^2 + R^2 = (2R - r)^2$$

$$2R^2 - 6Rr = 0$$

$$R(R - 3r) = 0$$

Considering $\triangle OBN$, perimeter

$$OB + BN + NO = R + (R + r) + (2R - r) = 8$$

$$\Rightarrow R = 2$$

$$r = \frac{1}{3}R = \frac{2}{3}$$

63. A

Sol. Square and add

$$24(\sin A \cos B + \cos A \sin B) = 12$$

$$\Rightarrow \sin(A + B) = \sin(\pi - C) = \frac{1}{2}$$

$$\Rightarrow \sin C = \frac{1}{2}$$

$$\Rightarrow C = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

$$\text{but if } C = \frac{5\pi}{6}, \text{ then } A < \frac{\pi}{6} \text{ so } 3\sin A + 4\cos B = \frac{3}{2} + 4 < 6$$

$$\Rightarrow C = \frac{\pi}{6} \text{ only.}$$

64. B

$$\text{Sol. } \frac{1}{4}(\sin^4 x + \cos^4 x) - \frac{1}{6}(\sin^6 x + \cos^6 x)$$

$$\Rightarrow \frac{1}{12}[3(\sin^4 x + \cos^4 x) - 2(\sin^6 x + \cos^6 x)]$$

$$\Rightarrow \frac{1}{12}[3((\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x) - 2\{(\sin^2 x + \cos^2 x)(1 - 3\sin^2 x \cos^2 x)\}]$$

$$\Rightarrow \frac{1}{12} \{3 - 6\sin^2 x \cos^2 x - 2(1 - 3\sin^2 x \cos^2 x)\}$$

$$\Rightarrow \frac{1}{12} = \frac{2}{24}$$

65. B

Sol. $\frac{\frac{a}{b} - 1}{\frac{a}{b} + 1} = \tan\left(\frac{A-B}{2}\right) \tan \frac{C}{2}$

$$\Rightarrow \frac{1 + \sqrt{3}}{\sqrt{3} + 3} = \tan\left(\frac{A-B}{2}\right) \tan \frac{C}{2} = \tan\left(\frac{A-B}{2}\right) \frac{1}{\sqrt{3}}$$

$$\Rightarrow \tan\left(\frac{A-B}{2}\right) = 1$$

$$\Rightarrow A - B = \frac{\pi}{2} \text{ and } A + B = \frac{2\pi}{3}$$

$$\Rightarrow A = 105^\circ, B = 15^\circ$$

66. D

Sol. $AO^2 = OB^2 = OC^2$

$$\alpha^2 + 4 = \beta^2 + 4 \Rightarrow \alpha = \pm \beta$$

$\Rightarrow \alpha = -\beta$ (if $\alpha = \beta$ then no triangle is formed)

ABC is right angled triangle at A

$\therefore (\alpha, 2)$ is orthocentre $\therefore \alpha = 3, \gamma = 2$

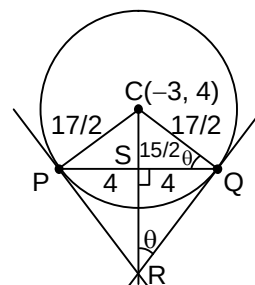
$\therefore$ vertices are $(3, 2), (3, -2)$ and $(-3, 2)$.

67. B

Sol. $\tan \theta = \frac{15/2}{4} = \frac{15}{8}$

In $\triangle RSQ$ $\tan \theta = \frac{4}{RS}$

$$\Rightarrow \frac{15}{8} = \frac{4}{RS} \Rightarrow RS = \frac{32}{15}$$



68. D

Sol. Image of vertex of P_1 in $3x - y + 10 = 0$ will be vertex of P_2

$$\frac{x_2 - 1}{3} = \frac{y_2 - 3}{-1} = \frac{-2(3 - 8 + 10)}{10}$$

$\therefore$ vertex of $P_2 = (-2, 9)$

Similarly focus of P_2 is $(-9, 3) \therefore \alpha = -9$ and $\beta = 3$

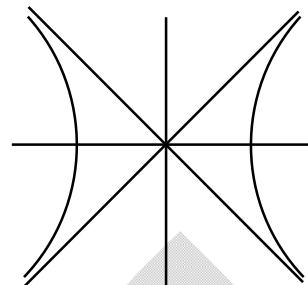
Slope of directrix $= -\frac{7}{6}$

Equation of directrix $y - 15 = -\frac{7}{6}(x - 5)$

$7x + 6y = 125 \therefore \gamma = 125$.

69. C

Sol. Equation of asymptotes $x - 3y = 0$ and $x + 3y = 0$
If tangents to different branches are drawn then
 $(\alpha^2 - 9\beta^2 - 9) < 0$ and $(\alpha - 3\beta)(\alpha + 3\beta) < 0$
i.e. $\alpha^2 < 9\beta^2 + 9$ and $\alpha^2 < 9\beta^2$.



70. B

Sol. $a^2 = \frac{1}{4}$, $b^2 = \frac{1}{9}$ let P be (x_1, y_1)

$$\text{Equation of normal } \frac{1x}{4x_1} - \frac{1y}{9y_1} = \frac{5}{36}$$

$$\text{It cuts y-axis at } \left(0, -\frac{5y_1}{4}\right) \Rightarrow \frac{-5y_1}{4} = \frac{-5}{8\sqrt{3}} \Rightarrow y_1 = \frac{1}{2\sqrt{3}} \therefore x_1 = \frac{1}{4}$$

$$\text{Thus } \alpha = \frac{5}{36}$$

$$\text{Area of } \triangle OAB = \frac{1}{2} \times \frac{5}{36} \times \frac{5}{8\sqrt{3}} = \frac{25}{576\sqrt{3}}$$

71. B

Sol. $x^2 + 24x = 25 \Rightarrow x = -25, 1$

$$\therefore P(1, 2\sqrt{6}) \text{ and } Q(1, -2\sqrt{6})$$

Equation of tangent at P and Q to the circle

$$x + 2\sqrt{6}y = 25 \text{ and } x - 2\sqrt{6}y = 25$$

$$\therefore R \text{ is } (25, 0)$$

Equation of tangent at P and Q to the parabola

$$2\sqrt{6}y = 12(x+1) \text{ and } -2\sqrt{6}y = 12(x+1)$$

$$\therefore S \text{ is } (-1, 0)$$

$$\text{Area of } \triangle PRS = \frac{1}{2} \times 2\sqrt{6} \times 26 = 26\sqrt{6}$$

$$R = \frac{abc}{4\Delta} = \frac{26 \times 2\sqrt{6} \times 10\sqrt{6}}{4 \times 26\sqrt{6}} = 5\sqrt{7}$$

72. D

Sol. $a^2 + b^2 = 18$ and $b^2 = 6$

$$\Rightarrow a^2 = 12 \Rightarrow 6 = 12(1 - e^2) \Rightarrow e = \frac{1}{\sqrt{2}}$$

73. B

Sol. $x(y - 1) = 8$

$$\text{Equation of directrix } x + (y - 1) = 2\sqrt{2} \times \sqrt{2}$$

$$\Rightarrow x + y = 5 \Rightarrow y = -x + 5 \text{ touches}$$

$$\frac{x^2}{30} - \frac{y^2}{30/k} = 1 \therefore 25 = 30 \times 1 - \frac{30}{k} \Rightarrow k = 6.$$

74. A

Sol. $P \equiv (|OP| \cos \theta, |OP| \sin \theta)$

$$Q \equiv (|OQ| \left(\cos \left(\theta \pm \frac{\pi}{2} \right), \sin \left(\theta \pm \frac{\pi}{2} \right) \right))$$

$$\Rightarrow \frac{1}{|OP|^2} = \frac{\sin^2 \theta}{b^2} + \frac{\cos^2 \theta}{a^2}$$

$$\Rightarrow \frac{1}{|OQ|^2} = \frac{\sin^2 \theta}{a^2} + \frac{\cos^2 \theta}{b^2}$$

$$\Rightarrow \frac{1}{|OP|^2} + \frac{1}{|OQ|^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

$$\Rightarrow |OP| \times |OQ| \text{ reaches } \frac{2a^2b^2}{a^2 + b^2}$$

75. B

Sol. Suppose there is a point $P(4\cos\theta, 3\sin\theta)$ on the ellipse, then distance of P to AB is

$$= \left| \frac{3(4\cos\theta) + 4(3\sin\theta) - 12}{5} \right|$$

$$= \frac{12}{5}(\cos\theta + \sin\theta - 1)$$

$$\leq \frac{12}{5}(\sqrt{2} - 1) \leq \frac{6}{5}$$

$$\text{But } AB = 5 \text{ so } \Delta APB \leq \frac{1}{2} \times 5 \times \frac{6}{5} = 3$$

So exactly two points.

76. C

Sol. $PF_1 + PF_2 = 2a = 6$

$$\frac{PF_1}{PF_2} = \frac{2}{1}; PF_1 = 4 \text{ and } PF_2 = 2$$

$$F_1F_2 = 2\sqrt{5}$$

$$\text{there } PF_1^2 + PF_2^2 = (F_1F_2)^2$$

$$16 + 4 = (2\sqrt{5})^2$$

So ΔPF_1F_2 is right angle triangle

$$\text{area} = \frac{1}{2} \times PF_1 \times PF_2 = \frac{1}{2} \times 4 \times 2 = 4$$

77. B

Sol. Since $\sqrt{2} + \sqrt{3} > \pi$, $\frac{\pi}{2} - \sqrt{2} < \sqrt{3} - \frac{\pi}{2} < \frac{\pi}{2}$

$$\Rightarrow \cos\left(\frac{\pi}{2} - \sqrt{2}\right) > \cos\left(\sqrt{3} - \frac{\pi}{2}\right)$$

$$\Rightarrow \sin(\sqrt{2}) > \sin(\sqrt{3})$$

$$\text{again } \cos(\sqrt{2}) > 0 \text{ and } \cos(\sqrt{3}) < 0$$

$$\cos\sqrt{2} - \cos\sqrt{3} > 0$$

$\Rightarrow$ given curve will be ellipse

Now $(\sin\sqrt{2} - \sin\sqrt{3}) - (\cos\sqrt{2} - \cos\sqrt{3})$

$$= 2\sqrt{2} \sin\left(\frac{\sqrt{2}-\sqrt{3}}{2}\right) \sin\left(\frac{\sqrt{2}+\sqrt{3}}{2} + \frac{\pi}{4}\right)$$

$$\text{and } -\frac{\pi}{2} < \frac{\sqrt{2}-\sqrt{3}}{2} < 0$$

$$\Rightarrow \sin\left(\frac{\sqrt{2}-\sqrt{3}}{2}\right) < 0 \text{ and } \frac{\pi}{2} < \frac{\sqrt{2}+\sqrt{3}}{2} < \frac{3\pi}{4}$$

$$\Rightarrow \sin\left(\frac{\sqrt{2}+\sqrt{3}}{2} + \frac{\pi}{4}\right) > 0$$

$$\Rightarrow 2\sqrt{2} \sin\left(\frac{\sqrt{2}-\sqrt{3}}{2}\right) \sin\left(\frac{\sqrt{2}+\sqrt{3}}{2} + \frac{\pi}{4}\right) < 0$$

$\Rightarrow$ ellipse is vertical ellipse.

78.

C

Sol. Assume AB is on the line $y = 2x - 17$ and coordinate of other two are on parabola are $C(x_1, y_1)$ and $D(x_2, y_2)$, then CD is on the line L whose equation is $y - 2x = b$ by solving with curve we get

$$x^2 - 2x - b = 0 \Rightarrow x_1, x_2 = 1 \pm \sqrt{b+1}$$

assume length of one side of square is a

$$\Rightarrow a^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 = 5(x_1 - x_2)^2 = 20(b+1) \quad \dots(1)$$

$$\text{Now } a = \frac{|17+b|}{\sqrt{5}} \quad \dots(2)$$

$$\Rightarrow b = 3 \text{ or } 63 \Rightarrow a^2 = 80 \text{ or } 1280.$$

79.

A

Sol. Equation of AB $\Rightarrow y = \sqrt{3}x$ by solving with parabola $y^2 = 8(x+2)$

$$3x^2 - 8x - 16 = 0$$

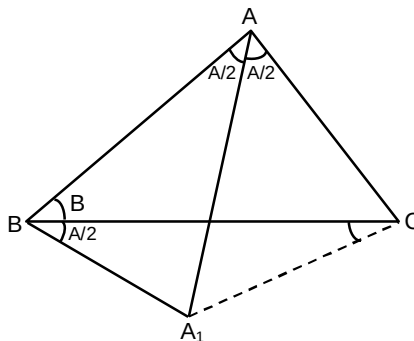
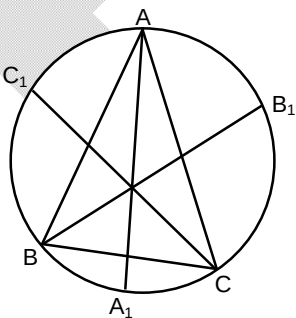
E is mid point of AB then x coordinate of E is $\frac{4}{3}$, then

$$|FE| = \frac{1}{\cos 60^\circ} \times \frac{4}{3} = \frac{8}{3}, |PF| = 2 \times |FE| = \frac{16}{3} \text{ where F is focus}$$

80.

A

Sol. In $\triangle ABC$



$$\angle B = \angle ABC + \frac{A}{2}$$

$$\frac{AA_1}{\sin\left(B + \frac{A}{2}\right)} = 2R = 2$$

$$AA_1 = 2 \sin\left(B + \frac{A}{2}\right) = 2 \cos\left(\frac{B-C}{2} - \frac{C}{2}\right)$$

$$\begin{aligned} \Rightarrow AA_1 \cos \frac{A}{2} &= 2 \cos\left(\frac{B-C}{2}\right) \cos\left(\frac{A}{2}\right) \\ &= \cos\left(\frac{\pi}{2} - C\right) + \cos\left(\frac{\pi}{2} - B\right) = \sin B + \sin C \end{aligned}$$

$$\Rightarrow \text{similarly } BB_1 \cos \frac{B}{2} = \sin A + \sin C$$

$$CC_1 \cos \frac{C}{2} = \sin A + \sin B$$

SECTION - B

81. 2022

Sol. Let $x_n = \tan a_n$

$$\Rightarrow \tan a_{n+1} = \frac{1 + \tan a_n}{1 - \tan a_n} = \tan\left(\frac{\pi}{4} + a_n\right)$$

$$\Rightarrow \tan a_{n+1} = \tan\left(\frac{\pi}{4} + a_n\right)$$

$$\Rightarrow a_{n+1} = a_n + \frac{\pi}{4} + n\pi$$

$$\Rightarrow a_{n+4} = a_n + k\pi; k \in \mathbb{I}$$

$$\Rightarrow x_{n+4} = \tan a_{n+4} = \tan a_n = x_n$$

$$\Rightarrow x_n \text{ is periodic for period 4}$$

$$\Rightarrow x_{2024} = x_0 = 2022$$

82. 36

Sol. Let assume $x = \cos^2 \alpha \cos^2 \beta$, $y = \cos^2 \alpha \sin^2 \beta$, $z = \sin^2 \alpha$

Now $x + y + z = 1$

$$\Rightarrow P = \sec^2 \alpha \sec^2 \beta + 4 \sec^2 \alpha \operatorname{cosec}^2 \beta + 9 \operatorname{cosec}^2 \alpha$$

$$\begin{aligned} \Rightarrow P &= (\tan^2 \alpha + 1)(\tan^2 \beta + 1) + 4(\tan^2 \alpha + 1)(\cot^2 \beta + 1) + 9(\cot^2 \alpha + 1) \\ &= 14 + 5\tan^2 \alpha + 9\cot^2 \alpha + (\tan^2 \beta + 4\cot^2 \beta)(1 + \tan^2 \alpha) \\ &\geq 14 + 5\tan^2 \alpha + 9\cot^2 \alpha + 2\tan \beta \cdot 2\cot \beta (1 + \tan^2 \alpha) \\ &= 18 + 9(\tan^2 \alpha + \cot^2 \alpha) \geq 18 + 9 \cdot 2 \tan \alpha \cot \alpha = 36 \end{aligned}$$

$$\text{Equality hold when } \cos^2 \alpha = \frac{1}{2}, \cos^2 \beta = \frac{1}{3}$$

$$\Rightarrow x = \frac{1}{6}, y = \frac{1}{3}, z = \frac{1}{2}$$

83. 2

Sol. $(1 - \cot 22^\circ)(1 - \cot 23^\circ)$

$$\Rightarrow \left(1 - \frac{\cos 22^\circ}{\sin 22^\circ}\right) \left(1 - \frac{\cos 23^\circ}{\sin 23^\circ}\right)$$

$$\begin{aligned} &\Rightarrow \left(\frac{\sin 22^\circ - \cos 22^\circ}{\sin 22^\circ} \right) \left(\frac{\sin 23^\circ - \cos 23^\circ}{\sin 23^\circ} \right) \\ &\Rightarrow \frac{\sqrt{2}(\sin(22^\circ - 45^\circ))\sqrt{2}\sin(23^\circ - 45^\circ)}{(\sin 22^\circ \sin 23^\circ)} \\ &\Rightarrow \frac{2\sin(-22^\circ)\sin(-23^\circ)}{(\sin 22^\circ)(\sin 23^\circ)} = 2 \end{aligned}$$

84.

2

Sol.

$$\frac{\sqrt{3}-1}{4} + \frac{\sqrt{3}+1}{4} = \sqrt{2}$$

$$\begin{aligned} &\Rightarrow \frac{\sin\left(\frac{\pi}{12}\right)}{\sin x} + \frac{\cos\left(\frac{\pi}{12}\right)}{\cos x} = 2 \\ &\Rightarrow \sin\left(\frac{\pi}{12}\right)\cos x + \cos\left(\frac{\pi}{12}\right)\sin x = 2\sin x \cos x \\ &\Rightarrow \sin\left(\frac{\pi}{12} + x\right) = \sin 2x \\ &\Rightarrow \frac{\pi}{12} + x = 2x \text{ or } \frac{\pi}{12} + x = \pi - 2x \\ &\Rightarrow x = \frac{\pi}{12} \text{ or } x = \frac{11\pi}{36} \end{aligned}$$

85.

1011

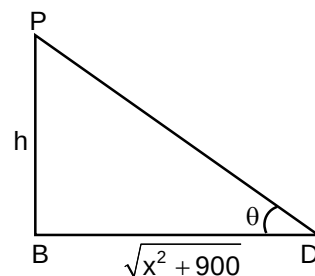
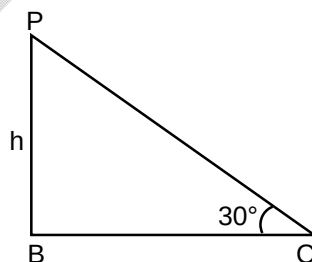
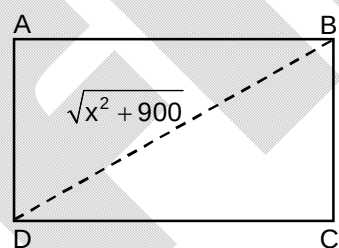
Sol.

Let P denote the desired product and $Q = \sin \alpha \cdot \sin 2\alpha \cdot \sin 3\alpha \dots \sin(1011\alpha)$
 then $2^{1011}PQ = \sin 2\alpha \cdot \sin 4\alpha \cdot \sin 6\alpha \dots \sin(2022)\alpha$
 $= (\sin 2\alpha \cdot \sin 4\alpha \dots \sin(1010\alpha)) (\sin(\pi - 1012\alpha)) \dots (\sin(\pi - 2020\alpha)) (\sin(\pi - 2022\alpha))$
 $\Rightarrow 2^{1011}PQ = \sin \alpha \cdot \sin 2\alpha \cdot \sin 3\alpha \dots \sin(1011\alpha) = Q$
 $\Rightarrow 2^{1011}PQ = Q$
 $P = \frac{1}{2^{1011}}$

86.

9

Sol.



$$\begin{aligned} \tan 30^\circ &= \frac{h}{x} \Rightarrow x = h\sqrt{3} \\ \tan \theta &= \frac{h}{\sqrt{x^2 + 900}} = \frac{8}{17\sqrt{3}} \\ \frac{h}{\sqrt{3h^2 + 900}} &= \frac{8}{17\sqrt{3}} \end{aligned}$$

$$289 \times 3 \times h^2 = 192h^2 + 64 \times 900$$

$$675h^2 = 64 \times 900$$

$$\Rightarrow h = \frac{16}{\sqrt{3}}$$

87. 21

Sol. Asymptotes $3(x-1) - 2(y+3) = 0$ and $3(x-1) + 2(y+3) = 0$

$$3x - 2y - 9 = 0 \text{ and } 3x + 2y + 3 = 0$$

It touches $x^2 + y^2 - 2x + k/13 = 0$

$$\therefore A\left(-\frac{5}{13}, -\frac{12}{13}\right) \text{ and } B\left(\frac{31}{13}, -\frac{12}{13}\right)$$

$$\therefore k = -23$$

88. 1

$$\text{Sol. } (\sin^2 \alpha)^3 + (\cos^2 \beta)^3 + (-1)^3 - 3\sin^2 \alpha \cos^2 \beta (-1) = 0$$

$$\Rightarrow \sin^2 \alpha + \cos^2 \beta - 1 = 0 \text{ or } \underbrace{\sin^2 \alpha = \cos^2 \beta = -1}_{\text{not possible}}$$

$$\Rightarrow \sin^2 \alpha = 1 - \cos^2 \beta$$

$$\Rightarrow \sin^2 \alpha = \sin^2 \beta$$

$$\alpha = \beta$$

89. 315

Sol. Let M is mid point of AB, $M = \left(\frac{a+b}{2}, 24\right)$

$$OM \perp MA, |OM| = \sqrt{3} MA$$

$$\Rightarrow \frac{a+b}{2} = 13\sqrt{3}, \frac{a-b}{2} = 8\sqrt{3}$$

$$\Rightarrow a = 21\sqrt{3} \text{ and } b = 5\sqrt{3}$$

90. 1

$$\text{Sol. } P(x, y) = (14\cos\theta - 5, 14\sin\theta + 12)$$

minimum value of $g(\theta)$ where

$$g(\theta) = (14\cos\theta - 5)^2 + (14\sin\theta + 12)^2$$

$$= 365 + 28(12\sin\theta - 5\cos\theta)$$

$$g(\theta) \geq 1$$

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – II
PAPER –1
TEST DATE: 11-12-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

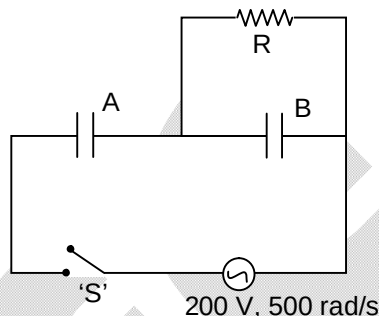
Physics

PART – I

Section – A (Maximum Marks: 24)

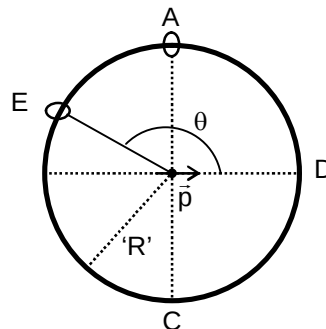
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. A resistor 'R' of resistance 2000Ω and two similar capacitor 'A' and 'B' of capacitance $1 \mu\text{C}$ each are connected with an A.C. source of 200 Volt and 500 rad/s. Now switch is closed at $t = 0$, then choose the correct option(s).



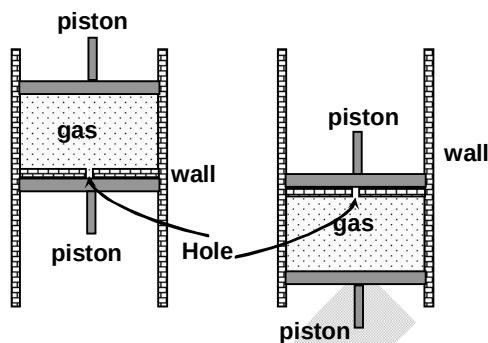
- (A) The phase difference between applied voltage and voltage across the capacitor A is $\tan^{-1}\left(\frac{1}{3}\right)$
- (B) The phase difference between current supplied by source and current in the capacitor 'B' is zero.
- (C) The current in the capacitor 'A' is $\frac{1}{5\sqrt{20}}$ ampere.
- (D) Power dissipated through resistor is 4 watt.

2. A ring of radius R is fixed and kept in a vertical plane. A small dipole of dipole moment $\vec{p}$ is fixed at the centre of ring. A small bead of mass 'm' and charge q is released from point A. Bead can slide on the ring. The inner diameter of bead and thickness of ring are nearly equal. Ring is insulating and frictionless. Ignore effect of gravity. Now choose the correct option(s).



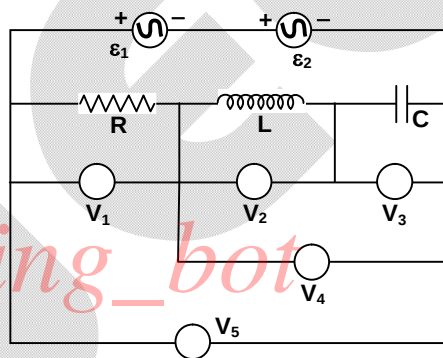
- (A) Bead will never reach at point D.
- (B) The normal force between bead and ring at 'E' is directly proportional to the charge on the bead.
- (C) The motion of bead is periodic.
- (D) Speed of bead at point 'E' is given by $\sqrt{\frac{-p(\cos\theta)q}{2\pi\epsilon_0 m r^2}}$

3. There is a fixed long vertical cylinder of good heat insulator with open ends, is separated from its surroundings by two insulated identical, frictionless pistons. An insulated separating wall is fixed between pistons. A very small hole is made at the middle of separating wall. An ideal gas is enclosed between the pistons at temperature of surrounding. Initially the lower piston is held up against the separating wall, when lower piston is released, it moves downward very slowly. Now choose the correct option(s).



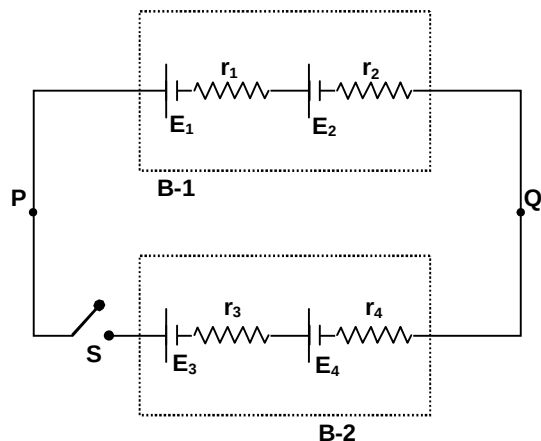
- (A) During the process the pressure of gas above the separating wall remains constant.
 (B) During the process the pressure of gas below the separating wall remains constant.
 (C) Initial and final temperature of gas remain same.
 (D) Initial and final temperature of gas is different.

4. A resistor $R = 500 \, \Omega$, an ideal inductor $L = 0.9 \, \text{H}$ and a capacitor $C = 2.5 \, \mu\text{F}$ are connected with two alternating source $e_1 = 100 \sin(1000t)$ and $e_2 = 100\sqrt{3} \cos(1000t)$. V_1, V_2, V_3, V_4 and V_5 are five ideal voltmeters and connected in the circuit as shown in the figure. Choose the correct option(s).



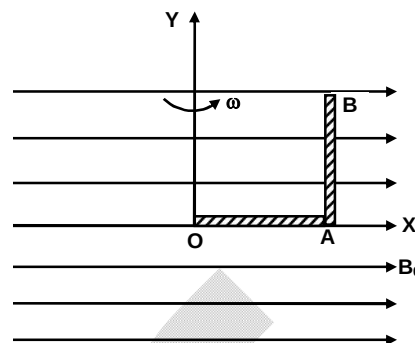
- (A) Phase difference between voltage of V_3 and V_5 is $\frac{\pi}{2} + \tan^{-1}\left(\frac{9}{5}\right)$
 (B) Reading of V_4 is 180 volt
 (C) Difference between reading of V_2 and V_1 is 80 volt.
 (D) Ratio of reading of V_4 and V_5 is $\frac{1}{\sqrt{2}}$

5. A battery B-1 is made by using two cells of emf E_1 and E_2 and internal resistance r_1 and r_2 respectively. Another battery B-2 is made by using different cells of emf E_3 and E_4 and internal resistance r_3 and r_4 respectively. Now they are connected with points P and Q as shown in the figure. All cells are rechargeable. Now switch is closed at $t = 0$, then choose the correct option(s).



- (A) The potential difference across both the battery will be equal.
 (B) Rate of heat dissipation in both batteries is equal.
 (C) One battery will charge the other battery.
 (D) Rate of loss in chemical energy of one battery is more than the rate of gain in chemical energy of the other.

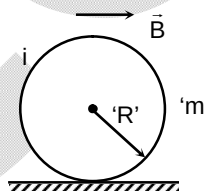
6. A L-shaped conducting rod OAB is rotating in a constant magnetic field of $B_0 = 100$ Tesla acting along x-axis. The length of rod OA = 20 cm and AB = 20 cm. If rod is rotating about y-axis with constant angular velocity of 5 rad/second then at a given instant.
- (A) Potential difference across OA is 10 Volt
 (B) Potential difference across OA is zero Volt
 (C) Potential difference across AB is 20 Volt
 (D) Potential difference across OB is 10 Volt



Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

7. A ring of mass $m = 10$ kg and radius $R = 5$ cm is kept at rest on a smooth horizontal surface in vertical plane as shown in the figure. The ring carries a current $i = 7$ ampere. A constant horizontal magnetic field $B = 100$ mT is switched on at $t = 0$.



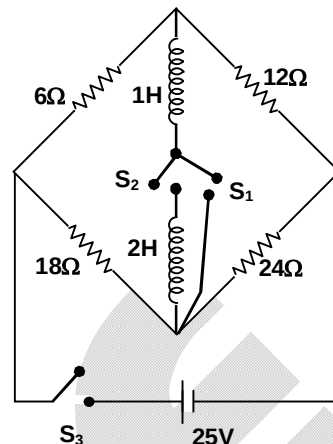
List –I describes the statement regarding ring and List-II gives their respective values.

List –I		List –II	
(I)	The magnetic moment of loop in S.I. units	(P)	0.0880
(II)	Torque acting on loop due to magnetic field in S.I. units at $t = 0$	(Q)	0.0044
(III)	Angular acceleration of loop in S.I. units at $t = 0$	(R)	0.0055
(IV)	Kinetic energy of ring when it has rotated through 90°	(S)	0.4400
		(T)	0.0550

Which one of the following options is correct ?

- (A) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ R
 (B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S
 (C) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ Q
 (D) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R

8. Four resistors 6Ω , 12Ω , 18Ω and 24Ω , two ideal inductors $1H$ and $2H$ and a battery of $25V$ are arranged in the given circuit with the help of three switches S_1 , S_2 and S_3 as shown in the figure. ($e^{-1} = 0.37$).



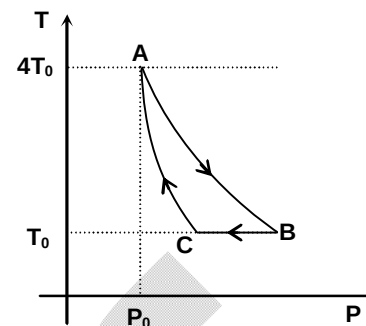
List -I describes statement regarding the given circuit and List-II gives their values. (either exactly or as a close approximation).

List -I		List -II	
(I)	When switch S_1 and S_3 closed simultaneously and S_2 remain opened. Then time constant of circuit in second.	(P)	1.35
(II)	When switch S_1 and S_3 remain closed for long time and S_2 remain opened then current through inductor in ampere at one time constant.	(Q)	0.07
(III)	When switch S_1 and S_3 remain closed for long time and S_2 remain opened then current in 12Ω resistance at one time constant.	(R)	0.056
(IV)	If S_1 and S_3 remain closed for long time and then S_1 is opened, S_2 is closed and S_3 remain closed then find current in the branch of inductor immediately after switching.	(S)	0.105
		(T)	0.167

Which one of the following options is correct ?

- (A) I $\rightarrow$ Q, II $\rightarrow$ T, III $\rightarrow$ S, IV $\rightarrow$ R
 (B) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R
 (C) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ S
 (D) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ S

9. One mole of an ideal diatomic gas undergoes a cyclic process as shown in Temperature-Pressure graph. Temperature and pressure of gas at 'A' is $4T_0$ and P_0 respectively and temperature in process BC is T_0 . The process AB is given by $TP^2 = \text{constant}$ and the process CA is given by $TP^4 = \text{constant}$.



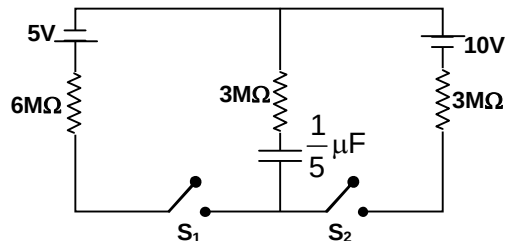
List -I describe the name of thermodynamics process involve in the cyclic process and List-II gives work done by gas in the process, change in internal energy of the gas in the process and heat given to the gas in the process.

List -I		List -II	
(I)	In the thermodynamics process $A \rightarrow B$	(P)	Work done by gas and heat given to gas is negative
(II)	In the thermodynamics process $B \rightarrow C$	(Q)	Work done by gas is negative but heat given to gas is positive
(III)	In the thermodynamics process $C \rightarrow A$	(R)	Work done by gas is positive but heat given to gas is negative
(IV)	In the cyclic process	(S)	Work done by gas and heat given to gas is positive
		(T)	Work done by gas is positive and change in internal energy of gas is negative

Which one of the following options is correct ?

- (A) I $\rightarrow$ P, II $\rightarrow$ T, III $\rightarrow$ S, IV $\rightarrow$ Q
 (B) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ R
 (C) I $\rightarrow$ P, II $\rightarrow$ S, III $\rightarrow$ S, IV $\rightarrow$ P
 (D) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ T

10. Three resistors of resistance $6M\Omega$, $3M\Omega$ and $3M\Omega$, one capacitor of capacitance $\frac{1}{5}\mu F$, two cells of potential difference 5V and 10V and two switches are connected in the circuit as shown in the figure. Both switches are closed simultaneously at $t = 0$. ($e^{-1} = 0.37$)



List-I describe the charges and currents in the circuit at a particular time and List-II gives possible values of charges and current.

List -I		List -II	
(I)	The maximum charge on the capacitor in μC	(P)	0.63
(II)	Charge on capacitor at $t = 1$ sec in μC	(Q)	1.54
(III)	Current in the branch of capacitor at $t = 1$ sec in μA	(R)	2.06
(IV)	Current passing through battery of 5V at $t = 1$ sec in μA	(S)	1.00
		(T)	0.37

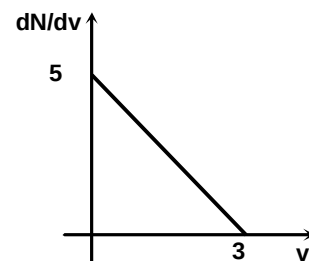
Which one of the following options is correct ?

- (A) I $\rightarrow$ P, II $\rightarrow$ T, III $\rightarrow$ R, IV $\rightarrow$ S
 (B) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ R
 (C) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ Q
 (D) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ T, IV $\rightarrow$ Q

Section - B (Maximum Marks: 24)

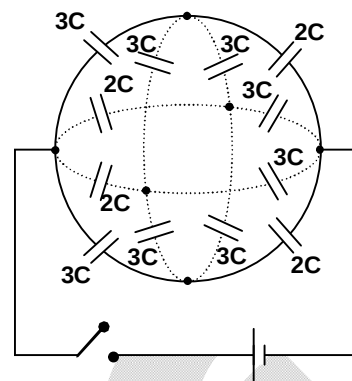
This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

11. Consider a gas kept in a closed container. 'N' is the no of molecules present in the container and v is the speed of molecules. Molecular speed distribution graph is shown in the figure with suitable S.I. units of $\frac{dN}{dv}$ and v . Find v_{rms}^2 of molecules in S.I. units.

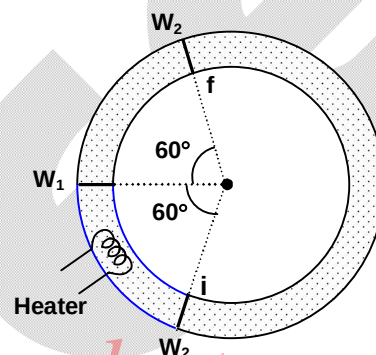


12. A semi-cylindrical shell (A) of diameter 15 cm and length $\frac{70}{22}$ meter is kept in another semi-cylindrical shell (B) of diameter 30 cm and same length such that the axis of both cylinder coincide with each other. A liquid of resistivity $100 \Omega m$ is filled in the space between them. The lower surface of smaller cylinder A is maintained at higher potential than the larger cylinder B. Find the electrical resistance between the surface of cylinder in Ω .

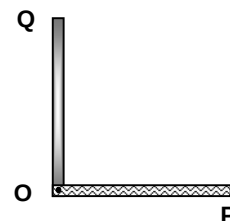
13. 12 capacitors are connected with a battery of potential difference V_0 as shown in the figure. Initially switch is open. Find total heat (in μJ) generated in the system after the switch is closed. (Given $C = \frac{1}{340} \mu\text{F}$, $V_0 = 17$ volt)



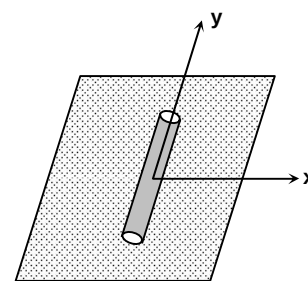
14. An insulated thin tube is fixed on horizontal surface. The total volume of tube is V_0 . Inside the tube, W_1 is fixed adiabatic wall and W_2 is movable adiabatic wall. An ideal gas of adiabatic coefficient $\gamma = \frac{3}{2}$ is filled to entire volume of tube in both chambers. The initial pressure in both chamber is P_0 and number of moles of gas in both chamber are same. A heater is fixed which supplies heat very slowly such that wall W_2 is moved from position i to f , at this position wall W_2 stays in equilibrium position even the heater is switched off. Neglect friction and any loss of heat to the surrounding. The total heat supplied by heater is $\mu P_0 V_0$. Find the value of μ . ($\sqrt{5} = 2.23$)



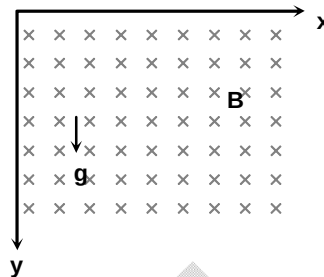
15. A L-shaped rod is made by using two different rods PO and QO. The length of rods PO and QO is 6.0 cm and 4.0 cm respectively and their coefficient of linear expansion is $1.2 \times 10^{-5} / ^\circ\text{C}$ and $3.0 \times 10^{-5} / ^\circ\text{C}$ respectively. The coefficient of linear expansion of the rod to be joined between point P and Q is $\lambda \times 10^5 / ^\circ\text{C}$. Find the value of λ so that no bending takes place anywhere.



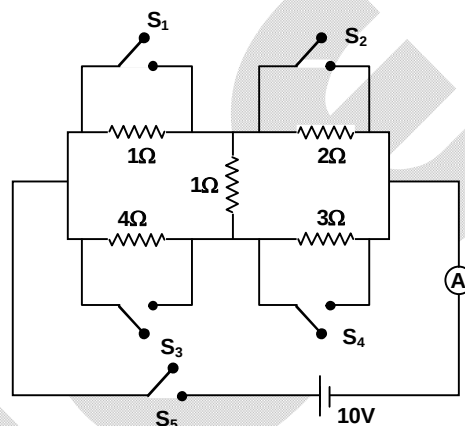
16. A uniform magnetic field $B = 4$ Tesla exist in the horizontal direction along x-axis. A conducting wire of mass 600 gm and length 5 cm is kept along y-axis on a rough horizontal surface. Conducting wire is carrying a current of magnitude 5 amp along y-axis. The minimum force required along x-axis to move the wire is F_1 . Now the direction of current in the wire is reversed then minimum force required along x-axis to move the wire is F_2 . Find the ratio of $\frac{F_1}{F_2}$.



17. A constant horizontal uniform magnetic field $B = 5 \times 10^6$ Tesla exist in a uniform vertical gravitational field $g = 10 \text{ m/s}^2$. Both fields are perpendicular to each other. A particle of mass $m = 50 \text{ gm}$ and charge $q = 2 \mu\text{C}$ is released in the space. Find the displacement (in cm) of particle along y-axis when velocity of particle is along x-axis first time.



18. Five resistors of resistance 1Ω , 1Ω , 2Ω , 3Ω and 4Ω , a battery of voltage 10V , an ideal ammeter and five switches S_1 , S_2 , S_3 , S_4 and S_5 are connected in a circuit as shown in the figure. When S_3 and S_5 are closed and S_1 , S_2 and S_4 remain opened then current read by ammeter is i_1 . When S_1 , S_3 and S_5 are closed and S_2 , S_4 remain opened then current read by ammeter is i_2 . The ratio of current $\frac{i_1}{i_2}$ is given by 11λ . Find the value of λ .



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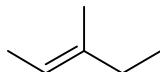
Chemistry

PART – II

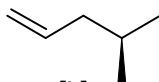
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

19. Choose the INCORRECT statement(s) about the following alkenes is/are



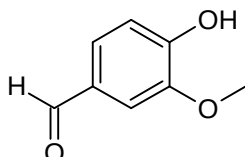
[K]



[L]

- (A) Alkene [K] is more stable than alkene [L].
 (B) Hydrogenation of achiral alkene [K] will give chiral product.
 (C) Hydrogenation of achiral alkene [L] will give chiral product.
 (D) The two alkenes [K] and [L] are structural isomers.

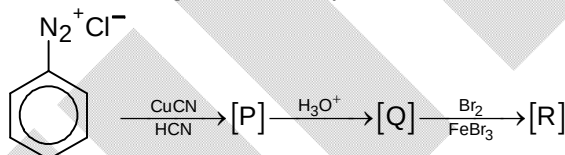
20. Choose the CORRECT statement (s) regarding Vanilin



Vanilin [P]

- (A) [P] can be obtained as a major product when 3,4-dihydroxy benzaldehyde is treated with one equivalent of NaOH and one equivalent of MeBr.
 (B) [P] can be obtained as a one of the product when 2-methoxy phenol is treated with chloroform and aq. KOH.
 (C) Position isomer of [P] can be obtained when 3,4-dihydroxy benzaldehyde is treated with one equivalent of NaOH and one equivalent of MeBr.
 (D) [P] can be obtained when 3,4-dihydroxy benzaldehyde is treated with two equivalent of NaOH and one equivalent of MeBr followed by acidic work up.

21. In the following reaction sequence:

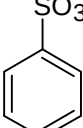


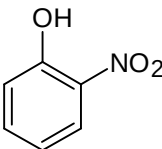
Choose the 'FALSE' statement(s):

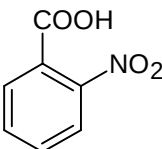
- (A) [R] is more acidic than [Q].
 (B) [R] is more reactive towards electrophilic aromatic substitution than [Q].
 (C) [Q] is more reactive towards electrophilic substitution than benzene.
 (D) [R] on reaction with sodalime form a compound which can undergo phenol formation with aq. NaOH treatment under the normal condition of pressure and temperature.

22. Which of the following compounds are soluble in NaHCO_3 ?

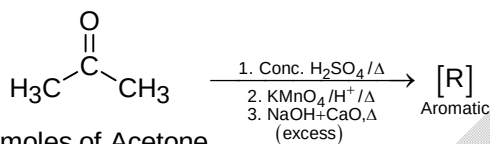
(A) Picric acid

(B) 

(C) 

(D) 

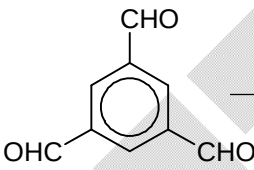
23.

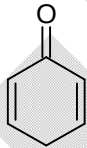


Out of the given reactions, which can give major product [R]?

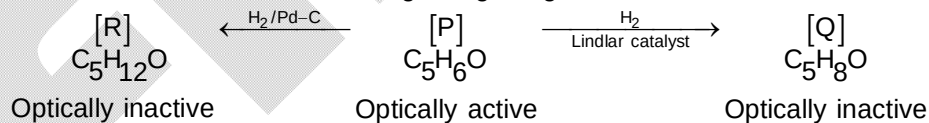
(A) $\text{HC}\equiv\text{CH} \xrightarrow[\text{Fe tube}]{\text{Red hot}}$

(B) $\text{n-hexane} \xrightarrow[773 \text{ K, 10-20 atm}]{\text{Cr}_2\text{O}_3-\text{Al}_2\text{O}_3}$

(C)  $\xrightarrow[\text{HCl}]{\text{Zn(Hg)}}$

(D)  $\xrightarrow[\text{3. Zn, dust}]{\text{1. dil. KOH, 2. H}_3\text{O}^+}$

24. Choose the correct statements regarding the given scheme



(A) [R] can be oxidized with PCC into optically active carbonyl compound.

(B) Oxidised product of [Q] with PCC can show tautomerism.

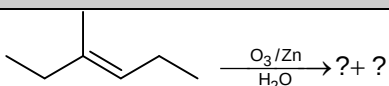
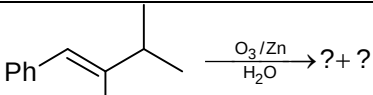
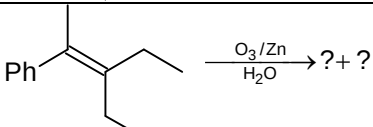
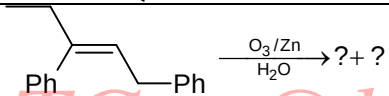
(C) [P] contains triple bond at 4th carbon (as per IUPAC)

(D) [Q] can give white turbidity immediately with Lucas test.

Section – A (Maximum Marks: 12)

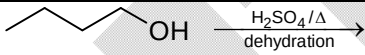
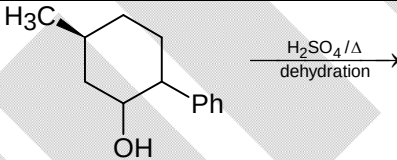
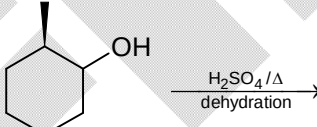
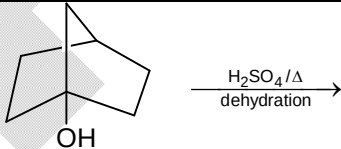
This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. Match each species in List – I with their properties in List-II and choose correct option

List – I		List – II (Products can be distinguished)	
(I)		(P)	Fehling's test
(II)		(Q)	Tollen's test
(III)		(R)	Iodoform test
(IV)		(S)	Schiff's test
		(T)	Benedict's test

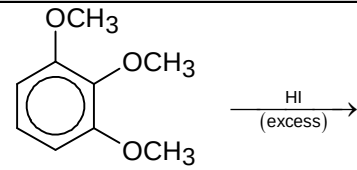
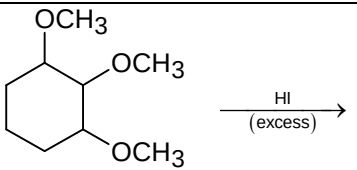
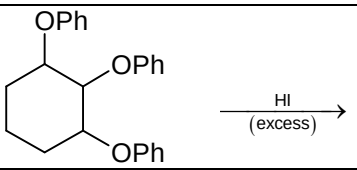
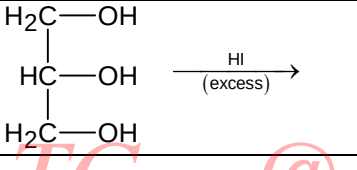
- (A) I→P, Q, R, S; II→P, Q, R; III→R, S; IV→P, Q, S, T
 (B) I→P, Q, R, S, T; II→Q, R, S; III→R; IV→P, Q, S, T
 (C) I→Q, R, S, T; II→Q, R; III→S; IV→P, Q, S
 (D) I→R, S, T; II→P, R; III→Q, S; IV→Q, S, T

26. Match each species in List – I with appropriate statement (s) of List – II and choose correct option

List – I		List – II	
(I)		(P)	At least one of the product shows optical isomerism.
(II)		(Q)	Major product contains even number of α – hydrogens (Hydrogens which can involve in hyperconjugation).
(III)		(R)	One of the product shows geometrical isomerism.
(IV)		(S)	Reactant can turn acidified dichromate to green.
		(T)	Will not undergo dehydration.

- (A) I→Q, R; II→Q, S, T; III→P, S; IV→T
 (B) I→Q, R, S; II→Q, R, S; III→P, Q; IV→R, S
 (C) I→P, Q, R, S; II→Q, R, S; III→P, R; IV→Q
 (D) I→Q, R, S; II→P, Q, R, S; III→P, S; IV→T

27. Match each species in List – I with appropriate statement(s) in List – II and choose correct options:

List – I		List – II	
(I)		(P)	More than 4 moles of HI will be consumed.
(II)		(Q)	Product will react with Na/dry ether.
(III)		(R)	Product formed contains 2° halide.
(IV)		(S)	CH ₃ I is one of the product.
		(T)	2-Iodopropane is a major product.

- (A) I→Q, R; II→P, Q, R; III→P, Q, R; IV→Q, S, T
 (B) I→P, T; II→Q, R, S; III→P, Q, R, T; IV→P, Q, R
 (C) I→P, Q; II→Q, S, T; III→P, Q, R; IV→Q, R, T
 (D) I→Q, S; II→P, Q, R, S; III→P, Q, R; IV→P, Q, R, T

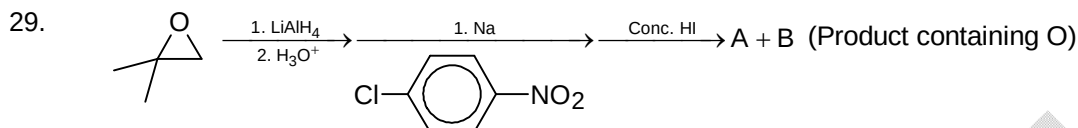
28. Match each species in List – I with List – II and choose correct option

List – I		List – II	
(I)	A carbohydrate which yields only glucose on hydrolysis	(P)	Sucrose
(II)	A carbohydrate which yields glucose and fructose on hydrolysis	(Q)	Lactose
(III)	A carbohydrate which yields glucose and galactose on hydrolysis	(R)	Amylose
(IV)	A carbohydrate which can reduce Fehling's solution	(S)	Mannose
		(T)	Cellulose

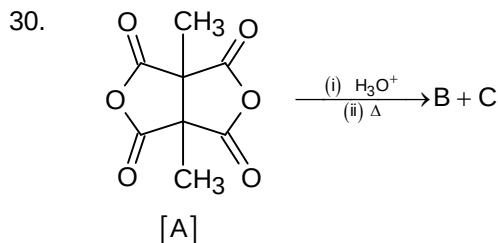
- (A) I→R, T; II→Q; III→P; IV→Q, S
 (B) I→T; II→Q; III→Q; IV→Q
 (C) I→R; II→P; III→Q; IV→P
 (D) I→R, T; II→P; III→Q; IV→Q, S

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.



In the above reaction, the ratio of number of atoms of O to C in the molecular formula of B is

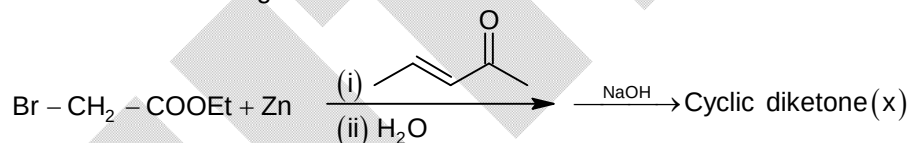


0.1 mol of (A) was reacted according to above reaction. 'C' is the gas produced which turns lime water milky. The amount of 'C' formed in gram is x. The value of $\frac{x}{2}$ is

31. 'A' is an optically active hydrocarbon with minimum molecular weight ('A' does not have any isotopic atoms like D and C^{14}).
If (a) Number of carbon atoms in A = x.
(b) Degree of unsaturation = y.

Calculate $\frac{x+y}{2}$

32. Consider the following reaction:



If (x) is finally treated with excess of $NaBH_4$ followed by acid work-up, the number of diols formed (including stereoisomers) is (y). Then, calculate the value of 10y.

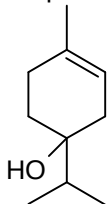
33. How many chloroethenes (including stereoisomers) are possible for molecular formula $C_2H_{4-n}Br_n$ (where 'n' can be an integer from 1 to 4) which can possess geometrical isomerism.

34. The number of co-polymer(s) in the below list is/are X:
(Nylon-6,6, Buna-N, Dacron, Buna-S, Teflon, PHBV)
The number of carbohydrate (s) which show mutarotation in the below list is/are Y:

(α - D - Glucopyranose, β - D - Glucopyranose, Lactose,
 α - D - Fructofuranose, β - D - Fructofuranose, Sucrose,
Maltose)

The value of $\frac{Y}{X}$?

35. Terpinen-4-ol is an active ingredient in tea tree oil has following structure



Which of the following statement(s) is/are correct about above compound?

1. It can rotate the plane polarized light.
 2. It reacts with Baeyer's reagent to form triol.
 3. On reaction with excess of NaBr and dilute H_2SO_4 , it can form a dibromo compound.
- (Note: Write your answer in three digits)
- i.e. If only statement-1 is correct, then 0.01 is correct answer.
 If only statement-2 is correct, then 0.02 is correct answer.
 If only statement-3 is correct, then 0.03 is correct answer.
 If statement-1 and 2 both are correct, then 0.12 is correct answer.
 If statement-1 and 3 both are correct, then 0.13 is correct answer.
 If statement-1, 2 and 3 are correct, then 1.23 is correct answer.

36. In Duma's method of estimation of nitrogen, 0.1840 g of an organic compound gave 30 ml of nitrogen collected over water at 287 K and 758 mm of Hg pressure. The percentage composition of nitrogen in the compound is.....[Given : aqueous tension at 287 K = 14 mm of Hg] [$R = 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}$]

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Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. To the parabola $x^2 = 4ay$, tangent is drawn at any point (x_1, y_1) other than the vertex. If tangents are drawn to the circle $x^2 + y^2 = 4a^2$ from any point on this tangent such that all the corresponding chords of contact pass through a fixed point (x_2, y_2) , then
 (A) $y_1, 2a, y_2$ are in Geometric Progression
 (B) $-4, \frac{x_1}{x_2}, \frac{y_1}{y_2}$ are in Geometric Progression
 (C) $\frac{x_1}{2}, 2a, x_2$ are in Geometric Progression
 (D) $x_1x_2 + y_1y_2 = 4a^2$
38. Two ellipse with same centre and same length of major axis are such that the foci of one are on the other. If their eccentricities are e_1 and e_2 , then
 (A) the angle θ between their axes is given by $\theta = \sin^{-1} \left(\frac{\sqrt{(e_1^2 - 1)(e_2^2 - 1)}}{e_1 e_2} \right)$
 (B) quadrilateral formed by joining the foci is a parallelogram
 (C) for $\theta = \frac{\pi}{4}$, $e_1^2 = \frac{2e_2^2 + 1}{e_2^2 - 2}$
 (D) none of these
39. Normal to the circle $x^2 + y^2 = 9$ divides the circle having centre at $\left(3\sqrt{2}, \frac{3}{2}\right)$ and radius 3 in the areas of ratio $(2\pi - 3\sqrt{3}) : (10\pi + 3\sqrt{3})$. Then the normal can be
 (A) $\sqrt{2}y = x$
 (B) $y = \sqrt{2}x$
 (C) $5y + \sqrt{2}x = 0$
 (D) $\sqrt{2}y + x = 0$
40. If $\alpha = p \cos^3 \theta \cdot \sin^2 \theta$, $\beta = p \sin^3 \theta \cdot \cos^2 \theta$ and $\frac{(\alpha^2 + \beta^2)^a}{(\alpha\beta)^b}$ ($a, b \in \mathbb{N}$) is independent of θ , then
 (a, b) is equal to
 (A) (4, 5)
 (B) (8, 10)
 (C) (5, 4)
 (D) (10, 8)
41. If one of the lines represented by $4x^2 + pxy + 6y^2 = 0$ is perpendicular to one of the lines given by $4x^2 + qxy - 6y^2 = 0$ and the other lines represented by them are coincident, then the value of $p - 2q$ is
 (A) -6
 (B) -8
 (C) 8
 (D) 6
42. The expression $\frac{x^3}{2} \operatorname{cosec}^2 \left(\tan^{-1} \left(\frac{x}{y} \right) \right) + \frac{y^3}{2} \sec^2 \left(\tan^{-1} \left(\frac{y}{x} \right) \right)$ is divisible by
 (A) $x + y$
 (B) $x^2 + y^2$
 (C) $x^3 + y^3$
 (D) $x^2 - y^2$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Match the following List-I with List-II

List-I		List-II	
(I)	The number of solutions of $x^4 - 2x^2 \sin^2 \frac{\pi x}{2} + 1 = 0$ is	(P)	4
(II)	The number of solutions of $16(x^2 + 1) + \pi^2 = \tan x + 8\pi x$, $x \in \left(-\frac{\pi}{2}, \frac{3\pi}{2}\right)$, $x \neq \frac{\pi}{2}$ is	(Q)	2
(III)	The numbers of pairs (x, y) satisfying the equation $\sin x + \sin y = \sin(x + y)$ and $ x + y = 1$ is	(R)	1
(IV)	The number of integral values of x in [0, 6] satisfying $\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x > 0$ is	(S)	3
		(T)	6

The correct option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (Q)
 (B) (I) $\rightarrow$ (S); (II) $\rightarrow$ (P); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (Q)
 (C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
 (D) (I) $\rightarrow$ (R); (II) $\rightarrow$ (T); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (S)

44. Let $t_0 = (\sin^{-1} x)^{\sin^{-1} x}$, $t_2 = (\sin^{-1} x)^{\cos^{-1} x}$, $t_3 = (\cos^{-1} x)^{\sin^{-1} x}$, $t_4 = (\cos^{-1} x)^{\cos^{-1} x}$ match the following List-I with List-II

List-I		List-II	
(I)	$x \in (0, \cos 1)$	(P)	$t_1 > t_2 > t_4 > t_3$
(II)	$x \in \left(\cos 1, \frac{1}{\sqrt{2}}\right)$	(Q)	$t_4 > t_3 > t_1 > t_2$
(III)	$x \in \left(\frac{1}{\sqrt{2}}, \sin 1\right)$	(R)	$t_4 > t_1 > t_2 > t_3$
(IV)	$x \in (\sin 1, 1)$	(S)	$t_2 > t_1 > t_4 > t_3$
		(T)	$t_3 > t_4 > t_1 > t_2$

The correct option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (S); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)
 (B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (T); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (P)
 (C) (I) $\rightarrow$ (R); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)
 (D) (I) $\rightarrow$ (P); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)

45. Match the following List-I with List-II

List-I		List-II	
(I)	The shortest distance between the parabola $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$ is	(P)	$\frac{2\sqrt{2}}{3}$
(II)	The triangle formed by the tangent to the parabola $y = x^2$ at the point whose abscissa is x_0 ($x_0 \in [1, 2]$), the y-axis and the straight line $y = x_0^2$ has the greatest area if x_0 is equal to	(Q)	$\frac{1}{4}$
(III)	An ellipse is inscribed in a circle and a point within the circle is chosen at random. If the probability that this point lies outside the ellipse is $\frac{2}{3}$, then the eccentricity of the ellipse is	(R)	$\frac{1}{2\sqrt{2}}$
(IV)	If $f(x) = x^3 + ax^2 + bx + c$, (a, b, c are rational numbers) and 2 roots of $f(x) = 0$ are eccentricities of a parabola and a rectangular hyperbola, then $\frac{1}{abc}$ equals	(S)	$\frac{1}{\sqrt{2}}$
		(T)	2

The correct option is:

(A) (I) $\rightarrow$ (S); (II) $\rightarrow$ (T); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)

(B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (T)

(D) (I) $\rightarrow$ (R); (II) $\rightarrow$ (T); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (Q)

46. Match the following List-I with List-II

List-I		List-II	
(I)	If the angle between lines belonging to the family of lines $(3x + y - 7) + \lambda(x + y - 3) = 0$ and $(x + 3y - 7) + \mu(x + 4y - 3) = 0$ which are farthest from origin (λ and μ are real parameters), is $\tan^{-1}\left(\frac{a}{b}\right)$ where a, b are coprime natural numbers, then $b - a$ is	(P)	11
(II)	If the roots of the equation $px^2 + qx + 10 = 0$ are not real and distinct where $p, q \in \mathbb{R}$ and α and β are the values of p and q respectively for which $5p + q$ is minimum, then the family of lines $\alpha(4x + 5\alpha y + 3) + \beta(4x + \beta y - 4) = 0$ are concurrent at (a, b) . $(a - b)$ is	(Q)	7
(III)	A circle is inscribed into a rhombus PQRS with one angle 120° . The distance from the centre of the circle to the nearest vertex is equal to 1. If A is any point on the circle, then $ AP ^2 + AQ ^2 + AR ^2 + AS ^2$ is	(R)	3
(IV)	If the area bounded by the circles $x^2 + y^2 = r^2$, $r = 1, 2$ and the rays given by $2x^2 - 3xy - 2y^2 = 0$, $y > 0$ is $\frac{\lambda\pi}{4}$ sq. units. Then the value of λ is	(S)	1
		(T)	5

The correct option is:

(A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (R); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (Q)

(B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

(C) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)

(D) (I) $\rightarrow$ (S); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (Q)

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. If $\sin^{-1}\alpha + \sin^{-1}\beta + \sin^{-1}\gamma = \pi$, then $\frac{\alpha\sqrt{1-\alpha^2} + \beta\sqrt{1-\beta^2} + \gamma\sqrt{1-\gamma^2}}{\alpha\beta\gamma}$ is equal to
48. If $b^2 + c^2 = 2023a^2$, where a, b, c are the side lengths of $\triangle ABC$ for usual notation, then $\frac{\cot A}{\cot B + \cot C}$ is equal to
49. For the hyperbola $\frac{x^2}{25} - \frac{y^2}{45} = 1$, the chord $x \cos \theta + y \sin \theta = \lambda$ subtends a right angle at the centre. Then the length of tangent to the circle, concentric with this hyperbola, to which the given chord is a tangent, from the point $P(5, 5\sqrt{2})$ is
50. Two circles S and S' pass through both the points $P(1, -2)$ and $Q(2, 3)$ such that the line $x - 3y + 9 = 0$ is a common tangent touching them at A and B respectively. If the possible coordinates of a point R are (α, β) such that PARB is a parallelogram and $\alpha = \frac{a}{b}$, $\beta = \frac{c}{d}$ (both in lowest forms) with a, b, c, d $\in \mathbb{I}$, then |abcd| is
51. For real numbers x and y satisfying $x^2 + 4y^2 - 6x + 4y + 9 = 0$. Then the absolute difference between the maximum and minimum values of $(8x - 12y)$ is
52. There are two families of lines in a plane $y = x + i$, $y = -x + i$, where $i = \{0, 1, 2, 3, 4, 5\}$. The number of squares having diagonal lengths of odd units, formed by these lines is
53. Points P, Q, R lie on the parabola $y^2 = 4ax$. Tangents to the parabola at these points taken in pairs, intersect at A, B, C. The ratio of the areas of $\triangle ABC$ and $\triangle PQR$ is
54. The sum of all possible values of α (in degrees) such that $\cos^3 3\alpha + \cos^3 5\alpha = 8 \cos^3 4\alpha \cos^3 \alpha$ where α is measured in degrees and $\alpha \in (100^\circ, 200^\circ)$, is

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – II
PAPER –2
TEST DATE: 11-12-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

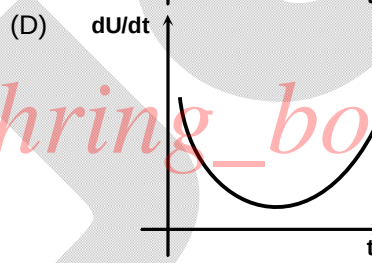
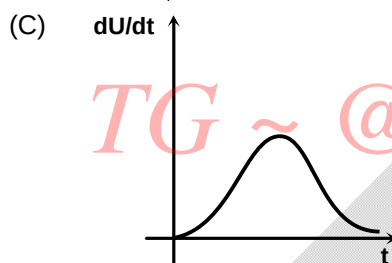
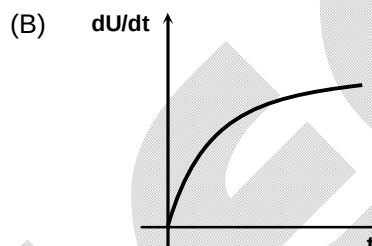
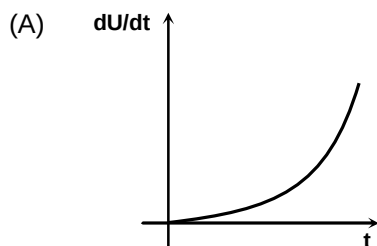
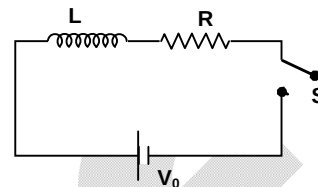
Physics

PART – I

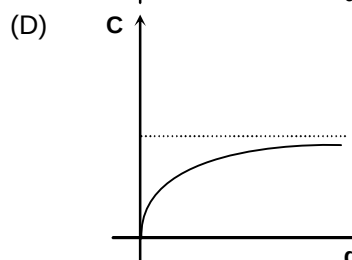
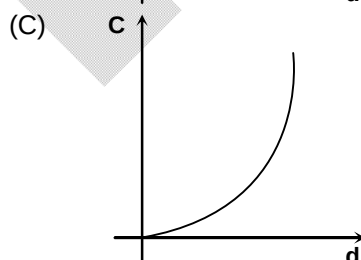
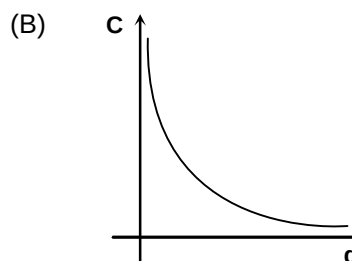
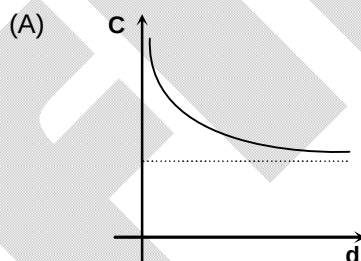
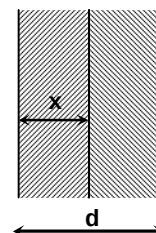
Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

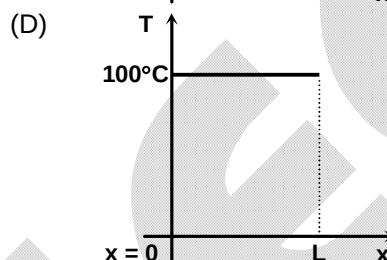
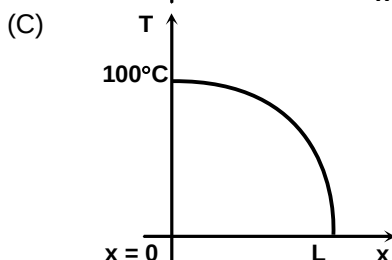
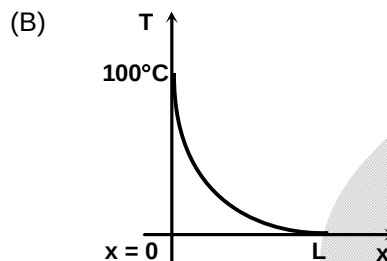
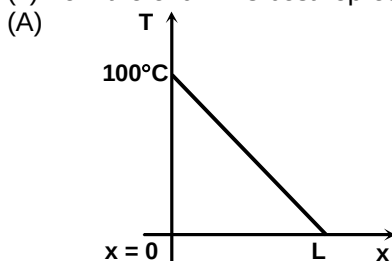
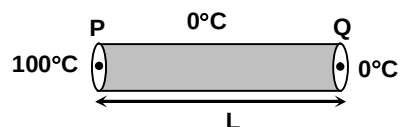
1. An ideal inductor and a resistor are connected in series with a battery of potential difference V_0 through a switch 'S'. The switch is closed at $t = 0$. The potential energy stored in inductor at time 't' is given by U. The rate of change of potential energy in inductor with time is best represented by



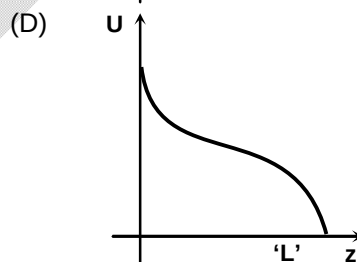
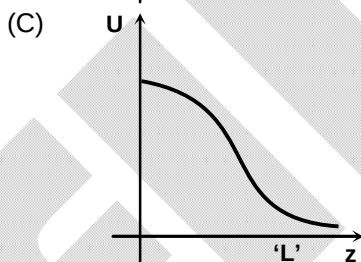
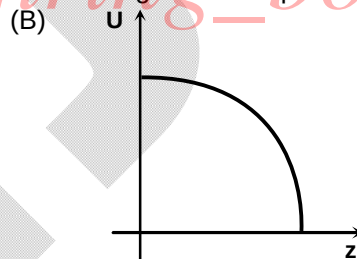
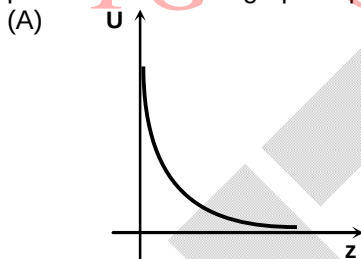
2. A dielectric is completely filled between the space of a parallel plate capacitor. The separation between plates of capacitor is 'd' and dielectric constant is given by $k = k_0 e^{\lambda x}$, where ' k_0 ' and ' λ ' are positive constants and x is distance from one plate of capacitor. If distance between plate (d) increases slowly then graph between capacitance of capacitor and distance between plate is given by



3. A conducting rod of length 'L' cross-sectional area 'A' and thermal conductivity 'k' is kept in the surrounding at 0°C . The end P is maintained at 100°C and Q at 0°C . The graph between temperature (T) of the rod and distance (x) from the end 'P' is best represented by



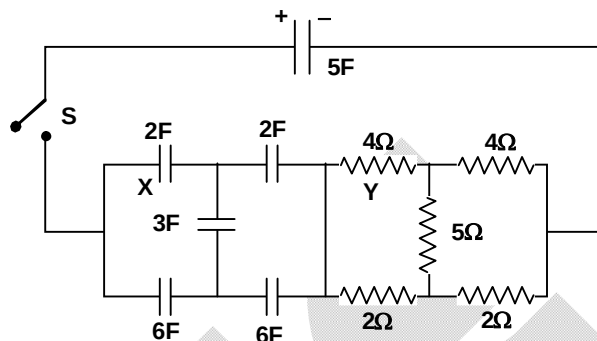
4. Four point charges $2q$, $2q$, $-q$ and $-q$ are kept at $(a, -a, 0)$, $(-a, +a, 0)$, $(a, a, 0)$ and $(-a, -a, 0)$ and fixed at their position. Another point charge q_0 is kept at origin and slightly displaced along positive z-axis. The graph of potential energy of the charge is best represented by



Section – A (Maximum Marks: 24)

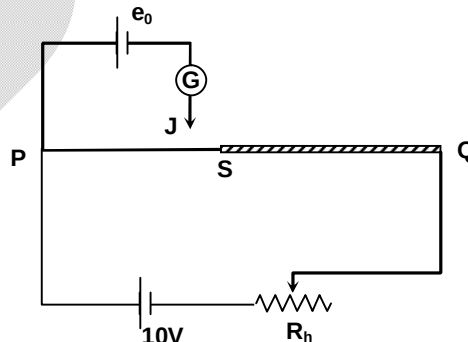
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A capacitor of capacity 5F is charged by a battery of 2 volt and then connected with 5 uncharged capacitors and 5 resistors with the help of a switch 'S' as shown in the figure. At $t = 0$, switch is closed. Choose the correct option(s).



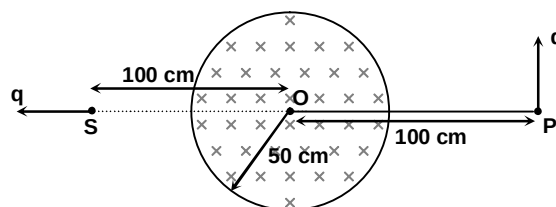
- (A) Charge on capacitor X at time t is given by $\frac{10}{9} \left(1 - e^{-\frac{27}{160}t} \right)$
- (B) Current in resistor Y at time t is given by $\frac{1}{4} \left(e^{-\frac{27}{160}t} \right)$
- (C) Total energy dissipated in the circuit is $\frac{50}{9}$ J
- (D) Total energy dissipated in the circuit is $\frac{40}{9}$ J

6. A potentiometer wire PQ is made by using two wires of different cross sectional area A and $2A$. The material of wire PS and SQ is same and their lengths are 5m each. A source battery of 10V and rheostat R_h are connected in the circuit as shown in the figure. Choose the correct option(s) regarding the balance point.



- (A) If a cell of emf $e_0 = 5$ V is used then balance point from the end P is at 7.5 m
- (B) If a cell of emf $e_0 = 5$ V is used then balance point from the end P is at 3.75 m
- (C) If a cell of emf $e_0 = 8$ V is used then balance point from the end P is at 7 m
- (D) If a cell of emf $e_0 = 8$ V is used then balance point from the end P is at 8.5 m

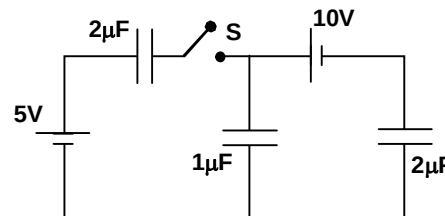
7. A cylindrical vertical magnetic field exist in a gravity free space. Magnetic field is increasing at a rate of 700 T/sec. The radius of cylindrical magnetic field is 50 cm. A point charge $q = 50 \mu\text{C}$ is moved very slowly by an external agent from two point 'P' and 'S' then



- (A) Work done by external agent is zero if charge particle is moved from point 'S' upto infinity along the line OS.
- (B) Work done by external agent is zero if charge particle is moved from point 'P' upto infinity perpendicular to the line OP.

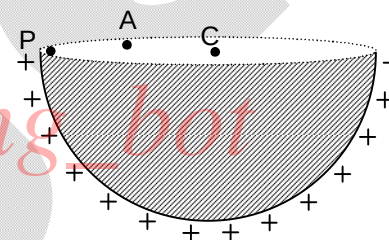
- (C) Work done by external agent is $\frac{55}{8} \text{ mJ}$ if charge particle is moved from point 'P' upto infinity perpendicular to the line OP.
- (D) Work done by external agent to move a charge particle from any point to infinity is path independent.

8. Three capacitors of capacity $2 \mu\text{F}$, $2 \mu\text{F}$, $1 \mu\text{F}$ and two ideal cells of potential difference 10V and 5V are connected in a circuit through a switch 'S'. Initially switch is opened for a long time and then it is closed. Choose the correct option(s) after switch is closed.



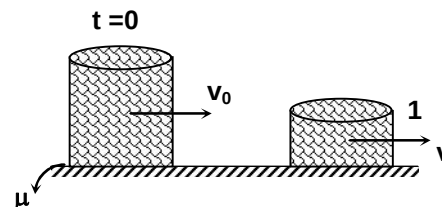
- (A) Charge passing through 10V cell is $\frac{4}{3} \mu\text{C}$
- (B) Work done by 5V battery is $10 \mu\text{J}$
- (C) Charge on $1 \mu\text{F}$ capacitor is $6 \mu\text{C}$
- (D) Total heat dissipated in the circuit after switching is $\frac{5}{3} \mu\text{J}$

9. A non conducting hemispherical shell of radius 'R' has uniform surface charge density σ . Point 'C' is center of hemispherical shell. Point 'A' and point 'P' lie on the flat surface of the shell. The distance between C and A is $R/2$ and between C and P is R . Choose the correct option(s).



- (A) The potential at point P is $\frac{\sigma R}{2\epsilon_0}$
- (B) The potential at point A is $\frac{\sigma R}{4\epsilon_0}$
- (C) Work done by electrostatic force if a charge ' q_0 ' is moved from P to A is $\frac{-q_0 \sigma R}{4\epsilon_0}$
- (D) Work done by electrostatic force if a charge ' q_0 ' is moved from P to A is $\frac{q_0 \sigma R}{4\epsilon_0}$

10. An ice in the shape of cylinder is kept on a rough horizontal surface as shown in the figure. An initial velocity v_0 is imparted to cylindrical ice. The initial mass of ice is m_0 and coefficient of friction between ice and horizontal surface is ' μ '. The initial temperature of ice 0°C and its latent heat of fusion is L . Choose the correct option(s).

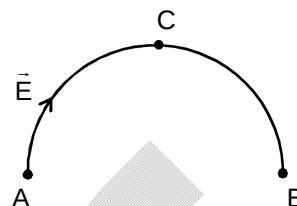


- (A) The velocity of cylindrical ice at time 't' is given by $(v_0 + \mu g t)$
- (B) The velocity of cylindrical ice at time 't' is given by $(v_0 - \mu g t)$
- (C) The mass of cylindrical ice at time 't' is given by $m_0 e^{-\frac{\mu g}{L} \left(v_0 t - \frac{\mu g L^2}{2} \right)}$
- (D) The mass of cylindrical ice at time 't' is given by $m_0 e^{-\frac{\mu g}{L} \left(\frac{v_0 t}{2} - \frac{\mu g L^2}{2} \right)}$

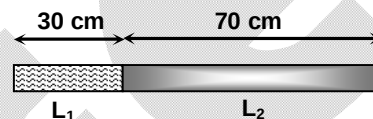
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

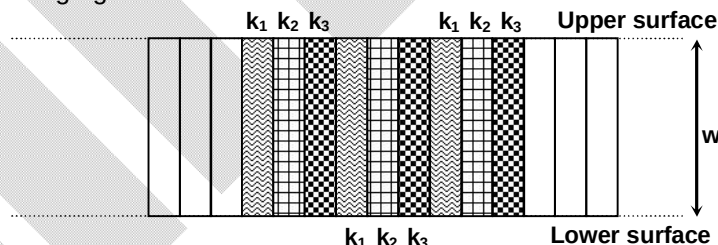
11. Semicircular electric field line is present in a space. The magnitude of electric field is constant at each point of electric field lines. A point charge is constrained to move along the field lines. The particle starts from rest and initially it is at point A. The magnitude of acceleration of the particle at C is a_1 and it is a_2 when it reaches at B. The ratio of a_1 and a_2 is given by $\sqrt{\frac{1+k_1}{1+k_2}}$. Find the value of $\frac{k_2}{k_1}$.



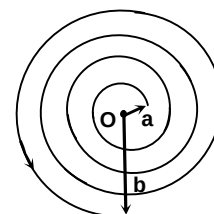
12. Two rods L_1 and L_2 of different materials are used to make a single composite rod as shown in the figure. The cross sectional area of both rods is same. At 25°C , the total length of combined rod is 1m, in which the length of L_1 is 30 cm. When temperature of combined rod is increased, at 125°C , the total length of combined rod increased by 1.91 mm. The coefficient of linear expansion of L_1 is $1.7 \times 10^{-5}/^\circ\text{C}$ and that of the L_2 is $k \times 10^{-5}/^\circ\text{C}$. Find the value of k .



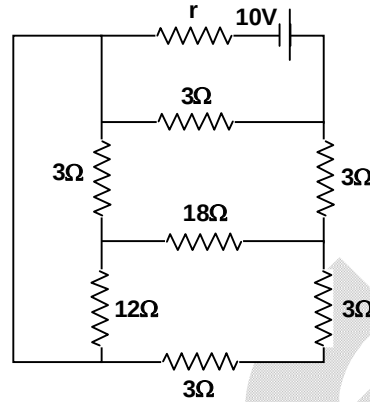
13. A particle of mass $m = \sqrt{227}$ gm and charge $q = 7\mu\text{C}$ is projected in a space where magnetic field is given by $\vec{B} = y^2\hat{i} + z^2\hat{j} + x^2\hat{k}$. The initial velocity of projection of particle is $\vec{v} = \hat{i} - 2\hat{j} + 3\hat{k}$ from (1, 2, 1). Find radius of curvature of particle at $t = 0$ in kilometer.
14. Three different type of glasses are used to make a large glass slab. The thermal conductivity of glass slabs are k_1 , k_2 and k_3 . The width of each glass slab is 'w' and cross-sectional area is A. The glass slabs are kept in a pattern as shown in the figure. The upper surface of glass slab is maintained at temperature T_1 and lower surface at T_2 . It is given that $k_1 = 1.10 \text{ W/mK}$, $k_2 = 2.82 \text{ W/mK}$, $k_3 = 2.08 \text{ W/mK}$, $w = 10 \text{ cm}$, $A = 1.3 \text{ m}^2$, $T_1 = 100^\circ\text{C}$, $T_2 = 0^\circ\text{C}$. Find equivalent thermal conductivity of large glass slab in W/mK.



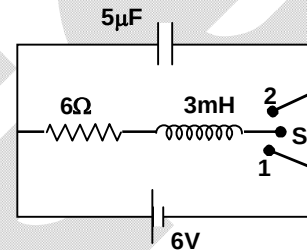
15. A conducting spiral of inner radius $a = 20 \text{ cm}$ and outer radius $b = 40 \text{ cm}$ is kept on a frictionless horizontal surface. The total number of turns is equal to 1000 which are wound tightly and insulated to each other. There exist a vertical magnetic field which is changing at a rate of $\frac{50}{\pi} \text{ mT/sec}$. If the resistance of conducting spiral is $\frac{14}{9} \Omega$, then find induced current in ampere.



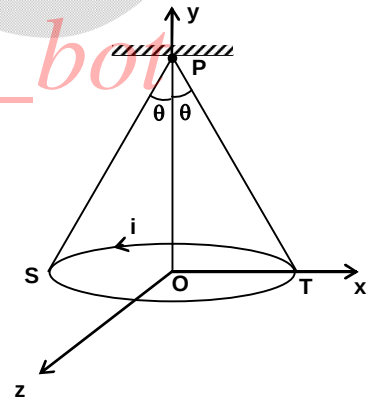
16. A battery of potential difference 10 V and internal resistance 'r' is connected to seven resistors as shown in the figure. If battery delivers maximum power to the circuit with internal resistance $r = \frac{4}{13} \lambda$, then find the value of λ .



17. A capacitor ($5\mu\text{F}$), a resistor (6Ω), an inductor (3mH), a cell (6V) and a switch are connected in a circuit as shown in the figure. Initially switch is kept at position-1 for a long time. The potential difference (in volt) across the ends of inductor immediately after the switch is shifted from position-1 to position-2 is



18. A ring of uniform mass density ($m = 100 \text{ gm}$) and radius ($R = 50 \text{ cm}$) is suspended by two strings PS and PT from the point P on y-axis in vertical plane. The ring is kept parallel to horizontal surface on x-z plane. Ring carries a current $\left(i = \frac{7}{22} \text{ amp}\right)$ as shown in the figure. There exist a magnetic field ($B = 1 \text{ Tesla}$) along negative x-axis. The tension in the string PS is T_1 and in the string PT is T_2 . Find the ratio $\frac{T_1}{T_2}$. (It is given that $\theta = 60^\circ$)



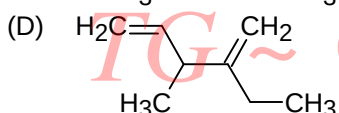
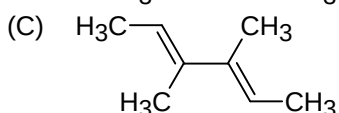
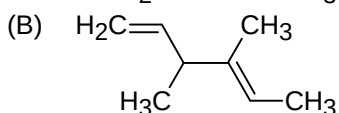
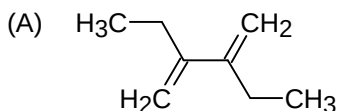
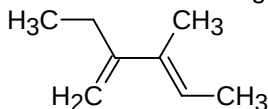
Chemistry

PART – II

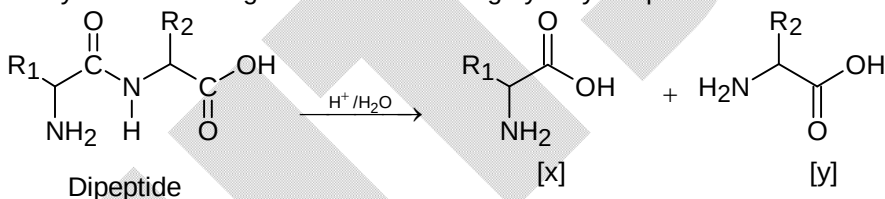
Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

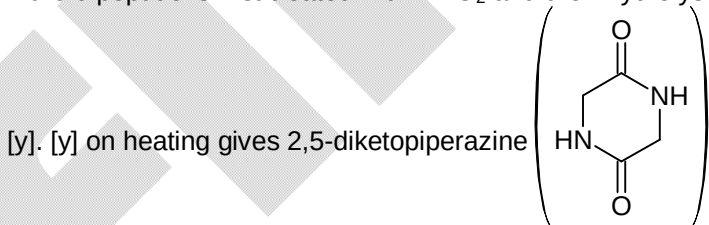
19. Which of the following is NOT the chain isomer of



20. Analyse the following data of the following hydrolyzed products and choose the correct option:

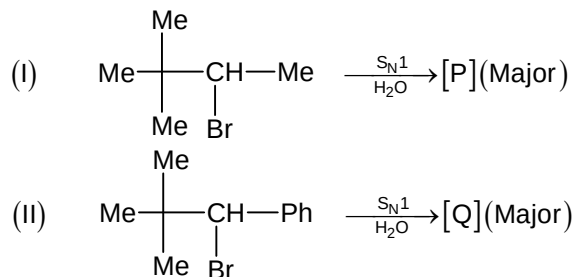


If the dipeptide is first treated with HNO_2 and then hydrolysis is carried out, we got lactic acid and

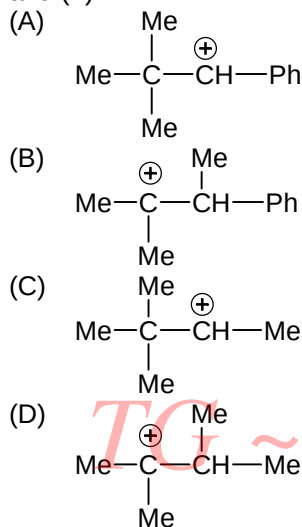


- (A) $\text{R}_1 = \text{H}$ and $\text{R}_2 = \text{CH}_3$
 (B) $\text{R}_1 = \text{CH}_3$ and $\text{R}_2 = \text{CH}_3$
 (C) $\text{R}_1 = \text{CH}_3$ and $\text{R}_2 = \text{H}$
 (D) $\text{R}_1 = \text{H}$ and $\text{R}_2 = \text{H}$

21.



Identify the INCORRECT intermediate carbocation, which does NOT form during the reaction (I) and (II)?



22.

Identify which test are positive (+) or negative (–) for β -D-glucopyranose?

[Note: Here positive means respond to test and negative means not respond to test]

1. 2, 4-DNP
2. $\text{NH}_2\text{OH} / \text{H}^+$
3. Tollen's test
4. NaHSO_3

(A)

1	2	3	4
+	+	+	–

(B)

1	2	3	4
+	+	–	–

(C)

1	2	3	4
–	+	+	–

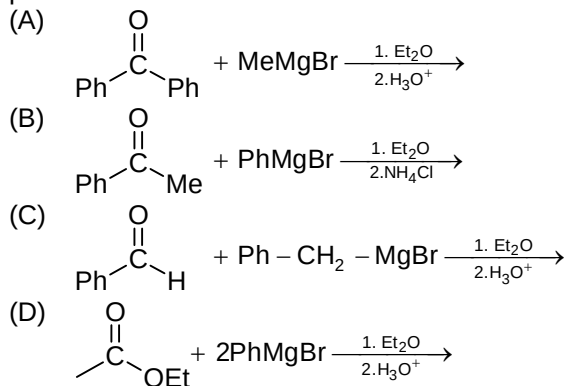
(D)

1	2	3	4
–	–	–	–

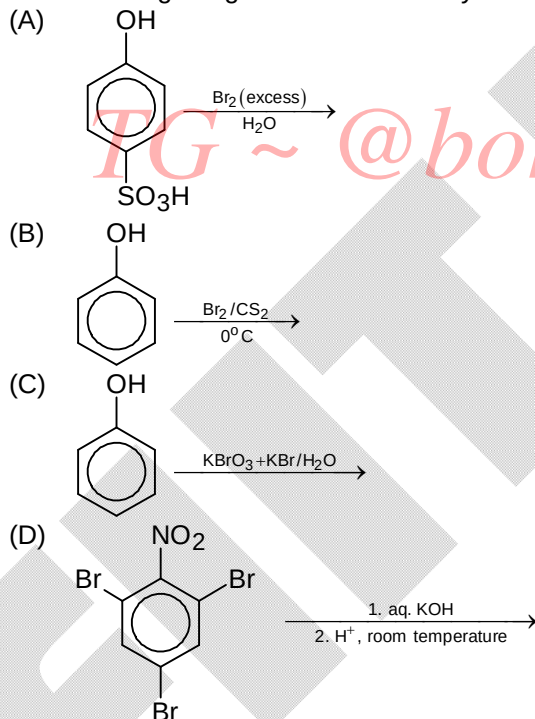
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

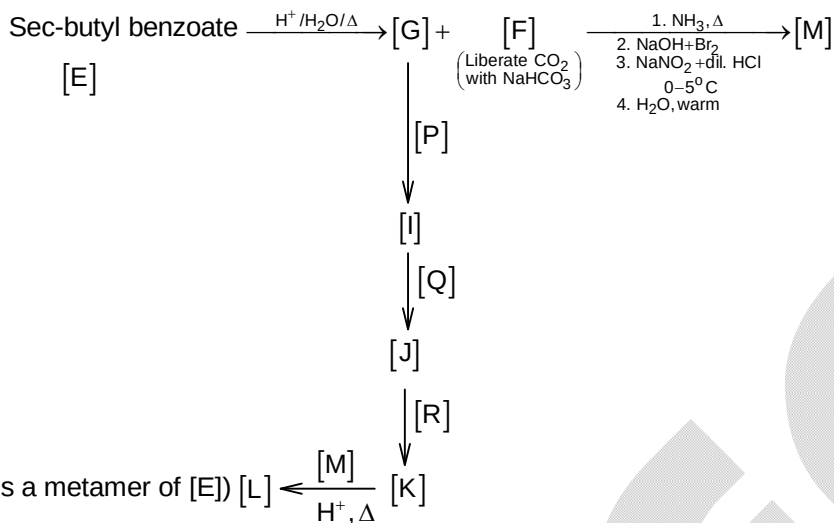
23. Which of the following reaction(s) will provide a synthesis of 1,1-diphenylethanol as a major product?



24. Which among the given reactions can yield the 2, 4,6-tribromo phenol as a major product?

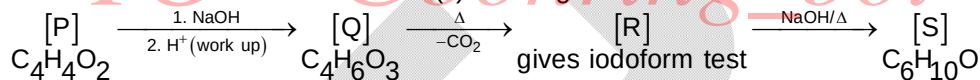


25. Choose the correct options regarding the given scheme?



- (A) $[P] = \text{SOCl}_2$ $[Q] = \text{Mg} / \text{dry ether}$ $[R] = 1. \text{HCHO}$ 2. H^+
(B) $[P] = \text{KBr} + \text{H}_2\text{SO}_4$ $[Q] = \text{AgCN} / \text{DMF}$ $[R] = \text{H}_3\text{O}^+ / \Delta$
(C) $[P] = \text{PBr}_3$ $[Q] = \text{Mg} / \text{dry ether}$ $[R] = 1. \text{CO}_2$ 2. H^+
(D) $[P] = \text{HBr}$ $[Q] = \text{NaCN} / \text{DMF}$ $[R] = \text{H}_3\text{O}^+ / \Delta$

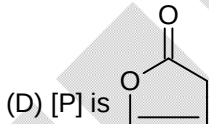
26. Choose the INCORRECT statement(s) for the given reaction scheme



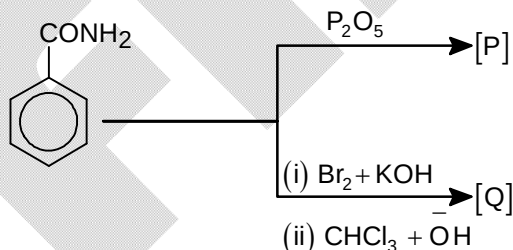
Unsaturated

Lactone

- (A) [S] is an α,β -unsaturated aldehyde.
(B) Four stereoisomers are possible for [S].
(C) Two structural isomers of [P] can satisfy the above reaction scheme.



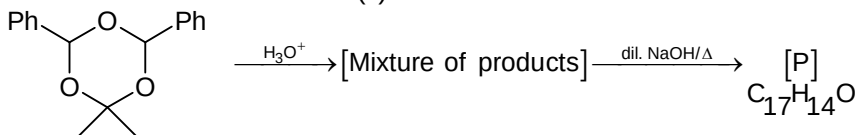
- 27.



Which statement(s) is/are CORRECT?

- (A) Reduced product of P and Q will be metamers to each other. (Consider LiAlH_4 as a reducing agent)
- (B) By dry distillation of hydrolysed product of P with Ca(OH)_2 gives benzophenone.
- (C) One of the hydrolysis product of Q reacts with $\text{NaNO}_2 + \text{HCl}$ followed by reaction with phenol in mild basic medium gives orange red dye.
- (D) Electrophile involved in the formation of Q is dichlorocarbene.

28. Choose the correct statement(s)

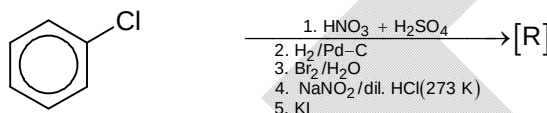
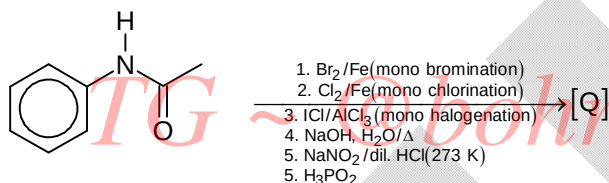
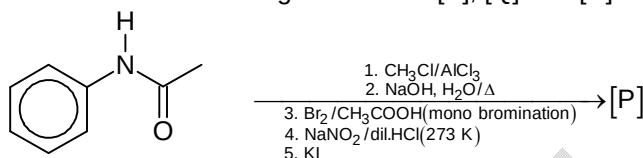


- (A) In mixture, one of the product shows positive iodoform test.
 (B) In mixture, one of the product shows positive Tollen's test and negative Fehling's test.
 (C) [P] on reaction with hydroxyl amine gives oxime that can have 4 stereoisomers.
 (D) Number of double bonds in [P] is 10.

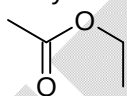
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

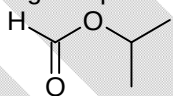
29. Consider the major organic product in each case and calculate the Sum of number of halogen atoms in [P], [Q] and [R]



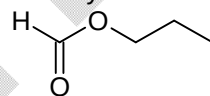
30. An organic compound reacts with excess of MeMgBr followed by hydrolysis to give two organic products of which one reacts with NaOI to give yellow precipitate and OTHER DOES NOT. How many of the following compounds can satisfy these conditions?



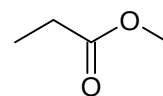
(I)



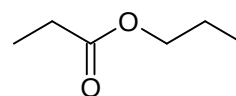
(II)



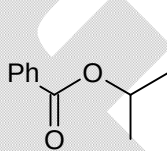
(III)



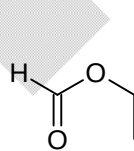
(IV)



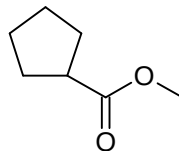
(V)



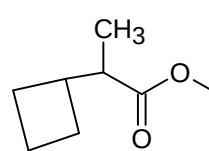
(VI)



(VII)

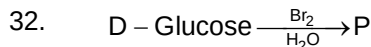


(VIII)



(IX)

31. Consider all possible isomeric ketones including stereoisomers of MW = 100. All these isomers are independently reacted with CH₃MgBr (Note : Stereoisomers are also reacted separately and consider no racemisation of alpha carbon of the carbonyl group during the reaction). The total number of ketones that gives a racemic product is/are



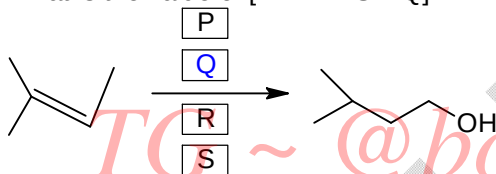
Number of chiral centre in compound P is

33. Number of correct statement(s) is/are

1. Chlorofluorocarbons are responsible for ozone layer depletion.
2. Green house effect is responsible for global warming.
3. Ozone layer does not permit infrared radiation from sun to reach the earth.
4. Acid rain is mostly because of oxides of nitrogen and sulphur.
5. Classical smog occurs in cool humid climate.
6. Classical smog is a mixture of smoke, fog and sulphur dioxide.
7. Classical smog is also called reducing smog.
8. Classical smog is also called oxidizing smog.
9. Ozone absorbs infrared radiations.
10. Ozone absorbs the intense ultraviolet radiations of the sun.

34. Choose the appropriate reagent sequence from the given reagent pool to complete the synthetic conversion. Make sure to use strictly four reagents, not more not less.

Note: You can simply mention the numerical values corresponding to the reagent in the boxes P, Q, R, S respectively and also consider major organic product in each step. What is the value of $[P + R + S - Q]$.

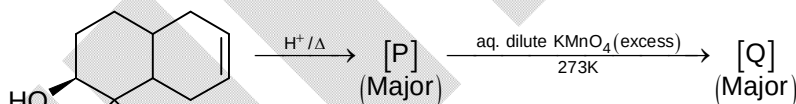


- | | | | |
|--|--|---|-----------------------------------|
| 1. HBr | 2. $\text{H}^+ / \text{H}_2\text{O}$ | 3. $\text{Hg}(\text{OAc})_2 / \text{H}_2\text{O}$ | 4. $\text{BH}_3 \cdot \text{THF}$ |
| 5. $\text{EtONa} / \text{EtOH} / \Delta$ | 6. $\text{HBr} / \text{ROOR} / \Delta$ | 7. $\text{conc. H}_2\text{SO}_4 / \Delta$ | |
| 8. NaBH_4 | 9. $\text{H}_2\text{O}_2, \text{aq. NaOH}$ | 10. $t\text{-BuOK} / \Delta$ | |

35. [A] is a monomer of natural rubber, on reaction with HBr (1 eq)/ CCl_4 at 298 K, it forms major product [B].

The number of allylic hydrogen of [A] is x and that of [B] is y, the value of $(y - x)$?

36.



Calculate the number of secondary hydroxyl groups in [Q]

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. In a triangle ABC let a, b, c denote the length of the sides BC, AC and AB respectively, if $8(b^2 + c^2) = 65a^2$, then the value of $\frac{\cot A}{\cot B + \cot C}$ is equal to
- (A) $\frac{57}{16}$ (B) $\frac{53}{8}$
(C) $\frac{57}{8}$ (D) $\frac{59}{16}$
38. Three circles C_1, C_2, C_3 with radii r_1, r_2, r_3 ($r_1 < r_2 < r_3$) respectively are given as $r_1 = 4$ and $r_3 = 9$. They are placed such that C_2 lies to the right of C_1 and touches it externally, C_3 lies to the right of C_2 and touches it externally. There exists two straight lines each of which is a direct common tangent simultaneously to all three circles, then r_2 is equal to
- (A) $\frac{13}{2}$ (B) 6
(C) $\frac{72}{13}$ (D) 7
39. Consider the ellipse $E = 9x^2 + 25y^2 = 225$ with foci S and S_1 where S lies on positive x-axis. Let the locus of the image of S with respect to any tangent to the ellipse be the curve C. If the line $ky - 2x + 22 = 0$ touches the curve C, then k is equal to
- (A) ± 1 (B) $\pm \sqrt{3}$
(C) $\pm \sqrt{5}$ (D) $\pm 4\sqrt{2}$
40. Let A and B be two distinct points on the parabola $y^2 = 8x$ such that a circle with AB as diameter passes through the vertex of the parabola. Tangents at A and B to the parabola intersect at point C. Another pair of tangents are drawn from point C to the parabola $y^2 = kx$. If angle between these tangents is 90° , then k is equal to
- (A) 1 (B) 2
(C) 32 (D) 16

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. Let $f(n) = \sum_{k=1}^n \cos\left(\frac{2k\pi}{7}\right)$ and $g(m) = \sum_{k=1}^m \cos\left(\frac{2^k\pi}{7}\right)$, then which of the following is/are CORRECT?
- (A) $f(3) = g(3)$ (B) $2f(3) = f(6)$
(C) $2g(3) = g(6)$ (D) $f(8) = g(8)$
42. Consider a circle whose radius is 2 and centre is origin. Tangents are drawn from a point $P(\alpha, 2)$ which touches the circle at $A(a_1, a_2)$ and $B(b_1, b_2)$. Another pair of tangents are drawn from the point $Q(\beta, -2)$ which touches the circle at $A(a_1, a_2)$ and $C(c_1, c_2)$. Then which of the following is/are CORRECT? ($\alpha, \beta > 0$)
- (A) $b_1 + b_2 + c_1 + c_2 = 0$ (B) $\alpha\beta = 4$
(C) If $\alpha = 4$, then $4a_2 + 3a_1 = 0$ (D) If $\alpha = \beta$, then $a_2 = 0$

43. Consider a parabola $P : x^2 = 4ky$ and a circle $C : x^2 + y^2 - 2\alpha x + 6y + 9 = 0$ (where $\alpha > 0$). The line of shortest distance between them cuts the parabola P at the end point of latus rectum in first quadrant, then which of the following is/are TRUE?
 (A) If $k = 1$, then $\alpha = 6$ (B) If $k = 1$, then $\alpha = -6$
 (C) If $k = 2$, then $\alpha = 12$ (D) If $\alpha = 20$, then $k = \frac{17}{3}$
44. In a ΔPQR , let p, q, r denote the length of the sides opposite to angles P, Q and R respectively, let s be the semiperimeter. If $s - p = \frac{s - q}{4} = \frac{s - r}{6}$ and area of the circumcircle of the triangle PQR is $\frac{175}{4\sqrt{66}}$, then which of the following is/are CORRECT?
 (A) Area of triangle PQR is $4\sqrt{66}$
 (B) Radius of incircle of triangle PQR is $2\sqrt{\frac{6}{11}}$
 (C) $\sin \frac{P}{2} \sin \frac{Q}{2} \sin \frac{R}{2} = \frac{12}{175}$
 (D) Radius of the excircle opposite to vertex P is $\frac{\sqrt{66}}{5}$
45. Define the collections $\{E_1, E_2, E_3, \dots\}$ of ellipse and $\{R_1, R_2, R_3, \dots\}$ of rectangles as follows:
 $E_1 : x^2 + 4y^2 = 32$
 $R_1 : \text{rectangle of largest area with sides parallel to the axes, inscribed in } E_1$
 $E_n : \text{ellipse } \frac{x^2}{a_n^2} + \frac{y^2}{b_n^2} = 1 \text{ of largest area inscribed in } R_{n-1}, n > 1$
 $R_n : \text{rectangle of largest area with sides parallel to axes, inscribed in } E_n, n > 1.$
 Then which of the following options is/are CORRECT?
 (A) $\sum_{n=1}^N (\text{area of } R_n) < 64$ for each positive integer N
 (B) The ratio of eccentricities of E_{10} and E_{12} is $1 : 4$
 (C) The length of major axis of E_6 is 2
 (D) The length of diagonal of R_6 is $\frac{\sqrt{5}}{2}$
46. Let the vertex P of ΔPQR move on the curve $xy = 4$. If the perpendicular bisectors of side PQ and PR are given by $y = 0$ and $x - y = 0$, then which of the following is/are CORRECT?
 (A) Locus of centroid of ΔPQR is $9xy = 18y^2 + 4$
 (B) Locus of centroid ΔPQR is $9xy = 18x^2 + 4$
 (C) Locus of orthocentre of ΔPQR is $xy = 2y^2 + 4$
 (D) Circumcentre of ΔPQR is origin

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. $f(\theta) = \sin^3 \theta + \frac{1}{3} \sin^3 (3\theta) + \frac{1}{9} \sin^3 9\theta + \frac{1}{27} \sin^3 27\theta + \dots$ upto 10 terms, then
 $4f(\theta) = k \sin \theta - \frac{\sin 3^m \theta}{3^n}$, then $k + n - m$ is

48. Consider the ellipse $E: \frac{x^2}{25} + \frac{y^2}{9} = 1$. Tangents are drawn at the ends points of a double ordinate passing through $(c, 0)$ (where $5 < c < 5$). The tangents intersect at G . If $\Delta(c)$ = area of ΔEFG , then for $1 \leq c \leq 3$ if minimum value of $\Delta(c)$ is k , then $\frac{5k}{8}$ is equal to
49. In a triangle PQR , $\angle P = 54^\circ$, $\angle R = 18^\circ$. If S is a point on PR such that area of triangle PSQ is $(\sqrt{5} - 1)$ times the area of triangle RSQ , then ratio of angle $\angle PQS : \angle RQS$ is
50. A circle with centre $(6, 12)$ and of variable radius cuts the hyperbola $x^2 - y^2 = 144$ at the points A, B, C and D . If the locus of the centroid of the ΔABC is $(x - \alpha)^2 - (y - \beta)^2 = \gamma^2$, then $|\alpha + \beta - \gamma|$ is equal to
51. Consider the parabola $y^2 = 4x$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$. Tangents are drawn from a point P to the parabola $y^2 = 4x$, the chord of contact of these tangents touches the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$. The locus of point P is a conic whose eccentricity is e , then $4e^2$ is equal to
52. An equilateral triangle ABC is inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. If the eccentric angles of the vertices of triangle ABC are α, β, γ respectively and the circumcentre of triangle ABC lies on the ellipse $\frac{9x^2}{7a^2} + \frac{9y^2}{7b^2} = 1$, then the value of $\cos \alpha \cos \beta + \cos \beta \cos \gamma + \cos \gamma \cos \alpha + \sin \alpha \sin \beta + \sin \beta \sin \gamma + \sin \gamma \sin \alpha$ is equal to
53. Let the director circle of the hyperbola $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be denoted by C_1 and let director circle of the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{3}$ be denoted C_2 . If the circle C_2 is also the director circle of C_1 and let e_1 and e_2 be the eccentricities of the ellipse E and hyperbola H , then $\frac{e_2^2}{e_1^2}$ is equal to ($a > b$)
54. Let a, b, c denote the lengths of the sides opposite to A, B and C in triangle ABC . If $2a^2 + 4b^2 + 9c^2 = 4ab + 6ac$, then $\cos B = \frac{p}{k}$ (where p and k are coprime), then $p - k$ is equal to

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – II
PAPER –1
TEST DATE: 11-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A, C, D

Sol. $x_C = \frac{1}{\omega_C} = 2000\Omega$... (i)

$$i^2 = (i_C)_A^2 + (i_R)^2$$
$$z' = 1000\sqrt{2}$$

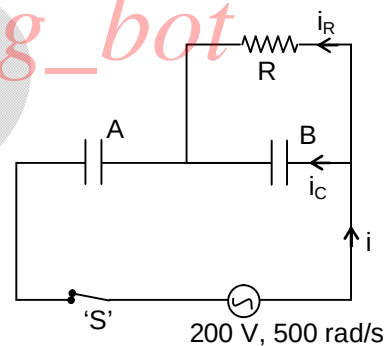
$$V^2 = (V_C)_B^2 + (V')^2 + 2V_C \cdot V' \cos 45^\circ$$

$$z = \sqrt{10} \cdot (10)^3$$

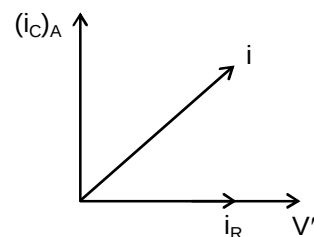
$$i = \frac{200}{\sqrt{10} \times 10^3} = \frac{1}{5\sqrt{10}}$$

$$i_R = i \cos 45^\circ$$

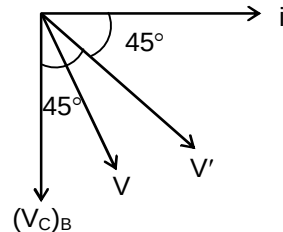
$$(i_C)_A = i \cos 45^\circ$$



... (ii)



... (iii)



$$\tan \alpha = \frac{V' \sin 45^\circ}{(V_C)_B + V' \cos 45^\circ} = \frac{1}{3} \quad \dots (iv)$$

$$\text{Power} = i^2 R = \left(\frac{1}{5\sqrt{20}} \right)^2 \cdot 2000 = 4 \text{ W}$$

2. A, C, D

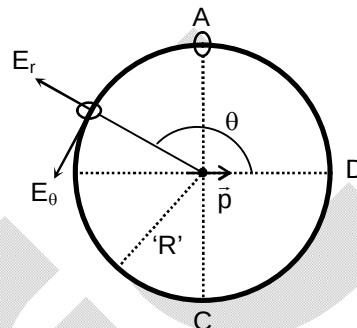
Sol. $E_r = \frac{2kp \cos \theta}{r^3}$

$$E_\theta = \frac{kp \sin \theta}{r^3}$$

From conservation of energy at A and E.

$$0 + 0 = \frac{1}{2}mv^2 + \frac{(kp \cos \theta)q}{r^2}$$

$$v = \sqrt{\frac{-2kp(\cos \theta)q}{mr^2}} = \sqrt{\frac{-p(\cos \theta)q}{2\pi\epsilon_0 mr^2}}$$



3. A, B, C

Sol. Since piston moves very slowly so they are at equilibrium at each instant

Above separately wall $P_1 = P_0 + \frac{mg}{A}$

Below separately wall $P_2 = P_0 - \frac{mg}{A}$

$$\Delta Q = 0$$

$$W + \Delta U = 0$$

$$W = -\Delta U = -nC_V \Delta T$$

$$P_2 V_2 - P_1 V_1 = -n \frac{f}{2} R (T_2 - T_1) = \frac{f}{2} [nRT_1 - nRT_2] = \frac{f}{2} [P_1 V_1 - P_2 V_2]$$

$$\left(1 + \frac{f}{2}\right) P_2 V_2 = \left(1 + \frac{f}{2}\right) P_1 V_1$$

$$P_1 V_1 = P_2 V_2, \text{ So, } T_1 = T_2$$

4. A, C, D

Sol. $e = e_1 + e_2 = 200 \sin \left(\omega t + \frac{\pi}{2} \right)$

$$e_0 = 200 \text{ volt}$$

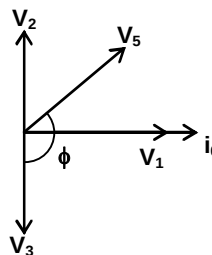
$$v_5 = \frac{200}{\sqrt{2}} = 100\sqrt{2}$$

$$z = \sqrt{(500)^2 + (500)^2} = 500\sqrt{2}$$

$$i_{\text{rms}} = \frac{100\sqrt{2}}{500\sqrt{2}} = \frac{1}{5}$$

$$V_1 = 500 \times \frac{1}{5} = 100 \text{ volt}$$

$$V_2 = 900 \times \frac{1}{5} = 180 \text{ volt}$$



$$V_3 = 400 \times \frac{1}{5} = 80 \text{ volt}$$

$$V_4 = V_2 - V_3 = 100 \text{ volt}$$

$$\phi = \frac{\pi}{2} + \tan^{-1}\left(\frac{9}{5}\right)$$

5. A, C, D

Sol. $i = \frac{|E_1 + E_2 - E_3 - E_4|}{r_1 + r_2 + r_3 + r_4}$

6. B, C

Sol. $dV = (\vec{v} \times \vec{B}) \cdot d\vec{\ell}$

So potential difference across OA is zero

$$\Rightarrow \text{Potential difference across OB} = \frac{1}{2} B_0 \omega (\sqrt{2} OA)^2 = \frac{1}{2} (100)(5)^2 (0.2)^2 = 20 \text{ Volt.}$$

So potential difference across AB is also 20 Volt.

7. A

Sol. $\mu = iA = 0.0550$

$$\tau = \vec{\mu} \times \vec{B} = 0.0055$$

$$|\tau| = I \alpha$$

$$\alpha = 0.0440$$

From conservation of energy

$$U_i + K_i = U_f + K_f$$

$$K_f = -U_f = 0.4400$$

8. B

Sol. $\tau = \frac{1}{\frac{6 \times 12}{6+12} + \frac{18 \times 24}{18+24}} = 0.07$

$\Rightarrow$ At steady state

$$R_{eq} = \frac{25}{2}$$

Current through battery = 2 ampere

Current through inductor = 1/6 ampere

$\Rightarrow$ At one time constant

$$\text{Current through inductor} = 0.63 \times \frac{1}{6} = 0.105 \text{ ampere}$$

From KVL, current through $12\Omega = 1.353$ ampere

$\Rightarrow$ For part (iv) after switching

$$\frac{1}{6} L = i_L (L + 2L)$$

$$i_L = \frac{1}{18} = 0.056 \text{ ampere}$$

9. C

Sol. A $\rightarrow$ B

$$TP^2 = \text{constant}$$

$$PV^{1/3} = \text{constant}$$

$$W_{A \rightarrow B} = \frac{P_1 V_1 - P_2 V_2}{x-1} = \frac{nR(T_1 - T_2)}{x-1} = -\frac{9}{2} nRT_0$$

$$\Delta U_{A \rightarrow B} = -3nRT_0$$

$$\Delta Q_{A \rightarrow B} = -\frac{15}{2} nRT_0$$

$C \rightarrow A$

$$TP^4 = \text{constant}$$

$$PV^{1/5} = \text{constant}$$

$$W_{C \rightarrow A} = \frac{nR(T_1 - T_2)}{x-1} = \frac{15}{4} nRT_0$$

$$\Delta U_{C \rightarrow A} = nR(T_A - T_C) > 0$$

$$\Delta Q_{C \rightarrow A} > 0$$

$B \rightarrow C$

$$\Delta U = 0$$

$$W_{B \rightarrow C} > 0$$

$$\Delta Q_{B \rightarrow C} > 0$$

10. D

Sol. For τ :

$$R_{eq} = \frac{6 \times 3}{6+3} + 3 = 5 \text{ M}\Omega$$

$$\tau = 5 \times \frac{1}{5} = 1 \text{ sec}$$

$\Rightarrow$ For $q_{\max}$ at steady state

$$\text{Current passing through } 10 \text{ V battery} = \frac{5}{3} \mu\text{A}$$

$$V_A + 5 - \frac{5}{3} \times 6 = V_B$$

$$V_A - V_B = 5 \text{ V}$$

$$q_{\max} = 1 \mu\text{C}$$

$\Rightarrow$ At time $t = 1 \text{ sec}$

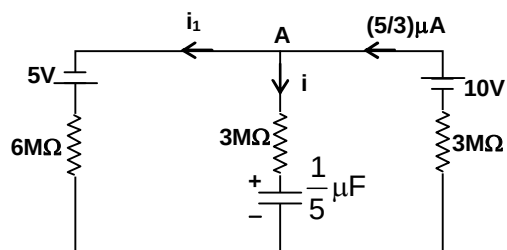
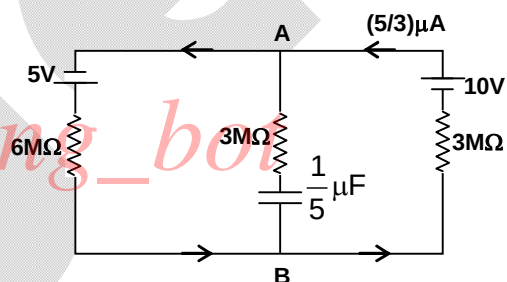
$$q = 1(1 - e^{-t}) = 0.63 \mu\text{C}$$

$$i = \frac{dq}{dt} = e^{-t} = 0.31 \mu\text{amp}$$

From KVL

$$\frac{0.63}{1/5} + (0.37)3 + 5 - i_1(6) = 0$$

$$\mu = 1.54 \mu\text{A}$$



Section - B

11. 1.50

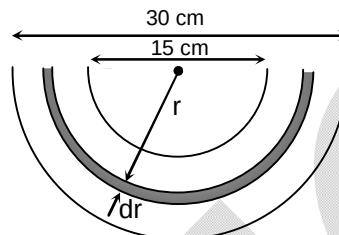
Sol.
$$v_{rms}^2 = \frac{\int v^2 dN}{\int dN} = \frac{\int v^2 \frac{dN}{dv} dv}{\int \frac{dN}{dv} dv}$$

$$\frac{\int_0^3 v^2 \left(-\frac{5}{3}v + 5 \right) dv}{\int_0^3 \left(-\frac{5}{3}v + 5 \right) dv} = 1.50$$

12. 6.93

Sol. $dR = \frac{\rho dr}{\pi r \ell}$

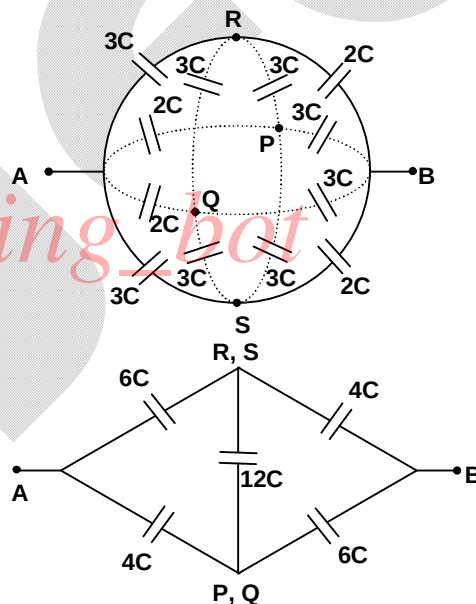
$$R = \frac{\rho}{\pi \ell} \int \frac{dr}{r} = \frac{\rho}{\pi \ell} \ln 2 = 6.93 \Omega$$



13. 2.10

Sol. P & Q and R & S are equipotential

$$C_{AB} = \frac{2(6C)(4C) + (6C)(12C) + (4C)(12C)}{6C + 4C + 2(12C)} = \frac{84}{17} C$$



$$\text{Heat generated} = \frac{1}{2} (C_{AB}) V_0^2 = 2.10 \mu J$$

14. 20.30

Sol. Let final pressure on both side of W_2 is P'

$$P' \left(\frac{V_0}{6} \right)^\gamma = P_0 \left(\frac{5V_0}{6} \right)^\gamma$$

$$P' = 5\sqrt{5}P_0$$

$$\text{Work done by gas} = \frac{5\sqrt{5}P_0 \left(\frac{V_0}{6} \right) - P_0 \left(\frac{5V_0}{6} \right)}{\gamma - 1} = \frac{5P_0 V_0}{3} (\sqrt{5} - 1)$$

$$\Delta U = nC_V \Delta T = n \left(\frac{R}{\gamma - 1} \right) \Delta T = 2nR \Delta T$$

$$= 2 \left(P' \frac{5V_0}{6} - P_0 \frac{V_0}{6} \right) = 2 \left(\frac{5\sqrt{5}P_0 5V_0}{6} - \frac{P_0 V_0}{6} \right)$$

$$= \frac{P_0 V_0}{3} (25\sqrt{5} - 1)$$

$$\Delta Q = \Delta U + W = 2P_0 V_0 (5\sqrt{5} - 1) = 20.3 P_0 V_0$$

15. 1.75

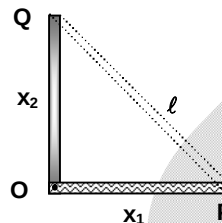
Sol. $\ell^2 = x_1^2 + x_2^2$

$$2\ell \frac{d\ell}{dt} = 2x_1 \frac{dx_1}{dt} + 2x_2 \frac{dx_2}{dt}$$

$$\ell^2 \left[\frac{1}{\ell} \frac{d\ell}{dt} \right] = x_1^2 \left(\frac{1}{x_1} \frac{dx_1}{dt} \right) + x_2^2 \left(\frac{1}{x_2} \frac{dx_2}{dt} \right)$$

$$\ell^2 (\alpha_{PQ}) = x_1^2 (\alpha_{PO}) + x_2^2 (\alpha_{QO})$$

$$\alpha_{PQ} = \frac{\alpha_{PO} x_1^2 + \alpha_{QO} x_2^2}{x_1^2 + x_2^2} = 1.7538 \times 10^{-5} / ^\circ\text{C}$$



16. 1.40

Sol. In first case

$$N = mg + iB\ell$$

$$F_1 = \mu(mg + iB\ell)$$

In second case

$$N = mg - iB\ell$$

$$F_2 = \mu(mg - iB\ell)$$

$$\frac{F_1}{F_2} = 1.40$$

17. 0.05

Sol. $mg - qBv_x = m \frac{dv_y}{dt}$

$$qBv_y = m \frac{dv_x}{dt}$$

Differentiate equation (i) with respect to time

$$0 - qB \frac{dv_x}{dt} = m \frac{d^2 v_y}{dt^2}$$

$$\frac{d^2 v_y}{dt^2} = -\frac{q^2 B^2}{m^2} v_y = -\omega^2 v_y$$

Solution of equation

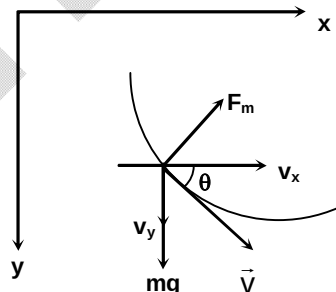
$$v_y = (v_y)_0 \sin \omega t$$

$$\frac{dv_y}{dt} = (v_y)_0 \omega \cos \omega t$$

at $t = 0$, $\frac{dv_y}{dt} = g$, so $(v_y)_0 = \frac{g}{\omega}$

Now from equation (i)

$$v_x = \frac{mg}{qB} (1 - \cos \omega t)$$



So, $v_y = 0$, at $\omega t = \pi$, $t = \frac{\pi}{\omega}$

$$v_x = \frac{mg}{qB} [1 - (-1)] = \frac{2mg}{qB}$$

From work energy theorem

$$mgy = \frac{1}{2}mv_x^2$$

$$y = \frac{2m^2g}{q^2B^2} = 0.05 \text{ meter}$$

18. 0.08

Sol. In first case

$$R_{eq} = \frac{\frac{5}{2} \times 3}{\frac{5}{2} + 3} = \frac{15}{11} \Omega$$

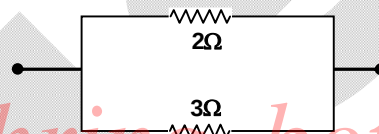
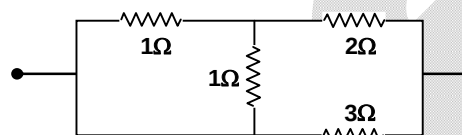
$$i_1 = \frac{V_0}{15/11} = \frac{11}{15} V_0$$

In second case

$$R_{eq} = \frac{2 \times 3}{2 + 3} = \frac{6}{5} \Omega$$

$$i_2 = \frac{V_0}{6/5} = \frac{5}{6} V_0$$

$$\frac{i_1}{i_2} = \frac{22}{25}$$



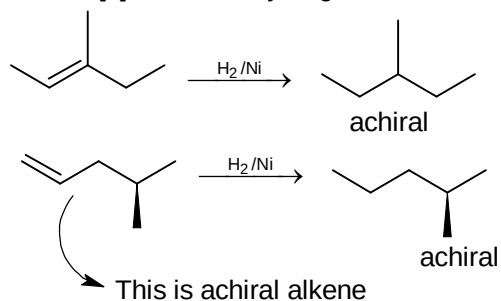
Chemistry

PART – II

Section – A

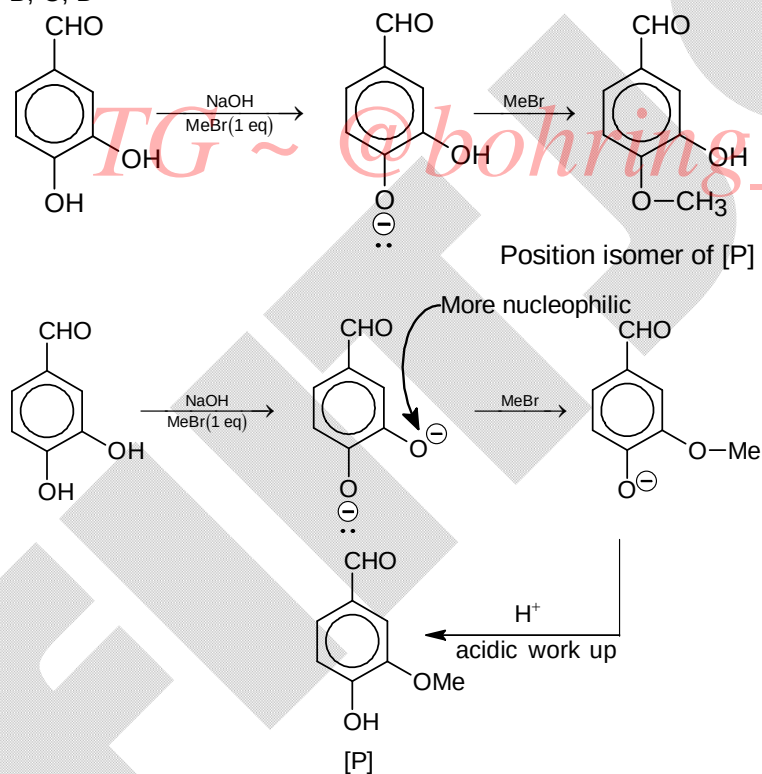
19. B, C

 Sol. Alkene [K] has 8α – Hydrogens.

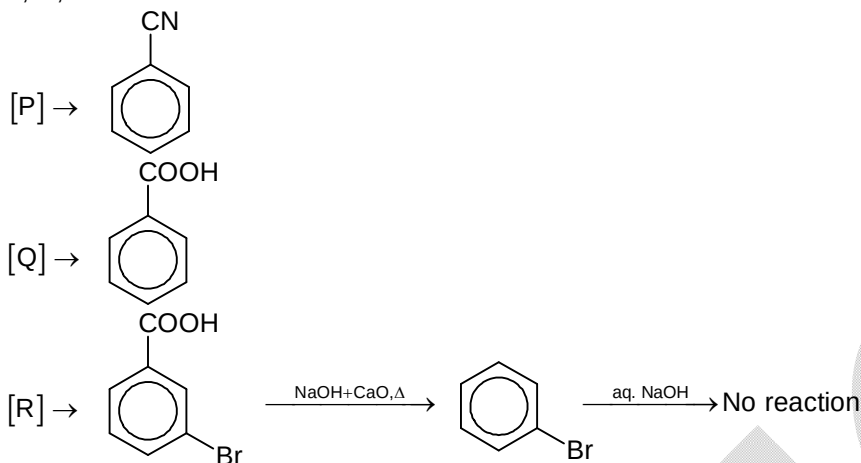
 Alkene [L] has 2α – Hydrogens .


20. B, C, D

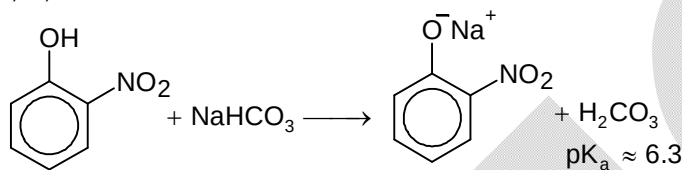
Sol.



21. B, C, D
Sol.

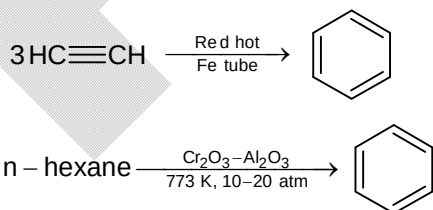
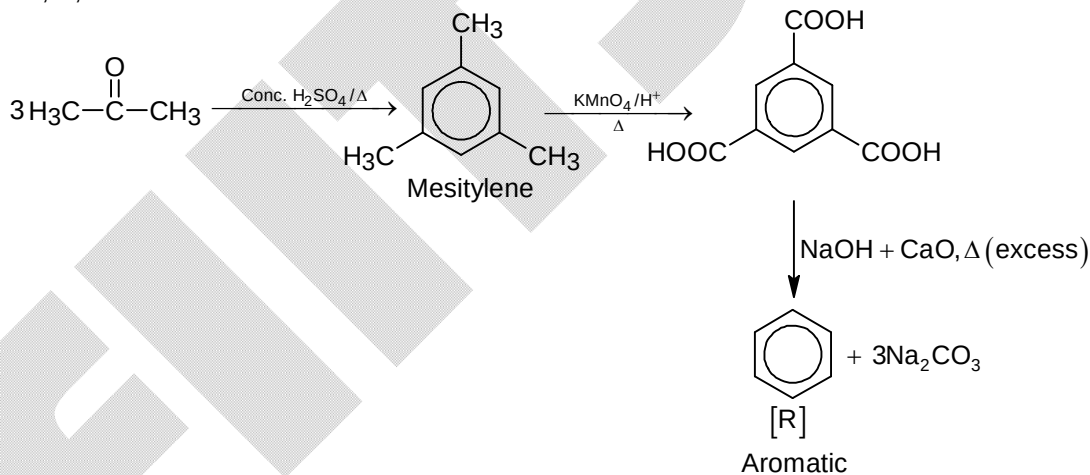


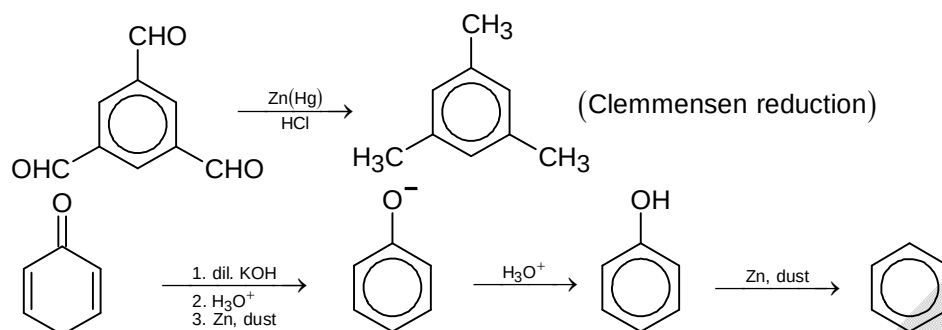
22. A, B, D
Sol.



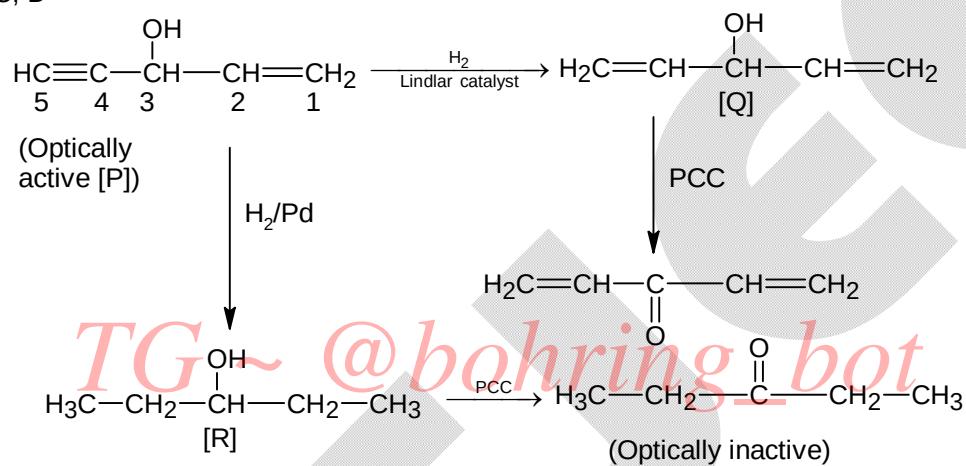
As H_2CO_3 is more acidic than o-nitrophenol, so forward reaction is not possible.

23. A, B, D
Sol.

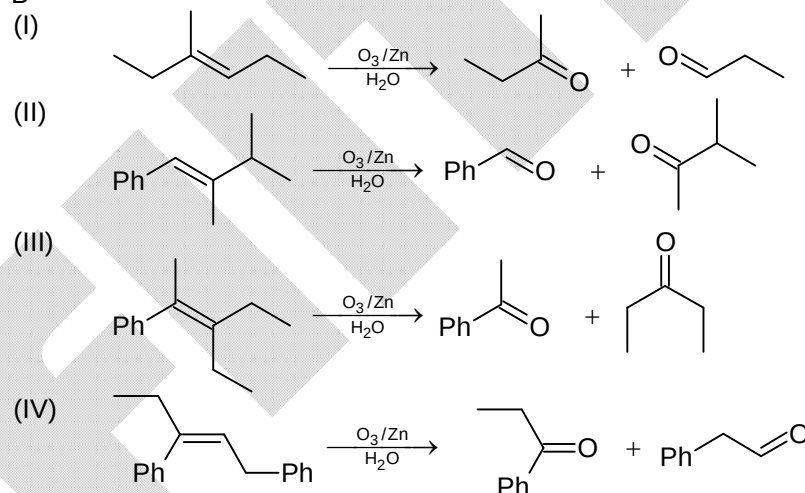




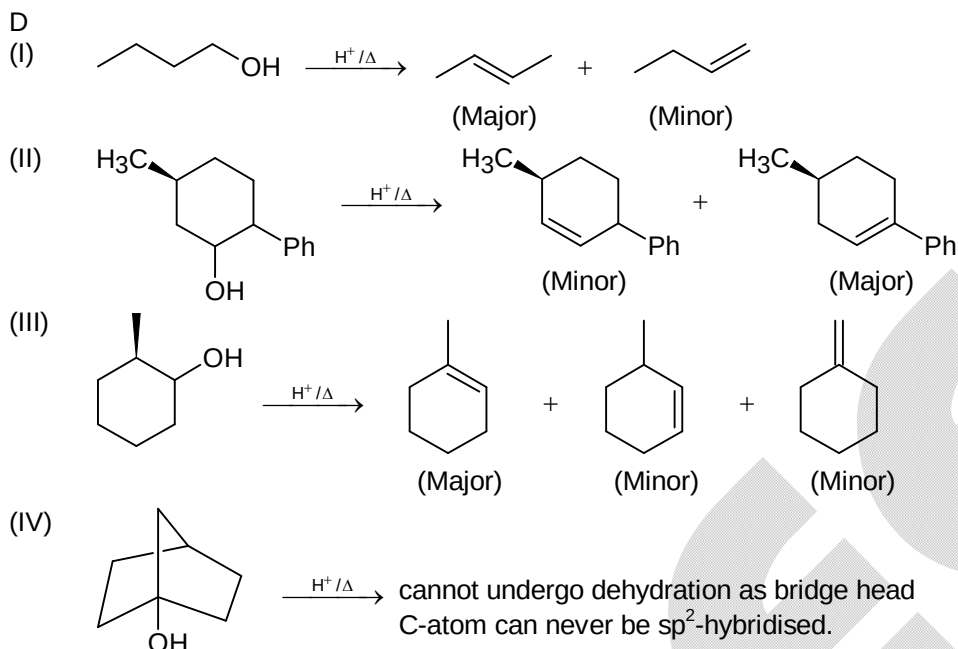
24. C, D
Sol.



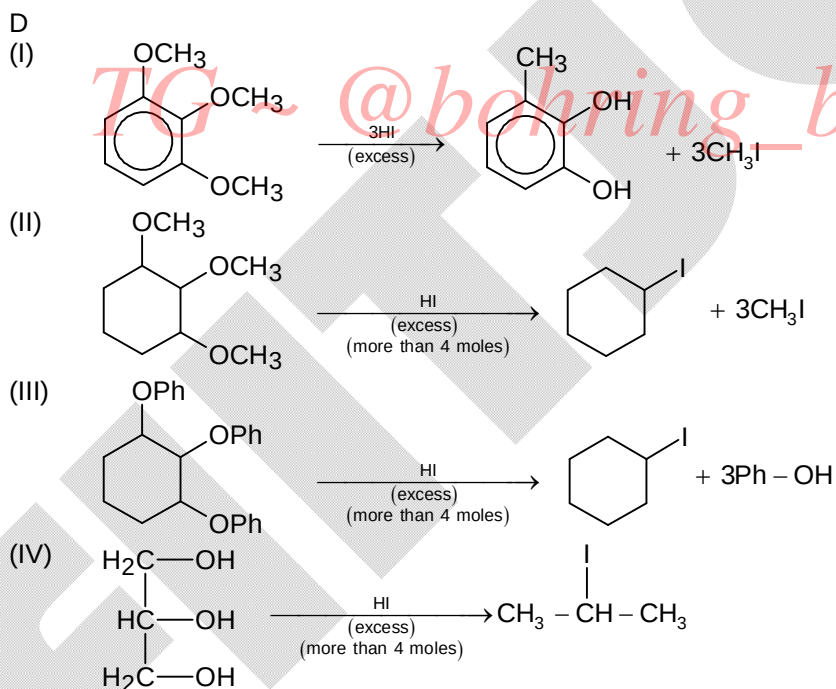
25. B
Sol. (I)



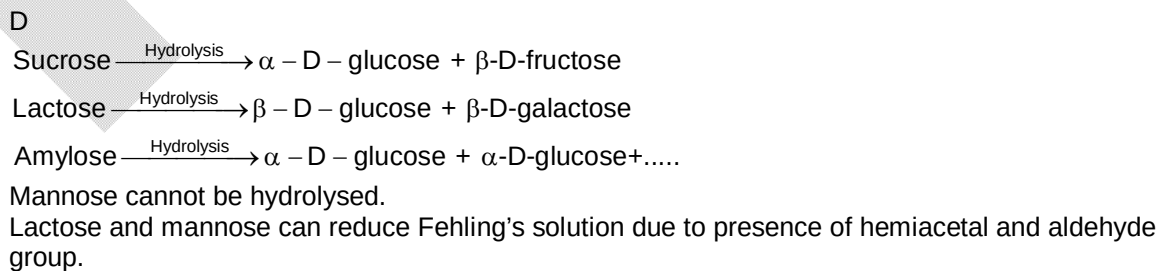
26.
Sol.



27.
Sol.



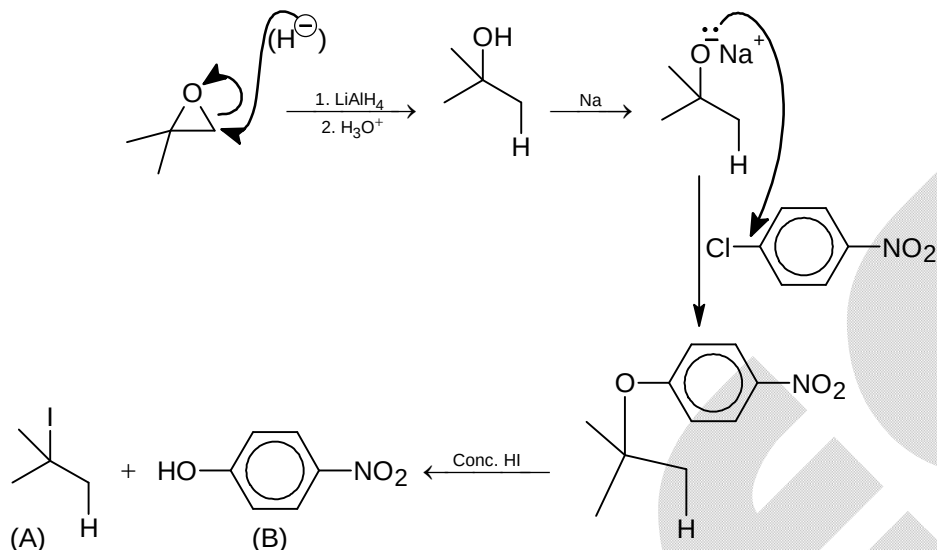
28.
Sol.



Section – B

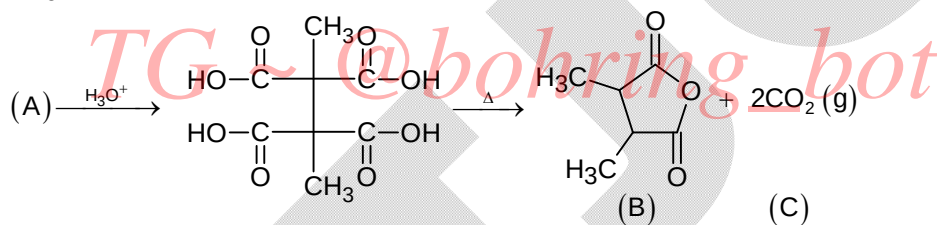
29. 0.50

Sol.



30. 4.40

Sol.



$$0.1 \text{ mol A} \rightarrow 0.2 \text{ mol C}$$

$$n_C = 0.2$$

$$w_C = 0.2 \times 44 \text{ (g)} = 8.8 \text{ (g)} = x$$

$$\frac{x}{2} = 4.40 \text{ grams.}$$

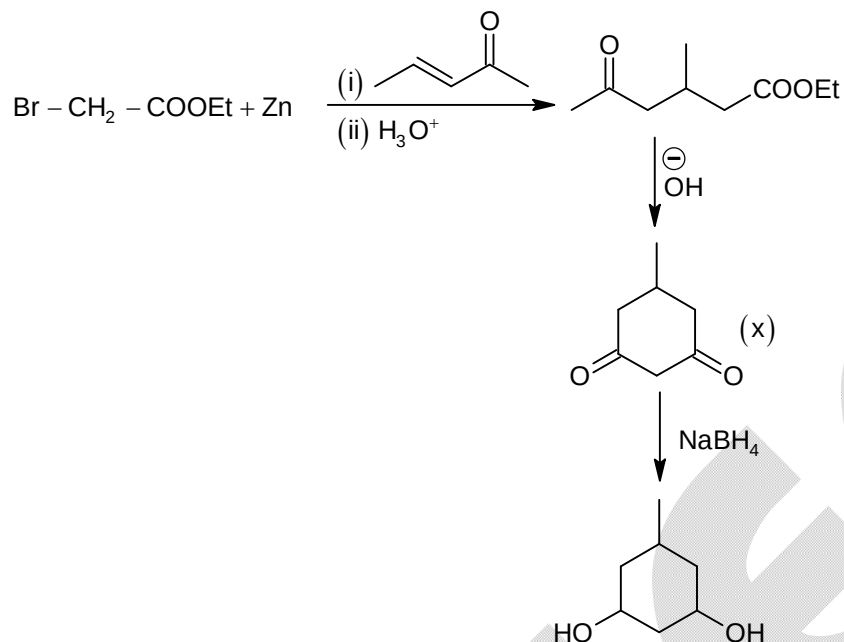
31. 3.50

Sol.

'A' is $CH_3 - CH = C = CH - CH_3$, optically active with minimum weight.

$$\therefore x = 5, y = 2.$$

32. 40.00
Sol.



Number of stereoisomers = 4.

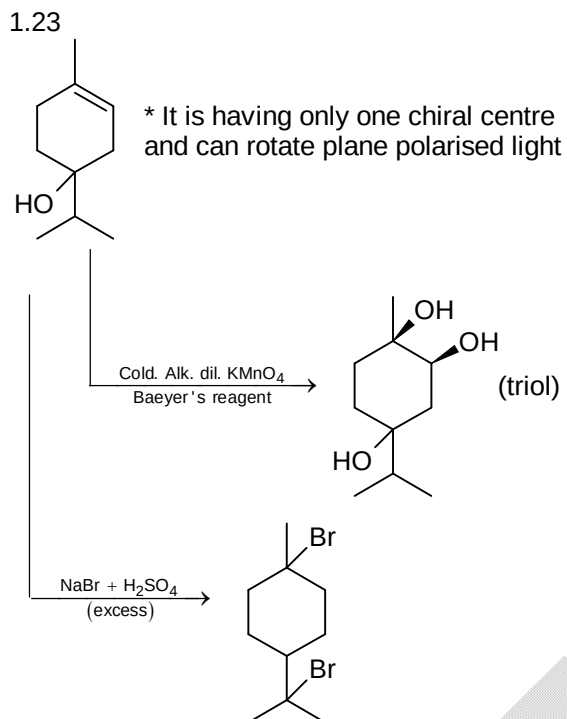
33. 2.00

Sol. $n = 1, \text{C}_2\text{H}_3\text{Br}_1$ (No stereoisomer)
 $n = 2, \text{C}_2\text{H}_2\text{Br}_2$ (Two geometrical isomers)
 $n = 3, \text{C}_3\text{HBr}_3$ (No stereoisomer)
 $n = 4, \text{C}_2\text{Br}_4$ (No stereoisomer)

34. 1.20

Sol. Co-polymer are Nylon-6,6, Buna-N, Dacron, Buna-S, PHBV. So, X = 5.
 Mutarotation is shown by α -D-G, β -D-G, Lactose, α -D-F, β -D-F, Maltose.
 So, Y = 6.

35.
Sol.



36.
Sol.

[18.95 – 18.99]
Aqueous tension at 287 K = 14 mm of Hg.
Hence, actual pressure = (758 – 14) = 744 mm of Hg.

$$\text{Number of moles of } \text{N}_2 = \frac{744 \times 30 \times 10^{-2}}{760 \times 0.0821 \times 287}$$

$$= 1.246 \times 10^{-3} \text{ mol}$$

$$\text{Mass of } \text{N}_2 = 1.246 \times 10^{-3} \times 28$$

$$= 34.888 \times 10^{-3} \text{ g}$$

$$\% \text{ mass} = \frac{34.888 \times 10^{-3}}{0.184} \times 100$$

$$= 18.96\%$$

Mathematics

PART – III

Section – A

37. C, D

Sol. Consider $(x_1, y_1) \equiv (2at, at^2) \Rightarrow$ Equation of tangent $\equiv tx = y + at^2$
 $\Rightarrow$ COC w.r.t $(\lambda, t\lambda - at^2)$ is $\lambda(x + ty) - (at^2y + 4a^2) = 0$
 $\Rightarrow (x_2, y_2) \equiv \left(\frac{4a}{t}, -\frac{4a}{t^2}\right)$

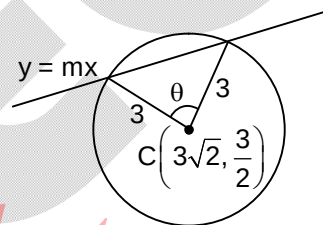
38. A, B

Sol. Here, $\cos \theta = \sqrt{\frac{1}{e_1^2} + \frac{1}{e_2^2} - \frac{1}{e_1^2 e_2^2}}$

39. B, C

Sol. $\Rightarrow \frac{\left(\frac{\pi(3)^2}{2\pi} \cdot \theta - \frac{1}{2}(3)^2 \sin \theta\right)}{\pi(3)^2 - \left(\frac{\pi(3)^2}{2\pi} \cdot \theta - \frac{1}{2}(3)^2 \sin \theta\right)} = \frac{(2\pi - 3\sqrt{3})}{(10\pi + 3\sqrt{3})}$

$\theta = \frac{\pi}{3}$ satisfies this $\Rightarrow m = -\frac{\sqrt{2}}{5}$ or $\sqrt{2}$



40. C, D

Sol. Here, $\frac{(\alpha^2 + \beta^2)^a}{(\alpha\beta)^b}$ is independent of θ if $4a = 5b$

41. A, D

Sol. Consider $4x^2 + pxy + 6y^2 = (y - m_1x)(y - m_2x)$, then $4x^2 + qxy - 6y^2 = \left(y + \frac{1}{m_1}x\right)(y - m_2x)$
 Comparing, $p = -10, q = -2$ or $p = 10, q = 2$

42. A, B

Sol. Given expression = $\frac{(x + y)(x^2 + y^2)}{2}$

43. C

Sol. (I) Here, $x^2 = \sin^2 \frac{\pi x}{2} = 1$

(II) $|\tan x| = (4x - \pi)^2 + 16$

(III) $x + y = 0$ or $x = 0 \Rightarrow (x, y) = \left(\pm \frac{1}{2}, \mp \frac{1}{2}\right), (0, \pm 1), (\pm 1, 0)$

(IV) The given equation becomes, $\frac{(\sin x)[(1 + \cos x) + (1 + \cos x)^2 + 3 \cos^2 x]}{3} > 0$

$\Rightarrow x \in (0, \pi)$

44.

B

Sol.

- (I) $\cos^{-1} x > 1 > \sin^{-1} x$
 (II) $1 > \cos^{-1} x > \sin^{-1} x$
 (III) $1 > \sin^{-1} x > \cos^{-1} x$
 (IV) $\sin^{-1} x > 1 > \cos^{-1} x$

45.

D

Sol.

- (I) The given parabolas are symmetric about the line $y = x$
 (II) Area of the given triangle = x_0^3
 (III) Probability (required) = $\frac{\pi a^2 - \pi ab}{\pi a^2} = \frac{2}{3} \Rightarrow a = 3b$
 (IV) Here, roots are $1, \sqrt{2}, -\sqrt{2}$

46.

B

Sol.

- (I) The two given families pass through A(2, 1) and B(19, -4) respectively
 $\Rightarrow m_{OA} = \frac{1}{2}, m_{OB} = -\frac{4}{19} \Rightarrow$ Slope of $\perp$ lines are -2 and $\frac{19}{4} \Rightarrow \theta = \tan^{-1}\left(\frac{27}{34}\right)$
 (II) Given family is always concurrent at $\left(-\frac{1}{6}, -\frac{7}{6}\right)$
 (III) $AP^2 + AQ^2 + AR^2 + AS^2 = \frac{7}{4} + \frac{7}{4} + \frac{3}{4} + \frac{27}{4} = 11$
 (IV) Given rays are $y = \frac{x}{2}$ and $y = -2x (\perp) \Rightarrow$ Required area = $\frac{4\pi - \pi}{4} = \frac{3\pi}{4}$

Section - B

47.

2.00

Sol.

If $A + B + C = \pi$, then $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

48.

1011.00

Sol.

$$\frac{\cot A}{\cot B + \cot C} = \frac{\cos A \cdot \sin B \cdot \sin C}{\sin^2 A}$$

49.

4.33

Sol.

By the concept of homogenization, $\lambda = \frac{15}{2}$

50.

34937.00

Sol.

Here, $R(\alpha, \beta)$ lies on PQ and midpoint of PR lies on AB
 $\Rightarrow \alpha = \frac{23}{7}, \beta = \frac{31}{7}$

51.

20.00

Sol.

The given ellipse is $(x-3)^2 + \frac{\left(y + \frac{1}{2}\right)^2}{\frac{1}{4}} = 1$

$$\Rightarrow x = 3 + \cos \theta, y = -\frac{1}{2} + \frac{1}{2} \sin \theta \Rightarrow 8x - 12y \in [20, 40]$$

52. 35.00

Sol. Required number of squares = $5^2 + 3^2 + 1^2 = 35$

53. 0.50

Sol. For $P(t_1)$, $Q(t_2)$, $R(t_3)$. Area of $\Delta PQR = |a^2(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)|$

And area of $\Delta ABC = \frac{a^2}{2} |(t_1 - t_3)(t_2 - t_1)(t_2 - t_3)|$

54. 906.00

Sol. Here, $\cos x = 0$, $\cos 3x = 0$, $\cos 4x = 0$, $\cos 5x = 0$

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FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – II
PAPER –2
TEST DATE: 11-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1.

C

Sol.

$$i = i_0(1 - e^{-t/\tau})$$

$$\frac{di}{dt} = \frac{i_0}{\tau} e^{-t/\tau}$$

$$U = \frac{1}{2} Li^2$$

$$\frac{dU}{dt} = \frac{1}{2} L \left(2i \frac{di}{dt} \right) = k e^{-t} (1 - e^{-t})$$

$$\text{at } t = 0, \frac{dU}{dt} = 0$$

$$t \rightarrow \infty, \frac{dU}{dt} = 0$$

2.

A

Sol.

$$dC = \frac{k \epsilon_0 A}{dx}$$

$$\frac{1}{C_{ev}} = \int \frac{1}{dC} = \int \frac{1}{k \epsilon_0 A} dx$$

$$= \int_0^d \frac{1}{k_0 e^{\lambda x} \epsilon_0 A} dx$$

$$C_{eq} = \frac{\lambda k_0 \epsilon_0 A}{1 - e^{-\lambda d}}$$

3. B

Sol. $H = -kA \frac{d\theta}{dx}$

The heat current decreases from the end P and Q through the rod hence slope of (T-x) graph also decreases.

4. C

Sol. $U = \frac{kq_0(2q)}{\sqrt{z^2 + 2a^2}}$

5. A, B, D

Sol. At time 't'

$$-\frac{q}{4} - i \times \frac{8}{3} + \frac{10 - q}{5} = 0$$

$$\int_0^q \frac{dq}{2 - \frac{9}{20}q} = \int_0^t \frac{3}{8} dt$$

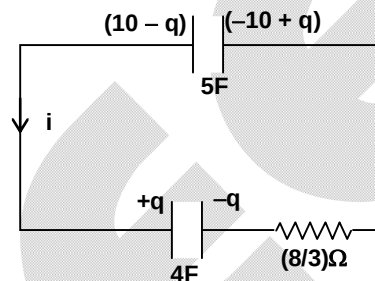
$$q = \frac{40}{9} \left[1 - e^{-\frac{27}{160}t} \right],$$

$$\frac{dq}{dt} = i = \frac{3}{4} e^{-\frac{27}{160}t}$$

$$U_i = \frac{(10)^2}{2 \times 5}$$

$$U_f = \frac{(40/9)^2}{2 \times 4} + \frac{(50/9)^2}{2 \times 5}$$

$$\Delta U = U_i - U_f = \frac{40}{9} \text{ J}$$



6. B, C

Sol. Current in the circuit

$$i = \frac{5}{R + \frac{R}{2}} = \frac{20}{3R}$$

$$\Rightarrow \text{Potential drop across PS} = iR = \frac{20}{3} \text{ V}$$

$$\text{Potential drop across SQ} = i \left(\frac{R}{2} \right) = \frac{10}{3} \text{ V}$$

$$\Rightarrow \text{Potential q radiant for PS} = \frac{20/3}{5} = \frac{4}{3} \text{ V/m}$$

$$\text{Potential q radiant for SQ} = \frac{10/3}{5} = \frac{2}{3} \text{ V/m}$$

$$\Rightarrow \text{Length required for 5V} = \frac{5}{4/3} \text{ V} = 3.75 \text{ m}$$

$$\text{Length required for 8V} = 5 + \frac{8 - \frac{20}{3}}{2/3} = 7 \text{ m}$$

7. A, C

Sol. When charge particle is moved from 'S' to infinity then induced electric field is perpendicular to the displacement hence work done by external agent is zero.

From P to infinity

Work done = $q(\text{change in potential difference})$

$$= q = \frac{\pi r^2 dB}{4 dt} = \frac{55}{8} \mu\text{J}$$

8. A, C, D

Sol. Before switch is closed

$$10 - q_0 - \frac{q_0}{2} = 0$$

$$q_0 = \frac{20}{3} \mu\text{C}$$

After switch is closed

From KVL,

$$5 - \frac{q - q_1}{2} - \frac{q_0 + q}{1} = 0$$

$$10 - \frac{q_0 + q}{1} - \frac{q_0 + q}{2} = 0$$

On solving,

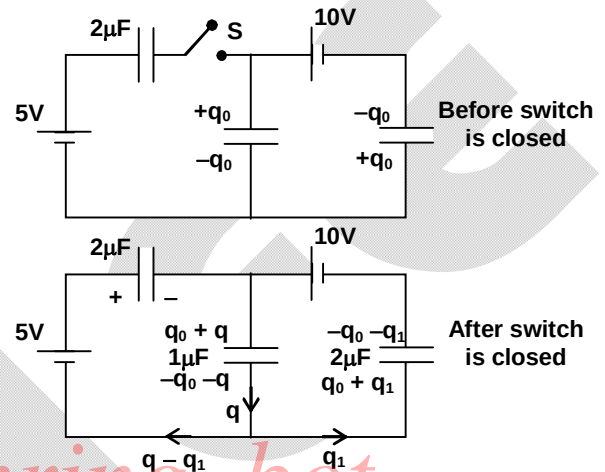
$$q = -\frac{2}{3} \mu\text{C}, q_1 = \frac{4}{3} \mu\text{C}$$

$$U_i = \frac{1}{2} \frac{(20/3)^2}{1} + \frac{1}{2} \frac{(20/3)^2}{2} = \frac{100}{3} \mu\text{J}$$

$$U_f = \frac{1}{2} \frac{(6)^2}{1} + \frac{1}{2} \frac{(8)^2}{2} + \frac{1}{2} \frac{(2)^2}{2} = 35 \mu\text{J}$$

$$W_B = 5(-2) + 10 \times \frac{4}{3} = \frac{10}{3} \mu\text{J}$$

$$\text{So, heat required} = \frac{5}{3} \mu\text{J}$$



9. A

Sol. Potential at 'C' = $\frac{kQ}{R} = \frac{\sigma R}{2\epsilon_0}$

Flat surface of hemispherical shell is on equipotential surface.

10. B, C

Sol. $f_f = \mu mg$

$$a = \mu g$$

$$\int_{v_0}^v dv = \int_0^t -\mu g dt$$

$$v - v_0 = -\mu gt$$

$$v = v_0 - \mu gt$$

$$\Rightarrow L \left(-\frac{dm}{dt} \right) = f_f v = \mu mg(v_0 - \mu gt)$$

$$\int_{m_0}^m \left(-\frac{dm}{m} \right) = \int_0^t -\frac{\mu g}{L} (v_0 - \mu g t) dt$$

$$\ln \frac{m}{m_0} = -\frac{\mu g}{L} \left(v_0 t - \frac{\mu g t^2}{2} \right)$$

$$m = m_0 e^{-\frac{\mu g}{L} \left(v_0 t - \frac{\mu g t^2}{2} \right)}$$

Section – B

11. 4

Sol. $\frac{1}{2} m v_C^2 = qE \cdot \frac{\pi R}{2} \quad \dots (i)$

$\frac{1}{2} m v_B^2 = qE \cdot \pi R \quad \dots (ii)$

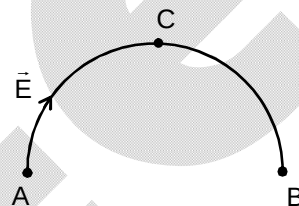
At C, $a_C = \frac{v_C^2}{R} = \frac{qE\pi}{m}$

At B, $a_C = \frac{v_B^2}{R} = \frac{2qE\pi}{m}$

$a_1 = \sqrt{\left(\frac{qE}{m} \right)^2 + \left(\frac{qE\pi}{m} \right)^2}$

$a_2 = \sqrt{\left(\frac{qE}{m} \right)^2 + \left(\frac{2qE\pi}{m} \right)^2}$

$\frac{a_1}{a_2} = \sqrt{\frac{1 + \pi^2}{1 + 4\pi^2}}$



12. 2

Sol. $\Delta \ell_1 = L_1 \alpha_1 \Delta T = \frac{30}{100} (1.7 \times 10^{-5}) \times 100 = 0.51 \text{ mm}$

$\Delta \ell_2 = (1.91 - 0.51) = 1.4 \text{ mm}$

So, $\Delta \ell_2 = L_2 \alpha_2 \Delta T$

$\alpha_2 = 2 \times 10^{-5} / ^\circ\text{C}$

13. 2

Sol. $\vec{B} = 4\hat{i} + \hat{j} + \hat{k}$

$\vec{v} = \hat{i} - 2\hat{j} + 3\hat{k}$

So, $\vec{F} = q(\vec{v} \times \vec{B}) = q(-5\hat{i} + 11\hat{j} + 9\hat{k})$

$|\vec{F}| = q\sqrt{227}$

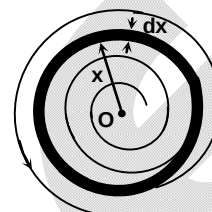
$R = \frac{|\vec{v}|^2}{a_C} = 2 \times 10^3 \text{ m} = 2 \text{ km}$

14. 2

Sol.
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$
$$k_{eq}A = k_1A_1 + k_2A_2 + k_3A_3$$
$$k_{eq} = \frac{k_1 + k_2 + k_3}{3} = 2 \text{ W/mK}$$

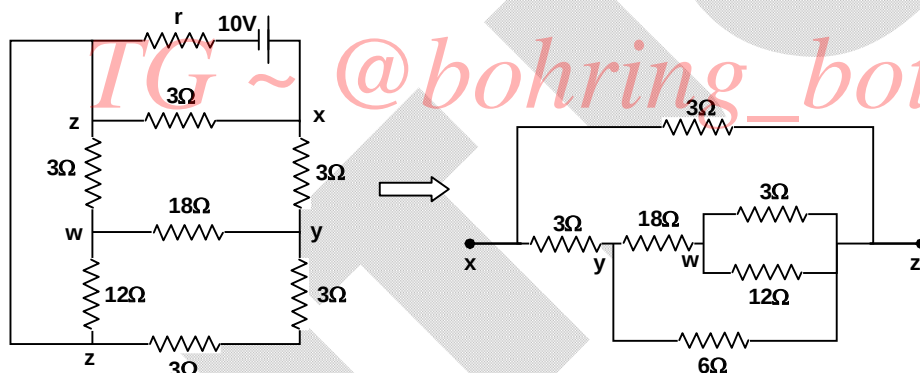
15. 3

Sol.
$$\text{emf} = \frac{d\phi}{dt} = \pi x^2 \frac{dB}{dt} (dN)$$
$$= \frac{N}{b-a} \pi \left(\frac{dB}{dt} \right) \int_a^b x^2 dx = \frac{14}{3} \text{ volt}$$
$$i = \frac{\text{emf}}{R} = 3 \text{ A}$$



16. 7

Sol. R_{eq} between points x and z
$$R_{eq} = \frac{28}{13} \Omega$$

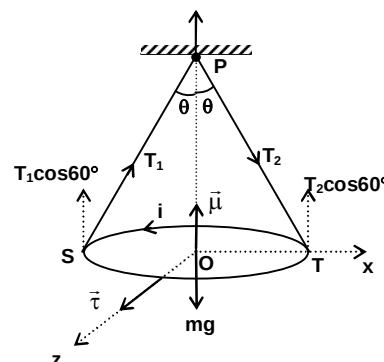


17. 6

Sol. Immediately after the switching, current in the branch of inductor will not change and charge on capacitor will be zero.

18. 3

Sol. $T_1 \cos 60^\circ + T_2 \cos 60^\circ = mg$
$$T_1 + T_2 = 2mG \quad \dots (i)$$
Taking torque about 'O'
$$(T_2 \cos 60^\circ)R + |\vec{\mu} \times \vec{B}| = (T_1 \cos 60^\circ)R \quad \dots (ii)$$
On solving $\frac{T_1}{T_2} = 3$



Chemistry

PART – II

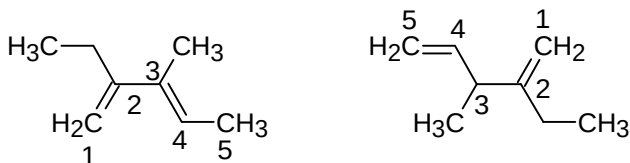
Section – A

19.

D

Sol.

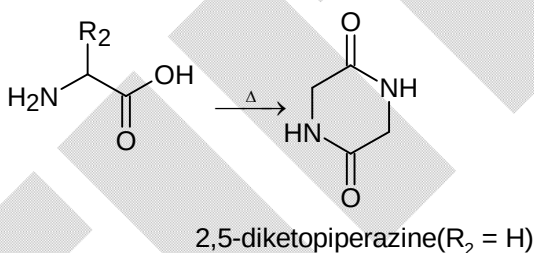
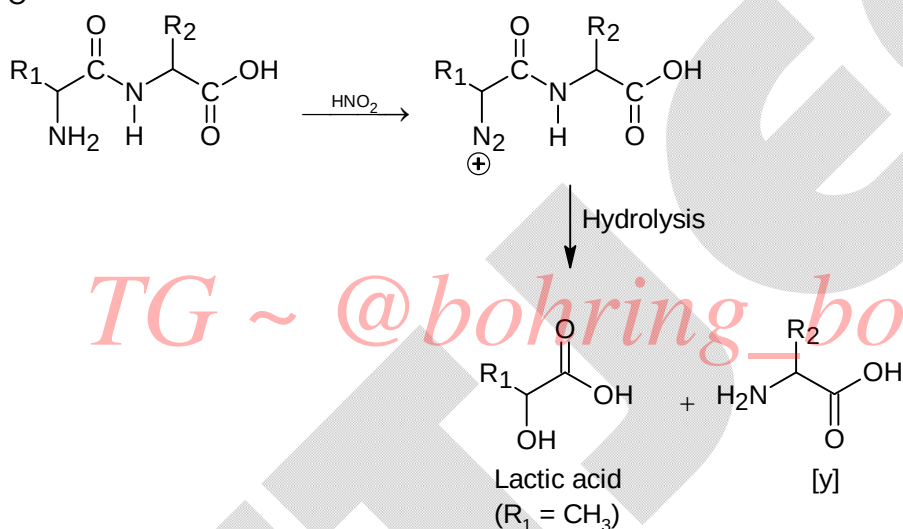
Option (D) has longest chain of 5C-atom which is same as in the compound given



20.

C

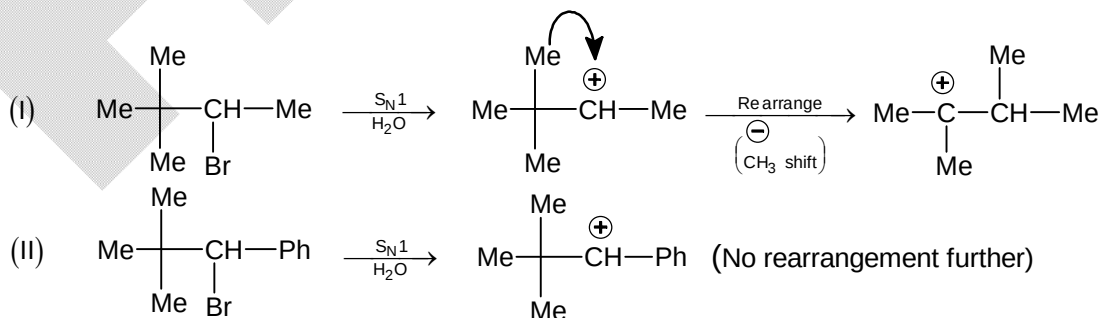
Sol.



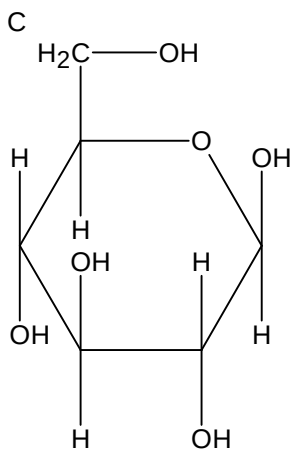
21.

B

Sol.



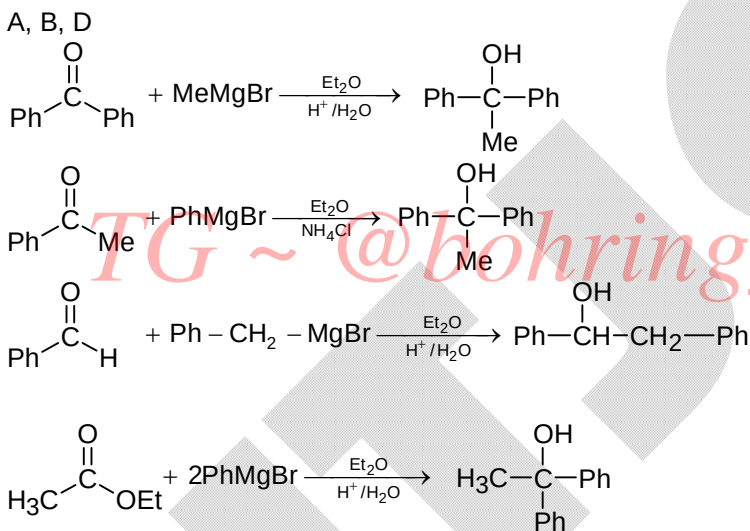
22.
Sol.



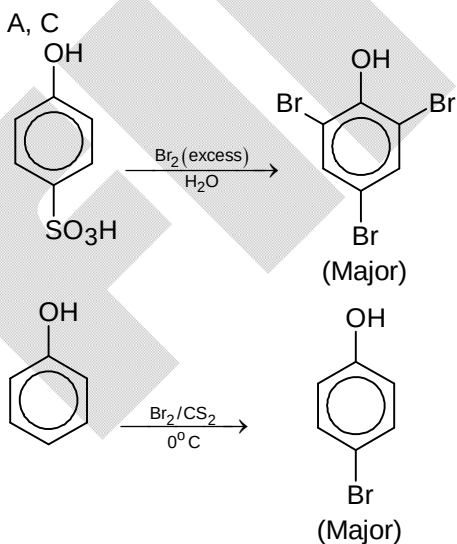
gives positive Tollen's test and $\text{NH}_2\text{OH}/\text{H}^+$ due to presence of Hemiacetal linkage.

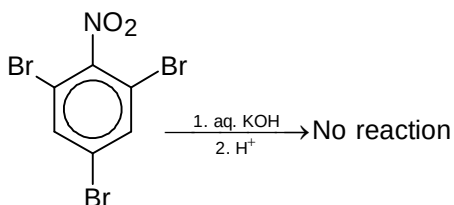
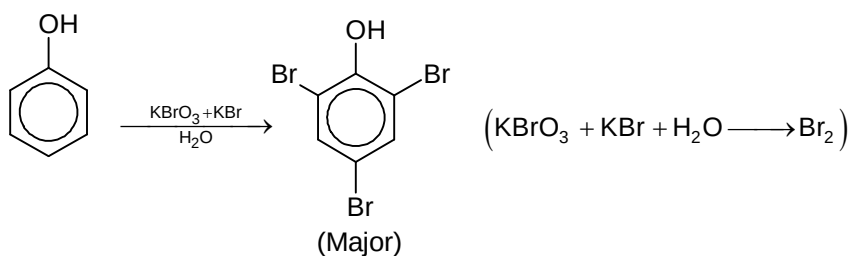
β - D - Glucopyranose

23.
Sol.

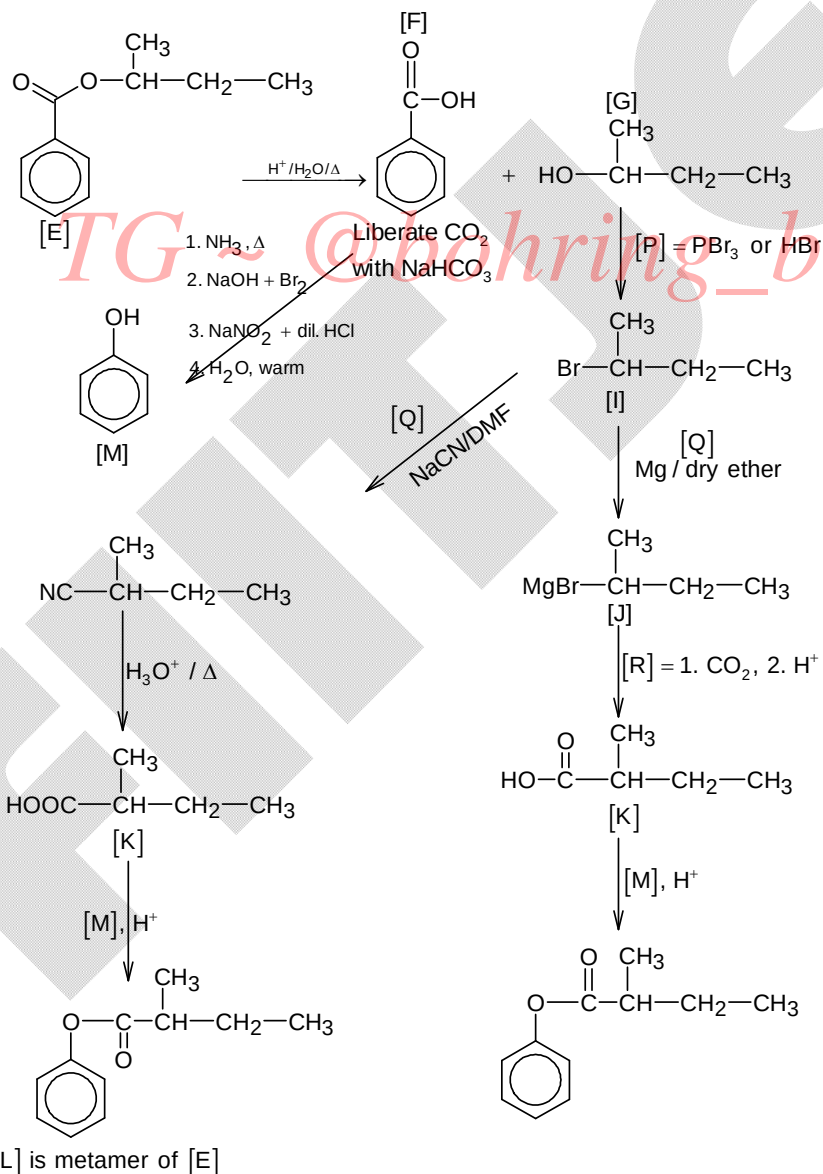


24.
Sol.

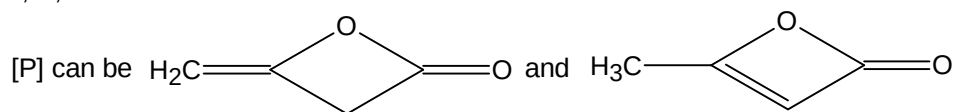




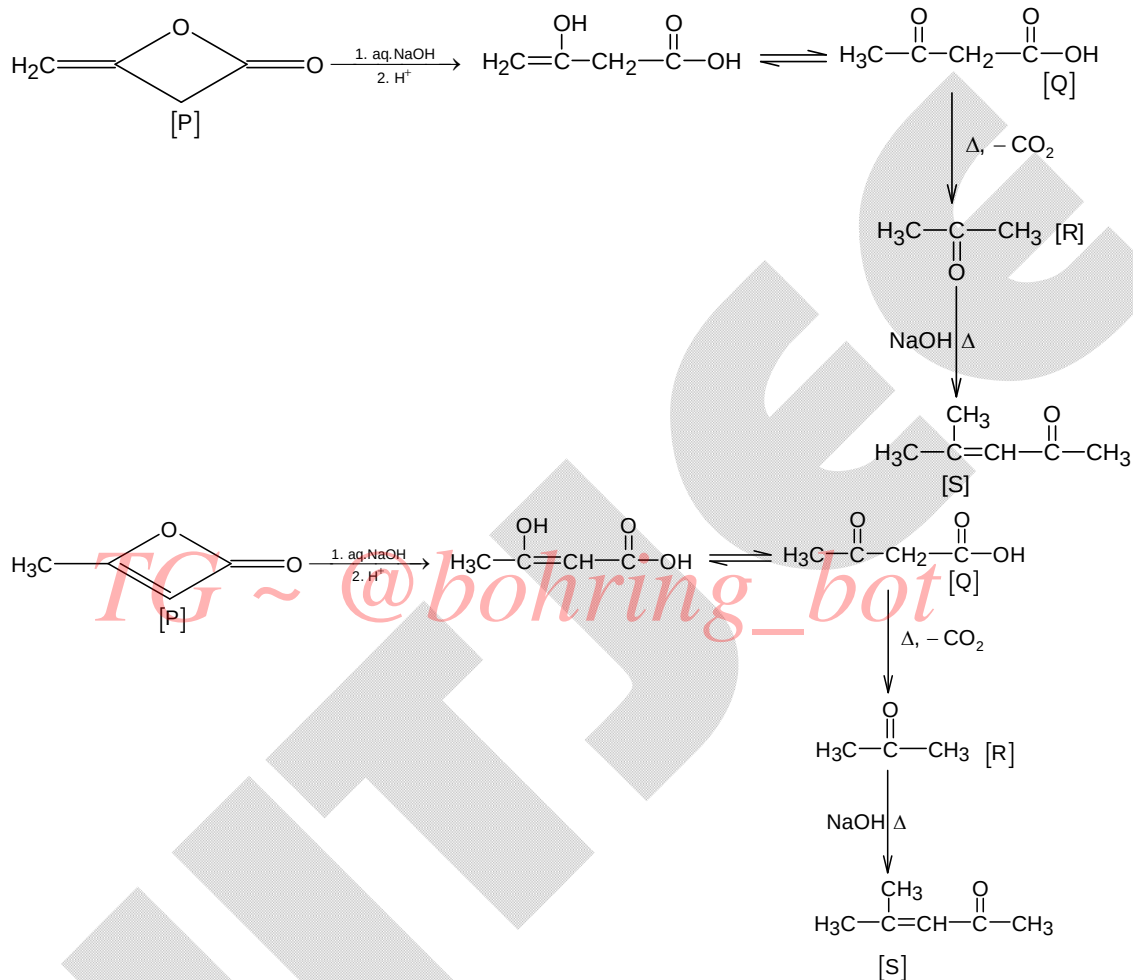
25. C, D
Sol.



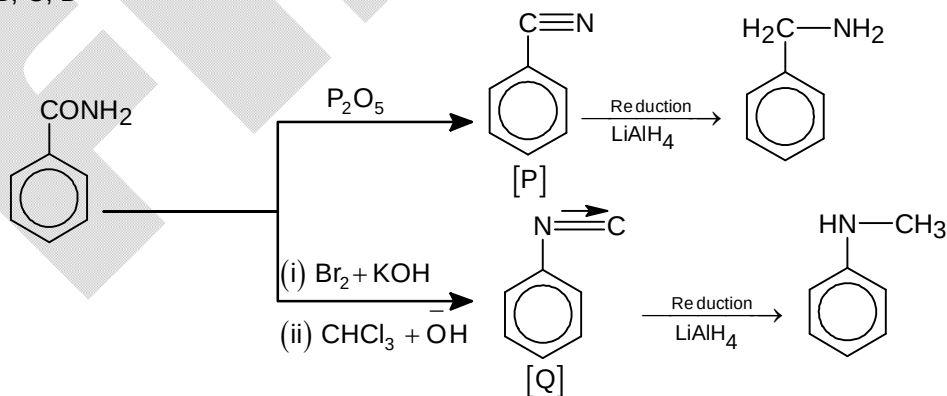
26. A, B, D
Sol.

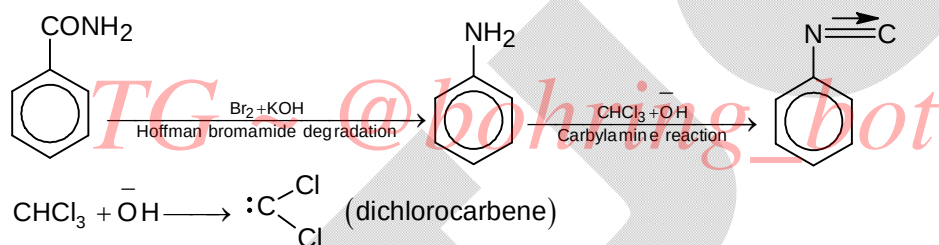
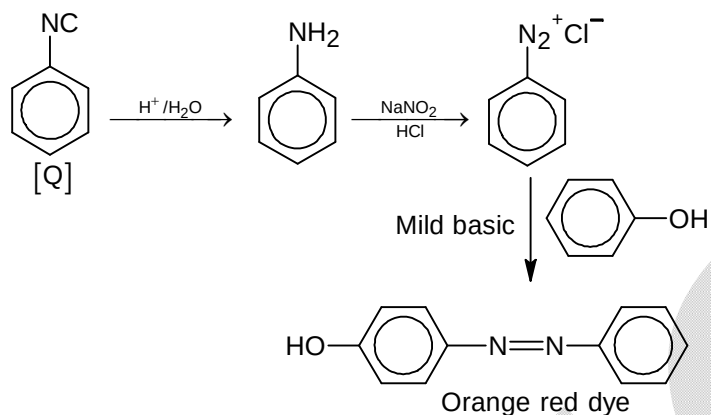
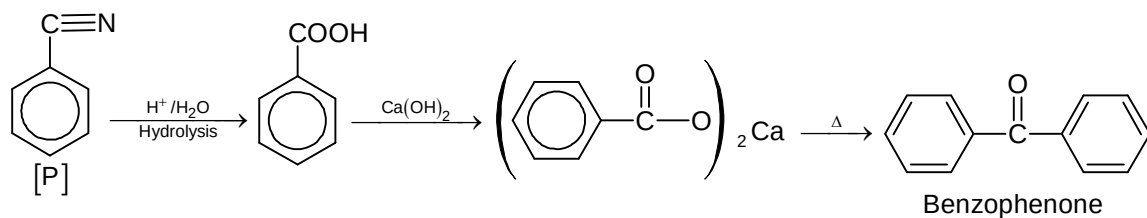


(Two structural isomers)

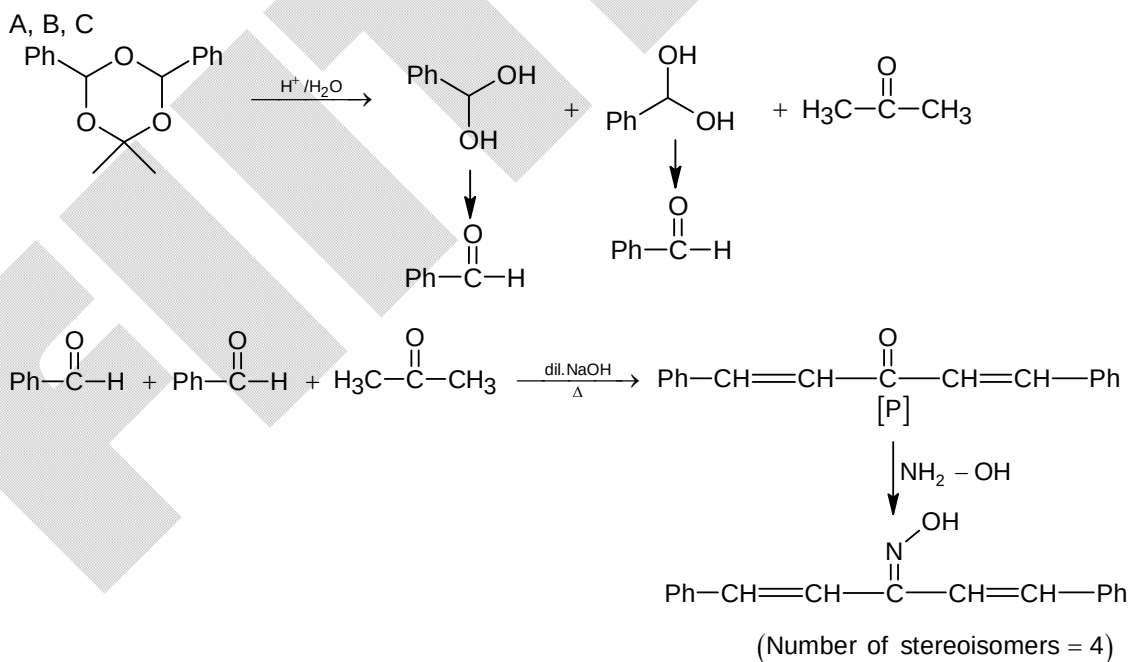


27. B, C, D
Sol.



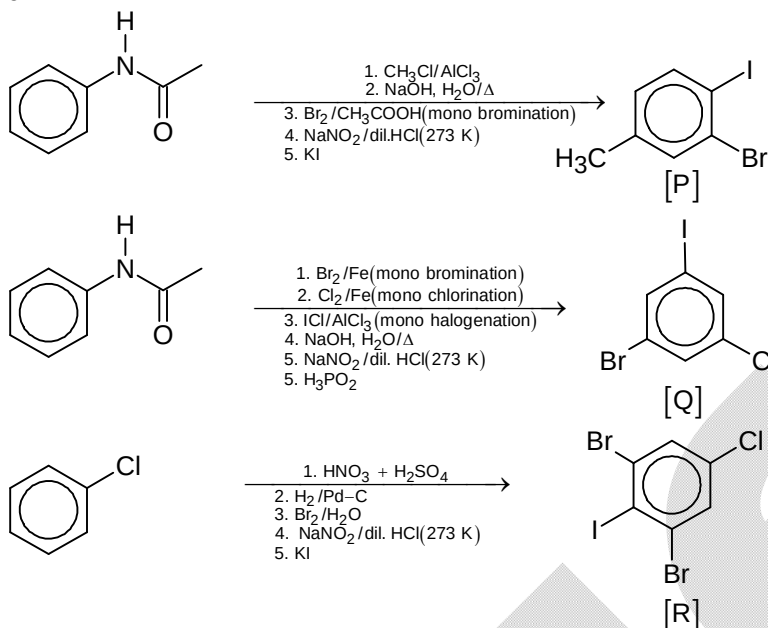


28.
Sol.



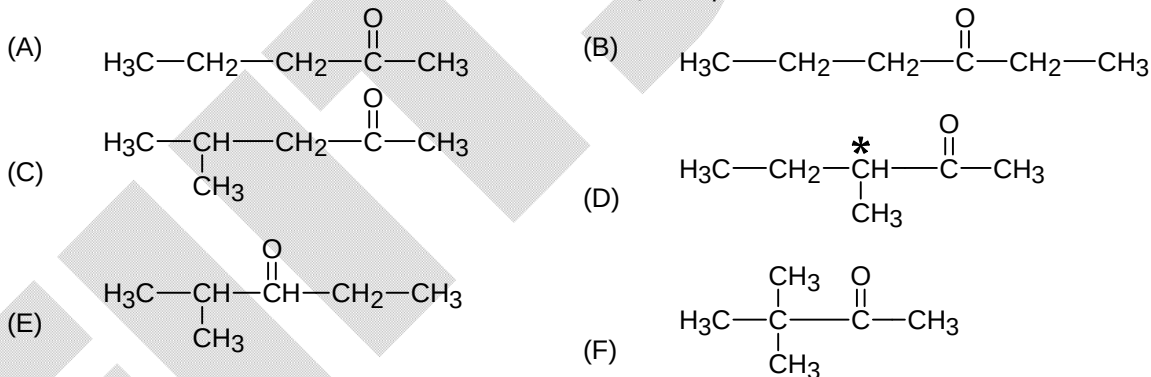
Section – B

29. 9
Sol.



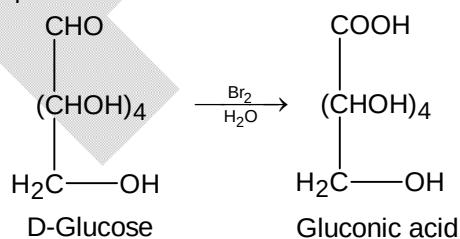
30. 3
Sol. I, III and VI compounds satisfy the conditions

31. 2
Sol. General formula of isomeric ketone of MW = 100 is $\text{C}_6\text{H}_{12}\text{O}$ possible structures:

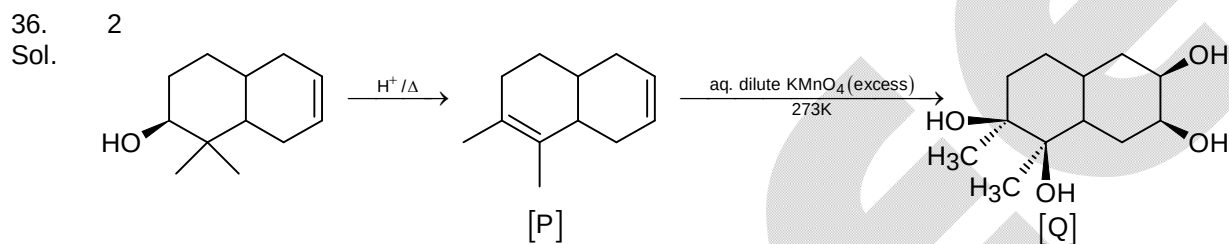
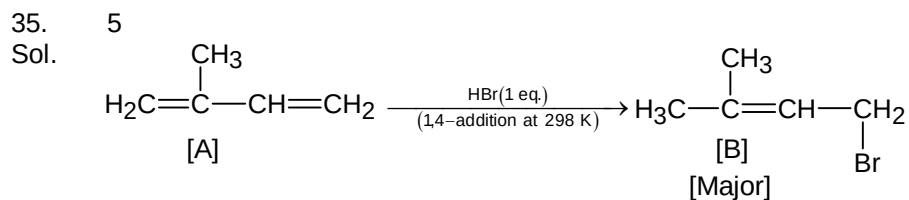
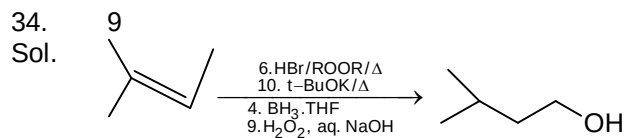


Compound B and E will give racemic mixture.

32. 4
Sol.



33. 7
Sol. Statements 1, 2, 4, 5, 6, 7 and 10 are correct.



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Mathematics

PART – III

Section – A

37. A

$$\text{Sol. } \frac{\frac{\cos A}{\sin A}}{(\cos B \sin C + \sin B \cos C)} = \frac{\cos A \sin B \sin C}{\sin^2 A} = \frac{\left(\frac{b^2 + c^2 - a^2}{2bc}\right) \frac{b}{2R} \times \frac{c}{2R}}{\frac{a^2}{4R^2}} = \frac{57a^2}{8 \times 2bc} \times \frac{bc}{a^2} = \frac{57}{16}$$

38. B

$$\text{Sol. } DE = AK = \sqrt{(r_1 + r_2)^2 - (r_2 - r_1)^2} = 2\sqrt{r_1 r_2}$$

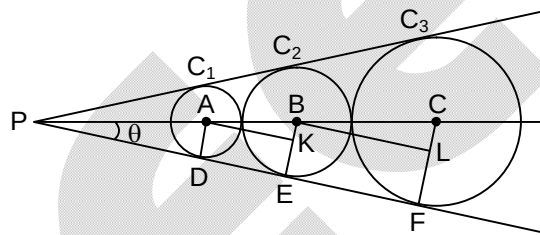
$$\text{Similarly, } BL = EF = 2\sqrt{r_2 r_3}$$

Let $\angle APD = \theta$ and $PD = x$, then

$$\tan \theta = \frac{AD}{PD} = \frac{BE}{PE} = \frac{CF}{PF}$$

$$\Rightarrow \frac{r_1}{x} = \frac{r_2}{x + 2\sqrt{r_1 r_2}} = \frac{r_3}{x + 2\sqrt{r_1 r_2} + 2\sqrt{r_2 r_3}} = k \text{ (say)}$$

$$r_1 = kx, r_2 = kx + 2k\sqrt{r_1 r_2}, r_3 = kx + 2k\sqrt{r_1 r_2} + 2k\sqrt{r_2 r_3}; \frac{r_2 - r_1}{2\sqrt{r_1 r_2}} = \frac{r_3 - r_2}{2\sqrt{r_2 r_3}} \Rightarrow r_2 = \sqrt{r_1 r_3} = \sqrt{4 \times 9} = 6$$



39. C

$$\text{Sol. } \frac{x^2}{25} + \frac{y^2}{9} = 1 \therefore S \text{ is } (4, 0)$$

Let image of S in any tangent be P(h, k),

Midpoint of SP will lie on auxiliary circle

($\because$ foot of perpendicular from focus on any tangent lies on auxiliary circle)

$$\therefore \left(\frac{h+4}{2}, \frac{k}{2}\right) \text{ lies on } x^2 + y^2 = 25 \therefore \text{locus of image is } (x+4)^2 + y^2 = 100$$

$$\text{It touches } ky - 2x + 22 = 0 \therefore \frac{|0+8+22|}{\sqrt{k^2+4}} = 10 \Rightarrow k^2 + 4 = 9 \Rightarrow k = \pm\sqrt{5}$$

40. C

$$\text{Sol. } \angle AOB = 90^\circ \text{ (as AB is diameter)}$$

$$\therefore \text{Slope of OA} \times \text{Slope of OB} = -1$$

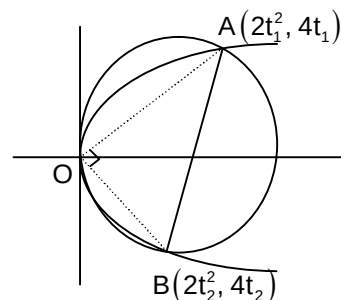
$$\frac{2}{t_1} \times \frac{2}{t_2} = -1 \Rightarrow t_1 t_2 = -4$$

Point of intersection of tangent at A and B is

$$C(2t_1 t_2, 2(t_1 + t_2)) = C(-8, 2(t_1 + t_2))$$

If perpendicular tangents are drawn from C to

$$y^2 = kx, \text{ then C must be on its directrix } \therefore k = 32$$



41. A, B, C

$$\text{Sol. } f(3) = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \text{ and } g(3) = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7}$$

$$\therefore \cos \frac{6\pi}{7} = \cos \frac{8\pi}{7} \therefore f(3) = g(3)$$

$$f(6) = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{10\pi}{7} + \cos \frac{12\pi}{7}$$

$$2\left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}\right) = 2f(3)$$

$$(\because \cos \theta = \cos(2\pi - \theta))$$

$$g(6) = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{16\pi}{7} + \cos \frac{32\pi}{7} + \cos \frac{64\pi}{7}$$

$$\left(\because \cos \frac{64\pi}{7} = \cos \frac{8\pi}{7}, \cos \frac{32\pi}{7} = \cos \frac{4\pi}{7}, \cos \frac{16\pi}{7} = \cos \frac{2\pi}{7}\right) \therefore g(6) = 2g(3)$$

42. A, B, C, D

Sol. B(0, 2) and C is (0, -2). Slope of PQ = $\frac{4}{\alpha - \beta}$

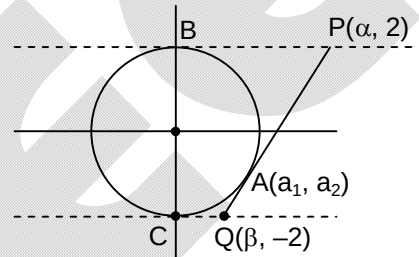
$$\text{Equation of PQ } y - 2 = \frac{4}{\alpha - \beta}(x - \alpha)$$

$$y(\alpha - \beta) - 2(\alpha - \beta) = 4x - 4\alpha$$

$$4x - y(\alpha - \beta) - 2(\alpha + \beta) = 0$$

$$\therefore \text{it touches } x^2 + y^2 = 4 \therefore \frac{|0 - 0 - 2(\alpha + \beta)|}{\sqrt{16 + (\alpha - \beta)^2}} = 2$$

$$\Rightarrow \alpha\beta = 4 \Rightarrow \text{If } \alpha = 4 \text{ slope of PQ} = \frac{4}{3} \therefore \text{Slope of OA} = -\frac{3}{4} = \frac{a_2}{a_1} = -\frac{3}{4}$$



43. A, D

$$\text{Sol. } X^2 = 4ky \Rightarrow 2x = 4k \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2k} \Rightarrow \left(\frac{dy}{dx}\right)_{(2k, k)} = 1$$

$$\therefore \text{Slope of normal at } (2k, k) = -1$$

$$\therefore \text{Equation of normal } y - k = -1(x - 2k) \Rightarrow x + y = 3k$$

$$\text{If } k = 1, \text{ equation of normal } x + y = 3$$

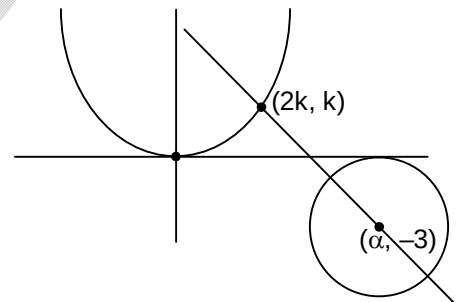
$$\therefore \text{Shortest distance lies along the common normal}$$

$$\therefore \text{It passes through } (\alpha, -3); \alpha - 3 = 3 \Rightarrow \alpha = 6$$

$$\Rightarrow \text{If } k = 2, x + y = 6 \text{ passes through } (\alpha, -3) \therefore \alpha = 9$$

$$\Rightarrow \text{If } \alpha = 20 \therefore x + y = 3k \text{ passes through } (20, -3)$$

$$\therefore k = \frac{17}{3}$$



44. B, C

$$\text{Sol. } s - p = \frac{s - q}{4} = \frac{s - r}{6} = k \text{ (say)} \therefore s = \frac{p + q + r}{2} \Rightarrow q + r - p = 2k; p + r - q = 8k; p + q - r = 12k$$

$$\Rightarrow p = 10k, r = 5k, q = 7k \therefore S = 11k \therefore \Delta = \sqrt{s(s-p)(s-q)(s-r)}$$

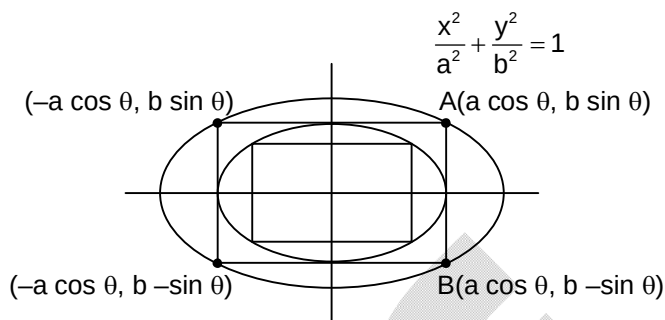
$$\Rightarrow \Delta = \sqrt{11k \times k \times 4k \times 6k} = 2\sqrt{66}k^2; \text{Circumradius} = \frac{pqr}{4\Delta} \Rightarrow \frac{175}{4\sqrt{66}} = \frac{5k \times 7k \times 10k}{4 \times 2\sqrt{66}k^2} \Rightarrow k = 1$$

$$\therefore \text{Area} = 2\sqrt{66} \text{ inradius} = 4(\text{Circumradius}) \sin \frac{P}{2} \sin \frac{Q}{2} \sin \frac{R}{2}$$

$$\Rightarrow 2\sqrt{11} = 4 \times \frac{175}{4\sqrt{66}} \times \sin \frac{P}{2} \sin \frac{Q}{2} \sin \frac{R}{2}$$

45. A, C, D

Sol. Area of inscribed rectangle
 $= 2a \cos \theta \times 2a \sin \theta = 2ab \sin 2\theta$
 (Here, $a = 4\sqrt{2}$; $b = 2\sqrt{2}$)
 Maximum value $= 2ab$ when $\theta = 45^\circ$
 $\therefore$ Area of $R_1 = 2ab$
 and sides of r_1 are $a\sqrt{2}$; $b\sqrt{2}$
 Let major axis and minor axis of E_2
 be $2a_2$, $2b_2$
 $\therefore 2a_2 = a\sqrt{2} \Rightarrow a_2 = \frac{a}{\sqrt{2}}$; $b_2 = \frac{b}{\sqrt{2}}$



$\therefore$ Area of $R_2 = 2 \times \frac{a}{\sqrt{2}} \times \frac{b}{\sqrt{2}} = ab$ and sides are of R_2 are a and b

:

Similarly area of $R_n = \frac{2ab}{2^{n-1}}$ and sides of R_n are $\frac{\sqrt{2}a}{(\sqrt{2})^{n-1}}$; $\frac{\sqrt{2}b}{(\sqrt{2})^{n-1}}$

Similarly major axis and minor axis of E_n are $\frac{2a}{(\sqrt{2})^{n-1}}$; $\frac{2b}{(\sqrt{2})^{n-1}}$

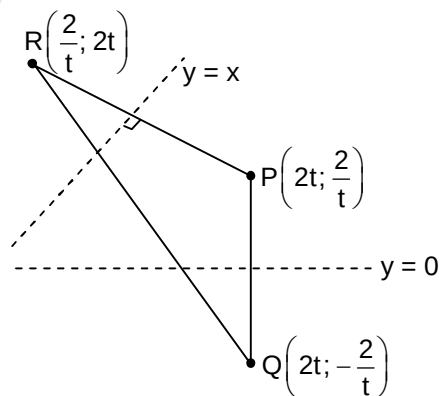
$\therefore$ Sum of areas of rectangles $R_1, R_2, R_3, \dots$ is $2ab + ab + \frac{ab}{2} + \frac{ab}{4} + \dots = \frac{2ab}{1 - \frac{1}{2}} = 4ab$

$\therefore$ Sum of areas is always less than $4 \times 4\sqrt{2} \times 2\sqrt{2} = 64$

Eccentricities does not change $\therefore E_{10} = E_{12}$

46. A, C, D

Sol. Centroid : $\left(\frac{4t + \frac{2}{t}}{3}, \frac{2t}{3} \right)$; $h = \frac{4t + \frac{2}{t}}{3}$; $k = \frac{2t}{3}$
 $\Rightarrow t = \frac{3k}{2} \therefore 3h = 6k + \frac{4}{3k} \Rightarrow 9hk = 18k^2 + 4$
 Circumcentre is $(0, 0)$
 $\therefore$ Orthocentre : $\left(4t + \frac{2}{t}, 2t \right)$
 $\therefore$ Locus $hk = 2k^2 + 4$



Section - B

47. 2

Sol. $T_r = \frac{1}{3^{r-1}} \sin^3(3^{r-1}\theta) = \frac{1}{4 \times 3^{r-1}} (3 \sin^3 3^{r-1}\theta - \sin 3^r \theta) = \frac{1}{4} \left(\frac{\sin 3^{r-1}\theta}{3^{r-2}} - \frac{\sin 3^r \theta}{3^{r-1}} \right) = \frac{1}{4} (V_r - V_{r+1})$
 $\therefore S_{10} = \frac{1}{4} (V_1 - V_2 + V_2 - V_3 + \dots + V_{10} - V_{11}) = \frac{1}{4} (V_1 - V_{11}) \Rightarrow f(\theta) = \frac{1}{4} \left(3 \sin \theta - \frac{\sin 3^{10} \theta}{3^9} \right)$

48. 8

Sol. Let E be $(5 \cos \theta, 3 \sin \theta) \Rightarrow 5 \cos \theta$

$$\Rightarrow c \Rightarrow \cos \theta = \frac{c}{5} \quad F \text{ is } (5 \cos \theta, -3 \sin \theta)$$

$$\text{Tangent at E: } \frac{x \cos \theta}{5} + \frac{y \sin \theta}{3} = 1$$

$$\text{Tangent at F: } \frac{x \cos \theta}{5} - \frac{y \sin \theta}{3} = 1 \therefore G \text{ is } \left(\frac{5}{\cos \theta}, 0 \right)$$

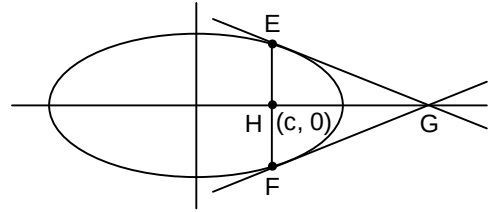
$$\text{Area of } \triangle EFG = \frac{1}{2} \times GH \times EF = \frac{1}{2} \times \left(\frac{5}{\cos \theta} - 5 \cos \theta \right) \times 6 \sin \theta$$

$$= \frac{1}{2} \times \left(\frac{25}{c} - c \right) \times 6 \left(\sqrt{1 - \frac{c^2}{25}} \right) = \frac{3}{5c} (25 - c^2)^{3/2} \Rightarrow \Delta'(c) = \frac{3}{5} \left(\frac{c \cdot \frac{3}{2} (25 - c^2)^{1/2} \times (-2c) - (25 - c^2)^{3/2}}{c^2} \right)$$

Which is always negative (-ve) $\therefore \Delta(c)$ is always decreasing

If $1 \leq c \leq 3$, then minimum value of $\Delta(c)$ is at $c = 3$

$$\Delta(3) = \frac{3}{5 \times 3} (16)^{3/2} = \frac{3}{5 \times 3} \times 2^6 = \frac{64}{5}$$



49. 5

Sol. $\therefore$ Area of $\triangle PSQ = (\sqrt{5} - 1)$ area of $\triangle RSQ$

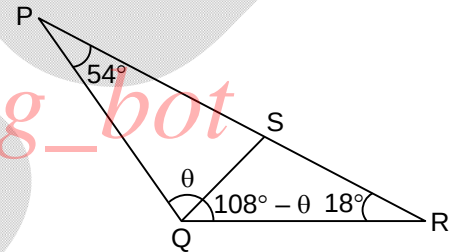
$$\frac{1}{2} \times PQ \times QS \times \sin \theta = (\sqrt{5} - 1) \times \frac{1}{2} \times QR \times QS \times \sin(108^\circ - \theta)$$

$$\therefore \frac{PQ}{\sin 18^\circ} = \frac{QR}{\sin 54^\circ}$$

$$\therefore \sin \theta \sin 18^\circ = (\sqrt{5} - 1) \sin 54^\circ \sin(108^\circ - \theta)$$

$$\therefore \sin \theta \frac{\sqrt{5} - 1}{4} = (\sqrt{5} - 1) \times \frac{\sqrt{5} + 1}{4} \times \sin(108^\circ - \theta) \Rightarrow \frac{\sin(108^\circ - \theta)}{\sin \theta} = \frac{1}{\sqrt{5} + 1} = \frac{\sqrt{5} - 1}{4}$$

$$\sin 108^\circ \cot \theta - \cos 108^\circ = \frac{\sqrt{5} - 1}{4} \Rightarrow \cot \theta = 0 \Rightarrow \theta = 90^\circ$$



50. 8

Sol. Let A be $(12 \sec \theta_1, 12 \tan \theta_1)$

B be $(12 \sec \theta_2, 12 \tan \theta_2)$

C be $(12 \sec \theta_3, 12 \tan \theta_3)$ and

D be $(12 \sec \theta_4, 12 \tan \theta_4)$

Let centroid of ABC be (h, k)

$$\Rightarrow h = 4 \sec \theta_1 + 4 \sec \theta_2 + 4 \sec \theta_3 \text{ and}$$

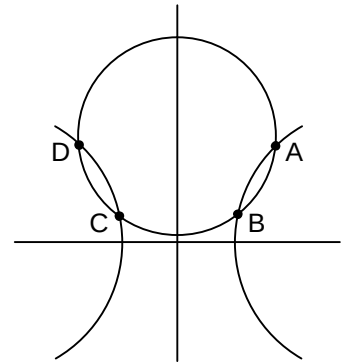
$$K = 4 \tan \theta_1 + 4 \tan \theta_2 + 4 \tan \theta_3$$

$$\frac{12 \sec \theta_1 + 12 \sec \theta_2 + 12 \sec \theta_3 + 12 \sec \theta_4}{4} = \frac{6 + 0}{2}$$

$$3h + 12 \sec \theta_4 = 12 \Rightarrow 4 \sec \theta_4 = 4 - h$$

$$\frac{12 \tan \theta_1 + 12 \tan \theta_2 + 12 \tan \theta_3 + 12 \tan \theta_4}{4} = \frac{12 + 0}{2}$$

$$3k + 12 \tan \theta_4 = 24 \Rightarrow 4 \tan \theta_4 = 8 - k \Rightarrow (h - 4)^2 - (k - 8)^2 = 16 \Rightarrow \alpha = 4, \beta = 8, \gamma = 4$$



51. 3

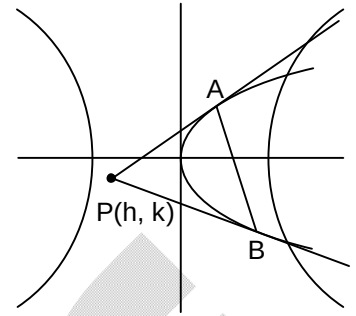
Sol. Equation of chord of contact AB $ky = 2(x + h)$

$$y = \frac{2x}{k} + \frac{2h}{k} \therefore \text{it touches } \frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$\frac{4h^2}{k^2} = \frac{9 \times 4}{k^2} - 16 ; h^2 + 4k^2 = 9$$

$$\Rightarrow \text{locus is } \frac{x^2}{9} + \frac{y^2}{9/4} = 1$$

$$\frac{9}{4} = 9(1 - e^2) \Rightarrow e = \frac{\sqrt{3}}{2}$$



52. 2

Sol. $\therefore$ It is equilateral $\Delta \therefore x_1 = \frac{a}{3}(\cos \alpha + \cos \beta + \cos \gamma)$; $y_1 = \frac{b}{3}(\sin \alpha + \sin \beta + \sin \gamma)$

$$\frac{9x_1^2}{a^2} + \frac{9y_1^2}{b^2} = (\cos \alpha + \cos \beta + \cos \gamma)^2 + (\sin \alpha + \sin \beta + \sin \gamma)^2$$

$$= 3 + 2(\cos \alpha \cos \beta + \cos \beta \cos \gamma + \cos \gamma \cos \alpha + \sin \alpha \sin \beta + \sin \beta \sin \alpha + \sin \gamma \sin \alpha)$$

53. 6

Sol. $\therefore C_1 : x^2 + y^2 = a^2 - b^2 \Rightarrow C_2 : x^2 + y^2 = \frac{a^2 + b^2}{3}$

Director circle of C_1 is $x^2 + y^2 = 2(a^2 - b^2) \Rightarrow 2(a^2 - b^2) = \frac{a^2 + b^2}{3}$

$$\Rightarrow 6a^2 - 6b^2 = a^2 + b^2 \Rightarrow 5a^2 = 7b^2$$

$$5a^2 = 7a^2(1 - e_1^2) \Rightarrow e_1^2 = \frac{2}{7} \text{ and } 5a^2 = 7a^2(e_2^2 - 1) \Rightarrow e_2^2 = \frac{12}{7}$$

54. 7

Sol. $(a - 2b)^2 + (a - 3c)^2 = 0 \Rightarrow a = 2b, a = 3c \Rightarrow b = \frac{a}{2}, c = \frac{a}{3}$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} \Rightarrow \cos B = \frac{a^2 + \frac{a^2}{9} - \frac{a^2}{4}}{2 \times a \times \frac{a}{3}} = \frac{31}{24}$$

FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Main)-2023

TEST DATE: 17-12-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

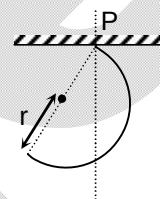
SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

- The wave length of light incident on a metal surface is reduced from 300 nm to 200 nm (both are less than threshold wavelength). What is the change in the stopping potential for photoelectrons emitted from the surface (take $h = 6.6 \times 10^{-34}$ J-sec)
(A) 0.0625 V (B) 1.0625 V
(C) 2.0625 V (D) 3.0625 V

- A uniform semicircular ring having mass m and radius r is hanging at one of its end (P) freely as shown in the given figure. The ring is slightly disturbed so that it oscillates in its own plane. The time period of oscillation of the ring is



- (A) $2\pi \sqrt{\frac{r}{g\left(1 + \frac{1}{\pi^2}\right)}}$ (B) $2\pi \sqrt{\frac{r}{g\left(1 - \frac{4}{\pi^2}\right)^{1/2}}}$
(C) $2\pi \sqrt{\frac{r}{g\left(1 - \frac{2}{\pi^2}\right)^{1/2}}}$ (D) $2\pi \sqrt{\frac{2r}{g\left(1 + \frac{4}{\pi^2}\right)^{1/2}}}$

- A number of little droplets of a liquid of surface tension T , density ρ , all of same radius r , combine to form a single drop of radius R . If the total energy released in this process is converted into kinetic energy of the bigger drop, velocity of the bigger drop is

- (A) $\sqrt{\frac{3T(R-r)}{R\rho}}$ (B) $\sqrt{\frac{6T(R-r)}{R\rho}}$
(C) $\sqrt{\frac{T(R-r)}{Rr}}$ (D) $\sqrt{\frac{2T(R-r)\rho}{R}}$

- In a screw gauge, 5 complete rotations of the screw cause it to move a linear distance of 0.25 cm and each rotation corresponds to one main scale division. There are 100 circular scale divisions. The thickness of a wire measured by this screw gauge gives a reading of 4 main scale divisions and 30 circular scale divisions. Assuming negligible zero error, what is the thickness (in mm) of the wire?

- (A) 2.15 (B) 3.15
(C) 1.25 (D) 2.25

- The refracting angle of a prism is A and the refractive index of the material of the prism is $\cot\left(\frac{A}{2}\right)$. The angle of minimum deviation is

- (A) $180^\circ - 3A$ (B) $180^\circ + 2A$
(C) $90^\circ - A$ (D) $180^\circ - 2A$

6. In a hypothetical system, a particle of mass m and charge $-3q$ is moving around a very heavy particle having charge q . Assume that Bohr's model is applicable to this system, then velocity of mass m in the first orbit is

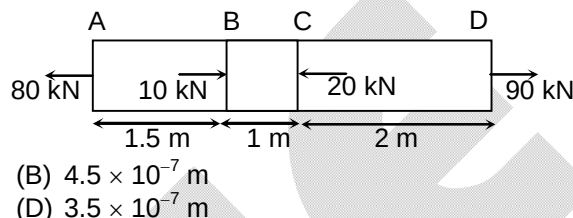
(A) $\frac{3q^2}{2\epsilon_0 h}$

(B) $\frac{3q^2}{4\epsilon_0 h}$

(C) $\frac{3q}{2\pi\epsilon_0 h}$

(D) $\frac{3q}{4\pi\epsilon_0 h}$

7. A steel rod of cross sectional area 1 m^2 is acted upon by forces shown in figure. The elongation of the length BC of the rod is ($Y = 2.0 \times 10^{11} \text{ N/m}^2$)



(A) $13 \times 10^{-7} \text{ m}$

(B) $4.5 \times 10^{-7} \text{ m}$

(C) $5 \times 10^{-7} \text{ m}$

(D) $3.5 \times 10^{-7} \text{ m}$

8. A body executes SHM of period 20 seconds. Its velocity is 5 cm/s , 2 seconds after it has passed the mean position. Find the amplitude (in cm) of the bob. ($\cos 36^\circ = 0.809$ and $\pi = \frac{22}{7}$)

(A) 15.67

(B) 17.67

(C) 19.67

(D) 21.67

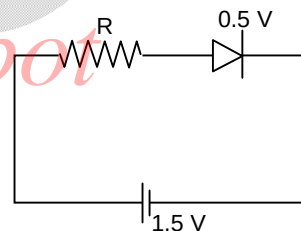
9. The diode used in the circuit shown in figure has a constant voltage drop of 0.5 V at all currents and a maximum power rating of 100 mW . What should be the value of the resistance R , connected in series with diode for obtaining maximum current?

(A) 5Ω

(B) 5.6Ω

(C) 6.7Ω

(D) 20Ω



10. A ray enters a glass slab at an angle θ from air and the refractive index varies along the slab as $\mu = \mu_0 - kt^2$, where t is the distance measured along the normal. μ_0 and k are positive constants. If the glass slab is sufficiently thick, how far along the normal will the ray go before it reflects back?

(A) $\frac{\mu_0 - \sin \theta}{k}$

(B) $\frac{\mu_0 + \sin \theta}{k}$

(C) $\sqrt{\frac{\mu_0 - \sin \theta}{k}}$

(D) $\sqrt{\frac{\mu_0 + \sin \theta}{k}}$

11. The electric field part of an electromagnetic wave in a medium is represented by $E_x = 0$

$$E_y = 2.5 \text{ N/C} \cos \left[(2\pi \times 10^6 \text{ rad/s})t - (\pi \times 10^{-2} \text{ rad/m})x \right]$$

and $E_z = 0$

Then the wave is

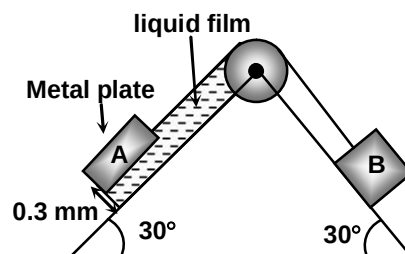
(A) moving along x direction with frequency 10^6 Hz and wavelength 100 m

(B) moving along x direction with frequency 10^6 Hz and wavelength 200 m

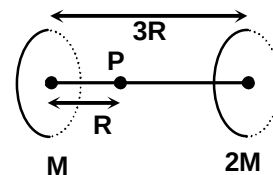
(C) moving along $-x$ direction with frequency 10^6 Hz and wavelength 200 m

(D) moving along y direction with a frequency $2\pi \times 10^6 \text{ Hz}$ and wavelength 200 m .

12. In a Fraunhofer diffraction experiment at a single slit using a light of wavelength 400 nm, the first minimum is formed at an angle 30° . The direction θ of the first secondary maximum is given by:
- (A) $\sin^{-1}\left(\frac{2}{3}\right)$ (B) $\sin^{-1}\left(\frac{3}{4}\right)$
 (C) $\sin^{-1}\left(\frac{1}{4}\right)$ (D) $\sin^{-1}\left(\frac{2}{5}\right)$
13. A cylindrical vessel filled with water upto a height h weighs w and is resting on a horizontal plane. The side of wall of the vessel has a plugged circular hole touching the bottom. The coefficient of friction between the bottom of the vessel and the plane is μ . The minimum diameter of the hole so that the vessel just begins to move on the floor after the removal of the plug is:
- (A) $\sqrt{\frac{\pi\mu W}{\rho hg}}$ (B) $\sqrt{\frac{\mu W}{2\pi\rho hg}}$
 (C) $\sqrt{\frac{\mu W}{\pi\rho hg}}$ (D) $\sqrt{\frac{2\mu W}{\pi\rho hg}}$
14. The transverse displacement $y(x, t)$ of a wave on a string is given by $y(x, t) = e^{-(ax^2 + bt^2 + 2\sqrt{ab}xt)}$. This represents a
- (A) wave moving along +x-axis with speed $\sqrt{\frac{a}{b}}$
 (B) wave moving along -x-axis with speed $\sqrt{\frac{b}{a}}$
 (C) standing wave of frequency $\sqrt{b}$
 (D) standing wave of frequency $\frac{1}{\sqrt{b}}$
15. A body is fired with a velocity of magnitude $\sqrt{gR} < v < \sqrt{2gR}$ at an angle of 30° with the radius vector of the earth. If at the highest point, the speed of the body is $\frac{v}{4}$, the maximum height attained by the body is equal to (R = radius of earth and use the appropriate assumptions)
- (A) $\frac{v^2}{8g}$ (B) R
 (C) $R\sqrt{2}$ (D) $2R$
16. A metal plate of area 0.10 m^2 is connected to a 0.04 kg mass via a string that passes over an ideal pulley (considered massless and frictionless). A liquid with a film of thickness of 0.3 mm is placed between the plate and the table. When the system is released the plate moves up with constant speed of 0.085 m/s . Find the approximate value of coefficient of viscosity of the liquid. (mass of metal plate is 0.02 kg , $g = 9.8 \text{ m/s}^2$)



17. Two rings having masses M and $2M$, respectively, having the same radius are placed coaxially as shown in the figure. If the mass distribution on both the rings is non-uniform, then the gravitational potential at point P is

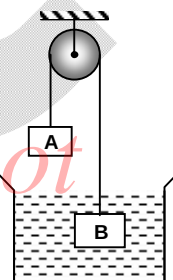


- (A) $-\frac{GM}{R} \left[\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{5}} \right]$
 (B) $-\frac{GM}{R} \left[1 + \frac{1}{\sqrt{2}} \right]$
 (C) zero
 (D) cannot be determined from the given information

18. A heavenly body is receding from earth such that the fractional change in λ is 1, then its velocity is

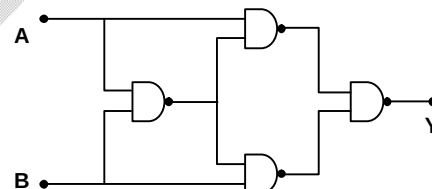
- (A) c
 (B) $\frac{3c}{5}$
 (C) $\frac{c}{5}$
 (D) $\frac{2c}{5}$

19. In the given figure $m_B = 3m$, density of liquid is ρ and density of block B is 2ρ . The system is released from rest so that block B moves up when in liquid and moves down when out of liquid with the same acceleration. Find the mass of block A.



- (A) $\frac{7}{4}m$
 (B) $2m$
 (C) $\frac{9}{7}m$
 (D) $\frac{9}{4}m$

20. Truth table for a system of four NAND gates as shown is



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(C)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(D)

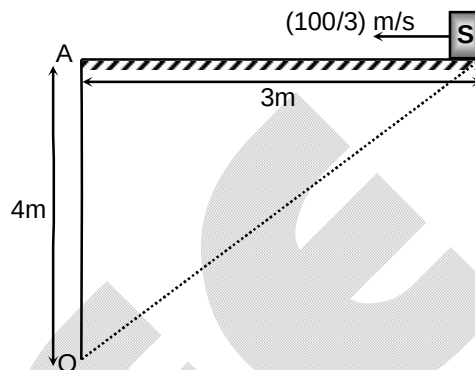
A	B	Y
0	0	1
0	1	0
1	1	0
1	1	1

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A source of sound S is travelling at $\frac{100}{3}$ m/s along a road towards a point A. A person standing at point O on a road perpendicular to AS hears a sound of frequency f' corresponding to the sound emitted at the instant when the source is 3m away from A. The distance of O from A is 4m. If the original frequency is 640 Hz, then find the value of f' in Hz. (velocity of sound is 340 m/s)



22. A metallic sphere of radius 21 cm is kept in the path of a parallel beam of light of large aperture of intensity $\frac{1}{110}$ W/m². The force exerted by beam on the sphere is $k \times 10^{-13}$ N. Find k. ($\pi = \frac{22}{7}$ and light is completely absorbed by sphere)
23. An astronomical telescope has objective and eyepiece of focal length 40 cm and 4 cm respectively. To view an object 200 cm away from the objective, at what distance the lenses must be separated (in cm)?
24. Cobalt-57 is radioactive, emitting β -particles. The half life for this is 270 days. If 100 mg of this is kept in an open container, then what mass (in mg) of Cobalt-57 will remain after 540 days.
25. A clear sheet of polaroid is placed on the top of similar sheet so that their axes make an angle $\sin^{-1}\left(\frac{3}{5}\right)$ with each other. The ratio of intensity of the emergent light to that of unpolarised incident light is $\frac{k}{25}$. Find k.
26. A narrow parallel beam of light is incident on a transparent sphere of refractive index $\frac{4}{n}$. If the beam finally gets focused at a point situated at a distance = $2 \times$ (radius of sphere) from the centre of the sphere, then find n.
27. A hydrogen like atom of atomic number Z is in excited state of quantum number $2n$. It can emit a maximum energy photon of 204 eV. If it makes a transition to the quantum state n, a photon of energy 40.8 eV is emitted. Calculate the atomic number Z. Ground state energy of hydrogen atom is -13.6 eV.
28. Image of an object approaching convex mirror of radius of curvature 20 m along its optical axis is observed to move from $\frac{25}{3}$ m to $\frac{50}{7}$ m in 30 sec. If the object is moving uniformly, then it's speed is n km/h. Find the value of n.

29. An open pipe is suddenly closed at one end with the result that the frequency of third harmonic of the closed pipe is found to be higher by 100 Hz than the fundamental frequency of the open pipe. Find the fundamental frequency of the open pipe in Hz.
30. In a double slit arrangement fringes are produced using light of wavelength $4800 \text{ \AA}$. One slit is covered by a thin plate of glass of refractive index 1.4 and the other with another glass plate of same thickness but of refractive index 1.7. By doing so the central bright shifts to original fifth bright fringe from centre. Thickness of glass plate is $k \text{ }\mu\text{m}$. Find the value of k .

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Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

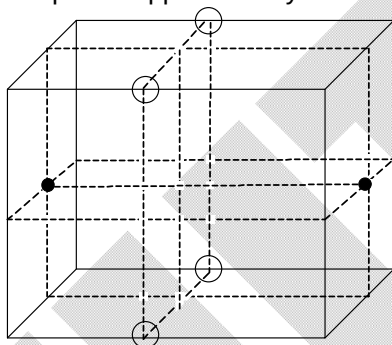
31. The products obtained when I_2 reacts with hot and concentrated aqueous NaOH are
 (A) IO^- and IO_3^-
 (B) IO_2^- and IO_3^-
 (C) I^- and IO^-
 (D) I^- and IO_3^-
32. Which of the following product is not produced during the hydrolysis of XeF_4 ?
 (A) XeO_3
 (B) O_2
 (C) XeO_2F_2
 (D) Xe
33. Four successive members of first row transition element are listed below. Which one of them unexpectedly have highest $E_{M^{3+}/M^{2+}}^0$ value?
 (A) Mn (Z = 25)
 (B) Fe (Z = 26)
 (C) V (Z = 23)
 (D) Cr (Z = 24)
34. In context of actinoids, which of the following statements is not correct?
 (A) The actinoids are radioactive elements which make their study more difficult.
 (B) There is a gradual decrease in the radii of the members with increasing atomic number in the series.
 (C) The magnetic properties of actinoids can not be explained easily, as they are more complex.
 (D) As compare to lanthanoid compounds, actinoid compounds are less basic.
35. Which of the following has a square planar geometry?
 (A) $[FeCl_4]^{2-}$
 (B) $[PdCl_4]^{2-}$
 (C) $[NiCl_4]^{2-}$
 (D) $[CoCl_4]^{2-}$
36. Which of the following statements is correct?
 (A) NO_2 is a reddish-orange gas and it does not play any role in photochemical smog.
 (B) SO_2 does not affect the formation of chlorophyll in plants.
 (C) CO combines with haemoglobin to form carboxyhaemoglobin which is more stable than oxyhaemoglobin.
 (D) SO_2 is more harmful than SO_3 .

37. The reaction that is involved in the ozone layer depletion mechanism in the stratosphere is
- (A) $\text{NO} + \frac{1}{2} \text{O}_2 \xrightarrow{h\nu} \text{NO}_2$
- (B) $\text{CH}_4 + \text{O}_2 \longrightarrow \text{CH}_2\text{O} + \text{H}_2\text{O}$
- (C) $\text{Cl}^\bullet + \text{CH}_4 \longrightarrow \dot{\text{C}}\text{H}_3 + \text{HCl}$
- (D) $\text{ClO}^\bullet + \text{O} \longrightarrow \text{Cl}^\bullet + \text{O}_2$
38. Which of the following chromates is not yellow in colour?
- (A) PbCrO_4
- (B) SrCrO_4
- (C) Ag_2CrO_4
- (D) BaCrO_4
39. Which of the following statements is correct?
- (A) Dettol is a mixture of chloroxynol and bithionol.
- (B) 2-3% solution of chloroform in alcohol-water is known as tincture of iodine.
- (C) Boric acid in concentrated aqueous solution is weak antiseptic for eyes.
- (D) 0.2% solution of phenol is an antiseptic while 1% solution of phenol is disinfectant.
40. Which of the following antibiotic is bacteriostatic?
- (A) Penicillin
- (B) Tetracyclin
- (C) Aminoglycosides
- (D) Ofloxacin
41. Multimolecular colloid is present in
- (A) Sol of starch
- (B) Soap solution
- (C) Sol of sulphur
- (D) Sol of protein
42. 100 ml of 0.6 M acetic acid is shaken with 4 g activated carbon. The final concentration of the solution after adsorption is 0.4 M. The amount of acetic acid (in g) adsorbed per gram of carbon is.....?
- (A) 0.6
- (B) 0.3
- (C) 1.2
- (D) 0.9
43. Which of the following statements is incorrect?
- (A) Average translational kinetic energy of a molecule in a gas at given temperature does not depend on its molecular mass.
- (B) Root mean square speed of a molecule in a gas at a give temperature is halved when its molar mass is increased four times.
- (C) Average translational kinetic energy of a molecule in a gas at a given temperature is doubled when its molar mass is increased twice.
- (D) Root mean square speed of a molecule in a gas is doubled when its temperature is increased four times.

44. When 100 ml of 1.0 M HCl was mixed with 100 ml of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of 5.8°C was measured for the beaker and its constants. If the enthalpy of neutralization of a strong acid with a strong base is $-58.0 \text{ kJ mol}^{-1}$, the calorimeter constant in (kJ K^{-1}) is.....

(Given: Heat capacity of solution is $4.2 \text{ J g}^{-1} \text{ K}^{-1}$ and density of solution is 1 g mL^{-1})

- (A) 0.84
(B) 1
(C) 0.42
(D) 0.16
45. Which of the following ores has carbonate anion?
(A) Diaspore
(B) Calamine
(C) Willemite
(D) Argentite
46. Extraction of aluminium from bauxite does not involve which of the following steps:
(A) Crushing followed by concentration of the ore is done by Leaching process.
(B) Extraction of aluminium from concentrated ore is done by Hall Heroult process.
(C) Extraction of aluminium from concentrated ore is done by carbon.
(D) In reduction process CaF_2 is mixed with concentrated ore to reduce the melting point of the mixture and brings conductivity.
47. The cubic unit cell structure of compound containing cation A and anion B is shown below. The size of cation is smaller than anion and anion-anions are in contact. The packing efficiency of compound approximately is



- (A) 73
(B) 55
(C) 85
(D) 25
48. The increase in the molar conductivity of HCl with dilution is due to
(A) Increase in the self ionization of water.
(B) Hydrolysis of HCl with dilution.
(C) Decrease in the self ionization of water.
(D) Decrease in the interionic forces by dilution.
49. The moles of $\text{K}_2\text{Cr}_2\text{O}_7$ are required to react completely with 2.4 mol of CuS in hot acidic medium
(A) 1.8
(B) 3.2
(C) 1.6
(D) 6.4

50. The value of n in the molecular formula $\text{Be}_n\text{Al}_2\text{Si}_6\text{O}_{18}$ is
 (A) 2
 (B) 5
 (C) 4
 (D) 3

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. The reaction of white phosphorous and excess of dry chlorine gas gives a compound Q. The number of moles of HCl produced by the complete hydrolysis of 1 mole of Q is _____
52. An aqueous solution is prepared by dissolving 0.2 mole of an ionic salt in 1.86 kg of water at 27°C . The salt is only 80% dissociated in the solution. The freezing point of the solution is -0.52°C . The number of ions present per formula unit of ionic salt is _____ (Given molal depression constant of water (K_f) is $1.86 \text{ K kg mol}^{-1}$ and freezing point of water is 0°C).
53. The electrode potential of the following cell is $-x$ at 300 K.
 $\text{Pt(s)} | \text{H}_2(\text{g})(4 \text{ atm}) / \text{CH}_3\text{COOH}(\text{aq})(10^{-1} \text{ M}) || \text{NaOH}(\text{aq})(10^{-3} \text{ M}) / \text{H}_2(\text{g})(1 \text{ atm}) / \text{Pt(s)}$.
 The value $1000x$ is _____
 [Given: $\frac{2.303 RT}{F}$ at 300 K = 0.06, $\log 2 = 0.3$, K_a for acetic acid is 10^{-5}]
54. A crystalline solid of a pure substance X has body centred cubic structure with a cell edge of 800 pm. The number of atoms present in 512 g of crystal X is $N \times 10^{22}$ if the density of the substance in the crystal is 8 g cm^{-3} . The value of N is _____
55. Reaction of aqueous solution of KI with ozone produced a paramagnetic product Y. The spin only magnetic moment (in B.M.) of Y in nearest integer is _____
56. 60 ml of $x \text{ M AgNO}_3$ required for complete precipitation of Br^- ions present in 15 ml of 0.02 M solution of $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{Br}_2$ as silver bromide. The value $100x$ is _____
57. The enthalpy change (ΔH) for the following reaction at temperature 25°C is $-2573 \text{ kJ mol}^{-1}$.
 $2\text{C}_2\text{H}_2(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 4\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$
 The average bond enthalpy of C – H bond is $x \text{ kJ mol}^{-1}$, if the average bond enthalpy of $\text{C} \equiv \text{C}$, $\text{O} = \text{O}$, $\text{C} = \text{O}$ and $\text{O} - \text{H}$ bonds are 814, 499, 724 and 640 kJ mol^{-1} respectively at temperature 25°C . The x is....
58. On heating acidified NaCl with orange solid gives orange-red vapours. On passing these orange red vapours through NaOH solution turns it into yellow solution. This yellow solution gives yellow precipitate when the solution of acetic acid and lead acetate is added to this solution. The oxidation state of the element which has highest oxidation state in yellow precipitate is _____

59. In the chemical reaction between stoichiometric quantities of KMnO_4 and MnSO_4 in weakly alkaline solution, the number of moles of MnO_2 formed is Y. If 4.2 moles of KMnO_4 are consumed. The value of 2Y is.....
60. Ammonia is formed by reaction of ammonium chloride and calcium hydroxide. What is the volume of ammonia gas in litres (L) produced at 600 K and 0.82 atm pressure when 10.7 g of ammonium chloride and 10 ml of 5.0 M calcium hydroxide are mixed together for the reaction? (Given atomic mass in g mol^{-1} H, N, Cl are respectively 1, 14 and 35.5 respectively, $R = 0.082 \text{ atm L mol}^{-1} \text{ K}^{-1}$)

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Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

61. A function f is defined on the complex number by $f(z) = (a + ib)z$, where a and b are positive number. This function has the property that the image of each point in the complex plane is equidistant from that point and the origin. If $|a + ib| = 8$, then b^2 is
- (A) $\frac{255}{4}$ (B) $\frac{245}{4}$
 (C) $\frac{235}{4}$ (D) $\frac{225}{4}$
62. The smallest prime that is the fifth term of an increasing arithmetic sequence all four preceding terms also a prime is
- (A) 31 (B) 29
 (C) 37 (D) not possible
63. The largest two digit prime factor of ${}^{200}C_{100}$ is
- (A) 61 (B) 59
 (C) 17 (D) 60
64. The sum of all positive integers n for which $n^2 - 19n + 99$ is a perfect square is
- (A) 36 (B) 37
 (C) 39 (D) 38
65. Initially A, B and C had a total of 444 peanuts. C had the most peanuts and A had the least. The numbers of peanuts that each person had formed a G.P. A eats 5 of his peanuts, B eats 9 of its peanuts and C eats 25. Now the three numbers of peanuts each has forms an A.P. Find the number of peanuts A had initially
- (A) 108 (B) 106
 (C) 300 (D) 144
66. There is a 40% chance of rain on Saturday and a 30% chance of rain on Sunday. However, it is twice as likely to rain on Sunday if it rains on Saturday than if it does not rain on Saturday. The probability that it rains at least one day this weekend is
- (A) $\frac{3}{10}$ (B) $\frac{3}{14}$
 (C) $\frac{37}{70}$ (D) $\frac{33}{70}$
67. For polynomial $P(x) = 1 - \frac{1}{3}x + \frac{1}{6}x^2$, define $Q(x) = P(x).P(x^3).P(x^5).P(x^7).P(x^9) = \sum_{i=0}^{50} a_i x^i$, then
- $\sum_{i=0}^{50} |a_i| =$
- (A) $\frac{275}{37}$ (B) $\frac{243}{32}$
 (C) $\frac{243}{39}$ (D) $\frac{245}{37}$

68. Let x, y and z be real numbers satisfying the system
 $\log_2(xyz - 3 + \log_5 x) = 5$
 $\log_3(xyz - 3 + \log_5 y) = 4$
 $\log_4(xyz - 3 + \log_5 z) = 4$
 then the value of $|\log_5 x| + |\log_5 y| + |\log_5 z|$ is
 (A) 376 (B) 378
 (C) 380 (D) 382
69. Let a and b be positive integers satisfy $\frac{ab+1}{a+b} < \frac{3}{2}$, then the maximum possible value of $\frac{a^3b^3+1}{a^3+b^3}$ is
 (A) $\frac{31}{9}$ (B) $\frac{65}{9}$
 (C) $\frac{65}{16}$ (D) $\frac{31}{5}$
70. In three dimensional space $f(x, y, z) = xy + xz$, then locus of all points which satisfy the equation $f(x, y, z) = 0$ is
 (A) pair of perpendicular lines
 (B) pair of perpendicular planes
 (C) pair of a line and a plane which are perpendicular to each other
 (D) pair of line and plane which are parallel to each other
71. Out of all possible 8 digit numbers formed using all the digits 0, 0, 1, 1, 2, 3, 4, 4 a number is randomly selected, probability that the selected number is odd is
 (A) $\frac{5}{7}$ (B) $\frac{5}{9}$
 (C) $\frac{5}{11}$ (D) $\frac{5}{14}$
72. If α, β, γ are the roots of $x^3 - 3x + 1 = 0$, then the equation whose roots are $\alpha - \frac{1}{\beta\gamma}, \beta - \frac{1}{\gamma\alpha}, \gamma - \frac{1}{\alpha\beta}$ is
 (A) $x^3 - 3x + 8 = 0$ (B) $x^3 - 6x + 8 = 0$
 (C) $x^3 - 9x + 8 = 0$ (D) $x^3 - 12x + 8 = 0$
73. Three Arithmetic mean, three Geometric mean and three Harmonic mean are inserted between 1 and 5. The cubic equation whose roots are 3rd A.M, 2nd G.M, and 1st H.M is
 (A) $x^3 - \frac{1}{4}(21 + 4\sqrt{5})x^2 + 5x - 5\sqrt{5} = 0$
 (B) $x^3 - \frac{1}{4}(21 + 4\sqrt{5})x^2 + \left(5 + \frac{21}{4}\sqrt{5}\right)x - 5\sqrt{5} = 0$
 (C) $4x^3 - (21 + 4\sqrt{5})x^2 + (15 + 21\sqrt{5})x - 20\sqrt{5} = 0$
 (D) $4x^3 - (21 + 4\sqrt{5})x^2 + (15 + 21\sqrt{5})x - 5\sqrt{5} = 0$
74. If lines $\frac{x-1}{2} = \frac{y-2}{x_1} = \frac{z-3}{x_2}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ lies in the same plane then for equation $x_1t^2 + (x_2 + 2)t + a = 0$
 (A) $2x_1 - x_2 = 1$ (B) sum of roots of above equation = -2
 (C) $2x_1 + x_2 = -4$ (D) sum of root is 0

75. Let $\vec{a}$ and $\vec{b}$ are two perpendicular unit vectors. If $\vec{c}$ is another unit vector equally inclined at angle θ to the vectors $\vec{a}$ and $\vec{b}$, then the set of exhaustive values of θ is
- (A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left[0, \frac{\pi}{4}\right]$
 (C) $\left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$ (D) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$
76. If $x + y + z = 1$ and $0 \leq x, y, z \leq 1$, then the maximum value of expression $27(xy + yz + zx - 2xyz)$ is equal to
- (A) 7 (B) 10
 (C) 8 (D) 9
77. Matrix A is given by $A = \begin{bmatrix} 6 & 11 \\ 2 & 4 \end{bmatrix}$, then the determinant of $A^{2005} - 6A^{2004}$ is
- (A) 2^{2006} (B) $(-11) 2^{2005}$
 (C) $-2^{2005} \cdot 7$ (D) $(-9) 2^{2004}$
78. Let a, b, c be distinct cube roots of unity and $\Delta = \begin{vmatrix} a^2 + b^2 & c^2 & c^2 \\ a^2 & b^2 + c^2 & a^2 \\ b^2 & b^2 & c^2 + a^2 \end{vmatrix}$, then
- (A) $\operatorname{Re}(\Delta) = 0$ (B) $\operatorname{Im}(\Delta) = 0$
 (C) $\operatorname{Re}(\Delta) + \operatorname{Im}(\Delta) = 0$ (D) $\operatorname{Re}(\Delta)\operatorname{Im}(\Delta) = 4$
79. An urn contains 4 green balls and 6 blue balls. A second urn contains 16 green balls and N blue balls. A single ball is drawn at random from each urn. The probability that both balls are of the same colour is 0.58, then $(N - 140)$ is
- (A) 5 (B) 4
 (C) 6 (D) 7
80. A and B are two square matrix (all the entries of A and B are real such that $A^2B = BA$ and if $(AB)^{10} = A^k B^{10}$ is true for all square matrices of same order for the value of k, then value of $k - 1020$ is
- (A) 5 (B) 4
 (C) 3 (D) 2

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $f(x) = x^4 + ax^3 + bx^2 + cx + d$ be a polynomial whose roots are all negative integer, if $a + b + c + d = 2009$, then d is
82. If x and y are real numbers such that $\frac{(x-4)^2}{4} + \frac{y^2}{9} = 1$, then the largest possible value of $\frac{x^2}{4} + \frac{y^2}{9}$ is

83. Let $p(x)$ be the polynomial of degree 4 with roots 1, 2, 3, 4 and leading coefficient 1. Let $q(x)$ be the polynomial of degree 4 with roots 1, $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ and leading coefficient 1, then $\lim_{x \rightarrow 1} \frac{p(x)}{q(x)}$ is equal to k , then $|k|$ is
84. Kiran is deciding whether to visit Russia or Ukrain for the holidays. She makes her decision by rolling a regular 6-sided die. If she gets 1 or 2 she goes to Ukrain, if she rolls a 3, 4 or 5 she goes to Russia. If she rolls a 6 she rolls again, the probability that she goes to Ukrain is p , then $5p$ is
85. There are real numbers a, b, c and d such that -20 is a root of $x^3 + ax + b = 0$ and -21 is a root of $x^3 + cx^2 + d = 0$. These two polynomials share a complex root $m + i\sqrt{n}$, where m, n are positive integers and $i = \sqrt{-1}$, then $\frac{n}{4m}$ is
86. If $A = \begin{bmatrix} a & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 1 & b \end{bmatrix}$, $B = \begin{bmatrix} 2 & -5 & d \\ -8 & -2 & 7 \\ c & 3 & -5 \end{bmatrix}$ and $B = \text{adj } A$ and $|B| =$
87. If the plane containing x -axis and the point $(1, 1, 1)$ is parallel to the line $\frac{x+1}{2k} = \frac{y-1}{k} = \frac{z}{-3}$, then the value of $|k|$ is
88. The number of ways of arranging the letters of the word **ALGEBRA** in which either vowels or consonants, but not both appear in alphabetical order is
89. If three successive terms of a G.P. with common ratio $r (r > 1)$ form the sides of a $\triangle ABC$ and $[r]$ denotes G.I.F., then $[r] + [-r] + 5$ is
90. Let a three term strictly increasing arithmetic sequence of integers are one such that the sum of the squares of the three terms equals the product of the middle term and the square of the common difference, then the sum of third term of such sequence is p , then $(p - 25)$ is

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C

Sol. $eV_1 = \frac{hc}{\lambda_1} - \phi$

$$eV_2 = \frac{hc}{\lambda_2} - \phi$$

$$e(V_2 - V_1) = hc \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)$$

$$\Rightarrow (V_2 - V_1) = \frac{hc}{e} \left(\frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2} \right)$$

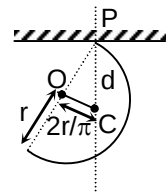
$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19}} \times \frac{100}{6 \times 10^{-5}} = 2.0625 \text{ volt}$$

2. D

Sol. Use $T = 2\pi \sqrt{\frac{I_P}{mgd}}$

$$\text{Where } d = \sqrt{r^2 + \left(\frac{2r}{\pi} \right)^2} \quad \dots(i)$$

$$I_0 = mr^2 = I_C + m \left(\frac{2r}{\pi} \right)^2 \quad \dots(ii)$$



$$\text{and } I_p = I_C + md^2$$

$$I_p = I_C + m \left\{ r^2 + \left(\frac{2r}{\pi} \right)^2 \right\} \quad \dots(iii)$$

From (ii) and (iii)

$$I_p = 2mr^2$$

$$T = 2\pi \sqrt{\frac{2mr^2}{mgr \left(1 + \frac{4}{\pi^2} \right)^{1/2}}}$$

$$T = 2\pi \sqrt{\frac{2r}{g \left(1 + \frac{4}{\pi^2} \right)^{1/2}}}$$

3.

Sol.

$$B \quad R^3 = Nr^3$$

$$\Rightarrow N = \frac{R^3}{r^3}$$

$$\text{Energy released} = T \times 4\pi(Nr^2 - R^2)$$

$$= 4\pi T \left(\frac{R^3}{r^3} r^2 - R^2 \right) = 4\pi TR^2 \left(\frac{R-r}{r} \right)$$

$$\text{So, } \frac{1}{2}mv^2 = 4\pi TR^2 \left(\frac{R-r}{r} \right)$$

$$\frac{1}{2} \times \frac{4}{3} \pi R^3 \rho v^2 = 4\pi TR^2 \left(\frac{R-r}{r} \right)$$

$$v = \sqrt{\frac{6T(R-r)}{R\rho}}$$

4.

A

$$\text{Sol. Least count} = \frac{0.25}{5 \times 100} \text{ cm} = 5 \times 10^{-4} \text{ cm}$$

$$\text{Reading} = 4 \times 0.05 \text{ cm} + 30 \times 5 \times 10^{-4} \text{ cm} \\ = 2.15 \text{ mm}$$

5.

D

$$\text{Sol. Since, } \mu = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

$$\cot \left(\frac{A}{2} \right) = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

$$\cos \left(\frac{A}{2} \right) = \sin \left(\frac{A + \delta_m}{2} \right)$$

$$\sin\left(\frac{\pi}{2} - \frac{A}{2}\right) = \sin\left(\frac{A + \delta_m}{2}\right)$$

$$\Rightarrow \delta_m = \pi - 2A$$

6. A

Sol. $\frac{mv^2}{r} = \frac{3q^2}{4\pi\epsilon_0 r^2} \Rightarrow mvr = \frac{3q^2}{4\pi\epsilon_0 v} \quad \dots(i)$

and $\frac{nh}{2\pi} = mvr \quad \dots(ii)$

using (i) and (ii) and putting $n = 1$

$$\frac{h}{2\pi} = \frac{3q^2}{4\pi\epsilon_0 v} \Rightarrow v = \frac{3q^2}{2\epsilon_0 h}$$

7. D

Sol. The bar is at equilibrium. The net force from right or left of a section of BC is 70 KN.

$$\Delta\ell = \frac{F\ell}{AY} = \frac{(70 \times 10^3) \times 1}{1 \times 2 \times 10^{11}}$$

$$\Delta\ell = 3.5 \times 10^{-7} \text{ m}$$

8. C

Sol. $T = 20 \text{ sec}$
at $t = 2$, $v = 5 \text{ cm/s}$
 $v = A\omega \cos \omega t$

$$v = A \times (2\pi f) \cos\left(\frac{2\pi}{T} t\right)$$

$$5 = A \times \left(\frac{2\pi}{20}\right) \cos\left(\frac{2\pi}{20} t\right)$$

$$\Rightarrow \frac{A\pi}{10} \cos 36^\circ = 5$$

9. A

Sol. The current through circuit $i = \frac{P}{V} = \frac{100 \times 10^{-3}}{0.5} = 0.2 \text{ A}$

Voltage drop across resistance $= 1.5 - 0.5 = 1 \text{ V}$

$$\Rightarrow R = \frac{1}{0.2} = 5\Omega$$

10. C

Sol. As t is increasing μ is decreasing, so refraction is taking place from denser to rarer medium as light is proceeding in the medium. Using continuous varying refractive index, let the ray reflect

back after travelling $t = t_0$ along the normal, then at this location, angle of refraction is $\frac{\pi}{2}$.

$$\text{So, } 1 \times \sin \theta = (\mu_0 - kt_0^2) \sin\left(\frac{\pi}{2}\right)$$

$$\Rightarrow t_0 = \sqrt{\frac{\mu_0 - \sin \theta}{k}}$$

11.

B

Sol. Compare the given equation with standard equation

$$E_y = E_0 \cos(\omega t - kx)$$

$$\therefore \omega = 2\pi \times 10^6$$

$$2\pi v = 2\pi \times 10^6$$

$$\Rightarrow v = 10^6 \text{ Hz}$$

$$k = \pi \times 10^{-2}$$

$$\text{or } \frac{2\pi}{\lambda} = \pi \times 10^{-2}$$

$$\text{or } \lambda = 200 \text{ m}$$

12.

B

Sol. For first diffraction minimum

$$a \sin \theta = \lambda$$

$$\Rightarrow a = \frac{\lambda}{\sin \theta}$$

For first secondary maximum, we use

$$a \sin \theta' = \frac{3\lambda}{2}$$

$$\sin \theta' = \frac{3\lambda}{2} \times \frac{1}{a} = \frac{3\lambda}{2} \times \frac{\sin \theta}{\lambda}$$

$$= \frac{3}{2} \times \sin 30^\circ = \frac{3}{4}$$

$$\theta' = \sin^{-1}\left(\frac{3}{4}\right)$$

13.

D

Sol. u = velocity of water through the hole = $\sqrt{2gh}$

R = rate of mass flow = ρAu

[A = area of the hole]

F = reaction force on the wall

$$= R(u - 0) = \rho Au^2$$

For cylinder to just move

$$\mu w = \rho Au^2 = \frac{\rho \pi d^2}{4} (\sqrt{2gh})^2$$

$$\therefore d = \sqrt{\frac{2\mu w}{\pi \rho h g}}$$

14.

B

Sol. General equation of travelling wave is

$$y = f(ax \pm bt) = f(x \pm vt)$$

$$\therefore y = e^{-(ax^2 + bt^2 + 2\sqrt{ab}xt)}$$

$$= e^{-(\sqrt{a}x + \sqrt{b}t)^2}$$

So, the wave will be travelling along $-x$ axis with speed $\sqrt{\frac{b}{a}}$

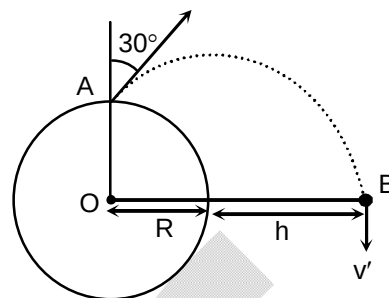
15.

B

Sol. By conservation of angular momentum of the body about O
 $(mv \sin 30^\circ)R = mv'(R+h)$

$$\frac{vR}{2} = \frac{v}{4}(R+h)$$

$$\Rightarrow h = R$$



16.

A

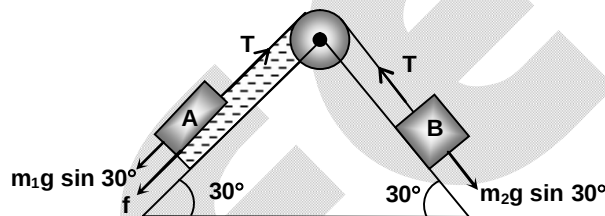
Sol. $T = m_2 g \sin 30^\circ$... (i)

$$T = m_1 g \sin 30^\circ + \frac{\eta A v}{\ell}$$

... (ii)

From (i) and (ii)

$$\eta = \frac{(m_2 - m_1)g \ell \sin 30^\circ}{vA} = 3.46 \times 10^{-3} \text{ Pa sec}$$



17.

A

Sol. As all the points on the periphery of either ring are at the same distance from point P, the potential at point P due to the whole ring can be calculated as $V = -\frac{GM}{\sqrt{R^2 + x^2}}$, where x is the

axial distance from the centre of the ring. This expression is independent of the fact whether the distribution of mass is uniform or non-uniform

$$\text{So, at P, } V = -\frac{GM}{R\sqrt{2}} - \frac{G2M}{R\sqrt{5}} = -\frac{GM}{R} \left[\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{5}} \right]$$

18.

A

Sol. $\frac{\Delta \lambda}{\lambda} = \frac{v}{c}$

$$1 = \frac{v}{c}$$

$$v = c$$

19.

D

Sol. Acceleration when block B is in the liquid

$$a_1 = \frac{m_A g - (m_B g - \text{upthrust})}{m_A + 3m}$$

$$= \frac{m_A g - \left(3mg - \frac{3m}{2\rho} \rho g \right)}{(m_A + 3m)} \uparrow$$

Acceleration when block B is outside of the liquid.

$$a_2 = \frac{3mg - m_A g}{m_A + 3m}$$

Given $a_1 = a_2$

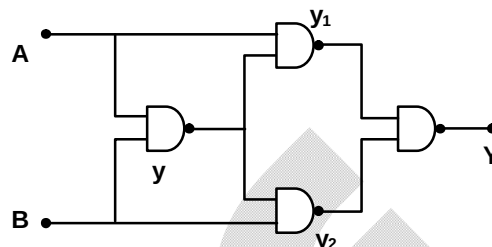
$$\Rightarrow m_A g - \frac{3}{2} mg = 3mg - m_A g$$

$$m_A = \frac{9}{4}m$$

20.

A

Sol. Output of the gates are marked as y, y₁ and y₂.



A NAND gate gives low output when both the inputs are high, otherwise it gives a high output. Now the truth table.

A	B	y	y ₁	y ₂	Y
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	0

SECTION - B

21. 680

Sol.

$$f' = f \left(\frac{v}{v - v_s \cos \theta} \right)$$

$$f' = 640 \left(\frac{340}{340 - \frac{100}{3} \cos \theta} \right)$$

$$f' = 640 \left(\frac{340}{340 - \frac{100}{3} \times \frac{3}{5}} \right)$$

$$\Rightarrow f' = 680 \text{ Hz}$$

22. 42

Sol.

$$F = (\text{Area}) \frac{I}{C}$$

$$\text{Effective area} = \pi R^2$$

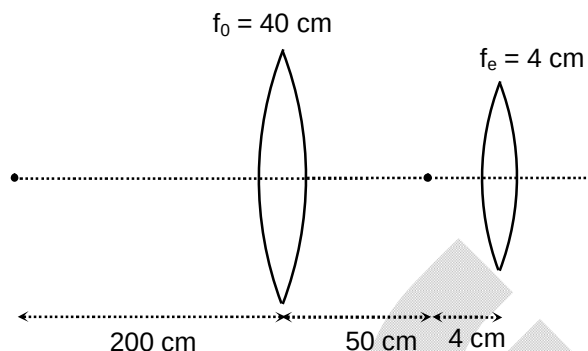
$$F = (\pi R^2) \frac{I}{C}$$

$$F = \frac{22}{7} \times (21 \times 10^{-2})^2 \times \frac{1}{110} \times \frac{1}{3 \times 10^8}$$

$$F = 42 \times 10^{-13} \text{ N}$$

23. 54

Sol. $\frac{1}{v} - \frac{1}{-200} = \frac{1}{40}$
 $v = 50 \text{ cm}$
 $d = 50 + 4 = 54 \text{ cm}$



24. 25

Sol. Here $T_{1/2} = 270 \text{ days}$

After 540 days, it means two half lives have been completed $n = 2$

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^n$$

$$\frac{m}{m_0} = \left(\frac{1}{2}\right)^n$$

$$\Rightarrow \frac{m}{100} = \frac{1}{4}$$

$$\Rightarrow m = 25 \text{ mg}$$

25. 8

Sol. If I is the intensity of the incident unpolarised light, the intensity transmitted by the first is $I/2$. This is the intensity of incident light on the second polaroid. Intensity transmitted by the second polaroid is $\frac{I}{2} \cos^2 \theta$, where θ is the angle between the axes.

Here $\sin \theta = 3/5$ therefore $\cos \theta = 4/5$

$$\frac{I}{2} \cos^2 \theta = \frac{I}{2} \left(\frac{4}{5}\right)^2 = \frac{8}{25} I$$

$\therefore \frac{8}{25}$ is the required ratio.

26. 3

Sol. from first refraction

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{n}{v} - \frac{1}{\infty} = \frac{n-1}{R}$$

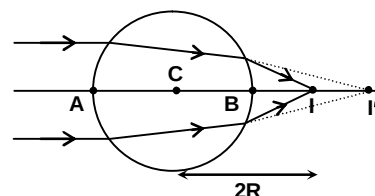
$$\frac{n}{v} = \frac{n-1}{R}$$

$$AI' = v = \frac{nR}{n-1}$$

for 2nd refraction

$$u = BI' = AI' - 2R$$

$$= \frac{nR}{n-1} - 2R$$



$$= \frac{(2-n)R}{(n-1)}$$

and $v = R$

For second surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{v} - \frac{n}{u} = \frac{1-n}{-R}$$

$$\frac{1}{R} - \frac{n(n-1)}{(2-n)R} = \frac{n-1}{R}$$

$$(2-n) - n(n-1) = (n-1)(n-2)$$

$$n = 4/3$$

27. 4

Sol. Let ground state energy (in eV) be E_1 ,

Then from given condition,

$$E_{2n} - E_1 = 204 \text{ eV}$$

$$\frac{E_1}{4n^2} - E_1 = 204 \text{ eV}$$

$$E_1 \left(\frac{1}{4n^2} - 1 \right) = 204 \text{ eV} \quad \dots(i)$$

and $E_{2n} - E_n = 40.8 \text{ eV}$

$$\frac{E_1}{4n^2} - \frac{E_1}{n^2} = 40.8 \text{ eV}$$

$$\Rightarrow E_1 \times -\frac{3}{4n^2} = 40.8 \text{ eV} \quad \dots(ii)$$

From (i) and (ii)

$$\frac{1 - \frac{1}{4n^2}}{\frac{3}{4n^2}} = 5$$

$$n = 2$$

Now from equation (ii)

$$E_1 = -\frac{4}{3}n^2(40.8)$$

$$\text{Or } E_1 = -217.6 \text{ eV}$$

$$\text{and } E_1 = -13.6 Z^2$$

Therefore

$$Z^2 = \frac{E_1}{-13.6} = \frac{-217.6}{-13.6} = 16$$

$$Z = 4$$

28. 3

$$\text{Sol. } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{3}{25} + \frac{1}{u} = \frac{1}{10}$$

$$\Rightarrow u = -50 \text{ m}$$

and for final position of the object

$$\frac{7}{50} + \frac{1}{u} = \frac{1}{10}$$

$$\Rightarrow u = -25 \text{ m}$$

Therefore, the shift in object = $50 - 25 = 25 \text{ m}$

$$\text{Speed} = \frac{25}{30} \text{ m/s} = 3 \text{ km/hr}$$

29. 200

Sol. Fundamental frequency of open pipe $f_0 = \frac{v}{2\ell}$

Third harmonic of the closed pipe $f_c = 3\left(\frac{v}{4}\right)$

$$\text{Given } 3\left(\frac{v}{4\ell}\right) = \frac{v}{2\ell} + 100$$

$$\text{Or } \frac{v}{2\ell} = 200 \text{ Hz}$$

30. 8

Sol. Shift $\Delta x = \frac{\beta}{\lambda}(\mu - 1)t$

$$\text{Shift due to one plate} = \Delta x_1 = \frac{\beta}{\lambda}(\mu_1 - 1)t$$

$$\text{Shift due to another plate} = \Delta x_2 = \frac{\beta}{\lambda}(\mu_2 - 1)t$$

$$\text{Net shift } \Delta x = (\Delta x_2 - \Delta x_1)$$

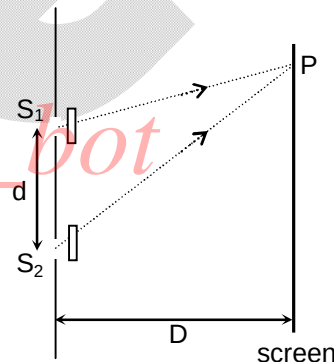
$$\Delta x = \frac{\beta}{\lambda}(\mu_2 - \mu_1)t \quad \dots(i)$$

$$\text{and also } \Delta x = 5\beta \quad \dots(ii)$$

$$\text{Hence } 5\beta = \frac{\beta}{\lambda}(\mu_2 - \mu_1)t$$

$$\Rightarrow t = \frac{5\lambda}{(\mu_2 - \mu_1)} = 8 \times 10^{-6} \text{ m}$$

$$t = 8 \mu\text{m}$$

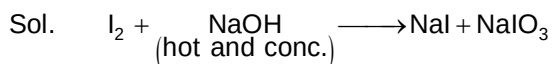


Chemistry

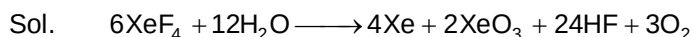
PART – B

SECTION – A

31. D



32. C



33. A

Sol. Because half filled electronic configuration is achieved in Mn^{2+} .

34. D

Sol. Actinoid compounds are more basic than lanthanoid compounds.

35. B

Sol. $[PdCl_4]^{2-}$ has square planar geometry.

36. C

Sol. SO_3 is more harmful than SO_2 . SO_2 affects the formation of chlorophyll.

37. D

Sol. $ClO^\bullet + O \longrightarrow Cl^\bullet + O_2$ reaction is involved in ozone layer depletion.

38. C

Sol. Ag_2CrO_4 is red in colour.

39. D

Sol. Dilute boric acid solution is weak antiseptic for eyes.

40. B

Sol. Bactericidal – Penicillin, Aminoglycosides and ofloxacin.
Bacteriostatic – Tetracyclin.

41. C

Sol. Multimolecular : Sol of sulphur
Macromolecular : Sol of protein, Sol of starch.

42. B

Sol. Moles of acetic acid adsorbed $= (0.6 \times 100 - 0.4 \times 100) \times 10^{-3}$
 $= 20 \times 10^{-3}$

Mass of acetic acid adsorbed (W) $= 20 \times 10^{-3} \times 60$ g

Mass of acetic acid adsorbed per gram activated charcoal $= \frac{1.2}{4}$
 $= 0.3$ g.

43. C

Sol. $KE = \frac{3}{2}RT$ $KE \propto T$

$$u_{\text{rms}} = \sqrt{\frac{3RT}{M}} \quad u_{\text{rms}} \propto \sqrt{T}$$

$$u_{\text{rms}} \propto \frac{1}{\sqrt{M}}$$

44. D

Sol. $+5.8 = C \times 5.8 + \frac{200 \times 4.2 \times 5.8}{1000}$

$$1 = C + 0.84$$

$$C = 0.16 \text{ kJ/K}$$

45. B

Sol. Calamine : ZnCO_3 Argentite : Ag_2S
 Willemite : Zn_2SiO_4 Diaspore : $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$

46. C

Sol. Extraction of aluminium from concentrated solution is not done by coke.

47. D

Sol. $PE = \frac{\frac{4}{3}\pi r_+^3 + \frac{4}{3}\pi r_-^3}{(2r_-)^3} \times 100$

$$= \frac{4 \times \pi}{3 \times 8} \times 100 \left[\left(\frac{r_+}{r_-} \right)^3 + 1 \right]$$

$$= \frac{\pi}{6} \times 100 \left[(0.732)^3 + 1 \right]$$

$$= \frac{\pi}{6} \times 100 \times 1.4$$

$$= 72.8$$

$$\approx 73$$

48. D

Sol. Due to decrease in the interionic forces.

49. B

Sol. $(n \times nf)_{\text{K}_2\text{Cr}_2\text{O}_7} = (n \times nf)_{\text{CuS}}$

$$n \times 6 = 2.4 \times 8$$

$$n = \frac{2.4 \times 8}{6} = 3.2$$

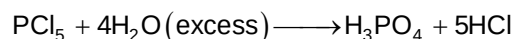
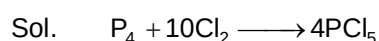
50. D

Sol. $n \times 2 + 2 \times 3 + 4 \times 6 - 18 \times 2 = 0$

$$n = 3$$

SECTION – B

51. 5



52. 3

Sol. $\Delta T_f = iK_f, m$

$$0.52 = i \times 1.86 \times \frac{0.2}{1.86}$$

$$i = 2.6$$

$$\alpha = \frac{i-1}{n-1}$$

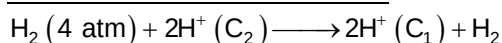
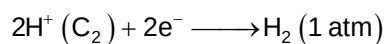
$$(n-1) = \frac{i-1}{\alpha} = \frac{2.6-1}{0.8} = \frac{1.6}{0.8} = 2$$

$$n-1 = 2$$

$$n = 3$$

53. 462

Sol. $H_2 (4 \text{ atm}) \longrightarrow 2H^+ (C_1) + 2e^-$



For acetic acid

$$\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{10^{-5}}{10^{-1}}} = 10^{-2}$$

$$[H^+] (C_1) = C\alpha = 10^{-1} \times 10^{-2} = 10^{-3}$$

$$\text{From NaOH } [H^+] (C_2) = 10^{-11}$$

$$E = \frac{0.06}{2} \log \frac{C_2^2 \times 4}{C_1^2 \times 1}$$

$$= 0.03 \log \left(\frac{10^{-22} \times 4}{10^{-6} \times 1} \right)$$

$$= 0.03 (\log 4 \times 10^{-16})$$

$$= 0.03 [-16 + 0.6]$$

$$= -0.03 \times 15.4$$

$$= -0.462$$

54. 25

Sol. $d = \frac{Z \times M}{a^3 \times N}$

$$N = \frac{Z \times M}{d \times a^3} = \frac{2 \times 512}{8 \times (8 \times 10^{-8})^3} = \frac{2 \times 512 \times 10^{24}}{8 \times (8)^3}$$

$$= 0.25 \times 10^{24} = 25 \times 10^{22}$$

55. 3

Sol. $2KI + H_2O + O_3 \longrightarrow 2KOH + I_2 + O_2$

Number of unpaired electrons in $O_2 = 2$

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

$$= \sqrt{2(2+2)} \text{ B.M.}$$

$$= \sqrt{8} \text{ B.M.}$$

$$= 2.83$$

$$\approx 3$$

56. 1

Sol. $60 \times x = 15 \times 0.02 \times 2$

$$x = 0.01 \text{ M}$$

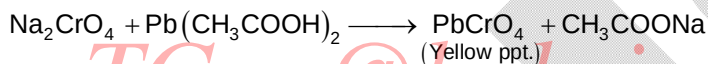
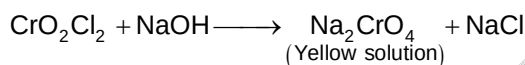
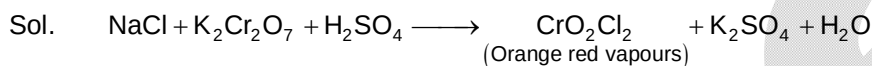
57. 414

Sol. $-2573 = [2 \times 814 + 4 \times H_{C-H}] + [5 \times 499] - [8 \times 724 + 4 \times 640]$

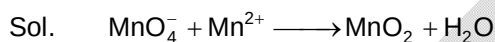
$$-2573 = [1628 + 4 \times H_{C-H} + 2495] - [5792 + 2560]$$

$$H_{C-H} = 414$$

58. 6



59. 21



$$n_f(\text{MnO}_2) = \frac{6}{5}, n_f(\text{MnO}_4^-) = 3$$

$$(n_f \times n)_{\text{MnO}_2} = (n_f \times n)_{\text{KMnO}_4}$$

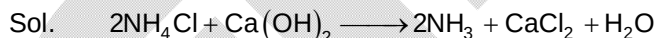
$$\frac{6}{5} \times n = 3 \times 4.2$$

$$n = 2.1 \times 5$$

$$n = 10.5$$

$$2Y = 2 \times 10.5 = 21.$$

60. 6



$$n_{\text{NH}_4\text{Cl}} = \frac{10.7}{53.5} = 0.2$$

$$n_{\text{Ca}(\text{OH})_2} = \frac{M \times V}{1000} = \frac{5 \times 10}{1000} = 0.05$$

Limiting reagent is $\text{Ca}(\text{OH})_2$

$$n_{\text{NH}_3} = 0.05 \times 2 = 0.1$$

$$PV = nRT$$

$$0.82 \times V = 0.1 \times 0.082 \times 600$$

$$V = 6 \text{ L}$$

Mathematics

PART – C

SECTION – A

61. A

Sol. $z = 1, f(1) = a + ib$

$\Rightarrow a + ib$ must fall on the line $\text{Re}(z) = a = \frac{1}{2}$ (By equidistant rule)

$$\Rightarrow |a + ib| = 8 \Rightarrow b^2 = \frac{255}{4}$$

62. B

Sol. Conceptual

63. A

Sol. If p is a prime such that $100 > p > \frac{200}{3}$, then p and $2p$ but not $3p$ belong to $[1, 2, \dots, 200]$.

While just p belongs to $(1, 2, \dots, 100)$, so p does not divide ${}^{200}C_{100}$ but consider 61 the largest prime $< \frac{200}{3}$

61^3 divides $200!$ but only 61 divides $100!$

Hence 61 divides ${}^{200}C_{100}$.

64. D

Sol. If $n^2 - 19n + 99 = x^2$
 $\Rightarrow n^2 - 19n + 99 - x^2 = 0$

$$n = \frac{19 \pm \sqrt{4x^2 - 35}}{2}$$

$$4x^2 - 35 = q^2$$

$$(2x + 9)(2x - 9) = 35 = 7 \times 5$$

$$\Rightarrow x = 3 \text{ or } 9$$

$$n = 1, 9, 10 \text{ or } 18$$

65. A

Sol. We have

$$ar - 9 - (a - 5) = a(r - 1) - 4 = ar^2 - 25 - (ar - 9)$$

$$\Rightarrow ar(r - 1) - 16$$

$$\text{We now have } ar^2 + (-2ar) + a = 12$$

$$\Rightarrow 3ar = 432, ar = 144$$

$$\Rightarrow r = \frac{4}{3} = 108.$$

66. C

Sol. Let x be the probability that it rains on Sunday given that it does not rain on Saturday.

$$\text{We then have } \frac{3}{5}x + \frac{2}{x}2x = \frac{3}{10} \Rightarrow x = \frac{3}{14}$$

$$\text{So probability that it does not rain on either day} = \left(1 - \frac{3}{14}\right) \cdot \frac{3}{5} = \frac{33}{70}$$

$$\text{So the probability that rains on at least one of the day is } 1 - \frac{33}{70} = \frac{37}{70}$$

67.

B

Sol.

All the coefficient of odd powered terms is an odd number of odd degree terms multiplied together and all coefficient of even powered terms has an even number of odd degree terms multiplied together. Since every odd degree term is negative and every even degree is positive so we sum is just $Q(-1) = P(-1)^5 = \frac{243}{32}$.

68.

B

Sol.

$$xyz - 3 + \log_5 x = 32 \quad \dots(i)$$

$$xyz - 3 + \log_5 y = 81 \quad \dots(ii)$$

$$xyz - 3 + \log_5 z = 256 \quad \dots(iii)$$

Add (i), (ii) and (iii) we have

$$xyz = 125$$

$$\Rightarrow \log_5 x = -90$$

$$\log_4 y = -41$$

$$\log_4 z = 134$$

69.

D

Sol.

$$\frac{ab+1}{a+b} < \frac{3}{2}$$

$$\Rightarrow 2ab + 2 < 3a + 3b$$

$$\Rightarrow (2a-3)(2b-3) < 5$$

So $(2a-3)$ and $(2b-3)$ each cannot be even or else a and b will not be integer

$$(2a-3)(2b-3) = (1, 1), (1, 3), (3, 1)$$

$$(a, b) = (2, 2), (2, 3), (3, 2)$$

$$\frac{a^3b^3+1}{a+b} = \frac{65}{16} \text{ or } \frac{31}{5}. \text{ So Ans is } \frac{31}{5}.$$

70.

B

Sol.

$$x(y+z) = 0$$

$$\Rightarrow x = 0, y+z = 0 \text{ pair of perpendicular plane.}$$

71.

D

Sol.

$$\text{Total number} = {}^7C_2 \cdot \frac{6!}{2!2!}$$

$$\text{Unit digit 3, } {}^6C_2 \cdot \frac{5!}{2!2!}$$

$$\text{Unit digit 1, } {}^6C_2 \cdot \frac{5!}{2!}$$

$$\text{Probability} = \frac{5}{14}$$

72.

D

Sol.

$$\alpha - \frac{1}{\beta\gamma} \Rightarrow \alpha \left(1 - \frac{1}{\alpha\beta\gamma} \right) = 2\alpha$$

So, equation whose roots are $2\alpha, 2\beta, 2\gamma$ are

$$\left(\frac{x}{2} \right)^3 - 3 \left(\frac{x}{2} \right) + 1 = 0$$

$$x^3 - 12x + 8 = 0$$

73. B

 Sol. $A_3 = 4$

$$G_2 = \sqrt{5}$$

$$H_1 = \frac{5}{4}$$

 So equation is $(x - A_3)(x - G_2)(x - H_1) = 0$

74. B

 Sol. Lines are coplanar if $\begin{vmatrix} 1 & 1 & 1 \\ 2 & x_1 & x_2 \\ 3 & 4 & 5 \end{vmatrix} = 0$

$$\Rightarrow 2x_1 - x_2 = 2$$

$$\text{Also sum of roots} = \frac{-(x_2 + 2)}{x_1} = \frac{-2x_1}{x_1} = -2.$$

75. D

 Sol. Let $\vec{c} = \lambda \vec{a} + \mu \vec{b} + \nu(\vec{a} \times \vec{b})$, then

$$\vec{a} \cdot \vec{c} = \lambda \Rightarrow \lambda = \cos\theta.$$

 Similarly $\mu = \cos\theta$.

$$\text{also, } |\vec{c}|^2 = \lambda^2 + \mu^2 + \nu^2 |\vec{a} \times \vec{b}|^2$$

$$\Rightarrow 1 = 2\lambda^2 + \nu^2 [|\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2] = 2\lambda^2 + \nu^2 \quad [\because \lambda = \mu]$$

$$\therefore \nu^2 = 1 - 2\lambda^2 = 1 - 2\cos^2\theta = -\cos 2\theta.$$

$$\text{Now } \nu^2 \geq 0 \Rightarrow \cos 2\theta \leq 0 \Rightarrow \frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}.$$

76. A

 Sol. $(1 - 2x)(1 - 2y)(1 - 2z) = 1 - 2 \sum x + 4 \sum xy - 8xyz$

$$= 4 \sum xy - 8xyz - 1$$

$$\therefore \sum xy - 2xyz = \frac{1 + (1 - 2x)(1 - 2y)(1 - 2z)}{4}$$

$$\frac{\sum (1 - 2x)}{3} \geq \{(1 - 2x)(1 - 2y)(1 - 2z)\}^{1/3}.$$

$$\Rightarrow (1 - 2x)(1 - 2y)(1 - 2z) \leq \frac{1}{27}$$

$$\Rightarrow xy + yz + zx - 2xyz \leq \frac{7}{27}.$$

 Equality holds for $x = y = z = \frac{1}{3}$

$$\therefore \text{maximum value} = \frac{7}{27}.$$

77. B

 Sol. $|A|^{2005} - 6A^{2004} = |A|^{2004} |A - 6I|$

$$= 2^{2004} \cdot \begin{vmatrix} 0 & 11 \\ 2 & -2 \end{vmatrix} = (-22) 2^{2004} = (-) 2 \times 11 \times 2^{2004} = (-11)(2)^{2005}.$$

78. B

Sol. $\Delta = \begin{vmatrix} 0 & -2b^2 & -2a^2 \\ a^2 & b^2 + c^2 & a^2 \\ b^2 & b^2 & c^2 + a^2 \end{vmatrix} \quad (R_1 \rightarrow R_1 - R_2 - R_3)$

$$= -2 \begin{vmatrix} 0 & b^2 & a^2 \\ a^2 & b^2 + c^2 & a^2 \\ b^2 & b^2 & c^2 + a^2 \end{vmatrix}$$

$$= -2 \begin{vmatrix} 0 & b^2 & a^2 \\ a^2 & c^2 & 0 \\ b^2 & 0 & c^2 \end{vmatrix} \quad (R_1 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1)$$

$$= 4a^2b^2c^2 = 4 \quad [\text{as } a, b, c \text{ are cube roots of unity}].$$

79. B

Sol. First find the probability both are green, then the probability both are blue and add the two probability. The sum should be equal to 0.58.

$$\Rightarrow \frac{4}{10} \times \frac{16}{16+N} + \frac{6}{10} \times \frac{N}{16+N} = \frac{29}{50}$$

$$\Rightarrow N = 144$$

80. C

Sol. $(AB)^2 = AB \cdot AB = A^3B^2$
 $(AB)^3 = (AB)^2 \cdot AB = A^3B^2 \cdot AB = A^7B^3$
 $(AB)^4 = (AB)^3 \cdot AB = A^7B^3 \cdot AB = A^{15}B^4$
 $\Rightarrow (AB)^{10} = A^{1023}B^{10}$

SECTION - B

81. 528

Sol. Let roots are $-x_1, -x_2, -x_3, -x_4$
 $f(x) = (x + x_1)(x + x_2)(x + x_3)(x + x_4)$
 $f(1) = (1 + x_1)(1 + x_2)(1 + x_3)(1 + x_4) = 1 + a + b + c + d = 2010$
 Since $2010 = 2 \cdot 3 \cdot 5 \cdot 6 \cdot 7$
 So d is the product of 4 roots $= (-1)(-2)(-4)(-66) = 528$

82. 9

Sol. Take
 $x = 4 + 2\cos\theta$
 $y = 3\sin\theta$

83. 24

Sol. Let $f(x) = x^4q\left(\frac{1}{x}\right)$ it has same root as 1, 2, 3, 4 as $p(x)$, but has same coefficient as $q(x)$. Just in reverse order, its leading coefficient is $q(0) = \frac{1}{24}$, limit will be -24.

84. 2

Sol. On each roll, the probability that Kiran decides to go to Russia is $\frac{3}{2}$ times the probability she decides to go to Ukrain, so the total probability that she goes to Russia is $\frac{3}{2}$ times the total probability that she goes to Ukrain. Since the total probability is 1 (they are the only two eventual outcomes) Kiran goes to Ukrain with probability $\frac{2}{5}$ and Russia with probability $\frac{3}{5}$.

85. 8

Sol. Let $z = m + \sqrt{n}i$

$$\bar{z} = m - \sqrt{n}i$$

i.e. $x^3 + ax + b$ has roots $-20, z, \bar{z}$ and $x^3 + cx^2 + b$ has roots $-21, z, \bar{z}$

$$\Rightarrow -20 + z + \bar{z} = 0$$

$$m = 10$$

$$-21z - 21\bar{z} + z\bar{z} = 0$$

$$-21(z + \bar{z}) + z\bar{z} = 0$$

$$\Rightarrow n = 320$$

86. 121

Sol. $A \cdot B = |A| \cdot |B|$

$$\Rightarrow |A| = -11 \therefore |B| = |A|^2 = 121$$

87. 3

Sol. The plane is $y - z = 0$

$$a_1b_1 + a_2b_2 + a_3b_3 = 0 \Rightarrow k = -3$$

88. 943

$$\text{Sol. } {}^7P_4 \times 1 + \frac{{}^7P_3}{2} + 1 - 2 = 943$$

89. 4

Sol. a, ar, ar^2 in G.P. $\Rightarrow ar^2 - ar < 1 \Rightarrow r^2 - r < 1 \Rightarrow r^2 - r - 1 < 0$

$$r = \frac{\sqrt{5}+1}{2}, [r] = 1, [-r] = -2$$

90. 6

Sol. Let $x - d, x, x + d$ one the sequence

$$\text{So, it is given that } (x - d)^2 + x^2 + (x + d)^2 = xd^2 \Rightarrow 3x^2 + 2d^2 - xd^2 = 0 \Rightarrow 3x^2 - xd^2 + 2d^2 = 0$$

$$\Rightarrow x = \frac{d^2 \pm \sqrt{d^4 - 24d^2}}{6} = \frac{d^2 \pm d\sqrt{d^2 - 24}}{6}$$

So, $d = 5$ and $d = 7$ are the only (+ve) integers value of d in possible, so that $\sqrt{d^2 - 24}$ a positive integer.

$$\text{At } d = 5, x = 5 \text{ and } x = \frac{10}{3}, d = 7, x = 14 \text{ so, } (5 + 5) + (7 + 14) = 31$$

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – III
PAPER –1
TEST DATE: 25-12-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

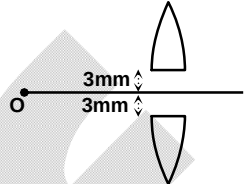
Physics

PART – I

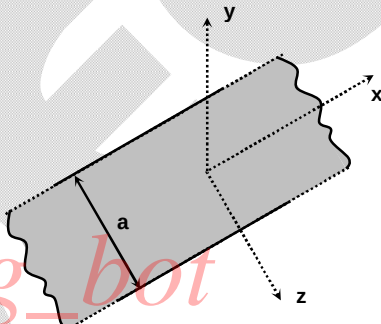
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

- A point object is placed at a distance of 0.4 meters from a convex lens of focal length 0.2 m. The lens is cut into two equal parts each of which is displaced vertically by 3 mm away from the optic axis as shown in figure. Mark the correct statements.



(A) Image is formed at a distance of 0.4 meter from the lens
 (B) Number of images formed are 2
 (C) Separation between images is 12 mm
 (D) Separation between images is 6 mm
- A thin metallic plate of uniform mass density is kept along x-axis symmetrically. The length of plate along x-axis is very large as compared to its width 'a' along z-axis. It is given that ' σ ' is mass per unit area. Then choose the correct option(s)

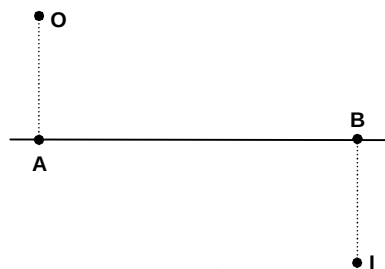


(A) The gravitational field at $\left(0, \frac{\sqrt{3}a}{2}, 0\right)$ is $\frac{2G\sigma\pi}{3}$.
 (B) The gravitational field at $\left(0, \frac{a}{2}, 0\right)$ is $G\sigma\pi$.
 (C) The ratio of gravitational field at $\left(0, \frac{a}{2\sqrt{3}}, 0\right)$ and $\left(0, \frac{\sqrt{3}a}{2}, 0\right)$ is 2.
 (D) The ratio of gravitational field at $\left(0, \frac{a}{2\sqrt{3}}, 0\right)$ and $\left(0, \frac{a}{2}, 0\right)$ is $\frac{4}{3}$.
- Choose the correct option(s).

(A) If Q value of an exothermic nuclear reaction is +15.76 MeV and the kinetic energy of products of reaction is 18.24 MeV then kinetic energy of incident particles is 2.48 MeV.
 (B) In a nuclear reaction, during α -decay from a nucleus at rest, energy released is shared between alpha particle and daughter nucleus is in the form of kinetic energy and its value for α -particle has more than that of daughter nucleus.
 (C) In a decay process of a radioactive element, the activity of the sample at any instant is directly proportional to the number of undecayed nuclei present in the sample at that time.
 (D) Photo electric effect supports quantum nature of light because there is a minimum frequency of light below which no photoelectrons are emitted.
- In a resonance tube experiment, a closed organ pipe of length 125 cm resonates when tuned with a tuning fork of frequency 340 Hz. If water is poured in the pipe then (Given $v_{\text{air}} = 340 \text{ m/s}$)

(A) Minimum length of water column to have the resonance is 50 cm
 (B) The distance between two successive nodes is 50 cm
 (C) Maximum length of water column to create the resonance is 100 cm
 (D) Maximum length of water column to create the resonance is 105 cm

5. A luminous point object is placed at O, whose image is formed at I as shown in figure. Line AB is the optical axis. Which of the following statement is/are correct?



- (A) If a lens is used to obtain the image, then it must be a converging lens and its optical center will be the intersection point of line AB and OI.
 (B) If a lens is used to obtain the image, then it must be a diverging lens and its optical center will be the intersection point of line AB and OI.
 (C) If a mirror is used to obtain the image then the mirror must be concave and object and image subtend equal angles at the pole of the mirror.
 (D) I is a real Image.
6. Point A (0, 0.1 cm) and B (12 cm, 0.5 cm) are object image pair (one of the point acts as object and the other point as image) x-axis is the principal axis of the mirror. Then this object image pair is:
 (A) Due to a convex mirror of focal length 2.5 cm
 (B) Due to a concave mirror having its pole at (2cm, 0)
 (C) Real virtual pair
 (D) Data is insufficient for (A) and (B)

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

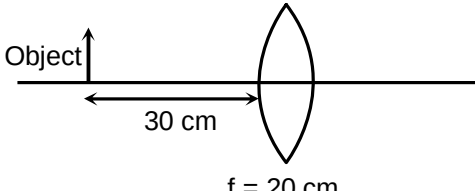
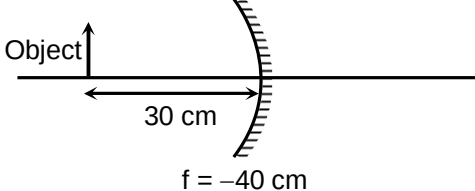
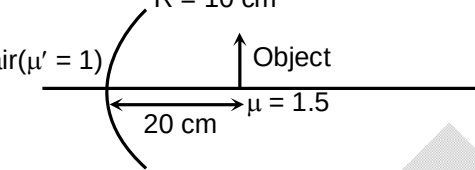
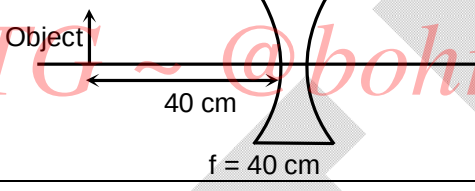
7. List –I describes physical quantities and List-II gives their dimensions.

List –I		List -II	
(I)	Magnetic flux	(P)	$MLT^{-3}A^{-1}$
(II)	Electrical conductivity	(Q)	$M^{-1}L^{-3}T^3A^2$
(III)	Electric field	(R)	$M^{-1}L^{-2}T^4A^2$
(IV)	Capacitance	(S)	$ML^2T^{-2}A^{-1}$
		(T)	$M^2L^2T^{-2}A^{-1}$

Which one of the following options is correct ?

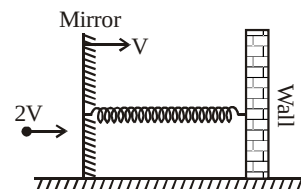
- (A) I → R, II → T, III → P, IV → P
 (B) I → S, II → T, III → P, IV → P
 (C) I → R, II → Q, III → Q, IV → P
 (D) I → S, II → Q, III → P, IV → R

8. List –I shows the object and optical device, List-II shows nature of the image formed for paraxial approximation. Match the List-I with nature of image in List-II.

List –I		List -II	
(I)		(P)	Erect image
(II)		(Q)	Inverted image
(III)		(R)	Magnified image
(IV)		(S)	Diminished image
		(T)	Real image

Which one of the following options is correct ?

- (A) I $\rightarrow$ Q,R,T II $\rightarrow$ P,S III $\rightarrow$ P,R IV $\rightarrow$ Q,R,T
 (B) I $\rightarrow$ P,S II $\rightarrow$ Q,R,T III $\rightarrow$ P,S IV $\rightarrow$ P,R
 (C) I $\rightarrow$ P,S II $\rightarrow$ P,R III $\rightarrow$ Q,R,T IV $\rightarrow$ P,S
 (D) I $\rightarrow$ Q,R,T II $\rightarrow$ P,R III $\rightarrow$ P,R IV $\rightarrow$ P,S
9. A plane mirror is tied to the free end of an ideal spring. The other end of the spring is attached to a wall. The spring with mirror is held vertically to the floor, can slide along it smoothly. When the spring is at its natural length, the mirror is found to be moving at a speed of V with respect to ground frame. An object is moving towards the mirror with speed $2V$ with respect to ground frame. Then, Match the following :

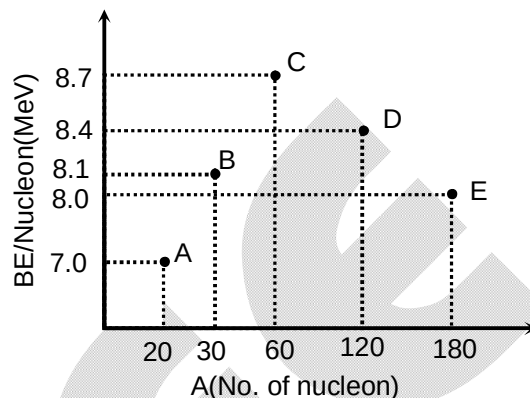


List –I		List –II	
(I)	Speed of image with respect to ground frame when spring is at natural length	(P)	V
(II)	Speed of image with respect to mirror when spring is at natural length	(Q)	0
(III)	Speed of image with respect to object when spring is at natural length	(R)	$2V$
(IV)	Speed of image with respect to ground frame when spring is at maximum compressed state	(S)	$3V$
		(T)	$4V$

Which one of the following options is correct ?

- (A) $I \rightarrow Q, II \rightarrow R, III \rightarrow P, IV \rightarrow S$
 (B) $I \rightarrow T, II \rightarrow R, III \rightarrow S, IV \rightarrow P$
 (C) $I \rightarrow Q, II \rightarrow P, III \rightarrow R, IV \rightarrow R$
 (D) $I \rightarrow T, II \rightarrow Q, III \rightarrow R, IV \rightarrow P$

10. Consider the position of different nuclei on the binding energy per nucleon graph as shown :



List -I (Reactions)		List -II [Q-values(MeV)]	
(I)	$2B \rightarrow C$	(P)	Zero
(II)	$2B + C \rightarrow D$	(Q)	102
(III)	$E \rightarrow C + D$	(R)	Non-zero
(IV)	$3A \rightarrow C$	(S)	36
		(T)	90

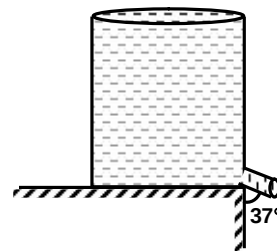
Which one of the following options is correct ?

- (A) $I \rightarrow P, S, II \rightarrow P, III \rightarrow R, T, IV \rightarrow Q, R$
 (B) $I \rightarrow R, S, II \rightarrow P, III \rightarrow P, T, IV \rightarrow Q, R$
 (C) $I \rightarrow S, T, II \rightarrow P, R, III \rightarrow R, T, IV \rightarrow Q, R$
 (D) $I \rightarrow R, S, II \rightarrow P, III \rightarrow R, T, IV \rightarrow Q, R$

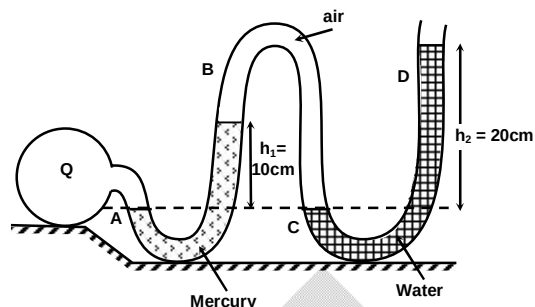
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

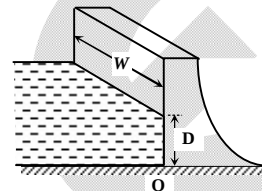
11. A cylindrical tank of cross-sectional area A filled with water upto certain height is placed on a rough horizontal surface having coefficient of friction $\mu = 0.5$. Water comes out of the tank through a small tube of cross-sectional area 'a' inclined at an angle $\theta = 37^\circ$ with the wall of the tank as shown in the figure. Assuming Bernoulli's theorem to be applicable here, find the minimum ratio of a/A so, that the frictional force between tank and horizontal surface is limiting. Neglect the mass of the tank.



12. A bulb Q is filled with air. A tube is connected to the bulb filled with mercury and water. A column of air is trapped between mercury and water as shown in the figure. The heights of mercury and water are shown in the figure. Find the pressure inside the bulb Q (in mega N/m²) assuming one end of tube opens to open atmosphere. Assume density of air to be small. (Take $g = 10 \text{ m/s}^2$ and atmospheric pressure 10^5 N/m^2)



13. Water stands upto depth D, behind the vertical face of dam of width W. Given that it exerts a resultant horizontal force = 3000 kN and torque about O tending to overturn is 10000 kN-m. Find the width (in m). (Taking density of water = 1000 kg/m³, $g = 10 \text{ m/s}^2$)



14. A solid cylindrical piece of wood of length $L = 0.5 \text{ m}$, cross-sectional area $A = 1 \times 10^{-2} \text{ m}^2$ and density $\rho = 600 \text{ kg/m}^3$ is floating with its axis vertical in a liquid of density $\rho_0 = 1200 \text{ kg/m}^3$. The wood is pressed down slowly into the liquid from its equilibrium position. Find the work done (in J) by the force pressing it until it is just immersed completely.

15. The resistance of a metal is given by $R = V/I$, where V is potential difference and I is the current. In a circuit the potential difference across resistance is $V = (12 \pm 0.4) \text{ V}$ and current in resistance $I = (2 \pm 0.2) \text{ A}$. What is the maximum percentage error in measurement of resistance?

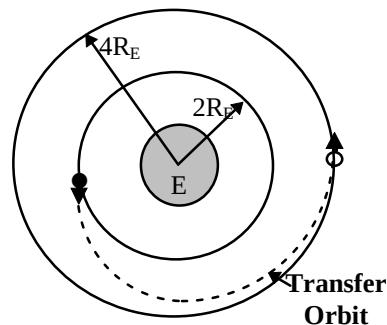
16. In a screw gauge, the graduation in the circular scale consists of 100 equal divisions and 10 complete rotation of the cap is equivalent to 5 mm on the main scale. While measuring diameter of a wire with screw gauge, a particular observation gives initial circular cap reading as 57 and after completing 3 complete rotation the final circular cap reading as 63. It is observed that while proceeding from initial reading to final reading the cap reading increases. Calculate the diameter of the wire (in mm) from the above data.

17. In the measurement of the focal length of a convex lens using the u-v method the following values were obtained:

u	35 cm	40 cm	50 cm	60 cm
v	46.2 cm	40.5 cm	33 cm	30.5 cm

Find the value of focal length by taking the average of the measurements. If the errors in u and v are 0.4 cm respectively, estimate the error in the measurement of focal length in a typical measurement (in cm).

18. An artificial satellite of mass m orbiting the earth in a circular orbit of radius twice the earth's radius R . It is to be taken in another circular orbit of radius $4R$. The transfer is accomplished through an elliptical orbit as shown in the figure. The energy needed to complete the job is calculated to be equal to $\frac{1}{k} \left(\frac{GMm}{R} \right)$, where M is the mass of the earth. Find the value of k.



Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

19. 500 ml aqueous solution is made by dissolving a certain amount of Na_2CO_3 and NaOH mixture. 100 ml of this solution require 40 ml, 0.1 M HCl solution for the colour change when phenolphthalein is used as indicator, while 60 ml 0.1 N HCl require for the colour change when methyl orange is used as indicator in a separate titration. The correct option(s) is/are
(A) Weight % of NaOH in the mixture is 27.4.
(B) Weight of NaOH in the mixture is 0.08 gm.
(C) Weight of Na_2CO_3 in the mixture is 1.06 gm.
(D) Total weight of mixture is 0.292 gm.
20. The possible product(s) obtained by the hydrolysis (partial or complete) of XeF_6 is/are
(A) XeO_3
(B) XeOF_4
(C) XeO_3F_2
(D) XeO_2F_2
21. The INCORRECT statement(s) among the following is/are
(A) Multilayer adsorption of any gas on a solid surface is physisorption.
(B) The plot of $\log(x/m)$ vs. $\log(P)$ in the Freundlich adsorption isotherm is straight line with negative slope.
(C) High pressure and high temperature is favourable condition for physisorption.
(D) The magnitude for heat of adsorption of chemisorption is lower than physisorption.
22. Which of the following is correct?
(A) van Arkel method is used for the purification of Ti.
(B) Poling is used for purification of Cu.
(C) Mond's process is used for the purification of Au.
(D) Cupellation is used for the purification of Ag.
23. A gaseous mixture of 1 : 2 molar ratio of CH_4 and He diffuses two times slower than a gas 'X' under identical condition. Which of the following statement(s) is/are true about the gas 'X'?
(A) The gas 'X' is O_2 .
(B) The gas 'X' diffuse $\sqrt{2}$ times faster than He gas under identical condition.
(C) The gas 'X' is lightest known molecule.
(D) The gas 'X' diffuses 4 times slower than H_2 gas under identical condition.
24. A compound has two isomeric liquid 'A' and 'B' with pure vapour pressure $P_A^\circ = 100$ torr and $P_B^\circ = 80$ torr at 298 K. The binary solution of liquid A and B is in equilibrium with its vapour. The molar ratio of A and B in its liquid form is found to be 1 : 2 at room temperature. Which of the following options are correct?
(A) Mass ratio of A and B in vapour phase is 5 : 8.
(B) Mole ratio of A and B in vapour phase is 1 : 2.
(C) Mole ratio of A and B in vapour phase is 5 : 8.
(D) Total vapour pressure of the solution is $\frac{260}{3}$ torr.

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. List-I contains processes List-II contains metal or their compound used as catalyst

List – I		List – II	
(I)	Ziglar-Natta catalyst	(P)	V_2O_5
(II)	Ostwald process	(Q)	$TiCl_4$
(III)	Contact process	(R)	Fe
(IV)	Haber-Bosch process	(S)	Pt
		(T)	$Al(C_2H_5)_3$

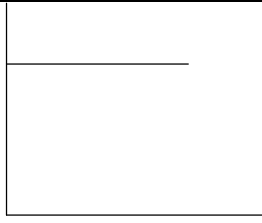
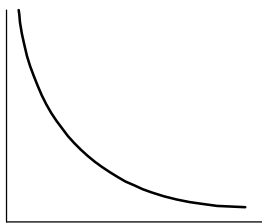
- (A) I→Q; II→P; III→T; IV→R
 (B) I→T; II→Q; III→R; IV→S
 (C) I→Q,T; II→R; III→P; IV→S
 (D) I→Q,T; II→S; III→P; IV→R

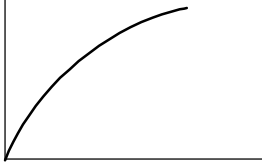
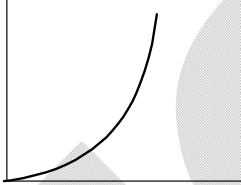
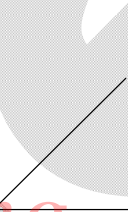
26. Match the List-I with List-II

List – I		List – II	
(I)	The process is mainly applicable for sulphide ore	(P)	Froth floatation
(II)	The process is mainly applicable for carbonates	(Q)	Calcination
(III)	Physical method of separation	(R)	Roasting
(IV)	Chemical method of separation	(S)	Leaching
		(T)	Concentration of the ore

- (A) I→S; II→P; III→P,S; IV→Q
 (B) I→P, R; II→Q; III→P; IV→S
 (C) I→Q,R; II→P; III→T; IV→P
 (D) I→R; II→T; III→Q; IV→S, R

27. Match the correct graph, if the process is taken place at constant temperature for the fixed amount of ideal gas

List – I		List – II	
(I)	P vs. $\frac{1}{V^2}$	(P)	
(II)	P vs. $\frac{1}{V}$	(Q)	

(III)	$\frac{1}{P^2}$ vs. V	(R)	
(IV)	PV vs. V	(S)	
		(T)	

- (A) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ Q; IV $\rightarrow$ P
 (B) I $\rightarrow$ S; II $\rightarrow$ T; III $\rightarrow$ R; IV $\rightarrow$ P
 (C) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ R
 (D) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ R

28. List-I contains metal complexes and List-II contains their properties

List - I		List - II	
(I)	$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	(P)	Central atom is sp^3d^2 hybridisation
(II)	$[\text{Co}(\text{CN})_6]^{3-}$	(Q)	Central atom is d^2sp^3 hybridized
(III)	$[\text{CuF}_6]^{4-}$	(R)	t_{2g} orbital contains 6 electron
(IV)	$[\text{MnF}_6]^{3-}$	(S)	Unsymmetrical e_g orbitation electronic configuration
		(T)	3-unpaired electron present

[Given atomic number of Cr = 24, Co = 27, Ni = 28, Mn = 25]

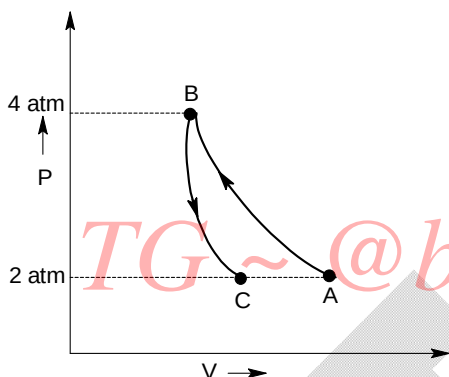
- (A) I $\rightarrow$ Q, T; II $\rightarrow$ Q, R; III $\rightarrow$ Q, R; IV $\rightarrow$ P, S
 (B) I $\rightarrow$ P, T; II $\rightarrow$ Q, R; III $\rightarrow$ R, S; IV $\rightarrow$ P, S
 (C) I $\rightarrow$ Q, S, T; II $\rightarrow$ Q, R; III $\rightarrow$ P, R, S; IV $\rightarrow$ P, T
 (D) I $\rightarrow$ Q, T; II $\rightarrow$ Q, R; III $\rightarrow$ P, S; IV $\rightarrow$ P, S

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

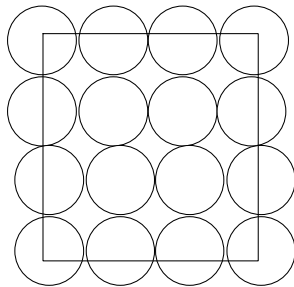
29. The EMF of the cell at 298 K
 $\text{Ag} | \text{AgBr(s)}, \text{KBr} (1 \text{ M}) || \text{AgNO}_3 (0.1 \text{ M}) / \text{Ag(s)}$ is 0.591 V. If in aq. solution AgNO_3 dissociate 80% and KBr dissociate 75%, the solubility of AgBr found to be $x \times 10^{-12}$ mol/litre. The value of x is

30. One mole of an ideal gas ($C_{p,m} = \frac{5}{2}R$) at 500 K follows path $A \rightarrow B$ (Reversible isothermal) and then $B \rightarrow C$ (Reversible adiabatic) as shown in graph. What will be the volume of gas (in litre) at point C?
 [Given $R = 0.082$, $2^{3/5} = 1.52$]

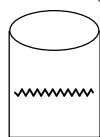


31. AgX crystallizes in rock salt like structure in which X^- occupy FCC lattice site. The density of AgX crystal found to be $\frac{40}{6.022} \text{ g/cm}^3$ and the edge length of the unit cell is 600 pm. What will be the mass of 1 mole unit cell if all the atoms along one body diagonal of each unit cell have been removed. [Given : Atomic weight of $\text{Ag} = 108 \text{ g/mol}$, $N_A = 6.022 \times 10^{23}$]
32. An excess amount of CaCO_3 is treated with 600 ml, 0.1 N HCl solution. The obtained CaCl_2 is used for the preparation of plaster of paris. The maximum amount of plaster of paris (in gram) that could be obtained is? [At. wt. of $\text{Ca} = 40$, $\text{S} = 32$, $\text{O} = 16$, $\text{C} = 12$, $\text{H} = 1$]
33. An aqueous solution of 0.2 M CH_3COOH is placed between parallel electrodes separated 1 cm apart with cross sectional area 1 cm^2 . The resistance of solution is found to be 50Ω . What will be the osmotic pressure of same acetic acid solution at 300 K in atm?
 [Given $\lambda_M^0 \text{CH}_3\text{COO}^- = 50 \text{ S cm}^2 \text{ mol}^{-1}$, $\lambda_M^0 \text{H}^+ = 350 \text{ S cm}^2 \text{ mol}^{-1}$, $R = \frac{1}{12} \text{ lit atm K}^{-1} \text{ mol}^{-1}$]

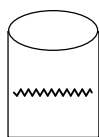
34. The % packing efficiency of two dimensional square close packed unit cell shown below is ($\pi = 3.14$)



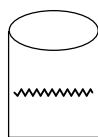
35. All four identical beakers containing $\text{H}_2\text{O}(\ell)$ (shown below) is heated separately to reach temperature 90°C , for this purpose w, x, y, and z Joule energy is required. If the magnitude of thermal energy required is $w < x < y < z$ then what is the value of $\frac{x}{y}$. (Assuming there is no loss of $\text{H}_2\text{O}(\ell)$ during heating)



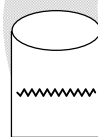
200 g
 40°C
(A)



100 g
 20°C
(B)



150 g
 50°C
(C)



300 g
 30°C
(D)

36. 45 gram of anhydrous oxalic acid is present in 500 ml solution. If 50 ml of this solution is shaken with 0.2 gram of wood charcoal, the concentration of oxalic acid become 0.6 molar. The amount of (in gm) oxalic acid adsorbed per gram of charcoal is [M.Wt. of C = 12, O = 16, H = 1]. [Assuming volume of solution remains same after adsorption]

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. Starting at $(-3, -3)$ a bug moves in the co-ordinate plane via a sequence of steps, each of length one, each step is left, right, up or down all four being equally likely, if P_n is the probability that object reaches $(-1, -1)$ first time in 'n' steps, then which of the following is/are CORRECT?
- (A) $P_4 = \frac{3}{128}$ (B) $P_5 + P_6 = \frac{3}{128}$
 (C) $P_4 + P_6 = \frac{3}{64}$ (D) $P_4 + P_5 = \frac{3}{64}$
38. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 4$, $|\vec{b}| = 3$ & $\vec{a} \cdot \vec{b} = 6\sqrt{3}$ and let $\vec{P}_1$ and $\vec{P}_2$ be two vectors such that $\vec{P}_1 = (\vec{a} \cdot \vec{b})\vec{b} - |\vec{a}|(\vec{a} \cdot \vec{b})^2 \vec{a}$, $\vec{P}_2 = \vec{a} - (\vec{a} \cdot \vec{b})\vec{b} + \vec{P}_1$ and vector $\vec{P}_3$ is parallel to vector $\vec{a} \times \vec{b}$ such that $|\vec{P}_3| = 6$, then which of the following is/are CORRECT?
- (A) $|\vec{P}_1| + |\vec{P}_2| = 1 + \sqrt{3}$ (B) $|\vec{P}_3 \cdot (\vec{P}_1 \times \vec{P}_2)| = 18\sqrt{3}$
 (C) $|\vec{P}_1| + |\vec{P}_2| = 4 + \sqrt{3}$ (D) $|\vec{P}_3 \cdot (\vec{P}_1 \times \vec{P}_2)| = 6\sqrt{3}$
39. If $\sum_{r=0}^{1000} \left(\frac{{}^{1000}C_r}{{}^{2019}C_r} \right) = \frac{m}{n}$ (where m and n are relatively prime positive integers) and $N = \left[\frac{m}{n} \right] + \left[\frac{2m}{n} \right] + \dots + \left[\frac{(n-1)m}{n} \right]$ (where $[.]$ represents greatest integer function), then which of the following is/are CORRECT?
- (A) $m + n = 100$ (B) N is divisible by 3
 (C) N is divisible by 5 (D) $m - n = 50$
40. If all the complex number $z = x + iy$, $x, y \in \mathbb{R}$, $i = \sqrt{-1}$, such that points of complex coordinates z , z^2 , z^3 , z^4 in this given order, are the vertices of a cyclic quadrilateral ($z \neq 0 + i0, 1 + i0$), then
- (A) $\arg z \in \left(0, \frac{2\pi}{3} \right)$ (B) $\arg z \in \left(\frac{2\pi}{3}, \frac{4\pi}{3} \right)$
 (C) $\frac{1+z+z^2}{z}$ is purely real (D) $\arg z \in \left(\frac{4\pi}{3}, 2\pi \right)$
41. Let M_1, M_2 be two matrices of order 3×3 , all the elements of both the matrices are real numbers and $(M_1)^2 + (M_2)^2 = O$ (where O is null matrix), then which of the following statements is/are CORRECT?
- (A) $\det(M_1) + \det(M_2) = 0$
 (B) There exist a non-null matrix M of order 3×3 , such $(MM_1 + MM_2)$ is invertible matrix
 (C) There exist infinitely many ordered pair of (a, b), $a, b \in \mathbb{R}$, such that $\det(aM_1 + bM_2) = 0$
 (D) There exist infinitely many matrix M of order 3×3 such that $M(M_1 + M_2)$ is non-invertible matrix

42. Let z_1, z_2, z_3 be three complex numbers of unit modulus such that $z_1 + z_2 + z_3 \neq 0$ and $z_1^2 + z_2^2 + z_3^2 = 0$, then which of the following is/are CORRECT?
- (A) $|z_1^3 + z_2^3 + z_3^3| = 1$ (B) $|z_1^{2021} + z_2^{2021} + z_3^{2021}| = 2$
 (C) z_1^2, z_2^2, z_3^2 are collinear (D) $|z_1^{2023} + z_2^{2023} + z_3^{2023}| = 0$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. For any finite set S , let $n(S)$ denote the number of elements in S , suppose A and B are two subset of $\{1, 2, 3, 4, 5\}$ that satisfy $n(A) \cdot n(B) = n(A \cap B) \cdot n(A \cup B)$ and $P(E)$ is the probability of occurrence of event E , then (where $n(A)$ is number of elements in the set A)

List-I		List-II	
(I)	$P(n(A) > n(B))$	(P)	$\frac{32}{211}$
(II)	$P(n(A) \leq n(B))$	(Q)	$\frac{211}{454}$
(III)	$P\left(\frac{n(A) = n(B)}{n(A) \geq n(B)}\right)$	(R)	$\frac{32}{454}$
(IV)	$P(n(A) = n(B))$	(S)	$\frac{243}{454}$
		(T)	$\frac{32}{243}$

The correct option is:

- (A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (Q)
 (B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (T); (II) $\rightarrow$ (R); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (Q)
 (D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (T); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

44. Consider the matrices $M = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$ such that $\det(M) = 1$, $a, b, c \in \mathbb{R}$, then

List-I		List-II	
(I)	There is only one matrix M	(P)	$a \in \{-1, 0, 1\}, b \in \{-2, -1, 0\}, c \in \{1, 2, 3\}$
(II)	There are two distinct matrices M	(Q)	$a \in \{-1, 0, 1\}, b \in \{-2, 0, 1\}, c \in \{-2, -1, 0\}$
(III)	There are more than 2, but finitely many matrices M	(R)	$a, b, c \in \{0, 1, 2\}$
(IV)	There are infinitely many matrices M	(S)	a, b, c all are negative integers
		(T)	$a, b, c \in \mathbb{Q}$

The correct option is:

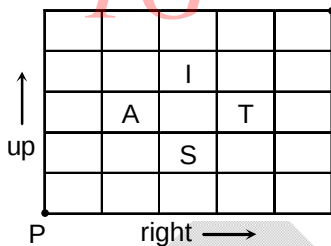
- (A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (T); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (B) (I) $\rightarrow$ (S); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (T)
 (C) (I) $\rightarrow$ (P); (II) $\rightarrow$ (R); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
 (D) (I) $\rightarrow$ (P); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (T)

45. If $z_1, z_2, \dots, z_8$ are the roots of the equation $(z^2 - 4z + 4)^4 = (9)^4$, such that $\arg z_1 < \arg z_2 < \dots < \arg z_8$. (where $\arg z \in [0, 2\pi)$). Let z_k is a complex number satisfying the equation $|z_k - z_4| = |z_8 - z_k|$, and let $L = |(z_k - z_2)(z_k - z_3)(z_k - z_4)(z_k - z_5)|$, then

List-I		List-II	
(I)	$z_k = 3 + i$	(P)	$L = 3^4$
(II)	$z_k = 5 + 3i$	(Q)	$L = \sqrt{\left(\frac{3\sqrt{2}-3}{3\sqrt{2}+3}\right)}(3^8 \cdot 2^4 - 3^8)$
(III)	$z_k = 2 + i0$	(R)	$L = \sqrt{\frac{1}{5} \cdot (3^8 - 2^8)}$
(IV)	$z_k = 0 + i0$	(S)	$L = \sqrt{\left(\frac{3-\sqrt{2}}{3+\sqrt{2}}\right)}(3^8 - 2^4)$
		(T)	$L = 2^4$

The correct option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)
 (B) (I) $\rightarrow$ (S); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)
 (C) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (D) (I) $\rightarrow$ (S); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)
46. A square of 5×5 unit² area is shown in the diagram, contain 25 square of 1×1 unit² area. A person can move on this grid (on the sides of small squares) either right or up, then



List-I		List-II	
(I)	Number of such paths from point P to point Q which do not touch the boundary of squares (1×1 unit ² area) contain alphabets A, I, T and S	(P)	73
(II)	Number of such paths from point P to point Q which do not crosses the diagonal PQ at any moment	(Q)	84
(III)	How many rectangles can be formed by joining small (1×1 unit ² area) so that none of rectangles contain alphabets A, I, T and S	(R)	218
(IV)	Number of such paths from P to Q, which always pass through the boundaries of the square which contain either (A and I) or (T and S)	(S)	144
		(T)	4

The correct option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)
 (B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
 (C) (I) $\rightarrow$ (T); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)
 (D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. Each unit square of a 3 by 3 unit grid is to be coloured either green or blue for each square, either colour is equally likely. If the probability of obtaining a grid that does not have any 2 by 2 green square is $\frac{p}{q}$, where $p, q \in \mathbb{N}$ and p, q are coprime, then the value of $|p - q|$
48. A person can take one step or two steps or three steps at a time. If "P" is the number of distinct ways he can cover 8 steps, then value of P is
49. Consider a polynomial $f(x) = a_1x^3 + a_2x^2 + a_3x + a_4$ such that $a_1, a_2, a_3, a_4 \in \{0, 1, 2, \dots, 8, 9\}$, how many such polynomial of degree three or less than three such that $a_1 - a_2 + a_3 - a_4 + 9 = 0$
50. If $A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$, $a, b, c, d \in \mathbb{R}$ such that $\det(A) + \text{Tr}(A) = 2$, then minimum value of $\det(A^2 + \det(A)A + \text{Tr}(A) \cdot I_2)$ is (where $\text{Tr}(A)$ is trace of A and I_2 is identity matrix of order 2×2)
51. Let $f(x)$ be a least degree unique polynomial with rational coefficients such that, when $f(x)$ divided by $(x^2 + x + 1)$, the remainder is $x + 2$ and when $f(x)$ divided by $(x^2 + 1)$, the remainder is $2x + 1$, then find the sum of squares of coefficients of the polynomial $f(x)$ is
52. In a tetrahedron OPQR, $\vec{p}, \vec{q}, \vec{r}$ are position vector of vertices P, Q and R respectively and O is origin. If $OP = QR = \alpha$, $OQ = PR = \beta$, $OR = PQ = \gamma$ and G_1 and G_2 are centroid of ΔPQR and ΔPOR respectively such that $OG_1 \perp QG_2$, then the value of $\frac{\alpha^2 + \gamma^2}{\beta^2}$ is
53. If $x^2 + px + q = 0$ has two distinct real roots and $(x^2 + 2x - 1)^2 + p(x^2 + 2x - 1) + q = 0$ has four distinct real roots $\alpha, \beta, \gamma, \delta$ such that $\alpha < \beta < \gamma < \delta$ and $\alpha, \beta, \gamma, \delta$ form an arithmetic sequence, then maximum values of $(p - q)$ is
54. If $m = e^{\frac{i\pi}{6}}$ and $n = e^{\frac{2\pi i}{3}}$, where $i = \sqrt{-1}$ two complex numbers, then number of ordered pair of (p, q) , $p, q \in \text{integers}$ and $1 \leq p, q \leq 100$ which satisfy the equation $(n)^q = i(m)^p$, is

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – III
PAPER –2
TEST DATE: 25-12-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 12)

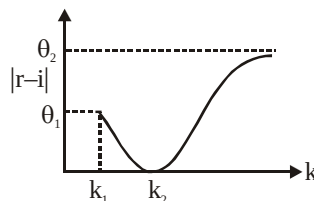
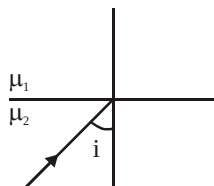
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

- The refracting angle of a prism is 60° . What should be the minimum value of the refractive index of the material from which the prism is to be made so that light cannot emerge from the second refracting surface.
 (A) $\frac{\sqrt{3}}{2}$ (B) 2
 (C) 1.5 (D) $\frac{2}{\sqrt{3}}$
- The de-Broglie wavelength of Helium atom of mass 'm' having energy corresponding to the mean thermal energy at a temperature of 'T' Kelvin (k_B is Boltzmann constant) is
 (A) $\frac{h}{\sqrt{6mk_B T}}$ (B) $\frac{h}{\sqrt{3mk_B T}}$
 (C) $\frac{h}{\sqrt{2mk_B T}}$ (D) $\frac{h}{\sqrt{mk_B T}}$
- If K_α radiation of copper $\text{Cu}(Z=29)$ has a wavelength of $1.52 \text{ \AA}$, the wavelength of the corresponding radiation of molybdenum $\text{Mo}(Z = 42)$ is
 (A) $0.71 \text{ \AA}$ (B) $1.71 \text{ \AA}$
 (C) $0.50 \text{ \AA}$ (D) $1.5 \text{ \AA}$
- The atomic ratio between the uranium isotopes ^{238}U and ^{234}U in a mineral is 1.8×10^4 . The activity of both the isotopes at the given moment is same. The half life of ^{238}U is $T_{1/2}(^{238}\text{U}) = 4.5 \times 10^9 \text{ yr}$, the half life of ^{234}U is
 (A) $4.05 \times 10^{13} \text{ yr}$ (B) $2.5 \times 10^5 \text{ yr}$
 (C) $2.00 \times 10^{13} \text{ yr}$ (D) $8.1 \times 10^{13} \text{ yr}$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

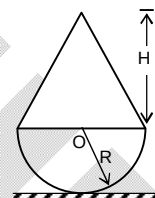
- The figure shown a ray of light incident at an angle $i = \pi/3$. For different values of μ_1 and μ_2 angle of refraction is difference. The plot drawn shows the variation of $|r - i|$ versus $\frac{\mu_1}{\mu_2} = k$, (r = angle of refraction).



- The value of k_1 is $\frac{2}{\sqrt{3}}$
- The value of $\theta_1 = \pi/6$
- The value of $\theta_2 = \pi/3$
- The value of k_2 is 1

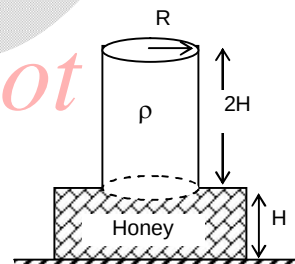
6. Two plane mirrors at an angle such that a ray incident on a mirror undergoes a total deviation of 240° after two reflections.
- (A) The angle between the mirrors is 60° .
 (B) The number of images formed by this system will be 5, if an object is placed symmetrically between the mirrors.
 (C) The number of images will be 5 if an object is kept un-symmetrically between the mirrors.
 (D) A ray will retrace its path after 2 successive reflections, if the angle of incidence with one mirror is 60° .

7. It is desired to construct a toy by attaching a solid cone of height H on the top of solid hemisphere as shown in the figure. The radius of the hemisphere and the base of cone is R . The whole system is kept on a rough horizontal surface. If the toy is slightly disturbed from the given equilibrium position it oscillates simple harmonically. Then mark the correct statements. (Given that both the cone and hemisphere are made of same material)



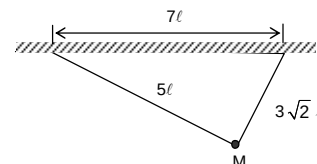
- (A) $H < \sqrt{3}R$
 (B) $H > \sqrt{3}R$
 (C) $\sqrt{3}R < H < 2R$
 (D) centre of mass of the combined system must not be above O.

8. A bottle is kept on the ground as shown in the figure. The bottle can be modelled as having two cylindrical zones. The lower zone of the bottle has a cross-sectional radius of $R\sqrt{2}$ and is filled with honey of density 2ρ . The upper zone of the bottle is filled with the water of density ρ and has a cross-sectional radius R . The height of the lower zone is H while that of the upper zone is $2H$. If now the honey and the water parts are mixed together to form a homogeneous solution (Assume that total volume does not change)



- (A) The pressure inside the bottle at the base will remain unaltered.
 (B) The normal reaction on the bottle from the ground will remain unaltered.
 (C) The pressure inside the bottle at the base will increase by an amount $(1/2)\rho gH$
 (D) The pressure inside the bottle at the base will decrease by an amount $(1/4)\rho gH$

9. A particle is suspended by two ideal strings as shown in the figure. Now mass m is given a small displacement perpendicular to the plane of triangle formed. Choose the correct statement(s).

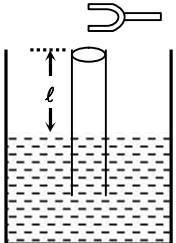


- (A) The period of oscillation of the system is $2\pi\sqrt{\frac{3\sqrt{3}l}{g}}$.
 (B) The period of oscillation of the system is $2\pi\sqrt{\frac{3l}{g}}$.
 (C) The period of oscillation of the system is independent of M .
 (D) If the distance between the suspension points was kept constant and the lengths of the strings were quadrupled then the period of the system will double.

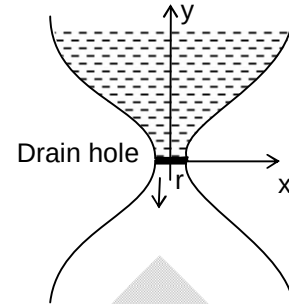
10. Two spherical bodies, each of mass M , are kept with their centres at a distance $2L$ apart. A particle of mass m is projected from the mid-point of the line joining their centres, perpendicular to the line. The gravitational constant is G . Which of the following statements is/are true?
- (A) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $4\sqrt{\frac{GM}{L}}$
- (B) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $2\sqrt{\frac{GM}{L}}$
- (C) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $\sqrt{\frac{2GM}{L}}$
- (D) The total mechanical energy of the mass m remains constant.

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.*

11. A man is looking down a 9 m deep tank filled with a transparent liquid of refractive index $\sqrt{3}$. The line of sight of the man makes an angle of 30° with the horizontal. What is the apparent depth of the tank (in meter) as seen by the man?
12. A tuning fork of known frequency is held at the open end of a long tube, which is dipped into water as shown in the figure. The tuning fork of frequency 165 Hz resonates in first and second harmonic when air columns have lengths $\ell_1 = 50 \pm 0.5$ cm and $\ell_2 = 150 \pm 0.1$ cm. Find the maximum possible error in measurement of speed of sound (in m/s) in air.
- 
13. A standing wave exists in a string of length 150cm and is fixed at both ends. The displacement amplitude of a point at a distance of 10cm from one of the ends is $5\sqrt{3}$ mm. The distance between the two nearest points, within the same loop and having displacement amplitude equal to $5\sqrt{3}$ mm, is 10cm. Find the mode of vibration of the string i.e. the harmonic produced.
14. A uniform inextensible string of length $L = 6$ m and linear density μ is suspended vertically and a transverse pulse is produced at the lower end. At the same time a point object is released from rest from the top of the string and falls freely. How far (in meter) from the top does the object passes the pulse?
15. A person standing between two parallel hills, fires a gun. He hears the first echo after 1.5sec and the second after 2.5 sec. The speed of sound is 332 m/s. When will he hear the third echo?
16. A point source of power 50π watts is producing sound waves. The velocity of sound is 330m/s, density of air is 1.0 kg m^{-3} . Then find the pressure amplitude (in Nm^{-2}) at a distance $r = \sqrt{330}$ m from the point source.

17. The shape of an ancient water clock jar is such that the water level descends at a constant rate at all times. The water level falls by λ m/s and the shape of the jar is given by $y \propto x^n$. Find the value of n if the radius r of the drain hole can be assumed to be very small.



18. During Searle's experiment, the zero of a vernier scale lies between 3.20×10^{-2} m and 3.25×10^{-2} m of the main scale divisions. The 20th division of the vernier scale exactly coincides with one of the main scale divisions. When an additional load of 2 kg is applied to the wire, the zero of the vernier scale still lies between 3.20×10^{-2} m and 3.25×10^{-2} m of the main scale but now the 45th division of the vernier scale coincides with one of the main scale divisions. The length of the thin metallic wire is 2 m and its cross-sectional area is 8.0×10^{-7} m². The least count of the vernier scale is 1.0×10^{-5} m. Find the maximum percentage error in the Young modulus of the wire.

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Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

19. The colourless inter halogen compound among the following is
 (A) IF_3
 (B) IBr
 (C) ICl
 (D) IF_7
20. What will be the value of entropy of mixing for 44 gm CO_2 gas, 28 g N_2 gas and 2 mole of Ar gas at 400 K. Considering ideal behaviour of gases.
 (A) $6R \ln \frac{1}{2}$
 (B) $\frac{3}{200} \ln 2$
 (C) $6R \ln 2$
 (D) $2400R \ln 2$
21. If the concentration (C) of a strong electrolyte is diluted 16 times. The magnitude of change in molar conductivity will be? (b is Debye, Huckel Onsagar constant)
 (A) $\frac{4}{3} b \sqrt{C}$
 (B) $\frac{b}{4} \sqrt{C}$
 (C) 0
 (D) $\frac{3}{4} b \sqrt{C}$
22. Which of the following is correct electronic configuration of ${}_{64}\text{Gd}^{2+}$ ion?
 (A) $[\text{Xe}] 4f^8 5d^0 6s^2$
 (B) $[\text{Xe}] 4f^7 5d^0 6s^1$
 (C) $[\text{Xe}] 4f^5 5d^1 6s^2$
 (D) $[\text{Xe}] 4f^7 5d^1 6s^0$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

23. The correct option(s) is/are
 (A) In 1 equivalent of H_2SO_4 , 2 equivalent of H-atom is present.
 (B) Volume strength of $\frac{1}{5} \text{N}$ H_2O_2 solution is 1.12.
 (C) 14 gm of ethene required, 48 gm of oxygen gas for complete combustion.
 (D) Number of moles of oxygen atom in 276 gm of N_2O_4 is 6.

24. Which of the following solution(s) is/are isotonic with 0.2 M NaCl ($\alpha = 1$) solution at identical temperature. (α = degree of dissociation)
- (A) 0.2 M MgCl_2 ($\alpha = 0.5$) solution.
 (B) 18 g of glucose present in 250 ml solution.
 (C) 0.5 mole of $\text{Al}(\text{NO}_3)_3$ ($\alpha = 0.4$) present in 4 litre solution.
 (D) 1 mole AlCl_3 ($\alpha = 0.2$) present in 4 litre solution.
25. The C – O bond length in $\text{BH}_3 \cdot \text{CO}$, $[\text{Mn}(\text{CO})_6]^+$, $[\text{V}(\text{CO})_6]^-$ and $[\text{Ti}(\text{CO})_6]^{2-}$ are w, x, y and z Å respectively. The correct option is/are
- (A) $x > w$
 (B) $y < z$
 (C) $z < x$
 (D) $y > w$
26. For 1 mole of an ideal gas
- $$\left(\frac{dP}{dT}\right)_V \cdot \left(\frac{dV}{dT}\right)_P =$$
- (A) $\frac{R}{T}$ (B) $\frac{1}{T}$
 (C) $\frac{R^2}{PV}$ (D) $\frac{R^2}{P^2}$
27. Which of the following statement(s) is/are true about PH_3 ?
- (A) It is a toxic gas.
 (B) It can be prepared by hydrolysis of white phosphorous with NaOH solution.
 (C) It is more basic than NH_3 .
 (D) The P – H bond in PH_3 is stronger than N – H bond in NH_3 .
28. The correct statement(s) is/are
- (A) NH_4^+ and Mg^{2+} cations (basic radical) belongs to zero group.
 (B) Chloride of Hg(I) and Hg(II) do not respond to chromyl chloride test.
 (C) In the brown ring test of NO_2^- , the FeSO_4 solution must be freshly prepared.
 (D) Hg^{+2} ion on reaction with excess KI produce yellow coloured HgI_2 precipitate.

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.*

29. How many of the following cations (basic radicals) do not belong to group-V.
 Ni^{+2} , Mg^{+2} , Sr^{+2} , Zn^{+2} , K^+ , Fe^{+3} , Ca^{+2} , Mn^{+2} , Pb^{+2}
30. The number of white coloured compounds among the following
 Ag_2CO_3 , Ag_2SO_3 , AgNO_2 , Ag_2S , AgI , Ag_2CrO_4 , PbS , ZnS , FeS , $\text{Fe}(\text{OH})_3$, BaCO_3 .
31. In crystal structure of Fe_3O_4 metal ions occupy void (Tetrahedral and octahedral) space. While O^{2-} ion present at FCC lattice. What will be total number of unpaired electron present in metal ion occupying tetrahedral voids per unit cell.

32. How many of the following ores contains CO_3^{2-} ion.
Cerrusite, Azurite, Calamine, Zincite, Siderite, Magnetite, Magnesite, Dolomite, Bauxite.
33. How many of the following species are coloured due to charge transfer
 $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$, V_2O_5 , $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$, $\text{K}_2\text{Cr}_2\text{O}_7$, MnO_4^- , $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$
34. Total number of correct statements among the following is
1. Pressure and density are intensive properties.
 2. Every isolated system is closed system and vice-versa.
 3. Entropy change for adiabatic reversible process is zero.
 4. A reaction is non-spontaneous at any temperature when enthalpy change is positive and entropy change is negative.
 5. For cyclic process internal energy change is zero.
 6. For isobaric process, volume (for fixed amount of an ideal gas) get doubled when temperature increases from 100°C to 200°C .
 7. The dissolution of NaCl in H_2O is endothermic though it is spontaneous process.
 8. Heat of combustion of ethane is -341.1 kcal/mole and heat of combustion of ethyne is -310 kcal/mole means ethane is a better fuel.
35. Total number of geometrical isomers for complex
 $[\text{Fe}(\text{en})(\text{F})(\text{Cl})(\text{Br})(\text{I})]^{2-}$ is (en = 1, 2-ethylenediamine).
36. How many of the following Antibiotics has inhibitory effect on microbes
Peniciline, Ertythromycin, Ofloxacin, Tetracyclin, Amonoglucosides, Chloramphenical.

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. A real number a is chosen randomly and uniformly from the interval $[-20, 18]$ the probability that the roots of the polynomial $x^4 + 2ax^3 + (2a - 2)x^2 + (-4a + 3)x - 2$ are all real is
 (A) $\frac{17}{19}$ (B) $\frac{13}{19}$
 (C) $\frac{21}{19}$ (D) $\frac{18}{19}$
38. Find the number of permutations x_1, x_2, x_3, x_4, x_5 of numbers 1, 2, 3, 4, 5 such that the sum of five products $x_1x_2x_3 + x_2x_3x_4 + x_3x_4x_5 + x_4x_5x_1 + x_5x_1x_2$ is divisible by 3
 (A) 16 (B) 80
 (C) 32 (D) 48
39. In the complex plane, let A be the set of solutions to $z^3 - 8 = 0$ and let B be the set of solution to $z^3 - 8z^2 - 8z + 64 = 0$, then the maximum distance between a point of A and a point of B is
 (A) $\sqrt{21}$ (B) $3\sqrt{21}$
 (C) $4\sqrt{21}$ (D) $2\sqrt{21}$
40. Let $f(x) = \{x\}[x]$; $g(x) = ax^2$ the sum of all real solutions of equation satisfying $f(x) = g(x)$ is 420. (Where a is (+ve) rational number), then a is equal to
 (Where $[.]$ and $\{.\}$ represents greatest integer function and fractional part function respectively)
 (A) $\frac{28}{900}$ (B) $\frac{29}{900}$
 (C) $\frac{31}{900}$ (D) $\frac{37}{900}$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. Let $P(x) = ax^2 + bx + c$ where a, b, c are real numbers and $a \neq 0$, the polynomial $P(x^2 + 4x - 7)$ has $x = 1$ as a root and at least one repeated roots, then the roots of polynomial $P(x^2 + 4x - 7)$ can be
 (A) $x = 1$ (B) $x = -5$
 (C) $x = -2$ (D) $x = 2$
42. If $b_n = \sum_{r=0}^n \frac{1}{n C_r}$, $n \geq 1$, then b_n can be written as
 (A) $b_n = \frac{1}{n} \cdot \sum_{r=0}^{n-1} \frac{r+1}{n-1 C_r} + 1$ (B) $b_n = \frac{1}{n} \cdot \sum_{r=0}^{n-1} \frac{r+1}{n-1 C_r}$
 (C) $b_n = \frac{n+1}{2n} \cdot b_{n-1} + 1$ (D) $b_n = \frac{n-1}{2n} \cdot b_{n-1} + 1$

43. Equation of two lines passes through point A and B whose position vectors are $\vec{a}$ and $\vec{b}$ is $\vec{r} = \vec{a} + \lambda \vec{\alpha}$ and $\vec{r} = \vec{b} + \mu \vec{\beta}$. The position vector of R and S will be, where RS is the shortest distance between the given lines is

(A) $\vec{a} + \frac{[(\vec{a} - \vec{b}) \cdot (\vec{\beta} \times (\vec{\beta} \times \vec{\alpha}))]}{|\alpha \times \beta|^2} \cdot \vec{\alpha}$

(B) $\vec{a} + \frac{[(\vec{a} - \vec{b}) \cdot (\vec{\beta} \times (\vec{\beta} \times \vec{\alpha}))]}{|(\alpha \times \beta)|^2} \vec{\alpha}$

(C) $\vec{b} + \frac{[(\vec{a} \times \vec{b}) \cdot (\vec{\alpha} \times (\vec{\beta} \times \vec{\alpha}))]}{|\alpha \times \beta|^2} \vec{\beta}$

(D) $\vec{b} + \frac{(\vec{a} - \vec{b}) \cdot (\vec{\alpha} \times (\vec{\beta} \times \vec{\alpha})) \vec{\beta}}{|\alpha \times \beta|^2}$

44. A and B are two events defined as follows:

A : It rains today with probability $P(A) = 40\%$. **B** : It rains tomorrow with probability $P(B) = 50\%$

Also, probability $P(\text{It rains today and tomorrow}) = 30\%$. Also, $E_1 : P\left(\frac{(A \cap B)}{(A \cup B)}\right)$,

$E_2 : P\left(\frac{\{(A \cap \bar{B}) \text{ or } B \cup \bar{A}\}}{A \cup B}\right)$, then which of the following is/are TRUE?

(A) A and B are independent

(B) $P\left(\frac{A}{B}\right) < P\left(\frac{B}{A}\right)$

(C) E_1 and E_2 are equiprobable

(D) $P\left(\frac{A}{A \cup B}\right) = P\left(\frac{B}{A \cup B}\right)$

45. Let A_n be an $n \times n$ matrix in which diagonal elements are 1, 2, 3, 4, n (i.e., $a_{11} = 1, a_{22} = 2, a_{33} = 3, \dots, a_{nn} = n$) and all other elements are equal to n

(A) A_n is singular for all n

(B) A_n is non-singular for all n

(C) $\det A_n = (-1)^n \cdot n!$

(D) $\det A_n = (-1)^{n+1} \cdot n!$

46. If $C_r = {}^nC_r$, then sum of the series $S = C_0^2 + \frac{C_1^2}{2} + \frac{C_2^2}{3} + \dots$ upto $(n + 1)$ terms, which of the following options is/are TRUE?

(A) $\frac{1}{(n+1)} \cdot {}^{2n+1}C_{n+1}$

(B) $\frac{1}{2(n+1)} \cdot {}^{2n+2}C_{n+1}$

(C) $\frac{1}{(n+1)} \cdot {}^{2n+1}C_n$

(D) $\frac{(2n+1)!}{((n+1)!)^2}$

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

47. Let $x_1 < x_2 < x_3$ be the three real roots of the equation $\sqrt{2014} x^3 - 4029x^2 + 2 = 0$, then $x_2(x_1 + x_3)$ is
48. There exists a unique strictly increasing sequence of non-negative integers $a_1 < a_2 < \dots < a_k$ such that $\frac{2^{289} + 1}{2^{17} + 1} = 2^{a_1} + 2^{a_2} + \dots + 2^{a_k}$, then $(k - 130)$ is
49. 10 chairs are arranged in a circle. Then the number of subset of this set of chairs that contain at least three adjacent chairs is N, then $(N - 580)$ is

50. Let a_n and b_n are real number such that $(2+i)^n = a_n + b_n i$ for all integers $n \geq 0$ where $i = \sqrt{-1}$, if $\sum_{n=0}^{\infty} \frac{a_n b_n}{7^n}$ is k , then $16k$ is equal to
51. Let the sequence is 4, 7, 1, 8, 9, 7, 6, for $n > 2$ the n th term of the sequence is the units digit of the sum of the two previous terms. Let S_n denote the sum of the first n terms of this sequence. The smallest value of n for which $S_n > 10,000$ is p , then $(p - 1990)$ is
52. Let ω and z be complex numbers such that $|\omega| = 1$ and $|z| = 10$. Let $\theta = \arg\left(\frac{\omega - z}{z}\right)$. Let $\frac{1}{m}$ is the maximum possible value of $\tan^2 \theta$, then $(m - 95)$ is
53. The number of rational number r , $0 < r < 1$, such that when r is written as a fraction in lowest terms the numerator and denominator have a sum of 1000 is N , then $(N - 197)$ is
54. Let A_n and B_n be square matrices of order 3, which are defined as $A_n = [a_{ij}]$ and $B_n = [b_{ij}]$ where $a_{ij} = \frac{2i+j}{3^{2n}}$ and $b_{ij} = \frac{3i-j}{2^{2n}} \forall i$ and j , $1 \leq i, j \leq 3$, if $\ell = \lim_{n \rightarrow \infty} T_r(3A_1 + 3^2 A_2 + 3^3 A_3 \dots + 3^n A_n)$ and $m = \lim_{n \rightarrow \infty} T_r(3B_1 + 2^2 B_2 + 2^3 B_3 \dots + 2^n B_n)$, then the value of $\frac{\ell + m}{3}$, where $T_r(A)$ represent trace of matrix A

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FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – III
PAPER –1
TEST DATE: 25-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A, B, C

Sol. Rays passing through optical centres pass unreflected. Two images are formed at I_1 and I_2
For a convex lens,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$u = -0.4\text{m}, f = 0.2\text{m}$$

$$= \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{0.4} = \frac{1}{0.2}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{0.2} = \frac{1}{0.4}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{0.4}$$

(A) $v = 0.4$ meter.

(B) Number of images formed = 2

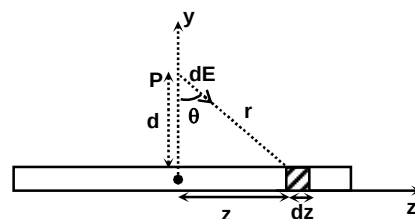
(C) Separation between images is $2\left(3 + 3 \times \frac{0.4}{0.4}\right) = 12\text{ mm}$

2. A, B, C, D

Sol. Gravitational field due to a long wire, $dE = \frac{2G\lambda}{r} = \frac{2G\sigma dz}{r}$

$$E_p = \int dE \cdot \cos\theta = \int \frac{2G\sigma dz}{r} \frac{d}{r} = 2G\sigma d \int_{-\frac{a}{2}}^{+\frac{a}{2}} \left(\frac{dz}{d^2 + z^2} \right)$$

$$E_p = 4G\sigma \tan^{-1}\left(\frac{a}{2d}\right)$$

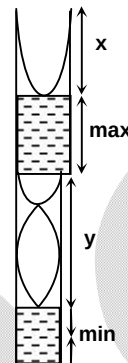


3. A, B, C, D

Sol. $Q = \Delta K$
and use basic concept

4. A, B, C

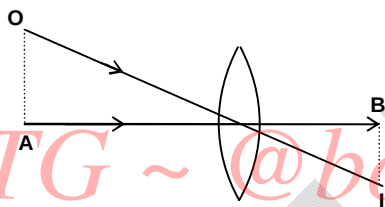
Sol. $\lambda = 1\text{m}$
 $\lambda/4 = x$
 So, $x = 25\text{ cm}$
 $\text{Max} = 125 - 25$
 $\text{Max} = 100\text{ cm}$
 $y = 3\lambda/4 = 75\text{ cm}$
 $\text{MIN} = 125 - 75 = 50\text{ cm}$
 $\lambda/2 = \text{distance between nodes} = 50\text{ cm}$



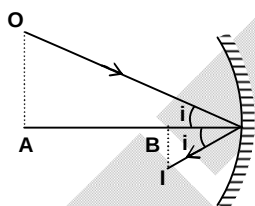
5. A, C, D

Sol.

(A)



(C)



(D) Image is inverted $\Rightarrow$ It should be real

6. A, B, C

Sol. Due to a convex mirror of focal length 2.5 cm and due to a concave mirror having its pole at (2cm, 0) real virtual pair

7. D

Sol. Use basic formulae for the given physical quantities to derive the dimensions.

8. D

Sol. $F = +20\text{ cm}$

$u = -30\text{ cm}$

$$v = \frac{uF}{u + F} = \frac{(-30)(20)}{-30 + 20} = \frac{+600}{-10} = +60\text{ cm}$$

$$m = \frac{v}{u} = \frac{60}{-30} = -2$$

Real, inverted and magnified

$F = -40\text{ cm}$

$u = -30\text{ cm}$

$$v = \frac{(-30)(-40)}{-30 + 40} = 120 \text{ cm}$$

$$m = -\left[\frac{v}{u}\right] = -\left[\frac{120}{-30}\right] = +4$$

virtual, erect and magnified

$$(R) \frac{u_2}{v} - \frac{u_1}{u} = \frac{u_2 - u_1}{R}$$

$$\frac{1}{v} - \frac{3/2}{-20} = \frac{1 - 3/2}{-10}$$

$$\frac{1}{v} + \frac{3}{40} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{3}{40} = \frac{2-3}{40} = -\frac{1}{40} \Rightarrow v = -40$$

$$m = +\frac{u_i}{u_r} \frac{v}{u} = \frac{+3/2}{1} \times \left[\frac{-40}{-20}\right] = +3$$

Virtual, erect, magnified

$$(S) v = \frac{uF}{u+F} = \frac{(-40)(-40)}{-40-40} = \frac{1600}{-80} = -20$$

$$m = \frac{v}{u} = \frac{-20}{-40} = +\frac{1}{2}$$

Virtual, erect, diminished.

9. C

Sol. Use momentum conservation during collision. In case of a plane mirror image and object will be equidistant from the mirror

10. D

Sol. $Q = (\Sigma B \cdot E)_{\text{Product}} - (\Sigma B \cdot E)_{\text{Reactant}}$ **Q**

Section – B

11. 0.24

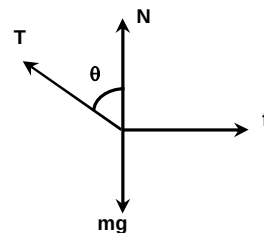
Sol. Thrust of the ejected water on the tank $T = \rho a u^2$ where u is the speed of the ejected water from tank

$$u = \sqrt{\frac{2gh}{1 - (a/A)^2}}$$

Now for the tank to slide, $T \sin \theta \geq \mu(mg - T \cos \theta)$

$$\Rightarrow T \geq \frac{\mu mg}{\sin \theta + \mu \cos \theta}$$

$$\Rightarrow \rho a u^2 \geq \frac{\mu A h \rho g}{\sin \theta + \mu \cos \theta} \Rightarrow \frac{2(a/A)}{1 - (a/A)^2} \geq \frac{\mu}{\sin \theta + \mu \cos \theta}$$



$$\Rightarrow \frac{2x}{1-x^2} \geq m \text{ where } x = \frac{a}{A} \text{ and } m = \frac{\mu}{\sin\theta + \mu \cos\theta}$$

$$\Rightarrow x \geq \frac{\sqrt{1+m^2}-1}{m} \text{ for } \theta = 37^\circ \text{ and } \mu = 0.5$$

$$m = 0.5 \text{ and } x \geq \frac{\sqrt{1+0.25}-1}{0.5}$$

$$\text{Hence } x_{\min} = 0.236$$

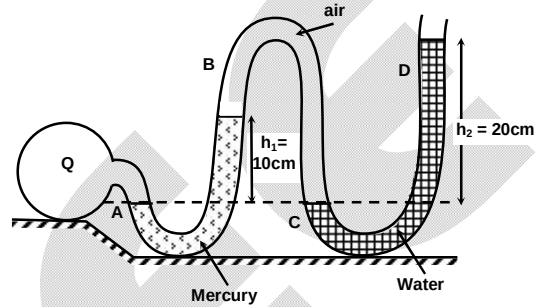
12. 0.12

Sol. $P_C = P_D + \rho_2 g h_2 = P_0 + \rho_2 g h_2$

$$P_A = P_B + \rho_1 g h_1$$

and $P_B = P_C$ (neglecting the density of air as compared to water or mercury)

$$\begin{aligned} \therefore P_A &= P_0 + \rho_2 g h_2 + \rho_1 g h_1 \\ &= 10^5 + 10(13.8 \times 0.1 + 1 \times 0.2) \times 10^3 \\ &= 10^5 + 10^4 (1.38 + 0.2) = 1.16 \times 10^5 \text{ N/m}^2 \end{aligned}$$



13. 6.00

Sol. Area of element = $w dy$

$$dF = \rho w g y dy$$

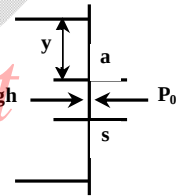
$$\text{Net force} = \frac{\rho w g D^2}{2} \dots (1)$$

$$d\tau = \rho w g y dy (D - y)$$

$$\tau = \frac{\rho w g D^3}{6} \dots (2)$$

Putting the value of F and τ in equation (1) and (2)

$$D = 10 \text{ m and } W = 6 \text{ m}$$



14. 3.75

Sol. Let V be the total volume of the piece of wood and V' be its volume inside the liquid. Since the wood piece is in equilibrium

Weight of the wood piece = buoyant force

$$\Rightarrow V \rho g = V' \rho_0 g$$

$$\Rightarrow A L \rho g = A L' \rho_0 g$$

• Now if the wood piece is pressed by distance x below the fluid.

$$\text{Buoyant force} = A(L' + x) \rho_0 g$$

$$\text{Net force} = A(L' + x) \rho_0 g - A L \rho g$$

$$= A L' \rho_0 g + A x \rho_0 g - A L \rho g$$

$$= A x \rho_0 g \text{ in the vertically upward direction (Since } A L' \rho_0 g = A L \rho g \text{)}$$

• Therefore the pressing force = $A x \rho_0 g$ in the vertically downward direction.

The work done by the pressing force for a small displacement dx

$$dW = F dx = A x \rho_0 g dx$$

Total work done by the pressing force

$$\begin{aligned} W &= \int dW = A \rho_0 g \int_0^{L/2} x dx \\ &= \frac{A \rho_0 g L^2}{8} = \frac{10^{-2} \times 1200 \times 10 \times (0.5)^2}{8} = 3.75 \text{ J} \end{aligned}$$

15. 13.33

Sol. $R = V/I = 12/2 = 6 \Omega$

$$\text{Error, } \frac{\Delta R}{R} \times 100 = \left(\frac{\Delta V}{V} + \frac{\Delta I}{I} \right) \times 100$$

$$= \left(\frac{0.4}{12} + \frac{0.2}{2} \right) \times 100$$

$$= 13.33\%$$

$$\therefore \text{Resistance} = (6 \pm 13.33\%) \Omega$$

16. 1.53

Sol. Pitch of the Screw gauge = $5/10 \text{ mm} = 0.5 \text{ mm}$

$$\text{Least count} = \frac{\text{Pitch}}{100} = \frac{0.5}{100} = 0.005 \text{ mm}$$

$$\text{Pitch scale reading} = \text{Pitch} \times N = 0.5 \times 3 = 1.5 \text{ mm}$$

$$\text{Difference of the cap reading} = F - I = 6$$

$$\text{Circular scale reading} = \text{Difference} \times \text{L.C} = 0.03 \text{ mm}$$

$$\text{Diameter} = \text{P.S.R.} + \text{C.S.R.} = (1.5 + 0.03) \text{ mm} = 1.53 \text{ mm}$$

17. 0.20

Sol. The focal length of the lens is given by

$$\frac{1}{f} = \frac{1}{|v|} + \frac{1}{|u|}$$

substituting the values, we get,

$$f = 19.9 \text{ cm}, 20.1 \text{ cm}, 19.9 \text{ cm}, 20.2$$

The average focal length $\approx 20.0 \text{ cm}$

The error in a typical measurement:

$$\frac{\delta f}{f^2} = \frac{\delta u}{u^2} + \frac{\delta v}{v^2}$$

$$\frac{\delta f}{f} = \left(\frac{0.4}{40^2} + \frac{0.4}{(40.5)^2} \right) \times 20 \approx 0.01$$

$$\delta f = 0.01 \times 20 = 0.2 \text{ cm}$$

18. 8.00

Sol. Initial orbital speed $v_1 = \sqrt{\frac{GM}{2R}}$... (i)

Final orbital speed $v_2 = \sqrt{\frac{GM}{4R}}$... (ii)

For elliptical orbit

From COAM

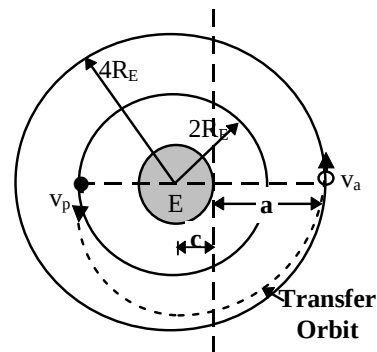
$$v'_1 \times 2R = v'_2 \times 4R$$

$$\Rightarrow v'_2 = \frac{v'_1}{2}$$

$$\text{From COE } \frac{1}{2} m v_1'^2 - \frac{GMm}{2R} = \frac{1}{2} m v_2'^2 - \frac{GMm}{4R}$$

$$\Rightarrow v'_1 = \sqrt{\frac{2GM}{3R}}$$

$$\text{So, energy required} = \frac{1}{2} m \left[\frac{2GM}{3R} - \frac{GM}{2R} \right] + \frac{1}{2} m \left[\frac{GM}{4R} - \frac{1}{4} \frac{2GM}{3R} \right] = \frac{GMm}{8R}$$



Chemistry

PART – II

Section – A

19. A, C

Sol. Suppose meq. of NaOH and Na_2CO_3 in the mixture are x and y respectively.
When phenolphthalein is used as indicator

$$x + \frac{y}{2} = 40 \times 0.1 = 4 \quad \dots (i)$$

When methyl orange is used as indicator

$$x + y = 60 \times 0.1 = 6 \quad \dots (ii)$$

Solving Eq. (i) and Eq. (ii), we get

$x = 2$ meq. and $y = 4$ meq.

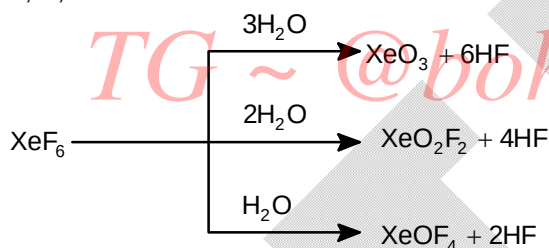
$$\therefore \text{Weight of NaOH in the mixture} = (2 \times 10^{-3} \times 40) \times 5 = 0.4 \text{ g.}$$

$$\text{Weight of Na}_2\text{CO}_3 \text{ in mixture} = (4 \times 10^{-3} \times 53) \times 5 = 1.06 \text{ g}$$

$$\text{Weight \% of NaOH in the mixture} = \frac{0.4}{0.4 + 1.06} \times 100 = 27.4\%$$

20. A, B, D

Sol.



21. B, C, D

Sol. Physisorption is multilayer

$\log (x/m)$ vs. $\log (P)$

Slope is +ve.

$\log (x/m)$

$\log (P)$

Physisorption favour at low temperature and high pressure.

$$|\Delta H_{\text{physisorption}}| < |\Delta H_{\text{chemisorption}}|$$

22. A, B, D

Sol. Mond's process is used for the purification of Ni.

23. B, C

Sol. $M_{\text{mix}} = \frac{1 \times 16 + 2 \times 4}{1 + 2} = 8$

$$\frac{r_x}{r_{\text{mix}}} = \sqrt{\frac{M_{\text{mix}}}{M_x}}$$

$$\Rightarrow 2 = \sqrt{\frac{8}{M_x}} \quad \therefore M_x = 2$$

$\therefore$ Gas is H_2 gas.

$$\frac{r_{\text{H}_2}}{r_{\text{He}}} = \sqrt{\frac{4}{2}} = \sqrt{2} \quad \therefore r_x = \sqrt{2} r_{\text{He}}$$

$\therefore$ (B) and (C) are correct statement.

24. A, C, D

Sol. From given data

$$X_A = \frac{1}{3} \quad X_B = \frac{2}{3}$$

Where, X_A and X_B are mole fraction of A and B in liquid phase.

If the Y_A and Y_B are mole fraction of A and B in vapour phase.

$$\frac{Y_A}{Y_B} = \frac{P_A^0 X_A / P_{\text{total}}}{P_B^0 X_B / P_{\text{total}}} = \frac{P_A^0 X_A}{P_B^0 X_B} = \frac{100 \times \frac{1}{3}}{80 \times \frac{2}{3}} = \frac{5}{8}$$

$$\frac{Y_A}{Y_B} = \frac{n_A}{n_B} = \frac{W_A / M_A}{W_B / M_B} = \frac{W_A}{W_B} \quad (\because A \text{ and } B \text{ are isomer})$$

$$\therefore \text{Mass ratio of A and B} = \frac{5}{8}$$

$$P_{\text{Total}} = P_A^0 X_A + P_B^0 X_B = 100 \times \frac{1}{3} + 80 \times \frac{2}{3} = \frac{260}{3} \text{ torr}$$

25. D

Sol. Ziegler Natta $\rightarrow \text{TiCl}_4 + \text{Al}(\text{C}_2\text{H}_5)_3$

Ostwald process $\rightarrow \text{Pt}$

Contact process $\rightarrow \text{V}_2\text{O}_5$

Haber process $\rightarrow \text{Fe}$

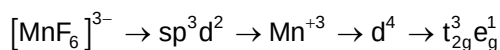
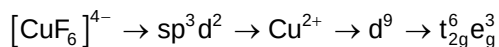
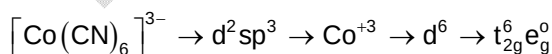
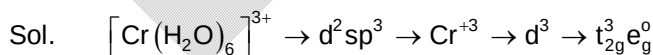
26. B

Sol. Conceptual

27. B

Sol. Application of Boyle's Law.

28. D



Section – B

29. 8.00

Sol.
$$[Ag^+]_{Anode} = \frac{K_{SP}}{[Br^-]} = \frac{K_{SP}}{1 \times 0.75} M$$

$$[Ag^+]_{Cathode} = 0.1 \times 0.8 = 0.08 M$$

$$E_{cell} = 0.0591 \log \frac{[Ag^+]_{Cathode}}{[Ag^+]_{Anode}}$$

$$0.591 = 0.0591 \log \left[\frac{0.08}{K_{SP} / 0.75} \right]$$

$$10 = \log \frac{0.75 \times 0.08}{K_{SP}}$$

$$\therefore 10^{+10} = \frac{0.75 \times 0.08}{K_{SP}}$$

$$\therefore K_{SP} = \frac{0.08 \times 0.75}{10^{10}} = 6 \times 10^{-12}$$

$$\text{Solubility} = \frac{K_{SP}}{[Br^-]} = \frac{6 \times 10^{-12}}{0.75} \text{ mol / lit.}$$

$$x \times 10^{-12} = 8 \times 10^{-12}$$

$$\therefore x = 8$$

30. 15.58

Sol.
$$V_A = \frac{RT}{P_A} = \frac{0.082 \times 500}{2} = 20.5 \text{ lit}$$

A → B is isothermal path

$$\therefore V_B = \frac{P_A V_A}{P_B} = \frac{21.5 \times 2}{4} = 10.25 \text{ litre}$$

B → C is adiabatic path

$$\frac{P_B}{P_C} = \left(\frac{V_C}{V_B} \right)^\gamma$$

$$\therefore C_P = \frac{5}{2} R$$

$$\frac{4}{2} = \left(\frac{V_C}{V_B} \right)^{\frac{5}{3}}$$

$$\therefore C_V = \frac{3}{2} R$$

$$\log 2 = \frac{5}{3} \log \frac{V_C}{V_B}$$

$$\gamma = \frac{5}{3}$$

$$\Rightarrow \log \frac{V_C}{V_B} = \log 2^{3/5}$$

$$\frac{V_C}{V_B} = 2^{3/5} = 1.52$$

$$\therefore V_C = 1.52 \times 10.25$$

$$= 15.58 \text{ litre}$$

31. 729.00

Sol.
$$M_{\text{AgX}} = \frac{\rho \times N_A \times a^3}{4} = \frac{6.022 \times 10^{23} \times (6 \times 10^{-8})^3}{4}$$

$$= 216 \text{ g / mole}$$

$$\therefore M_X = 216 - 108 = 108 \text{ g / mole}$$

If all atoms along one body diagonal have been removed, the formula of unit cell

$$= \text{Ag}_{4-1} \text{X}_{\frac{4-1}{4}} = \text{Ag}_3 \text{X}_{\frac{15}{4}}$$

$$\therefore \text{Mass of 1 mole of these unit cell} = 108 \times 3 + 108 \times \frac{15}{4} = 729.00$$

32. 4.35

Sol. Number of equivalent of HCl used = Number equivalent of CaCl_2 obtained = Number of equivalent of plaster of paris can form $= \frac{600 \times 0.1}{1000} = 0.06$

Equivalent weight of plaster of paris $\left(\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O} \right) = \frac{\text{Mol.wt.}}{2} = \frac{145}{2}$

$$\therefore \text{Maximum weight of } \text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O} \text{ that can be obtained} = 0.06 \times \frac{145}{2} = 4.35 \text{ g.}$$

33. 6.25

Sol.
$$\kappa = \frac{1}{R} \times \frac{\ell}{A} = \frac{1}{50} \times \frac{1}{1} = \frac{1}{50} \text{ S cm}^{-1}$$

$$\Lambda_m = \frac{\frac{1}{50} \times 1000}{0.2} = 100 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{100}{\lambda_{\text{H}^+}^0 + \lambda_{\text{CH}_3\text{COO}^-}^0} = \frac{100}{50 + 350} = \frac{1}{4}$$

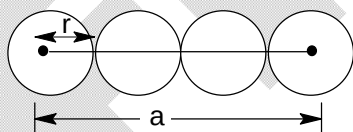
Van't Hoff factor $i = 1 + \alpha = 1.25$

$\pi = i.C.R.T.$

$$\pi = 1.25 \times 0.2 \times 300 \times \frac{1}{12} = 6.25 \text{ atm.}$$

34. 78.50

Sol. $a = 6r$



Effective number of atom in one unit cell $= 4 \times \frac{1}{4} + 8 \times \frac{1}{2} + 4 = 9$

$$\therefore \%P.E. = \frac{9 \times \pi r^2}{a^2} \times 100$$

$$= \frac{9 \times 3.14 \times r^2}{(6r)^2} \times 100$$

$$= \frac{9 \times 314r^2}{36r^2} = \frac{314}{4}$$

$$= 78.5\%$$

35. 0.70

Sol. Amount of thermal energy required = $w \times C_S \times \Delta T$

$$\text{Amount of thermal energy in beaker A} = 200 \times C_S \times (90 - 40) = 10000 C_S$$

$$\text{Amount of thermal energy in beaker B} = 100 \times C_S \times (90 - 20) = 7000 C_S$$

$$\text{Amount of thermal energy in beaker C} = 150 \times C_S \times (90 - 50) = 6000 C_S$$

$$\text{Amount of thermal energy in beaker D} = 300 \times C_S \times (90 - 30) = 18000 C_S$$

$$\therefore w = 6000 C_S, x = 7000 C_S, y = 10000 C_S, z = 18000 C_S$$

$$\therefore \frac{x}{y} = \frac{7000 C_S}{10000 C_S} = 0.7$$

36. 9.00

Sol. Weight of oxalic acid in 50 ml solution before adsorption = $\frac{45}{500} \times 50 = 4.5 \text{ gm}$

$$\text{Weight of oxalic acid in 50 ml solution after adsorption} = 0.6 \times \frac{50}{1000} \times 90 = 2.7 \text{ gm}$$

$$\therefore \text{Weight of oxalic acid adsorbed by 0.2 gm of charcoal} = 4.5 - 2.7 = 1.8 \text{ g}$$

$$\therefore \text{Weight adsorbed per gram} = 1.8 \times \frac{1}{0.2} = 9 \text{ gm.}$$

Mathematics

PART – III

Section – A

37. A, B, C

Sol. $P_4 = \frac{{}^4C_2}{4^4}$, $P_5 = 0$, $P_6 = \frac{2(60-12)}{4^6} = \frac{96}{4^6}$

38. A, D

Sol. $|\vec{P}_1| = \sqrt{3}$, $|\vec{P}_2| = 1$ and $\vec{P}_1 \cdot \vec{P}_2 = 0 \Rightarrow |\vec{P}_3 \cdot \vec{P}_1 \times \vec{P}_2| = |\vec{P}_3| |\vec{P}_1| |\vec{P}_2|$

39. C, D

Sol. $\frac{m}{n} = \frac{101}{51}$, $N = \frac{(m-1)(n-1)}{2}$

40. A, C, D

Sol. $\frac{(z^3 - z^2)}{(z - z^2)} \cdot \frac{(z - z^4)}{(z^3 - z^4)} \in \mathbb{R} \Rightarrow (z - \bar{z})(z\bar{z} - 1) = 0$ for given order z, z^2, z^3, z^4
 $\Rightarrow \arg z \in \left(0, \frac{2\pi}{3}\right) \cup \left(\frac{4\pi}{3}, 2\pi\right)$

41. A, C, D

Sol. $\det(M_1) = \det(M_2) = 0 \Rightarrow \det(aM_1 + bM_2) = 0 \forall a, b \in \mathbb{R}$

42. A, B

Sol. z_1, z_2, z_3 are the roots of the equation $z^3 - 2\lambda z^2 + 2\lambda^2 z - \lambda^3 = 0$

43. B

Sol. $n(A) > n(B) \rightarrow$ Total number of cases 211
 $n(A) = n(B) \rightarrow$ Total number of 32 cases

44. D

Sol. $(a + b + c)((a - b)^2 + (b - c)^2 + (c - a)^2) = 2$
 If $a, b, c \in \mathbb{Z} \Rightarrow (1, 0, 0), (0, 1, 0), (0, 0, 1)$ are solutions
 If $t \in \mathbb{Q} \Rightarrow a = c, 2a + b = \frac{1}{t^2}, a - b = t \Rightarrow$ infinite possibility

45. D

Sol. $|((z - 2)^8 - 3^8)| = |(z - z_1)(z - z_2)(z - z_3) \dots (z - z_8)|$

46. C

Sol. (I) There are 4 distinct path
 (II) There are $\frac{1}{3}({}^{10}C_5)$ distinct path
 (III) Count rectangle of all possible dimension = 73
 (IV) There are 218 distinct path

Section – B

47. 95.00

Sol. Let θ_i be the event that i is the upper left corner of required square

$$\Rightarrow P(\theta_1 \cup \theta_2 \cup \theta_3 \cup \theta_4) = \frac{95}{512} \Rightarrow P(\theta_1^c \cap \theta_2^c \cap \theta_3^c \cap \theta_4^c) = \frac{417}{512}$$

48. 81.00

Sol. Suppose there are T_n ways to cover n steps

$$T_n = T_{n-1} + T_{n-2} + T_{n-3}, T_1 = 1, T_2 = 2$$

49. 220.00

Sol. Let $p = 9 - a, q = 9 - a_3 \Rightarrow p + a_2 + q + a_4 = 9 \Rightarrow {}^{12}C_3 = 220$

50. 4.00

Sol. $|A|^2 + |A|A + \text{Tr}(A) \cdot I_2 = 4(2(|A| - 1)^2 + 1)$

51. 23.00

Sol. $P(x) = x^3 + 2x^2 + 3x + 3$

52. 3.00

Sol. $\left(\frac{\vec{p} + \vec{q} + \vec{r}}{3}\right) \cdot \left(\frac{\vec{p} + \vec{r}}{3} - \vec{q}\right) = 0 \Rightarrow |\vec{p}|^2 + |\vec{q}|^2 = 3|\vec{r}|^2$

53. 2.78

Sol. Let $\alpha = u, \beta = u + v, \gamma = u + 2v, \delta = u + 3v \Rightarrow p - q = -\frac{1}{9}(u+1)^2(u^2 + 2u - 9)$

54. 834.00

Sol. Let $im = n \rightarrow 9 \times 34$ cases

$im^5 = n^2 \rightarrow 8 \times 33$ cases

$im^9 = n^3 \rightarrow 8 \times 33$ cases

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – III
PAPER –2
TEST DATE: 25-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1.

B

Sol.

If $A > 2\theta_c$ then light does not emerge.

$$A = 60^\circ$$

θ_c should be less than 30°

$$\sin 30^\circ = \frac{1}{\mu}$$

$$\Rightarrow \mu = 2$$

minimum value of μ is 2.

2.

B

Sol.

$$\lambda = \frac{h}{mv}$$

$$\frac{1}{2}m\bar{v}^2 = \frac{3}{2}k_B T$$

$$v_{rms} = \sqrt{\bar{v}^2} = \sqrt{\frac{3k_B T}{m}}$$

$$\lambda = \frac{h}{mv_{rms}} = \frac{h}{\sqrt{3mk_B T}}$$

3.

A

Sol.

From Moseley's law,

$$\frac{1}{\lambda_{K_\alpha}} \propto (Z-1)^2$$

$$\frac{\lambda_{M_0}}{\lambda_{Cu}} = \frac{(Z_{Cu}-1)^2}{(Z_{M_0}-1)^2} = \frac{(28)^2}{(41)^2}$$

$$\lambda_{M_0} = \lambda_{Cu} \times \frac{(28)^2}{(41)^2} = 0.71 \text{Å}.$$

4. B

Sol. $N_{238}\lambda_{238} = N_{234}\lambda_{234}$

$$\frac{N_{238}}{T_{1/2}({}^{238}\text{U})} = \frac{N_{234}}{T_{1/2}({}^{234}\text{U})}$$

$$T_{1/2}({}^{234}\text{U}) = \frac{N_{234}}{N_{238}} \times T_{1/2}({}^{238}\text{U})$$

$$= \frac{1}{1.8 \times 10^4} \times 4.5 \times 10^9 \text{ y} = 2.5 \times 10^5 \text{ y}$$

5. B, C, D

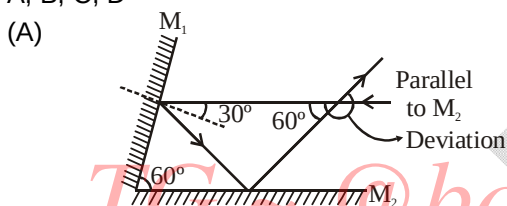
Sol. For the critical ray at the interface,

$$\mu_2 \sin i = \mu_1 \sin 90^\circ$$

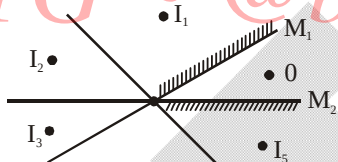
$$\theta_1 = (90^\circ - i) = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}.$$

6. A, B, C, D

Sol. (A)

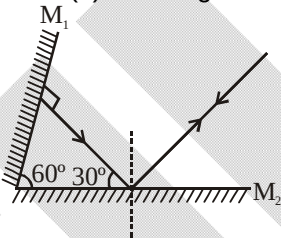


(B)



(C) same as (b) – 5 images.

(D)



7. A, D

Sol. For equilibrium, the system must be in a local minima of potential energy. This is possible only if the centre of mass of the system lies below O i.e. the centre of the hemisphere. In the critical case, the centre of mass of the system can coincide with O.

Now taking O as the origin, say H is the maximum principle height of the cone then:

$$\left(-\frac{3R}{8}\right) \cdot \frac{2}{3} \pi R^3 + \left(\frac{H}{4}\right) \cdot \frac{1}{3} \pi R^2 H = 0$$

$$\Rightarrow -\frac{R^2}{4} + \frac{H^2}{12} = 0 \quad \text{or} \quad H = \sqrt{3}R$$

8. B, C

Sol. Initial pressure at the bottom = $\rho g \times 2H + 2\rho \times g \times H = 4\rho gH$.

Final density of the homogenous mixture:

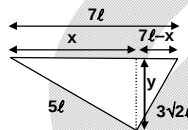
$$= \frac{\rho \times A \times 2H + 2\rho \times 2A \times H}{A \times 2H + 2A \times H} = \frac{3}{2}\rho$$

$$\text{Final pressure} = \frac{3}{2}\rho \times g \times 3H = \frac{9}{2}\rho gH.$$

9. B, C

Sol. $y^2 = (5\ell)^2 - x^2 = (3\sqrt{2}\ell)^2 - (7\ell - x)^2$

$$\Rightarrow x = 4\ell \Rightarrow y = 3\ell \Rightarrow T = 2\pi\sqrt{\frac{3\ell}{g}}$$



10. B, D

Sol. $\frac{1}{2}mv^2 + \left(-\frac{2GMm}{L}\right) = 0$

$$\Rightarrow v = 2\sqrt{\frac{GM}{L}}$$

Gravitational field is conservative, in which total mechanical energy (KE + PE) remains conserved. Hence, the kinetic energy imparted to the mass m is gradually reduced and gets converted into its potential energy, so that at every point of its flight the total mechanical energy remains constant.

Section – B

11. 1

Sol. $H_{\text{app}} = \frac{H}{\mu} \left(\frac{\cos i}{\cos r} \right)^3 = \frac{9}{\sqrt{3}} \times \left(\frac{1}{\sqrt{3}} \right)^3 = 1m$

12. 2

Sol. Let be λ is wavelength of sound in air

$$\ell_1 = \frac{\lambda}{4} \text{ and } \ell_2 = \frac{3\lambda}{4}$$

$$\lambda = 2(\ell_2 - \ell_1)$$

$$v = f\lambda = 165 \times 2(150 - 50) \times 10^{-2} = 330 \text{ m/s}$$

$$\frac{dv}{v} = \frac{|d\ell_2| + |d\ell_1|}{\ell_2 - \ell_1}$$

$$dv = \frac{0.5 + 0.1}{150 - 50} \times 330 = 1.98 \text{ m/s}$$

$v = (330 \pm 2) \text{ m/s}$, it is speed of sound in air.

13. 5

Sol. Let equation of the wave be

$$y = A \sin kx \cos \omega t$$

$$\therefore 5\sqrt{3} = A \sin(k \times 10)$$

... (1)

and $5\sqrt{3} = A \sin(k \times 20) \dots (2)$

From (1) and (2), and the given conditions,

$$20k = \pi - 10k \Rightarrow k = \frac{\pi}{30}$$

$$5\sqrt{3} = A \sin\left(\frac{\pi}{30} \times 10\right) \text{ using equation (1)}$$

$$\Rightarrow A = 10\text{mm}$$

$$\therefore k = \frac{2\pi}{\lambda} = \frac{\pi}{30},$$

$$\Rightarrow \lambda = 60\text{cm.}$$

$$\therefore L = n \frac{\lambda}{2}$$

$$\Rightarrow n = \frac{2L}{\lambda} = \frac{2 \times 150}{60} = 5$$

As the number of loops is 5, the string vibrates in the 4th overtone 5th harmonic.

14. 4

Sol. Tension in the string at a height y from the lower end
= Weight of the lower part (having y length) = $\mu y g$

$$\text{Wave speed at this point } v = \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow v = \sqrt{\frac{\mu y g}{\mu}} = \sqrt{y g}$$

$$\Rightarrow \frac{dy}{dt} = \sqrt{y g}$$

On integration we get,

$$y = \frac{gt^2}{4}$$

- Distance travelled by the body in time $t = \frac{1}{2}gt^2$

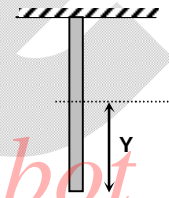
When the pulse & the body are at same level

$$\frac{gt^2}{4} + \frac{1}{2}gt^2 = L$$

$$\Rightarrow t = \sqrt{\frac{4L}{3g}}$$

Therefore, distance from the top

$$= \frac{1}{2}gt^2 = \frac{2L}{3}$$



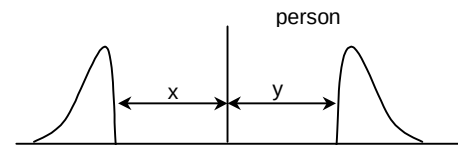
15. 4

Sol. $t_1 = \frac{2x}{v}$ & $t_2 = \frac{2y}{v}$

$$\therefore x + y = \frac{v}{2}(t_1 + t_2) = \frac{332}{2}[1.5 + 2.5] = 664 \text{ m.}$$

third echo will be heard after

$$t_3 = t_1 + t_2 = 1.5 + 2.5 = 4 \text{ sec.}$$



16. 5

Sol. $\frac{P_0^2}{2\rho V} = I = \frac{\text{Power}}{4\pi r^2}$

Put the values to get the answer.

17. 4

Sol. $(\pi r^2)\sqrt{2gy} = \pi x^2 \left(-\frac{dy}{dt} \right)$

$$\Rightarrow (r^2)\sqrt{2gy} = x^2 \lambda$$

$$\Rightarrow y \propto x^4$$

18. 4

Sol. $Y = \frac{FL}{A\ell}$

Here measurement is for ℓ only,

So, $\frac{\Delta Y}{Y} = \frac{\Delta \ell}{\ell}$

From observation, $\ell_1 = \text{MS} + 20(\text{LC})$, and $\ell_2 = \text{MS} + 45(\text{LC})$

Change in length $= \ell_2 - \ell_1 = 25 \times \text{LC}$, and the maximum permissible error in measurement of elongation in one LC.

$$\frac{\Delta Y}{Y} \times 100\% = \frac{1}{25} \frac{\text{LC}}{(\text{LC})} \times 100\% = 4\%$$

Chemistry

PART – II

Section – A

19.

D

Sol. $\text{IF}_3 \rightarrow \text{Yellow powder}$

$\text{IBr} \rightarrow \text{Black solid}$

$\text{ICl} \rightarrow \text{Red solid}$

$\text{IF}_7 \rightarrow \text{Colourless gas}$

20.

C

Sol. $\Delta S_{\text{mix}} = -n_{\text{total}} R \sum x_i \ln x_i$

$$n_{\text{CO}_2} = 1 \quad n_{\text{N}_2} = 1 \quad n_{\text{Ar}} = 2$$

$$\therefore \chi_{\text{CO}_2} = \frac{1}{4} \quad \chi_{\text{N}_2} = \frac{1}{4} \quad \chi_{\text{Ar}} = \frac{1}{2}$$

$$\Delta S_{\text{mix}} = -4R \left\{ \frac{1}{4} \ln \frac{1}{4} + \frac{1}{4} \ln \frac{1}{4} + \frac{1}{2} \ln \frac{1}{2} \right\}$$

$$= -4R \left\{ \frac{1}{2} \ln \frac{1}{4} + \frac{1}{2} \ln \frac{1}{2} \right\}$$

$$= -4R \times \frac{3}{2} \ln \frac{1}{2}$$

$$= -6R \ln \frac{1}{2} = 6R \ln 2$$

21.

D

Sol. $\Lambda_m = \Lambda_m^\circ - b\sqrt{C} \quad \dots (i)$

When it is diluted $C' = \frac{C}{16}$

$$\Lambda'_m = \Lambda_m^\circ - b\sqrt{\frac{C}{16}} \quad \dots (ii)$$

$$\therefore |\Lambda_m - \Lambda'_m| = \left| -b\sqrt{C} + \frac{b\sqrt{C}}{4} \right|$$

$$= \frac{3b\sqrt{C}}{4}$$

22.

D

Sol. ${}_{64}\text{Gd} \rightarrow [\text{Xe}] 4f^7 5d^1 6s^2$

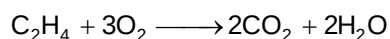


23.

B, C

Sol. 1 equivalent of $\text{H}_2\text{SO}_4 = 1$ equivalent of H – atom

$$V_{\text{H}_2\text{O}_2} = 5.6 \times \text{Normality} = 5.6 \times \frac{1}{5} = 1.12$$



$$\therefore \text{Moles of O}_2 \text{ required} = \frac{14}{28} \times 3 = \frac{3}{2}$$

$$\therefore \text{Weight of O}_2 = \frac{3}{2} \times 32 = 48 \text{ g}$$

$$\text{Moles of 'O' atom in 276 gm of N}_2\text{O}_4 = \frac{276}{92} \times 4 = 12 \text{ mole.}$$

24. A, B, D

Sol. For isotonic solution, osmotic pressure (π) should be same.

$$\pi = i.CRT$$

$$i.C \text{ for } 0.2 \text{ M NaCl } (\alpha = 1)$$

$$= [1 + (x + y - 1)\alpha]C$$

$$= [1 + (1 + 1 - 1)1]0.2 = 0.4$$

$$i.C \text{ for } 0.2 \text{ M MgCl}_2 (\alpha = 0.5)$$

$$= \{1 + (1 + 2 - 1) \times 0.5\} \times 0.2$$

$$= 0.4$$

$$i.C \text{ for } 18 \text{ g glucose in } 250 \text{ ml solution} = 1 \times \frac{18/180}{250/1000} = 0.4$$

$$i.C \text{ for } 0.5 \text{ mole of Al(NO}_3)_3 \text{ in } 4 \text{ litre } (\alpha = 0.4) = \{1 + (1 + 3 - 1)0.4\} \frac{0.5}{4} = 0.275$$

$$i.C \text{ for } 1 \text{ mole of AlCl}_3 (\alpha = 0.2) \text{ in } 4 \text{ litre solution} = \{1 + (1 + 3 - 1)0.2\} \times \frac{1}{4} = 0.4$$

$\therefore$ A, B and D are isotonic solution with 0.2 M NaCl ($\alpha = 1$).

25. A, B, D

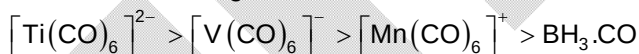
Sol. Strength of M – C back bonding

α electronic density of metal

α M – C bond order

α C – O bond length

M – C back bonding order



$\therefore$ Bond length order $z > y > x > w$.

26. A, C

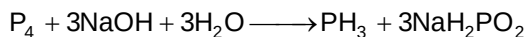
$$\text{Sol. } \left(\frac{dP}{dT}\right)_V = \frac{d}{dT} \left(\frac{RT}{V}\right)_V = \frac{R}{V}$$

$$\left(\frac{dV}{dT}\right)_P = \frac{d}{dT} \left(\frac{RT}{P}\right)_P = \frac{R}{P}$$

$$\therefore \left(\frac{dP}{dT}\right)_V \times \left(\frac{dV}{dT}\right)_P = \frac{R}{V} \times \frac{R}{P} = \frac{R^2}{PV} = \frac{R^2}{RT} = \frac{R}{T}$$

27. A, B

Sol. PH_3 – Toxic

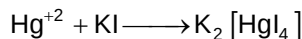


$\ddot{\text{N}}\text{H}_3$ is more basic than PH_3

N – H bond energy (389 kJ/mol) > P – H bond energy (318 kJ/mol)

28. B, C

Sol. Mg^{2+} belongs to Group-VI.



Option (A) and (D) are wrong.

Section – B

29. 7

Sol. Group – V cations are Ca^{+2} , Sr^{+2} , Ba^{+2} .

30. 5

Sol. White $\rightarrow \text{Ag}_2\text{CO}_3$, Ag_2SO_3 , AgNO_2 , ZnS , BaCO_3

Black $\rightarrow \text{Ag}_2\text{S}$, FeS , PbS

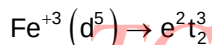
Yellow $\rightarrow \text{AgI}$

Reddish brown $\rightarrow \text{Fe}(\text{OH})_3$

31. 5

Sol. Fe_3O_4 is an example of inverse spinel, in which one Fe^{+3} ion present at tetrahedral void and one Fe^{+3} as well as one Fe^{+2} ion present at octahedral voids per unit cell.

$\therefore$ 1 Fe^{+3} ion is present at tetrahedral voids.



$\therefore$ Total 5 unpaired electrons.

32. 6

Sol. Cerrusite $\rightarrow \text{PbCO}_3$

Azurite $\rightarrow \text{Cu}(\text{OH})_2 \cdot 2\text{CuCO}_3$

Calamine $\rightarrow \text{ZnCO}_3$

Zincite $\rightarrow \text{ZnO}$

Siderite $\rightarrow \text{FeCO}_3$

Magnetite $\rightarrow \text{Fe}_3\text{O}_4$

Magnesite $\rightarrow \text{MgCO}_3$

Dolomite $\rightarrow \text{MgCO}_3$, CaCO_3

Bauxite $\rightarrow \text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$

33. 4

Sol. $\text{Fe}_4 [\text{Fe}(\text{CN})_6]_3$, $\text{K}_2\text{Cr}_2\text{O}_7$, MnO_4^- , V_2O_5 are coloured due to charge transfer.

34. 5

Sol. 1. Pressure and density are intensive property as it does not depends on amount.

2. Every close system is not isolated system.

3. $\Delta S = 0$ for adiabatic reversible process.

4. ΔG should be – ve for a spontaneous process.

5. For cyclic process $\Delta U = 0$.

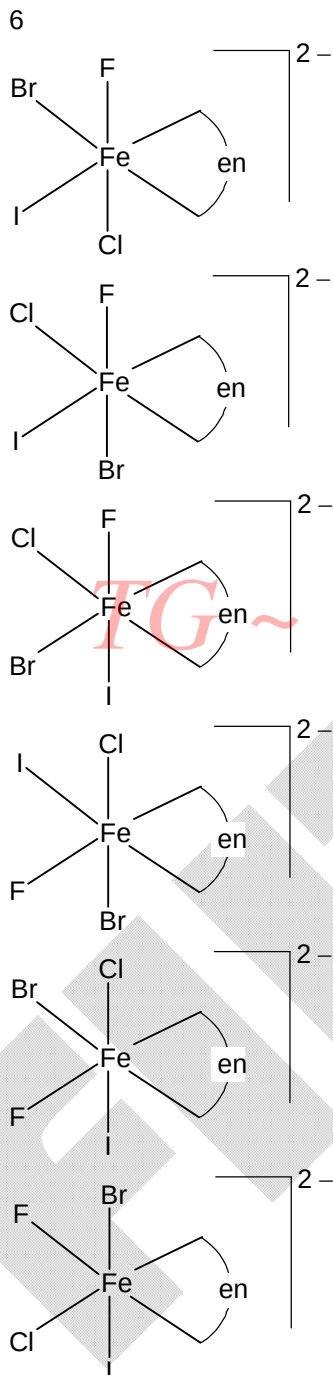
$$6. \frac{V_2}{V_1} = \frac{T_2}{T_1} = \frac{573}{473}$$

7. Calorific value of $C_2H_6 = \frac{341.1}{30} = 11.37 \text{ kcal / g}$. Calorific value of $C_2H_2 = \frac{310}{26} = 11.92 \text{ kcal / g}$.

∴ Ethyne is a better fuel.

1, 3, 4, 5 and 7 are correct statement.

35.
Sol.



36.
Sol.

3
Erythromycin, Tetracyclin and Chloramphenical has inhibitory effect on microbes.

Mathematics

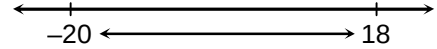
PART – III

Section – A

37. D

Sol. By rational root theorem turn the given polynomial into a degree = 2 polynomial with rational root
 $x = 1, -1, 2, -2$ are all possible root
 So, $x = -2, x = 1$. Polynomial is equal to 0
 So, polynomial becomes $(x - 1)(x + 2)(x^2 + (2a - 1)x + 1)$
 So, we only need to focus on the quadratic

$$(2a - 1)^2 - 4 \geq 0 \Rightarrow n(S) = 38 ; n(E) = 36 \therefore \text{Probability} = \frac{36}{38} = \frac{18}{19}$$



38. B

Sol. Let $x_3 = 3$, then $x_1x_2x_3 + x_2x_3x_4 + x_3x_4x_5 + x_4x_5x_1 + x_5x_1x_2 = 3(x_1x_2 + x_2x_4 + x_4x_5) + x_5x_1(x_2 + x_4)$
 So, the sum is divisible by 3 if $x_2 + x_4$ is divisible by 3, the possible sums of $(x_2 + x_4)$ can be
 $(1, 2), (2, 1), (1, 5), (5, 1), (2, 4), (4, 2), (4, 5)$ or $(5, 4)$ are we have 2 ways to designate x_1 and x_5
 for a total of $8 \times 2 = 16$
 So, desired = $16 \times 5 = 80$

39. D

Sol. $z^3 - 8 = 0 \Rightarrow (z - 2)(z^2 + 2z + 4) = 0 \Rightarrow z = 2$ and $z^2 + 2z + 4 = 0 \Rightarrow z = -1 \pm \sqrt{3}i$
 and $z^3 - 8z^2 - 8z + 64 = 0 \Rightarrow (z^2 - 8)(z - 8) = 0 \Rightarrow z = (2\sqrt{2}, -2\sqrt{2}, 8)$
 Maximum distance is from 8 and $-1 \pm \sqrt{3}i$ i.e. $2\sqrt{21}$

40. B

Sol. $[x]\{x\} = ax^2$

Case-I: $[x]\{x\} < 0$ where $x < 0$ while $ax^2 \geq 0 \forall x \in \mathbb{R}$ and $x = 0$ is one solution but that does not affect the sum of the solution

Thus we need to look at (+ve) solutions for $n \leq x < n + 1$

$[x] = n, \{x\} = x - n$ equation is $nx - n^2 = ax^2$

$$ax^2 - nx + n^2 = 0 \Rightarrow x = \frac{n \pm \sqrt{n^2 - 4an^2}}{2a} = n \left(\frac{1 \pm \sqrt{1 - 4a}}{2a} \right)$$

$$\text{For (+ve) roots we consider } x = \frac{n}{2a}(1 - \sqrt{1 - 4a})$$

Since, sum of all solutions is 420

$$\Rightarrow \sum \frac{n}{2a}(1 - \sqrt{1 - 4a}) = 420 \Rightarrow \left(\frac{1 - \sqrt{1 - 4a}}{2a} \right) \cdot \sum n = 420$$

$$\Rightarrow \left(\frac{1 - \sqrt{1 - 4a}}{2a} \right) \left(\frac{n(n+1)}{2} \right) = 420 \Rightarrow \left(\frac{1 - \sqrt{1 - 4a}}{2a} \right) \cdot 406 = 420$$

(Considering n is 28 i.e. $\frac{n(n+1)}{2} = 406$ at $n = 28$ which is nearer to 420)

$$\Rightarrow a = \frac{29}{900}$$

41. A, B, C

Sol. Let $Q(x) = P(x^2 + 4x - 7) \Rightarrow Q(1) = P(-2) = 0 \Rightarrow P(x) = a(x + 2)(x - \alpha)$
 Now, $Q(x) = P(x^2 + 4x - 7) = a(x^2 + 4x - 5)(x^2 + 4x - 7 - \alpha)$

$$\Rightarrow Q(x) = a(x-1)(x+5)(x^2 + 4x - (7 + \alpha))$$

As $Q(x)$ has at least one repeated root, this must be $x = 1$ or $x = -5$

$$Q(x) = a((x-1)(x+5))^2 \text{ or } x^2 + 4x - (7 + \alpha) = 0 \text{ has double roots if } D = 0, \alpha = -11$$

$$Q(x) = a(x-1)(x+5)(x+2)^2$$

42. A, C

Sol. $b_n = \sum_{r=0}^n \frac{1}{{}^nC_r} = \sum_{r=0}^{n-1} \frac{1}{{}^nC_r} + \frac{1}{{}^nC_n} = \sum_{r=0}^{n-1} \frac{(n-r)!r!}{n!} + 1 = \sum_{r=0}^{n-1} \left(\frac{(n-r)}{n} \right) \left(\frac{((n-1)-r)!r!}{(n-1)!} \right) + 1$

$$\Rightarrow \frac{1}{n} \cdot \sum_{r=0}^{n-1} \frac{(n-r)}{{}^{n-1}C_r} + 1 \quad \dots (1)$$

$$\Rightarrow \frac{1}{n} \cdot \sum_{r=0}^{n-1} \frac{n - (n-1-r)}{{}^{n-1}C_{n-1-r}} + 1$$

$$\Rightarrow \frac{1}{n} \cdot \sum_{r=0}^{n-1} \frac{r+1}{{}^{n-1}C_r} + 1 \quad \dots (2)$$

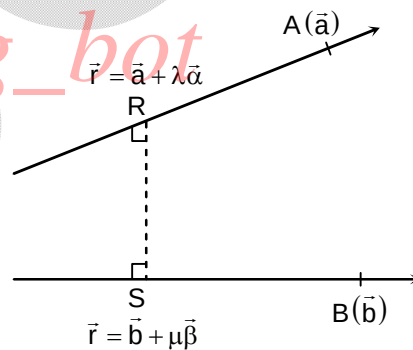
Adding equation (1) and (2), we get $2b_n = \frac{1}{n} \cdot \sum_{r=0}^{n-1} \frac{n-r+r+1}{{}^{n-1}C_r} + 2$

$$\Rightarrow \frac{n+1}{2n} \cdot \sum_{r=0}^{n-1} \frac{1}{{}^{n-1}C_r} + 1 \Rightarrow \frac{n+1}{2n} \cdot b_{n-1} + 1$$

43. B, D

Sol. Let position vector R is $\vec{a} + \lambda \vec{\alpha}$ and position vector of S is $\vec{b} + \mu \vec{\beta}$

Vector $\vec{RS} = (\vec{a} - \vec{b}) + \lambda \vec{\alpha} - \mu \vec{\beta}$ taking dot product of $\vec{RS}$ with $\vec{\alpha}$ and $\vec{\beta}$ we will get λ or μ and then the position vector R and S



44. B, D

Sol. $P(A) = 0.4, P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6$

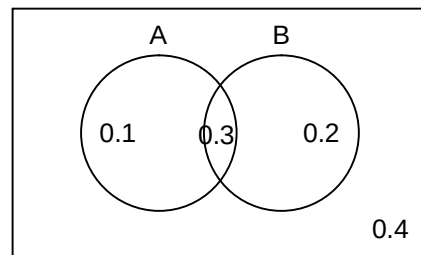
$$P(B) = 0.5, P(A \cap B) = 0.3$$

$$P(E_1) = \frac{P(A \cap B)}{P(A \cup B)} = \frac{0.3}{0.6} = \frac{1}{2}$$

$$P(E_2) = \frac{P(A \cap \bar{B}) + P(\bar{A} \cap B)}{P(A \cup B)} = \frac{0.1 + 0.2}{0.6} = \frac{1}{2}$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = \frac{3}{5}; P\left(\frac{B}{A}\right) = \frac{0.3}{0.4} = \frac{3}{4}$$

$$P\left(\frac{A}{A \cup B}\right) = \frac{P(A)}{P(A \cup B)} = \frac{0.4}{0.6} = \frac{2}{3}; P\left(\frac{B}{A \cup B}\right) = \frac{0.5}{0.6} = \frac{5}{6}$$



45. B, D

Sol. A_n is non-singular for all n and $\det A_n = (-1)^{n+1} \cdot n!$

46. A, B, C, D

Sol. For $r \geq 0$, $\frac{1}{r+1} \cdot C_2^2 = \frac{1}{(r+1)} \cdot \frac{n!}{r!(n-r)!} \cdot C_r = \frac{1}{(n+1)} \cdot \frac{((n+1)!)}{(r+1)!(n-r)!} \cdot C_r = \frac{1}{(n+1)} \cdot {}^{(n+1)}C_{r+1} \cdot {}^nC_r$

Thus, $S = \frac{1}{(n+1)} \cdot S_1$ where $S_1 = \sum_{r=0}^n {}^{n+1}C_{r+1} \cdot {}^nC_r = \frac{(2n+1)!}{n!(n+1)!}$

$\therefore S = \frac{1}{n+1} \cdot \frac{(2n+1)!}{n!(n+1)!} = \frac{(2n+1)!}{((n+1)!)^2} = \frac{1}{(2n+2)} \cdot {}^{2n+2}C_{n+1}$

Section – B

47. 2

Sol. Let $x = \frac{y}{\sqrt{2014}}$ i.e. is $\frac{y^3}{2014} - \frac{4029y^2}{2014} + 2 = 0 \Rightarrow y^3 - 4029y^2 + 4028 = 0$

$\Rightarrow (y-1)(y^2 - 4028y - 4028) = 0$ it is observed as $y = 1$, is middle root

Therefore, $x_2(x_1 + x_3) = \frac{1}{\sqrt{2014}} \times \frac{4028}{\sqrt{2014}} = 2$

48. 7

Sol. Let $2^{17} = x$ given equation is $\frac{x^{17} + 1}{x + 1} = x^{16} - x^{15} + x^{14} + \dots - x^1 + x^0$

Now, consider $(x^{16} - x^{15}) = x^{15}(x - 1) = x^{15}(2^{17} - 1)$

Since, $2^{17} - 1 = 2^{16} + 2^{15} + \dots + 1 \Rightarrow$ Total 17 terms

So, there are 8 such pair like $x^{14} - x^{13}$ and so on

So, total is $17 \times 8 + 1 = 137$, 1 is for x^0

49. 1

Sol. As a subset with less than 3 chairs cannot contain 3 adjacent chair, these are only 10 sets of 3 chairs so that they are all 3 adjacent there are 10 subsets of 4 chairs where all r adjacent and $10 \times 5 = 50$, where there are only 3

If there are 5 chairs, 10 have all 5 adjacent $10 \times 4 = 40$ have 4 adjacent and $10 \times {}^5C_2 = 100$ have 3 adjacent with 6 chairs in the subset

10 have all 6 adjacent, $10 \times 3 = 30$ have 5 adjacent, $10 \times {}^4C_2 = 60$ have 4 adjacent

$\frac{10 \times 3}{2} = 15$ have 2 groups of 3 adjacent chairs and $10 \times ({}^5C_2 - 3) = 70$ have 1 group of 3

adjacent chairs. All possible subsets with more than 6 chairs have at least 1 group of 3 adjacent. So adding all it is total = 581

50. 7

Sol. **First Solution:**

$(2+i)^n = (a_n + b_n i)$ square the given equality $(3+4i)^n = (a_n + b_n i)^2 = (a_n^2 - b_n^2) + 2a_n b_n i$

So, $a_n b_n = \frac{1}{2} \text{Im}(3+4i)^n \Rightarrow \sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} = \sum_{n=0}^{\infty} \frac{1}{2} \frac{\text{Im}(3+4i)^n}{7^n} = \frac{1}{2} \text{Im} \left(\frac{1}{1 - \frac{3+4i}{7}} \right) = \frac{7}{16}$

Second Solution:

$2+i = \sqrt{5} \left(\frac{2}{\sqrt{2}} + \frac{1}{\sqrt{5}} i \right) = \sqrt{5} (\cos \theta + i \sin \theta) \Rightarrow \theta = \tan^{-1} \frac{1}{2}$

$(2+i)^n = (\sqrt{5})^n (\cos n\theta + i \sin n\theta) \Rightarrow \sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} = \sum_{n=0}^{\infty} \frac{\cos n\theta \cdot \sin n\theta \cdot 5^n}{7^n}$

$$= \frac{1}{2} \sum \left(\frac{5}{7}\right)^n \cdot \sin 2n\theta = \frac{1}{2} \operatorname{Im} \sum \left(\frac{5}{7}\right)^n \cdot e^{i2n\theta} = \frac{1}{2} \operatorname{Im} \left[\sum_{n=0}^{\infty} \left(\frac{5}{7}\right)^n (e^{i2\theta})^n \right] \text{ is a Geometric Progression}$$

$$\Rightarrow \frac{1}{2} \operatorname{Im} \left[\frac{1}{1 - \frac{5}{7} \cdot e^{i2\theta}} \right] \Rightarrow \frac{1}{2} \operatorname{Im} \left(\frac{1}{1 - \frac{5}{7} (\cos 2\theta + i \sin 2\theta)} \right) = \frac{1}{2} \operatorname{Im} \left(\frac{7}{8} + \frac{7}{8} i \right) = \frac{1}{2} - \frac{7}{8} = \frac{7}{16}$$

51. 9

Sol. The given sequence 4, 7, 1, 8, 9, 7, 6, 3, 9, 2, 1, 3, 4, 7, 1
So, after 12th terms the separation of numbers starts and the sum of 12th terms is
4 + 7 + + 1 + 3 = 60

Since, $S_n > 10,000 \Rightarrow \frac{10,000}{60} = 166$ and $166 \times 12 = 1992$

So, $S_{1992} = 60 \times 166 = 9960 \Rightarrow 9960 + 4 + 7 + 1 + 8 + 9 + 7 = 9996 + 6 = 10,000$
 $\therefore S_{1999} > 10,000$

52. 4

Sol. Let $\omega = \cos \alpha + i \sin \alpha$; $z = 10(\cos \beta + i \sin \beta)$

$$\Rightarrow \frac{\omega - z}{z} = \frac{1}{10} \cos(\alpha - \beta) + \frac{1}{10} i \sin(\alpha - \beta) - 1. \text{ Let } x = \alpha - \beta$$

$$\tan \theta = \frac{\sin x}{\cos x - 10}, \text{ maximum occurs at } \cos x = \frac{1}{10}, \sin x = \pm \frac{\sqrt{99}}{10}$$

$$\therefore \tan^2 \theta = \frac{1}{99} \quad \text{TG} \sim @bohrring_bot$$

53. 3

Sol. Set of these rational numbers in from $\frac{1}{999}$ to $\frac{499}{501}$

There are 499 numbers between 501 and 999 $\Rightarrow$ 249 numbers are divisible by 2
 $\Rightarrow$ 99 numbers are divisible by 5 $\Rightarrow$ 49 numbers are divisible by 10
So, by inclusion/exclusion $499 - 249 - 99 + 49 = 200$

54. 7

Sol. $\ell = \lim_{n \rightarrow \infty} 18 \left(\frac{3}{3^2} + \frac{3^2}{4^4} + \frac{3^3}{3^6} + \dots \right) = 18 \times \frac{1}{3 \left(1 - \frac{1}{3} \right)} = 9$

$$m = \lim_{n \rightarrow \infty} 12 \left(\frac{2}{2^2} + \frac{2^2}{2^4} + \dots \right) = 12 \times \frac{1}{2 \left(1 - \frac{1}{2} \right)} = 12 \Rightarrow A_n = \frac{9+12}{3} = 7$$

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2023

TEST DATE: 28-12-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

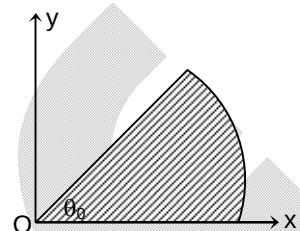
PART – A

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. Co-ordinates of the centre of mass of a sector of uniform circular disc of radius R and of mass m and subtends an angle θ_0 at its centre as shown in figure are



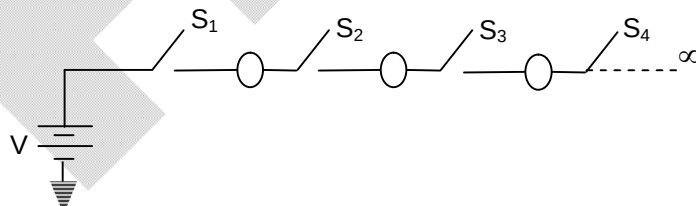
- (A) $\left[\frac{2R}{3} \frac{\sin \frac{\theta_0}{2}}{\frac{\theta_0}{2}}, \frac{2R}{3} \left(\frac{1 - \cos \frac{\theta_0}{2}}{\frac{\theta_0}{2}} \right), 0 \right]$
- (B) $\left[\frac{2R}{3} \frac{\sin \theta_0}{\theta_0}, \frac{2R}{3} \left(\frac{1 - \cos \theta_0}{\theta_0} \right), 0 \right]$
- (C) $\left[\frac{4R}{3} \frac{\sin \theta_0}{\theta_0}, \frac{4R}{3} \frac{\cos \theta_0}{\theta_0}, 0 \right]$
- (D) $\left[\frac{4R}{3} \frac{\sin \theta_0}{\theta_0}, \frac{2R}{3} \left(\frac{1 - \cos \theta_0}{\theta_0} \right), 0 \right]$

2. Two blocks A & B of mass ' m ' & $2m$ respectively are joined to the ends of an under formed massless spring of spring constant ' k '. They can move on a horizontal smooth surface. Initially A & B has velocities ' u ' towards left and ' $2u$ ' towards right respectively. Constant forces of magnitudes F and $2F$ are always acting on A and B respectively in the directions shown. The maximum extension in the spring during the motion is



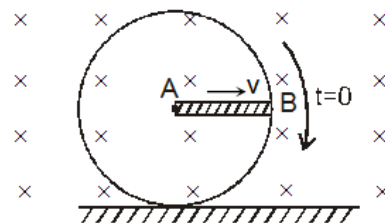
- (A) $x_{\max} = \frac{4F - \sqrt{16F^2 + 54mu^2k}}{3k}$
- (B) $x_{\max} = \frac{4F + \sqrt{16F^2 + 54mu^2k}}{3k}$
- (C) $x_{\max} = \frac{4F + \sqrt{16F^2 - 54mu^2k}}{3k}$
- (D) $x_{\max} = \frac{4F - \sqrt{16F^2 - 54mu^2k}}{3k}$

3. All spheres are identical of radius R and initially uncharged, S_1 is closed at $t = 0$, leaving all other switches open. After sufficiently long time S_1 is opened and S_2 is closed. The process is done till S_{n-1} is opened and S_n is closed. Net loss of energy of system after $t = 0$ if process is done for infinite number of spheres, is

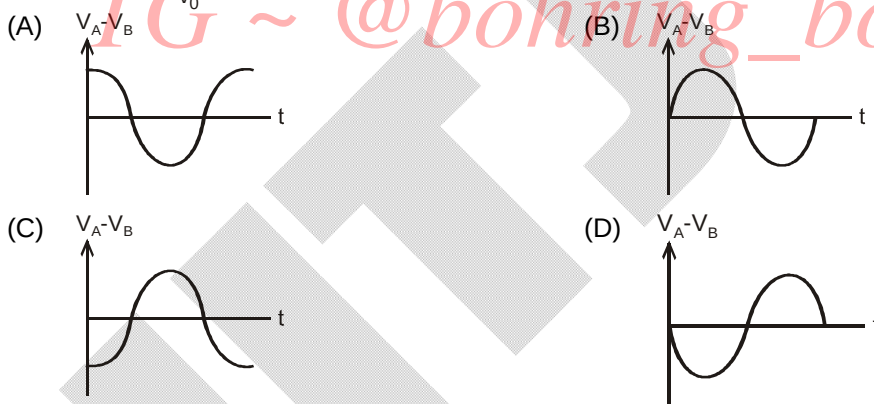
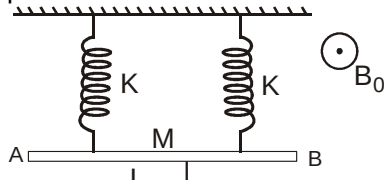


- (A) $\frac{20}{6} \pi \epsilon_0 R V^2$
- (B) $2 \pi \epsilon_0 R V^2$
- (C) $\frac{8}{3} \pi \epsilon_0 R V^2$
- (D) Zero

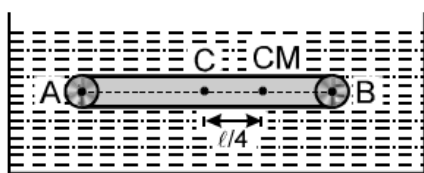
4. As shown in figure a conducting rod AB is attached to wooden disc of radius R and disc is in pure rolling with linear velocity v , find net emf developed across rod of length r as a function of time. Initial position of rod is shown in figure. Magnetic field is B and is directed into the plane.



- (A) $2BvR\left(1 + \sin \frac{v}{R}t\right)$ (B) zero
(C) $\frac{BvR}{2}\left(1 + 2\cos \frac{v}{R}t\right)$ (D) $\frac{BvR}{2}\left(1 - 2\sin \frac{v}{R}t\right)$
5. A horizontal conducting rod is kept in vertical plane is supported by two non-conducting springs as shown. A uniform magnetic field is directed outward rod initially at rest in equilibrium position and $t=0$ it is given velocity v_0 in downward direction. Which of the following graphs represent potential difference across to ends of the rod.

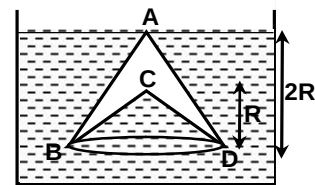


6. A non uniform cylinder of mass m , length ℓ and radius r is having its centre of mass at a distance $\ell/4$ from the centre and lying on the axis of the cylinder. The cylinder is kept in a liquid of uniform density ρ . The moment of inertia of the rod about the centre of mass is I . The angular acceleration of point A relative to point B just after the rod is released from the position shown in figure is

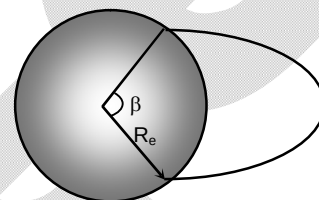


- (A) $\frac{\pi\rho g\ell^2r^2}{I}$ (B) $\frac{\pi\rho g\ell^2r^2}{4I}$
(C) $\frac{\pi\rho g\ell^2r^2}{2I}$ (D) $\frac{3\pi\rho g\ell^2r^2}{4I}$

7. A solid cone of uniform density and height $2R$ and the base radius R has a conical portion scooped out from its base with the same base radius but height R as shown in the figure. The solid cone is floating in a liquid of density r with vertex A touching the fluids surface. If atmospheric pressure is P_0 , the force acting on the surface BCD due to liquid is



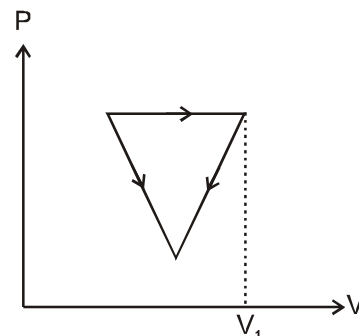
- (A) $\frac{\pi R^3 \rho g}{3} \left(2 + \frac{2P_0}{R\rho g} \right)$ (B) $\frac{\pi R^3 \rho g}{3} \left[5 + \frac{3P_0}{R\rho g} \right]$
 (C) $\frac{\pi R^3 \rho g}{3} \left[1 + \frac{3P_0}{R\rho g} \right]$ (D) $\frac{\pi R^3 \rho g}{3} \left[4 + \frac{3P_0}{R\rho g} \right]$
8. A particle is projected from the surface of the earth (of radius R_e and mass M_e) with speed equal to $\sqrt{\frac{GM_e}{R_e}}$, at certain angle from local horizontal as shown in the figure such that the angle subtended by arc between launching & landing site at earth's centre is $\beta = 90^\circ$. Calculate the maximum separation of the particle from the centre of the earth. Consider earth to be uniformly dense and air resistance to be absent.
- (A) $R_e \sqrt{2}$ (B) $R_e (\sqrt{2} + 1)$
 (C) $R_e \frac{(\sqrt{2} + 1)}{\sqrt{2}}$ (D) $R_e \frac{(\sqrt{2} + 1)}{(\sqrt{2} - 1)}$
9. A satellite is moved from one circular orbit around the earth, to another of lesser radius. Which of the following statement is true?
- (A) The kinetic energy of satellite increases and the gravitational potential energy of satellite-earth system increases;
 (B) The kinetic energy of satellite increases and the gravitational potential energy of satellite-earth system decreases;
 (C) The kinetic energy of satellite decreases and the gravitational potential energy of satellite-earth system decreases;
 (D) The kinetic energy of satellite decreases and the gravitational potential energy of satellite-earth system increases.



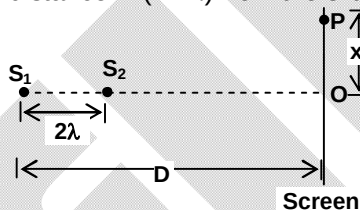
10. The diagram shows an equilateral cyclic process on a PV diagram. The ratio of maximum pressure to minimum pressure is 2. The maximum temperature in the process is T_1 . The work done by the two moles of ideal gas in the cyclic process is equal to

(A) $\frac{R^2 T_1^2}{\sqrt{3} V_1^2}$
 (C) $\frac{3 R^2 T_1^2}{V_1^2}$

(B) $\frac{R^2 T_1^2}{V_1^2}$
 (D) $\frac{3 R^2 T_1^2}{2 V_1^2}$

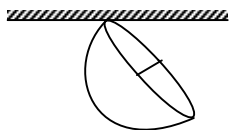


11. The surface temp of a small planet having a circular orbit round the sun with time period T assuming sun and the planet to be black bodies. Take radius of sun = R mass of sun = M , surface temp of the sun Q_0 is :
- (A) $\sqrt{\frac{R}{2r}}\theta_0$ (B) $\sqrt{\frac{R+r}{2}}\theta_0$
 (C) $\sqrt{\frac{R-r}{2}}\theta_0$ (D) $\sqrt{\frac{R}{r}}\theta_0$
12. In a sample of rock, the ratio of ^{206}Pb to ^{238}U nuclei is found to be 0.5. The age of the rock is. (Assume that all the Pb nuclides in the rock was produced due to the decay of Uranium nuclides and $T_{1/2}(^{238}\text{U}) = 4.5 \times 10^9$ year).
- (A) 2.25×10^9 year (B) $4.5 \times 10^9 \ln 3$ year
 (C) $4.5 \times 10^9 \frac{\ln(\frac{3}{2})}{\ln 2}$ year (D) $2.25 \times 10^9 \ln\left(\frac{3}{2}\right)$ year
13. A body of mass m_0 is placed on a smooth horizontal surface. The mass of the body is decreasing exponentially with disintegration constant λ . Assuming that the mass is ejected backwards with relative velocity u . initially the body was at rest. The velocity of it after time t :
- (A) $u \lambda t$ (B) $u e^{-\lambda t}$
 (C) $u e^{\lambda t}$ (D) none of these
14. A point object is moving with a velocity, $\vec{v}_0 = 2\hat{i} - 3\hat{j} + 4\hat{k}$ in front of a moving plane mirror whose normal is along x-axis. The mirror is moving with a velocity $\hat{i} - 4\hat{j} + 3\hat{k}$. The velocity vector of the image is given by
- (A) $15\hat{j} + 3\hat{k}$ (B) $-3\hat{j} + 4\hat{k}$
 (C) $\hat{j} + 2\hat{k}$ (D) $2\hat{i} + 3\hat{j} - 2\hat{k}$
15. Two coherent narrow slits emitting light of wavelength λ in the same phase are placed parallel to each other at a small separation of 2λ . The light is collected on screen which is placed at a distance D ($\gg \lambda$) from the slit S_1 as shown in figure.



- (A) The finite distance x such that the intensity at P is equal to the intensity at O is $\sqrt{3}D$
 (B) The finite distance x such that the intensity at P is equal to the intensity at O is $\sqrt{\frac{3}{2}}D$
 (C) There is no point on the screen at which the intensity at P is equal to intensity at O
 (D) The finite distance x such that the intensity at P is equal to the intensity at O is $\sqrt{2}D$

16. Time period of small oscillations of a uniform hemispherical shell of radius R and mass m is



- (A) $2\pi\sqrt{\frac{2\sqrt{5}R}{3g}}$ (B) $2\pi\sqrt{\frac{2\sqrt{5}R}{g}}$
 (C) $2\pi\sqrt{\frac{\sqrt{5}R}{2g}}$ (D) $2\pi\sqrt{\frac{\sqrt{5}R}{3g}}$

17. A siren is placed at a railway platform is emitting sound of frequency 10 KHz. A passenger sitting in a moving train A records frequency of 11 kHz, while the train approaches the siren. During his return journey in a different train B he records a frequency of 12 KHz while approaching the same siren. Ratio of velocity of train B to that of train A is

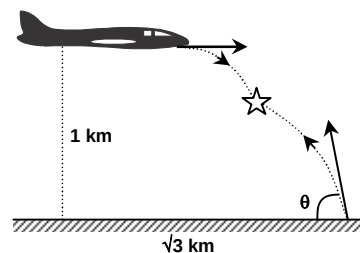
- (A) 2 (B) $\frac{242}{252}$
 (C) $\frac{6}{5}$ (D) $\frac{11}{6}$

18. The pressure of an ideal gas varies according to the law $P = P_0 - AV^2$ where P_0 and A are positive constants. The highest temperature that can be attained by the gas is

- (A) $\frac{P_0}{nR}\sqrt{\frac{P_0}{3A}}$ (B) $\frac{2P_0}{nR}\sqrt{\frac{P_0}{3A}}$
 (C) $\frac{2P_0}{3nR}\sqrt{\frac{P_0}{3A}}$ (D) $\frac{2P_0}{3nR}\sqrt{\frac{P_0}{A}}$

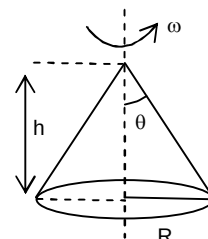
19. An aircraft is flying horizontally with a constant velocity = 200 m/s, at a height = 1 km above the ground. At the moment shown, a bomb is released from the aircraft and the cannon-gun below fires a shell with initial speed = 200 m/s, at some angle θ . For what value of ' θ ' will the projectile shell destroy the bomb in mid air?

- (A) $\theta = 15^\circ$ (B) $\theta = 30^\circ$
 (C) $\theta = 45^\circ$ (D) $\theta = 60^\circ$



20. 'Q' charge is uniformly distributed over the surface of a right circular cone of semi-vertical angle θ and height h . The cone is uniformly rotated about its axis at angular velocity ω . Magnetic moment associated with the cone is found to be

- (A) $\frac{Q\omega h^2 \tan^2 \theta}{2}$ (B) $Q\omega h^2 \tan^2 \theta$
 (C) $\frac{Q\omega h^2 \tan^2 \theta}{4}$ (D) $\frac{Q\omega h \tan \theta}{2}$



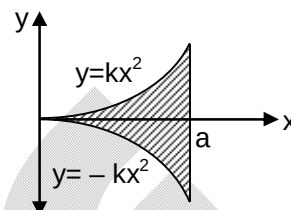
SECTION – B

(Numerical Answer Type)

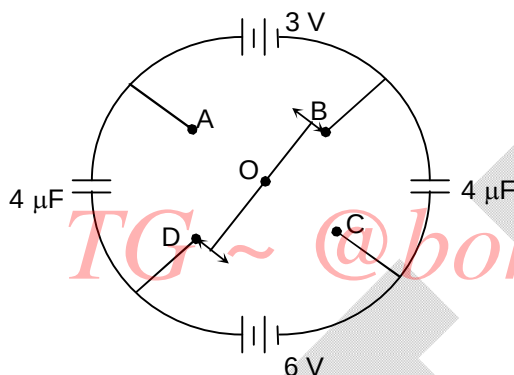
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A thin sheet of metal of uniform thickness is cut into the shape bounded by the line $x = a$ and $y = \pm kx^2$, as shown.

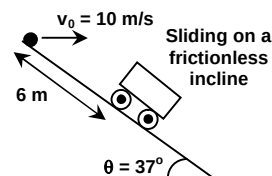
If the x-coordinate of the centre of mass is $\frac{na}{12}$ then find n.



22. In figure there is four way key at the middle. If key is thrown from situation BD to AC, then charge flowing through point is q_0 . Value of $\frac{q_0}{12}$ is

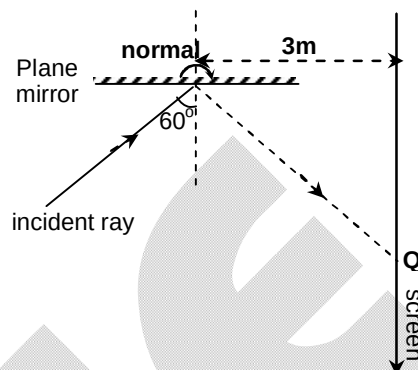


23. A fixed container of height 'H' with large cross-sectional area 'A' is completely filled with water. Two small orifices of cross-sectional area 'a' are made, one at the bottom and the other on the vertical side of the container at a distance H/2 from the top of the container. The time taken by the water level to reach a height of H/2 from the bottom of the container is $\frac{XA}{3a}(\sqrt{2} - 1)\sqrt{\frac{H}{g}}$. Find the value of X.
24. A wire forming a loop is dipped into soap solution and taken out, so that a film of soap solution is formed. A loop of 6.28 cm long thread is gently put on the film is pricked with a needle inside the loop. The thread loop takes the shape of a circle. Find the tension in the thread in (10^{-4} N). Surface tension of soap solution = 0.03 N/m.
25. A solid copper sphere (density ρ and specific heat C) of radius r at an initial temperature 200 K is suspended inside a chamber whose walls are almost at 0 K. the time required for the temperature of sphere to drop to 100 K is found to be $\frac{x}{72} \frac{r\rho C}{\sigma} \times 10^{-6}$ sec. The value of x is.
26. A ball is projected horizontally from an incline so as to strike a cart sliding on the incline. Neglect height of cart and point of projection of ball above incline. At the instance the ball is thrown, the speed of cart is 'v' (in m/s). Find 'v' so that the ball strikes the cart.

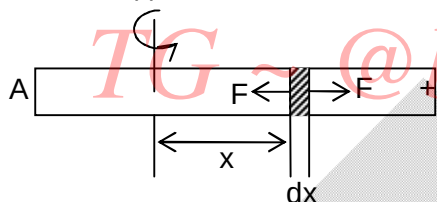


27. If the frequency of k_{α} x-ray emitted from the element with atomic number 31 is f , if the frequency of k_{α} X-ray emitted from the element with atomic number 51 is given by $\frac{25}{x} f$ then x will be ?

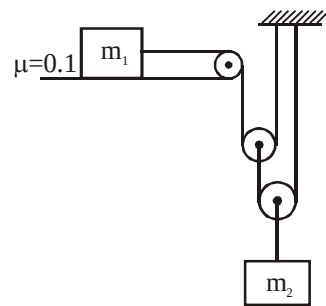
28. In the given figure, a ray is incident on the plane mirror as shown. The reflected ray strike the screen at Q. the mirror now starts, rotating about axis passing through P and perpendicular to plane of paper with angular speed $\pi/180$ rad/second. If the speed of point at which reflected ray strike the screen is $\frac{k\pi}{15}$ m/second at time $t = 15$ second, find the value of k .



29. A horizontally oriented copper rod of length l is rotated about a vertical axis passing through its middle. Frequency at which the rod ruptures when rotated at $\frac{1}{2\pi} \sqrt{\frac{X\sigma}{\rho l^2}}$. Breaking or breaking force of copper takes as σ and density of copper ρ . Find X .



30. In the figure, find the velocity of m_1 in ms^{-1} when m_2 falls by 9m. Given $m_1 = m$
 $m_2 = 2m$
 (take $g = 10 \text{ ms}^{-2}$)



Chemistry

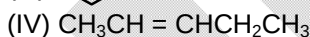
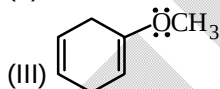
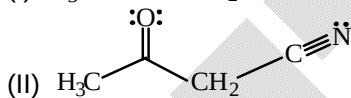
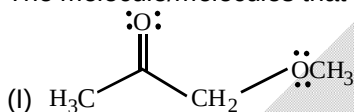
PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

31. The conductivity measurement of a coordination compound of Cobalt (III) shows that it dissociates into 3 ions in solution. The compound is
 (A) Hexaamminecobalt(III) chloride (B) Pentaamminesulphatocobalt(III) chloride
 (C) Pentaamminechloridocobalt(III) sulphate (D) Pentaamminechloridocobalt(III) chloride
32. Which is the major product obtained by hydrolysis of compound formed by reaction between formaldehyde and ethyl magnesium bromide?
 (A) Ethan-1-ol (B) Propan-2-ol
 (C) Propan-1-ol (D) 2-Methyl-propan-2-ol
33. The total number of isomeric linear dipeptide which can be synthesized from racemic alanine is
 (A) 1 (B) 2
 (C) 3 (D) 4
34. The charge carried by 1 millimole of M^{n+} ions is 193 coulombs. The value of n is
 (A) 1 (B) 2
 (C) 3 (D) 4
35. The molecule/molecules that has/have delocalised lone pair(s) of electrons is/are

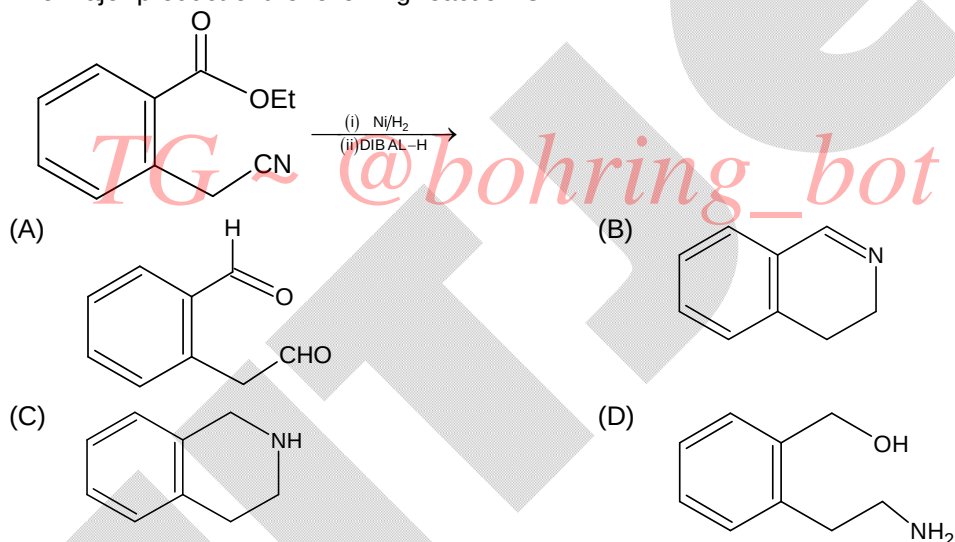


- (A) I, II and III
 (C) I and III

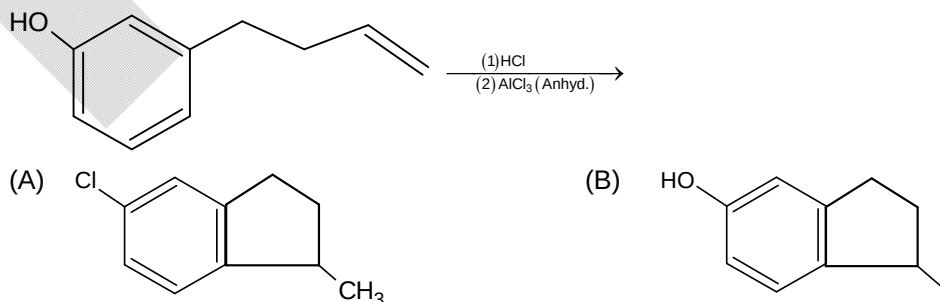
- (B) I, II and IV
 (D) only III

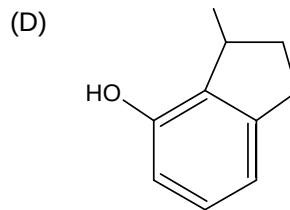
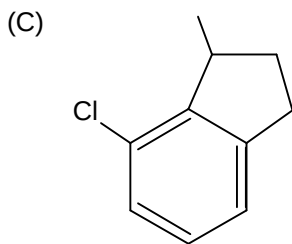
36. Which one of the following electronic arrangements is absurd?
 (A) $n = 3, \ell = 1, m = -1$ (B) $n = 3, \ell = 0, m = 0$
 (C) $n = 2, \ell = 0, m = -1$ (D) $n = 2, \ell = 1, m = 0$
37. In order to oxidise a mixture one mole of each of FeC_2O_4 , $\text{Fe}_2(\text{C}_2\text{O}_4)_3$, FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$ in acidic medium, the number of moles of KMnO_4 required is -
 (A) 3 (B) 2
 (C) 1 (D) 1.5
38. Which one of the following is likely to give a precipitate with AgNO_3 solution?
 (A) $(\text{CH}_3)_3\text{CCl}$ (B) CHCl_3
 (C) $\text{CH}_2=\text{CH}-\text{Cl}$ (D) CCl_4

39. 1 g of non-volatile non-electrolyte solute is dissolved in 100g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 5. The ratio of the elevation in their boiling points, is :
 (A) 5 : 1
 (B) 10 : 1
 (C) 1 : 5
 (D) 1 : 0.2
40. Among the following, the INCORRECT statement about colloids is :
 (A) They can scatter light.
 (B) They are large molecules and have high molar mass.
 (C) The range of diameters of colloidal particles is between 1 and 1000 nm.
 (D) The osmotic pressure of a colloidal solution is of higher order than the true solution for same weight of solute per litre solution.
41. The radius of the largest sphere which fits properly at the centre of the edge of body centred cubic unit cell is :
 (Edge length is represented by 'a') :-
 (A) 0.134 a
 (B) 0.027 a
 (C) 0.067 a
 (D) 0.047 a
42. The major product of the following reaction is:



43. Three complexes, $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ (I), $[\text{Co}(\text{NH}_3)_5\text{H}_2\text{O}]^{3+}$ (II) and $[\text{Co}(\text{NH}_3)_6]^{3+}$ (III) absorb light in the visible region. The correct order of the wavelength of light absorbed by them is:
 (A) (III) > (I) > (II)
 (B) (I) > (II) > (III)
 (C) (II) > (I) > (III)
 (D) (III) > (II) > (I)
44. The major product of the following reaction is :





45. The primary pollutant that leads to photochemical smog is :
 (A) Sulphur dioxide (B) Acrolein
 (C) Ozone (D) Nitrogen oxides
46. NaH is an example of :
 (A) Electron-rich hydride (B) Molecular hydride
 (C) Saline hydride (D) Metallic hydride
47. For the reaction of H_2 with I_2 , the rate constant is $2.5 \times 10^{-4} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 327°C and $1.0 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 527°C . The activation energy for the reaction, in kJ mol^{-1} is:
 ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)
 (A) 72 (B) 166
 (C) 150 (D) 59
48. Which of the following order is incorrect :
 (A) Ionic character – $\text{MCl} > \text{MCl}_2 > \text{MCl}_3$
 (B) Polarizability – $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$
 (C) Polarising power – $\text{Na}^+ < \text{K}^+ < \text{Mg}^{+2} < \text{Al}^{+3}$
 (D) Polarising power – $\text{Li}^+ < \text{Be}^{+2} < \text{B}^{+3}$
49. $(\text{CH}_3)_2\text{CHBr} \xrightarrow[\text{(ii) CuI}]{\text{(i) Li}} \text{A} \xrightarrow{(\text{CH}_3)_2\text{CHCH}_2\text{Br}} \text{B}$
 'B' is -
 (A) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$
 (B) $(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$
 (C) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{CH}_3)_2$
 (D) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$
50. Calculate the potential of an indicator electrode versus the standard hydrogen electrode, which originally contained 0.1 M MnO_4^- and 1.72 M H^+ and which was treated with Fe^{2+} necessary to reduce 90% of KMnO_4 to Mn^{2+}
 $E^0_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51 \text{ V}$
 (A) 1.4 V (B) 1.5 V
 (C) 1.6 V (D) 1.3 V

SECTION – B

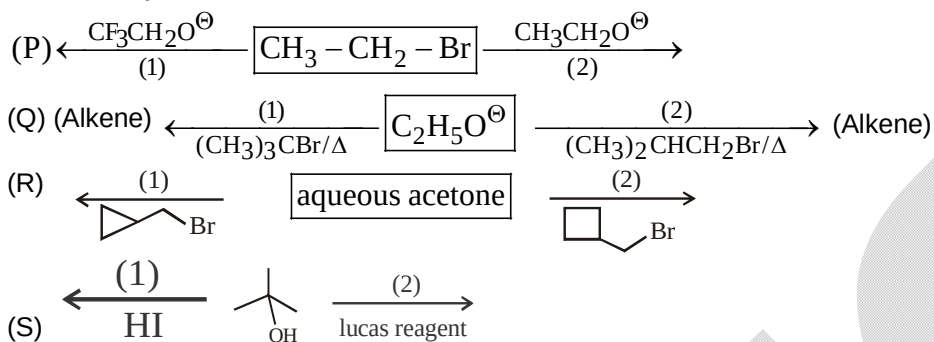
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

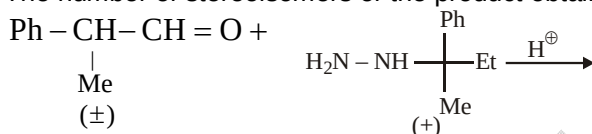
51. An adiabatic container fitted with a movable adiabatic piston (operating at 1 atm) is filled with 2 litre of $2\text{M H}_2\text{O}_2(\text{aq})$ solution at 300 K. If H_2O_2 dissociates following first order decay with a half life of 10min then value of [Magnitude of change in internal energy (in cal) $\div$ 100] in first 20 min will be

52. Number of compounds which are planar ?
 XeF_2 , ClF_3 , H_2O , $[\text{XeF}_5]^-$, I_3^- , BCl_3 , XeF_4 , SF_4 , PCl_5 , SF_6 , IF_7 .

53. In how many reactions the 1st reaction is faster than 2nd reaction?



54. The number of stereoisomers of the product obtained in the following reaction is :



55. Na_2SO_3 , NaCl , $\text{Na}_2\text{C}_2\text{O}_4$, Na_2HPO_4 , Na_2CrO_4 , NaNO_2 , CH_3COONa are separately treated with AgNO_3 solution. In how many cases white ppt. is/are obtained?
56. A sample consisting of chocolate-brown powder of PbO_2 is allowed to react with excess of KI and iodine liberated is reacted with N_2H_4 in another container. The volume of gas liberated from this second container at STP was measured out to be 1.12 litre. Find out volume of decimolar NaOH (in litre) required to dissolve PbO_2 completely. (Assume all reactions are 100% complete)
57. The ionization energy of the Be^{3+} ion with the electron in its ground state is 2.10×10^A kJ/mol. Find the value of A.
58. By how many times the absolute temperature of a gas when U_{rms} of a gas in a container of fixed volume is increased from 5×10^4 cm/sec to 10×10^4 cm/sec
59. A sample of HI(g) is placed in flask at a pressure of 0.2 atm. At equilibrium, the partial pressure of HI(g) is 0.04 atm. What is K_p for the given equilibrium ? $2\text{HI(g)} \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$
60. In 100 ml sample of hard water, 100 ml of (N/50) Na_2CO_3 was added and the mixture was boiled and filtered. The filtrate was neutralized with 60 ml of (N/50) HCl . Calculate the permanent hardness of water (in ppm). (sp. gr. of hard water = 1).

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. A triangle is formed by the lines whose combined equation is given by $(x + y - 4)(xy - 2x - y + 2) = 0$. The equation of its circumcircle is
 (A) $x^2 + y^2 - 5x - 3y + 8 = 0$ (B) $x^2 + y^2 - 3x - 5y + 8 = 0$
 (C) $x^2 + y^2 - 3x - 5y - 8 = 0$ (D) None of these
62. Two circles, each of radius 5, have a common tangent at (1,1) whose equation is $3x + 4y - 7 = 0$. Then their centres are
 (A) (4, -5), (-2, 3) (B) (4, -3), (-2, 5)
 (C) (4, 5), (-2, -3) (D) None of these
63. If p and q be the longest distance and the shortest distance respectively of the point (-7, 2) from any point (α, β) on the curve whose equation is $x^2 + y^2 - 10x - 14y - 51 = 0$, then G.M. of p and q is equal to
 (A) $2\sqrt{11}$ (B) $5\sqrt{5}$
 (C) 13 (D) None of these
64. Through the point (13, 31) a straight line is drawn to meet the axes of X and Y at Q and S respectively. If the rectangle OQRS is completed, the co-ordinates of R satisfy the equation.
 (A) $\frac{13}{x} + \frac{31}{y} = 1$ (B) $\frac{31}{x} + \frac{13}{y} = 1$
 (C) $\frac{13}{x} - \frac{31}{y} = 1$ (D) $\frac{31}{x} - \frac{13}{y} = 1$
65. The normal chord of parabola, which sub-tends right angle at vertex is inclined at an angle α to the axis of parabola. Find the value of α ?
 (A) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\tan^{-1}(\sqrt{7})$
 (C) $\frac{\pi}{3}$ (D) $\tan^{-1}(\sqrt{2})$
66. A ray of light travelling along the line $y = k$ from the positive direction of x-axis strikes a concave mirror, whose intersection with xy plane is parabola $y^2 = 4ax$. Find the equation of line along which the reflected ray will travel. Assume 'a' and 'k' are positive.
 (A) $(k^2 - 4a^2)y = 4ak(x - a)$ (B) $(k^2 - 4a^2)y = 4ak(x + a)$
 (C) $x = a$ (D) $x = 2a$
67. TP and TQ are tangents drawn to a parabola from the point T and P_1, P_2 and P_3 are the lengths of perpendicular from P, T and Q respectively on any tangent to the curve. Then
 (A) P_1, P_2, P_3 are in A.P (B) P_1, P_2, P_3 are in G.P
 (C) P_1, P_2, P_3 are in H.P (D) $\frac{1}{P_1} + \frac{1}{P_2} = \frac{a}{P_2}$ (4a is latus rectum)

68. Any point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is joined to its centre O. The locus of mid-point of OP is
- (A) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$ (B) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{4}$
 (C) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$ (D) $\frac{a^2}{x^2} + \frac{b^2}{y^2} = 1$
69. If the normal at any point P on an ellipse of eccentricity 'e' meets the major axis and minor axis at T and G respectively, then PT : PG is equal to
- (A) e (B) 1 : e
 (C) $(1 - e^2) : 1$ (D) $(1 - e^2) : e$
70. For a certain value of c, $\lim_{x \rightarrow -\infty} [(x^5 + 7x^4 + 2)^c - x]$ is finite & non zero. The value of c and the value of the limit is
- (A) 1/5, 7/5 (B) 0, 1
 (C) 1, 7/5 (D) none of these
71. If $\int (x^3 - 2x^2 + 5)e^{3x} dx = e^{3x} (Ax^3 + Bx^2 + Cx + D)$ then the statement which is incorrect is
- (A) $C + 3D = 5$ (B) $A + B + 2/3 = 0$
 (C) $3C + 2B = 0$ (D) $A + B + C = 0$
72. Let $I_1 = \int_0^{\pi/2} e^{-x^2} \sin(x) dx$; $I_2 = \int_0^{\pi/2} e^{-x^2} dx$; $I_3 = \int_0^{\pi/2} e^{-x^2} (1+x) dx$ and consider the statements
 I $I_1 < I_2$ II $I_2 < I_3$ III $I_1 = I_3$
 Which of the following is(are) true?
 (A) I only (B) II only
 (C) Neither I nor II nor III (D) Both I and II
73. If the function $f(x) = 2 \tan x + (2a + 1) \ln |\sec x| + (a - 2)x$ is increasing in $\left(0, \frac{\pi}{2}\right)$ then range of 'a' equal to
- (A) $(-\infty, 0]$ (B) $[0, 1]$
 (C) $[0, 3]$ (D) $[0, \infty)$
74. The maximum value of $\sec^{-1} \left(\frac{7 - 5(x^2 + 3)}{2(x^2 + 2)} \right)$ is:
- (A) $\frac{5\pi}{6}$ (B) $\frac{5\pi}{12}$
 (C) $\frac{7\pi}{12}$ (D) $\frac{2\pi}{3}$

75. Suppose $p, q, r \neq 0$ and system of equations
 $(p+a)x + by + cz = 0$
 $ax + (q+b)y + cz = 0$
 $ax + by + (r+c)z = 0$
 has a non-trivial solution, then value of $\frac{a}{p} + \frac{b}{q} + \frac{c}{r}$ is
 (A) -1 (B) 0
 (C) 1 (D) 2
76. If the system of equations $ax + hy + g = 0$, $hx + by + f = 0$ and $ax^2 + 2hxy + by^2 + 2gx + 2fy + c + t = 0$ has a unique solution, then
 (A) $t = 1$ (B) $t = 0$
 (C) $t = \frac{abc + 2fgh - af^2 - bg^2 - ch^2}{h^2 - ab}$ (D) $t = \frac{1}{h^2 - ab}$
77. Let $\vec{p}$ and $\vec{q}$ be the position vectors of P and Q respectively, with respect to O and $|\vec{p}| = p$, $|\vec{q}| = q$. The points R and S divide PQ internally and externally in the ratio 2 : 3 respectively. If $\vec{OR}$ and $\vec{OS}$ are perpendicular, then
 (A) $9p^2 = 4q^2$ (B) $4p^2 = 9q^2$
 (C) $9p = 4q$ (D) $4p = 9q$
78. If $\vec{d} = \lambda(\vec{a} \times \vec{b}) + \mu(\vec{b} \times \vec{c}) + \gamma(\vec{c} \times \vec{a})$ and $[\vec{a}, \vec{b}, \vec{c}] = 1$. Then $\lambda + \mu + \gamma$ is equal to
 (A) $\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c})$ (B) $\vec{a} + \vec{b} + \vec{c}$
 (C) 1 (D) none of these
79. Area of the region bounded by the curve $y = \frac{16 - x^2}{4}$ and $y = \sec^{-1}[-\sin^2 x]$ (where $[\cdot]$ denotes the greatest integer function) is
 (A) $\frac{1}{3}(4 - \pi)^{3/2}$ (B) $8(4 - \pi)^{3/2}$
 (C) $\frac{8}{3}(4 - \pi)^{3/2}$ (D) $\frac{8}{3}(4 - \pi)^{1/2}$
80. A contest consists of predicting the results (win, draw or defeat) of 10 football matches. The probability that one entry contains at least 5 correct answers is
 (A) $\frac{12585}{3^{10}}$ (B) $\frac{12385}{3^{10}}$
 (C) $\frac{9385}{3^{10}}$ (D) none of these

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If $\sin x + \cos x + \tan x + \cot x + \sec x + \operatorname{cosec} x = 7$ and $\sin 2x = a - b\sqrt{7}$, where a, b are rational, then $\frac{a+b}{10}$ is
82. If $\tan x - \tan^2 x = 1$, then the value of $\tan^4 x - 2\tan^3 x - \tan^2 x + 2\tan x + 1$ is
83. The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through $(4, 2)$ and length of latus rectum is $\frac{4}{3}$, the angle between asymptotes is θ then $\frac{6}{\pi}\theta$ is
84. Co-ordinates of points common to hyperbola $25x^2 - 9y^2 = 225$ and $25x + 12y = 45$ is (x_1, y_1) then $3|(x_1 + y_1)|$ is
85. For x satisfying $\log_{1/\sqrt{5}}(6^{x+1} - 36^x) \geq -2$, $x \in (-\infty, a] \cup [\log_b c, \infty)$ then $a + b + c$ is
86. S.D of a data is 6. When each observation is increased by 1, then the S.D of new data is
87. Let $u = \int_0^1 \frac{\ln(x+1)}{x^2+1} dx$ and $v = \int_0^{\pi/2} \ln(\sin 2x) dx$ then $4u + v =$
88. Let $f(x) = \text{Max.}(\cos x, x, 2x - 1)$ where $x \geq 0$. Then number of points of non-differentiability of $f(x)$, is equal to
89. If a_1 is the value of a for which function $f(x) = x^2 + \frac{a}{x}$ has a local minimum at $x = 2$ and a_2 is the value of a for which $f(x)$ has a point of inflection at $x = 1$, then $\left(\frac{a_1 + a_2}{3}\right)$ is equal to
90. If n positive integers are taken at random and multiplied together, then the probability that the last digit of the product is 2, 4, 6 or 8 is $\frac{a^n - b^n}{C^n}$ then $a + b + c$ is equal to

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2023

TEST DATE: 28-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B

Sol. Centre of mass will be on angle bisector and distance is $\frac{2}{3}R \frac{\sin \frac{\theta_0}{2}}{\frac{\theta_0}{2}}$

$$\text{x-coordinate} = \left(\frac{2}{3}R \frac{\sin \frac{\theta_0}{2}}{\frac{\theta_0}{2}} \right) \cos \frac{\theta_0}{2}$$

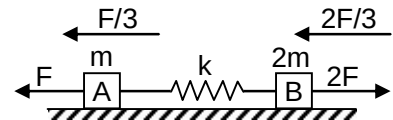
$$\text{y-coordinate} = \left(\frac{2}{3}R \frac{\sin \frac{\theta_0}{2}}{\frac{\theta_0}{2}} \right) \sin \frac{\theta_0}{2}$$

2. B

Sol. In frame of centre of mass $\rightarrow a_{\text{cm}} = \frac{F}{3m}$

Work energy theorem in centre of mass frame $\rightarrow$

$$\frac{4F}{3}x_1 + \frac{4F}{3}x_2 = \frac{1}{2}k(x_1 + x_2)^2 - \frac{1}{2}m(2u)^2 - \frac{1}{2}(2m)u^2$$



3. B

Sol. $\Delta H_1 = \frac{Q^2}{2} \left(\frac{1}{C} - \frac{1}{2C} \right)$

$$\Delta H_2 = \frac{Q^2}{2} \left(\frac{1}{2C} - \frac{1}{3C} \right)$$

.....
.....

$$\Delta H_n = \frac{Q^2}{2} \left(\frac{1}{nC} - \frac{1}{(n+1)C} \right)$$

$$\Delta H = \Delta H_1 + \Delta H_2 + \dots = \frac{Q^2}{2C} = \frac{C^2 V^2}{2C} = \frac{CV^2}{2}$$

$$= 2\pi\epsilon_0 R V^2$$

4. D

Sol. Add linear and Angular

$$EMF \frac{B\omega L^2}{2} = \frac{BvR}{2}$$

$$BvR \sin \omega t = B$$

5. A

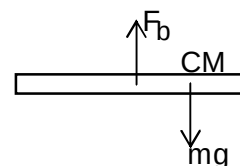
Sol. Velocity will be sinusoidal function of time.

6. B

Sol. Torque about CM;

$$F_b \frac{\ell}{4} = I\alpha$$

$$\alpha = \frac{\pi r^2 \ell^2 g \rho}{4I}$$



7. B

Sol. Upthrust on the complete solid cone $\Rightarrow B = \frac{\pi}{3} (2R)R^2 \rho g$

$$\Rightarrow F_{\text{Base}} - F_{\text{Csurface}} = \frac{2\pi R^2 \rho g}{3}$$

$$\Rightarrow F_1 = F_{\text{Csurface}} = [\rho_0 + \rho g(2R)] \pi R^2 - \frac{2}{3} \pi R^2 \rho g \dots (1)$$

For remaining cone

$$F_2 = B.F. + F_1$$

$$= \left[\frac{\pi}{3} R^2 \times 2R - \frac{\pi}{3} R^3 \rho \right] \rho g + P_0 (\pi R^2) + \frac{4}{3} \pi R^3 \rho g$$

$$F_1 = \frac{\pi}{3} R^3 \rho g \left[5 + \frac{3P_0}{R\rho g} \right]$$

8. C

Sol. Let length of semi major axis is a and eccentricity is e .

$$-\frac{GMm}{R_e} + \frac{1}{2}m\left(\sqrt{\frac{GM_e}{R_e}}\right)^2 = -\frac{GM_em}{2a} \Rightarrow a = R_e$$

As the separation between ends of semi minor axis and focus is equal to length of semi major axis we can say that launching and landing sites are ends of semi minor axis.

$$\Rightarrow R_e \cos 45^\circ = ae \Rightarrow e = 1/\sqrt{2}$$

$$\text{Then maximum separation will be } a + ae = R_e \frac{(\sqrt{2} + 1)}{\sqrt{2}}$$

9. B

Sol. In the circular motion around the Earth, the centripetal force on the satellite is a gravitational force. Therefore, $v^2 = GM/R$, where M is the mass of the Earth, R is the radius of the orbit of satellite and G is the universal gravitational constant. Therefore, the kinetic energy increases with the decrease in the radius of the orbit. The gravitational potential energy is negative and decreases with the decrease in radius.

10. A

Sol. In equilateral DABC given that $P_A = 2 P_C = P_1$

$$\tan 60^\circ = \frac{P_1/2}{v_1 - v}$$

$$v_1 - v = \frac{P_1}{2\sqrt{3}}$$

$$\text{Work done} = \text{area of triangle} = \frac{1}{2} 2x(v_1 - v) \times \frac{P_1}{2} = \frac{P_1^2}{4\sqrt{3}}$$

By ideal gas equation

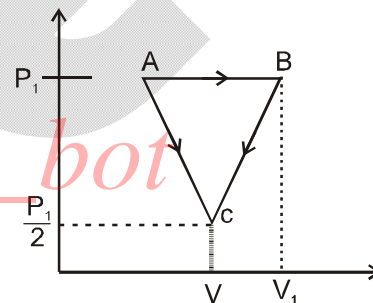
$$p_1 v_1 = nRT_1$$

$$p_1 = \frac{nRT_1}{V_1}$$

$$w = \frac{(nRT_1)^2}{4\sqrt{3}v_1^2} = \frac{n^2 R^2 T_1^2}{4\sqrt{3}v_1^2}$$

$N = 2$ moles

$$\text{Work done} = \frac{R^3 T_1^2}{\sqrt{3}v_1^2}$$



11. A

Sol. For circular orbit,

$$\frac{GMm}{r^2} = mw^2r \quad (r = \text{radius of planet's orbit})$$

$$w^2 = \frac{GM}{r^3} = \left(\frac{2\pi}{T}\right)^2 \Rightarrow r = \left(\frac{GMT}{4\pi^2}\right)^{3/2}$$

$$\text{Energy radiated by the sun per second} = s(4\pi R^2)q_0^4$$

$$\text{Energy received by the planet per second} = s(4\pi R^2)q_4^4 \left(\frac{\pi R_p^2}{4\pi r^2}\right)$$

In thermal equilibrium

$$\sigma \cdot (4\pi R^2) \theta_0^4 \left(\frac{\pi R_p^2}{4\pi r^2} \right) = \sigma \cdot (4\pi R_p^2) \theta^4$$

Energy received
by planet

$$\left(\theta = \sqrt{\frac{R}{2R}} \theta_0 \right)$$

Energy radiated
by planet

12.

C

Sol. Let N_0 be initial number of active nuclides of ^{238}U

$$\text{Given } \frac{N(\text{Pb})}{N(\text{U})} = \frac{1}{2} \Rightarrow \frac{N_0 - N_0 e^{-\lambda t}}{N_0 e^{-\lambda t}} = \frac{1}{2} \Rightarrow e^{-\lambda t} - 1 = \frac{1}{2}$$

$$\Rightarrow t = \frac{T_{1/2} \ln(1.5)}{\ln 2} = 4.5 \times 10^9 \frac{\ln\left(\frac{3}{2}\right)}{\ln 2} \text{ year}$$

13.

A

Sol. $m = m_0 e^{-\lambda t}$

Thrust force on the body is

$$F = u_r \left(\frac{dm}{dt} \right) = m \left(\frac{dv}{dt} \right)$$

14.

B

Sol. Along common normal $v_{1/0} = -2\hat{i}$

$$\vec{v}_{1/0} = \vec{v}_{1/g} - \vec{v}_{0/g} = 0 \text{ along C.N.}$$

Hence (B) is correct

15.

A

Sol. $S_1P - S_2P = 2\lambda \cos\theta = \lambda$, $\therefore \cos\theta = \frac{1}{2} \Rightarrow \tan\theta = x/D \Rightarrow x = D \tan 60^\circ$

$$x = D\sqrt{3}$$

16.

A

$$\text{Sol. } r = \frac{R}{2}$$

Using parallel axis theorem

$$I_x = I_{cm} + mr^2 \text{ twice}$$

17.

A

$$\text{Sol. } 11 = 10 \frac{V + V_A}{V}$$

$$12 = 10 \frac{V + V_B}{V}$$

Solve for ratio

$$\left(\frac{V_B}{V_A} \right)$$

18.

C

Sol.

$$PV = nRT$$

$$P = P_0 - AV^2$$

$$\Rightarrow (P_0 - AV^2)V = nRT$$

$$\Rightarrow P_0 - 3AV^2 = nR \left(\frac{dT}{dV} \right)$$

$$\text{For } T_{\max} \frac{dT}{dV} = 0$$

$$\Rightarrow V = \sqrt{\frac{P_0}{3A}} \Rightarrow P = P_0 - A \left(\sqrt{\frac{P_0}{3A}} \right)^2$$

$$\Rightarrow T_{\max} = \frac{2P_0}{3nR} \sqrt{\frac{P_0}{3A}}$$

$$PV = nRT$$

$$T = \frac{PV}{nR} = \frac{2P_0}{3nR} \sqrt{\frac{P_0}{3A}}$$

19.

D

Sol.

$$v_b = 200 \hat{i}$$

$$v_s = \begin{bmatrix} -200 \cos \theta \hat{i} + 200 \sin \theta \hat{j} \end{bmatrix}$$

$$v_{bs} = (200 + 200 \cos \theta) \hat{i} - 200 \sin \theta \hat{j}$$

$$+200 \sin \theta = \frac{1}{\sqrt{3}}$$

$$(200 + 200 \cos \theta)$$

20.

C

Sol.

$$d\mu = \int dq \left(\frac{\omega}{2\pi} \right) \pi r^2$$

SECTION - B

21.

9

Sol.

$$x_{cm} = \frac{\int dm \times x}{\int dm} = \frac{\int_0^a (\sigma \times 2kx^2) dx \times x}{\sigma \times \frac{2}{3} ka^3} = \frac{3}{4} a = \frac{9}{12} a$$

22.

6

Sol.

$$Q_0 = C(\epsilon_1 + \epsilon_2) \times 2$$

$$Q_0 = 72 \mu F$$

$$\frac{Q_0}{12} = \frac{72}{12} = 6 \mu C$$

23.

2

24. 3

Sol. For a small arc subtending angle $d\theta$ at the centre the inward force due to tension in thread is balanced by outward force due to surface tension

$$2T \sin\left(\frac{d\theta}{2}\right) = SRd\theta$$

$$\Rightarrow Td\theta = SRd\theta$$

$$T = SR$$

25. 7

Sol. $\frac{dQ}{dt} = mc\left(-\frac{dT}{dt}\right)$

$$E = \sigma AT^4$$

$$mc\left(-\frac{dT}{dt}\right) = \sigma A(T^4 - 0^4) = \sigma AT^4$$

$$\frac{dT}{T^4} = -\frac{\sigma A}{mc} dt$$

$$\int_{200}^{100} T^{-4} dT = -\frac{\sigma A}{mc} \int_0^t dt$$

$$\left| \frac{T^{-3}}{-3} \right|_{200}^{100} = -\frac{\sigma A}{mc} t$$

$$t = \frac{mc}{3\sigma A} \left[\frac{1}{T^3} \right]_{200}^{100}$$

$$= \frac{\left(\frac{4}{3}\pi r^3 \rho\right) c}{3\sigma \times 4\pi r^2} \left[\frac{1}{100^3} - \frac{1}{200^3} \right]$$

$$= \frac{7}{72} \frac{r\rho c}{\sigma} \times 10^{-6} \text{ sec.}$$

26. 4

Sol. $T = \frac{2v_0 \sin 37^\circ}{g \cos 37^\circ} = \frac{3}{2} \text{ sec}$

$$V_0 \cos 37^\circ \times \frac{3}{2} + \frac{1}{2} g \sin 37^\circ \times \left(\frac{3}{2}\right)^2 = 6 + V \times \frac{3}{2} + \frac{1}{2} g \sin 37^\circ \left(\frac{3}{2}\right)^2$$

$$V = 4$$

27. 9

Sol. $\sqrt{f} = a(Z-1)$

28. 2

Sol. When incident ray is fixed the angular velocity of reflected ray becomes twice in the same sense as that of reflected ray. After $t = 15$ sec, the mirror rotates 15° then reflected ray rotates 30° in the same sense.

$$\text{In } \triangle ABC, \cos 60^\circ = \frac{3}{r}$$

$$\text{Where } \omega = \frac{\pi}{180} \text{ rad / sec. } r = 6 \text{ m}$$

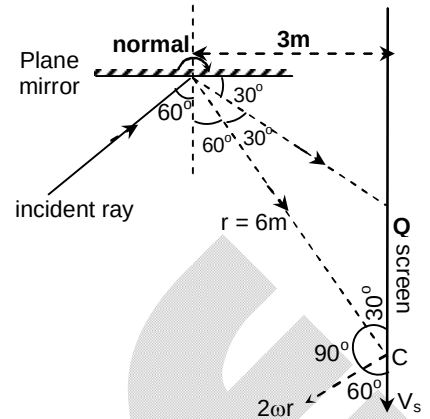
Then, there are two components of velocity of spot on the screen.

(a) radial component which increase the length of r i.e.,

$$V_s \sin 60^\circ = \frac{dr}{dt}$$

(b) perpendicular component

$$V_s \cos 60^\circ \Rightarrow V_s \times \frac{1}{2} = 2 \times \frac{\pi}{180} \times 6 \Rightarrow V_s = \left(\frac{2\pi}{15} \text{ m/s} \right)$$



29. 8

Sol. The stresses are zero at the free end and maximum at the axis. Therefore, the rod will rupture at the middle.

Let us consider the element of rod at a distance x from the axis, the mass of element

$$dm = \rho A dx$$

Applying Newton's second law,

$$F - (F + dF) = (\rho A dx) \omega^2 x$$

Where ω is the rotation speed.

$$F = -\int dF = -\rho A \omega^2 \frac{x^2}{2} + C$$

Rupture of rod will occur when $f = \sigma$

$$\therefore \frac{\rho \omega^2 l^2}{8} = \sigma$$

$$\Rightarrow \omega = \sqrt{\frac{8\sigma}{\rho l^2}} \Rightarrow n = \frac{1}{2\pi} \sqrt{\frac{8\sigma}{\rho l^2}}$$

30. 4

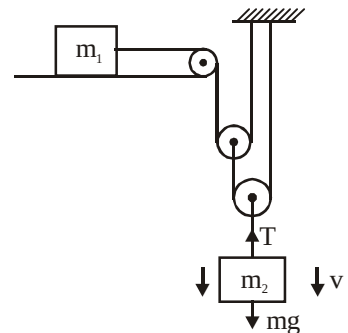
Sol. Applying work energy theorem

$$K_f - K_i = W$$

$$\frac{1}{2} m_2 v^2 + \frac{1}{2} m_1 (4v)^2$$

$$= m_2 g \times y - \mu m_1 g (4y)$$

$$v = 4 \text{ m/s.}$$

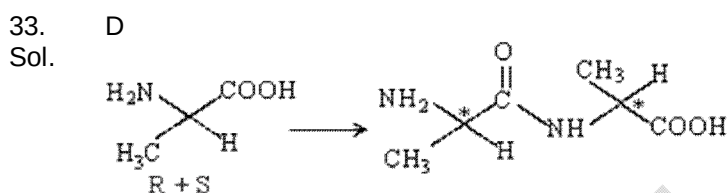
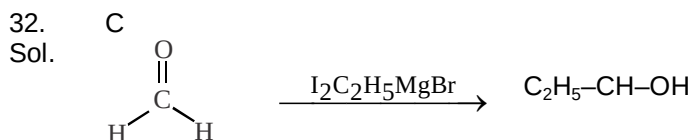


Chemistry

PART – B

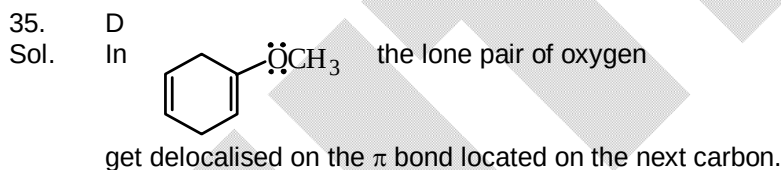
SECTION – A

31. D
Sol. Pentaammine chloride cobalt (III) chloride
 $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{+2} + 2\text{Cl}^-$
Gives 3 ions in aqueous solution.



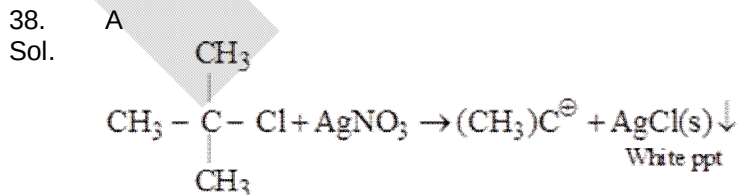
34. B
Sol. Charge on 1 millimole M^{n+} ions = 193 cb
$$= \frac{n \times 96500}{1000}$$

$$n = \frac{193 \times 1000}{96500} = 2$$



36. C
Sol. Quantum number set $n = 2$, $\ell = 0$, $m = -1$ it is not possible (not valid).

37. B
Sol. neq. KMnO_4
 $= \text{neq.} [\text{FeC}_2\text{O}_4 + \text{Fe}_2(\text{C}_2\text{O}_4)_3 + \text{FeSO}_4]$
or $n \times 5 = 1 \times 3 + 1 \times 6 + 1 \times 1$
 $\therefore n = 2$



Reason : Due to most stable carbocation formation tert-butyl chloride given the ppt immediately.

39. C

Sol. $\Delta T_b = K_b \times m$

$$\therefore \frac{\Delta T_b (A)}{\Delta T_b (B)} = \frac{K_b (A)}{K_b (B)} \text{ as } m_A = m_B$$

$$\therefore \frac{\Delta T_b (A)}{\Delta T_b (B)} = \frac{1}{5}$$

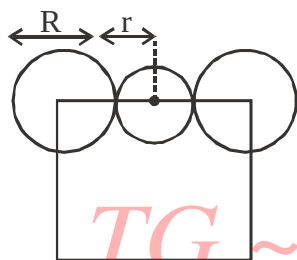
40. D

Sol. Colligative properties of colloidal solution are smaller than true solution.

41. C

Sol. $a = 2(R + r)$

$$\frac{a}{2} = (R + r) \dots (1)$$



$$a\sqrt{3} = 4R \dots (2)$$

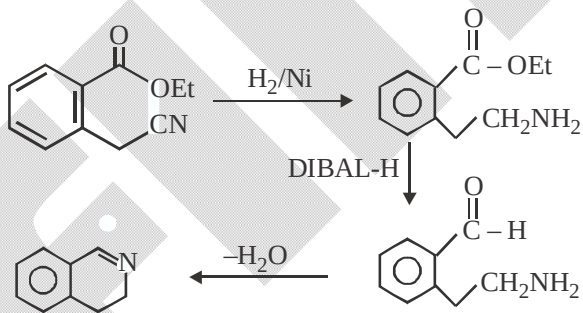
Using (1) & (2)

$$\frac{a}{2} = \frac{a\sqrt{3}}{4} = r ; a \left(\frac{2 - \sqrt{3}}{4} \right) = r$$

$$r = 0.067 a$$

42. B

Sol.



43. B

Sol.

A complex having strong field ligand has tendency to absorb light of highest energy. Among the three complexes $[\text{Co}(\text{NH}_3)_6]^{3+}$ will absorb radiation of highest energy and least wavelength. $[\text{Co}(\text{NH}_3)_5\text{H}_2\text{O}]^{3+}$ has field weaker than the above compound and therefore absorb radiation of lesser energy and more wavelength. $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ has the weakest field and therefore will absorb light of least energy and highest wavelength.

Strength of ligand $\text{NH}_3 > \text{H}_2\text{O} > \text{Cl}$.

44. B

Sol. Reagent based

45. D

Sol. Nitrogen oxides and hydrocarbons (unburnt fuel) are primary pollutant that leads to photochemical smog.

46. C

Sol. NaH is an example of ionic hydride which is also known as saline hydride.

47. B

 Sol. $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$

Apply Arrhenius equation

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303R} \left(\frac{1}{600} - \frac{1}{800} \right)$$

$$\log \frac{1}{2.5 \times 10^{-4}} = \frac{E_a}{2.303 \times 8.31} \left(\frac{200}{600 \times 800} \right)$$

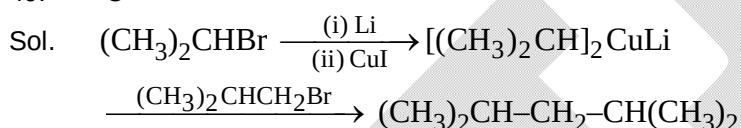
$$\therefore E_a \approx 166 \text{ kJ/mol}$$

48. C

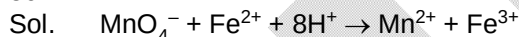
 Sol. Polarisation power $\propto +\text{ve charge}$

$$\propto 1 / \text{Size of ion}$$

49. C



50. B



$$\begin{array}{cc} 0.1\text{M} & 1.72\text{M} \end{array}$$

$$\begin{array}{ccc} 0.01\text{M} & 1.0\text{M} & 0.09\text{M} \end{array}$$

$$E^0_{\text{RP}(\text{MnO}_4^-/\text{Mn}^{2+})} > E^0_{\text{RP}(\text{H}^+/\text{H}_2)}$$

Permanganate electrode behave as cathode

$$E^{\circ}_{\text{cell}} = 1.51 \text{ V}$$

Using Nernsts equation : (n = 5)

$$E^0_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51 + \frac{0.059}{5} \log \frac{[\text{MnO}_4^-][\text{H}^+]^8}{[\text{Mn}^{2+}]}$$

$$= 1.51 - 0.011 = 1.499 \approx 1.5\text{V}$$

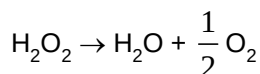
SECTION – B

51. 9

Sol. Initial moles of $\text{H}_2\text{O}_2 = 2 \times 2 = 4$
Moles of H_2O_2 left after 20 min.

$$= 4 \times \frac{1}{2^2} = 1 \text{ (2 half lives)}$$

Moles of H_2O_2 decomposed 20 min.
 $= 4 - 1 = 3$



Moles of O_2 formed during 20 min. = $3/2$

$$\text{Volume change} = \frac{\frac{3}{2} \times R \times T}{P}$$

$$q = 0; \Delta U = W$$

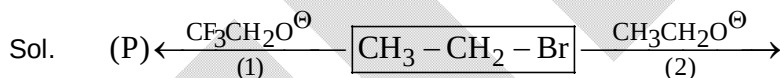
$$W = -P_{\text{ext}} (V_2 - V_1) = -P_{\text{ext}} \Delta V$$

$$= -\frac{3}{2} \times 2 \times 300 = -900 \text{ Cal.}$$

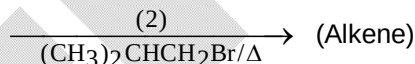
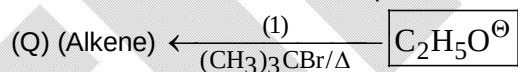
52. 7

Sol. (i) Planar molecules :
 XeF_2 , ClF_3 , H_2O , $[\text{XeF}_5]^-$, I_3^- , BCl_3 , XeF_4 .
(ii) SF_4 – See- Saw shape
 PCl_5 – Trigonal bipyramidal
 SF_6 – Square bipyramidal
 IF_7 – Pentagonal bipyramidal

53. 3

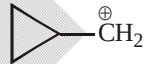


In reaction 2nd Nucleophile is stronger.



In reaction 1st 3° alkyl halide
 $1 > 2$

(R) In reaction 1st stable carbocation is formed



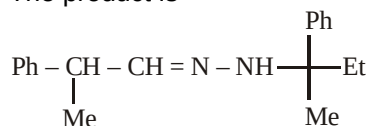
(Cyclopropyl methyl C^+)
 $1 > 2$

(S) In reaction 1st reagent in HI.

HI is more ionised, gives sufficient concentration of H^+ ion and I^+ is a good Nucleophile also.
 > 2

54. 4

Sol. The product is



I = (+, +) (E)

II = (+, +) (Z)

III = (-, +) (E)

IV = (-, +) (Z)

55. 5

 Sol. $\text{Na}_2\text{SO}_3 \xrightarrow{\text{AgNO}_3} \text{Ag}_2\text{SO}_3 \leftarrow \text{white ppt.}$
 $\text{NaCl} \xrightarrow{\text{AgNO}_3} \text{AgCl} \leftarrow \text{white ppt.}$
 $\text{Na}_2\text{C}_2\text{O}_4 \xrightarrow{\text{AgNO}_3} \text{Ag}_2\text{C}_2\text{O}_4 \leftarrow \text{white ppt.}$
 $\text{Na}_2\text{HPO}_4 \xrightarrow{\text{AgNO}_3} \text{Ag}_3\text{PO}_4 \leftarrow \text{yellow ppt.}$
 $\text{Na}_2\text{CrO}_4 \xrightarrow{\text{AgNO}_3} \text{Ag}_2\text{CrO}_4 \leftarrow \text{red ppt.}$
 $\text{NaNO}_2 \xrightarrow{\text{AgNO}_3} \text{AgNO}_2 \leftarrow \text{white ppt.}$
 $\text{CH}_3\text{COONa} \xrightarrow{\text{AgNO}_3} \text{CH}_3\text{COOAg} \leftarrow \text{white ppt.}$

56. 2

 Sol. $\text{PbO}_2 + 4\text{KI} \longrightarrow \text{PbI}_2 + \text{I}_2 + 2\text{K}_2\text{O}$
 $\text{N}_2\text{H}_4 + 2\text{I}_2 \longrightarrow \text{N}_2 + 4\text{HI}$
 $\text{moles of N}_2 \text{ liberated} = \frac{1.12}{22.4} = 5 \times 10^{-2} \text{ mole}$
 $\text{sol mole of I}_2 \text{ reacted} = 5 \times 10^{-2} \times 2 = 10^{-1} \text{ mole}$
 $\text{PbO}_2 + 2\text{H}_2\text{O} \longrightarrow \text{Pb(OH)}_4$
 $\text{Pb(OH)}_4 + 2\text{NaOH} \longrightarrow \text{Na}_2[\text{Pb(OH)}_6]$
 $\text{Na}_2\text{PbO}_2 \text{ mole of NaOH required} = 10^{-1} \times 2$
 $V \times \frac{1}{10} \times 10^3 = 2 \times 10^{-1}$
 $V = \frac{2}{10^{-3}} = 2000 \text{ mili litre} = 2 \text{ litre}$

57. 4

 Sol. The Be^{3+} ion has one electron in the 1s orbital ($n = 1$) in its ground state. Since beryllium has four protons in its nucleus ($Z = 4$), the energy of the electron in the 1s orbital of Be^{3+} can be calculated.

$$E = -\left(\frac{Z^2}{n^2}\right) (1310) \text{ kJ / mole}$$

$$= -\left(\frac{4^2}{1^2}\right) (1310) \text{ kJ / mole}$$

$$= 2.10 \times 10^4 \text{ kJ/mol}$$

58. 4

Sol. U_{rms} at temp $T_1 = \sqrt{\frac{3RT_1}{M}}$

$$U_{\text{rms}} \text{ at temp } T_2 = \sqrt{\frac{3RT_2}{M}}$$

$$\therefore \frac{U_{\text{rms}} \text{ at temp. } T_2}{U_{\text{rms}} \text{ at temp. } T_1} = \sqrt{\frac{T_2}{T_1}}$$

$$\text{or } \frac{10 \times 10^4}{5 \times 10^4} = \sqrt{\frac{T_2}{T_1}} \quad \text{or } \sqrt{\frac{T_2}{T_1}} = \frac{4}{1}$$

$\therefore$ Temperature is to be increased 4 times.

59. 4

Sol. We can write, $2\text{HI(g)} \rightleftharpoons \text{H}_2\text{(g)} + \text{I}_2\text{(g)}$

Initial pressure : 0.2 atm 0 0

Equilibrium partial pressure :

$$0.04 \text{ atm} \quad \frac{(0.2 - 0.04)}{2} \quad \frac{(0.2 - 0.04)}{2}$$

$$= 0.08 = 0.08$$

$$\text{Then, } K_p = \frac{[\text{H}_2][\text{I}_2]}{[\text{HI}]^2} = \frac{0.08 \times 0.08}{(0.04)^2} = 4.0$$

60. 400

Sol. Equivalents of Na_2CO_3 used

$$= (100 - 60) \times 10^{-3} \times \frac{1}{50}$$

Equivalent of Na_2CO_3 used = equivalent of Ca^{2+} or Mg^{2+} present in 100 ml of hard water

= equivalent of Ca^{2+} or Mg^{2+} present in 100 ml of hard water.

$$\text{The amount of } \text{CaCO}_3 \text{ in } 10^3 \text{ g of hard water} = \frac{40 \times 10^{-3} \times 10 \times 1 \times 50}{50} = 400 \times 10^{-3} \text{ g}$$

The amount of CaCO_3 in 10^6 g of hard water
= 400 g.

Permanent hardness of water = 400 ppm.

Mathematics

PART – C

SECTION – A

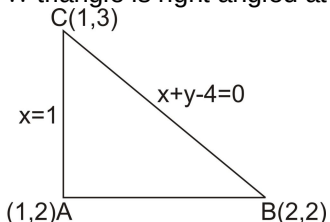
61. B

Sol. $(x + y - 4)(xy - 2x - y + 2) = 0$

$$\Rightarrow (x + y - 4)(x - 1)(y - 2) = 0$$

$\therefore$ sides of the triangle are $x + y - 4 = 0, x = 1, y = 2$

$\therefore$ triangle is right angled at $(1, 2)$.



Hypotenuse is $x + y = 4$, whose ends are $(1, 3), (2, 2)$.

$\therefore$ circumcentre is $\left(\frac{2+1}{2}, \frac{2+3}{2}\right)$ i.e. $\left(\frac{3}{2}, \frac{5}{2}\right)$.

$$\text{circumradius} = \frac{1}{2} \sqrt{(1-2)^2 + (3-2)^2} = \frac{1}{\sqrt{2}}$$

$\therefore$ circumcircle is $\left(x - \frac{3}{2}\right)^2 + \left(y - \frac{5}{2}\right)^2 = \frac{1}{2}$

$$\Rightarrow x^2 + y^2 - 3x - 5y + \frac{9}{4} + \frac{25}{4} - \frac{1}{2} = 0$$

$$\Rightarrow x^2 + y^2 - 3x - 5y + 8 = 0$$

62. C

Sol. Clearly (c) is the right choice.

$\therefore$ length of the $\perp$ from $(4, 5)$ upon $3x + 4y - 7 = 0$ is

$$\frac{12 + 20 - 7}{\sqrt{9 + 16}} = \frac{25}{5} = 5$$

Also for $(-2, -3)$.

63. A

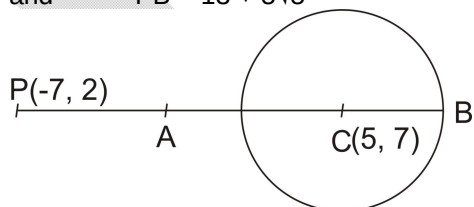
Sol. The centre C of the circle is $(5, 7)$

and radius = $\sqrt{5^2 + 7^2 + 51} = 5\sqrt{5}$

$$PC = \sqrt{(5+7)^2 + (7-2)^2} = 13$$

$\therefore PA = 13 - 5\sqrt{5}$

and $PB = 13 + 5\sqrt{5}$



$$\begin{aligned}\therefore p &= 13 + 5\sqrt{5} \\ q &= 13 - 5\sqrt{5} \\ \text{G.M. of } p, q &= \sqrt{pq} = \sqrt{169 - 125} = \sqrt{44} = 2\sqrt{11}.\end{aligned}$$

64.

A

Sol. Equation of any line through the given pt. (13, 31) is $y - 31 = m(x - 13)$.

It meets the x -axis at $Q\left(13 - \frac{31}{m}, 0\right)$ and y -axis at $S(0, 31 - 13m)$

$\therefore$ the co-ordinates of R are (h, k) , where $h = 13 - \frac{31}{m}, k = 31 - 13m$

$$\therefore (h - 13)(k - 31) = 31 \times 13$$

$$\Rightarrow hk = 31h + 13k \Rightarrow \frac{13}{h} + \frac{31}{k} = 1$$

Hence R satisfies $\frac{13}{x} + \frac{31}{y} = 1$.

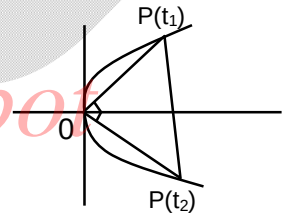
65.

D

Sol. $t_2 = -t_1 - \frac{2}{t_1}$ slope of OP = $\frac{2}{t_1} = M$
slope of OQ = $\frac{2}{t_2} = N$

$$\frac{2}{t_1} \times \frac{2}{t_2} = -1$$

$$\text{So, } t_1 = \pm\sqrt{2}$$



66.

A

Sol. The line $y = k$ intersects parabola $y^2 = 4ax$ at point $P\left(\frac{k^2}{4a}, k\right)$

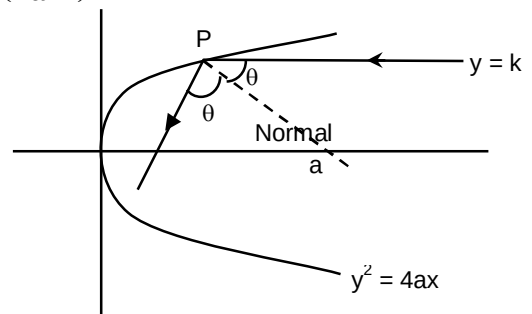
Slope of normal at P is $m = \frac{-k}{2a}$

Slope of incident ray is 0.

At slope of reflected ray is m_1 ,

$$\frac{-k}{2a} = \frac{m_1 - \left(\frac{-k}{2a}\right)}{1 + m_1\left(\frac{-k}{2a}\right)} \Rightarrow m_1 = \frac{4ak}{k^2 - 4a^2}$$

Equation of reflected ray is $(k^2 - 4a^2)y = 4ak(x - a)$.



67.

B

Sol. $P(t_1)$ $Q(t_2)$ $T(at_1t_2, a(t_1 + t_2))$

Let equation of tangent be $x = 0$

Hence

$$P_1 = at_1^2$$

$$P_2 = at_1t_2$$

$$P_3 = at_2^2$$

$$P_1P_3 = P_2^2.$$

68. B

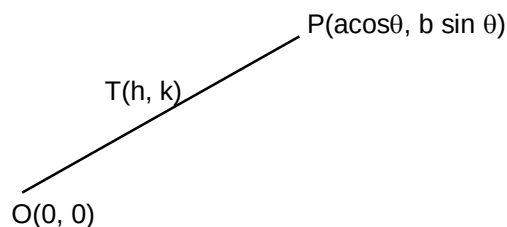
Sol.

$$2h = a \cos \theta \Rightarrow \cos^2 \theta = \left(\frac{2h}{a}\right)^2$$

$$2k = b \sin \theta \Rightarrow \sin^2 \theta = \left(\frac{2k}{b}\right)^2$$

$$\frac{4h^2}{a^2} + \frac{4k^2}{b^2} = 1$$

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{4}}$$



69. C

Sol.

A normal at P(a cos theta, b sin theta) is

a x sec theta - by cosec theta = a^2 - b^2

$$T\left(\frac{a^2 - b^2}{a} \cos \theta, 0\right)$$

$$G\left(0, \frac{a^2 - b^2}{a} \sin \theta\right)$$

$$PT^2 = \frac{b^2}{a^2} (a^2 \sin^2 \theta + b^2 \cos^2 \theta)$$

$$PG^2 = \frac{a^2}{b^2} (a^2 \sin^2 \theta + b^2 \cos^2 \theta)$$

$$PT : PG = \frac{b^2}{a^2} = 1 - e^2$$

70. A

Sol.

$$\lim_{x \rightarrow -\infty} \left(x^{5c} \left(1 + \frac{7}{x} + \frac{2}{x^5} \right)^c - x \right) = \lim_{x \rightarrow -\infty} x \left(x^{5c-1} \left(1 + \frac{7}{x} + \frac{2}{x^5} \right)^c - 1 \right)$$

This will be of the form $\infty \times 0$ only if

$5c - 1 = 0$ $c = 1/5$ substituting $c = 1/5$, I becomes

$$\begin{aligned} \text{Hence } I &= \lim_{x \rightarrow -\infty} x \left[(1+x)^{1/5} - 1 \right] \text{ where } X = \frac{7}{x} + \frac{2}{x^5} \\ &= \lim_{x \rightarrow -\infty} x \left[1 + \frac{X}{5} + \dots - 1 \right] = \lim_{x \rightarrow -\infty} x \left(\frac{7}{x} + \frac{2}{x^5} \right) \cdot \frac{1}{5} = \frac{7}{5} \end{aligned}$$

$$\text{Hence } c = \frac{1}{5} \text{ and } I = \frac{7}{5}$$

71. C

Sol.

$$[\text{Hint: } A = \frac{1}{3}, B = -1, C = \frac{2}{3}; D = \frac{13}{9}]$$

72. D

Sol.

[Sol. since $0 < \sin x < 1$ and $1 + x > 1$ in $(0, \pi/2)$

hence $I_3 > I_2 > I_1$

$\Rightarrow$ A and B are correct $\Rightarrow$ (D)]

73. D

Sol.

$$\text{Let } f(x) = 2 \tan x + (2a + 1) \ln |\sec x| + (a - 2)x$$

$$\Rightarrow f'(x) = 2 \sec^2 x + (2a + 1) \tan x + (a - 2) = 2 \tan^2 x + (2a + 1) \tan x + a$$

For $f(x)$ to be increasing, $f'(x) \geq 0 \forall x \in \left(0, \frac{\pi}{2}\right)$

$$\Rightarrow 2\tan^2 x + (2a+1)\tan x + a \geq 0 \forall x \in \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow (1+2\tan x)(\tan x + a) \geq 0 \forall x \in \left(0, \frac{\pi}{2}\right)$$

Hence $a \geq 0$. Ans

74. D

Sol. Find maximum of $\left(\frac{7-5(x^2+3)}{2(x^2+2)}\right)$

75. A

Sol. Since given system of equation has a non-trivial solution.

$$\therefore \begin{vmatrix} p+a & b & c \\ a & q+b & c \\ a & b & r+c \end{vmatrix} = 0$$

Operate $R_1 - R_2$ and $R_2 - R_3$, we get $\begin{vmatrix} p & -q & 0 \\ 0 & q & -r \\ a & b & r+c \end{vmatrix} = 0$

$$\Rightarrow p \begin{vmatrix} q & -r \\ b & r+c \end{vmatrix} + a \begin{vmatrix} -q & 0 \\ q & -r \end{vmatrix} = 0 \Rightarrow p(qr+qc+br)+a(qr)=0$$

$$\Rightarrow 1 + \frac{c}{r} + \frac{b}{q} + \frac{a}{p} = 0 \Rightarrow \frac{a}{p} + \frac{b}{q} + \frac{c}{r} = -1.$$

76. C

Sol. Clearly $ab - h^2 \neq 0$ [$\because$ first two equations are solvable for x, y]

Third equation can be written as

$$x(ax+hy+g) + y(hx+by+f) + gx+fy+c+t=0$$

Since any solution of first two must be sol. of the third equation.

$\therefore$ given system of equations can be written as

$$ax+hy+g=0$$

$$hx+by+f=0$$

$$gx+fy+c+t=0$$

Since the above system of equations are consistent

$$\therefore \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c+t \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} + \begin{vmatrix} a & h & 0 \\ h & b & 0 \\ g & f & t \end{vmatrix} = 0$$

$$\Rightarrow (abc + 2fgh - af^2 - bg^2 - ch^2) + t(ab - h^2) = 0$$

$$\Rightarrow t = \frac{abc + 2fgh - af^2 - bg^2 - ch^2}{(h^2 - ab)}$$

77. A

Sol. $\overline{OR} = \frac{2\vec{q} + 3\vec{p}}{2+3} = \frac{2\vec{q} + 3\vec{p}}{5}$

$$\overline{OS} = \frac{2\vec{q} - 3\vec{p}}{2-3} = \frac{2\vec{q} - 3\vec{p}}{-1}$$

Since $\overline{OR} \perp \overline{OS} \therefore \overline{OR} \cdot \overline{OS} = 0$

$$\therefore \frac{2\vec{q} + 3\vec{p}}{5} \cdot \frac{2\vec{q} - 3\vec{p}}{-1} = 0$$

$$\Rightarrow \frac{4|\vec{q}|^2 - 9|\vec{p}|^2}{-5} = 0 \Rightarrow 4q^2 - 9p^2 = 0 \Rightarrow 9p^2 = 4q^2$$

78. A

Sol. Now $\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) = \vec{d} \cdot \vec{a} + \vec{d} \cdot \vec{b} + \vec{d} \cdot \vec{c}$

$$= \mu(\vec{b} \times \vec{c} \cdot \vec{a}) + \gamma(\vec{c} \times \vec{a} \cdot \vec{b}) + \lambda(\vec{a} \times \vec{b} \cdot \vec{c})$$

$$= \lambda + \mu + \gamma \quad [\because (\vec{a}, \vec{b}, \vec{c}) = 1]$$

$$\text{Hence } \lambda + \mu + \nu = \vec{d} \cdot (\vec{a} + \vec{b} + \vec{c})$$

79. C

Sol. $0 \leq \sin^2 x \leq 1 \Rightarrow [-\sin^2 x] = 0 \text{ or } -1$

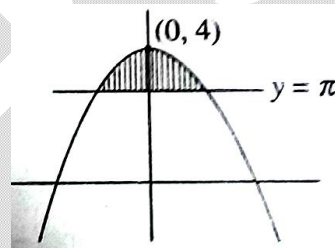
But $\sec^{-1}(0)$ is not defined

$$\Rightarrow y = \sec^{-1}[-\sin^2 x] = \sec^{-1}(-1) = \pi$$

$$\text{Solving with } y = \frac{16 - x^2}{4}, 16 - x^2 = 4\pi$$

$$\therefore x = \pm 2\sqrt{4 - \pi}$$

$$\text{Required area} = \int_{-2\sqrt{4-\pi}}^{2\sqrt{4-\pi}} \left(\frac{16 - x^2}{4} - \pi \right) dx = \frac{8}{3}(4 - \pi)^{3/2}$$



80. A

Sol. The total no. of ways of predicting the result is 3^{10} . The no. of ways in which exactly K results are correct is ${}^{10}C_K \cdot 2^{10-K}$

$\therefore$ the no. of favourable ways is

$${}^{10}C_5 \cdot 2^5 + {}^{10}C_6 \cdot 2^4 + {}^{10}C_7 \cdot 2^3 + {}^{10}C_8 \cdot 2^2 + {}^{10}C_9 \cdot 2 + {}^{10}C_{10} = 12585.$$

$$\text{Hence reqd. prob.} = \frac{12585}{3^{10}}$$

SECTION - B

81. 3

Sol. $\sin x + \cos x + \tan x + \cot x + \sec x + \operatorname{cosec} x = 7$

$$\Rightarrow (\sin x + \cos x) + \left(\frac{1}{\sin x \cos x} \right) + \frac{(\sin x + \cos x)}{\sin x \cos x} = 7$$

$$\Rightarrow (\sin x + \cos x) \left(1 + \frac{2}{\sin 2x} \right) = \left(7 - \frac{2}{\sin 2x} \right)$$

$$\Rightarrow (1 + \sin 2x) \left(1 + \frac{4}{\sin^2 2x} + \frac{4}{\sin 2x} \right) = 49 + \frac{4}{\sin^2 2x} - \frac{28}{\sin 2x} \quad (\text{squaring both sides})$$

$$\Rightarrow \sin^3 2x - 44 \sin^2 2x + 36 \sin 2x = 0$$

$$\Rightarrow \sin 2x = 22 - 8\sqrt{7}.$$

82. 4

Sol. $\tan^4 x - 2 \tan^3 x - \tan^2 x + 2 \tan x + 1$
 $\tan^4 x + \tan^2 x - 2 \tan^3 x + 2 \tan x - 2 \tan^2 x + 1$
 $= (\tan^2 - \tan x)^2 + 2(\tan x - \tan^2 x) + 1 = 4.$

83. 2

Sol. Use angle between asymptotes as $\theta = \tan^{-1} \frac{2ab}{b^2 - a^2}$ & Latusrectum = $\frac{2b^2}{a}$

84. 5

Sol. It is point of contact of tangent

85. 11

Sol. $x \in (-\infty, 0] \cup [\log_6 5, \infty)$

86. 6

Sol. S.D (and variance) of a data is not changed when each observation is increased (or decreased) by the same constant.

87. 0

Sol. $[u = \int_0^1 \frac{\ln(x+1)}{x^2+1} dx \quad \text{put } x = \tan \theta$
 $= \int_0^{\pi/4} \ln(1+\tan \theta) d\theta = \int_0^{\pi/4} \ln\left(1 + \frac{1-\tan \theta}{1+\tan \theta}\right) d\theta = \int_0^{\pi/4} \ln\left(\frac{2}{1+\tan \theta}\right) d\theta = \frac{\pi}{4} \ln 2 - u$
 $\therefore u = \frac{\pi}{8} \ln 2 \Rightarrow 4u = \frac{\pi}{2} \ln 2 \quad \dots(1)$

Again $v = \int_0^{\pi/2} \ln(\sin 2x) dx \quad (\text{put } 2x = t); v = \frac{1}{2} \int_0^{\pi} \ln(\sin t) dt = \int_0^{\pi/2} \ln(\sin t) dx$

$v = -\frac{\pi}{2} \ln 2 \quad \dots(2)$

$(1) + (2) \Rightarrow 4u + v = 0 \Rightarrow (C)]$

88. 2

Sol. Using Graph number of points of non differentiability = 2 Ans.

89. 5

Sol. $f'(x) = 2x - \frac{a}{x^2}$

$f'(2) = 4 - \frac{a}{4} = 0 \Rightarrow a = 16 \Rightarrow a_1 = 16$

Also, $f''(x) = 2 + \frac{2a}{x^3}$

$\therefore f''(1) = 2 + 2a = 0 \Rightarrow a = -1 \Rightarrow a_2 = -1$

Hence $\left(\frac{a_1 + a_2}{3}\right) = \frac{15}{3} = 5$ Ans.

90. 11

Sol. The last digit of the product will be 1, 2, 3, 4, 6, 7, 8 or 9 iff each of the n positive integer ends in any of these digits. Now the probability of an integer ending in 1, 2, 3, 4, 6, 7, 8 or 9 is $\frac{8}{10} = \frac{4}{5}$

$\therefore$ The prob. that the last digit of the product of n integers is 1, 2, 3, 4, 6, 7, 8 or 9 = $\left(\frac{4}{5}\right)^n$.

Next, the last digit of the product will be 1, 3, 7 or 9 iff each of the n + ve integers ends in 1, 3, 7 or 9 is $\frac{4}{10} = \frac{2}{5}$

$\therefore$ The prob. for the product of n + ve integer to end in 1, 3, 7 or 9 is $\left(\frac{2}{5}\right)^n$

Hence the prob. of the reqd. event = $\left(\frac{4}{5}\right)^n - \left(\frac{2}{5}\right)^n = \frac{4^n - 2^n}{5^n}$

TG ~ @bohring_bot

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – I
PAPER –1
TEST DATE: 28-12-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FOUR/FIVE** statements in **List-II** entries (P), (Q), (R), (S) OR (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

- A large insulating thick sheet of thickness $2d$ is charged with a uniform volume charge density ρ . A particle of mass m , carrying a charge q having a sign opposite to that of the sheet, is released from the surface of the sheet. The sheet does not offer any mechanical resistance to the motion of the particle. Find the oscillation frequency ν of the particle inside the sheet:

(A) $\nu = \frac{1}{2\pi} \sqrt{\frac{q\rho}{m\epsilon_0}}$ (B) $\nu = \frac{1}{2\pi} \sqrt{\frac{2q\rho}{m\epsilon_0}}$
 (C) $\nu = \frac{1}{4\pi} \sqrt{\frac{q\rho}{m\epsilon_0}}$ (D) $\omega = \frac{1}{2\pi} \sqrt{\frac{q\rho}{m\epsilon_0}}$
- A charged particles with velocity $\vec{v} = x\hat{i} + y\hat{j}$ moves in a magnetic field $\vec{B} = y\hat{i} + x\hat{j}$. The magnitude of magnetic force acting on the particle is F . Which one of the following statement(s) is/are correct?

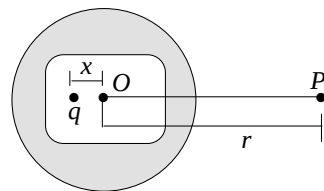
(A) No force will act on particle if $x = y$ (B) $F \propto (x^2 - y^2)$ if $x > y$
 (C) The force will act along z-axis if $x > y$ (D) The force will act along y-axis if $x > y$
- In a photoelectric effect experiment. If f is the frequency of radiations incident on the metal surface and I is the intensity of incident radiations, then which of the following is/are correct.

(A) If ' f ' is increased keeping ' I ' and work function constant, stopping potential and maximum kinetic energy of photoelectron increases.
 (B) If distance between cathode and anode is increased, stopping potential remains same.
 (C) If ' I ' is increased keeping ' f ' and work function constant, saturation current increases and stopping potential remains same.
 (D) If the work function is decreased keeping ' f ' and ' I ' constant then stopping potential will increase and maximum kinetic energy of photoelectrons also increases.
- $y(x, t) = \frac{0.6}{(2x - 5t)^2 + 5}$ is equation of moving pulse, where x and y are in meter t is in second, then:

(A) Pulse is moving in +x direction (B) It is a symmetric pulse
 (C) Its maximum displacement is 0.8m (D) In 1 second it will travel a distance of 2.5 m
- The speed (v) of a particle moving along a straight line, when it is at a distance (x) from a fixed point on the line, is given by $v^2 = 144 - 9x^2$. Select the correct option:

(A) Displacement of the particle $\leq$ distance moved by it
 (B) The magnitude of acceleration at a distance 3 units from the fixed point is 27 units
 (C) The motion is simple harmonic with $T = \frac{\pi}{3}$ units
 (D) The maximum displacement from the fixed point is 4 units

6. A point charge q is placed within the cavity of an electrically neutral conducting shell whose outer surface has spherical shape figure. Then:

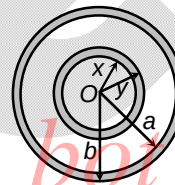


- (A) The potential V at point P lying outside the shell at a distance r from the centre O of the outer surface depends upon the value of x .
- (B) Potential at P does not depend upon the value of x .
- (C) A total charge q will be induced on the outer surface of the shell which will be distributed uniformly on the outer surface.
- (D) A total charge $-q$ will be induced on the inner surface of the shell, which will be distributed non-uniformly on the inner surface.

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FOUR/FIVE** statements in **List-II** entries (P), (Q), (R), (S) OR (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

7. In the figure shown a conducting spherical shell of inner radius x and outer radius y is concentric with a larger conducting spherical shell of inner radius a and outer radius b . The inner shell has a total charge $+3Q$ and the outer shell has a total charge $+5Q$. Let r be the distance of any point from the common centre O . Match List-I and List-II.

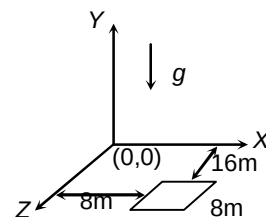


List-I		List-II	
(I)	Electric field strength is zero	(P)	On outer surface of the larger spherical shell
(II)	Electric field strength is non-zero	(Q)	On inner surface of the larger spherical shell
(III)	Magnitude of charge is $3Q$	(R)	On outer surface of the smaller spherical shell
(IV)	Charge is $+8Q$	(S)	For $a < r < b$

Which one of the following options is correct?

- (A) I $\rightarrow$ P, II $\rightarrow$ Q, R, III $\rightarrow$ Q, R, IV $\rightarrow$ R, S
- (B) I $\rightarrow$ S, II $\rightarrow$ P, Q, III $\rightarrow$ P, S, IV $\rightarrow$ R, Q
- (C) I $\rightarrow$ S, II $\rightarrow$ P, Q, R, III $\rightarrow$ Q, R, IV $\rightarrow$ P
- (D) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ Q, S, IV $\rightarrow$ P, Q, R

8. A square platform of side length 8 m is situated in x-z plane such that it is at 16 m from the x-axis and 8 m from the z-axis as shown in figure. A particle is projected with velocity $\vec{v} = (v_2\hat{i} + 25\hat{j})\text{m/s}$ relative to wind from origin and at the same instant the platform starts with acceleration $\vec{a} = (2\hat{i} + 2.5\hat{j})\text{m/s}^2$. Wind is blowing with velocity $v_1\hat{k}$. ($g = 10\text{ m/s}^2$)



List-I		List-II	
(I)	Possible values of v_2 (in m/s) so that particle hits the platform is/are	(P)	4
(II)	Possible value of v_1 (in m/s) so that particle hits the platform is/are	(Q)	6
(III)	Possible time of flight (in second) of the particle is/are	(R)	8
(IV)	Possible value of displacement/s (in m) of the particle in y-direction when v_2 has its minimum possible value is/are (till particle hit the platform)	(S)	20

Which one of the following options is correct?

- (A) I $\rightarrow$ Q, R, II $\rightarrow$ P, Q, III $\rightarrow$ P, IV $\rightarrow$ S
 (B) I $\rightarrow$ Q, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ S
 (C) I $\rightarrow$ S, II $\rightarrow$ P, Q, III $\rightarrow$ Q, R, IV $\rightarrow$ P, S
 (D) I $\rightarrow$ Q, R, II $\rightarrow$ P, III $\rightarrow$ Q, S, IV $\rightarrow$ P, Q, R

9. Match the statements in List-I with the results in List-II.

List-I		List-II	
(I)	A thin uniform spherical shell of surface area S has an initial temperature more than its surrounding atmosphere. Then magnitude of rate of change of its temperature with time	(P)	is independent of S
(II)	A soap bubble initially in equilibrium is given a charge Q , which distributes uniformly over its surface. The centre of the bubble is always fixed. For the duration the bubble having surface area S expands, the magnitude of electric potential at a fixed point always lying outside the bubble.	(Q)	depends on S
(III)	A container with open top and filled with ideal liquid is placed at rest on a smooth horizontal table. A small hole of area S is drilled at the bottom of a side wall of container. The magnitude of force exerted by escaping liquid on the container.	(R)	remains constant
(IV)	An infinitely long straight current carrying wire lies along the axis of a closed cylindrical surface of total surface area S in space. As the magnitude of current in the wire is continuously increased, the magnitude of the magnetic flux through the surface of this cylinder.	(S)	decreases with time

Which one of the following options is correct?

- (A) I $\rightarrow$ S, II $\rightarrow$ P, Q, III $\rightarrow$ R, S, IV $\rightarrow$ P, R
 (B) I $\rightarrow$ P, S, II $\rightarrow$ P, R, III $\rightarrow$ Q, S, IV $\rightarrow$ P, R
 (C) I $\rightarrow$ S, R, II $\rightarrow$ R, III $\rightarrow$ Q, S, IV $\rightarrow$ R, S
 (D) I $\rightarrow$ S, II $\rightarrow$ P, R, III $\rightarrow$ P, Q, IV $\rightarrow$ Q, R

10. Related to photoelectric effect, in List-I, some physical quantities change while in List-II effects of these changes are given. Match the entries of List-I with the entries of List-II.

List-I		List-II	
(I)	Intensity of incident light changes	(P)	$K_{\max}$ of emitted photoelectrons changes
(II)	Frequency of incident light changes	(Q)	Stopping potential changes
(III)	Target material changes	(R)	Saturation current changes
(IV)	Potential difference between the emitter and collector changes	(S)	Time delay in emission of photoelectrons changes

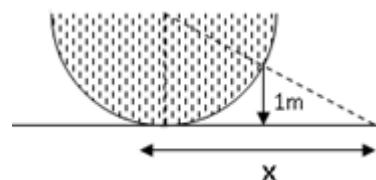
Which one of the following options is correct?

- (A) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ Q,S, IV $\rightarrow$ S
 (B) I $\rightarrow$ S, II $\rightarrow$ Q,S, III $\rightarrow$ P,Q, IV $\rightarrow$ S
 (C) I $\rightarrow$ R, II $\rightarrow$ P,Q, III $\rightarrow$ P,Q, IV $\rightarrow$ S
 (D) I $\rightarrow$ S, II $\rightarrow$ P,Q, III $\rightarrow$ Q,S, IV $\rightarrow$ R

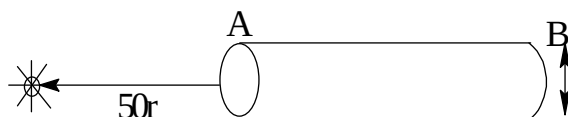
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

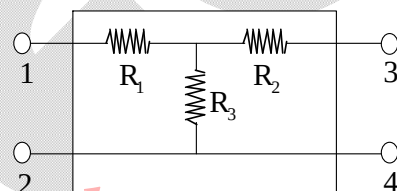
11. If resistance R_1 in resistance box is 300Ω , then the balanced length is found to be 75.0cm from end A. The diameter of unknown wire is 1mm and length of the unknown wire is 31.4cm . Find the specific resistance of the unknown wire ($\mu\Omega\text{-m}$)
12. A slab of glass of thickness 6cm and index 1.5 is placed somewhere in between a concave mirror and a point object, perpendicular to the mirror's optical axis. The radius of curvature of the mirror is 40cm . If the reflected final image coincides with the object, then find the distance of the object from the mirror (in cm).
13. A terrorist places a bomb at a horizontal distance of 6m from the foot of building of height 8m . When the bomb explodes its fragments fly in all directions with a velocity upto 30m/s . Find how long (in seconds) a man on the top of the building will be in danger. ($g=10\text{ms}^{-2}$)
14. A string is attached to a horizontal cylinder of radius $r = 4\text{cm}$ around which it passes several times. The free end hangs vertically and supports a particle. The vertical portion is equal to $l_0 = 28\text{cm}$. The particle is given a horizontal velocity u perpendicular to the axis of the cylinder and in such a sense as to wind up the string around the cylinder. What is the value of u (in m/sec) so that the string does not slacken in the subsequent motion?
15. A hemispherical tank of diameter 4m is filled with water. A very 'small' hole is punched at 1m above the bottom as shown in figure. Find x (in m) (distance at which water strikes the surface).



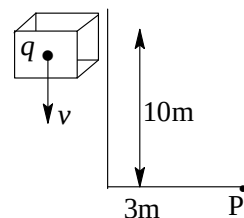
16. A cylindrical rod of length 64 cm and cross-section radius $r = (2/\sqrt{\pi})$ cm is placed at a distance $50r$ from an infrared point source S of power 1.25 kW as shown in the figure. The lateral surface of the rod is perfectly insulated from the surroundings. The cross-section A absorbs 80% of the incident energy, has temperature T_A in steady state. The surface B is radiating energy into space and the wavelength emitted by it with maximum energy density is $100,000 \text{ \AA}$. Find the value of T_A (temperature of end A in Kelvin) if conductivity varies with temp as $K = \frac{T}{T_A}$. Assume that the rate of flow of heat through the rod is steady. (Wein's constant = 0.003 K)



17. A concealed circuit (a black box) consisting of resistors has four terminals. If a voltage is applied between clamps 1 and 2 when clamps 3 and 4 are open, the power liberated is $P_1 = 40 \text{ W}$ and when clamps 3 and 4 are closed, the power liberated is $P_2 = 80 \text{ W}$. If the same source is connected to the clamps 3 and 4, the power liberated in the circuit when clamps 1 and 2 are open is $P_3 = 20 \text{ W}$. Determine the power P_4 (in Watt) consumed in the circuit when the clamps 1 and 2 are connected and the same voltage is applied between the clamps 3 and 4.



18. An elevator carrying a charge of 0.2 C , is moving down with a velocity of $4 \times 10^3 \text{ m/s}$. The elevator is 10 m from the bottom and 3 m horizontally from P as shown. What magnetic field (in μT) does it produce at point P ?



Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

19. Which of the following statement (s) is/are correct?
(A) Fe^{3+} and Mn^{2+} have equal paramagnetic character
(B) Cu_2Cl_2 and CuCl_2 are coloured.
(C) MnO_4^- is purple in colour because of unpaired electrons
(D) The magnetic moment of Fe^{2+} and Co^{3+} both are equal to $2\sqrt{6}$ B. M.
20. Which of the following statements are correct?
(A) The total spectral lines obtained from a single line during Zeeman effect is $(2I + 1)$.
(B) In the Lyman series as the energy liberated during transition increases then the distance between the spectral lines goes on decreasing.
(C) The highest probability of finding an electron in 1s orbital is in the vicinity of the circumference.
(D) The highest probability of finding an electron in 1s orbital is exactly at the middle between nucleus and circumference.
21. Which of the following is/are correct?
(A) Ammonium salts are more soluble than sodium salts
(B) SnCl_2 is more ionic than SnCl_4
(C) Calcium fluoride is more ionic than calcium chloride
(D) The formal charge on S atom in SO_2 is four
22. Which of the following statements are true for a gas -
(A) The mean free path is inversely proportional to the pressure of the gas at constant temperature
(B) The mean free path is proportional to temperature at constant pressure
(C) The collision frequency among the molecules of gas is proportional to temperature
(D) The velocity possessed by the largest fraction of molecules at a given temperature is known as root mean square velocity
23. Which of the following statements is/are false -
(A) Endothermic reactions are never spontaneous, at any temperature
(B) Exothermic reactions are always spontaneous at any temperature
(C) Exothermic reactions in which the entropy change for the system is negative are spontaneous at any temperature
(D) Reactions in which the entropy change for the system is negative are never spontaneous at any temperature
24. An aqueous solution contains either Hg_2^{2+} or Hg^{2+} the given solution gives green ppt with KI solution but no precipitate with H_2S . About the given aqueous solution which of the following is correct -
(A) It contains Hg_2^{2+}
(B) It contains Hg^{2+}
(C) With NH_3 solution it gives black precipitate
(D) None of these

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FOUR/FIVE** statements in **List-II** entries (P), (Q), (R), (S) OR (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. Match the following:

List-I		List-II	
(I)	$O < Te < Se < S$	(P)	Acidic character of oxides
(II)	$Be^+ < C^+ < B^+ < N^+ < F^+ < O^+ < Li^+$	(Q)	Electron Affinity
(III)	$Fe^{2+} < Fe^{3+} < Fe^{6+}$	(R)	Atomic radius
(IV)	$N < P < As < Sb$	(S)	Ionisation energy

Which one of the following options is correct?

- (A) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ R
 (B) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ P,Q,S, IV $\rightarrow$ R,Q
 (C) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ Q,R, IV $\rightarrow$ P,S
 (D) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ Q,S, IV $\rightarrow$ P,Q,R

26. Match the following:

List-I		List-II	
(I)	$XeOF_4$	(P)	no lone pairs of electron
(II)	XeO_4	(Q)	one lone pair of electrons.
(III)	XeF_4	(R)	more than one lone pairs of electrons
(IV)	IF_5	(S)	non planar

- (A) I $\rightarrow$ Q,S, II $\rightarrow$ P, S, III $\rightarrow$ R, IV $\rightarrow$ Q,S
 (B) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ P,Q,S, IV $\rightarrow$ R,Q
 (C) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ Q,R, IV $\rightarrow$ P,S
 (D) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ Q,S, IV $\rightarrow$ P,Q,R

27. Match the following:

List-I		List-II	
(I)	$[CoCl_3(NH_3)_3]$	(P)	show facial form
(II)	$[Cr(Ox)_3]^{3-}$	(Q)	cis form is optically active
(III)	$[CrCl_2(Ox)_2]$	(R)	Transform is optically inactive
(IV)	$[RhCl_3(Py)_3]$	(S)	show meridian form

- (A) I $\rightarrow$ Q,S, II $\rightarrow$ P, S, III $\rightarrow$ R, IV $\rightarrow$ Q,S

- (B) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ P,Q,S, IV $\rightarrow$ R,Q
 (C) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ Q,R,S, IV $\rightarrow$ P,S
 (D) I $\rightarrow$ P,S, II $\rightarrow$ Q, III $\rightarrow$ Q,R, IV $\rightarrow$ P,S

28. Match the following:

List-I		List-II	
(I)	Paramagnetic	(P)	N ₂ O ₃
(II)	Dark blue liquid in liquid or solid state	(Q)	NO ₂
(III)	Unstable gas due to auto decomposition.	(R)	NO
(IV)	Brown gas	(S)	N ₂ O ₅

- (A) I $\rightarrow$ Q,S, II $\rightarrow$ P, S, III $\rightarrow$ R, IV $\rightarrow$ Q,S
 (B) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ P,Q,S, IV $\rightarrow$ P,S
 (C) I $\rightarrow$ Q,R, II $\rightarrow$ P, III $\rightarrow$ P,S, IV $\rightarrow$ Q
 (D) I $\rightarrow$ P,S, II $\rightarrow$ Q, III $\rightarrow$ Q,R, IV $\rightarrow$ P,S

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

29. A mixture of CH₄ and C₂H₂ occupied a certain volume at a total pressure equal to 63 torr. The same gas mixture was burnt to CO₂ and H₂O (l). The CO₂ (g) alone was collected in the same volume and at the same temperature, the pressure was found to be 69 torr. What was the mole fraction of CH₄ in the original gas mixture?
30. N₂ gas is assumed to behave ideally. A given volume of N₂ originally at 373 K and 0.1013 MPa pressure is adiabatically compressed due to which its temperature rises to 673 K. What is final pressure (in MPa) of gas?
31. A saturated solution of iodine in water is 1.25×10^{-3} (M). In any saturated solution of I₂ concentration of I₂ is 1.25×10^{-3} (M). In 1 L of 0.1 (M) solution I⁻, it is seen 51.25×10^{-3} mole of I₂ can be maximum dissolved. In the aqueous solution of I⁻(aq), I₂ (aq) undergoes complex formation, $I_2(aq) + I^- \xrightleftharpoons{K_C} I_3^-(aq)$.
 What is the value of K_C?
32. The reaction, $N_2O_5(g) \rightleftharpoons 2NO_2(g) + \frac{1}{2}O_2(g)$, is started with initial pressure of N₂O₅ (g) equal to 600 torr. What fraction of N₂O₅ (g) decomposed when total pressure of the system is 960 torr?
33. At 298 K, the conductivity of a saturated solution of AgCl in water is $2.6 \times 10^{-6} \text{ ohm}^{-1} \text{ cm}^{-1}$. Given, $\lambda_m^\infty(\text{Ag}^+) = 63 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ & $\lambda_m^\infty(\text{Cl}^-) = 67 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 The solubility product of AgCl is $x \times 10^{-10}$, then value of 'x' is ____.

34. For a certain reaction of order n , the time for half change, $t_{1/2}$, is given by $t_{1/2} = \frac{[2 - \sqrt{2}]}{k} \times C_0^{1/2}$ where k is the rate constant and C_0 is the initial concentration what is n ____.
35. The density of solid argon (Ar = 40 g/mol) is 1.68 g/mL at 40 K. If the argon atom is assumed to be a sphere of radius = 1.50×10^{-8} cm, what % of solid Ar is apparently empty space?
36. Two liquids X and Y form an ideal solution. At 300 K, vapour pressure of the solution containing 1 mole of X and 3 moles of Y is 550 mm Hg. At the same temperature, if 1 mole of Y is further added to this solution, vapour pressure of the solution increases by 10 mm Hg. What is the sum of vapour pressure (in mm Hg) of X and Y in their pure states?

TG ~ @bohring_bot

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. Let A and B are commutative square matrices of order three such that A is symmetric and B is skew symmetric, if $C = (A + B)^T (A + B)(A - B)^{-1}$ (where X^T and $|x|$ denotes transpose and determinant value of x respectively), then
- (A) $A + B - C = 0$ (B) $A - B - C = 0$
 (C) $|B| = 0$ (D) $2A - C = C^T$
38. If $\lim_{n \rightarrow \infty} \left(\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2} \right) = \frac{\pi^2}{6}$ and $S = \lim_{n \rightarrow \infty} \left(\frac{1}{1^3 \cdot 2^3} + \frac{1}{2^3 \cdot 3^3} + \frac{1}{3^3 \cdot 4^3} + \dots + \frac{1}{n^3 \cdot (n+1)^3} \right)$, then $S + \pi^2$ is equal to
- (A) A non-integral number (B) An rational number
 (C) A transcendental number (D) An integer
39. $f(x)$ is a differentiable function such that $f(x) = x^2 + \int_0^x e^{-t} f(x-t) dt$, then
- (A) $f(x) = \frac{x^3}{3} + x$ (B) $f(x) = \frac{x^3}{3} + x^2$
 (C) $f'(x) = x^2 + 2x$ (D) $\int_{-2}^2 (f(x) - x^2) dx = 0$
40. Mr. A randomly picks 3 distinct numbers from the set $\{1, 2, 3, \dots, 9\}$ and arranges them in descending order to form a three digit number. Mr. B randomly picks 3 distinct numbers from the set $\{1, 2, 3, \dots, 8\}$ and also arranges them in descending order to form 3 digit numbers.
- (A) Probability that Mr. A's 3 digit number is always greater than Mr. B's 3 digit number is $\frac{1}{3}$
 (B) Probability that Mr. A and Mr. B has the same 3 digit numbers is $\frac{1}{84}$
 (C) Probability that Mr. A's 3 digit number is greater than Mr. B's 3 digit number is $\frac{37}{56}$
 (D) Probability that Mr. A's 3 digit number is greater than Mr. B's 3 digit number is $\frac{1}{3}$

41. Let $H : \frac{(x-1)^2}{9} - \frac{(y+2)^2}{4} = 1$ be the hyperbola with centre at C. Now if from any point 'Q' on the asymptotes a straight line be drawn perpendicular to the transverse axis to meet the hyperbola at P and P'. Again a tangent is drawn to the hyperbola at P which meet the asymptotes at L and G, then
- (A) $PQ \cdot QP' = 4$ (B) $PQ \cdot QP' = 9$
 (C) area of $\Delta CLG = 6$ (D) area of $\Delta CLG = 8$
42. If $I_1 = \int_0^{\frac{\pi}{2}} \frac{\sin(\sin x)}{\sin x} dx$; $I_2 = \int_0^{\frac{\pi}{2}} \frac{\sin x}{x} dx$; $I_3 = \int_0^{\frac{\pi}{2}} \frac{\sin(\tan x)}{\tan x} dx$, then which of the following is true
- (A) $I_1 > I_3$ (B) $I_2 > I_3$
 (C) $I_1 > I_2$ (D) $I_1 < I_2$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FOUR/FIVE** statements in **List-II** entries (P), (Q), (R), (S) OR (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Let $f(x) = \begin{cases} (15-3b)\{x\} - (b^2-4b-5)\operatorname{sgn}(x+1), & -\frac{\pi}{2} < x < 0 \\ k([x] + [-x]), & 0 \leq x \leq \pi \\ \frac{(a+2\cos x)(1+\tan x)}{\ln(1+\pi^2-2\pi x+x^2)}, & \pi < x < \frac{3\pi}{2} \end{cases}$

Where $[y]$, $\{y\}$ and $\operatorname{sgn}(y)$ denote greatest integer function, fractional part function and signum function of y respectively.

List-I		List-II	
(I)	If f is continuous in $\left(-\frac{\pi}{2}, 0\right)$, then the value of b is	(P)	0
(II)	If f is continuous at $x = \pi$, then value of $(a+k)$ is	(Q)	1
(III)	If f is continuous in $\left(-\frac{\pi}{2}, \pi\right)$, then value of $(b+k)$ is	(R)	5
(IV)	If f has exactly four points of discontinuity in $\left(-\frac{\pi}{2}, \frac{3\pi}{2}\right)$, then $(a+b+k)$ is equal to	(S)	6

Which one of the following options is correct?

- (A) $I \rightarrow R$, $II \rightarrow Q$, $III \rightarrow R$, $IV \rightarrow S$
 (B) $I \rightarrow P$, $II \rightarrow Q$, $III \rightarrow R$, $IV \rightarrow S$
 (C) $I \rightarrow S$, $II \rightarrow P$, $III \rightarrow Q, R$, $IV \rightarrow P$
 (D) $I \rightarrow Q$, $II \rightarrow P$, $III \rightarrow Q, S$, $IV \rightarrow P$

44. Match the following:

List-I		List-II	
(I)	The number of solution(s) of the equation $\sin^{-1}\left(\frac{1}{1+x^2}\right) = \operatorname{sgn}(x^2 + 1)$, is (are)	(P)	4
(II)	If the range of the function $f(x) = \cos^{-1}\left(\frac{x^2}{1+x^2}\right)$ is $(a, b]$, then $[a + b]$ is equal to	(Q)	1
(III)	If α and β are the roots of the $2x^2 - 3x - 2 = 0$, then $\frac{12}{17}\left(\tan^2\left(\sin^{-1}\frac{\alpha}{\sqrt{1+\alpha^2}}\right) + \tan^2\left(\cos^{-1}\frac{1}{\sqrt{1+\beta^2}}\right)\right)$ is	(R)	2
(IV)	If $f(x) = \sin\left(\cos^{-1}\left(\sin\left(\cos^{-1}x\right)\right) + \sin^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)$, then the value of $\sum_{x=1}^4 f\left(\frac{3x}{16}\right)$ is equal to	(S)	3

Which one of the following options is correct?

- (A) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ R
 (B) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ P
 (C) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ P
 (D) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ R

45. Consider a system of linear equation $3x + y - z = 0$, $x - \frac{py}{4} + z = 2$ and $2x - y + 2z = q$ where $p, q \in I$ and $p, q \in [1, 10]$, then identify the correct statement(s).

List-I		List-II	
(I)	Number of ordered pairs (p, q) for which system of equation has unique solution is	(P)	1
(II)	Number of ordered pairs (p, q) for which system of equation has no solution is	(Q)	9
(III)	Number of ordered pairs (p, q) for which system of equation has infinite solution is	(R)	91
(IV)	Number of ordered pairs (p, q) for which system of equation has atleast one solution is	(S)	90

Which one of the following options is correct?

- (A) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ R
 (B) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R
 (C) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ R
 (D) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ P

46. March the following:

List-I		List-II	
(I)	If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = \ln(h(x)) + g(x) + c$, where c is arbitrary constant and $f(x) = h(x) + g(x)$, then	(P)	$f'(0) = 3$
(II)	If $f(x)$ is a quadratic polynomial such that $f(0) = 1$ and $\int \frac{f(x)}{x^2(x+1)^3} dx$ is a rational function, then	(Q)	$f'(0) = 2$
(III)	If $\int \frac{\cos x(1+4\cos 2x)}{\sin x + 4\sin x \cos^2 x} dx = \frac{1}{2} \ln h(x) + \frac{1}{2} \ln f(x) + c$, then	(R)	$f(x)$ is neither even nor odd function
(IV)	If $\int \frac{x^4 + 1}{x(x^2 + 1)^2} dx = A \ln x + \frac{B}{1+x^2} + c$, where c is arbitrary constant and $f(x) = Ax^2 + 2Bx$, then	(S)	$f(0) = 0$

Which one of the following options is correct?

- (A) $I \rightarrow P, R, II \rightarrow Q, III \rightarrow R, IV \rightarrow S$
 (B) $I \rightarrow Q, II \rightarrow S, III \rightarrow P, Q, IV \rightarrow R, S$
 (C) $I \rightarrow S, II \rightarrow Q, R, S, III \rightarrow P, IV \rightarrow P, S$
 (D) $I \rightarrow Q, R, II \rightarrow P, Q, III \rightarrow S, IV \rightarrow Q, R, S$

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. A triangle is so placed that the midpoints of its sides are placed on the co-ordinate axes, if a , b and c are the sides of the triangle, then the equation of the triangle is $\frac{x}{x_1} + \frac{y}{y_1} + \frac{z}{z_1} = 1$ where

$$a^2 + b^2 + c^2 = k(x_1^2 + y_1^2 + z_1^2), \text{ then the value of } k \text{ is } \underline{\hspace{2cm}}$$

48. Let f be real function defined on R (the set of real numbers) such that $f'(x) = 100(x-1)(x-2)^2(x-3)^3 \dots (x-100)^{100}$, for all $x \in R$. If g is a function defined on R such that $\int_a^x e^{f(t)} dt = \int_0^x g(x-t) dt + 2x + 3$, If some of the all the values of x for which $g(x)$ has a local extremum be λ then find $\frac{\lambda}{3}$

49. If $\lim_{x \rightarrow 1} \frac{100}{1-x^{100}} - \frac{50}{1-x^{50}} = 5A$; find the value of A .

50. If S_1, S_2, S_3 denote the sums of first twenty terms of three non constant sequence in A.P., whose first terms are unity and common differences are in H.P. Then $\frac{2S_3S_1 - S_1S_2 - S_2S_3}{S_1 - 2S_2 + S_3}$ is equal to ____
51. If P is a point $(2, 4)$ on the parabola $y^2 = 8x$ and PQ is a focal chord, the coordinate of the mirror image of Q with respect to tangent at P are given by (α, β) then $(\beta - \alpha)$ is ____
52. If sum of the roots of the equation $\sqrt{2} \sin\left(x + \frac{\pi}{4}\right) (\sec^2 x - (1 + \sqrt{3}) \tan x + 2 + \sqrt{3}) = \sqrt{9 + 18 \sin x \cdot \cos x}$ in $[0, 2\pi]$ is $\frac{p}{q} \pi$ where $p, q \in N$, then find the least value of $|p - q|$
53. Let S be infinite sum of the series $2 + 3\cos x + 4\cos^2 x + 5\cos^3 x + \dots \infty$, where x satisfies the equation $|5\cos x + 4| + |5\cos x - 2| = 6$. If the least value of S is equal to $\left(\frac{a}{b}\right)$ where a and b are co-prime numbers, then the value of $(a + b)$ is:
54. If the coefficient of x^8 in the expansion of $(2 + 9x^3 + 6x^4 + x^5)^{10}$ is $5 \cdot 2^p \cdot 3^q$ where $p, q \in N$, then find the value of $(p + q)$

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – I
PAPER –2
TEST DATE: 28-12-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	-1	In all other cases.

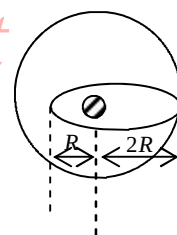
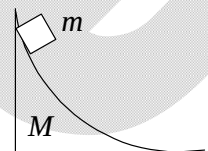
Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

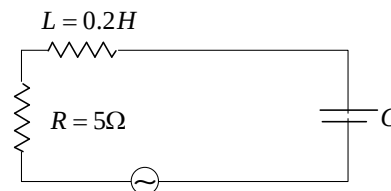
- In a process on a system, the initial pressure and volume are equal to the final pressure and volume:
 (A) The initial temperature must be equal to the final temperature
 (B) The net heat given to the system in the process must be zero
 (C) The net work done by the system in the process must be zero
 (D) None of these
- Block placed on wedge of mass M is released. All surface are smooth, then which of following is/are correct:
 (A) C.M. of mass will remain at rest
 (B) Acceleration of centre of mass is in downward direction
 (C) Normal force exerted by floor on wedge will be equal to wedge of weight at some point of its motion.
 (D) Normal force exerted by floor on wedge is always less than combined weight of block and wedge.
- A spaceship (which has negligible mass) is in an elliptical orbit about a planet. The minimum distance between the spaceship and the planet is R . At the point of maximum distance, spaceship is travelling at a speed v_0 . It then fires his thrusters so that it enters a circular orbit of radius $2R$. What is its new speed:
 (A) $v_0 \frac{\sqrt{3}}{2}$
 (B) $v_0 \sqrt{5}$
 (C) $v_0 \frac{\sqrt{3}}{5}$
 (D) $v_0 \sqrt{2}$
- A nucleus is kept at rest. It breaks into two parts which fly off with velocity in the ratio 27:8. Then the ratio of radii of the parts is
 (A) 27 : 8
 (B) 8 : 27
 (C) 2 : 3
 (D) 3 : 2

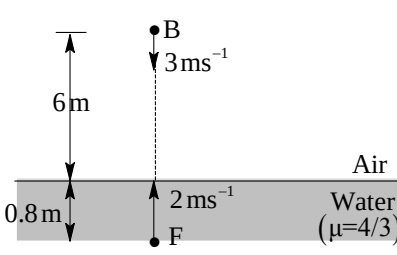


Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

- An alternating voltage $E = 6\sin 20t + 8\cos 20t$ is applied to series resonant circuit as shown. The correct statements is:
 (A) The capacitance is $12.5mF$
 (B) The resonant current in the circuit is $2A$
 (C) Power dissipated in the circuit is 10 watt
 (D) Power dissipated in the circuit is 20 watt



6. Two non-conducting solid spheres of radii R and $2R$, having uniform volume charge densities ρ_1 and ρ_2 respectively are touching each other. The net electric field at a distance $2R$ from the centre of the smaller sphere, along the line joining the centres of the spheres, is zero then the ratio $\frac{\rho_1}{\rho_2}$ can be:
- (A) -4 (B) $-\frac{32}{25}$
 (C) $+\frac{32}{25}$ (D) 4
7. If one of the slits of a standard Young's double-slit experiment is covered by a thin parallel sided glass slab so that it transmits only one-half the light intensity of the other, then:
- (A) The fringe pattern will get shifted toward the covered slit
 (B) The fringe pattern will get shifted away from the covered slit
 (C) The bright fringes will become less bright and the dark ones will become more bright.
 (D) The fringe width will remain unchanged.
8. According to Bohr's theory of the hydrogen atom, for the electron in the n th allowed orbit:
- (A) The linear momentum is proportional to $(1/n)$
 (B) The radius is proportional to n
 (C) The kinetic energy is proportional to $(1/n^2)$
 (D) The angular momentum is proportional to n
9. A fish F , in the pond is at a depth of 0.8 m from the water surface and is moving vertically upward with velocity 2 ms^{-1} . At the same instant, a bird B is at a height of 6 m from the water surface and is moving downward with velocity 3 ms^{-1} . At this instant, both are on the same vertical line as shown in figure. Which of the following statements are correct?
- 
- (A) Height of B , observed by F (from itself), is equal to 5.30 m
 (B) Depth of F , observed by B (from itself), is equal to 6.60 m
 (C) Velocity of B , observed by F (relative to itself), is equal to 5.50 ms^{-1}
 (D) Velocity of F , observed by B (relative to itself), is equal to 4.50 ms^{-1}
10. n moles of an ideal monatomic gas undergo a process in which the temperature changes with volume as $T=KV^2$. If the temperature of the gas changes from T_0 to $4T_0$ then:
- (A) Work done by the gas is $3nRT_0$ (B) Heat supplied to the gas is $6nRT_0$
 (C) Work done by the gas is $\frac{3}{2}nRT_0$ (D) Heat supplied to the gas is $\frac{3}{2}nRT_0$

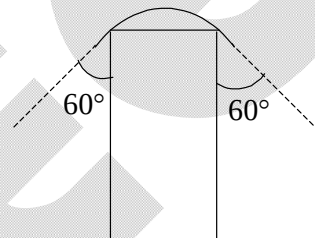
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

11. Sound of frequency 1000Hz from a stationary source is reflected from an object approaching the source at 30 m/s , back to stationary observer located at the source. The speed of sound in air is 330 m/s . The frequency of the sound heard by the observer is $K \times 400\text{Hz}$. Determine 'K'.

12. At an instant, the ratio of the amounts of radioactive substances is $2:1$. If their half lives be 12 & 16 hours respectively, then after 2 days the ratio of amounts of substances left over will be $1:x$ find x :

13. A soap bubble is being blown on a tube of radius 1 cm . The surface tension of the soap solution is 0.05 Nm^{-1} and the bubble makes 60° with the tube as shown in the figure. The excess of pressure over the atmosphere pressure in the tube is $2\beta\text{ Pa}$. Find the value of β ____.

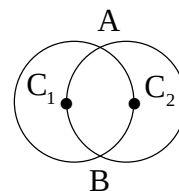


14. A uniform rod AB of mass M is placed in contact with a second rod BC of mass $m = M/2$ on a horizontal smooth table at right angles to each other as shown in figure. Impulse I is imparted to A kinetic energy generated in BC is $\frac{nI^2}{3m}$. Find the value of 'n' ____.



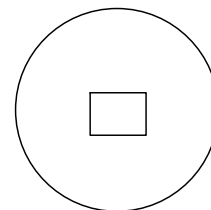
15. A ray of light is incident at an angle of 60° on one face of a prism which has refracting angle of 30° . The ray emerging out of the prism makes an angle of 30° with the incident ray. If the refractive index of the material of the prism is $\mu = \sqrt{a}$, find the value of a ____.

16. Two circular rings of identical radii and resistance of 36Ω each are placed in such a way that they cross each other's centre C_1 and C_2 as shown in figure. Conducting joints are made at intersection point A and B of the rings. An ideal cell of emf 20 V is connected across A and B. Find the power delivered by the cell (in 10^2W) ____.



17. A cubical block of mass M vibrates horizontally with amplitude of 4.0 cm and a frequency of 2.0 Hz . A small block of mass m is placed on the bigger block. In order that the smaller block does not slide on the bigger block, the minimum value of the coefficient of static friction between the two blocks is $(0.16x)$. Find the value of x . (Take $\pi^2=10$ and $g=10\text{ m/s}^2$)

18. Consider the two loops with common centre and on the same plane as shown in the figure. The large circular loop has radius $R = 10\text{ cm}$ and the small square loop has side $a = 1\text{ cm}$. There is a current circulating in the large loop given by $i(t) = i_0 t / \tau$ with $i_0 = \frac{2}{\pi} \times 100\text{ A}$, $\tau = 2 \times 10^{-2}\text{ sec}$. Emf (in micro volts) induced in the small loop is ____.



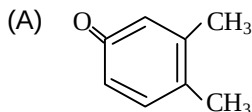
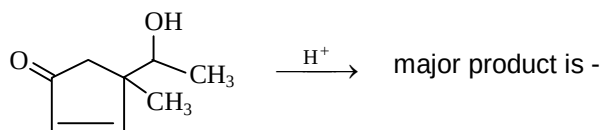
Chemistry

PART – II

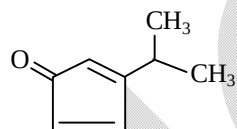
Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

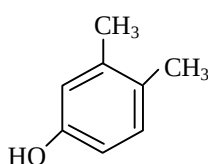
19.



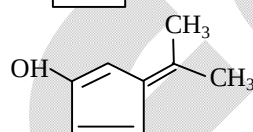
(B)



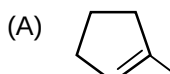
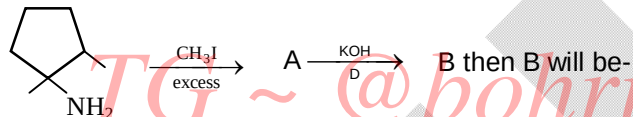
(C)



(D)



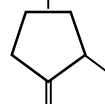
20.



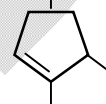
(B)



(C)



(D)



21.

If on a strand of DNA the base sequence is ATTGACGCAT then the sequence transcription on RNA would be -

(A) UAACUGCGUA

(B) AUUCUGCGUA

(C) UAACTGCGUA

(D) TAACTGCGTA

22.

The structure of a crystalline lattice AB is rock salt type where B atoms are occupying all the CCP positions but A atoms are occupying all the octahedral voids -

(A) If atoms lying on one tetrad axis of symmetry are removed then molecular formula of the compound become AB

(B) If atoms lying on one triad axis of symmetry are removed then molecular formula of the compound become A_4B_5

(C) If atoms lying on one diad axis of symmetry are removed then molecular formula of the compound become A_5B_8

(D) All of the above molecular formula are correct corresponding to the defects mentioned

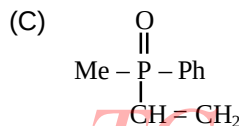
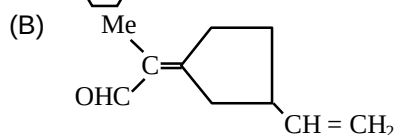
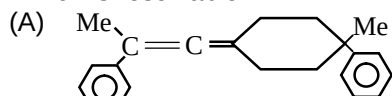
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

23. Identify the correct statement (s) regarding the oxidation of N_2H_5^+ by $\text{Cr}_2\text{O}_7^{2-}$ in acid medium.

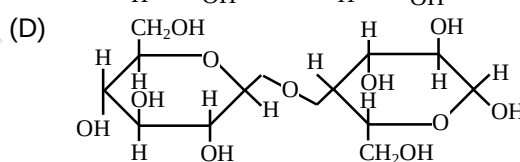
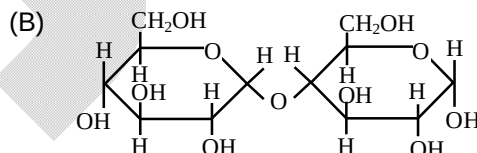
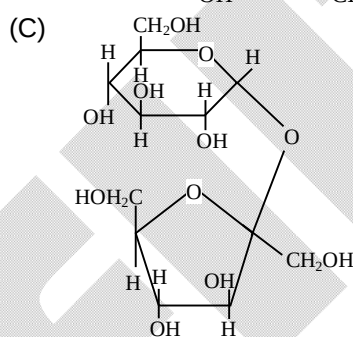
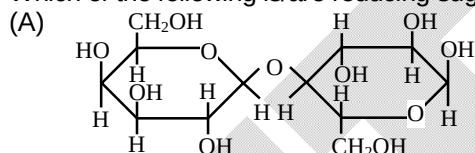
- (A) 10 ml 0.2 M dicromate is required for the complete oxidation of 12 ml 0.25 M N_2H_5^+
- (B) 10 ml 0.2 M dicromate is required for the complete oxidation of 10 ml 0.6 N N_2H_5^+
- (C) Number of moles of N_2 liberated per mole of dichromate is 1.5
- (D) One mole of N_2H_5^+ is 3 equivalents

24. Which is resolvable ?



(D) Ethylmethylphenyl amine at pH=2

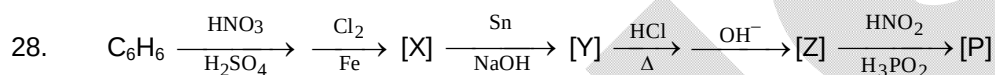
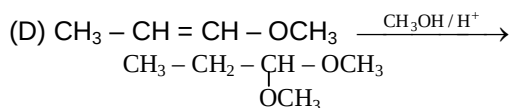
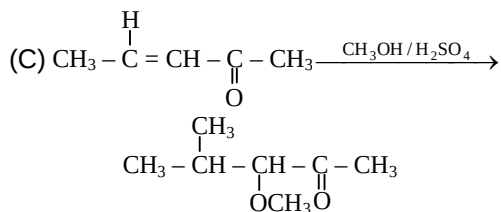
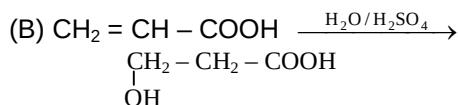
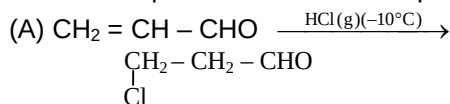
25. Which of the following is/are reducing sugars -



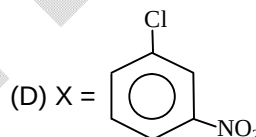
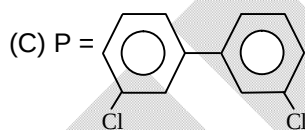
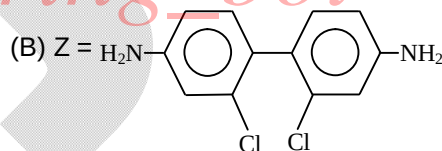
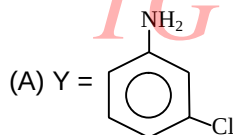
26. Which of the following statements is/are correct?

- (A) B_2H_6 reacts with excess NH_3 at low temperature to produce an ionic compound
- (B) B_2H_6 reacts with excess of NH_3 at higher temperature to form boron nitride solid
- (C) Boron nitride has polymeric layer structure
- (D) In borazine there are -ve and +ve charges on B and N respectively

27. The correct products have been reported in reactions –

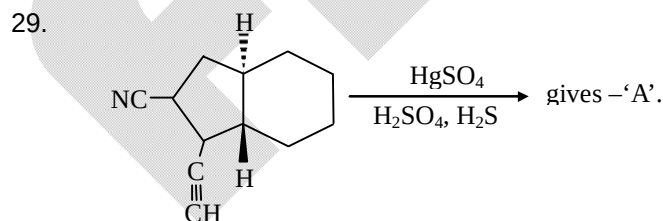


The correct statement(s) are/is –



Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

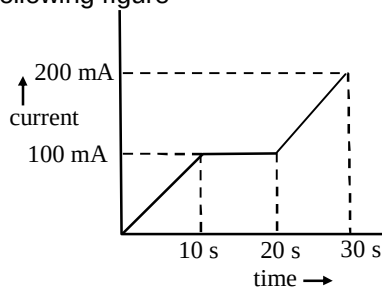


The degree of unsaturation of compound 'A' is?

30. What is the minimum pH required to prevent the precipitation of ZnS in a solution that is 0.01 M ZnCl_2 and saturated with 0.10 M H_2S ?

[Given $K_{\text{sp}} = 10^{-21}$, $K_{a_1} \times K_{a_2} = 10^{-20}$]

31. In a Cu-voltameter, mass deposited in 30s is m gm. If the time-current graph is shown in the following figure



The electrochemical equivalent of Cu is $\frac{m}{x}$. Then find the value of 'x' is ____.

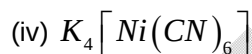
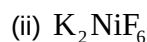
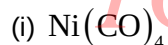
32. For a certain reaction the variation of rate constant with temperature is given by the equation

$$\ell n k_t = \ell n k_0 + \frac{(\ell n 3)t}{10} \quad (t \geq 0^\circ\text{C})$$

The value of temperature coefficient of the reaction is ____.

33. A compound MX_2 has observed and normal molar masses 65.6 and 164 respectively. The apparent degree of ionization of MX_2 is $15x\%$. Then find the value of 'x' is ____.

34. What is the highest oxidation state of nickel in the given complexes of Ni metal?



35. Benitoite is represented as $\text{BaTi}[\text{Si}_3\text{O}_n]$, the value of n is

36. In order to prepare a buffer of pH= 8.26, the amount of $(\text{NH}_4)_2\text{SO}_4$ required to be mixed with one litre of 0.1(M) NH_3 (aq) is $0.25x$ mole. The value of 'x' is ____.

(Given:- pK_b of $\text{NH}_4\text{OH} = 4.74$)

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. Maximum value of function $f(x) = (\sin^{-1}(\sin x))^2 - \sin^{-1}(\sin x)$ is:
 (A) $\frac{\pi}{4}(\pi + 2)$ (B) $\frac{\pi}{4}(\pi - 2)$
 (C) $\frac{\pi}{4}$ (D) None of these
38. For a parabola $x^2 - 4xy + 4y^2 - 32x + 4y + 16 = 0$, focus is:
 (A) (2,1) (B) (-2,1)
 (C) (-2,-1) (D) (2,-1)
39. Given three non – zero, non – coplanar vectors, and $\vec{a}, \vec{b}$ and $\vec{c}$ and $\vec{r}_1 = p\vec{a} + q\vec{b} + \vec{c}$ and $\vec{r}_2 = \vec{a} + p\vec{b} + q\vec{c}$ if the vectors $\vec{r}_1 + 2\vec{r}_2$ and $2\vec{r}_1 + \vec{r}_2$ are collinear then (p, q) is
 (A) (0, 0) (B) (1, -1)
 (C) (-1, 1) (D) (1, 1)
40. Solution of differential equation $f'(x) f''(x) = 3(f''(x))^2, (y = f(x))$ is:
 (A) $x = k_1 y^2 + k_2 y + k_3$ (B) $y = k_1 x^2 + k_2 + k_3$
 (C) $x = \sin y$ (D) None of these

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. Let $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{px}{n} \sum_{r=1}^n \frac{[r^2 - e^{-x} + r - 1]}{r(r+1)} \right) + \lambda, & x > 0 \\ q, & x = 0 \\ \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\{r^2 + r + e^x - 1\}}{r(r+1)}, & x < 0 \end{cases}$

Is differentiable in $\mathbb{R}$:

([.] is G.I.F. and { } is F.P. of x) Then:

- (A) $p=1$ (B) $q=1$
 (C) $p+q+\lambda=3$ (D) if g is inverse of F then $g'(1/2) = 2$

42. Consider $f(x) = 4x^4 - 24x^3 + 31x^2 + 6x - 8$ be a polynomial function and $\alpha, \beta, \gamma, \sigma$ are the roots. ($\alpha < \beta < \gamma < \sigma$). So:

(A) $\sigma^\beta + \frac{1}{\sigma^\alpha} + \sigma^\gamma + \gamma^\sigma = 36$

(B) $\sigma^\beta + \frac{1}{\sigma^\alpha} + \sigma^\gamma + \gamma^\sigma = 18$

(C) $\int_{2\alpha}^{2\beta} \frac{x^{\sigma+1} - 5x^{\gamma+1} + 2\beta|x| + 1}{x^2 + 4\beta|x| + 1} dx = 2 \ln 2$

(D) $\int_{2\alpha}^{2\beta} \frac{x^{\sigma+1} - 5x^{\gamma+1} + 2\beta|x| + 1}{x^2 + 4\beta|x| + 1} dx = \ln 2$

43. Consider the equation $z^2 - (3+i)z + (m+2i) = 0$ ($m \in \mathbb{R}$). If the equation has exactly one real and one-non-real complex root, then which of the following hold(s) good:

(A) Modulus of the non-real complex root is 2

(B) The value of m is 3

(C) Additive inverse of non-real root is $(-1-i)$

(D) Product of real root and imaginary part of non-real complex root is 2

44. $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{m^2 n}{(n \cdot 3^m + m \cdot 3^n)} = \frac{p}{q}$ (Where p & q are coprime) then:

(A) $p + q = 41$

(B) $p < q$

(C) p is perfect square

(D) q is perfect square

45. Let $p(x) = 4\sin^3 x - \sin x + 2\left(\sin \frac{x}{2} - \cos \frac{x}{2}\right)^2$, then which of the following is/are correct?

(A) Range of $p(x)$ is $[1, 3]$

(B) Range of $p(x)$ is $[0, 4]$

(C) Number of solution of $p(x) = 1$ in $[0, \pi]$ is 2

(D) Number of solution of $p(x) = 1$ in $[0, 2\pi]$ is 4

46. Let m, n be real numbers satisfying the relation:

$$\log_2(m^2 + n^2 + 1) + \log_3 \left(\frac{1 + 2\sin^2\left(\frac{2\pi}{7}\right)}{2 - \cos\left(\frac{4\pi}{7}\right)} \right) = \log_2 n + \log_2 (2m + 2 - n)$$

Identify which of the following statement(s) is/are correct?

(A) $m + n = 2$

(B) $\|m\| - \|n\| = 4$

(C) Area of the figure enclosed by $|x| + |y| = |m| + |n|$ is 8 square units.

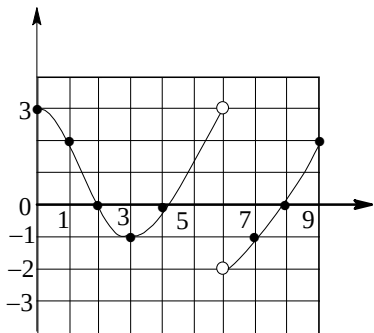
(D) Area of the figure enclosed by $|x| + |y| = |m| + |n|$ is 16 square units.

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

47. A and B are two square matrices such that $A^2B = BA$ and if $(AB)^{10} = A^k B^{10}$ then the value of $k - 1020$ is....

48. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$



If $f(x)$ has local minima at $x = x_1$ and $x = x_2$; f has inflection point at $x = k$; number of critical points of $y = f(x)$ is w then $x_1 + x_2 - k - w$ is.....

49. Number of values of $x \in [0, \pi]$ where $f(x) = [4 \sin x - 7]$ is non derivable is ([.] is G. I. F)

50. If the equation on reflection of $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$ about the line $x - y - 2 = 0$ is $16x^2 + 9y^2 + k_1x - 36y + k_2 = 0$ then $\frac{k_1 + k_2}{22}$ is

51. If $0 < \alpha < 30^\circ$, then find the value of $\log_2 \left(\frac{8 \cos(30^\circ - 2\alpha) - \frac{8}{\cot \alpha + \cot(30^\circ - \alpha)}}{\sqrt{3}} \right)$

52. Find the value of $\log_{\sqrt{5}} \left(\frac{\sqrt{(5\sqrt{5}+5)} \sqrt{(5\sqrt{5}+5)^2} \sqrt{(5\sqrt{5}+5)^3} \sqrt{\dots}}{(6+2\sqrt{5})^3 \sqrt{215-18^3} \sqrt{215-18^3} \sqrt{215-18^3} \sqrt{\dots}} \right)$

53. Three circles C_1, C_2, C_3 with radii r_1, r_2, r_3 ($r_1 < r_2 < r_3$) respectively are given as $r_1 = 2$, and $r_3 = 8$ they are placed such that C_2 lies to the right of C_1 and touches it externally, C_3 lies to the right of C_2 and touches it externally. There exist two straight lines each of which is a direct common tangent simultaneously to all the three circles then r_2 is equal to ____.

54. If K^{th} term is the greatest term in the square $a_n = \frac{n}{70+n^2}, n \in N$, then k is ____

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – I
PAPER –1
TEST DATE: 28-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. AD

2. ABC

3. ABCD

Sol. $\Delta E = \frac{hc}{\lambda}$

$$\Delta E = \frac{hc}{\lambda(\text{in nm})} (\text{in eV})$$

4. AD

5. BCD

6. BCD

7. C

8. A

9. B

10. C

Section – B

11. 250.00

12. 42.00

13. 2.96

Sol. Let u be the velocity and α the angle of projection for a fragment reaching the man. According to the questions the greatest value of u can be 20 m/s. If t be the time taken by this fragment to reach the man, then considering the horizontal and vertical motions of the fragment

$$8 + \frac{1}{2}gt^2 = u \sin \alpha t$$

$$g^2 t^4 - 4(u^2 - g \times 8)t^2 + 4 \times 100 = 0$$

The period in which the man will be in danger on account of this fragment

$$= t_1 - t_2 = \sqrt{(t_1^2 + t_2^2) - 2t_1 t_2} = \frac{2}{g} \{u^2 - 8g - 10g\}^{1/2}$$

14. 3.21

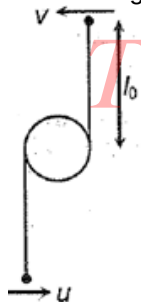
Sol. Using WET

$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$= -mg(l_0 + l_0 - \pi r) \dots \dots \dots (i)$$

$$\text{At highest point } v = \sqrt{(l_0 - \pi R)g}$$

Substituting in equation (i)



$$v = \sqrt{5gl_0 - 4r\pi g} \cong 3.21 \text{ m/sec}$$

15. 2.81

Sol. Velocity of water coming out from hole

$$v = \sqrt{2gh} = \sqrt{2g1}$$

$$y = u_y t + \frac{1}{2}a_y t^2$$

$$1 = \sqrt{2g}t \sin 30^\circ + \frac{1}{2} \times 10t^2$$

$$1 = \sqrt{2g} \frac{t}{2} + 5t^2$$

$$t = \frac{-2.24 \pm \sqrt{(2.24)^2 + 20}}{10} = 0.28 \text{ sec.}$$

$$x_1 = v_x t = \sqrt{2g} \cos 30^\circ \times 0.28 = 1.08 \text{ m}$$

$$x = x_1 + x_2 = 1.08 + 1.73 = 2.81 \text{ m}$$

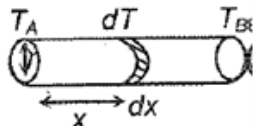
16. 500.00

Sol. Intensity of the source at the cross-section A

$$\Rightarrow I = \frac{P}{4\pi(50r)^2} = \frac{1.25 \times 10^3}{4\pi(50r)^2}$$

Power absorbed by the end A

$$T_B = \frac{0.003}{100000 \times 10^{-10}} = 300K$$



$$\int_0^j \frac{0.1T_A \cdot dx}{4 \times 10^{-4}} = - \int_{T_A}^{T_B} T dT$$

$$\Rightarrow T_A = 500K$$

17. 40.00

Sol. When (1) and (2) are connected and (3) and (4) are open

$$P_1 = \frac{v^2}{R_1 + R_2} \quad \dots(i)$$

When (3) and (4) are closed and (1) and (2) are connected

$$P_2 = \frac{v^2}{R_2 + \left(\frac{R_2 R_3}{R_2 + R_3} \right)} \quad \dots(ii)$$

When (3) and (4) are connected and (1) and (2) are open

$$P_3 = \frac{v^2}{R_2 + R_3} \quad \dots(iii)$$

When (3) and (4) are connected and (1) and (2) are closed

$$P_4 = \frac{v^2}{R + \left(\frac{R_1 R_3}{R_1 + R_3} \right)} \quad \dots(iv)$$

From (i) to (iv)

$$\frac{P_1}{P_3} = \frac{P_2}{P_4} \Rightarrow P_4 = 40W$$

18. 21.00

Sol. According to Biot-Savart law,

$$B = \frac{\mu_0}{4\pi} \frac{qv \sin \theta}{r^2}$$

$$B = \frac{10^{-7} \times 0.2 \times 4 \times 10^3 \times 3}{109 \times \sqrt{109}} = 21\mu T$$

Chemistry

PART – II

Section – A

19. AD

Sol. Both Fe^{3+} & Mn^{2+} have 5 unpair electrons
 Cu_2Cl_2 is colourless because Cu^+ has no unpair electrons

20. ABC

Sol. (A) The total lines obtained due to the splitting of a spectral line in the presence of magnetic effect is $(2l + 1)$ as the presence of orbitals which have specific orientation in the presence of external field take up certain new orientation. The number of orbitals are equal to $(2l + 1)$ and for each orbital one splitting takes place.

(B) As we move away from the nucleus the difference in the energy levels become lesser and lesser hence.

$$\text{For } 1^{\text{st}} \text{ line of Lyman } \Delta E = E_1 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\Delta E_1 = E_1 \left[\frac{3}{4} \right] - 1^{\text{st}} \text{ line; } \Delta E_2 = E_1 \left[\frac{8}{9} \right] - 2^{\text{nd}} \text{ line}$$

$$\Delta E_3 = E_1 \left[\frac{15}{16} \right] - 3^{\text{rd}} \text{ line; } \Delta E_4 = E_1 \left[\frac{24}{25} \right] - 4^{\text{th}} \text{ line;}$$

$$\Delta E_5 = E_1 \left[\frac{35}{36} \right] - 5^{\text{th}} \text{ line;}$$

$$\Delta E_2 - \Delta E_1 = E_1 \left[\frac{8}{9} - \frac{3}{4} \right] = E_1 \left[\frac{5}{36} \right]$$

$$\Delta E_3 - \Delta E_2 = E_1 \left[\frac{15}{16} - \frac{8}{9} \right] =$$

$$E_1 \left[\frac{135 - 128}{144} \right] = E_1 \left[\frac{7}{144} \right]$$

$$\Delta E_4 - \Delta E_3 = E_1 \left[\frac{24}{25} - \frac{15}{16} \right]$$

$$= E_1 \left[\frac{16 \times 24 - 15 \times 25}{25 \times 16} \right] = E_1 \left[\frac{9}{400} \right]$$

We find that $(\Delta E_2 - \Delta E_1) > (\Delta E_3 - \Delta E_2) > (\Delta E_4 - \Delta E_3)$

Hence the distance also become lesser and lesser.

(C) The distance from the nucleus for maximum probability of finding electrons is $0.53 \text{ \AA}$. This is not on the circumference of the orbital but is in the vicinity of the circumference.

21. ABC

Sol. Because of the H-bonding NH_4^+ is more extensively solvated than Na^+ , hence NH_4^+ salts is more soluble than Na^+ salts.

The polarising power of Sn^{4+} is greater than Sn^{2+} , hence SnCl_4 is more covalent

The polarisability of Cl^- is greater than F^- because of greater size.

Formal charge

$$= \left(\begin{array}{c} \text{no. of valence} \\ \text{electrons of the} \\ \text{atom} \end{array} \right) - \left(\begin{array}{c} \text{no. of} \\ \text{nonbonded} \\ \text{electrons} \end{array} \right) - \frac{1}{2} \left(\begin{array}{c} \text{no. of} \\ \text{bonded} \\ \text{electrons} \end{array} \right)$$

In SO_2 , formal charge of S = $6 - 2 - \frac{1}{2} \times 4 = 2$

22. AB

23. ABCD

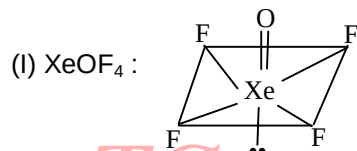
24. AC

Sol. $\text{Hg}_2^{2+} + 2\text{I}^- \rightarrow \text{Hg}_2\text{I}_2$ (green ppt) ;
 $\text{Hg}_2^{2+} + \text{H}_2\text{S} \rightarrow$ no ppt

25. B

26. A

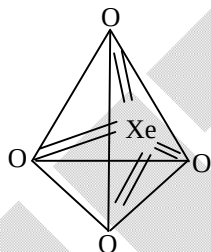
Sol.



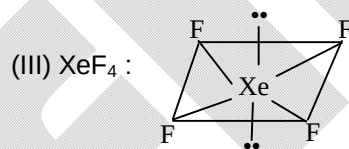
one ℓ p of electrons ; sp^3d^2 hybridisation ; square pyramidal

(I) $\rightarrow$ (Q,S)

(II) XeO_4 : tetrahedral and no ℓ p of electrons

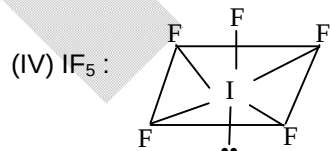


(II) $\rightarrow$ (P, S)



sp^3d^2 hybridisation ; planar ; two ℓ ps of electrons

(III) $\rightarrow$ R



sp^3d^2 hybridisation ; one ℓ p. of electrons

(IV) $\rightarrow$ (Q,S)

27. D

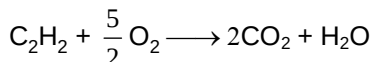
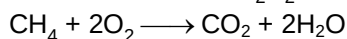
28. C

Section – B

29. 0.90

Sol. Let no. of moles of CH_4 present = n_1 mol.

Let no. of moles of C_2H_2 present = n_2 mol.



Total no. of moles at initial = $n_1 + n_2$

Total no. of moles at final = $n_1 + 2n_2$

At constant V & T,

$$\frac{P_1}{P_2} = \frac{\text{No. of moles at initial}}{\text{No. of moles at final}}$$

$$\frac{63}{69} = \frac{n_1 + n_2}{n_1 + 2n_2} \Rightarrow \frac{21}{23} = \frac{n_1 + n_2}{n_1 + 2n_2}$$

$$\Rightarrow \frac{n_2}{n_1} = \frac{2}{19} \Rightarrow \frac{n_1 + n_2}{n_1} = \frac{21}{19}$$

$$\therefore \frac{n_1}{n_1 + n_2} = \frac{19}{21}$$

30. 0.39

$$\text{Sol. } \Delta u = Q + W = 0 + \int -P_{\text{ext.}} dV = -P_{\text{ext.}} (V_2 - V_1)$$

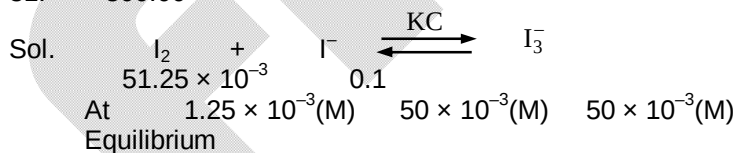
$$\Rightarrow C_V (T_2 - T_1) = -P_{\text{ext.}} \left(\frac{RT_2}{P_2} - \frac{RT_1}{P_1} \right);$$

$$\text{Here } P_{\text{ext.}} = P_2$$

$$\Rightarrow \frac{5}{2}R (673 - 373) = -673R + \left(\frac{373R}{0.1013} \right) P_2$$

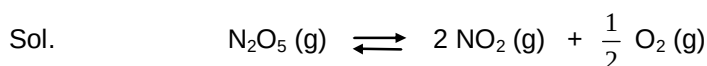
$$\Rightarrow P_2 = 0.3865 \text{ MPa}$$

31. 800.00



$$K_C = \frac{50 \times 10^{-3}}{50 \times 10^{-3} \times 1.25 \times 10^{-3}} = \frac{1000}{1.25} = 800$$

32. 0.40



Initially : 600 torr — —

At equilibrium: $(600 - p)$ torr $2p$ torr $\frac{p}{2}$ torr

Total press is $\left(600 + \frac{3p}{2}\right)$ torr

$$\therefore 600 + \frac{3p}{2} = 960$$

$$\text{or } \frac{3p}{2} = 360$$

$$\text{or } p = 240 \text{ torr}$$

$$\therefore \text{fraction of } \text{N}_2\text{O}_5 \text{ (g) decomposed} = \frac{240}{600} = 0.4$$

33. 4.00

Sol. $s = \frac{10^3 \times \kappa}{\lambda_m^\infty(\text{AgCl})} = \frac{10^3 \times 2.6 \times 10^{-6}}{(63+67)}$
 $= 2 \times 10^{-5} \text{ (M)}$
 $\therefore K_{sp} = (2 \times 10^{-5})^2 = 4 \times 10^{-10}$

34. 0.50

35. 64.36

Sol. Vol. of all atoms in 1.68 gm

$$\text{argon} = \frac{1.68}{40} \times N_A \times \frac{4}{3} \times \pi \times (1.5 \times 10^{-8})^3 = 0.3564$$

vol. of solid argon = 1 cm^3

$$\% \text{ empty space} = (1 - 0.3564) \times 100 = 64.36$$

36. 1000.00

Sol. Initially $P_m = P_X^\circ \cdot X_X + P_Y^\circ \cdot X_Y$

$$550 = P_X^\circ \left(\frac{1}{1+3}\right) + P_Y^\circ \left(\frac{3}{1+3}\right)$$

$$\text{or } P_X^\circ + 3P_Y^\circ = 2200 \quad \dots(i)$$

When 1 mole of Y is further added to it

$$P_m = P_X^\circ \cdot X_X + P_Y^\circ \cdot X_Y$$

$$560 = P_X^\circ \left(\frac{1}{1+4}\right) + P_Y^\circ \left(\frac{4}{1+4}\right)$$

$$\therefore P_X^\circ + 4P_Y^\circ = 2800 \quad \dots(ii)$$

By eqs. (i) and (ii)

$$P_X^\circ = 400 \text{ mm and } P_Y^\circ = 600 \text{ mm.}$$

Mathematics

PART – III

Section – A

37. ACD

Sol. $\therefore (A+B)(A-B) = (A-B)(A+B)$

$$\begin{aligned}
 \text{Now } C &= (A^T + B^T)(A+B)(A-B)^{-1} \\
 &= (A-B)(A+B)(A-B)^{-1} \\
 &= (A+B)(A-B)(A-B)^{-1} \\
 &= A+B
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } C^T &= A^T + B^T = A-B \\
 C + C^T &= 2A
 \end{aligned}$$

38. BD

Sol. Let $t_r = \left(\frac{1}{r(r+1)} \right)^3 = \left(\frac{1}{r} - \frac{1}{r+1} \right)^3 = \frac{1}{r^3} - \frac{1}{(r+1)^3} - \frac{3}{r(r+1)} \left(\frac{1}{r(r+1)} \right)$

$$= \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3 \left(\frac{(r+1)-r}{r(r+1)} \right)^2$$

$$t_r = \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3 \left(\frac{1}{r^2} + \frac{1}{(r+1)^2} \right) + 6 \left(\frac{1}{r} - \frac{1}{r+1} \right)$$

$$\sum_{r=1}^n t_r = \left(1 - \frac{1}{(n+1)^3} \right) - 3 \left(\frac{\pi^2}{6} + \frac{\pi^2}{6} - 1 \right) + 6 \left(1 - \frac{1}{n+1} \right)$$

$$\lim_{n \rightarrow \infty} \sum t_r = 10 - \pi^2$$

39. BCD

Sol. $f(x) = x^2 + \int_0^x e^{-t} f(x-t) dt$

$$= x^2 + \int_0^x e^{-(x-t)} \cdot f(x(x-t)) dt$$

$$= x^2 + e^{-x} \cdot \int_0^x e^t \cdot f(t) dt$$

$$\therefore e^x \cdot f(x) = x^2 \cdot e^x + \int_0^x e^t \cdot f(t) dt$$

Differentiate with respect to x , we get $f'(x) = x^2 + 2x$

$$\Rightarrow f(x) = \frac{x^3}{3} + x^2 + x, \text{ also } f(0) = 0$$

$$\Rightarrow c = 0$$

$$\Rightarrow f(x) = \frac{x^3}{3} + x^2$$

40. ABC

Sol. When 9 is not included $= \frac{{}^8C_3 \cdot {}^8C_3 - {}^8C_3}{2}$

$$\text{and required probability is } = \frac{1}{3} + \frac{{}^8C_3 \cdot {}^8C_3 - {}^8C_3}{2 \cdot {}^9C_3 \cdot {}^8C_3} = \frac{37}{56}$$

41. AB

Sol. By property: $PQ \cdot P'Q = b^2$ and area of $\triangle CLG = ab$

42. ABC

Sol. $0 < x < \frac{\pi}{2} \Rightarrow \frac{\sin x}{x}$ is decreasing and $\sin x < x < \tan x$

$$\Rightarrow \frac{\sin(\sin x)}{\sin x} > \frac{\sin x}{x} > \frac{\sin(\tan x)}{\tan x}$$

$$\Rightarrow I_1 > I_2 > I_3$$

43. A

44. B

Sol. (I) clearly, number of solutions is 2.

$$(II) f(x) = \cos^{-1}\left(\frac{x^2}{1+x^2}\right) = \cos^{-1}\left(1 - \frac{1}{1+x^2}\right)$$

$$t = 1 - \frac{1}{1+x^2} \Rightarrow t \in [0, 1)$$

$$\therefore \cos^{-1} t \in \left(0, \frac{\pi}{2}\right] = (a, b] \Rightarrow [a + b] = 1$$

$$(III) 12(\tan^2(\tan^{-1} \alpha) + \tan^2(\tan^{-1} \beta)) = 12(\alpha^2 + \beta^2)$$

$$= 12((\alpha + \beta)^2 - 2\alpha\beta)$$

$$= 12\left(\frac{9}{4} - 2.1\right) = 3$$

$$(IV) \sin(\cos^{-1} x) = \sin\left(\frac{\pi}{2} - \sin^{-1} x\right) = \cos(\sin^{-1} x) = t$$

$$f(x) = \sin(\cos^{-1} t + \sin^{-1} t) = \sin \frac{\pi}{2} = 1$$

$$\sum_{x=1}^4 f\left(\frac{3x}{16}\right) = 1+1+1+1 = 4$$

45. C

Sol. (a) $3x + y - z = 0 \dots (1)$

$$x - \frac{py}{4} + z = 0 \dots (2)$$

$$2x - y + 2z = q \dots (3)$$

Eqn. (2) $\times$ (2) – eqn. (3)

$$\Rightarrow \left(1 - \frac{p}{2}\right)y = 4 - q$$

For unique solution, $p \neq 2, q \in \mathbb{N} \Rightarrow$ number of ordered pairs (p, q) in $[1, 10]$ are 90.

For infinite solution, $p = 2$ and $q = 4 \Rightarrow$ exactly one ordered pair

For no solution, $p = 2$ and $q \neq 4 \Rightarrow$ number of ordered pairs (p, q) in $[1, 10]$ are 9

46. D

Section – B

47. 8.00

Sol. Let the intercept of plane on the axes be $(x_1, 0, 0), (0, y_1, 0), (0, 0, z_1)$

So, equation is $\frac{x}{x_1} + \frac{y}{y_1} + \frac{z}{z_1} = 1$

$$\therefore EF \parallel BC \text{ and } EF = \frac{1}{2}BC = 4(y_1^2 + z_1^2) = a^2$$

Similarly $4(x_1^2 + y_1^2) = c^2$ and $4(z_1^2 + x_1^2) = b^2$

$$\Rightarrow a^2 + b^2 + c^2 = 8(x_1^2 + y_1^2 + z_1^2)$$

$$\Rightarrow k = 8$$

48. 833.33

Sol. $\therefore \int_a^x e^{f(t)} dt = \int_0^x g(x-t) dt + 2x + 3$

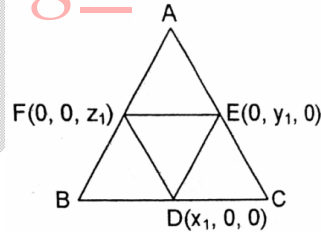
$$\Rightarrow \int_a^x e^{f(t)} dt = \int_0^x g(t) dt + 2x + 3 \quad (\text{Using King Property})$$

Differential both sides, we get

$$e^{f(x)} = g(x) + 2$$

$$\Rightarrow g(x) = e^{f(x)} - 2$$

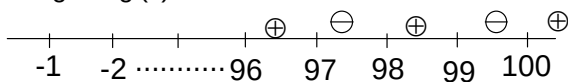
$$\Rightarrow g'(x) = e^{f(x)} \cdot f'(x)$$



$\therefore e^{f(x)}$ is always greater than zero.

$\therefore$ Sign of $g'(x)$ is same as sign of $f'(x)$.

$\therefore$ Sign of $g'(x)$



Clearly, local extreme (maximum or minimum) will occur at $x=99, 97, 95, \dots, 3, 1$

$$\therefore \text{Sum of all the values} = 1+3+5+\dots+99 = \frac{50}{2} [2 \times 1 + (50-1) \times 2] = 2500$$

49. 5.00

Sol. $\lim_{y \rightarrow 1} \frac{100y^{100}}{y^{100}-1} - \frac{50y^{50}}{y^{50}-1}$

$$\lim_{y \rightarrow 1} 100 - 50 - \left(\frac{100}{1-y^{100}} - \frac{50}{1-y^{50}} \right)$$

$$L = 50 - L \therefore L = \frac{50}{2} = 25$$

50. 20.00

Sol. $S_1 = \frac{20}{2} (2 + 19d_1)$

$$\frac{2S_3S_1 - (S_1 + S_3)}{S_1 + S_3 - 2S_2} = \frac{200(2 + 19d_1)(2 + 19d_3) - 100(2 + 19d_2)(4 + 19(d_1 + d_3))}{10(4 + 19(d_1 + d_3) - 4 - 38d_2)}$$

$$= 10 \left(\frac{8 + 76(d_1 + d_3) + 722d_1d_3 - 8 - 38(d_1 + d_3 + 2d_2) - 361d_2(d_1 + d_3)}{19(d_1 + d_3 - 2d_2)} \right)$$

$$= 10 \left(\frac{38(d_1 + d_3 - 2d_2) + 361(2d_1d_3 - d_2d_1 + d_3)}{19(d_1 + d_3 - 2d_2)} \right)$$

$$= 20$$

51. 10.00

52. 25.00

Sol. $(\sin x + \cos x) \left(\tan^2 x - (1 + \sqrt{3}) \tan x + 3 + \sqrt{3} \right) = 3 |\sin x + \cos x|$

Case - 1: $\sin x + \cos x \geq 0$

$$(\sin x + \cos x) \left(\tan^2 x - (1 + \sqrt{3}) \tan x + \sqrt{3} \right) = 0$$

$$\tan x = -1, 1, \sqrt{3}$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{7\pi}{4}$$

Case- 2 : $\sin x + \cos x \leq 0$

$$\tan x = -1$$

$$\therefore \text{Sum} = \frac{37\pi}{12}$$

53. 151.00

Sol. $S = 2 + 3\cos x + 4\cos^2 x + \dots \infty$

$$\frac{S \cdot \cos x = 2\cos x + 3\cos^2 x + \dots \infty}{S(1 - \cos x) = 2 + \cos x + \cos^2 x + \dots \infty}$$

$$= 1 + \frac{1}{1 - \cos x}$$

$$S = \frac{1}{1 - \cos x} + \frac{1}{(1 - \cos x)^2}$$

$$|5\cos x + 4| + |5\cos x - 2| = 6$$

$$\Rightarrow \cos x \in \left[-\frac{4}{5}, \frac{2}{5}\right]$$

For the least value of S, $\cos x = -4/5$

$$\therefore a + b = 151$$

54. 14.00

Sol. $(2 + x^3(3 + x)^2)^{10} \sim @bohrring_bot$

$$\downarrow$$

$${}^{10}C_r \cdot 2^{10-r} \cdot x^{3r} (3 + x)^{2r}$$

$$\downarrow$$

$${}^{2r}C_k \cdot 3^{2r-k} \cdot x^k$$

$$3r + k = 8 \Rightarrow k = 2, r = 2$$

$$\therefore \text{Coefficient of } x^8 = {}^{10}C_2 \cdot 2^8 \cdot {}^4C_2 \cdot 3^2 = 5 \cdot 2^9 \cdot 3^5$$

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – I
PAPER –2
TEST DATE: 28-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A

2. C

Sol. No horizontal force is on block C therefore it will have vertical path from ground frame when all surfaces are smooth.

3. A

Sol. By conservation of energy

$$\frac{-GMm}{R} + \frac{1}{2}mv'^2 = \frac{-GMm}{2R} + \frac{1}{2}mv_0^2 \dots\dots(i)$$

By conservation of angular momentum

$$mv_0 2R = mv' R$$

$$\Rightarrow v' = 2v_0 \quad (ii)$$

$$\Rightarrow v_0 = \sqrt{\frac{GM}{3R}} \quad (\text{using (i) \& (ii)})$$

For circular orbit

$$\frac{mv''^2}{2R} = \frac{GMm}{(2R)^2} \Rightarrow v'' = \sqrt{\frac{GM}{2R}}$$

$$v'' = \sqrt{\frac{3}{2}}v_0$$

4. C

Sol. $\frac{v_1}{v_2} = \frac{27}{8}$

So, $\frac{m_1}{m_2} = \frac{8}{27}$ (from conservation of momentum)

$$r \propto A^{1/3}$$

$$\frac{r_1}{r_2} = \left(\frac{8}{27}\right)^{1/3} = \frac{2}{3}$$

5. AC

Sol. $E = 10 \sin(20t + \phi) \quad \omega = 20$

$$20 = \frac{1}{\sqrt{0.2 \times C}} \Rightarrow C = \frac{1}{400 \times 0.2} = \frac{1}{80}$$

$$\cos \phi = \frac{6}{\omega} = \frac{3}{5}$$

$$= 12.5 \text{ Amf}$$

6. BD

7. ACD

8. ACD

9. BD

10. BC

TG ~ @bohring_bot

Section - B

11. 3

12. 1

13. 5

14. 5

15. 3

16. 1

17. 4

18. 2

Chemistry

PART – II

Section – A

19. C

20. C

21. A

Sol. Thymine base of DNA is replaced by uracil in RNA

22. D

23. AC

Sol. $3\text{N}_2\text{H}_5^+ + 2\text{Cr}_2\text{O}_7^{2-} + 13\text{H}^+ \longrightarrow 3\text{N}_2 + 4\text{Cr}^{3+} + 14\text{H}_2\text{O}$

24. BCD

25. ABD

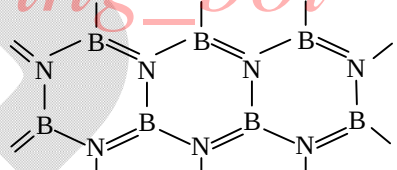
Sol. Sucrose with anomeric–OH groups in glycosidic linkage & thus non-Reducing

26. ABCD

Sol. $\text{B}_2\text{H}_6 + 2\text{NH}_3 \xrightarrow[\text{Temp.}]{\text{low}} \text{B}_2\text{H}_6 \cdot 2\text{NH}_3$ (ionic solid)

$\downarrow$ excess NH_3
 On further heating
 $(\text{BN})_x$
 Boron nitride

$[\text{H}_3\text{N} \rightarrow \text{BH}_2 \leftarrow \text{NH}_3]^+ [\text{BH}_4]^-$
 Ionic solid



$(\text{BN})_x$

27. ABD

28. CD

Section – B

29. 5

30. 1

Sol. $K_{sp} = [\text{Zn}^{2+}] [\text{S}^{2-}]$

$$[\text{S}^{2-}] = \frac{10^{-21}}{0.01} = 10^{-19}$$

$$\text{for } K_{a1} \cdot K_{a2} = \frac{[\text{H}^+]^2 [\text{S}^{2-}]}{[\text{H}_2\text{S}]}$$

$$10^{-20} = \frac{[\text{H}^+]^2 \times 10^{-19}}{0.1} \quad \square \quad [\text{H}^+] = 0.1 \quad \text{or } \text{pH} = 1$$

31. 3

Sol. Total current passed in 30 sec.

$$= \frac{1}{2} \times 100 \times 10 + 100 \times 10 + 100 \times 10 + \frac{1}{2}$$

$$\times 100 \times 10$$

$$= 3 \times 100 \times 10 \text{ mA} \times \text{s}$$

$$= 3000 \times 10^{-3} \text{ amp s} = 3 \text{ C}$$

$$W = \frac{EQ}{F} \text{ or } \frac{W}{Q} = \frac{E}{F} = Z = \frac{m}{3}$$

32. 3

Sol. $\frac{k_t}{k_0} = \mu^{\frac{t-0}{10}}$

$$\log_e k_t - \log_e k_0 = \frac{t}{10} \log (\mu)$$

$$\ell n k_t = \ell n k_0 + \left(\frac{\ell n(\mu)}{10} \right) t$$

$$\therefore \mu = 3$$

33. 5

Sol. $i = \frac{\text{Theo. Mw.}}{\text{exp. Mw}}$

$$= \frac{164}{65.6} = 2.5$$

$$a = \frac{1-i}{n-1}$$

$$= \frac{2.5-1}{3-1} = 0.75$$

$$a = 75 \%$$

34. 4

35. 9

36. 2

Sol. $\text{pOH} = \text{pK}_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_3]}$

$$5.74 = 4.74 + \log \frac{[\text{NH}_4^+]}{0.1}$$

$$\text{or } \log \frac{[\text{NH}_4^+]}{0.1} = 1$$

$$\text{or } \frac{[\text{NH}_4^+]}{0.1} = 10$$

$$\text{or } [\text{NH}_4^+] = 1 \text{ (M)}$$

$$\therefore (\text{NH}_4)_2\text{SO}_4 \text{ required} = 0.5 \text{ mole}$$

Mathematics

PART – III

Section – A

37. A

Sol. $(\sin^{-1} \sin x - 1/2)^2 - 1/4$

38. D

Sol. $(x - 2y - 4)^2 = 12(2x + y)$

39. D

40. A

Sol.
$$\frac{3f''(x)}{f'(x)} = \frac{f'''(x)}{f''(x)}$$
$$\Rightarrow 3 \ln f'(x) = \ln f''(x) + k$$
$$\Rightarrow \frac{d^2x}{dy^2} = c$$

41. ABCD

Sol.
$$f(x) = \begin{cases} px + \lambda, & x > 0 \\ q, & x = 0 \\ e^x, & x < 0 \end{cases}$$

42. AC

Sol. $f(x) = (2x-1)(2x+1)(x-2)(x-4)$

$$\alpha = -\frac{1}{2}$$

$$\beta = \frac{1}{2}$$

$$\gamma = 2$$

$$\sigma = 4$$

43. CD

Sol. If α is real root then $\alpha = 2, m = \pm 2$, non-real root = $1 + i$

44. ABC

Sol.
$$2S = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left(\frac{m \cdot n}{m \cdot 3^n + n \cdot 3^m} \right) \left(\frac{n}{3^m} + \frac{m}{3^n} \right)$$

45. AC

46. AC

Section – B

47. 3

48. 1

Sol. $x_1 = 4, x_2 = 8, k = 3, w = 4$

49. 7

50. 6

51. 2

52. 2

53. 4

54. 8

Sol. Let $f(x) = \frac{x}{70+x^2}$; $f'(x) = \frac{70-x^2}{(70+x^2)^2}$

So $f(x)$ increase in $(0, \sqrt{70})$ and $f(x)$ decrease in $(\sqrt{70}, \infty)$

So, greater term may be a_8 or a_9

But $\frac{a_8}{a_9} > 1$

$\Rightarrow k = 8$

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Main)-2023

TEST DATE: 07-01-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A

(Single Choice Answer Type)

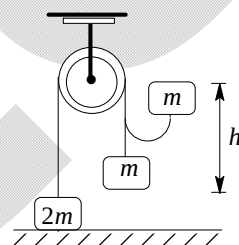
This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-5} cm. If the distance between the slits is 10^{-3} m, calculate the wavelength of the light used.

(A) 4500 Å (B) 3000 Å
(C) 9000 Å (D) 6000 Å

2. A mass $2m$ rests on a horizontal table. It is attached to a light inextensible string which passes over a smooth pulley and carries a mass m at the other end. If the mass m is raised vertically through a distance h and is then dropped, then the speed with which the mass $2m$ begins to rise is:

(A) $\sqrt{2gh}$ (B) $\frac{\sqrt{2gh}}{3}$
(C) $\frac{\sqrt{gh}}{2}$ (D) $\sqrt{gh}$



3. A particle starts moving with initial velocity from the origin of co-ordinates at time $t = 0$ and moves in the xy plane with a constant acceleration α in the y -direction. Its equation of motion is $y = \beta x^2$. Its velocity component in the x -direction is:

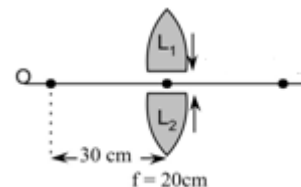
(A) α / β (B) $\sqrt{2\alpha / \beta}$
(C) $\alpha / 2\beta$ (D) $\sqrt{\alpha / 2\beta}$

4. By what percent the energy of a satellite has to be increased to shift it from an orbit of radius r to $\frac{3}{2}r$?

(A) 66.7% (B) 33.3%
(C) 15% (D) 20.3%

5. A point object O is placed at a distance of 0.3 m from a convex lens (focal length 0.2 m) cut into two halves each to which is displacement by 0.005 m thus making separation between them of 2×0.005 as shown in fig. Find the position of the image. Find the distance between the images.

(A) 0.003 m (B) 0.002 m
(C) 0.004 m (D) 0.001 m



6. The relation between U , p and V for an ideal gas in an adiabatic process is given by relation $U = a + bpV$. Find the value of adiabatic exponent (γ) of this gas:

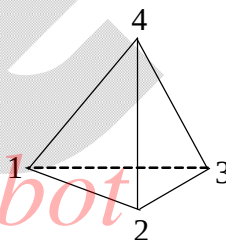
(A) $\frac{b+1}{b}$ (B) $\frac{b+1}{a}$
 (C) $\frac{a+1}{b}$ (D) $\frac{a}{a+b}$

7. A metal disc of radius a rotates with a constant angular velocity ω about its axis. The potential difference between the centre and the rim of the disc is ($m = \text{mass of electron}$, $e = \text{charge on electron}$):

(A) $\frac{m\omega^2 a^2}{e}$ (B) $\frac{1}{2} \frac{m\omega^2 a^2}{e}$
 (C) $\frac{e\omega^2 a^2}{2m}$ (D) $\frac{e\omega^2 a^2}{m}$

8. Six wires each of resistance r form a tetrahedron. The equivalent resistance between corners 1-2 and 1-3 are respectively:

(A) $\frac{r}{2}, \frac{r}{2}$ (B) r, r
 (C) $\frac{r}{2}, r$ (D) $r, \frac{r}{2}$

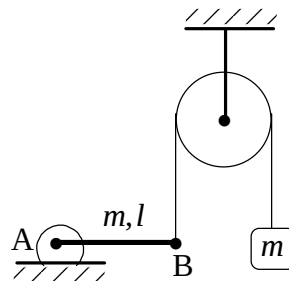


9. The ionization potential of Li^{++} ion is 122.4 V. Find the value of its first excitation potential.

(A) 91.8V (B) 45.9 V
 (C) 122.4V (D) 28.4 V

10. Uniform rod AB is hinged at the end A in a horizontal position as shown in the figure. The other end of the rod is connected to a block through a massless string as shown. The pulley is smooth and massless. Masses of the block and the rod are same and are equal to ' m '. Acceleration due to gravity is g . The tension in the thread and angular acceleration of the rod just after release of block from this position are:

(A) $\frac{3mg}{8}, \frac{g}{8l}$ (B) $\frac{5mg}{8}, \frac{3g}{8l}$
 (C) $\frac{mg}{8}, \frac{5g}{8l}$ (D) $\frac{7mg}{8}, \frac{7g}{8l}$



11. A radioactive sample has 6.0×10^{18} active nuclei at a certain instant. How many of these nuclei will still be in the same active state after two half-lives?

(A) 3×10^{18} (B) 1.5×10^{18}
 (C) 0.75×10^{18} (D) 6×10^{18}

12. A particle performs SHM of amplitude A along a straight line. When it is at a distance $\sqrt{3}/2 A$ from mean position, its kinetic energy gets increased by an amount $1/2 m\omega^2 A^2$ due to an impulsive force. Then its new amplitude becomes:

- (A) $\frac{\sqrt{5}}{2} A$ (B) $\frac{\sqrt{3}}{2} A$
 (C) $\sqrt{2} A$ (D) $\sqrt{5} A$

13. Find the constant b in Moseley's equation $\sqrt{\nu} = a(Z - b)$ from the following data.

Element	Z	Wavelength of K_{α} X-ray
Mo	42	71 pm
Co	27	178.5 pm

- (A) 1.43 (B) 1.11
 (C) 1 (D) 1.37

14. A 70 cm long sonometer wire is in unison with a tuning fork. If the length of the wire is decreased by 1.0 cm, it produces 4 beats per sec. with the same tuning fork. Find the frequency of the tuning fork -

- (A) 276 sec^{-1} (B) 272 sec^{-1}
 (C) 269 sec^{-1} (D) 280 sec^{-1}

15. Mr. Sanjay wants to calculate the current and power dissipated in an LCR series circuit. He connected 100Ω resistance to an AC source of 200 V and angular frequency 300 radian/sec. When he removed only the capacitance, the current was found to be lagging behind the voltage by 60° . While on removing the inductance he found, the current leading the voltage by 60° . Find the value of current and the power dissipated obtained by him

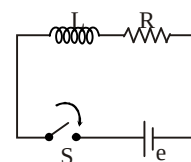
- (A) 200 Watt (B) 100 Watt
 (C) 400 Watt (D) 300 Watt

16. The length of a wire between the two ends of a sonometer is 105 cm. Where the two bridges should be placed so that the fundamental frequencies of the three segments are in the ratio of 1:3:15.

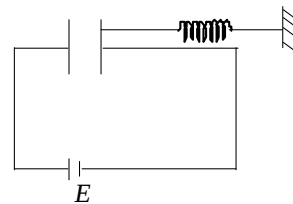
- (A) 75 cm, 25 cm, (B) 25 cm, 75 cm,
 (C) 75 cm, 100 cm, (D) None of these

17. In an LR circuit as shown in fig. when the switch is closed how much time will it take the current to grow a value $\frac{\eta \epsilon}{R}$ (where $\eta < 1$)

- (A) $(L/R) \ln \left| \frac{1}{\eta} \right|$ (B) $(L/R) \ln \left| \frac{1}{1-\eta} \right|$
 (C) $(2L/R) \ln \left| \frac{1}{\eta} \right|$ (D) None of these

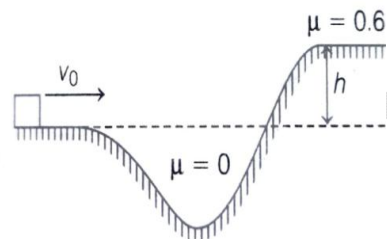


18. One plate of a capacitor is connected to a spring as shown in figure. Area of both the plates is A . In steady state, separation between the plates is $0.8d$ (spring was unstretched and the distance between the plates was d when the capacitor was uncharged). The force constant of the spring is approximately:



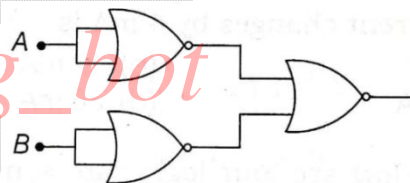
- (A) $\frac{4\epsilon_0 AE^2}{d^3}$ (B) $\frac{2\epsilon_0 AE}{d^2}$
 (C) $\frac{6\epsilon_0 E^2}{Ad^3}$ (D) $\frac{\epsilon_0 AE^3}{2d^3}$

19. In the figure, a block slides along a track from one level to a higher level, by moving through an intermediate valley. The track is frictionless until the block reaches the higher level. There a frictional force stops the block in a distance d . The block's initial speed v_0 is 6 m/s , the height difference h is 1.1 m and the coefficient of kinetic friction μ is 0.6 . The value of d is ($g = 10 \text{ m/s}^2$)



20. The combination of the gates shown in the figure produces:

- (A) NOR gate
 (B) OR gate
 (C) AND gate
 (D) XOR gate

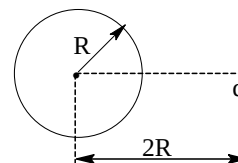


SECTION - B

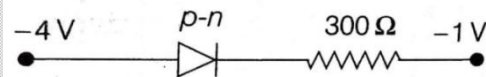
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A point charge is placed outside a spherical shell as shown in the figure. Find the electric potential (in kilo volt) on the surface of shell (Take: $q=1\mu\text{C}$ and $R=50\text{cm}$)



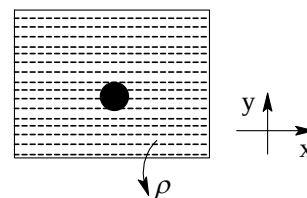
22. What is the current in the circuit shown below?



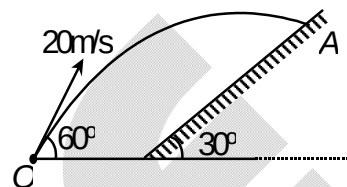
23. The pressure and volume of an ideal gas are related as $P = \frac{a}{1 + \left(\frac{V}{b}\right)^3}$, where 'a' and 'b' are

constants. The translational kinetic energy due to the thermal motion of the gas sample at volume $V=b$, (given that $ab=12 \text{ kJ}$) is $n \text{ kJ}$. Find n .

24. A small spherical ball of density same as that of liquid is released from rest in a vessel filled completely with a liquid and accelerating with acceleration $5\hat{i} + 5\hat{j} \text{ m/s}^2$ as shown in the figure. Find the initial acceleration of ball just after the release with respect to vessel.

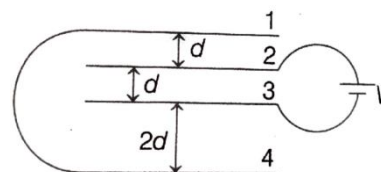


25. A ball is projected from the point O with velocity 20 m/s at an angle of 60° with horizontal as shown in figure. At highest point of its trajectory it strikes a smooth plane of inclination 30° at point A. The collision is perfectly inelastic. The maximum height from the ground attained by the ball is $\frac{75}{k}$ meter. Find the value of k ? ($g = 10 \text{ m/s}^2$)



26. The potential energy of a particle is determined by the expression $U = \alpha(x^2 + y^2)$, where α is a positive constant. The particle begins to move from a point with the co-ordinates (3, 3) only under the action of potential fields force. When it reaches the point (1, 1) its kinetic energy is $4K\alpha$. Find the value of K.
27. An ambulance blowing a siren of frequency 700Hz is travelling slowly towards vertical reflecting wall with a speed of 2 m/s. The speed of sound is 350 m/s. How many beats are heard per sec to the driver of the ambulance?
28. In YDSE, when a glass plate of refractive index 1.5 of thickness t is placed in the path of one of the interfering beams of wavelength λ , intensity at the position where central maximum occurred previously remain unchanged. If the minimum thickness of the glass plate is $k\lambda$. Find the value of k
29. If the ionization potential of hydrogen atom is 13.6eV, the energy required to remove from the third orbit of hydrogen atom is $k/2 \text{ eV}$. Find the value of k

30. Four identical metal plates are arranged as shown. Plates 1 and 4 are connected by a connecting wire. A battery of emf V volts is connected between plates 2 and 3. The electric field between plates 3 and 4 is $\frac{2V}{Kd}$. Find the value of K _____.



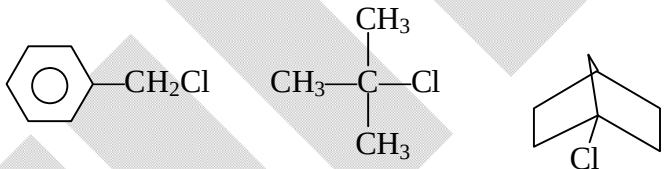
Chemistry

PART – B

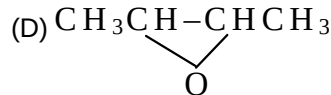
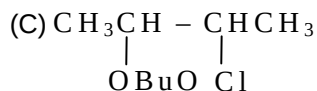
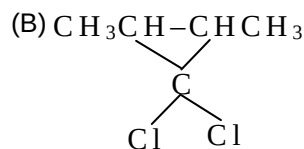
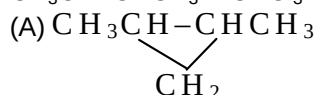
SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

31. Which of the following molecule has highest bond angle?
 (A) NH_3 (B) H_2O
 (C) Cl_2O (D) CH_4
32. Identify the least stable ion amongst the following
 (A) Li^- (B) Be^-
 (C) B^- (D) C^-
33. When sulphur is dissolved in oleum, a deep blue solution containing polyatomic sulphur cation is obtained. The formula of the cation present is
 (A) S_4^{2+} (B) S_8^{2+}
 (C) S_{19}^{2+} (D) S_{16}^{+2}
34. 
 I II III IV
 Increasing order of stability is
 (A) $\text{I} < \text{III} < \text{II} < \text{IV}$ (B) $\text{IV} < \text{III} < \text{II} < \text{I}$
 (C) $\text{III} < \text{IV} < \text{II} < \text{I}$ (D) $\text{II} < \text{IV} < \text{III} < \text{I}$
35. There is formation of precipitate almost instantly when alcoholic AgNO_3 reacts with

 (I) (II) (III)
 (A) I, II and III (B) I and II
 (C) II and III (D) I and III
36. In the detection of nitrogen, blue/green colour is due to formation of Prussian blue. It is:
 (A) $\text{Fe}_4^{\text{III}}[\text{Fe}^{\text{II}}(\text{CN})_6]$ (B) $\text{NaFe}^{\text{II}}[\text{Fe}^{\text{III}}(\text{CN})_6]$
 (C) $\text{Na}_4[\text{Fe}(\text{CN})_6]$ (D) $\text{Na}_3[\text{Fe}(\text{CN})_6]$
37. An alkane with the formula C_6H_{14} can be prepared by the hydrogenation of only two alkenes (C_6H_{12}). IUPAC name of the alkane is:
 (A) 2, 2-dimethylbutane (B) 2, 3-dimethylbutane
 (C) 2-methyl-pentane (D) n-hexane

38. $\text{CH}_3\text{CH}=\text{CHCH}_3 + \text{CHCl}_3 + t\text{-BuOK} \rightarrow \text{A}$. A is:

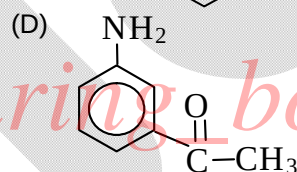
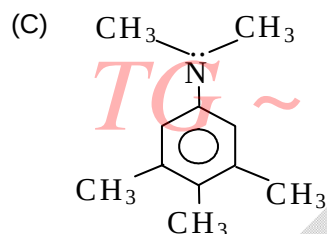
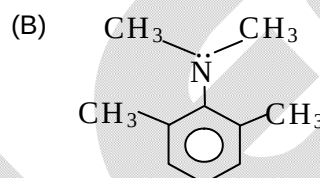
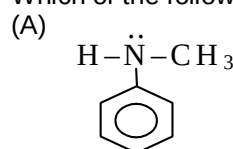


39. Glucose is added to 1 litre water to such an extent that $\Delta T_f / K_f$ becomes equal to $1/1000$, the wt. of glucose added is

(A) 180 g
(C) 1.8 g

(B) 18 g
(D) 0.18 g

40. Which of the following is most basic?



41. Cellulose on hydrolysis with dil. H_2SO_4 gives -

(A) α -D-glucose
(C) D-fructose

(B) β -D-glucose
(D) none of these

42. Which of the following is the example of condensation polymer

(A) Polystyrene
(C) Polyethylene

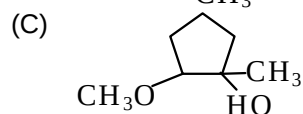
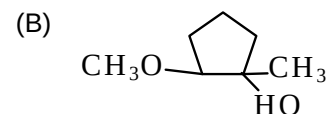
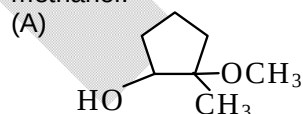
(B) Terylene
(D) Polypropene

43. Tin reacts with-

(A) Hot conc. HCl
(C) HgCl_2 on heating

(B) Conc. HNO_3
(D) All of these

44. If the starting material is 1-methyl-1, 2-epoxy-cyclopentane, of absolute configuration, decide which one compound correctly represent the product of its reaction with sodium methoxide in methanol.



(D) all of these

45. Pick out the incorrect statement for XeF_6 —
 (A) XeF_6 is hydrolysed partially to form XeOF_4
 (B) It reacts with SiO_2 to form XeF_4
 (C) On complete hydrolysis, it forms XeO_3
 (D) It acts as F^- acceptor when treated with alkali metal fluoride, but cannot act as F^- donor to form complexes.
46. An inorganic compound gives a white ppt. with a solution of AgNO_3 , a white ppt. with dil. H_2SO_4 and impart green colour to flame. The probable compound is
 (A) CuCl_2 (B) BaCl_2
 (C) $\text{Cu}(\text{NO}_3)_2$ (D) PbCl_2
47. When $\text{CH}_2 = \text{CH}-\text{COOH}$ is reduced with LiAlH_4 , the compound obtained will be
 (A) $\text{CH}_3-\text{CH}_2-\text{COOH}$ (B) $\text{CH}_2 = \text{CH}-\text{CH}_2\text{OH}$
 (C) $\text{CH}_3-\text{CH}_2-\text{CH}_2\text{OH}$ (D) $\text{CH}_3-\text{CH}_2-\text{CHO}$
48. In the electronic configuration of nitrogen has $1s^7$, it would have energy lower than that of the normal ground state configuration $1s^2 2s^2 2p^3$ because the electrons would be closer to the nucleus. Yet $1s^7$ is not observed because it violates
 (A) Heisenberg uncertainty principle (B) Hund's rule
 (C) Pauli's exclusion principle (D) Bohr postulates of stationary orbits
49. A mixture of calcium ethanoate and calcium methanoate is heated strongly. The expected product is
 (A) acetaldehyde (B) Ethyl methyl ketone
 (C) diethyl ketone (D) Propyl ketone
50. Among the following complexes which one will show co-ordination isomerism?
 (A) $[\text{Cr}(\text{NH}_3)_6][\text{CoCl}_6]$ (B) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_2$
 (C) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (D) $[\text{Cr}(\text{en})_2\text{Cl}_2]^+$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. A large irregularly-shaped closed tank is first evacuated and then connected to a 55 litre cylinder containing compressed nitrogen gas. The gas pressure in the cylinder, originally at 21.4 atm, falls to 1.5 atm after it is connected to the evacuated tank. Volume of the tank in litres is $x \times 10$, then the value of 'x' is ____.
52. Number of neutrons in 18 mL of water (density of water = 1 gm/mL) is $x \times 10^{23}$. The value of 'x' is _____. (where, $N_A = 6 \times 10^{23}$)
53. A solution containing H^+ and D^+ ions is in equilibrium with a mixture of H_2 and D_2 gases at 25°C . If partial pressure of both the gases are 1.0 atm, then $\log \frac{[\text{D}^+]}{[\text{H}^+]}$ is $x \times 10^{-2}$. The value of 'x' is (approximate value to nearest integer) _____. [$E^\circ(\text{D}^+/\text{D}_2) = -0.003\text{V}$]

54. The solubility product of AgI at 25°C is $1.0 \times 10^{-16} \text{ mol}^2 \text{ L}^{-2}$. The solubility of AgI in 10^{-x} N solution of KI at 25°C is approximately (in mol L^{-1}) is $1 \times 10^{-12} \text{ mol/L}$. Find the value of 'x'.
55. 0.7 g of $\text{Na}_2\text{CO}_3 \cdot x \text{ H}_2\text{O}$ is dissolved in 100 ml, 20 ml of which required 19.8 ml of 0.1 N HCl. The value of x is ____.
56. k for a zero order reaction is $2 \times 10^{-2} \text{ mol l}^{-1}\text{s}^{-1}$. If the concentration of the reactant after 25 s is 0.5 M, the initial concentration must have been
57. The temperature of 5 ml of a strong acid increases by 5K when 5 ml of a strong base is added to it. If 10 ml of each is mixed, temperature should increase byK
58. The coagulation of 10 cm^3 of gold sol is completely prevented by addition of 0.025g of starch to it. The gold number of starch is ____ (when 1ml 10% NaCl solution is added in the gold solution)
59. One mole of $\text{N}_2\text{O}_4(\text{g})$ at 300K is kept in a closed container under one atmosphere. It is heated to 600K the resultant pressure is 2.4 atm at equilibrium. Find out % decomposition of $\text{N}_2\text{O}_5(\text{g})$
60. In cubic ZnS lattice, if the radii of Zn^{2+} and S^{2-} ions are 0.83 Å and 1.74 Å, the lattice parameter (edge length, a) of cubic ZnS is (in nearest integer)

TG ~ @bohring_bot

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. A function $f : R \rightarrow R$ is defined as $f(x) = 3x^2 + 1$. Then $f^{-1}(x)$ is
- (A) $\frac{\sqrt{x-1}}{3}$ (B) $\frac{1}{3}\sqrt{x-1}$
- (C) f^{-1} does not exist (D) $\sqrt{\frac{x-1}{3}}$
62. $\lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin\left(\frac{\pi}{3} - x\right)}{2 \cos x - 1}$ is equal to:
- (A) $\frac{2}{\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$
- (C) $\sqrt{3}$ (D) $\frac{1}{2}$
63. Let $f(x) = \int e^x (x-1)(x-2) dx$. Then $f(x)$ decreases in the interval?
- (A) $(2, \infty)$ (B) $(-2, -1)$
- (C) $(1, 2)$ (D) $(-\infty, 1) \cup (2, \infty)$
64. If $I = \int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}}$ is equal to:
- (A) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + C$ (B) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + C$
- (C) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$ (D) None of these
65. The area enclosed between the curves $y = ax^2$ and $x = ay^2$ ($a > 0$) is 1 sq. unit , then the value of a is:
- (A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{2}$
- (C) 1 (D) $\frac{1}{3}$

66. The solution of the differential equation $\frac{dy}{dx} = (4x + y + 1)^2$ is:
 (A) $4x + y + 1 = 2 \tan(2x + y + C)$ (B) $4x + y + 1 = 2 \tan(x + 2y + C)$
 (C) $4x + y + 1 = 2 \tan(2y + C)$ (D) $4x + y + 1 = 2 \tan(2x + C)$
67. For $x \in \mathbb{R}$, the expression $\frac{x^2 + 2x + c}{x^2 + 4x + 3c}$ can take all real values if $c \in$:
 (A) $(1, 2)$ (B) $[-1, 1]$
 (C) $(0, 1)$ (D) $(-1, 0)$
68. The third term of a G.P. is 2. Then the product of the first five terms is:
 (A) 2^3 (B) 2^4
 (C) 2^5 (D) None of these
69. If the system of linear equations
 $x + 2ay + az = 0$
 $x + 3by + bz = 0$
 $x + 4cy + cz = 0$
 Has a non-zero solution, then a, b, c :
 (A) are in A.P. (B) are in G.P.
 (C) are in H.P. (D) satisfy $a + 2b + 3c = 0$
70. The number of points of intersection of the curves represented by
 $\arg(z - 2 - 7i) = \cot^{-1}(2)$ and $\arg\left(\frac{z - 5i}{z + 2 - i}\right) = \pm \frac{\pi}{2}$
 (A) 0 (B) 1
 (C) 2 (D) None of these
71. If all the roots of $z^3 + az^2 + bz + c = 0$ are of unit modulus, then:
 (A) $|a| \leq 3$ (B) $|b| \leq 3$
 (C) $|c| = 1$ (D) All of these
72. If M be a square matrix of order 3 such that $|M| = 2$, then $\left| \text{adj}\left(\frac{M}{2}\right) \right|$ equals to:
 (A) $\frac{1}{2}$ (B) $\frac{1}{4}$
 (C) $\frac{1}{8}$ (D) $\frac{1}{16}$
73. How many ways are there to arrange the letters of the word "GARDEN" with the vowels in alphabetical order?
 (A) 120 (B) 480
 (C) 360 (D) 240

74. If $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then $a_r - {}^nC_1 \cdot a_{r-1} + {}^nC_2 \cdot a_{r-2} - {}^nC_3 \cdot a_{r-3} + \dots + (-1)^r {}^nC_r \cdot a_0$ is equal to: (r is not multiple of 3)
- (A) 0 (B) nC_r
(C) a_r (D) 1
75. If $(1+x)^{2010} = C_0 + C_1x + C_2x^2 + \dots + C_{2010}x^{2010}$ then the sum of series $C_2 + C_5 + C_8 + \dots + C_{2009}$ equal to:
- (A) $\frac{1}{2}(2^{2010} - 1)$ (B) $\frac{1}{3}(2^{2010} - 1)$
(C) $\frac{1}{2}(2^{2009} - 1)$ (D) $\frac{1}{3}(2^{2009} - 1)$
76. The boy comes from a family of two children. What is the probability that the other child is his sister?
- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{2}{3}$ (D) $\frac{1}{4}$
77. If A be any event in sample space then the maximum value of $3\sqrt{P(A)} + 4\sqrt{P(\overline{A})}$ is:
- (A) 4 (B) 2
(C) 5 (D) Cannot be determined
78. Least positive integral value of 'a' for which $\log_{\left(\frac{x+1}{x}\right)}(a^2 - 3a + 3) > 0; (x > 0)$:
- (A) 1 (B) 2
(C) 3 (D) 4
79. The minimum value of $x^2 + y^2 + z^2$ if $ax + by + cz = p$, is:
- (A) $\left(\frac{p}{a+b+c}\right)^2$ (B) $\frac{p^2}{a^2 + b^2 + c^2}$
(C) $\frac{a^2 + b^2 + c^2}{p^2}$ (D) 0
80. The scalar triple product $\left[\vec{a} + \vec{b} - \vec{c} \quad \vec{b} + \vec{c} - \vec{a} \quad \vec{c} + \vec{a} - \vec{b}\right]$ is equal to:
- (A) 0 (B) $[\vec{a} \vec{b} \vec{c}]$
(C) $2[\vec{a} \vec{b} \vec{c}]$ (D) $4[\vec{a} \vec{b} \vec{c}]$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Find the number of integers in the domain of $f(x) = \frac{1}{\sqrt{\ln \cos^{-1} x}}$.
82. If $x = \cos \theta$ and $y = \sin^3 \theta$, then $\left| \frac{y d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right|$ at $\theta = \frac{\pi}{2}$ is:
83. For a positive integer n , let $I_n = \int_{-\pi}^{\pi} \left(\frac{\pi}{2} - |x| \right) \cos nx \, dx$. Find the value of $[I_1 + I_2 + I_3 + I_4]$ where $[.]$ denotes greatest integer function.
84. Find the number of positive integral values of k for which $kx^2 + (k-3)x + 1 < 0$ for atleast one positive x .
85. The value of $\sum_{r=1}^{\infty} \frac{8r}{4r^4 + 1}$ is equal to:
86. The sum of all digits of n for which $\sum_{r=1}^n r 2^r = 2 + 2^{n+10}$ is:
87. For a unique value of μ & λ , the system of equations given by

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 14 \\ 2x + 5y + \lambda z &= \mu \end{aligned}$$
 Has infinitely many solutions, then $\frac{\mu - \lambda}{4}$ is equal to:
88. A and B are two square matrices. Such that $A^2 B = BA$ and if $(AB)^{10} = A^k \cdot B^{10}$. Find the value of $k - 1020$.
89. The remainder when $\left(\sum_{k=1}^5 {}^{20}C_{2k-1} \right)^6$ is divided by 11, is:
90. If $\log_{16} \left(\log_{\sqrt[3]{5}} \left(\log_{\sqrt[3]{5}} (x) \right) \right) = \frac{1}{2}$; find x .

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Main)-2023

TEST DATE: 07-01-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. The fringe-width β is given by

$$\beta = \frac{\lambda D}{2d} \text{ and } \beta' = \frac{\lambda D'}{2d}$$

Where λ is the wavelength of light used, D is the distance of the screen from the two slits and 2 d is the separation between two slits.

$$\text{Now } \beta - \beta' = \frac{\lambda(D - D')}{2d}$$

$$\Rightarrow \lambda = \frac{(\beta - \beta')2d}{D - D'}$$

$$\begin{aligned} \lambda &= \frac{(3 \times 10^{-5})(10^{-3})}{5 \times 10^{-2}} \\ &= 0.6 \times 10^{-6} \text{ m} = 6000 \text{ \AA} \end{aligned}$$

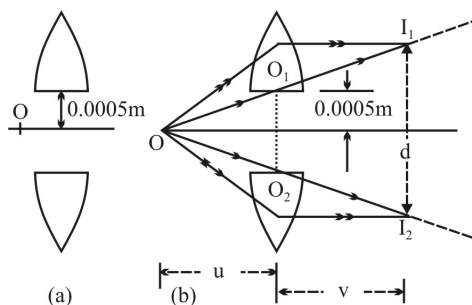
2. B

3. D

4. B

5. A

Sol. The image formation is shown in fig.



The rays through optical centres O_1 and O_2 pass undeflected. The image of O are formed at I_1 and I_2 due to upper and lower lenses respectively.

The number of images formed = two

For a convex lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Here $u = -0.3$ m and $f = 0.2$ m

$$\therefore \frac{1}{v} = \frac{1}{u} + \frac{1}{f} = -\frac{1}{0.3} + \frac{1}{0.2} = \frac{1}{0.6}$$

$u = 0.6$ m

Let d be the distance between the two images I_1 and I_2 . From $\triangle OO_1O_2$ and $\triangle II_1I_2$, we have

$$\frac{d}{O_1O_2} = \frac{u+v}{u}$$

$$\text{or } \frac{d}{0.0005 + 0.0005} = \frac{0.3 + 0.6}{0.3}$$

Solving we get $d = 0.003$ m

6. A

7. B

8. A

9. A

Sol. According to given data, ionization energy of Li^{++} ion is 122.4 eV

$\therefore$ Excitation energy of this ion

$$= 122.4 \left[\frac{1}{1^2} - \frac{1}{2^2} \right]$$

$$= 122.4 \left[1 - \frac{1}{4} \right]$$

$$= 122.4 \times \frac{3}{4} \text{ eV}$$

$$\therefore E_{\text{ex}} = 91.8 \text{ eV}$$

Hence, the value of the first excitation potential
= 91.8 V

10. B

11. B

Sol. In one half-life the number of active nuclei reduces to half the original number. Thus, in two half-lives the number is reduced to $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)$ of the original number. The number of remaining active nuclei is, therefore, $6.0 \times 10^{18} \times \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) = 1.5 \times 10^{18}$

12. C

13. D

Sol. Moseley's equation is

$$\sqrt{\nu} = a(Z - b).$$

$$\text{Thus } \sqrt{\frac{c}{\lambda_1}} = a(Z_1 - b) \quad \dots(i)$$

$$\text{and } \sqrt{\frac{c}{\lambda_2}} = a(Z_2 - b) \quad \dots(ii)$$

Using equations (i) and (ii), we get

$$\sqrt{c} \left(\frac{1}{\sqrt{\lambda_1}} - \frac{1}{\sqrt{\lambda_2}} \right) = a(Z_1 - Z_2)$$

$$\Rightarrow a = \frac{\sqrt{c}}{(Z_1 - Z_2)} \left(\frac{1}{\sqrt{\lambda_1}} - \frac{1}{\sqrt{\lambda_2}} \right)$$

Putting given values

$$a = 5.06 \times 10^7 \text{ Hz}$$

Dividing equation (i) by (ii), we get

$$\frac{\sqrt{\lambda_2}}{\sqrt{\lambda_1}} = \frac{Z_1 - b}{Z_2 - b}$$

$$\Rightarrow \sqrt{\frac{178.5}{71}} = \frac{42 - b}{27 - b}$$

$$\Rightarrow b = 1.37$$

14. A

Sol. Let the frequency of the fork be n . In the first case, the length of the wire is 70 cm (0.70 m).

Therefore,

$$n = \frac{1}{2 \times 0.70} \sqrt{\frac{T}{m}} \quad \dots(1)$$

On decreasing the length of the wire, its frequency will increase. Hence in the second case when the length is 69 cm = 0.69 m, then

$$n + 4 = \frac{1}{2 \times 0.69} \sqrt{\frac{T}{m}} \quad \dots(2)$$

dividing equation (1) by (2), we get

$$\Rightarrow \frac{n}{n+4} = \frac{0.69}{0.70}$$

$$\Rightarrow n \times 0.70 = (n + 4) \times 0.69$$

$$\Rightarrow n = \frac{4 \times 0.69}{0.01} = 276 \text{ sec}^{-1}$$

15. A

Sol. $\tan 60^\circ = \frac{\omega L}{R}$, $\tan 60^\circ = \frac{1/\omega C}{R}$.

$$\therefore \omega L = \frac{1}{\omega C}$$

$\therefore$ impedance of the circuit,

$$Z = \left[R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 \right]^{\frac{1}{2}} = R.$$

Current in the circuit,

$$i_0 = \frac{V_0}{Z} = \frac{V_0}{R} = \frac{200}{100} = 2 \text{ ampere.}$$

$$\text{Average power, } \bar{P} = \frac{1}{2} V_0 i_0 \cos \phi.$$

$$\text{But } \tan \phi = \frac{\omega L - 1/\omega C}{R} = 0, \quad \therefore \cos \phi = 1.$$

$$\therefore \bar{P} = \frac{1}{2} \times 200 \times 2 \times 1 = 200$$

16. C

Sol. From the law of length of stretched string, we have $n_1 \ell_1 = n_2 \ell_2 = n_3 \ell_3$

Here $n_1 : n_2 : n_3 = 1 : 3 : 15$

$$\therefore \frac{\ell_1}{\ell_2} = \frac{n_2}{n_1} = \frac{3}{1} \text{ and } \frac{\ell_1}{\ell_3} = \frac{n_3}{n_1} = 15/1$$

$$\ell_2 = \frac{\ell_1}{3} \text{ and } \ell_3 = \frac{\ell_1}{15}$$

The total length of the wire is 105 cm.

$$\text{Therefore } \ell_1 + \ell_2 + \ell_3 = 105$$

$$\text{or } \ell_1 + \frac{\ell_1}{3} + \frac{\ell_1}{15} = 105 \text{ or } \frac{21\ell_1}{15} = 105$$

$$\ell_1 = \frac{105 \times 15}{21} = 75 \text{ cm.}$$

$$\therefore \ell_2 = \frac{\ell_1}{3} = \frac{75}{3} = 25 \text{ cm}$$

$$\ell_3 = \frac{\ell_1}{15} = \frac{75}{15} = 5 \text{ cm}$$

Hence the bridge should be placed at 75 cm and $(75 + 25) = 100$ cm from one end.

17. B

Sol. We know that

$$i = \frac{\varepsilon}{R} (1 - e^{-t/\tau})$$

$$i = \eta \frac{\varepsilon}{R} \text{ (given)}$$

$$n \frac{\varepsilon}{R} = \frac{\varepsilon}{R} (1 - e^{-t/\tau})$$

$$\text{or } e^{-t/\tau} = 1 - \eta$$

or $t = \tau \ln \left| \frac{1}{1-\eta} \right|$

18. A

19. A

20. C

SECTION – B

21. 9

Sol. Field inside the shell is zero, so potential is uniform

$$V = \frac{kq}{2R} = 9 \text{ kilo volts}$$

22. 0

23. 9

24. 0

25. 4

26. 4

27. 8

28. 2

Sol. $t(\mu - 1) = \lambda \Rightarrow t(1.5 - 1) = \lambda \Rightarrow t = 2\lambda$

29. 3

Sol. Energy required removing the electron from the third orbit

$$= \frac{13.6}{3^2} \text{ eV} = 1.5 \text{ eV} = 1.5 \times 1.6 \times 10^{-19} \text{ J} = 3 \text{ eV}$$

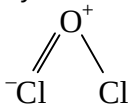
30. 6

Chemistry

PART – B

SECTION – A

31. C
Sol. Due to back bonding between lone pair of oxygen atom and vacant d-orbital of Cl atom, hybridization of oxygen atom changes from sp^3 to sp^2 , so the bond angle increases.



32. B
Sol. Be atom has its configuration $1s^2.2s^2$ so, B^- has its configuration $1s^2, 2s^2, 2p^1$ and therefore it will be most destabilized.

33. B

34. B
Sol. Stability order is $I > II > III > IV$, because I is neutral, while in II all atoms with complete octet system while III is more stable than IV because in case of IV oxygen is positive with incomplete octet system.

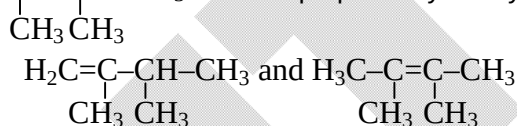
35. B

- Sol. In (III) carbocation cannot be formed because of the bridge head.

36. A

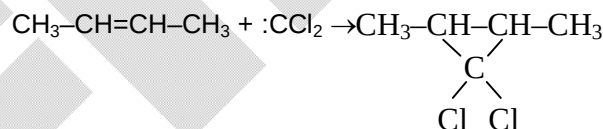
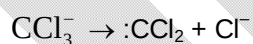
37. B

- Sol. $\text{CH}_3-\text{CH}-\text{CH}-\text{CH}_3$ It can be prepared by the hydrogenation of only two alkenes.



38. B

- Sol. $\text{CHCl}_3 + t\text{-BuO}^-\text{K}^+ \rightarrow t\text{-BuOH} + \text{CCl}_3^-$



39. D

- Sol. $\Delta T_f = k_f \times \text{Molality}$

$$\therefore \frac{\Delta T_f}{k_f} = \text{molality} = \frac{\text{number of moles of glucose}}{\text{mass of water (in kg)}} = \frac{1}{1000} = \frac{\text{weight of glucose (in g)}}{180 \times 1}$$

$$\therefore \text{weight of glucose (in g)} = 0.18 \text{ g}$$

40. B

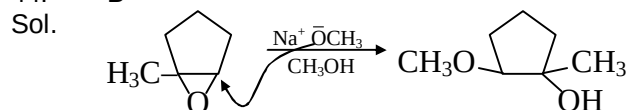
Sol. Steric hinderence of two $-\text{CH}_3$ groups makes $-\text{N} \begin{smallmatrix} \text{CH}_3 \\ \text{CH}_3 \end{smallmatrix}$ group out of plane of benzene ring. So, lone pair on nitrogen atom is not involved in resonance, therefore, this amine is most basic.

41. B

42. B

43. D

44. B



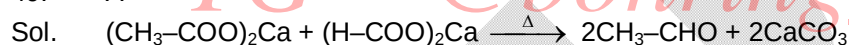
45. D

46. B

47. B

48. C

49. A



50. A

SECTION - B

51. 73

Sol. Let the volume of the tank be V litres.

As the number of moles of N_2 before and after connecting it to the tank will be same,

$$\frac{21.4 \times 55}{RT} = \frac{1.5 \times (55 + V)}{RT} \quad \left[n = \frac{PV}{RT} \right]$$

(moles of N_2 before connection) (moles of N_2 after connection)

$$\therefore V = 726 \text{ litres.}$$

52. 48

Sol. $18\text{mL water} \equiv 18\text{g water} = 1 \text{ mole water}$

$= N_A$ water molecules

$\therefore$ One water molecule contains 8 neutrons

$\therefore$ total number of neutrons in the given sample $= 8N_A$.

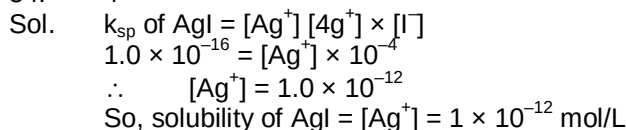
53. 5

Sol. At equilibrium, $E_{\text{cell}} = 0$, Cell reaction is $\frac{1}{2} \text{D}_2(\text{g}) + \text{H}^+ \rightarrow \text{D}^+ + \frac{1}{2} \text{H}_2(\text{g})$

$$\therefore E_{\text{cell}}^0 = 0.0591 \log \frac{[\text{D}^+]}{[\text{H}^+]} = 0.003$$

$$\therefore \log \frac{[D^+]}{[H^+]} = \frac{E_{\text{cell}}^{\circ}}{0.0591} = \frac{0.003}{0.0591} \cong 0.05$$

54. 4



55. 2

Sol. Total number of equivalents of $Na_2CO_3 \cdot xH_2O = \frac{\text{weight (in g)}}{\text{Equivalent weight}} = \frac{0.7}{53 + 9x}$

(n factor of $Na_2CO_3 \cdot xH_2O = 2$)

Total number of equivalents of HCl required for complete neutralization

$$= \left(\frac{0.1 \times 19.8}{1000} \right) \times 5$$

$$\text{so, } \frac{0.7}{53 + 9x} = \frac{0.1 \times 19.8 \times 5}{1000}$$

$$\Rightarrow x = 2$$

56. 1

Sol. For a zero order reaction,

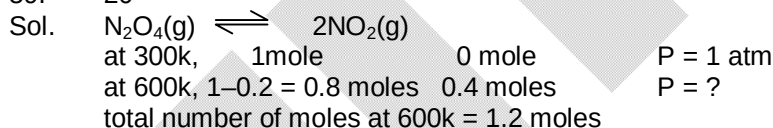
$$C_0 - C_t = Kt$$

$$\therefore C_0 = C_t + Kt = 0.5 + 2 \times 10^{-2} \times 25 = 1.0 \text{ M}$$

57. 5

58. 25

59. 20



$$\therefore P \propto nT$$

$$\therefore \frac{P_1}{P_2} = \frac{n_1 T_1}{n_2 T_2} \text{ where } n \text{ is number of moles and } T \text{ is the temperature in 'K'}$$

$$\therefore \frac{1}{P_2} = \frac{1 \times 300}{1.2 \times 600} \therefore P_2 = 2.4 \text{ atm}$$

60. 6

Sol. In ZnS lattice, S^{2-} ions are present as FCC with Zn^{2+} ions occupying alternate tetrahedral voids.

$$\therefore \frac{\sqrt{3} a}{4} = (r_{Zn^{2+}} + r_{S^{2-}})$$

$$a = \frac{4}{\sqrt{3}} (0.83 + 1.74) = 5.94 \text{ \AA} \approx 6$$

Mathematics

PART – C

SECTION – A

61. C

Sol. Function is many one so inverse does not exist.

62. B

Sol. Let $x - \pi/3 = t$, $x \rightarrow \pi/3 \Rightarrow t \rightarrow 0$

$$\lim_{t \rightarrow 0} \frac{\sin(-t)}{2 \cos(t + \pi/3) - 1}$$

63. C

Sol. $f(x)$ is decreasing, so $\Rightarrow f'(x) < 0$

64. C

Sol. $= \int \frac{\frac{1}{x^3} - \frac{1}{x^5}}{\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}} dx$, let $t = 2 - \frac{2}{x^2} + \frac{1}{x^4}$

65. D

66. D

67. C

Sol. $y = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$, apply nature of roots

68. C

Sol. Let terms a, ar, ar^2, ar^3, ar^4 given that $ar^2 = 2$, product of five terms
 $= a \cdot ar \cdot ar^2 \cdot ar^3 \cdot ar^4 = (ar^2)^5 = 2^5$

69. C

70. A

71. D

72. D

73. C

74. A

75. B

Sol. Put $x = 1, x = \omega$ & $x = \omega^2$

76. A

77. C

78. C

Sol. $x + \frac{1}{x} > 2$

$$\left. \begin{aligned} a^2 - 3a + 3 &> 1 \\ a^2 - 3a + 3 &> 0 \end{aligned} \right\} \cap$$

79. B

80. D

Sol. $[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c})$

SECTION - B

81. 2

Sol. $\ln \cos^{-1} x > 0$
 $-1 \leq x \leq 1$

82. 3

83. 4

84. 0

Sol. $k < 0$, & $D < 0$

85. 2

Sol. $\sum_{r=1}^{\infty} \frac{2 \cdot \{(2r^2 + 2r + 1) - (2r^2 + 1 - 2r)\}}{(2r^2 + 2r + 1) \cdot (2r^2 + 1 - 2r)}$

86. 9

87. 7

Sol. $\Delta = 0$, also $\Delta_x = \Delta_y = \Delta_z = 0$ and atleast one co-factor of $\Delta \neq 0$

88. 3

89. 4

90. 5

Sol. $\log_b a = x \Rightarrow a = b^x$

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Main)-2023

TEST DATE: 11-01-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A

(Single Choice Answer Type)

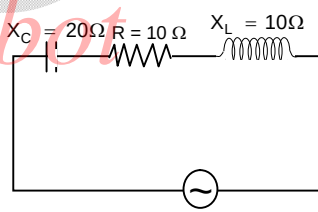
This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

- A uniform solid brass sphere is rotating with angular speed ω_0 about a diameter. If its temperature is now increased by 100°C . What will be its new angular speed (Given $\alpha_{\text{brass}} = 2.0 \times 10^{-5} \text{ per } ^\circ\text{C}$)

(A) $1.1 \omega_0$ (B) $1.01 \omega_0$
 (C) $0.996 \omega_0$ (D) $0.824 \omega_0$
- When a positively charged conductor is connected to earth,

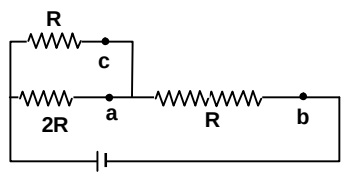
(A) electrons flow from the earth to the conductor
 (B) protons flow from the conductor to the earth
 (C) electrons flow from the conductor to the earth
 (D) any of (A) or (C) may happen
- In LCR circuit shown in figure, then choose incorrect option

(A) Current will lead the voltage
 (B) rms value of current is 20 A
 (C) Power factor of the circuit is $\frac{1}{\sqrt{2}}$
 (D) Voltage drop across resistance = voltage drop across inductor

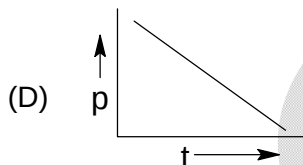
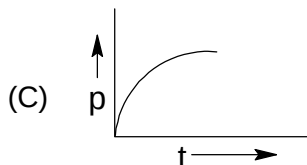
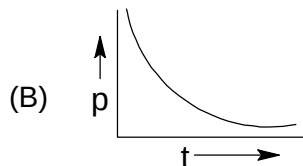
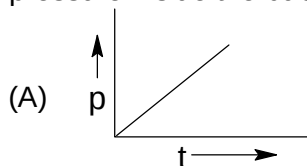

- The mass of a body is measured as 10.1 kg . The possible percentage error in the measurement is

(A) $\pm 1\%$ (B) $\pm 0.1\%$
 (C) $\pm 10\%$ (D) $\pm 0.01\%$
- Referring to the shown circuit, the current will be minimum in

(A) a
 (B) b
 (C) c
 (D) same in all the branches



6. A soap bubble is blown slowly at the end of a tube by a pump supplying air at a constant rate. Which one of the following graphs represents the correct variation of the excess of pressure inside the bubble with time?



7. There is a 90° bend in a pipe line in which water is flowing at speed v . The cross sectional area of pipe is A . The force exerted by pipe walls on water at bend is

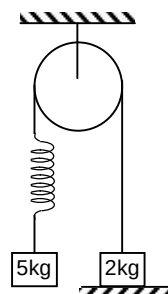
- (A) $\sqrt{3} A v^2 \rho$ (B) $\sqrt{2} A v^2 \rho$
(C) $A v^2 \rho$ (D) $\sqrt{5} A v \rho$

8. A uniform solid hemisphere of radius r is joined to uniform solid right circular cone of base of radius r . Both have same density. The centre of mass of the composite solid lies on the common face. The height (h) of the cone is

- (A) $2r$ (B) $\sqrt{3} r$
(C) $3r$ (D) $r\sqrt{6}$

9. System shown in figure is released from rest pulley and spring is massless and friction is absent everywhere. The speed of 5 kg block when 2 kg block leaves the contact with ground is ($K = 40 \text{ N/m}$ & $g = 10 \text{ m/s}^2$)

- (A) $\sqrt{2} \text{ m/s}$ (B) $2\sqrt{2} \text{ m/s}$
(C) 2 m/s (D) $4\sqrt{2} \text{ m/s}$

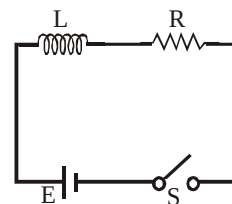


10. One half mole of a monoatomic ideal gas absorbs 1200 J of heat energy while performing 2196 J of work. The temperature of the gas increases by

- (A) 160°C (B) 80°C
(C) 40°C (D) -160°C

11. In the LR circuit as shown in figure, the switch is closed at $t = 0$. Mark the incorrect statement of the following:

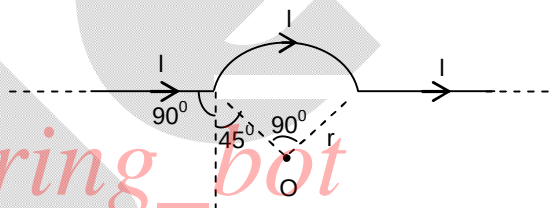
- (A) The current increases with time
(B) The current increases with a decreasing rate
(C) The voltage drop across the inductor increases
(D) The voltage drop across the resistor increases



12. A charged particle of unit mass and unit charge moves with velocity of $\vec{v} = (8\hat{i} + 6\hat{j})\text{m/s}$ in a magnetic field of $\vec{B} = 2\hat{k}\text{T}$. Choose the correct alternative
- (A) The path of the particle may be $x^2 + y^2 - 4x - 21 = 0$
- (B) The path of the particle may be $x^2 + y^2 - 4x - 5 = 0$
- (C) The path of the particle may be $x^2 + y^2 + 2x - 8 = 0$
- (D) The path of the particle may be $x^2 + y^2 - 2x - 15 = 0$

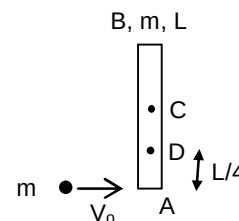
13. A source of sound is moving towards a stationary observer with a speed of 50 m/s. The observer measures the frequency of the source as 1000 Hz. The speed of sound is 350 m/s. The apparent frequency measured by the observer when the source is moving away after crossing the observer is
- (A) 750 Hz (B) 850 Hz
- (C) 1150 Hz (D) 1250 Hz

14. The magnetic field at the centre O of the arc in figure is (r is the radius of circular arc)



- (A) $\frac{\mu_0 I}{4\pi \times r} [\sqrt{2} + \pi]$ (B) $\frac{\mu_0 I}{2\pi r} \left[\frac{\pi}{4} + 1(\sqrt{2} - 1) \right]$
- (C) $\frac{\mu_0}{4\pi} \times \frac{I}{r} [\sqrt{2} + \pi]$ (D) $\frac{\mu_0}{4\pi} \times \frac{I}{r} \left[\sqrt{2} + \frac{\pi}{4} \right]$

15. A thin, uniform rod of mass M and length L is at rest on a smooth horizontal surface. A particle of same mass M collides with the rod at one end perpendicular to its length with velocity V_0 and sticks to it. C is the middle point of the rod and D is a point at a distance of L/4 from end A. The time taken by the rod to turn through 90° after collision is



- (A) $\frac{L}{V_0}$ (B) $\frac{\pi L}{V_0}$
- (C) $\frac{3\pi L}{5V_0}$ (D) $\frac{5\pi L}{12V_0}$
16. A string of length 1m and linear mass density 0.01 kg m^{-1} is stretched to a tension of 100 N. When both ends of the string are fixed, the three lowest frequencies for standing wave are f_1 , f_2 and f_3 . When only one end of the string is fixed, the three lowest frequencies for standing wave are n_1 , n_2 and n_3 . Then
- (A) $n_3 = 5n_1 = f_3 = 125\text{ Hz}$ (B) $f_3 = 5f_1 = n_2 = 125\text{ Hz}$
- (C) $f_3 = n_2 = 3f_1 = 150\text{ Hz}$ (D) $n_2 = \frac{f_1 + f_2}{2} = 75\text{ Hz}$

17. A simple pendulum is suspended from the ceiling of the lift. When the lift is at rest, its time period is T . With what acceleration should lift be acceleration upwards in order to reduce its time period to $T/2$?

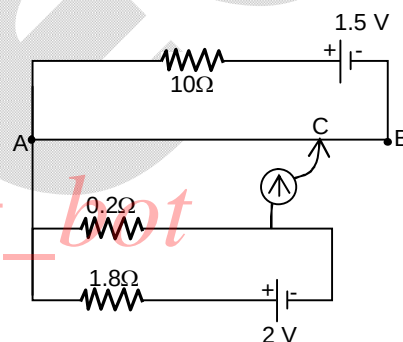
(A) $9g$ (B) g
(C) $2g$ (D) $3g$

18. A tunnel is dug along diameter inside the surface of earth and a particle is projected from the centre of tunnel. The minimum velocity of particle such that it escape out from the earth's gravitation field is (Radius of earth = R_e)

(A) $\sqrt{2gR_e}$ (B) $\sqrt{\frac{3}{2}gR_e}$
(C) $\sqrt{3gR_e}$ (D) $\sqrt{\frac{5}{2}gR_e}$

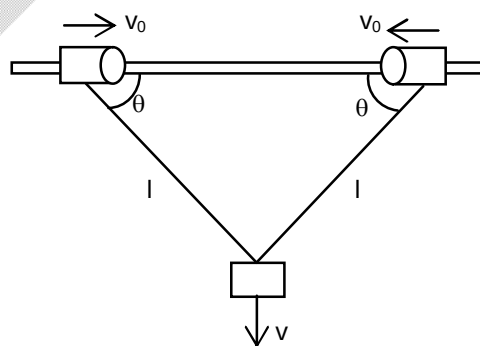
19. In the figure, AB is a uniform wire of length 100 cm whose resistance is $5\ \Omega$. If galvanometer reads zero, then AC =

(A) 10cm (B) 20cm
(C) 40cm (D) 80cm



20. In the shown figure two identical beads slide along a smooth horizontal rod as shown in figure. The relation between v and v_0 in the shown position will be

(A) $v = v_0 \cot \theta$
(B) $v = v_0 \sin \theta$
(C) $v = v_0 \tan \theta$
(D) $v = v_0 \cos \theta$



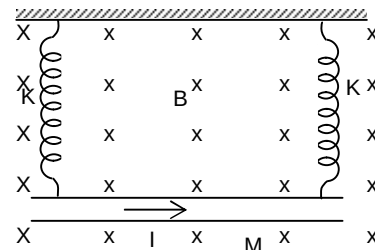
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. K_α wavelength emitted by an atom of atomic number $Z = 11$ is λ . Find the atomic number for an atom that emits K_α radiation with wavelength 4λ .

22. A metal rod of mass 10gm and length 25 cm is suspended on two springs as shown in figure. The springs are extended by 4 cm. When a 20 ampere current passes through the rod it rises by 1 cm. The magnetic field is $x \times 10^{-2}$ T ($g = 10 \text{ m/s}^2$). Find the value of 10x.



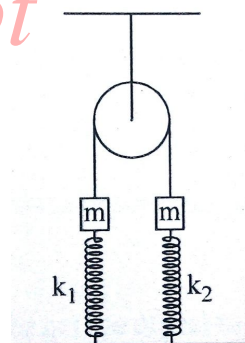
23. In a certain hypothetical radioactive decay process, species A decays into species B and species B decays into species C according to the reactions



The decay constant for species A is $\lambda_1 = 1 \text{ sec}^{-1}$ and that for species B is $\lambda_2 = 100 \text{ sec}^{-1}$. Initially 10^4 moles of species of A were present while there was none of B and C. It was found that species B reaches its maximum number at a time $t_0 = 2 \ln(10) \text{ sec}$. If maximum number of moles of B is N, then $10N$ is:

24. In YDSE experiment if the screen is shifted by a distance of 0.5 m away from the slit, 3rd maxima is shifted by 3×10^{-4} m. The slit width is 2×10^{-3} m. If the wavelength (in nm) used in the experiment is 'k', find the value of 'k'.

25. On the diagram, two blocks of equal mass are connected by an ideal string. The values of $m = 1 \text{ kg}$ and $k_2 = 100 \text{ N/m}$ are given. Initially, both springs are relaxed. Then the left block is slowly pulled down a distance 0.1 m and released. The maximum possible value of k_1 (in N/m) for which both blocks will have same magnitude of acceleration just after releasing, is k . Both spring have natural length of 2m . Then find the value of k .

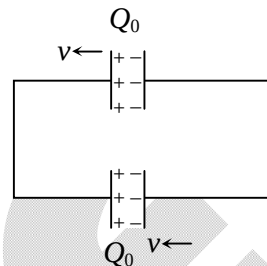


26. In YDSE distance between the slits plane and screen is 1m and distance between two slits is 5 mm. If slabs of thickness 2mm and 1.5mm having refractive index 1.5 and 1.4 are placed in front of two slits, find the shift (in mm) of central maxima
27. Suppose potential energy between electron and proton at separation r is given by $U = k \ln r$, where k is constant. For such hypothetical hydrogen atom, find the ratio of energy difference between energy levels ($n = 1$ and $n = 2$) and ($n = 2$ and $n = 4$) is

28. A concave mirror of focal length 15 cm is placed in air at a height of 45 cm above the water surface as shown. The axis of the mirror is vertical and the reflecting surface faces the water surface. A point object is placed on the axis of concave mirror at a distance 35 cm above the water surface. Find the position of image formed after two successive processes- first reflection from the concave mirror and then first refraction from the water (refractive Index of water $\frac{4}{3}$). The distance of image from water surface (in cm) is



29. A point mass of 1 kg collides elastically with a stationary point mass of 5 kg. After their collision, the 1 kg mass reverses its direction and moves with a speed of 2 ms^{-1} . Find the kinetic energy (in mJ) of a particle of mass 6 kg moving with same velocity as that of centre of mass of the above two particles of mass 1 kg and 5 kg.
30. Two identical capacitor connected as shown and having initial charge Q_0 . Separation between plates of capacitor is d_0 . Suddenly the left plate of upper capacitor and right plate of lower capacitor start moving with speed v towards left while other plate of capacitor remains fixed. (given $\frac{Q_0 v}{2d} = 1$ amp). Find the value of current (in amp) in the circuit.



TG ~ @bohring_bot

Chemistry

PART – B

SECTION – A

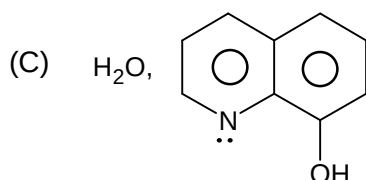
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

31. Molar heat capacity of an ideal gas at constant volume is given by $C_v = 2 \times 10^{-2}$ J (in Joule). If 3.5 mole of this ideal gas are heated at constant volume from 300 K to 400 K the change in internal energy will be
(A) 7 (B) 4
(C) 8 (D) 6
32. The compound X – CH_2COOH will be the strongest acid if X is:
(A) CH_3 (B) NO_2
(C) F (D) OH
33. $\text{BCl}_3 + \text{H}_2 + \text{C} \xrightarrow{1700-1800^\circ\text{C}}$ Product(s)
The product(s) of above reaction is/are
(A) $\text{CCl}_4 + \text{B}_2\text{H}_6$ (B) $\text{B}_4\text{C} + \text{HCl}$
(C) $\text{B} + \text{Cl}_2 + \text{CH}_4$ (D) $\text{B}_2\text{H}_6 + \text{CHCl}_3$
34. Atomic radii of Ti, Zr and Hf vary
(A) $\text{Ti} > \text{Zr} > \text{Hf}$ (B) $\text{Ti} < \text{Zr} < \text{Hf}$
(C) $\text{Ti} < \text{Hf} < \text{Zr}$ (D) $\text{Ti} < \text{Zr} = \text{Hf}$
35. Which of the following hydrocarbon can form one monochloro product with Cl_2/light ?
(A) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH} - \text{CH}_3 \\ | \\ \text{H}_3\text{C} \end{array}$ (B) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
(C) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$ (D) $\text{CH}_3\text{CH}_2\text{CH}_3$
36. Dehydration of alcohols produces
(A) alkanes (B) alkenes
(C) alkynes (D) aldehydes
37. The shape of $[\text{Co}(\text{NH}_3)_6]^{3+}$ is:
(A) Tetrahedral (B) Trigonal bipyramidal
(C) Octahedral (D) Square planar

38. AgNO_3 solution will not give white precipitate with:
 (A) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (B) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
 (C) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ (D) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
39. Which of the following compound exhibits tautomerism?
 (A) chloroethane (B) ethanol
 (C) ethoxyethane (D) nitroethane
40. Which of the following two reagents form silicates when react with silica (SiO_2)?
 (A) NaCl and Na_2S (B) NaOH and Na_2CO_3
 (C) Na_2SO_4 and NaNO_3 (D) NaBr and NaHSO_4

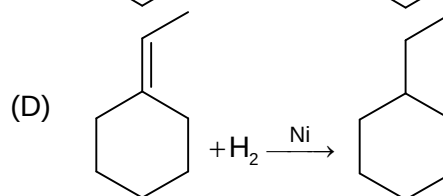
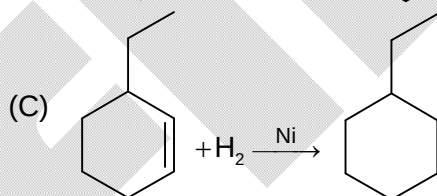
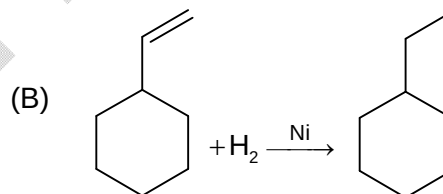
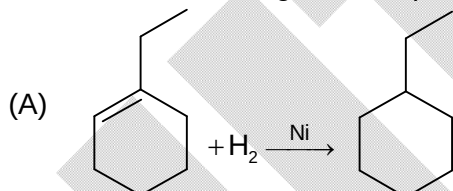
41. In which combinations, all the ligands are ambient in nature?
 (A) NO_2 , SCN and CN (B) EDTA , SCN



(D) All of these

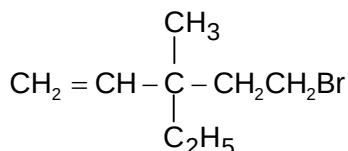
42. $\text{CH}_3\text{CH}=\text{CH}-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}=\text{CHCH}_3$
 How many stereoisomer(s) is/are possible for the above compound?
 (A) 2 (B) 3
 (C) 4 (D) 6

43. Which of the following reaction produces maximum heat?



44. Which of the following substance can decrease the ionization of CH_3COOH in water?
 $\text{CH}_3\text{COOH}(\text{l}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 (A) NaOH (B) NH_4Cl
 (C) K_2CO_3 (D) NaCl

45.



Which of the following substance converts the above optically active compound to optically inactive compound?

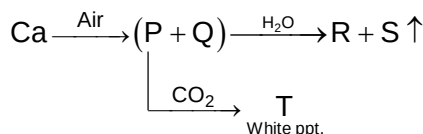
- (A) HBr (B) Br₂/CCl₄
(C) HBr/R₂O₂ (D) Br₂/H₂O

46.

Which substance is used for refining of Nickel in Mond's process?

- (A) I₂ (B) CO
(C) CO₂ (D) MnO

47.



Which of the following is 'U'?

- (A) CH₃COONH₄ (B) NH₂CONH₂
(C) NH₂COONH₄ (D) NH₂COOH

48.

Which of the following orbital has the maximum number of radial nodes?

- (A) 4s (B) 4p
(C) 4d (D) 4f

49.

In which case bond length of C – O bond is maximum

- (A) Carboxylate ion (B) Phenoxide ion
(C) p-nitro phenoxide ion (D) 2, 4 di-nitro phenoxide ion

50.

Which of the following overlap forms the strongest sigma bond?

- (A) 2s – 2s (B) 2p_x – 2p_x
(C) 3p_y – 3p_y (D) 2p_x – 3p_x

SECTION – B

(Numerical Answer Type)

This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51.

The energy of an electron of hydrogen atom is -13.6 eV when it is 0.53 Å away from the nucleus. When the hydrogen atom is supplied with 12.75 eV energy the electron moves to a higher orbit. If the distance of that orbit from the nucleus in is expressed as $\frac{x}{100} \text{ Å}$. What is the value of x?

52.

56.7 g anhydrous oxalic acid (H₂C₂O₄) completely decolourises 200 mL of acidified KMnO₄ solution. If the molarity of the KMnO₄ solution is expressed as $\left(\frac{20+x}{100}\right) \text{ M}$. What is the value of x?

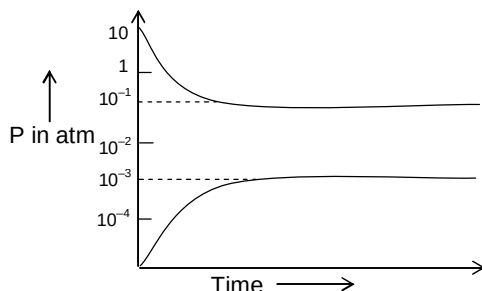
53. A large number of structural isomers are possible with formula $C_4H_{11}N$

If x = Number of primary amines,
 y = Number of secondary amines and
 z = number of tertiary amines, then

What is $(x + y + z)$

54. The temperature of a sample of diatomic ideal gas is increased by 4.5 times of its initial value. At this temperature the molecules are completely dissociated into atoms. How many times will the gas velocity increase from initial value due to the change?

55.



The plot of partial pressure of the product and reactant is given above. What is the value of ΔG° in L atm unit at temperature 'T'?

[Assume $2.303 RT = 4 \text{ L atm mol}^{-1}$]

56. AB_2 is 10% dissociated in water to A^{2+} and B^- . The boiling point of a 10.0 molal aqueous solution of AB_2 is _____ $^\circ\text{C}$. (Round off to the nearest integer). [Given: Molal elevation constant of water $K_b = 0.5 \text{ K kg mol}^{-1}$ boiling point of pure water = 100°C]
57. At 363 K, the vapour pressure of A is 21 kPa and that of B is 18 kPa. One mole of A and 2 moles of B are mixed. Assuming that this solution is ideal, the vapour pressures of mixture is _____ kPa. (Round off to the nearest integer)
58. A certain element crystallises in a bcc lattice of unit cell edge length $27/\text{AA}$. If the same element under the same conditions crystallizes in the fcc lattice, the edge length of the unit cell in AA will be _____ (Round off to the nearest integer)
 [Assume each lattice point as a single atom]
 [Assume $\sqrt{3} = 1.73$, $\sqrt{2} = 1.41$]
59. When light of wavelength 248 nm falls on a metal of threshold energy 3.0 eV, the de-Broglie wavelength of emitted electrons is _____ $\text{\AA}$.
 (Round off to the nearest integer)
 [Use: $\sqrt{3} = 1.73$, $h = 6.63 \times 10^{-34} \text{ Js}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$, $c = 3.0 \times 10^8 \text{ ms}^{-1}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$]
60. A certain orbital as $n = 4$ and $m_l = -3$. The number of radial nodes in this orbital is _____ (Round off to the nearest integer)

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. The area bounded by the x – axis, part of the curve $y = 1 + x^{-2}$ and the ordinates $x = 1$ and $x = 2$ is divided into equal parts by the ordinate at $x = a$ such that $a = \frac{3 + \sqrt{p}}{8}$, then the value of p is
 (A) 70 (B) 71
 (C) 72 (D) 73
62. $\lim_{y \rightarrow \infty} \frac{\sqrt{1 + \sqrt{1 + y^4}} - \sqrt{2}}{y^4} =$
 (A) $4\sqrt{2}$ (B) $\frac{1}{4\sqrt{2}}$
 (C) $\sqrt{2}$ (D) 0
63. If the parabola $y = ax^2 + bx + c$ has vertex at $(4, 2)$ and $\lambda \in [1, 3]$, and the maximum value of product abc is λ , then $\frac{|\lambda|}{24}$ is
 (A) 2 (B) 4
 (C) 6 (D) 8
64. Let P be a point on the line segment joining $A(5\cos\alpha, 5\sin\alpha)$ and $B(5\cos\beta, 5\sin\beta)$ such that $3PA = 2PB$ then the locus of P is if $|\alpha - \beta| = \frac{\pi}{3}$
 (A) $x^2 + y^2 = 13$ (B) $x^2 + y^2 = 19$
 (C) $x^2 + y^2 = 1$ (D) $x^2 + y^2 = 25$
65. Let z_1, z_2 and z_3 are three complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ and $\frac{z_1}{z_2 z_3} + \frac{z_2}{z_3 z_1} + \frac{z_3}{z_1 z_2} = -1$
 then the number of possible integral values of $|z_1 + z_2 + z_3|$ is
 (A) 3 (B) 2
 (C) 1 (D) 4

66. Let P be any point on the line $x - y + 3 = 0$ and A be a fixed point (3, 4). If the family of lines given by the equations $(3 \sec \theta + 5 \operatorname{cosec} \theta)x + (7 \sec \theta - 3 \operatorname{cosec} \theta)y + 11(\sec \theta - \operatorname{cosec} \theta) = 0$ are concurrent at a point B for all permissible value of θ , then maximum value of $|PA - PB|$ is
- (A) 3 (B) $2\sqrt{10}$
(C) $2\sqrt{34}$ (D) 5
67. Let $P(x)$ be a quadratic polynomial such that $P(1) = -1$, if $\int \frac{p(x) dx}{(2x-3)^2(3x-2)^2} = \frac{1}{5} \ln|f(x)| + C$ where C is constant of integration and $f(x)$ is a rational function such that $\lim_{x \rightarrow \infty} f(x) = \frac{4}{3}$, then value of $|f(1)|$ is
- (A) 1 (B) 2
(C) 3 (D) 4
68. Let f be a function defined by $f(x) = \begin{cases} (x-r)^2, & r-1 \leq x < r+1 \\ 1, & r+1 \leq x \leq r+2 \end{cases}$ Where $r = 3k$, $k \in \mathbb{I}$. Find the value of $\sqrt{\int_0^{45} f(x) dx}$
- (A) 1 (B) 3
(C) 5 (D) 7
69. If $f(x)$ is an even function which is also periodic with the period T and $\int_0^a f(x) dx = 3$ and $\int_{-T/2}^{3T/2} f(x) dx = 18$, then $\int_{-a}^{a+5T} f(x) dx$ is equal to
- (A) 31 (B) 43
(C) 51 (D) 71
70. Let $f(x) = 2 \tan^{-1} x$ and $g(x)$ be a differentiable function satisfying $g\left(\frac{x+2y}{3}\right) = \frac{g(x) + 2g(y)}{3} \forall x, y \in \mathbb{R}$ and $g'(0) = 1$ and $g(0) = 2$. The number of integers 'x' satisfying $f^2(g(x)) - 5f(g(x)) + 4 > 0$ where $x \in (-10, 20)$ is greater than or equal to
- (A) 5 (B) 6
(C) 7 (D) 8
71. Area bounded by the straight lines $x^2y - y^3 - x^2 + 5y^2 - 8y + 4 = 0$ is equal to (in sq. units)
- (A) 1 (B) 3
(C) 5 (D) 7

72. If $(2xy - y^2 - y)dx = (2xy + x - x^2)dy$ and $y(1) = 1$, then the value of $12|y(-1)|$ is
 (A) 12 (B) 13
 (C) 15 (D) 17
73. The point (2, 1) is translated parallel to the line $L : x - y = 4$ by $2\sqrt{3}$ units. If the new point Q lies in the third quadrant, then the equation of the line passing through Q and perpendicular to L is
 (A) $x + y = 2 - \sqrt{6}$ (B) $x + y = 3 - 3\sqrt{6}$
 (C) $x + y = 3 - 2\sqrt{6}$ (D) $2x + 3y = 1 - \sqrt{6}$
74. A circle C touches the line $x = 2y$ at the point (2, 1) and intersects the circle $C_1 : x^2 + y^2 + 2y - 5 = 0$ at two points P and Q such that PQ is a diameter of C_1 . Then the length of diameter of C is:
 (A) $7\sqrt{5}$ (B) 15
 (C) $\sqrt{285}$ (D) $4\sqrt{15}$
75. Three circles of radii a, b, c ($a < b < c$) touch each other externally. If they have x - axis as a common tangent, then
 (A) a, b, c are in A.P. (B) $\frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{c}}$
 (C) $\frac{1}{\sqrt{a}} = \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}}$ (D) $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are in A.P.
76. If pair of variable straight lines $x^2 + 4y^2 + \alpha xy = 0$ (α is real parameter) cuts the ellipse $x^2 + 4y^2 = 4$ at two points A and B, then the locus of point of intersection of tangents at A and B of ellipse
 (A) $x^2 - 4y^2 = 0$ (B) $4x^2 - y^2 = 0$
 (C) $x^2 - xy + 4y^2 = 0$ (D) $x^2 - xy - 4y^2 = 0$
77. If $\arg \frac{(z - (10 + 6i))}{(z - (4 + 2i))} = \frac{\pi}{4}$ then the perimeter of locus of z is:
 (A) $\sqrt{13} \left(\frac{\pi}{4} \right)$ (B) $3\sqrt{13} \pi$
 (C) $3\sqrt{13} \left(\frac{\pi}{4} \right)$ (D) $3\sqrt{26} \left(\frac{\pi}{2} \right)$

78. The sum of all values of $\theta \in \left(0, \frac{\pi}{2}\right)$ satisfy $\sin^2 2\theta + \cos^4 2\theta = \frac{3}{4}$ is

- (A) $\frac{\pi}{2}$ (B) π
 (C) $\frac{3\pi}{8}$ (D) $\frac{5\pi}{4}$

79. The value of $\frac{1}{3^2+1} + \frac{1}{4^2+2} + \frac{1}{5^2+3} + \dots$ to ∞ is?

- (A) $\frac{31}{13}$ (B) $\frac{13}{36}$
 (C) $\frac{31}{36}$ (D) $\frac{1}{36}$

80. Let $x = \sin 1^\circ$, then the value of the expression

$$\frac{1}{\cos 0^\circ \cos 1^\circ} + \frac{1}{\cos 1^\circ \cos 2^\circ} + \frac{1}{\cos 2^\circ \cos 3^\circ} + \dots + \frac{1}{\cos 44^\circ \cos 45^\circ}$$

is equal to

(A) x (B) $\frac{1}{x}$
 (C) $\frac{\sqrt{2}}{x}$ (D) $\frac{x}{\sqrt{2}}$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If $\binom{40}{0} + \binom{41}{1} + \binom{42}{2} + \dots + \binom{60}{20} = \frac{m}{n} \binom{60}{20}$, m and n are coprime, then maximum value of ' $m+n$ ' is equal to

82. Let $P(A \cap B) = \frac{1}{4}$ and $P(B) = \frac{1}{3}$, where $P\left(\frac{A \cap B}{A \cup B}\right) = \frac{k}{22}$, then ' k ' is equal to

83. The line which contains all points (x, y, z) which are of the form $(x, y, z) = (2, -3, 4) + \lambda(1, 2, -3)$ intersects the plane $2x + 3y - z = 13$ at the point P and intersects the yz – plane at Q . If the distance between P and Q is $a\sqrt{b}$, where $a, b \in \mathbb{N}$ and $a > 3$, then $\frac{a+b}{3}$ equals

84. Let S be the set $(2, 3, 5, 7, 11)$. Suppose A and B are two matrices of order 2 each with distinct elements belonging to set S . The probability that the matrix AB has at least one odd entry is

85. If $\cos^{-1}\sqrt{p} + \cos^{-1}\sqrt{1-p} + \cos^{-1}\sqrt{1-q} = \frac{3\pi}{4}$ then $q =$
86. Let $\sum_{i=0}^{100} \sum_{j=0}^{100} (C_i^2 + C_j^2 + C_i C_j) = a \cdot {}^{2n}C_n + 2^b$, where a, b, n are even natural numbers and C_r , represent the coefficient of x^r in the expansion of $(1+x)^{100}$ then $a + b + n = C_r$ represent
87. If maximum value of quadratic equation $y = ax^2 + bx + c$ is $\frac{1}{2}$ at $x = -2$, then $2c - 8a + 5$ is
88. The sum of possible value(s) of p for which at least one root of equation $x^4 - 8x^3 + 22x^2 - 24x + p = 0$ is of multiplicity 2, is
89. The sum of the following infinite series $1 + \left(\frac{\sqrt{2}-1}{2\sqrt{2}}\right) + \left(\frac{3-2\sqrt{2}}{8}\right) + \left(\frac{5\sqrt{2}-7}{16\sqrt{2}}\right) + \frac{17-12\sqrt{2}}{64} + \dots \infty$ is
90. If $2\tan^2 x - 5\sec x - 1 = 0$ has 7 different roots in $\left[0, \frac{n\pi}{2}\right]$, $n \in \mathbb{N}$, then greatest value of n is

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Main)-2023

TEST DATE: 11-01-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C
Sol. Variation of moment of inertia with temperature
 $I = I_0 (1 + 2\alpha \Delta T)$
C.O.A.M:
 $I_i \omega_i = I_f \omega_f$
2. D
Sol. Upon earthing, final potential of conductor becomes zero. Final charge may or may not be zero.
3. B
Sol. $\tan \theta = \frac{X_c - X_L}{X_R}$
and $\cos \theta = \text{power factor.}$
4. A
Sol. $\Delta m = 0.1 \text{ kg} ; \frac{\Delta m}{m} = \frac{0.1}{10}$
 $\therefore \text{Error} = \pm \frac{\Delta m}{m} \times 100 = \pm 1\%.$

5. A

Sol. $I_a = \frac{3V}{15R}, I_b = \frac{3V}{5R}; I_c = \frac{6V}{15R}.$

6. C

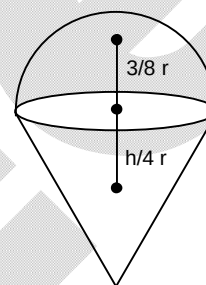
Sol. At first the pressure will increase rapidly and then it will become constant.

7. B

Sol. $F = \frac{dp}{dt} = (\sqrt{2} V) \frac{dm}{dt} = (\sqrt{2} V \rho) \frac{d(\text{volume})}{dt} = (\sqrt{2} V \rho) (AV)$

8. B

Sol. $\frac{3}{8}r \cdot \rho \frac{2}{3}\pi r^3 = \frac{h}{4} \cdot \rho \frac{1}{3}\pi r^2 h$
 $\Rightarrow h = \sqrt{3}r$



9. B

Sol. $Kx = 2 \times g = 20, x = \frac{1}{2}m$

$$5gx - \frac{1}{2}Kx^2 = \frac{1}{2}5V^2$$

$$V = 2\sqrt{2} \text{ m/s}$$

10. D

Sol. $\Delta q = \Delta u + \Delta w$

$$\Rightarrow \Delta U = -996 \text{ J} = \frac{nR}{\gamma - 1} \Delta T$$

$$\gamma = \frac{5}{3} \text{ for mono-atomic gas.}$$

11. C

Sol. $i = \frac{E}{R} \left(1 - e^{-\frac{tR}{L}} \right)$

12. A

Sol. $r = \frac{mv}{qB} = \frac{1 \times 10}{1 \times 2} = 5m$ And motion of particle will be in x – y plane.

13. A

Sol. $1000 = \left(\frac{350}{350 - 50} \right) f \dots (i)$

$$f_1 = \left(\frac{350}{350 + 50} \right) f \quad \dots(ii)$$

$$f_1 = 750 \text{ Hz}$$

14. B

$$\text{Sol. } B = 2 \left[\frac{\mu_0 I}{4\pi r \cos 45^\circ} \right] (\sin 90^\circ - \sin 45^\circ) + \frac{\mu_0 I}{4\pi r} \left(\frac{\pi}{2} \right)$$

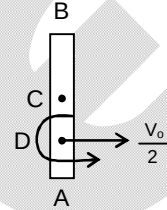
15. D

Sol. By COAM about point D (ie about new centre of mass)

$$MV_o \frac{L}{4} = \left[\frac{ML^2}{12} + \frac{ML^2}{16} + \frac{ML^2}{16} \right] \omega$$

$$\text{or, } \omega = \frac{6v_o}{5L}$$

$$\text{Now, } t = \frac{\theta}{\omega} = \frac{(\pi/2)}{\omega} = \frac{5\pi L}{12v_o}$$



16. D

$$\text{Sol. } f = \frac{m}{2\ell} \sqrt{\frac{T}{\mu}} ; n = \frac{2m+1}{4\ell} \sqrt{\frac{T}{\mu}} \quad \text{--- } @bohrring_bot$$

17. D

$$\text{Sol. } T = 2\pi \sqrt{\frac{\ell}{g_{\text{eff}}}} = 2\pi \sqrt{\frac{\ell}{g+a}}$$

18. C

$$\text{Sol. Potential energy of particle at the centre of earth is } U = -\frac{3}{2} \frac{GMm}{R_e}$$

$$\text{So, } V_e = \sqrt{\frac{3GM}{R_e}} = \sqrt{3gR_e}$$

19. C

$$\text{Sol. } V_{AC} = 2 \times \left(\frac{0.2}{1.8 + 0.2} \right) \quad \dots(i)$$

$$\text{Also, } V_{AC} = (1.5) \left(\frac{5}{15} \right) \left(\frac{\ell}{100} \right) \quad \dots(ii)$$

20. A

$$\text{Sol. } V_o \cos \theta = V \sin \theta$$

SECTION – B

21. 6

Sol. $\sqrt{f_1} = \sqrt{\frac{v}{\lambda_1}} = a(11-1)$ and $\sqrt{f_2} = \sqrt{\frac{v}{\lambda_2}} = a(Z-1)$

By dividing, $\sqrt{\frac{\lambda_2}{\lambda_1}} = \frac{10}{Z-1} \Rightarrow \sqrt{\frac{4}{1}} = \frac{10}{Z-1}$

$\Rightarrow Z = 6.$

22. 15

Sol. Initially the rod will be in equilibrium if

$2T_0 = Mg$ with $T_0 = kx_0$... (i)

when the current I is passed through the rod, it will experience a force

$F = BIL$ vertically up,

In equilibrium

$2T + BIL = Mg$ with $T = kx$... (ii)

from (i) & (ii)

$\frac{T}{T_0} = \frac{Mg - BIL}{Mg}$ i.e. $\frac{x}{x_0} = 1 - \frac{BIL}{Mg}$

or, $B = \frac{Mg(x_0 - x)}{ILx_0}$

Putting the values we get $B = 1.5 \times 10^{-2} T$.

23. 20

Sol. $\frac{dN_A}{dt} = -\lambda_1 N_A$, $\frac{dN_B}{dt} = 2\lambda_1 N_A - \lambda_2 N_B$,

$N_B = \text{maximum} \Rightarrow \frac{dN_B}{dt} = 0$

$\Rightarrow 2\lambda_1 N_A = \lambda_2 N_{B_{\max}}$

$\Rightarrow N_{B_{\max}} = \frac{2\lambda_1}{\lambda_2} N_A$

$\Rightarrow N_{B_{\max}} = \frac{2\lambda_1}{\lambda_2} N_0 e^{-\lambda_1 t} = 2.$

24. 400

Sol. Here 3rd maxima is shifted by 3×10^{-4} m. It indicates fringe width increases by 1×10^{-4} m.

Hence $\beta = \frac{\lambda(D+0.5)}{d} = \frac{\lambda D}{d} + 1 \times 10^{-4}$

or $\frac{0.5\lambda}{d} = 1 \times 10^{-4}$ m or $\lambda = \frac{2 \times 10^{-3} \times 1 \times 10^{-4}}{0.5} = 4 \times 10^{-7}$ m = 400 nm

25. 300

Sol. Let T be the tension in the ideal string and 'a' be the acceleration of the blocks at the instant of release. For the block on the left, the upward acceleration may be found from

$$T + k_1x - mg = ma$$

For the block on the right, the downward acceleration may be found from

$$k_2x + mg - T = ma$$

Adding the equations gives the acceleration of the blocks as

$$a = (k_1 + k_2)x/(2m)$$

However, subtracting the equations gives

$$T = mg - (k_1 - k_2)x/2$$

for maximum value of k_1T will be zero.

$$mg = \left(\frac{k_1 - k_2}{2} \right) x \quad ; \quad k_1 = 300.$$

26. 80

Sol. $D = 1\text{m}$, $d = 5 \times 10^{-3}\text{m}$

$$\text{shift} = \frac{D}{d} [t_1(\mu_1 - 1) - t_2(\mu_2 - 1)]$$

$$= \frac{1}{5 \times 10^{-3}} [2(1.5 - 1) - 1.5(1.4 - 1)] \times 10^{-3} = \frac{0.4}{5} = 0.08\text{m}$$

27. 1

Sol. $F = -\frac{dU}{dr} = -\frac{k}{r}, \quad \frac{k}{r} = \frac{mv^2}{r} \quad \dots (i)$

$$E_n = \frac{1}{2}mv^2 + k \ln r \quad \dots (ii)$$

$$mvr = \frac{nh}{2\pi} \quad \dots (iii)$$

$$\text{Solving these } E_n = \frac{k}{2} \left(1 + \ln \left(\frac{n^2 h^2}{4\pi^2 m k} \right) \right)$$

$$\text{required ratio} = \frac{E_2 - E_1}{E_4 - E_2} = 1.$$

28. 100

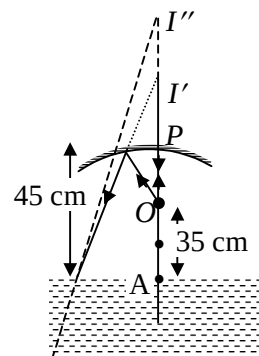
Sol. For the reflection at the concave mirror,

$$u = -10 \text{ cm}; \quad v = ?; \quad f = -15 \text{ cm}$$

From the mirror formula, we have

$$v = \frac{uf}{u - f} = \frac{(-10) \times (-15)}{-10 + 15} = \frac{150}{5} = +30 \text{ cm}$$

The positive sign indicates that the image is formed on the other side of the concave mirror,



Now, the image formed by the concave mirror serves as a virtual object for refraction at water surface which takes place from air to water. So,

$$\mu = \frac{\text{Apparent height}}{\text{Real height}}$$

$$\therefore \text{AI}'' = \text{Apparent height} = \mu \times \text{real height} = \frac{4}{3} \times 75 = 100 \text{ cm.}$$

29. 750

Sol. $e = \frac{v_2 - v_1}{u_1 - u_2}$

$$1 = \frac{v_2 - (-2)}{u_1}$$

$$u_1 = v_2 + 2$$

$$u_1 = 1(-2) + 5(u_1 - 2)$$

$$u_1 = -2 + 5u_1 - 10$$

$$u_1 = \frac{12}{4} = 3 \text{ m/s}$$

$$v_2 = 1 \text{ m/s}$$

$$\text{Kinetic energy of the centre of mass} = \frac{1}{2} \times (1+5) \times \left(\frac{3}{1+5} \right)^2 = \frac{3}{4} \text{ J}$$

30. 2

Sol. $\frac{q_1}{C_1} = \frac{q_2}{C_2}$; $q_1 + q_2 = 2Q_0$

$$C_1 = \frac{\epsilon_0 A}{d_0 + vt}; C_2 = \frac{\epsilon_0 A}{d_0 - vt}$$

$$\frac{q_1}{q_2} = \frac{d_0 - vt}{d_0 + vt}$$

$$q_2 \left(\frac{d_0 - vt}{d_0 + vt} \right) + q_2 = 2Q_0$$

$$q_2 \left[\frac{2d_0}{d_0 + vt} \right] = 2Q_0$$

$$q_2 = \frac{2Q_0}{2d_0} (d_0 + vt)$$

$$I = \frac{dq_2}{dt} = \frac{Q_0 v}{d_0} = 2 \text{ amp}$$

Chemistry

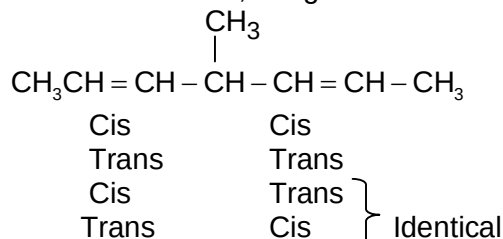
PART – B

SECTION – A

31. A
Sol. Using
 $\Delta U = nC_V \times \Delta T$
 $\Delta T = 100$
 $C_V = 2 \times 10^{-2}$
 $\Delta U = 7$
 $n = 3.5$
32. B
Sol. NO_2 is the strongest electron withdrawing group among the given groups or atoms.
33. B
Sol. $4\text{BCl}_3 + 6\text{H}_2 + \text{C} \longrightarrow \text{B}_4\text{C} + 12\text{HCl}$
 The reaction takes place in the preparation of bullet proof fabric B_4C .
34. D
Sol. This is due to lanthanoid contraction on 6th period.
35. B
Sol. It is most symmetrical.
36. B
Sol. $\text{RCH}_2\text{CH}_2\text{OH} \xrightarrow[\Delta]{\text{Conc. H}_2\text{SO}_4} \text{R}-\text{CH}=\text{CH}_2 + \text{H}_2\text{O}$
37. C
Sol. Octahedral complex with C.N number 6.
38. A
Sol. No free Cl^- in the ionisation sphere.
39. D
Sol.
40. B
Sol. $\text{SiO}_2 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{SiO}_3 + \text{H}_2\text{O}$
 $\text{SiO}_2 + \text{Na}_2\text{CO}_3 \longrightarrow \text{Na}_2\text{SiO}_3 + \text{CO}_2$
41. A
Sol. Having two possible donating site but donate from one only.

42. C

Sol. Due to double bond, the geometrical isomers are:



When the double bonds are Cis and trans, the molecule becomes optically active.

 $\therefore$ For trans Cis (two optical isomers are possible)

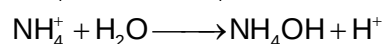
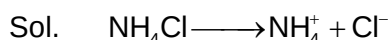
 $\therefore$ Total no. of stereoisomers are:

Geometrical (2) + Optical (2) = 4

43. B

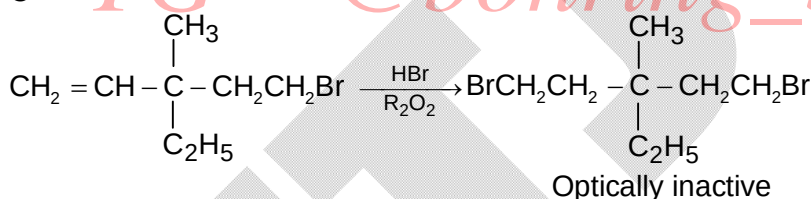
Sol. It is least substituted alkene. Hence it is unstable.

44. B

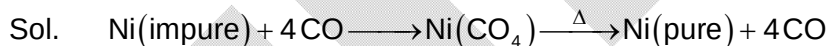

 H^+ shift the ionization reaction toward backward direction due to common ion effect.

45. C

Sol.



46. B



47. B

 Sol. P = CaO, Q = Ca_3N_2 , R = $\text{Ca}(\text{OH})_2$, S = NH_3 , T = CaCO_3 , U = NH_2CONH_2

48. A

 Sol. No. of radial nodes = $(n - l - 1)$

For 4s orbital, it is 3

For 4p orbital, it is 2

For 4d orbital, it is 1

For 4f orbital, it is 0

49. B

Sol. More double bond character, smaller bond length

50. B

Sol. Small size atomic orbitals undergo better overlap than larger atomic orbitals.

SECTION – B

51. 848

Sol. $E_n - E_1 = 12.75$

$$\text{or } \frac{-E_1}{n^2} - E_1 = 12.75 \quad [E_1 = -13.6 \text{ eV (given)}]$$

$$\therefore n = 4$$

$$r_n = \frac{n^2}{Z} a_0 = \frac{16}{1} (0.53) = 8.48 \text{ \AA}$$

$$\therefore \frac{x}{100} = 8.48, x = 848$$

52. 106

Sol. $M_{\text{eq}} \text{ of } \text{KMnO}_4 = M_{\text{eq}} \text{ of } \text{H}_2\text{C}_2\text{O}_4$

$$(V) \times (M)(n) = \frac{W}{E} \times 1000$$

$$\text{Or, } (200)(M)(5) = \frac{56.7}{\frac{90}{2}} \times 1000$$

$$\therefore M = 1.26$$

$$\therefore \frac{20+x}{100} = 1.26 \Rightarrow x = 106$$

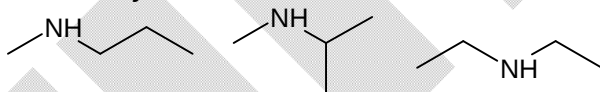
53. 7

Sol. $x = 3, y = 3, z = 1$

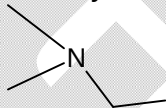
Primary amines are



Secondary amines are



Tertiary amine is



54. 3

Sol. $C_i = \sqrt{\frac{3RT}{M}}$

$$C_f = \sqrt{\frac{3R(4.5)T}{\frac{M}{2}}} = 3\sqrt{\frac{3RT}{M}} = 3C_i$$

You can take any velocity.

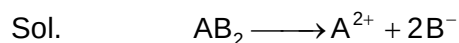
55. 8

Sol. $K_p = \frac{10^{-3}}{10^{-1}} = 10^{-2}$

$$\Delta G^\circ = -2.303 RT \log K_p$$

$$= -2.303 RT \log 10^{-2} = (-4)(2 \log 10) = 8$$

56. 106



$$t=0 \quad a \quad 0 \quad 0$$

$$t=t \quad a - a\alpha \quad a\alpha \quad 2a\alpha$$

$$nT = a - a\alpha + a\alpha + 2a\alpha$$

$$= a(1 + 2a\alpha)$$

$$\text{So, } i = 1 + 2\alpha$$

$$\text{Now } \Delta T_b = i \times m \times K_b$$

$$\Delta T_b = (1 + 2\alpha) \times m \times K_b$$

$$\alpha = 0.1, m = 10, K_b = 0.5$$

$$\Delta T_b = 1.2 \times 10 \times 0.5 = 6$$

$$\text{So boiling point} = 106$$

57. 19

Sol. According to Raoult's law: $P_T = X_A P_A^\circ + X_B P_B^\circ$ $P_T = 1.3 \times 21 + 2.3 \times 18 = 7 + 12$
 $P_T = 19 \text{ kPa.}$

58. 33

Sol. For BCC $\sqrt{3}a = 4r$

$$\text{so, } r = \frac{\sqrt{3}}{4} \times 27$$

$$\text{for FCC } a = 2\sqrt{2}r$$

$$= 2 \times \sqrt{2} \times \frac{\sqrt{3}}{4} \times 27$$

$$= \frac{\sqrt{3}}{\sqrt{2}} \times 27 = 33$$

59. 9

Sol. Incident energy of photon = Work function of metal + K.E. of photoelectron

$$h\nu = h\nu_0 + KE$$

$$\frac{6.63 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{248 \times 10^{-9} \text{ m} \times 1.6 \times 10^{-19} \text{ J eV}^{-1}} = 3.0 + \text{K.E.}$$

$$\text{K.E.} = 2.0 \text{ eV}$$

$$\lambda = \frac{h}{\sqrt{2m\text{K.E.}}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 2 \times 10^{-19} \times 1.6}}$$

$$= 8.68 \times 10^{-10} \text{ m} = 868 \text{ \AA}$$

60. 0

Sol. $n = 4$ and $m_\ell = -3$

Hence, ℓ value must be 3. Now number of radial nodes = $n - \ell - 1$
 $= 4 - 3 - 1 = 0$

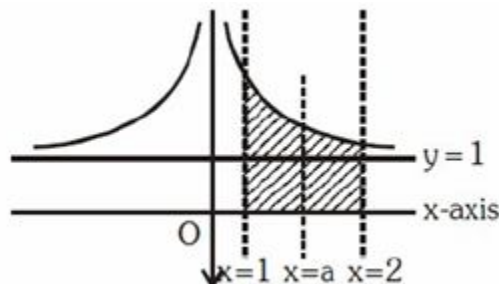
Mathematics

PART – C

SECTION – A

61. D

$$\begin{aligned} \text{Sol. } & \Rightarrow x - \frac{1}{x} \Big|_1^a = x - \frac{1}{x} \Big|_0^2 \\ & \Rightarrow 2 \left(a - \frac{1}{a} \right) = \frac{3}{2} \\ & \Rightarrow 4a^2 - 3a - 4 = 0 \\ & \Rightarrow a = \frac{3 + \sqrt{73}}{8} \end{aligned}$$



62. B

$$\begin{aligned} \text{Sol. } & \lim_{y \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + y^4}} - \sqrt{2}}{y^4} \\ & = \lim_{y \rightarrow 0} \frac{1 + \sqrt{1 + y^4} - 2}{y^4 \left(\sqrt{1 + \sqrt{1 + y^4}} + \sqrt{2} \right)} \\ & = \lim_{y \rightarrow 0} \frac{(\sqrt{1 + y^4} - 1)(\sqrt{1 + y^4} + 1)}{y^4 \sqrt{(1 + \sqrt{1 + y^4} + \sqrt{2})(\sqrt{1 + y^4} + 1)}} \\ & = \lim_{y \rightarrow 0} \frac{1 + y^4 - 1}{y^4 \left(\sqrt{1 + \sqrt{1 + y^4}} + \sqrt{2} \right) (\sqrt{1 + y^4} + 1)} \\ & = \lim_{y \rightarrow 0} \frac{1}{\left(\sqrt{1 + \sqrt{1 + y^4}} + \sqrt{2} \right) (\sqrt{1 + y^4} + 1)} = \frac{1}{4\sqrt{2}} \end{aligned}$$

63. C

$$\text{Sol. As given } -\frac{b}{2a} = 4 \text{ and } -\frac{D}{4a} = 2$$

$$\text{Hence, } E = abc = -16(a^2 + 8a^3)$$

$$\frac{dE}{da} < 0 \text{ for } a \in [1, 3]$$

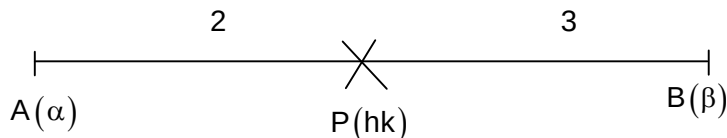
$$E_{\max} = -16(1^2 + 8 \cdot 1^3) = -144$$

$$\Rightarrow |\lambda| = 144$$

$$\frac{|\lambda|}{24} = 6$$

64. B

Sol. $\frac{PA}{PB} = \frac{2}{3}$



$$h = \frac{10 \cos \beta + 15 \cos \alpha}{5} = 2 \cos \beta + 3 \cos \alpha$$

$$k = 2 \sin \beta + 3 \sin \alpha$$

$$h^2 + k^2 = 13 + 12 \cos(\alpha - \beta)$$

$$x^2 + y^2 = 13 + 12 \cos(\alpha - \beta)$$

65. B

Sol. On solving, we have $z_1^3 + z_2^3 + z_3^3 + z_1 z_2 z_3 = 0$

$$\Rightarrow (z_1 + z_2 + z_3) \left((z_1 + z_2 + z_3)^2 - 3z_1 z_2 - 3z_2 z_3 - 3z_3 z_1 \right) = -4z_1 z_2 z_3$$

$$\Rightarrow (z_1 + z_2 + z_3)^3 = z_1 z_2 z_3 \left(3(z_1 + z_2 + z_3) \left(\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right) - 4 \right)$$

$$\Rightarrow |z_1 + z_2 + z_3|^3 = \left| 3(z_1 + z_2 + z_3) (\bar{z}_1 + \bar{z}_2 + \bar{z}_3) - 4 \right|$$

$$\Rightarrow x^3 = |3x^2 - 4| \Rightarrow x = 1, 2$$

66. B

Sol. $(3 \sec \theta + 5 \operatorname{cosec} \theta)x + (7 \sec \theta - 3 \operatorname{cosec} \theta)y + 11(\sec \theta - \operatorname{cosec} \theta) = 0$

$$\sec \theta(3x + 7y + 11) + \operatorname{cosec} \theta(5x - 3y - 11) = 0$$

Hence point B is (1, -2)

Now proceed.

67. B

Sol. $\frac{p(x)}{(2x-3)^2(3x-2)^2} = \frac{A}{2x-3} + \frac{B}{(2x-3)^2} + \frac{C}{3x-2} + \frac{D}{(3x-2)^2}$

$f(x)$ is rational function

$$B = D = 0 \text{ and } P(x) = \lambda(2x-3)(3x-2)$$

$$P(1) = -\lambda = -1 \Rightarrow \lambda = 1$$

$$\int \frac{dx}{(2x-3)(3x-2)} = \int \left[\frac{2}{2x-3} - \frac{3}{3x-2} \right] \frac{dx}{5}$$

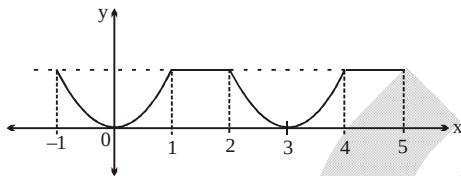
$$= \frac{1}{5} \ln \left| \frac{2x-3}{3x-2} \right| + C = \frac{1}{5} \ln \left| \frac{4x-6}{3x-2} \right| + C'$$

$$\begin{aligned} \therefore \lim_{x \rightarrow \infty} f(x) &= \frac{4}{3} \Rightarrow f(x) = \frac{4x-6}{3x-2} \\ &= |f(1)| = 2 \end{aligned}$$

68. C

Sol.

$$f(x) = \begin{cases} x^2 & -1 \leq x < 1 \\ 1 & 1 \leq x \leq 2 \\ (x-3)^2 & 2 < x < 4 \\ 1 & 4 \leq x \leq 5 \\ (x-6)^2 & 5 < x < 7 \\ 1 & 7 \leq x \leq 8 \end{cases} \quad \left. \begin{array}{l} \text{for } r = 0 \\ \text{for } r = 3 \\ \text{for } r = 6 \end{array} \right\}$$


 From the graph of $f(x)$, it is clear that $f(x)$ is periodic with period 3.

$$\text{Now } \int_0^{45} f(x) dx = \sqrt[3]{15 \int_0^3 f(x) dx} = \sqrt[3]{15 \int_{-1}^2 f(x) dx} = \sqrt[3]{15 \left[\int_{-1}^1 x^2 dx + \int_1^2 1 dx \right]} = \sqrt[3]{15 \left(\frac{2}{3} + 1 \right)} = \sqrt[3]{25} = 5$$

69. C

Sol.

$$\int_{-T/2}^{3T/2} f(x) dx = 18 \Rightarrow \int_0^{2T} f(x) dx = 18 \Rightarrow \int_0^T f(x) dx = 9$$

$$\begin{aligned} \int_{-a}^{a+5T} f(x) dx &= \int_{-a}^a f(x) dx + \int_a^{a+5T} f(x) dx \\ &= 2 \int_0^a f(x) dx + 5 \int_0^T f(x) dx = 2 \times 3 + 5 \times 9 = 51 \end{aligned}$$

70. D

Sol.

$$g\left(\frac{x+2y}{3}\right) = \frac{2f(y) + f(x)}{3}$$

 $\Rightarrow g(x)$ is linear

$$g(x) = mx + c$$

$$g'(x) = m = 1$$

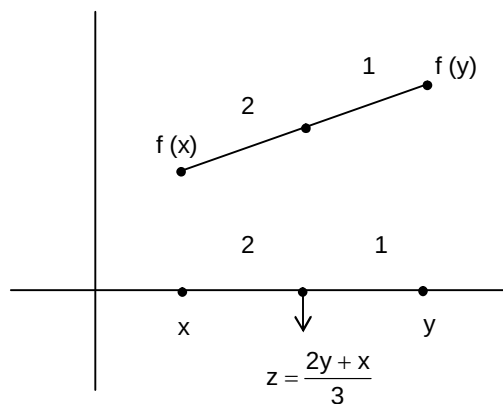
$$g(x) = c = 2$$

$$\Rightarrow g(x) = x + 2$$

$$f(g(x)) = 2 \tan^{-1}(x+2)$$

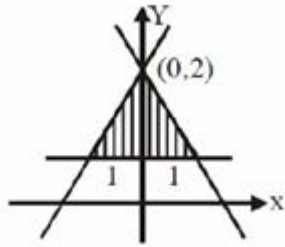
$$f^2(g(x)) - 5f(g(x)) + 4 > 0$$

$$(f(g(x)) - 1)(g(x) - 4) > 0$$



71. A

Sol. $x^2y - x^2 - y^3 + y^2 + 4y^2 - 4y - 4y + 4 = 0$
 $\Rightarrow x^2(y-1) - y^2(y-1) + 4y(y-1) - 4(y-1) = 0$
 $\Rightarrow (y-1)(x^2 - y^2 + 4y - 4) = 0$
 $\Rightarrow y = 1$ or $x^2 = (y-2)^2$
 $\Rightarrow y = 1, x = y - 2$ or $x = 2 - y$



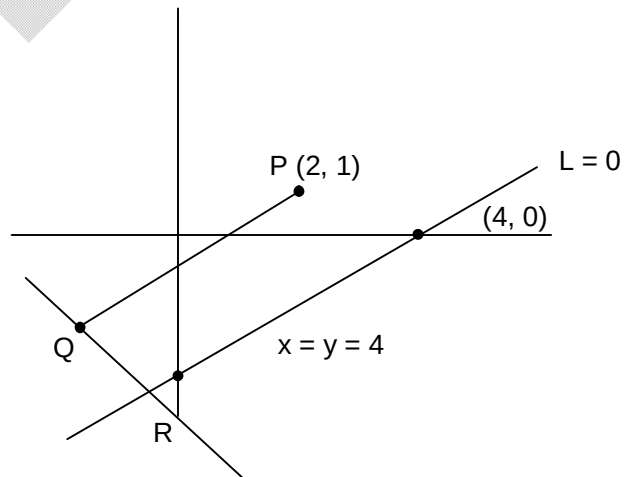
Shaded area $= \frac{1}{2} \times 2 \times 1 = 1$ sq. units.

72. A

Sol. $d(x^2y) = d(xy) + d(xy^2)$
 $\Rightarrow x^2y = xy + xy^2 + c$
 at $x = 1, y = 1 \Rightarrow c = -1$
 $\Rightarrow x^2y = xy + xy^2 - 1 \Rightarrow y = -y - y^2 - 1$
 $\Rightarrow y^2 + 2y + 1 = 0 \Rightarrow y = -1$
 $\therefore 12|y(-1)| = 12$

73. C

Sol. $x - y = 4$
 To find equation of R
 Slope of L = 0 is 1
 $\Rightarrow$ Slope of QR = -1
 Let QR is $y = mx + c$
 $y = -x + c$
 $x + y - c = 0$ distance of QR from
 (2, 1) is $2\sqrt{3}$
 $2\sqrt{3} = \frac{|2+1-c|}{\sqrt{2}}$
 $2\sqrt{6} = |3-c|$
 $c - 3 = \pm 2\sqrt{6} \Rightarrow c = 3 \pm 2\sqrt{6}$
 Line can be $x + y = 3 \pm 2\sqrt{6}$
 $x + y = 3 - 2\sqrt{6}$



74. A

Sol. $(x-2)^2 + (y-1)^2 + \lambda(x-2y) = 0$

$$C: x^2 + y^2 + x(\lambda - 4) + y(-2 - 2\lambda) + 5 = 0$$

$$C_1: x^2 + y^2 + 2y - 5 = 0$$

$$S_1 - S_2 = 0$$

(Equation of PQ)

$$(\lambda - 4)x - (2\lambda + 4)y + 10 = 0$$

Press through $(0, -1)$

$$\Rightarrow \lambda = -7$$

$$C: x^2 + y^2 - 11x + 12y + 5 = 0$$

$$= \frac{\sqrt{245}}{4}$$

$$\text{Diameter} = 7\sqrt{5}$$

75. C

Sol. Length of common tangent

$$\Rightarrow \sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2}$$

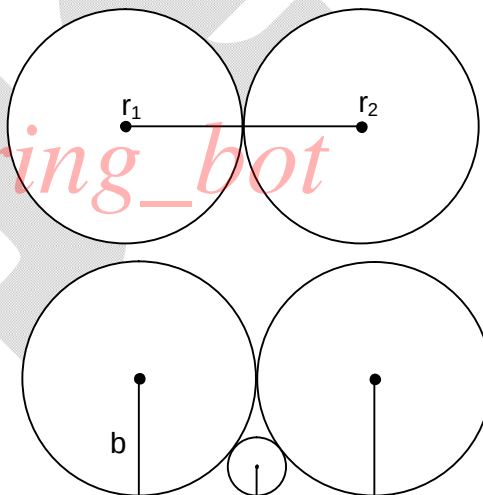
TG ~ @bohring_bot

$$\sqrt{(a+b)^2 + (a-b)^2} + \sqrt{(a+c)^2 - (a-c)^2}$$

$$= \sqrt{(b+c)^2 - (b-c)^2}$$

$$\sqrt{ab} + \sqrt{ac} = \sqrt{bc}$$

$$\frac{1}{\sqrt{c}} + \frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}}$$



76. A

Sol. Equation of AB is $\frac{xb}{4} + ky = 1$

Centroid equation of as $OB = 4$

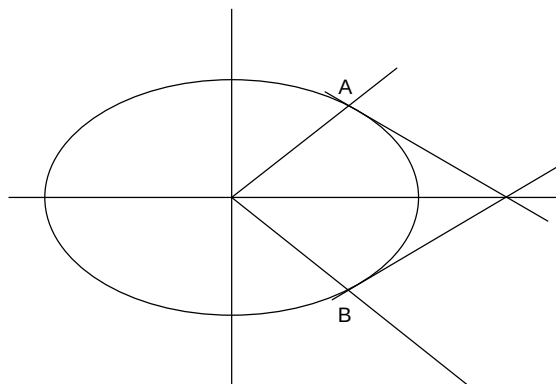
$$x^2 + 4y^2 - 4\left(\frac{xb}{4} + \frac{xy}{1}\right)^2 = 0 \quad \dots\dots\dots(i)$$

$$x^2 + 4y^2 + \alpha xy = 0 \quad \dots\dots\dots(ii)$$

$$\frac{h^2 - 4}{16} = \frac{k^2 - 4}{4} = \frac{br}{2h}$$

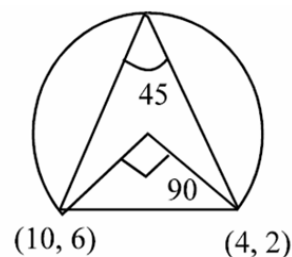
$$(h^2 - 4) = 4(k^2 - 1)$$

$$\text{Locus is } x^2 - 4y^2 = 0$$



77. D

Sol. $\sqrt{2r} = \sqrt{(10-4)^2 + (6-2)^2}$
 $= \sqrt{36+16} = 2\sqrt{13}$
 $r = \sqrt{26}$
 Perimeter $= \frac{3}{4} \times 2\pi r = \frac{3}{2} \pi \sqrt{26}$



78. A

Sol. $\sin^2 2\theta + \cos^4 2\theta = \frac{3}{4}, \theta \in \left(0, \frac{\pi}{2}\right)$
 $\Rightarrow 1 - \cos^2 2\theta + \cos^4 2\theta = \frac{3}{4}$
 $\Rightarrow 4\cos^4 2\theta - 4\cos^2 2\theta + 1 = 0$
 $\Rightarrow (2\cos^2 2\theta - 1)^2 = 0$
 $\Rightarrow \cos^2 2\theta = \frac{1}{2} = \cos^2 \frac{\pi}{4}$
 $\Rightarrow 2\theta = n\pi \pm \frac{\pi}{4}, n \in \mathbb{I}$
 $\Rightarrow \theta = \frac{n\pi}{2} \pm \frac{\pi}{8}$
 $\Rightarrow \theta = \frac{\pi}{8}, \frac{\pi}{2} - \frac{\pi}{8}$

79. B

Sol. $S_n = \sum_{n=3}^{\infty} \frac{1}{n^2 + n - 2}; = \sum \frac{1}{(n+2)(n-1)}$
 $= \sum \frac{1}{3} \left(\frac{1}{n-1} - \frac{1}{n+2} \right)$
 $= \frac{1}{3} \left[\left(\frac{1}{2} - \frac{1}{5} \right) + \left(\frac{1}{3} - \frac{1}{6} \right) + \left(\frac{1}{4} - \frac{1}{7} \right) + \left(\frac{1}{5} - \frac{1}{8} \right) + \left(\frac{1}{6} - \frac{1}{9} \right) + \dots \right]$
 $= \frac{1}{3} \left[\frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right] = \frac{13}{36}$

80. B

Sol. $\frac{1}{\sin 1} \left[\frac{\sin(1-0)}{\cos 0 \cos 1} + \frac{\sin(2-1)}{\cos 1 \cos 2} + \dots + \frac{\sin(45-44)}{\cos 44 \cos 45} \right]$
 $= \frac{1}{\sin 1} [\tan 1 - \tan 0 + \tan 2 - \tan 1 + \tan 3 - \tan 2 + \dots + \tan 45 - \tan 44]$
 $= \frac{1}{\sin 1} [1 - 0]$

$$= \frac{1}{\sin 1}$$

$$= \frac{1}{x}$$

SECTION – B

81. 101

Sol. ${}^{41}C_0 + {}^{41}C_1 + {}^{42}C_2 + \dots + {}^{60}C_{20}$

$${}^{42}C_1 + {}^{42}C_2 + \dots + {}^{60}C_{20}$$

$$\frac{61}{41} \cdot {}^{60}C_{20} = \frac{m}{n} \cdot {}^{60}C_{20}$$

82. 6

Sol. Let $P = P\left(\frac{(A \cap B)}{(A \cup \bar{B})}\right) = \frac{P(A \cap B)}{P(A \cup \bar{B})}$

$$\therefore A \cap B \subset A \cap \bar{B}$$

$$\Rightarrow P = \frac{P(A \cap B)}{1 - (P(B) - P(A \cap B))}$$

$$= \frac{\frac{1}{4}}{1 - \frac{1}{3} + \frac{1}{4}}$$

$$= \frac{3}{11} \Rightarrow k = 6$$

83. 6

Sol. Here equation of line is $\frac{x-2}{1} = \frac{y+3}{2} = \frac{z-4}{-3} = \lambda$ say(i)

Hence any point on the line (i) will be $p(\lambda + 2, \lambda - 3, -3\lambda + 4)$

Given line (i) intersect the given plane $2x + 3y - z = 13$ at the point P

$$\Rightarrow \lambda = 2$$

$$\therefore P = (4, 1, -2)$$

also equation (i) intersects the YZ – plane at Q i.e. $x = 0$

$$\therefore \lambda + 2 = 0 \Rightarrow \lambda = -2$$

$$\therefore Q = (0, -7, 10)$$

84. 1

Sol. Total ways in which A and B can be chosen $= ({}^5C_4 4!)^2 = (5!)^2$

Required probability $= 1 - P(A \text{ and } B \text{ does not contain } 2)$

$$= 1 - \frac{(4!)^2}{(5!)^2} = 1 - \frac{1}{25} = \frac{24}{25} = 0.96$$

85. 0

Sol. Given, $\cos^{-1} \sqrt{p} + \cos^{-1} \sqrt{(1-p)} + \cos^{-1} \sqrt{(1-q)} = \frac{3\pi}{4}$

$$\Rightarrow \cos^{-1} \sqrt{p} + \cos^{-1} \sqrt{1 - (\sqrt{p})^2} + \cos^{-1} \sqrt{1 - (\sqrt{q})^2} = \frac{3\pi}{4}$$

$$\Rightarrow \cos^{-1} \sqrt{p} + \sin^{-1} \sqrt{p} + \cos^{-1} \sqrt{(1-q)} = \frac{3\pi}{4}$$

$$\frac{\pi}{2} + \cos^{-1} \sqrt{(1-q)} = \frac{3\pi}{4}$$

$$\therefore q = \frac{1}{2}$$

86. 502

Sol. $\sum_{i=0}^{100} \sum_{j=0}^{100} (C_i^2 + C_j^2 + C_i C_j)$

$$\sum_{i=0}^{100} (101 C_i^2 + {}^{200}C_{100} + C_i 2^{100})$$

$$= a {}^{2n}C_n + 2^b$$

$$\Rightarrow a = 202 \quad n = 100, b = 200$$

87. 6

Sol. $y = ax^2 + bx + c$
 y is max. at $x = 2$

$$\Rightarrow -\frac{b}{2a} = -2 \Rightarrow b = 4a$$

$$\text{max value} = \frac{-D}{4a}$$

$$\frac{(b^2 - 4ac)}{4a} = \frac{1}{2}$$

$$4ac - b^2 = -2a$$

$$4ac - 16a^2 = -2a$$

$$2c - 8a = -1$$

$$\Rightarrow 2c - 8a + 5 = 6$$

88. 17

Sol. Let α be a double root of $f(x) = 0$

$$f'(x) = 4(x-1)(x-2)(x-3)$$

If $x = 1$ is double root, then $f(1) = 0 \Rightarrow p = 9$

If $x = 2$ is double root, then $f(2) = 0 \Rightarrow p = 8$

If $x = 3$ is double root, then $f(3) = 0 \Rightarrow p = 9$

$\therefore$ Sum of all possible values of $p = 17$

89. 2

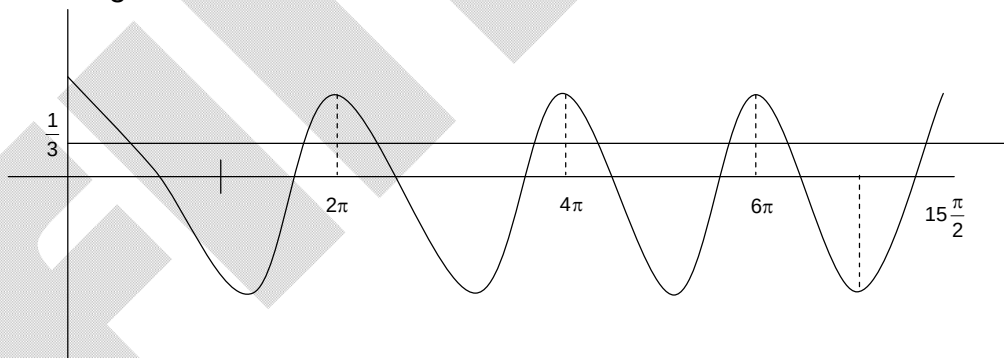
Sol. The given series is in the form $1 + a + a^2 + a^3 + \dots$ to ∞ where $a = \left(\frac{\sqrt{2}-1}{2\sqrt{2}}\right)$

$$\begin{aligned} S_n &= \frac{1}{1-a} \\ &= \frac{1}{1-\left(\frac{\sqrt{2}-1}{2\sqrt{2}}\right)} \\ &= \frac{2\sqrt{2}}{2\sqrt{2}-\sqrt{2}+1} \\ &= \frac{2\sqrt{2}}{\sqrt{2}+1} \\ &= 2\sqrt{2}(\sqrt{2}-1) \end{aligned}$$

90. 15

Sol. $2\sec^2 x - 5\sec x - 3 = 0$
 $(2\sec x + 1)(\sec x - 3) = 0$
 $\Rightarrow \sec x = 3$

$$\cos x = \frac{1}{3}$$



FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2023

TEST DATE: 15-01-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

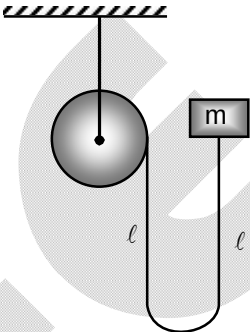
PART – A

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

- A block of mass m is attached to one end of a light string which is wrapped on a disc of mass $2m$ and radius R which is free to rotate about the horizontal axis passing through its centre of mass. The total length of the slack portion of the string is ℓ . The block is released from rest as shown. The velocity of the block just after the string becomes taut is (no slipping between disc and string)



(A) $\sqrt{g\ell}$ (B) $\sqrt{\frac{g\ell}{2}}$
(C) $\sqrt{\frac{2g\ell}{3}}$ (D) $\sqrt{\frac{3g\ell}{2}}$
- A potential of 2 V is applied across the faces of a pure germanium plate of area $4 \times 10^{-4} \text{ m}^2$ and of thickness $2 \times 10^{-3} \text{ m}$. Concentration of carriers in germanium at room temperature is $1.0 \times 10^6 \text{ m}^{-3}$, Mobility of electrons and holes are $0.7 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $0.3 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ respectively. The current produced in germanium plate at room temperature, is

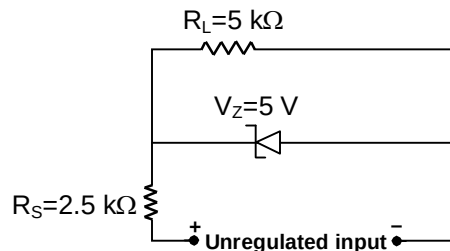
(A) $1.40 \times 10^{-10} \text{ A}$ (B) $1.40 \times 10^{-9} \text{ A}$
(C) $9.6 \times 10^{-10} \text{ A}$ (D) $6.4 \times 10^{-14} \text{ A}$
- The wavelength of the waves associated with a proton and a photon are the same. Therefore, the two have equal.

(A) mass (B) velocity
(C) momentum (D) kinetic energy
- One mole of an ideal diatomic gas undergoes a thermodynamic process, in which its molar heat capacity varies directly proportional to temperature as $C = \alpha T$, where α is a positive constant. Work done by the gas when it is heated from initial temperature T_0 to a final temperature $3T_0$ will be

(A) $4\alpha T_0^2$ (B) $(\alpha T_0 - R) \frac{3T_0}{2}$
(C) $(4\alpha T_0 - 5R) T_0$ (D) $(3\alpha T_0 - 5R) \frac{T_0}{2}$
- When a sample of atoms is irradiated by neutrons, radioactive atoms are produced at a constant rate R , which decay with decay constant λ . The number of radioactive atoms accumulated after an irradiation time t is given by

(A) $N(t) = Rte^{-\lambda t}$ (B) $N(t) = \frac{R}{\lambda} e^{-\lambda t}$
(C) $N(t) = \frac{R}{\lambda} (1 - e^{-\lambda t})$ (D) $N(t) = Rt(1 - e^{-\lambda t})$

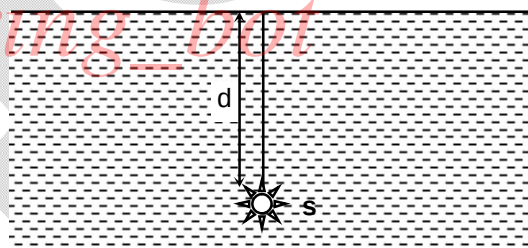
6. In the DC voltage regulator circuit shown, the zener breakdown voltage $V_Z = 5\text{ V}$. If the unregulated input varies between 11 V to 15 V, maximum zener current (in mA) is
- (A) 1 (B) 2
(C) 3 (D) 4



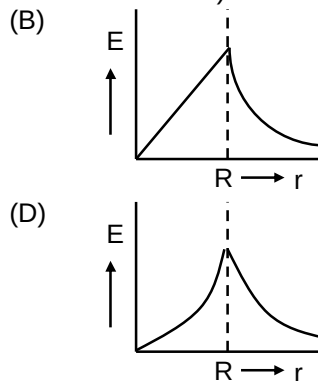
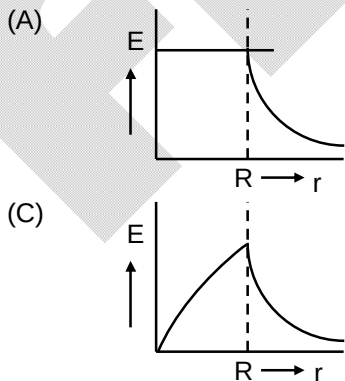
7. Radius of $^{32}_{32}\text{Ge}$ (germanium) nucleus is measured to be twice the radius of ^9_4Be nucleus. Number of neutrons in Ge are
- (A) 38 (B) 40
(C) 42 (D) 44

8. A SONAR system fixed in a submarine operates at a frequency 40 kHz. An enemy submarine moves towards the SONAR with a speed of 360 km/h. The frequency of sound is reflected by submarine is (Speed of sound in water = 1400 m/s)
- (A) $\frac{300}{7}$ kHz (B) $\frac{600}{13}$ kHz
(C) $\frac{560}{13}$ kHz (D) 40 kHz

9. A point source S of light is placed at a depth d below the surface of water in a large and deep lake. Maximum fraction of light that escapes in space above directly from water (refractive index = μ) surface is given by



- (A) $\frac{1}{2} - \frac{\sqrt{\mu^2 - 1}}{\mu}$ (B) $\frac{\sqrt{\mu^2 - 1}}{\mu}$
(C) $\frac{1}{2} - \frac{\sqrt{\mu^2 - 1}}{2\mu}$ (D) $\frac{1}{2} - \frac{\sqrt{\mu^2 - 1}}{\mu^2}$
10. In a sphere of radius R, the volume charge density ρ varies as $\rho \propto r$ (r is distance from centre). Variation of electric field E with r (For all values of $r : r \leq R$ and $r > R$) is best represented by

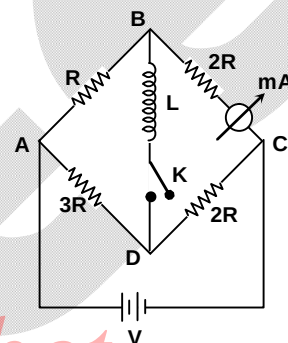


11. Binding energy per nucleon of Helium and deuterium nuclei are 7 MeV and 1.1 MeV, respectively. When two deuterium nuclei undergo fusion to form helium nucleus, energy released in the process is
 (A) 23.6 J (B) 37.76×10^{-13} MeV
 (C) 23.6 eV (D) 37.76×10^{-13} J

12. Current I flows through a long thin walled metallic cylinder of radius R with a thin longitudinal slit of width d_0 ($d_0 \ll R$) running parallel to the axis of the cylinder. The magnetic induction B produced at any point on the axis of the cylinder is approximately

- (A) $B = \text{zero}$ (B) $B = \frac{\mu_0 I}{2\pi R^2}$
 (C) $B = \frac{\mu_0 I d_0}{4\pi^2 R^2}$ (D) $B = \frac{\mu_0 I d_0}{2\pi R^2}$

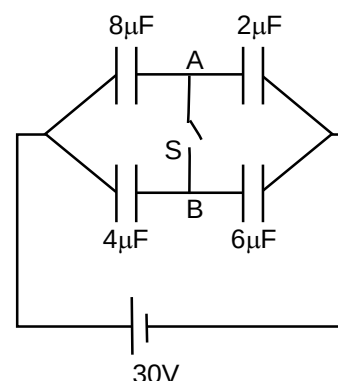
13. The reading of the ammeter, used in the electrical network shown below, is 20 mA, a long time after the key K is closed. The reading of the same ammeter, immediately after the key was closed was
 (A) zero (B) 16 mA
 (C) 25 mA (D) $\frac{70}{3}$ mA



14. Consider the diffraction pattern due to a single slit. The first maximum for a certain monochromatic light coincides with the first minimum for red light of wavelength 660 nm. The wavelength of the monochromatic light is
 (A) 660 nm (B) 550 nm
 (C) 440 nm (D) 330 nm

15. The energy of the characteristic X-ray photon in Coolidge tube comes from
 (A) the kinetic energy of striking electron.
 (B) the kinetic energy of the free electrons of the target.
 (C) the kinetic energy of the ions of the target.
 (D) the electronic transition of the target atom.

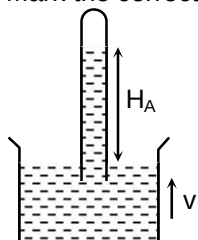
16. A system of capacitors $C_1 = 8\mu\text{F}$, $C_2 = 2\mu\text{F}$, $C_3 = 4\mu\text{F}$ and $C_4 = 6\mu\text{F}$ connected across a battery of emf $E = 30$ V is shown in figure. The charge that will flow, through the switch S , when it is closed is
 (A) $60\mu\text{C}$ A to B
 (B) $48\mu\text{C}$ A to B
 (C) $24\mu\text{C}$ B to A
 (D) $36\mu\text{C}$ B to A



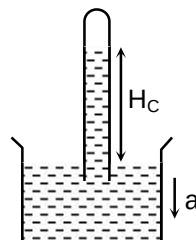
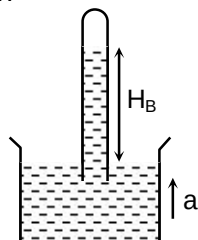
17. Four barometer A, B, C and D are shown in the figure. A is moving up with constant velocity, B is accelerated up, C is accelerated down and D is at rest. Atmospheric pressure is represented by 76 cm of mercury column.

(i) $H_A = 76$ cm, (ii) $H_B < 76$ cm, (iii) $H_C = 76$ cm, (iv) $H_D = 76$ cm

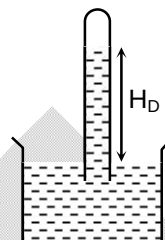
Mark the correct answer.



(A) (i), (ii), (iii), (iv)
(C) (i), (iii), (iv)



(B) (i), (ii), (iv)
(D) (ii), (iii), (iv)



18. A total charge Q is uniformly distributed over a non-conducting ring of radius r . There is a time varying magnetic field perpendicular to its plane and changing at the uniform rate of $\frac{dB}{dt}$.

The magnitude of torque experienced by the ring is

(A) $Qr^2 \left(\frac{dB}{dt} \right)$

(B) $Qr^3 \left(\frac{dB}{dt} \right)$

(C) $\frac{1}{2} Qr^2 \left(\frac{dB}{dt} \right)$

(D) $\frac{1}{2} Qr^3 \left(\frac{dB}{dt} \right)$

19. A circular loop of radius r is placed inside another circular loop of radius R ($R \gg r$). The loops are coplanar and concentric. The mutual inductance (M) of the system is proportional to

(A) $\frac{\mu_0 \pi r}{R}$

(B) $\frac{\mu_0 \pi r^2}{R}$

(C) $\frac{\mu_0 \pi R^2}{r}$

(D) $\frac{\mu_0 \pi r^2}{R^2}$

20. A sphere and cube have equal surface area are made of the same material. The two are heated to the same temperature and kept in identical surrounding. The ratio of their initial rates of cooling is

(A) $1 : 1$

(B) $\sqrt{\frac{\pi}{2}} : 1$

(C) $\sqrt{\frac{\pi}{3}} : 1$

(D) $\sqrt{\frac{\pi}{6}} : 1$

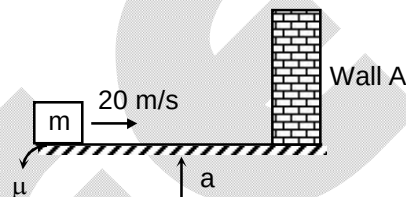
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. In a young's double slit experiment distance between the slits is $d = 1$ mm, Wavelength of light used is 600 nm and distance of screen from the plane of slits is $D = 1$ m. The minimum distance between two points is $\frac{k}{10}$ mm on the screen, where intensity is maximum. Find the value of k . (Assume both sources of equal power).

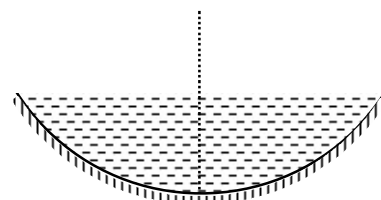
22. A block of mass m is projected from a point A with velocity 20 m/s on the rough surface having coefficient of friction μ to hit the wall B with velocity 10 m/s. With what velocity same mass m should be projected to hit the wall B with same velocity 10 m/s if the surface is moving now upward with an acceleration of $a = 4g$ as shown in the figure.



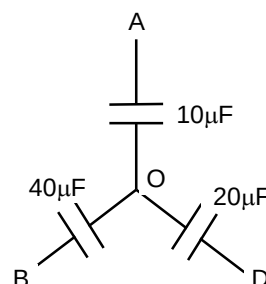
23. A sound source of constant frequency resonates with an open end organ pipe of length 30 cm and a close end organ pipe of length 23 cm (both of same diameter). Both pipes are sounding their first overtone. If velocity of sound is 340 ms^{-1} , then find the end correction in cm.

24. In an electromagnetic wave the amplitude of magnetic field is $3 \times 10^{-8} \text{ T}$. If the frequency of wave is 10^{12} Hz , then find the amplitude of associated electric field in S.I. unit.

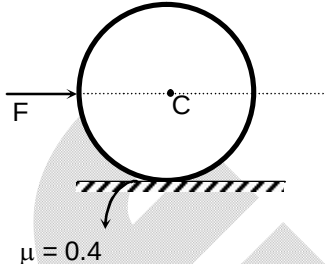
25. A concave mirror when placed in air has a focal length $f = 20$ cm. The mirror is now placed horizontally and filled with a thin layer of water having refractive index $\frac{4}{3}$. The object is placed at principal axis at a distance d from the mirror such that a real, inverted image coincides with the object. Find the value of d in cm.



26. Three uncharged capacitors of capacitances $C_1 = 10\mu\text{F}$, $C_2 = 40\mu\text{F}$ and $C_3 = 20\mu\text{F}$ are connected as shown in figure to one another at O and to points A, B and D at potentials $V_A = 100 \text{ V}$, $V_B = 50 \text{ V}$ and $V_D = 200 \text{ V}$ respectively. Calculate charge in $40 \mu\text{F}$ capacitor in μC .



27. The rate of flow of a certain liquid of viscosity η through a horizontal capillary of length ℓ and radius r is Q when the pressure difference between the inlet and outlet is equal to the atmospheric pressure. The rate of flow of the same liquid through another capillary of length 2ℓ and radius $2r$ when the pressure difference between inlet and outlet is 3 times the atmospheric pressure is kQ . Find the value of k .

28. A ball is projected from horizontal ground. It attains a maximum height H on its projectile path and there after strikes a stationary smooth vertical wall and falls on ground vertically below the point of maximum height. Assume the collision with wall to be perfectly elastic, the height of the point on the wall where the ball strikes is $\frac{3H}{k}$. Find the value of k .
29. A solid cylinder of mass $m = 10$ kg is rolling without slipping on a rough horizontal surface, under the action of a horizontal force $F = 48$ N such that the line of action of F passes through centre C of the cylinder. Coefficient of friction between cylinder and surface is $\mu = 0.4$. Calculate the frictional force (in newton) acting on the cylinder.
- 
30. Two small solid balls of masses m and $8m$ made up of same material and same density are tied at the two ends of a thin weightless thread. They are dropped from a balloon in air. The tension T of thread during fall, after the motion of balls has reached the steady state is $T = kmg$. Find the value of k .

TG ~ @bohring_bot

Chemistry

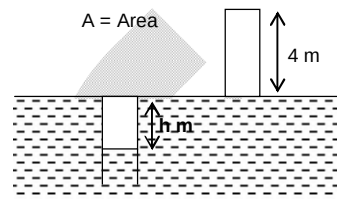
PART – B

SECTION – A

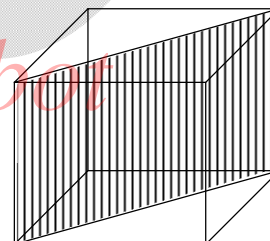
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

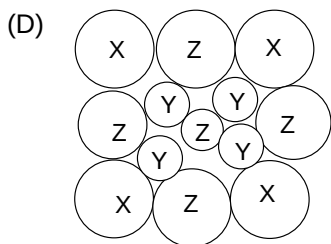
31. A 4 m long tube closed at one end is lowered vertically into water until the closed end is flushed with water surface (See figure). Calculate the water level height in the tube, h . [Barometric pressure = 1 atm = 10 m of hydrostatic water head; $T = 25^\circ\text{C}$, $d_{\text{H}_2\text{O}} = 1 \text{ g/mL}$. Neglect water vapour pressure]



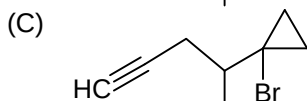
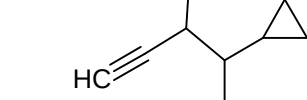
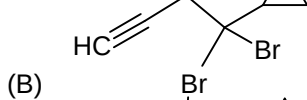
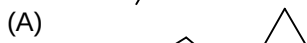
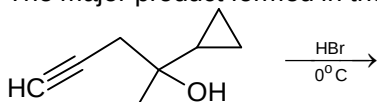
- (A) 1.71 m
(B) 3.06 m
(C) 1.53 m
(D) 0.855 m
32. The oxidation state of sulphur in the dithionous and dithionic acids, respectively are:
(A) +4, +6
(B) +4, +5
(C) +3, +5
(D) +3, +6
33. In a hypothetical solid, x atoms form CCP lattice. 'Y' atoms occupy all tetrahedral voids and 'Z' atoms occupy all octahedral voids. If a unit cell of crystal is cut by a plane shown in the figure, then the cross section of the plane looks like:



- (A)
- (B)
- (C)



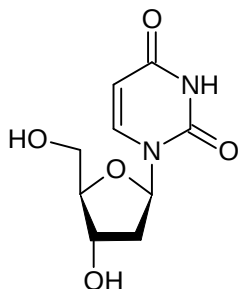
34. The major product formed in the following reaction is:



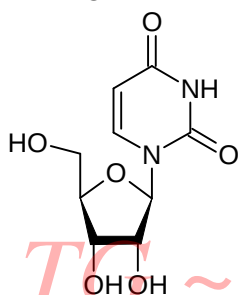
35. In the synthesis of polydimethyl siloxane, the chain forming, branching and terminating agent respectively, are
- (A) Me_2SiCl_2 , Me_3SiCl and MeSiCl_3
- (B) Me_2SiCl_2 , MeSiCl_3 and Me_3SiCl
- (C) MeSiCl_3 , Me_2SiCl_2 and Me_3SiCl
- (D) Me_2SiCl_2 , MeSiCl_3 and Me_4Si
36. Choose the correct statement(s) among the following
- (I) LiF is more soluble than LiClO_4 in water.
- (II) The standard reduction potential $[E^\circ]$ of Li is more negative than that of Na .
- (III) The heat of hydration of Li^+ (g) is greater than that of Na^+ (g).
- (A) I and II
- (B) I and III
- (C) II and III
- (D) III only
37. The two main pollutants which are released when fossil fuels are burnt are
- (A) O_3 and hydrocarbon
- (B) NO_2 and O_3
- (C) NO and NO_2
- (D) NO_2 and hydrocarbon

38. The role of BF_3 as an industrial polymerization catalyst is to generate
 (A) Carbanion
 (B) Carbocation
 (C) Organic radical
 (D) Carbon radical

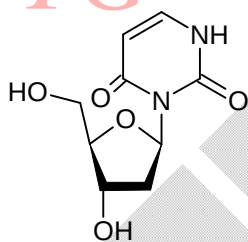
39. The correct structure of ribonucleoside uridine is:
 (A)



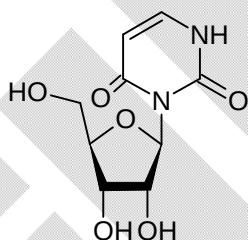
(B)



(C)

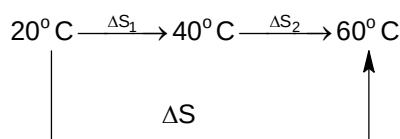


(D)



40. The number of degenerate spatial orbitals of a hydrogen like atom with principal quantum number $n = 5$ is
 (A) 25
 (B) 5
 (C) 50
 (D) 10

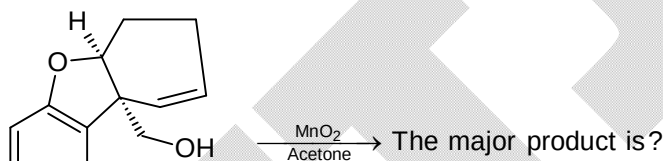
41. Consider the entropy changes in a system undergoing transformation, as depicted in the diagram below



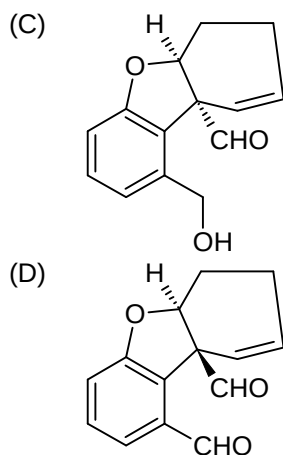
The correct statement among the following is

- (A) $\Delta S_1 = \Delta S_2$ and $\Delta S = \Delta S_1 + \Delta S_2$
 (B) $\Delta S_1 > \Delta S_2$ and $\Delta S \neq \Delta S_1 + \Delta S_2$
 (C) $\Delta S_1 < \Delta S_2$ and $\Delta S \neq \Delta S_1 + \Delta S_2$
 (D) $\Delta S_1 > \Delta S_2$ and $\Delta S = \Delta S_1 + \Delta S_2$
42. For the following complexes, the increasing order of magnetic moment (spin only value) is
 (a) $[\text{TiF}_6]^{-3}$ (b) $[\text{CrF}_6]^{-3}$ (c) $[\text{MnF}_6]^{-3}$ (d) $[\text{CoF}_6]^{-3}$
 (A) $d < a < b < c$
 (B) $c < a < d < b$
 (C) $b \approx a < d < c$
 (D) $a < b < c \approx d$
43. The stability of lyophobic colloids is a consequence of the
 (A) Electric double layer at the surface of the particle
 (B) van der Waal forces between the particles
 (C) Small size of particles
 (D) Shape of particles

44.



- (A)
- (B)



45. The rate constant of a second order reaction $A \longrightarrow B$ is k_2 . If the initial concentration of the reactant is x_0 and the concentration of the product at time t is a , then a linear function of t with the slope $k_2 x_0$ is :

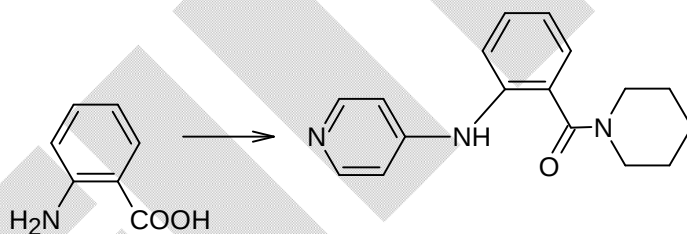
(A) $\ln\left(\frac{a}{x_0 - a}\right)$

(B) $\left(\frac{a}{x_0(x_0 - a)}\right)$

(C) $\left(\frac{a}{x_0 - a}\right)$

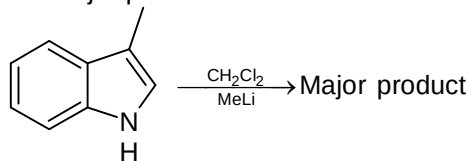
(D) $\ln\left(\frac{a}{x_0(x_0 - a)}\right)$

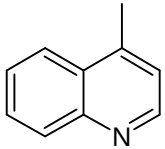
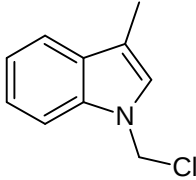
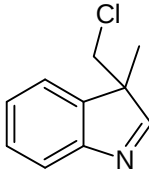
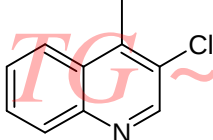
46. Correct sequence of reagent (i) to (iii) required for the conversion of A to B is:



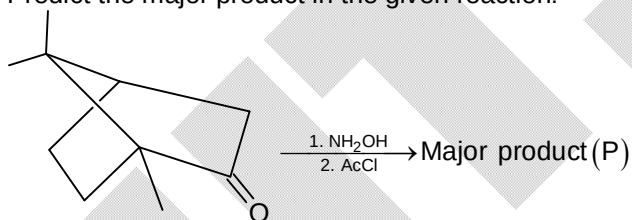
- (i) Thionyl chloride
 (ii) 4-chloropyridine
 (iii) Piperidine
 (A) i, ii and iii
 (B) i, iii and ii
 (C) ii, i and iii
 (D) iii, i and ii

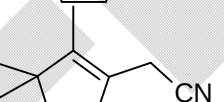
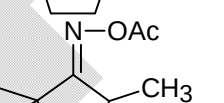
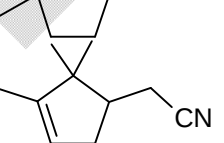
47. The major product formed in the following reaction is



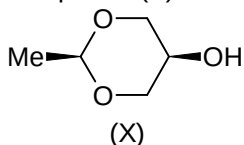
- (A) 
- (B) 
- (C) 
- (D) 

48. Predict the major product in the given reaction:



- (A) 
- (B) 
- (C) 
- (D) 

49. Among the structures given below, the one that corresponds to the most stable conformation of compound (X) be:



- (A)
- (B)
- (C)
- (D)

50. Allred – Rochow electronegativity of an element is:
- Directly proportional to the effective nuclear charge.
 - Directly proportional to the covalent radius.
 - Inversely proportional to the square of the covalent radius.
 - Directly proportional to the square of the effective nuclear charge.
- The correct answer is:
- (A) I and II
(B) I and III
(C) II and III
(D) I and IV

SECTION – B

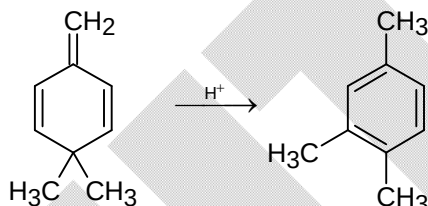
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

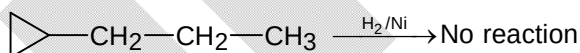
51. The degree of unsaturation in the product formed by heating mellitic acid ($C_6(COOH)_6$) is:
52. XH_2 (ion exchange resin) can replace Ca^{+2} ions in hard water as $XH_2 + Ca^{+2} \longrightarrow XCa + 2H^+$. If 1 L hard water after passing through XH_2 has a pH = 4 then hardness in parts per million of Ca^{+2} is
53. The theoretical efficiency of a hypothetical cell is about 84% which involves the following reaction
 $A(s) + B^{+2}(aq) \longrightarrow A^{+2}(aq) + B(s) \quad \Delta H = -285 \text{ kJ}$
 then, the standard EMF of the cell is:

54. An amount of 0.1 millimole of CdSO_4 is present in 10 mL acid solution of 0.08 M HCl. Now, H_2S is passed to precipitate all the Cd^{+2} ions. What would be the pH of solution after filtering off precipitate, boiling off H_2S and making the solution 100 mL by adding water?
55. The freezing point of an aqueous solution of KCN containing 0.2 mole/kg water was -0.8°C . On addition of 0.1 mole of $\text{Hg}(\text{CN})_2$ in the solution containing 1 Kg of water, the freezing point of the solution was -0.6°C . Assuming that the complex is formed according to the following equation:
 $\text{Hg}(\text{CN})_2 + m\text{CN}^- \longrightarrow \text{Hg}(\text{CN})_{m+2}^-$ and $\text{Hg}(\text{CN})_2$ is the limiting reagent, the value of 4 m is:
56. For the following molecules; if x is the total number of p-orbitals used in hybridisation of each central atom, then what is the value of $\frac{\sum x}{6}$?
 XeO_2F_2 , B_2H_6 , $[\text{NiCl}_4]^{-2}$, $[\text{PtCl}_4]^{-2}$, $\text{N}(\text{SiH}_3)_3$, H_3BO_3
57. The number of optically active stereoisomer possible for X are:
X : $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}(\text{OH}) - \text{CH}(\text{OH}) - \text{CH}_3$
58. The number of bridging carbonyls in $[\text{Mn}_2(\text{CO})_{10}]$ are:
59. The total number of black coloured sulphides in the following are:
 PbS , CuS , HgS , MnS , Ag_2S , NiS , CoS , SnS_2 , CdS
60. The sum of reaction numbers for which the major product is/are correctly mentioned is:

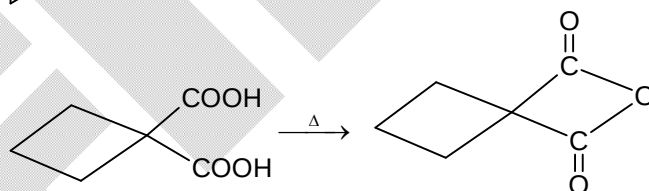
1.



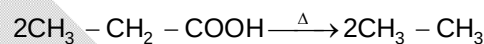
2.



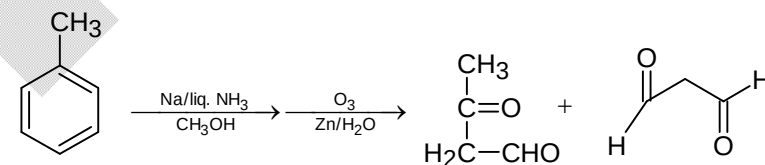
3.



4.



5.



Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

61. Let $A = \{x : x^2 \leq x + 20,000 ; x \in \mathbb{N}\}$
 $B = \{x = 3k + 2 ; k \in \mathbb{N}\}$
 $C = \{x = 4k ; k \in \mathbb{N}\}$
 The sum of the elements of the set $A \cap (B - C)$ is
 (A) 2443 (B) 2447
 (C) 2450 (D) 2453
62. The number of integral values of a for which the equation $x^4 - (a + 2)x^3 + 2ax^2 + 4(a - 2)x - 16 = 0$ has at least two positive roots; $a \in [-10, 10]$ is/are
 (A) 5 (B) 6
 (C) 7 (D) 16
63. Let $(1 + x + x^2)^{30} = \sum_{r=0}^{60} a_r x^r$. If $\alpha a_{21} = \beta a_{20} + \gamma a_{19}$, ($\alpha, \beta, \gamma \in \mathbb{N}$) then the value of $\alpha + \beta + \gamma$ can be
 (A) 62 (B) 72
 (C) 82 (D) 74
64. $\int e^{\frac{1}{2}\left(x^2 + \frac{1}{x^2}\right)} \left(\frac{x^4 + x^2 - 1}{x^2}\right) dx = f(x) + c$, then $\left(f(\sqrt{2})\right)^4$ is
 (A) e^5 (B) $2e^5$
 (C) $4e^5$ (D) $\sqrt{2}e^5$
65. If $1(50)^{49} + 2(51)^1(50)^{48} + 3(51)^2(50)^{47} + \dots + (50)(51)^{49} = k(50)^{49}$ then the number of factor of k of the form $4m + 1$, ($m \in \mathbb{W}$) is/are
 (A) 3 (B) 4
 (C) 5 (D) 7
66. If the pair of perpendicular lines $4x^2 + by^2 + 2\cos\theta xy + 12x + 2\sin^2\theta y + c = 0$, $\theta \in \left(\frac{3\pi}{2}, 2\pi\right)$ intersect on x-axis then the area of triangle formed by the given pair of lines and $y = 2$ is
 (A) $\sqrt{65}$ (B) $\frac{\sqrt{65}}{2}$
 (C) $\frac{\sqrt{65}}{4}$ (D) $2\sqrt{65}$
67. Consider two circles
 $C_1 : (x - 1)^2 + (y - 4)^2 - 16 = 0$ and $C_2 : (x - 13)^2 + (y - 9)^2 - 81 = 0$
 If a circle of radius r touches x-axis and C_1 and C_2 externally, then r is equal to
 (A) 1.44 (B) 1.48
 (C) 1.52 (D) 1.6

68. A line $y = m(x - 4)$ meets x-axis at P and the parabola $x^2 = 32y$ at $Q(x_1, y_1)$. The tangent to the parabola at $Q(x_1, y_1)$ meets x-axis at $R(x_2, 0)$ $0 < x_2 < 6$. If the area of triangle PQR assumes a local maximum, then the value of m is
- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$
(C) $\frac{1}{2}$ (D) $\frac{3}{4}$
69. Consider three curves
 $H : (x + a)y = \lambda; \lambda < 0, a > 0$
 $C : x^2 + y^2 - 21y + 109 = 0$
 $P : y^2 = bx; b > 0$
 Let the line $2x - y + 8 = 0$ touches the curve H, C and P at L, M & N respectively such that $LM = MN = \sqrt{45}$, then the value of $8\lambda + b + a$ is
- (A) 0 (B) 1
(C) -1 (D) 2
70. A line $L = 2x - y + 5 = 0$ is a tangent to the hyperbola $H \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that foot of perpendicular from the foci of hyperbola on line L is $\left(\frac{\sqrt{3} - 2\sqrt{5}}{\sqrt{5}}, \frac{2\sqrt{3} + \sqrt{5}}{\sqrt{5}} \right)$. If the hyperbola intersects an ellipse $E \equiv \frac{x^2}{25} + \frac{y^2}{\lambda^2} = 1$ orthogonally, then eccentricity of the ellipse is
- (A) $\sqrt{\frac{1}{5}}$ (B) $\sqrt{\frac{3}{5}}$
(C) $\frac{1}{5}$ (D) $\sqrt{\frac{2}{3}}$
71. Let $y = y(x)$ be a solution curve of the differential equation $\frac{dy}{dx} + \frac{y \cos x}{\sin x + \cos x} = \frac{2 \tan^2 x + \tan x + 2}{\sqrt{\sin x + \cos x}}$.
 If $y\left(\frac{\pi}{4}\right) = 2^{3/4}$, then $y^2\left(\frac{7\pi}{12}\right)$ is
- (A) $7\sqrt{6} + 4\sqrt{2}$ (B) $7\sqrt{2} + 4\sqrt{6}$
(C) $3\sqrt{6} + 7\sqrt{2}$ (D) $28\sqrt{2} + 16\sqrt{6}$
72. Let $P(z)$, $R(z^4)$ and $Q(z^2)$ be three points in Argand plane such that $PR + RQ = PQ$; ($z \neq 0, 1$) then z lies on
- (A) a circle centred at $\left(-\frac{1}{2}, 0\right)$ (B) a line segment having length $2\sqrt{3}$
(C) a line having slope $-\frac{1}{2}$ (D) a line parallel to y-axis
73. Let $\begin{vmatrix} a & \sqrt{5} & \sqrt{7} \\ \sqrt{3} & b & \sqrt{7} \\ \sqrt{3} & \sqrt{5} & c \end{vmatrix} = 0$, $(a \neq \sqrt{3}, b \neq \sqrt{5}, c \neq \sqrt{7})$ and $\frac{a}{a - \sqrt{3}} + \frac{b}{b - \sqrt{5}} + \frac{c}{c - \sqrt{7}} = \lambda$, if $a = 2\sqrt{3}$ then the point (b^2, c^2) may lie on the line
- (A) $5y - 21x = 0$ (B) $5y - 24x = 0$
(C) $5y - 63x = 0$ (D) $21y - 5x = 0$

74. In $\triangle ABC$, $\angle A = \tan^{-1}7$, $\angle C = \tan^{-1}\frac{4}{3}$. Let D be an interior point on side AC such that area of ABD is twice that of area of $\triangle BCD$. If $\angle ABD = \theta$, then the value of $\tan 2\theta$ is
- (A) $\frac{12}{5}$ (B) $\frac{5}{12}$
(C) $\frac{36}{77}$ (D) $\frac{77}{36}$
75. There are two townships A and B in a city containing 40% and 60% of the population respectively. 15% of the total population suffer from heart disease. $P(\text{person suffering from heart disease who lives in A}) = 6 P(\text{person suffering from heart disease who lives in B})$ where $P(E)$ denotes the probability of an event E. A person randomly diagnosed turns out to be free from heart disease, then the probability that he lives in B township is
- (A) $\frac{28}{85}$ (B) $\frac{38}{85}$
(C) $\frac{57}{85}$ (D) $\frac{47}{85}$
76. Consider the following frequency distribution
- | Class : | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 – 50 | 50 – 60 |
|-------------|--------|---------|---------|---------|---------|---------|
| Frequency : | 3 | a | b | 11 | 8 | 4 |
- If the mean is $\frac{277}{9}$ and median is $\frac{335}{11}$, then the value of $4a + b$ is
- (A) 40 (B) 38
(C) 50 (D) 48
77. Consider the following statements
P : Sachin is a topper
Q : Sachin is sincere
R : Sachin is not demotivated
The negation of the statement.
If Sachin is a sincere and he is not demotivated, then he is a topper
- (A) $P \rightarrow (Q \vee R)$ (B) $(\sim P) \wedge (Q \wedge R)$
(C) $P \rightarrow ((\sim Q) \vee (\sim R))$ (D) $P \rightarrow (Q \wedge R)$
78. The upper four-fifth portion of a vertical tower subtends an angle $\tan^{-1}\frac{8}{21}$ at a point A in the horizontal plane through its foot and at a distance 50m from the foot. If the angle subtended by the lower one fifth of tower at point A is β , then the height of the tower can be
- (A) 25 (B) 50
(C) 13 (D) 75
79. Let N be the number of four digits even numbers such that if 3 is one of the digit, then 5 is the succeeding digit. If N can be resolved as $P_1^\alpha P_2^\beta P_3^\gamma$ (P_1, P_2, P_3 are primes and α, β, γ are natural numbers then $P_1 + P_2 + P_3$ is
- (A) 11 (B) 31
(C) 29 (D) 9

80. Let $S_1 = \{z \in \mathbb{C} : |z - 2| \leq |\operatorname{Re} z + 2|\}$
 $S_2 = \{z \in \mathbb{C} : z(1 + i) + \bar{z}(1 - i) - 12 \leq 0\}$
 $S_3 = \{z \in \mathbb{C} : \operatorname{Re} z \geq 0, \operatorname{Im} z \geq 0\}$
 and $S = S_1 \cap S_2 \cap S_3$, then the maximum value of $|z - 20i|^2$; $z \in S$
 (A) 436 (B) 446
 (C) 456 (D) 384

SECTION - B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If the sum of values of θ in $(-3\pi, 3\pi)$ satisfying

$$\sum_{m=1}^{15} \sec\left(\theta + (m-1)\frac{\pi}{18}\right) \sec\left(\theta + m\frac{\pi}{18}\right) = (4 + 2\sqrt{3}) \operatorname{cosec} \frac{\pi}{18}$$
 is $\frac{k\pi}{10}$
 then the value of k is _____
82. If $2(\sin A - \sin^3 A) = \cos B$; $0 < A, B < \pi/2$, then the value of $\cos B = \sqrt{\frac{m}{n}}$, m & n are co-prime
 $2(\cos A + \cos^3 A) = \sin B$
 where $m + n$ is _____
83. Let $\frac{5}{6} \cos^{-1} \frac{\sqrt{3}}{\sqrt{3+\pi^2}} + \frac{1}{3} \sin^{-1} \frac{2\sqrt{3}\pi}{3+\pi^2} + \frac{1}{6} \tan^{-1} \frac{\sqrt{3}}{\pi} = a$
 and $\cos^{-1} \left[\frac{13}{40} \cos\left(\cot^{-1} \frac{5}{12}\right) + \frac{13}{32} \operatorname{sincos}^{-1} \left(\frac{5}{13}\right) \right] = b$
 then $\operatorname{cosec} \left(\int_b^a \frac{\tan x}{\sqrt{3}} dx \right)$ is ($[.]$ represents greatest integer function)
84. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ \sqrt{a} & 1 & 0 \\ a\sqrt{a} & \sqrt{b} & 1 \end{bmatrix}$; $a, b \in \mathbb{R}^+$. If for some $n \in \mathbb{N}$, $A^n = \begin{bmatrix} 1 & 0 & 0 \\ 72 & 1 & 0 \\ 3600 & 72 & 1 \end{bmatrix}$ then number of triangles formed by joining the vertices of n sided polygon having no side common with the polygon are _____
85. Consider the curves
 $C_1 : y^2 - x = 0$
 $C_2 : y - x^2 = 0$; $0 \leq x \leq \frac{\sqrt{3}}{2}$
 $C_3 : y = f(x)$; $f(x) < 0 \quad \forall x \in \left(0, \frac{\sqrt{3}}{2}\right)$
 From any point P on C_2 , lines are drawn parallel to the coordinate axes so as to intersect C_1 at Q and C_3 at R . If area of region $OPRO$ is twice of the area of region $OPQO$; (O being origin) then the value of $\left| 32f\left(\frac{1}{2}\right) \right|$ is _____

86. Let f be a differentiable function satisfying $(f(x) - 1)(f(y) - 1) = f(x + y) - 1 \quad \forall x, y \in \mathbb{R}; f(0) \neq 1, f'(0) = \ln 4$. If $\lim_{x \rightarrow 1} \frac{\sin(f(x) - 5) + 2 - 2^x}{x^3 - 7x^2 + ax + b} = \ln 8$, then the value of $a - b$ is

87. Let $f(x) = \begin{cases} 6a(\ln 2)^2 \tan^{-1}\left(\frac{b+x}{3}\right) & ; -1 < x < 0 \\ \frac{\ln 4}{3} & ; x = 0 \\ \frac{\frac{cx}{2^3} - 1}{x} & ; 0 < x < 1 \end{cases}$

If $f(x)$ is differentiable at $x = 0$, then the value of $\frac{b^2 + 9}{a}$ is

88. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable function such that $f(x) = x^5 + 2x^3 + 3x + 3$, $g(f(x)) = x$ and $h(g(g(x))) = x \quad \forall x \in \mathbb{R}$, then $h'(0)$ is

89. If L_1 is the line of intersection of the planes $x + y + z - 6 = 0$, $2x + 3y + z - 5 = 0$ and line L_2 is $\frac{(x-14)}{3} = \frac{y+6}{0} = \frac{z-2}{1}$. If P is the plane containing lines L_1 and L_2 which intersects the coordinate axes at A, B, C respectively, then volume of tetrahedron $OABC$ (O is origin) is V . The value of $90V$ is

90. Let $\hat{x}, \hat{y}$ and $\hat{z}$ be unit vectors such that $\hat{x} + \hat{y} - \hat{z} = \vec{p}$, $\hat{x} \times (\hat{y} \times \hat{z}) = \hat{q}$ and $(\hat{x} \times \hat{y}) \times \hat{z} = \vec{r}$. If $|\vec{p}| = \sqrt{7}$, $\vec{q} \cdot \hat{y} = 5$ and $\vec{p}$ is perpendicular to $\hat{x}$, then the number of positive integral solutions(s) of the equation $x_1 x_2 x_3 = ((\hat{x} \cdot \hat{y})(\hat{y} \cdot \hat{z})(\hat{z} \cdot \hat{x}))^4$ is

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1.

Sol.

B

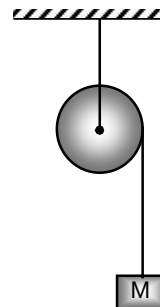
Applying conservation of angular momentum of system about centre of mass of disc

$$mvR = mv'R + \frac{2mR^2}{2}\omega$$

$$v' = R\omega$$

$$mvR = mv'R + mv'R$$

$$v' = \frac{v}{2} = \frac{\sqrt{2g\ell}}{2} = \sqrt{\frac{g\ell}{2}}$$



2.

Sol.

D

$$I = neA (\mu_e + \mu_h) E$$

$$= 1.0 \times 10^6 \times 1.6 \times 10^{-19} \times 4 \times 10^{-4} (1.0) \times \frac{2}{2 \times 10^{-3}} = 6.4 \times 10^{-14} \text{ A}$$

3.

Sol.

C

Basic concepts

4.

Sol.

C

$$C = C_V + \frac{dW}{ndT}$$

$$\int dW = \int_{T_0}^{3T_0} \alpha T dT - \int_{T_0}^{3T_0} \frac{5R}{2} dT \text{ for one mole}$$

$$W = (4\alpha T_0 - 5R)T_0$$

5. C

Sol. $\frac{dN}{dt} = R - \lambda N$

So $\int_0^N \frac{dN}{R - \lambda N} = \int_0^t dt \Rightarrow \ln(R - \lambda N) \Big|_0^N = -\lambda t$

So $\ln \frac{R - \lambda N}{R} = -\lambda t$

So $1 - \frac{\lambda N}{R} = e^{-\lambda t} \Rightarrow N = \frac{R}{\lambda} (1 - e^{-\lambda t})$

6. C

Sol. Zener current will be maximum, when $V = 15V$

$15 - (i \times 2.5k) = 5$

$i = 4 \text{ mA}$

$i_z = 3 \text{ mA}$

7. B

Sol. $\frac{R_{Ge}}{R_{Be}} = \left(\frac{X}{9}\right)^{1/3}$

$X = 72$

Number of neutron = $72 - 32 = 40$

8. B

Sol. frequency received by submarine

$n_1 = \left(\frac{1400 + 100}{1400}\right) \times 40 \times 10^3 = \left(\frac{15}{14}\right) \times 40 \times 10^3$

Frequency reflected by submarine

$n_2 = \left(\frac{1400}{1400 - 100}\right) n_1 = \left(\frac{15}{13}\right) \times 40 \times 10^3 \text{ Hz} = \frac{600}{13} \text{ kHz}$

9. C

Sol. $f = \frac{2\pi(1 - \cos \theta_C)}{4\pi}$

$f = \frac{2\pi}{4\pi} \left[1 - \sqrt{1 - \frac{1}{\mu^2}} \right] = \frac{1}{2} \left[1 - \sqrt{1 - \frac{1}{\mu^2}} \right]$

10. D

Sol. For ($r \leq R$)

$E \times 4\pi r^2 = \int_0^r cx 4\pi x^2 dx / \epsilon_0$

So $E \propto r^2$

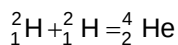
and For ($r \geq R$)

$E \propto \frac{1}{r^2}$

11.

D

Sol.



$$BE_{\text{H}} = 1.1 \text{ MeV}$$

$$BE_{\text{He}} = 7 \text{ MeV}$$

$Q_{\text{value}} = \text{Binding energy of product} - \text{Binding energy of reactants}$

$$= (4 \times 7 - 4 \times 1.1) \text{ MeV} = 23.6 \text{ MeV}$$

$$= 23.6 \times 1.6 \times 10^{-19} \times 10^6 = 37.76 \times 10^{-23}$$

12.

C

Sol.

$$B = \frac{\mu_0 dI}{2\pi R} = \frac{\mu_0 I d_0}{(2\pi R)(2\pi R)} = \frac{\mu_0 I d_0}{4\pi^2 R^2}$$

13.

D

Sol.

After long time,

$$I_A = \frac{2V}{7R} = 20 \text{ mA} \quad \dots(i)$$

Immediately after the key is closed,

$$I = \frac{V}{3R} = \frac{70}{3} \text{ mA}$$

14.

C

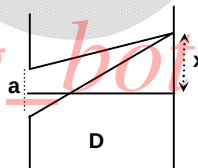
Sol.

For 1st secondary minima $\frac{ax}{D} = \lambda_1$

For 1st secondary maxima $\frac{ax}{D} = \frac{3\lambda_2}{2}$

$$\lambda_1 = \frac{3}{2}\lambda_2$$

$$\lambda = 440 \text{ Å}$$



15.

D

Sol.

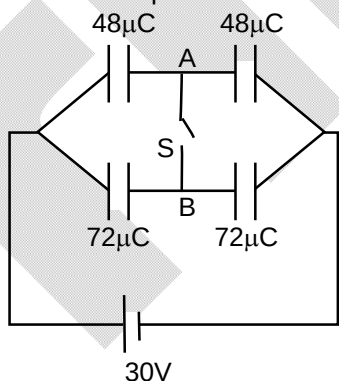
Basic Concepts

16.

A

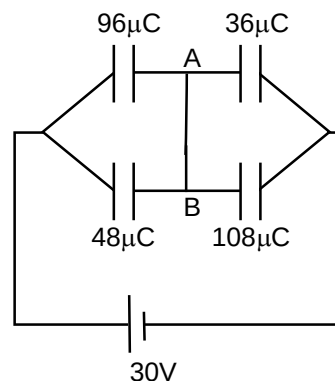
Sol.

When S is open



So charge flown = 60 μC
From A to B

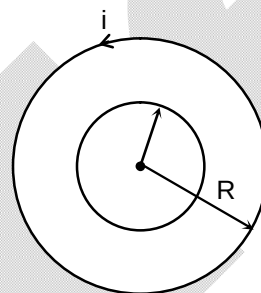
When S is closed



17. B
Sol. Basic concept

18. C
Sol. $E2\pi r = \frac{d}{dt}(B\pi r^2)$
 $E = \frac{r}{2} \frac{dB}{dt}$
 $\tau = QEr = \frac{Qr^2}{2} \frac{dB}{dt}$

19. B
Sol. $B = \frac{\mu_0 i}{2R}$
 $\phi = B\pi r^2 = \frac{\mu_0 i \pi r^2}{2R}$
Mutual inductance, $M = \frac{\phi}{i} = \frac{\mu_0 \pi r^2}{2R}$



20. D
Sol. Rate of cooling
 $-\frac{dT}{dt} = \frac{\epsilon \sigma A(T^4 - T_s^4)}{mC} = r$
 $\frac{r_{\text{sphere}}}{r_{\text{cube}}} = \frac{m_{\text{cube}}}{m_{\text{sphere}}} = \frac{\rho a^3}{\rho \frac{4}{3}\pi r^3}$
given, $4\pi r^2 = 6a^2$
 $\frac{r_{\text{sphere}}}{r_{\text{cube}}} = \frac{3\left(\frac{4\pi}{6}\right)^{3/2} r^3}{4\pi r^3} = \sqrt{\frac{\pi}{6}}$

SECTION - B

21. 6
Sol. Separation between two maxima $\beta = \frac{\lambda D}{d}$

22. 40
Sol. $v_0^2 = 4v_0^2 - (2)(\mu g)(d) \Rightarrow 2\mu g d = 3v_0^2$
 $v_0^2 = v^2 - (2)(5\mu g)(d) \Rightarrow v^2 = v_0^2 + 15v_0^2$
So $v = 4v_0$

23. 1
Sol. $\frac{2v}{2(\ell_0 + 1.2d)} = \frac{3v}{4(\ell_c + e)}$

So $4(23 + e) = 3(30 + 2e)$

So $e = 1 \text{ cm}$

24. 9

Sol. $E_0 = CB_0$
 $= 3 \times 10^8 \times 3 \times 10^{-8} = 9 \text{ V/m}$

25. 30

Sol. $\frac{1}{f_e} = -2\frac{1}{f_\ell} + \frac{1}{f_m}$,
 Where $\frac{1}{f_\ell} = \left(\frac{4}{3} - 1\right)\left(\frac{1}{40}\right) = \frac{1}{120}$
 So $\frac{1}{f_e} = -2\frac{1}{120} - \frac{1}{20} = -\frac{1}{15} \text{ cm}$
 So $R = 30 \text{ cm}$

26. 2000

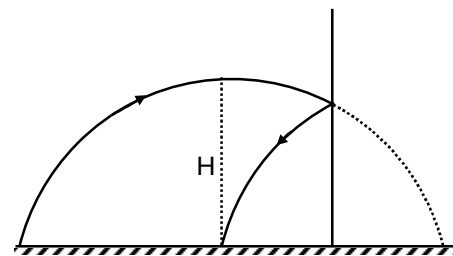
Sol. $(100 - v)10 + (200 - v)20 + (50 - v)40 = 0$
 So $v = 100 \text{ volt}$
 $Q = 40 \times 50 = 2000 \mu\text{C}$

27. 24

Sol. $\frac{dV}{dt} = \frac{\pi \Delta p r^4}{8\eta \ell}$
 $Q = \frac{\pi P_0 r^4}{8\eta \ell}$
 Now, $Q' = \frac{\pi 3P_0 (2r)^4}{8\eta \times 2\ell} = \frac{3\pi P_0 r^4}{\eta \ell} = 24Q$

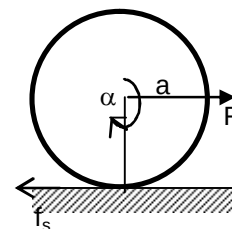
28. 4

Sol. $y = x \tan \theta \left(1 - \frac{x}{R}\right) \Rightarrow H = \frac{R}{2} \tan \theta \left(1 - \frac{1}{2}\right)$
 $\Rightarrow H = \frac{R \tan \theta}{4}$
 $y = \frac{3R}{4} \tan \theta \left(1 - \frac{3R}{4R}\right)$
 $= \frac{3R}{16} \tan \theta = \frac{3R}{16} \frac{4H}{R} = \frac{3H}{4}$



29. 16

Sol. $a = \alpha R$... (i)
 $F - f_s = ma$
 $f_s R = \frac{mR^2 \alpha}{2}$
 $f_s = \frac{ma}{2}$.. (ii)



$$F = \frac{3ma}{2}$$

$$\Rightarrow a = \frac{2F}{3m} \Rightarrow f_s = \frac{ma}{2} = \frac{F}{3}$$

30. 2

Sol. In steady state, $v = \text{constant}$

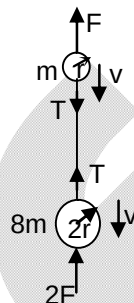
$$9mg = 3F$$

$$\Rightarrow F = 3mg \quad \dots (i)$$

$$T + mg = F$$

$$T = F - mg = 3mg - mg = 2mg$$

$$T = 2mg$$



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Chemistry

PART – B

SECTION – A

31. B

Sol. $P_1 V_1 = P_2 V_2$

$$10m \times (4) \times A = (10 + h)m \times h \times A$$

$$h = 3.06 \text{ m}$$

32. C

Sol. Dithionous acid $\rightarrow \text{H}_2\text{S}_2\text{O}_4$

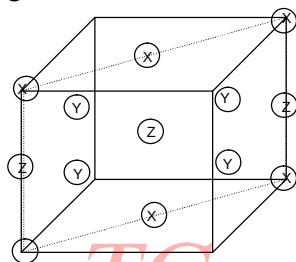
$$S = +3$$

Dithionic acid $\rightarrow \text{H}_2\text{S}_2\text{O}_6$

$$S = +5$$

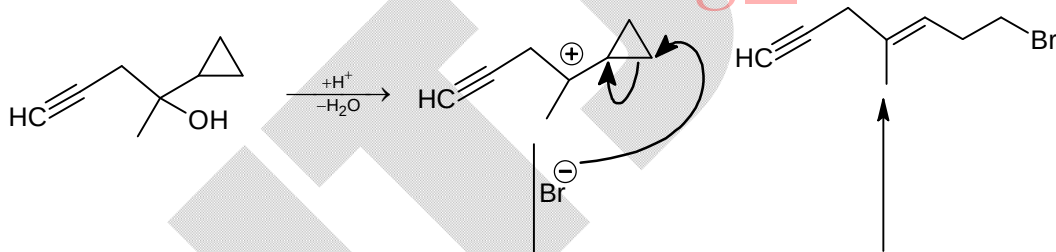
33. C

Sol.



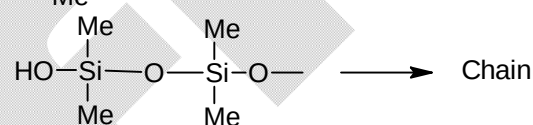
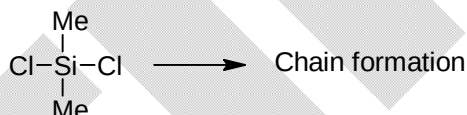
34. D

Sol.



35. B

Sol.



36. C

Sol. LiClO_4 is more soluble than LiF .

37. D

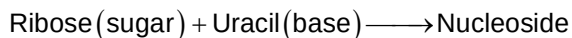
Sol. Factual

38. B

Sol. BF_3 is electron deficient and is used to generate electrophile in F.C. reactions. Therefore, acts as catalyst in industrial processes.

39. B

Sol. Uridine is glycosylated pyrimidine-containing uracil attached to ribose ring.



40. A

 Sol. Degenerate orbitals = n^2

$$= (5)^2 = 25$$

41. D

 Sol. $\Delta S \rightarrow$ State function $\Delta S = \Delta S_1 + \Delta S_2$

$$\Delta S = \frac{q}{T} \text{ since } q = \text{same; smaller temperature, higher entropy.}$$

42. D

 Sol. $\text{Ti}^{+3} \longrightarrow 3d^1 \longrightarrow t_{2g}^1$
 $\text{Cr}^{+3} \longrightarrow 3d^3 \longrightarrow t_{2g}^3$
 $\text{Mn}^{+3} \longrightarrow 3d^4 \longrightarrow t_{2g}^3 e_g^1$
 $\text{Co}^{+3} \longrightarrow 3d^6 \longrightarrow t_{2g}^4 e_g^2$

43. A

Sol. Factual

44. B

 Sol. MnO_2 oxidises unhindered primary allylic and benzylic alcohols to aldehyde.

45. C

$$\text{Sol. } \frac{1}{x_0 - a} - \frac{1}{x_0} = k_2 t$$

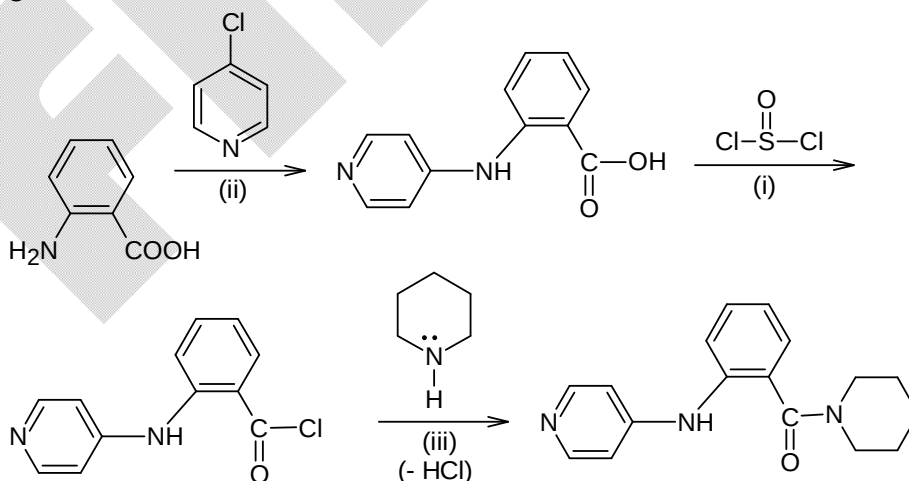
Rearranging

$$\frac{a}{x_0 - a} = x_0 k_2 t$$

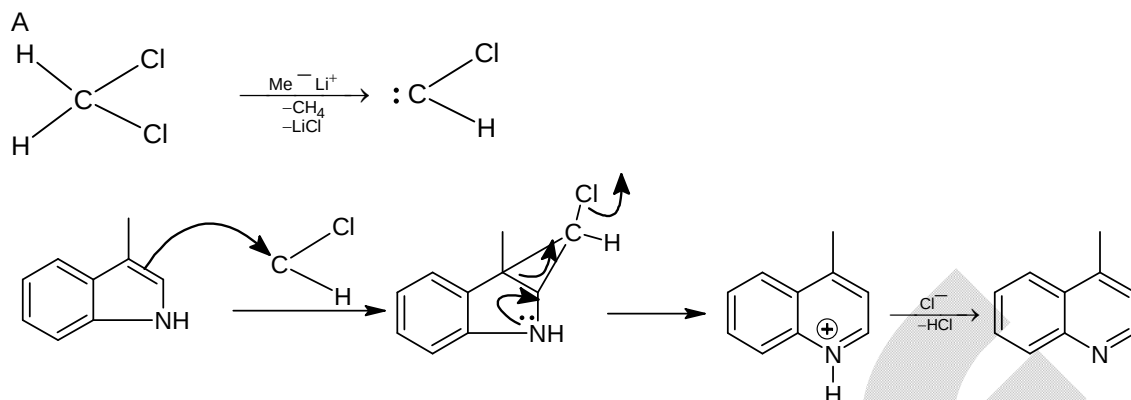
 $\therefore \frac{a}{x_0 - a}$ vs. t gives $x_0 k_2$ as slope.

46. C

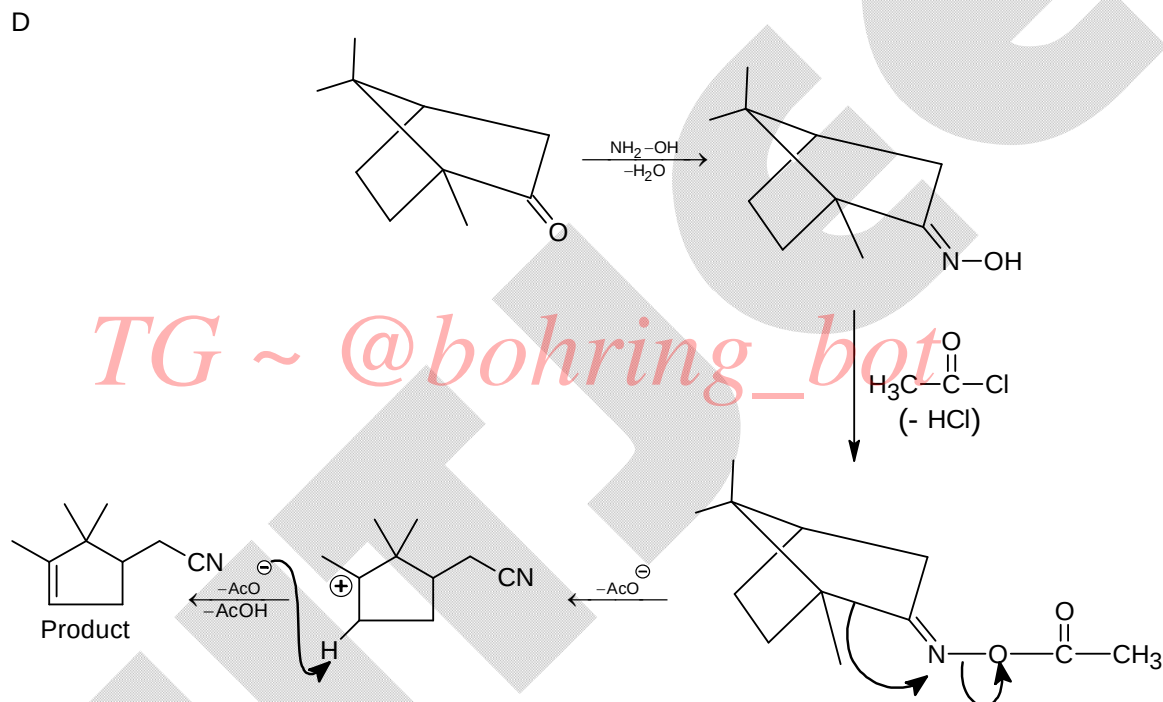
Sol.



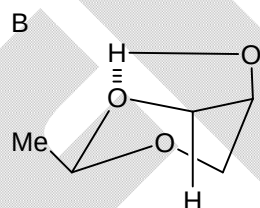
47.
Sol.



48.
Sol.



49.



Sol.

Hydrogen bonding and as 1-3 dihedral interaction between methyl and hydrogen.

50.

B

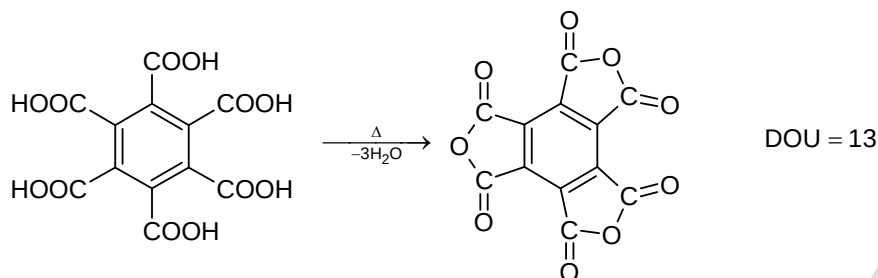
Sol.

$$x_A = \frac{0.744 + 0.359Z_{\text{eff}}}{r_A^2} \leftarrow \text{Alfred - Rochow scale}$$

SECTION - B

51. 13

Sol.



52. 2

Sol.

$$[H^+] = 10^{-4}$$

$$\therefore [Ca^{+2}] = \frac{10^{-4}}{2} M$$

$$\text{Weight of } Ca^{+2} = 20 \times 10^{-4} \text{ g in 1 L}$$

$$\therefore \text{Weight of } Ca^{+2} \text{ ion in } 10^6 \text{ mL } H_2O = \frac{20 \times 10^{-4} \times 10^6}{10^3} = 2$$

$$\therefore \text{Hardness} = 2 \text{ ppm.}$$

53. 1

Sol.

$$\text{Theoretical efficiency} = \frac{-\Delta G^0}{-\Delta H^0}$$

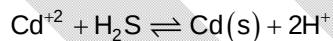
$$0.84 = \frac{-\Delta G^0}{285}$$

$$\Delta G^0 = -0.84 \times 285 \text{ kJ} = -nFE_{\text{cell}}^0$$

$$E_{\text{cell}}^0 = 1.24 \text{ V}$$

54. 2

Sol.



$$\text{Initial } 0.1 \times 10^{-3} \text{ moles}$$

$$0.08 \times 10 \times 10^{-3} \text{ moles}$$

$$\text{final } -$$

$$0.8 \times 10^{-3} + 0.2 \times 10^{-3}$$

$$= 10^{-3} \text{ moles}$$

$$\therefore [H^+]_{\text{final}} = \frac{10^{-3}}{100} \times 1000 = 0.01 \text{ M} = 10^{-2} \text{ M}$$

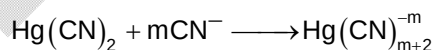
$$\text{pH} = 2.0$$

55. 8

Sol.

$$\text{For KCN: } \Delta T_f = iK_f \cdot m \Rightarrow 0.8 = K_f \cdot 0.2 \times 2$$

$$K_f = 2 \quad \dots (i)$$



$$\text{Initial } 0.1 \quad 0.2 \quad -$$

$$\text{Final } (0.2 - 0.1 \text{ m}) \quad 0.1$$

$$\text{Final effective molality} = (0.2 - 0.1 \text{ m}) + 0.1 + 0.2 = 0.5 - 0.1 \text{ m}$$

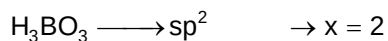
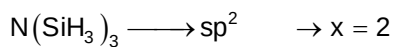
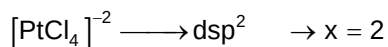
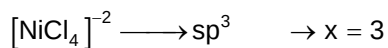
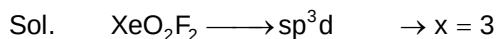
New: $\Delta T_f = K_f \cdot m$

$$0.6 = 2 \times (0.5 - 0.1 m)$$

$$m = 2$$

$$4m = 8$$

56. 3

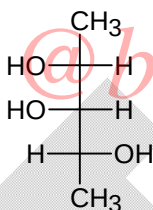
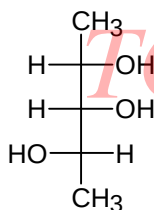


$$\sum x = 3 + 6 + 3 + 2 + 2 + 2 = 18$$

$$\therefore \frac{\sum x}{6} = 3$$

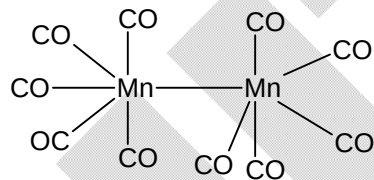
57. 2

Sol.



58. 0

Sol.

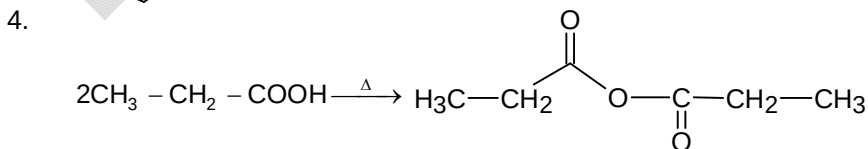
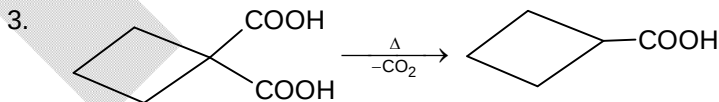
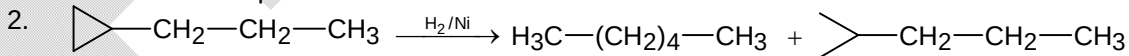


59. 6

Sol. Black coloured : PbS , CuS , HgS , Ag_2S , NiS and CoS .

60. 6

Sol. 1 and 5 have correct products.



Mathematics

PART – C

SECTION – A

61. B

Sol. $A = \{1, 2, 3, \dots, 141\}$

$A \cap (B - C) = \{5, 11, 14, 17, 23, 26, 29, \dots, 134, 137\}$

Sum of elements = $(5 + 11 + 17 + \dots + 137) + (14 + 26 + 38 + \dots + 134)$

62. C

Sol. $x^2 - ax + 4 = 0$ must have atleast one positive root

$\therefore \alpha\beta > 0$

$\therefore$ both roots must be positive

$\Rightarrow a \geq 4$

63. B

Sol. $30(1 + x + x^2)^{30} (1 + 2x) = \sum_{r=0}^{60} r a_r x^{r-1} (1 + x + x^2)$

Equating coefficient of x^{20} on both sides we have,

$30(a_{20} + 2a_{19}) = 21a_{21} + 20a_{20} + 19a_{19}$

$\Rightarrow 21a_{21} = 10a_{20} + 41a_{19}$

64. C

Sol. Use $\int (xg'(x) + g(x))dx = xg(x) + C_1$

65. C

Sol. $k = 2500 = 2^2 \cdot 5^4$

$\therefore$ number of factor = 5

66. B

Sol. $b = -4, c = 9$

$2\sin^2\theta = 3\cos\theta$

$\Rightarrow 2\cos^2\theta + 3\cos\theta - 2 = 0$

$\cos\theta = \frac{1}{2} \therefore \sin^2\theta = \frac{3}{4}$

$\therefore 4x^2 - 4y^2 + xy + 12x + \frac{3}{2}y + 9 = 0$

67. A

Sol. $r = \frac{r_1 r_2}{r_1 + r_2 + 2\sqrt{r_1 r_2}}$

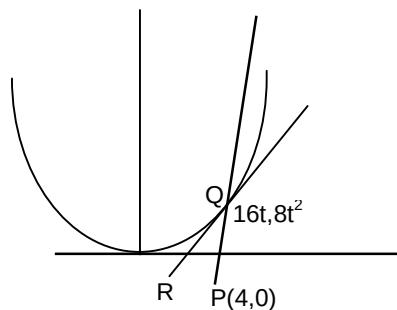
68. B

Sol. $A = \frac{1}{2} (4 - 8t)8t^2 = 16(t^2 - 2t^3)$

$\frac{dA}{dt} = 16(2t - 6t^2) = 0 \Rightarrow t = 0, \frac{1}{3}$

$\frac{d^2A}{dt^2} = 16(2 - 12t) < 0$ at $t = \frac{1}{3} \Rightarrow A$ is maximum

$\therefore Q\left(\frac{16}{3}, \frac{8}{9}\right)$ lies on the line $\therefore m = \frac{2}{3}$



69. A

Sol. M is (1, 10)
 $\therefore$ N is (4, 16)
 L is (-2, 4)

Equation of tangent to $(x + a)y = \lambda$ is

$$y = 2x + \frac{\lambda}{t} + 2(a - t); \lambda = -2t^2$$

$$\therefore 2a - 4t = +8 \Rightarrow a - 2t = 4$$

$$\text{and } (a - 2)4 = \lambda = -2t^2$$

$$\therefore (t + 2)^2 = 0$$

$$\Rightarrow t = -2$$

$$\therefore \lambda = -8 \text{ \& } a = 0$$

70. B

Sol. $4a^2 - b^2 = 25$

Equation of auxiliary circle is $x^2 + y^2 = 8$

$$\Rightarrow a^2 = 8, b^2 = 7$$

$$\text{Also } a^2 + b^2 = A^2 - \lambda^2$$

$$\Rightarrow \lambda^2 = 25 - 15 = 10$$

$$\therefore \text{eccentricity of ellipse } e = \sqrt{1 - \frac{10}{25}} = \sqrt{\frac{3}{5}}$$

71. D

Sol. IF = $e^{x/2} \sqrt{\sin x + \cos x}$

$$\therefore \text{solution is } ye^{x/2} \sqrt{\sin x + \cos x} = \int e^{x/2} (\tan x + 2 \sec^2 x) dx$$

$$ye^{x/2} \sqrt{\sin x + \cos x} = 2e^{x/2} \tan x + c$$

$$\therefore y\left(\frac{\pi}{4}\right) = 2^{3/4} \therefore c = 0$$

$$y = \frac{2 \tan x}{\sqrt{\sin x + \cos x}}$$

72. D

Sol. $\text{Arg}\left(\frac{z^4 - z}{z^2 - z}\right) = 0$

$$\Rightarrow \text{Arg}\left(\frac{z(z-1)(z^2+z+1)}{z(z-1)}\right) = 0; z \neq 0, 1$$

$$\Rightarrow z^2 + z + 1 \text{ is positive real number } (x + iy)^2 + x + iy + 1 > 0$$

$$\Rightarrow x^2 - y^2 + x + 1 > 0 \text{ and } y(1 + 2x) = 0$$

$$(i) y = 0, x^2 + x + 1 > 0 \Rightarrow x \in \mathbb{R}$$

$$(ii) x = -\frac{1}{2}, y^2 < \frac{3}{4} \Rightarrow y \in \left(-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$$

73. C

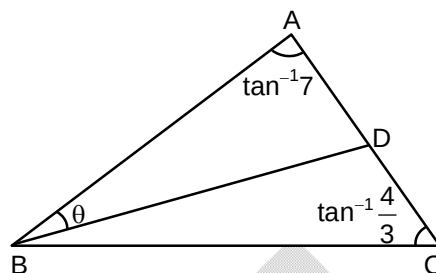
Sol. $\frac{a}{a-\sqrt{3}} + \frac{b}{b-\sqrt{5}} + \frac{c}{c-\sqrt{7}} = 2$

$$\text{If } a = 2\sqrt{3}, \text{ then } \frac{b}{b-\sqrt{5}} + \frac{c}{c-\sqrt{7}} = 0$$

$$\text{Possible value of } b = \frac{2\sqrt{5}}{3}, c = 2\sqrt{7}$$

74. D

$$\begin{aligned} \text{Sol. } \frac{\frac{1}{2} \sin\left(\tan^{-1} \frac{4}{3}\right) BD \sin \theta}{\frac{1}{2} \sin(\tan^{-1} 7) BD \sin\left(\frac{\pi}{4} - \theta\right)} &= 2 \\ \Rightarrow \frac{\frac{4}{5} \sin \theta}{\frac{7}{5\sqrt{2}} \left(\frac{\cos \theta - \sin \theta}{\sqrt{2}}\right)} &= 2 \Rightarrow \cot \theta = \frac{11}{7} \end{aligned}$$



75. C

$$\begin{aligned} \text{Sol. } P(A) &= \frac{2}{5}, P(B) = \frac{3}{5} \\ P(D) &= \frac{3}{20} \\ 6 \times \frac{2}{5}x + \frac{3}{5} \times x &= \frac{3}{20} \Rightarrow x = \frac{1}{20} \\ P\left(\frac{B}{D^c}\right) &= \frac{P(B \cap D^c)}{P(D^c)} = \frac{\frac{3}{5} \times \frac{19}{20}}{\frac{85}{100}} = \frac{57}{85} \end{aligned}$$

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76. A

$$\text{Sol. } \frac{\sum_{i=1}^n x_i f_i}{\sum_{i=1}^n f_i} = \frac{15a + 25b + 980}{26 + a + b} = \frac{277}{9} \quad \dots(1)$$

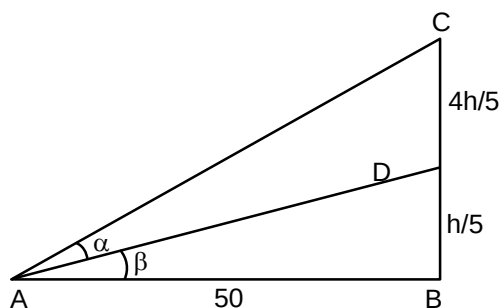
$$\begin{aligned} \text{Also } 30 + \frac{\frac{26 + a + b}{2} - (a + b + 3)}{11} \times 10 &= \frac{335}{11} \\ \Rightarrow a + b &= 19 \end{aligned} \quad \dots(2)$$

From (1) and (2) $a = 7, b = 12$.

77. B

78. A

$$\begin{aligned} \text{Sol. } \tan(\alpha + \beta) &= \frac{h}{50} \\ \Rightarrow \frac{h}{50} &= \frac{\frac{8}{21} + \frac{h}{250}}{1 - \frac{8}{21} \times \frac{h}{250}} = \frac{2000 + 21h}{5250 - 8h} \\ h &= 25 \end{aligned}$$



79.

B

Sol.

$$\boxed{3} \boxed{5} \dots\dots = 9 \times 5 = 45$$

$$\boxed{\quad} \boxed{3} \boxed{5} \quad = 8 \times 5 = 40$$

$$\dots\dots\dots = 8 \times 9 \times 9 \times 5 = 3240$$

$$N = 3325$$

80.

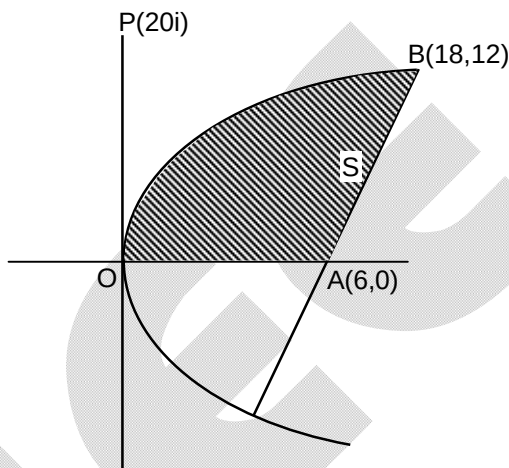
A

Sol.

$$PB = \sqrt{324 + 64} = \sqrt{388}$$

$$PA = \sqrt{36 + 400} = \sqrt{436}$$

$$OP = 20$$



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SECTION - B

81.

5

Sol.

$$\sum_{m=1}^{15} \frac{\sin\left(\theta + m\frac{\pi}{18} - \left(\theta + (m-1)\frac{\pi}{18}\right)\right)}{\cos\left(\theta + m\frac{\pi}{18}\right)\cos\left(\theta + (m-1)\frac{\pi}{18}\right)}$$

$$= \sum_{m=1}^{15} \tan\left(\theta + m\frac{\pi}{18}\right) - \tan\left(\theta + (m-1)\frac{\pi}{18}\right)$$

$$\Rightarrow \tan\left(\theta + \frac{5\pi}{6}\right) - \tan\theta = 4 + 2\sqrt{3}$$

$$\Rightarrow \cos\left(2\theta + \frac{5\pi}{6}\right) = 1$$

$$\Rightarrow \theta = n\pi - \frac{5\pi}{12}; n \in \mathbb{I}.$$

82.

59

Sol.

Squaring and adding

$$1 + \sin^6 A + \cos^6 A + 2(\cos^4 A - \sin^4 A) = 1/4$$

$$\Rightarrow 2 + 1 - 3\sin^2 A \cos^2 A + 2\cos 2A = 1/4$$

$$2 - 3/4 \sin^2 2A + 2 \cos 2A = 1/4$$

$$\Rightarrow 3 \cos^2 2A + 8 \cos 2A + 4 = 0$$

$$\cos 2A = -2/3$$

$$\sin A = \sqrt{\frac{5}{6}}$$

$$\therefore \cos B = 2\left(\sqrt{\frac{5}{6}} - \frac{5}{6}\sqrt{\frac{5}{6}}\right) = \sqrt{\frac{5}{54}}.$$

83. 13

Sol. $a = \frac{5\pi}{12}, b = \frac{\pi}{3}$

$$\int_{\pi/3}^{5\pi/12} \left[\frac{\tan x}{\sqrt{3}} \right] dx$$

$$= \int_{\pi/3}^{\tan^{-1} 2\sqrt{3}} 1 + \int_{\tan^{-1} 2\sqrt{3}}^{5\pi/12} 2dx = \cot^{-1} 2\sqrt{3} = \operatorname{cosec}^{-1} 13.$$

84. 546

Sol. Let $A = I + B$ where $B = \begin{bmatrix} 0 & 0 & 0 \\ \sqrt{a} & 0 & 0 \\ a\sqrt{a} & \sqrt{b} & 0 \end{bmatrix}$, $B^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \sqrt{ab} & 0 & 0 \end{bmatrix}$, $B^3 = 0$ (Null)

$$\therefore A^n = I + {}^nC_1 B + {}^nC_2 B^2$$

$$\Rightarrow A^n = \begin{bmatrix} 1 & 0 & 0 \\ n\sqrt{a} & 1 & 0 \\ na\sqrt{a} + \frac{n(n-1)}{2}\sqrt{ab} & n\sqrt{b} & 1 \end{bmatrix}$$

$$\Rightarrow n\sqrt{a} = n\sqrt{b} = 72 \Rightarrow a = b \text{ and } 72a + 36(72 - \sqrt{a}) = 3600$$

$$\Rightarrow \sqrt{a} = 4 \Rightarrow a = 16 \text{ and } n = 18.$$

85. 20

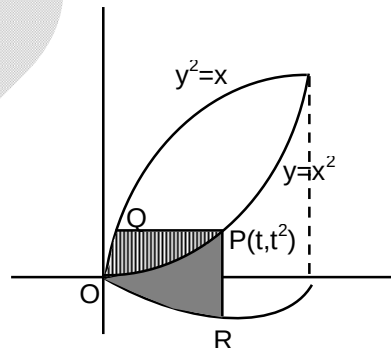
Sol. Let $P(t, t^2)$, $Q(t^4, t^2)$

$$\int_0^t (x^2 - f(x)) dx = 2 \int_0^{t^2} (\sqrt{y} - y^2) dy$$

$$\Rightarrow t^2 - f(t) = 2(t - t^4)2t$$

$$\Rightarrow f(t) = 4t^5 - 3t^2$$

$$f\left(\frac{1}{2}\right) = \frac{4}{32} - \frac{3}{4} = -\frac{20}{32}.$$



86. 20

Sol. $f(x) = 1 + 4^x$

$$\lim_{x \rightarrow 1} \frac{\sin(4^x - 4) + 2 - 2^x}{x^3 - 7x^2 + ax + b} = \ln 8$$

For $\left(\frac{0}{0}\right)$ form $a + b = 6$

$$\text{Also } \lim_{x \rightarrow 1} \frac{(\cos(4^x - 4))4^x \ln 4 - 2^x \ln 2}{3x^2 - 14x + a} = \ln 8$$

$$\Rightarrow \frac{6 \ln 2}{a - 11} = 3 \ln 2$$

$$\Rightarrow a - 11 = 2 \Rightarrow a = 13, b = -7$$

87. 81
Sol. For continuity at $x = 0$

$$\lim_{x \rightarrow 0} \frac{2^{\frac{cx}{3}} - 1}{\frac{cx}{3}} \cdot \frac{c}{3} = \frac{\ln 4}{3} \Rightarrow c = 2$$

For differentiability at $x = 0$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{6a(\ln 2)^2}{1 + \left(\frac{b+x}{3}\right)^2} \cdot \frac{1}{3} &= \lim_{x \rightarrow 0} \frac{2^{\frac{cx}{3}} - 1}{x} - \frac{\ln 4}{3} \\ \Rightarrow \frac{6a(\ln 2)^2}{3 \left(1 + \frac{b^2}{9}\right)} &= \lim_{x \rightarrow 0} \frac{3(2^{\frac{cx}{3}} - 1) - x \ln 4}{3x^2} \\ &= 2 \frac{(\ln 2)^2}{9} = \lim_{x \rightarrow 0} \frac{3 \left(2^{\frac{cx}{3}} \ln 2 \times \frac{c}{3}\right) - \ln 4}{6x} = \frac{2(\ln 2)^2}{9} \\ &= \frac{6a(\ln 2)^2 \times 9}{3(b^2 + 9)} = \frac{2(\ln 2)^2}{9} = \frac{b^2 + 9}{a} = 81 \end{aligned}$$

88. 1386
Sol. $h(g(g(x))) = x$
 $h'(g(g(x))) \cdot g'(g(x)) \cdot g'(x) = 1$; $g(x) = f^{-1}(x)$
 $h'(0) \Rightarrow g(g(x)) = 0 \Rightarrow g(x) = f(0) = 3 \Rightarrow x = f(3) = 309$

$$\text{Also } g'(f(x)) = \frac{1}{f'(x)}$$

$$g'(3) = \frac{1}{f'(0)} = \frac{1}{3}$$

$$g'(309) = \frac{1}{f'(3)} = \frac{1}{462}$$

$$\Rightarrow h'(0) = \frac{1}{g'(3)g'(309)}$$

89. 10648
Sol. L_1 is parallel to vector $2\hat{i} - \hat{j} - \hat{k}$ and it passes through point $(13, -7, 0)$.
Equation of plane P is $x + 5y - 3z + 22 = 0$

$$V = \frac{1}{6} \begin{vmatrix} -22 & 0 & 0 \\ 0 & -22 & 0 \\ 0 & 0 & 22 \end{vmatrix} = \frac{(22)^3}{90}$$

90. 225

Sol. Let $\hat{x} \cdot \hat{y} = a$, $\hat{y} \cdot \hat{z} = b$, $\hat{x} \cdot \hat{z} = c$

$$|\hat{x} + \hat{y} - \hat{z}| = |\vec{p}| \Rightarrow a - b - c = 2 \quad \dots(1)$$

$$\text{Also } c\hat{y} - a\hat{z} = \vec{q} \quad \dots(2)$$

$$\text{and } c\hat{y} - b\hat{x} = \vec{r} \quad \dots(3)$$

$$\therefore \vec{q} \cdot \hat{y} = 5 \quad \therefore c - ab = 5 \quad \dots(4)$$

$$\text{and } c - a = 1 \quad \dots(5)$$

$$\text{Hence } a = 1, b = -3, c = 2$$

TG ~ @bohring_bot

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2023

TEST DATE: 18-01-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

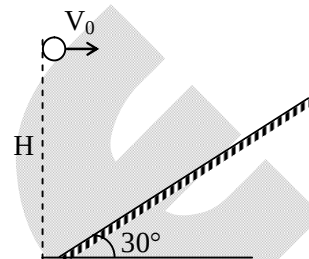
1. In the given figure, the angle of inclination of the inclined plane is 30° . A particle is projected horizontally from a height H above the foot of the inclined plane. Find the horizontal velocity V_0 so that the particle will retrace the path after hitting the inclined plane. Collision is perfectly elastic.

(A) $v_0 = \sqrt{\frac{2gH}{5}}$

(B) $v_0 = \sqrt{\frac{2gH}{7}}$

(C) $v_0 = \sqrt{\frac{gH}{5}}$

(D) $v_0 = \sqrt{\frac{gH}{7}}$



2. A signal of frequency 25 KHz and peak voltage 20 V is used to modulate a carrier wave of frequency 1.5 MHz and peak voltage 50 V. Choose the correct statement.

- (A) Modulation index is 0.4, side frequency bands are 1475 KHz and 1525 KHz
 (B) Modulation index is 0.4, side frequency bands are 2950 KHz and 3150 KHz
 (C) Modulation index is 2.5, side frequency bands are 1475 KHz and 1575 KHz
 (D) Modulation index is 2.5, side frequency bands are 2950 KHz and 3150 KHz

3. A ball of mass m falls vertically from a height $h = \frac{20}{3}m$ and

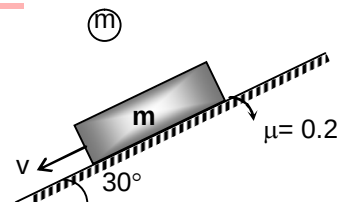
collides with a block of equal mass m moving down an inclined plane with a velocity v as shown in the figure. The coefficient of kinetic friction between the block and the surface is 0.2, while the coefficient of restitution (e) between the ball and the block is 0.5. There is no friction acting between the ball and the block. The velocity of the block decreases by

(A) 5 m/s

(B) 0

(C) 1 m/s

(D) 3 m/s



4. Air column in an open organ pipe of length $\ell = 30$ cm resonates with a sonic source emitting wave of frequency $f = 1100\text{Hz}$. If maximum pressure variation in the pipe over the atmospheric pressure due to standing wave is $0.1 P_0$, where P_0 is the atmospheric pressure. Find the difference of the maximum and minimum pressure in the tube at a distance 5 cm from the open end. Take speed of sound in air equal to 330 m/s and ignore the end correction.

(A) $\frac{\sqrt{3}}{20} P_0$

(B) $\frac{\sqrt{3}}{10} P_0$

(C) $\sqrt{3} P_0$

(D) $2 P_0$

5. At time $t = 0$, a horizontal disc starts rotating with angular acceleration 1 rad/sec^2 about an axis perpendicular to its plane and passing through its center. A small block is lying on this disc at a distance 0.5 m from center, coefficient of friction between surface of block and disc is 0.255. The block will start slipping on the disc at time t , is approximately equal to

(A) $2\sqrt{3} \text{ s}$

(B) $2\sqrt{2} \text{ s}$

(C) 6 s

(D) $\sqrt{5} \text{ s}$

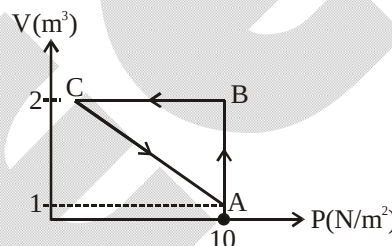
6. A thin spherical conducting shell of radius R has a charge q . Another charge Q is placed at the centre of the shell. Calculate electrostatic potential at a point P a distance $R/2$ from the centre of the shell.

(A) $\frac{2Q}{2\pi\epsilon_0 R} - \frac{2q}{4\pi\epsilon_0 R}$ (B) $\frac{Q}{2\pi\epsilon_0 R} + \frac{q}{4\pi\epsilon_0 R}$
 (C) $\frac{(q+Q)}{2\pi\epsilon_0} \frac{2}{R}$ (D) $\frac{2Q}{2\pi\epsilon_0 R}$

7. An ammeter reads upto 1 ampere. Its internal resistance is 0.81 ohm. To increase the range to 10A the value of the required shunt is –

(A) 0.3 Ω (B) 0.9 Ω
 (C) 0.09 Ω (D) 0.30 Ω

8. An ideal gas is taken through a cycle $A \rightarrow B \rightarrow C \rightarrow A$ as shown in figure. If the heat supplied in the cycle is 5J, then work done on the gas in the process $C \rightarrow A$ is in joules is
 (A) -5J (B) -10J
 (C) -15J (D) -20J

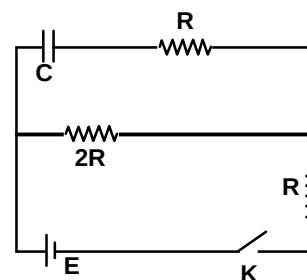


9. A space vehicle approaching a planet has a speed v , when it is very far from the planet. At that moment tangent of its trajectory would miss the centre of the planet by distance R . If the planet has mass M and radius r , what is the value of R in order that the resulting orbit of the space vehicle will just miss the surface of the planet?

(A) $R = \frac{r}{v} \left[v^2 + \frac{2GM}{r} \right]^{1/2}$ (B) $R = vr \left[1 + \frac{2GM}{r} \right]$
 (C) $R = \frac{r}{v} \left[v^2 + \frac{2GM}{r} \right]$ (D) $R = \frac{2GMv}{r}$

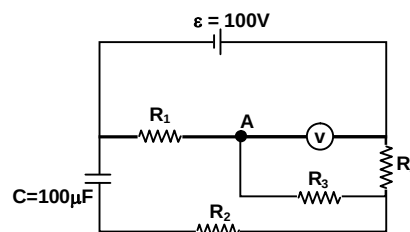
10. In the given circuit, the capacitor of capacitance C is charged by closing key K at $t = 0$. Find the time required to charge the capacitor up to maximum charge for the given circuit, if it were to be charged with the constant initial charging rate at $t = 0$ in the given circuit.

(A) $RC/3$ (B) $2RC/3$
 (C) $2RC/3$ (D) $5RC/3$



11. In the circuit shown in figure E is a battery of emf 100 V. Resistance $R_1 = 100\Omega$, $R_2 = 200\Omega$, $R_3 = 200\Omega$. The voltmeter resistance is 200Ω . What reading will the voltmeter show in steady state?

(A) 50/3 V (B) 400/7 V
 (C) 100/3 V (D) 50 V



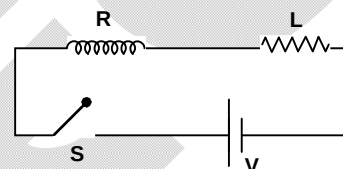
12. A ray of light travelling in a medium of refractive index μ is incident at an angle θ on a composite transparent plate consisting of 50 plates of refractive indices $1.01\mu, 1.02\mu, 1.03\mu, \dots, 1.50\mu$. The ray emerges from the composite plate into a medium of refractive index 1.6μ at angle x . Then

(A) $\sin x = \left(\frac{1.01}{1.5}\right)^{50} \sin \theta$ (B) $\sin x = \frac{5}{8} \sin \theta$
 (C) $\sin x = \frac{8}{5} \sin \theta$ (D) $\sin x = \left(\frac{1.5}{1.01}\right)^{50} \sin \theta$

13. The binding energy per nucleon of deuteron (${}^2_1\text{H}$) and helium nucleus (${}^4_2\text{He}$) is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is—

(A) 23.6 MeV (B) 26.9 MeV
 (C) 13.9 MeV (D) 19.2 MeV

14. An inductor of inductance $L = \frac{\tau R}{2}$ and a resistor of resistance R is connected to a battery of emf V as shown in the figure. The potential difference across the resistance at a time, $t = \tau \ln 2$ after the switch S is closed is (τ is constant)



(A) $\frac{V}{4}$ (B) $\frac{3V}{4}$
 (C) $\frac{V}{2}$ (D) $\frac{2V}{3}$

15. A resistance R draws P power when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes Z , the power drawn will be

(A) $P \left(\frac{R}{Z}\right)^2$ (B) $P \left(\frac{R}{Z}\right)$
 (C) $P \sqrt{\frac{R}{Z}}$ (D) P

16. At STP 44.8 liter of helium gas is confined in a cylindrical vessel by a movable piston. The gas is thermally insulated from surrounding, however, heat is supplied to the gas very slowly by an electric heater consequently piston moves slowly. In this process find the heat supplied to the gas by the heater to increase its temperature by 20°C . Ignore any dissipative force on the piston. (R is universal gas constant)

(A) $200 R$ (B) $100 R$
 (C) $60 R$ (D) $140 R$

17. Wavelength of K_α X-ray of an element A is λ_1 and wavelength of K_α X-ray element B is λ_2 . $\frac{\lambda_1}{\lambda_2}$ is

equal to $\frac{1}{4}$ and Z_1 and Z_2 are the atomic numbers of element A and B respectively. Then relation between Z_1 and Z_2 are given by

(A) $2Z_2 - Z_1 = 1$ (B) $Z_2 - 2Z_1 = 1$
 (C) $\frac{Z_2}{Z_1} = 4$ (D) $\frac{Z_1}{Z_2} = 4$

18. A radioactive element undergoes two different types of radioactive disintegration, one with disintegration constant λ_1 and the other with λ_2 . The half-life of the element is
- (A) $\frac{0.693}{\lambda_1 + \lambda_2}$ (B) $\frac{0.693}{(\lambda_1 + \lambda_2)/2}$
- (C) $0.693 \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$ (D) $\frac{0.693}{2} \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$
19. A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. Almost instantaneously, another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector ?
- (A) two photons of energy 10.2 eV
 (B) two photons of energy 1.4 eV
 (C) one photon of energy 10.2 eV and an electron of energy 1.4 eV
 (D) one electron having kinetic energy nearly 11.6 eV
20. A uniform string of mass $m = 200$ g and lengths $l = 5$ m is stretched along x-axis. One end at $x = 0$ is attached to a harmonic oscillator and the other end is fixed to a rigid support. Tension in the string is $T = 1600$ N. At the time $t = 0$, the oscillator starts producing transverse harmonic waves in the string. The y-coordinate of a particle in the string near the source varies with time as $y = 0.5$ cm $\sin(100\pi t)$, where t is the time in second. Find the y-co-ordinate of the mid point on the string at the instant when first disturbance emitted by the source reaches the fixed end.
- (A) $\frac{1}{2\sqrt{2}}$ cm (B) $\frac{1}{2\sqrt{2}}$ cm
 (C) $\frac{1}{\sqrt{2}}$ cm (D) $\frac{1}{2}$ cm

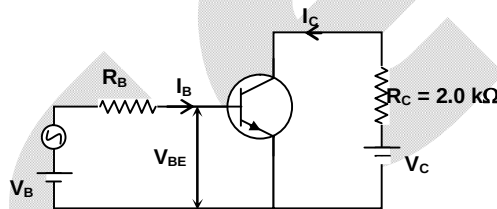
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. An electric lamp designed for operation on 110V AC is connected to a 220 V AC supply, through a choke coil of inductance 2H, for proper operation. The angular frequency of the AC is $100\sqrt{10}$ rad/s. If a capacitor is to be used in place of the choke coil, its capacitance must be (in μF)
22. A cylindrical block of mass $m = 200$ gm is suspended from a fixed support by a light spring of stiffness $k = 90$ N/m. The block is slightly pulled down from its equilibrium and then the pulling force is removed. Consequently, the block performs oscillatory motion. If damping constant of air is 40 g/s, the time (in sec) after which displacement amplitude reduces to the half of the initial value is nearly
23. An object is projected with a speed 10 m/s at an angle of 30° with the horizontal. The object breaks down into n equal fragments during its motion. One fragment is found to strike the ground at a distance of $\sqrt{3}$ m from the point of projection in the same azimuthal plane, in which the object is projected. If the centre of mass of the remaining fragments strikes the ground at distance of $7\sqrt{3}$ m from the point of projection, then the value of n is
24. A beam of light converges at a point P. A converging lens of focal length 25 cm is placed in the path of the beam before converging it at P at a distance of 20 cm from the point P. Now locate the point where beam will converge. Find the distance from the position of the lens.

25. All atoms in a sample of helium gas are in the first excited state. Photons emitted from these atoms due to de-excitation are incident on the surface of a metal having work function $\phi = 2.8$ eV. Find the stopping potential of the photoelectrons ejected from metal surface. The ionization energy of hydrogen atom is 13.6 eV.
26. The half life of the radioactive nuclei ${}^{60}_{26}\text{Co}$ is 5.3 year. The activity of a sample containing ${}^{60}_{26}\text{Co}$ nuclei is observed to be 8.0 mCi. Find the amount of ${}^{60}_{26}\text{Co}$ in the sample in microgram. (Given: $1\text{Ci} = 3.7 \times 10^{10}$ dPS)
27. When forward biasing across a p-n junction diode is increased from 0.65 V to 0.70 V, the diode current increases by 5mA. Find the dynamic resistance of the diode.
28. Circuit diagram of a common emitter amplifier is shown in the figure. The audio signal voltage across the collector resistance $R_C = 2.0$ k Ω is 2V. The current amplification factor of the transistor is 100. Find the value of R_B (in k Ω) in the series with V_B supply of 2V. given that DC base current is 10 times the signal current and $V_{BE} = 0.6$ V.
29. A 200 m long antenna is fixed on a 500 m tall tower. This antenna can transmit waves of wavelength nearly equal to?
30. A parallel beam of monochromatic light passes through a narrow slit of width $d = 0.2$ mm. The angular width of the central maximum is found to be 4×10^{-3} rad. Find the wavelength of the light in $\text{\AA}$.



Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

31. Substance which tends to decompose at their sublimation temperature can preferably purified by sublimation under the following pressure condition
(A) increased pressure
(B) reduced pressure
(C) pressure has no effect on the sublimation temperature of a substance
(D) at atmospheric pressure
32. Liquid A and liquid B form ideal solution. The ratio of the pure vapour pressure of A and B, respectively at 298 K is 1 : 4. Also, the ratio of their masses in liquid phase is 2 : 1. What will be the ratio of their masses in vapour phase?
(A) 1 : 2
(B) 1 : 8
(C) 8 : 1
(D) 2 : 1
33. In which of the following conditions, the electrical conductivity decreases?
(A) ZnO is heated in Zn vapour
(B) NaCl is heated in Na vapour
(C) Silicon is doped with arsenic
(D) Heating metallic conductor
34. Which of the following order is INCORRECT?
(A) $\text{Li} < \text{Be} < \text{B} < \text{C}$ (magnitude of effective nuclear charge)
(B) $\text{Li}^+ > \text{Na}^+ < \text{K}^+ < \text{Rb}^+$ (ionic radius in aqueous medium)
(C) $\text{N}^{3-} > \text{Ne} > \text{Mg}^{2+} > \text{Al}^{3+}$ (ionic radius)
(D) $\text{B} > \text{Al} < \text{Ga} > \text{In}$ (first ionization energy)
35. Which of the following orders of given character in the following oxides is CORRECT?
(A) $\text{V}_2\text{O}_3 < \text{V}_2\text{O}_4 < \text{V}_2\text{O}_5$ (basic character)
(B) $\text{CrO} < \text{Cr}_2\text{O}_3 < \text{CrO}_3$ (acidic character)
(C) $\text{CO}_2 > \text{SO}_3 > \text{Cl}_2\text{O}_7$ (acidic character)
(D) $\text{MgO} < \text{CaO} > \text{BaO}$ (basic character)
36. Which of the following statement about hydrogen peroxide is INCORRECT?
(A) The gas phase bond angles in H_2O_2 is less than that in crystalline phase.
(B) H_2O_2 can be prepared by adding ice-cold dilute H_2SO_4 (20%) on Na_2O_2 .
(C) The gas phase O – H bond length is greater than that of in crystalline phase.
(D) The dihedral angle in H_2O_2 in its gas phase is larger than that of bond angle.
37. Electrons in a group of degenerate orbitals can have five Z-components of angular momentum. Which sub-shell does this group of orbitals belong to:
(A) s
(B) p
(C) d
(D) s + d

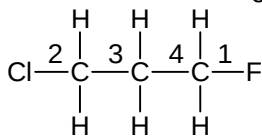
38. Two containers containing N_2 and O_2 gases respectively, are present in the same compartment of a train which is moving with 120 km/hr. The volume of first container is twice the size of second one. Find the ratio of the mean square speeds of N_2 gas and O_2 gas, respectively

(A) $\frac{8}{7}$ (B) $\frac{7}{8}$
(C) $\frac{1}{2}$ (D) $\frac{2}{1}$

39. He gas and N_2 gas both initially at the same temperature and pressure are compressed adiabatically and reversibly from a volume of V to $\frac{V}{3}$ (Assuming ideal behaviour of both the gases). Which gas will be at higher temperature?

(A) He
(B) N_2
(C) Both remain at same temperature
(D) None of these

40. Which of the following highlighted bond most likely break through homolytic bond cleavage?



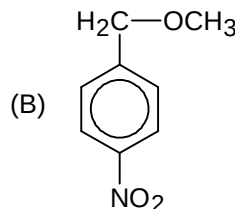
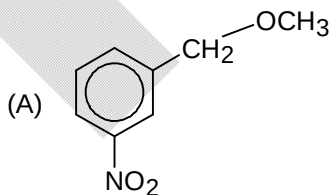
(A) 1 (B) 2
(C) 3 (D) 4

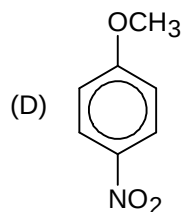
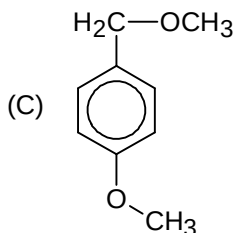
41. $\xrightarrow{\text{Reagent}}$ Product

With which reagent, the major product will not be syn addition product?

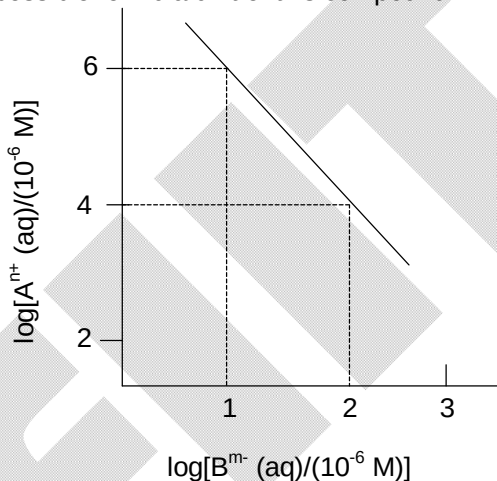
(A) (1) OsO_4 , (2) Na_2SO_3 , H_2O
(B) (1) $KMnO_4$, H_2O / OH^- (cold)
(C) (1) HBr , $(C_6H_5COO)_2$
(D) (1) B_2H_6 / THF, (2) H_2O_2 / $NaOH$

42. (Major product)





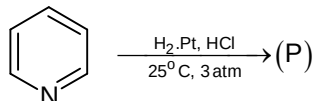
43. Which of the following is CORRECT?
- (A) In case of association, value of van't Hoff factor is more than unity while for dissociation it is less than unity.
- (B) Solid CO_2 directly sublimates upon heating at atmospheric pressure but it will first liquefy if this heating is done at sufficiently higher pressure.
- (C) Even at high pressure solid CO_2 always sublimates upon heating.
- (D) When placed in saline water (concentration $< 0.9\%$ m/V) blood cells (internal concentration = 0.9% m/V) shrink.
44. In which of the following reactions, the catalyst has been incorrectly marked?
- (A) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \xrightarrow{\text{Fe}(\text{s})} 2\text{NH}_3(\text{g})$ (Haber's process)
- (B) $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \xrightarrow{\text{Pt}(\text{s})} 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ (Ostwald's process)
- (C) $2\text{KClO}_3 \xrightarrow{\text{MnO}_2(\text{s})} 2\text{KCl} + 3\text{O}_2$
- (D) $\text{CH}_3\text{COOCH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \xrightarrow{\text{NaOH}} \text{CH}_3\text{COOH}(\text{aq}) + \text{CH}_3\text{OH}(\text{aq})$
45. A sparingly soluble salt A_mB_n shows following relation in its saturated aqueous solution. Find the possible formula unit of this compound



- (A) AB
(C) A_2B

- (B) AB_2
(D) A_3B

46.

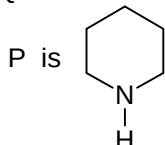


Pyridine [Q]

Which of the following statement is correct?

(A) Q and P both are aromatic

(B)

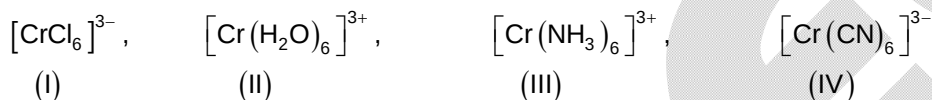


(C) P is acyclic amine compound

(D) Q is more basic than P

47.

Arrange the following coordination compounds in increasing order of wavelength of light that they absorbed



(A) I > II > III > IV

(B) IV > II > III > I

(C) I > III > II > IV

(D) IV > III > II > I

48.

 Order of octahedral ionic radii of M^{2+} for the first row transition elements

 (A) $\text{Ca} < \text{V} < \text{Mn} > \text{Zn}$

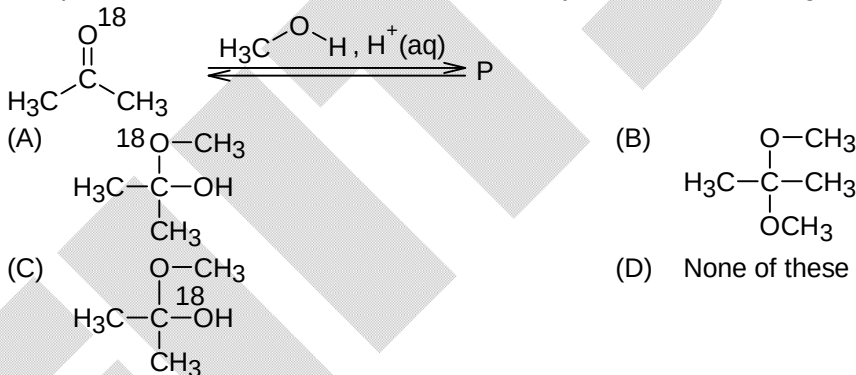
 (B) $\text{V} < \text{Mn} > \text{Ni} < \text{Zn}$

 (C) $\text{V} < \text{Mn} < \text{Ni} > \text{Zn}$

 (D) $\text{Fe} < \text{Co} < \text{Ni} > \text{Zn}$

49.

The product or intermediate which can not be synthesized from the given reaction



50.

Compound A upon strong heating produces a salt B and a colourless gas C. The gas C is also produced by heating hydrogen peroxide. Aqueous solution of salt B produces white precipitate when aqueous AgNO_3 solution is mixed into it. When gas D is passed through hot and concentrated solution of potassium hydroxide, then compound A is formed. In presence of moisture, gas D acts as an oxidizing agent.

Correct option is:

 (A) $\text{A} = \text{KNO}_3$

 (B) $\text{A} = \text{KClO}_3$

 (C) $\text{D} = \text{O}_2$

 (D) $\text{B} = \text{NaI}$

SECTION – B

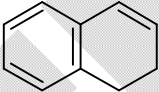
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. Find the total number of bonds in which there is involvement of hybridized orbitals in the following molecules/ions.
 CH_4 , ClF_3 , C_2H_4 , C_2H_2 , NO_2^+
52. 20 mL of '10 volume' hydrogen peroxide solution upon heating releases that much amount of oxygen at STP which is just sufficient to completely burn 'x mg' of ethene. Find the value of x (to the nearest integer).
53. The total loss in mass (in g) when mixture of 1 mole Li_2CO_3 and 2 moles $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ is heated.
54.
$$\begin{array}{c} \text{H} \quad \text{H} \\ | \quad | \\ \text{Cl}-\text{C}-\text{C}-\text{Cl} \\ | \quad | \\ \text{T} \quad \text{H} \end{array} \xrightarrow[\text{Ether}]{\text{Na}} \text{Product(s)}$$

(Product should not have more than 4 carbons)

From all the possible products, how many of them will be chiral?
55. During rusting of iron, iron converted to ferric oxide it has been found that 3.2 kg of iron rust has been formed. How many mole(s) of electrons does iron has given up to form this much rust? (Atomic mass : Fe = 56 u, O = 16 u)
56. 2 Moles of radioactive substance X undergo radioactive decay as shown below at 298 K temperature in a closed container of volume V lit. Half life period of X is 16 hrs. After 32 hr, 75% of X has been disintegrated and the pressure developed inside the container is 18.32 atm. Find the value of V to the nearest integer. $[R = 0.082 \text{ atm lit mol}^{-1} \text{ K}^{-1}]$

$${}_Z^AX(s) \longrightarrow {}_{Z-4}Y(s) + {}_2^4\text{He}(g)$$
57. Find the number of π electrons (which are in conjugation) because of which the following compound is aromatic?

58. The sum of the total number of lone pair electrons in the Lewis structure of I_3^- ion.
59. The CFCs which are generally called by the trade name Freons, lead to ozone layer depletion. Freon-12 has a percentage abundance of $4 \times 10^{-4} \%$ in troposphere. Find its partial pressure at sea level (in microbar). [Answer should be reported to nearest integer]
60. Acid catalysed hydrolysis of 0.2 mole of sucrose yields D(+) glucose and D (–) fructose. If the percentage dissociation of sucrose and the percentage yield of this reaction are 60% and 80%, respectively. Find the mass of fructose (in gram) that has been produced. [Answer should be reported to nearest integer]

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

61. For a data set having 100 observations, mean, median, mode are 50, 60, 70 respectively. If the largest 50 observations are increased by 10, then sum of new mean, median, mode equals
(A) 180 (B) 190
(C) 200 (D) 210
62. Maximum number of compound propositions out of
 $p \wedge q \wedge r$, $p \wedge \sim q \wedge r$, $p \wedge q \wedge \sim r$
 $\sim p \wedge q \wedge s$, $\sim p \wedge \sim q \wedge s$, $\sim p \wedge \sim q \wedge \sim s$
 $q \wedge r \wedge s$, $q \wedge \sim r \wedge s$, $\sim q \wedge \sim r \wedge \sim s$
 that can be made simultaneously false by an assignment of the truth values to p, q, r and s, is equal to
 (A) 6 (B) 7
 (C) 8 (D) 9
63. Let $f(x) = \begin{vmatrix} \cos x & \cos^2 x & \cos^4 x \\ \cos 3x & \cos^2 3x & \cos^4 3x \\ \cos 5x & \cos^2 5x & \cos^4 5x \end{vmatrix}$, then $\int_0^\pi f(x) dx$ equals
 (A) 9 (B) 18
 (C) 27 (D) none of these
64. Let R be a relation defined on first twenty natural numbers such that $R = \{(a, b) : \frac{a}{b} \text{ is a prime number}\}$. Consider the following statements :
 (I) R is reflexive and symmetric but not transitive
 (II) Range of R^{-1} has 20 elements
 (III) Domain of R^{-1} has 10 elements
 which of the statements are true ?
 (A) I, II, III (B) II, III
 (C) II (D) III
65. Value of $\sin 32^\circ \sin 88^\circ \sin 152^\circ$ equals
 (A) $\sin 6^\circ$ (B) $\frac{\sin 6^\circ}{4}$
 (C) $\cos 6^\circ$ (D) $\frac{\cos 6^\circ}{4}$
66. Let α, β be roots of the equation $x^2 - 3x - \sqrt{7} = 0$ and $\alpha^2 + \frac{1}{\beta}, \beta^2 + \frac{1}{\alpha}$ be the roots of the equation $x^2 + px + q = 0$, then for the equation $x^2 + p^2q^2x - (11q + p) = 0$, roots are
 (A) positive (B) negative
 (C) one positive and other negative (D) none of these

67. Let $A = \sum_{i=0}^{2023} \sum_{j=0}^{2023} \min\{ {}^{2023}C_i, {}^{2023}C_j \}$, $B = \sum_{i=0}^{2023} \sum_{j=0}^{2023} \max\{ {}^{2023}C_i, {}^{2023}C_j \}$, then $A + B$ equal
 (A) $2024 \cdot 2^{2023}$ (B) $2023 \cdot 2^{2023}$
 (C) $2024(2^{2023} - 1)$ (D) $2023(2^{2023} - 1)$
68. Let H_L and H_R be two branches of a hyperbola having A, B as endpoints of latus-rectum on H_L . If C be any point on H_R such that $\angle ACB$ is never obtuse, then maximum possible eccentricity of hyperbola is
 (A) $\sqrt{2}$ (B) $\sqrt{3}$
 (C) 2 (D) 3
69. If chord of contact of pair of tangents from points P and Q on $\frac{x^2}{9} + \frac{y^2}{16} = 1$ to the ellipse $\frac{x^2}{3} + \frac{y^2}{4} = 1$ is normal to the curve $2x^2 + 2y^2 - 4x - 4y + 1 = 0$, then sum of eccentric angles of P and Q is
 (A) $\frac{\pi}{2}$ (B) π
 (C) 2π (D) none of these
70. A rectangular hyperbola and ellipse have coincident end points of latus-rectum, then eccentricity of ellipse equals
 (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$
 (C) $\frac{1}{2}$ (D) none of these
71. If $\lim_{x \rightarrow 0} \frac{\sin x \sin 2x \sin 3x \dots \sin nx - x^n}{x \tan((1+x)(1+2x)(1+3x) \dots (1+2023x))}$ exists and is non-zero, then 'n' equals
 (A) 2022 (B) 2023
 (C) 2024 (D) no such value of n exists
72. Let $A_k = [a_{ij}]$ be square matrix of order 3 such that $a_{ij} = (i - j)^k$ for all $i, j = 1, 2, 3$. Then determinant value of $|A_1 + A_3 + A_5 + \dots + A_{2023}|$ equals
 (A) 3^{4046} (B) 3^{6069}
 (C) $3^{6069} - 3^{4046}$ (D) none of these
73. Which of the following functions is differentiable at $x = 0$?
 (A) $|\tan|x| - \cos|x||$ (B) $|\sin|x| - \tan|x||$
 (C) $|\cos|x| - \sin|x||$ (D) all of these
74. Water is being filled at rate of $\pi \text{ cm}^3/\text{s}$ in right circular conical vessel (vertex upwards) of height 5cm and diameter 8cm. When the height of water level is 3cm, the rate (in cm^2/s) at which the wet conical surface area of the vessel increases is
 (A) $\frac{2\pi}{3}$ (B) $\frac{\sqrt{41}}{12} \pi$
 (C) $\frac{\sqrt{41}}{18} \pi$ (D) none of these

75. Let $f(x) = x^7(x^3 + 2x^2 - x - 2) + (x^3 - x)(x + 3) + 2(x^2 - 1)$, then which of the following is **NOT** true?
 (A) $y = f(x)$ has 7 critical points
 (B) there are exactly 2 points in $(-1, 1)$ where $f'(x) = 0$
 (C) $y = f(x)$ does not have an extrema at $x \in I$
 (D) $y = f(x)$ has exactly 2 stationary point in $(-2, 1)$
76. Value of $\int_0^1 x^6 (x^3 - 1)^{2022} dx$ is
 (A) $\frac{3^{2022} (2022)!}{10.13.16.....6073}$
 (B) $\frac{3^{2022} (2022)!}{7.10.13.....6073}$
 (C) $\frac{3^{2022} (2022)!}{4.7.10.....6073}$
 (D) none of these
77. If area bounded by an ellipse with its auxiliary circle equals area bounded by its auxiliary circle and director circle, then eccentricity of ellipse equals
 (A) $\sqrt{2}\sin 18^\circ$
 (B) $\sqrt{2}\cos 18^\circ$
 (C) $\sqrt{2}\sin 36^\circ$
 (D) $\sqrt{2}\cos 36^\circ$
78. Ajay has to take medicine having 100 mg paracetamol at a gap of every 12 hours. The amount of paracetamol (salt) in Ajay's bloodstream decreases at a rate that is proportional at any time to the amount of salt in bloodstream at that time. If the amount of salt in bloodstream is halved in every 5 hours, then amount of salt in Ajay's bloodstream 15 hours after taking the medicine first time is (in mg)
 (A) $100\left(\frac{1}{2}\right)^3$
 (B) $100\left[\left(\frac{1}{2}\right)^3 + 1\right]$
 (C) $100\left[\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^{12/5}\right]$
 (D) $100\left[\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^{3/5}\right]$
79. Equation of plane passing through the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ at minimum possible distance from the point $(-1, -3, 1)$ is
 (A) $7x - 2y - 2z + 3 = 0$
 (B) $x + 3y + 10 = 0$
 (C) $3x + y - z = 2$
 (D) none of these
80. $[(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c})] \times (\vec{c} \times \vec{a}) \cdot \vec{b}$ equal (where $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors)
 (A) $[\vec{a} \vec{b} \vec{c}]^2 |\vec{b}|^2$
 (B) $[\vec{a} \vec{b} \vec{c}]^2 (\vec{b} \cdot \vec{c})$
 (C) $[\vec{a} \vec{b} \vec{c}]^2 (\vec{b} \cdot \vec{a})(\vec{c} \cdot \vec{b})$
 (D) none of these

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let z_1, z_2, z_3 be represented by A, B, C on Argand plane such that $z_1 = 3 + 3i$, $z_2 = 9 - 6\sqrt{2} + (6\sqrt{2} - 3)i$ and $\triangle ABC$ is right angled at C. If minimum and maximum possible value of Argument of z_3 is a, b respectively (Argument of $z_3 \in (-\pi, \pi]$, then $b - a$ is $\frac{\pi}{k}$ where k equals
82. Number of ordered pairs (x, y) satisfying $\frac{4^{\sin x} 16^{\sin y}}{(1 + 16^{\sin x})(1 + 256^{\sin y})} = \frac{1}{4}$, $3^{1 + \sqrt{\cos^2 x}} + 3^{1 + \cos y} = 10$; x, y $\in [0, 2\pi]$ is
83. Two families each having 4 members are to be seated around a circular table having alternate red and blue chairs. If probability that members of same family are seated together is p, then $35p$ equals
84. Number of 5-digit natural numbers such that the product of their digits equals number of ways to create a necklace out of 6 beads from 10 distinct beads is
85. Remainder when $(2022)^{2023}$ is divided by 11 is
86. Let $\{a_1, a_2, \dots, a_n\}$ be a sequence such that $a_1 = 0$, $a_2 = 1$, $a_3 = 2$ and $a_{n+3} = a_{n+2} + a_n - a_{n+1} \forall n \geq 1$, then value of $15 \sum_{n=1}^{\infty} \frac{a_n}{2^n}$ equals
87. Let A(0, 0), B(x_1, y_1) and C(x_2, y_2) form a triangle having circumcentre at $(3^{1/4}, 3^{1/4})$. If area of $\triangle ABC$ lies in (a, b], then $2(a + b)$ equals
88. If difference of distances of A(1, 1) and B(-3, 13) from any point R on $x + 2y - 1 = 0$ is minimum, then absolute value of five times sum of abscissa and ordinate of R equals
89. Sum of all solutions of the equation $\sin^{-1}[x] = \cos^{-1}\{x\}$ is (where [.] represents greatest integer function and { } represents fractional part function)
90. Number of skew symmetric matrices of order 3 with integer entries such that their squares equals identity matrix is

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2023

TEST DATE: 18-01-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1.
Sol.

A
x-axis along the inclined plane
y-axis perpendicular to the inclined plane.

$$v_{0x} = v_0 \cos 30^\circ = v_0 \frac{\sqrt{3}}{2}, \quad a_x = -g \sin 30^\circ = -\frac{g}{2}$$
$$v_{0y} = v_0 \sin 30^\circ = \frac{v_0}{2}, \quad a_y = -g \cos 30^\circ = -g \frac{\sqrt{3}}{2}$$

$$x_0 = 0, \quad y_0 = H \sin 30^\circ = H \frac{\sqrt{3}}{2}$$

When particle reaches the inclined plane, $y = 0$

$$\Rightarrow 0 = \frac{H\sqrt{3}}{2} - \frac{v_0}{2}t - \frac{1}{2}g\frac{\sqrt{3}}{2}t^2 \quad \dots(i)$$

If particle hits the inclined plane at 90° . It will retrace the path for this $v_x = 0$

$$\Rightarrow 0 = \frac{v_0\sqrt{3}}{2} - \frac{g}{2}t$$

$$\Rightarrow t = \frac{v_0\sqrt{3}}{g} \quad \dots(ii)$$

$$0 = \frac{H\sqrt{3}}{2} - \frac{v_0}{2} \times \frac{v_0\sqrt{3}}{g} - \frac{1}{2} \times \frac{g\sqrt{3}}{2} \times \frac{v_0^2 \times 3}{g^2}$$

$$\Rightarrow v_0 = \sqrt{\frac{2gH}{5}}$$

2. A

Sol. Modulation index = $\frac{A_m}{A_c} = \frac{20}{50} = 0.4$

Side frequency band is $f_c - f_m$ and $f_c + f_m$

3. D

Sol. For the ball along perpendicular to the inclined plane

$$\int N dt = mv_0(e+1) \quad \dots(i) \quad (v_0 = 10 \text{ m/s})$$

For the block along the surface of the inclined plane

$$-\int \mu N dt = m(v_f - v_i) \quad \dots(ii)$$

$$\Rightarrow -\mu \times mv_0(e+1) = m\Delta v$$

$$\Rightarrow \Delta v = -0.2 \times v_0 \times 1.5 = -3 \text{ m/s}$$

4. B

Sol. $\therefore f = n \frac{v}{2\ell}$

$$\Rightarrow 1100 = n \times \frac{330}{2 \times 30 \times 10^{-2}}$$

This shows that air column vibrates in 2nd harmonic. This implies that wavelength $\lambda = \ell = 30 \text{ cm}$.

$$\therefore \Delta P = 0.1 P_0 \sin kx \cos \omega t.$$

x is measured from open end.

At $x = 5 \text{ cm}$, i.e., at $x = \frac{\lambda}{6}$

$$\Delta P = 0.1 P_0 \sin\left(\frac{2\pi}{\lambda} \cdot \frac{\lambda}{6}\right) \cos \omega t$$

$$\Rightarrow P_{\max} - P_{\min} = 2 \times 0.1 P_0 \cdot \frac{\sqrt{3}}{2}$$

$$= \sqrt{3} \times 10^4 \text{ N/m}^2$$

5. D

Sol. $f_T = m\alpha r$

$$f_R = m\omega^2 r = m\alpha^2 t^2 r$$

Since $\sqrt{f_T^2 + f_R^2} = \mu mg$

$$\Rightarrow m\alpha r \sqrt{1 + \alpha^2 t^4} = \mu mg$$

$$\Rightarrow t = \frac{1}{\sqrt{\alpha}} \left[\left(\frac{\mu g}{\alpha r} \right)^2 - 1 \right]^{1/4} = \frac{\left[\left(\frac{0.255 \times 10}{1 \times 0.5} \right)^2 - 1 \right]^{1/4}}{\sqrt{\alpha}} = \sqrt{5} \text{ s}$$

6. B

Sol. $V = \frac{kQ}{\frac{R}{2}} + \frac{kQ}{R} = \frac{Q}{2\pi\epsilon_0 R} + \frac{Q}{4\pi\epsilon_0 R}$

7. C

Sol. $1A \times 0.081 \Omega = 9A \times R$
 $\Rightarrow R = 0.09 \Omega$

8. A

Sol. $\Delta Q_{AB} = \Delta W_{AB} + \Delta U_{AB}$
 $= 10 \times +nC_V \frac{10}{nR} = 10 + \frac{10C_V}{R}$
 $\Delta Q_{BC} = \Delta W_{BC} + \Delta U_{BC} = 0 + \left(\frac{2P_C - 20}{R} \right) C_V$
 $\Delta Q_{CA} = \Delta W_{CA} + \Delta U_{CA} = \Delta W_{CA} + \left(\frac{10 - 2P_C}{R} \right) C_V$
 $\Delta Q = \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA} = 5J$
 $\Rightarrow 10 + \frac{10C_V}{R} + \left(\frac{2P_C - 20}{R} \right) C_V + \Delta W_{CA} + \left(\frac{10 - 2P_C}{R} \right) C_V = 5J$
 $\Rightarrow \Delta W_{CA} = -5J$

9. A

Sol. $mvR = mv'r \quad \dots(i)$

and $\frac{1}{2}mv^2 = -\frac{GMm}{r} + \frac{1}{2}mv'^2$

$\Rightarrow v'^2 = v^2 + \frac{2GM}{r}$

$\Rightarrow \frac{v^2 R^2}{r^2} = v^2 + \frac{2GM}{r}$

$R = \frac{r}{v} \left[v^2 + \frac{2GM}{r} \right]^{1/2}$

10. D

Sol. $q_{\max} = \frac{2EC}{3}$

In the beginning current through the cell

$I_0 = \frac{3E}{5R}$

Therefore, $\frac{dq}{dt} = \frac{2}{3} I_0 = \frac{2E}{5R}$

Since, $\left(\frac{2E}{5R} \right) \times t = \frac{2EC}{3}$

$\Rightarrow t = \frac{5RC}{3}$

11.

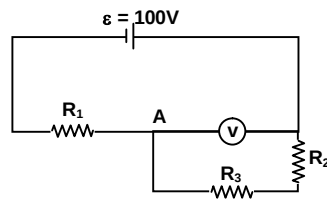
B

Sol. In the steady state current in the branch having capacitor is zero. The effective circuit is shown in the figure.
Current through the cell

$$I = \frac{3}{7} \text{ A}$$

Therefore current through the voltmeter is $I' = \frac{2}{3}I = \frac{2}{7} \text{ A}$

Therefore, voltmeter reading is $= I'R_V = \frac{400}{7} \text{ V}$



12.

B

Sol. From snell's law

$$\mu \sin \theta = 16\mu \sin x$$

$$\Rightarrow \sin x = \frac{5}{8} \sin \theta$$

13.

A

Sol. $Q = A_{\text{He}} \cdot BE_{\text{He}} \text{ (per nucleon)} - 2A_{\text{H}}(BE_{\text{H}} \text{ per nuclear})$
 $= 28 \text{ MeV} - 4.4 \text{ MeV} = 23.6 \text{ MeV}$

14.

B

Sol. $V_R = V \left(1 - e^{-\frac{tR}{L}} \right)$, at $t = \tau \ln 2$

$$V_R = \frac{3V}{4}$$

15.

A

Sol. Initially, $P = \frac{v_{\text{rms}}^2}{R} \Rightarrow v_{\text{rms}} = \sqrt{PR}$

$$\text{Finally, } P' = \frac{v_{\text{rms}}^2}{R} = \frac{v_{\text{rms}}^2}{\frac{R}{z^2}} = \frac{PR}{\frac{R}{z^2}} \times R = P \left(\frac{R}{z} \right)^2$$

16.

B

Sol. The gas expands isobarically, therefore, $\Delta Q = nC_p \Delta T$

At STP, 1 mole of a gas occupies the volume 22.4 liter. This implies that in the vessel 2 mole helium is present

$$\Rightarrow \Delta Q = 2 \times \frac{5R}{2} \times 20\text{J} = 100R$$

17.

A

$$\text{Sol. } \frac{\lambda_2}{\lambda_1} = \frac{(z_1 - 1)^2}{(z_2 - 1)^2} = 4$$

$$\Rightarrow 2z_2 - z_1 = 1$$

18.

A

Sol. The effective decay constant

$$\lambda = \lambda_1 + \lambda_2$$

Therefore effective half life

$$T_{1/2} = \frac{\ell n 2}{\lambda} = \frac{0.693}{\lambda_1 + \lambda_2}$$

19. D

Sol. Collective energy of both photons is 11.6 eV more than the ionization energy of a hydrogen atom.

20. B

Sol. $\therefore y(x, t) = 0.5 \text{ cm} \sin \left\{ 100\pi \left(t - \frac{x}{v} \right) \right\}$

Here, $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{1600}{40 \times 10^{-3}}} = 200 \text{ m/s}$

and $t = \frac{5}{200} \text{ s}$

$x = \frac{5}{2} \text{ m}$

$$y\left(x = \frac{5}{2}, t = \frac{5}{200} \text{ s}\right) = 0.5 \text{ cm} \sin \left\{ 100\pi \left(\frac{5}{200} - \frac{5}{400} \right) \right\}$$

$$= -\frac{0.5}{\sqrt{2}} \text{ cm}$$

TG ~ @bohring_bot

21. 5

Sol. Let the resistance be R,

Therefore, $I_{\text{rms}} = \frac{110}{R} \text{ A}$

With L, $I_{\text{rms}} = \frac{220}{\sqrt{R^2 + \omega^2 L^2}} = \frac{110}{R}$

$$\Rightarrow R^2 = \frac{4}{3} \times 10^5 \Omega^2$$

With C, $I_{\text{rms}} = \frac{220}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} = \frac{110}{R}$

$$\Rightarrow C = 5 \mu\text{F}$$

22. 7

Sol. $\therefore A = A_0 e^{\frac{-bt}{2m}}$

Where, A_0 = initial amplitude

b = damping constant

M = mass of the block

$$\Rightarrow \frac{A_0}{2} = A_0 e^{\frac{-bt}{2m}}$$

$$\Rightarrow t = \frac{\ell n 2}{b / 2m}$$

$$= \frac{0.693}{40 \text{ g/s}} \times 2 \times 200 \text{ g} = 7 \text{ s}$$

23. 3

Sol. $R_{cm} = \frac{u^2 \sin 2\theta}{g} = 5\sqrt{3} \text{ m}$

Let M be the mass of each fragment

$$\Rightarrow \frac{M\sqrt{3} + (n-1)M \times 7\sqrt{3}}{nM} = 5\sqrt{3}$$

$$\Rightarrow n = 3$$

24. 11

Sol. The point P acts like a virtual object for the lens.

Since, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\Rightarrow \frac{1}{25} = \frac{1}{v} - \frac{1}{20}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{25} + \frac{1}{20} = \frac{9}{100}$$

$$\Rightarrow v = 11.11 \text{ cm} \approx 11 \text{ cm}$$

25. 38

Sol. Energy of the incident photon

$$E = E_0 z^2 \times \frac{3}{4} = 40.8 \text{ eV}$$

Stopping potential, $v_0 = \frac{E - \phi}{e} = 38 \text{ V}$

26. 7

Sol. Since, $\lambda = \frac{cn^2}{T_{1/2}} = 4.14 \times 10^{-9} \text{ s}$

Activity, $\frac{dN}{dt} = \lambda N$

$$\Rightarrow N = \frac{\left(\frac{dN}{dt}\right)}{\lambda} = \frac{8 \times 3.7 \times 10^7}{4.14 \times 10^{-9}} = 7.133 \times 10^{10}$$

Therefore mass, $M = \frac{7.133 \times 10^{10}}{6.023 \times 10^{23}} \times 60 = 7.12 \times 10^{-6} \text{ g} \approx 7 \mu\text{g}$

27. 10

Sol. Since, $\frac{\Delta V}{\Delta I} = \frac{0.05 \text{ V}}{5 \times 10^{-3} \text{ A}} = 10 \Omega$

28. 14

Sol. The output AC voltage is 2.0 V. Therefore AC collector current, $I_C = \frac{2.0}{2000 \Omega} = 10^{-3} \text{ A}$

Since, amplification factor, $\beta = \frac{I_C}{I_B} = 100$

$$\Rightarrow I_B = \frac{I_C}{100} = 10^{-5} \text{ A}$$

DC base current is $10 \times 10^{-5} \text{ A} = 10^{-4} \text{ A}$

From the circuit, $V_B = I_B R_B + V_{BE}$

$$\Rightarrow R_B = \frac{V_B - V_{BE}}{I_{B(DC)}} = \frac{2 - 0.6}{10^{-4}} = 14 \text{ k}\Omega$$

29. 800

Sol. Length of the antenna, $\ell = \frac{\lambda}{4}$

$$\Rightarrow \lambda = 4\ell = 800 \text{ m}$$

30. 4000

Sol. Let θ be the angular diffraction.

$$\Rightarrow \sin \theta = \frac{n\lambda}{d}$$

$$\text{For small } \theta, \sin \theta = \theta = \frac{n\lambda}{d}$$

For 1st minima, $n = 1$

$$\Rightarrow \theta \text{ for first minima} = \frac{\lambda}{d}$$

Since, angular width of the central maximum is $2\theta = \frac{2\lambda}{d} = 4 \times 10^{-3} \text{ rad}$

$$\Rightarrow \lambda = d \times 2 \times 10^{-3} \text{ m} = 4 \times 10^{-7} \text{ m} = 4000 \text{ \AA}$$

Chemistry

PART – B

SECTION – A

31. B
Sol. Decreasing pressure usually decreases the sublimation temperature of the substance and thus preventing it from decomposition.
32. A
Sol. $\frac{Y_A}{Y_B} = \frac{X_A p_A^0}{X_B p_B^0}$
 $\frac{Y_A}{Y_B} = \left(\frac{m_A}{m_B}\right) \times \frac{M_B}{M_A}$; M_A = molar mass of A
 M_B = molar mass of B
 $\frac{m_A}{m_B}$ = ratio of their masses in vapour phase.
Similarly, $\frac{X_A}{X_B} = \left(\frac{m'_A}{m'_B}\right) \times \frac{M_B}{M_A}$
 $\frac{m'_A}{m'_B}$ = (Ratio of their masses in liquid phase)
33. D
Sol. Defect usually increases the conductivity, while heating metallic conductor decreases its conductivity due to increase in the vibration of metal ions which consequently increases the resistance.
34. B
Sol. In aqueous medium,
Size of ion is in the order of
 $Li^+ > Na^+ > K^+ > Rb^+$
35. B
Sol. In a particular group, lower member have high basic character of their oxide. For same element with different oxides (having different oxidation states), the oxides having high oxidation state of element have high acidic character.
36. C
Sol. (Structure of H_2O_2 , NCERT)
37. C
Sol. Z-components of angular momentum of an electron in an atomic orbital is governed by magnetic quantum number. For, d sub-shell, there are 5 Z-components possible.
38. A
Sol. Root mean square speed of any gas = $\left(\frac{3RT}{M}\right)^{\frac{1}{2}}$
Mean square speed = $\left(\frac{3RT}{M}\right)$
Both the gases are in same compartment, the temperature is same for both the gases.

$$\text{Required ratio} = \frac{3RT}{28} \times \frac{32}{3RT} = \frac{32}{28} = \frac{8}{7}$$

39. A

Sol. $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$ (for adiabatical change)

$$\left(\frac{T_2}{T_1}\right) = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = 3^{\gamma-1}$$

As γ is more for monoatomic gas He. Hence, temperature will also be more for it.

40. C

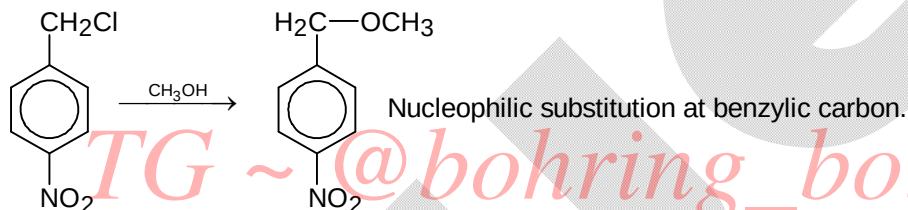
Sol. C—C bond has lowest electronegativity difference. For homolytic bond cleavage a bond should have low electronegative difference.

41. C

Sol. With HBr along with peroxide, the product can both have syn as well as anti addition.

42. B

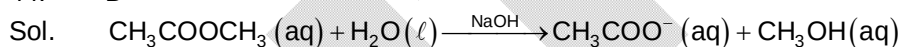
Sol.



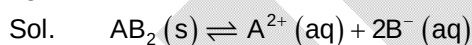
43. B

Sol. Value of van't Hoff factor is more than unity during dissociation. Solid CO_2 first liquefies upon heating at high pressure. Blood cells present in hypotonic solution will swell.

44. D



45. B



$$K_{\text{SP}} = [\text{A}^{2+} (\text{aq})][\text{B}^- (\text{aq})]^2$$

$$\text{i.e. } [\text{A}^{2+} (\text{aq})][\text{B}^- (\text{aq})]^2 = \text{Constant (C)}$$

$$C = K_{\text{SP}}$$

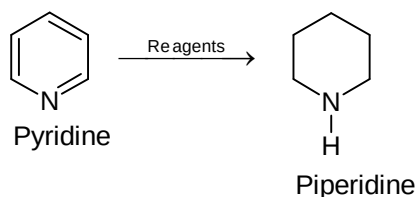
$$\log [\text{A}^{2+}] + 2 \log [\text{B}^-] = \log C$$

$$\log [\text{A}^{2+}] = \log C - 2 \log [\text{B}^-]$$

$$y = C - mx \quad (m = 2)$$

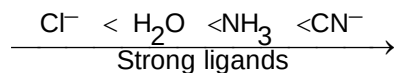
46. B

Sol.



47. A

Sol.



Stronger the ligand, greater will be the splitting and hence the compound will absorb light of high frequency (or low wavelength).

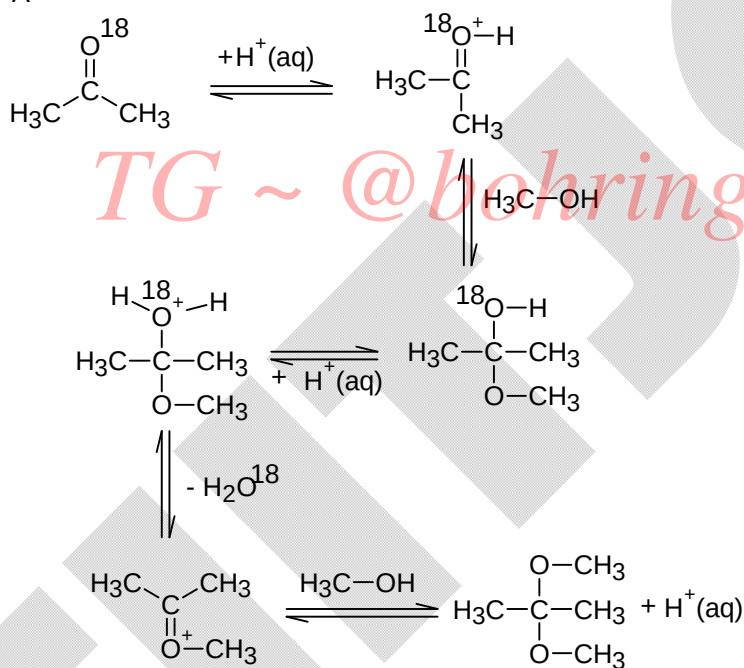
48. B

Sol.

Octahedral ionic radii of these elements in M^{2+} state is related with its stabilization due to CFSE.

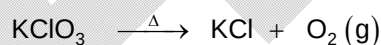
49. A

Sol.

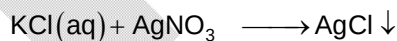


50. B

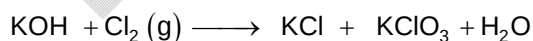
Sol.



(A) (B) (C)



(B)

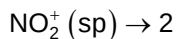
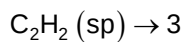
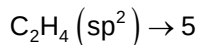
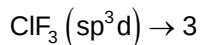
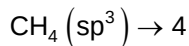


(D)

SECTION – B

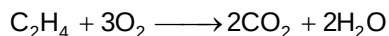
51. 17

Sol. Hybridisation is involved in sigma bond. Hybridisation and number of corresponding bonds, respectively are given as



52. 83

Sol. 20 mL of '10 volume' H_2O_2 solution produces 200 ml of O_2 at STP (or $\frac{200}{22400}$ mole of O_2)



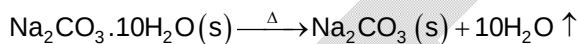
$$\frac{200}{22400} \text{ mole of } \text{O}_2 \text{ will burn } \left(\frac{1}{3} \times \frac{200}{22400} \times 28 \right) \text{ g of } \text{C}_2\text{H}_4$$

Required answer = 83 mg.

53. 404

Sol. $\text{Li}_2\text{CO}_3 (\text{s}) \xrightarrow{\Delta} \text{Li}_2\text{O} (\text{s}) + \text{CO}_2 \uparrow$

1 mole

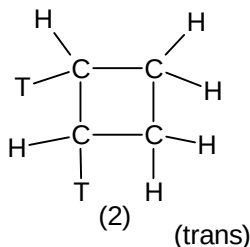
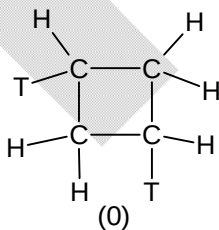
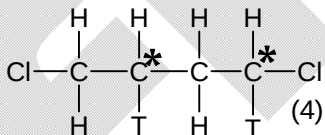
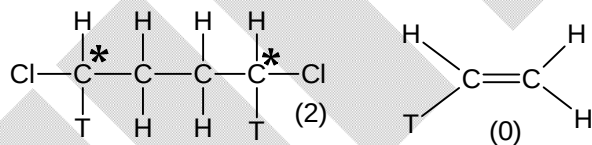


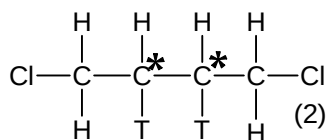
2 moles

$$\begin{aligned} \text{Required answer} &= 44 + 180 \times 2 \\ &= 44 + 360 \\ &= 404 \text{ g.} \end{aligned}$$

54. 10

Sol.





Mathematics

PART – C

SECTION – A

61. C
Sol. $55 + 65 + 80 = 200$
62. D
Sol. $p \rightarrow T, q \rightarrow F, r \rightarrow F, s \rightarrow T$
63. D
Sol. $\frac{1}{2} \int_0^\pi f(x) + f(\pi - x) dx = 0$
64. D
Sol. R is symmetric but neither reflexive nor transitive. Domain and range of R has 19 and 10 elements respectively.
65. D
Sol. $\sin 32^\circ \sin 88^\circ \sin 28^\circ = \frac{1}{4} \sin 84^\circ = \frac{\cos 6^\circ}{4}$
66. C
Sol. $-p = (\alpha + \beta)^2 - 2\alpha\beta + \frac{\alpha + \beta}{\alpha\beta} = 9 - \frac{11}{\sqrt{7}}$
 $q = \alpha^2\beta^2 + \alpha + \beta + \frac{1}{\alpha\beta} = 10 - \frac{1}{\sqrt{7}}$
67. A
Sol. $A + B = 2024({}^{2023}C_0 + {}^{2023}C_1 + \dots + {}^{2023}C_{2023}) = 2024.2^{2023}$
68. C
Sol. $\frac{b^2}{a} \leq a + ae \Rightarrow e^2 - 1 \leq 1 + e$
 $\Rightarrow e \in (1, 2]$
69. A
Sol. $\cos \theta + \sin \theta = 1 \Rightarrow \theta = 0, \frac{\pi}{2}$
70. B
Sol. $ae = AE, \frac{b^2}{a} = \frac{B^2}{A} \Rightarrow eE = 1 \Rightarrow e = \frac{1}{\sqrt{2}}$
71. D
Sol. Limit equals 0 for $n \geq 1$ and does not exist for $n < 1$
72. D
Sol. $A_1 + A_3 + \dots + A_{2023}$ is skew symmetric matrix of order 3 so its determinant equals 0.

73.

B

Sol. Check RHD and LHD at $x = 0$

74.

B

Sol. $\frac{dv}{dt} = -\pi r^2 \cot \theta \frac{dr}{dt}, \frac{ds}{dt} = -2\pi r \frac{dr}{dt} \operatorname{cosec} \theta$

$$\Rightarrow \frac{ds}{dt} = \frac{\sqrt{41}}{12} \pi$$

75.

D

Sol. $f(x) = (x+2)(x+1)(x-1)(x^7+x+1)$
There are 3 stationary points in $(-2, 1)$

76.

B

Sol. Let $I_n = \int_0^1 x^6 (x^3 - 1)^n dx$

$$= \left[\frac{x^7}{7} (x^3 - 1)^n \right]_0^1 - \frac{3n}{7} \int_0^1 x^9 (x^3 - 1)^{n-1} dx$$

$$= -\frac{3n}{7} (I_n + I_{n-1}) \Rightarrow I_n = -\frac{3n}{3n+7} I_{n-1}$$

$$I_{n-1} = -\frac{3(n-1)}{3n+4} I_{n-2}$$

...

...

...

$$I_2 = -\frac{3 \cdot 2}{13} I_1 \text{ where } I_1 = -\frac{3}{70}$$

77.

A

Sol. $\pi(a^2 - ab) = \pi(a^2 + b^2 - a^2) \Rightarrow \frac{b}{a} = \frac{\sqrt{5}-1}{2}$

$$\Rightarrow e = \sqrt{\frac{2\sqrt{5}-2}{4}} = \sqrt{2 \sin 18^\circ}$$

78.

D

Sol. Let amount of salt in bloodstream at time $t = x$

$$\Rightarrow \frac{dx}{dt} = -kx \Rightarrow x = ce^{-kt}$$

$$\text{where } c = 100 \text{ and } k = \frac{\ln 2}{5}$$

$$\Rightarrow x = 100e^{-\frac{\ln 2}{5}t} \text{ (for first 12 hours)}$$

$$\text{At } t = 12^+ \text{ hrs, initial amount is } 100 \left(1 + e^{-\frac{12}{5} \ln 2} \right)$$

$$\text{At } t = 15 \text{ hrs, } x = 100 \left(1 + e^{-\frac{12}{5} \ln 2} \right) e^{-\frac{3 \ln 2}{5}}$$

79. A

Sol. Plane contains (1, 2, 3) and (-1, -3, 1) and parallel to $2\hat{i} + 3\hat{j} + 4\hat{k}$

80. D

$$\begin{aligned} \text{Sol. } & [(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c})] \times (\vec{c} \times \vec{a}) \cdot \vec{b} = [\vec{a} \vec{b} \vec{c}] \vec{b} \times (\vec{c} \times \vec{a}) \cdot \vec{b} \\ & = [\vec{a} \vec{b} \vec{c}] ((\vec{b} \cdot \vec{a})\vec{c} - (\vec{b} \cdot \vec{c})\vec{a}) \cdot \vec{b} = 0 \end{aligned}$$

SECTION - B

81. 4

$$\text{Sol. } b = \frac{\pi}{2}, a = \frac{\pi}{4}$$

82. 3

$$\begin{aligned} \text{Sol. } & (4^{\sin x} + 4^{-\sin x})(16^{\sin y} + 16^{-\sin y}) = 4 \\ & \Rightarrow \sin x = \sin y = 0 \quad (\text{by A.M - G.M}) \\ & 3^{|\cos x|} + 3^{\cos y} = \frac{10}{3} \Rightarrow 3^{\cos y} = \frac{1}{3} \Rightarrow \cos y = -1 \\ & y = \pi, x = 0, \pi, 2\pi \end{aligned}$$

83. 2

$$\text{Sol. } p = \frac{2}{35} \quad \text{TG} \sim @bohrring_bot$$

84. 60

$$\begin{aligned} \text{Sol. } & x_1 x_2 x_3 x_4 x_5 = 9 \times 8 \times 7 \times 5 \times 5 \\ & \Rightarrow \text{total numbers} = \frac{5!}{2} = 60 \end{aligned}$$

85. 3

Sol. Using Binomial expansion remainder comes out to be 3.

86. 9

$$\begin{aligned} \text{Sol. } & S = \frac{1}{4} + \frac{2}{8} + \frac{1}{2} \left[S - \frac{1}{4} \right] + \frac{S}{8} - \frac{1}{4} S \\ & \Rightarrow S = \frac{3}{5} \Rightarrow 15S = 9 \end{aligned}$$

87. 9

$$\begin{aligned} \text{Sol. } & \text{Maximum area of } \triangle ABC = \frac{3\sqrt{3}}{4} \cdot 2\sqrt{3} = \frac{9}{2} \\ & \Rightarrow \triangle ABC \in \left(0, \frac{9}{2} \right] \end{aligned}$$

88. 18

$$\begin{aligned} \text{Sol. } & R \text{ lies on the line } \frac{y-7}{x+1} = \frac{1}{3} \text{ and on the line } x + 2y - 1 = 0 \\ & \Rightarrow R \text{ is } \left(-\frac{41}{5}, \frac{23}{5} \right) \end{aligned}$$

89. 1

Sol. If $[x] = 0 \Rightarrow 0 = \cos^{-1}x \Rightarrow x = 1$ (not possible)

If $[x] = 1 \Rightarrow \frac{\pi}{2} = \cos^{-1}\{x\} \Rightarrow \{x\} = 0 \Rightarrow x = 1$

If $[x] = -1 \Rightarrow -\frac{\pi}{2} = \cos^{-1}\{x\}$ not possible

90. 0

Sol. Let $A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} -a^2 - b^2 & -bc & -ac \\ -bc & -a^2 - c^2 & -ab \\ -ac & -ab & -b^2 - c^2 \end{bmatrix} \neq I \quad \forall a, b, c \in I$

TG ~ @bohring_bot

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VI

JEE (Main)-2023

TEST DATE: 19-01-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

- An ice block at 0°C is dropped from height 'h' above the ground. What should be the value of 'h' so that it melts completely by the time it reaches the bottom assuming the loss of whole gravitational potential energy is used as heat by the ice? [Given: $L_f = 80 \text{ cal/gm}$ and $g = 10 \text{ m/s}^2$]

(A) 33.6 m (B) 33.6 km
(C) 8 m (D) 8 km
- A Solid floats with $\frac{2}{3}$ of its volume immersed in a liquid and with $\frac{3}{4}$ of its volume immersed in another liquid. What fraction of its volume will be immersed if it floats in a homogenous mixture formed of equal volumes of the liquids?

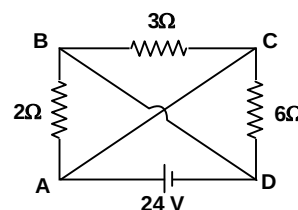
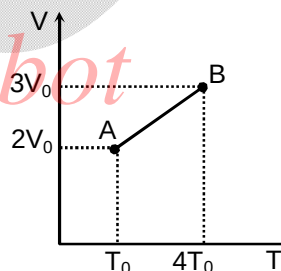
(A) $\frac{6}{11}$ (B) $\frac{8}{11}$
(C) $\frac{11}{16}$ (D) $\frac{12}{17}$
- One mole of ideal gas undergoes an expansion from state A to state B as shown in the V-T diagram. The pressure of gas:

(A) remains constant during the process.
(B) increases from A to B, $\frac{8}{3}$ times
(C) is same at A and B, but increases in between
(D) decreases from A to B, $\frac{3}{8}$ times
- To increase the resonant frequency in series L-C-R circuit,

(A) source frequency should be increased
(B) another resistance should be added in series with the first resistance
(C) another capacitor should be added in series with the first capacitor
(D) the source frequency should be decreased
- Time taken by light to travel in two different materials A and B of refractive indices μ_A and μ_B of same thickness is t_1 and t_2 , respectively. If $t_2 - t_1 = 5 \times 10^{-10} \text{ s}$ and the ratio of μ_A to μ_B is 1:2. Then, the thickness of material in metre is (Given, V_A and V_B are velocities of light in A and B materials, respectively.)

(A) $(5 \times 10^{-10} \times v_A) \text{ m}$ (B) $(5 \times 10^{-10}) \text{ m}$
(C) $(1.5 \times 10^{-10}) \text{ m}$ (D) $(5 \times 10^{-10} \times v_B) \text{ m}$
- Current flowing between points A and C is

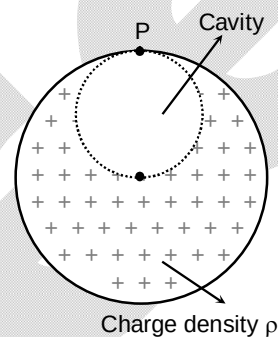
(A) zero (B) 4A
(C) 8A (D) 12A



7. If frequency of incident light in a photoelectric cell is doubled, then stopping potential increases by
 (A) 50%
 (B) 100%
 (C) more than 100%
 (D) any amount depending on work function of metal

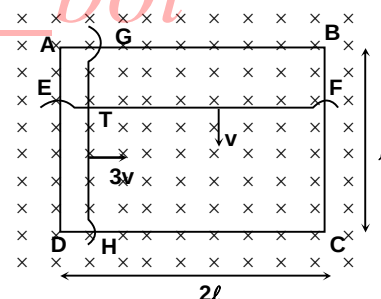
8. A charged particle enters in a region of a uniform magnetic field making 45° with direction of field. If P and R represent pitch and radius of helical path followed by particle, then $\frac{P}{R}$ equals to
 (A) π
 (B) 2π
 (C) 1
 (D) $\sqrt{\pi^2 + 1}$

9. A sphere of diameter d is removed from a uniform non-conducting sphere of charge density ρ and radius R thereby creating a cavity as shown. An electron (of charge $-e$ and mass m) is released at top point P of this cavity. Time after which the electron strikes the surface of the cavity is (Take, $d = R$ and gravity is absent)



- (A) $\sqrt{\frac{12\epsilon_0 m}{\rho e}}$
 (B) $\frac{1}{2} \sqrt{\frac{\rho e}{3\epsilon_0 m}}$
 (C) $\sqrt{\frac{2\epsilon_0 m}{\rho e}}$
 (D) $\sqrt{\frac{\epsilon_0 m}{2\rho e}}$

10. The figure here shows a fixed rectangular conducting frame ABCD of side ℓ and 2ℓ in a region of transverse uniform magnetic field B_0 . Two straight conducting sliders EF and GH are moving with respective speeds v and $3v$ as shown in figure. Initially, EF and GH are located very near to sides AB and AD respectively. Intersection contact of the sliders is represented by T. Emf induced in the loop ETHD when it is a square, is



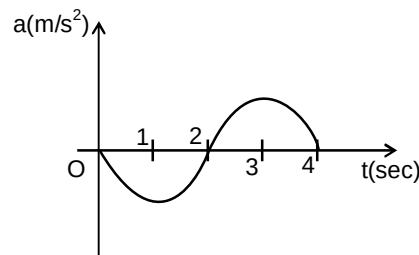
- (A) $2B_0 v \ell$
 (B) $1.5B_0 v \ell$
 (C) $B_0 v \ell$
 (D) $0.5 B_0 v \ell$

11. A particle experiences a force $\vec{F} = A r^2 \hat{r}$, where $\hat{r}$ is the unit vector along position vector $\vec{r}$. The dimensional formula of A is:

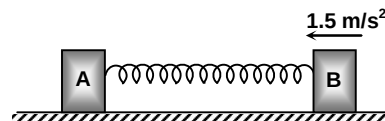
- (A) $[MLT^{-2}]$
 (B) $[ML^{-2}T^{-2}]$
 (C) $[ML^{-2}T^{-1}]$
 (D) $[ML^{-1}T^{-2}]$

12. Acceleration (a)-time (t) graph for a particle starting from rest at $t = 0$ is as given aside. The particle has maximum speed at:

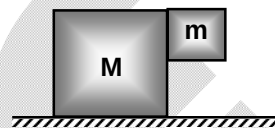
- (A) 1s
 (B) 2s
 (C) 3s
 (D) 4s



13. Two blocks A and B with mass 4 kg and 6 kg respectively are connected by a stretched spring of negligible mass as shown in figure. When the two blocks are released simultaneously the initial acceleration of B is 1.5 m/s^2 westward. The acceleration of A is
 (A) 1 m/s^2 westward
 (B) 2.25 m/s^2 eastward
 (C) 1 m/s^2 eastward
 (D) 2.75 m/s^2 westward

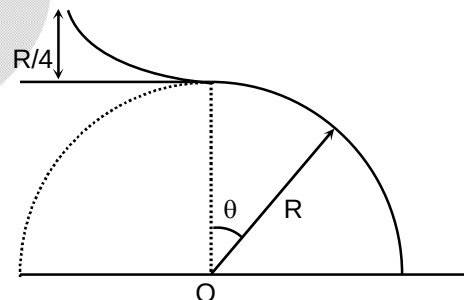


14. With what minimum acceleration, mass M must be moved on frictionless surface so that m remains stick to it as shown? The coefficient of friction between M and m is μ .



- (A) μg
 (B) $\frac{g}{\mu}$
 (C) $\frac{\mu mg}{M + m}$
 (D) $\frac{\mu mg}{M}$
15. In a circular motion of a particle, the tangential acceleration of the particle is given by $a_t = 9 \text{ m/s}^2$. The radius of the circle is 4 m. The particle was initially at rest. Time after which total acceleration of the particle makes an angle of 45° with the radial acceleration is
 (A) $\frac{1}{3} \text{ sec}$
 (B) $\frac{2}{3} \text{ sec}$
 (C) 1 sec
 (D) $\frac{4}{3} \text{ sec}$

16. A skier plans to ski a smooth fixed hemisphere of radius R. He starts from rest from a curved smooth surface of height $(R/4)$. The angle θ at which he leaves the hemisphere is:

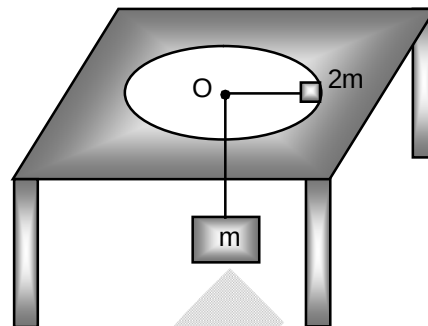


- (A) $\cos^{-1}\left(\frac{2}{3}\right)$
 (B) $\cos^{-1}\left(\frac{5}{\sqrt{3}}\right)$
 (C) $\cos^{-1}\left(\frac{5}{6}\right)$
 (D) $\cos^{-1}\left(\frac{5}{2\sqrt{3}}\right)$

17. Sachin (55 kg) and Kapil (65 kg) are sitting at the two ends of a boat at rest in still water. The boat weighs 100 kg and is 3.0 m long. Sachin walks down to Kapil and shakes hand. The boat gets displaced by:
 (A) zero m
 (B) 0.75 m
 (C) 3.0 m
 (D) 2.3 mm

18. A mass $2m$ rotating freely in a horizontal circle of radius $1m$ on a frictionless smooth table supports a stationary mass m , attached to the other end of the string passing through smooth hole O in table, hanging vertically. Find the angular velocity of rotation. (Given $g = 10 \text{ m/s}^2$)

- (A) $\sqrt{5} \text{ rad/s}$
 (B) $2\sqrt{5} \text{ rad/s}$
 (C) $\sqrt{10} \text{ rad/s}$
 (D) $\frac{\sqrt{5}}{2} \text{ rad/s}$



19. A bowling ball of mass m , which can be treated as a uniform rigid sphere, is rolling without slipping on a horizontal surface. The coefficient of static friction between the ball and the surface is μ_s , the coefficient of kinetic friction is μ_k , and the acceleration due to gravity is g . What is the force of friction acting on the ball?

- (A) zero
 (B) $\mu_s mg$
 (C) $\mu_k mg$
 (D) $\frac{2}{5} \mu_s mg$

20. A disc of radius R and mass M is pivoted at the rim and is set for small oscillations in vertical plane (in its plane). If simple pendulum has to have the same period as that of the disc, the length of the simple pendulum should be :

- (A) $\left(\frac{5}{4}\right)R$
 (B) $\left(\frac{2}{3}\right)R$
 (C) $\left(\frac{3}{4}\right)R$
 (D) $\left(\frac{3}{2}\right)R$

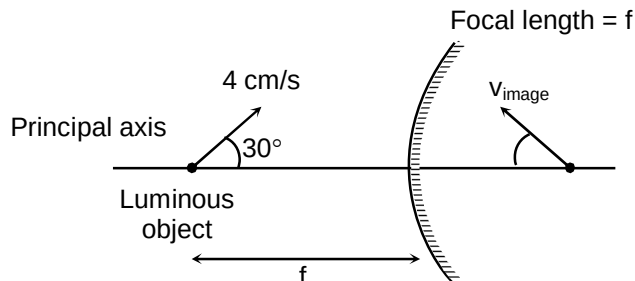
SECTION - B

(Numerical Answer Type)

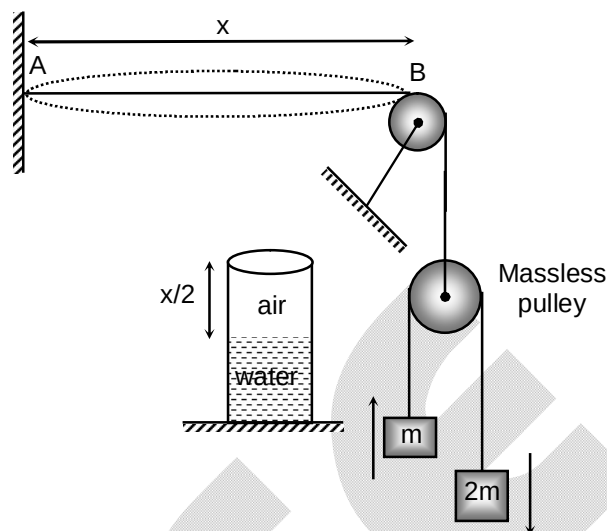
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A point charge of $\frac{1}{4\pi}$ coulomb is placed in front of an infinite imaginary plane. Electric flux through the plane is $4.5 \times 10^b \text{ N-m}^2/\text{C}$. Find b .

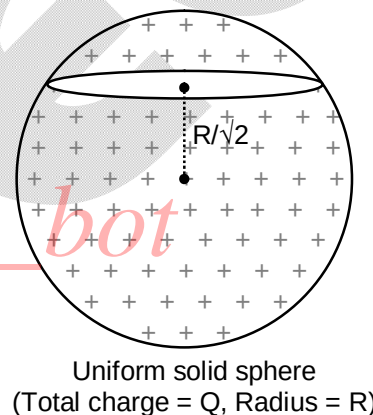
22. The speed of image of the object moving on axis of the mirror as shown is $\frac{\sqrt{7}}{k} \text{ cm/s}$. Find the value of k .



23. Consider the arrangement shown in the figure. A uniform wire (linear mass density 0.2 g/m) vibrating in its fundamental mode is in resonance with air column in a resonance tube vibrating in 1st overtone. The value of m is $\frac{54}{k} \text{ kg}$. Find the value of k . (Given that speed of sound in air is 400 m/s and $g = 10 \text{ m/s}^2$)



24. Consider the shown section at a perpendicular distance $\frac{R}{\sqrt{2}}$ from centre of the shown uniform solid sphere. Electric flux through the section is $\frac{Q}{k\sqrt{2}\epsilon_0}$. Find the value of k .



25. Average life time of a hydrogen atom excited to $n = 2$ state is 10^{-8} s . The number of revolutions made by the electron on the average before it jumps to ground state is close to 8×10^x . Find x . (Take radius of first Bohr's orbit $r_0 = 0.53 \text{ \AA}$ and speed of electron in first Bohr's orbit $2.2 \times 10^6 \text{ m/s}$)
26. If percentage errors in measurement of length of pendulum and time period of pendulum in an experiment to measure g are 1% and 2% respectively, then find maximum possible percentage error in measurement of g .
27. A vibration magnetometer has two identical bar magnets placed over each other such that they are mutually perpendicular and bisect each other. Time period of combination is $\sqrt[4]{32} \text{ s}$. Now if one of the magnet is removed, find time period of the remaining magnet in second.
28. 20 divisions on a vernier scale coincides with 19 divisions of main scale. If least count of the instrument is 0.05 mm , find the length of one Main Scale Division (MSD) in mm.
29. Two identical thin equiconvex lenses of focal length 15 cm and refractive index 1.5 are in contact with each other. The space between the lenses is filled with a liquid of refractive index 1.25 . Find the focal length of the combination in cm.
30. In a potentiometer arrangement, a cell gives a balancing point at 75 cm length of wire. This cell is now replaced by another cell of unknown emf. If the ratio of the emf's of two cells, is $3: 2$, the difference in the balancing length of the potentiometer wire in above two cases will be $x \text{ cm}$. Find x .

Chemistry

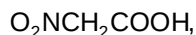
PART – B

SECTION – A

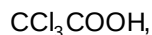
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

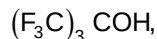
31. The correct order of decreasing acidity for following compound is



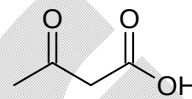
1



2



3

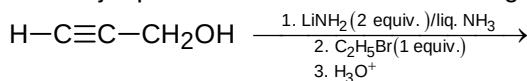


4

- (A) $1 > 3 > 2 > 4$
(C) $2 > 1 > 4 > 3$

- (B) $3 > 1 > 2 > 4$
(D) $3 > 1 > 4 > 2$

32. The major product obtained in the following reaction, is

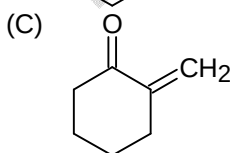
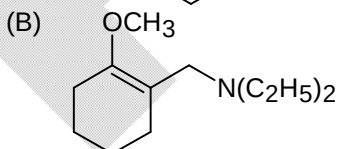
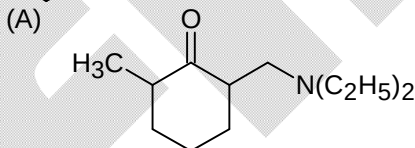
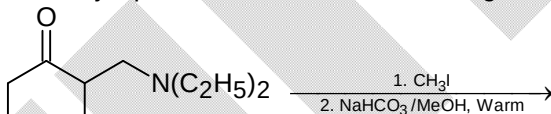


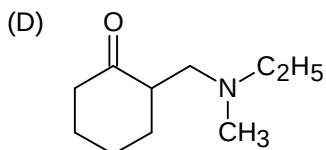
- (A) $\text{CH}_3\text{CH}_2-\text{C}\equiv\text{C}-\text{CH}_2\text{OH}$
(B) $\text{H}-\text{C}\equiv\text{C}-\text{CH}_2\text{OCH}_2\text{CH}_3$
(C) $\text{CH}_3\text{CH}_2-\text{C}\equiv\text{C}-\text{CH}_2\text{NH}_2$
(D) $\text{H}-\text{C}\equiv\text{C}-\text{CH}_2\text{NH}-\text{CH}_2\text{CH}_3$

33. The reaction of P_2O_5 with HNO_3 and HClO_4 , respectively, gives

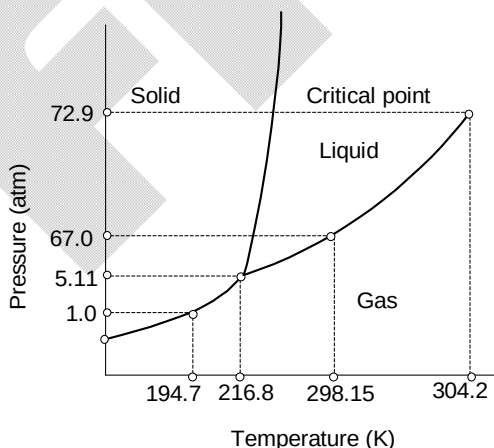
- (A) NO_2 and ClO_2
(B) N_2O_5 and Cl_2O_6
(C) N_2O_3 and Cl_2O_7
(D) N_2O_5 and Cl_2O_7

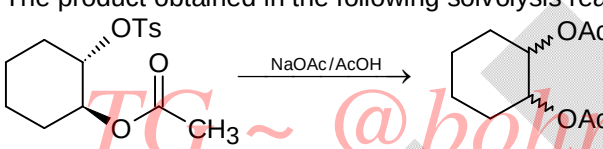
34. The major product formed in the following reaction, is





35. Select the correct statement(s) about drugs:
 (A) Chlordiazepoxide is suitable for relieving tension.
 (B) Barbiturates are sleep producing agents.
 (C) Cimetidine prevent the interaction of histamine with the receptors present in stomach wall.
 (D) All of the above are correct statements.
36. The correct electronic configuration and spin only magnetic moment Gd^{3+} (at. no. 64) are
 (A) $[Xe] 4f^7$ and 7.9 BM
 (B) $[Xe] 4f^7$ and 8.9 BM
 (C) $[Xe] 4f^6 5d^1$ and 7.9 BM
 (D) $[Rn] 5f^7$ and 7.9 BM
37. Among the following octahedral complexes, the one that has the highest enthalpy of hydration is
 (A) $[Ca(H_2O)_6]^{2+}$
 (B) $[Mn(H_2O)_6]^{2+}$
 (C) $[V(H_2O)_6]^{2+}$
 (D) $[Cr(H_2O)_6]^{2+}$
38. The Lewis acidity of BF_3 is less than BCl_3 even though fluorine is more electronegative than chlorine. It is due to
 (A) stronger $2p(B)-2p(F)$ σ – bonding
 (B) stronger $2p(B)-2p(F)$ π – bonding
 (C) stronger $2p(B)-3p(Cl)$ σ – bonding
 (D) stronger $2p(B)-3p(Cl)$ π – bonding
39. Consider the following phase diagram of CO_2 (not to scale). At equilibrium, the INCORRECT statement is:



- (A) At 200 K, on increasing the pressure from 1 to 50 atm, CO_2 gas condenses to liquid.
 (B) It is not possible to obtain liquid CO_2 from gaseous CO_2 below 5.11 atm.
 (C) Both liquid and gas phase CO_2 coexist at 29.15 K and 67 atm.
 (D) With increasing pressure, the melting point of solid CO_2 increases.
40. Among the following species, the one that has pentagonal shape is:
 (Given: atomic numbers of O, F, S, I and Xe are 8, 9, 16, 53 and 54, respectively)
 (A) XeOF_4 (B) IF_5
 (C) $[\text{SF}_5]^-$ (D) $[\text{XeF}_5]^-$
41. Which of the following is a viable particulate
 (A) Smoke (B) Algae
 (C) Mists (D) Dust
42. The number of unit cells in 117 g of NaCl is approximately
 (A) 6×10^{20} (B) 3×10^{23}
 (C) 6×10^{23} (D) 0.5×10^{24}
43. The product obtained in the following solvolysis reaction is

 enantiomerically pure compound
 (A) a racemic mixture of *trans* 1,2-diacetoxycyclohexane
 (B) enantiomerically pure *trans* 1,2-diacetoxycyclohexane
 (C) racemic *cis* 1,2-diacetoxycyclohexane
 (D) a mixture of *cis* and *trans* 1,2-diacetoxycyclohexane
44. A 0.1 M solution of $[\text{Cu}(\text{NH}_3)_4]^+$ is stirred with an excess of potassium cyanide sufficient to convert all the ammonium complex to the corresponding cuprocyanide complex $[\text{Cu}(\text{CN})_4]^{3-}$ and in addition to provide the solution with an excess of CN^- equal to 0.2 M. Calculate the maximum pH of the solution when the final solution is treated with hydrogen sulphide to maintain $[\text{H}_2\text{S}] = 0.1 \text{ M}$ and the precipitation of cuprous sulphide is prevented. The instability constant for $[\text{Cu}(\text{CN})_4]^{3-}$ is 6.4×10^{-15} , $K_{a, \text{overall}}$ of $\text{H}_2\text{S} = 1.6 \times 10^{-21}$, K_{sp} of $\text{Cu}_2\text{S} = 2.56 \times 10^{-27}$.
 (A) 4.0 (B) 10.0
 (C) 10.8 (D) 3.2
45. The compound which don't liberate coloured gas with conc. H_2SO_4 is
 (A) $\text{KCl} + \text{K}_2\text{Cr}_2\text{O}_7$ (B) KCl
 (C) KNO_2 (D) KBr
46. Suppose the mass of electron is decreased by 25%. How will it affect the Rydberg constant?
 (A) It remains unchanged.
 (B) It becomes one-fourth.
 (C) It reduces to 75% of its original value.
 (D) It is doubled.

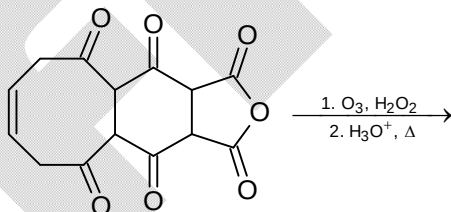
47. At moderate pressure, the compressibility factor for a gas is given as $Z = 1 + 0.35P - \frac{168}{T} \cdot P$, where P is in bar and T is in Kelvin. What is the Boyle's temperature of the gas?
 (A) 168 K (B) 480 K
 (C) 58.8 K (D) 575 K
48. Graph between $\log\left(\frac{x}{m}\right)$ and $\log P$ is a straight line at an angle 45° with intercept on y-axis, 0.3010. The amount (in g) of the gas absorbed per g of the adsorbent when pressure is 0.2 atm is (assume that the adsorption obey Freundlich isotherm)
 (A) 0.4 (B) 0.6
 (C) 0.8 (D) 0.2
49. The correct set of reagents for the following conversion is
-
- (A) (i) NaNH_2 / liq. NH_3 ; (ii) NaNO_2 / dil. HCl ; (iii) CuCN , heat
 (B) (i) HNO_3 / H_2SO_4 ; (ii) ZnHCl ; (iii) NaNO_2 / dil. HCl ; (iv) CuCN , heat
 (C) (i) Mg / ether, H_2O^+ ; (ii) $(\text{EtO})_2\text{CO}$; (iii) NH_4OH ; (iv) PCl_5
 (D) (i) Mg / ether, H_2O^+ ; (ii) HNO_3 / H_2SO_4 ; (iii) NaNO_2 / dil. HCl ; (iv) CuCN , heat
50. Phenol on reacting with Hinsberg's reagent gives:
 (A) Sulphone
 (B) Sulphanilic
 (C) Sulphonic ester
 (D) Sulphonal

SECTION – B

(Numerical Answer Type)

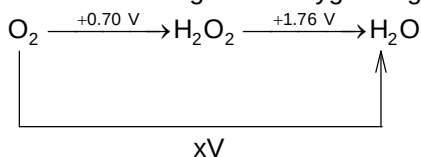
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. Number of carbonyl groups present in the final product of the following reaction sequence is.....

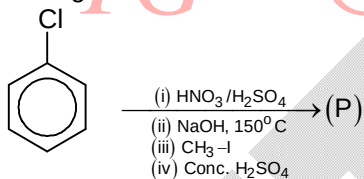


52. The rate constant of a first order reaction, $X \rightarrow Y$, is $1.6 \times 10^{-3} \text{ s}^{-1}$ at 300 K. Given that the activation energy of the reaction is 28 kJ mol^{-1} and assuming Arrhenius behaviour for the temperature dependence, the total time required to obtain 90% of Y at 350 K is _____s. (Use $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

53. The Latimer diagram of oxygen is given below. The value of $100x$ is _____ V.



54. On heating a sample of 25 g hydrated compound (molecular weight = 250 g/mol) in thermogravimetric analysis, 16 g of dehydrated compound remains. The number of water molecules lost per molecule of hydrated compound is _____.
55. The entropy change for the melting of 'x' moles of ice (heat of fusion is 80 cal g^{-1}) at 273 K and 1 atm pressure is 28.80 cal K^{-1} . The value of '50x' is _____
(Molecular weight of water = 8 g/mol)
56. At 10°C , the osmotic pressure of urea solution is 500 mm. The solution diluted and the temperature is raised to 100°C , when the osmotic pressure is found to be 94 mm. The extent of dilution (ratio of final to initial volume of solution) is _____
57. Among the following ores
Bauxite, Kaolinite, Malachite, Sphalerite, Calamine, Siderite, Magnetite, the number of principal ores of copper and zinc are _____
58. The number of nitrogen atoms present in the adenine is _____
59. In the given reaction



- The number of π bonds present in compound (P) is _____
60. How many different secondary halide names can be assigned to compounds with molecular formula $\text{C}_5\text{H}_{11}\text{Cl}$ (excluding stereoisomers)?

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

61. If P is any point on ellipse with foci S_1 and S_2 and eccentricity is $\frac{1}{2}$ such that $\angle PS_1S_2 = \alpha$, $\angle PS_2S_1 = \beta$, $\angle S_1PS_2 = \gamma$, then $\cot\left(\frac{\alpha}{2}\right), \cot\left(\frac{\gamma}{2}\right), \cot\left(\frac{\beta}{2}\right)$ are in
 (A) A.P. (B) G.P.
 (C) H.P. (D) not A.P., G.P. and H.P.
62. For each positive integer n, consider the point P with abscissa n on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line $y = x$, then $\lim_{n \rightarrow \infty} (n \cdot d_n)$ is
 (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$
 (C) $\frac{1}{\sqrt{2}}$ (D) 0
63. Latus-rectum of the conic satisfying the differential equation $xdy + ydx = 0$ and passing through the point (2, 8) is
 (A) $4\sqrt{2}$ (B) 8
 (C) $8\sqrt{2}$ (D) 16
64. Let a and b be real numbers and let $f(x) = a\sin x + b\sqrt[3]{x} + 4$, $\forall x \in \mathbb{R}$. If $f(\log_{10}(\log_3 10)) = 5$, then $f(\log_{10}(\log_3 3))$ is
 (A) 2 (B) 1
 (C) 3 (D) 4
65. Suppose f is an even periodic function with period 2, and that $f(x) = x$ for all x in the interval [0, 1]. The value of $f(3.14)$ is
 (A) 3.14 (B) 0.86
 (C) 1.57 (D) 1.3
66. $\int (x^2 + 1)((x + 1)e^x)^2 dx = A(f(x))^2 + C$ where C is constant of integration and $f(-1) = \frac{2}{e}$, then $2A + f(0)$ is
 (A) 3 (B) 2
 (C) 1 (D) 0
67. If $\sum_{i=1}^6 (\sin^{-1}(x_i) + \cos^{-1}(y_i)) = 9\pi$, then $\int_{\sum_{i=1}^6 x_i}^{\sum_{i=1}^6 y_i} x \ln(1 + x^2) \left(\frac{e^x}{1 + e^{2x}} \right) dx$ is equal to
 (A) 0 (B) $e^6 + e^{-6}$
 (C) $\ln\left(\frac{37}{12}\right)$ (D) $e^6 - e^{-6}$

68. Area bounded by $y = \frac{1}{x^2 - 2x + 2}$ and x-axis is
- (A) 2π sq. units (B) $\frac{\pi}{2}$ sq. units
(C) 2 sq. units (D) π sq. units
69. Let $f(x) = \frac{\sin(\pi x^4) + (x+2)^n \frac{\tan(\pi x)}{(x+1)}}{1 + (x+2)^n - x^4}$, then $\lim_{x \rightarrow -1} f(x)$ is
- (A) π (B) $\frac{22}{7}$
(C) 1 (D) $-\pi$
70. If $f(x) = \frac{\sqrt{x^2 + kx + 1}}{x^2 - k}$, the interval(s) of all possible values of k for which f is continuous for every $x \in \mathbb{R}$, is
- (A) $(-\infty, -2]$ (B) $[-2, 0)$
(C) $\mathbb{R} - (-2, 2)$ (D) $(-2, 2)$
71. Number of complex numbers satisfying the relation $|z + \bar{z}| + |z - \bar{z}| = 2$ and $|z + i| + |z - i| = 2$ is
- (A) 1 (B) 2
(C) 3 (D) 4
72. If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\det(A^n - I) = 1 - \lambda^n$, $n \in \mathbb{N}$, then the value of λ is
- (A) 2 (B) -2
(C) 3 (D) 4
73. The matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix}$ be a zero divisor of the polynomial $f(x) = x^2 - 4x - 5$, then the trace of matrix A^3 , is
- (A) 213 (B) 123
(C) 203 (D) 201
74. The relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$ on the set $A = \{1, 2, 3\}$ is
- (A) reflexive but not symmetric (B) reflexive but not transitive
(C) symmetric and transitive (D) neither symmetric nor transitive
75. A tower subtends angles d , 2α , 3α respectively at points A, B and C all lying on a horizontal line through the foot of the tower, then $\frac{AB}{BC}$ is equal to
- (A) $\frac{\sin(3\alpha)}{\sin(2\alpha)}$ (B) $1 + 2\cos(2\alpha)$
(C) $2\cos(2\alpha)$ (D) $\frac{\sin(2\alpha)}{\sin\alpha}$

76. The sum of squares of deviations for 10 observations taken from mean 50 is 250. The coefficient of variation is
 (A) 10% (B) 40%
 (C) 50% (D) none of these
77. If p and q be two statements, then $(\sim p \vee q) \wedge (\sim p \wedge \sim q)$ is a
 (A) tautology (B) contradiction
 (C) neither tautology nor contradiction (D) either tautology or contradiction
78. Let $y = \tan^{-1}\left(\frac{4x}{1+5x^2}\right) + \tan^{-1}\left(\frac{2+3x}{3-2x}\right)$ where $x \in \left(0, \frac{2}{3}\right)$. If $\frac{dy}{dx} = \frac{\alpha}{1+25x^2}$, then the value of α is
 (A) 3 (B) 4
 (C) 5 (D) 6
79. $I = \int_{e^{\pi/6}}^{e^{\pi/2}} \frac{\sin(\ln(\sin(\ln x))) \cdot \cos(\ln x)}{x \sin(\ln x)} dx$, then $\cos^{-1}(I + 1)$ is
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$
 (C) $\ln 2$ (D) $2\ln 2$
80. The largest value of $\frac{y}{x}$, where (x, y) is real number pair satisfying $(x - 3)^2 + (y - 3)^2 = 6$, is
 (A) $2\sqrt{3}$ (B) $2 + \sqrt{3}$
 (C) $3 + 2\sqrt{2}$ (D) $6 + 2\sqrt{3}$

SECTION - B

(Numerical Answer Type)

This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If $f(\theta) = \frac{1}{1 + (\cot \theta)^x}$ and $s = \sum_{\theta=1^\circ}^{89^\circ} f(\theta)$, then $[s]$ is (where $[.]$ denotes G.I.F)
82. Let $\vec{v}_1$ and $\vec{v}_2$ are two vectors such that $\vec{v}_1 = 2(\sin \alpha + \cos \alpha) \hat{i} + \hat{j}$ and $\vec{v}_2 = (\sin \beta) \hat{i} + (\cos \beta) \hat{j}$ where α and β satisfy the relation $2(\sin \alpha + \cos \alpha) \sin \beta = 3 - \cos \beta$, then the value of $3 \tan^2 \alpha + 4 \tan^2 \beta$ is
83. The line which contains all points (x, y, z) which are of the form $(x, y, z) = (2, -2, 5) + \lambda(1, -3, 2)$ intersects the plane $2x - 3y + 4z = 163$ at P and intersects the YZ plane at Q. If the distance PQ is $a\sqrt{b}$, where a, b $\in \mathbb{N}$ and $a > 3$ then $(a + b)$ is
84. If A and B are two events with $P(A) = 0.5$ and $P(A \cup B) = 0.8$. If $P(B) = p$ and A and B are mutually exclusive and $P(B) = q$, if A and B are independent then $\left|\frac{q}{p}\right|$ is
85. If $x > 0$ and $y > 0$ and $x^2 y^3 = 6$. If m be the least value of $3x + 4y$, then m is
86. If the point P(h, k) lies on the line $2x + 3y = 5$ such that $|PA - PB|$ is maximum where A(2, 3) and B(1, 2), then $3h + 2k$ is

87. If x, y, z be positive numbers such that $x + y + z = 10$. If maximum value of $x^2y^3z^5$ is N , then number of odd divisors is
88. The remainder, when $15^{23} + 23^{23}$ is divided by 19 is m , then m is
89. Let x, y be real variables satisfying $x^2 + y^2 + 8x - 10y + 40 = 0$. Let $a = \max\{(x + 2)^2 + (y - 3)^2\}$ and $b = \min\{(x + 2)^2 + (y - 3)^2\}$, then ab is
90. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = 3\pi$, then $(x + y - z)^{2022}$ is

TG ~ @bohring_bot

ALL INDIA TEST SERIES

FULL TEST – VI

JEE (Main)-2023

TEST DATE: 19-01-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1.

B

Sol.

Applying energy conservation:

$$mgh = mL_f$$

$$h = \frac{L_f}{g} = \frac{80 \text{ cal/gm}}{10 \text{ m/s}^2}$$

$$= \frac{80 \times 42 \times 1000 \text{ J/kg}}{10 \text{ m/s}^2} = \left(\frac{336 \times 10^3}{10} \right) \frac{\text{N-s}^2}{\text{kg}} = 33.6 \text{ km}$$

2.

D

Sol.

$$\frac{2}{3} v_{p1} g = v_{pg}$$

$$\frac{3}{4} v_{p2} g = v_{pg}$$

$$v_{p1} + v_{p2} = 2v_{pm}$$

$$\frac{\rho_1 + \rho_2}{2} = \rho_m$$

$$v' \left(\frac{\rho_1 + \rho_2}{2} \right) g = v_{pg}$$

$$\frac{v'}{2} \left(\frac{3\rho g}{2} + \frac{4\rho g}{3} \right) = v_{pg}$$

$$v' = \frac{12}{17} v$$

3. B

Sol. As $P = nR \frac{T}{V}$ so $P_A = nR \frac{T_0}{2V_0}$

$$\text{and } P_B = nR \frac{4T_0}{3V_0}$$

$$\Rightarrow \frac{P_B}{P_A} = \frac{8}{3}$$

4. C

Sol. Resonant frequency of an L-C-R circuit is given by

$$\omega_r = \frac{1}{\sqrt{LC}}$$

So, resonance frequency does not depend on resistance of circuit and frequency of supply. To increase resonance frequency, we have to reduce inductance or capacitance of circuit. As capacitance reduces when capacitors are connected in series

(i.e. $\frac{1}{C_{\text{series}}} = \frac{1}{C_1} + \frac{1}{C_2}$), another capacitor in series to increase resonant frequency.

5. A

Sol. As, refractive index,

$$\mu = \frac{\text{speed of light in air}}{\text{speed of light in medium}}$$

So, speed of light in a medium

$$v = \frac{c}{\mu}$$

Hence, for medium A and B,

$$v_A = \frac{c}{\mu_A} \text{ and } v_B = \frac{c}{\mu_B}$$

Time taken by light to cross some

thickness x is $\left(\text{speed} = \frac{\text{distance}}{\text{time}} \right)$

$$t_1 = \frac{x}{(c / \mu_A)} \text{ and } t_2 = \frac{x}{(c / \mu_B)}$$

$$\text{Hence, } t_2 - t_1 = \frac{\mu_B \cdot x}{c} - \frac{\mu_A \cdot x}{c}$$

$$\Rightarrow t_2 - t_1 = \frac{x}{c} (\mu_B - \mu_A)$$

$$\text{Now, given, } \frac{\mu_A}{\mu_B} = \frac{1}{2}$$

$$\Rightarrow \mu_B = 2\mu_A$$

$$\text{and } t_2 - t_1 = 5 \times 10^{-10} \text{ s}$$

substituting these values, we get

$$5 \times 10^{-10} = \frac{x}{c} (2\mu_A - \mu_A)$$

$$\Rightarrow x = 5 \times 10^{-10} \times \frac{c}{\mu_A}$$

But $\frac{C}{\mu_A} = v_A$

$\therefore x = (5 \times 10^{-10} \times v_A) \text{ m}$

6. D

Sol. Assume potential at point D is equal to zero. As we know potential of all points on a plane wire are equal, so potential at B will also be equal to 0 V. Also potential difference across cell is 24 V, so potential at A will be 24 V. Similarly, potential at C is also 24V.

Using Ohm's law for branch AB,

$$i_1 = \frac{V_A - V_B}{R_1} = \frac{24 - 0}{2} = 12 \text{ A}$$

For branch CB,

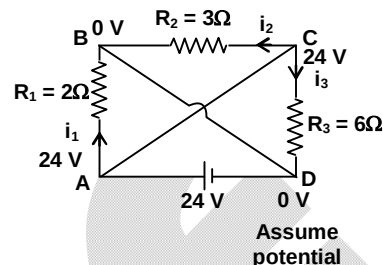
$$i_2 = \frac{V_C - V_B}{R_2} = \frac{24 - 0}{3} = 8 \text{ A}$$

For branch CD,

$$i_3 = \frac{V_C - V_D}{R_3} = \frac{24 - 0}{6} = 4 \text{ A}$$

By KCL for junction C,

$$i = i_2 + i_3 = 8 + 4 = 12 \text{ A}$$



7. C

Sol. As per Einstein's photoelectric equation

$$v_s = \frac{hf}{e} - \frac{\phi}{e} \quad \dots(i)$$

If f is doubled, new stopping potential,

$$v'_s = \frac{2hf}{e} - \frac{\phi}{e} = 2\left(\frac{hf}{e} - \frac{\phi}{e}\right) + \frac{\phi}{e} = 2v_s + \frac{\phi}{e} \quad [\text{from equation (i)}]$$

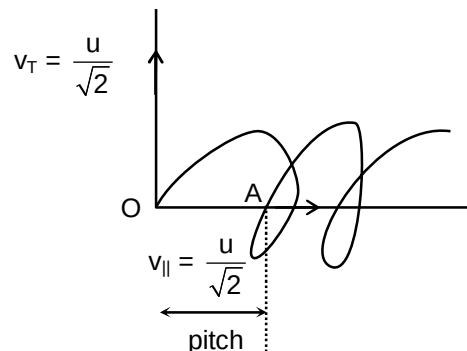
Which is more than double or an increase of more than 100%

8. B

Sol. When a charged particle enters a region of a uniform magnetic field, such that the angle it makes with the direction of $B \neq 90^\circ$, then it follows a helical path as shown below

Radius of helix,

$$R = \frac{mv_{\perp}}{qB} = \frac{mu}{\sqrt{2}qB} \quad \dots(i)$$



Where, $v_{\perp}$ = Component of velocity perpendicular to field.

Also, pitch, P = Displacement along field in one revolution (OA)

$$= v_{||} \times T = \frac{u}{\sqrt{2}} \times \frac{2\pi m}{qB} \quad (ii)$$

Dividing equation (ii) by equation (i), $\frac{P}{R} = 2\pi$

9. A

Sol. Field inside spherical cavity in a sphere = $\frac{\rho a}{3\epsilon_0}$

Where a = distance between centres of sphere and cavity.
Direction of field is along line joining centres as shown in figure.

So, field inside cavity will be $E = \frac{\rho \left(\frac{R}{2}\right)}{3\epsilon_0} = \frac{\rho R}{6\epsilon_0}$ (as shown)

Under influence of field, electron will accelerate down with acceleration,

$$a = \frac{eE}{m} = \frac{\rho Re}{6\epsilon_0 m} \quad \dots(i)$$

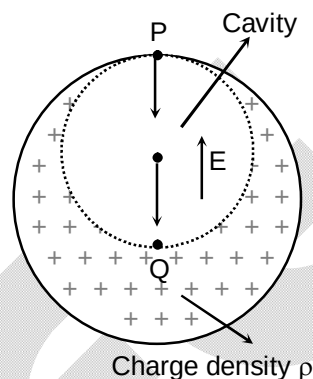
As electrons move from P to Q, striking the cavity at Q, its displacement will be equal to $PQ = R = s$

$\dots(ii)$

So, by using, $s = ut + \frac{1}{2}at^2$, we get

$$R = 0 + \frac{1}{2} \frac{\rho Re}{6\epsilon_0 m} t^2 \quad \text{[using equation (i) and (ii)]}$$

$$\Rightarrow t = \sqrt{\frac{12\epsilon_0 m}{\rho e}}$$



10. B

Sol. At time t , distances moved by sliders GH and EF will be $3vt$ and vt respectively as shown. Therefore, area of loop ETHD will be $A = 3vt(\ell - vt) = 3v\ell t - 3v^2 t^2$

$$\Rightarrow \frac{dA}{dt} = 3v\ell - 6v^2 t \quad \dots(i)$$

At the instant, the loop is square, its two sides will be equal. Therefore,

$$3vt = \ell - vt \Rightarrow t = \frac{\ell}{4v}$$

$\dots(ii)$

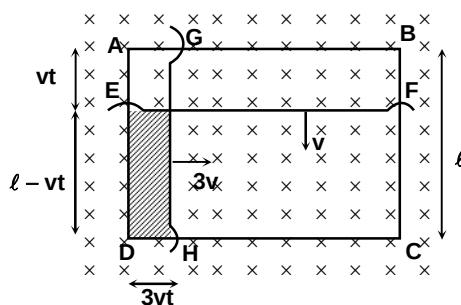
By Faraday's law, emf induced in the loop

$$\epsilon = \frac{d\phi}{dt} = \frac{d(B_0 \cdot A)}{dt} = B_0 \frac{dA}{dt}$$

$$= B_0(3v\ell - 6v^2 t) \text{ using equation (i)}$$

$$= B_0 \left[3v\ell - 6v^2 \left(\frac{\ell}{4v} \right) \right] \text{ using equation (ii)}$$

$$= \frac{3B_0 v \ell}{2} = 1.5 B_0 v \ell$$



11.

D

Sol. $[\vec{F}] = [Ar^2\hat{r}] = [Ar^2]$

$$[A] = \frac{[\vec{F}]}{[r^2]} = \frac{MLT^{-2}}{L^2} = [ML^{-1}T^{-2}]$$

12.

B

Sol. Between $t = 0$ and $t = 2$, acceleration has same sign which increases speed. At $t = 2$ sec. Direction of acceleration is reversed and speed starts decreasing.

13.

B

Sol. $T = M_B \times a_B$ but T is also equal to $M_A a_A$

$$\therefore M_B a_B = M_A a_A$$

$$\Rightarrow 6 \times 1.5 = 4 \times a_A$$

$$\Rightarrow a_A = 2.25 \text{ m/s}^2$$

14.

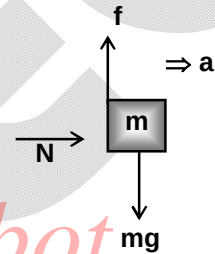
B

Sol. $f = mg$ and $N = mA$

$$\text{So, } \mu N = mg$$

$$\mu mA = mg \text{ (For } A_{\min}, f \text{ is max)}$$

$$\Rightarrow A = \frac{g}{\mu}$$



TG ~ @bohring_bot

15.

B

Sol. At 45° , $a_t = a_r = 9$

$$\frac{v^2}{r} = \frac{v^2}{4} = 9$$

$$v = 6 \text{ m/s}$$

$$\text{Now, } 6 = 0 + 9t$$

$$t = \frac{2}{3} \text{ sec}$$

16.

C

Sol. Conserving energy between points A and B,

$$mg \left[\frac{R}{4} + R(1 - \cos\theta) \right] = \frac{1}{2}mv^2$$

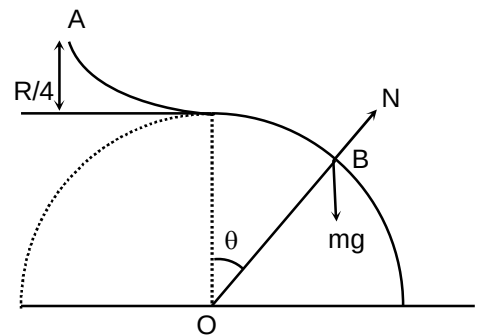
Also at point B, $N = 0$

$$\text{So, } mg \cos\theta = \frac{mv^2}{R}$$

$$\frac{1}{4} + (1 - \cos\theta) = \frac{1}{2} \cos\theta$$

$$\frac{5}{4} = \frac{3}{2} \cos\theta$$

$$\cos\theta = \frac{5}{6}$$



17. B

Sol. Kapil and the boat can be considered as one body of mass $m_b = (65 + 100) = 165$ kg. Note that the centre of mass of the system remains unchanged since no external force acts on the system. Let m_s be the mass of Sachin and Δx_b and Δx_s be the displacements of the combined body of mass m_b and Sachin respectively with reference to the centre of mass. Then use the equation $m_s \Delta x_s + m_b \Delta x_b = 0$, to get the answer.

18. A

Sol. $T = (2m)\ell\omega^2 = mg$

$$\omega = \sqrt{\frac{g}{2}} = \sqrt{5} \text{ rad/s}$$

19. A

Sol. If friction acts backwards then ω will increase and v_{cm} will decrease hence violating pure rolling condition and if friction acts in forward direction the ω will decrease and v_{cm} will increase again violating pure rolling condition hence no friction acts.

20. D

Sol. $2\pi\sqrt{\frac{\frac{3}{2}MR^2}{MgR}} = 2\pi\sqrt{\frac{\ell}{g}}$

TG ~ @bohring_bot

SECTION - B

21. 9

Sol. To find through infinite plane-1, let us construct an infinite plane-2 at same distance from q on other side. If electric flux through plane-1 is ϕ , then flux through plane-2 will be also ϕ .
By Gauss's Law

$$\phi_{\text{total}} = 2\phi = \frac{q}{\epsilon_0}$$

$$\phi = \frac{q}{2\epsilon_0} = \frac{1}{8\pi\epsilon_0} = \frac{9 \times 10^9}{2} = 4.5 \times 10^9$$

$$b = 9$$

• q

22. 2

Sol. Magnification of the object

$$m = \frac{f}{f-u} = \frac{f}{f-(-f)} = \frac{1}{2}$$

As velocity component of image along axis is given by

$$v_i = -m^2 v_0$$

$$\Rightarrow v_i = -\left(\frac{1}{2}\right)^2 4 \cos 30^\circ = -\frac{\sqrt{3}}{2} \text{ cm/s}$$

Also velocity component of image perpendicular to axis is given by

$$v'_i = m v_0 = \frac{1}{2} \cdot 4 \sin 30^\circ = 1 \text{ cm/s}$$

Therefore, net velocity of image is

$$v'_{\text{net}} = \sqrt{v_i^2 + v_i'^2} = \sqrt{\frac{3}{4} + 1} = \frac{\sqrt{7}}{2} \text{ cm/s}$$

23.

5

Sol.

Consider FBD of blocks.

For mass m , using $F = ma$, we get

$$T - mg = ma \quad \dots(i)$$

Similarly for mass $2m$, we get

$$2mg - T = 2ma \quad \dots(ii)$$

From Equation (i) and (ii),

$$T = \frac{4mg}{3} \quad \dots(iii)$$

From FBD of pulley,

$$T' = 2T = \frac{8mg}{3}$$

Therefore, frequency of vibration of wire in fundamental mode,

$$f_1 = \frac{1}{2\ell} \sqrt{\frac{T'}{\mu}} = \frac{1}{2x} \sqrt{\frac{8mg}{3\mu}}$$

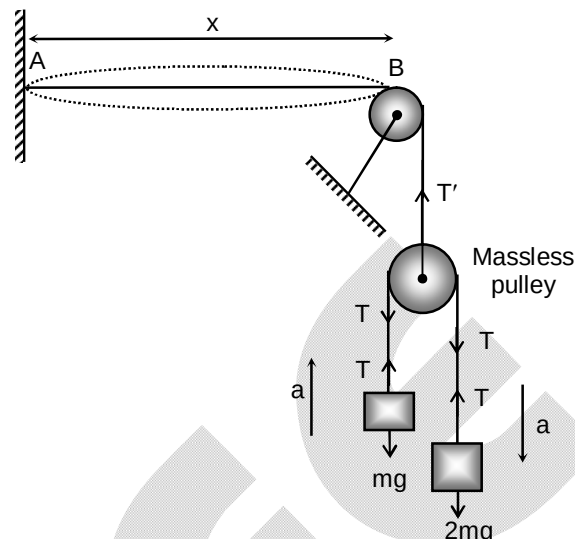
Also, frequency of vibration of air in 1st overtone (3rd harmonic)

$$f_2 = \frac{3v}{4\ell} = \frac{3v}{4x} = \frac{3v}{2x}$$

As, resonance implies frequencies are equal

$$\text{i.e., } f_1 = f_2 \Rightarrow \frac{1}{2x} \sqrt{\frac{8mg}{3\mu}} = \frac{3v}{2x} \quad [\text{using Eq. (v) and (vi)}]$$

$$\Rightarrow m = \frac{27\mu v^2}{8g} = \frac{27(0.2 \times 10^{-3})(400)^2}{8 \times 10} = \frac{54}{5} \text{ kg}$$



24.

8

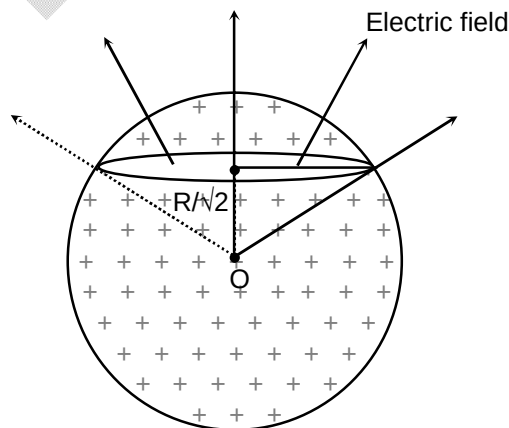
Sol.

Lets construct a cone as shown. By field picture, ϕ through lateral surface is zero.

$$\Rightarrow \phi_{\text{section}} = \phi_{\text{cone}} = \frac{q_{\text{inside}}}{\epsilon_0}$$

$$= \frac{\rho \cdot \frac{1}{3} \pi \left(\frac{R}{\sqrt{2}} \right)^2 \left(\frac{R}{\sqrt{2}} \right)}{\epsilon_0}$$

$$= \frac{\left(\frac{Q}{\frac{4}{3} \pi R^3} \right) \cdot \left(\frac{1}{3} \pi \frac{R^3}{2\sqrt{2}} \right)}{\epsilon_0} = \frac{Q}{8\sqrt{2}\epsilon_0}$$



25.

6

Sol.

Frequency of revolution of electron,

$$f = \frac{v}{2\pi r} = \frac{v_0 \frac{Z}{n}}{2\pi r_0 \frac{n^2}{Z}} = \frac{v_0 Z^2}{2\pi r_0 n^3}$$

Number of revolutions in time t will be

$$N = ft = \frac{v_0 z^2 t}{2\pi r_0 n^3}$$

As we know, Bohr's radius

$$r_0 = 0.53 \text{ \AA} = 0.53 \times 10^{-10} \text{ m}$$

$$v_0 = \text{Bohr's speed} = 2.2 \times 10^6 \text{ m/s}$$

Putting these values in Eq. (ii), we get

$$N = \frac{2.2 \times 10^6 \times 1^2 \times 10^{-8}}{2 \times 3.14 \times 0.53 \times 10^{-10} \times 2^3} \approx 8 \times 10^6$$

Comparing with given value, we get $x = 6$

26.

5

Sol. Time period of a pendulum is given by

$$T = 2\pi \sqrt{\frac{\ell}{g}}$$

$$\Rightarrow g = \frac{4\pi^2 \ell}{T^2}$$

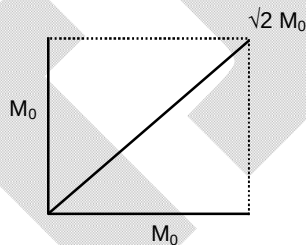
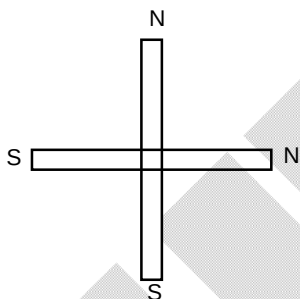
$$\Rightarrow \left(\frac{\Delta g}{g} \times 100 \right)_{\max} = \frac{\Delta \ell}{\ell} \times 100 + 2 \frac{\Delta T}{T} \times 100 = 1 + 2 \times 2 = 5\%$$

27.

2

Sol. For two perpendicular magnets of magnetic moment M_0 and moment of inertia each, equivalent magnetic moment,

$$M = \sqrt{2} M_0$$



and total moment of inertia

$$I = 2I_0$$

...(ii)

Also, time period of oscillation of magnet in magnetic field is given by

$$T = 2\pi \sqrt{\frac{I}{MB}}$$

... (iii)

Time period of the system, using

Equations (i), (ii) and (iii), we get

$$T = 2\pi \sqrt{\frac{2I_0}{\sqrt{2} M_0 B}} = 2\pi \sqrt{\frac{\sqrt{2} I_0}{M_0 B}}$$

...(iv)

Also, using Equation (iii) time period of single magnet,

$$T' = 2\pi \sqrt{\frac{I_0}{M_0 B}}$$

...(v)

Dividing Equation (v) by Equation (iv), we get

$$\frac{T'}{T} = \frac{1}{2^{1/4}} \Rightarrow T' = \frac{T}{2^{1/4}} = \frac{32^{1/4}}{2^{1/4}} = 16^{1/4} = 2\text{s}$$

28. 1

Sol. LC = 1 MSD - 1 VSD

$$\Rightarrow LC = 1 \text{ MSD} - \frac{19}{20} \text{ MSD} = \frac{\text{MSD}}{20}$$

$$\Rightarrow \text{MSD} = 20 \times LC = 20 \times 0.05 = 1 \text{ mm}$$

29. 10

Sol. Given, focal length of two thin biconvex lens, $f_1 = f_2 = f = 15 \text{ cm}$

Using lens Maker's formula for first lens,

$$\frac{1}{f_1} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{15} = (1.5 - 1) \left(\frac{1}{R} - \frac{1}{-R} \right) \Rightarrow \frac{1}{15} = \frac{1}{R}$$

$$\Rightarrow R = 15 \text{ cm}$$

$\therefore$ Focal length of combination is given as

$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} \quad \dots(i)$$

Where, $f_1 = f_2 = f = 15 \text{ cm}$

$$\text{and } \frac{1}{f_3} = (\mu_1 - 1) \left(\frac{1}{-R} - \frac{1}{R} \right) = (1.25 - 1) \left(\frac{-2}{R} \right) = -\frac{1}{30}$$

$$\Rightarrow f_3 = -30 \text{ cm}$$

Hence, from Equation (i), we get

$$\frac{1}{f_{eq}} = \frac{1}{15} + \frac{1}{15} - \frac{1}{30} = \frac{3}{30} = \frac{1}{10}$$

$$\Rightarrow f_{eq} = 10 \text{ cm}$$

30. 25

Sol. Comparison of emfs of two cells using the experiment of potentiometer is given as

$$\frac{e_1}{e_2} = \frac{l_1}{l_2}$$

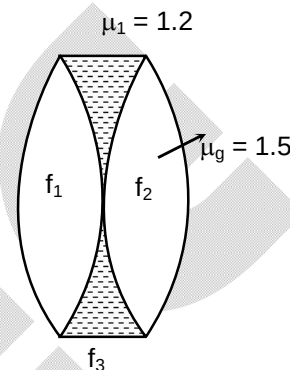
Where, l_1 and l_2 are their corresponding balancing length.

$$\therefore \frac{3}{2} = \frac{75}{l_2} \quad (\text{since, } l_1 = 75 \text{ cm given})$$

$$\Rightarrow l_2 = 50 \text{ cm}$$

$\therefore$ difference in balancing length

$$= l_1 - l_2 = 75 - 50 = 25 \text{ cm}$$



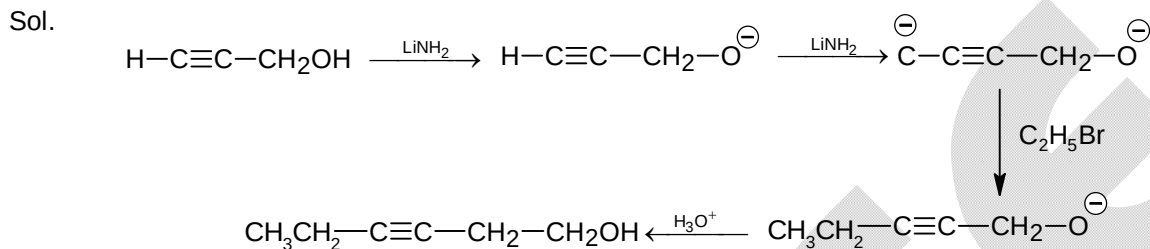
Chemistry

PART – B

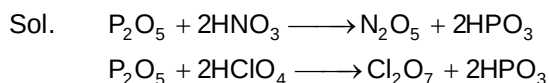
SECTION – A

31. C
Sol. Carboxylic acids are more acidic than alcohol. Electronic effects also influence acidity.

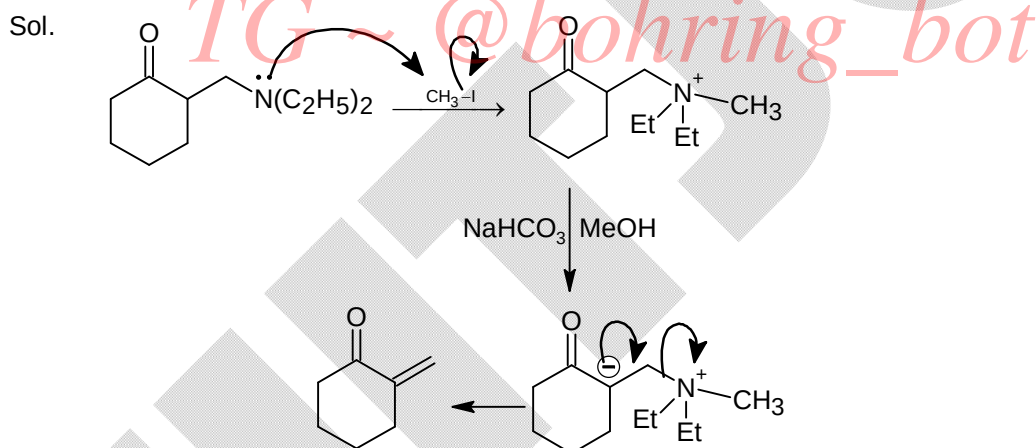
32. A



33. D



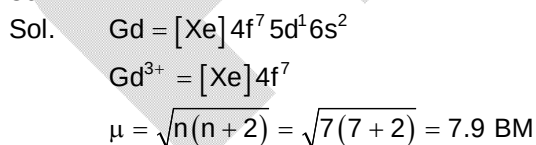
34. C



35. D

Sol. Fact

36. A



37. C

Sol. Highest the CFSE of compound, highest is the enthalpy of hydration.

38. B

Sol. Smaller the size of orbital extent of back bonding is more.

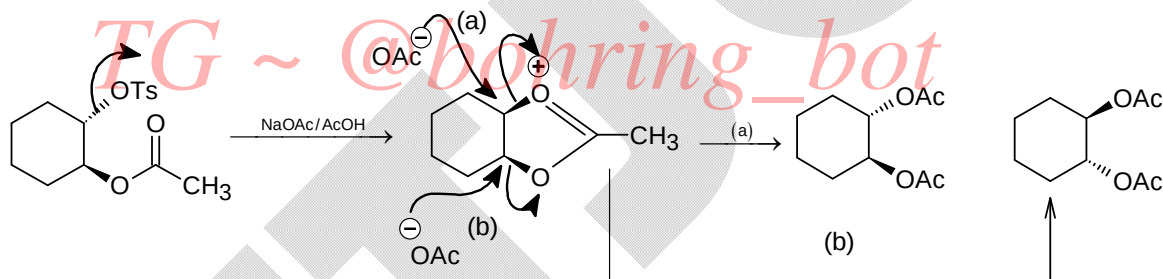
39. A
Sol. On increasing pressure CO_2 gas condenses to liquid.

40. D
Sol. $[\text{XeF}_5]^-$ is sp^3d^3 hybridized.

41. B
Sol. Viable particulate are minute living organisms that are dispersed in the atmosphere.

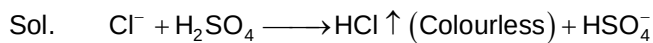
42. B
Sol. Molar mass of $\text{NaCl} = 58.5 \text{ g}$
Moles of $\text{NaCl} = \frac{117}{58.5} = 2$
In a unit cell = 4 NaCl
Total number unit cell = $\frac{1}{4} \times \text{Number of NaCl}$
 $= \frac{1}{4} \times 2 \times 6.02 \times 10^{23}$
 $= 3 \times 10^{23}$

43. A
Sol.



44. B
Sol. From the question,
 $[\text{Cu}(\text{CN})_4]^{3-} = 0.1 \text{ M}$ and $[\text{CN}^-] = 0.2 \text{ M}$
 $\therefore [\text{Cu}^+] = \frac{K_{\text{instab}} \cdot [\text{Cu}(\text{CN})_4]^{3-}}{[\text{CN}^-]^4} = \frac{6.4 \times 10^{-15} \times 0.1}{(0.2)^4}$
 $= 4 \times 10^{-13} \text{ M}$
Now, $[\text{S}^{2-}] = \frac{K_{\text{SP}}(\text{Cu}_2\text{S})}{[\text{Cu}^+]^2} = \frac{2.56 \times 10^{-27}}{(4 \times 10^{-13})^2}$
 $= 1.6 \times 10^{-2} \text{ M}$
 $\therefore [\text{H}^+] = \sqrt{\frac{K_a \times [\text{H}_2\text{S}]}{[\text{S}^{2-}]}} = \sqrt{\frac{1.6 \times 10^{-21} \times 0.1}{1.6 \times 10^{-2}}} = 10^{-10} \text{ M}$
and $\text{pH} = 10.0$.

45. B



46. C

Sol.
$$R = \frac{2\pi^2 m e^4}{(4\pi\epsilon_0)^2 h^3 c}$$

 $\Rightarrow R \propto m$

Here, m becomes $\left(m - \frac{m}{4} = \frac{3m}{4}\right)$, then R becomes $\frac{3}{4}R$.

47. B

Sol. At Boyle's temperature, $\frac{dz}{dp} = 0 \Rightarrow T = \frac{168}{0.35} = 480 \text{ K}$.

48. A

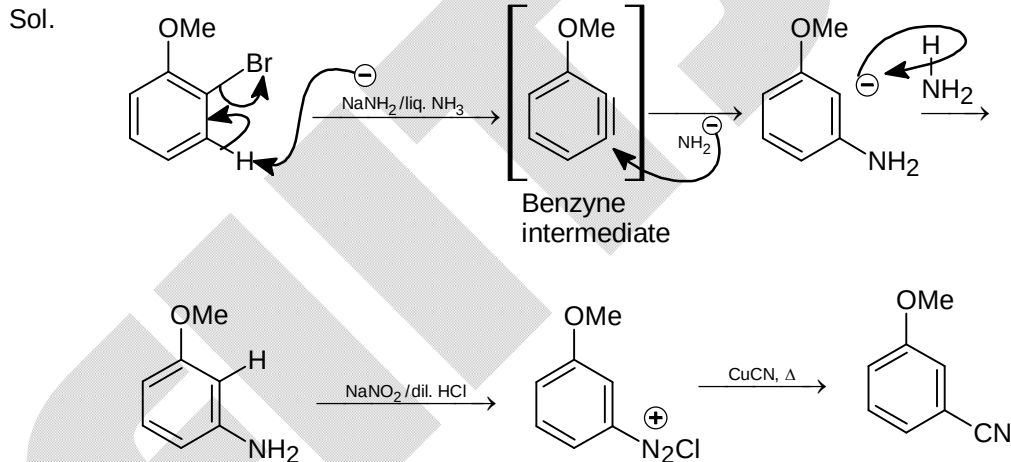
Sol. $\frac{x}{m} = K \cdot P^{\frac{1}{n}} \Rightarrow \log \frac{x}{m} = \log K + \frac{1}{n} \cdot \log P$

From question, $\log K = 0.3010 = \log 2 \Rightarrow K = 2$

And $\frac{1}{n} = \tan 45^\circ = 1 \Rightarrow n = 1$

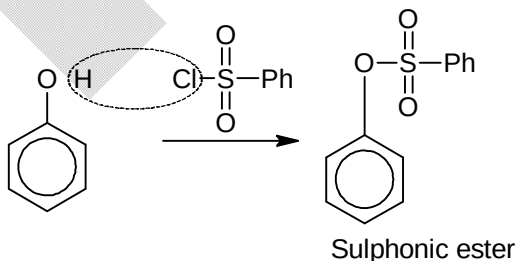
$\therefore \frac{x}{m} = 2 \times P = 2 \times 0.2 = 0.4$

49. A



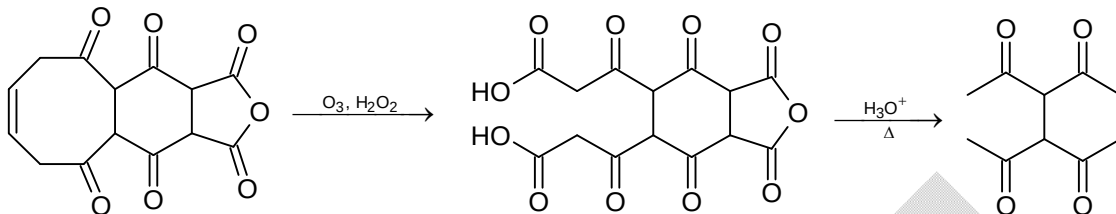
50. C

Sol. Benzene sulphonyl chloride ($\text{Ph} - \text{SO}_2\text{Cl}$) is known as Hinsberg reagent.



SECTION – B

51. 4
Sol.



52. 289

Sol.

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$T_1 = 300 \text{ K} \quad K_1 = 1.6 \times 10^{-3},$$

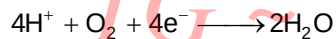
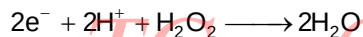
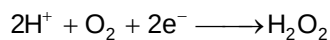
$$T_2 = 350 \text{ K}$$

$$t_{90\%} = \frac{2.303}{K_2} \log \frac{100}{10}$$

$$= 289 \text{ s.}$$

53. 123

Sol.



$$\Delta G_1^0 + \Delta G_2^0 = \Delta G_3^0$$

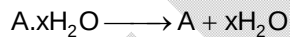
$$(-2 \times 0.70) + (-2 \times 1.76) = -4 \times x$$

$$E_3^0 = 1.23$$

$$100x = 100 \times 1.23 = 123$$

54. 5

Sol.



$$\frac{25}{250} = \frac{16}{M}$$

$$\frac{25}{250} = \frac{16}{M}$$

$$M = 160$$

$$x = 5.$$

55. 273

Sol.

$$\Delta S_{\text{fusion}} = \frac{\Delta H_{\text{fusion}}}{T} = \frac{80 \times 18 \times x}{273}$$

$$x = 5.46 \text{ mole}$$

$$50x = 273$$

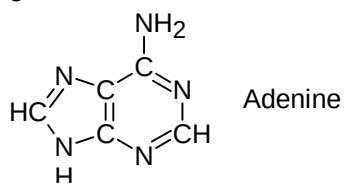
56. 7

Sol.

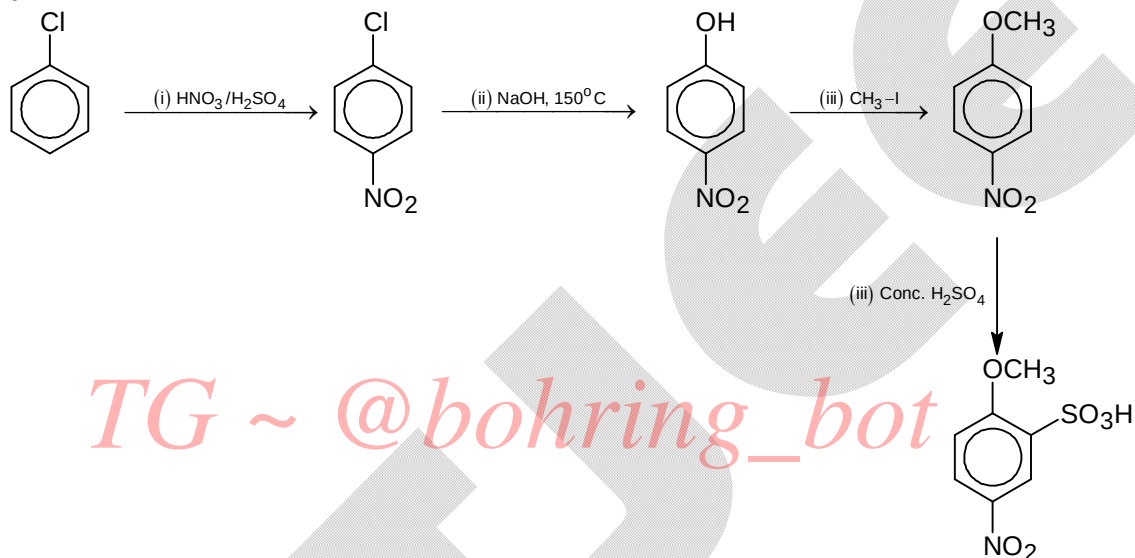
$$\frac{V_1}{V_2} = \frac{C_1}{C_2} = \frac{\pi_1 / T_1}{\pi_2 / T_2} = \frac{500 / 283}{94 / 373} = 7$$

57. 3
Sol. Malachite, Sphalerite and Calamine are the principal ores of copper and zinc.

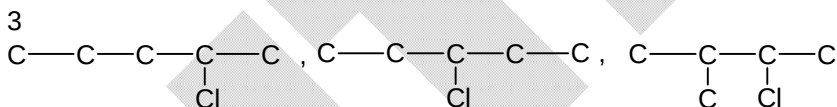
58. 5
Sol.



59. 6
Sol.



60. 3
Sol.



Mathematics

PART – C

SECTION – A

61. A

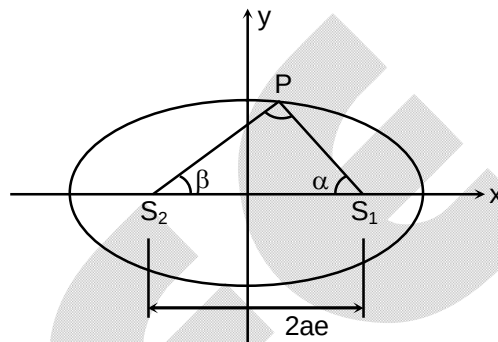
Sol.
$$\frac{2ae}{\sin(\alpha + \beta)} = \frac{S_1P}{\sin\beta} = \frac{S_2P}{\sin\alpha} = \frac{2a}{\sin\alpha + \sin\beta}$$

$$e = \frac{\sin(\alpha + \beta)}{\sin\alpha + \sin\beta}$$

$$\frac{1-e}{1+e} = \tan\left(\frac{\alpha}{2}\right)\tan\left(\frac{\beta}{2}\right) = \frac{1-\frac{1}{2}}{1+\frac{1}{2}} = \frac{1}{3}$$

$$\begin{aligned} \text{Also } \cot\left(\frac{\alpha}{2}\right) + \cot\left(\frac{\beta}{2}\right) + \cot\left(\frac{\gamma}{2}\right) \\ = \cot\left(\frac{\alpha}{2}\right)\cot\left(\frac{\beta}{2}\right)\cot\left(\frac{\gamma}{2}\right) = 3\cot\left(\frac{\gamma}{2}\right) \end{aligned}$$

$$\cot\left(\frac{\alpha}{2}\right) + \cot\left(\frac{\beta}{2}\right) = 2\cot\left(\frac{\gamma}{2}\right)$$



62. A

Sol.
$$d_n = \left| \frac{n - \sqrt{n^2 + 1}}{\sqrt{2}} \right|$$

$$\lim_{n \rightarrow \infty} (n \cdot d_n) = \frac{n}{\sqrt{2}(n + \sqrt{n^2 + 1})} = \frac{1}{2\sqrt{2}}$$

63. C

Sol.
$$\frac{dy}{y} + \frac{dx}{x} = 0 \Rightarrow xy = c$$

$$c = 16$$

$$xy = 16$$

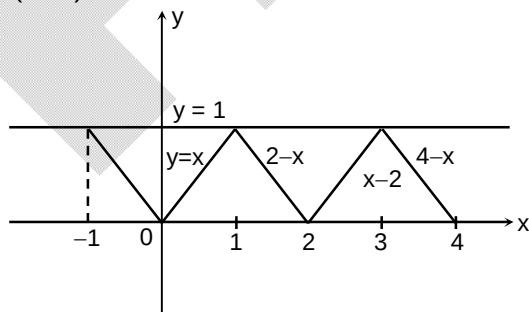
$$\text{L.R.} = 8\sqrt{2}$$

64. C

Sol.
$$f(x) + f(-x) = 8$$

65. B

Sol.
$$f(3.14) = 4 - 3.14 = 0.86$$



66. B

Sol. $(x^2 + 1)e^x = y \Rightarrow (x + 1)^2 \text{exdx} = dy$

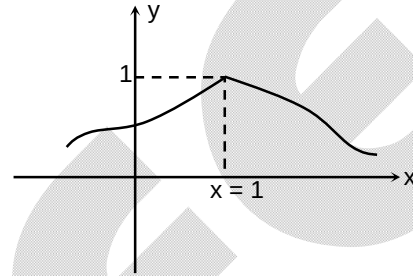
67. A

Sol. $x_i = 1 \Rightarrow \sum_{i=1}^6 x_i = 6$

$y_i = -1 \Rightarrow \sum_{i=1}^6 y_i = -6$

68. D

Sol. Req. area = $2 \int_1^{\infty} \frac{dx}{(x-1)^2 + 1} = \pi \text{ sq. units}$



69. A

Sol. $\lim_{x \rightarrow -1} f(x) = \pi$

70. B

Sol. $x^2 - k \neq 0$
 $k < 0$
 $x^2 + x + 1 \geq 0$

71. B

Sol. Number of solutions 2

72. A

Sol. $A^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$
 $\det(A^n - I) = (2^{n-1} - 1)^2 - (2^{n-1})^2 = 1 - 2^n$

73. B

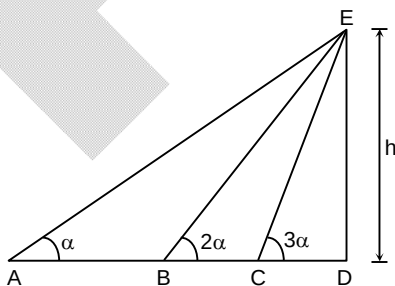
Sol. Trace $(A^3) = 123$

74. A

Sol. Reflexive, transitive but not symmetric

75. B

Sol.



76. A

Sol. $\sum (x_i - \bar{x})^2 = 250$

$$\sigma^2 = \frac{250}{10} = 25$$

Co-efficient of variation is $= \frac{\sigma}{\bar{x}} \times 100 = \frac{5}{50} \times 100 = 10\%$

77. C

78. C

Sol. $y = (\tan^{-1}(5x) - \tan^{-1}(x)) + (\tan^{-1}x + \tan^{-1}(2/3))$
 $\frac{dy}{dx} = \frac{5}{1+25x^2}$

79. C

Sol. $I = \int_{-\ln 2}^0 \sin t dt = \cos(\ln 2) - 1$

80. C

Sol. $y = mx$
 $\left| \frac{3m-3}{\sqrt{m^2+1}} \right| = \sqrt{6}$
 $m = 3 \pm 2\sqrt{2}$

IG ~ @bohring_bot

SECTION - B

81. 44

Sol. $f(\theta) = \frac{(\sin \theta)^x}{(\sin \theta)^x + (\cos \theta)^x}$
 $\sum_{\theta=1^\circ}^{89^\circ} f(\theta) = \frac{89}{2}$

82. 35

Sol. $\vec{v}_1 \cdot \vec{v}_2 = 3 \Rightarrow \cos^2 \theta = \frac{9}{5 + \sin 2\alpha} \leq 1$

$$\alpha = \frac{\pi}{4}$$

$$\theta = 0$$

$\Rightarrow \vec{v}_1$ and $\vec{v}_2$ are collinear

$$\therefore \frac{2(\sin \alpha + \cos \alpha)}{\sin \beta} = \frac{1}{\cos \beta}$$

$$\tan \beta = 2\sqrt{2}$$

$$3\tan^2 \alpha + 4\tan^2 \beta = 3 + 4(8) = 35$$

83. 23

Sol. $(\lambda + 2, -(3\lambda + 2), 2\lambda + 5)$ lies on $2x - 3y + 4z = 16 \Rightarrow \lambda = 7$
 $P = (9, -23, 19)$
 For Q : $x = 0$
 $\lambda + 2 = 0$
 $\therefore Q = (0, 4, 1)$
 $PQ = 9\sqrt{14}$

84. 2

 Sol. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$0.8 = 0.5 + p - 0 \Rightarrow p = 0.3 = \frac{3}{10}$$

$$P(A \cap B) = P(A) \cdot P(B)$$

$$0.8 = 0.5 + 9 - 0.59$$

$$0.3 = 0.5q \Rightarrow q = \frac{3}{5}$$

$$\frac{q}{p} = 2$$

85. 10

$$\text{Sol. } 3x + 4y = \frac{3x}{2} + \frac{3x}{2} + \frac{4y}{3} + \frac{4y}{3} + \frac{4y}{3}$$

$$AM \geq GM$$

$$\Rightarrow \frac{3x + 4y}{5} \geq 2$$

86. 4

 Sol. $|PA - PB| \leq AB$
 $|PA - PB|$ to be maximum, P must be the intersection of line joining A and B and $2x + 3y = 5$

$$\therefore h = \frac{2}{5}$$

$$k = \frac{7}{5}$$

$$3h + 2k = \frac{6 + 14}{5} = 4.$$

87. 24

$$\text{Sol. } \frac{2\left(\frac{x}{2}\right) + 3\left(\frac{y}{3}\right) + 5\left(\frac{z}{5}\right)}{10} \geq \left(\frac{x^2}{2^2} \cdot \frac{y^3}{3^3} \cdot \frac{z^5}{5^5}\right)^{\frac{1}{10}}$$

$$\Rightarrow x^2 y^3 z^5 \leq 2^2 \cdot 3^3 \cdot 5^5$$

$$N = 2^2 \cdot 3^3 \cdot 5^5$$

$$\text{Number of odd divisors} = (3 + 1)(5 + 1) = 24$$

88. 0

$$\begin{aligned} \text{Sol. } & 15^{23} + 23^{23} \\ &= (19 - 4)^{23} + (19 + 4)^{23} \\ &= 2[(19)^{23} + {}^{23}C_2(19)^{21}(4)^2 + \dots + {}^{23}C_{22}19(4)^{22}] \end{aligned}$$

89. 49

$$\begin{aligned} \text{Sol. } & (x + 4)^2 + (y - 5)^2 = 1 \\ & (-4 + \cos\theta, 5 + \sin\theta) \\ & (x + 2)^2 + (y - 3)^2 \\ &= (\cos\theta - 2)^2 + (\sin\theta + 2)^2 \\ &= 9 + 4(\sin\theta - \cos\theta) \\ & a = 9 + 4\sqrt{2} \\ & b = 9 - 4\sqrt{2} \end{aligned}$$

90. 1

$$\text{Sol. } x = y = z = -1$$

FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2023

TEST DATE: 20-01-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A

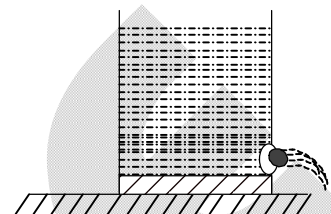
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. Liquid of density ρ is filled in the container of negligible mass having area A . Area of hole 'a' is very small ($a \ll A$). Co-efficient of friction between container and surface is equal to $\left(\frac{a}{A}\right)$. As

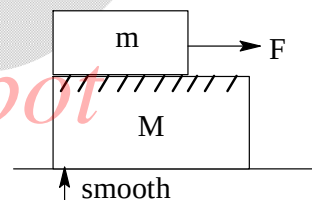
liquid is coming out of vessel, instantaneous acceleration of the container.

- (A) $\left(\frac{A}{a}g\right)$ (B) $\frac{3g}{2}$
(C) $\left(\frac{ag}{A}\right)$ (D) zero



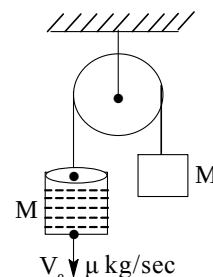
2. A plank having mass M is placed on smooth horizontal surface. Block of mass m is placed on it coefficient of friction between block and plank is $\mu_0 + kx$, where k is constant and x is relative displacement between block and plank. A force F is applied on block where $F = at$, where $a = 10$; t is in second. Find t_0 when relative motion will occur between block and plank

- (A) $\mu_0 M + \frac{\mu_0 M^2}{m}$ (B) $\mu_0 m + \frac{\mu_0 M^2}{m}$
(C) $\mu_0 m + \frac{\mu_0 m^2}{M}$ (D) $\mu_0 M + \frac{\mu_0 m^2}{M}$



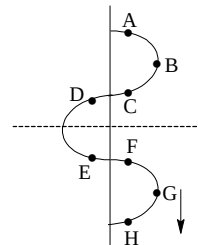
3. Velocity of water flow in downward direction relative to cylindrical tank is V_e acceleration at any time t will be

- (A) $\frac{\mu(V_e + gt)}{(2M - \mu t)}$ (B) $\frac{\mu V_e}{(2M - \mu t)}$
(C) $\frac{\mu gt}{(2M - \mu t)}$ (D) None of these

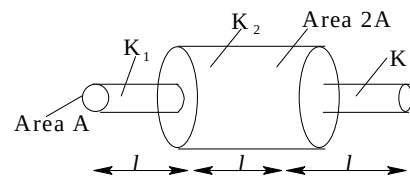


4. A transverse wave travelling along negative y-axis its snap shot is given at any instant of time t . Points which is having velocity in negative x-direction.

- (A) A,E,F (B) C,D,H
(C) B,G,H (D) A,H

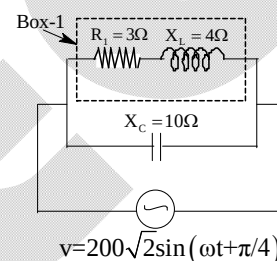


5. Equivalent thermal conductivity of figure given below will be? Given that length of each cylinder is l and area of cylinder having thermal conductivity K_1 & K_3 is A while middle cylinder is having K_2 .



- (A) $\frac{5}{2 \left[\frac{1}{K_1} + \frac{1}{2K_2} + \frac{1}{K_3} \right]}$
- (B) $\frac{1}{\left(\frac{1}{K_1} + \frac{1}{2K_2} + \frac{1}{K_3} \right)}$
- (C) $\frac{2}{5 \left[\frac{1}{K_1} + \frac{1}{2K_2} + \frac{1}{K_3} \right]}$
- (D) None of these

6. As shown in figure, Instantaneous current in branch having capacitor C will be : (given $\tan^{-1}(4/3) = 53^\circ$)

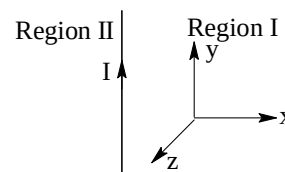


- (A) $20\sqrt{2} \sin(\omega t + 3\pi/4)$
- (B) $40\sqrt{2} \sin(\omega t + \pi/4)$
- (C) $60\sqrt{2} \sin(\omega t - \pi/4)$
- (D) None of above

7. A thermodynamic process obeys the following relation $2dQ = dU + 2dW$ where dQ , dU , dW has usual meaning. Then heat capacity for the process is: [Given di-atomic gas; R = gas constant]

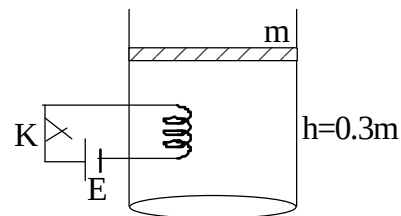
- (A) $\frac{5R}{2}$
- (B) $\frac{7R}{2}$
- (C) $\frac{3R}{5}$
- (D) infinite

8. A infinite length current carrying wire is placed in x, y plane parallel to y axis as shown in figure. (All charges are projected in region I)

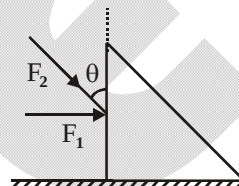


- (A) A positively charged particle projected along y axis will get deflected along z axis
- (B) A positively charged particle projected along x axis will get deflected along y axis
- (C) A negatively charged particle is projected along x axis will get deflected along $+y$ axis
- (D) A negatively charge particle is projected along x axis will get deflected along $+z$ axis

9. An insulated cylinder contains nitrogen gas which is sealed on the top by a heavy metal piston of mass m . Piston is free to move with negligible friction. Initial height of the piston is 0.3 m at temperature 27°C when it is given some heat with the help of an electrical circuit piston slowly rises to height 0.5 m above the bottom of the cylinder. During expansion of the gas, net heat absorbed by the nitrogen gas:
- (A) Equal to work done by nitrogen gas on the piston
 (B) Greater to work done by nitrogen gas on the piston
 (C) Less to work done by nitrogen gas on the piston
 (D) Will be always twice of the work done by the gas on the piston



10. A wedge of mass m , lying on a rough horizontal plane, is acted upon by a horizontal force F_1 and another force F_2 , inclined at an angle θ to the vertical. The block is in equilibrium, then minimum coefficient of friction between it and the surface is



- (A) $(F_2 \sin \theta) / (mg + F_2 \cos \theta)$
 (B) $(F_1 \cos \theta + F_2) / (mg - F_2 \sin \theta)$
 (C) $(F_1 + F_2 \sin \theta) / (mg + F_2 \cos \theta)$
 (D) $(F_1 \sin \theta - F_2) / (mg - F_2 \cos \theta)$
11. A charge particle is projected in a region of magnetic field and electric field. If mass of charged particle is m and its speed changes from V_0 to $2V_0$ [W_E = work done by electric field; W_B = work done by magnetic field]. Which statement is incorrect?

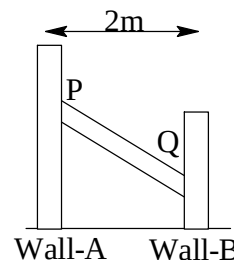
- (A) $W_B = \frac{1}{2} m v_0^2$
 (B) $W_B + W_E = \frac{3}{2} m v_0^2$
 (C) $W_E = \frac{3}{2} m v_0^2$
 (D) $W_E - W_B = \frac{3}{2} m v_0^2$

12. De-Broglie wavelength of the particle increases by 75% then kinetic energy of particle becomes

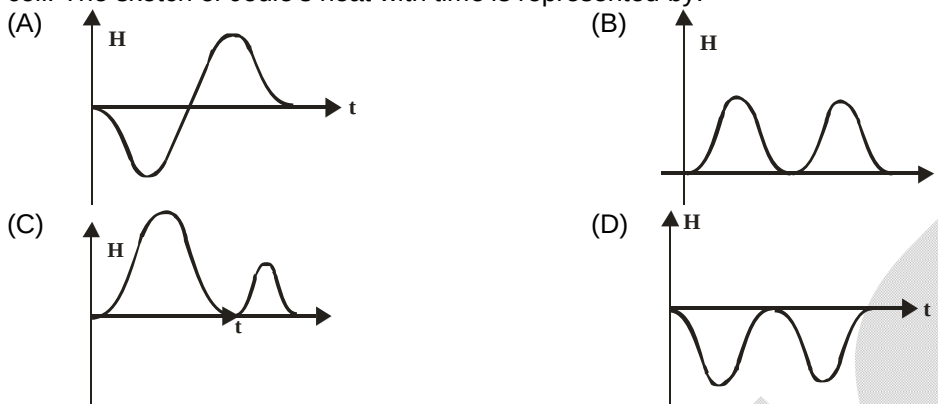
- (A) $\frac{16}{49}$ times
 (B) $\frac{9}{16}$ times
 (C) $\frac{16}{25}$ times
 (D) $\frac{4}{9}$ times

13. Two vertical walls are separated by a distance of 2 metres. Wall 'A' is smooth while wall B is rough with a coefficient of friction $\mu = 0.5$. A uniform rod is propped between them. The length of the longest rod that can be propped between the walls is equal to :

- (A) 2 metres
 (B) $2\sqrt{2}$ metres
 (C) $\sqrt{5}$ metres
 (D) $\frac{\sqrt{17}}{2}$ metres



14. A bar magnet is pulled along axis of a coil with uniform velocity with South Pole entering into the coil. The sketch of Joule's heat with time is represented by:



15. A parent radioactive nucleus A (decay constant λ_a) converts into a radio-active nucleus B of decay constant λ_b , initially, number of atoms of B is zero. At any time N_a , N_b are number of atoms of nuclei A and B respectively then maximum value of N_b .

(A) $\frac{\lambda_a N_a}{\lambda_b}$ (B) $\frac{\lambda_b N_a}{\lambda_a}$
 (C) $\frac{(\lambda_a + \lambda_b) N_a}{\lambda_b}$ (D) $\frac{\lambda_b}{(\lambda_a + \lambda_b)} N_a$

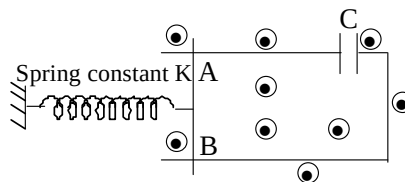
16. Fundamental note of a closed pipe of length l is f_0 at 0°C . First overtone of the same pipe at 273°C is (assume there is no effect of temperature on the dimension of the pipe).

(A) $3f_0$ (B) f_0
 (C) $(3\sqrt{2})f_0$ (D) $3/\sqrt{2} f_0$

17. A satellite is revolving round the earth at a height $(R/2)$ from surface of earth in circular orbit. The change in speed required for the satellite to escape to infinity for the satellite to escape to infinity is: [Take R as radius of earth; and M as mass of earth]

(A) $\sqrt{\frac{2GM}{3R}}$ (B) $\sqrt{\frac{GM}{3R}}(2 - \sqrt{2})$
 (C) $\sqrt{\frac{GM}{3R}}(\sqrt{2} - 1)$ (D) $\sqrt{\frac{4GM}{3R}}$

18. In the arrangement shown in the figure, uniform magnetic field B is upward to the plane of paper. Connector AB is smooth and conducting, having mass m and length l . Initially spring has extension X_0 . Spring is non-conducting (neglect induction in spring). Connector is released at $t = 0$. Maximum charge on the capacitor is:



(A) $X_0 \sqrt{\frac{KC}{1 + \frac{B^2 l^2 C}{m}}}$ (B) $X_0 \sqrt{\frac{KC}{1 + \frac{m}{B^2 l^2 C}}}$
 (C) $\sqrt{\frac{KC}{2 + \frac{B^2 l^2 C}{m}}}$ (D) $\sqrt{\frac{4KC}{1 + \frac{B^2 l^2 C}{2m}}}$

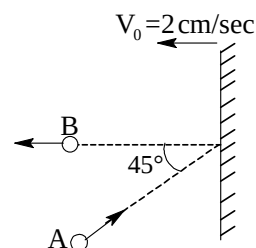
19. Mirror is moving towards the particle with speed 2 cm/sec. Speed of A and B are $10\sqrt{2}$ and 5 cm/sec respectively in the direction shown in figure. Magnitude of velocity of image of the particle B with respect to image of A.

(A) $\sqrt{325}$ cm/sec

(B) 15 cm/sec

(C) 13 cm/sec

(D) $\sqrt{269}$ cm/sec



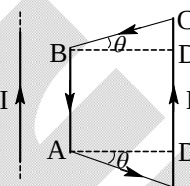
20. Loop ABCD and long straight wire are in the same plane. Side CD = 2AB. Distance between long straight wire and AB is equal to distance BD. Net force on the loop is:

(A) Towards the long straight wire

(B) Away from the long straight wire

(C) Parallel to the long straight wire

(D) Net force on the loop will be zero



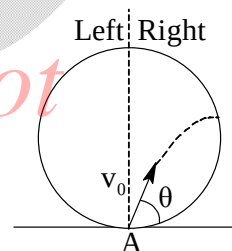
SECTION - B

(Numerical Answer Type)

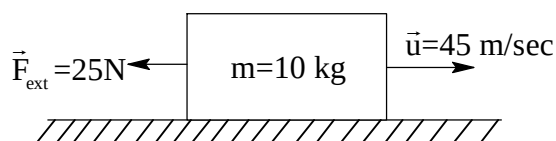
This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A hollow sphere of mass m and radius R is placed on smooth ground. A particle of mass m is projected with velocity v_0 and angle θ from lowest point A inside the sphere as shown in diagram. If particle strikes the sphere at a point which is on horizontal level of centre and at that moment particle is at highest point. The collision between particle and

sphere is elastic. The value of v_0 is $\frac{\sqrt{kRg}}{\sin\theta}$. Find the value of 'k'

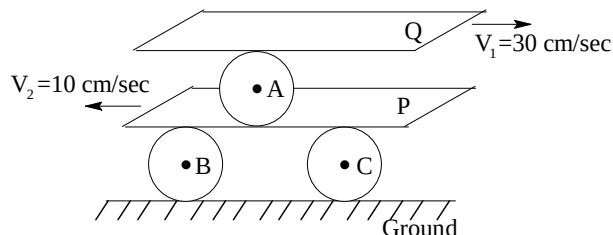


22. Initial velocity of the block $\vec{u} = 45\hat{i}$ while external force on it is $\vec{F} = 25(-\hat{i})$. Its coefficient of static and kinetic friction are 0.3 and 0.2 respectively then distance traveled by the block in 12 second is $25x$ m, then find the value of 'x' ($g = 10 \text{ m/s}^2$):

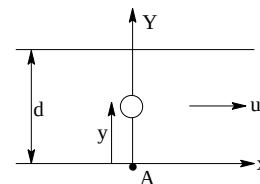


23. Calculate the time (in hrs) in which a layer of ice of thickness 10 cm will increase by 5 cm on the surface of a pond when temperature of the surrounding is -10°C . Coefficient of thermal conductivity of ice $K = 0.005 \left(\frac{\text{Cal}}{\text{cm} \cdot \text{sec} \cdot ^\circ\text{C}} \right)$ and density $\rho = 0.9 \text{ gram/cm}^3$. ($L = 80 \text{ cal/g}$)

24. All three spheres are identical having radius 10 cm. There is no slipping at any point of contact. Plank P and Q are moving in opposite direction with speed 10 and 30 cm/sec. Find the height of instantaneous axis of rotation of sphere A from the ground (in cm):



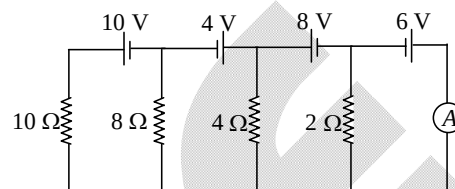
25. A river of width d is flowing with a velocity u . A person starts from point A. He always try to keep himself along y axis. Speed of man w.r.t. to river at any position is given by $v = k\sqrt{y}$ ($k \rightarrow +ve$ constant). Time taken by man to cross the river is $\frac{x\sqrt{d}}{4k}$, then find the value of $\frac{x}{d}$.



(Assume that at $t = 0$, $y = 0$)

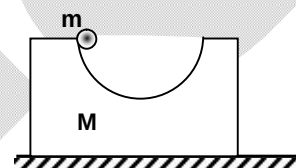
26. If the reading of the ideal ammeter connected in the given circuit is $n\left(\frac{13}{20}\right)$ ampere, find the value of n .

Assume that the cells have negligible internal resistance.



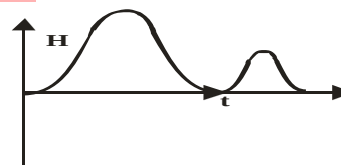
27. A semi-cylinder is cut out of a block and the block is kept on the horizontal surface. A small body of mass m is released at the top of the cylinder shaped hole. (Friction is negligible.)

If the block is able to move on the horizontal surface without friction, then the force exerted on the block by the small body at the lowest point is $7mg/2$. Find the ratio of M/m ?

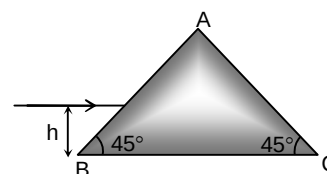


28. At what speed (in m/s) must a pebble be thrown from a height of $h = 3.6m$, and at an angle of 30° , measured from the horizontal, if it is to hit the ground at an angle of 45° . (Air drag can be neglected.) (Take $g = 10 m/s^2$)

29. A sphere is just immersed in a liquid. Find the ratio of hydrostatic force acting on the lower half and the upper half of the sphere. (Neglect atmospheric pressure)



30. A ray of light is incident parallel to BC at a height $h = 3.0$ cm from BC. Find the height (in cm) above BC at which the emergent ray leaves the surface AC. It is given that $\mu = \sqrt{2}$ and length $BC = 20$ cm. Take $\tan 15^\circ = 0.25$.



Chemistry

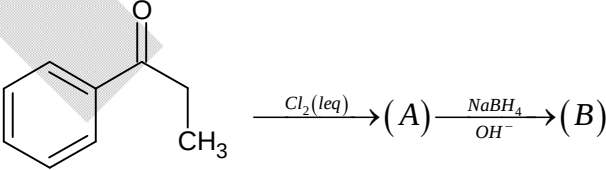
PART – B

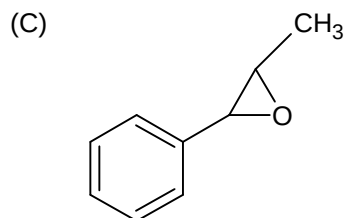
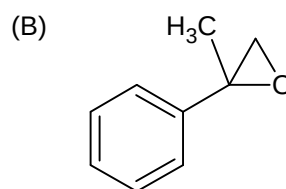
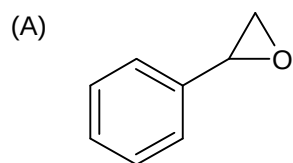
SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

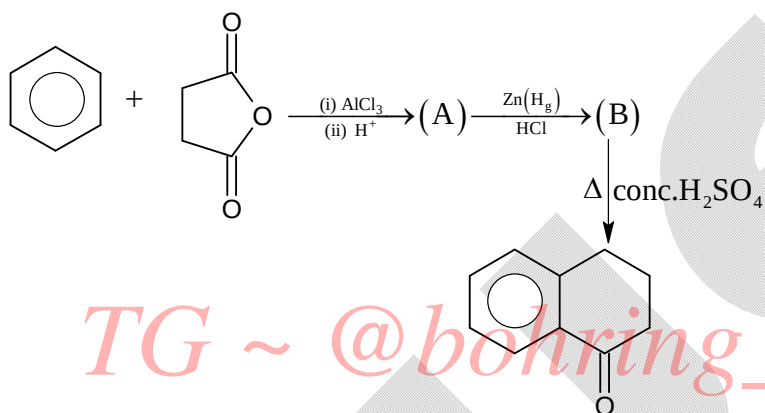
31. In the following statement, which combination of true (T) and false (F) options is correct?
 - (A) Ionic mobility is highest for I^- in water as compared to other halides.
 - (B) IF_5 is square pyramidal and IF_7 is pentagonal bipyramidal in shape
 - (C) Reactivity order is $F_2 < Cl_2 < Br_2 < I_2$
 - (D) Oxidising power order is $F_2 < Cl_2 < Br_2 < I_2$
 - (A) TTTF
 - (B) TTFT
 - (C) TTTT
 - (D) TTFF
32. According to VSEPR model, the shape of $[XeOF_5]^-$ is
 - (A) Octahedral
 - (B) Triagonal bipyramidal
 - (C) Square pyramidal
 - (D) Pentagonal monopyrarnidal
33. Identify the statement(s) which is not correct with respect to surface phenomenon.
 - (A) If on adding electrolyte in an emulsion, the conductivity increases then it will be oil in water type emulsion.
 - (B) Tyndall effect is observed when refractive indices of dispersed phase and dispersion medium differ largely.
 - (C) Macromolecular colloids are generally lyophobic in nature
 - (D) Gases which can react with adsorbents are generally chemisorbed
34. Xenon fluorides are very good oxidising and fluorinating agents. They also act as F^- donors and acceptors. When XeF_4 donates its fluoride to SbF_5 , then the states of hybridisation of central atom of cationic part and anionic part of product are:
 - (A) sp^3d, sp^3d^2
 - (B) sp^3d^2, sp^3d
 - (C) sp^3d^2, sp^3d^2
 - (D) dsp^2, sp^3
35. Correct order of bond angle is
 - (A) $(\bullet CH_3) > (\bullet CF_3)$
 - (B) $CH_3^- > CH_3^+$
 - (C) $CH_4 > CF_4$
 - (D) $CH_4 > (\bullet CH_3)$
36. The end product of the reaction is:





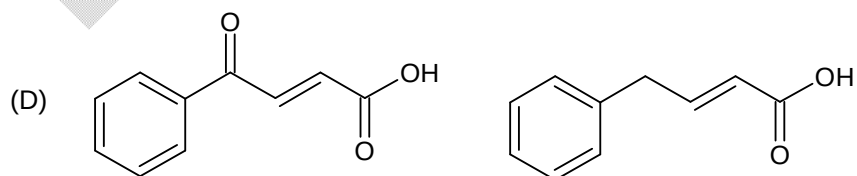
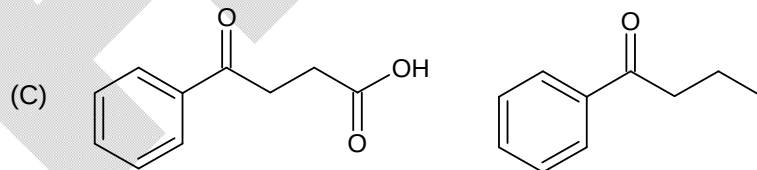
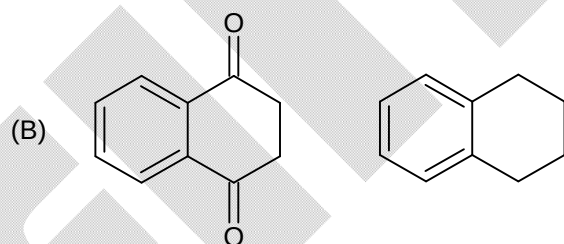
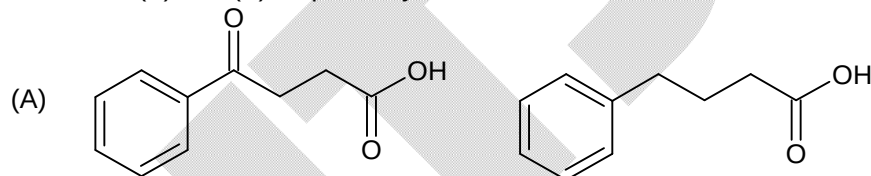
(D) None of these

37.



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Structure of (A) and (B) respectively are:



38. An aqueous solution of a metal ion (A) on reaction with KI gives a black brownish ppt (B) and this aqueous suspension on treatment with excess KI gives orange yellowish solution. Then A is
 (A) Pb^{2+} (B) Bi^{3+}
 (C) Hg_2^{2+} (D) Hg^{2+}
39. Which of the following compounds consist of a P – P linkage?
 (A) Hypo phosphoric acid (B) Pyrophosphorous acid
 (C) Dipolyphosphoric acid (D) Metaphosphoric acid
40. Which of the following elements of lanthanide series have highest tendency to show +2 oxidation state?
 (A) *Eu* (B) *Gd*
 (C) *Tb* (D) None of these
41. The specific conductance of a saturated solution of silver bromide is $K \text{ Scm}^{-1}$. The limiting ionic conductivity of Ag^+ and Br^- ions are x and y respectively. The solubility of silver bromide in g / L is (molar mass of $AgBr = 188$)

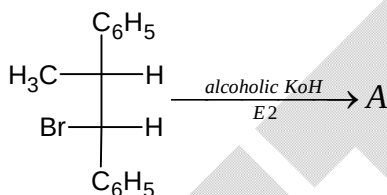
(A) $\frac{K \times 1000}{x - y}$

(B) $\frac{K}{x + y} \times 188$

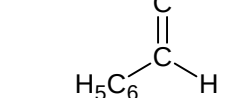
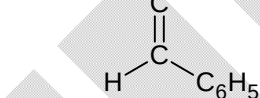
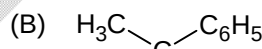
(C) $\frac{K \times 1000 \times 188}{x + y}$

(D) $\frac{x + y}{k} \times \frac{1000}{188}$

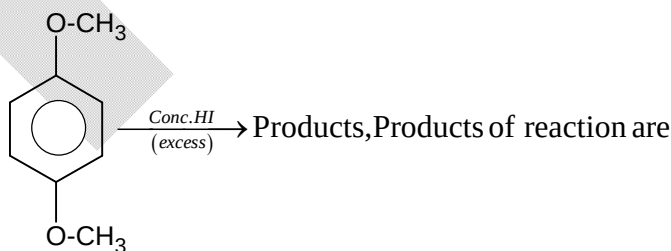
42.

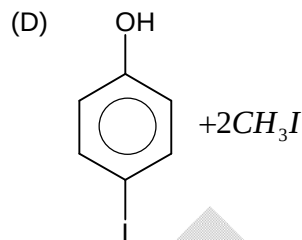
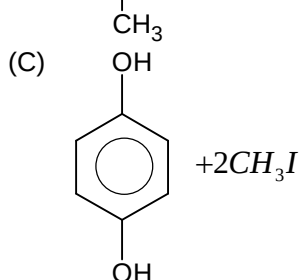
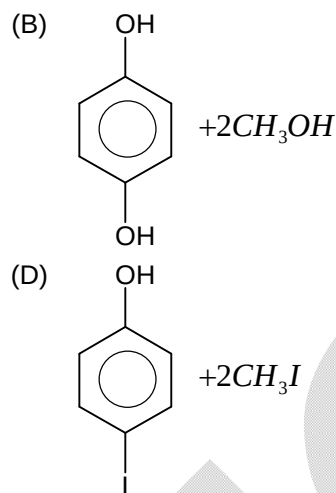
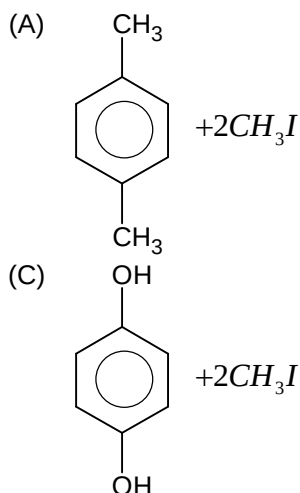


A is



43.





44. The activation energies of two reactions are E_1 & E_2 with $E_1 > E_2$. If temperature of reacting system is increased from T_1 (rate constant are k_1 and k_2) to T_2 (rate constant are k_1^1 and k_2^1) predict which of the following alternative is correct.

(A) $\frac{k_1^1}{k_1} = \frac{k_2^1}{k_2}$

(B) $\frac{k_1^1}{k_1} > \frac{k_2^1}{k_2}$

(C) $\frac{k_1^1}{k_1} < \frac{k_2^1}{k_2}$

(D) $k_1^1 < k_2^1$

45. The wavelength of the first Lyman lines of hydrogen, He^+ and Li^{2+} ions and λ_1, λ_2 & λ_3 . The ration of these wavelengths is

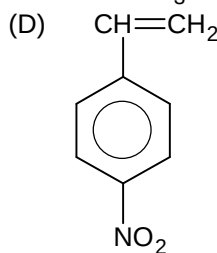
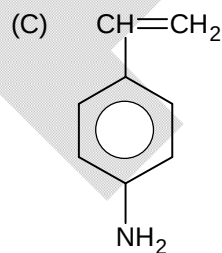
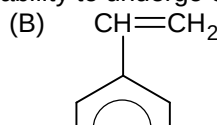
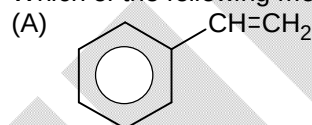
(A) 1 : 4 : 9

(B) 9 : 4 : 1

(C) 36 : 9 : 4

(D) 6 : 3 : 2

46. Which of the following monomers has greatest ability to undergo cationic polymerisation?

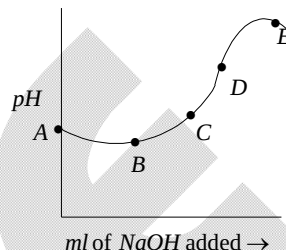


47. Two system $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ and $COCl(g) \rightleftharpoons CO(g) + Cl_2(g)$ are simultaneously in equilibrium in a vessel at constant volume. If some CO is introduced in the vessel, then at new equilibrium, the moles of

(A) PCl_5 increase (B) PCl_5 remain unchanged
(C) PCl_5 decrease (D) Cl_2 increase

48. The curve in the figure shows the variation of pH during the course of titration of weak acid HA with a strong base NaOH. At which point in the titration curve is the concentration of acid equal to that of its conjugate base.

(A) Point D (B) Point E
(C) Point C (D) Point B



49. The minimum mass of NaBr which should be added in 200 ml of 0.0004 M $AgNO_3$ solution just to start precipitation of $AgBr$. K_{sp} of $AgBr = 4 \times 10^{-13}$. (Br = 80)

(A) $1.0 \times 10^{-9} g$ (B) $2 \times 10^{-10} g$
(C) $2.06 \times 10^{-8} g$ (D) $1.03 \times 10^{-7} g$

50. In which of the following cases is the value of x maximum?

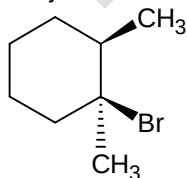
(A) $CaSO_4 \cdot xH_2O$
(B) $BaSO_4 \cdot xH_2O$
(C) $MgSO_4 \cdot xH_2O$
(D) All have same values of x as all metal ions belong to same group

SECTION – B

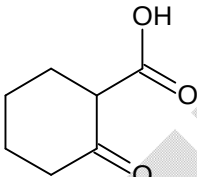
(Numerical Answer Type)

This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

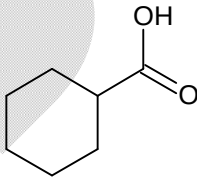
51. The number of incorrect statements of the following statements is:
- H^+ is the smallest size cation in the periodic table
 - Van der Waals radius of chlorine is more than covalent radius
 - Ionic mobility of hydrated Li^+ is greater than that of hydrated Na^+
 - He atom has the highest ionisation enthalpy in the periodic table
52. How many dichloride cyclopentanes (including stereoisomers) are obtained when cyclopentane reacts with excess chlorine at high temperature?
53. The number of alpha hydrogen's present in major product obtained when following substrate is subjected to $E2$ reaction will be:

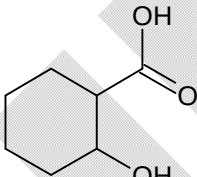


54. The number of paramagnetic complexes from given compounds:
- $(Ni(CN)_4)^{2-}$
 - $(NiCl_4)^{2-}$
 - $(CoCl_4)^{2-}$
 - $(CoF_6)^{2-}$
55. A certain amount of reducing agent reduces x mole of $KMnO_4$ & y mole of $K_2Cr_2O_7$ in different experiments in acidic medium. If the change in oxidation state of reducing agent is same in both experiments then $(x - y)$ is ____.
56. Number of electrons having $m_l = 0$ for sodium atom is ____.
57. The number of non polar molecules from the given molecules?
- XeF_4
 - BF_3
 - C_2F_4
 - PCl_2F_3
58. The number of following compounds which loses CO_2 on heating to $100^\circ C$
- (A)

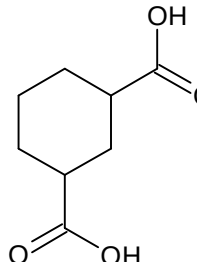
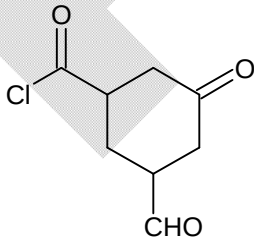


(B)


- (C)

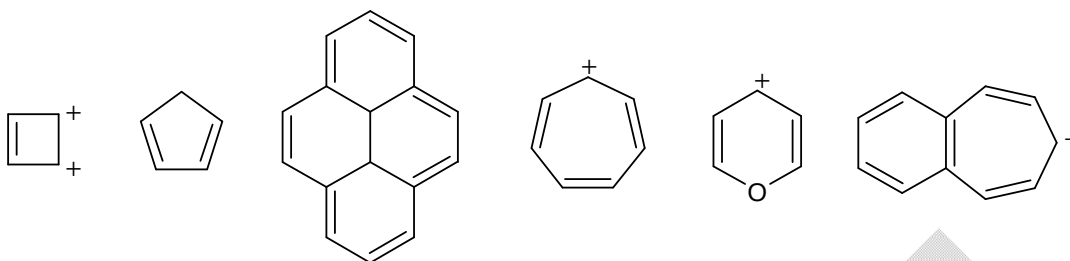


(D)


59. How many molecules of $RMgX$ are consumed in the below reaction
- 

$\xrightarrow[H_3O^+]{ RMgX }$

60. Amongst the following, total no of aromatic compounds are



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Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. If $a\alpha^2 + b\alpha + c = 3\alpha^2 - 4\alpha + 1$, $a\beta^2 + b\beta + c = 3\beta^2 - 4\beta + 1$ and $a\gamma^2 + b\gamma + c = 3\gamma^2 - 4\gamma + 1$ (where $\alpha, \beta, \gamma \in R$ and distinct) these sum of root of the equation $3ax^2 + 9bx + 7c = 0$.
 (A) 1 (B) 2
 (C) 3 (D) 4
62. If k denotes the sum of the imaginary parts of the roots of the equation $z^2 - 8(1-i)z + 63 - 16i = 0$ then k is
 (A) 6 (B) -6
 (C) 8 (D) -8
63. The sum $\sum_{k=0}^{49} (-1)^k {}^{99}C_{2k}$ equals
 (A) -2^{98} (B) 2^{98}
 (C) -2^{49} (D) 2^{49}
64. For differential equation $(x^4 y^2 - y)dx + (x^2 y^4 - x)dy = 0$
 (A) $(x^3 - y^3) + \frac{3}{xy} = c$ (B) $x^3 - y^3 + \frac{1}{xy} = c$
 (C) $x^3 + y^3 + \frac{3}{xy} = c$ (D) $x^3 + y^3 + \frac{1}{xy} = c$
65. Sum of roots of the equation $2^{333x-2} + 2^{111x+1} = 2^{222x+2} + 1$ is.
 (A) $\frac{111}{2}$ (B) $\frac{2}{111}$
 (C) 2 (D) 111
66. $\int \frac{e^{\tan^{-1} x}}{(1+x^2)} \left(\left(\sec^{-1} \sqrt{1+x} \right)^2 + \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \right) dx$ is ($x > 0$)
 (A) $e^{\tan^{-1} x} \cdot \tan^{-1} x + c$ (B) $e^{\tan^{-1} x} \cdot \frac{(\tan^{-1} x)^2}{2} + c$
 (C) $e^{\tan^{-1} x} \left(\sec^{-1} \sqrt{1+x^2} \right)^2 + c$ (D) $e^{\tan^{-1} x} \left(\cos^{-1} \sqrt{1+x^2} \right)^2 + c$

67. If $(x - 2y - 4)^2 = 24x + 12y$ then length of latus rectum is
- (A) $\frac{5}{\sqrt{12}}$ (B) $\frac{\sqrt{5}}{12}$
- (C) $\frac{12}{\sqrt{5}}$ (D) None of these
68. IF $P(a \sec \theta, b \tan \theta), Q(a \sec \phi, b \tan \phi)$ are the ends of a focal chord of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then $\tan \frac{\theta}{2} \tan \frac{\phi}{2}$ can be
- (A) $\frac{e-1}{e+1}$ (B) $\frac{e+1}{e-1}$
- (C) $\frac{1+e}{1-e}$ (D) None of these
69. Consider the function $f(x) = \frac{\sqrt{1+\cos x} + \sqrt{1-\cos x}}{\sqrt{1+\cos x} - \sqrt{1-\cos x}} (x \in (x, 2\pi))$ then $f(x)$ is
- (A) $\cot\left(\frac{\pi}{4} + \frac{x}{2}\right)$ (B) $\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$
- (C) $\cot\left(\frac{\pi}{4} - \frac{x}{2}\right)$ (D) None of these
70. In $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ the equation $\log_{\sin \theta} \cos 2\theta = 2$ has
- (A) No solution (B) One solution
- (C) 2 solutions (D) None of these
71. Let $f(x) = \cos x, g(x) = \begin{cases} \min\{f(t) : 0 \leq t \leq x\} & x \in [0, \pi] \\ \sin x - 1 & x > \pi \end{cases}$ then
- (A) $g(x)$ is discontinuous at $x = \pi$ (B) $g(x)$ is continuous for $x \in [0, \infty)$
- (C) $g(x)$ is differentiable at $x = \pi$ (D) $g(x)$ differentiable for $x \geq 0$
72. If $f(x)$ and $g(x)$ are two functions such that $f(x) = [x] + [-x]$ and $g(x) = \{x\} \forall x \in R$ and $h(x) = f(g(x))$ these incorrect statement is $[.]$ is G.I.F and $\{.\}$ is F Part of x
- (A) $f(x)$ and $g(x)$ are identical function (B) $f(x) = g(x)$ has no solution
- (C) $f(x) + h(x) > 0$ has no solution (D) $f(x) - h(x)$ is periodic function
73. Let $f(x)$ be a polynomial $f : R \rightarrow R$ such that $f(2x) = f'(2x) f''(x)$ these $f(x)$ is
- (A) One - One onto (B) One - One into
- (C) Many One - Onto (D) Many - One into

74. Consider three planes $2x + py + 6z = 8$, $x + 2y + qz = 5$ and $x + y + 3z = 4$ then 3 planes intersect at a point if
 (A) $p = 2, q \neq 3$ (B) $p \neq 2, q \neq 3$
 (C) $p \neq 2, q = 3$ (D) None of these
75. If a^2, b^2, c^2 are in AP then $\cot A, \cot B$ and $\cot C$ in
 (A) GP (B) AP
 (C) HP (D) None of these
76. R is relation over the set of integers and it is given by $(x, y) \in R \Leftrightarrow |x - y| \leq 1$. Then R is
 (A) Reflective and transitive (B) Reflective and symmetric
 (C) Symmetric and transitive (D) An equivalence relation
77. If $A = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$ and a, b, c are roots of the equation $x^3 + x^2 - 4 = 0$ then $A A^T$ is
 (A) I (B) $I + A$
 (C) A^2 (D) $A - I$
78. If $a > b > c$ and the system of equations $ax + by + cz = 0$, $bx + cy + az = 0$, $cx + ay + bz = 0$ has a non-trivial solution these both the roots of equation $at^2 + bt + c = 0$ are
 (A) Cereal (B) Negative
 (C) Positive (D) None of these
79. If A and B are non-zero square metrics of order 3 there incorrect statement is
 (A) $\text{adj}(AB) = \text{adj}A \cdot \text{adj}B$ (B) $(AB)^{-1} = B^{-1} \cdot A^{-1}$
 (C) $(AB)^T = B^T A^T$ (D) $AB = 0 \Rightarrow |A| = 0$ and $|B| = 0$
80. The only statement among the following that is tautology is
 (A) $A \cap (A \cup B)$ (B) $A \cup (A \cap B)$
 (C) $[A \cap (A \rightarrow B)] \rightarrow B$ (D) $B \rightarrow [A \cap (A \rightarrow B)]$

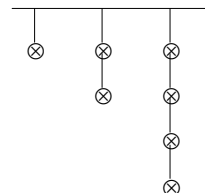
SECTION - B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. $\int_0^{\frac{\pi}{2}} \frac{1 - \sin 2x}{(1 + \sin 2x)^2} dx = \frac{a}{b}$ (a & b are coprime) these $a + b + ab$ is ____.

82. 7 clay balls (shown as $\otimes$). If N be the number of different ways these can be shot (one at a time). If no ball can be shot until the ball (s) below it, have been shot, then N is ____.



83. A regular pyramid on a square base has an edge 150m long and the length of the side of its base is 200m. Then height of pyramid is ____
84. 20 teachers of a school either teach maths or physics 12 of them teach maths while 4 teach both the subjects then the no. of teachers teaching only physics is ____

85. If $a_r = (\cos 2r\pi + \sin 2r\pi)^{1/9}$ then $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is ____.

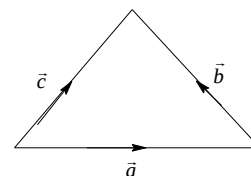
86. $[\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + [\sqrt{4}] + [\sqrt{5}] + \dots + [\sqrt{224}] - 1112$, is ([.] is G.I.F)

87. $2 \tan^{-1} x + \sin^{-1} \frac{2x}{1+x^2}$ ($x > 1$) is equal to $k\pi$ then find k is ____.

88. $\int_{-1}^3 \left(\tan^{-1} \left(\frac{x}{1+x^2} \right) + \tan^{-1} \left(\frac{x^2+1}{x} \right) \right) dx$ is equal to $k\pi$ then find k is ____.

89. Number of integral values of k for which the equation $4 \cos^{-1}(-|x|) = k$ has exactly two solutions, is ____.

90. If $\vec{a} = \lambda x\hat{i} + y\hat{j} + 4z\hat{k}$, $\vec{b} = y\hat{i} + x\hat{j} + (3y)\hat{k}$, $\vec{c} = -z\hat{j} - 2z\hat{k} - (\lambda + 1)x\hat{k}$ are sides of triangle then the value of α is λ is (where x, y, z are not all zero).



ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2023

TEST DATE: 20-01-2023

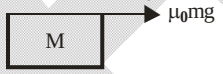
ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C
Sol. At any time
 $F_{\text{thrust}} - F_{\text{friction}} = ma$
 $\rho a v^2 - \mu m g = ma$
 $\rho a (2gh) - \left(\frac{a}{A}\right)(\rho A h)g = (\rho A h)a'$
 $2\rho a g h - \rho a g h = \rho A h a' ; \frac{a}{A}g = a'$

2. C
Sol. Let at time t_0 relative motion will occur
- 
- $\mu_0 m g = Ma \dots (1)$
- 
- $10 t_0 - \mu_0 m g = ma \dots (2)$
From (1) and (2) ;
 $t_0 = \mu_0 m + \frac{\mu_0 m^2}{M}$

3. A

Sol. $Mg - T = M \frac{dv}{dt} \dots (1)$

$$T + \mu v_e - (M_0 - \mu t)g = (M_0 - \mu t) \frac{dV}{dt} \dots (2)$$

From (1) and (2), we get

$$Mg + \mu v_e - (M - \mu t)g = (2M - \mu t) \frac{dv}{dt} \dots (3)$$

$$(\mu v_e + \mu gt) = (2M - \mu t) \frac{dv}{dt} \dots (4)$$

$$\frac{dv}{dt} = \frac{\mu(v_e + gt)}{(2M - \mu t)}$$

$$\therefore \frac{dv}{dt} = \frac{\mu(v_e + gt)}{(2M - \mu t)}$$

So, correct answer is (A)

4. A

5. A

Sol. $\frac{l}{K_1 A} + \frac{l}{2K_2 A} + \frac{l}{K_3 A} = \frac{l}{K A} + \frac{l}{2K_A} + \frac{l}{K A}$

$$\frac{5}{2K} = \left(\frac{1}{K_1} + \frac{1}{2K_2} + \frac{1}{K_3} \right)$$

$$K_{eq} = \frac{5}{2 \left(\frac{1}{K_1} + \frac{1}{2K_2} + \frac{1}{K_3} \right)}$$

6. A

Sol. Voltage across capacitor will lag current by 90° ;

$$\therefore i_c = \frac{200\sqrt{2}}{10} \sin \left(\omega t + \frac{\pi}{4} + \frac{\pi}{2} \right)$$

$$i = 20\sqrt{2} \sin \left(\omega t + \frac{3\pi}{4} \right)$$

$\therefore$ Correct answer is (A)

7. D

Sol. $2dQ = dU + 2dW$

$$dQ = dU + dW$$

$$dQ = dW$$

$$\Rightarrow dU = 0$$

Process is isothermal.

8. B

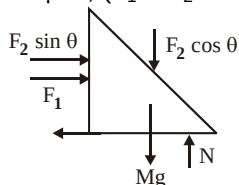
Sol. $F = q \mathbf{v} \times \mathbf{B}$

$$= q v_0 \mathbf{i} \times \mathbf{B} (-\mathbf{k}) = q v_0 B \mathbf{j}$$

9. B

10. C

Sol. $f = \mu N$; $(F_1 + F_2 \sin \theta) = \mu (mg + F_2 \cos \theta)$



$$\mu = \frac{F_1 + F_2 \sin \theta}{mg + F_2 \cos \theta}$$

11. C

Sol. $W_B = 0$

$$W_E = \Delta KE \quad ; \quad W_B + W_E = \Delta KE$$

12. A

Sol. K.E. = $p^2 / 2m$ and $p = h / \lambda$

13. D

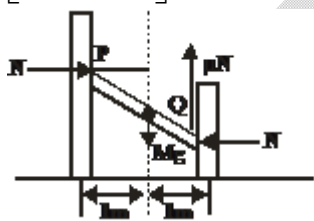
Sol. Drawing F.B.D. of rod PQ

$$\Sigma F_x = 0$$

$$\Sigma F_y = 0 \Rightarrow \mu N = mg \quad \dots (1)$$

$$\Sigma \tau \text{ about centre of mass} = 0$$

$$\left[N \times \frac{l}{2} \sin \theta \right] [2] = \mu N \times \frac{l}{2} \cos \theta \Rightarrow \tan \theta = \frac{\mu}{2}$$



$$\text{Hence } \sec \theta = \left(\sqrt{\mu^2 + 4} \right) / 2$$

$$\text{Thus, } \left(\frac{l/2}{1} \right) = \frac{\sqrt{\mu^2 + 4}}{2} \Rightarrow l = \sqrt{\mu^2 + 4}$$

$$\Rightarrow l = \sqrt{0.25 + 4} = \frac{\sqrt{17}}{2} \text{ metres}$$

14. B

Sol. $e = - \frac{d\phi}{dt}$

$$i = \frac{e}{R} \text{ where } R \text{ is resistance}$$

$$H = i^2 R t$$

The emf will induce from zero to some maximum value and then from maximum to zero.

15. A

Sol. N_b reaches maximum when activity of A will be equal to activity of B.

$$\text{i.e., } \lambda_a N_a = \lambda_b N_b \Rightarrow N_b = \frac{\lambda_a N_a}{\lambda_b}$$

16. C

Sol. $f_0 = \frac{c}{\lambda} = \frac{c}{4(l+e)}$ and $\frac{c}{c_0} = \sqrt{\frac{273+\theta}{273}}$

$$\therefore f'_0 = \frac{c_0}{4(l+e)} \sqrt{1 + \frac{\theta}{273}}$$

If temperature is same, first overtone will be $3f_0$

$$\therefore \text{at } 273^\circ\text{C first overtone is } 3\sqrt{2} f_0.$$

17. B

Sol. $V_0 = \sqrt{\frac{GM}{3R/2}} = \sqrt{\frac{2GM}{3R}}$

In order to escape to infinity let $(V - V_0)$ is the speed change required. Thus,

$$\frac{-GMm}{(3R/2)} + \frac{1}{2} m(v)^2 = 0$$

Thus, $v = \sqrt{\frac{4GM}{3R}}$

Thus speed change required $= v - v_0$

$$\Delta V = \sqrt{\frac{4GM}{3R}} - \sqrt{\frac{2GM}{3R}} ; \Delta V = \sqrt{\frac{GM}{3R}} (2 - \sqrt{2})$$

18. B

Sol. At any time

$$Kx - l/B = ma \quad \dots (1)$$

$$\frac{q}{C} = Bvl$$

$$I = B/Ca \quad \dots (2)$$

From (1) and (2)

$$kx = a(B^2 l^2 C + m)$$

$$a = \left(\frac{K}{B^2 l^2 C + m} \right) x \quad \dots (3)$$

Since motion of connector will be opposite to the displacement from equilibrium. Therefore equation (3) represents S.H.M.

$$\omega = \sqrt{\frac{K}{B^2 l^2 C + m}}$$

$$V_{\max} = X_0 \omega$$

$$\therefore q_{\max} = B l C V_{\max}$$

$$q_{\max} = B l C X_0 \omega$$

$$X_0 \sqrt{\frac{KC}{1 + \frac{m}{B^2 l^2 C}}}$$

19.

A

Sol. Velocity of image of particle B

$$\bar{v}_B = 5(\hat{i}) + 4(-\hat{i}) = \hat{i}$$

velocity of image of particle A

$$\begin{aligned} v_A &= 10(-\hat{i}) + 4(-\hat{i}) - 10\hat{j} \\ &= 14(-\hat{i}) - 10\hat{j} \end{aligned}$$

Relative velocity of image

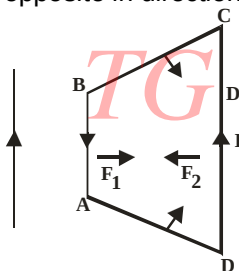
$$\bar{V}_{BA} = \bar{v}_B - \bar{v}_A = \hat{i} - [14(-\hat{i}) - 10\hat{j}] = 15\hat{i} + 10\hat{j}$$

$$|\bar{V}_{BA}| = \sqrt{325} \text{ cm/sec}$$

20.

B

Sol. Magnitude of force on AB (F_1) and magnitude of force on CD (F_2) will be equal in magnitude and opposite in direction.



Hence $\bar{F}_1 + \bar{F}_2 = 0$ resultant of $\bar{F}_3 + \bar{F}_4$ will be away from the long straight wire.

SECTION - B

21.

2

Sol. Conservation of energy

$$\frac{1}{2}mv_0^2 = mgR + \frac{1}{2}mv^2, v = v_0 \cos \theta$$

$$\frac{1}{2}v_0^2 = gR + \frac{v_0^2 \cos^2 \theta}{2}$$

$$v_0^2 [1 - \cos^2 \theta] = gR$$

$$v_0 = \frac{\sqrt{2gR}}{\sin \theta}$$

22.

9

Sol. Net retardation $a = \frac{F_{\text{ext}} + \mu_k mg}{m} = 4.5$

If body stop at time t, then

$$V = u + at$$

$$0 = 45 - 4.5t \Rightarrow t = 10 \text{ sec}$$

When block stops, F_{ext} will try to bring the block back ward while frictional force will oppose its motion, since block is stationary therefore at this moment, frictional force will be static. Whose maximum value will be $\mu_s mg = 30 \text{ N}$. Since static frictional force is self adjusting therefore it will be 25 N and block will not move after $t = 10 \text{ sec}$.

$$S = \frac{u^2}{2a} = 225 \text{ m}$$

23. 25

Sol. Time taken by the layer of ice to increase its thickness by $x = x_2 - x_1$

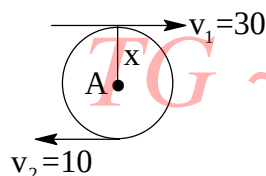
$$t = \frac{\rho L}{2K\theta} (x_2^2 - x_1^2) = \frac{.9 \times 80}{2 \times .005 \times 10} [(15)^2 - (10)^2]$$

$$t = 9 \times 10^4 \text{ sec} ; t = 25 \text{ Hours}$$

24. 25

Sol. Let x be the distance of axis of rotation from the top of sphere A (downward). Since angular velocity of the top and bottom point of sphere A will be same, therefore,

$$\omega = \frac{30}{x} = \frac{10}{(2R - x)}$$



$$\frac{30}{x} = \frac{10}{20 - x}$$

$$\Rightarrow x = 15$$

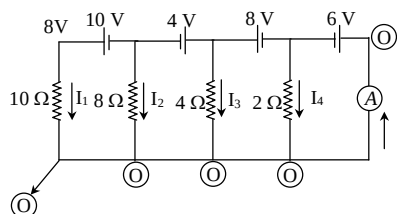
$$\begin{aligned} \therefore \text{Distance from ground} &= 2R + (2R - x) \\ &= 20 + 5 = 25 \text{ cm} \end{aligned}$$

25. 8

$$\text{Sol. } t = \int \frac{dy}{V_y} = \int_0^d \frac{dy}{Ky^{1/2}} = \frac{2\sqrt{d}}{k}$$

26. 3

$$\begin{aligned} \text{Sol. } I &= I_1 + I_2 + I_3 + I_4 = \frac{8}{10} - \frac{2}{8} + \frac{2}{4} - \frac{6}{2} \\ \frac{32 - 10 + 20 - 120}{40} &= \frac{-78}{40} \text{ Amp} \end{aligned}$$



27. 4

Sol. Momentum conservation along x

$$0 = mv_2 - Mv_1$$

Work energy theorem

$$mgR = \frac{1}{2}mv_2^2 + \frac{1}{2}Mv_1^2$$

from (i) and (ii)

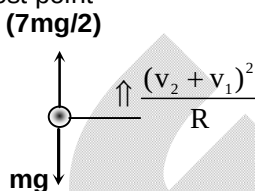
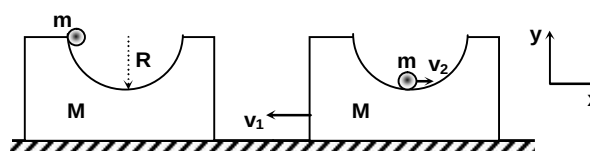
$$\Rightarrow 2mgR = mv_2^2 + \frac{m^2}{M}v_2^2$$

$$\sqrt{\frac{2gRM}{M+m}} = v_2 \text{ and } v_1 = \frac{m}{M} \sqrt{\frac{2gRM}{M+m}}$$

Making FBD of particle with respect wedge at lowest point

$$\frac{7mg}{2} - mg = \frac{m(v_1 + v_2)^2}{R}$$

Substituting the values, we get : $\frac{M}{m} = 4$



28. 12

Sol. $v_0 \cos 30^\circ = v \cos 45^\circ$

$$\Rightarrow v = v_0 \sqrt{\frac{3}{2}}$$

Now using conservation of energy

$$\frac{1}{2}m(v^2 - v_0^2) = mgh$$

$$\frac{3v_0^2}{2} - v_0^2 = 2gh \Rightarrow \frac{v_0^2}{2} = 2gh$$

$$\Rightarrow v = \sqrt{4gh} = \sqrt{4 \times 10 \times 3.6} = 12 \text{ m/s}$$

29. 5

Sol. $F_1 = \rho \left(\pi R^3 - \frac{2}{3} \pi R^3 \right) g = \frac{\rho \pi R^3}{3} g$, $F_2 = \rho \left(\pi R^3 + \frac{2}{3} \pi R^3 \right) g = \frac{5\rho \pi R^3}{3} g$

$$\therefore F_2/F_1 = 5$$

30. 1

Sol. $\sin 45^\circ = \mu \sin r \Rightarrow r = 30^\circ$

From $\triangle BQR$ and $\triangle CSR$

$$\frac{BQ}{SC} = \frac{BR}{RC}$$

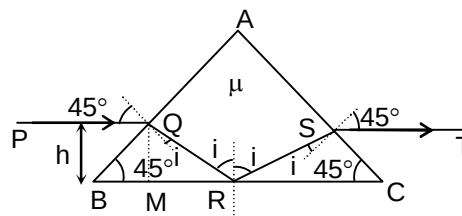
$$\therefore h' = 3 \left(\frac{RC}{BR} \right)$$

$$i = 45^\circ + r = 75^\circ$$

$$\text{So, } \frac{QM}{MR} = \tan 15^\circ, \text{ so } MR = 12 \text{ cm}$$

$$\text{So, } BR = 15 \text{ cm and } RC = 5 \text{ cm}$$

$$\text{So, } h' = 1 \text{ cm}$$



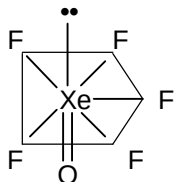
Chemistry

PART – B

SECTION – A

31. D
Sol. Fluorine is most reactive amongst halides and has highest oxidising power

32. D
Sol.

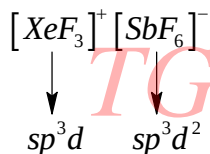


Hybridisation – sp^3d^3

Shape – pentagonal monopyramidal

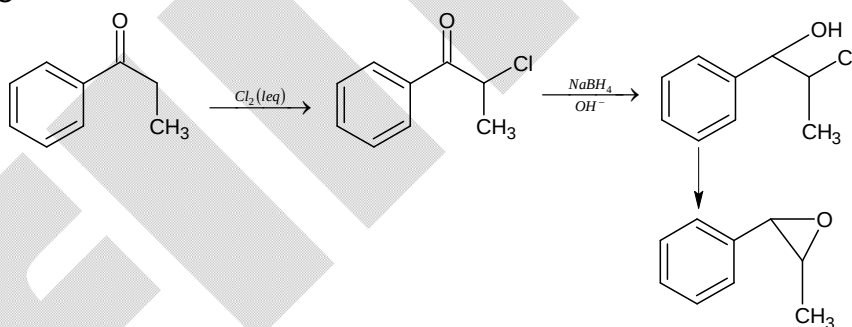
33. C
Sol. Macromolecular colloids like starch, gelatin are generally lyophilic.

34. A
Sol.

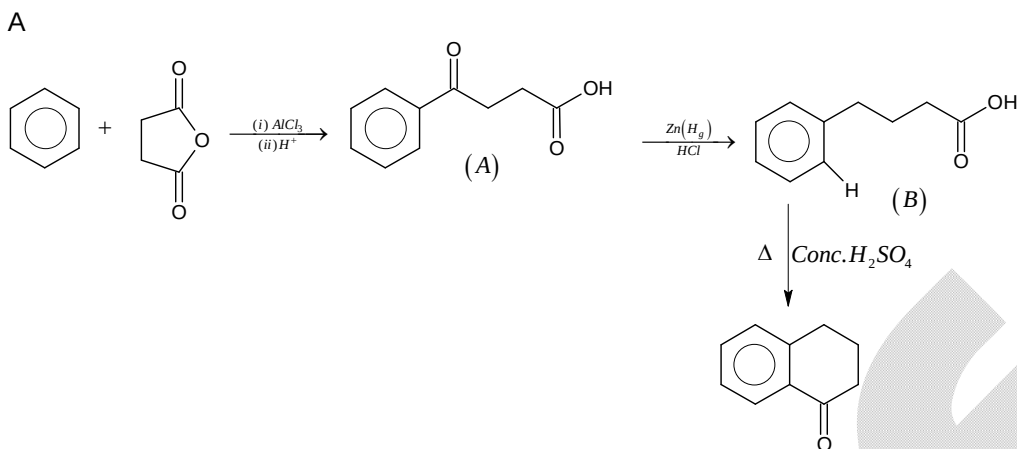


35. A
Sol. In $(\bullet CF_3)$, % p-character because of which bond angle decreases

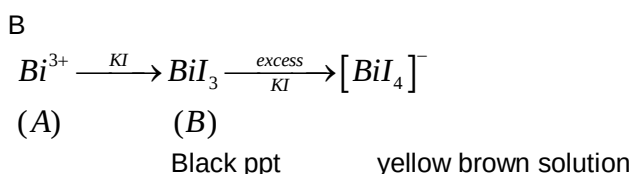
36. C
Sol.



37.
Sol.



38.
Sol.



39.
Sol.



40.
Sol.

A
Because of stability of half filled f-subshell configuration.

41.
Sol.

C

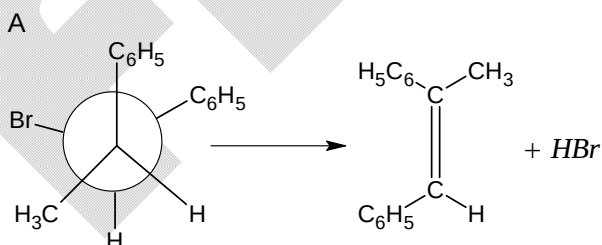
$$AgBr = x + y$$

$$AgBr = \frac{K \times 1000}{M}$$

M – molarity of AgBr solution

$$8 \text{ (solubility in g/L)} = \frac{K \times 1000}{(x + y)} \times 188$$

42.
Sol.



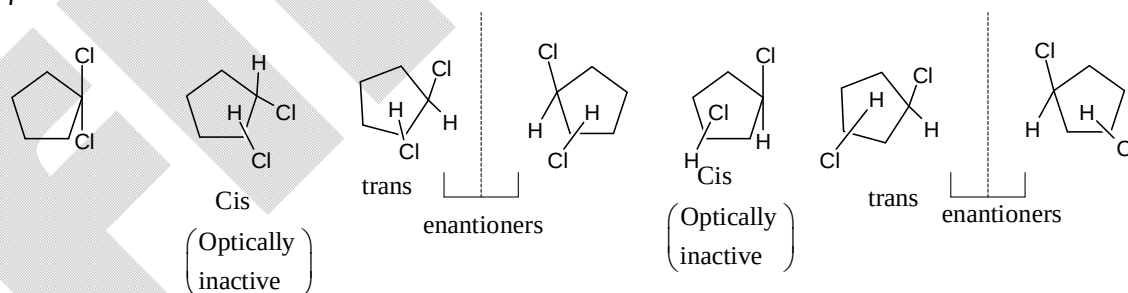
43.
Sol.

C
–OH group of phenol cannot be replaced by iodine as it does not undergo nucleophilic substitution reaction under normal conditions.

44. B
Sol. Greater the activation energy of a reaction, greater is the temperature dependence of rate constant of reaction.
45. C
Sol. $\frac{1}{\lambda} = RZ^2 \left(1 - \frac{1}{4}\right)$ for lyman line
46. C
Sol. Methoxy group stabilises the cationic intermediate by $+M$ effect
47. C
Sol. Cl_2 is present simultaneously in two equilibria. Decrease in its concentration shift both equilibria forward.
48. D
Sol. Point B represents exactly half neutralisation of acid. At this point concentration acid left and its conjugate base are equal.
49. C
Sol. $[Br^-] = \frac{4 \times 10^{-13}}{4 \times 10^{-4}} = 10^{-9} M$
Mass of $NaBr$ added of $200\text{ ml} = 103 \times \frac{10^{-9}}{1000} \times 200$
 $= 2.06 \times 10^{-8} g$
50. C
Sol. Greater the charge density, greater is hydration.

SECTION – B

51. 1
52. 7
Sol.



53. 10
54. 3
Sol. $(Ni(CN)_4)^{2-}$ is dsp^2 hybridised and diamagnetic

55. 1

Sol. Equivalents of $KMnO_4 = 5x$

Equivalents of $K_2Cr_2O_7 = 6y$

Equivalents of oxidising agent must be same $5x = 6y$

$$x - y = 6 - 5 \Rightarrow 1$$

56. 7

Sol.

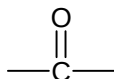
$Na - 1s^2$	$2s^2$	$2p^6$	$3s^1$
1↑	1↑	1↑ 1↑ 1↑	1↑
$m_l - 0$	0	$+1 \ 0 \ -1$	0

57. 3

Sol. In PCl_2F_3 , net dipole moment does not cancel out.

58. 1

Sol.



Group stabilise the carbanion character on α -carbon

59. 4

Sol. 2 molecules of $RMgX$ are consumed by acid chloride group as it undergoes addition elimination first and then addition. The other aldehydic & ketonic group consume one $RMgX$ molecule each.

60. 4

Sol. Structure containing a ring of $(4n + 2)\pi e^-$ s is aromatic

Mathematics

PART – C

SECTION – A

61. D

Sol. $a = 3, b = -4, c = 1$

62. D

63. C

Sol. Put $x = i$ and equate the real part

64. C

Sol. $x^2 dx + y^2 dy = \frac{d(xy)}{(xy)^2}$

65. B

Sol. Let $x = \frac{\log_2 y}{111}$ these $y^3 - 16y^2 + 8y - 4 = 0$

66. C

Sol. $\sec^{-1} \sqrt{1+x^2} = \tan^{-1} x, \cos^{-1} \frac{1-x^2}{1+x^2} = 2 \tan^{-1} x$

67. C

Sol. $\left(\frac{x-2y-4}{\sqrt{5}} \right)^2 = \frac{12}{\sqrt{5}} \left(\frac{2x+y}{\sqrt{5}} \right)$ (if is parabola)

68. C

Sol. $\tan \frac{\theta}{2} \tan \frac{\theta}{2} = \frac{1+e}{1-e}$ if focus is $(-ae, \theta)$

69. D

Sol. $f(x) = \tan \left(\frac{\pi}{4} - \frac{x}{2} \right)$

70. B

71. B

72. D

73. C

74. B

75. B

Sol. $b^2 + c^2 - a^2, c^2 + a^2 - b^2, a^2 + b^2 - c^2$ are in AP.

76. B
77. A
78. D
Sol. Both the roots opposite in sign
79. A
Sol. $\text{adj}(AB) = \text{adj}B \cdot \text{adj}A$
80. C

SECTION – B

81. 7
Sol. $\int_0^{\frac{\pi}{2}} \frac{(1 - \tan x)^2}{(1 + \tan x)^4} \sec^2 x \, dx = \frac{1}{3}$
82. 105
Sol. $\frac{7}{4 \cdot 2} = 105$
83. 50
Sol. $OM = \sqrt{(50\sqrt{5})^2 - (100)^2} = 50m$
84. 8
85. 0
Sol. $\Delta = 0$
86. 1023
Sol. $3 + 5 + 7 + \dots$ upto 32nd terms = 1023
87. 1
Sol. $\sin^{-1} \frac{\frac{2}{x}}{1 + \left(\frac{1}{x}\right)^2} = 2 \tan^{-1} \frac{1}{x}$
88. 2
Sol. $\int_1^3 \left(\tan^{-1} \left(\frac{x}{1+x^2} \right) + \tan^{-1} \left(\frac{x^2+1}{x} \right) \right) dx = 2\pi$
89. 6
90. 0
Sol. $\vec{a} + \vec{b} = \vec{c}$

FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2023

TEST DATE: 20-01-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A

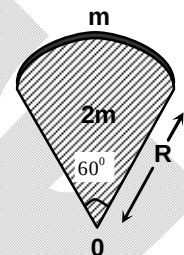
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

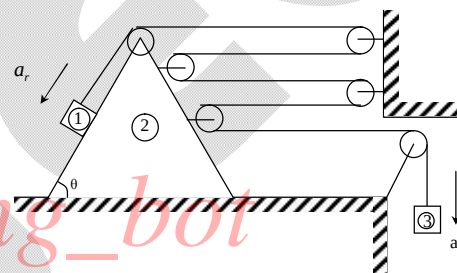
1. What is the distance of centre of mass of the combined system as shown in figure from 'O'. System consists of a uniform sector of mass $2m$ and radius R , fixed with a uniform arc of mass m and radius R along the edge of the sector. (System is making an angle $\theta = 60^\circ$ at their common centre)

(A) $\frac{5R}{3\pi}$
(C) $\frac{7R}{3\pi}$

(B) $\frac{4R}{3\pi}$
(D) none of these



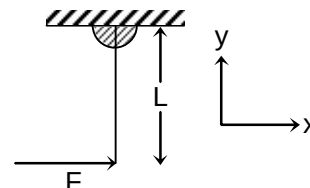
2. In the figure shown, all the three bodies 1, 2, 3 can move. If acceleration of body 1 w.r.t 2 is $a_r = 6 \text{ m/s}^2$ and acceleration of 3 w.r.t ground is $a = 4 \text{ m/s}^2$, then magnitude of acceleration wedge 2 w.r.t. ground is
- (A) 2 m/s^2
(B) 0.8 m/s^2
(C) 0.4 m/s^2
(D) 1 m/s^2



3. A rod of mass m , length L is kept in a vertical plane such that upper end is fixed. A force F is applied at bottom as shown. The hinge reaction in the given position will be :

(A) $F(\hat{i}) + mg(\hat{j})$
(C) $\frac{F}{2}\hat{i} + mg(\hat{j})$

(B) $mg(\hat{j})$
(D) $\frac{9F}{7}\hat{i} + mg(\hat{j})$



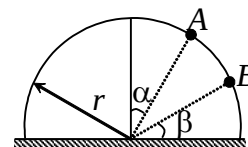
4. From a circle of radius a , an isosceles right angled triangle with the hypotenuse as the diameter of the circle is removed. The distance of the centre of mass of the remaining part from the centre of the circle is

(A) $3(\pi - 1)a$
(C) $\frac{a}{3(\pi - 1)}$

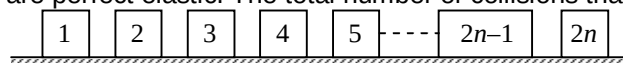
(B) $\frac{(\pi - 1)a}{6}$
(D) $\frac{a}{3(\pi + 1)}$

5. A particle moves from rest at A on the surface of a smooth circular cylinder of radius r as shown. At B it leaves the cylinder. The equation relating α and β is

(A) $3 \sin \alpha = 2 \cos \beta$
(B) $2 \sin \alpha = 3 \cos \beta$
(C) $3 \sin \beta = 2 \cos \alpha$
(D) $2 \sin \beta = 3 \cos \alpha$



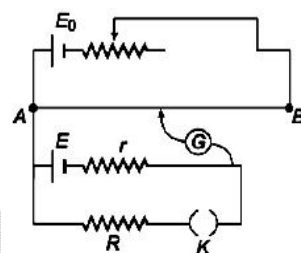
6. $2n$ identical cubical blocks are kept in a straight line on a horizontal smooth surface. The distance between the consecutive blocks is same. The blocks 1, 3, 5, ..., $(2n - 1)$ are given velocity v to the right whereas blocks 2, 4, 6, ..., $2n$ are given velocity to the left. All collisions between blocks are perfect elastic. The total number of collisions that will take place is



- (A) n (B) $n + 1$
 (C) $\frac{n(n+1)}{2}$ (D) $n(n+1)$

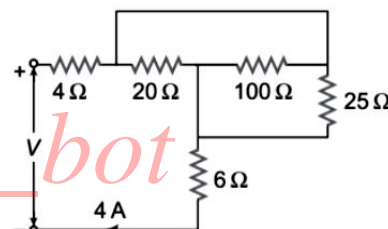
7. The given figure represents an arrangement of potentiometer for the calculation of internal resistance (r) of the unknown battery (E). The balance length is 70.0 cm with the key opened and 60.0 cm with the key closed. R is 132.40Ω . The internal resistance (r) of the unknown cell will be

- (A) 22.1Ω (B) 113.5Ω
 (C) 154.5Ω (D) 10Ω



8. In the circuit shown in figure, V must be

- (A) 50 V (B) 80 V
 (C) 100 V (D) 1290 V



9. A galvanometer of resistance 50Ω is connected to a battery of 3 V along with resistance of 2950Ω in series. A full scale deflection of 30 divisions is obtained in the galvanometer. In order to reduce this deflection to 20 divisions, the above series resistance should be

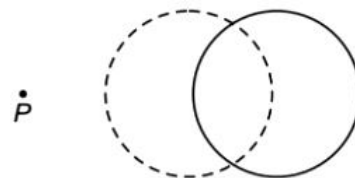
- (A) 4450Ω (B) 5050Ω
 (C) 5550Ω (D) 6050Ω

10. Two identical coaxial rings each of radius R are separated by a distance of $\sqrt{3}R$. They are uniformly charged with charges $+Q$ and $-Q$ respectively. The minimum kinetic energy with which a charged particle (charge $+q$) should be projected from the centre of the negatively charged ring along the axis of the rings such that it reaches the centre of the positively charged ring is

- (A) $\frac{Qq}{4\pi\epsilon_0 R}$ (B) $\frac{3Qq}{2\pi\epsilon_0 R}$
 (C) $\frac{Qq}{2\pi\epsilon_0 R}$ (D) $\frac{3Qq}{4\pi\epsilon_0 R}$

11. Figure shows a closed dotted surface which intersects a conducting uncharged sphere. If a positive charge is placed at the point P , the flux of the electric field through the closed surface

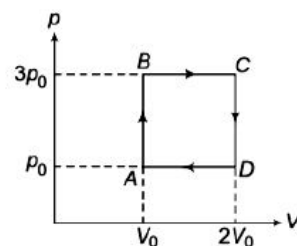
- (A) will remain zero
 (B) will become positive
 (C) will become negative
 (D) data insufficient



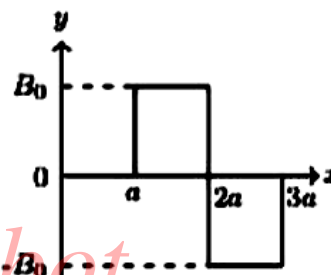
12. A gas is found to be obeyed the law $p^2V = \text{constant}$. The initial temperature and volume are T_0 and V_0 . If the gas expands to a volume $3V_0$, then the final temperature becomes
- (A) $\sqrt{3}T_0$ (B) $\sqrt{2}T_0$
 (C) $\frac{T_0}{\sqrt{3}}$ (D) $\frac{T_0}{\sqrt{2}}$

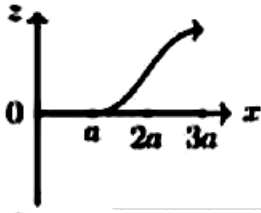
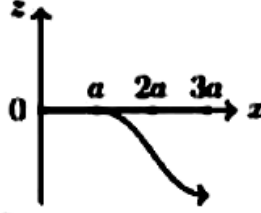
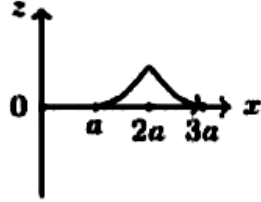
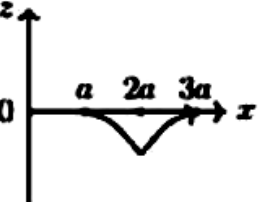
13. An ideal monoatomic gas is carried around the cycle ABCDA as shown in the figure. The efficiency of the gas cycle is

- (A) $\frac{4}{21}$ (B) $\frac{2}{21}$
 (C) $\frac{4}{31}$ (D) $\frac{2}{31}$



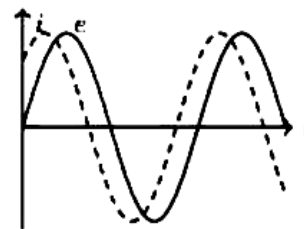
14. A magnetic field vector $\vec{B} = B_0\hat{j}$ exists in the region $a < x < 2a$ and vector $\vec{B} = -B_0\hat{j}$ in the region $2a < x < 3a$, where B_0 is a positive constant. A positive point charge moving with a velocity vector $\vec{v} = v_0\hat{i}$, where v_0 is a positive constant, enters the magnetic field at $x = a$. The trajectory of the charge in this region can be like



- (A) 
- (B) 
- (C) 
- (D) 

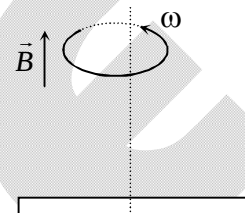
15. Two identical circular loops of metal wire are lying on a table without touching each other. Loop A carries a current which increases with time. In response, the loop B
- (A) remains stationary (B) is attracted by the loop A.
 (C) is repelled by the loop A. (D) rotates about its CM, with CM fixed.

16. When an AC source of emf $e = e_0 \sin(100t)$ is connected across a circuit, the phase difference between the emf e and the current i in the circuit is observed to be $\pi/4$, as shown in the diagram. If the circuit consists possibly R-C or R-L or L-C in series, find the relationship between the two elements.



- (A) $R = 1 \text{ k}\Omega$, $C = 10 \text{ }\mu\text{F}$
 (B) $R = 1 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$
 (C) $R = 1 \text{ k}\Omega$, $L = 10 \text{ H}$
 (D) $R = 1 \text{ k}\Omega$, $L = 1 \text{ H}$

17. A conducting rod of length $2l$ is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The emf induced between two ends of the rod is



- (A) $B\omega l^2$
 (B) $\frac{1}{2} B\omega l^2$
 (C) $\frac{1}{8} B\omega l^2$
 (D) zero

18. The energy that should be added to an electron, to reduce its de-Broglie wavelengths from 10^{-10} m to $0.5 \times 10^{-10} \text{ m}$, will be

- (A) four times the initial energy
 (B) thrice the initial energy
 (C) equal to the initial energy
 (D) twice the initial energy

19. The wavelength of a certain line in the x-ray spectrum for tungsten ($Z = 74$) is $200 \text{ }\text{\AA}$. What would be the wavelength of the same line for platinum ($Z = 78$)? The screening constant a is unity.

- (A) $179.76 \text{ }\text{\AA}$
 (B) $189.76 \text{ }\text{\AA}$
 (C) $289.76 \text{ }\text{\AA}$
 (D) $379.76 \text{ }\text{\AA}$

20. Two radioactive substances X and Y initially contain equal number of nuclei. X has a half life of 1 hour and Y has half life of 2 hours. After two hours the ratio of the activity of X to the activity of Y will be

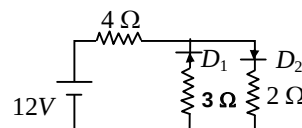
- (A) $1 : 4$
 (B) $1 : 2$
 (C) $1 : 1$
 (D) $2 : 1$

SECTION – B

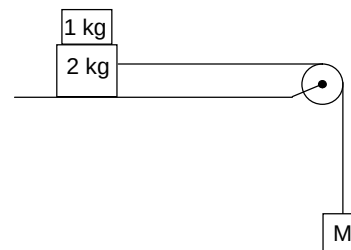
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. The circuit has two oppositely connected ideal diodes in parallel. What is the current flowing in the circuit?

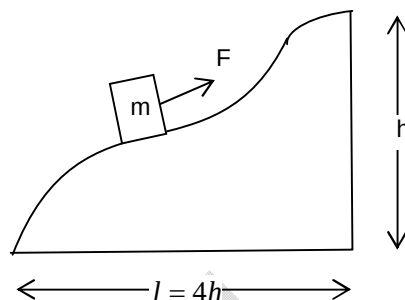


22. Find minimum value of mass suspended M for which 1 kg block start toppling. Assume 1 kg block does not slip over smooth 2 kg block. Length of 1 kg block is 4 cm and height is 8 cm . (ground is smooth).

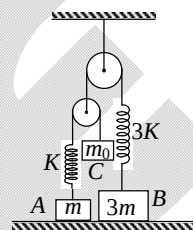


23. A body of mass 'm' is slowly hauled up a hill by an external force which is always acting along the surface as shown. The coefficient of friction between the body and the surface is $\mu = 0.5$. If work done by friction and gravity on the body are ΔW_{fr} and ΔW_g respectively during the process. Find

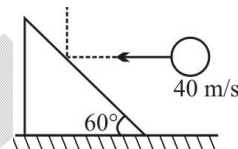
the ratio $\frac{\Delta W_{fr}}{\Delta W_g}$



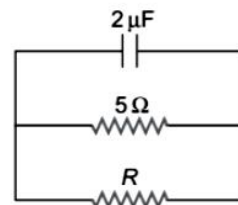
24. A spring mass system arrangement in vertical plane is as shown. Initially springs are in natural length and blocks A & B are resting on the ground. Determine the minimum mass (in kg) of block C so that block A leaves contact with the ground after releasing block C. (Take $m = 10$ kg)



25. Inclined surface of a smooth wedge of mass 6 kg makes an angle of 60° with horizontal as shown. A ball hits the wedge horizontally and elastically with a velocity of 40 m/s and moves vertically with respect to ground after the collision. Find the mass (in kg) of the ball if the wedge can also move on the smooth floor.



26. The peak power consumed by a resistive coil when connected to an AC source is 80 W. Find the energy consumed by the coil is 100 seconds which is many times larger than the time period of the source. (in kJ)
27. In YDSE experiment if the screen is shifted by a distance of 0.5 m away from the slit, 3rd maxima is shifted by 3×10^{-4} m. If the slit width is 2×10^{-4} m, find the wavelength used in the experiment (in nm).
28. When the voltage applied to an X-ray tube is increased from 10 kV to 20 kV the wavelength interval between the K_α line and the short wave cut off of the continuous X-ray spectrum increases by a factor 3. Find the atomic number of element of which the tube anti-cathode is made. (Rydberg's constant = 10^7 m^{-1})
29. A bus is moving towards a huge wall with a velocity of 5 m/s. the driver sounds a horn of frequency 200 Hz. What is the frequency of beats heard by a passenger of the bus, if the speed of sound in air is 330 m/s.
30. A capacitor of capacitance $2 \mu\text{F}$ is charged to a potential difference of 5 V. Now, the charging battery is disconnected and the capacitor is connected in parallel to a resistor of 5Ω and another unknown resistor of resistance R as shown in figure. If the total heat produced in 5Ω resistance is $10 \mu\text{J}$, then the unknown resistance R is equal to $10/n$. Find the value of n.



Chemistry

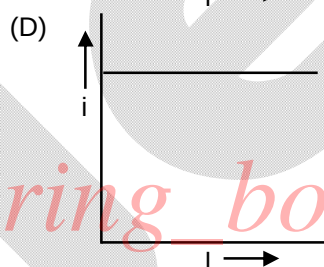
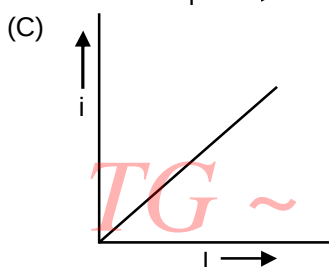
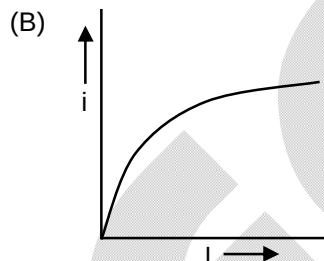
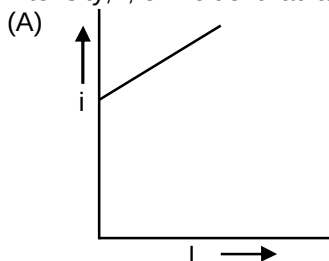
PART – B

SECTION – A

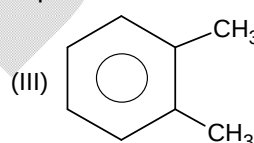
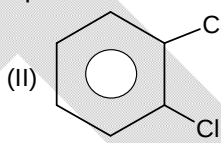
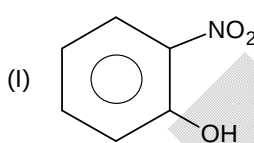
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

31. Which of the following graphs depicts correctly the variation of photoelectric current i , with the intensity, I , of incident radiation?



32. Which is the correct order of dipole moments of the compounds



- (A) I > II > III
(C) III > I > II

- (B) II > I > III
(D) III > II > I

33. The compressibility factor of helium as a real gas is

(A) unity

(B) $1 - \frac{a}{RTV}$

(C) $1 + \frac{Pb}{RT}$

(D) $\frac{RTV}{1-a}$

34. 20 ml of a solution containing Na_2CO_3 and NaHCO_3 requires 10 ml of 0.1 N H_2SO_4 for neutralization using phenolphthalein as an indicator. Methyl orange is then added when a further 10 ml of 0.2 N H_2SO_4 is required. Molar concentration of Na_2CO_3 and NaHCO_3 in the solution respectively are

- (A) 0.05 M and 0.05 M
(C) 0.05 M and 0.10 M

- (B) 0.1 M and 0.05 M
(D) 0.1 M and 0.1 M

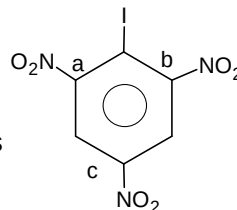
35. Time required for 99.9% completion of a first order reaction is nearly

- (A) 10 times that required for 25% of the reaction
(B) 10 times that required for 50% of the reaction
(C) 20 times that required for 50% of the reaction
(D) 100 times that required for 90% of the reaction

36. A 1.0 M with respect to each of metal halides AX_3 , BX_2 , CX_3 and DX_2 is electrolysed using platinum electrodes. If $E_{A^{3+}/A}^\circ = 1.50V$, $E_{B^{2+}/B}^\circ = +0.34V$, $E_{C^{3+}/C}^\circ = -0.74V$, $E_{D^{2+}/D}^\circ = -2.37V$, the correct sequence in which the various metals are deposited at the cathode, is:

(A) A, B, C, D
(B) D, C, B, A
(C) A, B, C
(D) C, B, A

37. 2, 4, 6- Trinitroiodobenzene has three C-N bonds

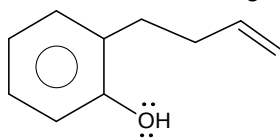


labelled as a, b and c. The

correct bond length of three bonds is

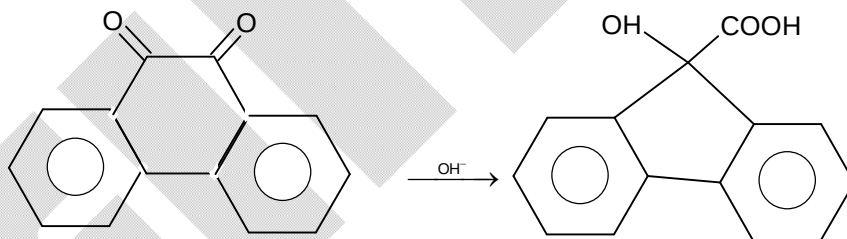
(A) $a = b = c$
(B) $a > b > c$
(C) $a = b > c$
(D) $a = b < c$

38. Which of the following ion is formed in the following reaction?



(A) (B)
(C) (D) All the three

- 39.



The above reaction is an example of

(A) Baeyer – Villiger rearrangement
(B) Benzilic acid rearrangement
(C) Cannizzaro reaction
(D) Pinacol – pinacolone rearrangement

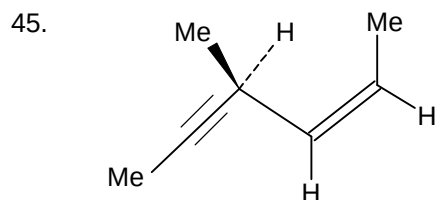
40. Oxide which can not be a reducing agent is

(A) NO_2
(B) CO_2
(C) ClO_2
(D) SO_2

41. The number of S – S bonds in sulphur trioxide trimer (S_3O_9) is

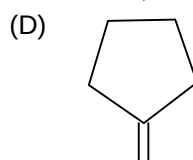
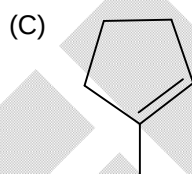
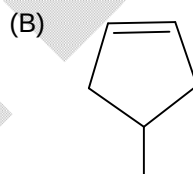
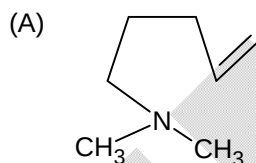
(A) 3
(B) 2
(C) 1
(D) 0

42. The number of geometrical isomers for octahedral $[\text{Co}(\text{NH}_3)_2\text{Cl}_4]^-$, square planar $\text{AuCl}_2\text{Br}_2^-$ and $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]^{2+}$ are
 (A) 2, 2, 2 (B) 2, 2, no isomerism
 (C) 3, 2, 2 (D) 2, 3, no isomerism
43. For the correct assignment of electronic configuration of a complex, the valence bond theory often requires the measurement of
 (A) molar conductance (B) optical activity
 (C) magnetic moment (D) dipole moment
44. Which of the following pairs can be distinguished by Fehling's solution?
 (A) Glucose and fructose (B) Glucose and sucrose
 (C) Methanal and ethanal (D) Hydroxypropanone and benzaldehyde

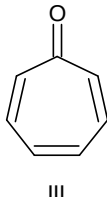
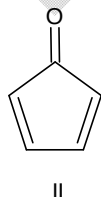
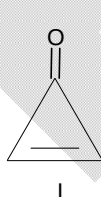


Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives
 (A) an optically active compound (B) an optically inactive compound
 (C) a racemic mixture (D) a diastereomeric mixture

46. $\xrightarrow[\text{(ii) Heat}]{\text{(i) H}_2\text{O}_2}$ Product P is



47. Which of the following is least stable?



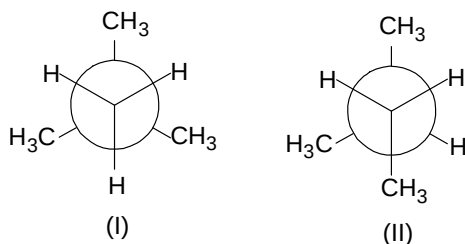
- (A) I
(C) III

- (B) II
(D) All are equally stable

48. A mixture of CH_4 , N_2 and O_2 is enclosed in a container of 1 litre capacity at 0°C . Total pressure of gaseous mixture is 2660 mm Hg. If the ratio of partial pressure of the gases is 1 : 4 : 2 respectively, the number of moles of oxygen present in the vessel is:

(A) $\frac{1}{22.4}$ (B) 1.0
(C) 0.1 (D) None of these

49.



(A) Conformational isomers (B) Stereoisomers
(C) Constitutional isomers (D) Identical

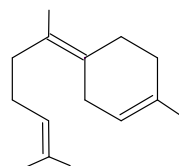
50. In zone refining method the molten zone
(A) moves to either side
(B) contains the purified metal only
(C) contains more impurity than the original metal
(D) consists of impurities only

SECTION - B
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

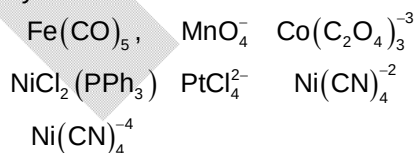
51. Consider the reaction: $\text{AB}_{2(g)} \rightleftharpoons \text{AB}_{(g)} + \text{B}_{(g)}$. If the initial pressure of AB_2 is 500 torr and equilibrium pressure is 600 torr, equilibrium constant K_p in terms of torr is

52. Reductive ozonolysis of a terpenoid of following structure gives how many different products



53. For H_2SO_4 , K_{a1} = infinite and $K_{a2} = 1.2 \times 10^{-2}$. The molarity of H_2SO_4 solution of pH 2.0 is 'xM'. The value of '1000000x' is

54. In how many of the following compound 'd' orbital (which contains no nodal plane) is used in hybridisation of central atom/ions?



55. The number of monohalogenated alkanes formed by the reaction of $\begin{array}{c} \text{H}_3\text{C} \\ \diagdown \\ \text{CH}-\text{CH}_3 \\ \diagup \\ \text{H}_3\text{C} \end{array}$ with Cl_2 in presence of light is
56. Iron forms a series of compounds of the type $\text{Fe}_x(\text{CO})_y$. In air, they are oxidised to Fe_2O_3 and CO_2 gas. After heating a 0.142 g sample in air, you isolate the CO_2 in a 1.50 L flask at 25°C . The pressure of the gas is 44.9 mm Hg. What will be the value of $\frac{y}{x}$ in the simplest empirical.
57. The edge length of the unit of LiCl (NaCl like structure) is 520 pm. Assuming that the lithium ion is small enough that the chloride ions are in contact, the ionic radius of chloride ion (in pm) is $\left(\frac{1}{\sqrt{2}} = 0.7\right)$
58. A certain metal 'M' forms an insoluble oxalate complex $\text{M}_4\text{O}_3(\text{C}_2\text{O}_4)_3 \cdot 12\text{H}_2\text{O}$. If 3.2 g of the complex is formed from 1 g of oxalic acid, what is the atomic mass of M?
59. For the reaction $\text{R}-\text{X} + \text{OH}^- \rightarrow \text{R}-\text{OH} + \text{X}^-$ the rate of reaction is given as, rate = $4.7 \times 10^{-5} [\text{R}-\text{X}][\text{OH}^-] + 0.24 \times 10^{-5} [\text{RX}]$ what percentage of $\text{R}-\text{X}$ react by $\text{S}_\text{N}2$ mechanism when $[\text{OH}^-] = 0.001$ molar.
60. Find the sum of optically active isomers of both $[\text{Pt}(\text{gly})_2\text{Cl}_2]$ and $[\text{Co}(\text{en})_2\text{Cl}_2]$.

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. The angle between the line $\frac{x-2}{2} = \frac{y+1}{-1} = \frac{z-3}{2}$ and the plane $3x + 6y - 2z + 5 = 0$ is
 (A) $\cos^{-1}\left(\frac{4}{21}\right)$ (B) $\sin^{-1}\left(-\frac{4}{21}\right)$
 (C) $\sin^{-1}\left(\frac{6}{21}\right)$ (D) $\sin^{-1}\left(\frac{4}{21}\right)$
62. The slope of the tangent to the curve $y = \int_x^{x^2} \cos^{-1} t^2 dt$ at $x = \frac{1}{(2)^{1/4}}$ is
 (A) 0 (B) $\pi/2$
 (C) $\left(\frac{(8)^{1/4}}{3} - \frac{1}{4}\right)\pi$ (D) $\left(\frac{(8)^{1/4}}{6} - \frac{1}{4}\right)\pi$
63. Area bounded by $f(x) = \max\{\sin x, \cos x\}$; $0 \leq x \leq \pi/2$, $x = \pi/2$ and the coordinate axes is equal to
 (A) $\sqrt{2}$ sq. units (B) 2 sq. units
 (C) $\frac{1}{\sqrt{2}}$ sq. units (D) None of these
64. The co-efficient of x^7 in the expansion of $(1 - x^4)(1 + x)^9$ is
 (A) 27 (B) -24
 (C) 48 (D) -48
65. The chords of contact of the pair of tangents drawn from each point on the line $2x + y = 4$ to the circle $x^2 + y^2 = 1$ pass through fixed point
 (A) (2, 4) (B) $\left(-\frac{1}{2}, -\frac{1}{4}\right)$
 (C) $\left(\frac{1}{2}, \frac{1}{4}\right)$ (D) (-2, -4)
66. The equation of the line through the intersection of the lines $2x + 3y + 4 = 0$ and $6x - 3y + 12 = 0$ and normal to the circle $x^2 + y^2 - 4x - 12 = 0$, is
 (A) $x = y$ (B) $x = 0$
 (C) $y = 0$ (D) None of these
67. The solution of the differential equation $x \frac{dy}{dx} = -\frac{y}{2} - \frac{\sin 2x}{2y}$ is given by
 (A) $xy^2 = \cos^2 x + c$ (B) $xy^2 = \sin^2 x + c$
 (C) $yx^2 = \cos^2 x + c$ (D) None of these

68. If $a, b, c > 0$ and $x, y, z \in \mathbb{R}$, then the determinant $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$ is equal to
- (A) $a^x + b^y + c^z$ (B) $a^{-x} b^{-y} c^{-z}$
 (C) $a^{2x} b^{2y} c^{2z}$ (D) 0
69. $\int_0^\infty \left[\frac{3}{x^2 + 1} \right] dx$ is equal to ([.] denotes the greatest integer function)
- (A) $\sqrt{2}$ (B) $\sqrt{2} + 1$
 (C) $3/\sqrt{2}$ (D) infinite
70. The value of $\int \frac{\ln x}{(1 + \ln x)^2} dx$ is
- (A) $\frac{x}{1 - \ln x} + c$ (B) $\frac{x \ln x}{1 + \ln x} + c$
 (C) $\frac{x}{(1 + \ln x)} + c$ (D) $\frac{\ln x}{x(1 + \ln x)} + c$
71. Domain of $\log_{1/2} \log_4 \log_3 [(x - 4)^2]$ is, [.] denotes the integer function.
- (A) $(-\infty, 2] \cup [6, \infty)$ (B) $(-\infty, 2] \cup [6, 8)$
 (C) $(2, 6)$ (D) $[2, 6]$
72. PQ is a vertical pole with end P on level ground. R is the mid-point of PQ. At a point A on the level ground, the portion QR subtends an angle α at A. If $AP = nPQ$, then $\tan \alpha =$
- (A) $\frac{n}{2n^2 + 1}$ (B) $\frac{n}{n^2 - 1}$
 (C) $\frac{n}{n^2 + 1}$ (D) None of these
73. The line $lx + my + n = 0$ will be a normal to the hyperbola $b^2x^2 - a^2y^2 = a^2b^2$ if
- (A) $\frac{a^2}{l^2} + \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$ (B) $\frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 - b^2)^2}{n^2}$
 (C) $\frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$ (D) None of these
74. The coordinates of the point on the parabola $y = x^2 + 7x + 2$, which is nearest to the straight line $y = 3x - 3$ are
- (A) $(-2, -8)$ (B) $(1, 10)$
 (C) $(2, 20)$ (D) $(-1, -4)$

75. If AFB is a focal chord of the parabola $y^2 = 4ax$ and $AF = 4$, $FB = 5$, then the latus-rectum of the parabola is equal to
 (A) $\frac{80}{9}$ (B) $\frac{9}{80}$
 (C) 9 (D) 80
76. $\sum_{n=1}^{\infty} \frac{n}{4n^4 + 1}$ equals to
 (A) 2 (B) 1
 (C) ∞ (D) $1/4$
77. If S denote the sum to infinity and S_n the sum of n terms of the series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$ such that $S - S_n < \frac{1}{300}$, then the least value of n is
 (A) 4 (B) 5
 (C) 6 (D) 7
78. The set of values of 'a' for which the equation $x^3 - 3x + a$ has three distinct real roots, is
 (A) $(-\infty, \infty)$ (B) $(-2, 2)$
 (C) $(-1, 1)$ (D) none of these
79. If the line $y = \sqrt{3}x$ cuts the curve $x^4 + ax^2y + bxy + cx + dy + 6 = 0$ at A, B, C and D, then OA.OB.OC.OD (where O is the origin) is
 (A) $a - 2b + c$ (B) $2c^2d$
 (C) 96 (D) 6
80. The equation of plane perpendicular to $2x + 6y + 6z - 1 = 0$ and passing through the points (2, 2, 1) and (9, 3, 6) is
 (A) $3x + 4y + 5z + 9 = 0$ (B) $3x + 4y - 5z + 9 = 0$
 (C) $3x + 4y - 5z - 9 = 0$ (D) $3x + 4y + 5z - 9 = 0$

SECTION - B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Shortest distance between lines $\frac{x-6}{1} = \frac{y-2}{-2} = \frac{z-2}{2}$ and $\frac{x+4}{3} = \frac{y}{-2} = \frac{z+1}{-2}$ is
82. The number of solutions of the equation $x^3 + 2x^2 + 5x + 2\cos x = 0$ in $[0, 2\pi]$ is
83. If the second, third and fourth terms in the expansion of $(a + b)^n$ are 135, 30 and $10/3$ respectively, then n =
84. If the equation $|z - z_1|^2 + |z - z_2|^2 = k$ represents the equation of a circle, where $z_1 \equiv 2 + 3i$, $z_2 \equiv 4 + 3i$ are the extremities of a diameter, then the value of k is
85. The sum of the distances of any point on the ellipse $3x^2 + 4y^2 = 12$ from its directrix is

86. Given that $g(x)$ is a function satisfying $g(x + y) = g(x) g(y) \quad \forall x, y \in \mathbb{N}$ and $g(1) = 3$, the value of $\sum_{x=1}^4 g(x)$ is equal to
87. Given a function $f(x)$ continuous $\forall x \in \mathbb{R}$ such that $\lim_{x \rightarrow 0} \left[f(x) + \log \left(1 - \frac{1}{e^{f(x)}} \right) - \log(f(x)) \right] = 0$, then $f(0)$ is
88. Let S be the set of 6-digit numbers $a_1 a_2 a_3 a_4 a_5 a_6$ (all digits distinct) where $a_1 > a_2 > a_3 > a_4 < a_5 < a_6$. Then $n(S)$ is equal to
89. Six identical coins are arranged in a row. The number of ways in which the number of tails is equal to number of heads is
90. Entries of a 2×2 determinant are chosen from the set $\{-1, 1\}$. The probability that determinant has zero value is $\frac{1}{\lambda}$ then λ is

TG ~ @bohring_bot

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2023

TEST DATE: 20-01-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C

Sol. For the sector, $y_{cm} = \frac{4R \sin(\phi/2)}{3\phi} = \frac{2R}{\pi} \left(\because \phi = \frac{\pi}{3} \right)$

For the arc, $y_{cm} = \frac{2R \sin(\phi/2)}{\phi} = \frac{3R}{\pi}$

2. A

Sol. If acceleration of block 2 is a_2 to the right, then

$$5a_2 = a_r + a = 6 + 4$$

$$a_2 = 2 \text{ m/s}^2$$

3. C

Sol. $\alpha = \frac{\tau}{I} = \frac{FL}{\frac{mL^2}{3}} = \frac{3F}{m\ell} \Rightarrow a_{cm} = \frac{\ell}{2} \alpha = \frac{3F}{2m}$

$$\Sigma F_x = ma_{cm}$$

$$N_x + F = \frac{3F}{2} \Rightarrow N_x = \frac{F}{2}$$

$$N_y = Mg$$

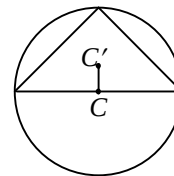
4. C

Sol. C.M. of triangle is at $\frac{a}{3}$ from centre of circular plate

$$m_{\text{triangle}} = \frac{m_{\text{circle}}}{\pi}$$

$$y_{CM} = \frac{M_y - m_y}{M - m}$$

$$\text{On solving, } \frac{a}{3(\pi - 1)}$$



5. C

Sol. $-\Delta U = \Delta K_e$

$$-[U_B - U_A] = \frac{1}{2}mv^2$$

Ball will leave when normal at B = 0

$$\therefore mg \sin \beta - N_B = \frac{mv_B^2}{R} \Rightarrow 0 \text{ at B} \Rightarrow mg \sin \beta = \frac{mv_B^2}{R}$$

6. D

Sol. Block number 1 and $2n$ will suffer one collision each, block number 2 and $2n - 1$ will suffer two collision each similarly block number n and $n + 1$ will suffer n collision each. So total number of collisions will be $2(1 + 2 + \dots + n) = n(n + 1)$

7. A

$$\text{Sol. } r = R \left(\frac{\ell_1}{\ell_2} - 1 \right) = 132.4 \left(\frac{70}{60} - 1 \right) \approx 22.1 \Omega$$

8. B

Sol. 100Ω , 25Ω and 20Ω are in parallel.

Their, net resistance is 10Ω

$$\therefore R_{\text{net}} = 4 \Omega + 10 \Omega + 6 \Omega = 20 \Omega$$

$$V = i R_{\text{net}} = 80 \text{ V}$$

9. A

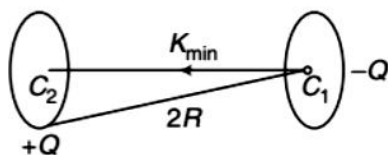
Sol. Current decreases $\frac{20}{30}$ times or $\frac{2}{3}$ times. Therefore, net resistance should become $\frac{2}{3}$ times.

$$\therefore R + 50 = \frac{3}{2}(2950 + 50)$$

Solving we get, $R = 4450 \Omega$

10. A

Sol.



$$K_{C_1} + U_{C_1} = K_{C_2} + U_{C_2}$$

$$K_{\text{min}} + qV_{C_1} = 0 + qV_{C_2} \quad \dots(i)$$

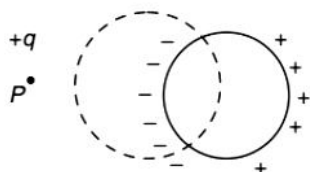
$$V_{C_1} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{2R} - \frac{Q}{R} \right)$$

$$V_{C_2} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{2} - \frac{Q}{2R} \right)$$

Substituting these values in Eq. (i), we can find $K_{\min}$.

11. C

Sol.



The induced charges on conducting sphere due to $+q$ charge at P are as shown in figure.

Now, net charge inside the closed dotted surface is negative. Hence, according to Gauss's theorem net flux is negative.

12. A

Sol. $p^2V = \text{constant}$

$$\therefore \left(\frac{nRT}{V} \right)^2 V = \text{constant}$$

$$\therefore T^2 \propto V$$

$$\text{or } T \propto \sqrt{V}$$

V is made three times. So, T will become $\sqrt{3}$ times.

13. A

Sol. $W_{\text{net}} = 2p_0V_0$

$$Q_{+ve} = Q_{ABC} = W_{ABC} + \Delta U_{ABC}$$

$$= \text{area under the graph} + nC_V\Delta T$$

$$= (3p_0V_0) + n \left(\frac{3}{2}R \right) (T_C - T_A)$$

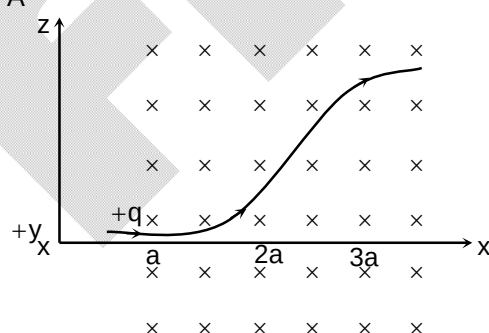
$$= (3p_0V_0) + \frac{3}{2}(p_CV_C - p_AV_A)$$

$$= 10.5 p_0V_0$$

$$\eta = \frac{W_{\text{net}}}{Q_{+ve}} = \frac{4}{21}$$

14. A

Sol.



15. C

Sol. Conceptual

16. A

$$\text{Sol. } \tan \phi = \frac{X_C}{R} \quad \frac{1}{\omega C} = R$$

$$\tan \frac{\pi}{4} = \frac{X_C}{R} \quad \frac{1}{\omega R} = C$$

$$X_C = R$$

i leads emf $\therefore$ circuit is LC

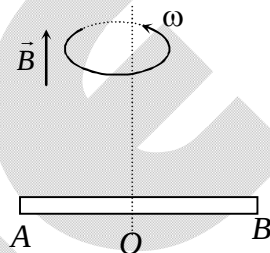
17. D

Sol. Potential difference between

$$O \text{ and } A \text{ is } V_A - V_O = \frac{1}{2} B l^2 \omega$$

$$O \text{ and } B \text{ is } V_B - V_O = \frac{1}{2} B l^2 \omega$$

$$\text{So } V_A - V_B = 0$$



18. B

$$\text{Sol. } \lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{E}}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}} \Rightarrow \frac{10^{-10}}{0.5 \times 10^{-10}} = \sqrt{\frac{E_2}{E_1}}$$

$$\Rightarrow E_2 = 4E_1$$

$$\text{Hence added energy} = E_2 - E_1 = 3E_1$$

19. A

Sol. Using Moseley's law, we get

$$\frac{\lambda_2}{\lambda_1} = \frac{(Z_1 - a)^2}{(Z_2 - a)^2}$$

$$\lambda_2 = \frac{200 \times (74 - 1)^2}{(78 - 1)^2} = 179.76 \text{ \AA}$$

20. C

$$\text{Sol. } {}^x\lambda = \frac{\ln 2}{t_{1/2}}$$

$${}^y\lambda = \frac{\ln 2}{t_{1/2}}$$

$$\frac{{}^x A}{{}^y A} = \frac{{}^x \lambda {}^x N_0 e^{-x\lambda t}}{{}^y \lambda {}^y N_0 e^{-y\lambda t}}$$

$$e^{\ln 2} = 2$$

SECTION - B

21. 2

 Sol. D_1 is reverse biased therefore it will act like an open circuit.

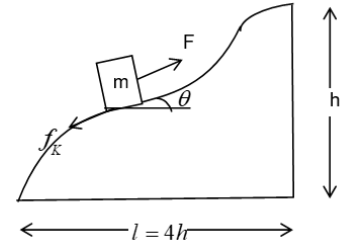
$$i = \frac{12}{6} = 2.00 \text{ A}$$

22. 3

Sol. Psueodo force and friction will make a force couple. Normal reaction and Mg will make a force couple.
When block is about to topple normal reaction will shift to edge.

23. 2

Sol. $\Delta W_f = -\int \mu mg \cos \theta ds$
 $= -\mu mg \int_0^l dx$
 $= -\mu mgl = 0.5mg4h = -2mgh$
 $\Delta W_g = -mgh$
 $\therefore \Delta W_{fr} / \Delta W_g = 2$



24. 5

Sol. $mg = kx$ (for m to leave the ground)
 $m_0gx = \frac{1}{2}kx^2$ or $m_0 = \frac{m}{2} = 5 \text{ kg}$

25. 4

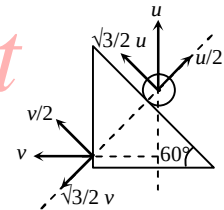
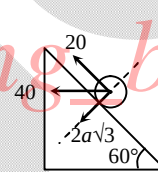
Sol. Conservation of momentum along horizontal

$$m40 = 6 \times v \quad \dots(i)$$

Newton's law of restitution

$$\frac{u}{2} + \frac{\sqrt{3}}{2}v = 20\sqrt{3} \quad \dots(ii)$$

Linear momentum of ball will remain conserved along inclined



Before Collision

$$\frac{\sqrt{3}}{2}u = 20$$

$$u = \frac{40}{\sqrt{3}} \quad \dots(iii)$$

From (i), (ii) and (iii)
 $m = 4 \text{ kg}$

26. 4

Sol. $P_{avg} = \frac{\epsilon_{10} i_0}{2} = \frac{80}{2} = 40$

$$P_{avg} = \frac{\text{Energy}}{\text{Time}}$$

27. 40

Sol. Here 3rd maxima is shifted by $3 \times 10^{-4} \text{ m}$. It indicates fringe width increases by $1 \times 10^{-4} \text{ m}$.

$$\text{Hence } \beta = \frac{\lambda(D+0.5)}{d} = \frac{\lambda D}{d} + 1 \times 10^{-4}$$

$$\text{or } \frac{0.5\lambda}{d} = 1 \times 10^{-4} \text{ m} \quad \text{or } \lambda = \frac{2 \times 10^{-3} \times 1 \times 10^{-4}}{0.5} = 4 \times 10^{-7} \text{ m} = 40 \text{ nm}$$

28. 30

Sol. $\lambda_{C_1} = \frac{hc}{eV} = \frac{12375}{10 \times 10^3} \text{Å} = 1.2375 \text{Å}$... (1)

$\lambda_{C_2} = 0.61875 \text{Å}$... (2)

$\frac{1}{\lambda_{K_\alpha}} = (Z-1)^2 \left\{ \frac{1}{1} - \frac{1}{4} \right\} \times 10^7$... (3)

It is given

$3 \times (\lambda_{K_\alpha} - 1.2375) = \lambda_{K_\alpha} - 0.61875$

$\therefore \lambda_{K_\alpha} = 1.54338 \text{Å}$... (4)

Putting this value in (3)

$Z-1 = 29$

$Z = 30$

29. 6

Sol. As source (horn of bus) is approaching stationary wall (say, listener), therefore, apparent frequency striking the wall is

$v' = \frac{v v}{v - v_s}$... (1)

Sound of this frequency will be reflected by the wall (now, source). The passenger is the listener moving towards source. Therefore, frequency heard by the listener $v'' = \frac{(v + v_L)v'}{v}$

using (1)

$$v'' = \frac{v + v_L}{v} \times \frac{v v}{v - v_s} = \frac{(v + v_L)v}{v - v_s}$$
$$= \frac{(330 + 5) \times 200}{330 - 5} = \frac{335}{325} \times 200$$

$v'' = 206 \text{Hz}$

$\therefore \text{Beat frequency} = (v'' - v)$

$= 206 - 200 = 6 \text{Hz}$

30. 3

Sol. Total heat product $= \frac{1}{2} CV^2$

$= \frac{1}{2} (2\mu\text{F})(5)^2$

$= 25 \mu\text{J}$

Now, this should distribute in inverse ratio of resistors, as they are in parallel.

$\therefore \frac{H_{5\Omega}}{H_R} = \frac{R}{5}$

or $H_{5\Omega} = \left(\frac{R}{R+5} \right) \text{ (Total heat)}$

or $10 = \left(\frac{R}{R+5} \right) (25)$

Solving this equation, we get

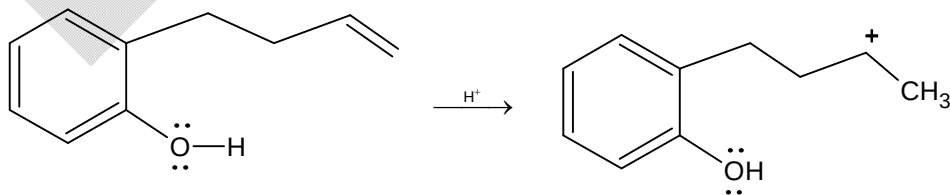
$R = \left(\frac{10}{3} \right) \Omega$

Chemistry

PART – B

SECTION – A

31. C
Sol. Intensity of incident radiations depends on no. of photons in the incident light and simultaneously no. of photoelectrons increases and photoelectric current increases
32. A
Sol. In all the three options groups are arranged at ortho position. So after that the value of dipole moment will depend on the electro negativity difference.
33. C
Sol. For helium is a very light gas for which attraction is ignored and do not have vanderwall constant 'a' in the equation of compressibility factor and for helium $Z > 1$.
34. A
Sol. $\text{Na}_2\text{CO}_3 \longrightarrow \text{NaHCO}_3 \quad x$
 $\text{NaHCO}_3 \longrightarrow \text{H}_2\text{CO}_3 \quad x$
 $\text{NaHCO}_3 \longrightarrow \text{H}_2\text{CO}_3 \quad y$
 With phenolphthalein as indication $10 \text{ mL} \times 1\text{N} \times 2$
 2 mill g. equation = x
 with methyl orange as indicator
 $10 \text{ mL} \times 0.2\text{N} \times 4 \text{ milli gram equation} = x + y$
 $y = 2 \text{ milli gram eq.}$
 $\text{Normality} = \frac{2 \text{ milligram eq.}}{20 \text{ mL}} = 0.1\text{N}$
 $N = nf \times \text{molarity } M = 0.05 \text{ M}$
35. D
Sol. For first order reaction, $t = \frac{2.303}{k} \log \frac{a}{a-x}$ $t_{99.9} = \frac{2.303}{k} \log \frac{100}{100-99.9} = \frac{2.303}{k} \times 3$
 $t_{50} = \frac{2.303}{k} \log \frac{2.303}{k} \log 2 = \frac{2.303}{k} \times 0.301 t_{99.9/t50} = 10 \text{ approx.}$
36. C
Sol. The cations will be liberated in the sequence of decreasing reduction potentials. Cations having E° value < -0.83 (reduction potential of water) will not be liberated from aqueous solutions
37. C
Sol. Iodine is a bulky group due to ortho effect mesomeric effect stops at ortho position & hence a and b will be longer than C bond
38. B
Sol.



More stable carbocation forms at this position

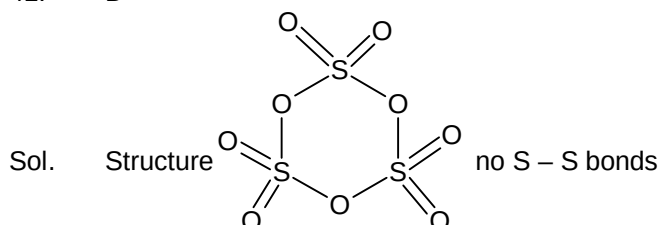
39. B

 Sol. Benzilic acid involves transformation of α -diketones to α -hydroxyl acid by means of OH^-

40. C

 Sol. CO_2 , in CO_2 carbon is already in its maximum oxidation state and cannot be oxidised further. While N, Cl, S are not in their maximum oxidation state.

41. D



42. B

Sol. $[\text{Co}(\text{NH}_3)_2\text{Cl}_4]^-$ = cis and trans isomerism
 $\text{AuCl}_2\text{Br}_2^-$ square planar cis and trans isomersim
 $[\text{Co}(\text{NO}_2)] [\text{NH}_3]^{2+}$ = No isomerism

43. C

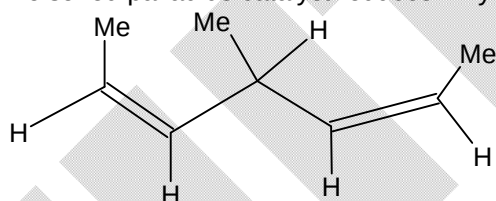
Sol. VBT is based on measurement of magnetic moment measurement. Based on that structure is being predicted.

44. B

Sol. Fehlings solution can separate reducing and non-reducing sugars.
 Glucose – Reducing sugar
 Sucrose – non-reducing sugar.

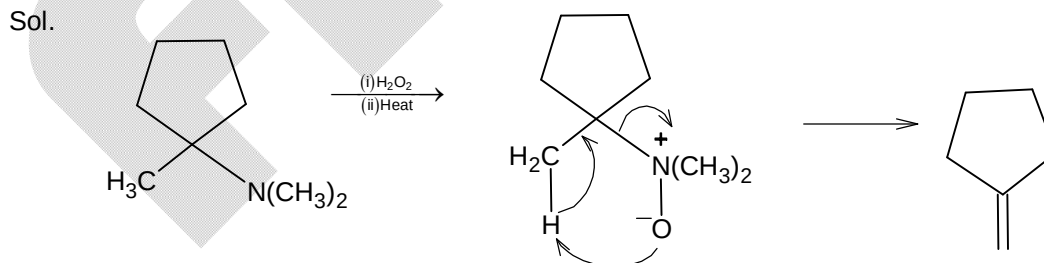
45. B

Sol. Poisoned palladium catalyst reduces Alkyne to cis-Alkene.



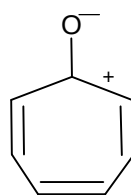
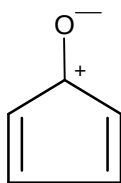
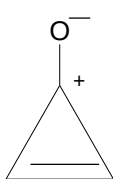
Optically inactive product

46. D



Less stable alkene is the major product

47. B
Sol.



Aromatic

Anti-aromatic

Aromatic

Anti-aromatic is the least stable

48. A

Sol. Partial pressure of $O_2 = \frac{2}{7} \times 2660 \text{ mm}$

Thus 1 L of O_2 is present at 0°C and 760 mm

So no. of O_2 molecules = $\frac{0.02 \times 10^{23}}{22.4}$

$PV = nRT$

No. of moles of $O_2 = \frac{1 \text{ atm} \times 1 \text{ L}}{0.0821^{-1} \text{ atm}} \times 273 \text{ K} = \frac{1}{22.4}$

49. C

Sol. They have different bond connectivity. Hence constitutional isomers.

50. C

Sol. Zone refining is based on the principal that impurities are more soluble in molten state than solid state.

Hence molten zone contains more impurities than the original metal.

SECTION - B

51. 25

Sol. $AB_2 \rightleftharpoons AB(g) + B(g)$

500

—

—

$500 - x$

x

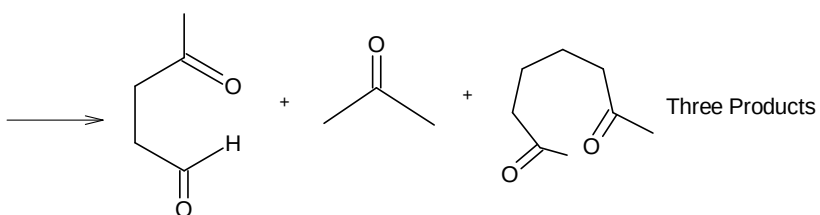
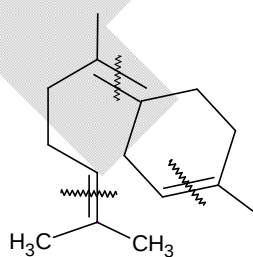
x

Require = 600 torr = $500 - x + x + x$, $x = 100 \text{ torr}$

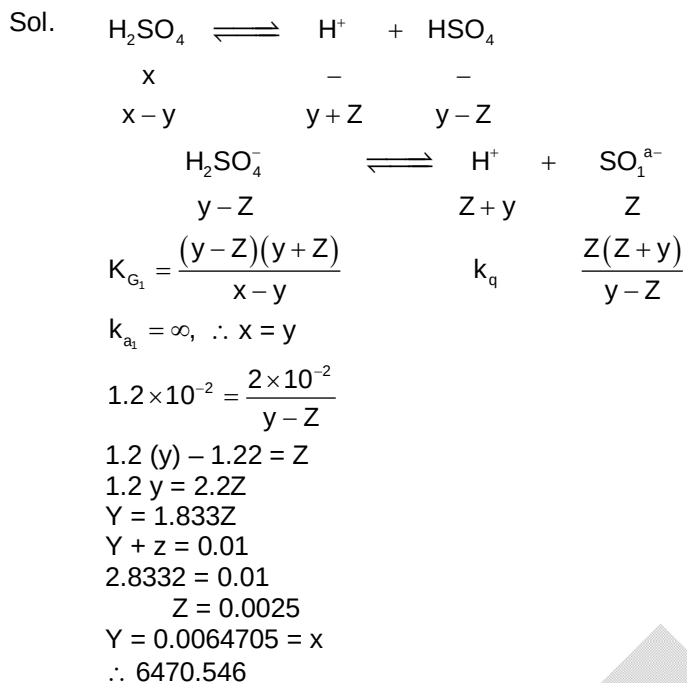
$K_p = \frac{P_{AB} \cdot P_B}{P_{AB_2}} = \frac{100 \times 100}{400} = 25 \text{ torr}$

52. 3

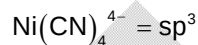
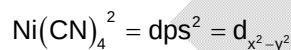
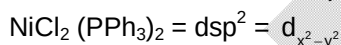
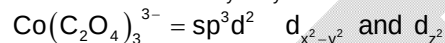
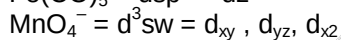
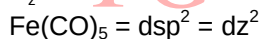
Sol.



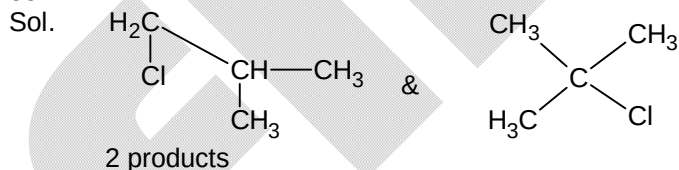
53. 6470



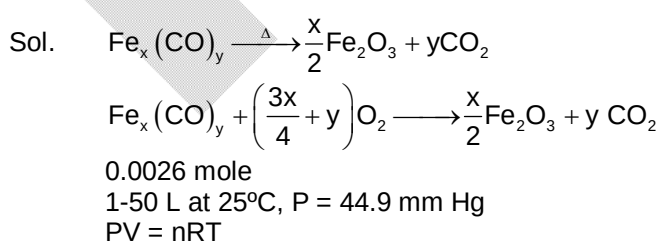
54. 2

 Sol. d_{z^2} orbital has no mola plane.


55. 2



56. 5



$$\frac{44.9}{760} \times 1.50L = n \times 0.0821 \times 298$$

$$= n = 0.036 \text{ mole}$$

$$0.142 = 56x + 0.0036 \times 28$$

$$x = \frac{0.0412}{56} = 0.00073$$

$$\frac{y}{x} = 4.9 \approx 5$$

57. 182

Sol. Fcc like structure has edge length

$$2r_+ + 2r_- = 520 \text{ ppm}$$

From radius ration

$$\frac{r^+}{r^-} = 0.414$$

$$2(0.414 r^-) + 2r^- = 520 \text{ pm}$$

$$2.85 r^- = 520$$

$$r^- = 182 \text{ pm}$$

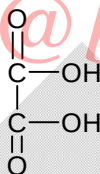
58. 84

Sol. Given wt. of complex = 3.2 gm

$$\text{Molecular wt. of complex} = 4M + 48 + 264$$

$$= 4M + 528 + 216$$

Molar mass of oxalic acid is



is 90 gm

$$\frac{3.2\text{g}}{4M + 528} \times = \frac{1\text{g}}{90}$$

$$M = 84 \text{ g/mol}$$

59. 2

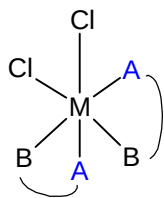
$$\text{Sol. Rate} = \underbrace{[4.7 \times 10^{-5}][R-x][OH^-]}_{S_N2} + \underbrace{0.24 \times 10^{-5}[RX]}_{S_N1}$$

$$\% \text{ of } S_N2 = \frac{4.7 \times 10^{-5}[RX] \times 10^{-3}M}{0.24 \times 10^{-5}[RX] + 4.7 \times 10^{-5}[RX]} \times 100$$

on solving answer = 2

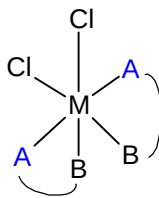
60. 8

Sol. $[\text{Pt}(\text{gly})_2\text{Cl}_2] = [\text{Pt}(\text{AB})_2\text{Cl}_2]$

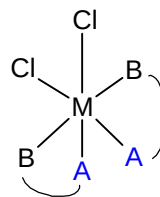


complete cis

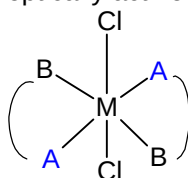
optically active = 2



optically active = 2



optically active = 2



optically inactive
optically inactive

& Cis - $[\text{Co}(\text{en})_2\text{Cl}_2]$ is optically active isomers = 2 while trans - $[\text{Co}(\text{en})\text{Cl}_2]$ is

TG ~ @bohring_bot

Mathematics

PART – C

SECTION – A

61. D

Sol. $\sin^{-1} \frac{|2 \times 3 + (-1) \times 6 + 2 \times (-2)|}{\sqrt{2^2 + (-1)^2 + 2^2} \cdot \sqrt{3^2 + 6^2 + (-2)^2}}$

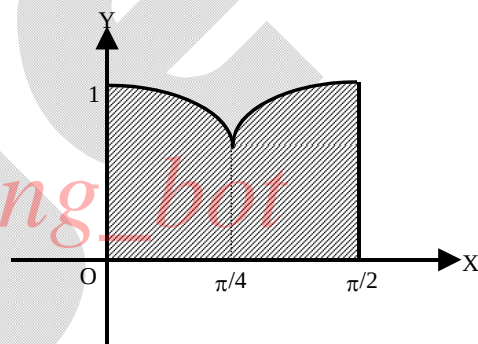
62. C

Sol. $\frac{dy}{dx} = \cos^{-1} x^4 x^2$
 $= -\cos^{-1} x^2$
 $= \frac{2\pi}{3} \times \frac{1}{2^{1/4}} - \frac{\pi}{4}$

63. A

Sol. $f(x) = \cos x$ for $0 \leq x \leq \pi/4$
 $= \sin x$ for $\pi/4 < x \leq \pi/2$

Required $= 2 \int_0^{\pi/4} \cos x dx = 2 \sin x \Big|_0^{\pi/4}$
 $= \sqrt{2}$ sq. units.



64. D

Sol. $(1 - x^4) (1 + {}^9C_1 x + {}^9C_2 x^2 + {}^9C_3 x^3 + \dots + {}^9C_9 x^9)$
 Coefficient of $x^7 = {}^9C_7 - {}^9C_3 = -48$

65. C

Sol. The chord of contact of tangents from (α, β) is
 $\alpha x + \beta y = 1$ (1)

Also, (α, β) lies on $2x + y = 4$, so that $2\alpha + \beta = 4$

$\Rightarrow \frac{\alpha}{2} + \frac{\beta}{4} = 1$

Hence, (1) passes through $\left(\frac{1}{2}, \frac{1}{4}\right)$.

66. C

Sol. Equation of line is $(2x + 3y + 4) + \lambda (6x - 3y + 12) = 0$... (1)
 (1) is normal to circle, therefore (1) passes through (2, 0)

$\Rightarrow \lambda = \frac{-1}{3}$ and required equation is $y = 0$.

67. A

Sol. $\Rightarrow 2y \frac{dy}{dx} x = -y^2 - \sin 2x$ $\Rightarrow y^2 + 2yx \frac{dy}{dx} = -\sin 2x$

$$\Rightarrow \frac{d}{dx}(xy^2) = \frac{d}{dx}(\cos^2 x) \quad \Rightarrow xy^2 = \cos^2 x + c$$

68. D

$$\text{Sol.} \quad \begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$$

Operating $C_1 \rightarrow C_1 - C_2$

$$\begin{vmatrix} 4a^x a^{-x} & (a^x - a^{-x})^2 & 1 \\ 4b^y b^{-y} & (b^y - b^{-y})^2 & 1 \\ 4c^z c^{-z} & (c^z - c^{-z})^2 & 1 \end{vmatrix} = 4 \begin{vmatrix} 1 & (a^x - a^{-x})^2 & 1 \\ 1 & (b^y - b^{-y})^2 & 1 \\ 1 & (c^z - c^{-z})^2 & 1 \end{vmatrix} = 0.$$

69. C

$$\text{Sol.} \quad 0 < \frac{3}{x^2 + 1} \leq 3, \quad \frac{3}{x^2 + 1} = 2 \Rightarrow x = \frac{1}{\sqrt{2}} \quad \text{and} \quad \frac{3}{x^2 + 1} = 1 \Rightarrow x = \sqrt{2}$$

$$I = \int_0^{1/\sqrt{2}} 2dx + \int_{1/\sqrt{2}}^{\sqrt{2}} 1 \cdot dx + \int_{\sqrt{2}}^{\infty} 0 \cdot dx = \sqrt{2} + \sqrt{2} - \frac{1}{\sqrt{2}} = 2\sqrt{2} - \frac{1}{\sqrt{2}} = \frac{3}{\sqrt{2}}$$

70. C

$$\text{Sol.} \quad \int \frac{\ln x}{(1 + \ln x)^2} dx$$

$$\text{Let } \ln x = t \Rightarrow dx = e^t dt$$

$$\int \frac{te^t}{(1+t)^2} dt = \int \frac{e^t(t+1-1)}{(1+t)^2} dt$$

$$= \int \frac{e^t(t+1-1)}{(1+t)^2} dt$$

$$= \int e^t \left\{ \frac{1}{t+1} - \frac{1}{(1+t)^2} \right\} dt$$

$$= \frac{e^t}{t+1} + c \quad \text{since} \quad \int e^x(f(x) + f'(x))dx = e^x f(x) + c$$

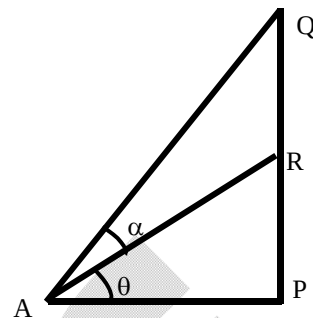
$$= \frac{e^{\ln x}}{1 + \ln x} + c = \frac{x}{1 + \ln x} + c.$$

71. A

$$\text{Sol.} \quad f(x) = \log_{1/2} \log_4 \log_3 [(x-4)^2] \Rightarrow \log_4 (\log_3 [(x-4)^2]) > 0 \Rightarrow \log_3 [(x-4)^2] > 1 \\ \Rightarrow [(x-4)^2] \geq 4 \Rightarrow (x-4)^2 - 4 \geq 0 \Rightarrow (x-2)(x-6) \geq 0 \\ \Rightarrow x \in (-\infty, 2] \cup [6, \infty).$$

72. A

Sol. $\tan(\theta + \alpha) = \frac{PQ}{AP} = \frac{1}{n}$
 $\tan \theta = \frac{PR}{AP} = \frac{(1/2)PQ}{AP} = \frac{1}{2n}$
 $\alpha = (\theta + \alpha) - \theta$
 $\Rightarrow \tan \alpha = \frac{\tan(\theta + \alpha) - \tan \theta}{1 + \tan(\theta + \alpha)\tan \theta}$
 $= \frac{\frac{1}{n} - \frac{1}{2n}}{1 + \frac{1}{n} \cdot \frac{1}{2n}} = \frac{n}{2n^2 + 1}$



73. C

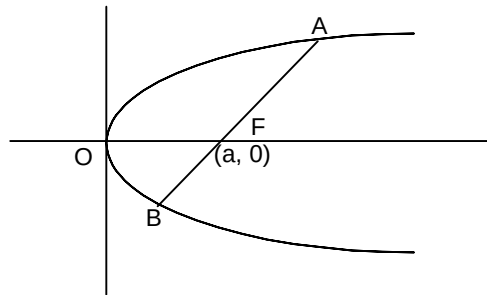
Sol. Normal to hyperbola will be
 $ax \cos \theta + by \cot \theta = a^2 + b^2$
 If this is same as $lx + my + n = 0$, comparing the coefficients,
 $\frac{a \cos \theta}{l} = \frac{b \cot \theta}{m} = \frac{a^2 + b^2}{-n}$
 $\sec \theta = -\frac{an}{l(a^2 + b^2)}, \tan \theta = \frac{-bn}{m(a^2 + b^2)}$
 $\sec^2 \theta - \tan^2 \theta = 1$
 $\Rightarrow \frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$

74. A

Sol. Any point on the parabola is $(x, x^2 + 7x + 2)$
 Its distance from the line $y = 3x - 3$ is given by
 $P = \frac{|3x - (x^2 + 7x + 2) - 3|}{\sqrt{9 + 1}} = \frac{|x^2 + 4x + 5|}{\sqrt{10}} = \frac{x^2 + 4x + 5}{\sqrt{10}}$ (as $x^2 + 4x + 5 > 0$ for all $x \in \mathbb{R}$)
 $\frac{dP}{dx} = 0 \Rightarrow x = -2$
 The required point is $(-2, -8)$.

75. A

Sol. $FA = 4, FB = 5$
 We know that $\frac{1}{a} = \frac{1}{AF} + \frac{1}{FB}$
 $\Rightarrow a = \frac{20}{9} \Rightarrow 4a = \frac{80}{9}$



76. D

$$\text{Sol. } \frac{n}{4n^4 + 1} = \frac{1}{4} \left[\frac{1}{2n^2 - 2n + 1} - \frac{1}{2n^2 + 2n + 1} \right]$$

$$\text{Hence } \sum_{n=1}^k \frac{n}{4n^4 + 1} = \frac{1}{4} \left[1 - \frac{1}{2k^2 + 2k + 1} \right] = \frac{1}{4} \text{ as } k \rightarrow \infty$$

77. C

$$\text{Sol. } S - S_n < \frac{1}{300}$$

$$\Rightarrow \frac{1}{1 - \frac{1}{3}} - \frac{1 - \frac{1}{3^n}}{1 - \frac{1}{3}} < \frac{1}{300} \Rightarrow \frac{3}{2} \left[1 - 1 + \frac{1}{3^n} \right] < \frac{1}{300}$$

$$\Rightarrow \frac{1}{3^n} < \frac{1}{300} \times \frac{2}{3} \Rightarrow \frac{1}{3^n} < \frac{1}{450} \Rightarrow 3^n > 450$$

$\therefore$ least value of $n = 6$

78. B

$$\text{Sol. Let } f(x) = x^3 - 3x + a$$

$$f'(x) = 3x^2 - 3.$$

For three distinct real roots (i) $f'(x) = 0$ should have two distinct real roots α and β and (ii) $f(\alpha)f(\beta) < 0$

Here $\alpha = 1, \beta = -1$.

Now $f(\alpha)f(\beta) < 0$

$$\Rightarrow (1 - 3 + a)(-1 + 3 + a) < 0 \Rightarrow (a - 2)(a + 2) < 0$$

$$\Rightarrow -2 < a < 2.$$

79. C

$$\text{Sol. The line } y = \sqrt{3}x \text{ can be written as } x = \frac{r}{2}, y = \frac{r\sqrt{3}}{2}.$$

If this line cuts the given curve, then

$$\frac{r^4}{16} + \frac{ar^3\sqrt{3}}{8} + \frac{br^2\sqrt{3}}{4} + \frac{cr}{2} + \frac{dr\sqrt{3}}{2} + 6 = 0.$$

$$\text{Therefore } OA \cdot OB \cdot OC \cdot OD = |r_1| |r_2| |r_3| |r_4| = |r_1 r_2 r_3 r_4| = 96$$

80. C

$$\text{Sol. Plane passing through } (2, 2, 1) \text{ is } a(x - 2) + b(y - 2) + c(z - 1) = 0$$

It passes through $(9, 3, 6)$

$$\Rightarrow 7a + b + 5c = 0 \quad \dots (1)$$

It is $\perp$ to $2x + 6y + 6z - 1 = 0$

$$\Rightarrow 2a + 6b + 6c = 0 \quad \dots (2)$$

$$(1) \text{ and } (2) \Rightarrow \frac{a}{-24} = \frac{b}{-32} = \frac{c}{40}$$

$$\Rightarrow \frac{a}{3} = \frac{b}{4} = \frac{c}{-5}$$

$\Rightarrow$ The required plane is

$$3(x - 2) + 4(y - 2) - 5(z - 1) = 0$$

$$\Rightarrow 3x + 4y - 5z - 9 = 0.$$

SECTION – B

81. 9

Sol. S. D =
$$\frac{\begin{vmatrix} 10 & 2 & 3 \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix}}{\sqrt{8^2 + 8^2 + 4^2}} = \frac{108}{12} = 9$$

82. 0

Sol. Let $f(x) = x^3 + 2x^2 + 5x + 2 \cos x$
 $\Rightarrow f'(x) = 3x^2 + 4x + 5 - 2 \sin x$
 $= 3\left(x + \frac{2}{3}\right)^2 + \frac{11}{3} - 2 \sin x$
 Now $\frac{11}{3} - 2 \sin x > 0 \quad \forall x \quad (\text{as } -1 \leq \sin x \leq 1)$
 $\Rightarrow f'(x) > 0 \quad \forall x \Rightarrow f(x)$ is an increasing function.
 Now $f(0) = 2$
 $\Rightarrow f(x) = 0$ has no solution in $[0, 2\pi]$.

83. 5

Sol. $T_2 = {}^nC_1 ab^{n-1} = 135 \quad \dots (1)$
 $T_3 = {}^nC_2 a^2 b^{n-2} = 30 \quad \dots (2)$
 $T_4 = {}^nC_3 a^3 b^{n-3} = 10/3 \quad \dots (3)$
 Dividing (1) by (2)

$$\frac{{}^nC_1 ab^{n-1}}{{}^nC_2 a^2 b^{n-2}} = \frac{135}{30}$$

$$\frac{n}{2} \cdot \frac{b}{a} = \frac{9}{2}$$

$$\frac{b}{a} = \frac{9}{n-1} \quad \dots (5)$$
 Dividing (2) by (3)

$$\frac{{}^nC_2 a^2 b^{n-2}}{{}^nC_3 a^3 b^{n-3}} = \frac{30}{10/3}$$

$$\frac{2}{n(n-1)(n-2)} \cdot \frac{b}{a} = 9$$

$$\therefore \frac{b}{a} = 3(n-2) \quad \dots (6)$$
 Eliminating a and b from (5) and (6)
 $\Rightarrow n = 5$

84. 4

Sol. $k = |z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2 \Rightarrow k = 4$

85. 8

Sol. $\frac{x^2}{4} + \frac{y^2}{3} = 1 \Rightarrow a = 2, b = \sqrt{3} \text{ and } e = \frac{1}{2}$

$$\text{Sum of distances} = \frac{2a}{e} = \frac{4}{1/2} = 8.$$

86. 120

Sol. $g(x) = g(x-1+1) = g(x-1)g(1)$
 $= g(x-2)[g(1)]^2 = \dots = [g(1)]^x = 3^x$
 $\Rightarrow \sum_{x=1}^4 3^x = 3^1 + 3^2 + 3^3 + 3^4 = 120$

87. 0

Sol. $\lim_{x \rightarrow 0} \left[f(x) + \log \left(1 - \frac{1}{e^{f(x)}} \right) - \log(f(x)) \right]$

$$= \lim_{x \rightarrow 0} \left[\log(e^{f(x)}) + \log \left(\frac{1 - \frac{1}{e^{f(x)}}}{f(x)} \right) \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \log \left(\frac{e^{f(x)} + 1}{f(x)} \right) = 0 \Rightarrow \lim_{x \rightarrow 0} \left(\frac{e^{f(x)} - 1}{f(x)} \right) = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) = 0 \Rightarrow f(0) = 0 \quad (\text{as } f(x) \text{ is a continuous function}).$$

88. 2100

Sol. First, 6 distinct digits can be selected in ${}^{10}C_6$ ways. Now the position of smallest digit in them is fixed i.e. position 4. Of the remaining 5 digits, two digits can be selected in 5C_2 ways. These two digits can be placed to the right of 4th position in one way only. The remaining three digits to the left of 4th position are in the required order automatically.
 So $n(S) = {}^{10}C_6 \times {}^5C_2 = 210 \times 10 = 2100$.

89. 20

Sol. Required number of ways = $\frac{6!}{3!3!} = \frac{720}{6 \times 6} = 20$

90. 2

Sol. $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc = 0 \Rightarrow$ whether $ad = 1, bc = 1$ or $ad = -1, bc = -1$

which occur in eight ways. Total number of 2×2 determinants from $\{-1, 1\}$ is 16. Thus required probability is $\frac{8}{16} = \frac{1}{2}$.

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
OPEN TEST – I
PAPER –1
TEST DATE: 05-02-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 24)

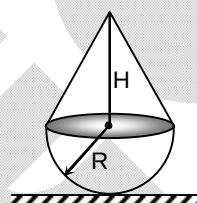
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. Two infinite long thin wire have linear charge density $-\lambda$ and $+\lambda$. They are parallel to z-axis passing through x-axis at point $x = -a$ and $x = +a$ respectively. The equipotential surface having potential $\frac{\lambda \ln(2)}{4\pi\epsilon_0}$ is

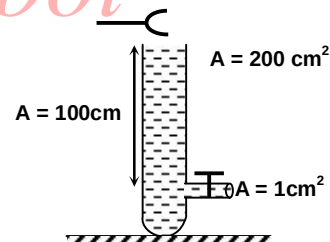
- (A) a sphere
(B) A cylinder
(C) having radius of $\sqrt{2}a$
(D) having radius $2\sqrt{2}a$

2. In the shown figure a solid hemisphere of radius R and a cone of radius height H is welded. The object formed is placed on rough horizontal surface. Density of material is same for hemisphere and cone. If it is slightly displaced from its equilibrium position. The correct statements are/is

- (A) It will oscillate if $H = R$
(B) It will have finite time period of $H = 2R$
(C) It will have finite time period if $H \leq R\sqrt{3}$
(D) It will not oscillate if $H = R\sqrt{2}$



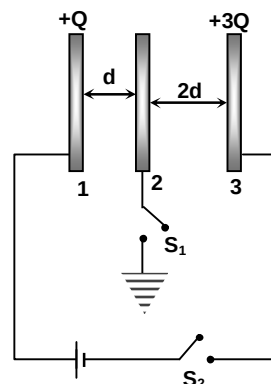
3. A tuning fork of frequency 1000 Hz is sounded over Resonance tube as shown in the figure. Speed of sound is $v = 320$ m/s and area of cross-section of tube is 200 cm^2 at the level of 100 cm a water outlet is attached. The water outlet has area of cross-section $a = 1 \text{ cm}^2$ water outlet is opened at time $t = 0$. Neglect the end effect of resonance tube.



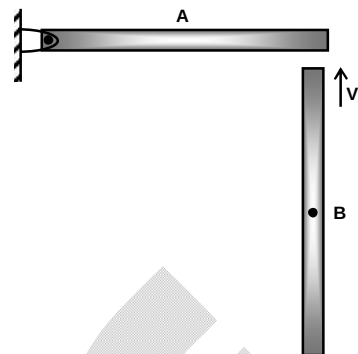
- (A) First resonance will occur when water level goes down by 8 cm from open end of tube.
(B) First resonance will occur when water level is 16 cm below open end of tube
(C) The time after which resonance occurs first time is nearly half minutes.
(D) The time between first two resonances is nearly 3 minutes.

4. Three conducting sheets 1, 2 and 3 are placed parallel to each other such that separation between them is negligible. Sheet-1 and sheet-3 are given charges $+Q$ and $+3Q$ respectively. Cell of emf 1 V is to be connected between sheet 1 and 3. (Take $\frac{\epsilon_0 A}{d} = 30 \text{ mF}$ and $Q = 3 \text{ mC}$) where A is area of sheets. First switch S_1 is closed and then switch S_2 is closed.

- (A) When switch S_1 is closed charge flow through S_1 is zero.
(B) When switch S_1 is closed charge flow through S_1 is 12 mC.
(C) When S_2 is closed charge flow through S_2 is 2 mC.
(D) When S_2 is closed charge flow through S_2 is 15 mC.

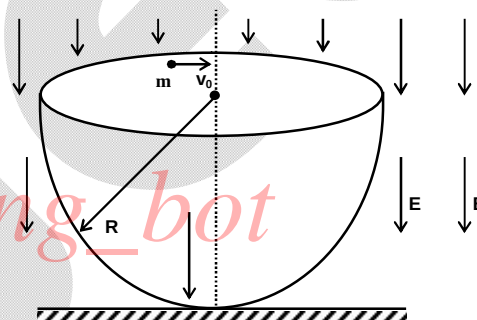


5. In figure shown rod A of mass $2m$ and length 2ℓ is kept on a horizontal frictional surface and hinged at point O such that it can rotate about vertical axis passing through O. Another identical rod B moving with a constant velocity v_0 collides with rod A at its end and sticks to it (both rods are perpendicular to each other)



- (A) angular velocity of system just after collision is $\frac{3v_0}{10\ell}$
- (B) Velocity of centre of mass at system just after collision is $9v_0\sqrt{\frac{2}{5}}$
- (C) Centre of mass of system is at a distance of $\sqrt{\frac{5}{2}}\ell$ from O.
- (D) Kinetic energy of system just after collision is $\frac{3mv_0^2}{5}$

6. A particle of mass m , charge q is given horizontal velocity v_0 at the rim of smooth, fixed hemispherical shell. The particle moves down and always is in contact with the inner surface of shell uniform electric field E_0 and magnetic field B_0 is present in vertical direction. The correct statements are:

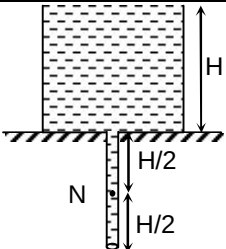


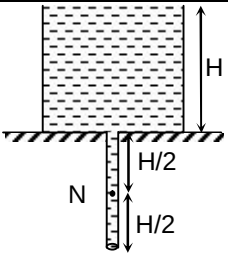
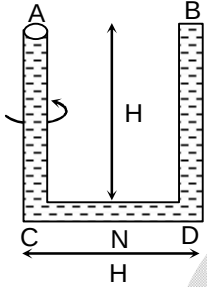
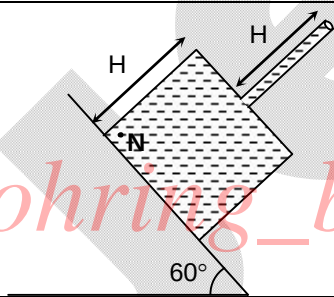
- (A) The particle moves down and then it starts moving upwards.
- (B) Conservation of energy can be applied.
- (C) Angular momentum of the particle along the vertical axis passing through centre of shell is conserved
- (D) Due to presence of torque. Angular momentum is not conserved

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

7. In List-I liquid of uniform density is filled in container and in the List-II gauge pressure at point N is given

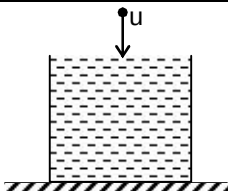
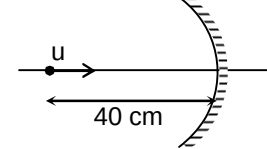
List -I			List -II	
(I)	Wide container of height H and a pipe of length H is shown in the figure. The end of the pipe is closed		(P)	ρgH

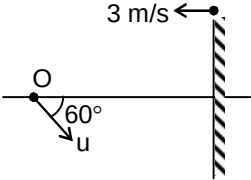
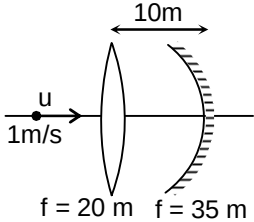
(II)	Wide container of height H is connected with a pipe of small area of cross section. Liquid is allowed to flow out from pipe. Assume Bernoulli's theorem is valid		(Q)	$\frac{5}{4}\rho gH$
(III)	ABCD is U shape tube in which liquid is filled fully and end B is closed with help of cap, while end A is open. It is rotated about AC with angular speed $\omega = \sqrt{\frac{2g}{H}}$. Point N is at mid point of horizontal part of the tube.		(R)	$\frac{-\rho gH}{2}$
(IV)	A thin pipe of height H , connected with a wide vessel of height H filled with liquid is released from rest on smooth inclined plane. Point N is at the edge point of bottom of vessel.		(S)	$\frac{\rho gH}{2}$
			(T)	$\frac{3}{2}\rho gH$

Which one of the following options is correct ?

- (A) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ T
 (B) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ Q, IV $\rightarrow$ P
 (C) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ R
 (D) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ R

8. In List-I motion of a particle is shown and in List-II relative speed between object and its image is given.

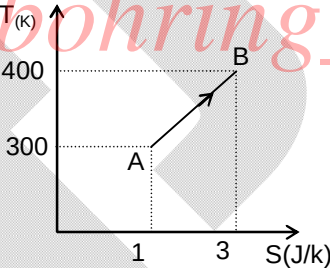
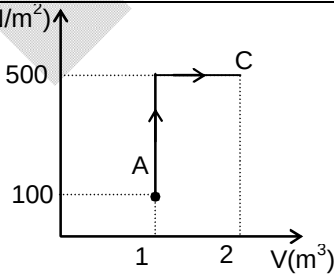
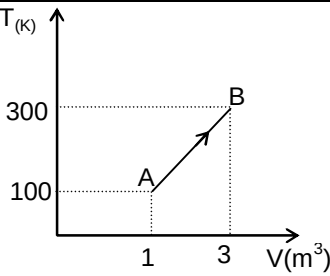
List -I			List -II
(I)	Object is moving towards water with velocity 3 m/s. Refractive index of water is $4/3$		(P) 10 m/s
(II)	Object O is moving towards concave mirror focal length 10 cm with velocity $u = 9$ m/s		(Q) 1 m/s

(III)	Object is moving towards plane mirror with velocity $u = 2$ m/s and mirror is also moving with velocity 3 m/s as shown in the figure.		(R)	0
(IV)	Observer O is moving toward lens with speed 1 m/s, which is at 15 m away from it. The separation between lens ($f = 20$ m) and mirror ($f = 35$ m) is 10 m.		(S)	8 m/s
			(T)	15 m

Which one of the following options is correct ?

- (A) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ S
 (B) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ R
 (C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ T
 (D) I $\rightarrow$ P, II $\rightarrow$ S, III $\rightarrow$ T, IV $\rightarrow$ R

9. In List-I, difference thermodynamics process are given and in List-II magnitude of possible heat exchange between system and surrounding are given due the process.

List-I		List-II	
(I)	2 mole monoatomic ideal gas is taken through process A $\rightarrow$ B which is shown in temperature (T) and entropy(s) diagram		(P) 1850J
(II)	2 mole ideal monoatomic gas is taken through process A $\rightarrow$ B $\rightarrow$ C as shown in P-V diagram		(Q) 8314J
(III)	Two mole monoatomic ideal gas is taken through the process A $\rightarrow$ B as shown in the T-V diagram		(R) 333J

(IV)	One mole monoatomic ideal gas is taken through the process ABC as shown in the figure, with the help of internal energy U and pressure P of the gas		(S)	400J
			(T)	700J

Which one of the following options is correct ?

- (A) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ Q
 (B) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ P, IV $\rightarrow$ Q
 (C) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ R
 (D) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ S

10. In List-I a body is given which can oscillate if slightly disturbed from its equilibrium position. In List-II time period of oscillation is given.

List -I			List -II	
(I)	A thin hemispherical shell of radius R is supported on the end of thin fixed vertical rod		(P)	$2\pi\sqrt{\frac{26R}{15g}}$
(II)	A thin semicircular ring of radius R is hanging from point O which is accelerating in horizontal direction with some acceleration so that the line joining two ends of the semicircular ring is vertical.		(Q)	$2\pi\sqrt{\frac{R}{g\left(1+\frac{4}{\pi^2}\right)}}$
(III)	Solid hemisphere of radius R is placed on rough sufficiently rough horizontal surface to support rolling.		(R)	$2\pi\sqrt{\frac{2R}{g\left(1+\frac{4}{\pi^2}\right)}}$
(IV)	A solid sphere of radius R is placed inside a fixed concave surface of radius $r = 6R$ friction is large between sphere and concave surface to support rolling.		(S)	$2\pi\sqrt{\frac{4R}{3g}}$
			(T)	$2\pi\sqrt{\frac{7R}{g}}$

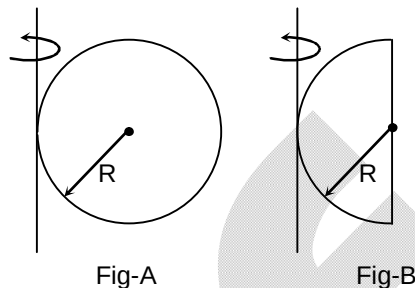
Which one of the following options is correct ?

- (A) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ P
 (B) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ Q
 (C) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ P
 (D) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ P, IV $\rightarrow$ T

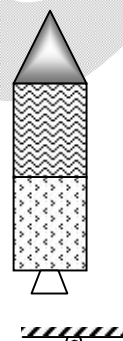
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

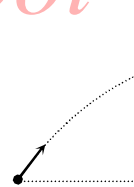
11. A circular metal disc has some moment of inertia about it tangent passing in its plane as shown in figure-A. Now half of the disc is folded over the other half as shown in the figure. B. Find the fractional change in moment of inertia about the same axis rotation (use value of $\pi = 3.14$)



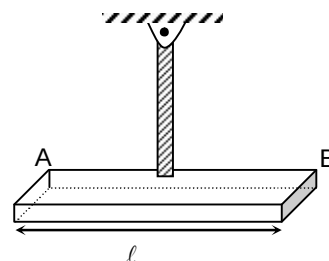
12. In year 2050 humans on earth as mastered in fusion reactor technology and due to atomic war possibility some of the people with all genetic material go on space journey for safer planet. Total mass of rocket system is 300000 kg thrust force is provided by radiation coming out of rocket system. Find the minimum rate of fuel consumption (in kg/s) so that rocket system can leave the earth. (Assume all radiation is coming out of nozzle of the rocket and it is in the form of parallel beam)



13. A thin rod of mass $m = 1\text{ kg}$ length $\ell = 20\text{ cm}$ is hanging from a high point. The point of suspension is capable of bearing only 10 N force. A particle of mass $m' = 1\text{ kg}$ is thrown from ground with velocity $v_0 = 20\text{ m/s}$ at angle $\theta = 45^\circ$ from horizontal. It collides with rod at its lowest point and sticks to it. The collision takes place when particle was moving horizontally. Find the number of rotation completed by rod before it falls on the ground. (Take value of $g = 10\text{ m/s}^2$ and $\pi = 3.14$)



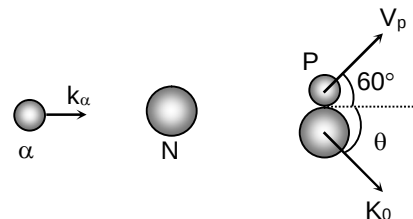
14. A bar AB of mass $m = 10\text{ kg}$ and length $\ell = 10\text{ m}$ is glued with a massless rod of length $L = 20\text{ m}$. The rod is free to rotate in vertical plane about horizontal axis passing at end O. An insect of mass $m = 0.02\text{ kg}$ lands smoothly on the bar at end A and runs towards end B such that the rod remains vertical. Find the time taken by insect to reach at end B in seconds.



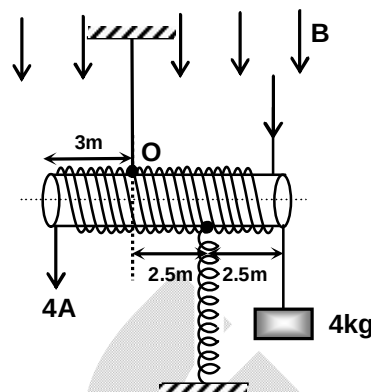
15. Consider the nuclear reaction, in which α particle collides with stationary nitrogen nucleus



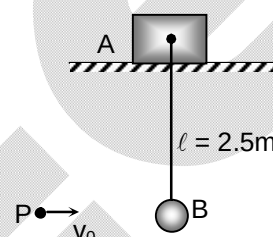
in which incoming α -particle has kinetic energy $k_\alpha = 4.0\text{ MeV}$ and outgoing proton going at angle 60° from incoming direction of α -particle has kinetic energy $k_p = 2.09\text{ MeV}$. The Q value of the reaction in MeV is



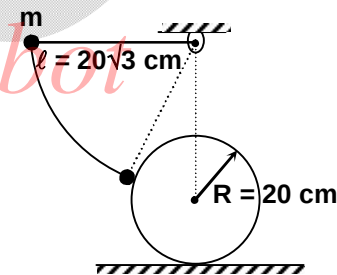
16. The figure shows a massless rod having current carrying coil of cross section area 10^{-2} m^2 , current $I = 4 \text{ A}$ and 20 turns. A spring of force constant $k = 100 \text{ N/m}$ is connected to rod at 2.5 m from point O. The other end of spring is connected to ground. Initially the spring is in relaxed state. At the end of rod a block of mass 4 kg is attached. When magnetic field of 5 T is switched on. The spring is extended. Find extension in the spring in steady state in meter (for small angle $\sin\theta \approx \theta$).



17. A block with horizontal nail has total mass 5 kg , can move on smooth horizontal surface. A small sphere B of mass 4 kg is hanging from the nail with the help of thin, mass less string of length $\ell = 2.5 \text{ m}$. A small particle of mass $m = 1 \text{ kg}$ moving horizontal with velocity v_0 , hits the block B and get stuck with it. The minimum value of v_0 (in m/s) such that block B can complete circular motion about nail in vertical plane is $N \times 10^2 \text{ m/s}$. The value of N is



18. A pendulum of mass m connected with string of length $\ell = 20\sqrt{3} \text{ cm}$. It is suspended above the centre of a sphere of radius 20 cm and mass m , at a distance 40 cm above centre of sphere. After collision pendulum stops. All surfaces are smooth. The coefficient of restitution is



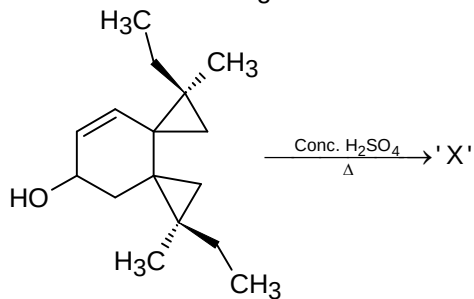
Chemistry

PART – II

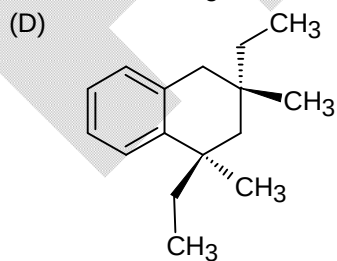
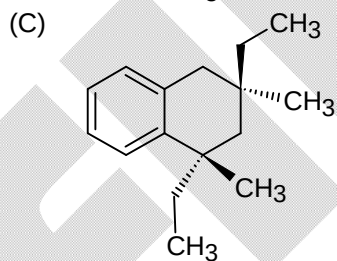
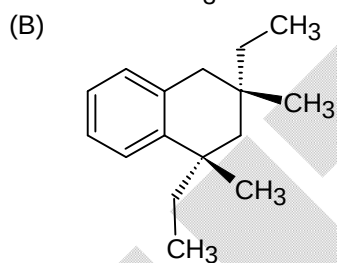
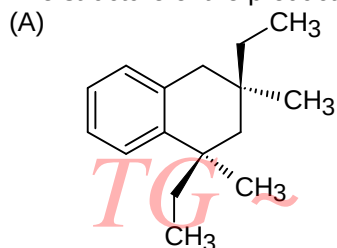
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

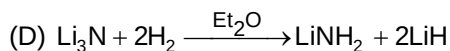
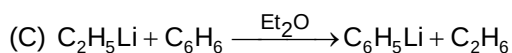
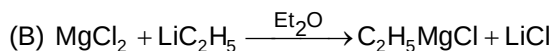
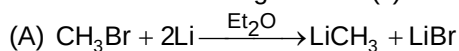
19. Consider the following reaction:



The structure of the product can be:



20. Which of the following reaction(s) can take place?



21. What happens when beryllium oxide is added to baryta water solution?

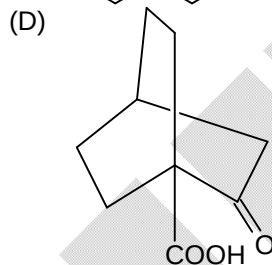
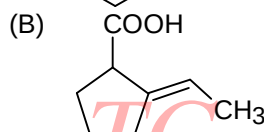
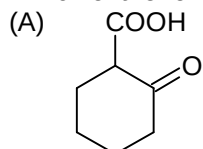
(A) BeO is insoluble in baryta water.

(B) BeO dissolves in baryta water.

(C) Basic character of the solution increases as the concentration of basic substance increases.

(D) pH of the solution decreases.

22. Which of the following compound(s) lose(s) CO_2 just on heating?



23. Which of the following statement(s) is/are correct?

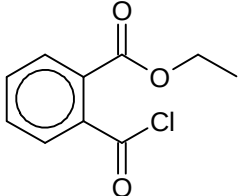
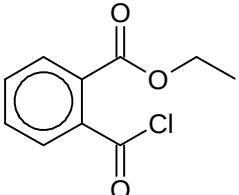
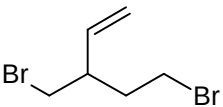
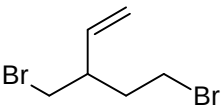
(A) Eight Cs^+ ions occupy the second nearest neighbour locations of a Cs^+ ion in CsCl crystal.

(B) Each sphere is surrounded by six voids in two dimensional hexagonal close packed layers.

(C) If the radius of cation and the anion are $0.3 \text{ \AA}$ and $0.4 \text{ \AA}$ respectively, then coordination number of the cation of crystal is 6.

(D) In AgCl, the silver ion is displaced from its lattice positions to an interstitial positions, such a defect is called Frankel defect.

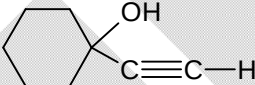
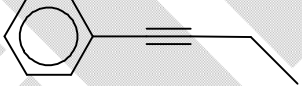
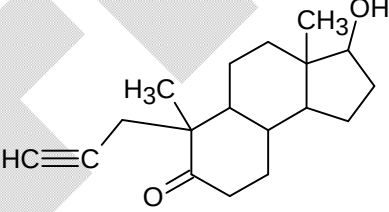
24. Which of the following is/are correct matched structure and IUPAC name?

- (A)  : 2-Carboxyethyl benzoyl chloride
- (B)  : Ethyl-2-(chloro-carbonyl) benzoate
- (C)  : 5-Bromo-3-(bromomethyl) pent-1-ene
- (D)  : 1, 4-dibromo-3-ethyl butane

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. Match the compounds in List-I with their characteristics in List-II and choose the answer from the codes given below:

List – I		List – II	
(I)		(P)	Reacts with Lindlar's catalyst ($H_2 / Pd - BaSO_4$).
(II)		(Q)	Trans alkene will form, when reacts with $Na / liq. NH_3$.
(III)		(R)	Produces hydrogen gas by means of sodium suspended in n-hexane.
(IV)		(S)	Degree of unsaturation is greater than two.
		(T)	Gives a white precipitate on reaction with $[Ag(NH_3)_2]NO_3$.

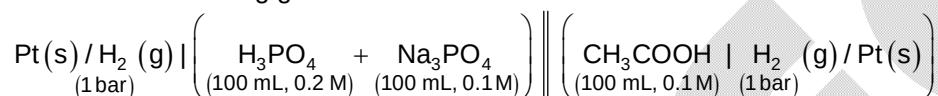
- (A) I→P, Q, S; II→P, Q, S, T; III→P, S, T; IV→Q, R, S, T
 (B) I→P, Q, S; II→P, R, S, T; III→P, Q, S; IV→P, R, S, T
 (C) I→P, Q, S, T; II→Q, R, S, T; III→P, Q, S; IV→Q, S, T
 (D) I→P, Q, S; II→P, R, S, T; III→Q, R, S; IV→P, R, S, T

26. Match the compounds given in List-I with their corresponding characteristics in List-II and choose the correct option from the codes given below:

List – I		List – II	
(I)	AlCl_3	(P)	Lewis acid
(II)	BCl_3	(Q)	Forms acidic solution in water
(III)	B_2H_6	(R)	Sublimes on heating
(IV)	Alum	(S)	Swells on heating
		(T)	Empty orbital is involved in hybridisation

- (A) I→P, Q; II→P, Q; III→P, Q, T; IV→Q, S
 (B) I→P, Q, R; II→P, Q, T; III→P, Q, T; IV→Q, S, T
 (C) I→Q, R, T; II→P, Q, T; III→P, Q; IV→Q, S
 (D) I→P, Q, R; II→P, Q; III→P, Q, T; IV→Q, S

27. Consider the following galvanic cell at 25°C.



For H_3PO_4 : $K_{a_1} = 10^{-4}$, $K_{a_2} = 10^{-8}$, $K_{a_3} = 10^{-12}$ at 25°C

For CH_3COOH : $K_a = 1 \times 10^{-5}$ at 25°C

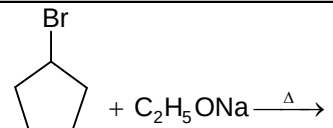
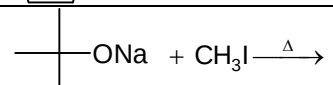
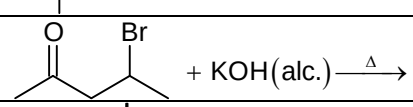
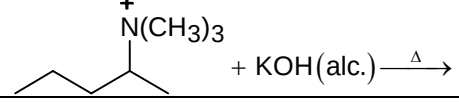
Now, some electrolyte is added in one of the compartment of the given cell in List-I and corresponding cell potential is given in List – II. Choose the correct option(s) from the codes given below:

[Take $\frac{2.303 RT}{F} = 0.06 \text{ V}$, $\log 2 = 0.3$]

List – I		List – II	
(I)	If no electrolyte is added in either of the compartment of cell, then	(P)	$E_{\text{cell}} = 0.06 \text{ V}$
(II)	If 100 mL, 0.1 M NaOH solution is added in the anode compartment of the original cell, then	(Q)	$E_{\text{cell}} = 0.282 \text{ V}$
(III)	If 50 mL, 0.1 M NaOH solution is added in the cathode compartment of original cell, then	(R)	$E_{\text{cell}} = 0.18 \text{ V}$
(IV)	If 100 mL, 0.1 M HCl solution is added in the anode compartment of original cell, then	(S)	$E_{\text{cell}} = 0.078 \text{ V}$
		(T)	E_{cell} decreases after addition with respect to its initial value.

- (A) I→R; II→Q; III→P, T; IV→S
 (B) I→Q; II→R, T; III→P; IV→S, T
 (C) I→R; II→Q; III→P, T; IV→S, T
 (D) I→P; II→Q, T; III→R, T; IV→S

28. Match the reactions given List-I with their corresponding characteristics in List-II and select the correct option from the codes given below:

List – I		List – II	
(I)		(P)	Major product is formed via E1cb mechanism
(II)		(Q)	Major product is formed via E ₂ or E' ₂ mechanism
(III)		(R)	Reaction proceeds with the formation of transition state
(IV)		(S)	Carbanion is formed as intermediate
		(T)	Major product is formed via S _N 2 mechanism

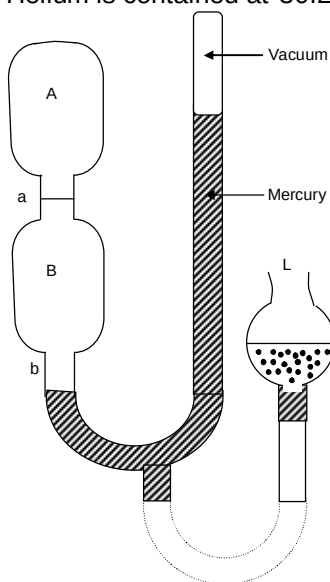
- (A) I→Q, R; II→R, T; III→P, S; IV→Q, S
 (B) I→Q, R; II→R, T; III→P, S; IV→P, S
 (C) I→R, T; II→Q, T; III→P, T; IV→P, S
 (D) I→Q, R; II→R, T; III→P, S; IV→Q, R

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

29. Consider the following statements:
- There are a maximum of 5 plane of symmetry in a molecule of SF₆.
 - Paracetamol gives positive FeCl₃ test.
 - Asprin is used as an antipyretic as well as analgesic.
 - About 60% of the solar energy reaching the earth is absorbed by earth surface, which increases its temperature.
 - Methane is one of the gas responsible for global warming.
 - SO₂ and NO₂ are major contributors of acid rain.
 - Clean water would have biochemical oxygen demand (BOD) value less than 5 ppm.
 - Water sample having nitrate content of 40 ppm is good for drinking purpose.
 - Azurite is a mineral of copper metal.
- Of the above statements, the ratio of number of correct statements to the number of incorrect statements is:

30. Helium is contained at 30.2°C in the system as shown in the figure.



Figure

The leveling bulb (L) can be raised so as to fill the lower bulb with mercury and force the gas into the upper part of the device. The volume of bulb 'A' to the mark 'a' is 100.5 cm^3 and volume of bulb 'B' between the marks 'a' and 'b' is 110 cm^3 . The pressure exerted by helium is measured by the difference between the mercury levels in the device and in evacuated arm of the manometer. When mercury level is at 'b', the pressure is 20.14 mm Hg . If the mass of helium in the container be $x \times 10^{-4}\text{ g}$ (scientific notation) ($R = 0.082\text{ L atm mol}^{-1}\text{ K}^{-1}$), then the value of 'x' is.....

31. Two moles of an ideal gas at 1 bar and 298 K are compressed at constant temperature by use of constant pressure of 5 bar. If the compression is driven by a 100 kg mass, how far (in 'metres') will it fall in the earth's gravitational field? ($g = 10\text{ m/s}^2$).
32. The iodine content of a solution was determined by titration with cerium (IV) sulphate in the presence of HCl, in which I^- is converted to ICl . A 200 mL sample of the solution required 15 mL of 0.05 N Ce^{4+} solution, calculate the iodine concentration in original solution in g/L?
33. Ratio of the total number of isomers of Benzene hexachloride to that of $\text{K}[\text{Co}(\text{NH}_3)_2(\text{F})_2(\text{Cl})_2]$ is:
34. Consider the following reactions
- $$\text{P}_4\text{S}_{10} + \text{H}_2\text{O} \longrightarrow (\text{X}) + (\text{Y})$$
- (Containing sulphur)
- $$\text{P}_4\text{S}_{10} \xrightarrow{\text{PCl}_5} (\text{Z})$$

Let, the sum of all atoms in one molecule each of 'X' and 'Y' be x, and total number of atoms in a molecule (Z) = y, then calculate the value of $\frac{x}{y}$

35. Let, the maximum number of electrons present in palladium atom ($Z = 46$) satisfying the condition $\ell + m_\ell > 0$ be 'x' and the total number of plane of symmetry in $\sigma_{2p_z}^*$ orbital be 'y', then the value of $\frac{x}{y}$ will be.....
36. Ratio of the total number of triangular faces to that of octagonal faces in a truncated solid cube is

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Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. Which of the following is/are CORRECT statement(s)?
- (A) The number of ways of arranging 5 boys and 5 girls in a queue if in front of each girl number of girls are more than or equal to the number of boys is k , then $\frac{k}{(5!)^2}$ is 42
- (B) Let $z_1, z_2, \dots, z_n$ be complex number such that $|z_1| = |z_2| = \dots = |z_n|$ and $\operatorname{Re}\left(\sum_{j=1}^n \sum_{k=1}^n \frac{z_j}{z_k}\right) = 0$ if and only if $\sum_{k=1}^n z_k = 0$
- (C) The number of ways of arranging 5 boys and 5 girls in a queue if in front of each girl number of girls are more than or equal to the number of boys is k , then $\frac{k}{(5!)^2}$ is 45
- (D) Let $a, b, c, x, y, z > 0$ and $ax + by + cz < d$, then maximum value of $xyz(d - ax - by - cz)$ is $\frac{d^4}{4^4 abc}$
38. Let $S_n = \sum_{r=1}^n t_r$ where $t_1 = 1$ and $t_r \geq 0, n \in \mathbb{N}$. If $S_n = \frac{1}{2}\left(t_n + \frac{1}{t_n}\right)$, then (Where $[.]$ represents greatest integer function)
- (A) $S_{100} = 10$
- (B) $\lim_{n \rightarrow \infty} (S_n t_n) = \frac{1}{2}$
- (C) $\left[\sum_{n=1}^{100} \frac{1}{S_n}\right] = 18$
- (D) $\lim_{n \rightarrow \infty} (S_n t_n) = 1$
39. $\sum_{r=0}^{10} \frac{{}^{10}C_r}{{}^{30}C_{10+r}}$ is equal to
- (A) $\frac{1}{{}^{30}C_{20} \cdot {}^{20}C_{10}} \sum_{r=0}^{10} {}^{10+r}C_r \cdot {}^{20+r}C_{10}$
- (B) $\frac{1}{{}^{30}C_{20} \cdot {}^{20}C_{10}} \sum_{r=0}^{10} {}^{10+r}C_r \cdot {}^{20-r}C_{10}$
- (C) $\frac{{}^{31}C_{10}}{{}^{30}C_{20} \cdot {}^{20}C_{10}}$
- (D) $\frac{{}^{40}C_{10}}{{}^{30}C_{20} \cdot {}^{20}C_{10}}$
40. Let $f(x) = \lim_{m \rightarrow \infty} \frac{\ln(3+x^2) - x^{2m} \sin(x^2)}{1+x^{2m}}$, then which of the following is/are CORRECT?
- (A) $f(x)$ is discontinuous at two points
- (B) There exists α such that $f(\alpha) = 1$
- (C) The equation $f(x) = 0$ has atleast one root in $(1, \infty)$
- (D) Minimum value of $f(x)$ is equal to $(-\sin 1)$

41. Let $F(x)$ be a six degree polynomial with leading coefficient k . If $y = F(x)$ touches, $f(x) = x + \alpha$ at $x = 1, 2$, and 3 , then which of the following is/are CORRECT?
- (A) If $\int_1^3 (F(x) - f(x)) dx = \frac{32}{105}$, then $k = 2$ (B) $F(x) - f(x)$ is divisible by $(x - 1)^2(x - 2)^2$
- (C) If $\int_1^3 (F(x) - f(x)) dx = \frac{32}{105}$, then $k = 1$ (D) $F(x) - f(x)$ is divisible by $(x - 2)(x - 3)^2$
42. Consider the 3 planes $\vec{r} \cdot \vec{n}_1 = a$, $\vec{r} \cdot \vec{n}_2 = b$ and $\vec{r} \cdot \vec{n}_3 = c$ where $abc \neq 0$, then which of the following is/are CORRECT?
- (A) If the above 3 planes have common line of intersection, then $a(\vec{n}_2 \times \vec{n}_3) + b(\vec{n}_3 \times \vec{n}_1) + c(\vec{n}_1 \times \vec{n}_2) = 0$
- (B) If the above 3 planes have a common line of intersection, then $a(\vec{n}_2 \times \vec{n}_3) + b(\vec{n}_3 \times \vec{n}_1) + c(\vec{n}_1 \times \vec{n}_2) \neq 0$
- (C) If $[\vec{n}_1 \vec{n}_2 \vec{n}_3] \neq 0$, then the above 3 planes will intersect at a point
- (D) If $\vec{n}_1, \vec{n}_2, \vec{n}_3$ are non coplanar vectors, then the above 3 planes will intersect at point with position vector = $\frac{1}{[\vec{n}_2 \vec{n}_3 \vec{n}_1]} \{c(\vec{n}_1 \times \vec{n}_2) + a(\vec{n}_2 \times \vec{n}_3) + b(\vec{n}_3 \times \vec{n}_1)\}$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Let $f(x) = \begin{cases} \sqrt{1-x^2} & ; -1 \leq x \leq 0 \\ 1+x^2 & ; 0 < x \leq 1 \\ 1+x^3 & ; x > 1 \end{cases}$ and $g(x)$ is a function such that $g(f(x)) = x$, $\forall x \geq -1$ and $f(g(x)) = x$, $\forall x \geq 0$, then match the following List-I with List-II

List-I		List-II	
(I)	The range of $y = g(x)$ is	(P)	$[-2, \infty)$
(II)	The domain of $y = g(x)$ is	(Q)	$[-1, \infty)$
(III)	The range of $y = f(f(f(g(x))))$ is	(R)	$[0, \infty)$
(IV)	The domain of $y = g(g(g(f(x))))$ is	(S)	$[1, \infty)$
		(T)	$[2, \infty)$

The correct option is:

- (A) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
 (B) (I) $\rightarrow$ (R); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
 (C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)
 (D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (T); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)

44. Match the following List-I with List-II

List-I		List-II	
(I)	The distance between the point A(-1, -5, -10) and point of intersection of $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ with the plane $x + 5y + z + 1 = 0$ is	(P)	$\frac{1}{3}$
(II)	The distance of the point P(-2, 3, -4) from the line $\frac{x+2}{3} = \frac{2y+3}{4} = \frac{3z+4}{5}$ measured parallel to the plane $4x + 12y - 3z + 1 = 0$ is	(Q)	2
(III)	In $\triangle ABC$ let P and Q be points on AB and BC respectively, such that $\frac{AP}{BP} = \frac{1}{3}$ and $\frac{CQ}{BQ} = \frac{3}{1}$. Let R be the point of intersection of AQ and CP, then area of $\triangle ABC$ is (given area of $\triangle BRC$ is 1 square unit)	(R)	$\frac{13}{9}$
(IV)	If $f(x) = \int_x^{x+\frac{\pi}{3}} \sin t dt$, $0 \leq x \leq \pi$ has local maximum at $x = k$, then $\frac{k}{\pi}$ is	(S)	13
		(T)	$\frac{17}{2}$

The correct option is:

- (A) (I) $\rightarrow$ (S); (II) $\rightarrow$ (T); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)
 (B) (I) $\rightarrow$ (S); (II) $\rightarrow$ (R); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (P)
 (C) (I) $\rightarrow$ (S); (II) $\rightarrow$ (P); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (T)
 (D) (I) $\rightarrow$ (S); (II) $\rightarrow$ (T); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (P)

45. Let a double differentiable function $g(x)$, satisfies the relation $g(x) g''(x) = (g'(x))^2 + g(x) g'(x)$, $g(0) = g'(0) = 1$, then match the following List-I with List-II

List-I		List-II	
(I)	The number of solution of the equation $g(x) = e^x$ is	(P)	1
(II)	$\lim_{x \rightarrow -\infty} (3e)^{g(x)}$ is	(Q)	5
(III)	$\lim_{x \rightarrow 0} \frac{4(g(x)-1)}{x}$ is	(R)	4
(IV)	The number of real roots of the equation $g(x) = x$	(S)	3
		(T)	0

The correct option is:

- (A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (T); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S)
 (B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (T)
 (C) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)
 (D) (I) $\rightarrow$ (P); (II) $\rightarrow$ (R); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)

46. Match the following List-I with List-II

List-I		List-II	
(I)	Let $2g(x) = \int_0^x (x-t)^2 f(t) dt$ and $\int_0^1 f(t) dt = 3$, then $g''(1)$ is	(P)	8
(II)	Let $g(x) = e^x + x + 1$, then $\int_2^{e+2} g^{-1}(x) dx$ is	(Q)	$\frac{1}{2}$
(III)	Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $g\left(\frac{x+y}{2}\right) = \frac{g(x)+g(y)}{2} \quad \forall x, y \in \mathbb{R}$. If $g(0) = 1$, $g'(0) = 1$, then area bounded by $y = g(x)$ and co-ordinate axes is	(R)	4
(IV)	If the area bounded by $ x \leq 2$, $ y \leq 2$, $2xy \leq x + y \leq x^2 + y^2$ is A, then $A + \frac{\pi}{2} - \frac{1}{2} \ln 15$ is	(S)	3
		(T)	$\frac{3}{2}$

The correct option is:

(A) (I) $\rightarrow$ (S); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

(B) (I) $\rightarrow$ (S); (II) $\rightarrow$ (T); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (Q)

(C) (I) $\rightarrow$ (S); (II) $\rightarrow$ (T); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (P)

(D) (I) $\rightarrow$ (S); (II) $\rightarrow$ (T); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (R)

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. The number of polynomials $P(x)$ with integral coefficient such that $P(a) = b$, $P(b) = c$ and $P(c) = a$ is (Where a , b and c are distinct integers)

48. Let a tangent to the parabola with focus $S(1, 2)$, is $x + y = 1$ at A . If tangent intersects directrix and tangent at vertex at B and C respectively, then the value of $AC \cdot BC$ is equal to

49. Let AB and CD be the direct common tangents to $x^2 + y^2 = 4$ and $(x - 3)^2 + (y - 4)^2 = 9$. Let both the circles intersect at the point P . Let the Circumradius of the $\triangle CPD$ be r , then $[r]$ is (Where $[.]$ represents greatest integer function)

50. The variable chords of the ellipse $\frac{x^2}{4} + y^2 = 1$ which subtends right angle at $\left(\sqrt{2}, \frac{1}{\sqrt{2}}\right)$ pass through a fixed point P , the length of the tangent drawn from P to $x^2 + y^2 = \frac{1}{4}$ is $\sqrt{\frac{p}{q}}$, then $p + q$ is equal to (Where HCF of (p, q) is 1)

51. The value of $1 + \sum_{k=1}^8 \frac{\cos k\theta}{\cos^k \theta}$ for $\theta = 30^\circ$ is p , then $\frac{\sin 9\theta}{p \sin \frac{\pi}{6} \left(\cos \frac{\pi}{6}\right)^8}$ is equal to

52. Let P and Q be invertible matrices of order 3×3 with real entries such that $P^{-1} + Q^{-1} = (P + Q)^{-1}$. If $|P| = 5$, then $|Q|$ is equal to
53. Two persons A and B fire at the target. A fires 9 times and B fires 10 times. If probability of their success is $\frac{1}{2}$, then the probability that number of successful hit of B is more than that of A is
54. Let $N = \prod_{r=1}^{45} \sin((2r-1)^\circ)$, then $\sqrt{2} \cdot 2^{45} N$ is equal to

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FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
OPEN TEST – I
PAPER –2
TEST DATE: 05-02-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

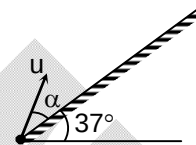
Physics

PART – I

Section – A (Maximum Marks: 12)

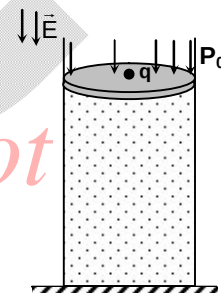
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. A particle is projected from the foot of smooth, fixed inclined plane, which is at an angle of 37° from horizontal. After its elastic collision with the plane it moves in vertical direction. Its initial velocity is $u = \sqrt{388}$ m/s. ($g = 10$ m/s²)



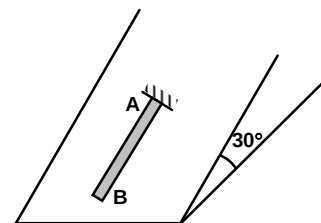
- (A) The time after which particle returns to its initial position is 6s
 (B) The time after which particle returns to its initial position is 4s
 (C) The angle of initial velocity with incline plane is $\tan^{-1}\left(\frac{3}{4}\right)$
 (D) The angle of initial velocity with incline plane is $\tan^{-1}\left(\frac{5}{9}\right)$

2. Below shown is heat insulating piston cylinder arrangement having ideal gas of adiabatic exponent γ enclosed. A point charge q having negligible mass is attached to centre of freely movable piston of mass M . The system is kept in uniform electric field $\vec{E}$ as shown. Atmospheric pressure is P_0 and when the piston is in equilibrium volume is V_0 . On disturbing the piston from equilibrium position (Assume system to have cross-sectional area as A)



- (A) The system executes S.H.M. with time period $T = 2\pi \sqrt{\frac{M V_0}{\left(\frac{qE + Mg}{A} + P_0\right) \gamma A^2}}$
 (B) The system does not execute S.H.M.
 (C) The system will not execute S.H.M. if magnitude of charge is increased.
 (D) The system executes S.H.M. with time period $T = 2\pi \sqrt{\frac{M V_0}{P_0 \gamma A^2}}$

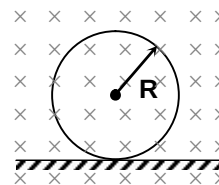
3. In the shown figure, A thin rod AB of length ℓ is placed on smooth inclined plane having angle 30° with horizontal. It is hinged at end A such that it is free to move parallel to the plane. Its linear mass density increases linearly as $\frac{dm}{dx} \propto x$, where x is distance from end



A. It is slightly displaced from its equilibrium position then it oscillates about an axis passing through A and perpendicular to inclined plane. The time period of oscillations is

- (A) $2\pi \sqrt{\frac{3\ell}{2g}}$ (B) $2\pi \sqrt{\frac{3\ell}{4g}}$
 (C) $2\pi \sqrt{\frac{2\ell}{3g}}$ (D) $2\pi \sqrt{\frac{\ell}{2g}}$

4. A uniform non-conducting sphere of mass m , radius R is placed on rough horizontal surface. Uniform horizontal magnetic field is present in circular region co-centric with centre of the sphere. Sphere is uniformly charged with total charge Q on its surface. Magnetic field is suddenly switched off. The sphere starts moving on the surface and finally it starts pure rolling with velocity.

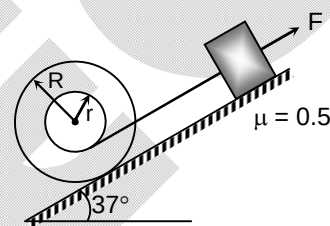


- (A) $\frac{QBR}{m}$ (B) $\frac{QBR}{6m}$
 (C) $\frac{5QBR}{21m}$ (D) $\frac{QBR}{2m}$

Section – A (Maximum Marks: 24)

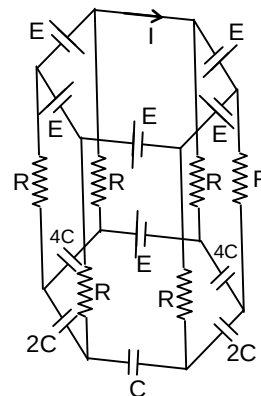
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. In the shown figure a spool of mass 10 kg of outer and inner radius $R=2\text{m}$ and $r=1\text{m}$ respectively, moment of inertia about its axis 10 kg-m^2 is placed on rough inclined surface where coefficient of friction is $\mu = 0.5$. A block of mass 4 kg is attached with a string which is wrapped on the spool. Force $F = 106\text{ N}$ is applied up the inclined plane as shown in the figure. The correct statements are (no slipping between spool and string) [take $g = 10\text{ m/s}^2$]



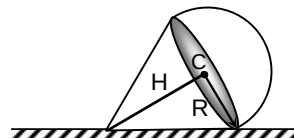
- (A) Acceleration spool is 2 m/s^2 down the plane
 (B) Acceleration spool is 2 m/s^2 up the plane
 (C) Acceleration of the block is 1 m/s^2 up the plane
 (D) Acceleration of the block is 1 m/s^2 down the plane

6. Hexagonal prism wire frame is made in which 6 sources of equal emf $\varepsilon = 5\text{V}$ with equal internal resistance $r = 1\Omega$. Six equal resistance each equal to $R = 2\Omega$ and four capacitors of capacitance as shown in the figure. Value of capacitance $C = 2\mu\text{F}$ then the correct statement(s) is/are



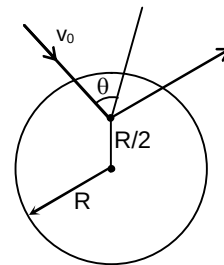
- (A) Value of current $I = 6\text{A}$
 (B) Value of current $I = 5\text{A}$
 (C) Value of charge on capacitor $4C = 16\mu\text{C}$
 (D) The value of charge on capacitor $C = 0$

7. A right circular cone of height $H = R\sqrt{3}$ is glued with hemisphere of radius R . Both are made from same material of uniform density. The object made is rolled on rough horizontal surface such that speed of centre of hemisphere is v_0 . The correct statements are

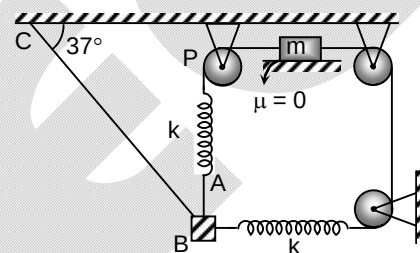


- (A) The ratio of time period of top about cones axis to the vertical axis is $2 : 3$
 (B) The ratio of time period of top about cones axis to the vertical axis is $1 : 2$
 (C) The frictional force acting on this top is $\frac{5}{2} \frac{mv_0^2}{R}$ where m is mass of the top.
 (D) The frictional force acting on this top is $\frac{2}{3} \frac{mv_0^2}{R}$, where m is mass of the top.

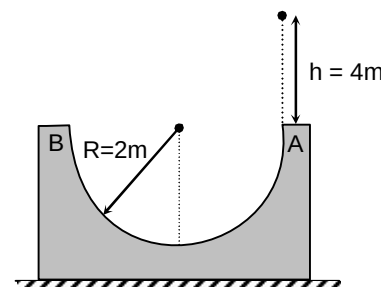
8. A disc of radius $R = 20$ cm and mass m is lying (in X-Y plane) on smooth horizontal surface. A particle of mass m moving with velocity $\vec{v}_0 = (5\hat{i} - 10\hat{j})$ m/s collides with the disc at a distance $\frac{R}{2}$ from centre of the disc. The coefficient of restitution $e = 1$ and coefficient of friction between particle and disc is $\mu = 0.2$. Assume slipping during collision and the vertical plane in which particle is moving is at distance $R/2$ from centre of disc. The correct statements are ($g = 10$ m/s²)
- (A) Angular velocity of disc after collision is 20 rad/s
- (B) Linear velocity of the centre of disc is 4 m/s
- (C) The ratio of time of flight of particle after collision and time period of disc is $\frac{20}{\pi}$
- (D) The ratio of time of flight of particle after collision to time period of disc is $\frac{\pi}{20}$



9. In the shown figure a block of mass $m = \frac{1}{2}$ kg is suspended in equilibrium with the help of ideal string and two ideal spring of force constant $k = 100$ N/m. Now string AP is cut. Pulleys are massless, frictionless, AP is vertical and other spring is horizontal. Just after cutting the string ($g = 10$ m/s²)
- (A) Tension in string BC is 3N
- (B) Tension in string BC is $\frac{37}{7}$ N
- (C) Acceleration of the suspended block is $\frac{32}{7}$ m/s².
- (D) Acceleration of the suspended block is 8 m/s².



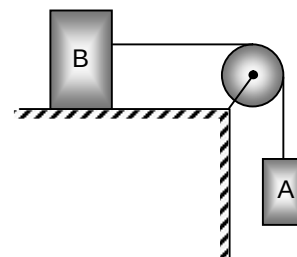
10. In the shown figure a block of mass 10 kg is placed on smooth horizontal surface. It has smooth hemispherical surface of radius $R = 2$ m. A particle of mass 5 kg is released from rest from height of 4m above point A, such that it smoothly lands on spherical surface at point A. Neglect air drag. The correct option(s) are ($g = 10$ m/s²)
- (A) Particle will move up from end B upto height 4m
- (B) Particle will move up from end B upto height less than 4m
- (C) maximum vertical acceleration of the particle is 90 m/s²
- (D) maximum vertical acceleration of particle is 40 m/s²



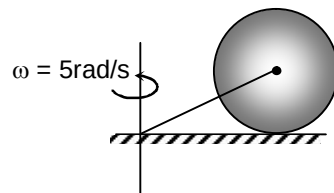
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

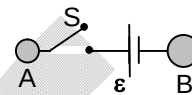
11. A block B of mass $M_B = \frac{\pi}{e}$ kg placed on rough horizontal surface is connected with block A with the help of string (ideal). The string passes over fixed horizontal cylinder. The coefficient of friction between block B and ground as well as between string and cylinder is $\mu = \frac{7}{11}$. Find the minimum mass of block A in kg so that block B starts sliding. (Take value of $\pi = \frac{22}{7}$ and $e = 2.72$)



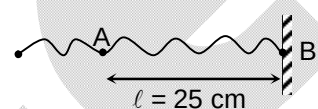
12. A sphere of mass 1.8 kg and radius 40 cm is radially pierced by a massless stick up to its centre. Length of stick is 2m. The free end of stick is pivoted on smooth ground and sphere is given velocity so that it slides on the ground such that its centre moves in a circle around pivot with angular velocity $\omega = 5 \text{ rad/s}$. The normal force acting on sphere due to ground in newton is 9P. Find the value of P. ($g = 10 \text{ m/s}^2$)



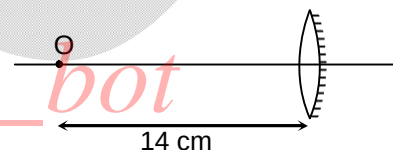
13. Two spherical conductors A and B of radius 9m and 18m respectively are placed at large distance. They are connected to each other through a cell of emf $\varepsilon = 300 \text{ volt}$ as shown in the figure. Initially they are chargeless. The loss of electric energy when switch is ON is $10P \text{ } \mu\text{J}$. The value of P is



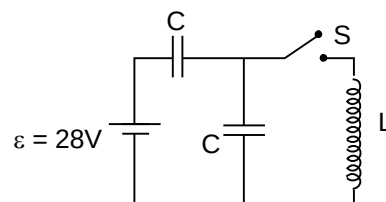
14. A wave $y = 10 \sin\left(200\pi t - \frac{4\pi}{100}x\right)$ (where x and y are in cm and t is in second) is travelling in string in which tension is 100 N. The wave is reflected from a fixed support and stationary wave is produced. Assuming no loss of energy due to reflection and air drag, the energy in 25 cm length of string is $k\pi^2 \text{ Joules}$. Find the value of k.



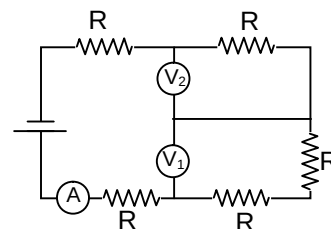
15. Object O is placed at a distance of 14 cm from an equi-convex lens of refractive index $\mu = \frac{3}{2}$ and radius of curvature of lens surface is 32 cm. One of its surface is silvered another convex lens of focal length 24 cm is placed between object and silvered lens such that the final image coincides with the object. The distance between object and the lens in cm is



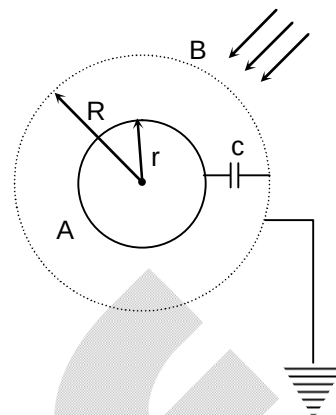
16. In the shown figure two uncharged capacitors each of equal capacitance $C = 2\text{mF}$ are connect to cell of emf 28V. The circuit was in this state from long time. Now switch S is closed with inductor with inductance $L = 16\text{mH}$. The maximum current in inductor in ampere will be



17. In the figure two identical voltmeter having their reading $V_1 = 15 \text{ V}$ and $V_2 = 10\text{V}$. The reading of ideal ammeter is $I = 3\text{A}$. All the five resistance have equal resistance R. The value of R is Ω is



18. A conducting solid sphere of radius $r = 2 \text{ cm}$ is surrounded by a conducting shell which is made up of fine wire mesh with radius $R = 4 \text{ cm}$ as shown in the figure. The shell and sphere are joined by a capacitor of capacitance $C = 2 \mu\text{F}$. Work function of the material of shell A and B is 1 eV and 4 eV respectively. Light falls on this system having frequency $\nu = 7.27 \times 10^{14} \text{ Hz}$. The energy stored in the capacitor C in steady condition (in μJ) is



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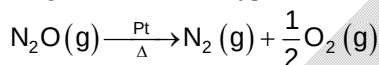
Chemistry

PART – II

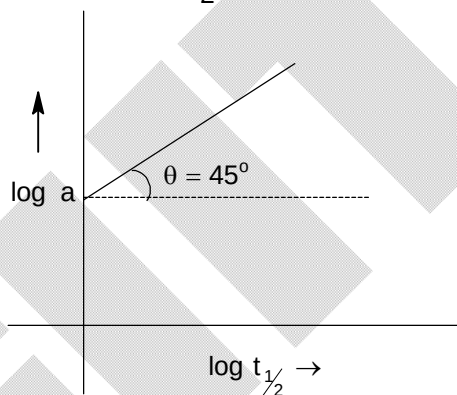
Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

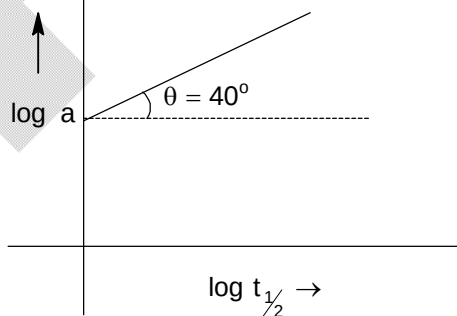
19. In a AB_xL_y (sp^3d hybridisation of 'A') have linear shape. What will be the shape of molecule if 'A' forms AB_yL_x type molecule?
(B = Bond pair of electron, L = Lone pair of electrons and A = Central atom).
(A) T-shape (B) See – saw
(C) Trigonal bipyramidal (D) Linear
20. If only gases remain after 12 g carbon is treated with 49.26 litre of air at 300 K and 2 atm pressure (Assume 20% by volume of O_2 , 79% by volume of N_2 and 1% by volume of CO_2 is present in air). Total amount of heat released in this reaction is.....
(ΔH_f for CO and CO_2 are $-26 \text{ kcal mol}^{-1}$ and $-94 \text{ kcal mol}^{-1}$)
(No reactants are left unreacted after the reaction)
(A) -96.8 kcal (B) -66.8 kcal
(C) -23.4 kcal (D) -33.4 kcal
21. Which of the following is correct graph for the reaction given below (where 'a' represents initial conc. of reactant and $t_{1/2}$ represents its half-life time) [Assume the rate constant of the reaction to be greater than unity]

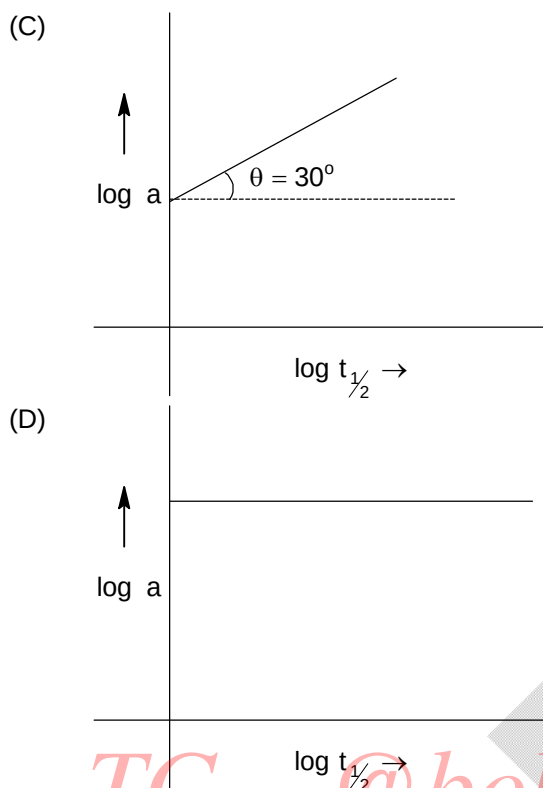


(A)



(B)





22. Fructose has a specific rotation of (-92.4°) . An aqueous solution of fructose has an observed angle of rotation of (-27.7°) when placed in a polarimeter tube 10 cm long. How much amount of fructose is dissolved in 100 ml of aqueous solution?
- (A) 0.299 g (B) 3.33 g
(C) 2.99 g (D) 29.97 g

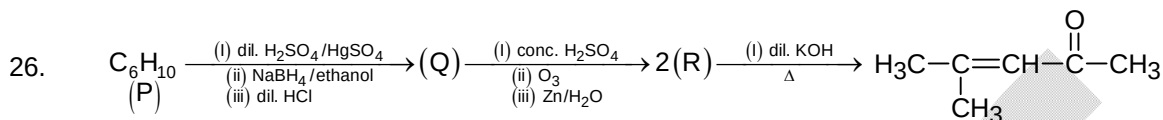
Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

23. A metal sulphide crystallizes with cubic close packed array of S^{2-} ions. Cations are Zn^{2+} and Al^{3+} ions. Only half of the octahedral voids and $\left(\frac{1}{8}\right)^{th}$ of the tetrahedral voids are occupied by cations. The correct conclusion regarding this solid is/are:
- (A) Empirical formula of the solid is $ZnAl_2S_4$.
(B) Empirical formula of the solid is $Zn_5Al_2S_8$.
(C) Co-ordination number of Zn^{2+} ions is 4.
(D) Co-ordination number of Al^{3+} ions is 6.
24. A solution is 0.01 M in $(NH_4)_2SO_4$ and 0.02 M in NH_4OH ($K_b = 2 \times 10^{-5}$). Which of the following is/are true about this solution?
- (A) The solution has buffer property.
(B) pH of this solution is $(9 + 2 \log 2)$
(C) pH of this solution is $(9 + \log 2)$
(D) Adding water to this solution increases pH.

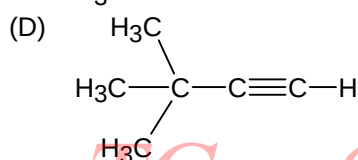
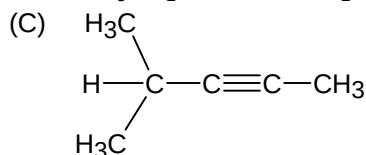
25. Which of the following option(s) is/are correct regarding XeF_6 ?

- (A) It acts as Lewis acid when it reacts with RbF .
- (B) It undergoes complete hydrolysis to give XeO_3 .
- (C) It fluorinates silica (SiO_2) to give XeOF_4 .
- (D) When it reacts with HCl , gives HF , XeCl_2 and F_2 .

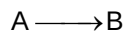


Which of the following is/are NOT the possible structure of (P)?

- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-\text{C}\equiv\text{C}-\text{H}$
- (B) $\text{CH}_3\text{CH}_2-\text{C}\equiv\text{C}-\text{CH}_2\text{CH}_3$



27. For the reaction:



The rate expression is given below:

$$\text{rate} = \frac{2[\text{A}]}{3 + [\text{A}]}$$

Where $[\text{A}]$ is the concentration (mol/L) of reactant left at any time 't (in minutes)' and $[\text{A}]_0$ represents initial concentration of the reactant 'A' ($\ln 2 = 0.7$)

Now, identify the correct statement(s) regarding the above reaction:

- (A) Plot of half-life time (Y-axis) versus $[\text{A}]_0$ (X-axis) is a straight line with slope 0.25 and intercept 1.05 on Y-axis.
- (B) If initial concentration of A is 8 M, then it takes 5.1 minutes to reduce its concentration to 2 M.
- (C) The reaction follows first order kinetics if concentration of 'A' is very small as compared to 3.
- (D) If concentration of A is much greater than 3, then the plot of $t_{1/2}$ (Y-axis) versus $[\text{A}]_0$ (X-axis) is a straight line passing through origin.

28. Cyclohexene on heating with conc. H_2SO_4 gives a major product (A) and a minor product (B) having greater number of α -hydrogen atoms as compared to the other minor product (B'). (A) on reductive ozonolysis gives (C) while (B) on reductive ozonolysis gives (D). (C) on heating with dil. KOH gives (E) while (D) on heating with dil. Alkali gives (F).

Now, choose the correct option(s) from the following:

- (A) (E) on reductive ozonolysis gives (C) and new product which on heating with conc. NaOH followed by acidification gives hydroxy cyclopentanecarboxylic acid.
- (B) Degree of unsaturation of (F) is 4.
- (C) Ratio of degree of unsaturation of (D) to that of (C) is 1.5.
- (D) (F) consists of two rings and both rings are of same size.

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

29. If total number of plane of symmetry in a molecule of TEREPHTHALIC ACID be 'x' and total number of elements of symmetry (excluding axis of symmetry) in a molecule of MESO-2,3-dichlorocyclobutane be 'y' and total number of stereocentres in pent-3-en-2-ol be 'z', then the value of (x + y + z) will be.....
30. Consider the following set of reactions:
- (i) $\text{Ag}_2\text{S} + \text{NaCN} \longrightarrow [\text{Ag}(\text{CN})_x]^{-n}$
- (ii) $\text{Au} + \text{NaCN} + \text{O}_2 \xrightarrow{\text{H}_2\text{O}} [\text{Au}(\text{CN})_y]^{-n}$
- (iii) $[\text{Ag}(\text{CN})_x]^{\pm n} \xrightarrow{\text{Zn}} \text{Ag} \downarrow + [\text{Zn}(\text{CN})_z]^{-m}$
- (iv) $[\text{Au}(\text{CN})_y]^{\pm n} \xrightarrow{\text{Zn}} \text{Au} \downarrow + [\text{Zn}(\text{CN})_z]^{-m}$
- (Where m, n, x, y and z, all are positive integers.)
What is the value of (x + y + z – 2n + m)?
31. 'P' gram of urea is dissolved in 500 g of water. If the difference in boiling point and freezing point of the solution is 100.46 K, then what is the value of (P)?
[For water, $K_b = 0.5 \text{ K kg mol}^{-1}$ and $K_f = 1.8 \text{ K kg mol}^{-1}$]
32. How many of the following molecules have locant-2 for bromine atom [Locant = Figure to indicate the position of bromine atom] as per IUPAC nomenclature?
- I

II

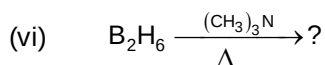
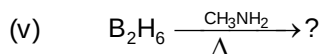
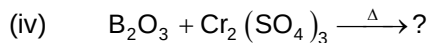
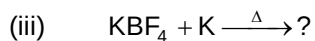
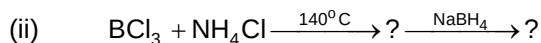
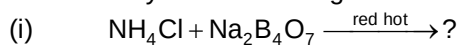
III

IV

V

VI
33. How many of the following statement(s) is/are correct?
- (i) Sodium amalgam is better reducing agent than hydrogen.
- (ii) Sodium carbonate can be decomposed to its oxides on gentle heating.
- (iii) On adding excess of NaOH to salt solution of ZnCl_2 , soluble product is formed.
- (iv) Copper sulphate on treatment aq. Na_2CO_3 solution gives basic copper carbonate.
- (v) In case of alkali metals, for performing flame test, chlorides are preferred over carbonates.
34. Volume strength of H_2O_2 solution is 22.4. 5 mL of this H_2O_2 solution is just sufficient to react with 0.40 gm impure sample of H_2S . If the mass % of pure H_2S in the impure sample is 17x, then find the value of 'x'. [Assume impurities are inert]

35. in how many of the following reactions 'BORON' is formed?



36. 6.0 g of He^4 having average velocity 4×10^2 m/s is mixed with 12 g of Ne^{20} having same average velocity. If the average kinetic energy per mole of the mixture is 'x' kJ, then find the value of 10x?

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Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. Consider the 12 face diagonals of a cube. If two diagonals are selected at random, then the probability that they are skew lines is
 (A) $\frac{3}{11}$ (B) $\frac{6}{11}$
 (C) $\frac{1}{2}$ (D) $\frac{5}{11}$
38. A new sequence is obtained from the sequence of the positive integers 1, 2, 3, by deleting all the perfect squares. Then the 2023rd term of the new sequence is
 (A) 2066 (B) 2067
 (C) 2068 (D) 2069
39. Let $x \in \left[-\frac{5\pi}{12}, -\frac{\pi}{3}\right]$. Then the maximum value of $y = \tan\left(x + \frac{2\pi}{3}\right) - \tan\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$ is
 (A) $\frac{12\sqrt{2}}{5}$ (B) $\frac{11\sqrt{2}}{6}$
 (C) $\frac{11\sqrt{3}}{6}$ (D) $\frac{11\sqrt{3}}{5}$
40. If $x^a = y^b = z^c$, where a, b, c are unequal positive numbers and x, y, z are in Geometric Progression, then $a^3 + c^3$ is
 (A) $> 2b^3$ (B) $> 2c^3$
 (C) $< 2b^3$ (D) $< 2c^3$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. Let A, B and C be the points (4, 1), (-1, -6) and (-3, 2) respectively. The point (x, y) in inside triangle ABC or on its perimeter, then
 (A) maximum value of $4x - 3y$ is 14 (B) maximum value of $4x - 3y$ is 18
 (C) minimum value of $4x - 3y$ is -14 (D) minimum value of $4x - 3y$ is -18
42. A variable point z_1 in the complex plane moves according to the equation $|z_1 - z_0| = |z_1|$, where z_0 is a fixed point ($z_0 \neq 0$). A second variable point z moves according to the equation $z_1 z = -1$. Then the locus of point z is
 (A) a circle with centre $-\frac{1}{z_0}$ (B) a circle with centre $\frac{1}{z_0}$
 (C) a circle with radius $\frac{1}{|z_0|}$ (D) a circle with radius $\frac{2}{|z_0|}$

43. Let $4x - y = 0$ and $y = 0$ be two lines and $C(6, 4)$ be a point in the xy plane. A variable line through C intersect the given lines at A and B respectively. P be the point of intersection of the given lines, then
- (A) if area of ΔPAB is minimum, then A is $\left(\frac{3}{2}, 6\right)$
 (B) if area of ΔPAB is minimum, then A is $(2, 8)$
 (C) if $PA + PB$ is minimum, then B is $\left(\frac{15}{2}, 0\right)$
 (D) if $PA + PB$ is minimum, then B is $(10, 0)$
44. The solution set of the inequality $\sqrt{\log_2 x - 1} + \frac{1}{2} \log_{\frac{1}{2}} x^3 + 2 > 0$ may be
- (A) $[2, 3]$ (B) $(2, 3]$
 (C) $[2, 4]$ (D) $(2, 4]$
45. Let $f : \mathbb{Z} \rightarrow \mathbb{R}$ be defined as $f(x + 2) = f(x) + 2(x + 1)$ if x is even and $f(x + 2) = f(x) + 1$ if x is odd and $f(1) = 1, f(2) = 5$, then
- (A) $f(6) = f(41)$ (B) $f(7) < f(8)$
 (C) $f(x)$ is increasing (D) $f(10) + f(11) = 59$
46. Consider the equation $3 \cot^2 x + \lambda \cot x + 3 = 0, x \in (0, 2\pi) - \{\pi\}$. Which of the following options is/are CORRECT?
- (A) Sum of the solutions of the equation $= 3\pi$ if $\lambda = 7$
 (B) Sum of the solutions of the equation $= 3\pi$ if $\lambda = -8$
 (C) Sum of the solutions of the equation $= 5\pi$ if $\lambda = 9$
 (D) Sum of the solutions of the equation $= 5\pi$ if $\lambda = -10$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer**, ranging from 0 to 9 both inclusive.*

47. The number of real solution of the equation $(x^{2024} + 1)(1 + x^2 + x^4 + \dots + x^{2022}) = 2024x^{2023}$ is
48. The complex numbers z_1 and z_2 satisfy $|z_1| = |z_1 + z_2| = 3$ and $|z_1 - z_2| = 3\sqrt{3}$. If the value of $\log_3 |(z_1 \bar{z}_2)^{2023} + (\bar{z}_1 z_2)^{2023}|$ is $578k$, then the value of k is
49. The maximum value of $f(x) = \sqrt{x^4 - 3x^2 - 6x + 13} - \sqrt{x^4 - x^2 + 1}$ is k , then $[k]$ is (where $[.]$ represents greatest integer function)
50. Let P be any point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$. If F_1 and F_2 be the foci of the hyperbola and I be the in-centre of the $\Delta PF_1 F_2$. If locus of I is $\lambda y^2 + 2x + \mu = 0$, then the value of $|\lambda + \mu|$ is equal to
51. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $|\vec{a}| = 1, |\vec{b}| = 2$, and $|\vec{c}| = 4$. If $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 62$, then the value of $|2\vec{a} + 3\vec{b} + 4\vec{c}|$ is equal to
52. Let $f(x) = \int_x^1 \ln(1 - t^2) dt, x \in (0, 1)$ and $\alpha = \int_0^1 f(x) dx$, then the value of $4\alpha + 5$ is equal to

53. Let tangent at (α_1, β_1) on the curve $y = x^4 - 2x^2 - x$ touch the curve again at (α_2, β_2) , then the value of $|\alpha_1| + |\alpha_2| + |\beta_1| + |\beta_2|$ is equal to
54. Let A and C be two points on the line $\frac{x-1}{3} = \frac{y-2}{2} = \frac{z-3}{1}$ and B & D be two point on the line $\frac{x-2}{5} = \frac{y-1}{2} = \frac{z-2}{3}$ such that AB is perpendicular to both the given lines and the length of CD is $3\sqrt{3}$. If V is the maximum volume of the tetrahedron ABCD, then $[V]$ is equal to (where $[.]$ represents greatest integer function)

TG ~ @bohring_bot

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
OPEN TEST – I
PAPER –1
TEST DATE: 05-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B, D

Sol.

$$E = \frac{2k\lambda}{r}$$

$$\Rightarrow \int_0^V dV = - \int_{r_0}^r E dr$$

$$\Rightarrow V = -2k\lambda(\ln r - \ln r_0)$$

$$V = -2k\lambda \ln(r) + C$$

$$V_P = V_{-\lambda} + V_{+\lambda} = 2k\lambda \ln(r_-) - 2k\lambda \ln(r_+) = 2k\lambda \ln\left(\frac{r_-}{r_+}\right)$$

$$\vec{r}_- = (x+a)\hat{i} + y\hat{j}$$

$$\vec{r}_+ = (x-a)\hat{i} + y\hat{j}$$

$$V_P = 2k\lambda \ln \sqrt{\frac{(x+a)^2 + y^2}{(x-a)^2 + y^2}}$$

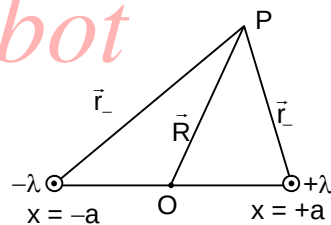
$$\text{for } V_P = \frac{\lambda}{4\pi\epsilon_0} \ln 2$$

$$\Rightarrow \frac{(x+a)^2 + y^2}{(x-a)^2 + y^2} = 2$$

$$(x+a)^2 + y^2 = 2(x-a)^2 + 2y^2$$

$$(x-3a)^2 + y^2 = 8a^2$$

$\Rightarrow$ Equipotential surface is cylinder with radius $2\sqrt{2}a$



2. A, C

Sol.

For oscillation to occur it must be in the stable equilibrium i.e. centre of mass of the system must be below the centre of hemisphere.

3. A, C, D

Sol. $1000 = \frac{320}{4\ell_1} \Rightarrow \ell_1 = 8 \text{ cm}$

$1000 = \frac{3 \times 320}{4\ell_2} \Rightarrow \ell_2 = 24 \text{ cm}$

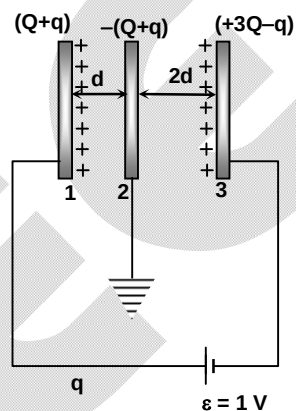
$-A \frac{dh}{dv} = 2\sqrt{2gh} \Rightarrow -\int \frac{dh}{\sqrt{h}} = \frac{a}{A} \sqrt{2g} \int dt$

4. B, D

Sol. Charge flown through S_1 is $Q + 3Q = 12 \text{ mC}$

$0 + \frac{(Q+q)}{\epsilon_0 A} d - \epsilon - \frac{(3Q-q)2d}{\epsilon_0 A} = 0$

By solving $q = 15 \text{ mC}$



5. A, C, D

Sol. Conserve angular momentum about O. $2mv_0\ell = I\omega$

Where $I = \frac{40m\ell^2}{3}$

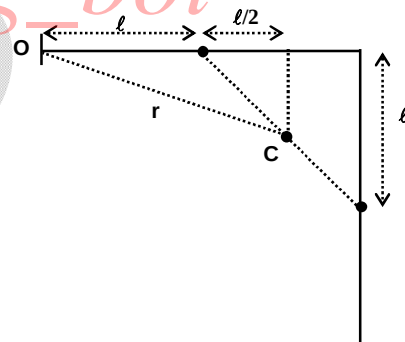
$\therefore \omega = \frac{3v_0}{10\ell}$

$r = \sqrt{\frac{\ell^2}{4} + \left(\ell + \frac{\ell}{2}\right)^2}$

$r = \sqrt{\frac{5}{2}}\ell$

$v_c = r_c\omega = \sqrt{\frac{5}{2}}\ell \times \frac{3v_0}{10\ell} = \frac{3}{2\sqrt{10}}v_0$

$\text{K.E.} = \frac{1}{2}I\omega^2 = \frac{3}{5}mv_0^2$



6. A, B, C, D

Sol. Torque along vertical axis is zero

7. B

Sol. (I) $P_N = P_0 + \rho g \left(\frac{3H}{2} \right)$

$P_N - P_0 = 3\rho g \frac{H}{2}$

$$(II) \quad V = \sqrt{2g(2H)} = 2\sqrt{gH}$$

$$P_0 + \rho gH = P_N - \rho g \frac{H}{2} + \frac{1}{2} \rho (4gH)$$

$$P_N - P_0 = \rho gH + \frac{\rho gH}{2} - 2\rho gH = -\frac{\rho gH}{2}$$

$$(III) \quad P_N = P_0 + \rho gh + \frac{1}{2} \rho \omega^2 \left(\frac{H}{2} \right)^2$$

$$P_N - P_0 = \rho gH + \frac{1}{8} \rho \omega^2 H^2$$

$$\omega^2 H = 2g$$

$$\therefore P_N - P_0 = \frac{5}{4} \rho gH$$

$$(IV) \quad P_N = P_0 + \rho(g \cos 60^\circ) 2H$$

$$P_N - P_0 = \rho gH$$

8. B

Sol. (I) $v_l = \frac{4}{3} \times 3 = 4$

$$v_r = 4 - 3 = 1 \text{ m/s}$$

$$(II) \quad \frac{dv}{dt} = -\left(\frac{v}{u}\right)^2 \frac{du}{dt}$$

$$\Rightarrow v_l = 1 \text{ m/s and } v_r = 10 \text{ m/s}$$

$$(III) \quad v_{l,m} = 4$$

$$\Rightarrow v_{l,g} = 7 \Rightarrow v_r = 8 \text{ m/s}$$

(IV) final image is at the position of object hence magnification is 1

$$\Rightarrow v_l = v_0 = 1 \text{ m/s}$$

$$\Rightarrow v_r = 0$$

9. C

Sol. (I) $Q = \int T ds = \text{Area between T-s diagram} = 700 \text{ J}$

$$(II) \quad \Delta Q = n \frac{3}{2} R (T_f - T_i) + n \frac{5}{2} R (T_f - T_i)$$

$$= \frac{3}{2} (nRT_f - nRT_i) + \frac{5}{2} (nRT_f - nRT_i)$$

$$= \frac{3}{2} (500 - 100) + \frac{5}{2} (1000 - 500) = 1850 \text{ J}$$

(III) Process is $T = 100 \text{ V} \Rightarrow \frac{T}{V} = \text{constant}$ i.e. pressure = constant

$$\therefore Q = nC_p \Delta T = 2 \frac{5}{2} R(300 - 100) = 1000R$$

(IV) $A \rightarrow B$ is $V = \text{constant}$ and $B \rightarrow C$, $P = \text{constant}$

$$\Delta Q = n \frac{3}{2} R(T_B - T_A) + n \frac{5}{2} R(T_C - T_B) = (1000 - 500) + \frac{5}{3}(500 - 1000)$$

10. D

Sol. (I) $-mg \frac{R}{2} \theta = \frac{2}{3} mR^2 \alpha \Rightarrow \alpha = -\left(\frac{3g}{4R}\right) \theta$

$$T = 2\pi \sqrt{\frac{4R}{3g}}$$

(II) $maR = mg \frac{2R}{\pi}$

$$a = \frac{2g}{\pi}$$

$$-mg_{\text{eff}} \sqrt{R^2 + \frac{4R^2}{\pi^2}} \sin \theta = 2mR^2 \alpha$$

$$g_{\text{eff}} = \sqrt{a^2 + g^2} = g \sqrt{1 + \frac{4}{\pi^2}}$$

$$\therefore \alpha = -\left[\frac{g}{2R} \left(1 + \frac{4}{\pi^2}\right)\right] \theta \Rightarrow T = 2\pi \sqrt{\frac{2R}{g \left(1 + \frac{4}{\pi^2}\right)}}$$

(III) $\tau_p = I_p \alpha$

$$-mgd \sin \theta = I_p \alpha$$

$$d = \frac{3R}{8}, I_p = \frac{13}{20} mR^2$$

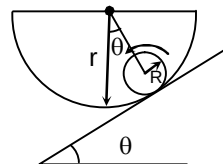
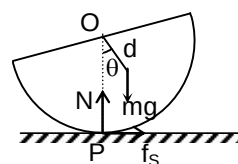
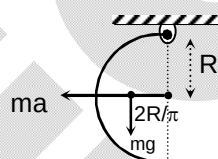
$$\alpha = -\left(\frac{mgd}{I_p}\right) \theta$$

$$T = 2\pi \sqrt{\frac{I_p}{mgd}} \Rightarrow T = 2\pi \sqrt{\frac{26R}{15g}}$$

(IV) $a = \frac{g \sin \theta}{1 + \frac{I_c}{mR^2}}$

$$a = -\frac{5g}{7} = -\frac{5g}{7} \frac{x}{r - R} = -\frac{g}{7R} x$$

$$T = 2\pi \sqrt{\frac{7R}{g}}$$



Section – B

11. 0.68

Sol. $I_1 = \frac{MR^2}{4} + MR^2 \quad \dots(i)$

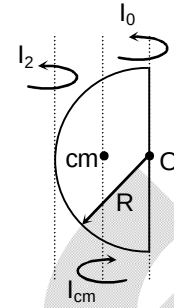
$I_2 = I_{cm} + M\left(R - \frac{4R}{3\pi}\right)^2 \quad \dots(ii)$

$I_0 = I_{cm} + M\left(\frac{4R}{3\pi}\right)^2$

$I_0 = \frac{MR^4}{4}$

Solving the equations

$\left| \frac{I_2 - I_1}{I_1} \right| = 0.68$



12. 0.01

Sol. $F = n \frac{h}{\lambda}$
and $F = mg$

$\therefore \text{Power } P = n \frac{hc}{\lambda}$

$\therefore mg = \frac{P}{c}$

$P = mgc$

Also, $P = \frac{dm}{dt} c^2$

$\therefore \frac{dm}{dt} = \frac{mg}{c} = 0.01 \text{ kg/s}$

13. 19.11

Sol. $mv_0 \cos \theta = 2mv_c$

$\Rightarrow v_c = \frac{v_0 \cos \theta}{2}$

Conservation of angular momentum

$mv_0 \cos \theta \frac{\ell}{4} = \left(\frac{m\ell^2}{16} + \frac{m\ell^2}{12} + \frac{m\ell^2}{16} \right) \omega$

$\omega = \frac{6v_0 \cos \theta}{5\ell}$

$\theta = \omega t \text{ and } N = \frac{\theta}{2\pi} = \frac{3v_0^2 \sin 2\theta}{10\pi g \ell}$

14. 3.14

Sol.

$$f_s = ma$$

$$N = mg$$

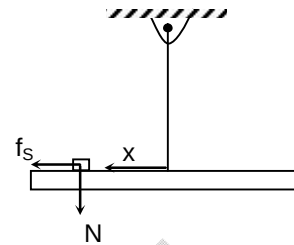
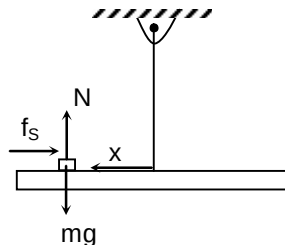
$$\tau_0 = 0$$

$$Nx - f_s \ell = 0$$

$$f_s = \frac{mgx}{\ell}$$

$$a = -\frac{g}{\ell} x$$

$$\text{Time} = \pi \sqrt{\frac{\ell}{g}} = \pi \text{ sec}$$



15. 1.19

Sol.

$$\sqrt{2 \times 4k_\alpha} = \sqrt{2 \times 1 \times k_p} \cos 60^\circ + \sqrt{2 \times 17k_0} \cos \theta$$

$$\sqrt{2 \times 1 \times k_p} \sin 60^\circ = \sqrt{2 \times 17 \times k_0} \sin \theta$$

Solving $k_0 = 0.71 \text{ MeV}$

$$\therefore Q = (k_\alpha - k_p - k_0) = 1.19 \text{ MeV}$$

16. 0.80

Range (0.79 to 0.82)

Sol. For rotation equilibrium net torque is zero

$$-NIAB + 4g \times 5 \times 10^{-2} + (100x) \times 2.5 \times 10^{-2} = 0$$

$$x = 0.8 \text{ m}$$

17. 0.75

Sol. Minimum speed of sphere w.r.t. block at highest point is $v_{\min} = 5 \text{ m/s}$.

Conservation of momentum

$$1v_0 = (1+4)v = 5v_1 + 5(v_1-5)$$

Conservation of energy

$$\frac{1}{2} \times 5v^2 = 5g(2 \times 2.5) + \frac{1}{2} 5v_1^2 + \frac{1}{2} 5(v_1-5)^2$$

18. 0.75

Sol. Velocity of pendulum before collision

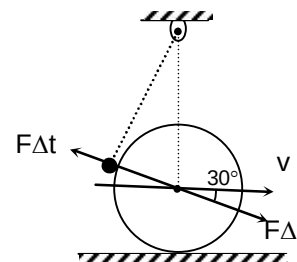
$$v_0 = \sqrt{2g\ell \cos 30^\circ} = \sqrt{6} \text{ m/s}$$

$$F\Delta t = mv_0$$

$$F\Delta t \cos 30^\circ = mv$$

$$\therefore mv_0 \frac{\sqrt{3}}{2} = mv \Rightarrow v = \frac{v_0 \sqrt{3}}{2}$$

$$e = \frac{v \frac{\sqrt{3}}{2}}{v_0} = \frac{3}{4}$$

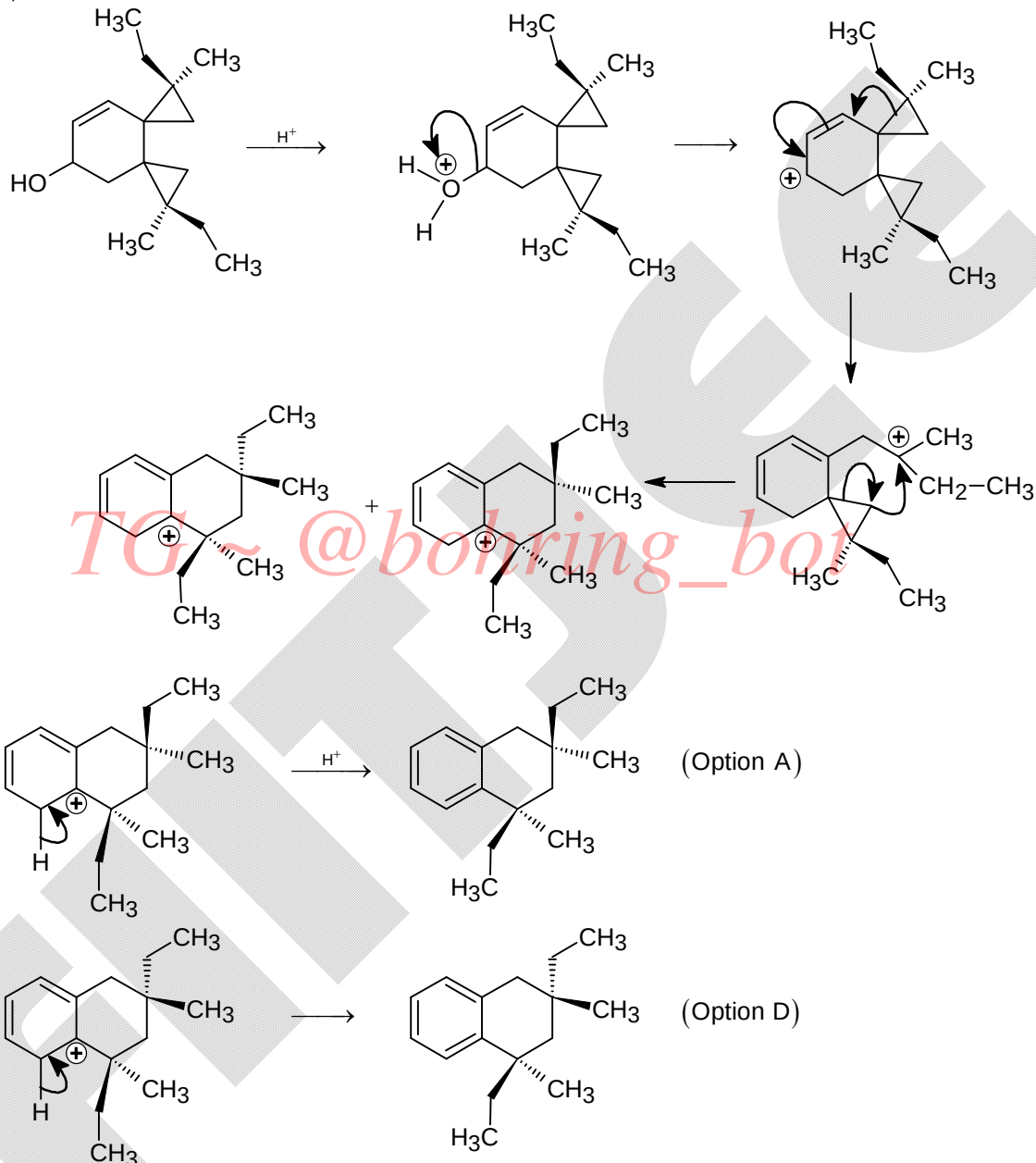


Chemistry

PART – II

Section – A

19. A, D
Sol.

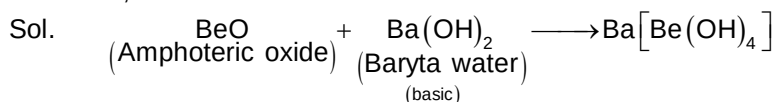


20. A, B, C, D

Sol. The driving force behind these reactions (A and B) is the formation of $LiCl$ or $LiBr$ (very large lattice energy and insoluble in organic solvents in which the reaction is performed)

- Benzene is more acidic than C_2H_6 , so the reaction (C) can occur.
- Li_3N converts into amides and hydrides by reaction with hydrogen.

21. B, D



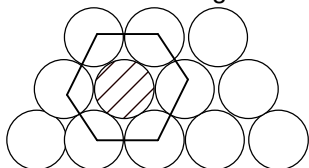
22. A, B, C

Sol. β -keto acids, gem-dioic acids and β, γ -unsaturated acids decarboxylates on simple heating. (D) is however a β -keto acid but in transition state, there will be a double bond at bridge head position which is not stable. So, (D) will not decarboxylate on heating.

23. B, D

Sol. (A) Second nearest neighbours of Cs^+ are 6 and not 8 (in CsCl crystal).

(B)



There are 6 triangular voids around a sphere in two dimensional hcp layer.

(C) $\frac{r_+}{r_-} = \frac{0.3}{0.4} = 0.75$ which indicates that cations are present in cubic void of a simple cubic unit cell, so its coordination number is 8 and not 6.

(D) Definition of Frenkel defect.

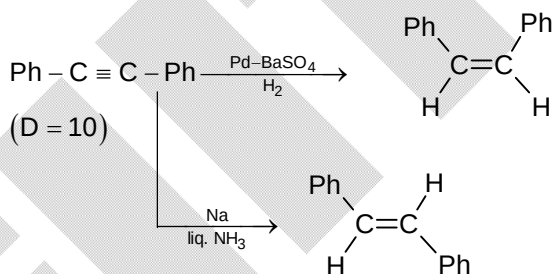
24. B, C

Sol. $\text{—}\overset{\text{C}}{\overset{\text{O}}{\parallel}}\text{—Cl}$ has less priority than ester, so, (A) is wrong but (B) is correct.

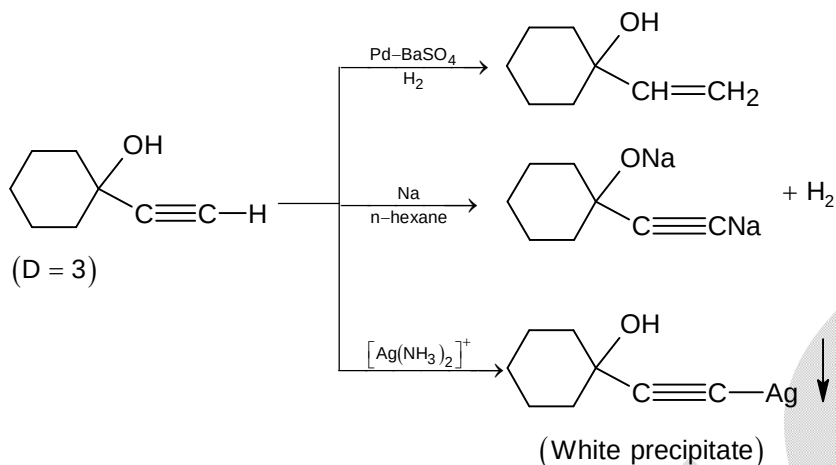
Double bond has more priority than 'Br'.

25. B

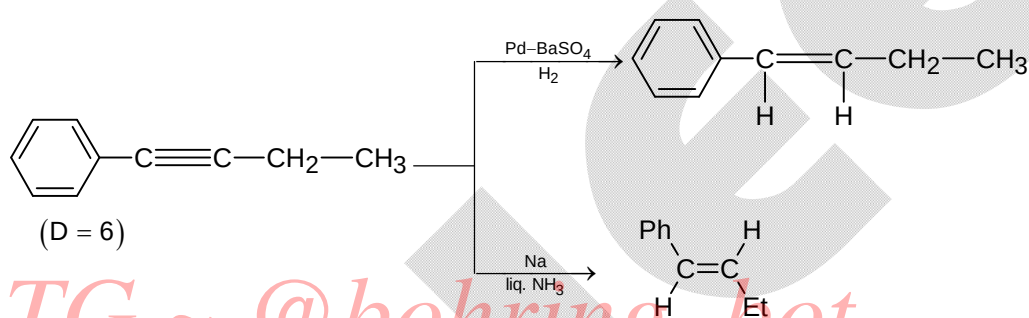
Sol. (I)



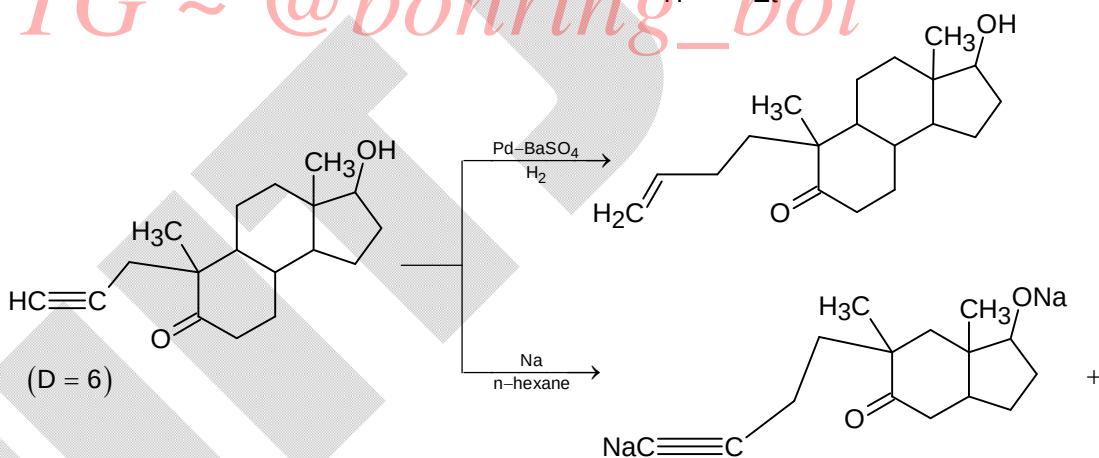
(II)



(III)



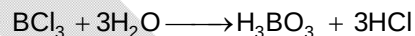
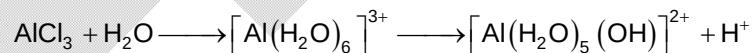
(IV)



26.

D

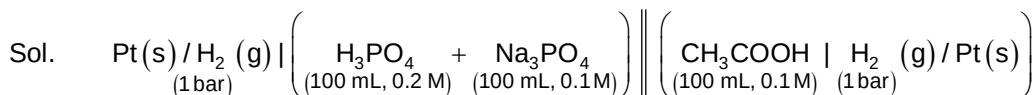
Sol.



Alums contains Al^{3+} or Cr^{3+} or Fe^{3+} ions whose aqueous solution is strongly acidic.

Also, in B_2H_6 , there is 3c-2e bond formation in which vacant 2p-orbital of 'B' is also involved in sp^3 -hybridisation.

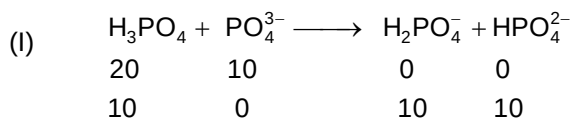
27. C



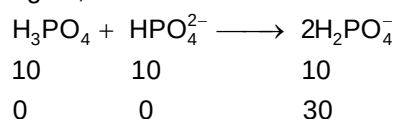
From Nernst Equation

$$E_{\text{cell}} = -0.06 \times \log \frac{[\text{H}^+]_a}{[\text{H}^+]_c}$$

$$\text{or } E_{\text{cell}} = [(\text{pH})_a - (\text{pH})_c] \times 0.06 \quad \dots (1)$$

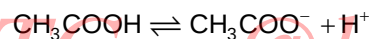


Again,



$$\therefore (\text{pH})_a = \frac{\text{pK}_{a1} + \text{pK}_{a2}}{2} = \frac{4 + 8}{2} = 6$$

Also, at cathode

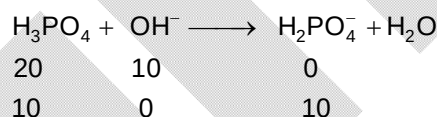
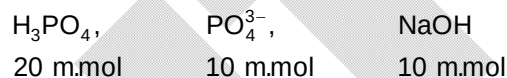


$$\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{10^{-5}}{0.1}} = 10^{-2}, \quad [\text{H}^+]_c = 10^{-3}$$

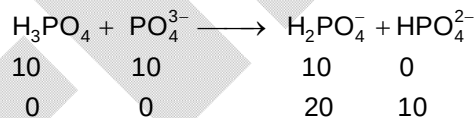
$$\therefore (\text{pH})_c = 3$$

$$E_{\text{cell}} = 0.06[6 - 3] = 0.18 \text{ V}$$

(II) If 100 mL, 0.1 M NaOH is added in anode compartments



Also,



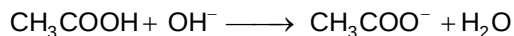
So, it is a buffer.

$$(\text{pH})_a = \text{pK}_{a2} + \log \frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]} = 8 + \log \frac{10}{20}$$

$$\therefore (\text{pH})_a = 7.7$$

$$E_{\text{cell}} = 0.06[7.7 - 3] = 0.282 \text{ V}$$

(III) 50 mL, 0.1 M NaOH is added in cathode compartment



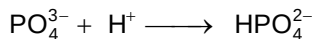
10	5	0
5	0	5

$\therefore$ P it is also a buffer.

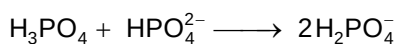
$$(\text{pH})_c = 5 + \log \frac{5}{5} = 5$$

$$\therefore E_{\text{cell}} = 0.06 [6 - 5] = 0.06 \text{ V}$$

(IV) 100 mL, 0.1 M HCl is added in anode compartment



10	10	0
0	0	10



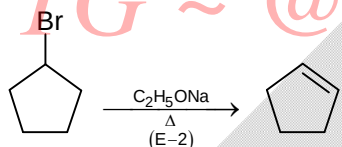
20	10	0
10	0	20

$$\therefore (\text{pH})_a = 4 + \log 2 = 4.3$$

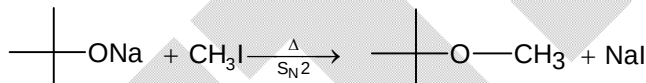
$$E_{\text{cell}} = 0.06 [4.3 - 3] = 0.06 \times 1.3 = 0.078 \text{ V}$$

28.
Sol.

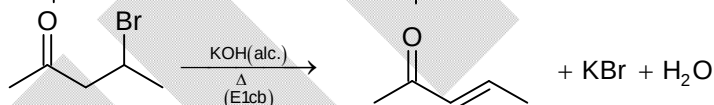
D
(I)



(II)



(III)



(IV)



Section - B

29.
Sol.

3.50
Except statements (i) and (iv), all other statements are correct.

(i) SF_6 molecule has a maximum of 9 planes of symmetry (and not 5).

(iv) About **75%** (and not 60%) of the solar energy reaching the earth is absorbed by earth surface which increases its temperature.

30.

8.97
[Range : 8.96 – 8.98]

Sol.

$$PV = nRT$$

$$PV = \left(\frac{W_{\text{He}}}{4} \right) \times R \times T$$

$$\frac{20.14}{760} \times \frac{(110 + 100.5)}{1000} = \frac{W_{\text{He}}}{4} \times 0.082 \times (273 + 30.2)$$

$$W_{\text{He}} = 8.97 \times 10^{-4} \text{ g} = x \times 10^{-4} \text{ g}$$

$$\therefore x = 8.97$$

31. 19.82

[Range : 19.81 – 19.83]

Sol. $W = -P_2 (V_2 - V_1)$

$$= -P_2 \left(\frac{nRT}{P_2} - \frac{nRT}{P_1} \right)$$

$$= -nRT \left(1 - \frac{P_2}{P_1} \right)$$

$$= -2 \times 8.314 \times 298 \left(1 - \frac{5}{1} \right)$$

$$W = 19820.576 \text{ J}$$

$$\text{Now, } |W| = mgh$$

$$h = \frac{W}{mg}$$

$$h = \frac{19820.576}{100 \times 10} = 19.82 \text{ 'm'}$$

32. 0.24

[Range : 0.23 – 0.25]

Sol. meq. of iodine in 200 mL solution = meq. of $\text{Ce}^{4+} = 15 \times 0.05 = 0.75$

$$\therefore \text{Normality of iodine} = \frac{0.75}{200}$$

$$\text{Eq. weight of } \text{I}^- = \frac{127}{2}$$

$$\text{Iodide ion concentration} = \frac{0.75}{200} \times \frac{127}{2} = 0.238 \text{ g/L}$$

33. 1.50

Sol. Total number of isomers of B.H.C. = 9

$$\text{Total number of isomers of } \text{K} [\text{Co}(\text{NH}_3)_2 (\text{F})_2 (\text{Cl})_2] = 6$$

34. 2.20

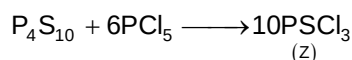
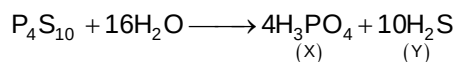
Sol. $\text{X} = \text{H}_3\text{PO}_4$, $\text{Y} = \text{H}_2\text{S}$,

$\text{Z} = \text{PSCl}_3$

$$x = 8 + 3 = 11$$

$$y = 5$$

$$\frac{x}{y} = \frac{11}{5} = 2.20$$



35. 9.33

[Range : 9.33 – 9.34]

Sol. Pd(46) : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^0$

↓	↓	↓	↓	↓
4	4	8	4	8

So, $x = 28$

$y = 3$

36. 1.33

[1.33 – 1.34]

Sol. Number of triangular faces in truncated cube = 8

Number of octagonal faces in truncated cube = 6

Each corner of the cube will become a triangular face and each square face will become an octahedral face in a truncated solid cube.

TG ~ @bohring_bot

Mathematics

PART – III

Section – A

37. A, B, D

Sol. Apply $AM \geq GM$
 $d - ax - by - cz > 0$
 $ax, by, cz, d - ax - by - cz$
 Apply $AM \geq GM$

38. A, B, C

Sol. $S_n^2 = S_{n-1}^2 + 1 \Rightarrow S_n = \sqrt{n}$

39. B, C

Sol. $\sum_{r=0}^{10} \frac{{}^{10}C_r}{{}^{30}C_{10+r}} = \frac{1}{{}^{30}C_{20}} \cdot {}^{20}C_{10} \sum_{r=0}^{10} {}^{10+r}C_r \cdot {}^{20-r}C_{10} = \text{coefficient of } x^{10} \text{ in } (1-x)^{-22}$

40. A, B, C

Sol. $f(x) = \begin{cases} -\sin x^2 & ; \quad x < -1 \\ \frac{\ln 4 - \sin 1}{2} & ; \quad x = -1 \\ \ln(3+x^2) & ; \quad -1 < x < 1 \\ \frac{\ln 4 - \sin 1}{2} & ; \quad x = 1 \\ -\sin(x^2) & ; \quad x > 1 \end{cases}$

41. A, B, D

Sol. $F(x) - f(x) = k(x-1)^2(x-2)^2(x-3)^2$

42. A, C, D

Sol. The plane $(\vec{r} \cdot \vec{n}_1 - a) + \lambda(\vec{r} \cdot \vec{n}_2 - b) = 0$ is identical with $\vec{r} \cdot \vec{n}_3 = c$

43. A

Sol. Domain of $g(x)$ is $[0, \infty)$ and range of $g(x)$ is $[-1, \infty)$

44. D

Sol. (I) The point of intersection is $(2, -1, 2)$

(II) Line drawn from P intersects the line $\frac{x+2}{3} = \frac{2y+3}{4} = \frac{3z+4}{5}$ measured parallel to the plane

$$4x + 12y - 3z + 1 = 0 \text{ at } \left(4, \frac{5}{2}, 2\right)$$

(III) $\frac{AR}{RQ} = \frac{4}{9}$

(IV) $f'(x) = \left| \sin\left(x + \frac{\pi}{3}\right) \right| - |\sin x| = 0 \Rightarrow x = \frac{\pi}{3} \text{ or } \frac{5\pi}{6} \text{ at } x = \frac{\pi}{3} \text{ local maxima exists}$

45. B

Sol. $\frac{d}{dx} \left(\frac{g(x)}{g'(x)} \right) = \frac{-g(x)}{g'(x)} \Rightarrow g(x) = \frac{1}{e} \cdot e^{e^x}$

46. C

Sol. (I) Expand the square

(II) $\int_0^1 g(x) dx + \int_{g(0)}^{g(1)} g^{-1}(x) dx = 1 \cdot g(1) - 0 \cdot g(0)$

$\int_0^1 (e^x + x + 1) dx + \int_2^{e+2} g^{-1}(x) dx = (e+2) \times 1 - 0 \times 2$

$\int_2^{e+2} g^{-1}(x) dx = (e+2) - \left[e^x + \frac{x^2}{2} + x \right]_0^1 = (e+2) - \left\{ \left(e + \frac{1}{2} + 1 \right) - e^0 \right\} = e+2 - e - \frac{1}{2} = \frac{3}{2}$

(III) $g(x) = ax + b$

(IV) Area bounded $= \frac{1}{2} \times 4 \times 4 - \int_{\frac{2}{3}}^2 \left(2 - \frac{x}{2x-1} \right) dx - \pi \left(\frac{1}{2} \right)^2$

Section - B

47. 0.00

Sol. $P(a) = b, P(b) = c$ and $P(c) = a$

$\Rightarrow \frac{b-c}{a-b} = I_1, \frac{c-a}{b-c} = I_2$ and $\frac{a-b}{c-a} = I_3$

48. 2.00

Sol. $AC \cdot BC = CS^2$

49. 2.00

Sol. Circle touches each other externally

50. 33.00

Sol. $P \left(\frac{3\sqrt{2}}{5}, \frac{-3}{5\sqrt{2}} \right) \Rightarrow \ell = \sqrt{\frac{36}{50} + \frac{9}{50} - \frac{1}{4}} = \sqrt{\frac{9}{10} - \frac{1}{4}} = \sqrt{\frac{36-10}{40}} = \sqrt{\frac{13}{20}}$

51. 1.00

Sol. Let $z = \frac{\cos \theta + i \sin \theta}{\cos \theta}$; $z^k = \frac{\cos k\theta + i \sin k\theta}{\cos^k \theta}$

$\sum_{k=0}^{n-1} z^k = \frac{1-z^n}{1-z} = \frac{1 - \frac{\cos n\theta + i \sin n\theta}{\cos^n \theta}}{1 - \frac{\cos \theta + i \sin \theta}{\cos \theta}} = \frac{\sin n\theta}{\sin \theta \cos^{n-1} \theta} + i \frac{\cos^n \theta - \cos n\theta}{\sin \theta \cos^{n-1} \theta}$

So, $\text{Re} \left(\sum_{k=0}^{n-1} z^k \right) = \frac{\sin n\theta}{\sin \theta \cos^{n-1} \theta} = \frac{\sin n \frac{\pi}{6}}{\sin \frac{\pi}{6} \left(\cos \frac{\pi}{6} \right)^{n-1}}$

52. 5.00

Sol. $P^{-1} + Q^{-1} = (P + Q)^{-1} \Rightarrow (P^{-1} + Q^{-1})(P + Q) = I \Rightarrow I + P^{-1}Q + Q^{-1}P + I = I$

Let $P^{-1}Q = A$, then $A + A^{-1} + I = 0 \Rightarrow A^2 + A + I = 0$

$A^3 = I \Rightarrow |A| = 1$

$|P^{-1}Q| = 1 \Rightarrow |P^{-1}||Q| = 1 \Rightarrow \frac{1}{|P|} \times |Q| = 1 \therefore |Q| = 5$

53. 0.50

Sol. The person B will score more if

Case-I: In first 9 chances A and B have same score and B is successful in 10th chance

Case-III: In first 9 chances B has more score.

54. 2.00

Sol. Multiply and divide by $\sin 2^\circ \sin 4^\circ \sin 6^\circ \dots \sin 88^\circ$

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FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
OPEN TEST – I
PAPER –2
TEST DATE: 05-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1.
Sol.

A

At $t = T$

$$Y = 0 \Rightarrow T = \frac{2u \sin \alpha}{g \cos 37^\circ}$$

$$\Rightarrow T = \frac{u \sin \alpha}{4}$$

Also at $t = T$, v is at 37° with normal on the plane i.e.

$$\tan 37^\circ = \frac{V_x}{-V_y}$$

$$\frac{3}{4} = \frac{u \cos \alpha - \frac{3}{5}gT}{-\left(u \sin \alpha - \frac{4}{5}gT\right)}$$

By solving $\tan \alpha = \frac{4}{9}$

$$\text{Also, } T = \frac{1}{4} \sqrt{388} \sin \alpha = \sqrt{\frac{388}{4}} \times \frac{4}{\sqrt{97}} = 2 \text{ sec}$$

Velocity of particle just before it collides

$$v_y = u_y + a_y T = -8$$

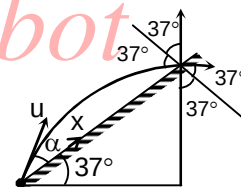
$$v_x = u_x + a_x T = 6$$

$$\therefore v = \sqrt{8^2 + 6^2} = 10 \text{ m/s}$$

Velocity after collision = 10 m/s

$$\therefore T' = \frac{2 \times 10}{g} = 2 \text{ sec}$$

Total times in returning is $2 + 2 + 2 = 6 \text{ sec}$



2.

A

Sol. $PA = qE + P_0A + Mg$

$$PV_0^\gamma = \text{constant}$$

After displacing piston through small distance x in downward directions pressure changes. Let new pressure be P'

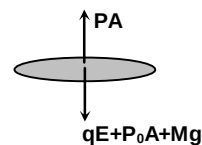
$$\text{So, } PV_0^\gamma = P'(V_0 - AX)^\gamma$$

$$PV_0^\gamma = P'V_0^\gamma \left(1 - \frac{\gamma AX}{V_0}\right)$$

$$\Rightarrow P = P' \left(1 - \frac{\gamma AX}{V_0}\right)$$

$$Mg + \frac{qE}{A} + P_0 = P' \left(1 - \frac{\gamma AX}{V_0}\right)$$

$$\omega = \sqrt{\frac{\left(\frac{qE + Mg}{A} + P_0\right) \gamma A^2}{MV_0}}$$



3.

A

Sol. $\frac{dm}{dx} = kx$

$$\Rightarrow \int_0^m dm = k \int_0^\ell x dx \Rightarrow k = \frac{2m}{\ell^2}$$

$$dl = x^2 dm = x^2 \frac{2m}{\ell^2} x dx$$

$$l = \frac{2m}{\ell^2} \int_0^\ell x^3 dx = \frac{m\ell^2}{2}$$

$$X_{cm} = \int x dm = \frac{2\ell}{3}$$

$$-(mg \sin 30^\circ) \frac{2\ell}{3} \theta = \frac{m\ell^2}{2} \alpha$$

$$\Rightarrow \alpha = -\frac{2g}{3\ell} \theta$$

$$T = 2\pi \sqrt{\frac{3\ell}{2g}}$$

4.

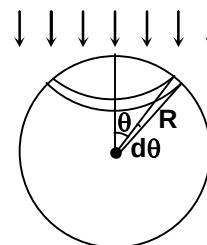
C

Sol. $E \cdot 2\pi r = \pi r^2 \frac{dB}{dt}$

$$\Rightarrow E = \frac{r}{2} \frac{dB}{dt}$$

$$rdq \cdot E = \frac{r^2}{2} dq \frac{dB}{dt}$$

$$d\tau = \frac{dB}{dt} \frac{r^2}{2} dq$$



$$\int d\tau = -\frac{dB}{dt} \int_0^{\pi/2} \frac{1}{2} R^2 \sin^2 \theta \cdot 2\pi R \sin \theta R d\theta \sigma \quad (\because r = R \sin \theta)$$

$$\tau = \frac{dB}{dt} \frac{4}{3} \sigma \pi R^4$$

$$\int \tau dt = \frac{4\sigma \pi R^4}{3} \int_B dB$$

$$I\omega - 0 = \frac{4}{3} B \sigma \pi R^4$$

Conservation of angular momentum

$$I\omega = I \frac{v}{R} + mvR$$

$$\Rightarrow v = \frac{I\omega R}{I + mR^2} = \frac{5QBR}{21m}$$

5. A, D

Sol. $F - T - \mu mg \cos \theta - mg \sin \theta = ma_1$

$$\Rightarrow 106 - T - 40 = 4a_1 \quad \dots(i)$$

$$10g \sin \theta - T - f_s = 10a_c$$

$$\Rightarrow 60 - T - f_s = 10a_c \quad \dots(ii)$$

$$f_s R + rT = I\alpha$$

$$\Rightarrow 2f_s + T = 10\alpha \quad \dots(iii)$$

$$r\alpha - a_c = a_1 \quad \dots(iv)$$

$$\text{and } a_c = R\alpha \quad \dots(v)$$

Solving the equations $a_c = 2 \text{ m/s}^2$, $a_1 = -1 \text{ m/s}^2$

6. A, C, D

Sol. $\varepsilon - i_1 r + \varepsilon - i_1 r + \varepsilon - i_1 r + \varepsilon - i_1 r + \varepsilon - i_1 r = 0$

$$\Rightarrow i_1 = \frac{\varepsilon}{r} = 5A$$

$$\varepsilon - i_2 r - i_2 R - i_2 R = 0 \Rightarrow i_2 = \frac{\varepsilon}{r + 2R} = 1A$$

$$I = i_1 + i_2 = 6A$$

Charges on capacitors $2C$, C is zero since current in resistance R is zero

$$\therefore \varepsilon - i_2 R = \frac{q}{4C}$$

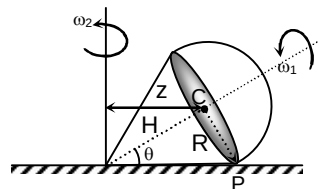
$$q = 16\mu C$$

7. B, D

Sol. Velocity of point P is $\omega_2 \times 2R - \omega_1 R = 0$

$$\frac{T_1}{T_2} = \frac{\omega_2}{\omega_1} = \frac{1}{2}$$

$$f_s = \frac{mv_0^2}{z} = \frac{2}{3} \frac{mv_0^2}{R}$$



8. A, B, C

Sol. Velocity of particle after collision in vertical direction is 10 m/s

$$\mu N \Delta t = \mu \times 20 \text{ m}$$

$$\text{For disc, } \mu N \Delta t = mv_c$$

$$\therefore v_C = 4 \text{ m/s}$$

$$\mu N \Delta t \frac{R}{2} = \frac{MR^2}{2} (\omega - 0)$$

$$\Rightarrow \omega = 20 \text{ rad/s}$$

9. B, C

Sol. Initial tension in spring $kx = \frac{4mg}{7}$

FBD of block just after string is cut

Since, $v = 0$

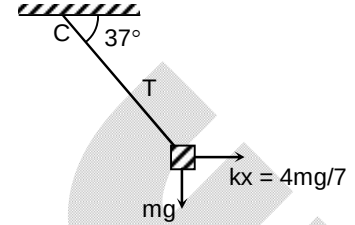
$\therefore$ acceleration along string is zero

$$T = k \cos 37^\circ + mg \sin 37^\circ = \frac{37}{7} N$$

Acceleration is only perpendicular to spring

$$ma = mg \cos 37^\circ - kx \sin 37^\circ$$

$$\Rightarrow a = \frac{32}{7} \text{ m/s}^2$$



10. A, C

Sol. Using conservation of energy and momentum when particle leaves end B, velocity of the block will be zero.

At lower most point vertical acceleration is $\frac{v^2}{R}$

$$\therefore 5v_1 = 10v_2$$

$$\text{and } 5 \times g \times 6 = \frac{1}{2} \times 5v_1^2 + \frac{1}{2} \times 10 \times v_2^2$$

$$\text{Solving equations } v_1 = \sqrt{80}, v_2 = \sqrt{20}$$

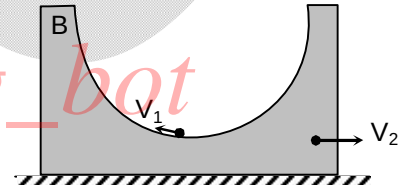
$$v_{P,B} = \sqrt{80} + \sqrt{20}$$

$$a_{P,B} = \frac{v_{P,B}^2}{R} = \frac{(\sqrt{80} + \sqrt{20})^2}{2}$$

$$\vec{a}_{P,g} = \vec{a}_{P,B} + \vec{a}_{B,g}$$

$$\vec{a}_{B,g} = 0 \text{ (when particle is at lowest position)}$$

$$\therefore \vec{a}_{P,g} = \vec{a}_{P,B} = 90 \text{ m/s}^2$$



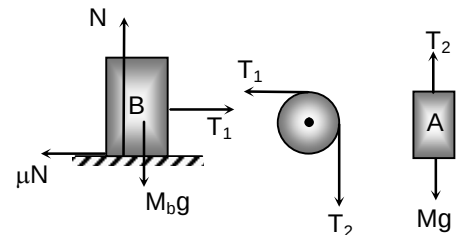
Section - B

11. 2

Sol. $T_1 = \mu M_B g$
 $T_2 = mg$
 due to friction on pull
 in limiting case, when block B is about to slide

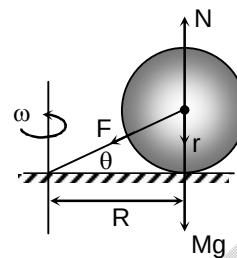
$$T_2 = T_1 e^{\mu \frac{\pi}{2}}$$

Solving equation $m = 2 \text{ kg}$



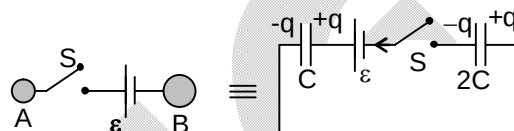
12. 4

Sol. $F \cos \theta = m\omega^2 R$
 $F \sin \theta + mg = N$
 $\therefore N = mg + m\omega^2 r = 36 \text{ N}$



13. 3

Sol. $\frac{q}{C} - \varepsilon + \frac{q}{2C} = 0$
 $q = \frac{2}{3} C\varepsilon$
 Loss energy $H = q\varepsilon - \Delta U$
 $= \frac{3}{4} \frac{q^2}{C}$, where $C = 4\pi\varepsilon_0 \times 9 = 30 \mu\text{J}$

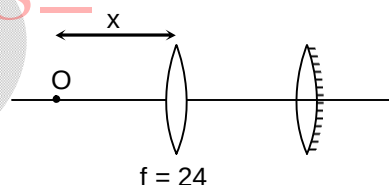


14. 4

Sol. Standing wave $y = 2A \sin kx \cos \omega t$
 Has energy $E = \mu \omega^2 A^2 \ell$

15. 6

Sol. $\frac{1}{v} - \frac{1}{-x} = \frac{1}{24}$
 $\Rightarrow \frac{1}{v} = \frac{1}{24} - \frac{1}{x}$
 $\frac{1}{v} = -\frac{1}{\left(\frac{24-x}{24x}\right)}$



Object distance for silvered lens is $(14-x) + \frac{24x}{(24-x)}$ for image to be on object O, this distance

must be equal to equivalent radius of mirror.

For (Reflecting lens is effectively mirror)

$$-\frac{2}{R_{eq}} = 2 \left(\frac{3}{2} - 1 \right) \left(\frac{1}{32} - \frac{1}{-32} \right) - \frac{3}{-32}$$

$$\Rightarrow R_{eq} = -16 \text{ cm}$$

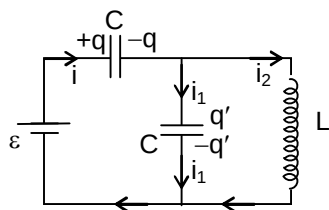
$$\therefore 16 = (14-x) + \frac{24x}{(24-x)}$$

$$\Rightarrow x = 6 \text{ cm}$$

16. 7

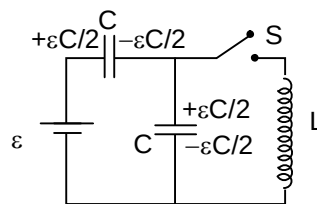
Sol. Using Kirchhoff's Law

$$\varepsilon - \frac{q}{C} - \frac{q'}{C} = 0 \quad \dots(i)$$



$$-\frac{q}{C} + L \frac{di_2}{dt} = 0$$

$$\varepsilon - \frac{q}{C} - L \frac{di_2}{dt} = 0$$



$$\dots(ii)$$

$$\dots(iii)$$

Solving these equations and at $t = 0$, $i_2 = 0$ and $i_2 = \varepsilon \sqrt{\frac{C}{2L}} \sin \omega t$

$$\therefore i_{2(max)} = \varepsilon \sqrt{\frac{C}{2L}} = 7A$$

17. 5

Sol. Consider some resistance to the voltmeter and applying Kirchhoff's law

18. 4

Sol. $K_{e_{max}} = h\nu - \phi = \frac{6.6 \times 10^{-34} \times 7.27 \times 10^{14}}{1.6 \times 10^{-19}} - 1 = 2 \text{ eV}$

Maximum potential difference is 2V

$$U = \frac{1}{2} CV^2$$

Chemistry

PART – II

Section – A

19. A

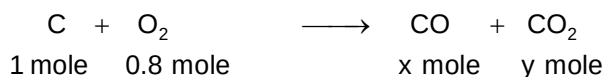
Sol. AB_2L_3 has linear shape but AB_3L_2 has T-shape.

20. B

Sol. Total moles of air = $\frac{PV}{RT} = \frac{2 \times 49.26}{0.0821 \times 300} = 4$ mole

$$\therefore n_{O_2} = 4 \times 0.2 = 0.8 \text{ mole}$$

Moles of 'C' = 1



Apply POAC for C-atom

$$1 \times 1 = x \times 1 + y \times 1 \text{ or } x + y = 1 \quad \dots (1)$$

Also, apply POAC for O-atoms,

$$0.8 \times 2 = x \times 1 + y \times 2 \text{ or } x + 2y = 1.6 \quad \dots (2)$$

On solving Equation (1) and (2), we get

$$y = 0.6 \text{ and } x = 0.4$$

So, total amount of heat produced

$$= -0.4 \times (26) - 0.6 \times (94)$$

$$= -10.4 - 56.4$$

$$= -66.8 \text{ kcal}$$

21. A

Sol. Since, decomposition of N_2O on hot platinum surface follows ZERO order kinetics:

$$\text{So, } t_{1/2} = \frac{a}{2k_0}$$

$$\text{or } a = (2k_0) \times t_{1/2}$$

$$\log a = \log(2k_0) + \log t_{1/2}$$

$$y = C + mx$$

$$m = 1 \text{ or } \tan \theta = 1 \text{ or } \theta = 45^\circ$$

$$C = \log 2k_0 = +ve$$

22. D

$$\text{Sol. } \alpha = \frac{\theta}{\ell(\text{dm}) \times C(\text{g/ml})}$$

$$-92.4^\circ = \frac{-27.7^\circ}{(1 \text{ dm}) \times C}$$

$$C = \frac{27.7}{1 \times 92.4} = 0.2997 \text{ g/ml}$$

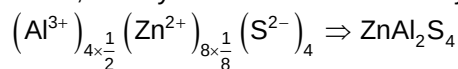
So, mass of fructose dissolved in 100 mL solution

$$= 0.2997 \times 100$$

$$= 29.97 \text{ g.}$$

23. A, C, D

Sol. Since, the crystal should be electrically neutral. So, empirical formula of the solid should be:


 Since, Zn^{2+} ions are present in tetrahedral void, so, its coordination number = 4.

 Also, Al^{3+} ions are present in octahedral void, so, its coordination number = 6.

24. A, C

 Sol. Since, the solution contains a weak base (NH_4OH) and salt of this weak base with strong acid, i.e. $(\text{NH}_4)_2\text{SO}_4$. So, it is a basic buffer.

By Henderson's Equation:

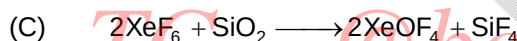
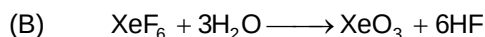
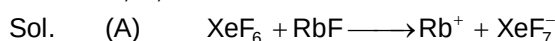
$$\text{pOH} = \text{pK}_b + \log \frac{2 \times 0.01}{0.02} = \text{pK}_b$$

$$\therefore \text{pH} = 14 - \text{pOH}$$

$$\text{pH} = 14 - (5 - \log 2) = 9 + \log 2$$

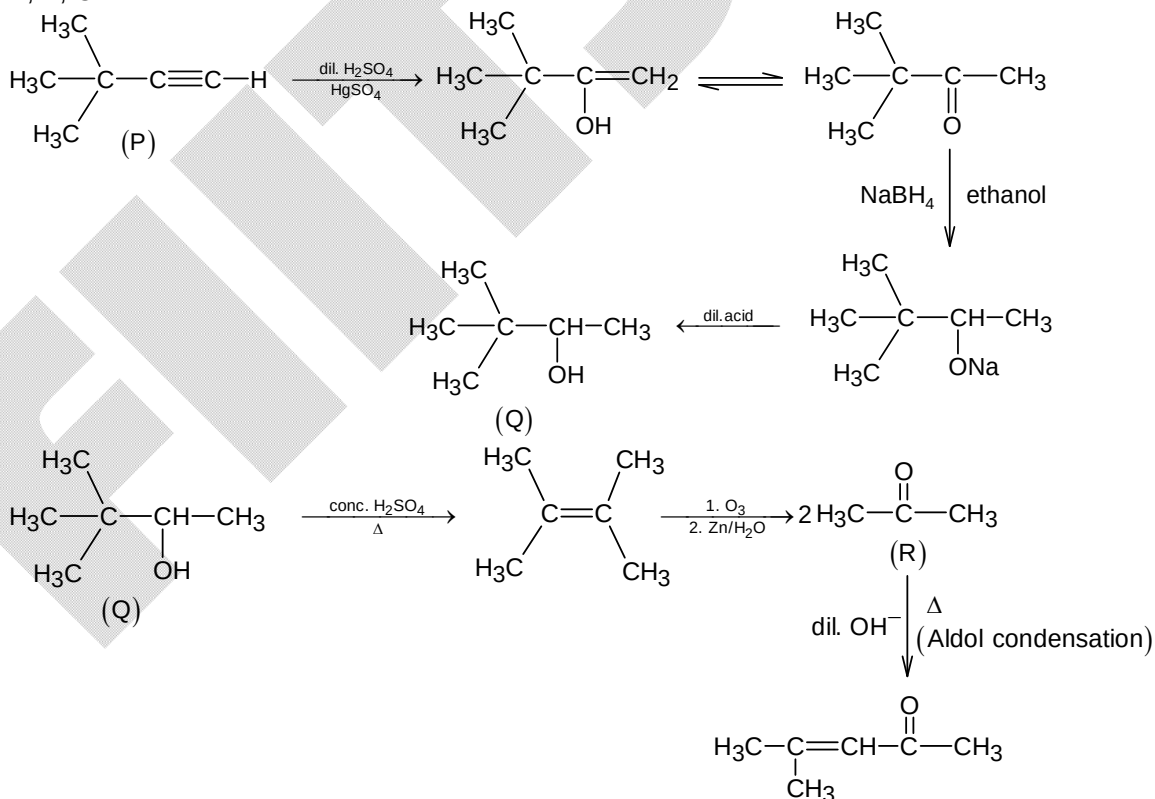
Since, it's a buffer, so dilution does not affect the pH.

25. A, B, C



26. A, B, C

Sol.



27. A, B, C, D

Sol.
$$r = \frac{2[A]}{3 + [A]}$$

So, integrated rate equation of the above reaction is

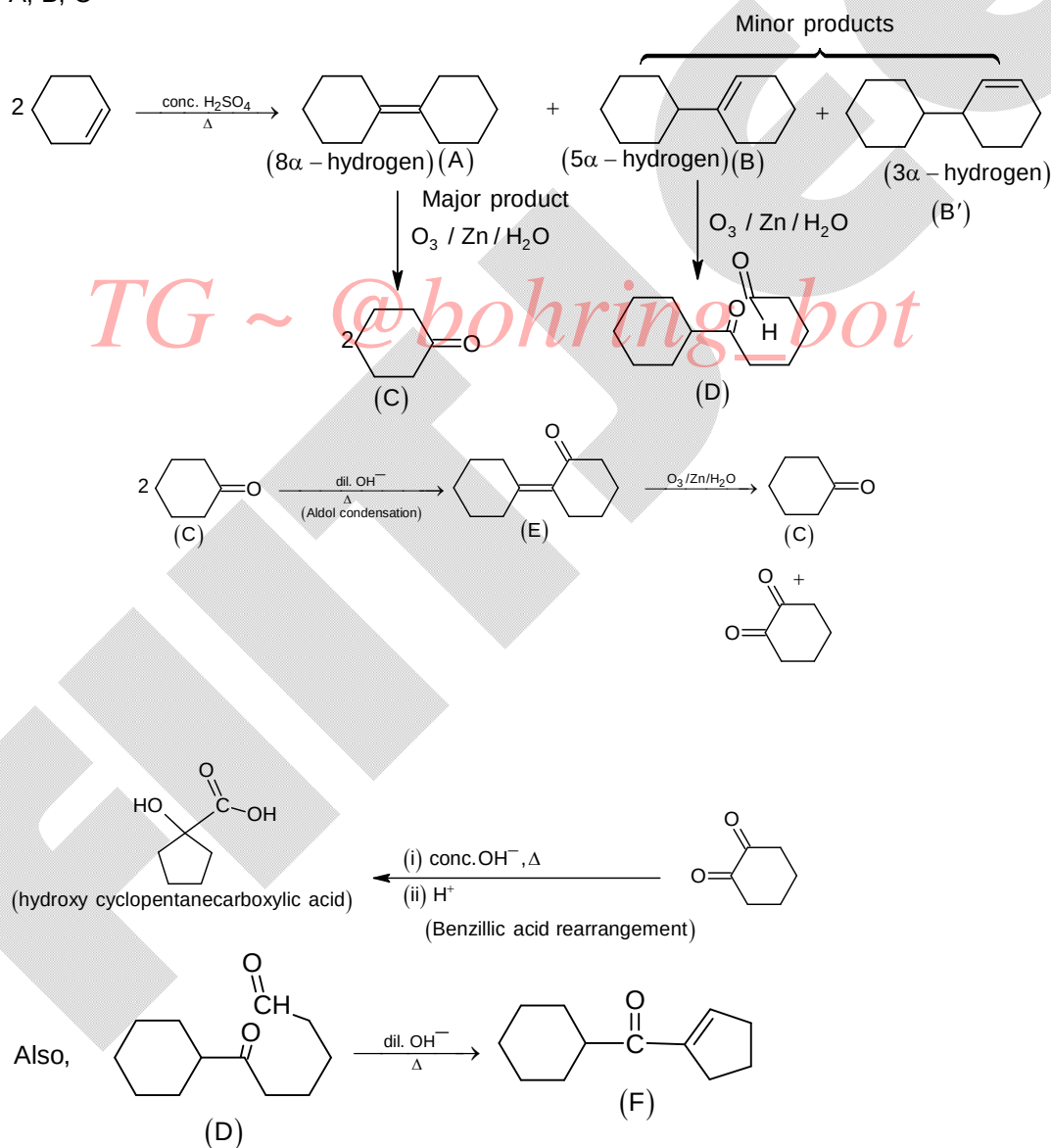
$$t = \frac{1}{2} \left[[A]_0 - [A] + 3 \ln \frac{[A]_0}{[A]} \right]$$

So, $t_{1/2} = \frac{1}{4} [A]_0 + \frac{3}{2} \ln 2$

$$t_{3/4} = \frac{3[A]_0}{8} + 3 \ln 2 = \frac{3}{8} \times 8 + 3 \times 0.7 = 5.1 \text{ minutes}$$

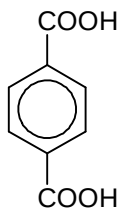
28. A, B, C

Sol.



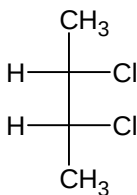
Section – B

29. 8
Sol.



has 3 planes of symmetry, i.e. $x = 3$

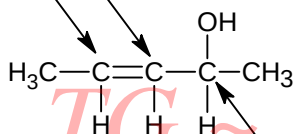
(Terephthalic acid)



It has one plane of symmetry and one centre of symmetry. So, $y = 2$.

(Meso-2,3-dichlorobutane)

Stereocentres



(Pent-3-en-2-ol)

Stereocentres

So, $z = 3$.

30. 8

Sol. $x = 2$, $y = 2$, $z = 4$.
 $n = 1$, $m = 2$

31. 6

Sol. Freezing point in Kelvin = $(-1.8 m + 273)$ K

Boiling point in Kelvin = $(373 + 0.5 m)$ K

Difference of boiling point and freezing point = $100 + 2.3 m = 100.46$

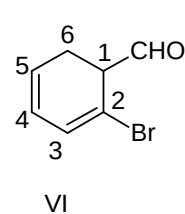
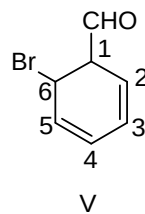
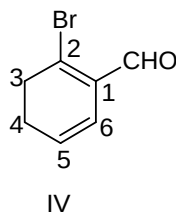
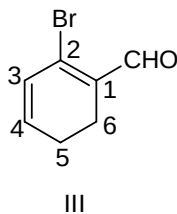
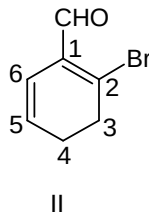
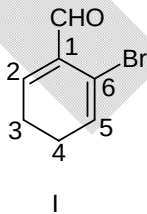
$\therefore M = 0.2$

Now, $0.2 = \frac{P \times 1000}{60 \times 500}$

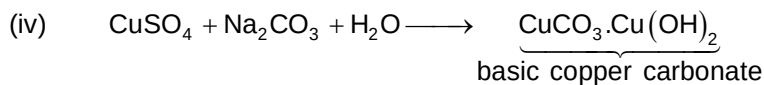
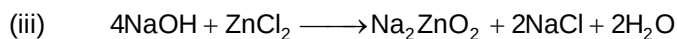
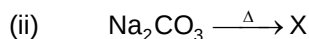
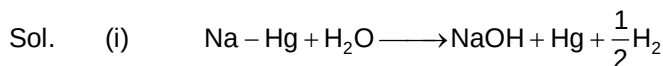
$\therefore P = 6$ gram

32. 4

Sol.



33. 4



(v) In case of alkali metals, for performing flame test, chlorides are preferred over carbonates as chlorides are more volatile.

34. 5

Sol. $N_{\text{H}_2\text{O}_2} = \frac{22.4}{5.6} = 4$

Let, mass of pure H_2S present in impure sample = 'w' gm

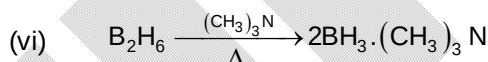
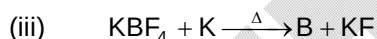
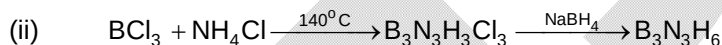
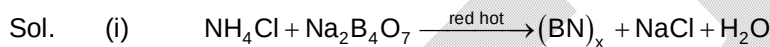
Then $4 \times 5 \times 10^{-3} = \frac{w}{\left(\frac{34}{2}\right)} \Rightarrow w = 0.34 \text{ gm}$

So, % of pure H_2S in impure sample = $\frac{0.34}{0.40} \times 100 = 85\%$

$\therefore 17x = 85$

$\therefore x = 5$

35. 1



36. 8

Sol. Average velocity = $\sqrt{\frac{8RT}{\pi M}} = 4 \times 10^2$

$\therefore RT = 2\pi M \times 10^4$

Total kinetic energy of 'He' = $\frac{3}{2}nRT = \frac{3}{2} \times \frac{6}{4} \times RT = \frac{9RT}{4}$

Total kinetic energy of 'He' = $\frac{9}{4} \times 2\pi M \times 10^4$

= $\frac{9}{4} \times 2\pi \times (4 \times 10^{-3}) \times 10^4$

= 180π Joule

Total kinetic energy of 'Ne²⁰' = $\frac{3}{2}nRT = \frac{3}{2} \times \frac{12}{20} \times RT$

$$\begin{aligned} &= \frac{9}{10} \times 2\pi M \times 10^4 \\ &= \frac{9}{10} \times 2\pi \times (20 \times 10^{-3}) \times 10^4 \\ &= 360\pi \text{ Joule} \end{aligned}$$

$$\text{Average kinetic energy of mixture per mole} = \frac{360\pi + 180\pi}{1.5 + 0.6}$$

$$\begin{aligned} &= \frac{540}{2.1} \times 3.14 \\ &= 807.42 \text{ J} \\ &= 0.80742 \text{ kJ} \\ \text{So, } x &= 0.8 \\ \therefore 10x &= 8. \end{aligned}$$

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Mathematics

PART – III

Section – A

37. D
Sol. Favorable cases: For first diagonal we have 12 choices then for the second to be skew we have 5 choices. So number of favorable choices is $\frac{12 \times 5}{2} = 30$
Required probability = $\frac{30}{{}^{12}C_2} = \frac{5}{11}$
38. C
Sol. Since $[\sqrt{2066}] = [\sqrt{2067}] = [\sqrt{2068}] = [\sqrt{2069}] = 45$ and $2023 + 45 = 2068$
39. C
Sol. Let $z = -x - \frac{\pi}{6}$, then $z \in \left[\frac{\pi}{6}, \frac{\pi}{4}\right]$ and $2z \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$
Then, $y = \cot z + \tan z + \cos z = \frac{2}{\sin 2z} + \cos z$
Since, both $\frac{2}{\sin 2z}$ and $\cos z$ are monotonic decreasing in this case
Then $y_{\max} = \frac{2}{\sin \frac{\pi}{3}} + \cos \frac{\pi}{6} = \frac{11\sqrt{3}}{6}$
40. A
Sol. $x^a = y^b = z^c = \lambda$, now x, y, z are in Geometric Progression
 $\Rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$ therefore a, b, c are in Harmonic Progression
By G.M. > H.M.; $\sqrt{ac} > b$ for a^3, b^3, c^3
A.M. > G.M.; $\frac{a^3 + c^3}{2} > (\sqrt{ac})^3 > b^3$
41. A, D
Sol. Draw variable lines of slope $4x - 3y$ that passes through the points on or inside the triangle
42. A, C
Sol. $\left|\frac{1}{z} + z_0\right| = \frac{1}{|z|}$ multiply both side by $\left|\frac{z}{z_0}\right|$, we have $\left|z - \left(-\frac{1}{z_0}\right)\right| = \frac{1}{|z_0|}$
Circle with centre $-\frac{1}{z_0}$ and radius $\frac{1}{|z_0|}$, however we have to exclude the point $z = 0$ from the locus
43. B, D
Sol. If $PA + PB$ is minimum $\Rightarrow$ Area of $\triangle PAB$ is minimum
 $\Rightarrow C$ is mid-point of $AB \Rightarrow$ Equation of AB is $x + y = 10$

44. A, B, C

Sol. Since, $\log_2 x - 1 \geq 0$; Let $\sqrt{\log_2 x - 1} = t$, we have $t - \frac{3}{2}t^2 + \frac{1}{2} > 0$ and $t \geq 0$
 Solution of the above inequality is $0 \leq t < 1 \Rightarrow 0 \leq \log_2 x - 1 < 1 \Rightarrow 2 \leq x < 4$

45. A, B, D

Sol. $f(x) = \frac{x^2}{2} + 3$ if x is even and $f(x) = \frac{x+1}{2}$ if x is odd

46. B, C

Sol. If $\lambda \geq 6$, then sum of solutions = 5π and if $\lambda \leq -6$, then sum of solutions = 3π

Section - B

47. 1

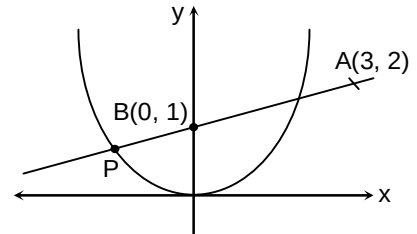
Sol. $\left(x + \frac{1}{x^{2023}}\right)(1 + x^2 + x^4 + \dots + x^{2022}) = 2024$
 $\Rightarrow (x + x^3 + x^5 + \dots + x^{2023}) + \left(\frac{1}{x^{2023}} + \frac{1}{x^{2021}} + \dots + \frac{1}{x}\right) = 2024$
 $\Rightarrow$ By A.M., G.M. inequality, we get
 $(x + x^3 + x^5 + \dots + x^{2023}) + \left(\frac{1}{x} + \frac{1}{x^3} + \dots + \frac{1}{x^{2023}}\right) \geq 2 \times 1012 = 2024$
 Where the equality holds if and only if $x = \frac{1}{x}, x^3 = \frac{1}{x^3}, \dots, x^{2023} = \frac{1}{x^{2023}} \Rightarrow x = 1$

48. 7

Sol. $|z_1| = |z_2| = 3$ and $z_1 \bar{z}_2 + \bar{z}_1 z_2 = -9 \Rightarrow |z_1 \bar{z}_2| = 9$
 Let $z_1 \bar{z}_2 = 9(\cos \theta + i \sin \theta)$, then $\bar{z}_1 z_2 = 9(\cos \theta - i \sin \theta)$
 $\Rightarrow 18 \cos \theta = -9 \Rightarrow \cos \theta = -\frac{1}{2}$
 $z_1 \bar{z}_2 = -9\omega$ and $\bar{z}_1 z_2 = -9\omega^2$ where ω is cube root of unity other than 1
 $\log_3 |(z_1 \bar{z}_2)^{2023} + (\bar{z}_1 z_2)^{2023}| = 2 \times 2023 = 4046$

49. 3

Sol. $f(x) = \sqrt{(x-3)^2 + (x^2-2)^2} - \sqrt{x^2 + (x^2-1)^2}$
 Let P be any point on the parabola $y = x^2$ and A(3, 2), B(0, 1)
 By Δ inequality $|PA - PB| \leq AB$
 $|PA - PB| \leq \sqrt{10}$
 $\therefore k = \sqrt{10}$



50. 6

Sol. Locus of I is the tangent at the vertex $\Rightarrow$ Locus is $x = \pm 3$

51. 8

Sol. $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 62 \Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -10 \Rightarrow \vec{c} = -4\vec{a} = -2\vec{b}$

52. 3

Sol. $\ln(1-t^2) = \sum_{r=1}^{\infty} \left(-\frac{t^{2r}}{r} \right) \Rightarrow \int_0^1 f(x) dx = \sum_{r=1}^{\infty} \frac{-1}{2r(r+1)} = -\frac{1}{2}$

53. 4

Sol. Let $y = mx + c$ be the tangent $\Rightarrow x^4 - 2x^2 - x - (mx + c) = (x^2 + ax + b)^2$
 $\Rightarrow a = 0, b = -1, m = -1, c = -1$
 $\Rightarrow x + y + 1 = 0$ is tangent to the curve at $(-1, 0)$ and $(1, -2)$
 $\Rightarrow |\alpha_1| + |\alpha_2| + |\beta_1| + |\beta_2| = 4$

54. 3

Sol. A and B are $(1, 2, 3)$ and $(2, 1, 2)$ respectively

If $AC = d_1$ and $BC = d_2$, then volume of tetrahedron $= \frac{1}{6} \cdot AB d_1 d_2 = \frac{\sqrt{3}}{6} d_1 d_2$

Also, $CD^2 = d_1^2 + d_2^2 + 3 \Rightarrow d_1^2 + d_2^2 = 24$

Using AM, GM, $\frac{d_1^2 + d_2^2}{2} \geq d_1 d_2 \Rightarrow d_1 d_2 \leq 12 \Rightarrow V = 2\sqrt{3}$

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FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – III
PAPER –1
TEST DATE: 08-02-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

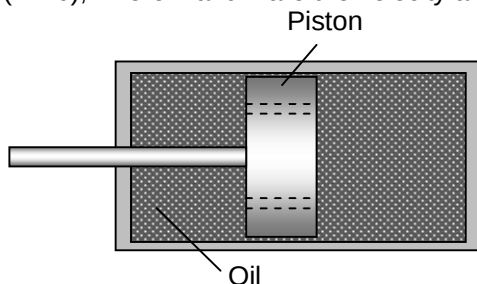
Physics

PART – I

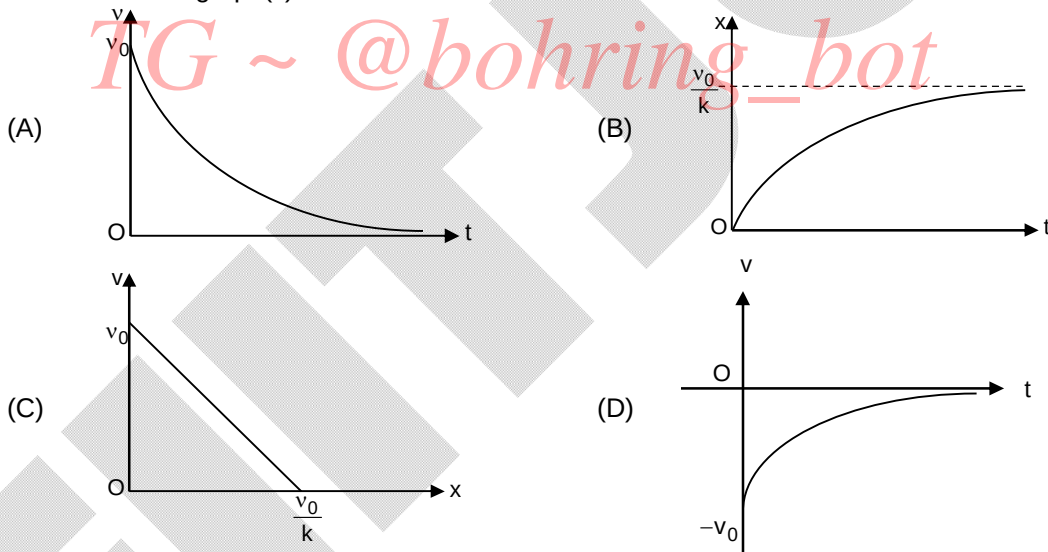
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

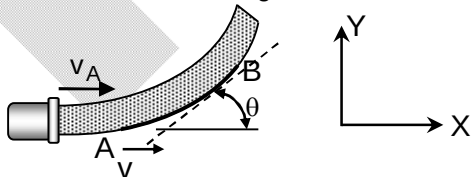
1. The brake mechanism used to reduce recoil in certain types of guns consist essentially of a piston attached to the barrel and moving in a fixed cylinder filled with oil. As the barrel recoils with an initial velocity v_0 , the piston moves and oil is forced through orifices in the piston, causing the piston and the barrel to decelerate at a rate proportional to their velocity; that is, $a = -kv$, ($k > 0$), where v and x are the velocity and position of barrel at time t .



Choose correct graph(s):



2. A nozzle discharges a stream of water of cross – sectional area A with a velocity v_A . The stream is deflected by a single blade which moves to the right with a constant velocity V . Assuming that the water moves along the blade at constant speed, Choose the correct option(s);

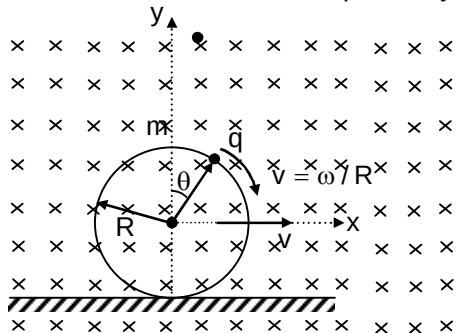


- (A) The Y component of the force exerted by the blade on the steam is $A\rho(V_A - V)^2 \sin\theta$.
 (B) The X component of the force exerted by the blade on the steam is $A\rho(V_A - V)^2 (1 - \cos\theta)$

(C) The X component of the force exerted by the blade on the steam is $A\rho(V_A - V)^2 \cos \theta$

(D) The velocity V for which maximum power is developed is $V = \frac{V_A}{3}$

3. A ring of mass m and radius R is set into pure rolling on horizontal rough surface, in a uniform magnetic field of strength B as shown in the figure. A point charge (q) of negligible mass is attached to rolling ring. Friction is sufficient so that it does not slip at any point of its motion. (θ is measured in clockwise from positive y-axis)



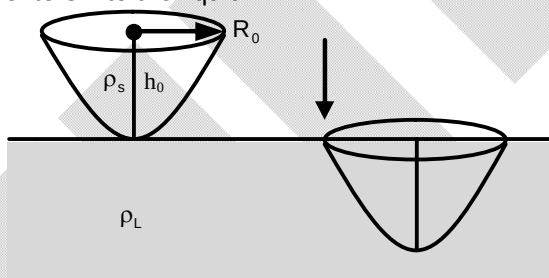
(A) Ring will continue to move with constant velocity.

(B) The value of friction acting on ring is $Bqv \cos \theta$.

(C) The value of friction acting on ring is $Bqv \sin \theta$.

(D) Ring will lose contact with ground if v is greater than $\left(\frac{mg}{2qB} \right)$

4. A solid paraboloid of base radius R_0 and height h_0 is having uniform volume mass density of ρ_s is Inverted and placed just above the surface as shown, of a liquid having volume mass density of ρ_L . When paraboloid is released from rest it has been observed that when it becomes completely submerged in the liquid the paraboloid comes to the rest again. There exist uniform gravity of g . Assume that the liquid body is very large so that liquid level is not changing as the paraboloid enters into the liquid.



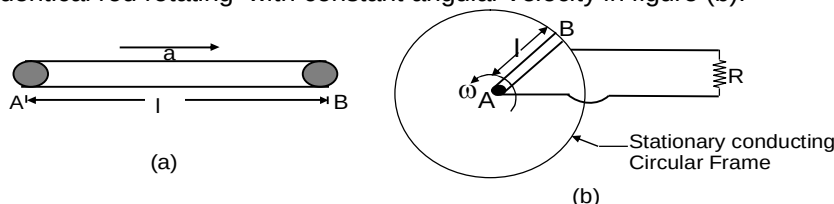
(A) The phenomenon is true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.

(B) The phenomenon is not true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.

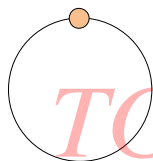
(C) The ratio of $\frac{\rho_L}{\rho_s}$ is 2.

(D) The ratio of $\frac{\rho_L}{\rho_s}$ is 3.

5. consider a metal rod of length L that is given a uniform acceleration as shown in figure (a) and an identical rod rotating with constant angular velocity in figure (b).



- (A) If V_A and V_B are potentials of end A and B respectively, then $V_A < V_B$ in figure (a)
 (B) If V_A and V_B are potentials of end A and B respectively, then $V_A > V_B$ in figure (b)
 (C) Electric field inside rod has magnitude $\frac{ma}{e}$ in figure (a)
 (D) Electric field inside rod has magnitude $\frac{m\omega^2 r}{e}$ in figure (b)
6. A bead is fixed on a rigid ring of radius r and negligible mass as compared to that of the bead. Assembly thus formed is held motionless on a horizontal floor with the plane of the ring vertical and the bead on the top. Coefficient of friction between the ring and the floor is μ . Now the assembly is given a slight push so that the ring starts rolling. Acceleration of free fall is g . Choose the correct statements.



- (A) The angle rotated by the ring before it begins to slide is $\tan^{-1}(\mu)$
 (B) The speed of centre of the ring before it begins to slide is $\mu\sqrt{gR}$
 (C) The angle rotated by the ring before it begins to slide is $2\tan^{-1}(\mu)$
 (D) The speed of the centre of the ring before it begins to slide is $\mu\sqrt{2gR}$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

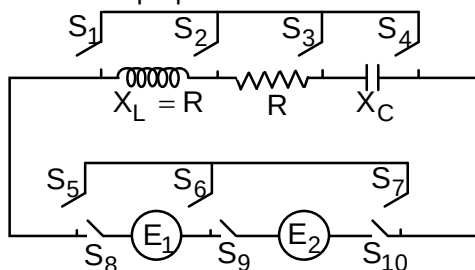
7. The ground state energy of hydrogen atom is -13.6 eV. A sample of hydrogen atoms in its ground state is irradiated with photons of energy 10.2 eV. The radiations from the above sample is then incident on two sample containing excited He^+ ions and excited Li^{2+} ions respectively. Both the samples absorb the radiations. (use $hc = 12400$ eV Å)

List-I		List-II	
(I)	No. of spectral lines obtained in the Spectra of Li^{2+}	(P)	6
(II)	No. of spectral lines obtained in the Spectra of He^+	(Q)	15
(III)	Smallest wavelength in spectra of He^+ is $(48.8) \times 10^{\circ}$ Å then X (Approx.) is	(R)	5
(IV)	Smallest wavelength in spectra of Li^{2+} is $(10.4) \times 10^{\circ}$ Å then Y (Approx.) is	(S)	10
		(T)	18

Codes:

	I	II	III	IV
(A)	R	P	P	T
(B)	Q	P	P	Q
(C)	Q	P	R	S
(D)	T	P	R	Q

8. Consider a given series LCR circuit as shown in figure. There are 10 switches. In list-I switches which are closed, is given and corresponding value is asked. In list-II these values are given. Choose the proper value from list-II to list-I using the codes given below.



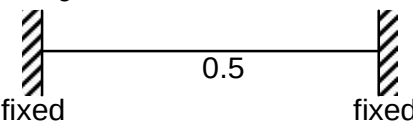
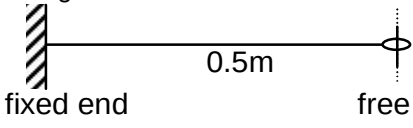
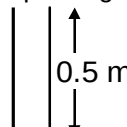
$$E_1 = E_0 \sin \omega t \quad E_2 = E_0 \sin(\omega t + \phi)$$

List-I		List-II	
(I)	It is observed that effective current in the circuit in case if either S_8, S_6 and S_7 or if S_5, S_6, S_9, S_{10} or if S_8, S_9 and S_{10} are closed (Rest open) remains same. Value of ϕ is:	(P)	$\frac{2\pi}{5}$
(II)	Switch S_8, S_9 & S_{10} are closed. Now its observed if rest switches are opened or switches S_3, S_4 are closed (rest opened) effective value of current in the circuit is same, value of $\frac{4\pi X_C}{5R}$ is	(Q)	$\frac{\pi}{2}$
(III)	Switch S_8, S_6, S_7 are closed. Now effective current if S_1, S_2 closed (Rest opened) is I_1 and If S_3, S_4 closed (Rest opened) is I_2 , value of $\pi \left(\frac{I_1}{I_2} \right)^2$ is	(R)	$\frac{2\pi}{3}$
(IV)	Switch S_5, S_6, S_9 and S_{10} are closed. Peak value current if rest are opened is I_3 and if S_2, S_3 closed (Rest opened) is I_4 . Value of $\pi \left(\frac{I_3}{I_4} \right)^2$ is	(S)	$\frac{8\pi}{5}$
		(T)	$\frac{\pi}{3}$

Codes:

	I	II	III	IV
(A)	Q	S	P	T
(B)	Q	T	P	S
(C)	R	P	T	Q
(D)	R	S	P	Q

9. In each of the four situations of List I, a stretched string or an organ pipe is given along with the required data. In case of strings the tension in strings is $T = 102.4 \text{ N}$ and the mass per unit length of string is 1 g/m . Speed of sound in air is 320 m/s . Neglect end corrections. The frequencies of resonance are given in List II. Match each situation in List I with the possible resonance frequencies given in List II.

List-I		List-II	
(I)	String fixed at both ends 	(P)	320 Hz
(II)	String fixed at one end and free at other end 	(Q)	480 Hz
(III)	Open organ pipe 	(R)	640 Hz
(IV)	Closed organ pipe 	(S)	800 Hz
		(T)	1280 Hz

Codes:

	I	II	III	IV
(A)	P	R	Q	S
(B)	P	S	R	Q
(C)	R	Q	S	P
(D)	Q	S	P	R

10. There exists a uniform electric field in space. Potential of few points in space is given below

Coordinates	(0, 0, 0)	(a, 0, 0)	(0, b, 0)	(0, 0, c)
Potential	1 V	2 V	3 V	4 V

Match the points in List I with their potential in List II

List-I		List-II	
(I)	(a, b, 0)	(P)	4V
(II)	(a, b, c)	(Q)	7V
(III)	(-a, -b, -c)	(R)	0V
(IV)	(0, b, c)	(S)	6V
		(T)	-5V

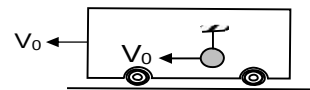
Codes:

	I	II	III	IV
(A)	R	T	Q	S
(B)	P	Q	T	S
(C)	S	Q	P	R
(D)	T	P	Q	R

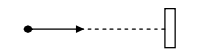
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

11. A pendulum with massless string of length l and bob of mass m is attached inside a railway carriage which is moving with speed v_0 as shown. Carriage is brought to rest uniformly by applying brakes. When carriage comes to rest pendulum string is horizontal. Minimum magnitude of retardation of carriage such that bob completes vertical circular motion is (in ms^{-2}). ($g = 10 \text{ ms}^{-2}$)



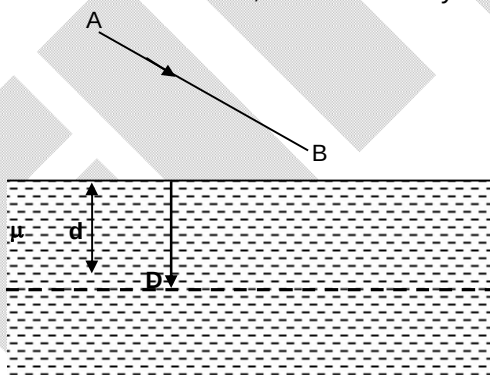
12. A small boat is travelling downstream with velocity 5 m/s relative to river flowing with speed $7/3 \text{ m/s}$. It encounters a log floating in the river at some distance from itself. It turns by an angle of 37° and just crosses the edge of the log after 8 sec . Find time (in seconds) taken by boat to just cross the edge of log if log is kept fixed.



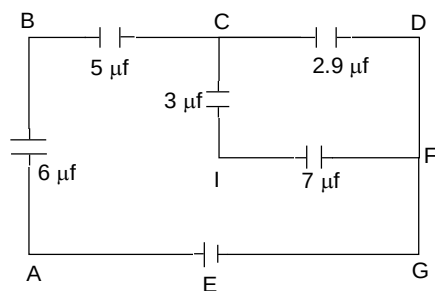
13. In YDSE, if sources are incoherent, the intensity on screen is $13I_0$. When these sources are coherent then minimum intensity on screen is I_0 . The maximum intensity produced by these coherent sources on screen is $______ \text{ W/m}^2$ [given $I_0 = 1 \text{ W/m}^2$]

14. The electrostatic flux through a rectangular area of $\frac{7}{188} \text{ m}^2$ lying in a plane $2x + 3y + 6z = 10$ due to an electric field $(8\hat{i} + 6\hat{j} + 10\hat{k}) \text{ V/m}$ will be in $(\text{V} \cdot \text{m})$

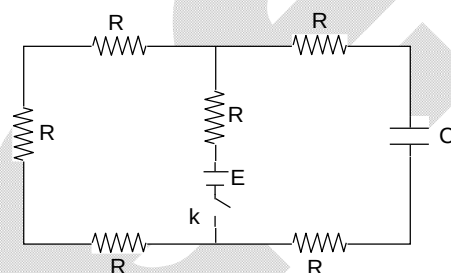
15. A diver D is still under water ($\mu = \frac{4}{3}$) at a depth $d = 10 \text{ m}$. A bird is diving along line AB at a constant velocity in air. When the bird is exactly above the diver he sees it at a height of 50 m from himself and velocity of the bird appears to be inclined at 45° to the horizontal. At what distance from the diver, the bird actually hits the water surface. (in meter)



16. Consider the circuit shown. If in steady state $V_A - V_C = 1$ volt, find $11(V_I - V_G)$ in steady state.



17. In the given circuit initially the capacitor is uncharged. At $t = 0$ the key k is closed. The time constant of circuit is n times of RC . Find the value of ' n '.



18. The pitch of a screw gauge is 1 mm and its cap is divided into 100 divisions. When nothing is placed between its studs, the zero of the circular scale lies 6 divisions below the reference line. When a wire is placed between its studs, the main scale reading is 4 mm and 56th division of circular scale coincides with the reference line. The length of the wire is 4.20 cm. Find the curved surface area (*in cm^2*) of the wire in true significant figures.

Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

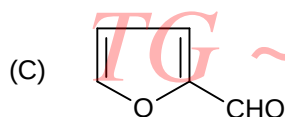
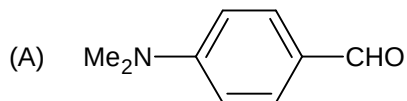
19. The radial wave equation for hydrogen atom is:

$$\psi = \frac{1}{16\sqrt{4}} \left(\frac{1}{a_0} \right)^{3/2} \left[(x-1)(x^2 - 8x + 12) \right] e^{-x/2}$$

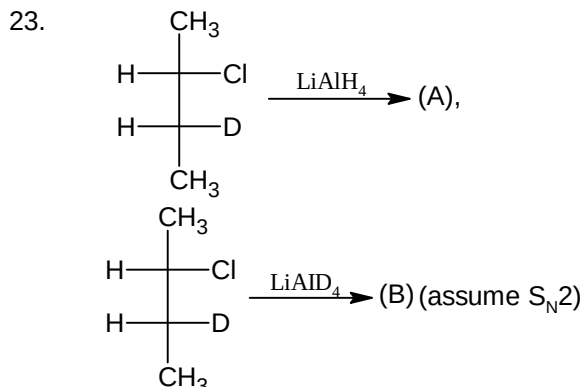
where $x = 2r / a_0$, a_0 = radius of first Bohr's orbit.

Select the correct statement(s)

- (A) minimum position of node will be $0.5a_0$
 (B) maximum position of node will be $3a_0$
 (C) There will be 3 radial nodes
 (D) There will be 2 radial nodes
20. How many of the following will give Cannizzaro reaction?



21. On the basis of MO theory, which is correct?
 (A) The bond order for C_2 molecule is two and both bond are π bond
 (B) The bond order for C_2 molecule is two and one bond is π and one is σ
 (C) The bond order for C_2 molecule is two and both bond are σ bond
 (D) Total bonding e^- in C_2 is 8
22. Which of the following statement(s) is (are) correct about slag?
 (A) The calcium silicate, CaSiO_3 is obtained in slag formation zone in the extraction of iron from haematite ore.
 (B) The chemical composition of "slag" formed during the smelting process in the extraction of copper is FeSiO_3
 (C) In blast furnace/Bessemer converter, the upper layer of molten liquid (i.e. molten metal) is of slag.
 (D) The slag is fusible matter.



Select the correct is/are statement(s)

- (A) A is optically active
 (B) B is optically active
 (C) A is optically active, but B is optically inactive
 (D) A is optically inactive, B is optically active
24. Which of the following orders is/are correct?
- (A) $\text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$ – Boiling point
 (B) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$ – Thermal stability
 (C) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$ – Basic character
 (D) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$ – Bond Angle

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

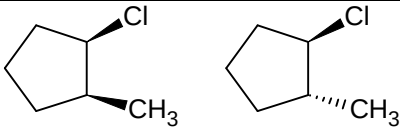
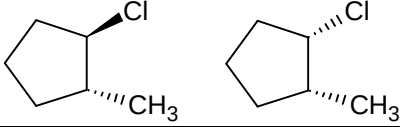
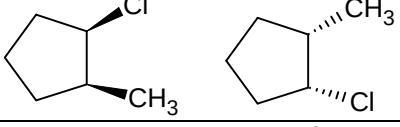
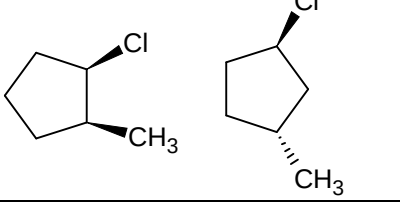
25. Consider the following list:

List - I			List - II		
	Electrode			Anode	Cathode
(I)	Fused NaCl	Graphite & mercury	(P)	$\text{O}_{2(g)}$	$\text{Na}_{(g)}$
(II)	Conc. NaCl	Pt	(Q)	$\text{O}_{2(g)}$	$\text{H}_2(g)$
(III)	Dil. NaCl	Pt	(R)	$\text{Cl}_{2(g)}$	$\text{H}_2(g)$
(IV)	Dil. $\text{NaCl}_{(aq)}$	Mercury	(S)	$\text{Cl}_{2(g)}$	$\text{Na}_{(s)}$
			(T)	$\text{O}_{2(g)}$	$\text{Cl}_{2(g)}$

Codes:

	I	II	III	IV
(A)	S	R	Q	P
(B)	T	R	Q	P
(C)	S	T	Q	P
(D)	S	R	T	P

26. Match List-I with List-II:

List - I (Molecules)		List - II (Relationship)	
(I)		(P)	Identical
(II)		(Q)	Enantiomer
(III)		(R)	Diastereomer
(IV)		(S)	Structural isomerism
		(T)	Homomers

Codes:

- (A) I R II T III P IV S
 (B) I R II R III T III S
 (C) I R II R III P III T
 (D) I R II R III P III S

27. Consider the following lists:

List - I		List - II	
(I)	$\text{CH}_2 = \text{CH} - \text{CHO} \xrightarrow{\text{NaBH}_4, \text{alc}}$	(P)	Acetal
(II)	$\text{C}_6\text{H}_5\text{CHO} + \text{Ph} - \text{NH}_2 \xrightarrow{\text{pH}(4-5)}$	(Q)	Schiff's base
(III)	$\text{C}_6\text{H}_5\text{COCH}_3 + \text{CH}_3 - \text{CH}_2 - \text{NH} - \text{CH}_3 \xrightarrow{\text{pH}(4-5)}$	(R)	Unsaturated alcohol
(IV)	$\text{RCHO} + 2\text{RCH}_2\text{OH} \xrightarrow{\text{H}^+}$	(S)	Enamine
		(T)	Imine

Which one is correct?

Codes:

- (A) I R II Q III S IV P
 (B) I T II Q III S IV P
 (C) I R II T III S IV P
 (D) I R II Q III T IV P

28. Consider the following lists:

List - I		List - II	
(I)	$\text{Me}_3\text{C} - \text{Cl} + \text{EtO}^-\text{Na}^+$	(P)	$\text{Me}_3\text{C} - \text{OEt}$
(II)	$\text{EtCl} + \text{Me}_3\text{CO}^-\text{Na}^+$	(Q)	Alcohol
(III)	$\text{PhCl} + \text{MeO}^-\text{Na}^+$	(R)	Anisole
(IV)	$\text{PhO}^-\text{Na}^+ + \text{MeCl}$	(S)	No result
		(T)	$\text{Ph} - \text{O} - \text{Ph}$

Which one is correct?

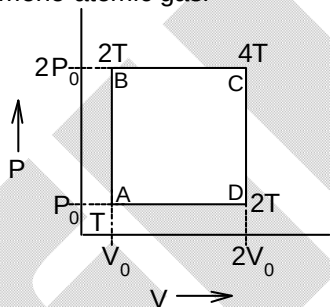
Codes:

- | | | | | |
|-----|---|----|-----|----|
| | I | II | III | IV |
| (A) | P | Q | S | R |
| (B) | S | P | Q | R |
| (C) | Q | P | S | R |
| (D) | Q | P | T | R |

Section - B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

29. With strong oxidizing agent, the oxidation product of $\text{K}_4[\text{Fe}(\text{CN})_6]$ is K_2O , Fe_2O_3 , NO_3^- and CO_3^{2-} , then what will be the 'n' factor of $\text{K}_4[\text{Fe}(\text{CN})_6]$.
30. For the acid indicator (HIn) thymol blue, pH is 2.176 when 60% of the indicator is in ionized form. Find the % indicator in unionized form in the solution with $[\text{H}^+] = 3.33 \times 10^{-3}\text{M}$.
31. Find out the efficiency (in %) of the following cyclic process $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ for 1-mole mono-atomic gas.



32. A solution having a vapour pressure of 520 mm of Hg contains o-nitrophenol and p-nitrophenol with mole fraction of 0.6 and 0.4 respectively. One of the above has $P^\circ = 400$ mm of Hg. The vapour pressure of o-nitrophenol is
33. A hydrogen like species with atomic number 'Z' is in excited state 'n' and emits photons of energy 25.7 and 8.7 eV when makes a transitions to 1st and 2nd excited state respectively. Ionisation energy (eV) of specie is

34. What should be the relative humidity below which $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ would efflorescent at 25°C ? For the reaction
- $$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}_{(s)} \rightleftharpoons \text{CuSO}_4 \cdot 3\text{H}_2\text{O}_{(s)} + 2\text{H}_2\text{O}_{(g)}$$
- at 25°C K_p is $1.0 \times 10^{-4} \text{ atm}^2$ and vapour pressure of water is 30.4 torr. (Given: Relative humidity is the ratio of amount moisture in air to the maximum amount of moisture that could exist in air at specific temperature)
35. Potassium has a bcc structure with nearest neighbours distance 4.52 Å. Its atomic weight is 39. Its density (in kg m^{-3}) will be:
36. Two moles of an ideal triatomic gas $\left(\gamma = \frac{4}{3}\right)$ at 16 bar and 300 K are expanded reversibly upto a final pressure of 1 bar. How much work (in J (magnitude)) is done by the gas?

TG ~ @bohring_bot

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. If $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ are roots of equation $x^5 - 5x^4 - 1 = 0$, then
- (A) $\sum_{r=1}^5 \frac{1}{\alpha_r^4} = -\frac{1}{20}$ (B) $\sum_{r=1}^5 \frac{1}{\alpha_r^4} = -20$
- (C) $\prod_{r=1}^5 \left(\frac{1}{\alpha_r^4} + 5 \right) = 1$ (D) $\prod_{r=1}^5 \left(\frac{1}{\alpha_r^4} + 5 \right)^3 = \frac{1}{5}$
38. If three non-zero and non-collinear vectors $\hat{i} + a\hat{j} - \hat{k}$, $2\hat{i} - \hat{j} + 3\hat{k}$ and $a\hat{i} - \hat{j} + 3\hat{k}$ are coplanar, then 'a' is/are
- (A) $\frac{1}{3}$ (B) 2
- (C) $-\frac{1}{3}$ (D) -2
39. If $3f(x) + 4f(3-x) = 3x^2 - 8$ and $\int_{\alpha}^{\beta} f(x) dx \leq 0$, where α, β ($\alpha < \beta$) are roots of $x^2 - 3x + a = 0$, then 'a' can be
- (A) $\frac{-1}{101}$ (B) $\frac{1}{101}$
- (C) $\frac{2}{301}$ (D) $\frac{401}{301}$
40. If the curves $y = \lambda x^2 + \lambda x + \frac{1}{24}$ and $x = \lambda y^2 + \lambda y + \frac{1}{24}$ are tangent to each other, then λ can be
- (A) $\frac{2}{3}$ (B) $\frac{3}{2}$
- (C) $\frac{13 + \sqrt{601}}{12}$ (D) $\frac{-13 + \sqrt{601}}{12}$
41. $\lim_{n \rightarrow \infty} \frac{1}{n^2} \left[(n+1) \left(n + \frac{1}{2} \right) \left(n + \frac{1}{2^2} \right) \dots \left(n + \frac{1}{2^{n-1}} \right) \right]^n$ is 'a', then
- (A) $[a] = 5$ (B) $[a] = 7$
- (C) $(a) = 6$ (D) $(a) = 8$
- Here (.) represents least integer part and [.] represents greatest integer part.

42. a, b, c are distinct complex numbers with $|a| = |b| = |c| = 1$ and z_1, z_2 be the roots of the equation $az^2 + bz + c = 0$ with $|z_1| = 1$. Also, let P and Q represents the complex numbers z_1 and z_2 in the complex plane with $\angle POQ = \theta$, where O is origin. Then,

(A) $b^2 = ac, \theta = \frac{2\pi}{3}$

(B) $\theta = \frac{2\pi}{3}, PQ = \sqrt{3}$

(C) $b^2 = ac, PQ = 2\sqrt{3}$

(D) $2b^2 = ac, \theta = \frac{\pi}{3}$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Let N be the number of words which can be formed using all the letters of the word "DARJEELING" so that there are at least two consonants between any two vowels. Then match List I with List II:

List - I		List - II	
(I)	If N is divisible by 2^n ($n \in N$), then n can be	(P)	1
(II)	If N is divisible by 6^P ($P \in N$), then P must be less than	(Q)	2
(III)	Number of odd divisors of N is greater than	(R)	3
(IV)	Number of zeroes at the end of N is less than	(S)	4
		(T)	5

Codes:

	I	II	III	IV
(A)	PQR	QRST	PQRST	PST
(B)	QRT	PQT	PQRS	PQRST
(C)	PQRST	ST	PQRST	QRST
(D)	PQST	PQRST	PQRT	PQR

44. Given $y = ax^2 + bx + c$ is a quadratic expression for $x \in [0, 2]$, y falls completely in $[0, 1]$ and $-\frac{b}{2a} \notin (0, 2)$. Then, for $a \neq 0$, match the following:

List - I		List - II	
(I)	The value of c may be	(P)	0
(II)	The value of $ 2a + b $ may be	(Q)	1
(III)	If $\frac{\lambda}{a}$ is a root of $ax^2 + bx + c = 0$, then λ may be	(R)	$\frac{1}{2}$
(IV)	The sum of roots of $ax^2 + bx + c = 0$ is $(p^2 + p + 3)$ for $p \in R$, then the value of $ a $ can't be	(S)	$\frac{4}{3}$
		(T)	2

Codes:

	I	II	III	IV
(A)	PQ	R	PR	QST
(B)	PQR	PR	T	ST
(C)	ST	Q	PQ	RS
(D)	R	ST	QST	PQ

45. Match the equations in List I to solution in List II:

List - I		List - II	
(I)	$\frac{2}{\pi} \tan\left(\frac{\pi x}{2}\right) + \frac{1}{2x} = \left(\frac{\pi+2}{\pi}\right) e^{-k}$ for $0 < x < 1$	(P)	Only one solution for $k \in (0, 1/e)$
(II)	$\ln x = kx$ has	(Q)	Only one solution for $k \in \left(-\frac{1}{e}, 0\right)$
(III)	$xe^x = k$ has	(R)	Two solutions for $k \in \left(-\frac{1}{e}, 0\right)$
(IV)	$x \ln x = k$ has	(S)	Two distinct solutions for $k \in \left(0, \frac{1}{e}\right)$
		(T)	No solution for $k \in \left(0, \frac{1}{e}\right)$

Codes:

	I	II	III	IV
(A)	PQ	T	S	PR
(B)	T	PT	PR	QS
(C)	QS	Q	PT	R
(D)	RT	QS	PR	PR

46. Match the locus of 'z' in List I from List II:

List - I		List - II	
(I)	Straight line	(P)	If $z^2 \bar{z} + \bar{z}^2 z = (z + \bar{z})$; then z lies on
(II)	Circle	(Q)	If $\ z - 3\ - \ z + 3\ \leq 6$, then z lies on
(III)	Parabola	(R)	If $z^3 \bar{z} + z \bar{z}^3 - 2(z^2 + \bar{z}^2 - 4(z ^2 - 2))$, then z lies on
(IV)	Hyperbola	(S)	$(z^2 + \bar{z}^2) - 4 \operatorname{Re}(z) \times ((z - \bar{z})^2 - (z + \bar{z})) = 0$, then z lies on
		(T)	$z^2 + \bar{z}^2 = z \bar{z}$, then z lies on

Codes:

	I	II	III	IV
(A)	PQ	PR	S	QR
(B)	Q	PR	QS	T
(C)	QR	QR	S	P
(D)	P	T	Q	S

Section - B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. If $\cos^{-1}(2 - x^2) + \sin^{-1}(2 - x^2) + \tan^{-1} x \geq \frac{2\pi}{3}$. Given the solution set, $\alpha \leq x \leq \beta$, then $\left(\alpha + \frac{1}{\beta^2}\right)$ is equal to

48. If $y = \frac{x}{x^2 + \frac{x}{x^2 + \dots \infty}}$, then $\int \frac{(y - x^2) dx}{(x^2 + y)(x + y^2)} = f(y)$. Then, $f'(4)$ is equal to

49. If $(9 \sin x + 4 \operatorname{cosec} x) = 2k + 4$ has only three solutions, then $k =$

50. If $\sqrt{5}x - 2y + 5 = 0$ is a tangent at any point P on the parabola whose focus is $\equiv (-\sqrt{5}, 2)$ & length of latus rectum is 4 units; intersect axis at A & normal at point P intersect axis at B, then the circum radius of ΔPAB is
51. If $\frac{1}{e-1} \sum_{n=1}^{50} \frac{1}{(n+1)!} \int_{-n}^n e^{|x|} \frac{dx}{1+e^{\sin x}} = 1 - \frac{1}{(ab)!}$
where a and b are prime, then $|a-b|$ is equals to
52. Let $f(x) = x^3 - x^2 - x + 1$, $g(x) = a^2 - 2$ (where a is real parameter) and $f(x) = g(x)$ has three real and distinct roots, then $f(x) = [a^2]$ has real root having the integral part (where $[.]$ is G.I.F) equal to
53. If $\frac{3x^2+1}{\sqrt{x^4+x^2}} \geq \max f(x)$, where $f(x)$ is constant for $0 < x \leq \sqrt{2}$, then minimum value of $\int_0^{\sqrt{2}} \frac{3x^2+1}{\sqrt{x^4+x^2}} dx$ is
54. Let x, y, z are selected from the set of natural numbers. If the probability that $x^2 + y^2 + z^2$ is divisible by 5 is $\frac{p}{q}$ (where p, q are coprime), then p + q is equal to

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – III
PAPER –2
TEST DATE: 08-02-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

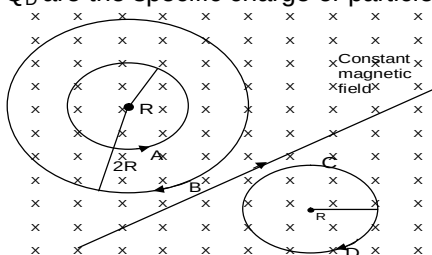
Physics

PART – I

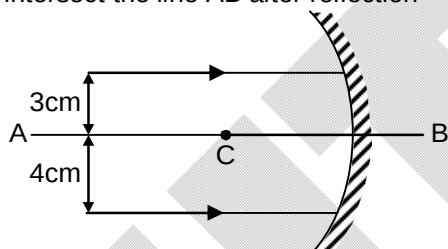
Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. Four particles A, B, C and D of masses m_A , m_B , m_C and m_D respectively, follow the paths shown in the figure, in a uniform magnetic field. Each particle moving with same speed. Q_A , Q_B , Q_C and Q_D are the specific charge of particles A, B, C and D respectively :

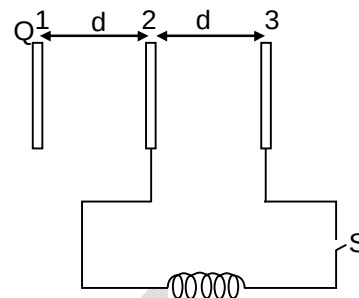


- (A) $Q_A < Q_B < Q_C < Q_D$
 (B) $Q_D < Q_B < Q_C < Q_A$
 (C) Charge on the particle B and particle D is of different nature
 (D) Work done by magnetic force on the particle C is minimum as compared to other particle
2. Two rays are incident on a spherical concave mirror of radius $R = 5$ cm and rays are parallel to a line AB which passes through the centre C. The rays are at perpendicular distances 3 cm and 4 cm from AB respectively as shown. Find the distance between the points at which these rays intersect the line AB after reflection –



- (A) $\frac{25}{24}$ cm
 (B) $\frac{24}{25}$ cm
 (C) zero
 (D) $\frac{1}{3}$ cm
3. A neutron collides head-on with a stationary hydrogen atom (${}_1\text{H}^1$) in ground state, then choose the correct statement: (Assume that mass of neutron and mass of ${}_1\text{H}^1$ atom is same)
- (A) If kinetic energy of the neutron is less than 13.6 eV, collision must be elastic
 (B) If kinetic energy of the neutron is less than 13.6 eV, collision may be inelastic
 (C) Inelastic collision may take place only when initial kinetic energy of neutron is less than 13.6 eV
 (D) None of these

4. Three identical large plates are fixed at separation of d from each other as shown. The area of each plate is A . Plate 1 is given charge $+Q$ while plates 2 and 3 are neutral and are connected to each other through coil of inductances L and switch S . If resistance of all connected wires is neglected the maximum current flow through coil after closing the switch is ($C = \epsilon_0 A / d$) (neglect fringe effect)

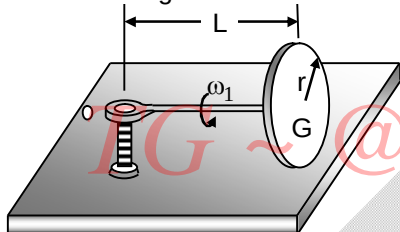


- (A) $\frac{Q_0}{\sqrt{LC}}$ (B) $\frac{Q_0}{\sqrt{2LC}}$
 (C) $\frac{2Q_0}{\sqrt{LC}}$ (D) $\frac{Q_0}{2\sqrt{LC}}$

Section – A (Maximum Marks: 24)

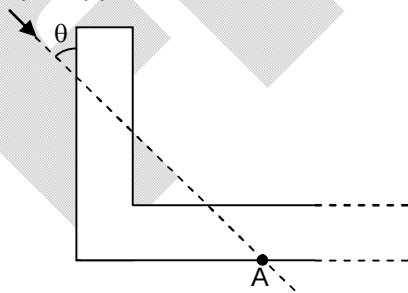
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A homogeneous disk of radius r and mass m is mounted on an axle OG of length L and negligible mass. The axle is pivoted at the fixed point O , and the disk is constrained to roll on a horizontal floor. Knowing that the 'disk' rotates counterclockwise at the rate ω_1 about the axle OG ,



Choose the correct option(s)

- (A) The angular velocity of the disk $\omega_1 \hat{i} - \frac{\omega_1 r}{L} \hat{j}$
 (B) The angular velocity of the disc $-\left(\frac{\omega_1 r}{L}\right) \hat{j}$
 (C) The X component of angular momentum of disc about 'O' is $\frac{1}{2}mr^2\omega_1$
 (D) The Z component of angular momentum of disc about 'O' is zero
6. Two glass slabs of same thickness and same refractive index are joined to form an 'L' shape as shown. A ray is incident on the combination as shown (in the plane of paper)., Choose correct option(s):



- (A) Total deviation will zero for all values of θ .
 (B) Total deviation will be zero for $\theta = 45^\circ$ only.
 (C) The ray will emerge from point A for all values of θ .
 (D) The ray will emerge from point A for $\theta = 45^\circ$ only.

7. If the two slits of double slit experiment were moved symmetrically apart with small relative velocity v and the distance between screen and mid-point of slits is fixed and equal to D . Consider a point P on the screen at a distance x from central maxima then ($x \ll D$):

(A) Rate of change of number of fringes between central maxima and point P changes with respect to time is $\frac{xv}{\lambda D}$

(B) Number of fringes contained between central maxima and point P increases with time

(C) Fringe width decreases as time passes

(D) Fringe width increases as time passes

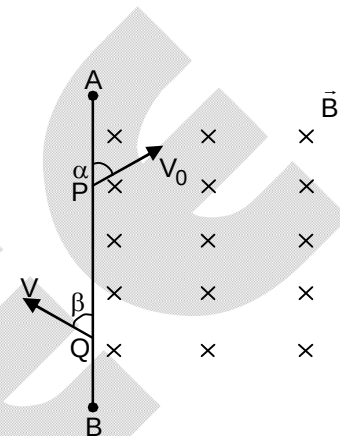
8. There exists a uniform magnetic field perpendicular to plane of the paper (inward) right side of the line AB . A particle of charge $-q$ and mass m enters uniform magnetic field B (perpendicular to paper inward) at P with a velocity V_0 at an angle α and leaves the field at Q with velocity V at angle β as shown in figure.

(A) $\alpha = \beta$

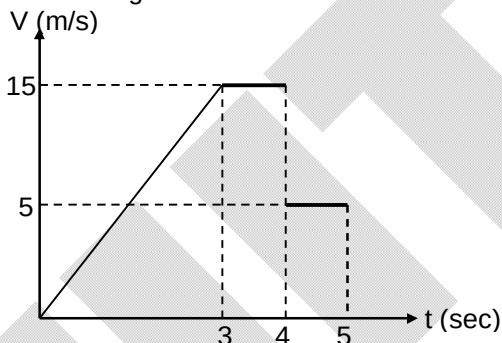
(B) $V = V_0$

(C) $PQ = \frac{2mV_0 \sin \alpha}{Bq}$

(D) The particle remains in the field for time $t = \frac{2m(\pi - \alpha)}{Bq}$



9. The figure shows the velocity as a function of the time for an object with mass 10 kg being pushed along a frictionless horizontal surface by external horizontal force. At $t = 3$ s, the force stops pushing and the object moves freely. It then collides head on and sticks to another object of mass 25 kg.



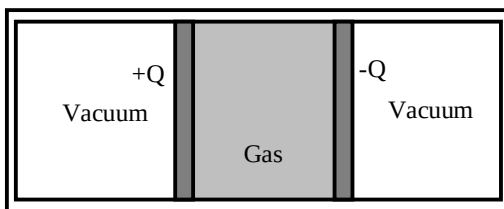
(A) External force acting on the block of mass 10 kg is 50 N from $t = 0$ to $t = 3$ sec.

(B) Speed of the 2nd particle just before the collision is 1 m/s.

(C) Before collision both bodies are moving in the same direction.

(D) Before collision, bodies are moving in opposite direction.

10. A ideal gas having f degrees of freedom is kept inside a thermally insulated vessel, the vessel has two large pistons each with charge of $+Q$ and $-Q$. Assume charges are uniformly distributed. Pistons can move without friction inside the vessel. At an instant the charge of pistons is increased k times with the help of an external agent. Then which of the following statements are correct

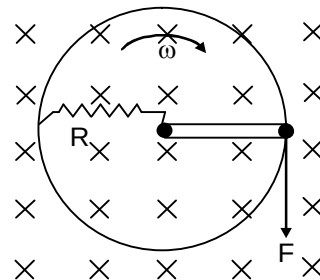


- (A) The Ratio of final to initial temperature of gas in equilibrium is $\frac{K^2 + f}{2 + f}$.
- (B) The Ratio of final to initial temperature of gas in equilibrium is $\frac{2K^2 + f}{2 + f}$.
- (C) Work done by the agent in the process is $(K^2 - 1)nRT_1$ where T_1 is initial temperature.
- (D) Work done by the agent in the process is $(K^2 - 1)\frac{Q^2 L_1}{A\epsilon_0}$ where L_1 is initial separation between pistons.

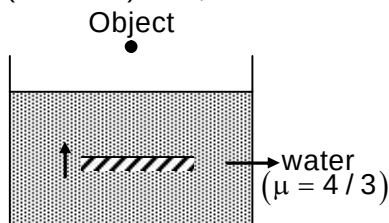
Section – B (Maximum Marks: 24)

This section contains EIGHT (08) numerical based questions. The answer to each question is a Single Digit Integer, ranging from 0 to 9 both inclusive.

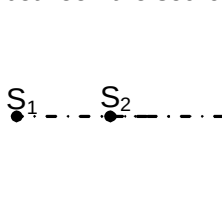
11. A thin spherical soap bubble has surface tension S . Its surface is uniformly charged with charge Q , its volume is V . It is found that when $Q^2 = 16n\pi\epsilon_0 SV$, the excess pressure inside the bubble becomes zero. Find n .
12. A circular disc of diameter d lies horizontally inside a metallic hemispherical bowl of radius a . The disc is just visible to an eye looking over the edge. The bowl is now filled with a liquid of refractive index $\mu = \sqrt{3}$. Now, the whole of the disc is just visible to the eye in the same position, if $\frac{d}{a}$ is equal to:
13. A metallic ring of mass 2 kg and radius 1m with a uniform metallic spoke of same mass 2kg and length 1 m is rotated about its axis with angular velocity 1 rev/sec. in a perpendicular uniform magnetic field B of magnitude 10T as shown in figure. If the central end of the spoke is connected to the rim of the wheel through a resistor R magnitude $\pi\Omega$ as shown. The resistor does not rotate, its one end is always at the center of the ring and other end is always in contact with the ring. A force F as shown is needed to maintain constant angular velocity of the spoke then, fill $\frac{F}{25}$ (The ring and the spoke has zero resistance)



14. Mirror in the arrangement shown in figure is moving up with speed 8 cm/sec. Speed of final image (in cm/sec) is $4x$, calculate x .



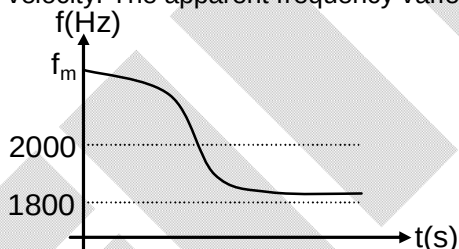
15. Two coherent monochromatic point sources S_1 and S_2 are placed in front of an infinite screen as shown in figure. Wavelength of the light emitted by both the sources is λ . Initial phase difference between the source is zero.



Initially $S_1S_2 = 2.5\lambda$ and the number of bright circular rings on the screen is n_1 . If the distance S_1S_2 is increased and made 5.7λ the number of bright circular rings becomes n_2 . The difference $n_2 - n_1$ is:

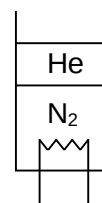
16. If the frequency of K_α x-ray emitted from the element with atomic number 31 is f , if the frequency of K_α x-ray emitted from the element with atomic number 51, is given by $\frac{25}{x}f$, then x is (assume that screening constant for K_α is 1)

17. A stationary observer receives a sound from a source of frequency 2000 Hz moving with constant velocity. The apparent frequency varies with time as shown in the figure.



The value of f_m is $(2300 - 10x)$ Hz. Find the value of x . (Take speed of sound = 300 m/s and neglect the time taken by sound to reach the stationary observer).

18. 5 moles of Nitrogen gas are enclosed in an adiabatic cylindrical vessel. The piston itself is a rigid light cylindrical container containing 3 moles of Helium gas. There is a heater which gives out a power 100 cal/second to the nitrogen gas. A power of 30 cal/sec is transferred to Helium through the bottom surface of the piston. The rate of increment of temperature of the nitrogen gas (in kelvin/sec) is (assuming that the piston moves slowly):



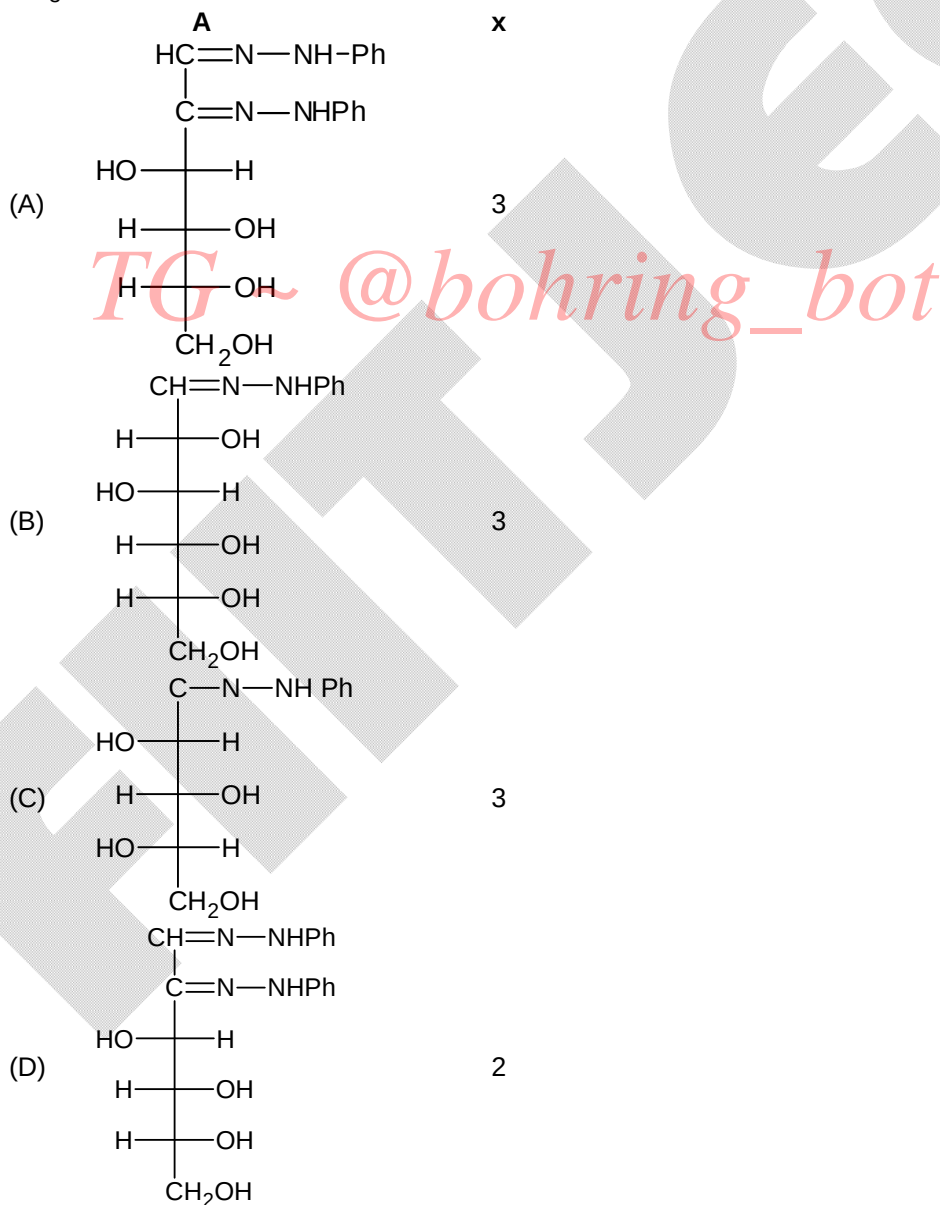
Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

19. The magnetic moment of $[\text{Mn}(\text{CN})_6]^{3-}$ is 2.8 BM while the magnetic moment of $[\text{MnBr}_4]^{2-}$ is 5.9 BM. The geometries of the complex ions respectively are:
 (A) Tetrahedral, tetrahedral (B) Octahedral, octahedral
 (C) Octahedral, tetrahedral (D) Tetrahedral, octahedral
20. D-glucose on treatment with x mol of PhNHNH_2 gives 1 mol of (A), 1 mol of PhNH_2 and 1 mol of NH_3 . (A) and x are:



21. Initially 3 moles of 'A' was taken in a 1L container. What would be the approx. moles of A left in the container when the following equilibrium is established.

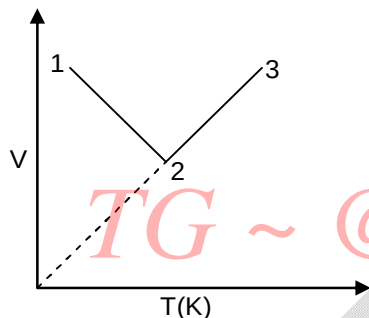


- (A) 1.67×10^{-4} (B) 3.33×10^{-8}
 (C) 1.25×10^{-10} (D) 5×10^{-6}
22. Which of the following cationic detergents?
 (A) Sodium lauryl sulphate (B) Cetyl trimethyl ammonium bromide.
 (C) Sodium dodecylbenzene sulphonate (D) Glyceryl oleate

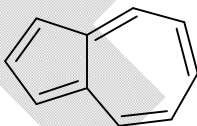
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

23. Following graph is constructed for the fixed amount of the gas. Which of the following is/are correct?



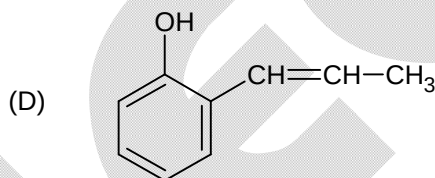
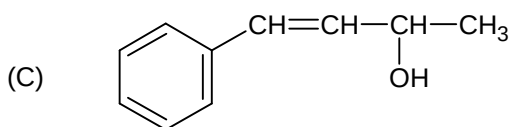
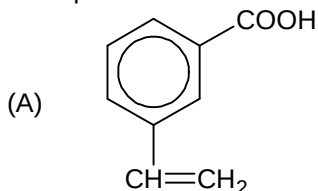
- (A) From 1 → 2 pressure will increase
 (B) From 2 → 3 pressure remains constant
 (C) Gas pressure at (3) is greater at state (1)
 (D) For pressure the correct order is $P(3) > P(2) > P(1)$
24. The compound(s) which (have) peroxo linkage is (are):
 (A) $H_2S_2O_3$ (B) H_2SO_5
 (C) $H_2S_2O_7$ (D) $H_2S_2O_8$
25. Which one is classified as addition polymer?
 (A) Acrylonitrile (B) Dacron
 (C) Neoprene (D) Teflon
26. Which of the following is /are true about azulene?



Azulene

- (A) Both rings are aromatic in nature
 (B) It is soluble in polar solvents
 (C) Attack of electrophile occurs at five membered ring
 (D) Attack of nucleophile occurs at seven membered ring

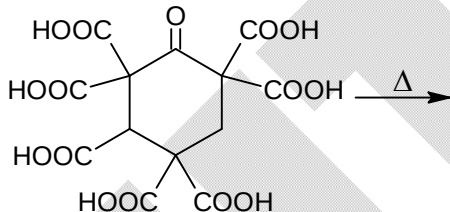
27. Which one of the following statements about acidic/ basic strength is/are correct?
 (A) HCOOH is more acidic than $\text{Ph}-\text{COOH}$
 (B) Acidic strength order is as $\text{CH}_3\text{OH} > \text{H}_2\text{O} > \text{C}_2\text{H}_5-\text{OH}$
 (C) Fumaric acid is stronger acid than Maleic acid
 (D) Pyridine is more basic than pyrrole
28. Compound 'P' ($\text{C}_{10}\text{H}_{12}\text{O}$) evolves H_2 gas with Na metal. It reacts with Br_2/CCl_4 to give 'Q' ($\text{C}_{10}\text{H}_{12}\text{Br}_2\text{O}$). With I_2/NaOH it forms iodoforms and an acid 'R' ($\text{C}_9\text{H}_8\text{O}_2$). 'P' has geometrical and optical isomers. The structure of 'P' and 'R' should be



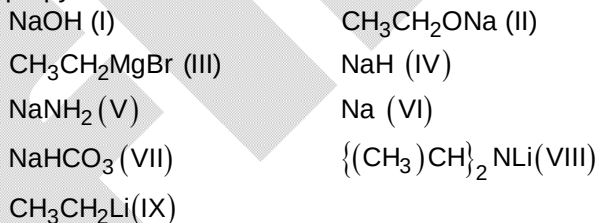
Section - B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

29. The number of moles of CO_2 that will be evolved in the following is



30. How many of the reagents from the lists below would give effereverence when treated with propyne?



31. How many of the following are reducing sugars?



32. The formation constant of complex HgI_4^{2-} is $1.0 \times 10^{(n \times 5)}$
 Given: $E^\circ_{(\text{Hg}^{2+}/\text{Hg})} = 0.85$ and $E^\circ_{(\text{HgI}_4^{2-}/\text{Hg}, \text{I}^-)} = -0.05 \text{ V}$.
 The value of 'n' is
33. 30 ml of 0.2 M NaOH is added to 50 ml 0.2 M CH_3COOH is give 80 ml of the solution. $\frac{10}{x}$ mL of 0.2 M NaOH required to make the pH of the solution 5.00. The value of 'x' is
 Given: The ionization constant of $\text{CH}_3\text{COOH} = 2 \times 10^{-5}$.
34. Iron exists as BCC at 10°C while exist as FCC at 910°C . What would be the percentage decrease in volume of iron at 910°C from 10°C ?
35. Among the following how many of the molecule is linear?
 CO_2 , NO_2^- , SO_2 , N_3^- , C_2H_2 , I_3^- , XeF_2 , BeCl_2 , CS_2
36. For the reaction: $3\text{A}_{(\text{g})} \longrightarrow 2\text{B}_{(\text{g})}$, the rate of formation of 'B' at 298 K is represented as $\ln\left(\frac{d[\text{B}]}{dt}\right) = -0.04 + 2\ln[\text{A}]$. The order of reaction is

TG ~ @bohiring_bot

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. If $(\alpha - \beta)$, $(\alpha + \beta)$, $\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$ are in A.P. for $\alpha > 0 > \beta$ and $|\alpha| > |\beta|$, then for quadratic equation $ax^2 + bx - 1 = 0$
- (A) $a \in (-2, \infty)$ (B) $a \in (-\infty, 2)$
 (C) $a \in (-\infty, 0)$ (D) $a \in (0, \infty)$
38. $\lim_{n \rightarrow \infty} \sum_{r=3}^n \frac{(r^3 - 8)}{(r^3 + 8)} =$
- (A) $\frac{1}{3}$ (B) $\frac{2}{5}$
 (C) $\frac{3}{8}$ (D) $\frac{2}{7}$
39. Let P be a point (not origin) lying on the parabola $x^2 = y$. The normal line to the parabola at P will intersect the parabola at another point Q. The minimum possible value for area bounded by the line PQ and the parabola is
- (A) $\frac{2}{3}$ (B) $\frac{3}{4}$
 (C) $\frac{4}{3}$ (D) $\frac{3}{5}$
40. If $\frac{(\tan^2 x)}{(1 + \tan x - \sqrt{1 + \tan x})^2} < \frac{\sec^2 x + 3 \tan x + 17}{(1 + \tan x)^3}$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, then number of solutions of $\tan x + x^2 = 1$ is/are
- (A) 0 (B) 1
 (C) 2 (D) Infinite

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. Which is/are correct?
- (A) $\int_0^{\pi} \frac{\sin^3 x dx}{(1+x)} = 3 \int_0^{\pi} \sin 2x \cdot (x+1) \left\{ \ln \left(\frac{x+1}{e} \right) \right\} dx$ (B) $\int_0^{\pi} \frac{\sin^3 x dx}{2x - \pi} = \int_0^{\pi/2} \frac{\cos^3 x}{x} dx$
 (C) $\int_0^{\pi} \frac{\sin^3 x dx}{(1+x)} + \int_0^{\pi} \frac{\cos^3 x}{x} dx = 0$ (D) $\int_0^{\pi} \frac{\sin^3 x dx}{(1+x)} = 3 \int_0^{\pi} \sin 2x (x+1) \ln(x+1) dx$

42. If α, β are roots of $x^2 - x - 1 = 0$, then quadratic whose roots are either $\left(\frac{\alpha^6 - 1}{\alpha^3}\right)$ or $\left(\frac{\beta^6 - \beta^3 - 1}{\beta^3}\right)$ is/are
 (A) $x^2 - 5x + 4 = 0$ (B) $x^2 - 7x + 12 = 0$
 (C) $x^2 - 5x + 6 = 0$ (D) $x^2 - 9x + 14 = 0$
43. If $f(n) = \sum_{k=1}^{n+1} \frac{(k+1)dx}{x(x+1)}$ and $g(x) = (x+1)\ln\left(\frac{x+2}{x+1}\right) + f(x)$, then
 (A) $g'(0) = 2$ (B) $g''(0) = -1$
 (C) $g'(1) = \frac{1}{2}$ (D) $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{2e^{g(r)} - 1 + n}\right) = \log_e 2$
44. If P moves on a circle which touches both the axis and angle bisector of $\angle APB$ for two different positions of P in the same segment are given by $x + y = 3$ and $x - y = 1$, then radius of circle can be _____ (AB is given chord)
 (A) 1 (B) 3
 (C) 4 (D) 5
45. Let $f: \mathbb{R} \rightarrow (0, 1)$ be a continuous function, then which of the following pair of vectors are linearly dependent for some $x \in (0, 1)$?
 (A) $\vec{a} = (f(x))\hat{i} + 2\hat{j}$, $\vec{b} = x^2\hat{i} + 3\hat{j}$ (B) $\vec{a} = (f(x))\hat{i} + 3\hat{j}$, $\vec{b} = x^2\hat{i} + 2\hat{j}$
 (C) $\vec{a} = \left(\int_0^{1-x} f(t)dt\right)\hat{i} + 3\hat{j}$, $\vec{b} = x\hat{i} + 2\hat{j}$ (D) $\vec{a} = \left(\int_0^{1-x} f(t)dt\right)\hat{i} + 2\hat{j}$, $\vec{b} = x\hat{i} + 3\hat{j}$
46. Matrices of order 2×2 are formed by using the elements of the set $A = \{-2, -1, 0, 1, 2\}$. Then, probability that matrix is either symmetric or skew-symmetric, is greater than
 (A) $\frac{1}{10}$ (B) $\frac{2}{15}$
 (C) $\frac{3}{5}$ (D) $\frac{3}{10}$

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

47. Number of integral values of a for which $\frac{2x^2 + ax + 2}{x^2 + x + 1} = \frac{e^y}{[y+1]}$ is true $\forall x \in \mathbb{R}$ and $y \geq 0$ (where $[.]$ denotes G.I.F), is
48. If $\det(A) = \begin{vmatrix} \alpha & 0 \\ 2 & \bar{\alpha} \end{vmatrix}$, where α is complex number, $\bar{\alpha}$ is its conjugate and $\det(2A^2 - 2A) = 144$, then $\text{tr}(A)$ is either m or n, then $(m + n) = \underline{\hspace{2cm}}$.
49. If $f(x) = \frac{\alpha x + 1}{x - \beta} \forall x \in \mathbb{R} - \{\beta\}$ & $\alpha\beta \neq 1$ & $\alpha \neq 1$ is a self-inverse function such that $\frac{f(4)}{4} = \frac{f(12)}{12} = f\left(\frac{1+\beta}{1-\alpha}\right)$, then “ β ” is equal to

50. If $f(x) = x + 1$ & $g^{-1}(x) = x^3 + x + 1$, let $\text{gof}(x) = h(x)$, then $\lim_{x \rightarrow 1} \frac{h^{-1}(x) - h^{-1}(1)}{x - 1} =$
51. Let α be a complex number satisfying $i\alpha^3 + \alpha^2 - \alpha + i = 0$ and β be a complex number satisfying $|\beta - 1| = 4$, then maximum value of $|\alpha + \beta|$ is
52. Let A be a square matrix of order 3 such that transpose of inverse of A is A itself, then $|\text{adj}(\text{adj}A)|$ is equal to
53. If $p = \sin^4 \alpha + \cos^4 \alpha + 1$ and $q = \cos^6 \beta + \sin^6 \beta + 1$ ($\alpha, \beta \in \mathbb{R}$), then the value of $[(p+q)_{\max}] + [(p+q)_{\min}]$ is (where $[\cdot]$ is G.I.F)
54. Let $n = x_1 x_2 x_3 \dots x_9$ is formed using the digit $1, 2 \dots 9$. Each digit is being used once only. If n_1 is number of possible n for which $1 < |x_3 - x_4| < 5$ and n_2 is number of possible n for which $|x_3 - x_4| = 1$, then $\left[\frac{n_1}{n_2} \right]$ is equal to (where $[\cdot]$ is G.I.F)

TG ~ @bohring_bot

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – III
PAPER –1
TEST DATE: 08-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. ABC

Sol. $-kv = \frac{dv}{dt}$, $\int \frac{dv}{v} = -k \int dt$ $\int_t^t dt$

$$\ln \frac{v}{v_0} = -kt \quad v = v_0 e^{-kt}$$

$$v_0 e^{-kt} = \frac{dx}{dt}$$

$$\int_0^x dx = v_0 \int_0^t e^{-kt} dt$$

$$x = \frac{v_0}{k} [e^{-kt}]_0^t = -\frac{v_0}{k} (e^{-kt} - 1), \quad x = \frac{v_0}{k} (1 - e^{-kt})$$

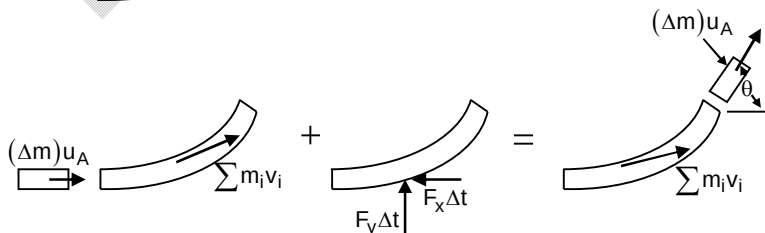
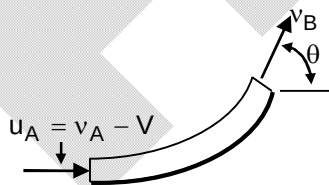
$$-kv = v \frac{dv}{dx}, \quad dv = -k dx$$

$$\int_{v_0}^v dv = -k \int_0^x dx$$

$$v - v_0 = -kx, \quad v = v_0 - kx$$

2. ABD

Sol.



$$\rightarrow x \text{ components: } (\Delta m)u - F_x \Delta t = (\Delta m)u \cos \theta$$

$$+ \uparrow y \text{ components: } +F_y \Delta t = (\Delta m)u \sin \alpha$$

Substituting $\Delta m = A\rho(v_A - V)\Delta t$ and $u = v_A - V$, we obtain

$$F_x = A\rho(v_A - V)^2(1 - \cos \theta) \leftarrow \text{ and } F_y = A\rho(v_A - V)^2 \sin \theta \uparrow$$

$$\text{Power} = F_x V = A\rho(v_A - V)^2(1 - \cos \theta)V$$

$$\frac{d(\text{power})}{dV} = A\rho(v_A^2 - 4v_A V + 3V^2)(1 - \cos \theta) = 0$$

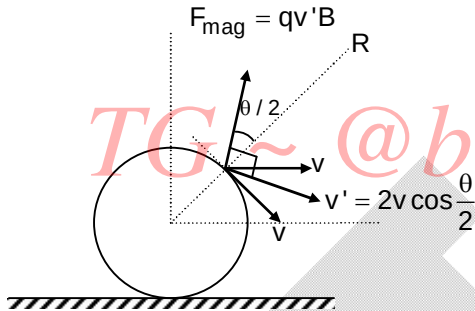
$$V = v_A \quad V = \frac{V_A}{3} \text{ For maximum power } V = \frac{V_A}{3}$$

3. ACD

Sol. $f = qv'B \sin \frac{\theta}{2} = q \left\{ 2v \cos \left(\frac{\theta}{2} \right) \right\} B \sin \frac{\theta}{2} = Bqv \sin \theta$

Since, speed at topmost point is maximum

$$\therefore N = mg - 2Bqv$$



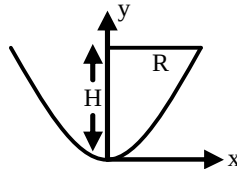
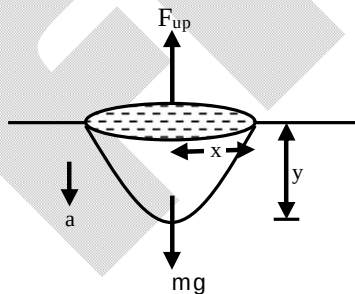
$\Rightarrow$ To loose the contact, $N = 0$

$$\Rightarrow v = \frac{mg}{2qB}$$

4. AD

Sol. Let $Y = ax^2$ (A parabola)

$$Y = \frac{H}{R^2} x^2 \text{ (from given data)}$$



$$v = \frac{dy}{dt}$$

$$mg - \rho_L \cdot \frac{1}{2} \pi x^2 y \cdot g = m \cdot \frac{dv}{dt}$$

$$\frac{dv}{dt} = g - \frac{\pi x^2 y g \rho_L}{2 \rho_s \frac{\pi R^2 H}{2}} \Rightarrow g - \frac{\rho_L}{\rho_s} \frac{x^2}{R^2} \times \frac{yg}{H}$$

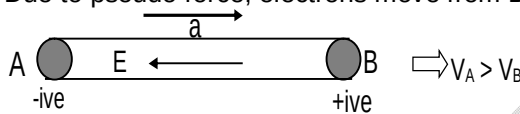
$$v \cdot \frac{dv}{dy} = g - \frac{\rho_L}{\rho_s} \frac{gy^2}{H^2}$$

$$\int_0^0 v dv = \int_0^H \left(g - \frac{\rho_L}{\rho_s} \frac{gy^2}{H^2} \right) dy$$

$$0 = gH - \frac{g\rho_L}{3\rho_s} H$$

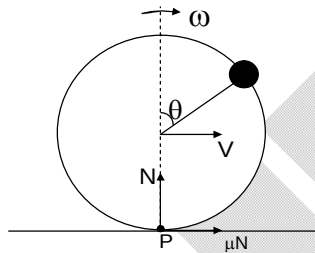
$$\frac{\rho_L}{\rho_s} = 3$$

5. ABCD
Sol. Due to pseudo force, electrons move from B to A.



In figure (b) electron moves from A to B
Region I Region II

6. BC
Sol.



V_C is Horizontal only.

$$\vec{V}_P = \vec{V}_{PC} + \vec{V}_C$$

$$\vec{V}_P = 0$$

Mechanical energy conservation,

$$mgR(1 - \cos \theta) = \frac{1}{2} m V_{\text{bead}}^2 \quad V_{\text{bead}} = 2v \cos \theta / 2$$

$$gR \times 2 \sin^2 \theta / 2 = 2v^2 \cos^2 \theta / 2$$

$$V = \sqrt{gR \tan \theta / 2}$$

$$\vec{\tau}_{\text{em}} = I_{\text{cm}} \alpha = 0 \quad \text{As } I_{\text{cm}} = 0$$

$$NR \sin \theta = \mu N (R + R \cos \theta)$$

$$\tan \theta / 2 = \mu$$

$$\theta = 2 \tan^{-1}(\mu)$$

7. C

Sol. For He^+ absorption occur from level $(2 \rightarrow 4)$

For Li^{2+} absorption occur from level $(3 \rightarrow 6)$

8. D

Sol. (I) If 1st case $E = E_1 = E_0 \sin \omega t$

If 2nd case $E = E_2 = E_0 \sin(\omega t + \phi)$

If 3rd case $E = E_1 + E_2 = E_0' \sin(\omega t + \delta)$

All three cases effective value is same $\Rightarrow E_0' = E_0$

$$\Rightarrow \phi = \frac{2\pi}{3}$$

(II) Case : (2) When S_3, S_4 closed $Z_1 = \sqrt{2}R$

Case : (2) When all opened $Z_2 = \sqrt{R^2 + (X_C - X_L)^2}$

$$\Rightarrow X_C - X_L = R \Rightarrow X_C = 2R \Rightarrow \frac{X_C}{R} = 2$$

$$\Rightarrow \frac{4\pi X_C}{5R} = \frac{8\pi}{5}$$

(III) When S_1, S_2 closed $Z_1 = \sqrt{5}R$

When S_3, S_4 closed $Z_2 = \sqrt{2}R$

$$\frac{I_1}{I_2} = \sqrt{\frac{2}{5}} \Rightarrow \pi \left(\frac{I_1}{I_2} \right)^2 = \frac{2\pi}{5}$$

(IV) When all opened $Z_1 = \sqrt{2}R$

When S_2, S_3 Closed $Z_2 = R$

$$\frac{I_3}{I_4} = \frac{1}{\sqrt{2}} \Rightarrow \pi \left(\frac{I_3}{I_4} \right)^2 = \frac{\pi}{2}$$

9. B

Sol. (I) The fundamental frequency in the string.

$$f_0 = \frac{\sqrt{T/\mu}}{2\ell} = \frac{\sqrt{102.4}}{\sqrt{1 \times 10^{-3}}} \times \frac{1}{2 \times 0.5} \text{ Hz} = 320 \text{ Hz.}$$

Other possible resonance frequencies are f_A and $f_0 = 320 \text{ Hz}, 640 \text{ Hz}, 960 \text{ Hz}.$

(II) The fundamental frequency in the string.

$$f_0 = \frac{\sqrt{T/\mu}}{4\ell} = \frac{320}{4 \times 0.5} = 160 \text{ Hz.}$$

Other possible resonance frequencies are

$$f_B = 160 \text{ Hz}, 480 \text{ Hz}, 800 \text{ Hz.}$$

(III) The fundamental frequency in both ends open organ pipe is

$$f_0 = \frac{v}{2\ell} = \frac{320}{2 \times 0.5} = 320 \text{ Hz.}$$

Other possible resonance frequencies are

$$f_C = 320 \text{ Hz}, 640 \text{ Hz}, 960 \text{ Hz}$$

(IV) The fundamental frequency in one end open organ pipe is

$$f_0 = \frac{v}{4\ell} = \frac{320}{4 \times 0.5} = 160 \text{ Hz.}$$

Other possible resonance frequencies are

$$f_D = 160 \text{ Hz}, 480 \text{ Hz}, 800 \text{ Hz.}$$

10. B

Sol. $E_x = \frac{-1}{a}$ $E_y = \frac{-2V}{b}$ $E_z = \frac{-3V}{c}$
 $\Delta V = -\vec{E} \cdot \vec{d}$

Section – B

11. 25.00

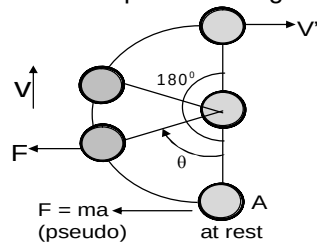
Sol. Acceleration of train = $a \rightarrow$

Motion of bob w.r.t. train:

Initially $\vec{V}_{\text{bob/train}} = 0$

From A $\rightarrow$ B, pseudo force $F = ma$ and has speed v' at highest position

$\leftarrow$ acts on bob. Let bob turns through 180°



W/E theorem A $\rightarrow$ B :

$$FR = \frac{1}{2}mv^2 + mgh$$

$$\frac{mv^2}{R} = 2(ma - mg) \dots\dots(i)$$

B $\rightarrow$ C : Now train has stopped, $a = 0$,
 No pseudo force acts on bob.

Conservation of Energy

$$mgR + \frac{1}{2}mv'^2 = \frac{1}{2}mv^2$$

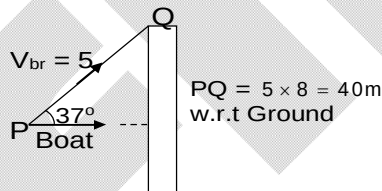
$$\frac{mv'^2}{R} = \frac{mv^2}{R} - 2mg$$

$$T + mg = 2(ma - mg) - 2mg \text{ [From eq. (i)]}$$

$$\text{On solving } a \geq 25m/s^2$$

12. 6.00

Sol.

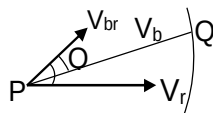


$$\vec{V}_b = \vec{V}_{br} + \vec{V}_r$$

Velocity must be along PQ

$$v_{br} \sin \theta = v_r \sin 37$$

$$\sin \theta = 7/25$$



$$t = \frac{40}{v_{br} \cos \theta + v_r \cos 37^\circ} = 6 \text{ sec.}$$

13. 25.00

Sol. $I_1 + I_2 = 13I_0$
 $(\sqrt{I_2} - \sqrt{I_1})^2 = I_0$
 $I_1 + I_2 - 2\sqrt{I_1 I_2} = I_0$
 $2\sqrt{I_1 I_2} = 12I_0$
 $I_{\max} = 13I_0 + 12I_0 = 25I_0 = (5)^2 I_0$

14. 0.50

Sol. $\phi = [\vec{E} \cdot \vec{n}] A = \left[\frac{8 \times 2 + (6 \times 3) + (10 \times 6)}{\sqrt{2^2 + 3^2 + 6^2}} \right] \frac{7}{47}$

15. 41.23

Sol. $\tan \alpha = \frac{v_y}{v_x}$

16. 1.80

Sol. $V_I - \frac{q_1}{7} - V_G = 0$
 $V_{IG} = \frac{1.80}{11}$

17. 2.75

Sol. Equivalent resistance across capacitor (after shorting the battery) = $\frac{11R}{4}$

Voltage across capacitor in steady state = $\frac{3E}{4}$

$$q = \frac{CE}{4} \left(1 - e^{\frac{-4t}{RC}} \right)$$

18. 5.94

Sol. Least count = $\frac{P}{N} = \frac{1\text{mm}}{100} = 0.01 \text{ mm}$

Diameter of the wire is

$$d = 4\text{mm} + 56 \times \text{L.C} - 6 \times \text{L.C}$$

$$d = 4\text{mm} + 50 \times \text{L.C}$$

$$d = 4\text{mm} + 50 \times 0.01\text{mm}$$

$$d = 4.50\text{mm}$$

$$d = 0.450 \text{ cm}$$

The curved surface area of the wire is

$$S = \pi dl$$

$$S = \frac{22}{7} \times 0.450 \times 4.2 = 5.94 \text{ cm}^2$$

Chemistry

PART – II

Section – A

19. ABC

Sol. $(x-1)(x^2 - 8x + 12) = 0$

$x = 1, 2, 6$

$x = \frac{2r}{a_0} \mid r = \frac{a_0}{2}, a_0, 3a_0$

Minimum position

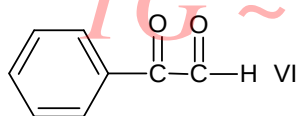
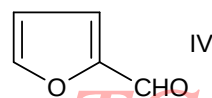
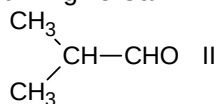
$0.5 a_0$

maximum position

$3a_0$

20. BCD

Sol. It will give Cannizzaro because it fails to give self aldol



These will give Cannizzaro reaction.

21. AD

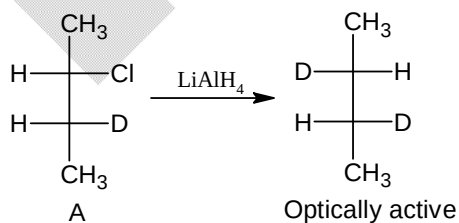
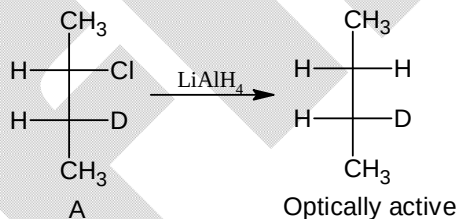
Sol. $\text{C}_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \underbrace{\pi 2p_x^2 \pi 2p_y^2}_{\text{forms } \pi \text{ bond}}$

22. BCD

Sol. Fact

23. AB

Sol.



24. ABCD

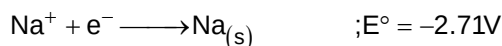
- Sol. (A) NH_3 has higher boiling point than AsH_3 and PH_3 on account H-bonding. SbH_3 has highest boiling point on account of highest molecular weight.
 (B) As the size of element increases, the strength of M-H bond decreases and thus stability decreases.
 (C) Basicity decreases down the group. (Drago rule).
 (D) (Drago rule). HEH angle ($^\circ$): $\text{NH}_3 = 107.8$; $\text{PH}_3 = 93.6$, $\text{AsH}_3 = 91.8$; $\text{SbH}_3 = 91.3$.

25. A

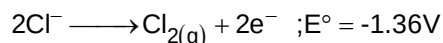
Sol. Electrolysis of NaCl

- (I) Fused NaCl
 (II) Conc. $\text{NaCl}_{(\text{aq})}$
 (III) Dil. $\text{NaCl}_{(\text{aq})}$
 (IV) NaCl (mercury electrode)
 (I) Fused NaCl:

At cathode:



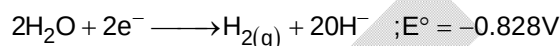
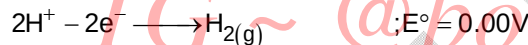
At anode:



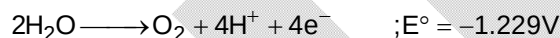
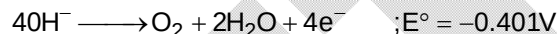
$$\text{Voltage} = -2.71 - 1.36 = -4.07\text{V}$$

- (II) Conc. $\text{NaCl}_{(\text{aq})}$

At cathode:



At anode:



(due to over voltage and high conc. of Cl^-)

At cathode: $\text{H}_{2(\text{g})}$

At anode: $\text{Cl}_{2(\text{g})}$, [O_2 also but percentage may be low]

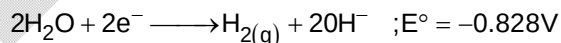
To increase the kinetics of reactions a higher voltage (as from required) is provided, due to this Cl^- start to oxidized at the anode, So

Cl_2 – higher voltage

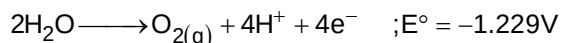
O_2 – lower voltage

- (III) Dil. $\text{NaCl}_{(\text{aq})}$

At cathode:



At anode:



At cathode: $\text{H}_{2(\text{g})}$

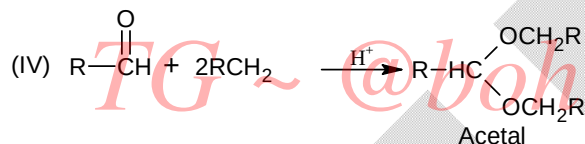
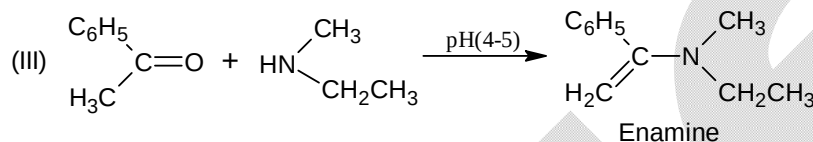
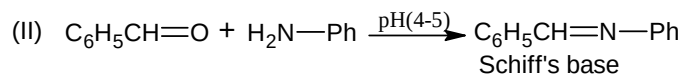
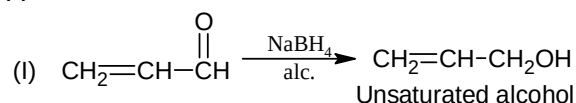
At anode: $O_{2(g)}$

(IV) $NaCl_{(aq)}$ (Mercury electrode):

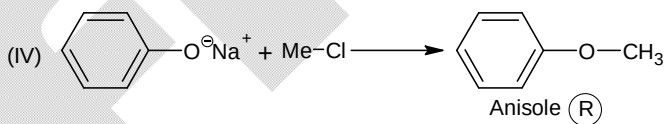
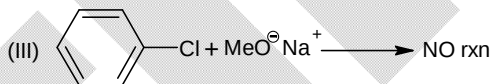
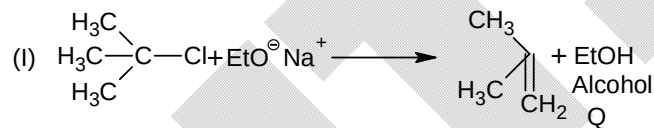
As Na makes an amalgam with Hg so the kinetics favoured the occurrence of Na rather than $H_{2(g)}$.

26. D
Sol. Fact

27. A
Sol.



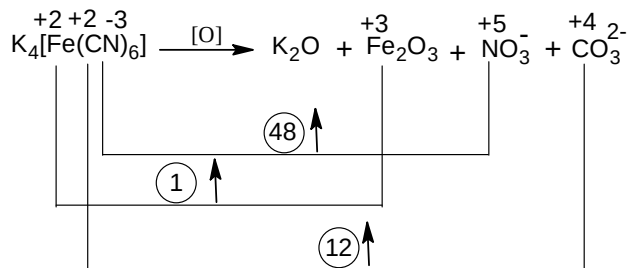
28. C
Sol.



Section – B

29. 61.00

Sol.

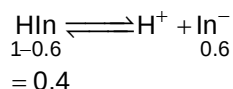


Total $\uparrow$ in oxidation number = $48 + 12 + 1 = 61$

So 'n' factor = 61

30. 25.00

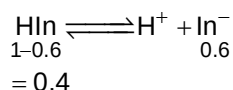
Sol.



$$pH = pK_{In} + \log \frac{[In^-]}{[HIn]}$$

$$2.176 = pK_{In} + \log \frac{0.6}{0.4}$$

$$pK_{In} = 2 \Rightarrow K_{In} = 10^{-2}$$



$$\frac{3.33 \times 10^{-3} \times C\alpha}{C(1-\alpha)} = 10^{-2}$$

$$\frac{\alpha}{1-\alpha} = 3 \Rightarrow \alpha = 3 - 3\alpha$$

$$\Rightarrow \alpha = 0.75$$

$$\text{So } [HIn] = 1 - 0.75 = 0.25$$

$$\%[HIn] = 25\%$$

31. 15.38

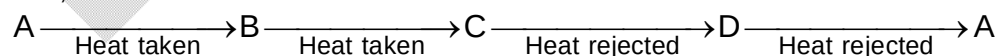
Sol.

Total work done in the cyclic process

= Area under PV

= $P_0 V_0$

Now,



$$\begin{array}{l}
 = 1 \times \frac{3R}{2} (2T - T) \quad \left| \quad = 1 \times \frac{3R}{2} (4T - 2T) \right| \\
 \text{at constant volume} \quad \left| \quad \text{at constant pressure} \right|
 \end{array}$$

$= \frac{3}{2}RT$	$\frac{5}{2} \times 2RT$	
$= 1.5RT$	$5RT$	
$= 1.5P_0V_0$	$5P_0V_0$	At point A $P_0V_0 = 1RT$

Total Heat taken

$$= 6.5 P_0 V_0$$

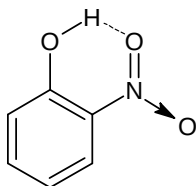
$$\eta = \frac{\text{work done}}{\text{Heat taken}} = \frac{P_0 V_0}{6.5 P_0 V_0} = 0.1538 \quad \text{i.e. } 15.38\%.$$

32, 600,00

Sol. o-nitrophenol

p-nitrophenol

Note:



Due to intramolecular H Bonding V.P of O nitrophenol will be higher. If one have V.P = 400 & the V.P of mixture is 520 mm of Hg, i.e. the V.P of other will be higher than 400, so if P is the V.P then it will of O nitrophenol

O nitrophenol P nitrophenol
P 400

$$520 = P \times 0.6 + 400 \times 0.4$$

$$P = \frac{520 - 160}{0.6} = \frac{360}{0.6} = 600 \text{ mm of Hg.}$$

33. 122.40

Sol. $25.7 = 13.6 z^2 \left| \frac{1}{2^2} - \frac{1}{n^2} \right| \dots (i)$

$$8.7 = 13.6 z^2 \left[\frac{1}{3^2} - \frac{1}{n^2} \right] \dots\dots(ii)$$

Now (i) $\div$ (ii)

$$\frac{25.7}{8.7} = \frac{(n^2 - 4)9n^2}{4n^2(n^2 - 9)}$$

$$\frac{25.7}{8.7} = \frac{9n^2 - 36}{4n^2 - 36}$$

$$2.95 \times 4n^2 - 2.95 \times 36 = 9n^2 - 36$$

$$n^2(11.816 - 9) = 36 \times 1.95$$

$$n^2 = 25 \Rightarrow n = 5$$

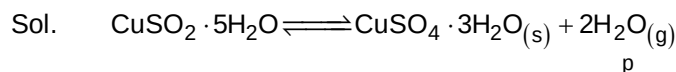
Now,

$$25.7 = 13.6z^2 \left[\frac{1}{4} - \frac{1}{25} \right]$$

$$z = 3$$

$$\text{Ionisation energy} = 13.6 \times 3^2 = 122.4 \text{ eV}$$

34. 25.00



$$K_p = p^2$$

$$10^{-4} = p^2$$

$$p = 10^{-2} \text{ atm}$$

$$= 10^{-2} \times 760 = 7.6 \text{ mm of Hg}$$

$$\text{Relative humidity} = \frac{7.6}{30.4} \times 100 = 25\%$$

35. 915.00

Sol. In BCC nearest neighbor distance

$$= \frac{\sqrt{3}a}{2}$$

$$\frac{\sqrt{3}a}{2} = 4.52$$

$$a = \left(\frac{2 \times 4.52}{\sqrt{3}} \right) \text{ \AA}$$

$$d = \frac{ZM}{NV} = \frac{2 \times 39}{6 \times 10^{23} \times \left(\frac{2 \times 4.52}{\sqrt{3}} \right)^3 \times 10^{-24}} = \frac{2 \times 39}{6 \times 14.2} = 0.915 \text{ g/cc}$$

$$= 915 \text{ kg/m}^3$$

36. 7482.60

Sol. Given: $P_1 = 16 \text{ bar}$, $T_1 = 300 \text{ K}$, $P_2 = 1 \text{ bar}$, $T_2 = ?$

To calculate T_2 use:

$$TP^{\left(\frac{1-\gamma}{\gamma} \right)} = \text{const}$$

$$T_1 P_1^{\left(\frac{1-\gamma}{\gamma} \right)} = T_2 P_2^{\left(\frac{1-\gamma}{\gamma} \right)}$$

$$300 \times 16^{\left(\frac{1-\gamma}{\gamma} \right)} = T_2 \times 1^{\left(\frac{1-\gamma}{\gamma} \right)}$$

$$T_2 = 300 \times 16^{\left(\frac{3}{4} - 1 \right)}$$

$$T_2 = \frac{300}{2} = 150 \text{ K}$$

$$W_{\text{rev, adia}} = nC_V (T_2 - T_1)$$

$$= 2 \times 3R (150 - 300)$$

$$= -7482.6 \text{ J}$$

Mathematics

PART – III

Section – A

37. BC

Sol. $x^4(x-5)=1$

$$\Rightarrow x-5 = \frac{1}{x^4}$$

$$\therefore \sum \frac{1}{\alpha_r^4} = \sum \alpha_r - 25 = 5 - 25 = -20$$

Similarly, $\left(\frac{1}{\alpha_r^4} + 5\right) = \alpha_r$

38. A

Sol. $\therefore \begin{vmatrix} 1 & a & -1 \\ 2 & -1 & 3 \\ a & -1 & 3 \end{vmatrix} = 0$

$$\Rightarrow a = \frac{1}{3} \text{ or } 2$$

But for $a = 2$, two vectors are parallel.

39. D

Sol. $\therefore I = \int_{\alpha}^{\beta} f(x) dx$

$$\Rightarrow I = \int_{\alpha}^{\beta} f(\alpha + \beta - x) dx = \int_{\alpha}^{\beta} f(3-x) dx$$

So, $7I = \int_{\alpha}^{\beta} (3x^2 - 8) dx$

$$= \beta^3 - \alpha^3 - 8(\beta - \alpha)$$

$$= (\beta - \alpha) \{ \beta^2 + \alpha^2 + \beta\alpha - 8 \}$$

$$= \sqrt{9-4a} \{ 9 - \alpha\beta - 8 \} = \sqrt{9-4a} (1-a) \leq 0$$

$$\Rightarrow 1 \leq a \leq \frac{9}{4}$$

40. AB

Sol. If $\lambda x^2 + (\lambda - 1)x + \frac{1}{24} = 0$ has equal and real roots,

$$\Rightarrow (\lambda - 1)^2 = 4\lambda \cdot \frac{1}{24}$$

$$\Rightarrow 6\lambda^2 - 13\lambda + 6 = 0$$

$$\therefore \lambda = \frac{13 \pm \sqrt{169 - 144}}{2 \times 6} = \frac{2}{3} \text{ or } \frac{3}{2}$$

41. BD

Sol.
$$\lim_{n \rightarrow \infty} \left(\frac{1}{n^n} \right)^n \cdot (n^n)^n \left[\left(1 + \frac{1}{n} \right) \left(1 + \frac{1}{2n} \right) \left(1 + \frac{1}{2^2 n} \right) \dots \right]^n$$

$$= e^1 \cdot e^{\frac{1}{2}} \cdot e^{\frac{1}{2^2}} \dots \dots \dots \infty$$

$$= e^{1 + \frac{1}{2} + \frac{1}{2^2} + \dots} = e^{\frac{1}{1 - \frac{1}{2}}} = e^2 \approx 7.3441$$

42. AB

Sol. $|z_1| = |z_2| = 1$ & $|z_1 + z_2| = 1$

$$\therefore |z_1 + z_2| = |z_1| |1 + e^{i\theta}| = 1$$

$$|z_1 - z_2| = |z_1| |1 - e^{i\theta}| = 1$$

$$\therefore z_2 = z_1 \omega$$

$$\therefore z_1 + z_2 = z_1 (1 + \omega) = \frac{-b}{a}$$

$$\& z_1 z_2 = z_1^2 \omega = \frac{c}{a}$$

$$\Rightarrow \left(\frac{b^2}{a^2} \right) = \frac{c}{a} \Rightarrow b^2 = ac$$

$$1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta = 1$$

$$2 + 2 \cos \theta = 1$$

$$\cos \theta = \frac{-1}{2} \Rightarrow \theta = \frac{2\pi}{3}$$

43. C

Sol. AEEI (4 vowels) and DRJLNG (6 consonants)

$$\therefore N = \frac{4!}{2!} \times 6 = 12 \times 720$$

$$= 2^6 \times 3^3 \times 5^1$$

44. A

Sol. Either increasing or decreasing part lie in $[0, 2] \rightarrow [0, 1]$

$$\text{Then } y(0) = c = 1 \text{ or } 0$$

$$\text{and } y = x(ax + b) + c \Rightarrow y(2) = 2(2a + b) + c = 1 \text{ or } 0$$

$$\therefore |2a + b| = \frac{1}{2}$$

$$\text{If } \frac{\lambda}{a} \text{ is root, } a \times \frac{\lambda^2}{a^2} + b \times \frac{\lambda}{a} + c = 0 \Rightarrow \frac{\lambda^2}{a} + \frac{b\lambda}{a} + 1 = 0 \dots \dots (i)$$

$$\therefore 4a + 2b + 1 = 0$$

$$1 + \frac{b}{2a} + \frac{1}{4a} = 0 \dots \dots (ii)$$

$$\text{From (i) and (ii), } \lambda = \frac{1}{2}$$

$$\text{Sum of roots} = -\frac{b}{a} = \left(\frac{2a + \frac{1}{2}}{a} \right) = p^2 + p + 3 \geq \frac{11}{4}$$

$$\therefore 4a + 2b + 1 = 0 \quad \Rightarrow 2 + \frac{1}{2a} \geq \frac{11}{4}$$

$$-b = \left(2a + \frac{1}{2} \right) \quad \text{For } a > 0, \quad \frac{1}{2a} \geq \frac{3}{4} \Rightarrow a \leq \frac{2}{3}$$

$$\text{For } a < 0, \quad a \geq \frac{2}{3} \Rightarrow |a| \leq \frac{2}{3}$$

$$|a| \leq 0.66$$

45. D

Sol. $\therefore f(x) = \frac{2}{\pi} \tan\left(\frac{\pi x}{2}\right) + \frac{1}{2x}$ has point of minima at $x = \frac{1}{2}$

$$\text{So, range} \in \left[\frac{\pi+2}{\pi}, \infty \right)$$

$$\therefore e^{-k} \geq 1$$

$$\Rightarrow e^k \leq 1 \Rightarrow k \leq 0$$

$$\text{So, for } k \in \left(-\frac{1}{e}, 0 \right), \text{ we have two solutions.}$$

In the same manner use graphs.

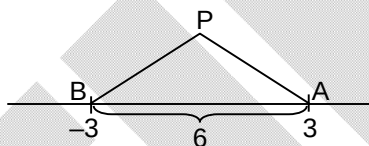
46. A

Sol. $\therefore z^2 \bar{z} + \bar{z}^2 z = (z + \bar{z})$

$$\Rightarrow z \bar{z} (z + \bar{z}) = (z + \bar{z})$$

$$\Rightarrow x^2 + y^2 = 1 \text{ or } x = 0$$

circle straight line



$$|PA - PB| = 6$$

means, P lies on straight line

& $|PA - PB| < 6$ shows it is hyperbola.

Section - B

47. 1.33

Sol. $\therefore 1 \leq |x| \leq \sqrt{3} \quad \& \quad \tan^{-1} \geq \frac{2\pi}{3} - \frac{\pi}{2} = \frac{\pi}{6}$

$$\Rightarrow x \geq \frac{1}{\sqrt{3}} \quad \therefore \frac{1}{3} \leq x \leq \sqrt{3}.$$

$$\therefore \alpha + \frac{1}{\beta^2} = 1 + \frac{1}{3} = \frac{4}{3}.$$

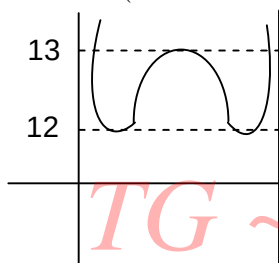
48. 0.25

Sol. $\therefore y = \frac{x}{x^2 + y} \Rightarrow y^2 + x^2 y - x = 0$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{-(2xy - 1)}{(x^2 + 2y)} \Rightarrow \frac{dy}{dx} = \frac{-\left(2x \cdot \frac{x}{x^2 + y} - 1\right)}{(x^2 + y + y)} = -\frac{(2x^2 - x^2 - y) / x^y y}{\left(\frac{x}{y} + y\right)} \\ &= \frac{-y(x^2 - y)}{(x^2 + y)(x + y^2)} \Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{(y - x^2)}{(x^2 + y)(x + y^2)} \\ &= \frac{d}{dx}(\ln y) = \frac{(y - x^2)}{(x^2 + y)(x + y^2)} \Rightarrow \ln y + c = \int \frac{(y - x^2) dx}{(x^2 + y)(x + y^2)} \end{aligned}$$

49. 4.50

Sol. Graph of $(9 \sin x + 4 \operatorname{cosec} x)$ is



$$\therefore 12 < 2k + 4 \leq 13$$

$$\Rightarrow 8 < 2k \leq 9$$

$$4 < k \leq \frac{9}{2}$$

$$\therefore k = \frac{9}{2}$$

50. 1.78

Sol. $(\text{distance from focus})^2 = (\text{Circumradius}) \times \frac{1}{4}(\text{L.R.}) \cdot \left(\frac{4}{3}\right)^2 = R \times \frac{1}{4} \times 4 \Rightarrow R = \frac{16}{9}$

51. 14.00

Sol. $\int_{-n}^n e^{|x|} \frac{dx}{1 + e^{\sin x}}$

$$\begin{aligned} &= \int_0^n e^{|x|} dx. \\ &= n \int_0^1 e^{x-[x]} = n(e - 1) \\ \sum_{n=1}^{50} \frac{n}{(n+1)!} &= 1 - \frac{1}{(3 \times 17)!} \\ \text{So } |a - b| &= 14 \end{aligned}$$

52. 1.00

Sol. $x^3 - x^2 - x + 1 = (x+1)(x-1)^2$, $f(x) = g(x)$ have 3 real and distinct roots

$$\Rightarrow 0 < a^2 - 2 < 1$$

$$\Rightarrow [a^2] = 2$$

Now $x^3 - x^2 - x + 1 = 2$

has solution in (1, 2)

$\Rightarrow$ the integral part of root is 1.

53. 4.00

Sol.
$$\frac{3x^2 + 1}{\sqrt{(x^4 + x^2)}} = \frac{(3x^2 + 1)}{\sqrt{x^2(x^2 + 1)}}$$

$$= \frac{2x^2 + (x^2 + 1)}{x\sqrt{x^2 + 1}}$$

$$= \frac{2x}{\sqrt{x^2 + 1}} + \frac{\sqrt{x^2 + 1}}{x}$$

$$\geq 2\sqrt{2}$$

Equality for $2x^2 = x^2 + 1$

$$x^2 = 1$$

$$x = 1$$

$$\therefore \int_0^{\sqrt{2}} \frac{3x^2 + 1}{\sqrt{x^4 + x^2}} dx \geq 2\sqrt{2}(\sqrt{2} - 0) = 4$$

54. 147.00

Sol. Required probability = $\frac{176}{10 \times 10 \times 10} = \frac{22}{125}$

So, $p + q = 147$

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – III
PAPER –2
TEST DATE: 08-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

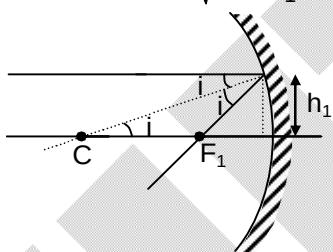
PART – I

Section – A

1. B
Sol. Since B & revolve in opposite since, thus they are oppositely charged. From direction of motion of the charges $Q_A \rightarrow +ve$ $Q_B \rightarrow -ve$ $Q_C \rightarrow neutral$

2. A

Sol. $CF_1 = \frac{R}{2\cos i} = \frac{R^2}{2\sqrt{R^2 - h_1^2}}$ and $(h_1 = 3 \text{ cm})$



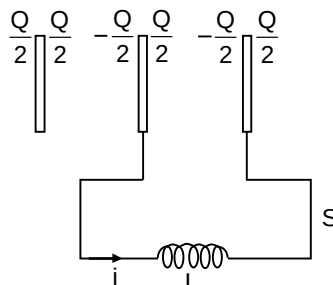
$$CF_2 = \frac{R^2}{2\sqrt{R^2 - h_2^2}} \text{ and } (h_2 = 4 \text{ cm}) \Rightarrow CF_2 - CF_1 = \frac{25}{24} \text{ cm}$$

3. A

Sol. If maximum loss in kinetic energy is greater than first excitation energy inelastic collision may take place.

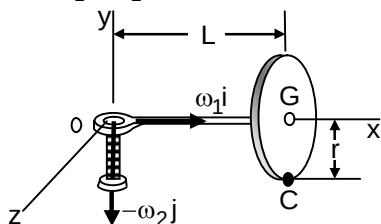
4. D

Sol. $\left(\frac{Q}{2}\right)^2 = \frac{1}{2} Li_0^2$
 $\Rightarrow i_0 = \frac{Q}{2\sqrt{LC}}$



5. ACD

Sol. $\omega = \omega_1 \mathbf{i} - \omega_2 \mathbf{j}$



$$v_C = \omega \times r_C = 0$$

$$(\omega_1 \mathbf{i} - \omega_2 \mathbf{j}) \times (L\mathbf{i} - r\mathbf{j}) = 0$$

$$(L\omega_2 - r\omega_1)\mathbf{k} = 0$$

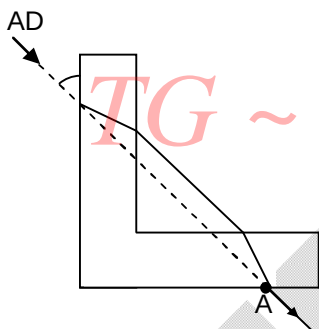
$$\omega = \omega_1 \mathbf{i} - (r\omega_1 / L)\mathbf{j}$$

$$H_x = I_x \omega_x = \left(\frac{1}{2} m r^2 \right) \omega_1$$

$$H_z = I_z \omega_z = \left(mL^2 + \frac{1}{4} m r^2 \right) 0 = 0$$

$$\omega_2 = \omega_1 / L$$

6. Sol.



Option B and C are incorrect.

7. ABC

Sol. Let N be the number of fringes within the length x, then we have,

$$\beta N = x \Rightarrow \frac{D\lambda}{d} N = x \Rightarrow N = \frac{xd}{\lambda D}$$

At any time t

$$N = \frac{x}{\lambda D} (d + vt)$$

$$\frac{dN}{dt} = \frac{xv}{\lambda D}$$

8. ABCD

Sol. $r = \frac{mV_0}{qB}$, and $PQ = 2r \sin \alpha = \frac{2mV_0}{qB} \sin \alpha$

$$\alpha = \beta$$

$$\text{Time taken (t)} = \frac{2\pi r}{V}$$

9. ABC

Sol. From graph in time from $t = 0$ to $t = 3$ sec.

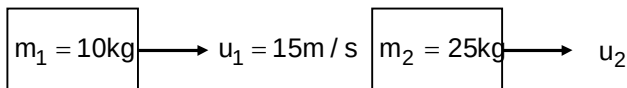
Acceleration of object of mass $m_1 = 10$ kg is

$$a = \frac{15-0}{3} = 5 \text{ m/s}^2$$

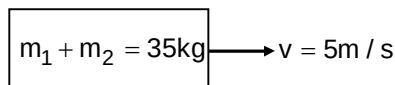
$\therefore$ Force on object of mass m_1 from $t = 0$ to $t = 3$ sec. ... (i)

$$= 10 \times 5 = 50 \text{ N}$$

Before and after collision at $t = 4$ sec, the velocities of blocks are as shown.



Before collision



After collision

$\therefore$ Initial momentum of system

$$= m_1 u_1 + m_2 u_2 = 150 + 25 u_2$$

Final momentum of system

$$= (m_1 + m_2) v = 35 \times 5 = 175$$

From conservation momentum

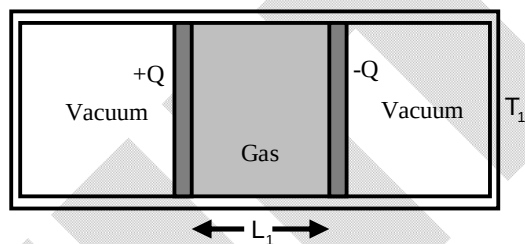
$$\therefore 150 + 25 u_2 = 175$$

or $u_2 = +1 \text{ m/s}$

$\therefore$ speed of second particle just before collision is 1 m/s and before collision both blocks move in same direction.

10. BCD

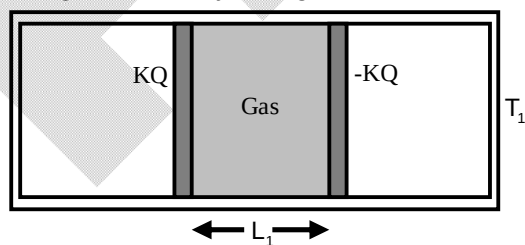
Sol. Stage 1



$$P_1 A L_1 = n R T_1 \quad \frac{Q^2}{2A\epsilon_0} = P_1 A = \frac{n R T_1}{L_1} \quad (\text{Force Balancing})$$

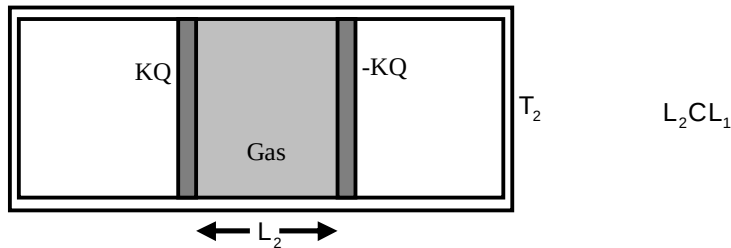
Stage 2

Charge is suddenly changed



System is not in equilibrium

Stage 3



Finally system in equilibrium

$$P_2 A L_2 = n R T_2 \quad \frac{K^2 Q^2}{2 A \epsilon_0} = P_2 A = \frac{n R T_2}{L_2}$$

Energy of the system is conserved between stage 2 and 3.

$$\Rightarrow \frac{1}{2} \epsilon_0 E_2^2 A L_1 + \frac{f}{2} n R T_1 = \frac{1}{2} \epsilon_0 E_2^2 A L_2 + \frac{f}{2} n R T_2$$

On solving all the above equation

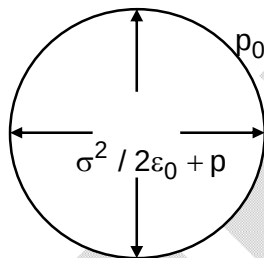
$$\frac{T_2}{T_1} = \frac{2K^2 + f}{2 + f} \quad w_{\text{ext}} = (K^2 - 1) \frac{Q^2 L_1}{2 A \epsilon_0} = (K^2 - 1) n R T_1$$

Section – B

11. 6

Sol. $V = \frac{4}{3} \pi r^3 \dots (1)$

$$\sigma = \frac{Q}{4 \pi r^2} \dots (2)$$



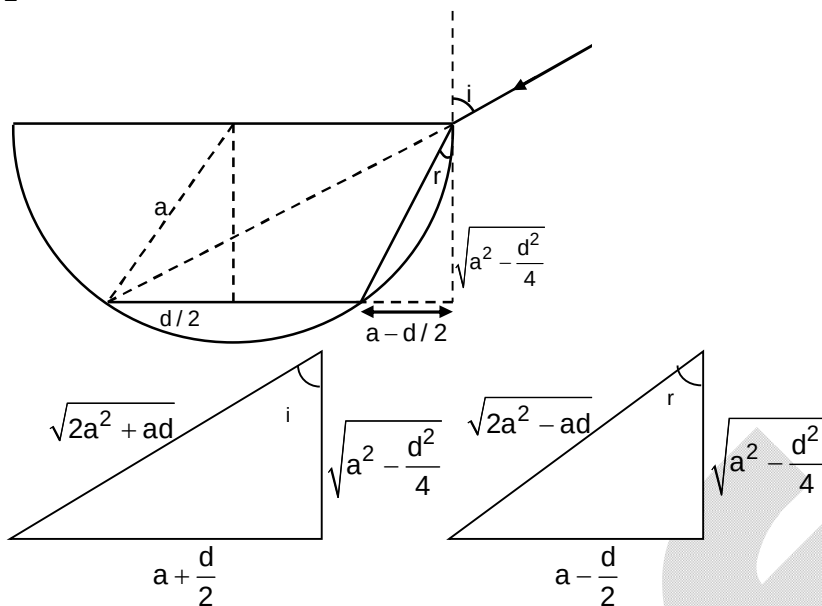
$$\frac{\sigma^2}{2 \epsilon_0} + p - p_0 = \frac{4s}{r} \dots (3)$$

$$p - p_0 = 0 \dots (4)$$

$$\Rightarrow \frac{1}{2 \epsilon_0} \cdot \frac{Q^2}{16 \pi^2 r^4} = \frac{4s}{r}$$

$$\Rightarrow n = 6$$

12. 1
Sol.



$$\sin i = \mu \sin r$$

$$\Rightarrow \frac{d}{a} = 1$$

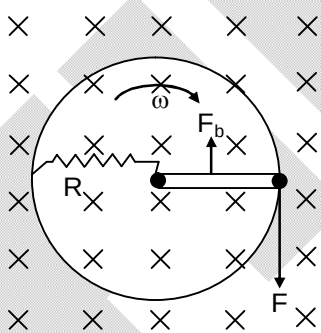
13. 2
Sol.

$$F_b = BIL, (\because L = r = 1\text{m})$$

Induced current:
$$I = \frac{(B\omega r^2 / 2)}{R}$$

$$\therefore F_b = B \left(\frac{B\omega r^2}{2R} \right) r = \frac{B^2 \omega r^3}{2R}$$

To maintain constant angular velocity:

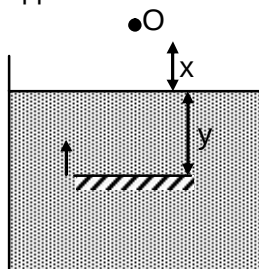


$$F(r) = F_b (r/2)$$

$$\Rightarrow F = \frac{F_b}{2} = \frac{B^2 \omega r^3}{4R} = \frac{(10)^2 \times 2\pi \times (1)^3}{4 \times \pi} = 50\text{N}$$

14. 3

Sol. Apparent distance of mirror from O



$$= x + \frac{y}{\mu}$$

Distance of final image from O

$$= 2 \left(x + \frac{y}{\mu} \right)$$

Velocity of image

$$= 2 \left(\frac{dx}{dt} + \frac{1}{\mu} \frac{dy}{dt} \right) = \frac{2}{\mu} \times 8 = \frac{2}{4/3} \times 8 = 12 \text{ cm/s}$$

15. 3

Sol. For $S_1S_2 = 2.5\lambda$ max path different = 2.5λ

Min path different = 0

Between 2.5λ and 0 lie 2λ and $\lambda \Rightarrow$ two circular bright fringes

$$n_1 = 2$$

For $S_1S_2 = 5.7\lambda$, max. path different = 5.7λ

Min path different = 0

Between 5.7λ and 0 lie $5\lambda, 4\lambda, 3\lambda, 2\lambda, \lambda \Rightarrow$ Five circular bright fringes.

$$\Rightarrow n_2 = 5$$

$$\therefore n_2 - n_1 = 5 - 2 = 3$$

16. 9

Sol. $\sqrt{f} = a(Z-1)$

$$\sqrt{f} = a(31-1)$$

$$\sqrt{\frac{25f}{x}} = a(51-1)$$

$$x = 9$$

17. 5

Sol. $f_m = \frac{v}{(v - v_s \cos \theta)_{\min}} \times f$

$$1800 = \frac{v}{(v + v_s \cos \theta)_{\max}} \times f$$

On solving,

$$f_m = 2250 \text{ Hz}$$

$$= 2300 - 50$$

18. 2

Sol. Net power given to N_2 gas = $100 - 30 = 70$ cal/s

The nitrogen gas expands isobarically

$$\therefore \frac{dq}{dt} = n C_p \frac{dT}{dt} \quad \text{or } 70 = 5 \times \frac{7}{2} R \frac{dT}{dt}$$

$$\therefore \frac{dT}{dt} = 2 \text{ K / sec}$$

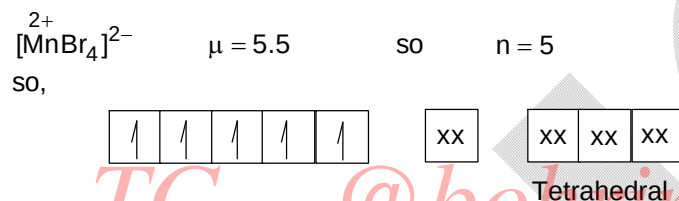
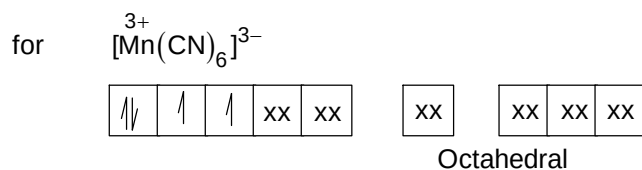
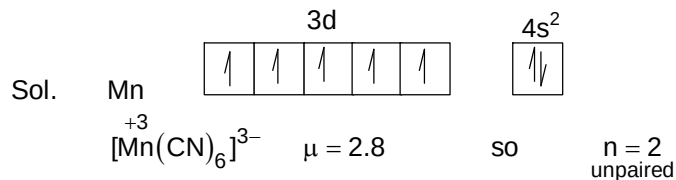
TG ~ @bohring_bot

Chemistry

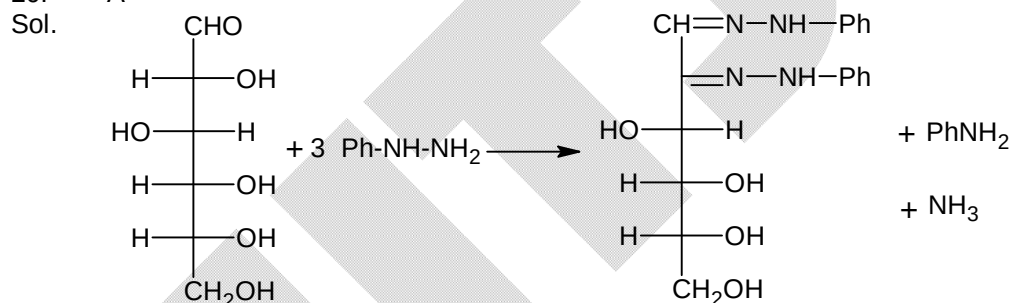
PART – II

Section – A

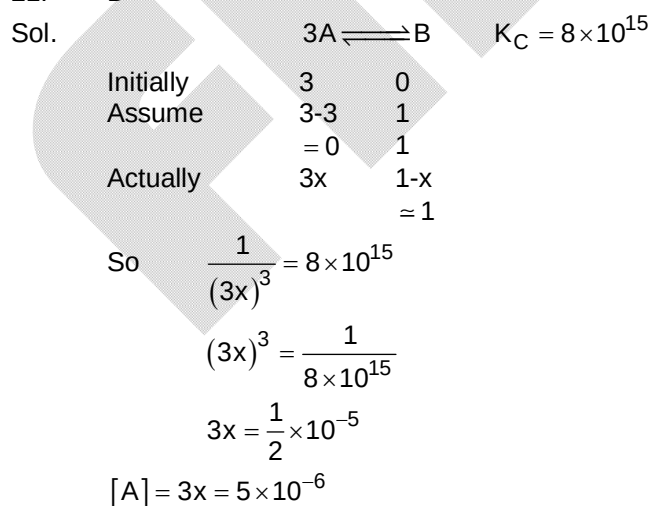
19. C



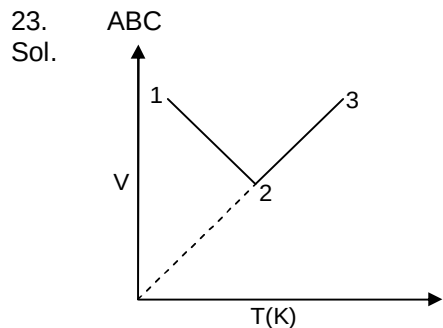
20. A



21. D



22. B
Sol. Fact.



(A) for $1 \rightarrow 2$

$V \downarrow$ & $T \uparrow$ so

By $PV = nRT$

$$P = \frac{nRT}{V}$$

P will $\uparrow$

(B) for $2 \rightarrow 3$

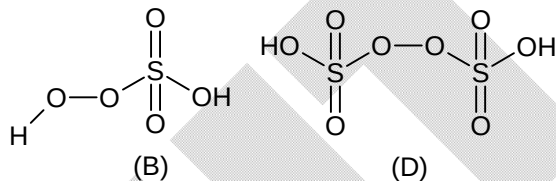
$V \propto T$

So P will be constant

(C) at 1 & 3, T is higher at 3 so

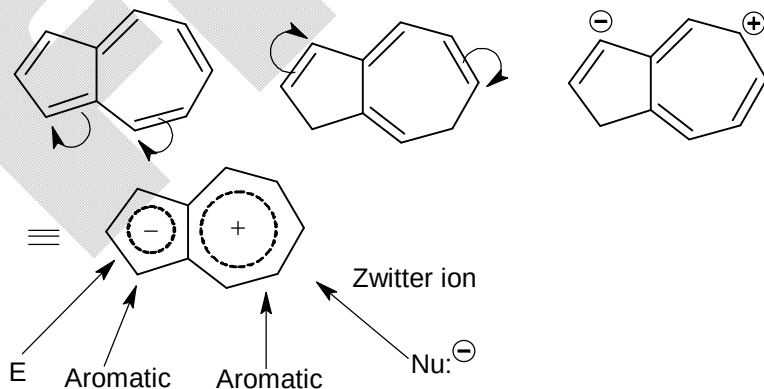
$P_{(3)} > P_{(1)}$

24. BD
Sol.



25. ACD
Sol. Dacron is a condensation polymer. Teflon, neoprene and acrylonitrile are addition polymers. Hence option (A, C and D) is the answer.

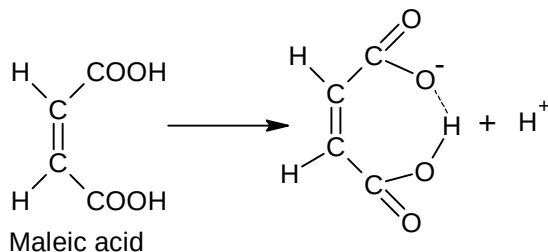
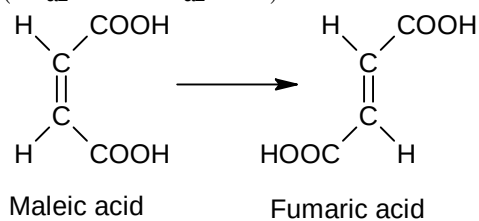
26. ABCD
Sol.



27. ABD

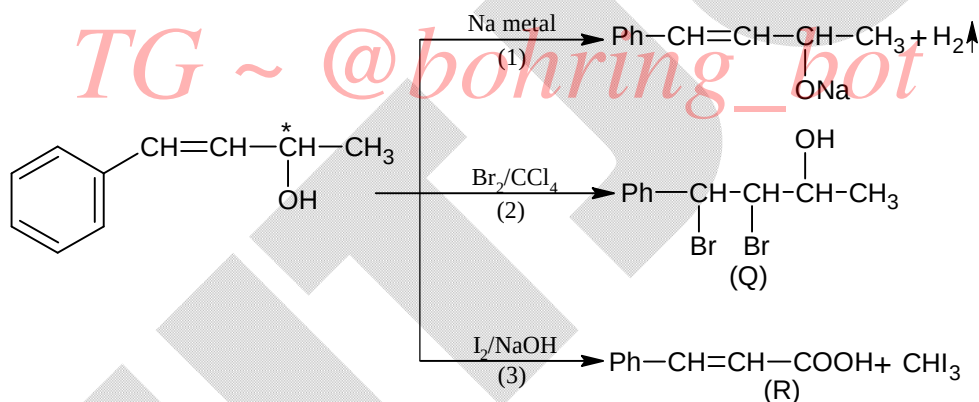
Sol. Maleic acid is stronger acid than fumaric acid

Maleic acid form cyclic ring after ionization while fumaric acid cannot do so that's why maleic acid ($pK_{a1} = 1.9$, $pK_{a2} = 6.1$) is more stronger acid than fumaric acid ($pK_{a1} = 3.0$, $pK_{a2} = 4.4$).



28. BC

Sol.

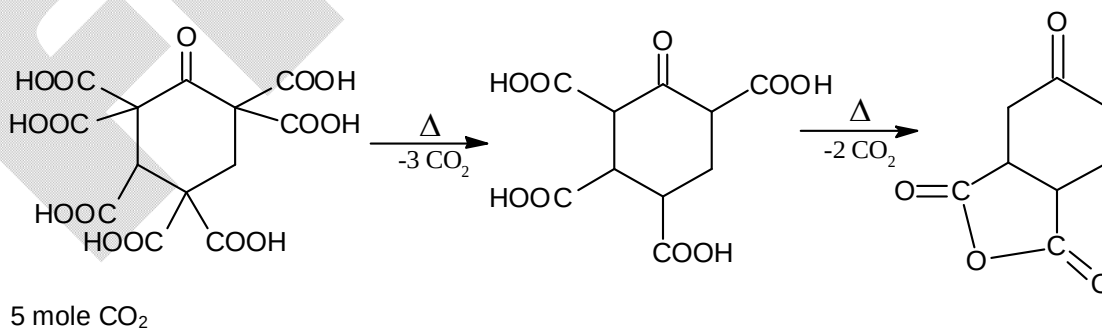


'P' has optical and geometrical isomers.

Section – B

29. 5

Sol.



30. 6
Sol. The following will release H_2 gas with propyne
 CH_3CH_2MgBr (III), NaH (IV), $NaNH_2$ (V), Na (VI), $\{(CH_3)CH\}_2NLi$ (VIII), CH_3CH_2Li (IX)

31. 6
Sol. Glucose Fructose Mannose Maltose Cellobiose Lactose

32. 6
Sol. $Hg^{2+} + 2e^- \longrightarrow Hg$ $\Delta G_1^0 = -2F(0.85)$

$Hg + 4I^- \longrightarrow HgI_4^{2-} + 2e^-$ $\Delta G_2^0 = -2F(0.05)$

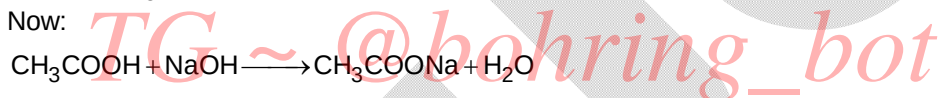
So

$Hg^{2+} + 4I^- \longrightarrow HgI_4^{2-}$ $\Delta G^0 = -2F(0.90)$

$$0.90 = \frac{0.06}{2} \log K_f$$

$$K_f = 1.0 \times 10^{30}$$

33. 3
Sol. m.m. of $NaOH = 30 \times 0.2 = 6$ m.m
m.m. of $CH_3COOH = 50 \times 0.2 = 10$ m.m
Now:



$$\begin{array}{ccc} 10 & 6 & 0 \\ 10 - 6 = 4 & 6 - 6 = 0 & 6 \end{array}$$

$$pH = -\log(2 \times 10^{-5}) + \log \frac{6}{4}$$

$$= 4.87$$

Suppose 'v' mL of $NaOH$ is added then

$$\text{m.m of } CH_3COONa = (6 + v \times 0.2)$$

$$\text{m.m of } CH_3COOH = (4 - v \times 0.2)$$

$$5 = -\log(2 \times 10^{-5}) + \log \left(\frac{6 + 0.2v}{4 - 0.2v} \right)$$

$$0.3010 = \log \left(\frac{6 + 0.2v}{4 - 0.2v} \right)$$

$$\text{So } \left(\frac{6 + 0.2v}{4 - 0.2v} \right) = 2 \Rightarrow v = 3.33 \text{ mL}$$

34. 8

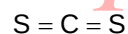
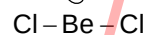
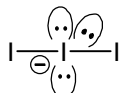
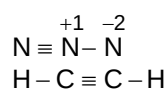
$$\text{Sol. \% decrease in vol} = \frac{V_{bcc} - V_{fcc}}{V_{bcc}} \times 100$$

$$= \frac{\frac{1}{d_{bcc}} - \frac{1}{d_{fcc}}}{\frac{1}{d_{bcc}}} \times 100$$

$$\begin{aligned}
 &= \frac{N_A (4r)^3}{2 \times 56 \times 3\sqrt{3}} - \frac{N_A (4r)^3}{4 \times 56 \times 2\sqrt{2}} \\
 &= \frac{N_A (4r)^3}{2 \times 56 \times 3\sqrt{3}} \\
 &= \frac{1}{3\sqrt{3}} - \frac{1}{2 \times 2\sqrt{2}} \\
 &= \frac{1}{3\sqrt{3}} \\
 &= \frac{0.46}{5.65} \times 100 \approx 8\%
 \end{aligned}$$

35. 7

Sol. $O = C = O$



are linear

36. 2

Sol. $+\frac{1}{2} \frac{d[B]}{dt} = k[A]^n$

Taking ln on both side

$$\ln\left(\frac{1}{2}\right) + \ln\left(\frac{d[B]}{dt}\right) = \ln k + n \ln[A]$$

$$\ln\left(\frac{d[B]}{dt}\right) = \left(\ln k - \ln\left(\frac{1}{2}\right)\right) + n \ln[A]$$

So $n = 2$

Mathematics

PART – III

Section – A

37. D

Sol. $\therefore 2(\alpha + \beta) = (\alpha - \beta) + \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

$$\Rightarrow \left(2 - \frac{1}{\alpha\beta}\right) = \frac{\alpha - \beta}{\alpha + \beta}$$

$$\Rightarrow (2 + a) = \frac{\alpha - \beta}{\alpha + \beta} > 0 \Rightarrow a + 2 > 0 \Rightarrow a > -2$$

But $\alpha\beta < 0 \Rightarrow \frac{a}{-1} < 0 \Rightarrow a > 0$

38. D

Sol. $\therefore \frac{(3^3 - 8)}{(3^3 + 8)} \times \frac{(4^3 - 8)}{(4^3 + 8)} \times \frac{(5^3 - 8)}{(5^3 + 8)} \times \dots \times \frac{(n^3 - 8)}{(n^3 + 8)}$

Use: $n^3 - 8 = (n - 2)(n^2 + 2n + 4)$

and $n^3 + 8 = (n + 2)(n^2 - 2n + 4)$

We get, product = $\frac{1.2.3.4.(n^3 - 3)(n^2 + 2n + 4)}{(n - 1)n.(n + 1)(n + 2).7.12}$

For $n \rightarrow \infty$, its value = $\frac{1.2.3.4}{7.12} = \frac{2}{7}$

39. C

Sol. Let $P(t, t^2)$ and $Q(s, s^2)$.

So, equation of normal at P is $x + 2 + y = t + 2t^3$

$\therefore \text{Area} = \int_s^t \left(t^2 + \frac{1}{2} - \frac{x}{2t} - x^2\right) dx = \frac{4}{3} \left(t + \frac{1}{4t}\right)^3 \geq \frac{4}{3} \quad (\text{AM} \geq \text{GM})$

(Here, take $t > 0$)

40. B

Sol. Let $1 + \tan x = t^2$

$$\frac{(t^2 - 1)^2}{(t^2 - t)^2} < \frac{t^4 + t^2 + 16}{t^4}$$

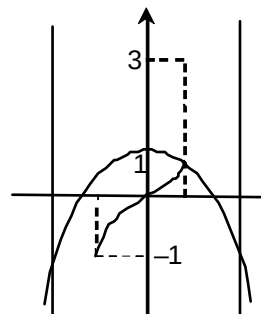
$\Rightarrow t^3 < 8 \Rightarrow t < 2$

But $t > 0$ and $t \neq 1$

$\Rightarrow \tan x \in (-1, 0) \cup (0, 3)$

$\Rightarrow -\frac{\pi}{4} < x < 0 \text{ or } 0 < x < \tan^{-1} 3$

Only one solution on graph.



41. AC

Sol.
$$\int_0^{\pi} \frac{\sin^3 x dx}{x+1} = \int_0^{\pi} \sin^3 x \cdot \frac{1}{(1+x)} dx$$

$$= 0 - 3 \int_0^{\pi} \sin^2 x \cdot \ln(x+1) dx$$

$$= -3 \left[0 - \int_0^{\pi} \sin 2x \cdot (x+1) \{ \ln(x+1) - 1 \} dx \right]$$

$$\& \int_0^{\pi} \frac{\sin^3 x dx}{2x - \pi} = 2 \int_0^{\pi/2} \frac{\sin^3 x}{2x - \pi} dx = 2 \int_0^{\pi/2} \frac{\cos^3 x}{\pi - 2x - \pi} dx$$

$$= - \int_0^{\pi/2} \frac{\cos^3 x}{x} dx$$

42. ABC

Sol. $\therefore \alpha^2 - \alpha - 1 = 0 \Rightarrow \alpha^6 - \alpha^3 - 1 = 3\alpha^3$

$$\Rightarrow \alpha^6 - 1 = 4\alpha^3$$

$$\Rightarrow \frac{\alpha^6 - 1}{\alpha^3} = 4$$

Similarly,

$$\beta^2 - \beta - 1 = 0$$

$$\beta^6 - \beta^3 - 1 = 3\beta^3$$

$$\frac{\beta^6 - \beta^3 - 1}{\beta^3} = 3$$

43. BCD

Sol. $\therefore f(n) = (n+1) \ln \left(\frac{n+1}{n+2} \right) + \ln \left(\frac{n+1}{2} \right)$

$$\Rightarrow g(n) = \ln \left(\frac{n+1}{2} \right) \Rightarrow g(x) = \log_e \left(\frac{x+1}{2} \right)$$

$$\Rightarrow g'(x) = \frac{2}{x+1} \cdot \frac{1}{2} \Rightarrow g'(0) = 0$$

$$g''(x) = -\frac{1}{(x+1)^2} \Rightarrow g''(0) = -1$$

$$\& \lim_{n \rightarrow \infty} \sum_{r=1}^{r=n} \left(\frac{1}{2 \cdot e^{\log_e \left(\frac{r+1}{2} \right)} - 1 + n} \right) = \sum_{r=1}^{r=n} \frac{1}{r+1-1+n} = \sum_{r=1}^{r=n} \frac{1}{n} \left(\frac{r}{n} + 1 \right)$$

44. AD

Sol. $\therefore (2, 1)$ lies on circle

$$\therefore (2-r)^2 + (1-r)^2 = r^2 \text{ for touching circles on axis.}$$

$$\Rightarrow r^2 - 6r + 5 = 0$$

$$\Rightarrow r = 1 \text{ or } 5$$

45. BCD

Sol. (A) $\frac{f(x)}{x^2} = \frac{2}{3} \Rightarrow g(x) = 3f(x) - 2x^2$

$$g(0) > 0 \text{ \& } g(1) > 0 \text{ or } < 0$$

So, can't say.

(B) $g(x) = 2f(x) = 3x^2$

$$\therefore g(0) > 0 \text{ \& } g(1) < 0$$

Similarly, (C) and (D).

46. AB

Sol. Probability = $\frac{5^3 + 5 - 1}{5^4} = 0.203$

Section – B

47. 5

Sol. $y \geq 0$

$$\Rightarrow \frac{e^y}{[y+1]} \leq 1$$

$$\Rightarrow \frac{2x^2 + ax + 2}{x^2 + x + 1} \geq 0$$

$$\Rightarrow x^2 + x(a-1) + 1 \geq 0 \forall x \in \mathbb{R}$$

$$\Rightarrow -1 \leq a \leq 3$$

So, no. of integer $a = 5$

48. 7

Sol. $\because |A| = \alpha \bar{\alpha} = |\alpha|^2$ and $|A - I| = \begin{vmatrix} \alpha - 1 & 0 \\ 2 & \bar{\alpha} - 1 \end{vmatrix}$

$$|A - I| = \alpha \bar{\alpha} - \alpha - \bar{\alpha} + 1 = |\alpha - 1|^2$$

$$\Rightarrow |A - I| = |\alpha|^2 - \text{tr}(A) + 1$$

Given, $|2A^2 - 2A| = 144$

$$\Rightarrow |A^2 - A| = 36$$

$$|A| |A - I| = 36$$

$$\Rightarrow |\alpha|^2 |\alpha - 1|^2 = 36$$

$$|\alpha(\alpha - 1)| = 6 \text{ or } -6$$

$$\alpha^2 - \alpha = 6 \text{ or } -6$$

$$\alpha = 3 \quad \text{or} \quad -2 \quad \text{or} \quad \alpha = \frac{+1 \pm i\sqrt{2}.3}{2 \times 1}$$

$$\bar{\alpha} = 3 \quad \text{or} \quad -2 \quad \text{or} \quad \bar{\alpha} = \frac{+1 \mp i\sqrt{2}.3}{2 \times 1}$$

$$\alpha + \bar{\alpha} = 6 \quad \text{or} \quad -4 \quad \text{or} \quad 1$$

49. 8

Sol. $\therefore \beta = \frac{f(4) + f(12)}{2}$
 $= \frac{4 + 12}{2} = 8$

50. 4

Sol. $f^{-1}(x) = x - 1$ then $h^{-1}(x) = (g \cdot f(x))^{-1}$
 $\therefore \frac{d}{dx}(h^{-1}(x)) = 3x^2 + 1 = f^{-1} \cdot g^{-1}(x)$ (reversal law)
 $= f^{-1}(x^3 + x^3 + x + 1)$
 $= x^3 + x$
 $(h^{-1}(1))' = 3 + 1 = 4$

51. 6

Sol. $|\alpha + \beta| \leq |\alpha| + |\beta|$
 Now,
 $i\alpha^3 + \alpha^2 - \alpha^2 - \alpha + i = 0$
 $\Rightarrow \alpha^2(i\alpha + 1) + i(i\alpha + 1) = 0$
 $\Rightarrow |\alpha| = 1$
 so $|\alpha + \beta| \leq 1 + 5 \leq 6$

52. 1

Sol. Given that $A^{-1}T = A = (A^T)^{-1} = AA^T = I$
 $\Rightarrow |A| = \pm 1$
 Now, $|\text{adj}(\text{adj}A)| = |A|^{2^2} = 1$

53. 4

Sol. $p \in \left[\frac{3}{2}, 2\right], q \in \left[\frac{1}{4}, 1\right]$
 $\Rightarrow \frac{7}{4} \leq p + q \leq 3$
 $\Rightarrow [(p+q)_{\max}] + [(p+q)_{\min}]$
 $= 3 + 1 = 4$

54. 2

Sol. $n_1 = 2(7 + 6 + 5)(7!) = 36(7!)$
 $n_2 = 16(7!)$
 $\left[\frac{n_1}{n_2}\right] = \left[\frac{36}{16}\right] = 2$

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – IV
PAPER –1
TEST DATE: 08-02-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

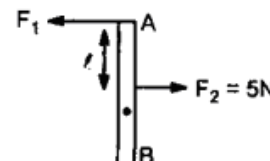
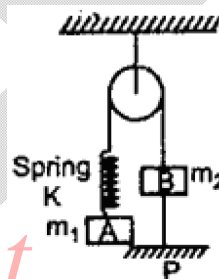
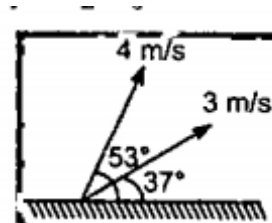
Physics

PART – I

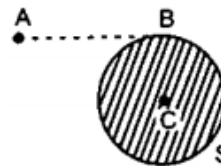
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

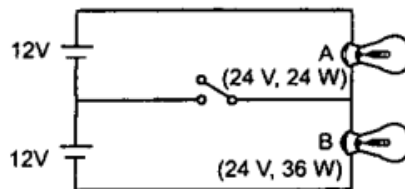
- Two particles are projected with speed 4 m/s and 3 m/s simultaneously from same point as shown in the figure. Then:
 (A) Their relative velocity is along vertical direction.
 (B) Their relative acceleration is non-zero and it is along vertical direction.
 (C) They will hit the surface simultaneously.
 (D) Their relative velocity is constant and has magnitude 1.4 m/s.
- In the system shown in figure $m_1 > m_2$. System is held at rest by thread BP. Just after the thread BP is burnt:
 (A) Magnitude of acceleration of both blocks will be equal to $\left(\frac{m_1 - m_2}{m_1 + 2m_2} \right) g$
 (B) Acceleration of m_1 will be equal to zero.
 (C) Acceleration of m_2 will be upwards
 (D) Magnitudes of acceleration of two blocks will be non-zero and unequal.
- A body of mass m was slowly hauled up the rough hill by a force which at each point was directed along tangent to the hill. Work done by the force
 (A) Is independent of shape of trajectory
 (B) Depends upon vertical component of displacement but independent of horizontal component.
 (C) Depends upon both the components of displacement.
 (D) Does not depend upon coefficient of friction.
- A thin uniform rod AB of mass 1 kg move translationally with acceleration $a = 2\text{m/s}^2$ due to two antiparallel force as shown. If $\ell = 20\text{ cm}$ then:
 (A) $F_1 = 3\text{ N}$
 (B) $F_1 = 5\text{ N}$
 (C) Length of rod is 1 m
 (D) Length of rod is 80 cm



5. S is solid neutral conducting sphere. A point charge q of 1×10^{-6} C is placed at point A. Point C is the centre of sphere and AB is a tangent. $BC = 3\text{m}$ and $AB = 4\text{m}$:
- (A) The electric potential of the conductor is 1.8 kV.
 (B) The electric potential of the conductor is 2.25 kV.
 (C) The electric potential at B due to induced charges on the sphere is -0.45 kV.
 (D) The electric potential at B due to induced charges on the sphere is 0.45 kV.



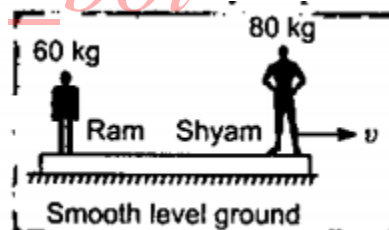
6. Two light bulbs shown in the circuit have ratings A(24 V, 24 W) and B(24 V and 36 W) as shown. When the switch is closed:
- (A) the intensity of light in bulb A increases.
 (B) the intensity of light in bulb A decreases.
 (C) the intensity of light in bulb B increases.
 (D) the intensity of light in bulb B decreases.



Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

7. Two men of mass 60 kg and 80 kg stand on a plank of mass 20 kg. Both of them can jump with a velocity of 1 m/s relative to the plank. In each event shown in List-I, find the velocity of plank after the event. Sign of right direction is (+ve) and (–ve) for left direction.

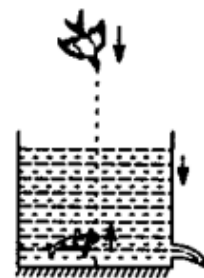


List-I		List-II	
(I)	Ram alone jumps to the left	(P)	$-\frac{17}{40}\text{m/s}$
(II)	Shyam alone jumps to the right	(Q)	$-\frac{1}{2}\text{m/s}$
(III)	Ram jumps to left and Shyam jumps to right simultaneously	(R)	$\frac{3}{8}\text{m/s}$
(IV)	Ram jumps to left and after that Shyam jumps to right.	(S)	$-\frac{1}{8}\text{m/s}$
		(T)	$\frac{1}{3}\text{m/s}$

The correct option is:

- (A) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ P
 (B) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ S
 (C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ P
 (D) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ Q

8. A bird in air is diving vertically over a tank with speed 6 cm/s. Base of tank is silvered. A fish in the tank is rising upward along the same line with speed 4 cm/s. Water level is falling at rate of 2 cm/s. (take: $\mu_{\text{water}} = 4/3$)

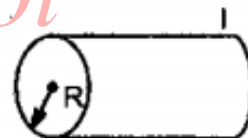


List-I		List-II	
(I)	Speed of the image of fish as seen by the bird directly (Apparent)	(P)	12
(II)	Speed of the image of fish formed after reflection from the mirror as seen by the bird.	(Q)	4
(III)	Speed of image of bird relative to the fish looking upwards	(R)	9
(IV)	Speed of image of bird relative to the fish looking downwards in the mirror	(S)	3
		(T)	5

The correct option is:

- (A) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ P
 (B) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ Q
 (C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ P
 (D) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ Q

9. A long straight conductor of radius R carries a current I as shown. The current density j varies as a function of radius according to $j = br$ where b is a constant. Match the quantities in List-I with those in List-II.



List-I		List-II	
(I)	B_{inside}	(P)	Maximum
(II)	B_{surface}	(Q)	$\frac{\mu_0 b r^2}{3}$
(III)	B_{outside}	(R)	$\frac{2\pi b r^3}{3}$
(IV)	I (total current inside wire)	(S)	$\frac{\mu_0 b R^3}{3r}$
		(T)	$\frac{\mu_0 b r^2}{3}$

The correct option is:

- (A) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ P
 (B) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ Q
 (C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S
 (D) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ R

10. List-I shows the cylindrical region of radius r where a downward magnetic field $\vec{B}$ exists, and $\vec{B}$ is increasing at the rate of $\frac{dB}{dt}$. A rod PQ is placed in different situation as shown. Match the list-I with the correct statement in list-II regarding the induced e.m.f. in rod.

List-I		List-II	
(I)		(P)	Induced e.m.f. in rod PQ is $\frac{1}{2}r^2\theta\frac{dB}{dt}$
(II)		(Q)	Induced e.m.f. in rod PQ is less than $\frac{1}{2}r^2\theta\frac{dB}{dt}$
(III)		(R)	End P is positive with respect to point Q
(IV)		(S)	End Q is positive with respect to point P
		(T)	End P and Q are at same potential

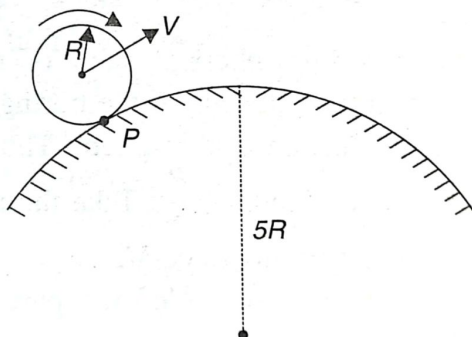
The correct option is:

- (A) I $\rightarrow$ RS, II $\rightarrow$ T, III $\rightarrow$ TP, IV $\rightarrow$ PS
 (B) I $\rightarrow$ RQ, II $\rightarrow$ S, III $\rightarrow$ PT, IV $\rightarrow$ QP
 (C) I $\rightarrow$ QS, II $\rightarrow$ PR, III $\rightarrow$ SP, IV $\rightarrow$ QR
 (D) I $\rightarrow$ QR, II $\rightarrow$ PS, III $\rightarrow$ RP, IV $\rightarrow$ QS

Section – B (Maximum Marks: 24)

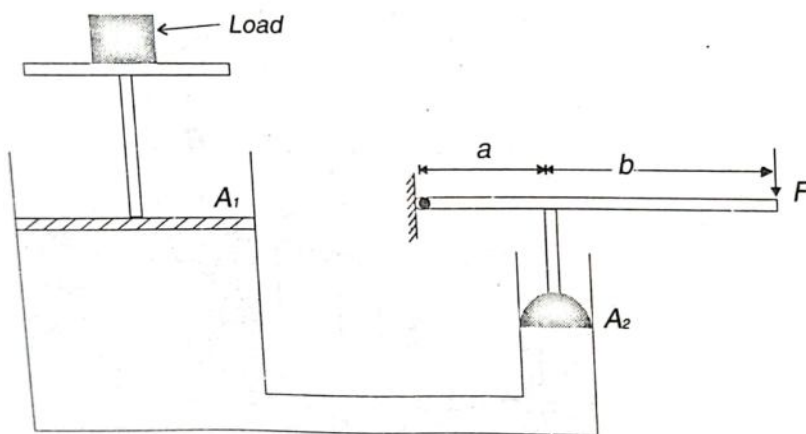
This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

11. A convex surface has a uniform radius of curvature equal to $5R$. A wheel of radius R is rolling without sliding on it with a constant speed V . Find the acceleration of the point (P) of the wheel which is in contact with the convex surface.

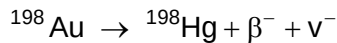


($V = 6$ meter/second, $R = 4$ meter)

12. A man can jump up to a height of $h_0 = 1$ m on the surface of the earth. What should be the radius of a spherical planet (in km) so that the man makes a jump on its surface and escapes out of its gravity? Assume that the man jumps with same speed as on earth and the density of planet is same as that of earth. Take escape speed on the surface of the earth to be 11.2 km/s and radius of earth to be 6400 km. ($g = 10$ m/sec²)
13. The figure shows a schematic layout of a hydraulic jack. The load to be raised weighs $20,000$ N. The area of cross section of the two pistons are $A_1 = 50$ cm² and $A_2 = 10$ cm². The force (F) is applied at the end of a light lever bar as shown in the figure. Lengths a and b are 4 cm and 36 cm respectively. Find the force (F) required to raise the load slowly.



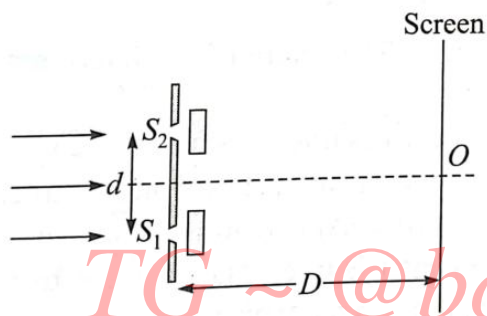
14. Consider the β -decay



where ^{198}Hg represents a mercury nucleus in an excited state of energy 1.088 MeV above the ground state. What can be maximum kinetic energy of the electron emitted (in kilo electron volt). The atomic mass of ^{198}Au is 197.968233 u and that of ^{198}Hg is 197.966760 u. $1 \text{ u} = 931 \text{ MeV}/c^2$.

15. Two circular aperture S_1 and S_2 of radii 1 mm and 2 mm. A parallel beam of light of intensity $\frac{7}{22} \text{ Wm}^{-2}$ and wavelength 600 nm is incident normally as shown in the figure.

A transparent film of thickness 200 nm having refractive index $\mu_1 = 2$ is placed in front of S_1 , and other transparent film of thickness 200 nm and refractive index 1.5 is placed in front of other slit. Find the resultant intensity (in μW) at the centre of screen.



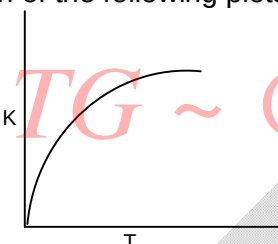
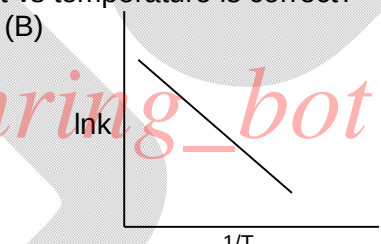
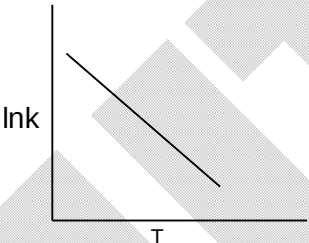
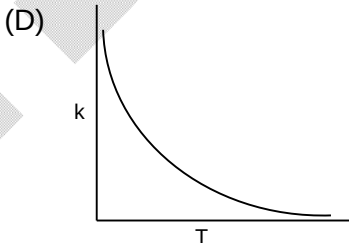
16. A vessel of volume 0.2 m^3 contains hydrogen gas at temperature 300 K and pressure 1 bar. Find the heat (in Kcal) required to raise the temperature to 400 K. The molar heat capacity of hydrogen at constant volume is 5 cal/mol K. ($R = 8.31$)
17. Steel wire of length 'L' at 40°C is suspended from the ceiling and then a mass 'm' is hung from its free end. The wire is cooled down from 40°C to 30°C to regain its original length 'L'. The coefficient of linear thermal expansion of the steel is $10^{-5} / ^\circ\text{C}$. Young's modulus of steel is 10^{11} N/m^2 and radius of the wire is 1 mm. Assume that $L \gg$ diameter of the wire. Then the value of 'm' in kg is nearly. ($g = 10 \text{ m/sec}^2$)
18. A tuning fork of frequency 200 Hz is in unison with a sonometer wire. How many beats are heard in 30 s if the tension is increased by 1%.

Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

19. Silicate having one monovalent corner oxygen atom in each tetrahedron unit is
 (A) Sheet silicate (B) Cyclic silicate
 (C) Single chain silicate (D) Double chain silicate
20. Given $\Delta H_f^\circ (\text{C}_2\text{H}_6, \text{g}) = -85 \text{ kJ / mole}$,
 $\Delta H_f^\circ (\text{C}_3\text{H}_8, \text{g}) = -104 \text{ kJ / mole}$,
 $\Delta H_{\text{sub}}^\circ (\text{C}, \text{s}) = 718 \text{ kJ / mole}$ & B.E. of $(\text{H} - \text{H}) = 436 \text{ kJ/mole}$.
 Then, in kJ/mole, the
 (A) C – C bond enthalpy is 218 (B) C – H bond enthalpy is 414
 (C) C – C bond enthalpy is 345 (D) C – H bond enthalpy is 448
21. Which of the following plots of rate constant vs temperature is correct?
 (A)  (B) 
 (C)  (D) 
22. The correct statement regarding Ytterbium is/are
 (A) it contains no unpaired electrons
 (B) Yb^{2+} contains a full-filled electronic configuration
 (C) it produces H_2 gas when reacts with HCl
 (D) radius of Yb^{3+} is larger than Yb^{2+} ion
23. Which of the following carbide(s) form CH_4 when react(s) with water?
 (A) Be_2C (B) CaC_2
 (C) Mg_2C_3 (D) Al_4C_3

24. Which of the following statement(s) is/are false?
- (A) All adiabatic processes are isentropic (or isentropic) process
- (B) The heat of vaporization of water at 100°C is 40.6 kJ/mole . When 9g of water vapour condenses to liquid at 100°C of 1 atm , then $\Delta S_{\text{system}} = 54.42 \text{ J/K}$
- (C) When $(\Delta G_{\text{system}})_{T,P} < 0$; the reaction must be exothermic
- (D) $dG = Vdp - S dT$ is applicable for closed system, both PV and non – PV work

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. Answer the following by appropriately matching the lists

List – I		List – II	
(I)		(P)	Nucleophilic substitution
(II)		(Q)	Elimination
(III)		(R)	Nucleophilic addition
(IV)		(S)	Esterification with acetic anhydride
		(T)	Dehydrogenation

Which of the following is the correct combination?

- (A) I → PQT, II → PST, III → RS, IV → P (B) I → PST, II → PQT, III → PRS, IV → PS
- (C) I → QT, II → QR, III → T, IV → S (D) I → SU, II → PQR, III → ST, IV → ST

26. Answer the following by appropriately matching the lists

List - I		List - II	
(I)	Work done in isobaric process	(P)	$-P_{\text{ext}} \times \Delta V$
(II)	Work done during free expansion	(Q)	$nRT \ln \frac{P_1}{P_2}$
(III)	Work done in reversible adiabatic process	(R)	$nRT \ln \frac{V_1}{V_2}$
(IV)	Work done in adiabatic irreversible process	(S)	$-nR(T_{\text{final}} - T_{\text{initial}})$
		(T)	Zero

Which of the following is the correct combination?

- (A) I $\rightarrow$ PS, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ PS
 (B) I $\rightarrow$ T, II $\rightarrow$ PQ, III $\rightarrow$ P, IV $\rightarrow$ Q
 (C) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ PS, IV $\rightarrow$ QR
 (D) I $\rightarrow$ P, II $\rightarrow$ QT, III $\rightarrow$ QR, IV $\rightarrow$ ST

27. Match the following list-I with list-II.

List - I (Electronic configuration)		List - II (Corresponding elements)	
(I)	$[\text{Xe}]4f^{14}5d^{10}6s^2$	(P)	s-block elements
(II)	$[\text{Rn}]5f^{14}6d^{17}s^2$	(Q)	Transition elements
(III)	$[\text{Xe}]4f^{14}5d^{10}6s^26p^67s^2$	(R)	d-block elements
(IV)	$[\text{Xe}]4f^{14}5d^26s^2$	(S)	Representative elements
		(T)	Inner transition elements

- (A) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ QS, IV $\rightarrow$ PR
 (B) I $\rightarrow$ RS, II $\rightarrow$ RS, III $\rightarrow$ P, IV $\rightarrow$ Q
 (C) I $\rightarrow$ R, II $\rightarrow$ T, III $\rightarrow$ PS, IV $\rightarrow$ QR
 (D) I $\rightarrow$ P, II $\rightarrow$ QT, III $\rightarrow$ QR, IV $\rightarrow$ ST

28. Answer the following by appropriately matching the lists

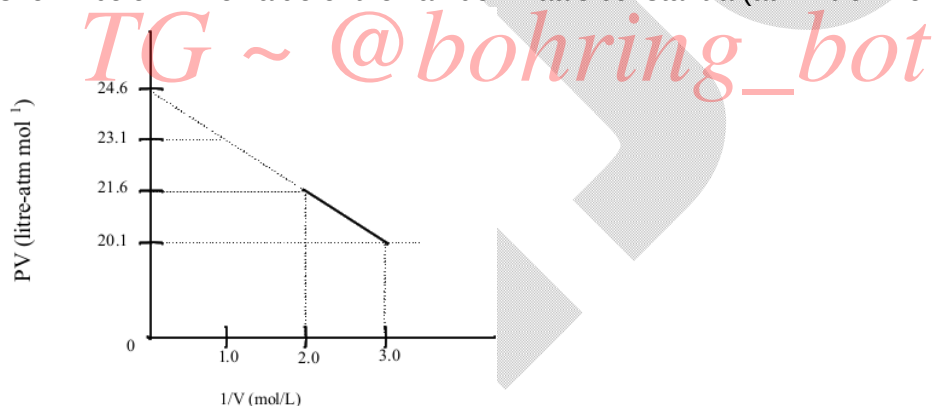
List - I		List - II	
(I)	Gas evolved by the action of dilute H_2SO_4 on a sulphite	(P)	Colour of acidified KMnO_4 is discharged
(II)	Gas evolved by the action of dilute H_2SO_4 carbonate	(Q)	Acidified dichromate solution is turned green
(III)	Gas evolved by heating an ammonium salt with NaOH	(R)	Nessler's reagent gives a brown precipitate
(IV)	Gas evolved by the action of dilute H_2SO_4 on a sulphide	(S)	Bartya water turns milky
		(T)	Alkaline nitroprusside turns violet

- (A) I $\rightarrow$ PS, II $\rightarrow$ ST, III $\rightarrow$ RS, IV $\rightarrow$ PQ
 (B) I $\rightarrow$ PQS, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ PQT
 (C) I $\rightarrow$ PQ, II $\rightarrow$ RS, III $\rightarrow$ PQ, IV $\rightarrow$ PQT
 (D) I $\rightarrow$ ST, II $\rightarrow$ QRS, III $\rightarrow$ QR, IV $\rightarrow$ ST

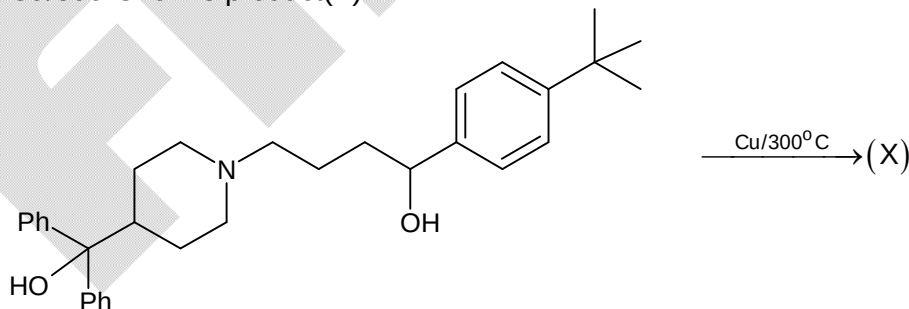
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

29. A sample of camphor used in a method for determining of molar mass had melting point of 450 K. The melting point of a solution containing 0.5 gm camphor and 0.04 gm of an unknown non-volatile non-electrolyte compound was 430 K. The molar mass of camphor is 152 and its $\Delta H_{\text{fusion}} = 1.52 \text{ Kcal mol}^{-1}$. The molar mass of unknown compound in gm/mole is.....
($R = 2 \text{ cal mol}^{-1} \text{ K}^{-1}$).
30. What volume of 0.001 M HCl solution be added into 100 mL of 0.001 NaOH to change the pH by one unit?
31. The molar conductance at infinite dilution of CaCl_2 , $\text{Ca}(\text{CH}_3\text{COO})_2$ and HCl are respectively 271.6, 200.8 and $425.95 \text{ S cm}^2 \text{ mol}^{-1}$. What will be the molar conductance of CH_3COOH in $\text{S cm}^2 \text{ mol}^{-1}$ unit?
32. For one mole of a vander Waals gas when $b = 0$ and $T = 300 \text{ K}$, the PV vs $1/V$ plot is shown below. The value of the van der Waals constant a ($\text{atm. litre}^2 \text{ mol}^{-2}$) is



33. Seldane contains a chemical named terfenadine. Reaction of this chemical with $\text{Cu}/300^\circ\text{C}$ forms product(X).



How many pi-bond(s) is/are present in (X)?

34. How much gram of Fe^{2+} ions can be completely oxidized into Fe^{3+} ions by 104.3 litre of acidified KMnO_4 solution of 0.1 M concentration, according to the following reaction.
- $$\text{MnO}_4^- + \text{Fe}^{2+} + \text{H}^+ \longrightarrow \text{Mn}^{2+} + \text{Fe}^{3+} + \text{H}_2\text{O}$$
35. What is the percentage of ferric ion in the non-stoichiometric solid having formula $\text{Fe}_{0.96}\text{O}$.
36. Aluminium crystallises in a cubic system with unit cell edge length equal to $4.0 \text{ \AA}$. Its density is $\frac{45}{16} \text{ g/cm}^3$, then the atomic radius of Al atom in $\text{\AA}$ is

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Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. Let V be the volume of the parallelepiped formed by the vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$, If a_r, b_r, c_r where $r = 1, 2, 3$ are non-negative real numbers and $\sum_{r=1}^3 (a_r + b_r + c_r) = 3L$, then which of the following may hold
- (A) $L^3 > V$ (B) $L^3 < V$
 (C) $L^3 = V$ (D) Nothing can be said
38. The real parameters m and n are such that the graph of the function $f(x) = \sqrt[3]{8x^3 + mx^2 - nx}$ has the horizontal asymptote $y = 1$ then
- (A) $|m + n| = 10$ (B) $|m + n| = 14$
 (C) $|m - n| = 10$ (D) $|m - n| = 2$
39. If $\int \frac{x^4 + 1}{x^6 + 1} dx = \alpha \arctan \frac{P(x)}{Q(x)} + C$, $\alpha \in \mathbb{Q}$,
- (A) $\alpha = \frac{1}{3}$ (B) $P(1) = 0$
 (C) $P(-1) = 0$ (D) $Q(1) = -4$
40. Let a and b be positive real numbers and $f(a, b) = \int_a^b \frac{e^{\frac{x}{a}} - e^{\frac{b}{x}}}{x} dx$, then
- (A) $f(1, 2) = 0$ (B) $f(3, 1) = 2$
 (C) $f(2, 1) = 0$ (D) $f(1, 3) = 2$
41.
$$\frac{1}{\cot 9^\circ - 3 \tan 9^\circ} + \frac{3}{\cot 27^\circ - 3 \tan 27^\circ} + \frac{9}{\cot 81^\circ - 3 \tan 81^\circ} + \frac{27}{\cot 243^\circ - 3 \tan 243^\circ} =$$
- (A) $10 \tan 9^\circ$ (B) $10 \cot 81^\circ$
 (C) $10 \cot 9^\circ$ (D) $10 \tan 81^\circ$

42. Point M moved on the circle $(x-4)^2 + (y-8)^2 = 20$. Then it broke away from it and moving along a tangent to the circle, cuts the x -axis at the point $(-2, 0)$. The coordinates of a point on the circle at which the moving point broke away is:

- (A) $\left(-\frac{3}{5}, \frac{46}{5}\right)$ (B) $\left(-\frac{2}{5}, \frac{44}{5}\right)$
 (C) $(6, 4)$ (D) $(3, 5)$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04) Matching List Type Questions**. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Match the following list-I with list-II.

List- I		List- II	
(I)	Let $f(x) = \begin{cases} x^{3/5} & \text{if } x \leq 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$ then the number of critical points on the graph of the function is	(P)	5
(II)	Number of real solution of the equation, $\log_2^2 x + (x-1)\log_2 x = 6 - 2x$, is	(Q)	4
(III)	The number of values of c such that the straight line $3x + 4y = c$ touches the curve $\frac{x^4}{2} = x + y$ is	(R)	3
(IV)	If $f(x) = \int_x^{x^2} (t-1)dt$, $1 \leq x \leq 2$, then global maximum value of $f(x)$ is	(S)	2
		(T)	1

Which is correct option?

- (A) $I \rightarrow Q$ (B) $II \rightarrow S$
 (C) $III \rightarrow P$ (D) $IV \rightarrow T$

44. Match the following list-I with list-II.

List- I		List- II	
(I)	If $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{1-\cos x}}$ is L then $\ln(L)$ equals k. The value of $\frac{1}{k^2}$ is equal to	(P)	0
(II)	The value of $\lim_{n \rightarrow \infty} \left(\frac{n^2}{n-1} \right)^{\tan \frac{1}{\sqrt{n}}}$ is	(Q)	1
(III)	Number of solution of the equation $8 \sin^4 x + 8 \cos^4 x = 5$ in the interval $0 < x < 2\pi$ is	(R)	3
(IV)	Let $f(x)$ be a non – constant polynomial function and $g(x) = x(x-1)(x-2)f(x) $. If $g(x)$ is differentiable $\forall x \in \mathbb{R}$, then minimum number of distinct roots of $f(x) = 0$ is	(S)	8
		(T)	9

Which is correct option?

(A) I $\rightarrow$ Q

(B) II $\rightarrow$ S

(C) III $\rightarrow$ P

(D) IV $\rightarrow$ R

45. The number of 10 letter permutations comprising 4a's, 3b's and 3c's such that

List- I		List- II	
(I)	a' s are separated and all b' s are together is equal to	(P)	18
(II)	a' s are separated and exactly two b' s are together is equal to	(Q)	20
(III)	no two adjacent letters are identical is equal to	(R)	150
(IV)	no two b's are together, no two c' s are together and all 'a' s are	(S)	180
		(T)	248

Which is correct option?

(A) I $\rightarrow$ Q

(B) II $\rightarrow$ T

(C) III $\rightarrow$ P

(D) IV $\rightarrow$ Q

46. In a tournament, there are twelve players $S_1, S_2, \dots, S_{12}$ and divided into six pairs at random. From each game a winner is decided on the basis of a game played between the two players of the pair. Assuming all the players are of equal strength, then

List- I		List- II	
(I)	Probability that S_2 is among the losers is	(P)	$\frac{5}{22}$
(II)	Probability that exactly one of S_3 and S_4 is among the losers is	(Q)	$\frac{1}{2}$
(III)	Probability that both S_2 and S_4 are among the winners is	(R)	$\frac{6}{11}$
(IV)	Probability that S_4 and S_5 not playing against each other is	(S)	$\frac{10}{11}$
		(T)	$\frac{3}{11}$

Which is correct option?

(A) I $\rightarrow$ P

(B) II $\rightarrow$ Q

(C) III $\rightarrow$ R

(D) IV $\rightarrow$ S

Section - B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. Let N be the integer next above $(\sqrt{3} + 1)^{2018}$. The greatest integer 'p' such that $(16)^p$ divides N is equal to
48. The coefficient of $x^{\frac{n^2+n-14}{2}}$ in $(x-1)(x^2-2)(x^3-3)(x^4-4)\dots(x^n-n), n \geq 30$ is equal to
49. If α, β, γ are such that $\alpha + \beta + \gamma = 2, \alpha^2 + \beta^2 + \gamma^2 = 6, \alpha^3 + \beta^3 + \gamma^3 = 8$, then $\alpha^4 + \beta^4 + \gamma^4$ is
50. Find the global minimum of the function
 $f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = x^4 + 6x^2y^2 + y^4 - \frac{9}{4}x - \frac{7}{4}y$
51. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that f' is continuous on $\mathbb{R}$ with $f'(3) = 18$. Define $g_n(x) = n \left\{ f\left(x + \frac{5}{n}\right) - f\left(x - \frac{2}{n}\right) \right\}$. Then $\lim_{n \rightarrow \infty} g_n(3)$ equals _____

52. Find the area of the region bounded by the curves $y = x^2$, $y = |2 - x^2|$ and $y = 2$, which lies to the right of the line $x = 1$.
53. Find the maximum value of $S = (1 - x_1)(1 - y_1) + (1 - x_2)(1 - y_2)$ if $x_1^2 + x_2^2 = y_1^2 + y_2^2 = 8$
54. Let $[x] =$ greatest integer less than or equal to x . If all the values of x such that the product $\left[x - \frac{1}{2}\right]\left[x + \frac{1}{2}\right]$ is prime, belongs to the set $[x_1, x_2) \cup [x_2, x_4)$, find the value of $x_1^2 + x_2^2 + x_3^2 + x_4^2$

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FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – IV
PAPER –2
TEST DATE: 08-02-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

Physics

PART – I

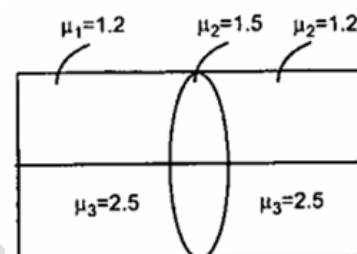
Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. A thin lens of refractive index 1.5, and focal length in air 20 cm is placed inside, a large container containing two immiscible liquids as shown. If an object is placed at an infinite distance close to principle axis (common intersection line of the layers), the distance between images

(A) 25 cm
(C) 65 cm

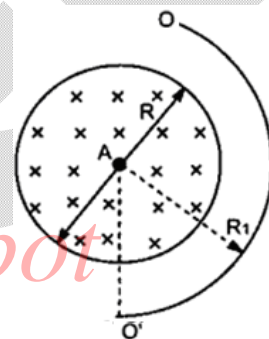
(B) 45 cm
(D) 85 cm



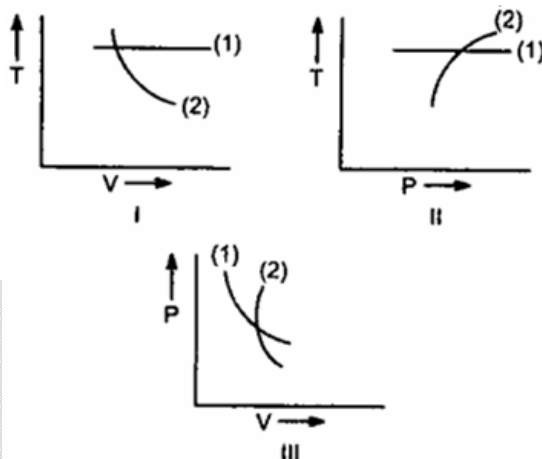
2. There is a uniform magnetic field B in circular region of radius R as shown in figure, whose magnitude changes at the rate of dB/dt . The e.m.f. induced across the ends of a circular concentric conducting arc of radius R_1 having an angle θ as shown ($\angle OAO'$) is:

(A) $\frac{\theta}{2\pi} R_1^2 \frac{dB}{dt}$
(C) $\frac{\theta}{2\pi} R^2 \frac{dB}{dt}$

(B) $\frac{\theta}{2} R^2 \frac{dB}{dt}$
(D) none of these



- 3.



If (I) represents isothermal and (II) represents adiabatic, which of the graphs given above in respect of an ideal gas are correct?

(A) I and II
(C) I and III

(B) II and III
(D) I, II and III

4. A travelling wave is given by

$$y = \frac{0.8}{3x^2 + 12xt + 12t^2 + 4}$$

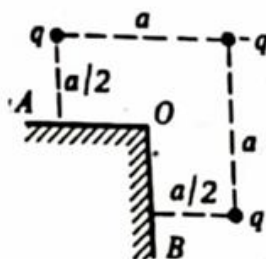
Where x and y are in m and t is in sec, then velocity and amplitude of wave will be

- (A) 2 m/s, 0.2 m (B) 4 m/s, 0.2 m
(C) 2 m/s, 0.4 m (D) none of these

Section – A (Maximum Marks: 24)

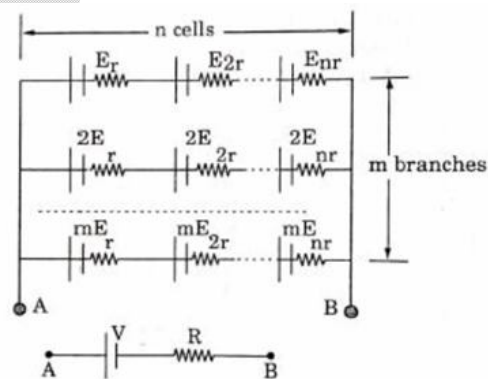
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. Three unlike point charges are arranged as shown in given figure, where AOB is the right angle formed by two conducting half-planes. The magnitude of each of the charges is $|q|$ and the distances between them are shown in the figure. Total charge induced on the conducting half-planes and the force acting on the charge $-q$ is



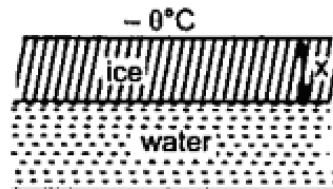
- (A) charge = $-q$
(B) charge = $-3q$
(C) Force = $\left(\frac{2\sqrt{2}-1}{4\pi\epsilon_0}\right) \frac{q^2}{2a^2}$
(D) Force = $\left(\frac{4\sqrt{2}+1}{4\pi\epsilon_0}\right) \frac{q^2}{2a^2}$

6. Consider an arrangement of cells shown in the figure. There are m -branches and in each branch there are n -cells. Suppose this combination of cells can be replaced by a single battery of emf V and internal resistance R , then V and R .



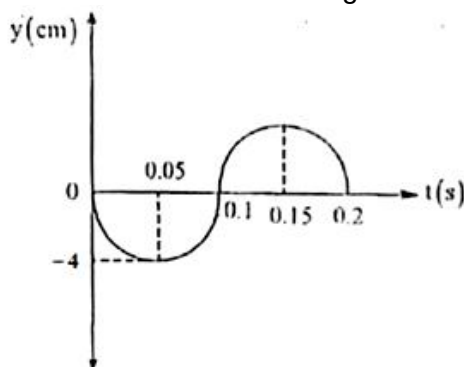
- (A) $V = \frac{n(m+1)E}{2}$
(B) $V = \frac{m(n+1)E}{2}$
(C) $R = \frac{n(n+1)r}{2m}$
(D) $R = \frac{m(m+1)r}{2n}$

7. Consider the shown case of a freezing lake due to negative environmental temperature ($- \theta^\circ\text{C}$). Thickness (x) of ice layer is small in comparison to depth of lake. Rate of increase in x will be greater:



- (A) if environmental temperature increases.
(B) for larger thickness of ice layer.
(C) if environmental temperature decreases.
(D) for smaller thickness of ice layer.

8. For a certain transverse standing wave on a long string, an antinode is formed at $x = 0$ and next to it, a node is formed at $x = 0.10$ meter, the displacement $y(t)$ of the string particle at $x = 0$ is shown in figure.



- (A) Transverse displacement of the particle at $x = 0.05$ m and $t = 0.05$ sec is $-2\sqrt{2}$ cm
 (B) Transverse displacement of the particle at $x = 0.04$ m and $t = 0.025$ sec is $-2\sqrt{2}$ cm
 (C) Speed of travelling wave that interfere to produce this standing wave is 2 m/s.
 (D) The transverse velocity of the string particle at $x = \frac{1}{15}$ m and $t = 0.1$ sec is 20π cm/sec.
9. A particle performing simple harmonic motion undergoes initial displacement of $A/2$ (where A is the amplitude of simple harmonic motion) in 1s. At $t = 0$, the particle may be at the extreme position or mean position. The time period of the simple harmonic motion can be
 (A) 6 s (B) 2.4 s
 (C) 12 s (D) 1.2 s
10. A sky lab of mass m is first launched from the surface of earth (radius R) in a circular orbit of radius $2R$ (from centre of earth) and then it is shifted from this circular orbit to another circular orbit of radius $3R$.
 (A) Minimum energy required to place the lab in first orbit is equal to $\frac{3}{4}mgR$.
 (B) Minimum energy required to shift lab from first orbit to second orbit is $\frac{mgR}{12}$.
 (C) Energy of sky lab in second orbit is greater than its energy in first orbit.
 (D) Energy of sky lab in second orbit is less than its energy in first orbit.

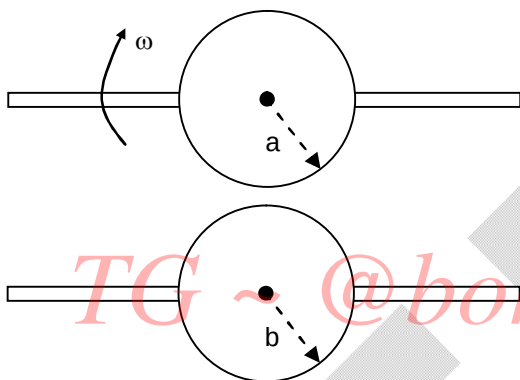
Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.*

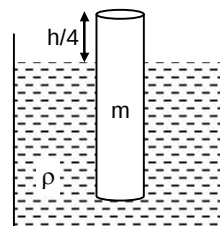
11. A glass of refractive index 1.5 is coated with a thin layer of thickness t , of refractive index 1.8. Light of wavelength λ travelling in air is incident normally on the layer. It is partly reflected at the upper and at the lower surfaces of the layer. The two reflected rays interfere. If $\lambda = 648$ nm, obtain the least value of t (in 10^{-8} m) for which the rays interfere constructively.

12. A Bohr hydrogen atom undergoes a transition $n = 5$ to $n = 4$ and emits a photon of frequency f . Frequency of circular motion of electron in $n = 4$ orbit is f_4 . The ratio $\frac{f}{f_4}$ is found to be $\frac{18}{5m}$. State the value of m .

13. Two uniform spheres shown in given figure rotate separately on two parallel horizontal axis passing through their centres. The upper sphere has an angular speed ω_0 and the lower sphere is at rest. Now the two spheres are moved together so that their surfaces touch. After a short time the two spheres, now in contact, are rotating without slipping. Find the final rate of rotation of upper sphere in rad/s [given moment of inertia of the upper sphere $I_1 = 4 \text{ kg m}^2$, moment of inertia of lower sphere $I_2 = 2 \text{ kg m}^2$, radius of upper sphere $a = 2 \text{ m}$, radius of lower sphere $b = 1 \text{ m}$, $\omega_0 = 12 \text{ rad/s}$].

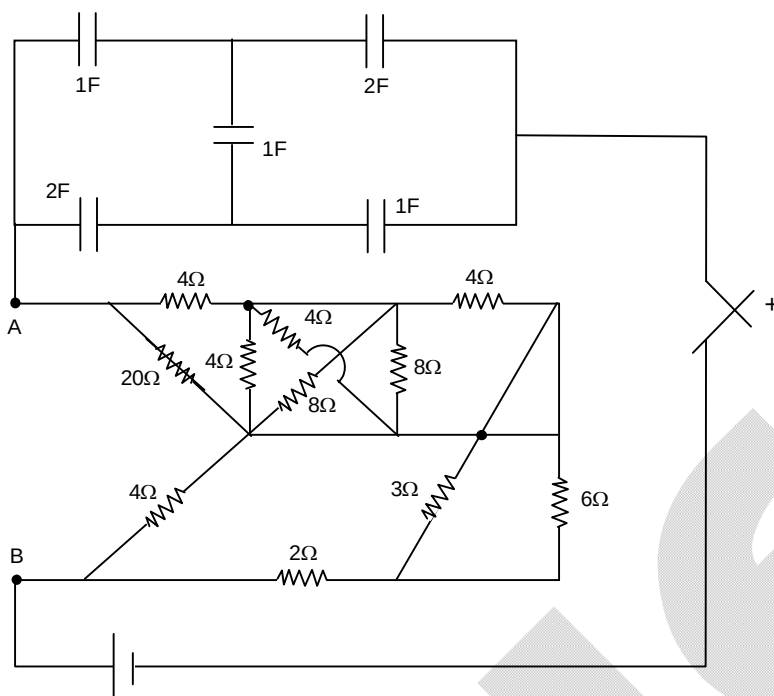


14. A solid cylinder of height h and mass m is floating in a liquid of density ρ as shown in the figure. Find the acceleration of the vessel (in m/s^2) containing liquid for which the relative downward acceleration of the completely immersed cylinder w.r.t. vessel becomes equal to one-third of that of the vessel acceleration in magnitude. (Take $g = 10 \text{ m/s}^2$)

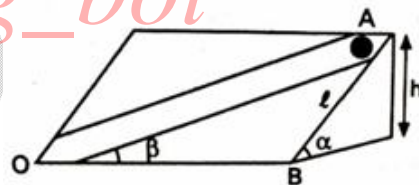


15. A barometer contains two uniform capillaries of radii $1.44 \times 10^{-3} \text{ m}$ and $7.2 \times 10^{-4} \text{ m}$. If the height of liquid in narrow tube is 0.2 m more than that in wide tube. The true pressure difference is $310 \times N$. Then the value of ' N ' is _____. Density of liquid = 10^3 kg/m^3 , surface tension = $72 \times 10^{-3} \text{ N/m}$ and $g = 9.8 \text{ m/s}^2$. (Angle of contact = 0)
16. Two electrode each of radius $a = \frac{1}{\pi}$ are embedded a large distance apart in a homogeneous conducting medium of resistivity $\rho = 1$. Resistance between the electrodes is $\frac{4}{n}$. Then the value of ' n ' is _____.

17. Time constant in given network. $\tau = 2.1 \times n$. Then the value of 'n' is _____.



18. Plane of height $h = 1.6$ meter and inclination $\alpha = 30^\circ$ to the horizontal has a smooth groove cut in it inclined at an angle $\beta = 45^\circ$ to the line of greatest slope as shown in figure. Time that a particle would take to move top point A to bottom point O in groove, starting from rest at the top is $\frac{3.2}{k}$. Then value of 'k' is ($g = 10 \text{ m/sec}^2$)



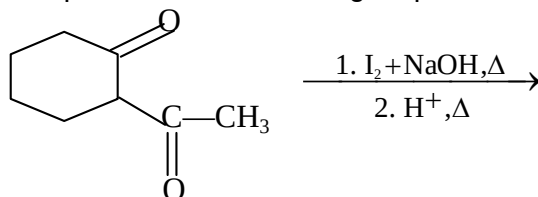
Chemistry

PART – II

Section – A (Maximum Marks: 12)

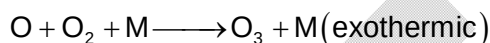
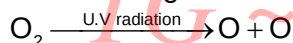
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

19. End product of the following sequence of reaction is



- (A) Yellow ppt. of CHI_3 ,
- (B) Yellow ppt. of CHI_3 ,
- (C) Yellow ppt. of CHI_3 ,
- (D) Yellow ppt. of CHI_3 ,

20. The following reactions takes place in stratosphere



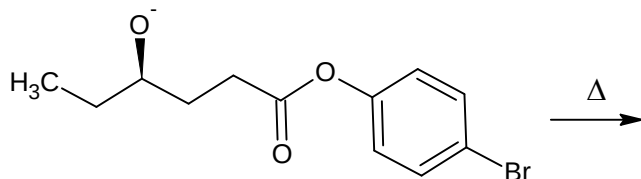
What is the role of 'M' in above reaction?

- (A) It is used as a catalyst
 (B) It absorbs some heat energy and prevents decomposition of O_3
 (C) It converts ground state O_2 to excited O_2^*
 (D) It reduces the reactivity of nascent oxygen
21. The temperature of a definite amount of an ideal monoatomic gas becomes four times in an isentropic process. Which of the following is correct relation between the final and initial parameters of gas?
- (A) $V_f = 8V_i$ (B) $V_f = 16V_i$
 (C) $P_f = 32P_i$ (D) $P_f = P_i$
22. 0.1 M acetic acid solution is titrated against 0.1 M NaOH solution. What would be the difference in pH between $\frac{1}{4}$ and $\frac{3}{4}$ stages of neutralization of acid.
- (A) $2 \log \frac{3}{4}$ (B) $2 \log \frac{1}{4}$
 (C) $\log \frac{1}{3}$ (D) $2 \log 3$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

23.



Which of the statements regarding the product formed in the above reaction is correct?

- (A) If Br is replaced with NO_2 rate of reaction increases
- (B) Inversion of configuration takes place.
- (C) Nucleophilic acyl substitution takes place.
- (D) Configuration at chiral carbon does not change

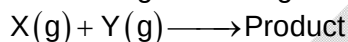
24.

Which of the following statements is/are true about an azeotropic mixture?

- (A) An azeotropic mixture boils at constant temperature
- (B) The composition of an azeotropic mixture changes on distillation
- (C) An azeotropic solution of two liquids has a boiling point lower than that of either of them when it shows positive deviation from the Raoult's law
- (D) An azeotropic solution of two liquids has a boiling point higher than that of either of them if it shows positive deviation from Raoult's law

25.

The following data are given for the reaction



No. of expt.	Initial concentration of 'X' in 'M'	Initial concentration of 'Y' in 'M'	Rate of reaction in M s^{-1} unit
1.	0.1	0.1	2×10^{-8}
2.	0.2	0.2	4×10^{-8}
3.	0.4	0.2	8×10^{-8}

Choose the correct statement(s) from the following

- (A) The order of reaction with respect to X is one
- (B) The overall order of reaction is one
- (C) The rate constant of the reaction is $2 \times 10^{-7} \text{ s}^{-1}$
- (D) The half-life of 'Y' does not depend on its concentration

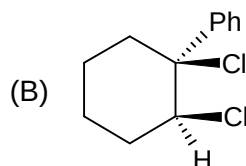
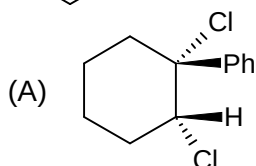
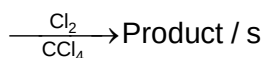
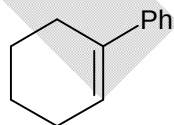
26.

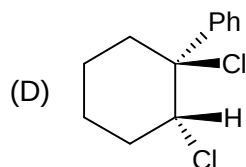
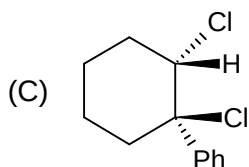
Which of the following substance(s) produce(s) gases when treated with water?

- (A) CaC_2
- (B) Mg_3N_2
- (C) Na_2S
- (D) KNO_3

27.

What is/are the product of reaction?

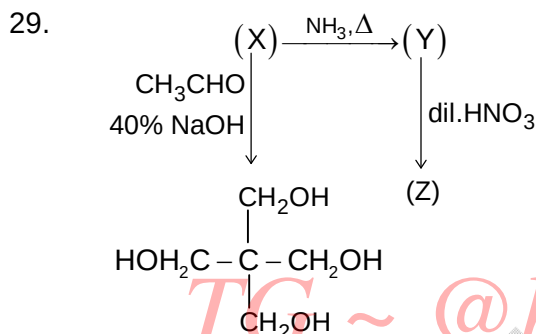




28. Which of the following complex(es) show(s) geometrical isomerism?
 (A) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (B) $[\text{Pt}(\text{NH}_2\text{CH}_2\text{COO})_2]$
 (C) $[\text{PtCl}_2\text{P}(\text{C}_2\text{H}_5)_3]_2$ (D) $\text{K}_2[\text{Fe}(\text{CN})_4(\text{NH}_3)_2]$

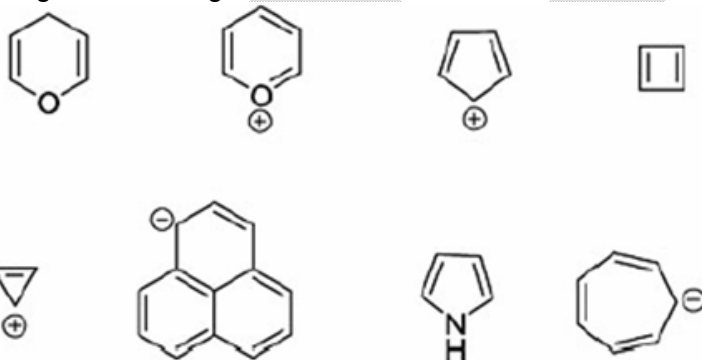
Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.



Total how many sp^2 hybridised N are present in (Z)

30. Among the following



The total number of aromatic species is

31. The Mo–Mo bond order in $[(\eta^5\text{-C}_5\text{H}_5)\text{Mo}(\text{CO})_2]_2$ which obeys the 18-electron rule is ____.

32. $\text{X(s)} \xrightarrow{\text{HCl}} \text{Y} \uparrow \xrightarrow[\text{NaOH}]{\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]} \text{Purple solution gas}$

Y has been allowed to react with following species in neutral/acidic medium

(a) FeCl_3 (b) CuSO_4 (c) BaCl_2 (d) SO_2 (e) $\text{Cr}_2\text{O}_7^{2-}$ (f) CH_3COONa (g) Hg^{2+}

The calculate the value of $\text{P} + \text{Q} - \text{R}$

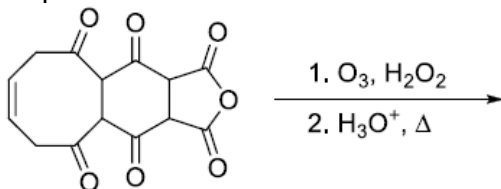
P : Number of species which undergoes redox reaction with gas Y

Q: Number of species with which gas Y undergoes precipitation

R: Number of species with which gas Y produce no observable change

33. On heating a sample of 25 mg hydrated compound (molecular weight = 250 g/mol) in thermogravimetric analysis, 16 mg of dehydrated compound remains. The number of water molecules lost per molecule of hydrated compound is _____
(Molecular weight of water = 18 g/mol)

34. Number of carbonyl groups present in the final product of the following reaction sequence is



35. A tetrapeptide, made up of natural amino acids, has alanine as the N-terminal residue which is coupled to a chiral amino acid. Upon complete hydrolysis, the tetrapeptide gives glycine, alanine, phenylalanine and leucine. The number of possible sequences of the tetrapeptide is
36. The number of possible stereoisomers for cyclononene is _____

TG ~ @bohring_bot

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. Let A and B be 2×2 matrices with real entries satisfying $(AB - BA)^n = I_2$ for some positive integer n, then n is
 (A) even (B) odd
 (C) prime (D) any integer
38. Let A & B be two $n \times n$ matrices that do not commute and for which there exist non zero real numbers p, q, r such that $pAB + qBA = I_n$ and $A^2 = rB^2$ then
 (A) $p + q = 0$ (B) $p - q = 0$
 (C) $p^2 + q = 0$ (D) $p = 2q$
39. The solution of the differential equation $\frac{d^2y}{dx^2} = \sin 3x + e^x + x^2$ when $y_1(0) = 1$ and $y(0) = 0$ is
 (A) $-\frac{\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$ (B) $-\frac{\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x$
 (C) $-\frac{\cos 3x}{3} + e^x + \frac{x^4}{12} + \frac{1}{3}x + 1$ (D) none of these
40. Consider a circle of diameter AB and center O, and the tangent t at B. A variable tangent to the circle with contact point M intersects t at P. The locus of the point Q where the line OM intersects the parallel through P to the line AB is
 (A) straight line (B) parabola
 (C) ellipse (D) hyperbola

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. Variable circle is described to pass through point (1, 0) and tangent to the curve $y = \tan(\tan^{-1} x)$. The locus of the centre of the circle is a parabola whose:
 (A) length of the latus rectum is $2\sqrt{2}$
 (B) axis of symmetry has the equation $x + y = 1$
 (C) vertex has the co – ordinates $\left(\frac{3}{4}, \frac{1}{4}\right)$
 (D) none of these

42. Line through $P(a, 2)$ meets the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D and meets the coordinate axes at B and C so that PA, PB, PC, PD are in G.P., then possible values of a can be:
 (A) 5 (B) 8
 (C) 10 (D) -7
43. Let α be a real number, then the integral, $(n \in \mathbb{I})$

$$I(\alpha) = \int_{-1}^1 \frac{\sin \alpha \, dx}{1 - 2x \cos \alpha + x^2} =$$

 (A) $\frac{\pi}{2}$ if $\alpha \in [2n\pi, (2n+1)\pi]$ (B) $\frac{\pi}{2}$ if $\alpha \in ((2n+1)\pi, (2n+2)\pi)$
 (C) $-\frac{\pi}{2}$ if $\alpha \in ((2n+1)\pi, (2n+2)\pi)$ (D) $\frac{\pi}{2}$ if $\alpha \in (2n\pi, (2n+1)\pi)$
44. Let $u = ax + by + a\sqrt[3]{b} = 0$, $v = bx - ay + b\sqrt[3]{a} = 0$, where $a, b \in \mathbb{R}$ be two straight lines. The equations of the bisectors of the angles formed by $k_1u - k_2v = 0$ and $k_1u + k_2v = 0$ for non zero real k_1 and k_2 are:
 (A) $u = 0$ (B) $k_2u + k_1v = 0$
 (C) $k_2u - k_1v = 0$ (D) $v = 0$
45. If $f: \mathbb{R} \rightarrow \mathbb{R}$, such that $f(x) = \frac{(x^2 - x + 1)^3}{x^6 - x^3 + 1}$, then
 (A) minimum value of $f(x)$ is 1 (B) minimum value of $f(x)$ is $\frac{\sqrt{3}}{1+\sqrt{3}}$
 (C) maximum value of $f(x)$ is 1 (D) maximum value of $f(x)$ is 9
46. Let z, iz and $z + iz$ represent three vertices of a triangle then,
 (A) area of Δ is given by, Area = $\frac{1}{2}|z|^2$ (B) area of Δ is given by, Area = $|z|^2$
 (C) orthocentre of Δ is given by $z + iz$ (D) None of these

Section – B (Maximum Marks: 24)

This section contains EIGHT (08) numerical based questions. The answer to each question is a Single Digit Integer, ranging from 0 to 9 both inclusive.

47. Tangent are drawn from any point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to the circle $x^2 + y^2 = 9$.
 If the locus of the mid – point of the chord of contact is $a(x^2 + y^2)^2 = bx^2 - cy^2$, then the value of $a^2 + b^2 + c^2 - 7870$ is equal to

48. Let $\vec{b} = 4\vec{i} + 3\vec{j}$ and $\vec{c}$ be two vectors perpendicular to each other in the xy – plane. If $\vec{r}_i, i = 1, 2, \dots, n$ are the vectors in the same plane having projection 1 and 2 along $\vec{b}$ and $\vec{c}$ respectively $\sum_{i=1}^n |\vec{r}_i|^2$ is equal to k then $\frac{k}{4}$ equals to
49. If the value of greater of $\sin x + \tan x$ and $2x \left(0 < x < \frac{\pi}{2} \right)$ at $\frac{\pi}{4}$ is $g\left(\frac{\pi}{4}\right)$ then $g\left(\frac{\pi}{4}\right) - \frac{1}{\sqrt{2}}$ is equal to
50. Let A & B be 2×2 matrices with integer entries such that $A, A + B, A + 2B, A + 3B$ and $A + 4B$ are all invertible matrices whose inverse have integer entries, then find $|\det(A + 5B)|$
51. Let $\langle a_n \rangle$ be an arithmetic sequence such that arithmetic mean of $a_1, a_3, a_5, \dots, a_{97}, a_{99}$ is 1. Find the value of $\left| \sum_{r=1}^{50} (-1)^{\frac{r(r+1)}{2}} \cdot a_{2r-1} \right|$
52. Let $f(x) = ax^2 + bx + c$, where $a \neq 0, a, b, c$ are integers and $f(1) = 1, 6 < f(3) < 8$ and $18 < f(5) < 22$. Then the number of solutions of equation $f(x) = e^x$ is _____
53. If z_1, z_2 are complex numbers, then the maximum value of $\frac{z_1 \bar{z}_2 + \bar{z}_1 z_2 + z_1 z_2 + \bar{z}_1 \bar{z}_2}{|z_1 z_2|}$ is equal to _____.
54. Let $f(x) = ax^2 + bx + c, (a < b)$ and $f(x) \geq 0 \forall x \in \mathbb{R}$. If the minimum value of $\frac{a+b+c}{b-a}$ is k then value of k is

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – IV
PAPER –1
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ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

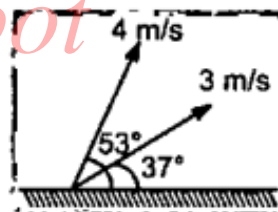
1. AD

Sol. So, velocity of first particle
 $= 3 \cos 37^\circ \hat{i} + 3 \sin 37^\circ \hat{j}$
 $= \frac{12}{5} \hat{i} + \frac{9}{5} \hat{j}$

Velocity of second particle
 $= 4 \cos 53^\circ \hat{i} + 4 \sin 53^\circ \hat{j}$
 $= \frac{12}{5} \hat{i} + \frac{16}{5} \hat{j}$

So, relative horizontal velocity is zero. So their relative velocity is in vertical direction only. Since, both particles are moving under gravity, so their relative acceleration is zero.

Their relative velocity $= \frac{16}{5} - \frac{9}{5} = \frac{7}{5} = 1.4 \text{ m/s}$



2. BC

Sol. Just after BP is cut.

For block A no force has changed.

$\therefore$ acceleration of $m_1 = 0$

for m_2 downward force is being reduced

$\therefore m_2$ will move upwards.

3. AC

Sol. $dw = \vec{f} \cdot d\vec{s}$

Since, body is hauled slowly, so

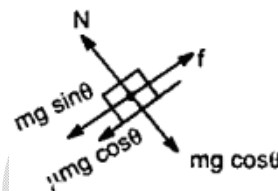
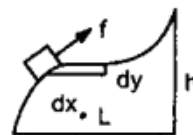
$$f = mg \sin \theta + \mu mg \cos \theta$$

$$W = \int (mg \sin \theta + \mu mg \cos \theta) ds$$

$$= \int mg ds \sin \theta + \int \mu mg ds \cos \theta$$

$$= \int mg dy + \int \mu mg dx$$

$$= mgh + \mu mgL$$



4. AC

Sol. $5 - F_1 = 1 \times 2 \Rightarrow F_1 = 3 \text{ N}$

Taking torque about CM:

$$5x = 3(\ell + x) \Rightarrow 2x = 3 \times 20 \Rightarrow x = 30 \text{ cm}$$

$$\text{Length of rod} = 2 \times (\ell + x) = 100 \text{ cm} = 1 \text{ m}$$

5. AC

Sol. $AC = 5 \text{ m}$

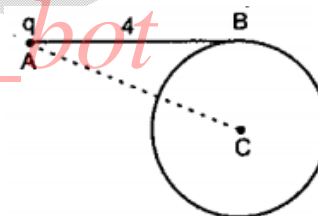
$$V = \frac{kq}{AC} = \frac{9 \times 10^9 \times 1 \times 10^{-6}}{5}$$

$$= 1.8 \times 10^3 = 1.8 \text{ kV}$$

$$V_B = (V_B)_{\text{due to } q} + (V_B)_i$$

$$(V_B)_i = -0.45 \text{ kV}$$

So, (A) and (C) are correct.



6. BC

Sol. For a bulb $R = \frac{V^2}{W}$

$$\Rightarrow R_B \leq R_A$$

when switch is open $I_A = I_B$

$$P_A = R_A I_A^2$$

$$P_B = R_B I_B^2$$

$$\Rightarrow P_B < P_A \text{ and } V_A > V_B$$

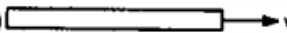
$$\Rightarrow V_A > 12 \text{ V and } V_B < 12 \text{ V}$$

After closing the switch

$$V_A = V_B = 12 \text{ V}$$

7. A

Sol.

(I) $60(1-v)$ 

$$P_i = P_f \Rightarrow 60(1-V) = 100 V$$

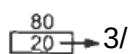
$$\Rightarrow V = \frac{3}{8} \text{ opposite to velocity of Ram i.e. } \frac{3}{8} \text{ m/s towards right.}$$

(II) $80 V = 80(1-V)$

$$\Rightarrow V = \frac{1}{2} \text{ m/sec left.}$$

(III) $80(1+V) + 60(-1+V) + 20V = 0$

$$V = -\frac{1}{8} \text{ m/s}$$

(IV) After jump of Ram 

$$\text{Now } (80+20)\frac{3}{8} = 80(1+V) + 20V$$

$$V = -\frac{17}{40} \text{ m/s}$$

8. B

Sol.

(I) Velocity of fish in air $= 4 \times \frac{3}{4} = 3\uparrow$

Velocity of fish w.r.t. bird $= 3 + 6 = 9\uparrow$

(II) Velocity of image of fish after reflection from mirror in air $= 4 \times \frac{3}{4} = 3\downarrow$

w.r.t. bird $= -3 + 6 = 3\uparrow$

(III) Velocity of bird in water $= 6 \times \frac{4}{3} = 8\downarrow$

w.r.t. fish $= 8 + 4 = 12\downarrow$

(IV) Velocity of bird in water after reflection from mirror $= 8\uparrow$

w.r.t. fish $= 8 - 4 = 4\uparrow$

9. D

Sol.

$$I = \int_0^r 2\pi r dr b r = \frac{2\pi b r^3}{3}$$

Use ampere law for B.

10. C

Sol.

$$\oint \vec{B} \cdot d\vec{\ell} = -A \left(\frac{dB}{dt} \right)$$

Section – B

11. 7.50

Sol. The centre of the wheel is moving with constant speed on a circular path of radius $6R$.

Hence, it has a centripetal acceleration of $a_c = \frac{V^2}{6R}$ directed towards the centre of curvature of the convex surface.

With respect to the centre of the wheel, the contact point has acceleration equal to $\frac{V^2}{R}$ directed towards the centre of the wheel.

$\therefore$ Acceleration of the contact point in reference frame of ground is

$$a_p = \frac{V^2}{R} - \frac{V^2}{6R} = \frac{5V^2}{6R}.$$

12. 2.56

Sol. For a jump of $h_0 = 1$ m on the earth, speed required is given by

$$\frac{1}{2}mV^2 = mgh_0 \Rightarrow V = \sqrt{20} \text{ m/s}$$

Escape speed on the surface of a planet is

$$V_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{8\pi GR^2\rho}{3}}$$

$$\therefore \frac{V_{\text{esc-planet}}}{V_{\text{esc-earth}}} = \frac{R_{\text{planet}}}{R_{\text{earth}}}$$

We want $V_{\text{esc}}^{\text{planet}} = \sqrt{20} \text{ m/s}$

And it is given that $V_{\text{esc}}^{\text{planet}} = 11.2 \text{ km/s}$

$$R_{\text{planet}} = \frac{\sqrt{20} \times (6400 \text{ km})}{11200} = 2.56 \text{ km}$$

13. 400.00

Sol. If force exerted on piston of area A_2 is F_2 then, the force acting on the other piston will be

$$= \frac{F_2}{A_2} \cdot A_1 \quad [\text{Pascal's law}]$$

$$= 5F_2 \quad \therefore \left[\frac{A_1}{A_2} = 5 \right]$$

To raise the load $5F_2 = 20000 \Rightarrow F_2 = 4000 \text{ N}$

Since lever bar is light, net torque on it (about the hinge) must be zero.

$$\therefore F_2 a = F(a + b)$$

$$\therefore F = \frac{F_2 a}{a + b} = \frac{4000 \times 4}{4 + 36} = 400 \text{ N}$$

14. 283.40

Sol. The product nucleus ^{198}Hg is in excited state and possesses extra 1.088 MeV energy. If ^{198}Hg would had been in ground state, the kinetic energy available to electron and antineutrino must have

$$Q = (m_{\text{Au}} - m_{\text{Hg}}) 931 \text{ MeV}$$

$$= (197.968233 - 197.966760) \text{ 931 MeV}$$

$$= 1.3714 \text{ MeV}$$

Since ^{198}Hg is in excited state, actual kinetic energy available to electron and antineutrino is

$$K = (1.3714 - 1.088) \text{ MeV}$$

$$= 0.2834 \text{ MeV}$$

As β -ray and antineutrino has continuous spectrum starting from zero value, therefore, this is also the maximum kinetic energy of the electron emitted.

15. 7.00

Sol. $\Delta x = (\mu_1 - 1)t_1 - (\mu_2 - 1)t_2$

$$= (2 - 1)200 - (1.5 - 1)200$$

$$= 200 - 100 = 100 \text{ nm}$$

$$\therefore \phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{600} \times 100 = \frac{\pi}{3}$$

$$\therefore P = P_1 + P_2 + 2\sqrt{P_1 P_2} \cos \theta$$

$$= I_0 \pi r_1^2 + I_0 \pi r_2^2 + 2\sqrt{I_0 \pi r_1^2 \times I_0 \pi r_2^2} \cos \frac{\pi}{3}$$

$$= 7 \text{ } \mu\text{W}$$

16. 4.01

Sol. As here volume of gas remains constant,

$$(\Delta Q)_v = \mu C_v \Delta T, \text{ Here } C_v = 5 \text{ cal/mol K}$$

$$\text{And } \Delta T = (400 - 300) = 100 \text{ K}$$

$$\text{And so for ideal gas } PV = \mu RT,$$

$$\mu = \frac{(10)^5 \times (0.2)}{8.31 \times 300} = 8.0224$$

$$(\Delta Q)_v = \mu \times 5 \times 100 = 4.01 \text{ kcal.}$$

17. 3.14

Sol. $\frac{F}{A} = Y \frac{\Delta L}{L} \Rightarrow \frac{mg}{A} = Y(\alpha \Delta \theta)$

$$m = \frac{AY\alpha(\Delta \theta)}{g} = \frac{\pi r^2 Y \alpha(\Delta \theta)}{g}$$

$$= \frac{\pi (10^{-3})^2 \times 10^{11} \times 10^{-5} \times 10}{10} = \pi$$

18. 30.00

Sol. $f \propto \sqrt{T}$ for strings.

On increasing the tension by 1%

$$f' = \sqrt{1.01 T}$$

$$\frac{f'}{f} = \frac{\sqrt{1.01 T}}{\sqrt{T}} = (1 + 0.01)^{\frac{1}{2}} = 1 + \frac{1}{200}$$

$$\text{Beat frequency, } f' - f = f \left(\frac{f'}{f} - 1 \right) = 1$$

$$\text{Number of beats in 30 seconds} = 1 \times 30 = 30.$$

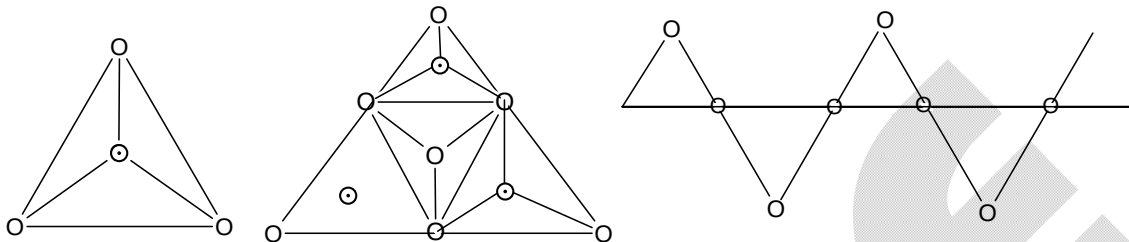
Chemistry

PART – II

Section – A

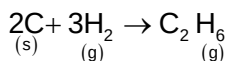
19. A

Sol.



20. BC

Sol.



$$\Delta H_f^0 = [2 \times \Delta H_{\text{sub}}^0 + 3 \times \text{B.E.}(\text{H}-\text{H})] - [\text{B.E.}(\text{C}-\text{C}) + 6 \times \text{B.E.}(\text{C}-\text{H})]$$

$$\Rightarrow -85 = [(2 \times 718) + (3 \times 436)] - (x + 6y)$$

$$\therefore x + 6y = 2829 \quad \dots\dots 1$$

Similarly for C_3H_8 (g)

$$2x + 8y = 4002 \quad \dots\dots 2$$

$$\text{Solving (1) \& (2), } x = 345 \\ y = 414$$

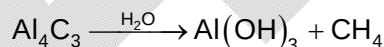
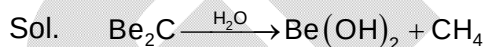
21. AB

$$\text{Sol. } k = A e^{-E_a/RT} \text{ or } \ln k = \ln A - \frac{E_a}{RT}$$

22. ABC

Sol. The outermost electronic configuration of Yb is $4f^{14}6s^2$
 $\therefore \text{Yb}^{2+}$ has full-filled $4f^{14}$ configuration radius of $\text{Yb}^{3+} < \text{Yb}^{2+}$

23. AD



24. ABCD

$$\text{Sol. } \Delta S = \frac{\Delta H}{T} = \frac{-40,600}{373} = -108.84 \text{ J/K for one mole}$$

$$\text{for } \frac{9}{18} \text{ mole, } \Delta S = \frac{-108.84}{2} = 54.42 \text{ J/K}$$

25. A
Sol. I – can undergo Nu^- substitution, elimination
II – can undergo Nu^- substitution, esterification, dehydrogenation & oxidation.
III – can undergo Nu^- addition, esterification and oxidation
IV – can undergo Nu^- substitution
26. A
Sol. Fact based
27. C
Sol. Elements having filled d-orbitals are not true transition elements.
28. B
Sol. $\text{SO}_3^{2-} + \text{dil.H}_2\text{SO}_4 \longrightarrow \text{H}_2\text{O} + \text{SO}_2 + \text{SO}_4^{2-}$
 $\text{CO}_3^{2-} + \text{dil.H}_2\text{SO}_4 \longrightarrow \text{H}_2\text{O} + \text{CO}_2 + \text{SO}_4^{2-}$
 $\text{NH}_4^+ + \text{NaOH} \longrightarrow \text{Na}^+ + \text{H}_2\text{O} + \text{NH}_3$
 $\text{S}^{2-} + \text{H}_2\text{SO}_4 \longrightarrow \text{H}_2\text{S} + \text{SO}_4^{2-}$

Section – B

29. 162.00
Sol. $K_f \text{ (of camphor)} = \frac{R(T_f^0)^2 \times M_2}{1000 \times \Delta H_f^0} = \frac{2 \times 152 \times (450)^2}{1000 \times 1.52 \times 1000} = 40.5$
 $\Delta T_f = i \times K_f \times m$
 $450 - 430 = 1 \times 40.5 \times \frac{0.04 \times 1000}{M_{\text{solute}} \times 0.5}$
 $M_{\text{solute}} = \frac{0.04 \times 1000 \times 40.5}{0.5 \times 20} = 162$
30. 81.81
Sol.

	$\text{NaOH} + \text{HCl} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$			
Millimoles	0.1	.001 x	0	0
	-0.001 x	-0.001x		

 $t = t \quad \frac{0.1 - .001}{x} \quad x$
 $10^{-3} \text{ M NaOH have pH} = 11$
New pH after adding acid be 10
Hence, $10^{-4} \text{ M } (100 + x) = 0.1 - .001 x$
 $100 + x = \frac{0.1 - 0.001x}{10^{-4}}$
 $\therefore 11x = 1000 - 100 = 900$
 $\therefore x = \frac{900}{11} = 81.81 \text{ mL}$

31. 390.55

Sol. $\Lambda_m^0 \text{CH}_3\text{COOH} = \frac{(\Lambda_m^0 \text{Ca}(\text{CH}_3\text{COO})_2 + 2\Lambda_m^0 \text{HCl} - \Lambda_m^0 \text{CaCl}_2)}{2}$
 $= \frac{200.8 + 2(425.95) - 271.6}{2} = 390.55$

32. 1.50

Sol. Vander waals equation for 1 mole of real gas, when $b = 0$

$$\Rightarrow \left(P + \frac{a}{V^2}\right)(V) = RT$$

$$\therefore PV = -a \times \frac{1}{V} + RT$$

$$y = mx + c$$

$$\text{Slope} = \tan(\pi - \theta) = -a$$

$$\text{So, } \tan \theta = a = \frac{21.6 - 20.1}{3 - 2} = 1.5$$

33. 11.00

Sol. The product contains three phenyl group and two multiple bonds

$$\therefore \text{Total number of pi-bonds} = (3 \times 3) + 2 = 11$$

34. 2920.40

Sol. Meq of $\text{MnO}_4^- = \text{Meq of Fe}^{2+}$

$$V \times M \times n = \frac{W}{E} \times 1000$$

$$\text{or, } 104.3 \times 1000 \times 0.1 \times 5 = \frac{W}{56/1} \times 1000$$

$$\text{On solving } w = 2920.4 \text{ g}$$

35. 8.33

Sol. The average O.S of iron in $\text{Fe}_{0.96}\text{O}$ is $\frac{200}{96}$

Let the % of Fe(II) be x

$$x \times (+3) + (100 - x) \times (+2) = \frac{100 \times 200}{96}$$

$$3x + 200 - 2x = 208.33$$

$$\therefore x = 8.33$$

36. 1.41

Sol. $d = \frac{Z \times M}{N_a \times a^3}$

$$\frac{45}{16} = \frac{Z \times 27}{6.02 \times 10^{23} \times (4 \times 10^{-8})^3}$$

$$\Rightarrow Z = 4 \text{ (so fcc)}$$

$$r = \frac{\sqrt{2}a}{4} = 1.41 \text{ \AA}$$

Mathematics

PART – III

Section – A

37. AC

Sol. $V = \left| \vec{a} \cdot (\vec{b} \times \vec{c}) \right| \leq \sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2} \sqrt{c_1^2 + c_2^2 + c_3^2} \dots\dots(i)$

$$\left[(a_1 + a_2 + a_3)(b_1 + b_2 + b_3)(c_1 + c_2 + c_3) \right]^{1/3} \text{ (Using A.M. } \geq \text{ G.M.)}$$

$$L^3 \geq (a_1 + a_2 + a_3)(b_1 + b_2 + b_3)(c_1 + c_2 + c_3)$$

$$\text{now } (a_1 + a_2 + a_3)^2 = a_1^2 + a_2^2 + a_3^2 + 2(a_1a_2 + a_2a_3 + a_3a_1)$$

$$(a_1 + a_2 + a_3)^2 \geq a_1^2 + a_2^2 + a_3^2$$

$$(a_1 + a_2 + a_3) \geq \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$\text{similarly } (b_1 + b_2 + b_3) \geq \sqrt{b_1^2 + b_2^2 + b_3^2}$$

$$c_1 + c_2 + c_3 \geq \sqrt{c_1^2 + c_2^2 + c_3^2}$$

$$L^3 \geq \left[(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)(c_1^2 + c_2^2 + c_3^2) \right]^{1/2}$$

$$L^3 \geq V$$

38. BC

Sol. We supposed to find m and n such that $\lim_{x \rightarrow \infty} \sqrt[3]{8x^3 + mx^2} - nx = 1$ or

$$\lim_{x \rightarrow \infty} \sqrt[3]{8x^3 + mx^2} - nx = 1.$$

We compute

$$\sqrt[3]{8x^3 + mx^2} - nx = \frac{(8 - n^3)x^3 + mx^2}{\sqrt[3]{(8x^3 + mx^2)^2} + nx\sqrt[3]{8x^3 + mx^2} + n^2x^2}.$$

$8 - n^3$ must be equal to 0

$$n = 2$$

$$\text{Now } f(x) = \frac{m}{\sqrt[3]{\left(8 + \frac{m}{x}\right)^2} + 2\sqrt[3]{8 + \frac{m}{x}} + 4}.$$

We see that $\lim_{x \rightarrow \infty} f(x) = \frac{m}{12}$. For this to be equal to 1, m must be equal to 12. Hence the answer to the problem is $(m, n) = (12, 2)$.

39. ABCD

Sol.
$$\int \frac{x^4 + 1}{x^6 + 1} dx = \int \frac{x^4 - x^2 + 1}{x^6 + 1} dx + \int \frac{x^2}{x^6 + 1} dx = \int \frac{1}{x^2 + 1} dx + \int \frac{1}{3} \frac{(x^3)'}{(x^3)^2 + 1} dx$$

$$= \arctan x + \frac{1}{3} \arctan x^3.$$

To write the answer in the required form we should have

$$3 \arctan x + \arctan x^3 = \arctan \frac{P(x)}{Q(x)}$$

Applying the tangent function to both sides, we deduce

$$\frac{\frac{3x - x^3}{1 - 3x^2} + x^3}{1 - \frac{3x - x^3}{1 - 3x^2} \cdot x^3} = \tan \left(\arctan \frac{P(x)}{Q(x)} \right).$$

From here $\arctan \frac{P(x)}{Q(x)} = \arctan \frac{3x - 3x^5}{1 - 3x^2 - 3x^4 + x^6}$, and hence

$P(x) = 3x - 3x^5, Q(x) = 1 - 3x^2 - 3x^4 + x^6$. The final answer is

$$\frac{1}{3} \arctan \frac{3x - 3x^5}{1 - 3x^2 - 3x^4 + x^6} + C.$$

40. AC

Sol. Denote the value of the integral by I. With the substitution $t = \frac{ab}{x}$ we have

$$I = \int_a^b \frac{e^{\frac{b}{t}} - e^{\frac{t}{a}}}{\frac{ab}{t}} \cdot \frac{-ab}{t^2} dt = - \int_a^b \frac{e^{\frac{t}{a}} - e^{\frac{b}{t}}}{t} dt = -I.$$

Hence, $I = 0$.

41. AB

Sol.
$$3 \tan 3x = \frac{3(3 \tan x - \tan^3 x)}{1 - 3 \tan^2 x} = \frac{3 \tan^3 x - 9 \tan x}{3 \tan^2 x - 1}$$

$$= \frac{8 \tan x}{3 \tan^2 x - 1}.$$

Hence

$$\frac{1}{\cot x - 3 \tan x} = \frac{\tan x}{1 - 3 \tan^2 x} = \frac{1}{8} (3 \tan 3x - \tan x) \text{ for all } x \neq k \frac{\pi}{2}, k \in \mathbb{Z}.$$

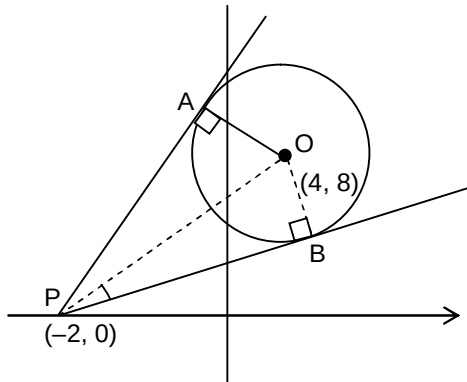
It follows that the left – hand side telescopes as

$$\frac{1}{8} (3 \tan 27^\circ - \tan 9^\circ + 9 \tan 81^\circ - 3 \tan 27^\circ + 27 \tan 243^\circ - 9 \tan 81^\circ + 81 \tan 729^\circ - 27 \tan 243^\circ)$$

$$= \frac{1}{8} (81 \tan 9^\circ - \tan 9^\circ) = 10 \tan 9^\circ.$$

42. BC

Sol.



$$\sin \theta = \frac{2\sqrt{5}}{10} = \frac{1}{\sqrt{5}}$$

Slopes of PA and PB are $\tan(\alpha \pm \theta)$ where $\tan \alpha = \frac{8}{6} = \frac{4}{3}$

$$= \frac{\frac{4}{3} + \frac{1}{2}}{1 - \frac{4}{3} \cdot \frac{1}{2}}, \frac{\frac{4}{3} - \frac{1}{2}}{1 + \frac{4}{3} \cdot \frac{1}{2}}$$

$$= \frac{11}{2}, \frac{5}{10}$$

$$\therefore A, B \equiv \left(4 + 2\sqrt{5} \left(\frac{-11}{5\sqrt{5}} \right), 8 + 2\sqrt{5} \left(\frac{2}{5\sqrt{5}} \right) \right),$$

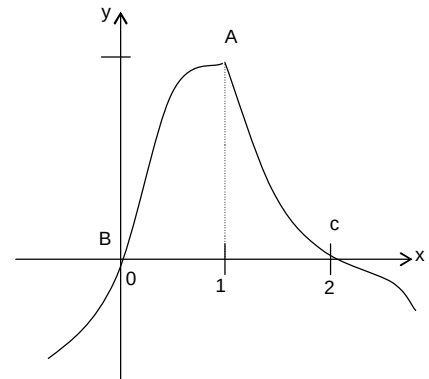
$$\left(4 - 2\sqrt{5} \left(\frac{-1}{\sqrt{5}} \right), 8 - 2\sqrt{5} \left(\frac{2}{\sqrt{5}} \right) \right) = \left(\frac{-2}{5}, \frac{44}{5} \right)$$

$$= (6, 4)$$

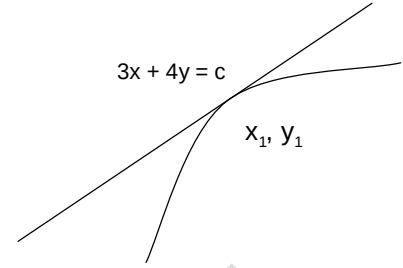
43. B

Sol. (I) A, B, C are the 3 critical points of $y = f(x)$.

$f''(x) = 0$ for $x = 2$ and fails to exist at $x = 0$.



(II) $x = \frac{1}{4}$ and 2. Make a quadratic in $\log_2 x$ and interpret the result.



$$(III) \frac{dy}{dx} = -1 + 2x_1^3 = -\frac{3}{4} \Rightarrow x_1 = \frac{1}{2}$$

$$\Rightarrow \frac{1}{32} = \frac{1}{2} + y_1 \text{ or } y_1 = -\frac{15}{32} \Rightarrow c = -\frac{3}{8}$$

(IV) $f'(x) = 2x^3 - 3x + 1$ this is always positive in $(1, 2)$.

$\therefore$ Increasing in $[1, 2]$

$\therefore f(2)$ will be the greatest value.

44. D

Sol. Let $L = \lim_{n \rightarrow \infty} \left(\frac{n^2}{n-1} \right)^{\tan \frac{1}{\sqrt{n}}} (\infty^0)$

$$\ln L = \lim_{n \rightarrow \infty} \tan \frac{1}{\sqrt{n}} \ln \left(\frac{n^2}{n-1} \right) = \lim_{n \rightarrow \infty} \frac{2 \ln n - \ln(n-1)}{\cot \left(\frac{1}{\sqrt{n}} \right)} \left(\frac{\infty}{\infty} \right)$$

Put $n = t^2 = \lim_{t \rightarrow \infty} \frac{4 \ln t - \ln(t^2 - 1)}{\cot \left(\frac{1}{t} \right)} = \lim_{t \rightarrow \infty} \frac{2(t^2 - 2)}{t(t^2 - 1) \operatorname{cosec}^2 \left(\frac{1}{t} \right) \left(\frac{1}{t^2} \right)}$

$$= \lim_{t \rightarrow \infty} \frac{2(t^2 - 2)}{t(t^2 - 1)} \left(\frac{\sin \frac{1}{t}}{\frac{1}{t}} \right)^2 = 0$$

$$L = 1 \Rightarrow (Q)$$

$$(III) \sin 2x = \pm \frac{\sqrt{3}}{2}$$

$$2x \in (0, 4\pi)$$

Hence, 8 solutions.

(IV) Since $g(x)$ is differentiable $\forall x \in \mathbb{R}$

$\therefore f(x)$ must have the factor $x(x-1)(x-2)$ atleast once.

$\therefore$ minimum 3 roots of $f(x) = 0 \Rightarrow (R)$

45. A

Sol. xbbbxccxcxcx

$$\text{Number of ways} = \frac{4!}{3!} \times {}^5C_4 = 20$$

(II) 2b, 1b; 2c, 1c or 2b, 1b; 1c, 1c, 1c or 2b, 1b; 3c

(same way starting with c)

cbbcbcc, cbcbbcc

bbcccb, bccbb

$$\text{Number of ways} = 12 \times {}^6C_4 = 180$$

$$\text{(III) } bcbcbcc = {}^7C_4$$

$$bccbcb \text{ or } bcbccb = 2 \times {}^6C_3$$

$$bbccbc \text{ or } bccbbc \text{ or } bbcbcc \text{ or } bcbbcc = 4 \times {}^5C_2$$

$$bccbb \text{ or } bbcccb = 2 \times {}^4C_1$$

$$bbbccc = {}^4C_4$$

$$\text{Total ways} = 2(35 + 40 + 40 + 8 + 1) = 248$$

(IV) bcbcbcc, cbcbbcc

bccbcb, cbbcbcc

bcbccb, cbcbbc

bcbccb, cbcbbc

$$\text{number of ways} = {}^7C_1 \times 2 + 4 = 18$$

46. D

$$\text{Sol. (I) } P = \frac{1 \times {}^{11}C_5}{{}^{12}C_6} = \frac{1}{2}$$

$$\text{(II) } P = \frac{{}^2C_1 \times {}^{10}C_5}{{}^{12}C_6} = \frac{6}{11}$$

$$\text{(III) } P = \frac{{}^{10}C_4}{{}^{12}C_5} = \frac{5}{22}$$

$$\text{(IV) } P = \frac{10}{11}$$

Section – B

47. 252.00

Sol. $x = (\sqrt{3} + 1)^{2018}$ and $y = (\sqrt{3} - 1)^{2018}$

$$[x] + \{x\} + y = (\sqrt{3} + 1)^{2018} + (\sqrt{3} - 1)^{2018}$$

$$\begin{aligned} [x] + 1 &= 2^{1009} \left[(2 + \sqrt{3})^{1009} + (2 - \sqrt{3})^{1009} \right] \\ &= 2^{1009} \times 2 \left[{}^{1009}C_0 2^{1009} + {}^{1009}C_2 2^{1007} 3^1 + \dots + {}^{1009}C_{1006} 2^3 3^{503} + {}^{1009}C_{1008} 2 \times 3^{504} \right] \\ [x] + 1 &= 2^{1011} \underbrace{\left[{}^{1009}C_0 2^{1008} + \dots + {}^{1009}C_{1006} 2^2 3^{503} + {}^{1009}C_{1008} 3^{504} \right]}_{\text{odd}} \end{aligned}$$

$$[\because \{x\} + y \in (0, 2) \Rightarrow \{x\} + y = 1]$$

$\Rightarrow N$ is divisible by 2^{1011} , hence divisible by $(16)^{252}$

48. 13.00

Sol. Coefficient of $x^{\frac{n(n+1)}{2}-7} = -7 + (1 \cdot 6 + 2 \cdot 5 + 3 \cdot 4) - (1 \cdot 2 \cdot 4) = 13$

49. 18.00

Sol. We have $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\beta\gamma + \gamma\alpha + \alpha\beta)$

$$\Rightarrow 4 = 6 + 2(\beta\gamma + \gamma\alpha + \alpha\beta)$$

$$\Rightarrow \beta\gamma + \gamma\alpha + \alpha\beta = -1.$$

$$\text{Also, } \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma$$

$$= (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \beta\gamma - \gamma\alpha - \alpha\beta)$$

50. -1.59

Sol. The critical point of f are solutions to the system of equations

$$\frac{\partial f}{\partial x}(x, y) = 4x^3 + 12xy^2 - \frac{9}{4} = 0,$$

$$\frac{\partial f}{\partial y}(x, y) = 12x^2y + 4y^3 - \frac{7}{4} = 0$$

divide the two equations by 4 and then add, respectively, subtract them, we obtain

$$x^3 + 3x^2y + 3xy^2 + y^3 - 1 = 0 \quad \text{and} \quad x^3 - 3x^2y + 3xy^2 - y^3 = \frac{1}{8}.$$

We write these as

$$(x + y)^3 = 1 \quad \text{and} \quad (x - y)^3 = \frac{1}{8},$$

from which we obtain $x + y = 1$ and $x - y = \frac{1}{2}$. We

find a unique critical point $x = \frac{3}{4}, y = \frac{1}{4}$. The minimum of f is attained at this point, and

$$\text{it is equal to } f\left(\frac{3}{4}, \frac{1}{4}\right) = -\frac{51}{32}.$$

51. 126.00

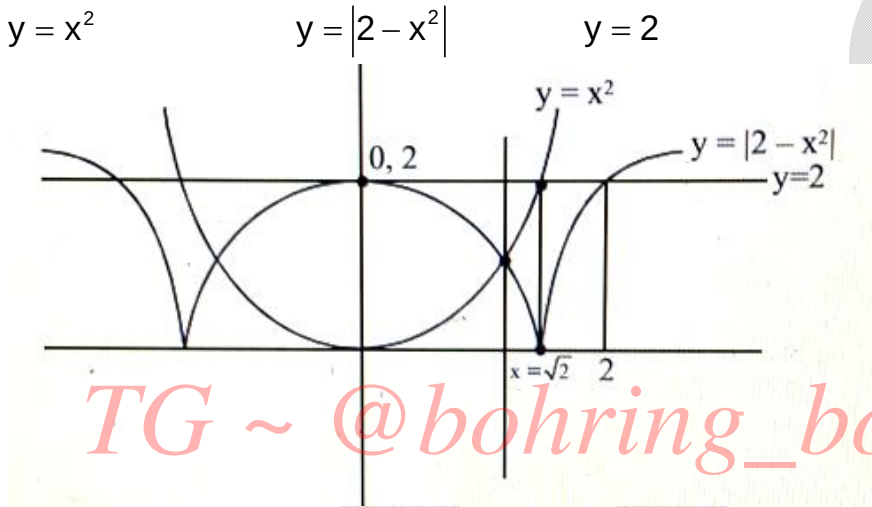
Sol. We have, $\lim_{n \rightarrow \infty} n g_n(3) = \lim_{n \rightarrow \infty} n \left(f\left(3 + \frac{5}{n}\right) - f\left(3 - \frac{2}{n}\right) \right)$

$$= 5 \lim_{h \rightarrow 0^+} \frac{f(3+h) - f(3)}{h} + 2 \lim_{h \rightarrow 0^+} \frac{f(3) - f(3-h)}{h}$$

$$= 5f'(3) + 2f'(3) = 5 \times 18 + 2 \times 18 = 90 + 36 = 126.$$

52. 1.01

Sol. $y = x^2$



Required area $\int_1^{\sqrt{2}} [x^2 - (2 - x^2)] dx + \int_{\sqrt{2}}^2 [2 - (x^2 - 2)] dx$

$$= \frac{20}{3} - 4\sqrt{2} = 1.01$$

53. 18.00

Sol. The points (x_1, x_2) and (y_1, y_2) lie on the circle of radius $2\sqrt{2} = c$ centered at the origin. We can write $(x_1, x_2) = (c \cos \phi, c \sin \phi)$ and $(y_1, y_2) = (c \cos \theta, c \sin \theta)$.

Then

$$S = 2 - c(\cos \phi + \sin \phi + \cos \theta + \sin \theta) + c^2(\cos \phi \cos \theta + \sin \phi \sin \theta)$$

$$= 2 + c\sqrt{2} \left(-\sin\left(\phi + \frac{\pi}{4}\right) - \sin\left(\theta + \frac{\pi}{4}\right) \right) + c^2 \cos(\phi - \theta) \text{ to 1 by choosing}$$

$$\phi = \theta = \frac{5\pi}{4}. \text{ The maximum of } S \text{ is } 2 + 2c\sqrt{2} + c^2 = (c + \sqrt{2})^2.$$

54. 11.00

Sol. $\left[x - \frac{1}{2} \right] = 1$ or $\left[x + \frac{1}{2} \right] = 1$

$$x - \frac{1}{2} \in [1, 2) \quad x + \frac{1}{2} \in [1, 2)$$

$$x \in \left[\frac{3}{2}, \frac{5}{2} \right) \dots (1) \quad x \in \left[\frac{1}{2}, \frac{3}{2} \right) \dots (2)$$

$$(1) \cup (2)$$

$$x \in \left[\frac{1}{2}, \frac{3}{2} \right) \cup \left[\frac{3}{2}, \frac{5}{2} \right)$$

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = \frac{1}{4} + \frac{9}{4} + \frac{9}{4} + \frac{25}{4} = 11$$

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FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – IV
PAPER –2
TEST DATE: 08-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1.
Sol.

C

For upper portion:

$$f_1 = \frac{\mu_2 - 1}{\mu_1} (f_a) = \frac{1.5 - 1}{1.2 - 1} \times 20 = 40 \text{ cm}$$

$$\begin{aligned} \text{For lower portion, } f_1 &= \frac{\mu_2 - 1}{\mu_3} (f_a) \\ &= \frac{1.5 - 1}{2.5 - 1} \times 20 = -25 \text{ cm} \end{aligned}$$

$\therefore$ Object is at infinity, images will be form at focal points.

Hence the distance between two images will = $40 + |-25| = 65 \text{ cm}$

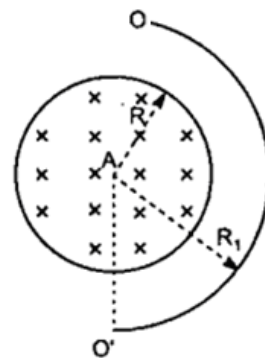
2. B

Sol.

$$E = \frac{R^2}{2R_1} \left(\frac{dB}{dt} \right)$$

$$\text{e.m.f} = \frac{\theta}{2\pi} \frac{R^2}{2R_1} \left(\frac{dB}{dt} \right) 2\pi R_1$$

$$= \frac{\theta}{2} R^2 \left(\frac{dB}{dt} \right)$$



3. D

Sol. As $\gamma > 1$

$$\text{for } TV^{\gamma-1} = \text{constant} \quad \frac{dT}{dv} < 0$$

$$\text{and for } T^{\gamma} = KP^{\gamma-1}$$

$$\frac{dT}{dP} > 0$$

4. A

$$\text{Sol. } y = \frac{0.8}{3(x+2t)^2 + 4}$$

$$v = 2 \text{ m/s}$$

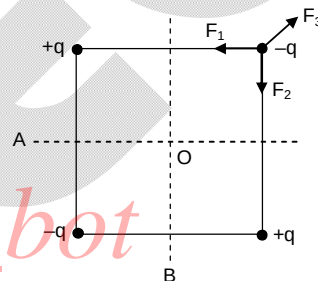
$$A = 0.2 \text{ m}$$

5. AC

Sol. AOB is zero potential surface

Induce charge = $-q$

$$F = \frac{kq^2}{a^2} \sqrt{2} - \frac{kq^2}{2a^2}$$



6. AC

$$\text{Sol. } I_1 \left[\frac{n(n+1)}{2} r \right] + I_x = nE \quad \text{where } x \text{ external resistance}$$

$$I_m \left(\frac{n(n+1)}{2} r \right) + I_x = mnE$$

$$I_1 + I_2 + I_3 + \dots + I_m = I$$

$$\Rightarrow I = \frac{\frac{n(m+1)E}{2}}{x + \frac{n(n+1)}{2m} r} = \frac{V}{x+R}$$

7. CD

$$\text{Sol. } \frac{dQ}{dt} = \frac{KA(\Delta T)}{x} = \frac{KA[0 - (-\theta)]}{x}; \quad \frac{dQ}{dt} = \frac{KA\theta}{x}$$

8. ACD

$$\text{Sol. } y(x, t) = -\cos kx \sin \omega t$$

$$y(0.05, 0.05) = -4 \cos \left(\frac{2\pi}{0.4} 0.05 \right) \sin \left(\frac{2\pi}{0.2} 0.005 \right)$$

$$= -4 \cos \left(\frac{\pi}{4} \right) \sin \left(\frac{\pi}{2} \right) = -2\sqrt{2} \text{ cm}$$

$$v = f\lambda = \frac{0.4}{0.2} = 2 \text{ m/s}$$

$$\begin{aligned} \frac{\partial y}{\partial t} &= -A \omega \cos(kx) \cos(\omega t) \\ &= -4 \left(\frac{2\pi}{0.2} \right) \cos \left(\frac{2\pi}{0.4} \frac{1}{15} \right) \cos \left(\frac{2\pi}{0.2} 0.1 \right) \end{aligned}$$

$$\frac{\partial y}{\partial t} = 20 \pi \text{ cm/sec} = \text{particle velocity}$$

9. AC

Sol. In SHM, time taken to move from $x = 0$ to $x = \frac{A}{2}$ is $\frac{T}{12}$ and time taken to move from extreme to $x = \frac{A}{2}$ is $\frac{T}{6}$.

$$\text{If } \frac{T}{12} = 1 \Rightarrow T = 12$$

$$\text{If } \frac{T}{6} = 1 \Rightarrow T = 6.$$

10. ABC

Sol. $E_s = -\frac{GMm}{R}$; $E_1 = -\frac{GMm}{2(2R)}$; $E_2 = -\frac{GMm}{2(3R)}$

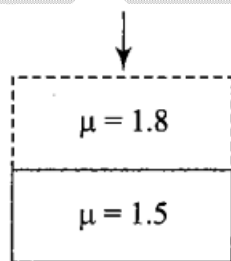
$$E_1 - E_s = \frac{3}{4} mgR$$

$$E_2 - E_1 = \frac{mgR}{12}$$

Section – B

11. 9

Sol. Path difference between rays reflected from upper and lower faces of layer = $2\mu t \cos r = 2\mu t$ (for normal incidence). But there is change in path of $\lambda/2$ of light at upper surface due to reflection from denser medium. So actual path difference is $2\mu t - \lambda/2$.



For constructive interference $2\mu t - \frac{\lambda}{2} = n\lambda$

$$t = \frac{(2n+1)\lambda}{4\mu}. \text{ For least thickness } n = 0.$$

$$\therefore t_{\min} = \frac{\lambda}{4\mu} = \frac{648}{4 \times 1.8} \text{ nm} = 90 \text{ nm}$$

12. 5

Sol. $E_n = -\frac{mZ^2e^4}{8\epsilon_0^2n^2h^2}$, so $hf = +\frac{mZ^2e^4}{8\epsilon_0^2h^2} \left[\frac{1}{16} - \frac{1}{25} \right]$

$$\therefore f = \frac{mZ^2e^4}{8\epsilon_0^2h^3} \left[\frac{9}{16 \times 25} \right]$$

and frequency $f_4 = \frac{Z^2e^2m}{4\epsilon_0^2n^3h^3} = \frac{Z^2e^4m}{4\epsilon_0^2(4)^3h^3}$

$$\therefore \frac{f}{f_4} = \frac{18}{25}, \text{ so } m = 5.$$

13. 4

Sol. When the two spheres come in contact, they exert equal and opposite impulse.

$$\int F_a dt = I_1(\omega_0 - \omega_1) \quad \dots(i)$$

$$\int F_b dt = I_2\omega_2 \quad \dots(ii)$$

Finally $\omega_1 a = \omega_2 b \quad \dots(iii)$

$$I_1(\omega_0 - \omega_1) = I_2 \frac{a^2}{b^2} \omega_1$$

$$\omega_1 = \frac{I_1\omega_0}{I_1 + \frac{a^2}{b^2}I_2} = 4 \text{ rad/s.}$$

14. 5

Sol. Let the volume of the cylinder be V. When the cylinder is floating, upthrust = weight. Hence,

$$\rho \left(\frac{3}{4} V \right) g = mg \Rightarrow V = \frac{4m}{3\rho}$$

Let the acceleration of the particle vessel be A (upwards). In the reference frame of the vessel, the acceleration of the cylinder is A/3.

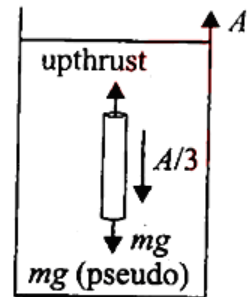
$$\therefore mg + mA - \text{upthrust} = m \left(\frac{A}{3} \right)$$

$$mg + mA - \rho V g' = m \left(\frac{A}{3} \right)$$

where $g' = g + A$ = effective value of g for upthrust.

$$\therefore mg + mA - \rho V (g + A) = m \left(\frac{A}{3} \right)$$

$$\Rightarrow mg - \frac{4}{3} m(g + A) = -m \frac{2}{3} A$$



$$A = \left(-\frac{g}{2}\right) \text{ upwards}$$

The acceleration of the vessel should be $\frac{g}{2} = \frac{10}{2} = 5 \text{ ms}^{-2}$ (downwards).

15. 6

Sol. Let the pressures in wide and narrow limbs be P_1 and P_2 , respectively. If R_1 and R_2 be the radii of meniscus in wide and narrow limb, pressure just below the meniscus of wide tube = $P_1 - \frac{2T}{R_1}$ and pressure just below the meniscus of narrow limb = $P_2 - \frac{2T}{R_2}$.

Therefore, difference of these pressures

$$\left(P_1 - \frac{2T}{R_1}\right) - \left(P_2 - \frac{2T}{R_2}\right) = h \rho g$$

Therefore, true pressure difference,

$$P_1 - P_2 = h \rho g - 2T \left(\frac{1}{R_2} - \frac{1}{R_1} \right)$$

For the water and glass surface, taking the angle of contact θ to be zero, we have

$R_1 = \frac{r_1}{\cos \theta} = r_1$ and $R_2 = \frac{r_2}{\cos \theta} = r_2$ where r_1 and r_2 are radii of wide and narrow limbs, respectively.

$\therefore$

$$P_1 - P_2 = h \rho g - 2T \left(\frac{1}{r_2} - \frac{1}{r_1} \right) = 0.2 \times 10^3 \times 9.8 - 2 \times 72 \times 10^{-3} \times \left(\frac{1}{7.2 \times 10^{-4}} - \frac{1}{1.44 \times 10^{-3}} \right)$$

$$= 1.96 \times 10^3 - 0.10 \times 10^3 = 1.86 \times 10^3 \text{ N/m}^2 = 1860 \text{ N/m}^2$$

$$\therefore N = 6.$$

16. 8

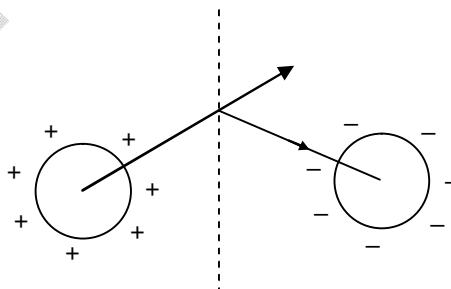
Sol. $J = \sigma E = \frac{\sigma q \ell}{2\pi \epsilon_0 \ell r^3}$

$$I = \int_0^\infty J 2\pi x dx = \frac{q}{\epsilon_0 \epsilon \rho}$$

$$\Delta V = \frac{q}{2\pi \epsilon_0 \epsilon a} = IR$$

$$\Rightarrow R = \frac{\rho}{2\pi a} = 0.50$$

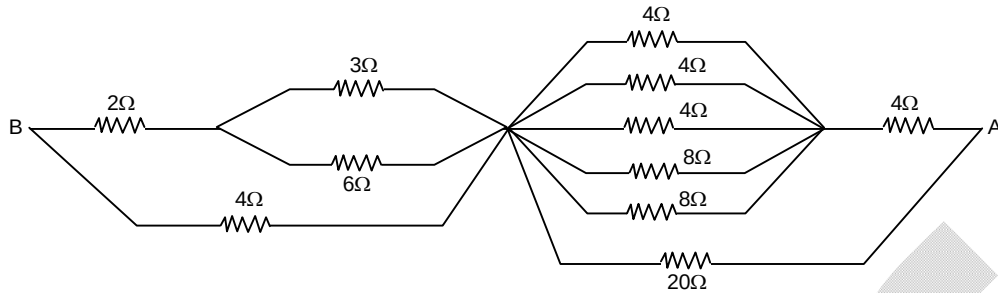
$$\therefore n = 8.$$



17. 4

Sol. $C_{eq} = \frac{2C_1C_2 + C_2C_3 + C_3C_1}{C_1 + C_2 + 2C_3} = \frac{7}{5}$

Where $C_1 = 1$, $C_2 = 2$, $C_3 = 1$



$$R_{eq} = 6$$

$$\tau = RC = 8.4$$

$$\therefore n = 4.$$

18. 2

Sol. Component of acceleration along incline is $g \sin \alpha$ and effective acceleration along groove is $a = \sin \alpha \cos \beta$.

From figure $\ell = \frac{h}{\sin \alpha} = OB$

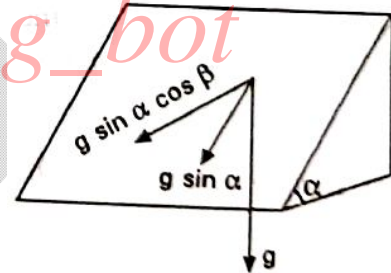
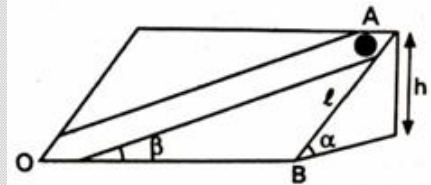
OA is the groove making an angle $\beta = 45^\circ$ with AB on the inclined surface.

From figure, $\frac{OB}{OA} = \cos \beta$

or $OA = \frac{h}{\sin \alpha \cos \beta}$

Since acceleration along groove is constant, we may use equation $s = \frac{1}{2}at^2$.

$$\therefore k = 2.$$



Chemistry

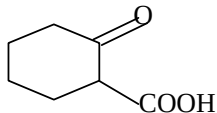
PART – II

Section – A

19. C

Sol.

Intermediate is



Which loses CO_2 on heating (β -keto acid)

20. B

Sol. M absorbs some heat energy and prevents decomposition of O_3

21. C

Sol. $T_f = 4T_i$

$$\Delta S = nC_v \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i}$$

$$0 = n \times \frac{3R}{2} \ln 4 + nR \ln \frac{V_f}{V_i}$$

$$\frac{3}{2} \log 4 = \log \frac{V_i}{V_f}$$

$$4^{3/2} = \frac{V_i}{V_f}$$

$$8V_f = V_i$$

$$0 = \Delta S = nC_p \ln \frac{T_f}{T_i} + nR \ln \frac{P_i}{P_f}$$

$$4^{5/2} = \frac{P_f}{P_i}$$

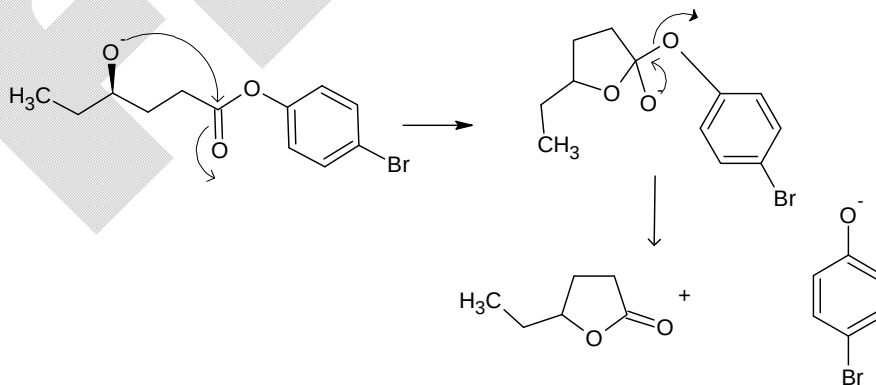
$$32P_i = P_f$$

22. D

Sol. At $\frac{1}{4}$ th and $\frac{3}{4}$ th neutralization points the solution is buffer solution. Upon substitution we can get the ans.

23. ACD

Sol.



24. AC

Sol. Azeotropic mixture has fixed composition.

25. ABC

Sol. $\text{Rate} = k[X][Y]^0 \Rightarrow 2 \times 10^{-8} = k[0.1][0.1]^0 \Rightarrow k = 2 \times 10^{-7} \text{s}^{-1}$

26. ABC

Sol. $\text{CaC}_2 + 2\text{H}_2\text{O} \longrightarrow \text{Ca}(\text{OH})_2 + \text{C}_2\text{H}_2$

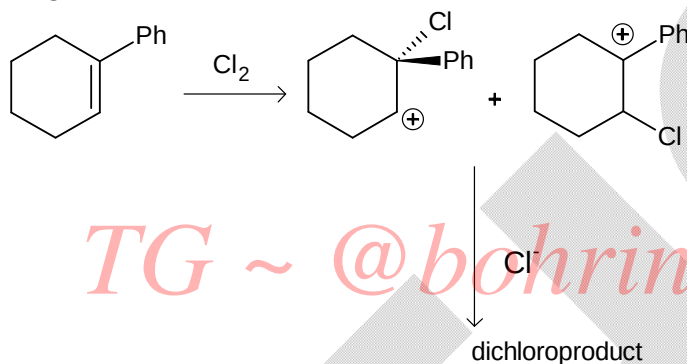
$\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \longrightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$

$\text{Na}_2\text{S} + 2\text{H}_2\text{O} \longrightarrow 2\text{NaOH} + \text{H}_2\text{S}$

$\text{KNO}_3 + \text{H}_2\text{O} \longrightarrow \text{No reaction}$

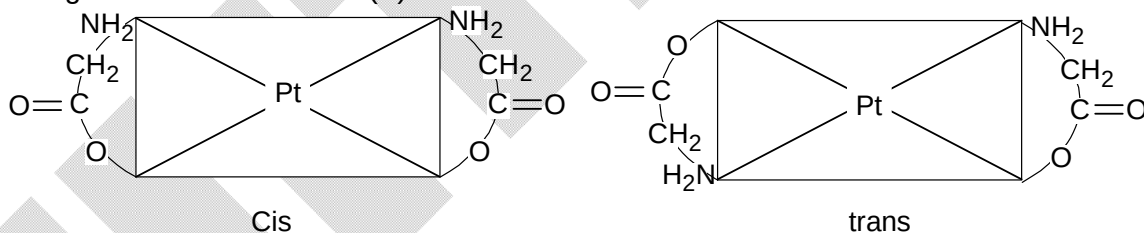
27. ABCD

Sol.

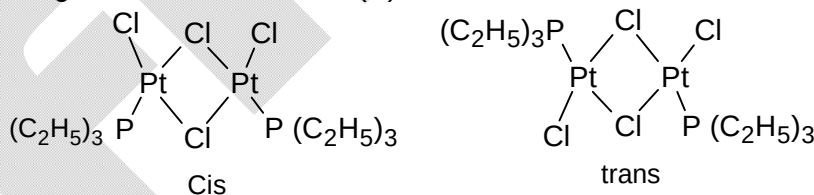


28. BCD

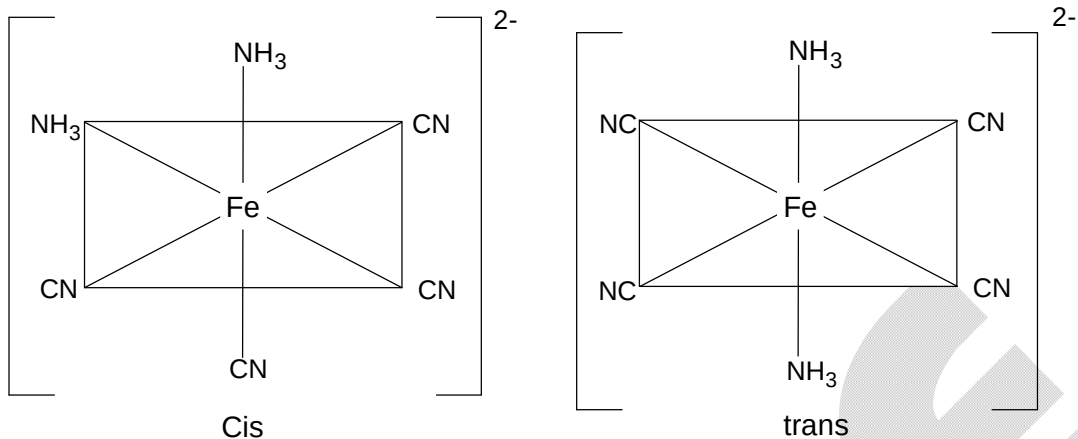
Sol. The geometrical isomers of (B) are



The geometrical isomers of (C) are



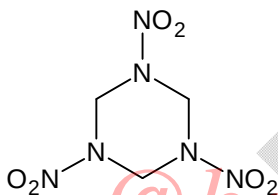
The geometrical isomers of (D) are



Section – B

29. 6

Sol. Compound (Z) is



30. 4

Sol. $(4n + 2)\pi e^-$ rule

31. 3

Sol. $\eta^5 C_2H_5 - 5e^-$
 $2 CO - 4e^-$
 $Mo - 6e^-$
 $Mo - Mo - 3e^-$

32. 3

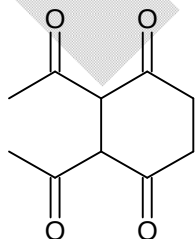
Sol. $S^{2-} + HCl \rightarrow H_2S \xrightarrow[NaOH]{Na_2[Fe(CN)_5NO]} Na_4[Fe(CN)_5(NOS)]$

33. 5

Sol. $\frac{25 \times 10^{-3} (g)}{250} = \frac{16 \times 10^{-3} (g)}{250 - x18}$
 $X = 5$

34. 4

Sol.



35. 4

Sol. (1) ala-phe-leu-gly
(2) ala-phe-gly-leu
(3) ala-leu-phe-gly
(4) ala-leu-gly-phe

36. 3

Sol. It exists in two geometrical isomeric forms in which trans isomer is optically active.

TG ~ @bohring_bot

Mathematics

PART – III

Section – A

37. A

Sol. Since for two square matrices A & B the trace of $AB - BA$ is zero.

$$\therefore \text{let } AB - BA = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

$$\Rightarrow (AB - BA)^2 = k I_2$$

where $k = a^2 + bc$

and odd power of $AB - BA$ is equal to a multiple of this matrix.

i.e. odd power it can not be equal to I_2

$\therefore$ 'n' is even.

38. B

Sol. $A^2 B^2 = r B^4$

$$B^2 A^2 = r B^4$$

$$\Rightarrow A^2 B^2 = B^2 A^2$$

$$\& r B^2 A = r A B^2 = A^3$$

Multiplying $pAB + qBA = I_n$ on the right and then on the left by B, we obtain

$$pAB^2 + qBAB = B$$

$$pBAB + qB^2A = B$$

$$\text{also we have } B^2A = AB^2$$

$$\text{on subtraction } (p - q)(AB^2 - BAB) = 0$$

$$\text{if } p \neq q \text{ then } AB^2 = BAB$$

$$\therefore (p + q)AB^2 = (p + q)B^2A = B$$

$$\Rightarrow (p + q)A^2B^2 = AB$$

$$\& (p + q)B^2A^2 = BA$$

$$\Rightarrow AB = BA$$

which is a contradiction

$$\therefore p = q$$

39. A

Sol. Integrating the given differential equation, we have $\frac{dy}{dx} = \frac{-\cos 3x}{3} + e^x + \frac{x^3}{3} + C_1$ but

$$y_1(0) = 1 \text{ so } 1 = \left(\frac{-1}{3}\right) + 1 + C_1 \Rightarrow C_1 = \frac{1}{3}.$$

$$\text{Again integrating, we get } y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x + C_2$$

$$\text{but } y(0) = 0 \text{ so } 0 = 0 + 1 + C_2 \Rightarrow C_2 = -1. \text{ Thus } y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$$

40. B

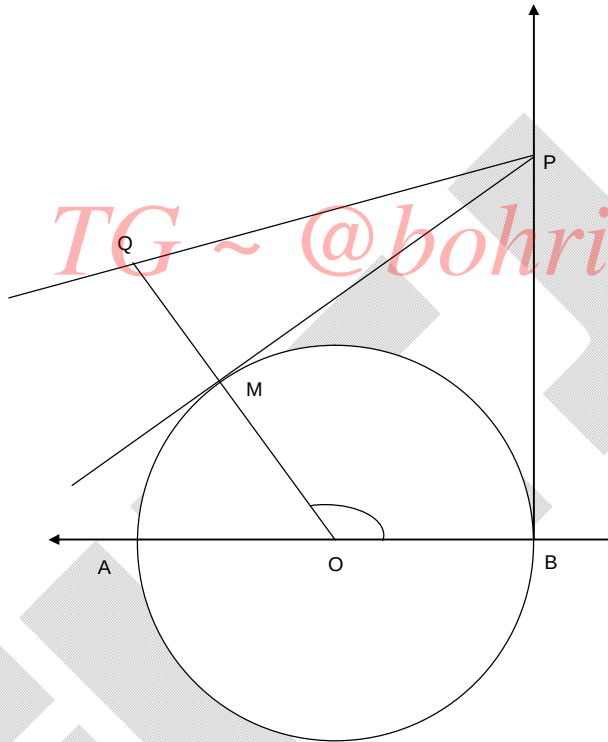
Sol. Let the radius of the circle is equal to 1. Set the origin at B with BA the positive x – semi axis and t the y – axis. If $\angle BOM = \theta$, then $BP = PM = \tan \frac{\theta}{2}$. In triangle PQM, PQ

$$= \frac{\tan \frac{\theta}{2}}{\sin \theta}. \text{ So the coordinates of Q are } \left(\frac{\tan \frac{\theta}{2}}{\sin \theta}, \tan \frac{\theta}{2} \right) = \left(\frac{1}{1 + \cos \theta}, \frac{\sin \theta}{1 + \cos \theta} \right).$$

The x and y coordinates are related as follows:

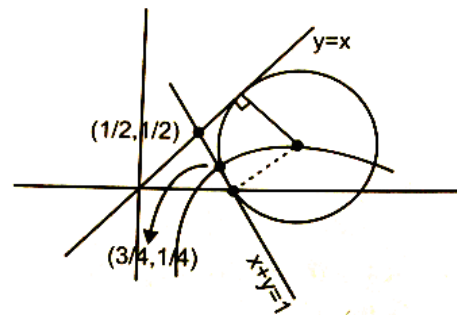
$$\left(\frac{\sin \theta}{1 + \cos \theta} \right)^2 = \frac{1 - \cos^2 \theta}{(1 + \cos \theta)^2} = \frac{1 - \cos \theta}{1 + \cos \theta} = 2 \frac{1}{1 + \cos \theta} - 1.$$

Hence the locus of Q is the parabola $y^2 = 2x - 1$.



41. BC

Sol. $4a = 2 \left(\frac{|0-1|}{\sqrt{2}} \right) = \frac{2}{\sqrt{2}} = \sqrt{2}$



42. BCD

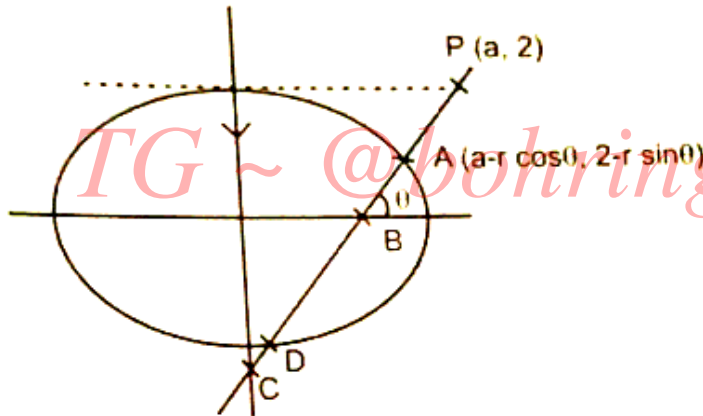
Sol. Put $(a - r \cos \theta, 2 - r \sin \theta)$ to equn. of ellipse

$$\Rightarrow \frac{a^2 + r^2 \cos^2 \theta - 2ar \cos \theta}{9} + \frac{4 + r^2 \sin^2 \theta - 4r \sin \theta}{4} - 1 = 0$$

$$= \frac{4a^2}{4 \cos^2 \theta + 9 \sin^2 \theta}$$

Put $(a - r \cos \theta, 2 - r \sin \theta)$ to equation $xy = 0$

$$\Rightarrow (PA)(PD) = \frac{\frac{a^2}{9}}{\frac{\cos^2 \theta}{9} + \frac{\sin^2 \theta}{4}}$$



$$\Rightarrow 2a + r^2 \sin \theta \cos \theta - (a \sin \theta + 2 \cos \theta)r = 0$$

$$\Rightarrow (PB)(PC) = \frac{2a}{\sin \theta \cos \theta}$$

$$PA \cdot PD = PB \cdot PC$$

$$\Rightarrow \frac{4a^2}{4 \cos^2 \theta + 9 \sin^2 \theta} = \frac{2a}{\sin \theta \cdot \cos \theta}$$

$$\Rightarrow a = \frac{4 \cot \theta + 9 \tan \theta}{2}$$

$$\Rightarrow |a| \geq \sqrt{4 + 9} = 5$$

$$\Rightarrow a \in (-\infty, -5] \cup [5, \infty)$$

43. CD

Sol. If α is a multiple of π , then $I(\alpha) = 0$. Otherwise, use the substitution

$x = \cos \alpha + t \sin \alpha$. The indefinite integral becomes

$$\int \frac{\sin \alpha dx}{1 - 2x \cos \alpha + x^2} = \int \frac{dt}{1+t^2} = \arctan t + C.$$

$$\therefore I(\alpha) = \arctan\left(\frac{1-\cos \alpha}{\sin \alpha}\right) - \arctan\left(\frac{-1-\cos \alpha}{\sin \alpha}\right),$$

where the angles are to be taken between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$. But

$$\frac{1-\cos \alpha}{\sin \alpha} \times \frac{-1-\cos \alpha}{\sin \alpha} = -1.$$

Hence the difference between these angles is $\pm \frac{\pi}{2}$. Notice that the sign of the integral is

the same as the sign of α . Hence $I(\alpha) = \frac{\pi}{2}$ if $\alpha \in (2k\pi, (2k+1)\pi)$ and $-\frac{\pi}{2}$ if

$\alpha \in ((2k+1)\pi, (2k+2)\pi)$ for some integer k .

44. AD

Sol. $k_1u - k_2v = 0$ (i)

$k_1u + k_2v = 0$ (ii)

$\therefore$ equations of bisectors of the angles formed by lines (i) and (ii) are

$$\begin{aligned} & \frac{k_1u - k_2v}{\sqrt{(ak_1 - bk_2)^2 + (k_1b + ak_2)^2}} \\ &= \frac{\pm(k_1u + k_2v)}{\sqrt{(k_1a + bk_2)^2 + (k_1b - ak_2)^2}} \\ &\Rightarrow k_1u - k_2v = \pm(k_1u + k_2v) \quad \text{.....(iii)} \end{aligned}$$

(i) by taking positive sign in (iii), we get

$$k_1u - k_2v = k_1u + k_2v$$

$$2k_2v = 0 \Rightarrow v = 0$$

(ii) by taking negative sign in (iii), we get

$$u = 0$$

45. BD

Sol. After we bring the function into the form $f(x) = \frac{\left(x - 1 + \frac{1}{x}\right)^3}{x^3 - 1 + \frac{1}{x^3}}$, let $x + \frac{1}{x} = s$

$$h(s) = \frac{(s-1)^3}{s^3 - 3s - 1} = 1 + \frac{-3s^2 + 6s}{s^3 - 3s - 1} \text{ over the domain } (-\infty, -2] \cup [2, \infty). \text{ Setting the}$$

first derivative equal to zero yields the equation $3(s-1)(s^3 - 3s^2 + 2) = 0$.

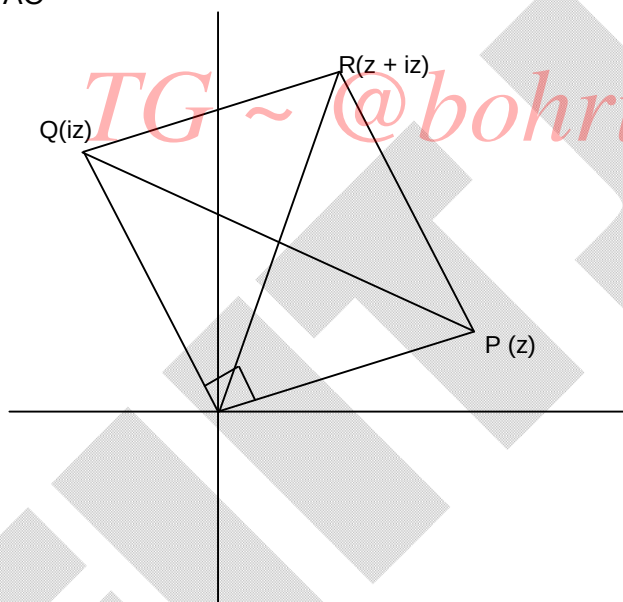
The roots are $s = 1$ (double root) and $s = 1 \pm \sqrt{3}$. Of these, only $s = 1 + \sqrt{3}$ lies in the domain of the function.

We compute

$$\lim_{x \rightarrow \pm\infty} h(s) = 1, h(2) = 1, h(-2) = 9, h(1 + \sqrt{3}) = \frac{\sqrt{3}}{2 + \sqrt{3}}.$$

46. AC

Sol.



Section – B

47. 3

Sol. Equation of the chord of contact of a point $P(3 \sec \theta, 2 \tan \theta)$ on the hyperbola with respect to the circle is $(3 \sec \theta)x + (2 \tan \theta)y = 9$ (1)

Let $M(h, k)$ be the mid point of (1), then equation of (1) in terms of the mid – point is $hx + ky = h^2 + k^2$ (2)

Since (1) and (2) represent the same line.

$$\sec \theta = \frac{3h}{h^2 + k^2}, \tan \theta = \frac{9k}{2(h^2 + k^2)}$$

$\Rightarrow$ locus of (h, k) is

$$\Rightarrow \frac{9x^2}{(x^2 + y^2)^2} - \frac{81y^2}{4(x^2 + y^2)^2} = 1$$

or $4(x^2 + y^2)^2 = bx^2 - cy^2$

$\Rightarrow a = 4, b = 36, c = 81$

$a^2 + b^2 + c^2 = 16 + 1296 + 6561 = 7873$

48. 5

Sol. Let $r = \lambda b + \mu c$ and $c = \pm(xi + yj)$. Since c and b are perpendicular, we have

$4x + 3y = 0 \quad \Rightarrow c = \pm x \left(i - \frac{4}{3}j \right)$

$\pm 1 = \text{proj. of } r \text{ on } b = \frac{r \cdot b}{|b|} = \frac{(\lambda b + \mu c) \cdot b}{|b|} = \frac{\lambda b \cdot b}{|b|}$

$[\because b \cdot c = 0]$

$= \lambda |b| = 5\lambda$. Hence $\lambda = \frac{1}{5}$

Also, $\pm 2 = \text{proj. of } r \text{ on } c = \frac{r \cdot c}{|c|}$
 $= \frac{(\lambda b + \mu c) \cdot c}{|c|} = \mu |c| = \frac{5}{3} \mu x$

Thus, $\mu x = \pm \frac{6}{5}$. Therefore,

$r = \frac{1}{5}(4i + 3j) + \frac{6}{5} \left(i - \frac{4}{3}j \right) = \pm(2i - j)$

$r = \frac{1}{5}(4i + 3j) - \frac{6}{5} \left(i - \frac{4}{3}j \right) = \pm \left(-\frac{2}{3}i + \frac{11}{5}j \right)$

Thus there are four such vectors

$\sum_{i=1}^4 |r_i|^2 = 2|2i - j|^2 + 2 \left| -\frac{2}{3}i + \frac{11}{5}j \right|^2 = 20$

49. 1

Sol. Let $f(x) = \sin x + \tan x - 2x$. Then $g(x) = f'(x)$

$= \cos x + \sec^2 x - 2$

$\Rightarrow g'(x) = -\sin x + 2\sec^2 x \tan x$

$= \sin x \left(-1 + \frac{2}{\cos^3 x} \right)$

$= \sin x \left(\frac{1 - \cos^3 x + 1}{\cos^3 x} \right)$

Since for $0 < x < \frac{\pi}{2}$, we have $0 < \cos^3 x < 1$, g is an increasing function. Hence $g(x) > g(0)$, i.e. $\cos x + \sec^2 x - 2 > 0$. Therefore f is an increasing function, so $f(x) > f(0)$ for $0 < x < \frac{\pi}{2}$. Hence $\sin x + \tan x > 2x$. Thus $g\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} + 1$

50. 1

Sol. The inverse of 2×2 matrices $C = [C_{ij}]_{2 \times 2}$ with integer entries is a matrix with integer entries if and only if $|C| = \pm 1$

$$\left\{ C^{-1} = \frac{\text{adj}(C)}{|C|} \& |C^{-1}| = \frac{1}{|C|} \right\}$$

So lets take a polynomial.

$$P(x) = |A + xB|$$

as per the problem $P(0), P(1), P(2), P(3), P(4) \in \{-1, 1\}$

three of these must be same
and $P(x)$ has degree at most 2.

$\therefore$ it is constt.

$$\therefore |A + xB| = \pm 1, \forall x$$

$$\therefore |A + 5B| = \pm 1$$

51. 2

Sol. $a_1 + a_3 + \dots + a_{99} = 50$

$$\Rightarrow a + (a + 2d) + (a + 4d) + \dots + (a + 98d) = 50$$

$$\Rightarrow 50a + 2d(1 + 2 + \dots + 49) = 50$$

$$\Rightarrow 50a + \frac{2d(50)(49)}{2} = 50$$

$$\Rightarrow a + 49d = 1$$

$$= -a_1 - a_3 + a_5 + a_7 - a_9 - a_{11} + \dots + a_{93} + a_{95} - a_{97} - a_{99}$$

(26 negative terms and 24 positive)

$$\Rightarrow |-a_1 - a_{99}| = |-2a - 98d| \Rightarrow |-2| \Rightarrow 2$$

52. 1

Sol. Given $a + b + c = 1$ (1)

$$9a + 3b + c = 7 \quad \dots\dots\dots(2)$$

$$18 < 25a + 5b + c < 22 \quad \dots\dots\dots(3)$$

$$\Rightarrow \text{From above (1), (2) and (3)} \quad 4 < 7a - b - c < 8$$

$$4 < 7a - b + a - 1 < 8$$

$$5 < 8a < 9$$

$$a = 1, b = -1, c = 1$$

$$a = 1, b = -1, c = 1$$

For question (2) $h(x) = \ln(x^2 - x + 1) - x$

53. 4

Sol. Let $Z_1 = z_1 e^{i\theta_1}$ and

$$z_2 = r_2 e^{i\theta_2} \Rightarrow \frac{\bar{z}_1 \bar{z}_2 + \bar{z}_1 z_2 + z_1 \bar{z}_2 + \bar{z}_1 \bar{z}_2}{|z_1 z_2|} = 2 \cos(\theta_1 + \theta_2) + 2 \cos(\theta_1 + \theta_2)$$

54. 3

Sol. $\frac{a+b+c}{b-a} = \frac{a+b-a+a+c}{b-a}$

$$= 1 + \frac{2a+c}{b-a}$$

Now: $f(x) = ax^2 + bx + c$

Given $f(x) \geq 0 \forall x \quad f(-2) \geq 0$

$$4a - 2b + c \geq 0$$

$$2a + c \geq 2(b-a)$$

$$\frac{2a+c}{b-a} \geq 2$$

$$\therefore \frac{a+b+c}{b-a} \geq 3$$

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – V
PAPER –1
TEST DATE: 08-02-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 06, 19 – 24, 37 – 42): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A (07 – 10, 25 – 28, 43 – 46): This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

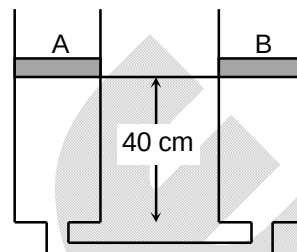
Physics

PART – I

Section – A (Maximum Marks: 24)

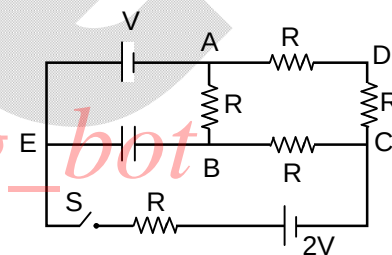
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. Two vertical conducting cylinders A and B of different cross sections are connected by a thin tube as shown. An ideal gas is trapped in the cylinders by two airtight pistons of masses 1.0 kg and 2.0 kg respectively. Initially the pistons are at the same height 40cm above the base of the cylinders. If an additional mass of 10 g were gently placed on the piston in the cylinder A, which of the following statements correctly describe the new steady state ultimately reached.



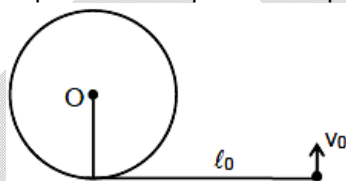
- (A) Pressure of the gas has been increased.
 (B) Pressure of the gas remains unchanged.
 (C) Height difference between the pistons becomes 30 cm.
 (D) Height difference between the pistons becomes 60 cm.

2. Five equal resistances, a capacitor and two batteries are connected in a circuit as shown in the figure. The capacitor is initially uncharged and each battery has negligible internal resistance. It is given that $R = 1\Omega$, $C = 1\mu F$ and $V = 8$ volt. On the basis of above information answer the following questions, if switch 'S' is closed at $t = 0$. (Assume $e^{-1} = 0.37$, if required)



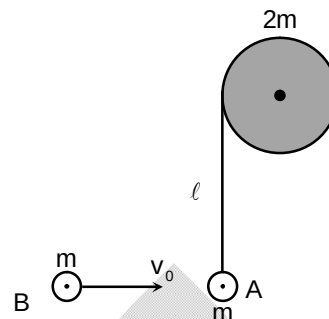
- (A) Current in the branch of capacitor at $t = 5/8 \mu s$ is (approximately) 6 A
 (B) Current in the branch AB at $t = 5/8 \mu s$ is (approximately) 1.7 A
 (C) Potential difference between points A and C at $t = 5/8 \mu s$ is (approximately) 2.5 V
 (D) Potential difference between points A and C at $t = 5/8 \mu s$ is (approximately) 6 V

3. A point mass m is attached to an ideal string, which in turns is wrapped on a vertical pole. Point mass is kept on a smooth horizontal surface and string is horizontal. A horizontal velocity is imparted to the point mass perpendicular to length of string.



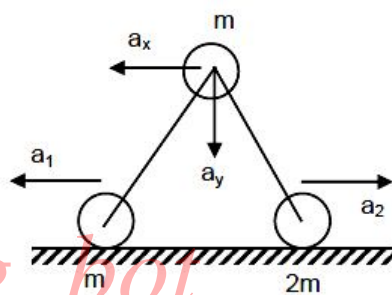
- (A) Speed of mass will become twice of initial value if length of string between the point mass and the pole becomes half of the original value.
 (B) Kinetic energy of mass increases.
 (C) Work done by tension on the point mass is zero.
 (D) Energy of the mass remains constant.

4. In the adjacent figure, a uniform disc of mass $2m$ and radius $\ell/2$ is lying at rest on a smooth horizontal surface. A particle 'A' of mass m is connected to a light string of length ℓ , whose other end is attached to the circumference of the disc. Initially string is just taut and tangential to the disc, particle A is at rest. In the same horizontal plane another particle B of same mass m moving with velocity v_0 perpendicular to string collides elastically with A. Just after impact which of the following statements will be true



- (A) Tension in the string is $\frac{2mv_0^2}{5\ell}$ (B) Acceleration of the centre of the disc is $\frac{v_0^2}{5\ell}$
 (C) Initial radius of curvature of 'A' is $\frac{5\ell}{2}$ (D) Acceleration of the centre of the disc is $\frac{2v_0^2}{5\ell}$

5. Three masses are joined using pin joints (can move freely). Length of both rods are identical and system starts from rest. Accelerations of the balls are shown in figure. Angle between the rods at this moment is 90° . The horizontal surface is smooth. Choose the correct option(s).



- (A) $2a_2 = a_1 + a_x$
 (B) $2a_y = a_1 + a_2$
 (C) $2a_x = a_1 - a_2$
 (D) $3a_1 = 5a_2$

6. A particle of mass m is subjected to two forces

$$\vec{F}_1 = f(r) \frac{\vec{r}}{|\vec{r}|} \text{ and } \vec{F}_2 = -\alpha \vec{v}$$

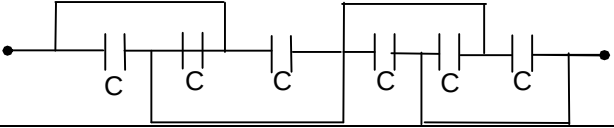
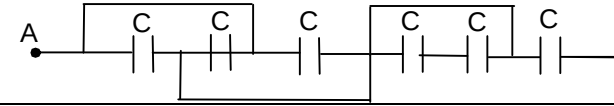
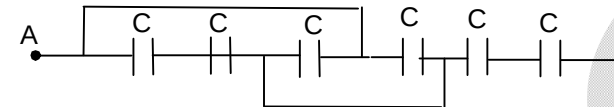
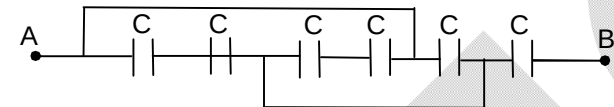
Where $f(r)$ is a real valued function depends on the distance of particle from origin, $\vec{r}$ is position vector of particle and α is positive constant. If at $t = 0$, $\vec{v}$ vector is in x-y plane and angular momentum of the particle about the origin is L_0 . Choose the correct option(s).

- (A) Plane of motion of particle does not change.
 (B) Angular momentum of particle about the origin is given by $L_0 e^{\frac{-\alpha t}{m}}$
 (C) Angular momentum of particle about the origin is remains constant.
 (D) The value of angular momentum about the origin will depend on $f(r)$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

7. Match List I with List II:

List I (Circuit)		List II (C_{eq})	
(I)		(P)	$\frac{3C}{4}$
(II)		(Q)	$\frac{5C}{12}$
(III)		(R)	$\frac{2C}{3}$
(IV)		(S)	$\frac{3C}{2}$

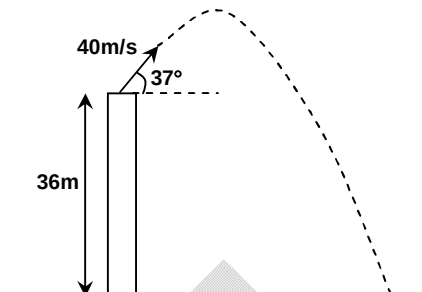
- (A) I – S; II – Q; III – R; IV – P
 (B) I – S; II – P; III – Q; IV – R
 (C) I – S; II – P; III – R; IV – Q
 (D) I – S; II – Q; III – P; IV – R

8. One mole of an ideal gas whose adiabatic exponent equals γ , undergoes a process $P = P_0 + \alpha/v$, where P_0 and α are positive constants. Its volume increases from v_1 to v_2 .

List-I		List-II	
(I)	Heat capacity of gas as a function of its volume	(P)	$\frac{P_0(V_2 - V_1)}{\gamma - 1}$
(II)	Internal energy increment of a gas	(Q)	$\frac{P_0(v_2 - v_1)}{\gamma - 1} + \alpha \ln \frac{v_2}{v_1}$
(III)	The work performed by it	(R)	$P_0(v_2 - v_1) + \alpha \ln \frac{v_2}{v_1}$
(IV)	The amount of heat transferred	(S)	$\frac{\gamma R}{\gamma - 1} + \frac{\alpha R}{p_0 v}$

- (A) I – S; II – P; III – R; IV – Q
 (B) I – P; II – Q; III – R; IV – S
 (C) I – P; II – S; III – R; IV – Q
 (D) I – S; II – P; III – Q; IV – S

9. Consider a situation in which a projectile is fired at 40 m/s from the top of a tower in absence of air resistance $t = 0$.



List-I		List-II	
(I)	Time at which particle moves at right angles with the initial direction	(P)	2.4 s
(II)	time at which kinetic energy is least	(Q)	6 s
(III)	Time at which it strikes the ground	(R)	20/3 s
(IV)	Time when the $\vec{v}$ make 45° with horizontal	(S)	5.6 s
		(T)	7.2 s

- (A) (I) $\rightarrow$ (R); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (Q) (B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (R); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (S)
 (C) (I) $\rightarrow$ (P); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (R); (IV) $\rightarrow$ (S) (D) (I) $\rightarrow$ (R); (II) $\rightarrow$ (P); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (S)

10. A neutron moving with kinetic energy K collides with a hydrogen atom at rest in ground state. Assume mass of neutron to be equal to that of hydrogen atom. Match the kinetic energy of neutron with the possible phenomena mentioned in column II. Energy of hydrogen in ground state is -13.6 eV

List-I		List-II	
(Kinetic energy of colliding neutron)			
I	$K = 10.2$ eV	(P)	Elastic collision may take place
II	$K = 13.6$ eV	(Q)	Inelastic collision may take place
III	$K = 20.4$ eV	(R)	Perfectly inelastic collision may take place
IV	$K = 22.0$ eV	(S)	Hydrogen atom may get ionised

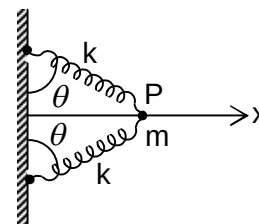
- (A) I-P; II-Q; III-R; IV-S
 (C) I-P; II-P; III-P; IV-Q

- (B) I-P; II-P; III-P, R; IV-P, Q
 (D) I-P; II-P; III-R; IV-Q

Section – B (Maximum Marks: 24)

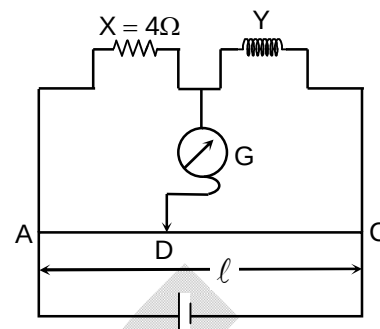
This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

11. Two identical springs each of stiffness k are welded at a point P. If a particle of mass m is welded at P. The period of oscillation of the particle in the direction of x is $\frac{\pi}{\sin \theta} \sqrt{\frac{Ym}{k}}$. Find Y

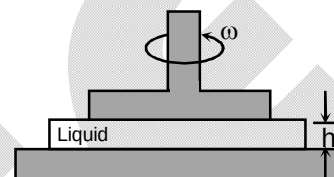


12. A body executing SHM with time period T and amplitude A . When it is at $\frac{A}{2}$ from equilibrium position moving toward right. It receives an impulse which doubles its velocity without changing direction. Time taken by the block to reach the right extreme position is $t = T \cos^{-1} \left(\frac{K}{\sqrt{13}} \right)$. Find K .

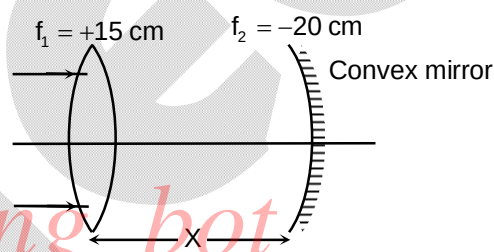
13. Figure shows a meter bridge wire AC has a uniform cross-section. The length of wire AC is 100 cm. X is a standard resistor of $4\ \Omega$ and Y is a coil. When Y is immersed in melting ice the null point is at 40 cm from point A. When the coil Y is heated to 100°C , a $12\ \Omega$ resistor has to be connected in parallel with Y in order to keep the bridge balanced at the same point.



14. A circular disc of a diameter d is slowly rotated in a liquid of large viscosity μ at a small distance h from a fixed surface. An expression for torque T necessary to maintain an angular velocity ω is $\frac{\mu\pi d^4\omega}{Xh}$. Find the value of X .



15. A set of parallel rays incident on the bi-convex lens. If after two refractions and one reflection final image formed at 15 cm left of lens, the value of $\frac{X}{9}$ is



16. Three SHM in the same direction in having same amplitude and same period are superposed. If each differ in phase from the next by 45° . Then, find the energy associated with resulting motion taking energy associated with each of given SHM as 1 J.
17. Two rods of different materials having coefficient of thermal expansion α_1 , α_2 and Young's modulus Y_1 , Y_2 are fixed between two massive walls. The rods are heated such that they undergo the same increase in temperature. If $\alpha_1 : \alpha_2 = 2 : 3$, then thermal stresses in rods are equal, then the ratio $Y_1 : Y_2$ is _____.
18. A student determines a dimensionless quantity,

$$B = \frac{e^n}{Z\epsilon_0 hc}$$

Where, e = electric charge,
 ϵ_0 = permittivity of vacuum,
 h = Plank's constant
 and c = speed of light.
 Find the value of n .

Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

19. An isotone of $^{76}_{32}\text{Ge}$ is

(A) $^{77}_{32}\text{Ge}$

(C) $^{77}_{34}\text{Se}$

(B) $^{77}_{33}\text{As}$

(D) $^{78}_{34}\text{Se}$
20. For a compound AB crystallises in a rock salt structure with edge length a of the unit cell,

(A) the minimum distance between A and A is $a/\sqrt{2}$

(B) the minimum distance between B and B is $a/\sqrt{2}$

(C) the minimum distance between A and B is $a/2$

(D) the density of the crystal is given by $\rho = \frac{4}{a^3} \frac{M_{AB}}{N_A}$
21. Which of the following statements are not correct

(A) For an ideal binary liquid solution, the plot of $1/x_A$ versus $1/y_A$ (where x_A and y_A are the mole fractions of A in liquid and vapour phases, respectively) is linear.

(B) According to Raoult's law, the relative lowering of vapour pressure is equal to the mole fraction of solvent in the solution.

(C) The cryoscopic constant K_f is property of solvent.

(D) The value of van't Hoff factor for CH_3COOH in benzene will be 2
22. Which of the following statements are correct?

(A) Linear overlap of two atomic p-orbitals leads to a sigma bond

(B) The bond angle $\text{H} - \text{N} - \text{H}$ in NH_3 is greater than the bond angle $\text{H} - \text{As} - \text{H}$ in AsH_3 .

(C) Anhydrous HCl is a bad conductor of electricity but aqueous HCl is a good conductor

(D) o-nitrophenol is steam volatile whereas p-nitrophenol is not.
23. Which of the following reactions give an alkyne?

(A) sodium fumarate $\xrightarrow{\text{Kolbe's electrolysis}}$

(B) $\text{CH}_3\text{CBr}_2\text{CHBr}_2 \xrightarrow{\text{Zn/alcohol}/\Delta}$

(C) $\text{CH}_3\text{CH}_2\text{CHBr}_2 \xrightarrow{\text{NaNH}_2/\Delta}$

(D) $\text{CH}_3\text{CHBr} - \text{CHBr} \xrightarrow{\text{NaNH}_2}$
24. Which of the following reactions give n-propyl alcohol as the major product?

(A) $\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{H}_2\text{O} \xrightarrow{\text{H}^+}$

(B) $\text{HCHO} \xrightarrow[\text{(ii) hydrolysis}]{\text{(i) CH}_3\text{CH}_2\text{MgBr}}$

(C) $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{\text{cold(aq)KMnO}_4}$

(D) $\text{CH}_2 = \text{CH} - \text{CHO} + \text{H}_2 \xrightarrow{\text{Ni}}$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

25. Match the entries in List-I for the spectral series of hydrogen atom with those given in List-II. The symbols n_1 and n_2 are the quantum level in the expression

List -I		List-II	
(I)	Lyman series	(P)	$n_1 = 2$ and $n_2 = 3, 4, 5, \dots$
(II)	Balmer series	(Q)	$n_1 = 4$ and $n_2 = 5, 6, 7$
(III)	Paschen series	(R)	$n_1 = 5$ and $n_2 = 6, 7, \dots$
(IV)	Pfund series	(S)	$n_1 = 1$ and $n_2 = 2, 3, 4, \dots$
		(T)	$n_1 = 3$ and $n_2 = 4, 5, 6, \dots$

Which of the following is correct match

- (A) I–S;
(C) III–R
(B) II–Q
(D) IV – T

26. Match the list -I and List- II

List -I		List-II	
(I)	Electrophilic substitution	(P)	$\text{C}_6\text{H}_5\text{N}_2\text{Cl} + \text{C}_6\text{H}_5\text{OH} \xrightarrow{\text{NaOH}}$
(II)	Nucleophilic substitution	(Q)	Toluene + NBS $\longrightarrow$
(III)	Free radical substitution	(R)	Dow's process
(IV)	Coupling reaction	(S)	Phenol $\longrightarrow$ picric acid
		(T)	$\text{C}_6\text{H}_5\text{ONa} + \text{CH}_3\text{Br} \longrightarrow$

Which of the following is correct match

- (A) I–R,
(C) III–Q
(B) II–P
(D) IV–T

27. Match the list -I and List- II

List -I		List-II	
(I)	$2p\pi - 3 p\pi$	(P)	$\text{F}_2\text{C} = \text{O}$
(II)	$3p\pi - 3 d\pi$	(Q)	$\text{F}_3\text{P} = \text{O}$
(III)	$2p\pi - 3 d\pi$	(R)	$\text{F}_3\text{P} = \text{S}$
(IV)	$2p\pi - 2 p\pi$	(S)	$\text{F}_2\text{S} = \text{O}$
		(T)	$\text{F}_2\text{C} = \text{S}$

Which of the following is correct match

- (A) I–R,
(C) III–Q
(B) II–T
(D) IV–P

28. List-II give the values of γ ($-\text{C}_{p,m}/\text{C}_{v,m}$) at high temperature of gases which are shown in column I. Identify the gas with the corresponding γ values.

List -I		List-II	
(I)	Monatomic	(P)	1.154
(II)	Diatomic	(Q)	1.286
(III)	Linear Triatomic	(R)	1.167
(IV)	Nonlinear Triatomic	(S)	1.407
		(T)	1.667

Which of following is correct match

- (A) I–Q
(C) III–P
(B) II–T
(D) IV–R

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

29. The ionization potential of hydrogen atom is 13.6 eV. The energy in electron volt required to remove an electron in the $n = 2$ state of the hydrogen atom is
30. The density of a gas is 1.5 g L^{-1} at STP. Its density at 27°C and 1000 Torr would be
31. Ammonium chloride crystallizes in a body-centred cubic lattice with edge length of 387 pm. If the size of NH_4^+ is 154 pm, the size of Cl^- ion would be
32. A container of volume 'V' litre contains an equilibrium mixture which consists of 2 mol each of PCl_5 , PCl_3 and Cl_2 . The equilibrium pressure is 3 atm and temperature is 'T' kelvin. A certain amount of Cl_2 gas is now introduced into the container keeping the temperature and pressure constant, until the equilibrium volume becomes '2V' litre. How many moles of Cl_2 were added?
33. The conductivity of a saturated solution of silver oxalate is $4.5 \times 10^{-5} \text{ S cm}^{-1}$. If its $K_{sp} = 1.35 \times 10^{-11} \text{ M}^3$, the molar conductivity of the saturated solution would be
34. How many grams of potassium perchlorate (KClO_4) must be mixed with 108 g of water in order to obtain a solution, which has a mole fraction of KClO_4 equal to $\frac{1}{7}$?
35. Calculate effective nuclear charge on last electron of Cl. (using Slater rule)
36. The number of isomers possible for complex $\text{K}_2[\text{PtClBr}_2\text{SCN}]$:

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

37. A point $P(\sqrt{3}, 1)$ moves on the circle $x^2 + y^2 = 4$ and after covering a quarter of the circle leaves it tangentially. The equation of a line along which the point moves after leaving the circle is
 (A) $y = \sqrt{3}x + 4$ (B) $\sqrt{3}y = x + 4$
 (C) $\sqrt{3}y = x - 4$ (D) $y = \sqrt{3}x - 4$
38. A chord PP' of a parabola cuts the axis of the parabola at O. The feet of the perpendiculars from P and P' on the axis are M and M' respectively. If V is the vertex then VM, VO, VM' are in
 (A) AP (B) GP
 (C) HP (D) none of these
39. α , β and γ are three angles given by $\alpha = 2\tan^{-1}(\sqrt{2}-1)$, $\beta = 3\sin^{-1}\frac{1}{\sqrt{2}} + \sin^{-1}\left(-\frac{1}{2}\right)$ and $\gamma = \cos^{-1}\frac{1}{3}$. Then
 (A) $\alpha > \beta$ (B) $\beta > \gamma$
 (C) $\alpha < \gamma$ (D) none of these
40. $\int_4^8 (\sqrt{\sqrt{16x-64}+x} + \sqrt{x-\sqrt{16x-64}}) dx$
 (A) $2 \int_4^6 (\sqrt{\sqrt{16x-64}+x} + \sqrt{x-\sqrt{16x-64}}) dx$ (B) $2 \int_8^{10} (\sqrt{\sqrt{16x-64}+x} + \sqrt{x-\sqrt{16x-64}}) dx$
 (C) $2 \int_7^9 (\sqrt{\sqrt{16x-64}+x} + \sqrt{x-\sqrt{16x-64}}) dx$ (D) 16
41. If M is the skew symmetric matrix of order $n \times n$, then
 (A) $\det(M - I) = \det(M + I)$ (B) $\det(M - I) = -\det(M + I)$ if n is odd
 (C) $\det(M - I) = \det(M + I)$ if n is even (D) $\det(I - M)(I + M)^{-1} = 1$, $\det(I + M) \neq 0$
42. If the unit vectors $\vec{a}$ and $\vec{b}$ are inclined at an angle 2θ such that $|\vec{a} - \vec{b}| < 1$ and $0 \leq \theta \leq \pi$, then θ lies in the interval
 (A) $[0, \pi/6]$ (B) $(5\pi/6, \pi]$
 (C) $(\pi/6, \pi/2]$ (D) $[\pi/2, 5\pi/6]$

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.

43. Match the List– I & List–II

List-I		List-II	
I	A straight line with negative slope passing through (1, 4) meets the coordinate axes at A and B. The minimum length of OA + OB, O being the origin, is	P	$5\sqrt{2}$
II	If the point P is symmetric to the point Q(4, -1) with respect to the bisector of the first quadrant, then the length of PQ is	Q	$3\sqrt{2}$
III	On the portion of the straight line $x + y = 2$ between the axis a square is constructed away from the origin, with this portion as one of its sides. If d denotes the perpendicular distance of a side of this square from the origin then the maximum value of d is	R	$9/2$
IV	If the parametric equation of a line is given $x = 4 + \lambda/\sqrt{2}$ and $y = -1 + \sqrt{2}\lambda$, where λ is a parameter, then the intercept made by the line on the x-axis is	S	9

(A) I → R; II → P; III → Q; IV → S

(B) I → S; II → P; III → Q; IV → R

(C) I → S; II → R; III → Q; IV → P

(D) I → P; II → S; III → Q; IV → R

44. Match the List–I & List–II

List-I		List-II	
I	The smallest integer greater than $\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi}$ is	P	10
II	Let $3^a = 4$, $4^b = 5$, $5^c = 6$, $6^d = 7$, $7^e = 8$, and $8^f = 9$. Then the value of the product (abcdef) is	Q	3
III	Characteristic of the logarithm of 2008 to the base 2 is	R	1
IV	If $\log_2(\log_2(\log_3 x)) = \log_2(\log_3(\log_2 y)) = 0$, then the value of (x - y) is	S	2

(A) I → R; II → S; III → P; IV → Q

(B) I → S; II → Q; III → P; IV → R

(C) I → Q; II → S; III → P; IV → R

(D) I → Q; II → R; III → P; IV → S

45. Match the following List-I with List–II

List-I		List-II	
I	$\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{e^{x^2} - e^x + x}$ equals to	P	1
II	If the value of $\lim_{x \rightarrow 0^+} \left(\frac{(3/x) + 1}{(3/x) - 1} \right)^{1/x}$ can be expressed in the form of $e^{p/q}$, where p and q are relative prime, then (p + q) is equal to	Q	2
III	$\lim_{x \rightarrow 0} \frac{\tan^3 x - \tan x^3}{x^5}$ equals to	R	4
IV	$\lim_{x \rightarrow 0} \frac{x + 2 \sin x}{\sqrt{x^2 + 2 \sin x + 1} - \sqrt{\sin^2 x - x + 1}}$ equals to	S	5

(A) I → R; II → S; III → P; IV → Q

(B) I → Q; II → S; III → S; IV → P

(C) I → R; II → P; III → S; IV → Q

(D) I → Q; II → S; III → P; IV → R

46. Let Π be the plane parallel to y-axis and containing the points (1, 0, 1) and (3, 2, -1). $A \equiv (4, 0, 0)$ and $B \equiv (6, 0, -2)$ are two points and $P \equiv (x_0, y_0, z_0)$ is a variable point on the plane $\Pi = 0$. Match List – I with List – II and select the correct answer using the code given below the list.

List - I		List - II	
I	If the equation of the plane $\Pi = 0$ is $x + ay + bz = c$, then $ a + b + c $ is	P	16
II	If $(PA + PB)$ is minimum, then $ 4x_0 + y_0 + 2z_0 $ is	Q	12
III	If $ PA + PB \in [0, \sqrt{N}]$, then N is	R	3
IV	If the reflection of the line AB in the plane $\Pi = 0$ is $\frac{x-2}{1} = \frac{y-\alpha}{0} = \frac{z+\beta}{-1}$, then $(\alpha^4 + \beta^4)$	S	8
		T	10

 (A) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ S

 (B) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ P

 (C) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ P

 (D) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ P

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

47. In the coordinate plane, the region M consists of all the points (x, y) satisfying the inequalities $y \geq 0$, $y \leq x$ and $y \leq 2 - x$ simultaneously. The region N, which varies with the parameter t, consists of all the points (x, y) satisfying the inequalities $t \leq x \leq t+1$ and $0 \leq t \leq 1$ simultaneously. The area of $M \cap N$ is $-k_1 t^2 + k_2 t + \frac{1}{2}$, then $k_1 + k_2$ is _____.
48. The slope of the tangent to the curve represented by $x = t^2 + 3t - 8$ and $y = 2t^2 - 2t - 5$ at the point M(2, 1) is $\left(\frac{\lambda}{7}\right)$, then value of λ is ?
49. Let $(1 + 3x + 4x^2) = a_0 + a_1x + a_2x^2 + \dots + a_{20}x^{20}$. Then value of a_3 is –
50. Find the value of $f\left(\frac{\pi}{6}\right)$, where $f(\theta) = \begin{vmatrix} \cos^2 \theta & \cos \theta \sin \theta & -\sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta & \cos \theta \\ \sin \theta & -\cos \theta & 0 \end{vmatrix}$
51. If α, β, γ are such that $\alpha + \beta + \gamma = 4$, $\alpha^2 + \beta^2 + \gamma^2 = 6$, $\alpha^3 + \beta^3 + \gamma^3 = 8$, then the value of $[\alpha^4 + \beta^4 + \gamma^4]$ must be equal to (where [.] denotes the greatest integer function)
52. Find the Coefficient of x^{103} in $(1 + x + x^2 + x^3 + x^4)^{199} (x - 1)^{201}$.
53. If $\left(1 + \frac{1}{x}\right)^{x+1} = \left(1 + \frac{1}{2021}\right)^{2021}$, then $|x|$ is equal to
54. If two events A and B are such that $P(\bar{A}) = 0.3$, $P(B) = 0.4$, $P(A \cap \bar{B}) = 0.5$, then the value of $P(B/(A \cup \bar{B}))$ is

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – V
PAPER –2
TEST DATE: 08-02-2023

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 – 10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

Physics

PART – I

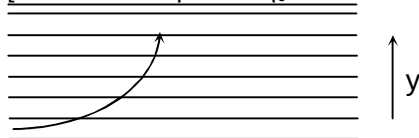
Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. The refractive index of the medium within a certain region, $x > 0$, $y > 0$, changes with y . A thin light ray travelling in the x -direction strikes the medium at right angles and moves through the medium along a circular arc.

How does the refractive index depend on y ?

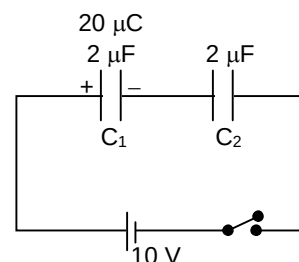
[R : Radius of path & η_0 : Initial refractive index]



- (A) $\frac{2R}{R-y} \eta_0$ (B) $\frac{R}{R-y} \frac{\eta_0}{2}$
 (C) $\frac{R+y}{R} \eta_0$ (D) $\frac{R}{R-y} \eta_0$
2. A mass is suspended from a wire and is pulled along the length of wire, resulting in oscillations of time period T_1 . The same mass is next attached to wire of the same material and length but double the cross-sectional area. The time period this time is T_2 . Then T_1/T_2 is equal to
 (A) 1 : 2 (B) 2 : 1
 (C) 1 : $\sqrt{2}$ (D) $\sqrt{2}$: 1

3. Initially, C_1 is charged but C_2 is uncharged. When switch is closed, charges on C_1 and C_2 are

- (A) 20 μC , 0
 (B) 10 μC , 10 μC
 (C) 30 μC , 10 μC
 (D) 0, 20 μC



4. A standing wave is produced in a vapour of an element of atomic mass 254 (g mol^{-1}) at 400 K. Nodes are found to be 6.77 cm apart when frequency of source was 1000 Hz. The vapour used for the experiment is ($R = 8.31 \text{ JK}^{-1}\text{mol}^{-1}$)
 (A) monatomic (B) diatomic
 (C) polyatomic (D) none of these

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A particle of mass m is moving in a field where the potential energy is given by $U(x) = U_0(1 - \cos ax)$, where U_0 and a are constants and x is the displacement from mean position. Then (for small oscillations)

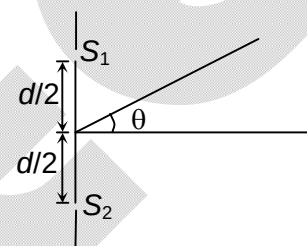
(A) the time period is $T = 2\pi\sqrt{\frac{m}{aU_0}}$

(B) the speed of particle is maximum at $x = 0$

(C) the amplitude of oscillations is $\frac{\pi}{a}$

(D) the time period is $T = 2\pi\sqrt{\frac{m}{a^2U_0}}$

6. In an interference arrangement similar to Young's double slit experiment, the slits S_1 and S_2 are illuminated with coherent microwave sources, each of frequency 10^6 Hz. The sources are synchronized to have zero phase difference. The slits are separated by a distance $d = 150.0$ m. The intensity $I(\theta)$ is measured as a function of θ , where θ is defined as shown. If I_0 is maximum intensity, then $I(\theta)$ for $0 \leq \theta \leq 90^\circ$ is given by



(A) $I(\theta) = \frac{I_0}{2}$ for $\theta = 30^\circ$

(B) $I(\theta) = \frac{I_0}{4}$ for $\theta = 90^\circ$

(C) $I(\theta) = I_0$ for $\theta = 0^\circ$

(D) $I(\theta)$ is constant for all values of θ

7. A uniform square plate of mass m and edge a initially at rest starts rotating about one of the edge under the action of a constant torque τ . Then at the end of the 5th sec after start

(A) angular momentum is equal to 5τ

(B) kinetic energy is equal to $\frac{75\tau^2}{ma^2}$

(C) angular momentum is equal to 2.5τ

(D) kinetic energy is equal to $\frac{75\tau^2}{2ma^2}$

8. A particle of mass m start moving at $t = 0$ due to a force $F = F_0 \cos \omega t$, where F_0 and ω are constant. Particle will stop first time at $t = t_0$ and distance travel in this time interval is S , then

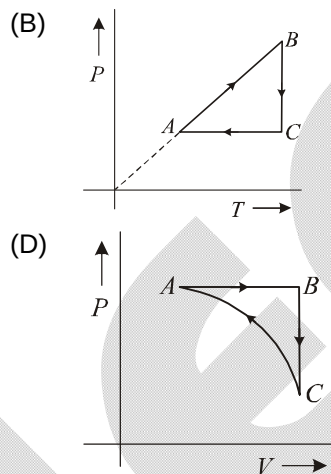
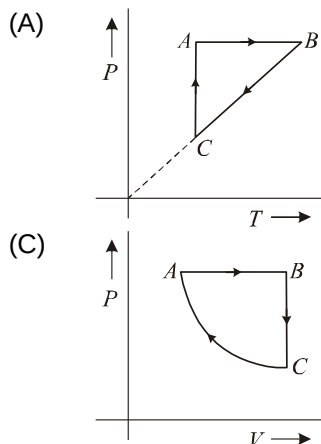
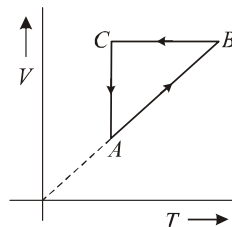
(A) $T_0 = \frac{\pi}{\omega}$

(B) $t_0 = \frac{\pi}{2\omega}$

(C) $S = \frac{2F_0}{m\omega^2}$

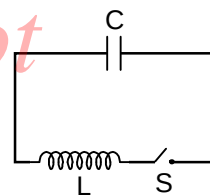
(D) $S = \frac{F_0}{m\omega^2}$

9. An ideal gas undergoes a thermodynamic cycle as shown in figure. Which of the following graphs represents the same cycle?



10. In the given circuit, initially the charge on capacitor is Q_0 . At $t = 0$, the switch S is closed. Which of the following statement(s) is/are correct?

- (A) Maximum current through the inductor is $\frac{Q_0}{\sqrt{LC}}$.
- (B) Charge on the capacitor is zero at $t = \frac{\pi}{2} \sqrt{LC}$.
- (C) Current through the inductor is maximum at $t = \frac{\pi}{2} \sqrt{LC}$.
- (D) Maximum magnetic energy can be stored in the inductor is $\frac{Q_0^2}{2C}$.

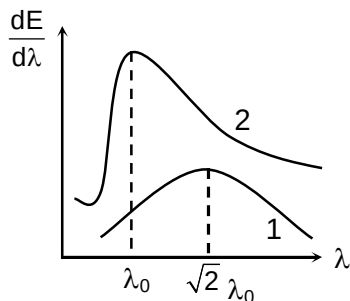


Section – B (Maximum Marks: 24)

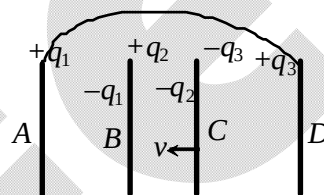
This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

11. A vertical hollow cylinder contains an ideal gas. The gas is enclosed by a 5 kg movable piston with area of cross-section $5 \times 10^{-3} \text{ m}^2$. The gas is slowly heated from 300 K to 350 K and the piston rises by 0.1 m. The piston is now clamped at this position and the gas is cooled back to 300 K. The difference between the heat energy added during heating process and heat energy lost during cooling process is $11x$. Find value of x . [Take $p_0 = 10^5 \text{ N/m}^2$]

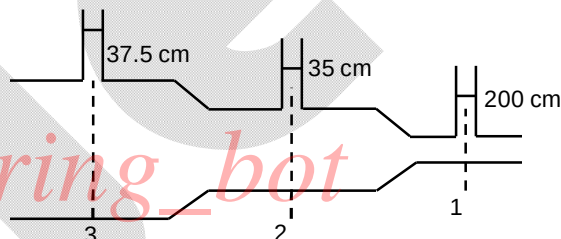
12. Graph 1 and 2 represent the spectral emissive power of a body at temperature T_1 and T_1 . The ratio of area under curve 2 and curve 1 is.



13. In the diagram there are four conducting plates A , B , C and D placed parallel to each other at equal separation L . If plate C starts moving towards plate B with velocity v . Find the current (in mA) flowing in the wire connecting A and D . (assume all other plates to be fixed)
(Given: $q_2 = 2\mu\text{C}$, $q_3 = 3\mu\text{C}$, $v = 30 \text{ m/s}$, $L = 0.05 \text{ m}$)

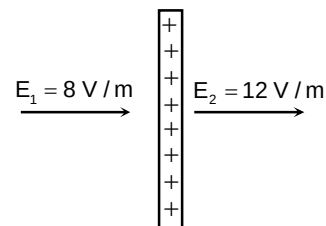


14. Water is flowing in varying cross-section pipe. The areas of cross-sections 1, 2 and 3 are 1 cm^2 , 2 cm^2 and $A \text{ cm}^2$, respectively. Water levels are shown in different vertical tubes of manometers. The speed of water at cross-section 3 is $\frac{1}{\sqrt{x}} \text{ m/s}$.



The value of x is

15. If the K_α radiation of Mo ($Z = 42$) has a wavelength of $0.71 \text{ \AA}$, calculate wavelength of the corresponding radiation of Cu. i.e. K_α for Cu ($Z = 29$). Find the value of nearest integer value (in $\text{\AA}$) of wavelength.
16. A particle is suspended by a light vertical inelastic string of length ℓ from a fixed support. At its equilibrium position, it is projected horizontally with a speed $\sqrt{6g\ell}$. What is the ratio of the tension in the string in its horizontal position to that in string, when the particle is vertically above the point of support?
17. The electric field on two sides of a charged plate is shown in the figure. If the charge density on the plate is $n\epsilon_0$, then n is



18. A hydrogen like atom (atomic number z) is in a higher excited state of quantum number n . The excited state by successively emitting two photons of energy 10.2 eV and 17.0 eV , respectively. Alternative, the atom from the same excited by successively emitting two photons of energies 4.25 eV and 5.95 eV , respectively. Determine the value of n . (Take, ionisation energy of H-atom = 13.6 eV)

Chemistry

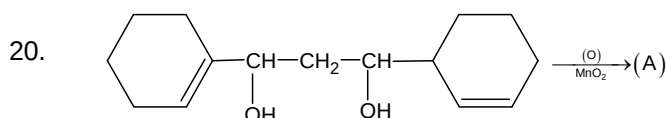
PART – II

Section – A (Maximum Marks: 12)

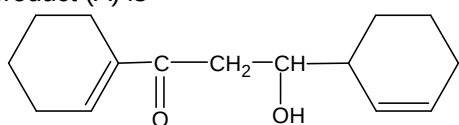
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

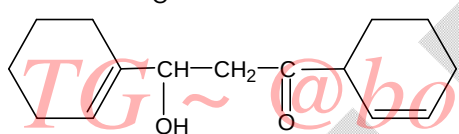
19. For $[\text{HgI}_3]^-$ ion geometry and magnetic behaviour is respectively:

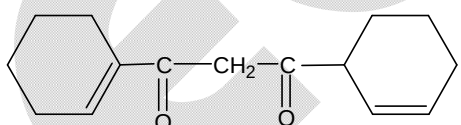
- (A) triangular bipyramideal, paramagnetic
- (B) triangular pyramidal, diamagnetic
- (C) triangular planar, paramagnetic
- (D) triangular planar, diamagnetic

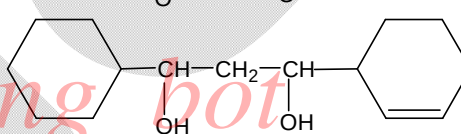


The product (A) is

- (A) 

(C) 

(B) 

(D) 

21. For most reactions, rate of a reaction increases about two times when its temperature is raised by 10 K. This fact is due to

- (A) increase in number of collisions per unit volume
- (B) increase in number of molecules having energy E_A or more
- (C) lowering of activation energy
- (D) lowering of pre-exponential factor in Arrhenius equation

22. At constant volume, for a fixed amount of a gas the pressure of the gas increase with rise of temperature due to

- (A) increase in average molecular speed
- (B) increased rate of collision amongst molecules
- (C) increase in molecular attraction
- (D) decrease in mean free path

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

23. Which of the following statements are correct?

- (A) Copper metal can reduce Fe^{2+} in acidic medium.
- (B) The absolute value of standard electrode potential can be determined experimentally
- (C) In a Daniell cell, electrons flow from zinc electrode to copper electrode outside the cell
- (D) Cell voltage is an intensive quantity.

24. Which of the following show the correct order of nucleophilicity?

- (A) $\text{CH}_3\text{S}^- > \text{CH}_3\text{O}^-$

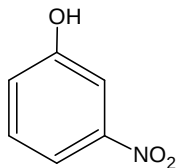
(C) $\text{CN}^- > \text{Cl}^-$

(B) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_3\text{P}$

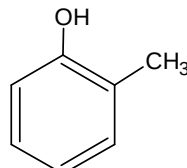
(D) $\text{OC}_2\text{H}_5 > \text{OH}$

25. Which of the following is/are more acidic than phenol?

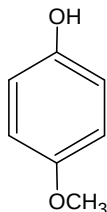
(A)



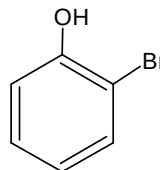
(B)



(C)



(D)



26. Select molecule(s) in which all the π -orbitals are in same plane.

(A) C_6H_6

(B) $O = C = C = C = O$

(C) $H_2C = C = CH_2$

(D) $CH_2 = CH - CH = CH_2$

27. In which of the following Pauli rule is violated?

(A) $1s^2, 2s^2, 2p^7$

(B) $1s^1, 2s^1, 2p^6$



$1s^2 \quad 2s^2 \quad 2p^6 \quad 1s^2 \quad 2s^2 \quad 2p^4$

28. Synergic bonding in metal carbonyl complexes [M = Metal]:

(A) decreases C – O bond strength

(B) increases M – C bond length

(C) increases bond order for M – C bond

(D) increases paramagnetism of complex

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

29. Find the total number of electron in Cr which have $(n \times l) = \text{zero}$

30. Total number of exchanges possible for $3d^4$ electronic configuration are?

31. Total number of bonds in PCl_5 which are at 90° to each other, are

32. Consider octahedral complex $K_2[Ni(EDTA)]$.

Total no. of 5 member rings in compound = X

Total number of geometrical isomers possible for compound = Y

Represent your answer as a total of X and Y.

33. Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The amount of Mohr's salt required per mole of dichromate is

34. The number of atoms in a single face-centred cubic unit cell is

35. Total no. of geometrical isomers possible for 1, 2, 4-tribromocyclohexane are?

36. Total no. of stereoisomers possible for 1, 2-dimethylcyclopropane are?

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

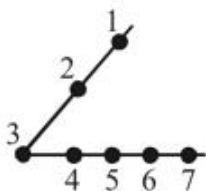
37. If the area bounded by $f(x) = \sqrt{1+x^2}$ ($x > 0$) the line $y = x$, y axis and $y = -x + a$ ($a > 1$) is k , then area bounded by the graph of $f^{-1}(x)$, the line $y = x$ between the lines $y = -x + 1$ and $y = -x + a$ is
- (A) $\frac{k-1}{4}$ (B) $\frac{2k-1}{2}$
 (C) k (D) $\frac{4k-1}{4}$
38. In a acute angle triangle ABC, the foot of the perpendicular from A divide the opposite side into two parts of length 3 and 17 units and $\tan A = \frac{22}{7}$, the $\left(\frac{\text{Area of } \triangle ABC}{22}\right)$ is
- (A) $\frac{5}{2}$ (B) 5
 (C) 10 (D) 20
39. The digits of three digit number N are in A.P. If sum of the digits is 15 and the number obtained by reversing the digits of the number is 594 less the original number, then $\frac{1000}{N-252}$ is equal to :
- (A) $\frac{5}{6}$ (B) $\frac{5}{3}$
 (C) 0.06 (D) 0.03
40. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar, is
- (A) $\sqrt{3}:1$ (B) $1:3$
 (C) $1:\sqrt{3}$ (D) $\sqrt{3}:2$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. Let $\vec{a}, \vec{b}, \vec{c}$ be noncoplanar vectors and $\vec{p} = \frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]}$, $\vec{q} = \frac{\vec{c} \times \vec{a}}{[\vec{b} \vec{c} \vec{a}]}$, $\vec{r} = \frac{\vec{a} \times \vec{b}}{[\vec{c} \vec{a} \vec{b}]}$ then
- (A) $\vec{p} \cdot \vec{a} = 1$ (B) $\vec{p} \cdot \vec{a} + \vec{q} \cdot \vec{b} + \vec{r} \cdot \vec{c} = 3$
 (C) $\vec{p} \cdot \vec{a} + \vec{q} \cdot \vec{b} + \vec{r} \cdot \vec{c} = 0$ (D) none of these

42. 7 points are given in a plane (as shown in adjacent figure), then using these 7 points –



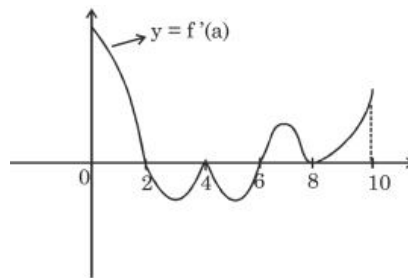
- (A) Number of lines that can be formed is 10.
 (B) Number of triangles that can be formed is 24.
 (C) Number of quadrilateral that can be formed is 6.
 (D) Maximum number of circles that can be formed is 24, where each circle contains atleast three of 7 points
43. The equations of four circles are $(x \pm a)^2 + (y \pm a)^2 = a^2$. The radius of a circle touching all the four circles is
 (A) $(\sqrt{2} - 1)a$
 (B) $2\sqrt{2}a$
 (C) $(\sqrt{2} + 1)a$
 (D) $(2 + \sqrt{2})a$
44. If $f(x)$ is a cubic polynomial such that $f(0) + 2 = 0$ and $f'''(0) = 6$ & $f(x) = 0$ posses three positive integral roots then
 (A) $f'(1) = 0$
 (B) $2f'(0) + f''(0) = 0$
 (C) $2f'(0) + f''(0) = 2$
 (D) $f'(2) = 0$
45. If $x + [y] + \{z\} = 1.1$
 $[x] + \{y\} + z = 2.2$
 $\{x\} + y + [z] = 3.3$ then
 (A) $x + y + z = 3.3$
 (B) $y - 2x = 1$
 (C) $2(z + 1) = 5y$
 (D) $\{x\} + \{y\} + \{z\} = 0.3$
46. If the sides a , b & c of a triangle ABC are in G.P. then $\angle B$ can never be
 (A) $\frac{\pi}{6}$
 (B) $\frac{5\pi}{6}$
 (C) $\frac{\pi}{4}$
 (D) $\frac{2\pi}{3}$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer**, ranging from 0 to 9 both inclusive.*

47. If $S = 3.\sqrt{9}.\sqrt{\sqrt{27}}.\sqrt{\sqrt{\sqrt{81}}}....\infty$, then sum of digits in S is
48. Let $f(x) = (1 + x) \ln(1 + x) - x - \frac{x^2}{4}$, $\forall x > -1$ and $f'(x) > 0 \forall x \in (0, \alpha)$, then value of $[\alpha]$ is (where denotes greatest integer function)

49. Let $f(x)$ be a derivable function such that $f(0) = 1$, $f(2) = 2$, $f(4) = 1$ and $f(6) = -4$. The graph of $y = f'(x)$ is given below. Then find the total number of points of local maximum/local minimum of $y = f(x)$ in $(0, 10)$



50. The projection length of a variable vector $x\hat{i} + y\hat{j} + z\hat{k}$ on the vector $\vec{p} = \hat{i} + 2\hat{j} + 3\hat{k}$ is 6. Let ℓ be the minimum projection length of the vector $x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ on the vector $\vec{p}$, then the value of $\sqrt[3]{\ell^2 + 15^2}$ is
51. Non-zero vectors $\vec{a}, \vec{b}, \vec{c}$ satisfy $\vec{a} \cdot \vec{b} = 0$, $(\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0$ and $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$. If $\vec{a} = \mu\vec{b} + 4\vec{c}$ then the value of μ is
52. 3 couples have to be seated around a circle. Let p be the probability that no couple is together then the value of $30p$ is
53. If $f(0) = 1$, $f(2) = 3$, $f'(2) = 5$ then the value of the definite integral $\int_0^1 x f''(2x) dx$ is
54. A special die is so constructed that the probabilities of throwing 1, 2, 3, 4, 5 and 6 are $(1 - k)/6$, $(1 + 2k)/6$, $(1 - k)/6$, $(1 + k)/6$, $(1 - 2k)/6$ and $(1 + k)/6$ respectively. If two such dice are thrown and the probability of getting a sum equal to 9 lies in $\left[\frac{1}{9}, \frac{2}{9}\right]$. Then find the number of integral solutions of k .

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – V
PAPER –1
TEST DATE: 08-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B, D
Sol. $\frac{1000}{S_1} = \frac{2000}{S_2} \Rightarrow S_2 = 2S_1$
 $S_1x + S_2y = (S_1 + S_2)40$
 $S_1x + 2S_1y = (S_1 + 2S_1)40$
 $x + 2y = 120$

2. A, B, C

- Sol. time constant of capacitor (τ) = $\frac{5RC}{8}$

Maximum charge on capacitor at steady state (q_0) = $5CV / 4$

Charge on capacitor at any time

$$q = \frac{5CV}{4} \left(1 - e^{-\frac{8t}{5RC}} \right) \dots (i)$$

Current in the branch of capacitor at any time

$$i_1 = \frac{2V}{R} e^{-\frac{8t}{5RC}}$$

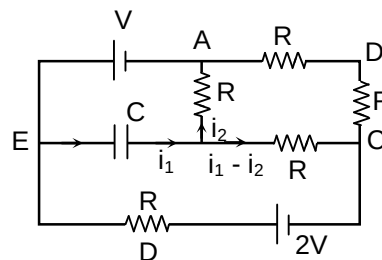
$$i_1 = 1.7 \text{ amp}$$

$$\text{Now, } V_A + i_2 R (i_1 - i_2) R = V_C$$

$$V_A - V_C = 2.52 \text{ volt}$$

3. C, D

- Sol. The force on the particle is not the central force.



4. A, B

Sol. For the disc (2m)

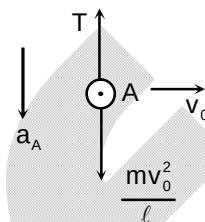
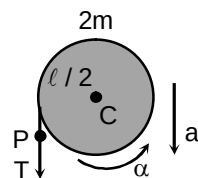
$$T = 2ma \quad \dots(i)$$

$$T \times \frac{\ell}{2} = I_C \alpha \quad \dots(ii)$$

For A (w.r.t. 'P')

$$T + m[a + R\alpha] = \frac{mv_0^2}{\ell}$$

$$\text{Initial R.O.C. of A} = \frac{V_0^2}{(T/m)}$$



5. A, B, C, D

$$\text{Sol. } a_x + a_y = a_1 \quad \dots(i)$$

$$a_y - a_x = a_2 \quad \dots(ii)$$

$$a_{cm(x)} = 0$$

$$2a_2 + a_1 + a_x \quad \dots(iii)$$

6. A, B

$$\text{Sol. } L = mv\theta r \quad \dots(i)$$

$$\frac{dL}{dt} = -\alpha v_0 r$$

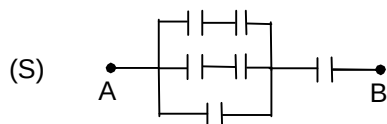
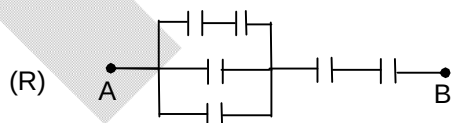
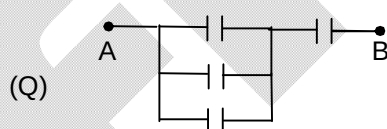
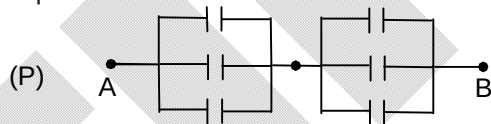
$$\Rightarrow \frac{dL}{dt} = -\frac{\alpha L}{m} \quad \text{from (i)}$$

$$\Rightarrow \int_{L_0}^L \frac{dL}{L} = -\frac{\alpha}{m} \int_0^t dt$$

$$L = L_0 e^{-\frac{\alpha t}{m}}$$

7. B

Sol. Eq. circuit of case P



8. A

Sol. (I) $C = C_v + C_v + \frac{RT}{v} \frac{dv}{dT} = C_v + \frac{Pdv}{dT}$ (for one mole gas)

$$\text{we have } p = P_0 + \frac{\alpha}{v}, \text{ or } \frac{RT}{v} = P_0 + \frac{\alpha}{v}$$

$$\text{or } RT = P_0 v + \alpha.$$

$$\text{Therefore } RdT = P_0 dv, \text{ so, } \frac{dv}{dt} = \frac{R}{P_0}$$

$$\text{Hence } C = C_v + \left(P_0 + \frac{\alpha}{v} \right) \frac{R}{P_0} = \frac{R}{\gamma-1} + \left(1 + \frac{\alpha}{P_0 v} \right) R$$

$$= \left(R + \frac{R}{\gamma-1} \right) + \frac{\alpha R}{P_0 v} = \frac{\gamma R}{\gamma-1} + \frac{\alpha R}{P_0 v}$$

(II) work done given by

$$W = \int_{v_1}^{v_2} \left(P_0 + \frac{\alpha}{v} \right) dv = P_0 (v_2 - v_1) + \alpha \ln \frac{v_2}{v_1}$$

$$\text{(III) } \Delta U = C_v (T_2 - T_1) = C_v \left(\frac{P_2 v_2 - P_1 v_1}{R} \right) = \frac{1}{\gamma-1} [p_2 v_2 - p_1 v_1]$$

$$= \frac{P_0 (v_2 - v_1)}{\gamma-1}$$

$$\text{(IV) } Q = DV + W = \frac{\gamma P_0 (v_2 - v_1)}{\gamma-1} + \alpha \ln \frac{v_2}{v_1}$$

9. D

Sol. (A) $(8-2t)\hat{j} - (8-2t)\hat{j}$

$$\text{(B) } S = \frac{8 \times 8}{2 \times 2} = 16$$

$$\text{(C) } v = 8i - 8j, |v| = 8\sqrt{2}$$

$$\text{(D) } 0 = 8t - \frac{1}{2} 2t^2, t = 8$$

10. B

Sol. If energy required by transition or ionization is Q then kinetic energy K required for colliding. Particle is

$$K = Q \left(1 + \frac{M_1}{M_2} \right)$$

$$\text{here } M_1 = M_2$$

$$\therefore K = 2Q$$

Minimum kinetic energy required for neutron to excite the atom = 2 × excitation energy.

Possible values of excitation energy = 10.2 eV, 12.1 eV,

Ionisation energy = 13.6 eV

Section – B

11. 2.00

Sol. $\frac{\delta F}{\delta x} = k_{\text{eff}}$ (numerically)

Let us find δx (static deformation of the system).

Since the particle is in equilibrium,

$$\delta F = 2\delta F_s \sin \theta$$

By Pythagoras theorem, $x^2 + y^2 = l^2$

Taking differentials of both sides, we have $2x\delta x + 2y\delta y = 2l\delta l$

Since y is constant, $\delta y = 0$

$$\delta l = \frac{x}{l} \delta x$$

$$\frac{\delta F}{\delta x} = 2k \sin^2 \theta$$

$$k_{\text{eff}} = 2k \sin^2 \theta$$

Substituting k_{eff} , we have $T = \frac{\pi}{\sin \theta} \sqrt{\frac{2m}{k}}$

12. 1.00

Sol. Velocity = $\omega \sqrt{A^2 - x^2}$
 $= \frac{2\sqrt{3}\pi A}{T}$ (doubles)

$$\frac{2\sqrt{3}\pi A}{T} = \frac{2\pi}{T} \sqrt{A^2 - x^2}$$

$$A_1 = \frac{\sqrt{13}}{2} A$$

$$\cos \theta = \frac{A/2}{A_1} = \frac{1}{\sqrt{13}} \quad \left(\theta = \cos^{-1} \left(\frac{1}{\sqrt{13}} \right) \right)$$

$$\cos^{-1} \left(\frac{1}{\sqrt{13}} \right) = \omega t$$

$$t = T \cos^{-1} \left(\frac{1}{\sqrt{13}} \right)$$

13. 1.00

Sol. According to given condition,

$$\frac{i \times 4}{i \times R_0} = \frac{AD}{DC} = \frac{40}{60} = \frac{2}{3}$$

$$\Rightarrow R = 12$$

Now, $R = R_0(1 + \alpha \Delta T)$

$$12 = 6(1 + \alpha 100)$$

$$\Rightarrow \alpha = 1 \times 10^{-2}$$

$$\text{So, } x = 1$$

14. 4.00

Sol. Consider an element of disc at a radius r and having a width dr .
Linearly velocity at this radius $= \omega r$

$$\text{Shear stress, } \tau = \mu \frac{du}{dy}$$

Assuming, the gap h to be small, so that the velocity distribution may be assumed linear

$$\tau = \mu \times \frac{v}{h} = \mu \frac{\omega r}{h}$$

$$\text{Viscous force, } dF = \tau \times \text{Area}$$

$$= \tau \times 2\pi r dr$$

Torque dT on the element,

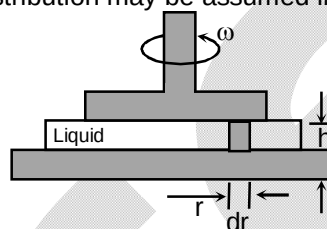
$$dT = dF \times r = \tau 2\pi r^2 dr$$

$$\text{or } dT = \frac{\mu \omega r}{h} \times 2\pi r^2 dr = \frac{2\pi \mu \omega r^3 dr}{h}$$

Total torque,

$$T = \int_0^{d/2} \frac{2\pi \mu \omega r^3 dr}{h} = \frac{\mu \pi d^4 \omega}{4h}$$

Thus, $x = 4$

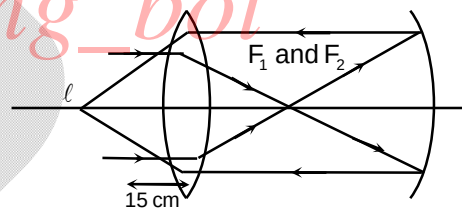


15. 4.00

Sol. The condition is possible only when the foci of lens and mirror coincide. Then, only the final image will be at 15 cm left to lens

$$\text{So, } x = 15 + 21 = 36 \text{ cm}$$

$$\text{Hence, } \frac{x}{9} = \frac{36}{9} = 4 \text{ cm}$$



16. 5.82

$$\text{Sol. } y_{\text{net}} = y_1 + y_2 + y_3$$

$$= a[\sin \omega t + \sin(\omega t + 45^\circ) + \sin(\omega t + 90^\circ)]$$

$$= (\sqrt{2} + 1)a \sin(\omega t + 45^\circ)$$

Energy ratio is

$$\frac{E_{\text{resultant}}}{E_{\text{ring}}} = \left(\frac{A}{a} \right)^2 = (\sqrt{2} + 1)^2$$

$$= (3 + 2\sqrt{2})$$

$$\therefore E_{\text{resultant}} = 5.82 \text{ J}$$

17. 1.50

Sol. Thermal stress $= Y\alpha\Delta\theta$
As, thermal stresses are equal

$$y_1 \alpha_1 \Delta\theta_1 = y_2 \alpha_2 \Delta\theta_2$$

$$\Rightarrow \frac{Y_1}{Y_2} = \frac{\alpha_2}{\alpha_1} = \frac{3}{2}$$

18. 2.00

Sol. Dimension of $B = [M^0 L^0 T^0]$

Dimension of $e = [AT]$

Dimension of $\epsilon_0 = [A^2 M^{-1} L^{-3} T^4]$

Dimension of $h = [ML^2 T^{-1}]$

Dimension of $c = [LT^{-1}]$

Equating,

$$[M^0 L^0 T^0] = \frac{[AT]^n}{[A^2 M^{-1} L^{-3} T^4][ML^2 T^{-1}][LT^{-1}]}$$

$$\Rightarrow [M^0 L^0 T^0] = [A^{n-2} L^0 T^{n-2}]$$

$$\therefore n - 2 = 0 \text{ or } n = 2$$

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Chemistry

PART – II

Section – A

19. B, D

Sol. Isotones have same number of neutrons

20. A, B, C, D

Sol. The $d_{A-A} = d_{B-B} = \sqrt{2} a/2$ and $d_{AB} = a/2$. There will be four molecules of AB in a unit cell. Hence. Its density will be given as $\rho = \frac{4 \times M_{AB}}{a^3 N_A}$.

21. B, D

Sol. RLVP = mole fraction of solute

Van't Hoff factor for CH_3COOH in benzene will be less than 1, because of association of CH_3COOH

22. A, B, C, D

Sol. Factual

23. A, B, C, D

Sol. Factual

24. B, D

Sol. (A) $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{\text{H}^+/\text{H}_2\text{O}} \text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$

(B) $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow[\text{cold}]{(\text{aq})\text{KMnO}_4} \text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \underset{\text{OH}}{\text{CH}_2}$

25. C

Sol. Factual

26. C

Sol. Factual

27. D

Sol. Factual

28. C

Sol. Factual

Section – B

29. 3.40

Sol. I.P = $R_H Z^2$, $13.6 = R_H$

$$\Delta E = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = 13.6 \left(\frac{1}{2^2} - \frac{1}{\infty^2} \right) = 13.6 \times \frac{1}{4} = 3.4$$

30. 1.80

Sol. $\rho = \frac{PM}{RT}$ or $M = \frac{\rho \times RT}{P} = \frac{1.5 \times R \times 273}{1}$
 $\rho = \frac{1000 \times 1.5 \times R \times 273}{760 \times R \times 300} = 1.8$

31. 181.14

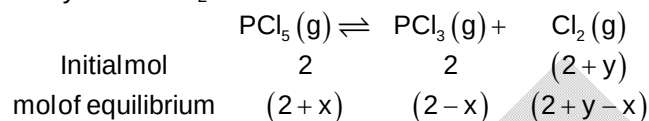
Sol. for bcc, $\sqrt{3} \times a = 2(r_+ + r_-)$
 $1.732 \times 387 = 2(154 + r_{Cl^-})$
 $r_{Cl^-} \approx 181.14$

32. 6.67

Sol. $P_{PCl_5} = P_{Cl_3} = P_{Cl_2} = \frac{2}{6} \times 3 = 1 \text{ atm}$

$$K_p = \frac{1 \times 1}{1} = 1$$

Let "y" mol of Cl_2 were added



Total mol = 12, because total volume is twice of initial volume

Mol of $PCl_5 + PCl_3 = 2 + x + 2 - x = 4$

Mol of $Cl_2 = 12 - 4 = 8$

$$PCl_5 = \frac{(2+x) \times 3}{12} = \frac{2+x}{4}, P_{PCl_3} = \frac{2-x}{4}, P_{Cl_2} = \frac{8 \times 3}{12} = 2$$

$$\frac{\left(\frac{2-x}{4}\right) \times 2}{\frac{2+x}{4}} = 1.0 \Rightarrow x = \frac{2}{3}$$

$$2 + y + \frac{2}{3} = 8$$

$$y = \frac{20}{3} = 6.67$$

33. 300.00



$$S = \left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}} = \left(\frac{1.35 \times 10^{-11}}{4}\right)^{\frac{1}{3}} = (3.375 \times 10^{-12})^{\frac{1}{3}} = 1.5 \times 10^{-4}$$

$$\Lambda_m = \frac{k \times 1000}{S}$$

$$\Lambda_m = \frac{4.5 \times 10^{-5} \times 10^3}{1.5 \times 10^{-4}} = 300 \text{ S cm}^2 \text{ mol}^{-1}$$

34. 138.50

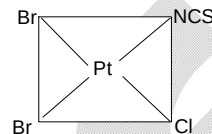
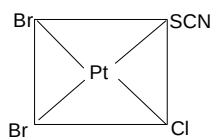
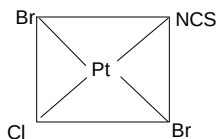
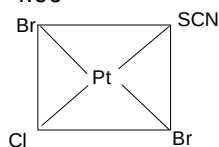
Sol. $X_{H_2O} = 1 - \frac{1}{7} = \frac{6}{7}$

$$\frac{X_{\text{KClO}_4}}{X_{\text{H}_2\text{O}}} = \frac{1}{7} \times \frac{7}{6} = \frac{1}{6} = \frac{n(\text{KClO}_4)}{n(\text{H}_2\text{O})}$$

It mean 6 mol H₂O (108g H₂O) contain 1 mol KClO₄ = 138.5g KClO₄

35. 6.10
Sol. Factual

36. 4.00
Sol.



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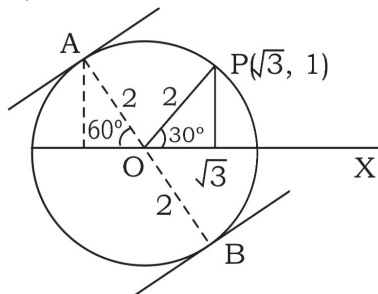
Mathematics

PART – III

Section – A

37. B, C

Sol.

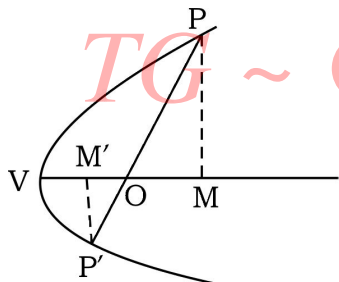


Clearly, $A = (-2 \cos 60^\circ, 2 \sin 60^\circ)$ and $B = (2 \cos 60^\circ, -2 \sin 60^\circ)$.

The tangent at A is $x(-2 \cos 60^\circ) + y(2 \sin 60^\circ) = 4$ and the tangent at B is $x(2 \cos 60^\circ) + y(-2 \sin 60^\circ) = 4$.

38. B

Sol.



Let the parabola be $y^2 = 4ax$ and $p = (at_1^2, 2at_1)$, $P' = (at_2^2, 2at_2)$

$$\therefore VM = at_1^2, VM' = at_2^2 \text{ and } VO = k, \text{ where } \begin{vmatrix} a_1^2 & 2at_1 & 1 \\ a_2^2 & 2at_2 & 1 \\ k & 0 & 1 \end{vmatrix} = 0$$

($\because p, p', O$ are collinear)

$$\Rightarrow k + at_1t_2 = 0$$

$$\therefore VM \cdot VM' = (at_1t_2)^2 = k^2 = OV^2.$$

39. B, C

$$\text{Sol. } \alpha = 2 \tan^{-1}(\sqrt{2} - 1) = 2 \tan^{-1} \tan \frac{\pi}{8} = 2 \times \frac{\pi}{8} = \frac{\pi}{4} = \cos^{-1} \frac{1}{\sqrt{2}}.$$

$$\beta = 3 \cdot \frac{\pi}{4} - \frac{\pi}{6} = \frac{7\pi}{12}.$$

$$\therefore \beta > \alpha. \text{ Also } \frac{1}{3} < \frac{1}{\sqrt{2}} \Rightarrow \cos^{-1} \frac{1}{3} > \cos^{-1} \frac{1}{\sqrt{2}} = \frac{\pi}{4}.$$

$$\text{So } \gamma > \alpha.$$

Again $\cos^{-1} \frac{1}{3}$ belongs to the first quadrant and β is in the second quadrant.

$$\therefore \beta > \gamma.$$

40. A, B, D

Sol. $\int_4^8 (|\sqrt{x-4}+2|+|\sqrt{x-4}-2|) dx = \int_4^8 4 dx = 16$

Similarly option A = B = D = 16

41. B, C, D

Sol. $\det(M-I) = \det(M-I)^T$
 $= \det(M^T - I)$
 $= \det(-M-I) = \begin{cases} \det(M+I), & \text{if } n \text{ is even} \\ -\det(M+I), & \text{if } n \text{ is odd} \end{cases}$

42. A, B

Sol. Since, $\vec{a}$ and $\vec{b}$ are unit vectors, we have

$$|\vec{a}-\vec{b}| = \sqrt{(\vec{a}-\vec{b})^2}$$

$$\therefore \sqrt{(\vec{a})^2 + (\vec{b})^2 - 2\vec{a} \cdot \vec{b}} = \sqrt{1+1-2\cos 2\theta} = 2|\sin \theta|$$

Therefore, $|\vec{a}-\vec{b}| < 1$

$$\Rightarrow 2|\sin \theta| < 1$$

$$|\sin \theta| < \frac{1}{2}$$

$$\Rightarrow \theta \in \left[0, \frac{\pi}{6}\right) \quad \text{or} \quad \left(\frac{5\pi}{6}, \pi\right]$$

43. B

Sol. I.

$$OA = 1 + 4 \cot \theta$$

$$OB = 4 + \tan \theta$$

$$OA + OB = 5 + 4 \cot \theta + \tan \theta \geq 5 + 2\sqrt{4 \cot \theta \tan \theta}$$

$$= 5 + (2 \times 2) = 9$$

II.

The reflection of $P(4, -1)$ on $y = x$ is $Q(-1, 4)$.

$$\text{Hence, } PQ = \sqrt{(4+1)^2 + (-1-4)^2} = \sqrt{50} = 5\sqrt{2}.$$

III. $AB = 2\sqrt{2}$

$$OC = \sqrt{2}$$

The maximum value of d is

$$OF = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$$

IV.

$$\text{The given line is } x = 4 + \frac{1}{\sqrt{2}} \left(\frac{y+1}{\sqrt{2}} \right) \text{ or } y = 2x - 9$$

Hence, the intercept made by the x -axis is $9/2$.

44. C

Sol. I. We have $\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} = \log_\pi 3 + \log_\pi 4 = \log_\pi 12$

But $\pi^2 < 12 < \pi^3$, we have $2 < \log_\pi 12 < 3$.

II. $3^a = 4$; $a = \log_3 4$

Similarly, $b = \log_4 5$ etc.

Hence, $abcde = \log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9 = \log_3 9 = 2$

III. We have to find characteristic of $\log_2 2008$.

We know that $\log_2 1024 = 10$ and $\log_2 2048 = 11$, therefore $10 < \log_2 2008 < 11$

Hence, it has characteristic 10.

IV. $\log_2 (\log_2 (\log_3 x)) = 0 \Rightarrow \log_2 (\log_3 x) = 1 \Rightarrow \log_3 x = 2$

$\Rightarrow x = 9$

Similarly, we have $\log_3 (\log_2 y) = 1$

$\Rightarrow \log_2 y = 3$ or $y = 8$

Therefore, $x - y = 1$.

45. A

Sol. I. $I = \lim_{x \rightarrow 0} \frac{e^{x^2} - 1 + x - e^x + 1}{2 \frac{\sin^2 x}{x^2} \cdot x^2} = \frac{1}{2} \left[\lim_{x \rightarrow 0} \frac{e^{x^2} - 1}{x^2} + \lim_{x \rightarrow 0} \frac{x - e^x + 1}{x^2} \right] = \frac{1}{2} \left[1 - \lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2} \right]$

II. $I = \lim_{x \rightarrow 0} \left(\frac{3+x}{3-x} \right)^{1/x} = e^{\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{3+x}{3-x} - 1 \right)} = e^{\lim_{x \rightarrow 0} \frac{2x}{x(3-x)}} = e^{2/3} \Rightarrow 2 + 3 = 5$

III. $\lim_{x \rightarrow 0} \frac{(\tan^3 x - x^3) - (\tan x^3 - x^3)}{x^5}$
 $= \lim_{x \rightarrow 0} \frac{\tan^3 x - x^3}{x^5} - \lim_{x \rightarrow 0} \frac{\tan^3 x - x^3}{x^5}$
zero (by expansion)
 $= \lim_{x \rightarrow 0} \frac{(\tan x - x)(\tan^2 x + x \tan x + x^2)}{x^3} = \frac{1}{3} \times 3 = 1$

IV. Rationalising given

$\lim_{x \rightarrow 0} \frac{(x + 2 \sin x) [\sqrt{(x^2 + 2 \sin x + 1)} + \sqrt{\sin^2 x - x + 1}]}{(x^2 + 2 \sin x + 1) - (\sin^2 x - x + 1)}$
 $= 2 \cdot \lim_{x \rightarrow 0} \frac{x + \sin 2x}{x^2 - \sin^2 x + 2 \sin x + x}$
 $= 2 \cdot \lim_{x \rightarrow 0} \frac{1 + \frac{\sin 2x}{x}}{x - \frac{\sin^2 x}{x} + 2 + 1} = 2 \left(\frac{1+2}{3} \right) = 2$

46. C

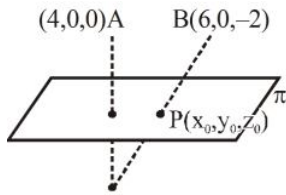
Sol. Let the plane be $\alpha x + \beta z + 1 = 0$

Pass through $(1, 0, 1)$; $(3, 2, -1)$

$\therefore \alpha = -\frac{1}{2}$; $\beta = -\frac{1}{2}$

(I) $\pi : x + z = 2$

(II)



Both A and B are on same side of π .

Reflection of A in plane is $A'(2, 0, -2)$

Equation of line $A'B = \vec{r} = 6\hat{i} - 2\hat{k} + \lambda(4\hat{i})$

$$\text{For } P : 6 + 4\lambda + 0 - 2 = 2 \Rightarrow \lambda = \frac{-1}{2}$$

$$\therefore P(4, 2, -2)$$

$$\therefore |4x_0 + y_0 + 2z_0| = 12$$

(III) Now $|PA - PB|_{\min} = 0$

$|PA - PB|_{\min}$ will approach

$$AB = \sqrt{4+0+4} = \sqrt{8}$$

$$\therefore |PA - PB| \in [0, \sqrt{8}).$$

$$\text{(IV) Also, } A' \text{ will lie on } \frac{x-2}{1} = \frac{y-\alpha}{0} = \frac{z+\beta}{-1}.$$

$$\Rightarrow \frac{2-2}{1} = \frac{0-\alpha}{0} = \frac{2+\beta}{-1}$$

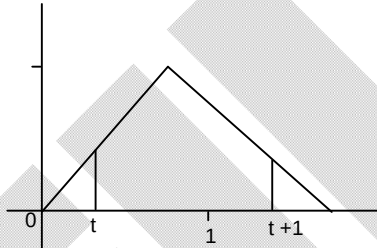
$$\Rightarrow \alpha = 0, \beta = 2$$

$$\therefore \alpha^4 + \beta^4 = 16$$

Section – B

47. 2.00

Sol. The region M is the large triangle in the diagram below, and the region N is the infinite strip between the lines $x = t$ and $x = t+1$ where $0 < t < 1$



Hence $M \cap N$ is the pentagon. Its area is given by

$$\frac{1}{2} \left(\sqrt{2}^2 - t^2 - (2 - (t+1))^2 \right) = -t^2 + t + \frac{1}{2}$$

48. 6.00

$$\text{Sol. } t^2 + 3t - 8 = 2 \Rightarrow t = 2, -5$$

$$2t^2 - 2t - 5 = -1 \Rightarrow t = 2, -1$$

So, at $t = 2$

$$\frac{dy}{dx} = \left(\frac{\frac{dy}{dt}}{\frac{dx}{dt}} \right)_{t=2} = \frac{6}{7}$$

49. 4320.00

Sol. a_3 is coefficient of x^3

$$\frac{10!}{8!} \times 3 \times 4 + \frac{10!}{7!3!} \times 3^3 = 4320$$

50. 1.00

Sol. Applying $C_1 \rightarrow C_1 - \sin \theta C_3$ and $C_2 \rightarrow C_2 + \cos \theta C_3$, we get

$$f(\theta) = \begin{vmatrix} 1 & 0 & -\sin \theta \\ 0 & 1 & \cos \theta \\ \sin \theta & -\cos \theta & 0 \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - \sin \theta R_1 + \cos \theta R_2$, we get

$$f(\theta) = \begin{vmatrix} 1 & 0 & -\sin \theta \\ 0 & 1 & \cos \theta \\ \sin \theta & 0 & \sin^2 \theta + \cos^2 \theta \end{vmatrix} = 1$$

Thus, $f\left(\frac{\pi}{6}\right) = 1$

51. 7.00

Sol. We have

$$(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$\Rightarrow 16 = 6 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$\therefore \alpha\beta + \beta\gamma + \gamma\alpha = 5$$

Also, $\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma = (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha)$

$$\Rightarrow 8 - 3\alpha\beta\gamma = 4(6 - 5)$$

$$\text{or } 3\alpha\beta\gamma = 4$$

$$\text{or } \alpha\beta\gamma = \frac{4}{3}$$

Now, $(\alpha^2 + \beta^2 + \gamma^2)^2 = \sum \alpha^4 + 2\sum \beta^2 \gamma^2$

$$= \sum \alpha^4 + 2\{(\sum \beta\gamma)^2 - 2\alpha\beta\gamma \sum \alpha\}$$

$$(6)^2 = \sum \alpha^4 + 2\left\{25 - 2 \cdot \frac{4}{3} \cdot 4\right\}$$

$$\sum \alpha^4 = 36 - 50 + \frac{64}{3} = \frac{64}{3} - 14 = \frac{22}{3}$$

$$\therefore [\alpha^4 + \beta^4 + \gamma^4] = \left[\frac{22}{3}\right] = 7$$

52. 0.00

Sol. Coefficient of x^{103} in $(1 + x + x^2 + x^3 + x^4)^{199} (x - 1)^{201}$

Coefficient of x^{103} in $-(1 - x)^2 (1 - x^5)^{199}$

Coefficient of x^{103} in $-(1 - 2x + x^2) (1 - {}^{199}C_1 x^5 + {}^{199}C_2 x^{10} - {}^{199}C_3 x^{15} + \dots)$

Coefficient of $x^{103} = 0$

53. 2022.00

Sol. For $x > 0$ not possible

For $x < 0$ put $x = -y$

$$\left(1 - \frac{1}{y}\right)^{1-y} = \left(\frac{y-1}{y}\right)^{1-y} = \left(\frac{y}{y-1}\right)^{1-y} = \left(\frac{2022}{2021}\right)^{2021}$$

$$\Rightarrow y = 2022$$

$$x = -2022$$

$$|x| = 2022$$

54. 0.25

$$\begin{aligned}\text{Sol. } P(B | (A \cup \bar{B})) &= \frac{P(B \cap (A \cup \bar{B}))}{P(A \cup \bar{B})} = \frac{P(A) - P(A \cap \bar{B})}{P(A) + P(\bar{B}) - P(A \cap \bar{B})} \\ &= \frac{0.7 - 0.5}{0.7 + 0.6 - 0.5} = \frac{0.2}{0.8} = \frac{1}{4}\end{aligned}$$

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FIITJEE
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JEE (Advanced)-2023
FULL TEST – V
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ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1.

Sol.

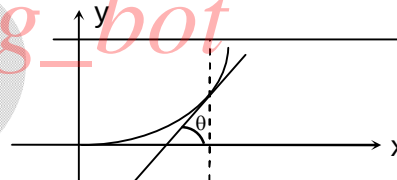
D

Equation of circle is

$$(x-0)^2 + (y-R)^2 = R^2 \quad \dots(1)$$

$$\frac{dy}{dx} = \tan \theta \quad \dots(2)$$

$$\text{By Snell's law } \sin(90 - \theta) \times \eta = 1 \times \eta_0 \quad \dots(3)$$



2.

Sol.

D

$$F = -\left(\frac{Ay}{L}\right) \Delta L = K \Delta L$$

$$\Rightarrow K = \frac{Ay}{L}$$

$$T = 2\pi \sqrt{\frac{m}{K}}$$

3.

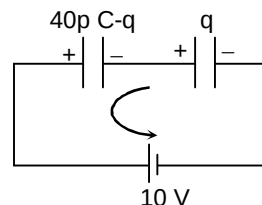
Sol.

A

Apply KVL

$$\frac{40-q}{2} - 10 - \frac{q}{2} = 0$$

Solve for q



4.

Sol.

B

$$\text{Velocity, } v = \sqrt{\frac{YRT}{M}}$$

Find Y

5. B, C, D

Sol. $U = U_0(1 - \cos ax)$, $F = \frac{-dU}{dx} = -aU_0 \sin ax$

$$m \frac{d^2 x}{dt^2} = -a^2 U_0 x \quad (\text{For small oscillations})$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{a^2 U_0}}$$

Potential energy is maximum at $x = \frac{\pi}{a}$

amplitude is $\frac{\pi}{a}$

Speed of particle is maximum at $x = 0$ ($\because$ PE is minimum)

6. A, C

Sol. (i) for $\theta = 30^\circ$

$$\lambda = \frac{c}{v} = \frac{3 \times 10^8}{10^6} = 300 \text{ m}, d = 150 \text{ m}$$

$$\therefore \delta = \frac{2\pi}{\lambda} (d \sin \theta) = \frac{2\pi}{(300)} (150) \left(\frac{1}{2}\right) = \frac{\pi}{2}$$

$$I(\theta) = I_0 \cos^2 \left(\frac{\delta}{2}\right) = I_0 \cos^2 \left(\frac{\pi}{4}\right) = \frac{I_0}{2}$$

(ii) for $\theta = 90^\circ \Rightarrow \delta = \left(\frac{2\pi}{300}\right) (150)(1) = \pi$

$$\therefore I(\theta) = I_0 \cos^2 \left(\frac{\delta}{2}\right) = I_0 \cos^2 \left(\frac{\pi}{2}\right) = 0$$

(iii) for $\theta = 0^\circ$, $\delta = 0$

$$\therefore I(\theta) = I_0$$

7. A, D

Sol. Change in angular momentum = angular impulse

$$L = 5\tau$$

$$I = \frac{ma^2}{3}$$

$$K = \frac{L^2}{2I} = \frac{75\tau^2}{2ma^2}$$

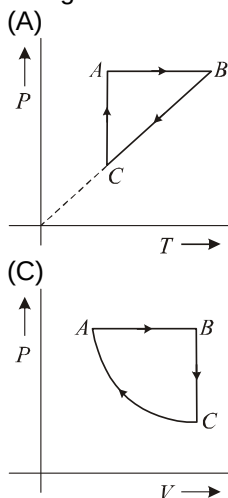
8. A, C

Sol. $a = \frac{F_0}{m} \cos \omega t$, $\int_0^{t_0} dv = \frac{F_0}{m} \int_0^{t_0} \cos \omega t dt$, $t_0 = \frac{\pi}{\omega}$

and $\int_0^v dv = \frac{F_0}{m} \int_0^t \cos \omega t dt$, $v = \frac{F_0}{m\omega} \sin \omega t$, $\int_0^s ds = \frac{F_0}{m\omega} \int_0^{t_0} \sin \omega t dt$

$$S = \frac{F_0}{m\omega^2} [-\cos \omega t]_0^{t_0} = \frac{F_0}{m\omega^2} [2] = \frac{2F_0}{m\omega^2}$$

9. A, C
Sol. Using $PV = nRT$



10. A, B, C, D
Sol. Charge on capacitor at time t is
 $q = Q_0 \cos \omega t$

Where $\omega = \frac{1}{\sqrt{LC}}$.

At $\omega t = \frac{\pi}{2}, q = 0$

$\Rightarrow t = \frac{\pi}{2} \sqrt{LC}$

By conservation of energy

$$\frac{Q_0^2}{2C} = \frac{1}{2} L i_{\max}^2$$

$\Rightarrow i_{\max} = \frac{Q_0}{\sqrt{LC}}$

Section – B

11. 5
Sol. $(\Delta Q)_p = nC_p \Delta T$

When piston is clamped; heat is rejected and volume becomes constant

$$(\Delta Q)_v = nC_v \Delta T$$

$$\text{Difference} = nC_p \Delta T - nC_v \Delta T = \Delta W = P \Delta V$$

$$P = P_0 + \frac{Mg}{A} \text{ and } \Delta V = (A \times 0.1)$$

$$\Rightarrow (\Delta Q)_p - (\Delta Q)_v = \left(P_0 + \frac{Mg}{A} \right) \times A \times 0.1$$

$$= 55 \text{ J}$$

$$\Rightarrow x = 5$$

12. 4

Sol. $\sqrt{2} \lambda_0 T_1 = \lambda_0 T_2$

$$\frac{T_2}{T_1} = \sqrt{2} \quad ; \quad \frac{A_2}{A_1} = \left(\frac{T_2}{T_1} \right)^4 = 4$$

13. 1

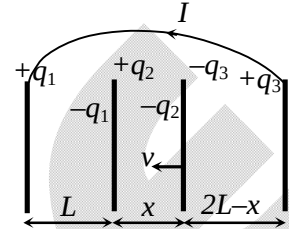
Sol. $\frac{2q_1 L}{2A\epsilon_0} + \frac{2q_2 x}{2A\epsilon_0} - \frac{2q_3(2L-x)}{2A\epsilon_0} = 0 \quad \dots (i)$

$$I = \frac{dq_1}{dt} = \frac{-dq_3}{dt} = \frac{dq_2}{dt} \quad \dots (ii)$$

From (i), $q_1 L + q_2 x - 2q_3 L + q_3 x = 0$

$$\Rightarrow L \frac{dq_1}{dt} + x \frac{dq_2}{dt} + q_2 \frac{dx}{dt} - 2 \frac{dq_3}{dt} L + x \frac{dq_3}{dt} + q_3 \frac{dx}{dt} = 0$$

$$\Rightarrow I = \frac{v}{3L} (q_2 + q_3) = 1 \text{ Ma}$$



14. 2

Sol. Apply Bernoulli's equations at pipe (1), (2) and (3)

$$p_3 + \frac{1}{2} \rho v_3^2 = p_2 + \frac{1}{2} \rho v_2^2 = p_1 + \frac{1}{2} \rho v_1^2$$

$$p_0 + \rho gh_3 + \frac{1}{2} \rho v_3^2 = p_0 + \rho gh_2 + \frac{1}{2} \rho v_2^2 = p_0 + \rho gh_1 + \frac{1}{2} \rho v_1^2$$

$$gh_3 + \frac{v_3^2}{2} = gh_2 + \frac{v_2^2}{2} = gh_1 + \frac{v_1^2}{2} \quad \dots (i)$$

$$v_3 A_3 = v_2 A_2$$

$$v_3 \times A = v_2 \times 2 \quad \dots (ii)$$

$$v_2 \times 2 = v_1 \times 1 \quad \dots (iii)$$

From Equation (i), we get

$$gh_2 + \frac{v_2^2}{2} = gh_1 + \frac{v_1^2}{2}$$

$$10 \times \frac{35}{100} + \frac{v_2^2}{2} = 10 \times \frac{20}{100} + \frac{v_1^2}{2}$$

$$\frac{15}{10} = \frac{v_1^2 - v_2^2}{2}$$

$$= \frac{4v_2^2 - v_2^2}{2}$$

$$= \frac{3v_2^2}{2}$$

$$v_2 = 1 \text{ m/s}, v_1 = 2 \text{ m/s}$$

From equation (i), we get

$$gh_3 + \frac{v_3^2}{2} = gh_2 + \frac{v_2^2}{2}$$

$$= 10 \times \frac{37.5}{100} + \frac{v_3^2}{2}$$

$$= 10 \times \frac{35}{100} + \frac{1}{2}$$

$$\frac{v_3^2}{2} = \frac{1}{2} + \frac{35}{10} - \frac{37.5}{10} = \frac{1}{2} - \frac{2.5}{10} = \frac{1}{2} - \frac{25}{100}$$

$$\frac{v_3^2}{2} = \frac{1}{4} \Rightarrow v_3^2 = \frac{1}{2}$$

$$\Rightarrow v_3 = \frac{1}{\sqrt{2}}, X = 2$$

15. 2

Sol.

$$\sqrt{v} = a(Z - b)$$

$$\sqrt{v} = a(Z - 1)$$

$$\Rightarrow (Z - 1)^2 \propto v$$

$$\text{or } (Z - 1)^2 \propto \frac{1}{\lambda}$$

$$\Rightarrow \frac{(Z_{\text{Mo}} - 1)^2}{(Z_{\text{Cu}} - 1)^2} = \frac{\lambda_{\text{Cu}}}{\lambda_{\text{Mo}}}$$

$$\Rightarrow \lambda_{\text{Cu}} = \lambda_{\text{Mo}} \frac{(Z_{\text{Mo}} - 1)^2}{(Z_{\text{Cu}} - 1)^2}$$

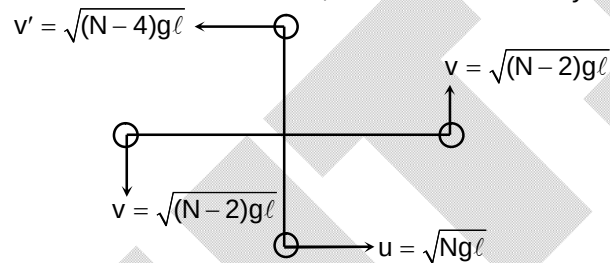
$$= 0.71 \times \left(\frac{41}{28} \right)^2 = 1.52 \text{ \AA}$$

$$\lambda = 2 \text{ \AA}$$

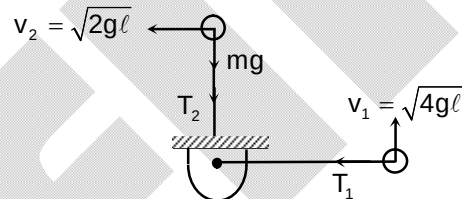
16. 4

Sol.

For vertical circular motion, student should always remember that



For given, $u = \sqrt{6g\ell}$



At horizontal position,

$$T_1 = \frac{mv_1^2}{\ell} = 4mg \quad \dots(i)$$

At highest point,

$$T_2 + mg = \frac{mv_2^2}{\ell} = 2mg$$

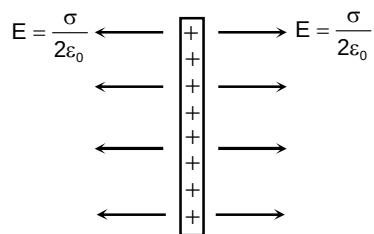
$$T_2 = mg \quad \dots(ii)$$

From above equation, we get

$$\frac{T_1}{T_2} = 4$$

17. 4

Sol. The electric field produced by thin charged sheet is given by $\frac{\sigma}{2\epsilon_0}$.



[Where, σ = surface charge density]

$$E_0 + E = 12 \text{ V/m} \quad \dots(i)$$

$$E_0 - E = 8 \text{ V/m} \quad \dots(ii)$$

Solving Equations (i) and (ii), we get

$$E = 2 \text{ V/m}$$

$$\text{Now, electric field due to plate} = \frac{\sigma}{2\epsilon_0} = 2$$

$$s = 4e_0$$

$$\therefore n = 4$$

18. 6

Sol. From the given conditions, we get

$$\begin{aligned} E_n - E_2 &= (10.2 + 17) \text{ eV} \\ &= 27.2 \text{ eV} \quad \dots(i) \end{aligned}$$

$$\begin{aligned} E_n - E_3 &= (4.25 + 5.95) \text{ eV} \\ &= 10.2 \text{ eV} \quad \dots(ii) \end{aligned}$$

Subtracting Equation (ii) from Eq. (i), we get

$$E_3 - E_2 = 17.0 \text{ eV}$$

$$Z^2(13.6) \left[\frac{1}{4} - \frac{1}{9} \right] = 17.0$$

$$Z^2(13.6) \left(\frac{5}{36} \right) = 17.0$$

$$Z^2 = 9 \text{ or } Z = 3$$

$$Z^2(13.6) \left[\frac{1}{4} - \frac{1}{n^2} \right] = 27.2$$

$$3^2(13.6) \left[\frac{1}{4} - \frac{1}{n^2} \right] = 27.2$$

$$\frac{1}{4} - \frac{1}{n^2} = 0.222$$

$$\Rightarrow \frac{1}{n^2} = 0.0278 \Rightarrow n^2 = 36 \Rightarrow n = 6$$

Chemistry

PART – II

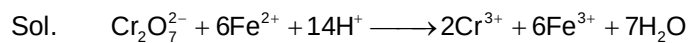
Section – A

19. D
Sol. sp^2 hybridization of Hg and Hg^{2+} has d^{10} configuration
20. A
Sol. MnO_2 oxidise allylic alcohols
21. B
Sol. Factual
22. A
Sol. Factual
23. C, D
Sol. Factual
24. A, C, D
Sol. Down the group nucleophilicity increases
25. A, D
Sol. $-NO_2$ and $-Br$ increases acidic character of phenol by dispersing $-ve$ charge of phenoxide ion – OCH_3 and $-CH_3$ decreases acidic character by intensification of $-ve$ charge of phenoxide ion.
26. A, D
Sol. Factual
27. A, C
Sol. p-subshell can have maximum 6 electron and an orbital can have maximum 2 electron with opposite spin.
28. A, C
Sol. Factual

Section – B

29. 7
Sol. for s-subshell, $l = 0$, hence for all electrons present in s-orbitals, $n \times l = 0$
 $Cr = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5, 4s^1$
In Cr there are 7 electrons present in s-orbitals
30. 6
Sol. Factual
31. 6
Sol. Triangular bipyramidal molecules have six, 90° angles
32. 5
Sol. edta make five rings of 5 member, $x = 5$ and octahedral $[M(edta)]$ complex can not show geometrical isomerism, $y = 0$

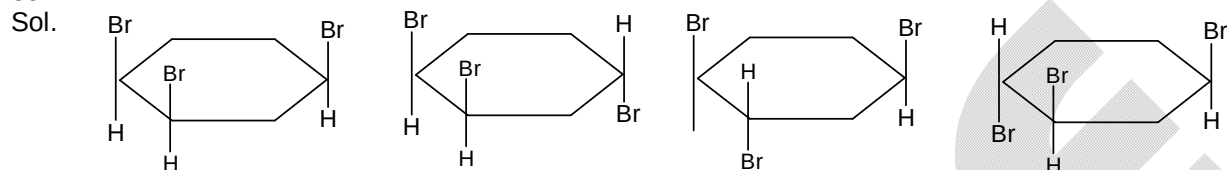
33. 6



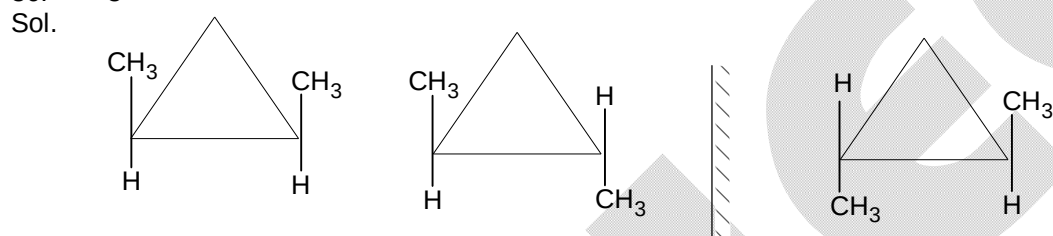
34. 4

Sol. $Z = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$

35. 4



36. 3



TG ~ @bohrring_bot

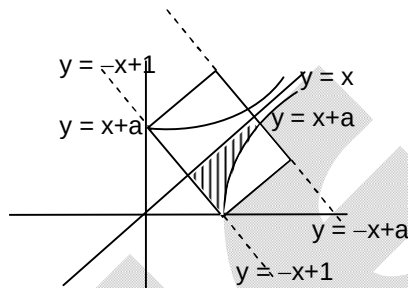
Mathematics

PART – III

Section – A

37. D

Sol. Required area of shaded region is $k - \frac{1}{4}$

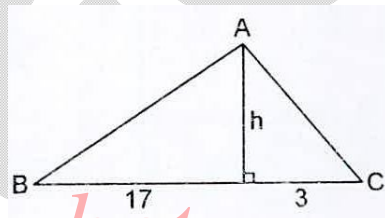


38. B

Sol. $B + C = \pi - A$
 $\tan(B + C) = -\tan A$

$$\frac{h\left(\frac{1}{17} + \frac{1}{3}\right)}{1 - \frac{h^2}{51}} = -\frac{22}{7} \Rightarrow \left(\frac{20h}{51 - h^2}\right) = -\frac{22}{7}$$

Area of triangle ABC $\Delta = -\frac{1}{2} \times 20 \times 11 = 110$



39. B

Sol. Let three digits of N be :
 Hundred's digit = $a - d$, ten's digit = a , and unit's digit = $a + d$

$$(a - d) + a + (a + d) = 15 \Rightarrow a = 5$$

$$N = 100(5 - d) + 50 + (5 + d) \quad \dots(1)$$

Number obtained by reversing digits of N is

$$N_1 = 100(5 + d) + 50 + (5 - d) \quad \dots(2)$$

$$\text{Now, } 594 = N - N_1 = 10(-2d) + 2d$$

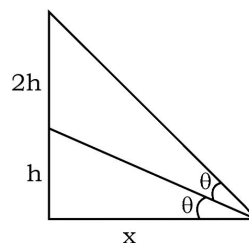
$$\Rightarrow 2d = -6 \Rightarrow d = -3$$

$$\text{Thus, } N = 852 \text{ and } \frac{1000}{N - 252} = \frac{1000}{600} = \frac{5}{3}$$

40. C

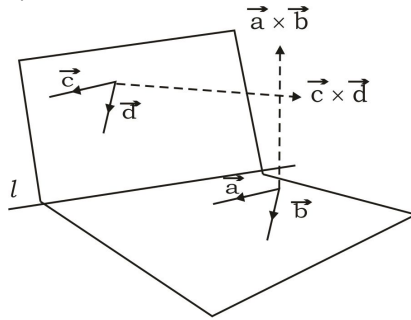
Sol. Let $\frac{h}{x} = a$. Then $\tan \theta = a$, $\tan 2\theta = 3a$.

$$\therefore \frac{2a}{1 - a^2} = 3a \Rightarrow a = \frac{1}{\sqrt{3}}$$



41. A, B

Sol.



$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \text{a vector perpendicular to } \vec{a} \times \vec{b} \text{ and } \vec{c} \times \vec{d}$$

If l is the line of intersection of the two planes of $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ then $l \perp \vec{a} \times \vec{b}$ and $l \perp \vec{c} \times \vec{d}$. So, $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ is along l which is equally inclined to $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$

42. A, B, C, D

Sol.

(A) ${}^7C_2 - {}^3C_2 - {}^5C_2 + 2 = 10$

(B) ${}^7C_3 - {}^3C_3 - {}^5C_3 = 24$

(C) There is no use of point '3'

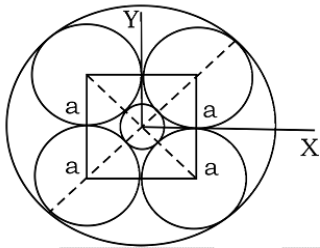
${}^6C_4 - {}^4C_4 - {}^4C_3 \cdot {}^2C_2 = 6$

(D) ${}^7C_3 - {}^3C_3 - {}^5C_3 = 24$

43. A, C

Sol.

The four circles are as shown in the figure. The smallest circle touching all of them has the radius $= \sqrt{2}a - a$ and the greatest circle touching all of them has the radius $= \sqrt{2}a + a$.



44. A, C

Sol.

Consider $f(x) = x^3 + ax^2 + bx - 2$

$\alpha \beta \gamma = 2$

$\alpha, \beta, \gamma \in \mathbb{N} \Rightarrow$ two of them are 1 & other is 2

$\Rightarrow a = -(\alpha + \beta + \gamma) = -4 = \frac{f''(0)}{2}$

$b = (\alpha\beta + \beta\gamma + \alpha\gamma) = 5 = f'(0)$

Hence $f(x) = x^3 - 4x^2 + 5x - 2$

Again $\therefore 1$ is a repeated root of $f(x) = 0$

$\Rightarrow f'(1) = 0$

45. A, B, C, D

Sol.

$x + [y] + \{z\} = 1.1$

...(1)

$$[x] + \{y\} + z = 2.2 \quad \dots(2)$$

$$\{x\} + y + [z] = 3.3 \quad \dots(3)$$

$$(1) + (2) + (3)$$

$$\Rightarrow 2(x + y + z) = 6.6$$

$$\Rightarrow x + y + z = 3.3 \quad \dots(4)$$

$$(4) - (1)$$

$$\{y\} + [z] = 2.2 \Rightarrow \{y\} = 0.2 \text{ \& } [z] = 2$$

$$(4) - (2)$$

$$\{x\} + [y] = 1.1 \Rightarrow \{x\} = 0.1, [y] = 1$$

$$(4) - (3)$$

$$[x] + \{z\} = 0 \Rightarrow [x] = 0 \text{ \& } \{z\} = 0$$

$$\begin{cases} x = 0.1 \\ y = 1.2 \\ z = 2 \end{cases}$$

46. B, D

Sol. $b^2 = ac$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{a^2 + c^2 - b^2}{2b^2}$$

$$\therefore \cos B = \frac{a^2 + c^2}{2ac} - \frac{1}{2}$$

$$\Rightarrow \cos B + \frac{1}{2} = \frac{a^2 + c^2}{2ac} \Rightarrow \cos B + \frac{1}{2} \geq 1 \Rightarrow \cos B \geq \frac{1}{2}$$

$$B \in (0, \pi/3)$$

Section – B

47. 9

Sol. $S = 3 \cdot 3^{2/2} \cdot 3^{3/4} \cdot 3^{4/8} \dots \infty$

$$S = 3^{1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \dots \infty}$$

$$S' = 1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \dots \infty \quad \dots(1)$$

$$\frac{S'}{2} = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \dots \infty \quad \dots(2)$$

Subtract (2) from (1)

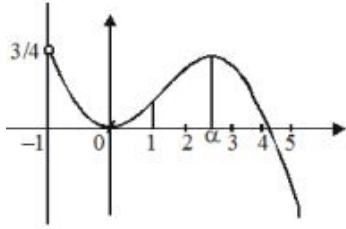
$$\frac{S'}{2} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \infty$$

$$S' =$$

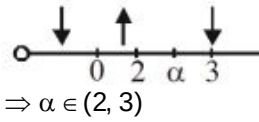
$$S = 3^4 = 81$$

48. 2

Sol. $f(x) = (1+x) \ln(1+x) - x - \frac{x^2}{4}$

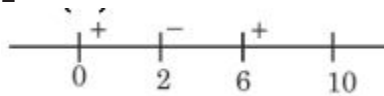


$$\lim_{x \rightarrow -1} f(x) = 0 + 1 - \frac{1}{4} = \frac{3}{4} \text{ and } f(0) = 0$$



49. 2

Sol.



50. 9

Sol.

$$\text{Projection length} = |\vec{a} \cdot \vec{p}|$$

$$\text{So, } \frac{|x + 2y + 3z|}{\sqrt{14}} = 6$$

$$\Rightarrow |x + 2y + 3z| = 6\sqrt{14}$$

$$\Rightarrow |(\hat{x}\hat{i} + \sqrt{2}\hat{y}\hat{j} + \sqrt{3}\hat{z}\hat{k}) \cdot (\hat{i} + \sqrt{2}\hat{j} + \sqrt{3}\hat{k})| = 6\sqrt{14}$$

$$\Rightarrow (x^2 + 2y^2 + 3z^2)(1 + 2 + 3) \cos^2 \theta = (6\sqrt{14})^2$$

$$\Rightarrow \frac{x^2 + 2y^2 + 3z^2}{\sqrt{14}} \geq 6\sqrt{14} \Rightarrow l = 6\sqrt{14}$$

$$\text{So, } (l^2 + 15^2)^{1/3} = (504 + 225)^{1/3} = (729)^{1/3} = 9.$$

51. 0

Sol.

$$\vec{c} = \frac{\vec{a} - \mu\vec{b}}{4} \text{ and } \vec{a} \cdot \vec{b} = 0$$

$$\text{Now, } (\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0 \Rightarrow (\vec{b} - \vec{a}) \cdot \left(\vec{b} + \frac{\vec{a} - \mu\vec{b}}{4} \right) = 0$$

$$\Rightarrow (4 - \mu)b^2 = a^2 \quad (\because \mu < 4) \dots (i)$$

$$\text{Again } 4|\vec{b} + \vec{c}|^2 = |\vec{b} - \vec{a}|^2 \Rightarrow 4 \left| \frac{(4 - \mu)\vec{b} + \vec{a}}{4} \right|^2 = |\vec{b} - \vec{a}|^2$$

$$\Rightarrow 4 \left(\frac{4 - \mu}{4} \right)^2 b^2 + \frac{a^2}{4} = b^2 + a^2 \Rightarrow ((4 - \mu)^2 - 4)b^2 = 3a^2 \dots (ii)$$

$$(i) \& (ii) \text{ we get } \frac{(4 - \mu)^2 - 4}{4 - \mu} = \frac{3}{1} \Rightarrow \mu^2 - 5\mu = 0$$

$$\Rightarrow \mu = 0 \text{ or } 5 \text{ but as } \mu < 4, \text{ so, } \mu = 0.$$

52. 8

Sol.
$$p = \frac{5! - {}^3C_1(4!)(2!) + {}^3C_2(3!)(2!)(2!) - {}^3C_3(2!)(2!)(2!)}{5!} = \frac{120 - 144 + 72 - 16}{120} = \frac{4}{15}$$

So $30p = 8$

53. 2

Sol.
$$\int_1^2 x f''(2x) dx = x \frac{f'(2x)}{2} - \int \frac{f'(2x)}{2} dx = x \frac{f'(2x)}{2} - \frac{f(2x)}{4}$$

$$\int_0^1 x f''(2x) dx = \left[\frac{xf'(2x)}{2} - \frac{f(2x)}{4} \right]_0^1 = \left(\frac{f'(2)}{2} - \frac{f(2)}{4} \right) - \left(0 - \frac{f(0)}{4} \right)$$

$$\left(\frac{5}{2} - \frac{3}{4} \right) - \left(0 - \frac{1}{4} \right) = 2$$

54. 1

Sol. Let E_1, E_2, E_3, E_4, E_5 and E_6 be the events of occurrence of 1, 2, 3, 4, 5 and 6 on the dice respectively, and let E be the event

$$\therefore P(E_1) = \frac{1-k}{6}; P(E_2) = \frac{1+2k}{6}; P(E_3) = \frac{1-k}{6}$$

$$P(E_4) = \frac{1+k}{6}; P(E_5) = \frac{1-2k}{6}; P(E_6) = \frac{1+k}{6} \text{ and } \frac{1}{9} \leq P(E) \leq \frac{2}{9}$$

Then, $E \equiv \{(3, 6), (6, 3), (4, 5), (5, 4)\}$

$$\begin{aligned} \text{Hence, } P(E) &= P(E_3 E_6) + P(E_6 E_3) + P(E_4 E_5) + P(E_5 E_4) \\ &= P(E_3)P(E_6) + P(E_6)P(E_3) + P(E_4)P(E_5) + P(E_5)P(E_4) \\ &= 2P(E_3)P(E_6) + 2P(E_4)P(E_5) \end{aligned}$$

{Since E_1, E_2, E_3, E_4, E_5 and E_6 are independent}

$$\begin{aligned} &= 2 \left(\frac{1-k}{6} \right) \left(\frac{1+k}{6} \right) + 2 \left(\frac{1+k}{6} \right) \left(\frac{1-2k}{6} \right) \\ &= \frac{1}{18} [2 - k - 3k^2] \end{aligned}$$

Since, $\frac{1}{9} \leq P(E) \leq \frac{2}{9}$

$$\therefore -\frac{1}{3} \leq k \leq 0$$

$\therefore$ Set of integral value of $k = \{0\}$

$\therefore$ Number of integral solution of k is 1